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CHAPTER 1

# Introduction to the Kontinua Sequence

The purpose of this book is to help you along the long and difficult trek to becoming a modern problem solver. As you explore this path, you will learn how to use the tools of math, computers, and science.

If this path is so arduous, it is only fair to ask why you should bother in the first place. There are big problems out there that will require expert problem solvers. Those people will make the world a better place, while also enjoying interesting and lucrative careers. We are talking about engineers, scientists, doctors, computer programmers, architects, actuaries, and mathematicians. Right now, those occupations represent about 6% of all the jobs in the United States. Soon, that number is expected to rise above 10%. On average, people in that 10% of the population are expected to have salaries twice that of their non-technical counterparts.

Solving problems is difficult. At some point on this journey, you will see people who are better at solving problems than you are. You, like every other person who has gone on this journey, may think "I have worked so hard on this, but that person is better at it than I am. I should quit." *Don't*.

Instead, remember these two important facts. First, solving problems is like a muscle. The more you do, the better you get at it. It is OK to say "I am not good at this yet." That just means you need more practice.

Second, you don't need to be the best in the world. 10 million people your age can be better at solving problems than you, *and you can still be in the top 10% of the world*. If you complete this journey, there will be problems for you to solve and a job where your problem-solving skills will be appreciated.

*Where do we start?*

The famous physicist Richard Feynman once asked, "If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence was passed on to the next generation of creatures, what statement would contain the most information in the fewest words?"

His answer was "All things are made of atoms—little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling

upon being squeezed into one another."

*That* seems like a good place to start.

CHAPTER 2

# Matter and Energy

The universe is made of matter and energy. Current models posit that the universe is approximately 68% dark energy, 27% dark matter, and 5% ordinary matter. Everything you can see and touch is part of the small part of the universe made of ordinary matter. Most science deals with ordinary matter and its interactions; highly trained theoretical physicists are currently debating the nature and effects of dark matter and dark energy.

What is this ordinary matter made of? All things (including the air around you) are made of *atoms*. Atoms are incredibly tiny — there are more atoms in a drop of water than there are drops of water in all the oceans. It is not easy to visualize the scale, but computers allow us to make simulations that can give us an idea. See this cool website by Cary and Michael Huang shows different size by scales! It may help to start at a length of $10^{-15.5}$ to see protons and neutrons for scale.

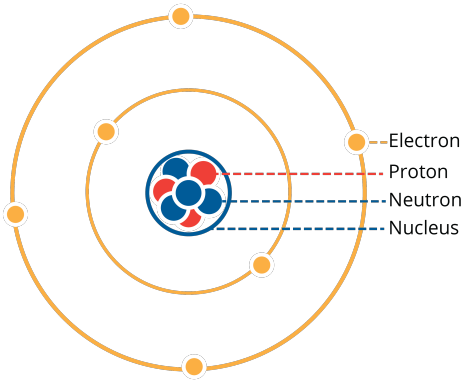



Figure 2.1

Every atom has a *nucleus* that contains *protons* and *neutrons*. Orbiting around the nucleus is a cloud of *electrons*. However, the mass of the atom comes mainly from the protons and neutrons, since they are about 2000 times as massive as an electron! These three particles, protons, neutrons, and electrons, are called *subatomic particles*. (See figure 2.1.)

Studies such as particle physics, nuclear physics, and astrophysics analyze how these particles interact.

Notice how these atoms are mostly empty space in between the subatomic particles. The force, or attractive pull, keeping them together is called *electromagnetic force*. What keeps these charged atoms in their orbits is the **electromagnetic field**.

The charges of every particle create an *electric field* around it. The orbiting and movement of an subatomic particles causes its *magnetic field*. *This is why electric current* (*moving electrons in a wire*) *produces a magnetic force. The faster the particles move, the stronger the field.*

## 2.1 Atoms and Their Models

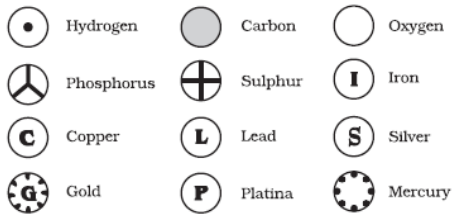



Figure 2.2

Over the history of science, there have been many ideas about the structure of atoms. The first model of the atom was rudimentary; it was modeled as one large, solid, uniform, and neutral object. Then British physicist J.J. Thomson discovered that atoms could be split into a light, negatively charged particle and a heavier, positively charged particle (we now know this is the nucleus, the dense grouping of protons and neutrons in the center of an atom).

Suddenly, the atom went from neutral and indivisible to made of different types of charged particles. Further experiments by Ernest Rutherford showed the atom to be mainly empty space, further updating scientists' model of the atom. Subsequently, Niels Bohr explained the phenomena of spectral lines (we will discuss this further in Sequence 2) by proposing that electrons orbit at specific distances from the nucleus.

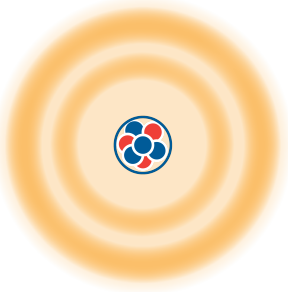


Figure 2.3

This is likely the model you are most familiar with seeing, and it is the one we will use most often in this text. The first figure shown in this chapter is a Bohr model: it shows the protons and neutrons in the nucleus, and models the electrons as moving in discrete orbits around the nucleus.

However, the Bohr model is slightly inaccurate. While it is a convenient model for thinking about atoms, in reality, electrons don't neatly orbit the nucleus. Scientists don't know exactly where an electron will be in relation to the nucleus, but they do know where it is most likely to be. They use a cloud that is thicker in the center but fades out at the edges to represent an electron's position (see figure 2.3). This cloud represents the probability distribution of finding an electron in a particular region around the nucleus. The figure is only in two dimensions, but the actual electron cloud is three-dimensional. The shape of the cloud depends on the energy level of the electron, and different energy levels have different shapes and sizes of clouds.

While the cloud model is more accurate, we will use the Bohr model as it allows you to easily and quickly assess the number and arrangement of electrons.

### 2.1.1 Classifying Atoms

We classify atoms by the numbers of protons they have. An atom with one proton is a hydrogen atom, an atom with two protons is a helium atom,

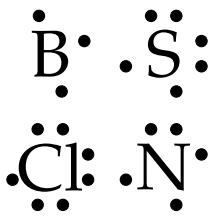



Figure 2.4

and so forth (refer to Table 10.1 on page 75)). We say that hydrogen and helium are *elements* because the classification of elements is based on the proton number. And we give each element an atomic symbol. Hydrogen gets H, helium gets He, oxygen gets O, carbon gets C, and so on. You can see an element's symbol on the periodic table.

An atom will typically have the same number of electrons as proton (with an overall electric charge of 0), but when it gains or loses electrons, it becomes an *ion* with a net electric charge. **Ionization** is the addition or removal of certain particles to reach a charge imbalance. Atoms can gain or lose electrons through various processes, such as chemical reactions or ionizing radiation. An atom likes to gain or lose electrons to achieve a full outer electron shell, typically with 8 electrons (the octet rule). When an atom gains electrons, it becomes a negatively charged ion, or *anion*. When it loses electrons, it becomes a positively charged ion, or *cation*. .

### 2.1.2 When Atoms Combine

When atoms of different elements combine, they make *compounds*. Compounds are substances made up of more than one element. Compounds can be *molecules* or *crystal lattices*. In the next section you'll learn *why* these different structures form.

There are many kinds of compounds. You know a few:

- Table salt is crystals made of $Na^+$ and $Cl^-$ ions: a sodium atom that lost an electron and a chlorine atom that gained an electron
- Baking soda, or sodium bicarbonate, is $NaHCO_3$.
- $O_2$ is the oxygen molecules that you breathe out of the air (air, a blend of gases, is mostly $N_2$.).
- Common quartz is $SiO_2$: silicon dioxide

The subscripts indicate what quantities of the elements are present in the compound. Each number indicates the quantity of the preceding element. A drop of water, $H_2O$, has two hydrogen atoms and one oxygen atom. Often times we'll see these numbers used to determine the ratio of elements in a compound. To do this, we compare the quantities of each element to find their simplest whole-number ratio. This is called the *empirical formula* of the compound.

**Example**: What is the ratio of elements present in Epsom salt?

**Solution**: Epsom salt, chemical name magnesium sulfate, has the chemical formula $MgSO_4$.

Therefore, the ratio of Mg:S:O is 1:1:4.

### *Exercise 1* Ratios of Atoms in Molecules

Give the elemental ratio for each compound.

*Working Space*

1. methane, $CH_4$
2. copper (II) sulfate, $CuSO_4$
3. glucose, $C_6H_{12}O_6$

*Answer on Page 1875*

## 2.2 Types of Matter

One way to classify matter is by the types of chemical bonds that hold a material's atoms together. The nature of these bonds, in turn, affects the properties of the material. For now, all you need to know is there are three types of chemical bonds: *metallic*, *covalent*, and *ionic*. Materials held together with the same type of bonds tend to have similar properties. For example, copper, bronze, iron, and steel (all containing metallic bonds) are all shiny, ductile, malleable, and good conductors of heat and electricity. On the other hand, Epsom salt and table salt form large crystals, have very high melting points, and are poor conductors of electricity in their pure form. These two substances (Epsom and table salt) both contain ionic bonds.

### 2.2.1 Covalent Compounds

Water is an example of a covalent compound: it is made of two hydrogen atoms covalently bonded to one oxygen atom (see figure 2.5). The result is a water molecule. The different atoms cluster together because they share electrons in their clouds. This is the nature of a *covalent bond*: it is formed when atoms share electrons. Sometimes, electrons are shared evenly, but in water, they are shared unevenly. Oxygen is better at attracting electrons to itself than hydrogen, and so the shared electrons spend more time on the oxygen atom than the hydrogen atoms. As a result, the oxygen side of a water molecule has a slight negative charge, while the hydrogen atoms have a slight positive charge. These slight charges are represented with a lower case Greek letter delta, $\delta$. When electrons are shared unevenly, we call this a *polar* covalent bond, because there are positive and negative poles at either

end of the bond.

Whether electrons are shared evenly or unevenly is based on the elements' relative *electronegativities*. Electronegativity is simply a measure of how strongly an atom can attract electrons to itself. In general, elements on the right side of the periodic table are more electronegative than elements on the left side. When covalent bonds form between two elements of similar electronegativities, the electrons are shared evenly. We call this a *non-polar* covalent bond. Oil is an example of a non-polar covalent substance. Different oils have different ratios and structures, but all oils are made mainly of carbon and hydrogen, which have similar electronegativities. What happens when you try to mix oil and water? They don't mix well! This is due to the difference between their bond types. Polar substances, like water, mix best with other polar substances, while non-polar substances, like oils, mix best with non-polar substances. You'll learn more about why this is in Sequence 2.

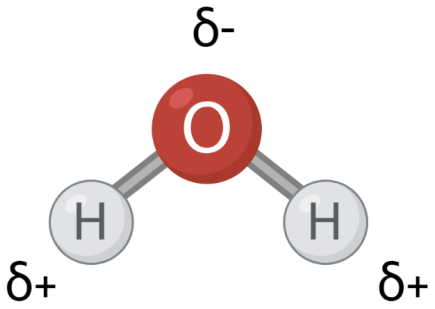
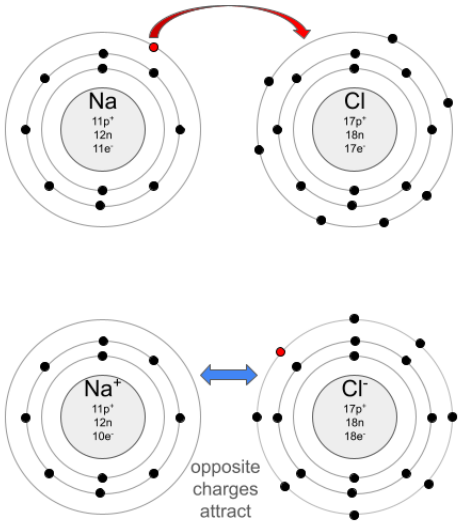



Figure 2.5

For both polar and non-polar covalent bonds, the electrons are held tightly to the nuclei, even if they are shared among atoms. Those electrons don't move to another molecule: they move around within the molecule they are already a part of. Since electrons don't flow freely in covalent substances, they are also poor conductors of electricity. Covalent compounds also tend to have lower melting and boiling points compared to ionic compounds.

### 2.2.2 Ionic Compounds




Ionic bonds are the electrical attraction between opposite-charged ions. When a neutral atom gains or loses an electron it becomes an *ion* (a charged atom), and oppositely-charged ions are attracted to each other. Which atom gets the electron and which loses it is based on their electronegativities: the more electronegative atom steals one or more electrons from the less electronegative atom.

Let's examine how a simple ionic compound is formed: sodium chloride, also known as table salt, is made up of sodium and chlorine atoms (see figure 2.6). When sodium and chlorine come in contact with each other, electrons move from the sodium to the chlorine, making a sodium *cation* (positively-charged ion) and a chloride *anion* (negatively-charged ion). Yes — chlor*ide* is correct! When naming an anion, the ending of the element name changes to *-ide*. Once the sodium cation and chloride anion are formed, their opposite charges attract them to each other, like north and south magnet poles.

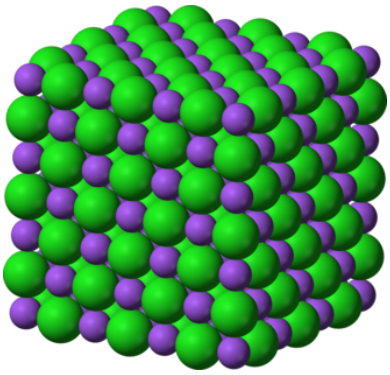


Figure 2.7

When there are many, many sodium and chloride ions around, they spontaneously arrange themselves in a pattern, giving ionic compounds their characteristic crystal structure (see figure 2.7). Because the electrons are tightly held by each ion, ionic substances don't conduct electricity well as solids. The atomic crystal lattice also determines the shape of the macroscopic crystals. Salt crystals are generally cubic, while other crystals (like quartz) form hexagonal prisms. You'll learn how to predict the atomic and macroscopic crystal structure of different compounds in Sequence 2.

**Modeling Ionic Compounds**

You can represent ions with Lewis dot diagrams by adding or subtracting electrons from the model. Since electrons are negative, anions have gained electrons while cations have lost them. Additionally, the ion is in brackets and the overall charge is indicated outside the top right corner of the brackets.

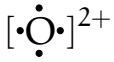
**Example**: Create Lewis dot diagrams for $Na^+$, $F^-$, $O^{2+}$, and $Mg^-$.

**Solution**: Sodium is in column 1, so a neutral sodium atom has 1 valence electron. A +1 charge means it has lost 1 electron, leaving zero.

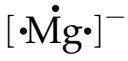
$$[Na]^+$$

Fluorine is in column 17 and has 7 valence electrons when neutral. The anion $F^-$ has gained one electron, for a total of 8.

Neutral oxygen has 6 valence electrons, therefore $O^{2+}$ has 4.



Neutral magnesium has 2 valence electrons, so $Mg^{-}$ has 3 valence electrons.



To make Lewis dot diagrams of ionic compounds, you show both ions in the ratio given in the formula. For example, a Lewis dot diagram of $MgCl_2$ would show one magnesium cation and two chloride anions.

### 2.2.3 Metallic Compounds

You may already know that metals are excellent conductors of electricity and heat. This is a consequence of their metallic bonds. In pure metals and alloys, the outermost layer of electrons can move freely from one atom to the next. As a result, at the atomic level, metals are best characterized as a lattice of cations surrounded by a "sea of electrons" (see figure fixme metallic bonding figure)

The free-flowing sea of electrons in pure and alloyed metal means the cation lattice can be rearranged without breaking the metallic bond. As a result, metals are ductile (able to be drawn out into a wire) and malleable (able to be hammered into a new shape without cracking or breaking). (fixme figure showing deformation of cation lattice in bending metal)

Metals can be pure, like copper or iron, or *alloys*, like bronze or steel. *Alloys* are mixtures between two or more elements where at least one element is a metal. Steel is an alloy of iron and carbon; bronze is an alloy of copper and tin. Alloying a metal changes its properties because of the change in the cation lattice (see figure FIXME pure vs alloyed lattice and structural changes).

## 2.3 Energy and Work

Look around you: things are moving, warming up, lighting up, and changing. No physical property changes without *work* being done. What is work? No, we are not talking about

having a job. Work is defined as a change in Energy. Every object has some type of energy. When you climb a flight of stairs, you feel tired. That tiredness is evidence that energy has left your body and gone somewhere else—into lifting your body upward. When the Sun warms your skin, energy is traveling through space and into you. When a moving car comes to a stop, its motion doesn't simply disappear; energy is transferred to the brakes and the surrounding air, mostly as heat.

Energy Transformations are all around us:

1. Feeling less tired after consuming calories from food
2. The light and the feelin of heat from the Sun
3. A ball speeding up while rolling down a hill
4. Being out of breath after climbing 5 flights of stairs
5. A car stopping when brakes are applied

Some types of energy are easy to visualize, while others are not. Energy is what moves from one object to another when work is being done. There are two main types of energy: energy that *stored* and energy that is motion. Most times, energy that is stored is transformed into energy in motion. This stored energy is called *Potential Energy* and the energy in motion is referred to as *Kinetic Energy*.

When you lift something, the energy stored in your body (in the form of sugar and fat) is transferred to the object, accelerating it upwards. Your body continues to transfer energy as you lift the object against gravity. When you've lifted it as high as you can, most of the energy your body lost (we call this "burning Calories" colloquially) is stored as *Potential Energy*, due to the object's height. Potential energy is the stored energy an object has due to its position or configuration, rather than its motion. *Kinetic Energy* is the term we use to describe the energy an object possesses due to its motion; you'll learn more about different forms of energy in 3, and in Sequence 2.

Another example of energy is a simple circuit connecting a battery and a light bulb. The battery has stored potential energy. When the circuit is complete, the potential energy in the battery is transferred to electrons in the light bulb, causing them to move and gain kinetic energy. In the light bulb's filament (we are referencing old, non-LED light bulbs here!), the electrons encounter resistance, which slows them down. The energy the electrons lose in this process is being transformed into light and heat, lighting your room.

The Work-Energy theorem explains the relationship between work and energy, and we will introduce the theorem and use it to explain energy transfer in a subsequent chapter.

## 2.4 Mass-Energy Equivalence

You've probably seen the equation

$$E = mc^2$$

$E$ is energy, $m$ is mass, and $c$ is the speed of light in a vacuum (about $3 \times 10^8 \frac{m}{s}$). So far, we've been discussing matter and energy separately. This equation shows that matter can be converted to energy, and vice-versa. This is the source of the light energy emitted by the sun.

The Sun is mostly hydrogen. At the very center, it is so hot and dense that the nuclei of hydrogen atoms are fused together to form helium atoms, a process called the *proton-proton chain reaction*. The actual reaction involves several steps and is more complicated, but the overall process can be summarized as:

$$4H \rightarrow He$$

If every hydrogen atom has a mass of approximately $1.6735575 \times 10^{-27}$ kg and every helium atom has a mass of approximately $6.6464731 \times 10^{-27}$ kg, how much energy is released when one atom of helium is created? First, notice that one helium has less mass than 4 hydrogen atoms:

$$4 \times \left(1.6735575 \times 1-^{27}\right) - 6.6464731 \times 10^{-27} = 4.77569 \times 10^{-29}$$

Now, we can use $E = mc^2$ to find out how much energy is equivalent to $4.77569 \times 10^{-29}$ kg:

$$E = \left(4.77569 \times 10^{-29}\text{kg}\right)\left(2.99792458 \times 10^8 \frac{m}{s}\right)^2$$

$$E \approx 4.292 \times 10^{-12}\text{joules}$$

All of these numbers are very small and hard to visualize. We could ask this: if 1 kilogram of hydrogen (about enough to fill a standard beach ball) were fused to make helium, how much energy would be released? For every kilogram of hydrogen that enters the proton-proton chain reaction, about 0.02854 kilograms of mass are converted to energy (the mass of about 5 quarters).

$$E = (0.02854\text{ kg})\left(2.99792458 \times 10^8 \frac{m}{s}\right)^2 \approx 2.5647 \times 10^{15}\text{ joules}$$

This is more than 700,000,000 kWh (kilowatt hours); the average US household uses only 30 kWh per day. So, fusing one kilogram of hydrogen releases enough energy to power an average US home every day for *over 65,000 years*! This relatively huge release of energy is why scientists continue to research nuclear fusion energy sources. The nuclear power plants currently running around the world today rely on *nuclear fission*: the splitting apart of atoms, the opposite process of *nuclear fusion*. Nuclear fission releases much, much less energy per kilogram of input material than nuclear fusion, and thus stable, affordable nuclear fusion power plants remain a "holy grail" of scientific research.

## 2.5 Conclusion

We have seen that the universe is made of dark energy, dark matter, and ordinary matter. Ordinary matter is made of atoms, which can be classified based on their number of protons. Atoms combine in different ways to make compounds, and the manner of combination (ionic, covalent, or metallic bonding) affects the macroscopic properties of the substance. Energy allows matter to do work, and work is the transfer of energy.

Matter and energy do share a fundamental property: they are both conserved. Neither matter nor energy can be created or destroyed. This means the total amount of ordinary matter and energy in the universe is constant. This great scientific truth *something about its application* In the next chapter,

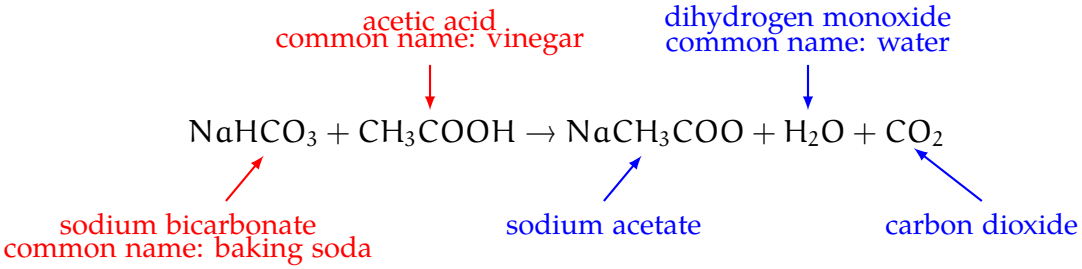
Chapter 3

# Conservation of Mass and Energy

One of the most fundamental laws in science is the conservation of mass and energy. This law states that mass (matter) and energy cannot be created or destroyed. This means the total matter and energy in the universe always stays the same.

## 3.1 Conservation of Mass

Since matter cannot be created or destroyed, a chemical reaction does not change the mass of the *reactants* as they form *products*. Consider the reaction between vinegar and baking soda, which produces carbon dioxide, water, and sodium acetate:




The reactants are labeled with red, and the products with blue. This equation is *balanced*: that is, it shows the same number of each element on each side of the arrow. This shows that the atoms are not created or destroyed during a chemical reaction, they are only rearranged. Take a minute to count up each element on each side. You should find there are 5 hydrogens, 1 sodium, 3 carbons, and 5 oxygens on each side.

Now, let's look at an *unbalanced* chemical reaction. As you know, hydrogen and oxygen combine to form water. Additionally, hydrogen and oxygen both exist as *diatomic gases*. When we say "oxygen gas" or "hydrogen gas", we mean the diatomic molecules, $O_2$ and $H_2$, respectively. Here is an unbalanced chemical reaction between hydrogen gas and oxygen gas to form water:

$$H_2 + O_2 \rightarrow H_2O$$

How do we know this equation is unbalanced? Count up the elements: there are two oxygen atoms on the reactant side, but only one on the product side. This violates the conservation of matter: that oxygen atom cannot just disappear!

### 3.1.1 Balancing Chemical Reactions

We solve this by *balancing* the chemical reaction: adjusting the number of products and reactants to comply with the Law of Conservation of Matter. You'll learn strategies for balancing chemical reactions in Sequence 2, but for now we'll briefly balance this chemical reaction so that it complies with the Law of Conservation of Matter. You may be tempted to simply add a lone oxygen atom to the products side:

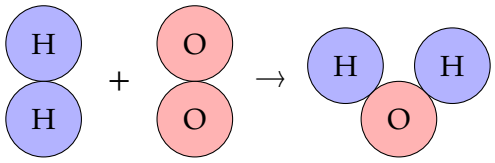
$$H_2 + O_2 \rightarrow H_2O + O$$

The major reason this is incorrect is that oxygen does not exist as a lone atom - as discussed above - so it doesn't make sense to have a lone oxygen as a product. So maybe we should add a molecule of oxygen gas to both sides?

$$H_2 + O_2 \rightarrow H_2O + O_2$$

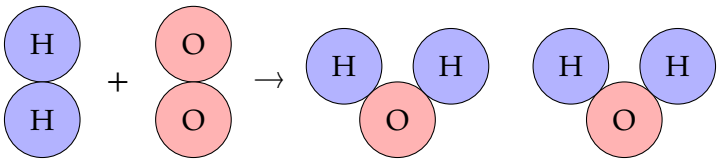
Well, now we have the same problem we started with: the oxygens are unbalanced. When balancing chemical reactions, we can only add *whole molecules* that are already in the reaction. Let's take another look at our unbalanced reaction with some molecular models for visualization:

$$H_2 + O_2 \rightarrow H_2O$$



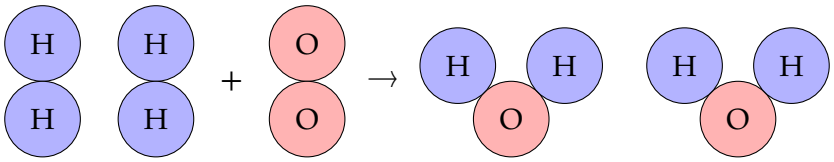

You can clearly see we need more oxygens on the product side. Since we can only add *whole molecules*, our only option is to add another water. We do this by adding a coefficient of 2 in front of $H_2O$ in our equation, which indicates 2 water molecules (just like 2x means two x's):

$$H_2 + O_2 \rightarrow 2H_2O$$




We've fixed our oxygen problem: now there are two oxygen atoms on both sides. But now we have a hydrogen problem: there are 2 on the reactant side and 4 on the product side. We can address this by adding another hydrogen gas molecule to the reactant side:

$$2H_2 + O_2 \rightarrow 2H_2O$$




And now we have the same number of hydrogens and oxygens on each side! Notice we have all the same reactants and products that we started with, but now in ratios that reflect the conservation of matter.

A final note: if atoms are in parentheses followed by a subscript, the subscript applies to every atom in the parentheses. For example, zinc nitrate, $Zn(NO_3)_2$ is made of 1 zinc, 2 nitrogens, and 6 oxygens.

## *Exercise 2* Balanced and Unbalanced Reactions

Classify the following chemical reactions as balanced or unbalanced. If it is unbalanced, state what element(s) are not conserved.

*Working Space*

1. $NiCl_2 + 2NaOH \rightarrow Ni(OH)_2 + 2NaCl$
2. $HgO \rightarrow Hg + O_2$
3. $BaSO_4 + 2C \rightarrow 2BaS + CO$
4. $Cd(NO_3)_2 + H_2S \rightarrow CdS + 2HNO_3$

*Answer on Page 1875*

## 3.2 Conservation of Energy

Just like matter, energy is also conserved: it cannot be created or destroyed, only change forms. You'll learn more about the types of energy in a subsequent chapter, Work and Energy. The transformation of energy from one type to another drives our modern world: your phone transforms electrical potential energy into light and sound energy, a nuclear power plant transforms nuclear energy to electrical energy, and your car transforms chemical potential energy (in the gasoline) into kinetic energy (motion).

### 3.2.1 Friction, Heat, and Energy "Loss"

Imagine rolling a ball across a flat surface: you have given the ball some kinetic energy in its motion. If the kinetic energy were conserved, the ball would keep rolling at the same speed forever, as long as it was on a flat surface. Experience tells us this isn't what happens: the ball will eventually come to a stop. Why doesn't this violate the Conservation of Energy?

Friction is the force that opposes motion: whenever you slide two objects past each other, friction transforms kinetic energy into heat. Rub the palms of your hands together. You should feel warmth, a product of the friction between your hands. As the ball in the example above rolls, it also experiences friction between itself and the ground. The friction slowly transforms the kinetic energy of the ball into heat, causing the ball to lose kinetic energy. When all of the ball's kinetic energy is transformed to heat, the ball comes to a rest. So, the kinetic energy of the ball wasn't destroyed and didn't disappear: it became heat.

In fact, nearly all energy in the universe will eventually be transformed to heat, resulting in the inevitable "heat death" of the universe. Here is a short video about heat, entropy, and the heat death of the universe: https://www.youtube.com/watch?v=gOWt_Hq3yrE/.

### 3.2.2 Energy Conservation in Falling Objects

When an object is positioned above the ground, it has *gravitational potential energy*. The potential energy is proportional to the mass of the object, the strength of the gravitational field, and the object's height above the ground:

$$E_p = mgh$$

where $m$ is the mass of the object in kg, $g$ is the acceleration due to gravity ($9.8 m/s^2$ on Earth), and $h$ is the height above the ground in meters.

**Example**: What is the gravitational potential energy of a 5.0 kilogram bowling ball in the hand of a bowler (approximately 1.2 meters high)?

**Solution**: The mass is 5.0 kg, the height is 1.2 m, and we assume the bowler is on Earth, so g is 9.8 $m/s^2$:

$$E_p = (5.0kg)\left(9.8m/s^2\right)(1.2m) = 59J$$

Here is a new unit: *joules*, represented by a capital J. A joule is a unit of energy, and it is the same as $\frac{kg \cdot m^2}{s^2}$.

If the bowler were to drop that ball, it would lose potential energy and gain *kinetic energy*. The kinetic energy of an object is proportional to its mass and the square of its speed:

$$E_k = \frac{1}{2}mv^2$$

where mass is in kg and speed (v) is in m/s. You can quickly check that the units still come out to a joule! If there is no air resistance, then as an object falls all of its potential energy is converted to kinetic energy. (If you'd like to explore what happens when friction is accounted for, you can play with this PhET simulation: https://phet.colorado.edu/en/simulations/energy-skate-park.)

**Example**: If the bowling ball were dropped, what would its speed be right before it hits the ground? (Neglect air resistance.)

**Solution**: Ignoring air resistance, according to the Law of Conservation of Energy, the kinetic energy of the ball right before it hits the ground must be equal to the gravitational potential energy of the ball right before its release. Therefore:

$$mgh = \frac{1}{2}mv^2$$

We can eliminate an m from both sides and rearrange to solve for v:

$$v = \sqrt{2gh} = \sqrt{2\,(9.8m/s^2)\,(1.2m)} = \sqrt{23.52\frac{m^2}{s^2}} = 4.8\frac{m}{s}$$

The ball will have a speed of 4.8 m/s just before it hits the ground.

## *Exercise 3* Kinetic and Potential Energy

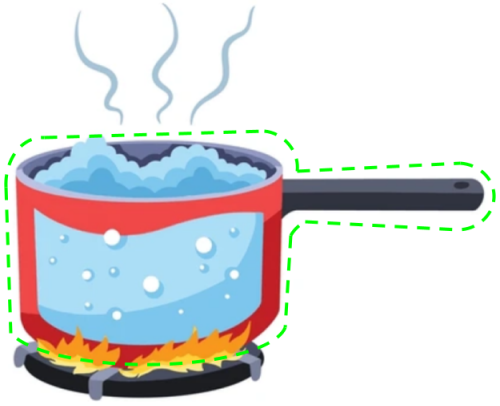
*Working Space*

1. How fast should you toss a ball straight up if you want it to reach your friend on the second floor (3.7 meters above you)?
2. Your little brother is teasing you from a treehouse 5.0 meters off the ground. If your slingshot can shoot a pebble at 15 m/s, can you hit your little brother?
3. A 63-kg roller-blader rolls down a hill. If the hill is 8.0 meters high and she reaches the bottom of the hill with a speed of 11.2 m/s, how much energy was converted to heat through friction?

*Answer on Page 1875*

## 3.3 Types of Systems

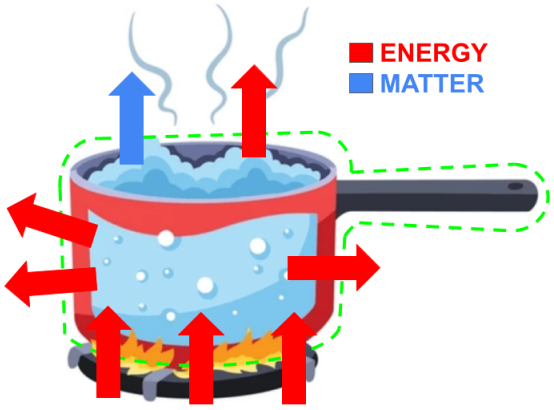
We classify systems based on the flow of matter and energy. A *system* is a set of interconnected elements. Your body is a system, as is a television or a boiling pot of water. Scientists define systems by separating the parts of the system from the rest of the universe, usually called "the surroundings". You can represent this separation with a dashed line. Here is a diagram defining a pot of boiling water as a system:



Everything inside the dashed line is the system: the pot and the water boiling in it. Everything outside the dashed line is the surroundings: the stove, the air around the pot, etc. Defining a system is *arbitrary*: there isn't one hard and fast definition of a system. Think of your school: you could look at the system of a classroom, the system of a hallway and all the classes connected to it, or the entire school building. How you define a system depends on what you're studying.

### 3.3.1 Open Systems

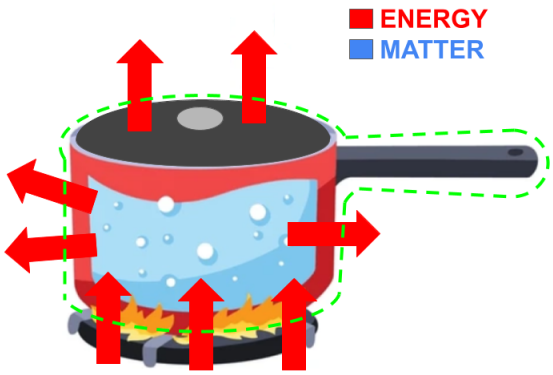
An *open system* allows for matter and energy to cross the imaginary boundary between the system and its surroundings. The uncovered pot of boiling water is an example of an open system. Energy enters the system as heat from the stove, and leaves the system as heat in the steam rising from the pot. Notice that steam rising: see how it crosses the imaginary boundary? The steam is matter *leaving the system*. Since matter and energy can cross the boundary between the system and its surroundings, the uncovered boiling pot is an open system.



The system also loses energy through the sides of the pot: if you touched the pot, it would feel hot, which means heat energy can also leave the system through the sides of the pot. This is due to the collision between air particles (the surroundings) and the outside of the pot (the system). With every collision, a little heat is transferred from the pot to the air. This is why your hot drink gets cold if you leave it out, even if you don't add any ice.

### 3.3.2 Closed Systems

A *closed system* allows for the transfer of energy but not the transfer of matter. If we put a lid on this pot, it would become a closed system.




Notice that the difference between an open and closed system is the flow of *matter*: open systems allow for the movement of matter, while closed systems do not. However, energy can still enter and leave closed systems. Sealed containers that aren't insulated are good examples of closed systems - a car with the windows up and doors closed.

### 3.3.3 Isolated Systems

An *isolated system* does not allow for the flow of matter *or* energy in or out of the system. There is no such thing as a truly isolated system: in reality, a small amount of energy can be transferred even through the best thermal insulators. However, for well-insulated systems, it can be a good approximation to model that system as isolated. A simple example would be a sealed, well-insulated coffee thermos. The transfer of heat energy between the coffee in the thermos and the thermos' surroundings is so slow that we can ignore that small amount of transfer and approximate the thermos as an isolated system.

### 3.3.4 Classifying Systems

To quickly categorize a system as open, closed, or isolated, ask yourself two questions:

1. Can matter enter or leave the system?
2. Can energy enter or leave the system?

If your answer to the first question is yes, you know automatically the system is open. If no, move to the second question. If energy can enter or leave, the system is closed. If not, the system is isolated. Sometimes, textbooks and exams will describe a system as "well-insulated". This is directing you to assume any transfer of energy between the system and surroundings is negligible, and that you should treat the system as an isolated system.

## *Exercise 4* Open, Closed, and Isolated Systems

Classify each system as open, closed, or isolated. Justify your answer.

1. The human body
2. Earth
3. Your cell phone
4. A well-insulated cooler with the lid sealed
5. A well-insulated cooler with the lid open
6. A bottle of soda before it is opened to be drunk

*Working Space*

*Answer on Page 1875*

CHAPTER 4

# Mass, Weight, and Gravity

Mass is a measure of the amount of matter in an object. Weight is the force of gravity on that object. An object's mass is the same no matter where it is in the universe: the amount of "stuff" in an object does not depend on its location. However, an object's weight *does* change: the same object has a different weight on the Moon than it does on Earth or Venus. Since the force of gravity on the Moon is about $1/6^{\text{th}}$ as on Earth, you would weigh $1/6^{\text{th}}$ as much on the Moon, but your mass would be the same.

## 4.1 Kilograms versus Pounds

In elementary school, you probably learned that 1 kilogram is about 2.2 pounds. This is only true *on Earth*. In everyday language, we may use kilograms and pounds, or mass and weight, interchangeably; scientists do not.

| **Mass** | **Weight** |
|---|---|
| measure of *amount* | measure of *force* |
| units include grams, kilograms, milligrams | units include pounds, Newtons, dynes |
| a *scalar* measurement | a *vector* measurement |
| does *not* depend on location | does depend on location |

On Earth, a 1 kilogram mass weighs 2.2 pounds. That same mass weighs 0.37 pounds on the Moon. You learned to "convert" between pounds in kilograms in elementary school because everywhere on Earth, a 1-kg mass weighs 2.2 lbs. The conversion doesn't apply to other locations in the universe because mass and weight are different.

### 4.1.1 Scalars and Vectors

One of the fundamental differences between mass and weight is the type of measurement: scalar versus vector. *Scalar* measurements are just a counting number: they tell the amount of magnitude. Mass is a scalar measurement: it tells how much matter is in an object. Energy is also a scalar measurement, as well as temperature and density. *Vectors*, on the other hand, also have a *direction*. A complete measure of a force gives the magnitude and direction. So, your weight isn't just 185 pounds: it's 185 pounds *downward*. Velocity is a speed plus a direction, such as 35 mph east. We will expand on this later in the vectors chapter!

## 4.2 Mass and Gravity

When we discuss weight, we usually mean the gravitational force between an object and the largest nearby object. My weight on Earth, a spacecraft's weight on the Moon, or a rover's weight on Mars are all examples. But planets and moons don't have some special property that makes them emit gravity. Rather, *all objects with mass are gravitationally attracted to each other*. But, compared to the mass of the Earth, the mass of all the other objects around you (a table, your family, your house or apartment complex) is very, very small.

The force of the gravitational attraction between two objects is proportional to the product of their masses, and inversely proportional to their distance squared. This means that as objects get farther away, the force decreases. If you double the distance, the force quarters. Quadruple the distance, the force is $1/16^{\text{th}}$ as much. This is why you are more attracted to the Earth than you are to distant stars, even though they have much more mass than the Earth.

**Newton's Law of Universal Gravitation**

Two masses ($m_1$ and $m_2$) that are a distance of $r$ from each other are attracted toward each other with a force of magnitude:

$$F = G\frac{m_1 m_2}{r^2} \tag{4.1}$$

where G is the universal gravitational constant. If you measure the mass in kilograms and the distance in meters, G is about $6.674 \times 10^{-11}$. That will get you the force of the attraction in newtons.

## *Exercise 5* Gravitation between two planets

A small moon orbits a planet at a distance of $4.0 \times 10^6$ m from the planet's center. The mass of the planet is $6.0 \times 10^{24}$ kg, and the mass of the moon is $8.0 \times 10^{22}$ kg.

1. Using Newton's Law of Universal Gravitation, calculate the gravitational force between the planet and the moon.
2. If the moon were moved to a distance of $8.0 \times 10^6$ m from the planet's center (twice the original distance), determine the new gravitational force.

*Working Space*

*Answer on Page 1876*

### *Exercise 6* Gravity

*Working Space*

The Earth's mass is about $6 \times 10^{24}$ kilograms.

Your spacecraft's mass is 6,800 kilograms.

Your spacecraft is also about 100,000 km from the center of the Earth. (For reference, the moon is about 400,000 km from the center of the Earth.)

What is the magnitude of the force of gravity that is pulling your spacecraft and the Earth toward each other?

*Answer on Page 1877*

## 4.3 Mass and Weight

Gravity pulls on things proportional to their mass, so we often ignore the difference between mass and weight.

The weight of an object is the force due to the object's mass and gravity. When we say, "This potato weighs 1 pound," we actually mean "This potato weighs 1 pound on Earth." That same potato would weigh about one-fifth of a pound on the moon (see figure 4.1).

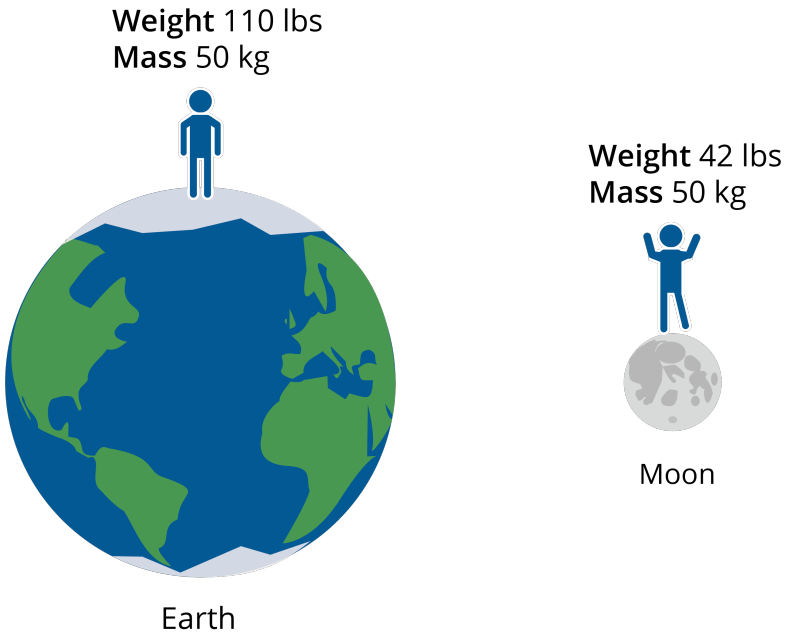



Figure 4.1: Mass is a measure of all the matter in an object. Weight is a measure of the force of gravity on that object. Mass is not location-dependent, while weight is.

However, that potato has a mass of 0.45 kg no matter where it is in the universe.

We talked about the Law of Universal Gravitation. We can simplify it for our calculations to derive the weight of objects near to the surface of the earth.

Starting with Equation 4.1, for objects near Earth's surface, the radius is essentially constant and equal to Earth's radius $R_E$. Substituting $r = R_E$ gives us:

$$F = G\frac{Mm}{R_E^2} \tag{4.2}$$

The only thing variable in this equation is $m$, so the object's mass can be factored out, leaving:

$$F = m\left(G\frac{M}{R_E^2}\right) \tag{4.3}$$

The quantity in parentheses involves only constants: the gravitational constant G, Earth's mass M, and Earth's radius $R_E$. Since this value stays the same for all objects near the surface, we define it as the gravitational field strength:

$$g = G\frac{M}{R_E^2} \tag{4.4}$$

Substituting this definition back into the force equation gives:

$$F = mg \tag{4.5}$$

where $m$ is the mass of some object, and $g = 9.8\ m/s^2$ is the gravitational constant, a constant found by $G\frac{M}{R_E^2}$.

The steps in sequence are shown below:

$$\begin{aligned} F &= G\frac{Mm}{r^2} \\ &= G\frac{Mm}{R_E^2} \\ &= m\left(G\frac{M}{R_E^2}\right) \\ g &= G\frac{M}{R_E^2} \\ F &= mg \end{aligned}$$

Equation 4.5 is the equation we will use a standard for the force of gravity, $F_g$, for most objects on Earth. Objects undergoing projectile motion, objects at rest, and objects hung by tension all have a force of gravity acting on them, [1]

## *Exercise 7* May the Force be with You 1

A hammer is falling along the side of a building, close to the surface of the Earth. What is it's acceleration?

We won't ask it yet, but think to yourself: why doesn't the Earth move as much to the hammer as the hammer moves to the Earth?

*Working Space*

*Answer on Page 1877*

[1] You may occasionally use 10 for g to simplify expressions if the problem says to do so, but use 9.8 otherwise.

### *Exercise 8* May the Force be with You 2

A coin is dropped. It is measured that the force of gravity on the coin is .0198 N. What is the mass of the coin?

*Working Space*

*Answer on Page 1877*

## 4.4 Gravitational Field

When larger bodies are under consideration for Newton's Universal Law of Gravity (Equation 4.1), the gravitational force on both objects increases linearly. Bigger masses lead to greater forces between them. This is most noticeable with planets and moons; moons and planets keep each other in-orbit through the gravitational force between them.

However, despite the force increasing proportionally with mass, the resulting acceleration of each body depends inversely on its mass, so more massive bodies experience much smaller accelerations for the same interaction force. This is why the falling hammer from Exercise 7 doesn't "pull the Earth" towards it in any sense; since the acceleration is $a_{\text{hammer}} = \frac{F}{m_{\text{hammer}}}$, and the mass of the hammer is approximately 1 kg, the hammer accelerates a large amount. Even the force on the earth is the same, the acceleration $a_{\text{Earth}} = \frac{F}{m_{\text{Earth}}}$ is significantly less because the mass of the Earth is so great. The hammer and Earth both accelerate toward their common center of mass, the *center of mass* for the combination of the hammer and Earth system. That center lies almost exactly at Earth's center because Earth's mass dominates. See more about this in Section 4.5.

See the following below figure below:

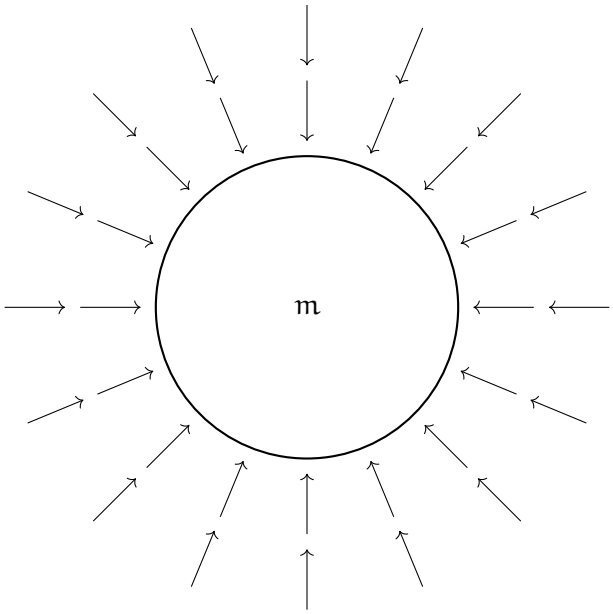



Figure 4.2: A figure of the direction of force vectors for a gravitational field.

All vectors in a gravitational field point to the center of mass of that field. The larger the body, as we said, the greater the force, and thus the greater magnitude of vectors. Thus, a gravitational field around a mass M is a *vector field*, a term you will hear in vector calculus sections, or an assignment of a vector to every point in space. Every point has a force assignment based on the **radius** between the two objects.

The gravitational field at a point in space is defined as the force per unit mass experienced by a small test mass placed at that point:

$$\vec{g} = \frac{\vec{F}}{m}$$

You'll notice $\vec{g}$ is the same as the acceleration! This is true, the gravitational field at a point is the acceleration any small test mass would have there. Therefore, points farther away from the body creating a gravitational field are smaller in magnitude. For symmetric and spherical bodies, this is true for all points revolved around the entire spherical body.

## 4.5 Center of Mass

The *center of mass* of a body is the point where its mass is evenly balanced in all directions, as if all the mass were concentrated at that single point. When we talk about *Free Body Diagrams* in a future chapter, we will assume the entire body's center of mass is located at a point that we use to analyze forces from. For an object with different densities at different points — for example, a hammer with a dense metal head and a lighter wooden handle

— the center of mass is found by averaging the positions of all parts, giving more weight to the heavier regions. You will see the term *center of mass* throughout this sequence; we will assume you understand how center of mass is calculated, but typically will not fully determine it.

For a one-dimensional object on some line, the center of mass can be found using the formula:

**1-Dimensional Center of Mass**

$$x_{CM} = \frac{m_1 x_1 + m_2 x_2 + \cdots + m_n x_n}{m_1 + m_2 + m_n}$$

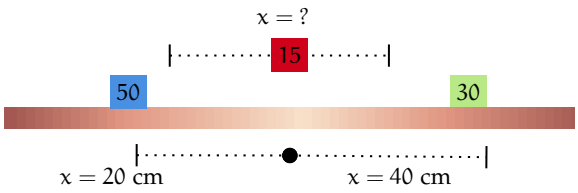
where $x_{CM}$ and $x_1 \ldots x_n$ are the distance from *a chosen origin* location. Notice how this is similar the formula for averages.

We can rewrite this using summation notation:

$$x_{CM} = \frac{\Sigma m_i x_i}{\Sigma m_i}$$

## *Exercise 9* Balancing a seesaw

Three masses are placed on a seesaw. $M_1$ has a mass of 30 lbs and is placed 40 cm to the right the equilibrium point of the seesaw. $M_2$ has a mass of 50 lbs and is placed 20 cm to the left of the equilibrium point. Where does $M_3$, which weighs 15 lbs, need to go in order for the seesaw to balanced? Assume the seesaw plank itself is uniformly distributed.




*Working Space*

*Answer on Page 1877*

In higher dimensions, it is easier to use integration to calculate the coordinates of the center of mass of some body, especially when it is not uniform (equally distributed). Although we haven't introduced integration formally, it is important to begin thinking about these topics:

**2D and 3D Center of Mass**

$$\mathbf{r} = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$$

$$dm = \rho(x, y, z)\,dV$$

$$2D: \quad M = \iint_V \rho(\mathbf{r})\,dV, \qquad 3D: \quad M = \iiint_V \rho(\mathbf{r})\,dV$$

where $M$ is the net mass, $dm$ is some infinitesimal weight at point $\mathbf{r}$, the position vector within $V$. $V$ is the dimensions of some the 3d boundary, and $\rho(x, y, z)$ is the density at the infinitesimal point $(x, y, z)$

The individual components can be found by:

$$x_{CM} = \frac{1}{M} \iiint_V x\,\rho(x, y, z)\,dV$$

$$y_{CM} = \frac{1}{M} \iiint_V y\,\rho(x, y, z)\,dV$$

$$z_{CM} = \frac{1}{M} \iiint_V z\,\rho(x, y, z)\,dV$$

# Chapter 5

# Introductory Motion Concepts

## 5.1 Position, Velocity, Acceleration

When studying motion, it is important to have ways to describe various aspects of an object.

*Position* is a location of an object relative to a coordinate system. The coordinate system, reference points, and *origin* is typically decided by you, the viewer. An object's position at one point in time can be $n$-dimensions, but typically we talk about 1D, 2D, and 3D.

- 1D can be thought of moving along a straight line $(x)$
- 2D can be represented on coordinate plane $(x, y)$
- 3D can be represented by 3 components $(x, y, z)$

The total distance traveled in a 1D plane is a total amount traveled. The *displacement* is net distance from the *starting location* or *origin*. Typically, a *change in position* is represented as a $\Delta x$ (in horizontal direction) and $\Delta y$ (in vertical direction). This distance can be thought of as a vector, with a directional component and magnitude. You may also see displacement as $\mathbf{s}$.

Average *Velocity*, usually represented as $\vec{\mathbf{v}}$ or $\mathbf{v}$ is defined as the net displacement over a certain period of time. Because displacement is a vector quantity, so is average velocity.

Average speed, however, does not contain directional information. Like the speedometer in a car, speed only contains information about change in displacement over time, $\frac{\Delta x}{\Delta t}$, making it a *scalar*. The magnitude of velocity is speed, and velocity is speed with a given direction (north, 48° south-east, etc).

The speedometer, however, tells us *instantaneous* speed, such that $\vec{\mathbf{v}} = \frac{d\mathbf{s}}{dt}$, the *derivative* of position. This gives us an instantaneous change in position, and as the $dt$, or difference in time, decreases, we get closer to the instantaneous speed.

Similarly, acceleration, $\mathbf{a}$, is the *instantaneous* change in velocity per difference in time, $dt$. This allows us to determine the instantaneous velocity using calculus: $\mathbf{a} = \frac{d\mathbf{v}}{dt}$ or equivalently, $\mathbf{a} = \frac{d^2\mathbf{s}}{dt^2}$ using second derivatives. This book will reintroduce these concepts later, but for now, know that all three quantities are related.

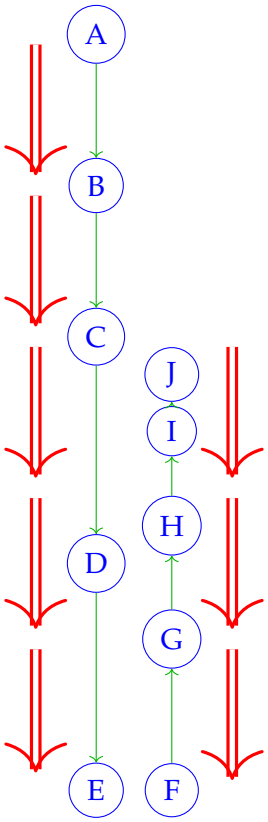
Typically, people will talk about velocity and acceleration at a *specific instant in time*, such as "at 3 seconds, the velocity is 5 m/s with a change of 2 m/s every second".

## 5.2 1D Motion Diagrams

Motion Diagrams are diagrams used to represent motion in a 1D plane, such as a ball being dropped from a height and the bouncing up to a fractional height. Imagine taking evenly spaced snapshots or blinking once every second as the ball changes position. A motion diagram captures this sequence, showing how the ball changes position over time, and whether it speeds up or slows down. A motion diagram contains usually 3 types of symbols:

- $\cdot$ for position representation
- $\rightarrow$ for velocity vectors
- $\Longrightarrow$ for acceleration vectors

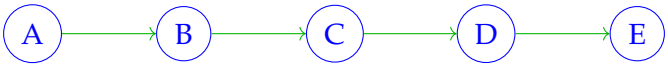
Below is the scenario of the ball being dropped and bouncing back up. Note that the upwards motion is shifted just for clarity. Here, red represents acceleration, green represents velocity, and blue represents equally spaced position dots.



Notice that in the above example, acceleration is constant throughout the motion, represented by the equally sized red arrows.

If there is no change in velocity, there is no acceleration vector. In this case, the would be no red arrow vectors.

Here is another motion diagram, representing an object moving forwards with constant velocity.



Chapter 6

# Force, Mass, and Acceleration

To talk about forces, we first need to understand the concept of forces. A force is a push or pull on an object which (normally) influences its direction or speed. Forces are *vectors*, which means they have a direction and a magnitude, and it means that forces can be added together, even when in opposite directions. This leads us to Newton's Laws of Motion, which describe the relationship between forces and motion.

## 6.1 Newton's First Law

Newton's First Law of Motion states that an object at rest will stay at rest, and an object in motion will stay in motion with a constant velocity (same speed and direction) unless acted upon by a *net external force*. This means that if there is no net force on an object, it will not *accelerate*.

For example, a car that stops pressing the gas on a flat road will eventually stop due to friction, which is a force that acts opposite to the direction of motion. If there were no friction, the car would continue moving at a constant speed forever.

This law is sometimes also referred to as the law of *inertia*, which is the tendency of objects to resist changes in their motion.

## 6.2 Mass, Acceleration, and Newton's Second Law

Each atom has a mass, which means everything made up of those atoms has mass as well (and that's pretty much everything!). We measure mass in grams. A paper clip is about 1 gram of steel. An adult human can have a mass of 70,000 grams, so for larger things, we often talk about kilograms, which is 1000 grams.

The first interesting thing about mass is that objects with more mass require more force to accelerate. For example, pushing a bicycle so that it accelerates from a standstill to jogging speed in 2 seconds requires much less force than pushing a train so that it accelerates at the same rate.

**Newton's Second Law of Motion**

The force necessary to accelerate an object of mass $m$ at an acceleration of $a$ is given

by:

$$F = ma$$

This is said as "the force is equal to the mass times the acceleration."

This is known as Newton's Second Law of Motion.[1]

What are the units here? We already know that mass is measured in kilograms. We can measure velocity in meters per second, but that is different from acceleration. So if we want to go from 0 to 5 meters per second (that's jogging speed) in two seconds, that is a change in velocity of 2.5 meters per second every second. We would say this acceleration is $2.5 m/s^2$.

### 6.2.1 Velocity versus Acceleration

Acceleration is the change in velocity. If an object is speeding up or slowing down, it is accelerating. In everyday language, we often use *decelerate* to indicate slowing down, but in physics you can use the word accelerate (slowing down is just negative acceleration). Since velocity is a vector (it has a magnitude and direction), changing direction is *also acceleration*. On the other hand, an object moving at a constant velocity (same speed, same direction) is *not accelerating*!

[1] This is a simplified version of Newton's Law of Gravitation. The formula $F = G\frac{m_1 m_2}{r^2}$ simplifies very close to $F = ma$ for any object within the Earth's atmosphere.

### *Exercise 10* Is it accelerating?

State whether the described object is accelerating or not.

1. A satellite orbiting the Earth at a constant speed.
2. A car moving due west at a constant 30 mile per hour.
3. A child coming to a stop on their bicycle.
4. A roller coaster going around a loop at a constant speed.
5. A roller coaster speeding up as it goes down the initial hill.
6. A book sitting on a table.

*Working Space*

*Answer on Page 1878*

It's a common misconception that all objects in motion are accelerating. If an object is moving with a constant velocity (same speed, same direction), then it is not accelerating.

### 6.2.2 Calculating Acceleration

When an object is speeding up or slowing down, we can calculate the acceleration by dividing the change in velocity by the time it takes to make that change.

**Calculating Acceleration**

The acceleration of an object from an initial velocity, $v_i$, to a final velocity, $v_f$, over a period of time, $t$, is given by:

$$a = \frac{v_f - v_i}{t}$$

Notice that if the velocity does not change, then $v_f - v_i = 0$ and the acceleration is also zero.

**Example**: Your car can go from zero to 60 mph in 3 seconds. What is the acceleration in $m/s^2$?

**Solution**: First, let's convert from the imperial units of miles per hour to the SI units of meters per second. You can do this using a search engine, but we will show how to do it by hand below. (You will learn more about this method in the Units chapter).

$$\frac{60 \text{ miles}}{1 \text{ hour}} \cdot \frac{1.61 \text{ km}}{1 \text{ mile}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ hour}}{3600 \text{ seconds}} \approx \frac{26.82 \text{ m}}{s}$$

Now we have the starting velocity (0 m/s), the ending velocity (26.82 m/s), and the time (3 s), and we can find the acceleration:

$$a = \frac{v_f - v_i}{t} = \frac{26.82 \frac{m}{s} - 0 \frac{m}{s}}{3s} \approx 8.94 \frac{m}{s^2}$$

### 6.2.3 Determining Force

What about measuring force? Newton decided to name the unit after himself: The force necessary to accelerate one kilogram at $1 m/s^2$ is known as *a newton*. It is often denoted by the symbol N.

$$1N = 1 \frac{kg \cdot m}{s^2}$$

**Example**: If the car in the above example has a mass of 1500 kg, how much force does the engine use to accelerate the car?

**Solution**: We have already found the car's acceleration: 8.94 $m/s^2$. With the mass and acceleration, we can use Newton's Second Law to find the force needed to accelerate the car:

$$F = m \cdot a = 1500 \text{ kg} \cdot 8.94 \frac{m}{s^2} = 13410 \text{ N}$$

### *Exercise 11* Acceleration

*Working Space*

While driving a bulldozer, you come across a train car (with no brakes and no locomotive) sitting on a track in the middle of a city. The train car has a label telling you that it has a mass of 2,400 kg. There is a time-bomb welded to the interior of the train car, and the timer tells you that you can safely push the train car for 120 seconds. To get the train car to where it can explode safely, you need to accelerate it to 20 meters per second. Fortunately, the track is level and the train car's wheels have almost no rolling resistance.

With what force, in newtons, do you need to push the train for those 120 seconds?

*Answer on Page 1878*

## 6.3 Net Force

So far, we've looked at examples where only one force is acting on an object. In reality, there are usually multiple forces acting on an object. For example, the engine pushes your car forward while friction pulls it backwards. Or your chair is pushing up on you while gravity pulls you down. How then can we describe the motion of an object if more than one force is acting on it?

We can rearrange Newton's Law:

$$a = \frac{F_{net}}{m}$$

This means that an object's acceleration is directly proportional to the *net force* acting on the object and inversely proportional to the object's mass. The *net force* is the vector sum of all the forces acting on an object. The vector sum just means we have to take the direction of the force into account. Usually, up and right are positive while down and left are negative. You can see some examples in figure 6.1.

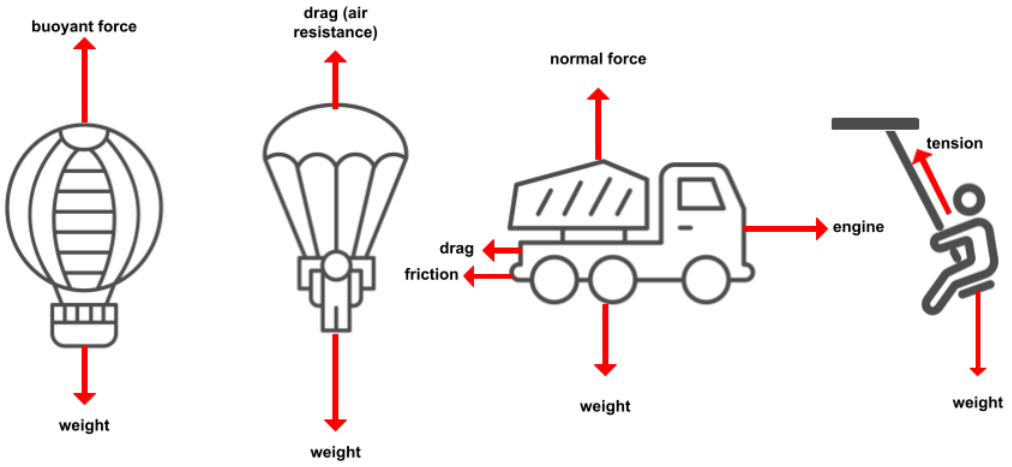
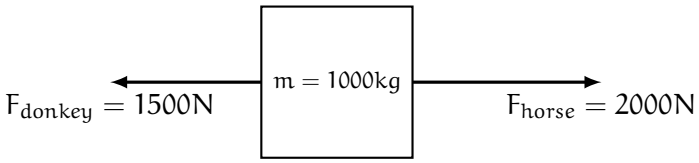



Figure 6.1: The hot air balloon has an upwards net force. The parachuter has a downwards net force. The truck has a rightward net force. The swinger has a leftward net force.

For now, we'll only look at parallel forces (up/down or left/right). You'll learn to use vector addition to combine orthogonal (at right angles) and skew (at angles other than parallel or right) forces in a later chapter.

**Example**: A donkey and a horse are each pulling a cart. The donkey pulls to the left with a force of 1500 N, while the horse pulls to the right with a force of 2000 N. What is the net force on the cart? If the cart has a mass of 1000 kg, in what direction and with what magnitude will the cart accelerate?

**Solution**: We begin by drawing a diagram (this is good practice - while a diagram may not be necessary for such a simple question, it will be very useful as we examine more complex scenarios in future chapters).




It is customary to take right as positive, so the horse applies a positive force while the donkey applies a negative force to the cart. Therefore, the net force is:

$$F_{net} = F_{horse} - F_{donkey} = 2000N - 1500N = 500N$$

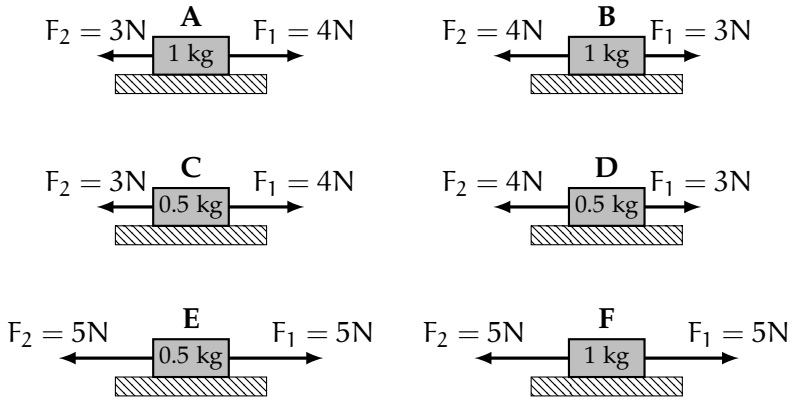
Since the net force is positive, it points to the right and the cart will accelerate to the right at a rate of:

$$a = \frac{F_{net}}{m} = \frac{500N}{1000kg} = 0.5\frac{m}{s^2}$$

## *Exercise 12* Net Force and Acceleration

*Working Space*

Rank the acceleration of the boxes shown below from greatest to least. All surfaces are frictionless and each box starts at rest. Take left as negative, and negative accelerations are less than a zero acceleration.

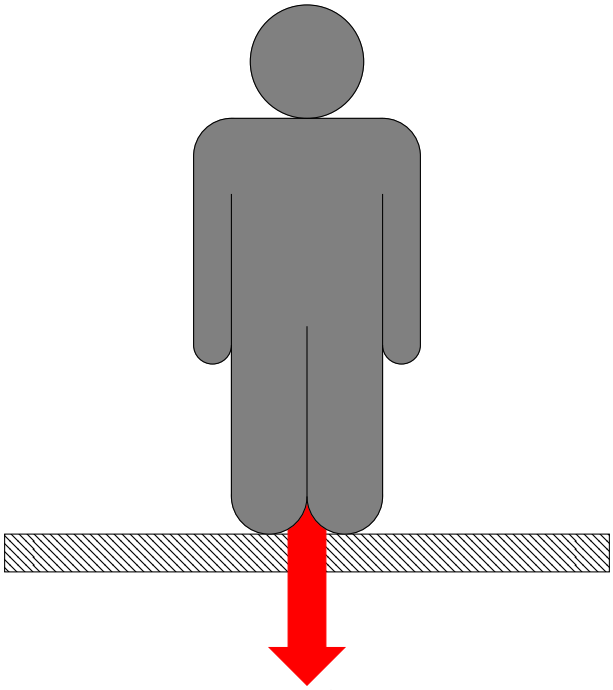



*Answer on Page 1878*

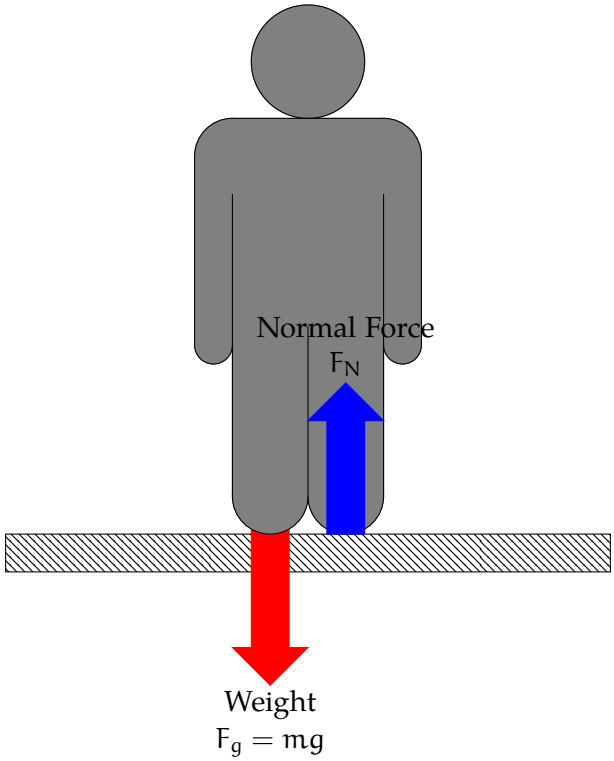
## 6.4 Normal Force and Apparent Weight

If you were to stand on a scale while riding an elevator, you would see the scale fluctuate as the elevator accelerates up and down (if you have a bathroom scale and live in a building with an elevator, you can try this yourself!). The reading on the scale is your *apparent weight* and is equal to the normal force between you and the floor.

What is a *normal force*? First, the word "normal" doesn't have the colloquial meaning of average or usual. In mathematics, "normal" means perpendicular. A normal force is perpendicular to the *contact* surface between two objects. Let's look at a person standing at rest. We know that gravity is pulling down on the person:



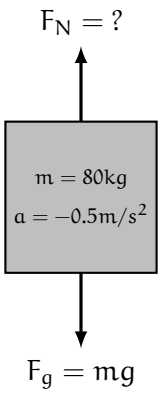

Since the person is not accelerating, there must be some other force acting on them in the upwards direction that balances out the person's weight (recall that if $a = 0 m/s^2$, then it must be true that $F_{net} = 0N$). In this case, the balancing force is the *normal force* of the floor pushing up on the person's feet:




When you step on a bathroom scale, it is actually measuring the normal force! When you are not accelerating, your weight and the normal force between you and the scale are equal, so the scale gives you an accurate measure of your weight. Let's look at what happens if you are accelerating, as in an elevator.

**Example**: Maria has a mass of 80. kg. If the elevator in her apartment building initially accelerates at 0.50 $m/s^2$, what is her apparent weight as the elevator begins to move down?

**Solution**: We begin with a diagram:




Her apparent weight is the normal force. Seeing that the net force acting on Maria is given

by $F_N - F_g$, we can use Newton's Second Law to find $F_N$:

$$F_{net} = F_N - F_g = ma$$
$$F_N - (80\text{kg})\left(9.8\frac{m}{s^2}\right) = (80\text{kg})\left(-0.5\frac{m}{s^2}\right)$$
$$F_N = (80\text{kg})\left(9.8\frac{m}{s^2} - 0.5\frac{m}{s^2}\right) = 744\text{N}$$

Maria's apparent weight is 744 newtons when the elevator is accelerating downwards.

## *Exercise 13* Moving Elevators

A 700-N person is standing on a scale in an elevator. Rank the reading on the scale from greatest to least for the following scenarios:

1. The elevator has an initial upward velocity of 2 m/s and an upward acceleration of 3 $m/s^2$
2. The elevator is initially at rest and has an upward acceleration of 3 $m/s^2$
3. The elevator has an initial downward velocity of 2 m/s and an upward acceleration of 5 $m/s^2$
4. The elevator is initially at rest and has a downward acceleration of 9.8 $m/s^2$
5. The elevator has an initial upward velocity of 2 m/s and a downward acceleration of 5 $m/s^2$
6. The elevator has an initial downward velocity of 2 m/s and is not accelerating

*Working Space*

*Answer on Page 1879*

Let's talk about the effect of normal force on the net force. On standard flat ground, the Normal Force will be equal to the gravitational force (on Earth), assuming the object is stationary. This normal force also influences another contact force, friction, which we will talk about in the next chapter. If an object is on an incline such as a ramp, the normal force is *perpendicular* to the ramp.

## 6.5 Newtons and Joules

You now know the fundamental units for newtons and joules, which hints at the relationship between force and energy:

$$N = \frac{kg \cdot m}{s^2} \quad J = \frac{kg \cdot m^2}{s^2} = N \cdot m$$

A joule is equivalent to a newton times a meter. So, multiplying a force (newtons) by a displacement (meters) tells you how the energy (joules) of the object changes. This relationship is described by the Work-Kinetic Energy Theorem, which we will explore in the next chapter.

## 6.6 Newton's Third Law

Now we can discuss Newton's Third Law, which states that every action has an equal but opposite reaction. In other words, if body A exerts a force on body B, then body B simultaneously exerts a force of equal magnitude and opposite direction on body A. Mathematically, we can express this as:

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$$

Any time we use Newton's Second Law, we can combine this with the Third Law to state that *the sum of the forces equals zero* in some action-reaction pair, assuming that the system is isolated. For example, if two boxes (1 and 2) are applying a force on each other, we can say that (no matter their mass values), $F_{\text{Box 1 on 2}} = -F_{\text{Box 2 on 1}}$. This applies to physics concepts like air resistance and drag, tension forces of conjoined boxes or train cars, and gravitation of planets, all of which we will discuss in future Sequences.

Note the following:

- Action-reaction pairs do *not* cancel when applying Newton's Second Law to a single body.
- The forces are always of the same type (both contact, both gravitational, etc).
- They occur simultaneously; there is no time delay.

Like we said above, gravity and the normal force are usually action-reaction pairs. Similarly, a force applied on a box always has an equal but opposite force on the root cause of the force. Two objects connected by a massless rope have the same tension force on each other ($F_{1 \text{ on } 2} = -F_{2 \text{ on } 1}$) **Example:** Consider a swimmer pushing the wall of a pool. What forces are in play?

**Solution:** While the swimmer exerts a force $\mathbf{F}_{\text{wall}}$ on the wall, the wall exerts the reaction $-\mathbf{F}_{\text{wall}}$ on the swimmer, propelling the swimmer forward. Although the wall does not move due to mass and acceleration differences, the forces are equal and opposite.

CHAPTER 7

# Friction

Imagine there is a large and heavy steel box resting in the middle of a floor, and you push it hard enough to get it moving. If you stop pushing, will it continue to glide gracefully across the floor?

Probably not. Unless the floor is very slippery for some reason, the box will come to a halt immediately after you stop pushing. We would say that it is stopped by the force of *friction*.

What is really happening? The kinetic energy of the box is being converted into heat between the bottom of the box and floor. As the bottom of the box and the floor get warmer, the speed of the box decreases.

The amount of friction is proportional to the force with which the box is pressing against the floor — so you should expect a box that is twice as heavy to experience twice as much frictional force.

In other words, the frictional force is proportional to the normal force. On a level floor, the normal force is parallel and equal to the force of gravity acting on the box. On a slope, the normal force points perpendicular to the surface of the slope. We will be talking about sloped surfaces in a later chapter, so focus on level surfaces for now.

(FIXME: picture here)

The amount of friction is also determined by the materials that are sliding against each other. For example, if the floor is ice, the frictional force will be less than if the floor is made of wood. Written mathematically, we can express the force of friction as

$$F_f = \mu N$$

where $F_f$ is the force of friction, N is the normal force (you may see written as R , $F_n$ , or just N), and $\mu$ is a coefficient that depends on the materials in contact with each other.

If you are pushing the box with a force of F and it is moving but neither accelerating nor decelerating, then the force you are applying is exactly balanced by the frictional force. If the box is pressing against the floor with a force of N, then we say the *coefficient of friction* is given by the ratio between the weight of the steel box and the floor:

$$\mu = \frac{F_f}{N}$$

## *Exercise 14* Bicycle Stopping

*Working Space*

You are riding your bicycle on a flat, horizontal at 11 meters per second when you suddenly slam on the brakes and lock up the wheels.

You weigh 55 kg.

When any piece of rubber is skidding across a dry road, the coefficient of friction will be about 0.7.

Answer the following questions:

- How much kinetic energy do you have when you engage the brakes?
- As you skid, how much frictional force is decelerating you?
- For how many meters will you slide?

*Answer on Page 1879*

Notice that the force of friction is not determined by how much of the tire is touching the ground. The coefficient of friction of the two materials and the normal force is all you need to compute the friction.

## 7.1 Static vs Kinetic Friction Coefficients

Let's return to the box on the floor we discussed earlier. As you start to push it, it will sit still until your force is greater than the force of friction. However, once it starts moving, the force of friction seems to be less.

Between the two materials, there are actually two different friction coefficients:

- Static friction coefficient: The coefficient you use to figure out how much force you need to get the box to start to move.
- Kinetic friction coefficient: The coefficient you use once the box is sliding against the floor.

The kinetic friction coefficient is always less than the static friction coefficient:

- *Static*, $\mu_s$: When the car is parked with its brakes on, it has a friction coefficient of about 1.0.
- *Kinetic*, $\mu_k$: For a car skidding on a dry road, the friction coefficient is about 0.7.

It is important to note once the force of static friction is overcome, there will be a force of kinetic friction less than the static friction force.

## *Exercise 15* Rocket Sled

*Working Space*

You built a rocket sled with steel runners that is resting on a flat, level wooden floor. The sled weighs 50 kg and you weigh 55 kg.

Before you get on the sled, you try pushing it around the floor. You find that you can get it to move from a standstill if you push it with a force of 270 N. Once it is moving, you can keep it moving at the same speed using a force of 220 N.

**What are $\mu_s$ and $\mu_k$ of your sled's runners on your wooden floor?**

Next, you get on the sled and gradually increase the thrust of the rocket mounted on the sled until it starts to move. You then keep the thrust constant.

**How much force was the rocket exerting on you and the sled when it started to move?**

**How fast do you accelerate now that the sled is moving?**

*Answer on Page 1879*

## 7.2 Skidding and Anti-Lock Braking Systems

When a car goes through a curve, the friction of the tire on the road is what changes the direction of the car's travel. Even though the wheel is turning, this is the *static friction coefficient* because the surface of the tire is not sliding across the road. The part of the tire touching the road at any instant is at rest relative to the road. The force of friction is the so-called *centripetal force* keeping the object on the road.

If you go into the curve too fast, the tire may not have enough friction to turn the car. In this case the car will start to slide sideways. Now, the friction between the tire and road uses the kinetic coefficient. In other words, you have significantly less friction than you had before you started to skid.

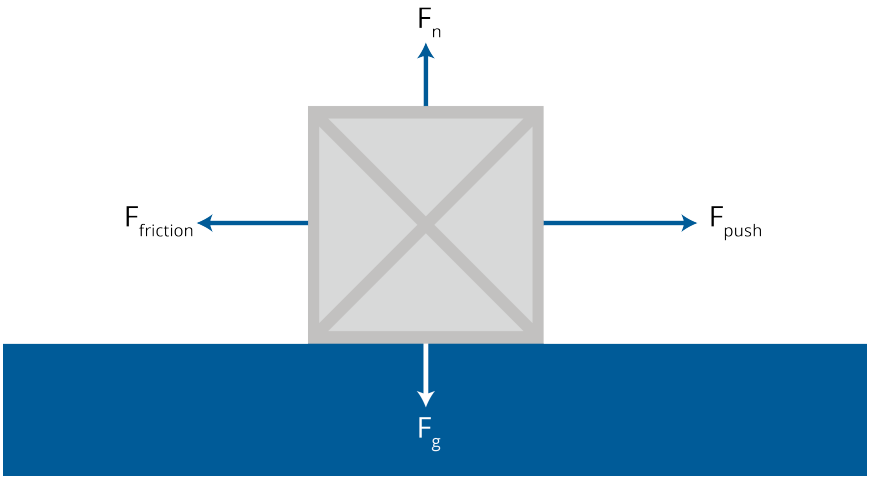
When you are driving a car, the force of friction that your tires create is your friend. It lets you steer, accelerate, and stop.

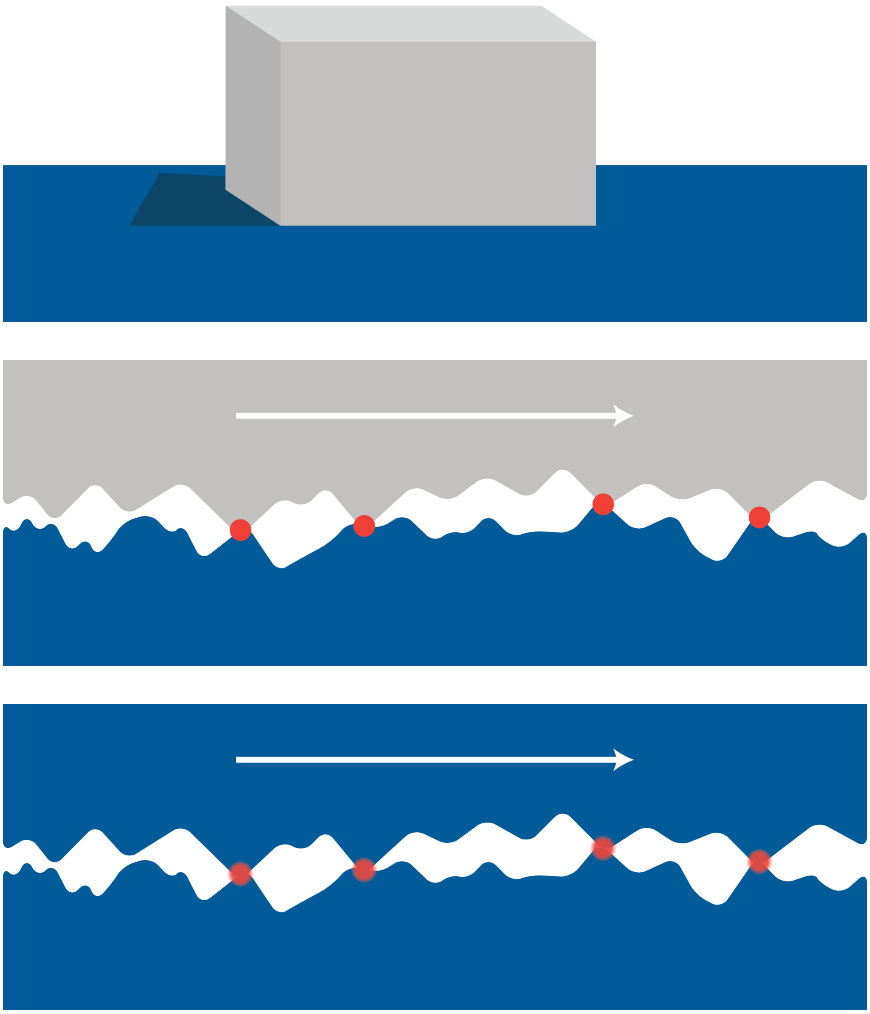
In older cars, if you panicked and slammed on the brakes, you would probably lock up the wheels: they would stop turning suddenly. And the surface of the tire would begin to slide across the pavement. At that moment, two problems occurred:

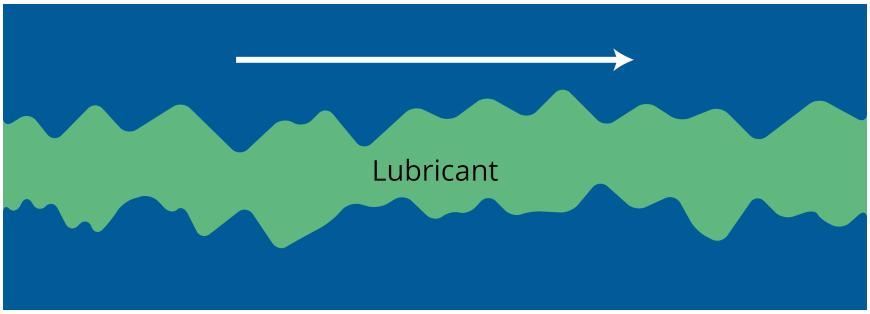
- You don't stop as quickly, because now the friction between your tires and the road is based on the kinetic friction coefficient instead of the static friction coefficient.
- You can't steer the car. Steering only happens because the wheels are turning in a particular direction.

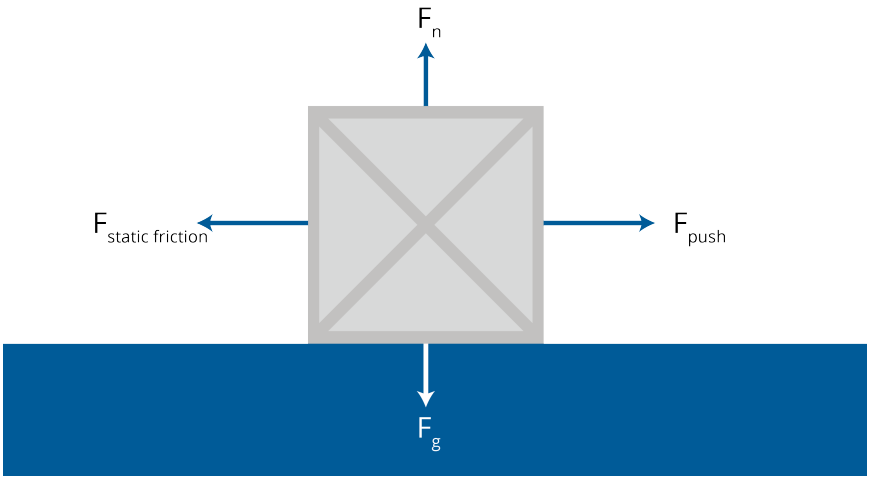
To prevent this problem, car companies developed the anti-lock brake system, or ABS.

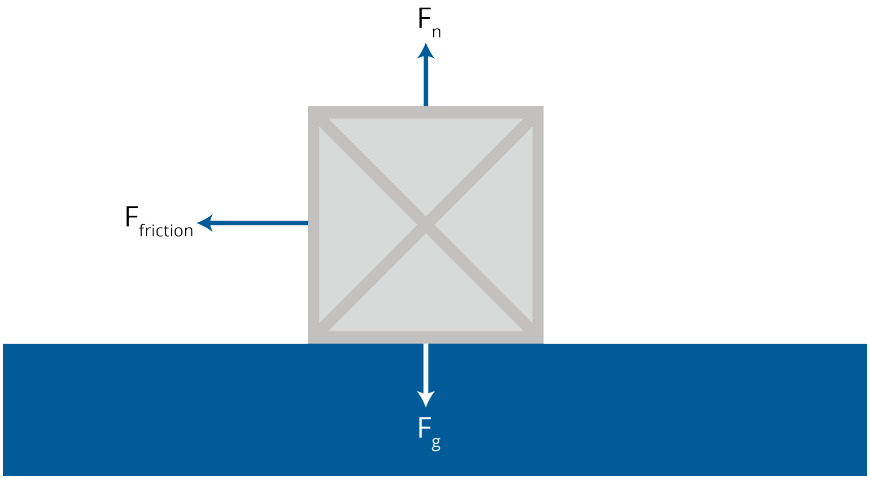
FIXME: More here.






Lubricant


F n
F static friction
F push
F g


F n
F friction
F g

CHAPTER 8

# Units and Conversions

Accurate measurements are at the heart of good data and good problem solving. Engineers need to be able to describe many different types of phenomena, such as distance, sound, light, force, and more.

At this point, you are working with a lot of units: grams for weight, joules for energy, newtons for force, meters for distance, seconds for time, and so on. For each type of measurement, there are several different units. For example, distance can be measured in feet, miles, and light-years.

**Some Equivalencies**

| | |
|---|---|
| **Distance** | |
| 1 mile | 1.6093 kilometers |
| 1 foot | 0.3048 meters |
| 1 inch | 2.54 centimeters |
| 1 light-year | $9.461 \times 10^{12}$ kilometers |
| **Volume** | |
| 1 milliliter | 1 cubic centimeter |
| 1 quart | 0.9461 liters |
| 1 gallon | 3.7854 liters |
| 1 fluid ounce | 29.6 milliliters |
| **Mass** | |
| 1 pound | 0.4535924 kilograms |
| 1 ounce | 0.4535924 grams |
| 1 metric ton | 1000 kilograms |
| **Force** | |
| 1 newton | 1 kilogram meter per $\text{sec}^2$ |
| **Pressure** | |
| 1 pascal | 1 newton per square meter |
| 1 bar | 0.98692 atmosphere |
| 1 pound per square inch | 6897 pascals |
| **Energy** | |
| 1 joule | 1 newton meter |
| 1 calorie | 4.184 joules |
| 1 kilowatt-hour | $3.6 \times 10^6$ joules |

(You don't need to memorize these! Just remember that this page is here.)

In the metric system, prefixes are often used to express a multiple. Here are the common prefixes:

**Common Prefixes for Metric Units**

| | |
|---|---|
| giga | $\times 10^9$ |
| mega | $\times 10^6$ |
| kilo | $\times 10^3$ |
| milli | $\div 10^3$ |
| micro | $\div 10^6$ |
| nano | $\div 10^9$ |

(These are worth memorizing. Here's a mnemonic for the most common ones: "King Henry Doesn't Usually Drink Chocolate Milk" Or Kilo ($\times 10^3$), Hecto ($\times 10^2$), Deca ($\times 10^1$), Unit (for example: gram) ($\times 10^0$), Deci ($\times 10^{-1}$), Centi ($1 \times 0^{-2}$), Milli ($\times 10^{-3}$).)

## 8.1 Conversion Factors

Here is a really handy trick to remembering how to do conversions between units.

Often, you will be given a table like the one above, and someone will ask you "How many miles are in 0.23 light-years?" You know that 1 mile = 1.6093 kilometers and that 1 light-year is $9.461 \times 10^{12}$ kilometers. How do you do the conversion?

The trick is to treat the two parts of the equality as a fraction that equals 1. In other words, you think:

$$\frac{1 \text{ miles}}{1.6093 \text{ km}} = \frac{1.6093 \text{ km}}{1 \text{ miles}} = 1$$

and

$$\frac{1 \text{ light-years}}{9.461 \times 10^{12} \text{ km}} = \frac{9.461 \times 10^{12} \text{ km}}{1 \text{ light-years}} = 1$$

We call these fractions *conversion factors*.

Now, your problem is

$$0.23 \text{ light-years} \times \textit{ Some conversion factors} = ? \text{ miles}$$

Note that when you multiply fractions together, things in the numerators can cancel with things in the denominator:

$$\left(\frac{31\pi}{47}\right)\left(\frac{11}{37\pi}\right) = \left(\frac{31\cancel{\pi}}{47}\right)\left(\frac{11}{37\cancel{\pi}}\right) = \left(\frac{31}{47}\right)\left(\frac{11}{37}\right)$$

When working with conversion factors, you will do the same with the units:

$$0.23 \text{ light-years}\left(\frac{9.461 \times 10^{12} \text{ km}}{1 \text{ light-years}}\right)\left(\frac{1 \text{ miles}}{1.6093 \text{ km}}\right) =$$
$$0.23 \cancel{\text{light-years}}\left(\times\frac{9.461 \times 10^{12} \cancel{\text{km}}}{1 \cancel{\text{light-years}}}\right)\left(\frac{1 \text{ miles}}{1.6093 \cancel{\text{km}}}\right) = \frac{(0.23)(9.461 \times 10^{12})}{1.6093} \text{ miles}$$

## *Exercise 16* Simple Conversion Factors

*Working Space*

How many calories are in 4.5 kilowatt-hours?

*Answer on Page 1880*

## 8.2 Conversion Factors and Ratios

Conversion factors also work on ratios. For example, if you are told that a bug is moving 0.5 feet every 120 milliseconds, what is that in meters per second?

The problem then is

$$\frac{0.5 \text{ feet}}{120 \text{ milliseconds}} = \frac{? \text{ m}}{\text{second}}$$

So you will need conversion factors to replace the "feet" with "meters" and to replace "milliseconds" with "seconds":

$$\left(\frac{0.5 \cancel{\text{ feet}}}{120 \cancel{\text{ milliseconds}}}\right)\left(\frac{0.3048 \text{ meters}}{1 \cancel{\text{ feet}}}\right)\left(\frac{1000 \cancel{\text{ milliseconds}}}{1 \text{ second}}\right) = \frac{(0.5)(0.3048)(1000)}{120} \text{ m/second}$$

### *Exercise 17* Conversion Factors

*Working Space*

The hole in the bottom of the boat lets in 0.1 gallons every 2 minutes. How many milliliters per second is that?

*Answer on Page 1880*

## 8.3 When Conversion Factors Don't Work

Conversion factors only work when the units being converted are proportional to each other. Gallons and liters, for example, are proportional to each other: If you have $n$ gallons, you have $n \times 3.7854$ liters.

Degrees celsius and degrees fahrenheit are *not* proportional to each other. If your food is $n$ degrees celsius, it is $n \times \frac{9}{5} + 32$ degrees fahrenheit. You can't use conversion factors to convert celsius to fahrenheit.

## 8.4 SI units

In standardized sciences and studies, there has a set of units for all scientists to commonly use. This is done so that scientific research, other papers, and different international communities can easily communicate their research with one another. This system is referred to as **SI Units**, or the **international system of units**.

Here is the very basic table for SI units:

| | |
|---|---|
| Time | second (s) |
| Length | meter (m) |
| Mass | kilogram (kg) |
| Electric Current | ampere (A) |
| Temperature | Kelvin (K) |
| Number of substance | mole (mol) |
| Luminous intensity | candela (cd) |

All other units can be derived from these 7 base units from the SI system.

Often, you will see units represented in the form of exponents. Let's look at velocity as an example. Velocity is often described as **meters** *per* **second**. This can be written as a fraction, or with seconds to a negative power:

$$v = \frac{m}{s} = m \cdot s^{-1}$$

Similarly, we can represent acceleration (meters per second per second) with exponents

$$a = \frac{m}{s^2} = m \cdot s^{-2}$$

### 8.4.1 Converting from SI units

There are many common units that are often used in place of SI units. Here is another table.

| Physical Quantity | Name | Symbol | Expression in Base Units |
|---|---|---|---|
| Force | Newton | N | $kg \cdot m \cdot s^{-2}$ |
| Energy/Work | Joule | J | $kg \cdot m^2 \cdot s^{-2}$ |
| Power | Watt | W | $kg \cdot m^2 \cdot s^{-3}$ |
| Frequency | Hertz | Hz | $s^{-1}$ |
| Electric Charge | Coulomb | C | $A \cdot s$ |

These units can all be derived manually, and it is good to know how to! Once you learn acceleration and force, try deriving the units for Newtons yourself!

# Chapter 9

# The Greek Alphabet

If you do anything involving math or physics, you will be encountering Greek letters on a regular basis. Here is a table for your reference:

| Capital | Lower | Pronounced | Capital | Lower | Pronounced |
|---|---|---|---|---|---|
| A | α | Alpha | N | ν | Nu |
| B | β | Beta | Ξ | ξ | Xi (”ku-ZY”) |
| Γ | γ | Gamma | O | o | Omicron |
| Δ | δ | Delta | Π | π | Pi |
| E | ϵ | Epsilon | P | ρ | Rho |
| Z | ζ | Zeta | Σ | σ | Sigma |
| H | η | Eta | T | τ | Tau |
| Θ | θ | Theta | Υ | υ | Upsilon |
| I | ι | Iota | Φ | ϕ | Phi |
| K | κ | Kappa | X | χ | Chi (”Kai”) |
| Λ | λ | Lambda | Ψ | ψ | Psi (”Sigh”) |
| M | μ | Mu | Ω | ω | Omega |

## 9.1 Greek Letters in Math and Science

These are important to know, for a large variety of reasons. Let’s look at the most common physics and math uses:

| Symbol | Common meaning |
|---|---|
| π | circle constant, approximately 3.14159 |
| θ | an angle |
| Δ | change in a quantity |
| Σ | sum of many terms |
| λ | wavelength |
| μ | coefficient of friction, mean, or micro- |
| ρ | density |
| ω | angular velocity |
| Ω | ohms, a unit of electrical resistance |

You will see a lot of equations that reference Greek letters, using them as *placeholders*, *constants*, or *variables* for an unknown quantity. For example, there area of a circle is given by

$$A = \pi r^2$$

where $r$ is the radius of the circle. In this case, $\pi$ is a constant, and $r$ is a variable. In other cases, you might see the physics equation

$$\Delta x = v\Delta t + \frac{1}{2}a\Delta t^2$$

This is the equation for the change in position under constant acceleration, where $\Delta x$ is the change in position, $v$ is the initial velocity, $a$ is the acceleration, and $\Delta t$ is the change in time. In this case, $\Delta x$, $v$, $a$, and $\Delta t$ are all variables.

One final example that we will explore later is the equation for wavelength in terms of frequency and the speed of light:

$$\lambda = \frac{c}{f}$$

where $\lambda$ is the wavelength, $c$ is the speed of light, and $f$ is the frequency. In this case, $\lambda$ and $f$ are variables, while $c$ is a constant. We can then translate lambda into other equations as needed.

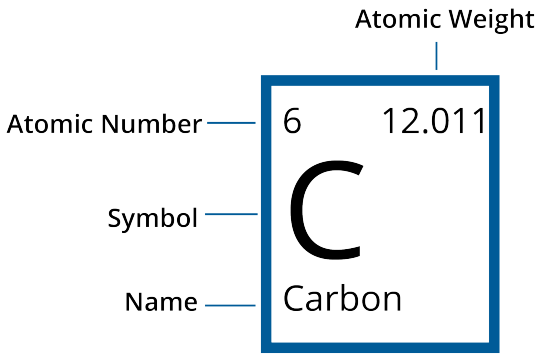
Generally, we will define the Greek letters when providing an equation. However, it is important to be familiar with the Greek alphabet, so that you can understand what the symbols mean when you see them in equations.

Chapter 10

# Atomic and Molecular Mass

## 10.1 Reading a Periodic Tile

Let's look at the different information shown on a periodic tile:

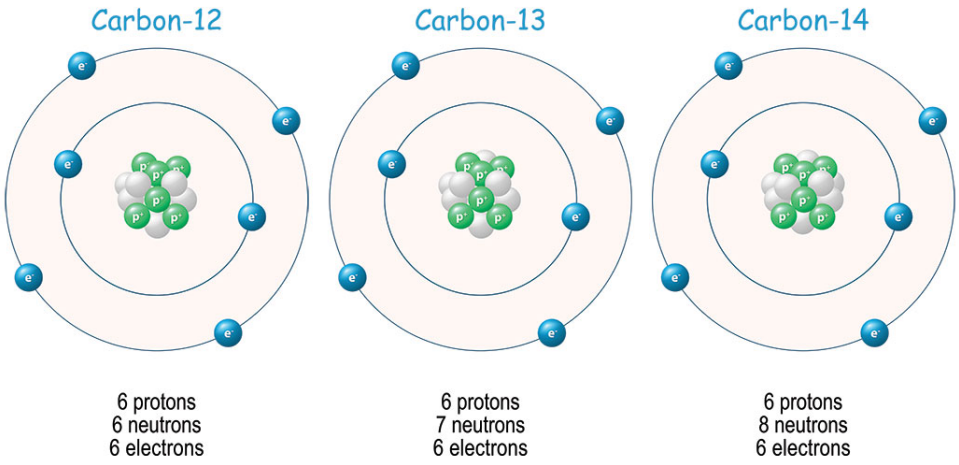



The four things we learn from a periodic tile are:

1. the symbol: as discussed in the previous chapter, each element has a unique symbol. Element symbols are used when showing the structure of a molecule and modeling chemical reactions.
2. the atomic number: this is also unique for each element. Take a look at the periodic table a few pages forward. Every tile has a unique atomic number, and the tiles are laid out in a generally increasing atomic number (you'll learn why the periodic table is arranged this way in Sequence 2).
3. the atomic weight: this is the average mass of all the atoms of that element in existence. Just like your overall grade in a class is the weighted average of all the individual grades you earned, atomic weight is the weighted average of the masses of all the individual atoms of that element. This is also sometimes referred to as atomic mass.

4. the name: not all periodic tables show the name of an element on its tile. This is why it is useful to know the symbols of common elements.

Recall from the previous chapter that we classify atoms by the number of protons they have. What this means is that if we want to know what element an atom is, we have to look at the number of protons. Take a look at the three carbon atoms below and note what is the same and what is different among them:




These different versions of carbon all have 6 protons, which is also carbon's atomic number. This isn't a coincidence: the atomic number *is* the number of protons in every atom of an element. If I tell you an atom has 4 protons, you would find atomic number 4 and see that the element is beryllium. To know how many protons an oxygen atom has, you would find its tile and see that it has atomic number 8.

Ok, so now we know atoms of the same element have the same number of protons, and that number is given by the element's atomic number. The difference between these carbon atoms explains the other number on a periodic tile: the atomic mass.

A proton and a neutron have about the same mass. An electron, on the other hand, has much less mass: One neutron weighs about the same amount as 2000 electrons. This means that the mass of any object comes mostly from the protons and neutrons in the nucleus of its atoms.

We know how many protons an atom has by what element it is, but how do we know the number of neutrons?

## 10.2 Mass of Atoms and Molecules

As you've seen, a periodic tile for an element tells us the average mass of an atom of that element in Daltons or amus (atomic mass units). The average mass of a carbon atom is 12.011 amu, and the average mass of an iron atom is 55.845 amu. Using the periodic table, determining the average mass of an atom is straightforward. What about molecules?

Consider water: $H_2O$. It is made of 2 hydrogen atoms, each with an average mass of 1.008 amu, and one oxygen atom, with an average mass of 15.999 amu. To find the mass of the molecule, called *molecular mass*, you simply add the masses of each of the atoms in the molecule. So, the molecular mass of water is 1.008 amu + 1.008 amu + 15.999 amu = 18.015 amu.

### *Exercise 18* Determining Molecular Mass

Find the molecular mass, in amu, of the following substances:

*Working Space*

1. $CH_4$
2. $CuSO_4$
3. $C_6H_{12}O_6$

*Answer on Page 1880*

## 10.3 Mole Concept

An atomic mass unit is a very, very, very small unit; we would much rather work in grams. Grams are a large enough unit that you can develop a natural sense for how much a gram is. Additionally, while you can't see a single carbon atom with your eyes, you can see 10 grams of carbon (about enough to fill a pen cap). To convert between the very, very, very small unit of amus to the tangible unit of grams, we use *Avogadro's Number* (sometimes called *Avogadro's Constant*).

Since 1 amu is defined as $1/12^{\text{th}}$ of the mass of a carbon-12 atom, carbon-12 by definition has a mass of 12 amu. Additionally, Avogadro's number is the number of carbon atoms in 12.000 grams of pure carbon-12. This amount is called *a mole*. If you have 12 doughnuts, that's a dozen doughnuts. If you have 20 donuts, you have a score of donuts. 500 donuts: a ream of donuts. If you have $6.02214076 \times 10^{23}$ doughnuts, you have a *mole* of doughnuts.

This isn't really a practical measurement, as a mole of doughnuts would be about the size of the earth. We use moles for small things like molecules. However, a mole is a not an abbreviation for a molecule. For a better idea about how large of a number Avogadro's number is, you can watch this video: https://www.youtube.com/watch?v=TEl4jeETVmg. A mole of carbon-12 has a mass of 12.000 g, but a mole of natural carbon (which includes all the isotopes of carbon) has a mass of 12.011 g. The mole is defined such that one mole of an element is the same mass in grams as one atom is in amus. Let's say you want to know how much a mole of NaCl weighs. From the periodic table, you see that Na has an atomic mass of 22.990 atomic mass units, and Cl has 35.453 atomic mass units. One atom of NaCl has a mass of $22.990 + 35.453 = 58.443$ atomic mass units. This means a mole of NaCl has a mass of 58.443 grams. Handy, right? This is called the *molar mass*. It is the mass of one mole of a substance, and is given in units of g/mol (grams per mole). The molar mass of NaCl is 58.443 g/mol. The molar mass of carbon is 12.011 g/mol. Using dimensional analysis and the molar mass, you can determine the mass of a given number of moles of a substance.

**Example**: What is the mass of 2 moles of copper?

**Solution**: The conversion we will use is 1 mol Cu = 63.546 g Cu.

$$\frac{2 \text{ mol Cu}}{} \times \frac{63.546 \text{ g Cu}}{1 \text{ mol Cu}} = 127.092 \text{ g Cu}$$

Therefore, 2 moles of copper has a mass of 127.092 grams.

You can also find the molar mass of a molecule, like methane. Just like with elements, a mole of a molecule has the same mass in grams as a single molecule has in amus.

**Example**: What is the mass of 3.5 moles of methane?

**Solution**: Methane ($CH_4$) has a molecular mass of 16.043 amu, which means 1 mole of methane has a mass of 16.043 grams.

$$\frac{3.5 \text{ mol } CH_4}{} \times \frac{16.043 g CH_4}{1 \text{ mol } CH_4} = 56.151 \text{ g } CH_4$$

You can also use the molar mass to determine how many moles of a substance there are in a given mass of that substance.

**Example**: A standard AAA battery contains about 7.00 g of zinc. How many moles of zinc are in a AAA battery?

**Solution**: Zinc's molar mass is 65.38 g/mol.

$$\frac{7.00 \text{ g Zn}}{} \times \frac{1 \text{ mol Zn}}{65.38 \text{ g Zn}} \approx 0.107 \text{ g Zn}$$

In summary, a mole of a substance contains approximately $6.02 \times 10^{23}$ particles (atoms or molecules) of that substance and has a mass equal to the molecular mass in grams.

**The Mole Concept**

For a substance, X, with a molar mass of x g/mol,

$$1 \text{ mol X} = 6.02 \times 10^{23} \text{ particles of X} = \text{x g of X}$$

## *Exercise 19* Grams, Moles, Molecules, and Atoms

*Working Space*

Complete the table.

| Substance | num. of particles | num. of moles | grams |
|---|---|---|---|
| $NaHCO_3$ | | | 35 |
| HCl | | 1.2 | |
| $KH_2PO_4$ | $12.5 \times 10^{24}$ | | |

*Answer on Page 1881*

## *Exercise 20* Burning Methane

*Working Space*

Natural gas is mostly methane ($CH_4$). When one molecule of methane burns, two oxygen molecules ($O_2$) are consumed. One molecule of $H_2O$ and one molecule of $CO_2$ are produced.

If you need 200 grams of water, how many grams of methane do you need to burn?

(This is how the hero in "The Martian" made water for his garden.)

*Answer on Page 1881*

If you fill a balloon with helium, it will have two different kinds of helium atoms. Most of the helium atoms will have 2 neutrons, but a few will have only 1 neutron. We say that these are two different *isotopes* of helium. We call them helium-4 (or $^4He$) and helium-3 (or $^3He$). Isotopes are named for the sum of protons and neutrons the atom has: helium-3 has 2 protons and 1 neutron.

A hydrogen atom nearly always has just 1 proton and no neutrons. A helium atom nearly always has 2 protons and 2 neutrons. So, if you have a 100 hydrogen atoms and 100 helium atoms, the helium will have about 4 times more mass than the hydrogen. We say "Hydrogen is about 1 atomic mass unit (amu), and helium-4 is about 4 atomic mass units."

What, precisely, is an atomic mass unit? It is defined as 1/12 of the mass of a carbon-12 atom. Scientists have measured the mass of helium-4, and it is about 4.0026 atomic mass

units. (By the way, an atomic mass unit is also called a *dalton*.)

Now you are ready to take a good look at the periodic table of elements. Here is the version from Wikipedia:

# Periodic Table of Elements

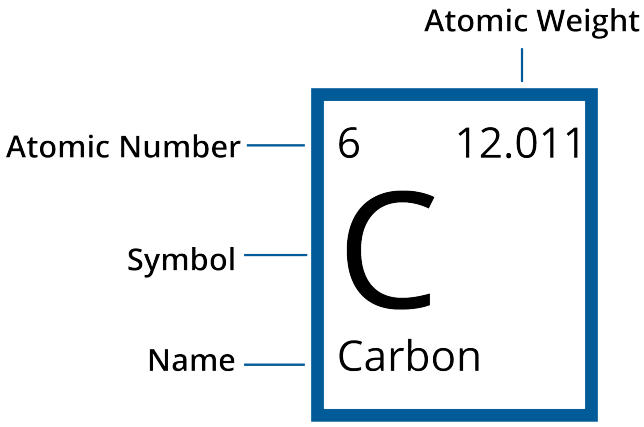
| IA | IIA | IIIB | IVB | VB | VIB | VIIB | VIIIB | VIIIB | VIIIB | IB | IIB | IIIA | IVA | VA | VIA | VIIA | VIIIA |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 **H** Hydrogen 1.01 | | | | | | | | | | | | | | | | | 2 **He** Helium 4.00 |
| 3 **Li** Lithium 6.94 | 4 **Be** Beryllium 9.01 | | | | | | | | | | | 5 **B** Boron 10.81 | 6 **C** Carbon 12.01 | 7 **N** Nitrogen 14.01 | 8 **O** Oxygen 16.00 | 9 **F** Fluorine 19.00 | 10 **Ne** Neon 20.18 |
| 11 **Na** Sodium 22.99 | 12 **Mg** Magnesium 24.31 | | | | | | | | | | | 13 **Al** Aluminum 26.98 | 14 **Si** Silicon 28.09 | 15 **P** Phosphorus 30.97 | 16 **S** Sulfur 32.06 | 17 **Cl** Chlorine 35.45 | 18 **Ar** Argon 39.95 |
| 19 **K** Potassium 39.10 | 20 **Ca** Calcium 40.08 | 21 **Sc** Scandium 44.96 | 22 **Ti** Titanium 47.87 | 23 **V** Vanadium 50.94 | 24 **Cr** Chromium 52.00 | 25 **Mn** Manganese 54.94 | 26 **Fe** Iron 55.85 | 27 **Co** Cobalt 58.93 | 28 **Ni** Nickel 58.69 | 29 **Cu** Copper 63.55 | 30 **Zn** Zinc 65.38 | 31 **Ga** Gallium 69.72 | 32 **Ge** Germanium 72.63 | 33 **As** Arsenic 74.92 | 34 **Se** Selenium 78.97 | 35 **Br** Bromine 79.90 | 36 **Kr** Krypton 83.80 |
| 37 **Rb** Rubidium 85.47 | 38 **Sr** Strontium 87.62 | 39 **Y** Yttrium 88.91 | 40 **Zr** Zirconium 91.22 | 41 **Nb** Niobium 92.91 | 42 **Mo** Molybdenum 95.95 | 43 **Tc** Technetium (98) | 44 **Ru** Ruthenium 101.07 | 45 **Rh** Rhodium 102.91 | 46 **Pd** Palladium 106.42 | 47 **Ag** Silver 107.87 | 48 **Cd** Cadmium 112.41 | 49 **In** Indium 114.82 | 50 **Sn** Tin 118.71 | 51 **Sb** Antimony 121.76 | 52 **Te** Tellurium 127.60 | 53 **I** Iodine 126.90 | 54 **Xe** Xenon 131.29 |
| 55 **Cs** Cesium 132.91 | 56 **Ba** Barium 137.33 | 57 - 71 **Lanthanides** | 72 **Hf** Hafnium 178.49 | 73 **Ta** Tantalum 180.95 | 74 **W** Tungsten 183.84 | 75 **Re** Rhenium 186.21 | 76 **Os** Osmium 190.23 | 77 **Ir** Iridium 192.22 | 78 **Pt** Platinum 195.08 | 79 **Au** Gold 196.97 | 80 **Hg** Mercury 200.59 | 81 **Tl** Thallium 204.38 | 82 **Pb** Lead 207.20 | 83 **Bi** Bismuth 208.98 | 84 **Po** Polonium (209) | 85 **At** Astatine (210) | 86 **Rn** Radon (222) |
| 87 **Fr** Francium (223) | 88 **Ra** Radium (226) | 89 - 103 **Actinides** | 104 **Rf** Rutherfordium (285) | 105 **Db** Dubnium (268) | 106 **Sg** Seaborgium (271) | 107 **Bh** Bohrium (270) | 108 **Hs** Hassium (277) | 109 **Mt** Meitnerium (276) | 110 **Ds** Darmstadtium (281) | 111 **Rg** Roentgenium (280) | 112 **Cn** Copernicium (285) | 113 **Nh** Nihonium (284) | 114 **Fl** Flerovium 289 | 115 **Mc** Moscovium (288) | 116 **Lv** Livermorium (293) | 117 **Ts** Tennessine (294) | 118 **Og** Oganesson (294) |

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 57 **La** Lanthanum 138.91 | 58 **Ce** Cerium 140.12 | 59 **Pr** Praseodymium 140.91 | 60 **Nd** Neodymium 144.24 | 61 **Pm** Promethium (145) | 62 **Sm** Samarium 150.36 | 63 **Eu** Europium 151.96 | 64 **Gd** Gadolinium 157.25 | 65 **Tb** Terbium 158.93 | 66 **Dy** Dysprosium 162.50 | 67 **Ho** Holmium 164.93 | 68 **Er** Erbium 167.26 | 69 **Tm** Thulium 168.93 | 70 **Yb** Ytterbium 173.05 | 71 **Lu** Lutetium 174.97 |
| 89 **Ac** Actinium (227) | 90 **Th** Thorium 232.04 | 91 **Pa** Protactinium 231.04 | 92 **U** Uranium 238.03 | 93 **Np** Neptunium (237) | 94 **Pu** Plutonium (244) | 95 **Am** Americium (243) | 96 **Cm** Curium (247) | 97 **Bk** Berkelium (247) | 98 **Cf** Californium (251) | 99 **Es** Einsteinium (252) | 100 **Fm** Fermium (257) | 101 **Md** Mendelevium (258) | 102 **No** Nobelium (259) | 103 **Lr** Lawrencium (262) |

Figure 10.1: Periodic table from Wikipedia.

There is a square for each element. In the middle, you can see the atomic symbol and the name of the element. In the upper-right corner is the atomic number — the number of protons in the atom.

In the upper-left corner is the atomic mass in atomic mass units.




Look at the atomic mass of boron. About 80% of all boron atoms have six neutrons. The other 20% have only 5 neutrons. This difference is why most boron atoms have a mass of about 11 atomic mass units, but some have a mass of about 10 atomic mass units. The atomic mass of boron is equivalent to the average mass of a boron atom: 10.811.

## *Exercise 21* Mass of a Water Molecule

*Working Space*

Using the periodic table, what is the average mass of one water molecule in atomic mass units?

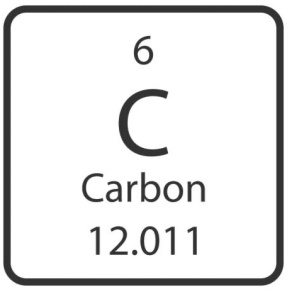
*Answer on Page 1882*

## 10.4 xfer from intro chapter

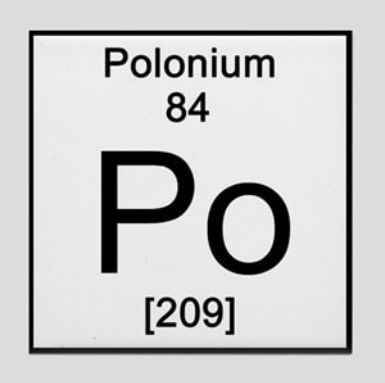
fixme integrate into this chapter

### 10.4.1 Reading the Periodic Table

The Periodic Table organizes what we know about the structure of different elements. Each element has its own block or tile on the Periodic Table, and the information on the tile tells us about the structure of that atom. Take a look at the tile for carbon:



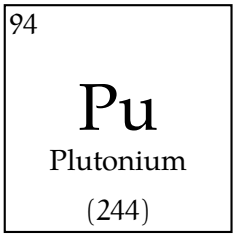

The letter (or letters, as is the case for other elements) is the atomic symbol for the element. There are two key numbers: the atomic number and the average atomic mass. For carbon, the atomic number is 6 and the average atomic mass is 12.011. The atomic number tells us how many protons there are in the nucleus of any atom of carbon. Since every element has a unique number of protons, every element has a unique atomic number. All carbon atoms have 6 protons. The other number is the average atomic mass - it tells us the weighted average of the mass of all the carbons in the universe. When the average atomic mass is in a whole number, as it is for polonium, it means that the element is very unstable. As a result, the mass given is the mass of the most stable isotope (we'll talk more about stability and isotopes below). On some periodic tables, the mass number of the most stable isotope will be in parentheses or brackets. In summary, if the larger number is a whole number, it is the mass number; if it is a decimal (even if the decimal ends in .00), it is the average atomic mass, which we will discuss further below.



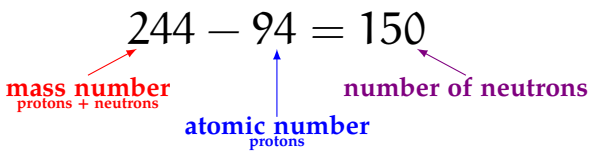

The Royal Society of Chemistry has a very useful interactive periodic table: periodictable.rsc.org. We can use the periodic tile for an element to determine the number of protons, electrons, and most common number of neutrons for a neutral atom of that element (we'll explain why the periodic tile tells us the "most common number of neutrons" below).

**Example**: State the atomic symbol for and the number of protons, neutrons, and electrons in a neutral atom of plutonium.

**Solution**: The plutonium tile on your periodic table should look something like this:




[The information may be arranged differently, but you should at least see the symbol and two numbers.] As you can see, the atomic symbol for plutonium is Pu. Since its atomic number is 94, we know every atom of plutonium has 94 protons. To know the number of electrons, we will take advantage of the fact that the question is asking about a *neutral* atom. This means there are the same number of positive charges as negative charges. So, since there are 94 protons, a neutral atom of plutonium must have 94 electrons (each proton has a +1 charge and each electron has a -1 charge). Lastly, let's determine the number of neutrons. The other number, 244, is the mass number. It represents the total number of protons and neutrons in the nucleus. Since we know plutonium has 94 protons, we can find the number of neutrons by subtracting the atomic number from the mass number:




Therefore, an atom of plutonium has 150 neutrons. Now let's address how to find the number of neutrons when the periodic tile shows an average atomic mass, instead of a mass number. This occurs when there is more than one "version" of an element. In the case of plutonium, there is only one version, which is why the periodic tile shows a mass number instead of an average atomic mass. To learn about average atomic mass, we will use carbon as an example.

Have you heard of carbon-14 dating? The phrase "carbon-14" refers to a rare type of carbon that decays radioactively. By seeing how much carbon-14 has decayed, scientists can estimate the age of organic materials, such as bone or ash. Carbon-14 is a radioactive isotope (or version) of carbon. The 14 refers to the mass number - the total amount of protons and neutrons in the nucleus (sometimes, we shorten the isotope name by just using the atomic symbol, in this case C-14). Isotopes are versions of an element with different numbers of neutrons. The atomic number is the same for them all - they all have the same number of protons. But the different number of neutrons causes different isotopes to have different masses. Examine the models of carbon-12, carbon-13, and carbon-14 below. What is different between them? What is the same?

You should have noticed that all three atoms have 6 protons and 6 electrons, while they have differing numbers of neutrons. The most common isotope of carbon is carbon-12, with 6 protons and 6 neutrons in its nucleus. Carbon-14, on the other hand, has 8 neutrons, which makes the nucleus unstable, leading to radioactive decay. The average atomic mass is the weighted average of all the carbon atoms in existence. Since the vast majority of carbon is carbon-12, the average atomic mass is very close to 12. You cannot determine the mass number of an individual atom from the periodic table; it only tells you the average of all the isotopes. However, especially for light atoms (atoms in the first two rows of the periodic table), you can usually determine the mass number of the most common isotope by rounding the average atomic mass to the nearest whole number.

**Example**: Germanium has atomic symbol Ge. State the number of protons, number of electrons, and most common number of neutrons in a neutral atom of germanium.

**Solution**: Examining the periodic table, we see that germanium has an atomic number of 32, which means a neutral atom of germanium has 32 protons and 32 electrons. The average atomic mass is 72.630, which rounds up to 73. So, the most common isotope of germanium is Ge-73, which has $73 - 32 = 41$ neutrons.

## *Exercise 22* Determining Numbers of Subatomic Particles

Use a periodic table to complete the table below (assume neutral atoms):

*Working Space*

| Element Name | Atomic Symbol | Protons | Most Common Number of Neutrons | Electrons |
|---|---|---|---|---|
| | Fr | | | |
| | | | | 33 |
| Erbium | | | | |
| | | 48 | | |

*Answer on Page 1882*

## 10.5 Heavy atoms aren't stable

When you look at the periodic table, there are a surprisingly large number of elements. You might be told to "Drink milk so that you can get the calcium you need." However, no one has told you "You should eat kale so that you get enough copernicium in your diet."

Copernicium, with 112 protons and 173 neutrons, has only been observed in a lab. It is highly radioactive and unstable (meaning it decays). A copernicium atom usually lives for less than a minute before decaying.

The largest stable element is lead, which has 82 protons and between 122 and 126 neutrons. Elements with lower atomic numbers than lead, have at least one stable isotope, while elements with higher atomic numbers than lead don't.

Bismuth, with an atomic number of 83, is *almost* stable. In fact, most bismuth atoms will live for billions of years before decaying!

Chapter 11

# Chemical Reactions

## 11.1 Chemical Reactions

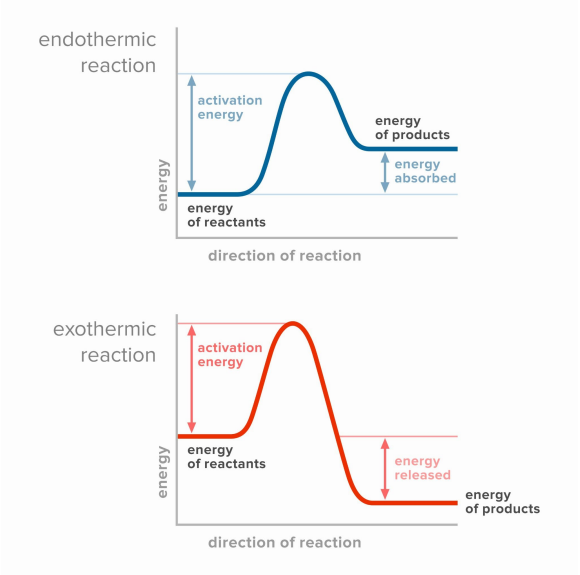



Figure 11.1

Hydrogen and oxygen have a special tendency to bond with themselves. Almost never will you find them unbonded. Oxygen is most often found as $O_2$ and hydrogen as $H_2$. If you mix these together in a 2:1 ratio of hydrogen to oxygen and light a match, they will rearrange themselves into water molecules. This is called a *chemical reaction*. In any chemical reaction, the atoms are rearranged into new molecules.

When a chemical reaction happens, some bonds are broken and new bonds are formed. The breaking and forming of bonds requires energy. The amount of energy required to break bonds is different for different bonds. The amount of energy released when new bonds are formed is also different for different bonds. Some chemical reactions (like the burning of hydrogen gas described above) are *exothermic* — that is, they give off more energy than they absorb. Burning hydrogen gas happens quickly and gives off a lot of energy. If you have enough, it will make quite an explosion! Other chemical reactions are *endothermic* — they consume energy.Photosynthesis, the process by which plants consume energy from the sun to make sugar from $CO_2$ and $H_2O$ requires an endothermic chemical reaction. Energy stored in chemical bonds is called *chemical energy*, which is a form of *potential energy*.

Examine the diagrams in figure 11.1. The x-axis represents time - time passes as we move from left to right across the diagram. At the far left, the energy of the reactants (the ingredients that go into the reaction) is shown. At the far right, the energy of the products (what is made in the chemical reaction) is shown. Look at the endothermic reaction diagram (the blue one). Based on the relative energies of the reactants and products, do you expect an endothermic reaction to release or absorb heat? Absorb! Since the products have more energy, they must have absorbed energy, in the form of heat, from the surroundings.

Now, the red diagram shows an exothermic reaction: the products have less energy than the reactants. Since energy is never created or destroyed, where did the energy go? It is released as heat. So, exothermic reactions release heat. What does this look and feel like in real life? If an exothermic reaction were happening in a glass beaker, you would feel warmth if you held the beaker. The heat is leaving the beaker and entering your hand, which feels warm. What about an endothermic reaction? Many students think that since an endothermic reaction absorbs heat, it must be getting hot. This is incorrect: *exothermic* reactions feel hot. If an endothermic reaction were happening in a beaker and you touched the beaker, it would feel **cold**. Why? Well, if the reaction is absorbing heat, then heat must be leaving it surroundings (your hand) and entering the reaction (this heat energy is turned into chemical energy that is stored in the new chemical bonds that are forming). So your hand feels cold.

Chapter 12

# Buoyancy

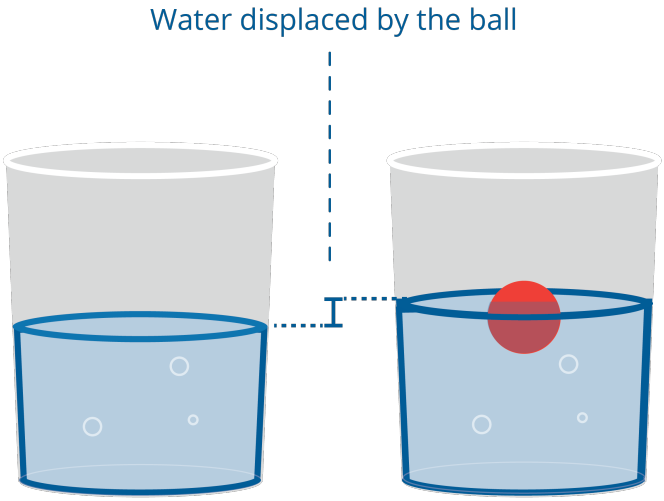
The word buoyancy probably brings to mind images of floating in water. Before we dive in, let's zoom out for a better understanding of everything buoyancy entails. You may be thinking: I want to be a computer programmer, why do I need to know about buoyancy? You might be surprised! This topic is much bigger than it seems at first glance.

Buoyancy concerns the ways in which liquids and gasses interact with gravity. The concept of buoyancy is connected to fundamental concepts about how the universe works. The *buoyant force*, as it is known in engineering, is an important concept that has wide ranging applications. A big part of engineering is moving stuff around, and understanding buoyancy helps us solve problems where we need to move things in and through fluids. Even if you don't have plans to build a robotic submarine, these are incredibly useful ideas to be familiar with. We will start exploring the topic with familiar scenarios around boats and water.

When you put a boat into water, it will sink into the water until the mass of the water it displaces is equal to the mass of the boat. We think of this in terms of forces. Gravity pulls the mass of the boat down; the *buoyant force* pushes the boat up. A boat dropped into the water will bob up and down at first before reaching an *equilibrium* where the two forces are equal.

The buoyant force pushes things up, fighting against the force of gravity. The force is equal to the weight of the fluid being displaced. For example, a cubic meter of freshwater has a mass of about 1000kg. If you submerge anything with a volume of one meter in freshwater on earth, the buoyant force will be about 9800 newtons (mass $\times$ gravity).

For some things, like a block of styrofoam, this buoyant force will be sufficient to carry it to the surface. Once it reaches the surface, it will continue to rise (displacing less water) until the mass of the water it displaces is equal to its mass. And then we say "It floats!"




For some things, like a block of lead, the buoyant force is not sufficient to lift it to the surface, and then we say "It sinks!"

This is why a helium balloon floats through the air. The air that it displaces weighs more than the balloon and the helium itself. (It is easy to forget that air has a mass, but it does.)

### *Exercise 23* Buoyancy

You have an aluminum box that has a heavy base, so it will always float upright. The box and its contents weigh 10 kg. Its base is 0.3 m x 0.4 m. It is 1m tall.

When you drop it into freshwater ($1000 kg/m^3$), how far will it sink before it reaches equilibrium?

*Working Space*

*Answer on Page 1882*

## 12.1 The Mechanism of Buoyancy: Pressure

As you dive down in the ocean, you will experience greater and greater pressure from the water. And if you take a balloon with you, you will gradually see it get smaller as the water pressure compresses the air in the balloon.

Let's say you are 3 meters below the surface of the water. What is the pressure in Pascals (newtons per square meter)? You can think of the water as a column of water crushing down upon you. The pressure over a square meter is the weight of 3 cubic meters of water pressing down.

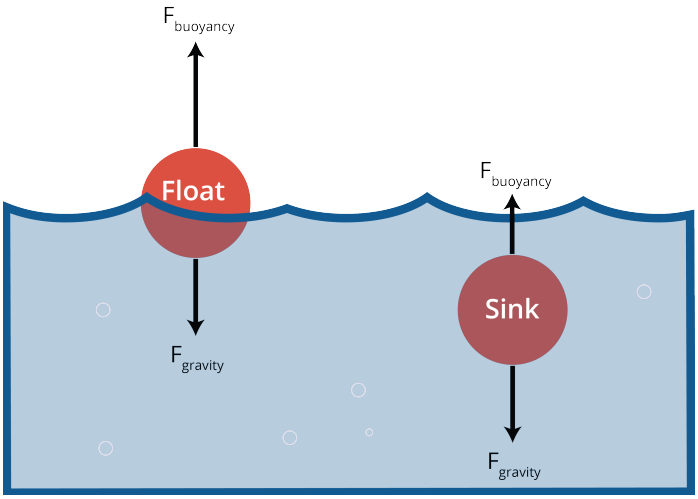
$$p = (3)(1000)(9.8) = 29,400 \text{ Pa}$$

This is called *hydrostatic pressure*. The general rule for hydrostatic pressure in Pascals $p$ is

$$p = dgh$$

where $d$ is the density of the fluid in kg per cubic meter, $g$ is the acceleration due to gravity in $m/s^2$, and $h$ is the height of the column of fluid above you.

So where does buoyant force come from? Basically, the pressure pushing up on the deepest part of the object is higher than the pressure pushing down on the shallowest part of the object. That is where buoyancy comes from.



### *Exercise 24* Hydrostatic Pressure

*Working Space*

You dive into a tank of olive oil on Mars. How much more hydrostatic pressure does your body experience at 5 meters deep than it did at the surface?

The density of olive oil is about 900 kg per square meter. The acceleration due to gravity on Mars is 3.721 $m/s^2$.

*Answer on Page 1882*

## 12.2 The Mechanism of Buoyancy: Density

Keep in mind that although the pressure is increasing as you go deeper, the buoyant force will *not increase*, because the buoyant force is always equal to the weight of the fluid that is displaced, regardless whether that is 1 meter or 100 meters underwater.

Due to the added minerals, saltwater is denser than freshwater. This causes objects to float better in the sea than they do in a river. Lipids, like fats and oils, are less dense than water, allowing them to float on top of a glass of water. When you're facing a grease fire, you're told not to put water on it. That's because the water sinks below the grease, then boils, throwing burning grease everywhere.

Chapter 13

# Heat

All mass in the universe has heat, whether you're looking at a block of dry ice (frozen $CO_2$, $-78.5°$ C) or the surface of the sun ($5,600°$ C). As long as the mass is above absolute zero — the coldest possible temperature in the universe — there is some amount of heat in it.

## 13.1 How Heat Works

As you heat up an object, you are imparting energy into it. Where does this energy go? The atoms take this energy and they begin to move, vibrating and bumping into each other, causing the heat to spread throughout. Every time the atoms collide and bounce off of each other, they emit a tiny amount of energy as light. In most cases, that light is in the infrared spectrum, but in extreme cases can be visible, such as with molten lava or hot metal.

As objects interact, they either put heat into colder objects or take heat from warmer objects. That's what allows you to heat up anything in the first place. The hot air from a stove or bunsen burner interacts with the pan or test tube you're heating, passing the air's heat on. How could you model this?

## 13.2 Specific Heat Capacity

If you are heating something, the amount of energy you need to transfer to it depends on three things: the mass of the thing you are heating, the amount of temperature change you want, and the *specific heat capacity* of that substance.

**Energy in Heat Transfer**

The energy moved in a heat transfer is given by

$$E = mc\Delta_T$$

where $m$ is the mass, $\Delta_T$ is the change in temperature, and $c$ is the specific heat capacity of the substance.

(Note that this assumes there isn't a phase change. For example, this formula works nicely on warming liquid water, but it gets more complicated if you warm the water past its boiling point.)

Can we guess the specific heat capacity of a substance? It is very, very difficult to guess the specific heat of a substance, so we determine it by experimentation.

For example, it takes 0.9 joules to raise the temperature of solid aluminum one degree Celsius. So we say "The specific heat capacity of aluminum is 0.9 J/g °C."

The specific heat capacity of liquid water is about 4.2 J/g °C.

Let's say you put a 1 kg aluminum pan that is 80° C into 3 liters of water that is 20° C. Energy, in the form of heat, will be transferred from the pan to the water until they are at the same temperature. We call this "thermal equilibrium".

What will the temperature of the water be?

To answer this question, the amount of energy given off by the pan must equal the amount of energy absorbed by the water. They also need to be the same temperature at the end. Let T be the final temperature of both.

3 liters of water weighs 3,000 grams, so the change in energy in the water will be:

$$E_W = mc\Delta_T = (3000)(4.2)(T - 20) = 12600T - 252000 \text{ joules}$$

The pan weighs 1000 grams, so the change in energy in the pan will be::

$$E_P = mc\Delta_T = (1000)(0.9)(T - 80) = 900T - 72000 \text{ joules}$$

The total energy stays the same, so $E_W + E_P = 0$. This means you need to solve

$$(12600T - 252000) + (900T - 72000) = 0$$

And find that the temperature at equilibrium will be

$$T = 24°C$$

### *Exercise 25* Thermal Equilibrium

*Working Space*

Just as you put the aluminium pan in the water as described above, someone also puts a 1.2 kg block of copper cooled to 10 ° C. The specific heat of solid copper is about 0.4 J/g °C.

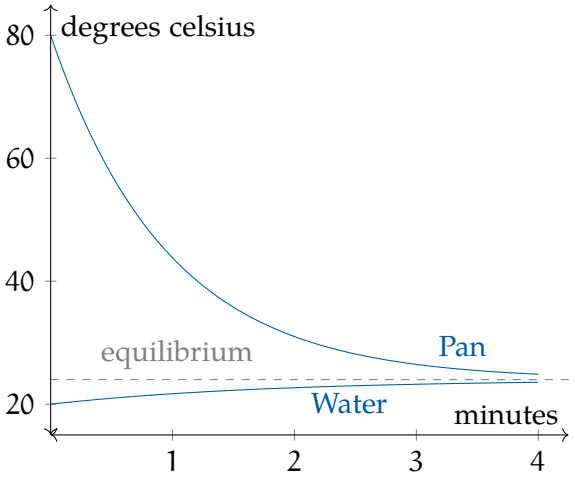
What is the new temperature at equilibrium?

*Answer on Page 1883*

## 13.3 Getting to Equilibrium

When two objects with different temperatures are touching, the speed at which they exchange heat is proportional to the differences in their temperatures. As their temperatures get closer together, the heat exchange slows down.

In our example, the pan and the water will get close to equilibrium quickly, but they may never actually reach equilibrium.



## *Exercise 26* Cooling Your Coffee

*Working Space*

You have been given a ridiculously hot cup of coffee and a small pitcher of chilled milk.

You need to start chugging your coffee in three minutes, and you want it as cool as possible at that time. When should you add the milk to the coffee?

*Answer on Page 1883*

## 13.4 Specific Heat Capacity Details

For any given substance, the specific heat capacity often changes a great deal when the substance changes state. For example, ice is 2.1 J/g °C, whereas liquid water is 4.2 J/g°C.

Even within a given state, the specific heat capacity varies a bit based on the temperature and pressure. If you are trying to do these sorts of calculations with great accuracy, you will want to find the specific heat capacity that matches your situation. For example, I might look for the specific heat capacity for water at 22°C at 1 atmosphere of pressure (atm).

Chapter 14

# Work and Energy

In this chapter, we are going to talk about how engineers define work and energy. It frequently takes force to get work done. Let's start with thinking about the relationship between force and energy. As we learned earlier, Force is measured in newtons, and one newton is equal to the force necessary to accelerate one kilogram at a rate of $1 m/s^2$.

When you lean on a wall, you are exerting a force on the wall, but you aren't doing any work. On the other hand, if you push a car for a mile, you are clearly doing work, as in Figure 14.1. Work, to an engineer, is the force you apply to something, as well as the distance that something moves, in the direction of the applied force. We measure work in *joules*. A joule is one newton of force applied over one meter.

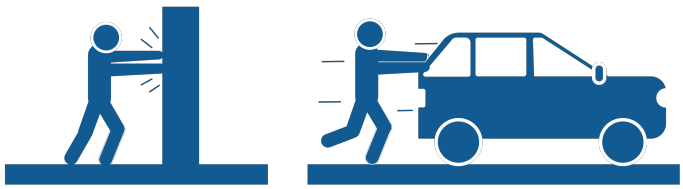


Figure 14.1: Pushing on a wall does no work while pushing a car does work! Why?

For example, if you push a car with a force of 10 newtons for 12 meters, you have done 120 joules of work.

**Formula for Work**

If a force applied is *constant*, the formulas for work is:

$$W = F \cdot d$$

where $W$ is the work in joules $(N \cdot m)$, $F$ is the *force* in newtons, and $d$ is the distance in meters.

If the force is not in the same direction as the distance, we can use the cosine of the angle between the force and the distance:

$$W = F \cdot d \cdot \cos(\theta)$$

where $\theta$ is the angle between the force and the distance. If the formula for work is *not constant*, the formula becomes:

$$W = \int \mathbf{F} \cdot \mathbf{dr}$$

where **dr** is an infinitesimal amount of displacement. An example of this kind of force is like a spring force, where the force varies with displacement.

## *Exercise 27* Book done by a constant force

*This question is from an AP Physics C Review Book.* You slowly lift a 2 kg book at a *constant velocity* through a distance of 3 m. How much work is done on the book?

*Working Space*

*Answer on Page 1883*

The work-energy theorem (or work-energy principle) states that the net work done on an object is equal to the **change in its energy**. In other words, if you do work on an object, you are changing its energy. This is derived from Newton's second law of motion, covered in Chapter 13.

$$W = \Delta E$$

Work is how energy is transferred from one thing to another. When you push the car, you also burn sugars (energy of the body) in your blood. That energy is then transferred to the car after it has been pushed uphill.

Thus, we measure the energy something consumes or generates in units of work: joules, kilowatt-hours, horsepower-hours, foot-pounds, BTUs (British Thermal Unit), and calories.

Let's go over a few different forms that energy can take.

## 14.1 Forms of Energy

In this section we are going to learn about several different types of energy:

- Heat
- Electricity
- Chemical Energy
- Kinetic Energy
- Gravitational Potential Energy

There are also other forms of energy such as spring potential energy, which we will cover in the oscillations chapter.

### 14.1.1 Heat

When you heat something, you are transferring energy to it. The BTU is a common unit for heat. One BTU is the amount of heat required to raise the temperature of one pound of water by one degree. One BTU is about 1,055 joules. In fact, when you buy and sell natural gas as fuel, it is priced by the BTU.

When we talk about heat energy, we will commonly be talking about *friction*. When a force does frictional work, it is output onto as a change in heat energy within the system.

### 14.1.2 Electricity

Electricity is the movement of electrons. When you push electrons through a space that resists their passage (like a light bulb), energy is transferred from the power source (like a battery) into the source of the resistance.

Let's say your lightbulb consumes 60 *watts* of electricity, and you leave it on for 24 hours. We would say that you have consumed 1.44 kilowatt hours, or 3,600,000 joules. This comes into play when rating house electricity systems, lightbulbs, or electric cars. The electric energy stored in a battery of an electric car gets converted to kinetic energy and thermal energy taken out by friction.

### 14.1.3 Chemical Energy

As mentioned earlier, some chemical reactions consume energy and some produce energy. This means energy can be stored in the structure of a molecule. When a plant uses photosynthesis to rearrange water and carbon dioxide into a sugar molecule, it converts the energy in the sunlight (solar energy) into chemical energy. Remember that photosynthesis is a process that consumes energy. Therefore, the sugar molecule has more chemical energy than the carbon dioxide and water molecules that were used in its creation.

Recall the balanced chemical equation for photosynthesis is:

$$6CO_2 + 6H_2O + \text{energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

In our diet, we measure this energy in *kilocalories*. A calorie is the energy necessary to raise one gram of water one degree Celsius, and is about 4.19 joules. This is a very small unit. An apple has about 100,000 calories (100 kilocalories), so people working with food started measuring everything in kilocalories.

(You may find it helpful that the chapter on unit conversions appears earlier in this text, so that the calorie-joule relationship can be placed in proper context.)

Here is where things get tricky: People who work with food got tired of saying "kilocalories", so they just started using "Calorie" to mean 1,000 calories. This has created a great deal of confusion over the years. So if the C is capitalized, "Calorie" probably means kilocalorie.

### 14.1.4 Kinetic Energy

A mass in motion has energy. For example, if you are in a moving car and you slam on the brakes, the energy from the motion of the car is converted into heat in the brake pads and under the tires. This stored energy of motion is known as *kinetic energy*.

But how much energy does a moving object actually have?

**Formula for Kinetic Energy**

$$KE = \frac{1}{2}mv^2$$

where KE is the kinetic energy in joules, $m$ is the mass in kilograms, and $v$ is the speed in meters per second.

This equation tells us two important things:

- Kinetic energy increases *linearly* with mass. A heavier object has more energy at the same speed.
- Kinetic energy increases *quadratically* with velocity. This means that doubling the speed results in *four times* the kinetic energy.

Why does the velocity term appear squared? One way to understand this is to consider the work required to accelerate an object. To speed something up, a force must act over a distance. As the object moves faster, each additional meter of acceleration takes less time but still increases the kinetic energy significantly. The mathematical result of this relationship is the $v^2$ dependence. Kinetic energy ultimately tells us how powerful an object is. Real life calculations involve calculating car crash preventions, motor speeds, rocket propulsions, battery requirements, and gyroscopic forces.

## *Exercise 28* Kinetic and Thermal Energy

A $1,200$ kg car traveling at $22\text{m/s}$ comes to a stop using its brakes. Assume all the initial kinetic energy is converted into thermal energy in the brakes and tires.

(a) How much thermal energy, Q is produced?

(b) If the break system has a 15 kg and a specific heat of $c = 460(\text{J/kg}^\circ\text{C})$, estimate the temperature increase of the brakes.

Also, you are given that the change in heat, $\Delta T = \frac{Q}{m_{brake}c}$.

*Working Space*

*Answer on Page 1883*

### 14.1.5 Gravitational Potential Energy

When you lift something heavy onto a shelf, you are giving it *potential energy*. The amount of energy that you transferred to it is proportional to its weight and the height that you lifted it.

**Formula for Gravitational Potential Energy**

The formula for gravitational potential energy is

$$E = mgh$$

where $E$ is the energy in joules, $m$ is the mass of the object you lifted, $g$ is acceleration due to gravity, and $h$ is the height that you lifted it.

On earth, then, gravitational potential energy is given by

$$E = (9.8)mh$$

since objects in free-fall near Earth's surface accelerate at $9.8 m/s^2$.

There are various kinds of potential energy. For example, when you draw a bow in order to fire an arrow, you have given that bow potential energy. When you release it, the potential energy is transferred to the arrow, which expresses it as kinetic energy. When you compress a spring, the energy put into the spring is proportional to the spring constant and distance it is compressed, $\frac{1}{2}kx^2$. We will dive deeper into this in the springs and oscillations chapter.

## 14.2 Conservation of Energy

The first law of thermodynamics says "Energy is neither created nor destroyed."

Energy can change forms. Your cells consume chemical energy to give gravitational potential energy to a car you push up a hill. A falling object converts potential energy into kinetic energy, causing it to speed up. However, the total amount of energy in a closed system stays constant.

The sum of mechanical energy (usually Kinetic and Potential) is always constant:

$$W_{net,ext} = \Delta E_{sys} = \Delta K + \Delta U + \Delta E_{th}$$

Choosing a specific system or environment will benefit your problem, especially if the net external work done is $0$. Problems that involve energy transformation are similar to accounting problems, the work before and after must be equal, and must equal some constant net value.

## *Exercise 29* The Energy of Falling

*Working Space*

A 5 kg cannonball falls off the top of a 3 meter ladder. As it falls, its gravitational potential energy is converted into kinetic energy. How fast is the cannonball traveling just before it hits the floor?

*Answer on Page 1884*

## *Exercise 30* Frictional Losses

*Working Space*

A 4.0 kg block is released from rest at the top of a rough incline of height 2.5 m. By the time it reaches the bottom, its speed is 4.2 m/s.

Determine the total mechanical energy lost to thermal energy.

You may use 10 $m/s^2$ for g

*Answer on Page 1884*

### *Exercise 31* Blast Off!

A rocket sled of mass 150 kg uses a small fuel-powered booster that releases $1.2 \times 10^5$ J of chemical energy. During the burn, 65% of the energy is converted into useful kinetic energy for the sled, the rest is lost as heat or sound. Assume the sled travels on frictionless, smooth surface.

Calculate the final speed of the rocket sled.

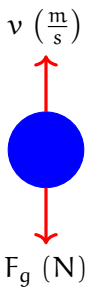
*Working Space*

*Answer on Page 1884*

## 14.3 Work and Kinetic Energy

As stated above, the work-energy theorem tells us that the change in an object's kinetic energy is equal to the work done on that object. For now, we will only consider examples where the force and the direction of motion are parallel or perpendicular. When you learn about vectors, we will expand this to include forces that are skew to the direction of motion.

Consider what happens when you toss a ball in the air: once the ball leaves your hand, the only force acting on it is gravity. Initially, the ball is moving upwards while gravity points downwards:




Intuitively, we know that the ball will slow down (lose kinetic energy) as it moves upwards:

$$\Delta KE < 0$$

Since $W = \Delta KE$, we also know that gravity must be doing *negative work*. Whenever the direction of the force is opposite the direction of the motion, the work done by that force is negative.

## *Exercise 32* A ball thrown upwards

If the ball has a mass of 0.5 kg, how much kinetic energy does it lose as it moves upwards by 1 m?

*Working Space*

*Answer on Page 1884*

## *Exercise 33* How far will you slide?

You are playing softball and have to slide into home. If you sprint at a maximum of 10 m/s and the force of friction between you and the ground is 0.3 times your weight, how far from the base can you start your slide and still reach home?

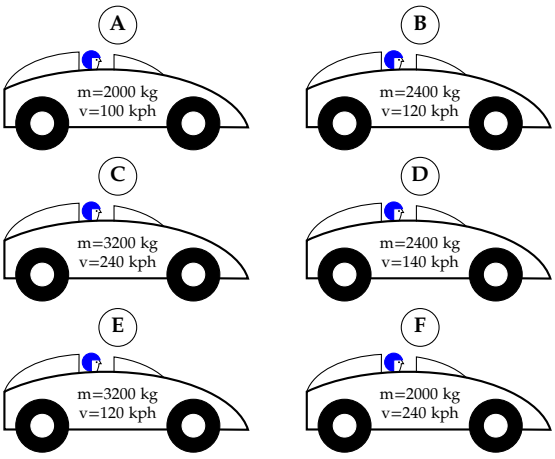
*Working Space*

*Answer on Page 1885*

## *Exercise 34* Ranking Stopping Force

*Working Space*

In drag racing, cars can reach speeds of 150 miles per hour (approximately 240 kilometers per hour). In order to be able to stop quickly and safely, drag racing cars are built with parachutes that deploy at the end of the race. Consider a drag race where cars of different masses reach different maximum speeds. There is 100 meters between the finish line and the fence surrounding the race track. If all the race cars deploy their parachutes at the finish line while going their maximum speed, rank the force needed from the parachute to stop each car in the required distance from least to greatest:




*Answer on Page 1885*

### 14.3.1 Forces that do no work

If the object you are pushing doesn't move. or the applied force is perpendicular to the direction of motion, that force does no work. Let's look at a few examples:

### Pushing Against an Immobile Object

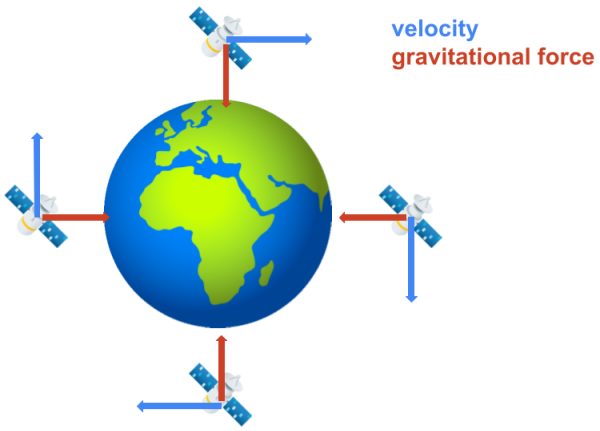
At the beginning of the chapter, we said that when you push on a wall, you don't do any work. Why is this? Well, if the wall is a good wall (that is, strong enough to not be pushed over by a person), the wall won't move while you push on it. This means the distance over which your push is applied is zero, and therefore the work done ($F \cdot d = F \cdot 0 = 0$) is zero joules.

### Walking Across a Room with a Book

Imagine holding a book flat on your hands and walking at a constant velocity. Your hand is applying an upwards force to the book, but the book is moving horizontally. This means the force and direction of motion are *perpendicular*. Recall from the beginning of the chapter than if the force and distance are not parallel, then the work is given by $W = F \cdot d \cos(\theta)$. (When the vectors are parallel, $\theta = 0$ and $\cos(\theta) = 1$, while when the vectors point in opposite directions, $\theta = 180°$ and $\cos(\theta) = -1$.) When the vectors are perpendicular, then $\theta = 90°$ and $\cos(\theta) = 0$. Therefore, $W = 0$ as well and the upward force of your hands does no net work.

### Circular Motion

We will discuss circular motion further in a subsequent chapter. For now, know that constant-speed circular motion is caused by a constant-magnitude force that always points to the center of the circle the object is moving in. For example, you can take a weight on the end of a string and spin it. The tension in the string spins the weight, and the string always points from the object to your hand (the center of the weight's circular path). For a satellite, that force is gravitational attraction to the Earth.



As a result, the force changes the *direction*, but not the *magnitude* of the satellite's velocity. Let's re-examine the equation for kinetic energy:

$$\text{KE} = \frac{1}{2}mv^2$$

Since the velocity is squared, the direction of motion doesn't affect the kinetic energy (a ball moving at 5 m/s upwards has the same kinetic energy as if the ball were moving at 5 m/s downwards). So, a force that causes circular motion doesn't change a circling object's kinetic energy, and therefore does no work (as expected when force and direction of motion are perpendicular)!

## 14.4 Efficiency and Power

Although energy is always conserved as it moves through different forms, scientists aren't always that good at controlling it.

In terms of an equation, efficiency is the ratio of the useful energy output to the total energy input. It is usually expressed as a percentage.

**Formula for Efficiency**

$$\text{Efficiency} = \frac{\text{Useful Energy Output}}{\text{Total Energy Input}} \times 100\%$$

where the useful energy output is the energy that is actually used to do work or complete a task, and the total energy input is the total energy consumed by the system.

A machine is considered 100% efficient only if all the input work is converted into useful output work, with no energy lost to heat, friction, or sound. 100% efficient process don't exist in real-life: every process loses some useful energy to heat.

For example, when a car engine consumes the chemical energy in gasoline, only about 20% of the energy consumed is used to turn the wheels. Most of the energy is actually lost as heat. If you run a car for a while, the engine gets very hot, as does the exhaust coming from the tailpipe.

A human is about 25% efficient. Most of the loss is in the heat produced during the chemical reactions that turns food into motion.

In general, if you are trying to increase efficiency in any system, the solution is usually easy to identify by the heat that is produced. Reduce the heat, increase the efficiency.

Light bulbs are an interesting case. To get the same amount of light of a 60 watt incandescent bulb, you can use an 8 watt LED or a 16 watt fluorescent light. This is why we say that the LED light is much more efficient. If you run both, the incandescent bulb will consume 1.44 kilowatt-hours; the LED will consume only 0.192 kilowatt-hours.

In addition to light, the incandescent bulb is producing a lot of heat. If it is inside your house, what happens to the heat? It warms your house.

In the winter, when you want light and heat, the incandescent bulb is 100% efficient!

Of course, this also means the reverse is true. In the summer, if you are running the air conditioner to cool down your house, the incandescent bulb is worse than just "inefficient at making light" — it is actually counteracting the air conditioner!

## *Exercise 35* Unit Costs

A machine is rated to produce 500 bolts per hour, but in actual operation it runs at only 82% efficiency due to loss and scraps. The machine requires $24 of electricity per hour to run.

(a) What is the actual output of bolts per hour?

(b) What is the unit cost (cost per bolt) of the production?

*Working Space*

*Answer on Page 1886*

Another measure of Work and Energy is *Power*. Power is the rate at which work gets done (or energy gets transferred). Suppose I do work in 1000 J of work in 10 minutes but you do the same amount of work in 5 minutes. Because you did the same amount of work but in a quick amount of time, you were more *powerful*.

**Formula for Power**

$$\text{Power} = \frac{\text{Work}}{\text{Time}} \implies P = \frac{W}{t}$$

Power is represented by the unit watt (W), defined as one joule per second (J/s).

Since power is a rate, we can also express it using calculus:

$$P = \frac{dW}{dt}$$

Equivalently, for work done by constant forces, this simplifies to $P = \frac{F \cdot r}{t} = F \cdot v$, since

Power is measured in the unit **watt**, where one watt is one Joule per second: $1W = 1\,J/s$

## *Exercise 36* Hydraulic Piston Problem

*Working Space*

A hydraulic piston pushes a crate horizontally across the floor with a constant force of 450 N. During operation, the piston moves the crate 1.8 meters in 0.75 seconds.

(a) How much work does the piston do on the crate?

(b) What is the power output of the piston during this motion?

*Answer on Page 1886*

## *Exercise 37* Spring Power

*Working Space*

A compressed spring launches a small block. The spring delivers 120 W of power while releasing its stored energy over 0.40 seconds. The spring constant is $k = 800$ N/m.

How far was the spring compressed?

*Answer on Page 1886*

## *Exercise 38* Elevator Lift

An elevator motor lifts a 750 kg cabin vertically upward a distance of 18 m in 12 s. The motor consumes electrical energy at a rate of 18 kW.

(a) Compute the total electrical energy consumed during the lift. Note that 1 kW = 1000 watts = 1000 J/s

(b) Determine the gravitational potential energy gained by the elevator.

(c) Calculate the efficiency of the motor during this lift.

*Working Space*

*Answer on Page 1887*

CHAPTER 15

# Simple Machines

As mentioned earlier, physicists define work as the force applied times the distance over which it is applied. For example, if you push your car 100 meters with a force of 17 newtons, you have done 1700 joules of work.

Humans have long needed to move heavy objects, so many centuries ago, we developed simple machines to reduce the amount of force necessary to perform such tasks. These include:

- Levers
- Pulleys
- Inclined planes (ramps)
- Screws
- Gears
- Hydraulics
- Wedges

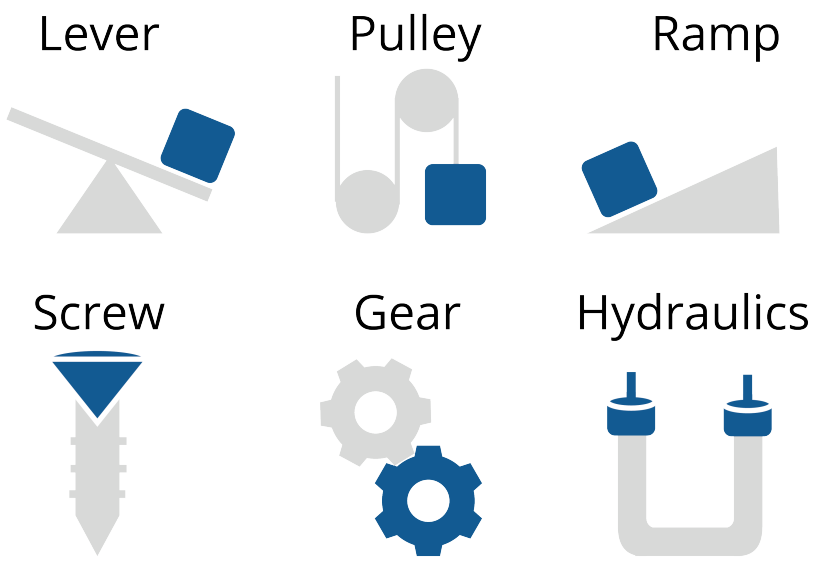



Figure 15.1: The 6 main simple machines: Levers, Pulleys, Ramps, Hydraulics, Screws, and Gears.

While these machines can reduce the force needed, they do not change the total amount of work that must be done. For instance, if the force is reduced by a factor of three, the distance over which the force must be applied *increases* by the same factor.

## 15.1 Mechanical Advantage

Mechanical advantage is the ratio between the force output by the machine and the force the user puts into the machine:

$$MA = \frac{F_{out}}{F_{in}} = \frac{Load}{Effort}$$

Since the input force is *applied* to the simple machine, sometimes the input force is called an applied force and abbreviated as $F_a$. For example, you only need to apply a relatively little force to your car's brakes in order for the hydraulic braking system to apply enough force to your tires to stop them spinning (we'll examine this further below).

### 15.1.1 What does it mean to work hard?

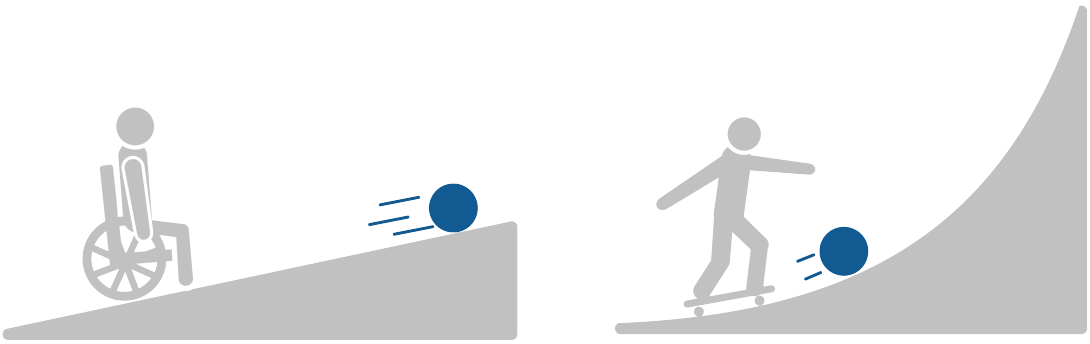
Humans use simple machines to "make work easier", but what does this mean in a physics sense? Does using a machine actually decrease the amount of work the user has to do?

When we say a task is easier, we usually mean *we have to apply less force*. You might say that it is "less work" to push something up a shallow incline than up a steep incline. But does the person pushing *actually do less work* (in a physics sense), or does that work simply require a smaller force? We'll answer this question by examining the physics of incline planes below, and the results will be true for all simple machines.

Ideally we want a higher Mechanical Advantage, which means we are putting in less input force for a greater output force.

## 15.2 Inclined Planes

Inclined planes, or ramps, allow you to roll or slide objects to a higher level. Steeper ramps require less mechanical advantage. For instance, it is much easier to roll a ball up a wheelchair ramp than a skateboard ramp.

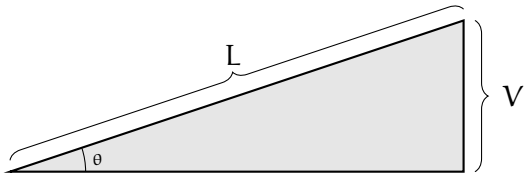


Assuming the incline has a constant steepness, the mechanical advantage is equal to the ratio of the length of the inclined plane to the height it rises.

If friction is neglected, the force required to push a weight up the inclined plane is given by:

$$F_A = \frac{V}{L} F_g \tag{15.1}$$

where $F_A$ is the applied force, $L$ is the length of the inclined plane, $V$ is the vertical rise, and $F_g$ is the gravitational force acting on the mass.




(We will discuss sine function later, but in case you're familiar with it, note that:

$$\frac{V}{L} = \sin\theta \tag{15.2}$$

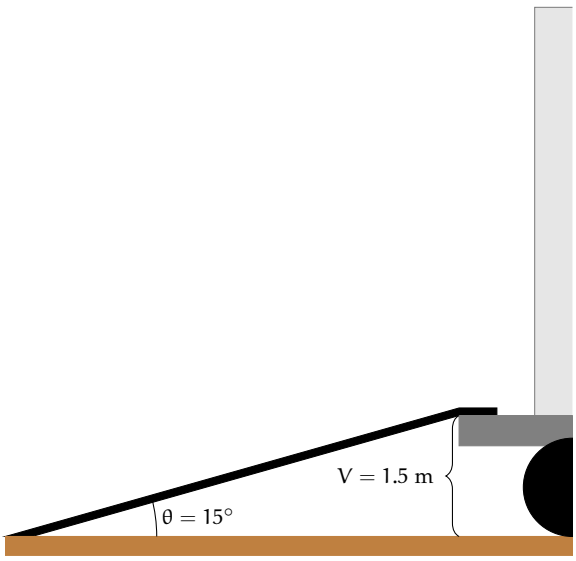
where $\theta$ is the angle between the inclined plane and the horizontal surface.)

Let's compare the force needed and work done when pushing a load up a ramp versus just lifting it vertically. Consider a family on moving day: there's a hand trolley loaded with 200 N (about 45 pounds) of boxes. If the bed of the moving truck is 1.5 m high, how much work would it take to lift the boxes straight up into the truck? What about with a ramp?

First, let's look at how much force and work is needed if you were to lift the entire 200 N load straight up into the air. You'd need to apply 200 N of force upwards for a distance of 1.5 m:

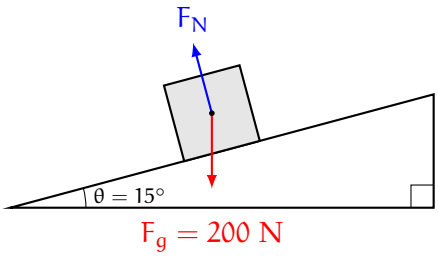
$$W = F \cdot d = (200\text{ N})\,(1.5\text{ m}) = 300\text{ J}$$

So, without a ramp, you would have to apply 200 N and do 300 J of work. Suppose your moving truck comes with a ramp that has an incline of 15 degrees:

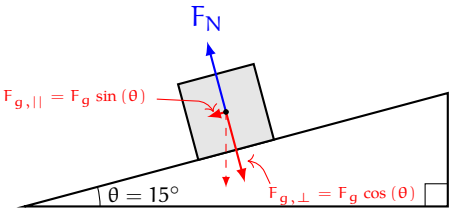


Since $\sin(\theta) = V/L$, we know that $L = V/\sin(\theta)$. You can use a calculator or search engine to find that the sine of $15° \approx 0.26$. Therefore, the length of the ramp is approximately 5.8 meters. How much force does it take to move the load of boxes up the ramp? Intuitively, we know it is less force. We can use a *free body diagram* to determine the minimum force needed to push the box up the ramp. (A free body diagram is a simplified model showing all the forces acting on an object. You'll learn to create and use your own free body diagrams in a later chapter. For now, just follow along.)

Before you push it, there are two forces acting on the loaded hand trolley: its weight ($F_g$) and the normal force between the trolley and the ramp ($F_N$):

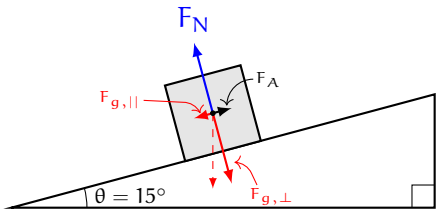



Notice that the normal force is perpendicular to the ramp! We want to know how much force it takes to push the load up the ramp, so we will "split" the weight force vector into two parts: one part parallel to the ramp ($F_{g,||}$) and one part perpendicular ($F_{g,\perp}$):




We did this by treating the weight vector as the hypotenuse of a right triangle with legs perpendicular and parallel to the ramp. You'll learn how to do this and why it works in the chapter on vectors. For now, just trust that the part of the hand trolley's weight that is perpendicular to the ramp is $F_g \cos(\theta)$ and the part that is parallel to the ramp is $F_g \sin(\theta)$.

What force do you need to overcome to push the hand trolley up the ramp? Just the part of the weight that is parallel to the ramp! You'll need to apply an equal force in the opposite direction (up the ramp) to move the hand trolley:




So, we know that you are pushing with an applied force of $F_A = F_{g,||} = F_g \sin(\theta)$. Therefore, the work you would do pushing the hand trolley up the ramp is:

$$F_A \cdot L = F_g \sin(\theta) \cdot \left(\frac{V}{\sin(\theta)}\right) = F_g \cdot V = 300 \text{ J}$$

Therefore, when using a ramp, you still perform the same amount of work! This is a key property of simple machines: *the work done doesn't change*.

So what makes it "easier" to use a ramp to lift the hand trolley? The fact that you need to apply less *force* to move the hand trolley ($F_A < F_g$). Now, let's look at the mechanical advantage of the ramp. In this case, the mechanical advantage is given by:

$$MA = \frac{F_g}{F_A}$$

Substituting for $F_A$, we see that:

$$MA = \frac{F_g}{F_g \sin(\theta)} = \frac{1}{\frac{V}{L}} = \frac{L}{V}$$

So for a ramp whose length is L and vertical rise is V, the mechanical advantage is equal to the length divided by the rise.

**Ramps**

For a ramp, the mechanical advantage is equal to $\frac{L}{V}$ and the force needed to push an object with weight $W$ up the ramp is given by $W \cdot \frac{V}{L} = W \cdot \sin(\theta)$, where L is the length of the ramp, V is the vertical rise of the ramp, and $\theta$ is the angle the ramp forms with the (horizontal) ground.

We have a whole inclined planes chapter coming up, but this was just a brief overview of their function as a mechanic

### *Exercise 39* Ramp

You need to lift a barrel of oil with a mass of 136 kilograms. You can apply a force of up to 300 newtons. You need to get the barrel onto a platform that is 2 meters high. What is the shortest length of inclined plane you can use?

*Working Space*

*Answer on Page 1887*

## 15.3 Levers

A lever pivots on a fulcrum. To decrease the necessary force, the load is placed closer to the fulcrum than where the force is applied.

Physicists also discuss the concept of *torque* created by a force. When you apply force to a lever, the torque is the product of the force you exert and the distance from the point of rotation.

Torque is typically measured in newton-meters (N·m).

To balance two torques, the products of force and distance must be equal. Thus, assuming the forces are applied in the correct direction, the equation becomes:

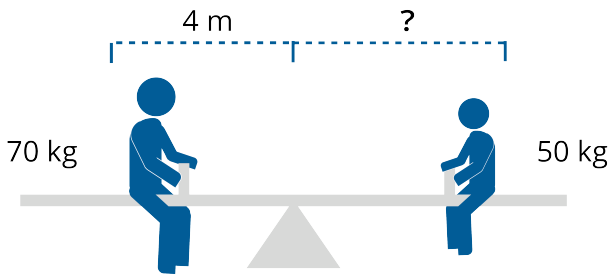
$$R_L F_g = R_A F_A \tag{15.3}$$

where $R_L$ and $R_A$ represent the distances from the fulcrum to where the load's weight and the applied force are exerted, respectively, and $F_g$ and $F_A$ are the magnitudes of the forces.

### *Exercise 40* Lever

Paul, whose mass is 70 kilograms, sits on a see-saw 4 meters from the fulcrum. Jan, whose mass is 50 kilograms, wishes to balance the see-saw. How far should Jan sit from the fulcrum?

*Working Space*




*Answer on Page 1887*

## 15.4 Gears

Gears are rotating parts of machines that transmit torque or other types of rotational motion through intertwined teeth. Often gears are meshed together; two or more meshed gears are referred to as a gear train.

Gears have teeth that mesh with each other. When you apply torque to one gear, it transfers torque to the other. The resulting torque is increased or decreased depending on the ratio of the number of teeth on the gears. We will go in depth on torque in a future chapter, but know that torque depends on the radius of the gear, and the radius of a gear is proportional to the number of teeth, the tooth count directly controls the torque.

### 15.4.1 Involute Gears

Gear teeth have a very unique shape. The vast majority of modern gears use what is known as an ***involute gear profile***. An involute gear is one whose tooth shape is derived

from the *involute of a circle*. This curve is generated by imagining a fixed string being unwrapped from the circumference of a circle. The path traced by the end of the string forms the involute.

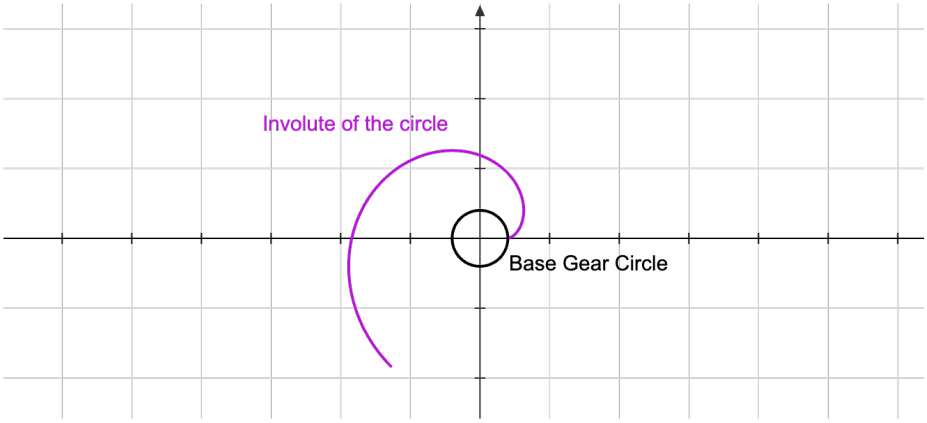



Figure 15.2: The involute of a circle shown. The purple line is revolved for $4\pi$ radians or 2 revolutions (for clarity).

Although the involute curve resembles a spiral, it is not an Archimedean spiral. Instead, its geometry is defined by the constant unwrapping of the string, which gives the involute several important mechanical properties.

The primary advantage of the involute profile shape is that it ensures smooth and reliable torque transfer between two or more meshing gears. When two involute gears engage, their teeth make contact at a single point that moves along a straight line known as the **line of action**. This line is tangent to the base circles of both gears. As the gears rotate, the point of contact travels along this line while maintaining a constant pressure angle, which allows the gears to transmit motion at a constant speed ratio even if the center distance between them varies slightly.

Some more gear construction terminology:

**The pressure angle** The acute angle between the line of action and a normal to the line connecting the gear centers. May vary depending on the involute shape, but most commonly $20°$.

**Addendum** The difference between the pitch circle and its tooth tip circle.

**Dedendum** The difference between the radius of the pitch circle of a gear and its root circle

**Tooth Height** The distance between its root circle and the tip are called the tooth height (h).

**Total Height of Tooth** The total height of the gear is the sum of the addendum (ha = 1.00m) and the dedendum (hf = 1.25m).

FIXME Possible gear diagram in tikz with labeled everything



Figure 15.3: Two gears meshing together diagram (not directly conjoined for clarity).

For part of the motion, Gear One pushes Gear Two down. Once the contact point passes the midpoint of the line of action, Gear One pushes up on Gear Two. This causes a constant transfer of torque.

FIXME Diagram of this

A parametric function, defined by $x(\theta)$ and $y(\theta)$ is given by

$$\begin{bmatrix} x(\theta) \\ y(\theta) \end{bmatrix} = r \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} + r\theta \begin{bmatrix} \sin(\theta) \\ -\cos(\theta) \end{bmatrix}$$

For gears, a large torque makes it easier to start rotating an object or to continue rotating it against resistance, such as climbing a hill or accelerating a heavy load.

**Power**, on the other hand, measures how quickly work is being done. In rotational motion, power depends on both torque and rotational speed. A system can produce high torque at low speed, or lower torque at high speed, and still deliver the same power.

Gears allow a machine to adjust this balance between torque and speed without significantly changing the power being transmitted (neglecting losses). When gears increase torque, they reduce rotational speed, and visa versa.

A familiar example is riding a bicycle up a hill. When climbing, it becomes difficult to pedal in a high gear because the torque required at the pedals is large. Shifting to a lower gear increases the torque applied to the rear wheel for the same pedaling effort, making it easier to climb. However, the pedals must turn more times for each rotation of the wheel,

so the bicycle moves more slowly. The rider's power output remains roughly the same, but it is delivered as higher torque at lower speed.

The same principle applies to the gears in a car. When starting from rest or climbing a steep incline, a low gear is used to provide high torque at the wheels, allowing the vehicle to overcome inertia and gravity. At highway speeds, higher gears are used to reduce torque and increase speed, improving efficiency and reducing engine wear. In both cases, the engine produces power, and the transmission adjusts how that power is distributed between torque and rotational speed.

In mechanical systems, gears therefore do not create power; instead, they shape how power is delivered.

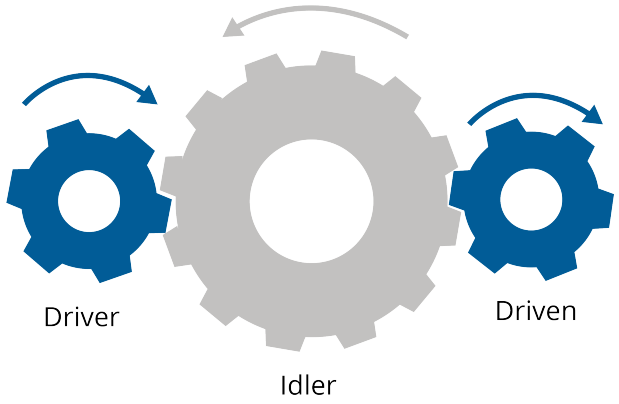



Figure 15.4: .

### 15.4.2 Teeth

If $N_A$ is the number of teeth on the gear you are turning with a torque of $T_A$, and $N_L$ is the number of teeth on the gear it is turning, the resulting torque is:

$$T_L = \frac{N_A}{N_L} T_A$$

### *Exercise 41* Gears

In a bicycle, the goal is not always to gain mechanical advantage, but to spin the pedals slower while applying more force.

You like to pedal your bike at 70 revolutions per minute. The chainring connected to your pedals has 53 teeth. The circumference of your tire is 2.2 meters. You want to ride at 583 meters per minute.

How many teeth should the rear sprocket have?

*Working Space*

*Answer on Page 1888*

## 15.5 Hydraulics

In a hydraulic system, such as a car's braking system, you exert force on a piston filled with fluid. The fluid transmits this pressure into another cylinder, where it pushes yet another piston that moves the load. The pressure at each end of the hydraulic system must be the same.

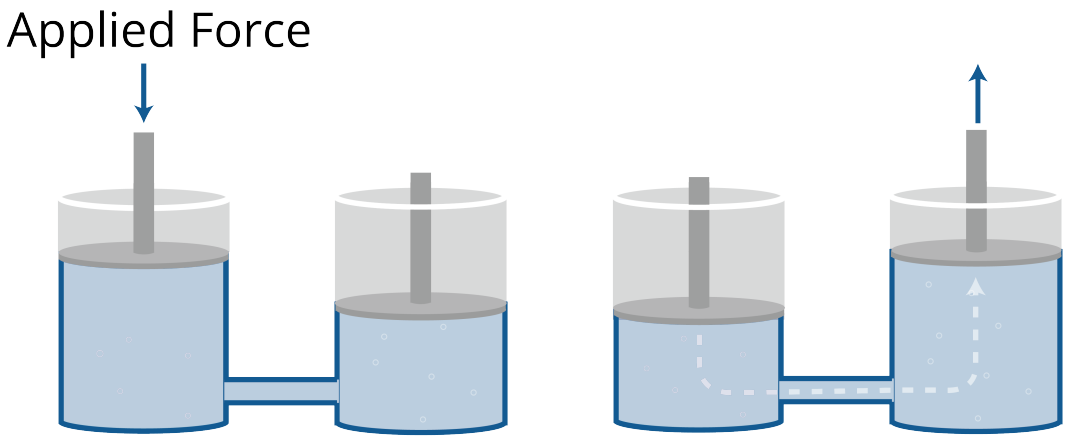



Figure 15.5: A diagram showing the transfer of fluid from one container to the other.

*Pressure* is force applied to an area; it is calculated by dividing the force by the area. The pressure in the fluid is typically measured in pascals (Pa), which is equivalent to $N/m^2$. We will use pascals for this calculation.

To calculate the pressure you create, divide the force applied $F_a$ by the area of the piston head $A$. To determine the force on the other piston, multiply the pressure by the area of the second piston.

$$P = \frac{F_{a_1}}{A_1} = \frac{F_{a_2}}{A_2} \tag{15.4}$$

### *Exercise 42* Hydraulics

Your car has disc brakes. When you apply 2,500,000 pascals of pressure to the brake fluid, the car stops quickly. As the car designer, you want this to require only 12 newtons of force from the driver's foot.

What should the radius of the master cylinder (the piston the driver pushes) be?

*Working Space*

*Answer on Page 1888*

## 15.6 Pulleys

Pulleys are anything that changes the direction of a force, typically by using a wheel and a rope. A single pulley can make things easier by allowing you to pull down instead of pushing up. This lets you use your body weight to help you pull rather than just your arm strength.

By attaching multiple pulleys, you can do more and actually reduce the force needed to lift a load. For example, if you use two pulleys, you can reduce the force needed to lift a load by half. However, you will have to pull twice as much rope to lift the load the same distance.

Each additional pulley you add to the system adds another segment of rope that supports the load. The mechanical advantage of a pulley system is equal to the number of rope segments supporting the load.

$$MA = N$$

where N is the number of rope segments supporting the load.

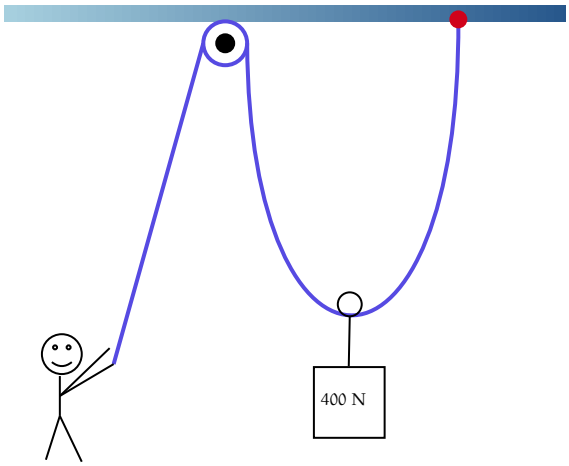



Figure 15.6: A 400 N weight being distributed by one pulley.

Here, a rope is carrying a 400 N weight. Attached to the weight is a movable pulley, while the top pulley is fixed to the ceiling. Assuming an ideal (massless) rope and frictionless pulleys, the tension is the same everywhere in the rope. The movable pulley is supported by two rope segments, so the upward force on the load is 2T. For the load to be held or lifted at constant speed,

$$2T = 400\ \text{N} \quad \Rightarrow \quad T = 200\ \text{N}$$

Assuming an ideal (massless) rope and frictionless pulleys, the tension is the same everywhere in the rope, so there is 200 N throughout the rope. Since there are two ropes supporting the load, the Mechanical Advantage is 2.

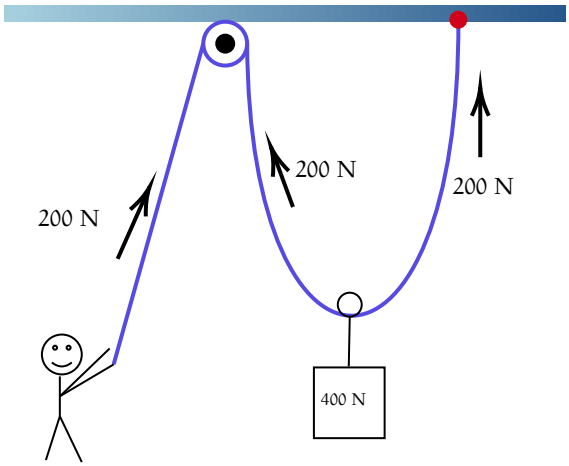



Figure 15.7: The tension in the rope is 200 N.

However, this does come at a cost! The weight of the load is 400 N so the pulley system must provide a total upward force of 400 N on the load. However, both sections of rope

(although technically the same rope) pull at 200 N for a total of 400 N.

The work done, however, is still the same! When the person pulls the rope down by 2 m, each supporting rope segment shortens by 1 m, so the load rises by 1 m. This is all ignoring friction and assuming constant speed. Work done by the person:

$$W = \mathrm{F} \times \mathrm{d} = 200\,\mathrm{N} \times 2\,\mathrm{m} = 400\,\mathrm{J}$$

Work gained by the load:

$$W = 400\,\mathrm{N} \times 1\,\mathrm{m} = 400\,\mathrm{J}$$

This satisfies our work equation

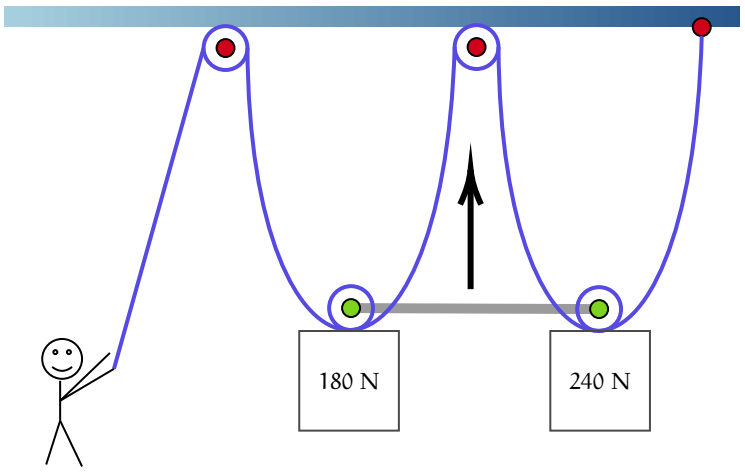
$$\mathrm{Work}_{\mathrm{in}} = \mathrm{Work}_{\mathrm{out}} \tag{15.5}$$

Note that we can then rewrite our mechanical advantage as a proportion of distance:

$$\mathrm{MA} = \frac{\mathrm{d}_{\mathrm{effort}}}{\mathrm{d}_{\mathrm{load}}} \tag{15.6}$$

### *Exercise 43* Pulleys

*Working Space*




A person is pulling 2 weights tensioned by 4 pulleys. There are 2 fixed pulleys (shown in red) and 2 movable pulleys (shown in green). The two movable pulleys are fixed such that they move together vertically. Calculate the input force that is needed to begin lifting the weights.

*Answer on Page 1888*

## 15.7 Wedges

A wedge is a triangular shaped simple machine that converts forces from one direction to another. Often, it is from a vertical direction to a horizontal one.

Similar to an inclined plane, wedges have steep angles to manipulate the properties of triangles. The force applied to the top portion of a wedge is transferred acting perpendicular to the sloped sides of the wedge.

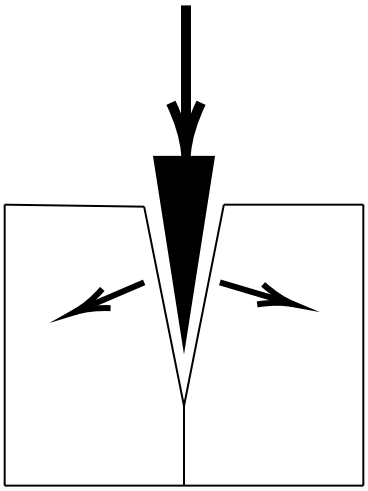


Figure 15.8: A wedge splitting a block of wood, with the forces acting perpendicular.

Similar to the inclined plane, the Mechanical Advantage is the proportion of the vertical rise or width, and the length of the sloped portion of the triangle.

$$\text{MA} = \frac{\text{L}}{\text{V}} \tag{15.7}$$

The smaller the angle of a wedge, the greater the ratio of the length of its slope to its width, and the more mechanical advantage it will yield. This is a concept that comes directly from trigonometry.

## *Exercise 44* Wedges 1

An axe is used to split wood. The length of the wedge is 12 cm and the thickness is 3 cm. Calculate the mechanical advantage.

*Working Space*

*Answer on Page 1889*

## *Exercise 45* Wedges 2

Explain why a flathead-screwdriver tip is a poor option for cutting a steak compared to a knife.

*Working Space*

*Answer on Page 1889*

Chapter 16

# Basic Statistics

You live near a freeway, and someone asks you, "How fast do cars on that freeway drive?"

You say, "Pretty fast."

They reply, "Can you be more specific?"

So, you pull out your radar gun you happen to always keep on you, and tell them, "That one is going 32.131 meters per second."

To which they say, "I don't want to know about that specific car. I want to know about all the cars."

So, you spend the day beside the freeway measuring the speed of every car that goes by. And you get a list of a thousand numbers, including:

| | | |
|---|---|---|
| 30.462 m/s | 29.550 m/s | 29.227 m/s |
| 37.661 m/s | 27.899 m/s | 28.113 m/s |
| 24.382 m/s | 35.668 m/s | 43.797 m/s |
| 31.312 m/s | 37.637 m/s | 30.891 m/s |

There are 12 numbers here. We say that there are 12 *samples*.

## 16.1 Mean

We often talk about the *average* of a set of samples, which is the same as the *mean*. To get the mean, sum up the samples and divide that number by the number of samples.

The numbers in that table sum to 388.599. If you divide that by 12, you find that the mean of those samples is 32.217 m/s.

We typically use the greek letter $\mu$ ("mu") to represent the mean.

**Definition of Mean**

If you have a set of samples $x_1, x_2, \ldots, x_n$, the mean is:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

This may be the first time you are seeing a summation ($\sum$). The equation above is equivalent to:

$$\mu = \frac{1}{n}(x_1 + x_2 + \ldots + x_n)$$

## *Exercise 46* Mean Grade

*Working Space*

Teachers often use the mean for grading. For example, if you took six quizzes in a class, your final grade might be the mean of the six scores. Find the mean of these six grades: 87, 91, 98, 65, 87, 100.

*Answer on Page 1889*

If you tell your friend, "I measured the speed of 1000 cars, and the mean is 31.71 m/s", your friend will wonder, "Are most of the speeds clustered around 31.71? Or are they all over the place and just happen to have a mean of 31.71?" To answer this question, we use variance.

## 16.2 Variance

**Definition of Variance**

If you have $n$ samples $x_1, x_2, \ldots, x_n$ that have a mean of $\mu$, the *variance* is defined to be:

$$v = \frac{1}{n}\sum_{i=1}^{n} (x_i - \mu)^2$$

That is, you figure out how far each sample is from the median, you square that, and then you take the mean of all those squared distances.

| $x$ | $x - \mu$ | $(x - \mu)^2$ |
|---|---|---|
| 30.462 | -1.755 | 3.079 |
| 29.550 | -2.667 | 7.111 |
| 29.227 | -2.990 | 8.938 |
| 37.661 | 5.444 | 29.642 |
| 27.899 | -4.318 | 18.642 |
| 28.113 | -4.104 | 16.839 |
| 24.382 | -7.835 | 61.381 |
| 35.668 | 3.451 | 11.912 |
| 43.797 | 11.580 | 134.106 |
| 31.312 | -0.905 | 0.818 |
| 37.637 | 5.420 | 29.381 |
| 30.891 | -1.326 | 1.757 |
| $\sum x = 386.599$ | | $\sum (x - \mu)^2 = 323.605$ |
| mean = 32.217 | | variance = 26.967 |

Thus, the variance of the 12 samples is 26.967. The bigger the variances, the farther the samples are spread apart; the smaller the variances, the closer samples are clustered around the mean.

Notice that most of the data points deviate from the mu by 1 to 5 m/s. Isn't it odd that the variance is a big number like 26.967? Remember that it represents the average of the squares. Sometimes, to get a better feel for how far the samples are from the mean, we use the square root of the variance, which is called *the standard deviation*.

The standard deviation of your 12 samples would be $\sqrt{26.9677} = 5.193$ m/s.

The standard deviation is used to figure out if a data point is an outlier. For example, if you are asked, "That car that just sped past. Was it going freakishly fast?" You might respond, "No, it was within a standard deviation of the mean." or "Yes, its speed was 2 standard deviations more than the mean. They will probably get a ticket."

A singular $\mu$ usually represents the mean. $\sigma$ usually represents the standard deviation. So $\sigma^2$ represents the variance.

### *Exercise 47* Variance of Grades

*Working Space*

Now, find the variance for your six grades. As a reminder, they were: 87, 91, 98, 65, 87, 100.

What is your standard deviation?

*Answer on Page 1889*

## 16.3 Median

Sometimes you want to know where the middle is. For example, you want to know the speed at which half the cars are going faster and half are going slower. To get the median, you sort your samples from smallest to largest. If you have an odd number of samples, the one in the middle is the median. If you have an even number of samples, we take the mean of the two numbers in the middle.

In our example, you would sort your numbers and find the two in the middle:

24.382
27.899
28.113
29.227
29.550
**30.462**
**30.891**
31.312
35.668
37.637
37.661
43.797

You take the mean of the two middle numbers: $(30.462 + 30.891)/2 = 30.692$. The median speed would be 30.692 m/s.

Medians are often used when a small number of outliers majorly skew the mean. For example, income statistics usually use the median income because a few hundred billion-

aires raise the mean a great deal.

### *Exercise 48* Median Grade

*Working Space*

Find the median of your six grades: 87, 91, 98, 65, 87, 100.

*Answer on Page 1889*

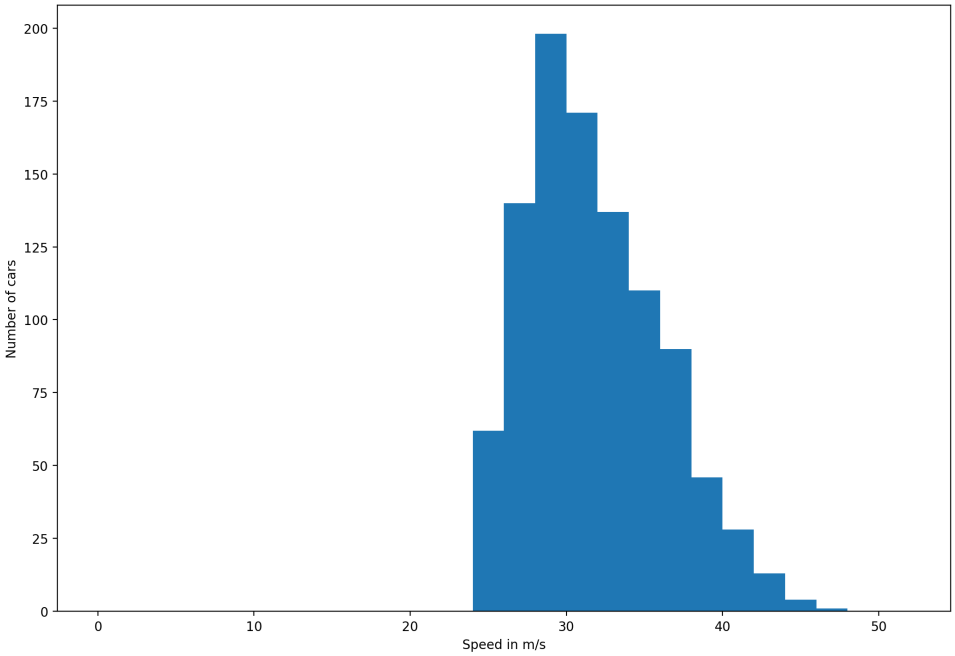
## 16.4 Histograms and Bell Curves

A histogram is a bar chart that shows how many samples are in each group. In our example, we group cars by speed. Maybe we count the number of cars going between 30 and 32 m/s. Next, we count the cars going between 32 and 34 m/s. Finally, we make a bar chart from that data.

Your 1000 cars would break up into these groups:

| | |
|---|---|
| 0 - 2 m/s | 0 cars |
| 2 - 4 m/s | 0 cars |
| 4 - 6 m/s | 0 cars |
| ... | ... |
| 20 - 22 m/s | 0 cars |
| 22 - 24 m/s | 0 cars |
| 24 - 26 m/s | 65 cars |
| 26 - 28 m/s | 160 cars |
| 28 - 30 m/s | 175 cars |
| 30 - 32 m/s | 168 cars |
| 32 - 34 m/s | 150 cars |
| 34 - 36 m/s | 114 cars |
| 36 - 38 m/s | 79 cars |
| 38 - 40 m/s | 52 cars |
| 40 - 42 m/s | 20 cars |
| 42 - 44 m/s | 12 cars |
| 44 - 46 m/s | 4 cars |
| 46 - 48 m/s | 1 cars |
| 48 - 50 m/s | 0 cars |

Next, we make a bar chart from that:




Often, a histogram will tell the story of the data. Here, you can see that no one is going less than 24 m/s, but a lot of people travel at 30 m/s. There are a few people who travel over 40 m/s, but there are also a couple of people who drive much faster than anyone else.

If we look closely at the shape of our histogram, we notice something interesting: it has a smooth, rounded hill in the middle and it tapers off on both sides. This is a common pattern in statistics, and it often suggests that the data follows what's called a **normal distribution**, also known as a **bell curve**.

A **bell curve** is a continuous curve that models how data is distributed when:

- Most values are near the average (mean), located at the peak of the curve,
- Fewer values are far from the mean, the tails of the curve,
- The distribution is roughly symmetric.

In our case, the majority of cars are traveling around 30–32 m/s. There are fewer cars going slower or faster than that. If we collected even more data (say, 10,000 cars), the histogram would start to resemble a smooth bell curve even more closely. The units of the x-axis (speed) would still be the same, but the y-axis would represent the *probability density* of finding a car at a certain speed, rather than just the count of cars in each speed

range.

The peak of the bell curve represents the **mean speed**, and the spread of the curve is related to the **standard deviation** — a measure of how spread out the speeds are. Most data falls within:

- 1 standard deviation of the mean ($\mu \pm \sigma$): about 68% of the cars,
- 2 standard deviations: about 95%,
- 3 standard deviations: about 99.7%.

This means that if we know the mean speed and the standard deviation, we can predict how many cars will be within certain speed ranges. If you have ever heard of Six Sigma methodology, that means data falls within six standard deviations of the mean, which is a very high level of quality control (3.4 defects per million measurements!).

So when we say a dataset "looks like a bell curve," we mean that it has a predictable and symmetric structure, often meaning it is a reliable set of data with few outliers and consistent results.

## 16.5 Root-Mean-Squared

Scientists have a mean-like statistic that they love. It is named quadratic mean, but most just calls it Root-Mean-Squared, or RMS.

**Definition of RMS**

If you have a list of numbers $x_1, x_2, \ldots, x_n$, their RMS is

$$\sqrt{\frac{1}{n}\left(x_1^2 + x_2^2 + \ldots + x_n^2\right)}$$

You are taking the square root of the mean of squares of the samples, thus the name Root-Mean-Squared.

Using your 12 samples:

| $x$ | $x^2$ |
|---|---|
| 30.462 | 927.933 |
| 29.550 | 873.203 |
| 29.227 | 854.218 |
| 37.661 | 1418.351 |
| 27.899 | 778.354 |
| 28.113 | 790.341 |
| 24.382 | 594.482 |
| 35.668 | 1272.206 |
| 43.797 | 1918.177 |
| 31.312 | 980.441 |
| 37.637 | 1416.544 |
| 30.891 | 954.254 |
| Mean of $x^2$ | 1064.875 |
| RMS | 32.632 |

Why is RMS useful? Let's say that all cars had the same mass $m$, and you need to know what the average kinetic energy per car is. If you know the RMS of the speeds of the cars is $v_{rms}$, the average kinetic energy for each car is

$$k = \frac{1}{2}mv_{rms}^2$$

(You don't believe me? Let's prove it. Substitute in the RMS:

$$k = \frac{1}{2}m\sqrt{\frac{1}{n}\left(x_1^2 + x_2^2 + \ldots + x_n^2\right)}^2$$

The square root and the square cancel each other out:

$$k = \frac{1}{2}m\frac{1}{n}\left(x_1^2 + x_2^2 + \ldots + x_n^2\right)$$

Use the distributive property:

$$k = \frac{1}{n}\left(\frac{1}{2}mx_1^2 + \frac{1}{2}mx_2^2 + \ldots + \frac{1}{2}mx_n^2\right)$$

That is all the kinetic energy divided by the number of cars, which is the mean kinetic energy per car. Quod erat demonstrandum! (That is a Latin phrase that means "which is what I was trying to demonstrate". You will sometimes see "QED" at the end of a long mathematic proof.)

Now you are ready for the punchline: Kinetic energy and heat are the same thing. Instead of cars, heat is the kinetic energy of molecules moving around. More on this soon.

Video: Mean, Median, Mode: https://www.youtube.com/watch?v=5C9LBF3b65s

CHAPTER 17

# Basic Statistics in Spreadsheets

When you completed the problems in the last section, you probably noticed how long it took to compute statistics like the mean, median, and variance by hand. Luckily, computers were designed to free us from these sorts of tedious tasks. The most basic tool for automating calculations is the spreadsheet program.

The first spreadsheet program (VisiCalc) was introduced in 1979 as a tool for finance people to play "what if" games. For example, a company might make a spreadsheet that told them how much more profit they would make if they changed from using an expensive metal to using a cheaper alloy.

There are lots of spreadsheet programs, including Microsoft's Excel, Google Sheets, and Apple's Numbers. Any spreadsheet program will work; they are all very similar. The instructions and screenshots here will be from Google Sheets — a free spreadsheet program you use through your web browser.

## 17.1 The Barrel Problem

In honor of this history, let's start by studying a business question: You have a friend who dreams of quitting her job to become a cooper. (A cooper makes barrels that are used for aging wine and whiskey.) According to her:

- It costs $45 dollars in materials to build one barrel.
- A barrel sells for $100 dollars.
- The workshop/warehouse she wants to rent costs $2000 per month.
- Taxes take 20% of her profits.
- She needs to make $4000 monthly after taxes.

She has asked you, "How many barrels do I need to make each month?"

## 17.2 Solving It Symbolically

Many problems can be solved two ways: symbolically or numerically. To solve this problem symbolically, you would write out the facts as equations or inequalities, then do

symbol manipulations until you ended up with an answer. In this case, you would let $b$ be the number of barrels and create the following inequality:

$$(1.0 - 0.2)\,(b(100 - 45) - 2000) \geq 4000$$

You would simplify it:

$$(0.8)\,(55b - 2000) \geq 4000$$

And simplify it more:

$$44b - 1600 \geq 4000$$

If that is true, then:

$$44b \geq 5600$$

And if that is true, then:

$$b \geq \frac{1400}{11}$$

1400/11 is about 127.27, so she needs to make and sell 128 barrels each month.

That is a perfect answer, and we didn't need a spreadsheet at all. However:

- As problems get larger and more realistic, it gets much more difficult to solve them symbolically.
- As soon as you say "Yes, you need to make and sell 128 barrels each month." Your friend will ask "What if I make and sell 200 barrels? How much money will I make then?"

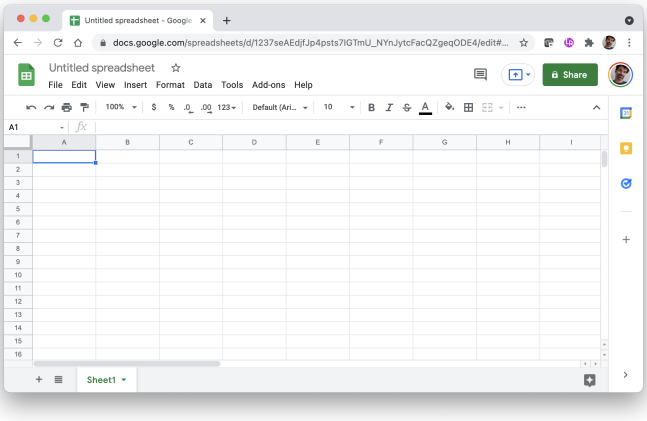
So, we use a spreadsheets to solve the problem numerically.

## 17.3 Your First Spreadsheet

Let's first make an initial spreadsheet.

In whatever spreadsheet program you are using, create a new spreadsheet document.

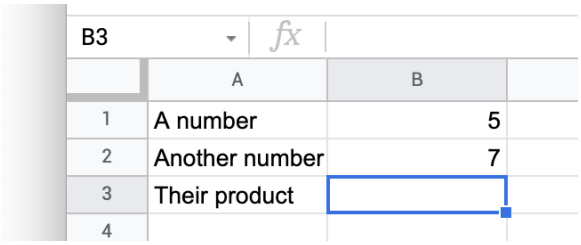
A spreadsheet is essentially a grid of cells. In each cell, you can put data (like numbers or text) and formulas.



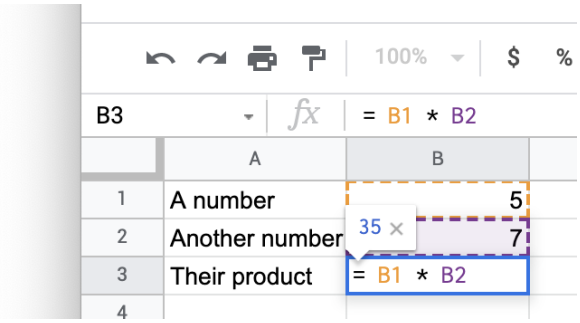

Let's put some labels in the column:

- Select the first cell (A1) and type "A number".
- Select the cell below it (A2) and type "Another number".
- Select the cell below that one (A3) and type "Their product".
- In the next column, type the number 5 in B1 and 7 in B2.

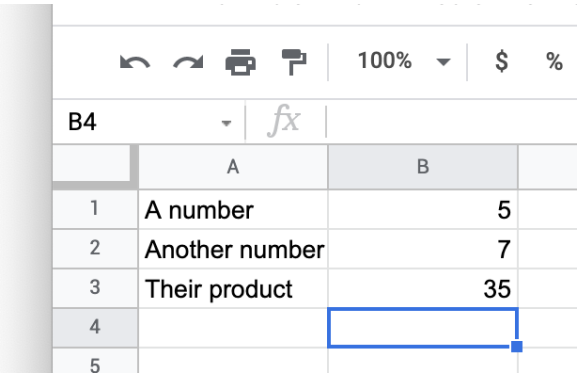
It should look like this:




Now, put a formula in cell B3. Select B3, and type "= B1 * B2". The spreadsheet knows this is a formula because it starts with '='. It will look like this as you type:




When you press Return or Tab, the spreadsheet will remember the formula, but display its value:



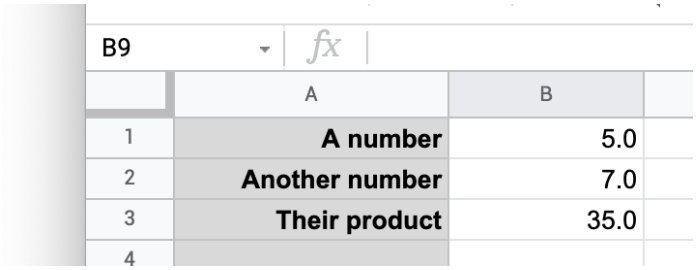

If you change the values of cell B1 or B2, the cell B3 will automatically be recalculated. Try it.

## 17.4 Formatting

Every spreadsheet lets you change the formatting of your columns and cells. They are all a little different, so play with your spreadsheet a little now. Try to do the following:

- Set the background of the first column to light gray.
- Right-justify the text in the first column.
- Make the text in the first column bold.
- Make the numbers in the second column have one digit after the decimal point.

It should look something like this:




That's a spreadsheet. You have a grid of cells, each of which can hold a value or a formula that uses values from other cells. The cells containing formulas automatically update as you edit the values in the other cells.

## 17.5 Comma-Separated Values

A large amount of data is exchanged in a file format called *Comma-Separated Values* or just CSV. Each CSV file holds one table of data. It is a text file, and each line of text corresponds to one row of data in the table. The data in each column is separated by a comma. The first line of a CSV is usually the names of the columns. A CSV might look like this:

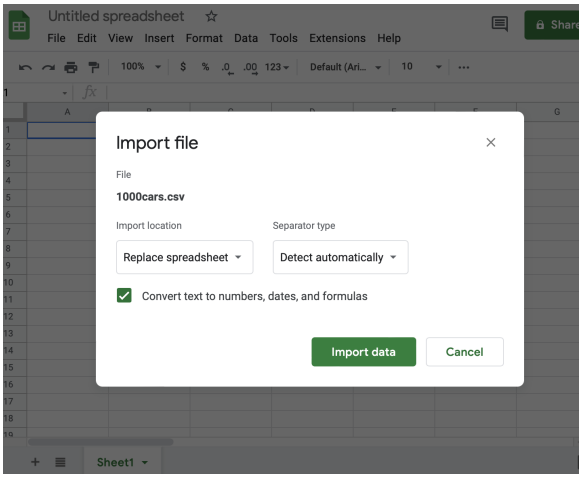
```
studentID,firstName,lastName,height,weight
1,Marvin,Sumner,260,45.3
2,Lucy,Harris,242,42.2
3,James,Boyd,261,44.2
```

In your digital resources for this module, you should have a file called `1000cars.csv`. It is a CSV with only one column called "speed". The first few lines look like this:

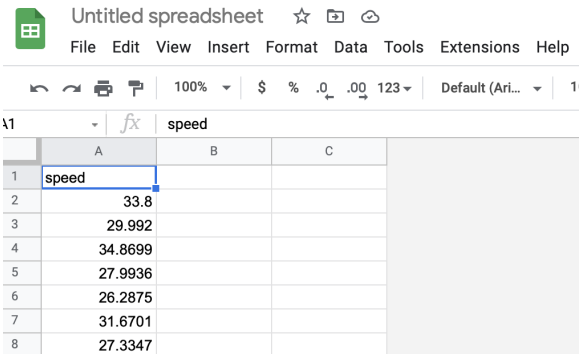
```
speed
33.8000
29.9920
34.8699
27.9936
```

There is a title line and 1000 data lines.

Import this CSV into your spreadsheet program. In Google Sheets, it looks like this:



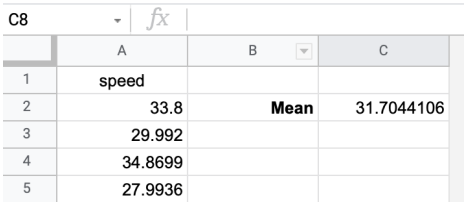

You should see a long, long column of data appear. (Mine goes from cell A2 through A1001.)




## 17.6 Statistics in Spreadsheets

Let's take the mean of all 1000 numbers. In cell B2, type in a label: "Mean". (Feel free to format your labels as you wish. Bolding is recommended.)

In cell C2, enter the formula "=AVERAGE(A2:A1001)". When you press return, the cell will show the mean: 31.70441, if done correctly.




Notice that by specifying that the function `AVERAGE` was to be performed on a range of cells: cells A2 through (":") A1001.

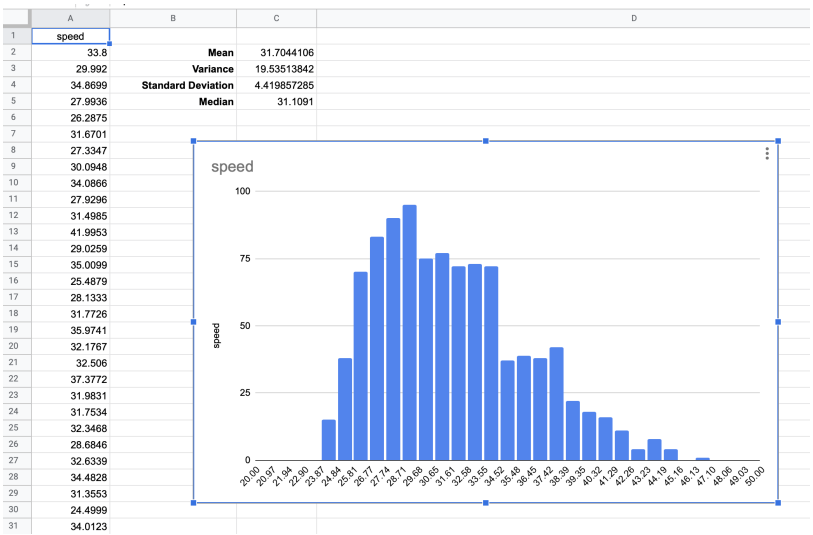
Do the calculations for variance, standard deviation, and median.

- The function for variance is `VAR`.
- The function for standard deviation is `STDEV`.
- The function for median is `MEDIAN`.

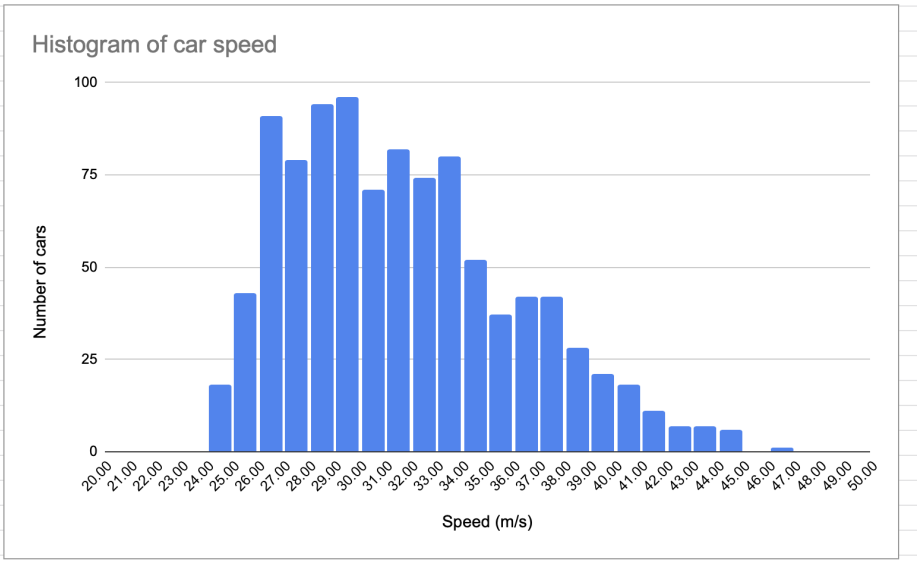
| | A | B | C |
|---|---|---|---|
| 1 | speed | | |
| 2 | 33.8 | **Mean** | 31.7044106 |
| 3 | 29.992 | **Variance** | 19.53513842 |
| 4 | 34.8699 | **Standard Deviation** | 4.419857285 |
| 5 | 27.9936 | **Median** | 31.1091 |
| 6 | 26.2875 | | |
| 7 | 31.6701 | | |

## 17.7 Histogram

Most spreadsheets have the ability to create a histogram. In Google Sheets, you select the entire range A2:A1001 by selecting the first cell and then shift-clicking the last. Next, you choose Insert→Chart. In the inspector, change the type of the chart to a histogram (at the bottom under "other"). This will get you a basic histogram.




Play with the formatting to see how unique you can make data. Here is an example:




## *Exercise 49* RMS

*Working Space*

In your spreadsheet, calculate the quadratic mean (the root-mean-squared) of the speeds.

You will need the following three functions:

- `SUMSQ` returns the sum of the squares of a range of cells.
- `COUNT` returns the number of cells in a range that contains numbers.
- `SQRT` returns the square root of a number.

*Answer on Page 1890*

## 17.8 The Return of the Barrel Problem

Let's get back to our example. Put labels in the A column:

- Barrels produced (per month)
- Materials cost (per barrel)
- Sale price (per barrel)
- Pre-tax earnings (per month)
- Taxes (per month)
- Take home pay (per month)

Format them any way you like. It should look something like this:

| | A |
|---|---|
| 1 | **Barrels Produced** (per month) |
| 2 | **Materials cost** (per barrel) |
| 3 | **Sale price** (per barrel) |
| 4 | **Rent** (per month) |
| 5 | **Pretax Earnings** (per month) |
| 6 | **Taxes** (per month) |
| 7 | **Take home pay** (per month) |
| 8 | |

In the B column, the first four cells are values (not formulas):

- 115 formatted as a number with no decimal point
- 45 formatted as currency
- 100 formatted as currency
- 2000 formatted as currency

It should look something like this:

| | A | B |
|---|---|---|
| 1 | **Barrels Produced** (per month) | 115 |
| 2 | **Materials cost** (per barrel) | $45.00 |
| 3 | **Sale price** (per barrel) | $100.00 |
| 4 | **Rent** (per month) | $2,000.00 |

The next three cells in the B column will have formulas:

- B1 * (B3 - B2) - B4
- 0.2 * B5

- B5 - B6

It should look something like this:

| | A | B |
|---|---|---|
| 1 | **Barrels Produced** (per month) | 115 |
| 2 | **Materials cost** (per barrel) | $45.00 |
| 3 | **Sale price** (per barrel) | $100.00 |
| 4 | **Rent** (per month) | $2,000.00 |
| 5 | **Pretax Earnings** (per month) | $4,325.00 |
| 6 | **Taxes** (per month) | $865.00 |
| 7 | **Take home pay** (per month) | $3,460.00 |
| 8 | | |

Now you can share this spreadsheet with your friend, and she can put different values into the cells for what-if games, such as "If I can get my materials cost down to $42 per barrel, what happens to my take home pay?"

Sometimes it is nice to show a range of values for a variable or two. In this case, it might be nice to show your friend what the numbers look like if she produces 115, 120, 125, 130, 135, or 140 barrels per month.

We have one column, and now we need six. How do we duplicate cells?

1. Click B1 to select it, then shift-click on B7 to select all seven cells.
2. Copy them. (Depending on what program you are using, there is likely a menu item for this.)
3. Click C1 to select it
4. Paste them.

| | A | B | C |
|---|---|---|---|
| 1 | **Barrels Produced** (per month) | 115 | 115 |
| 2 | **Materials cost** (per barrel) | $45.00 | $45.00 |
| 3 | **Sale price** (per barrel) | $100.00 | $100.00 |
| 4 | **Rent** (per month) | $2,000.00 | $2,000.00 |
| 5 | **Pretax Earnings** (per month) | $4,325.00 | $4,325.00 |
| 6 | **Taxes** (per month) | $865.00 | $865.00 |
| 7 | **Take home pay** (per month) | $3,460.00 | $3,460.00 |
| 8 | | | |

We want the first cell in the new column to be 120. You could just type in 120, but let's do something more clever. Put a formula into that cell: = B1 + 5. Now, the cell should show 120.

Why did we put in a formula? When we duplicate this column, this cell will always have 5 more barrels than the cell to its left.

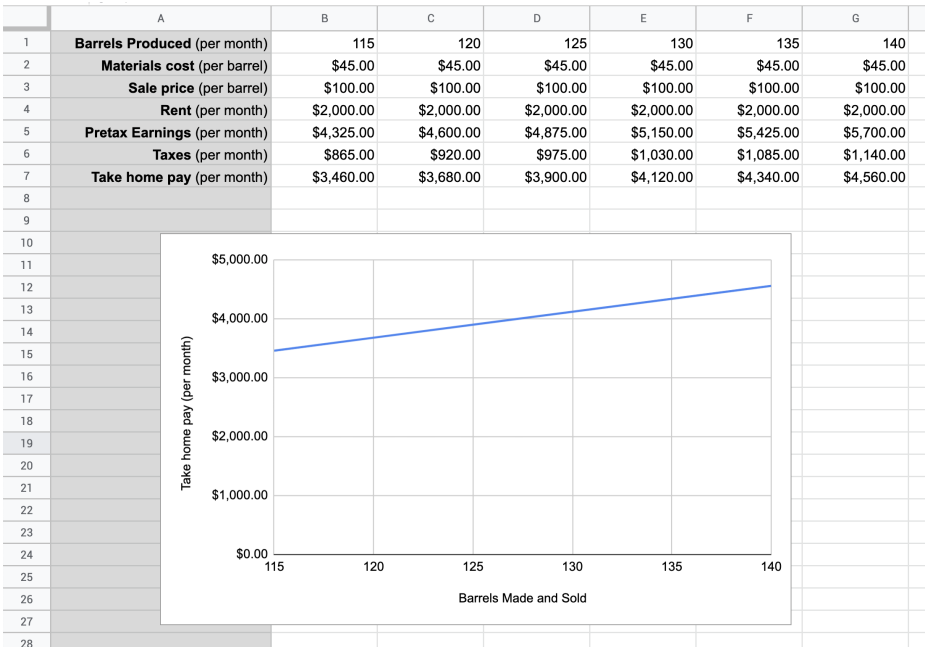
Next, let's duplicate the second column a few times. The easy way to do this is to select the cells as you did before, then drag the lower-right corner to the right until column G is in the selection. When you end the drag, the copies will appear:

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | **Barrels Produced** (per month) | 115 | 120 | 125 | 130 | 135 | 140 |
| 2 | **Materials cost** (per barrel) | $45.00 | $45.00 | $45.00 | $45.00 | $45.00 | $45.00 |
| 3 | **Sale price** (per barrel) | $100.00 | $100.00 | $100.00 | $100.00 | $100.00 | $100.00 |
| 4 | **Rent** (per month) | $2,000.00 | $2,000.00 | $2,000.00 | $2,000.00 | $2,000.00 | $2,000.00 |
| 5 | **Pretax Earnings** (per month) | $4,325.00 | $4,600.00 | $4,875.00 | $5,150.00 | $5,425.00 | $5,700.00 |
| 6 | **Taxes** (per month) | $865.00 | $920.00 | $975.00 | $1,030.00 | $1,085.00 | $1,140.00 |
| 7 | **Take home pay** (per month) | $3,460.00 | $3,680.00 | $3,900.00 | $4,120.00 | $4,340.00 | $4,560.00 |

Nice, right? Now your friend can easily see how many barrels correspond to how much take-home pay. But do you know what would be even more helpful? A graph.

## 17.9 Graphing

Graphing is a little different on every different platform. Here is what you want the graph to look like:



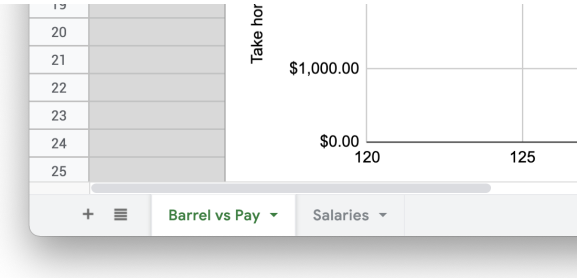

On Google Sheets:

1. Select cells B7 through G7.
2. Choose the menu item Insert -> Chart.
3. Choose the chart type (Line)

4. Add the X-axis to be B1 through G1.
5. Under the Customize tab, Set the label for the X-axis to be "Barrels Made and Sold".
6. Delete the chart title (which is the same as the Y-axis label).

## 17.10 Other Things You Should Know About Spreadsheets

Your spreadsheet document can have several "Sheets". Each has its own grid of cells. The sheet has a name; usually, you call it something like "Salaries". When you need to use a value from the "Salaries" sheet in another sheet, you can specify "Salaries!A2" — that is, cell A2 on sheet "Salaries". To flip between the sheets, there is usually a tab for each at the bottom of the document.




By default, the cell references are relative. In other words, when you write a formula in cell H5 that references the value in cell G4, the cell remembers "The cell that is one up and one to the left of me." Thus, if you copy that formula into B9, now that formula reads the value from A8.

If you want an absolute reference, you use $. If H5 references $G$4, G4 will be used no matter where on the sheet the formula is copied to.

You can use the $ on the row or column. In $A4, the column is absolute and the row is relative. In A$4, the row is absolute and the column is relative.

## 17.11 Challenge: Make a spreadsheet

You have a company that bids on painting jobs. Make a spreadsheet to help you do bids. Here are the parameters:

- The client will tell you how many square meters of wall needs to be painted.
- Paint costs $0.02 per square meter of wall
- On average, a square meter of wall takes 0.02 hours to paint.

- You can hire painters at $15 per hour.
- You add 20% to your estimated costs for a margin of error and profit.

Make a spreadsheet such that when you type in the square meters to be painted, the spreadsheet tells you how much you will spend on paint and labor. It should also tell you what your bid should be.

CHAPTER 18

# Introduction to Python

In this chapter we will discuss the very basics of Python. No we are not talking about snake anatomy, but the programming language Python. Python is a high-level, interpreted programming language known for its readability and versatility. It is widely used in various fields such as web development, data analysis, artificial intelligence, scientific computing, and more. It can be used for data parsing, visualization, and even machine learning. Python's syntax is designed to be easy to read and write, making it an excellent choice for beginners and experienced programmers alike. Throughout this chapter, we encourage you to write out the provided lines of code yourself! This will reinforce skills like computer literacy, Python syntax and troubleshooting, and of course, typing speed.

## 18.1 Getting Started with Python

To get started with Python, you need to have it installed on your computer. You can download the latest version of Python from the official website: `https://www.python.org/downloads/`. Follow the installation instructions for your operating system.

Once you have python installed, you can write and run Python code using various methods:

- **Interactive Shell:** You can open a terminal or command prompt and type `python` or `python3` to start an interactive Python shell where you can type and execute Python commands line by line.
- **Script Files:** You can write your Python code in a text file with a `.py` extension and run it using the command `python filename.py` or `python3 filename.py` in the terminal.
- **Integrated Development Environments (IDEs):** There are several IDEs available for Python, such as PyCharm, VSCode, and Jupyter Notebooks, which provide a more user-friendly environment for writing and running Python code. We will assume you will use VSCode, as it is widely used among computer scientists and developers around the world, and has direct integration to GitHub.

We are going to install VSCode as our IDE for this course. You can download it from `https://code.visualstudio.com/`. After installing VSCode, you will also need to install the Python extension for VSCode, which can be found in the Extensions Marketplace within VSCode.

### 18.1.1 The Console and Running Python Programs

When working with Python, it is important to understand the difference between *writing code* and *executing code*. Writing code simply means typing instructions into a file using a text editor or IDE. Executing code means telling Python to read those instructions and perform them.

The *console*, also called the *terminal* or *command prompt*, is a text-based interface where you can run programs and view their output. Any text printed using the `print()` function will appear in the console.

When you run a Python program, Python reads your file from top to bottom and executes each line in order. This is an important idea that will come up often: **Python code is executed sequentially, one line at a time**.

### 18.1.2 Running a Python File

To run a Python file, you must use the console. In VSCode, you can open the integrated terminal by selecting `Terminal` → `New Terminal` from the menu. Make sure your terminal is open in the same folder as your Python file.

If your file is named `hello.py`, you can run it by typing:

```
python hello.py
```

or, on some systems:

```
python3 hello.py
```

After pressing Enter, Python will execute the file and display any output in the console. If there are errors in your code, Python will stop execution and display an error message instead. You may also see a green play button at the top of your window, you could also run your program with that!

At this point, it is helpful to remember:

- Writing code does nothing by itself
- Code only runs when you explicitly execute the file
- Output from `print()` appears in the console

## 18.2 Output `print`

Let's talk more about the `print()` statement. It can take a string, multiple variables, or a combination of strings and variables. We are going to introduce the syntax first, and then expand on the definitions later.

```
x = "This is a variable."
print("This is a string.", x)
```

This should output `This is a string. This is a variable`. You can also do multiple variables:

```
x = "Good"
y = "Morning,"
z = "Chicago"
print(x, y, z)
```

This will output `Good Morning, Chicago`. Alternatively, printing can have addition within it:

```
x = "Good"
y = "Morning,"
z = "Chicago"
print(x+y+z)
```

This will output `GoodMorning,Chicago`. Notice the difference!

If you want to combine numbers and strings,

```
x = "it is"
y=78
z="degrees outside"
print(x+y+z)
```

***This will not work!*** Instead, we should use commas:

```
x = "it is"
y=78
z="degrees outside"
print(x, y, z)
```

This will correctly output "it is 78 degrees outside"

By default, a space is added in between each argument (where the commas would be).

We can change what separates each variable by the `sep=` argument.

```
x = "it is"
y=78
z="degrees outside"
print(x, y, z, sep="..!..")
```

This will output `it is..!..78..!..degrees outside`

By default, Python includes a new line at the end of each print statement. We can alter this to anything else by using the `end=` argument:

```
print("Good Morning, Chicago.", end="!!")
```

Outputting `Good Morning, Chicago.!!` to the console, with no new line character inserted. Note that the next print statement will continue on the same line unless it contains a new line character.

```
print("Hello World!", end=" ")
print("I will print on the same line.")
```

We have used all this terminology like strings and variables a lot, let's dive a bit deeper into it!

## 18.3 Datatypes and Variables

There are many different datatypes in Python, and all of them can be passed to `print` function. Here's the main few:

**Strings** Strings are just a sequence of characters. Anything closed between quotes gets included in the string, such as "This is the Python Chapter for the Kontinua Sequence!!%& × ()-+=1234567890 &"

**Integers** Integers are any whole number, positive negative or 0. Theoretically, integer values can go on for infinity, but it is important to understand they take up space in computer memory.

**Floats** Floats, or floating point values, are numbers that include decimal places. Division can be done between integers and floats but may result in a different value.

**Booleans** Booleans only two values: `True` or `False`. The output of conditional statements

(such as if's or while's) are booleans. They only answer Yes and No questions, and are important on deciding between two possible paths in code.

**Sequence Types** Sequences are consecutive lists or pairs of other data types can be represented in various forms: `list`, `set`, and `tuple`. Values are accessed by positions.

**Mapping Types** The main mapping type is a `dict`, short for dictionary. Mapping types store values as key–value pairs instead of positions. You access items using a key, not an index.

Let's create some of these in our code:

```
x = 5
y = 5.6
z = True
a = [x, y, z]
```

A *variable* is a container for information, usually containing one of the datatypes established above. In Python, variables are assigned using the assignment operator =, with the name of the variable on the left side of the operator and the value it is assigned to on the right side.

Python has specific naming rules for variables. The name

- Must start with a letter or underscore
- Cannot contain spaces
- Are case-sensitive

In our code above, `x` is a *variable* containing the number 5, an Integer. `y` contains the float 5.6, while `z` is a boolean with the value `True`. We then created a list containing the variables `x`, `y`, and `z`.

A unique feature of Python is that variables can change their datatypes even after assignment. We can see the datatype of an object using the `type()`

```
x = 10
print(x)
print(type(x))

x = "Hello!"
print(x)
print(type(x))
```

Output:

```
10
<class 'int'>
Hello!
<class 'str'>
```

Notice that, because of that feature, the datatype of a variable is not defined beforehand. This is a core difference between Python and other programming languages like Java or C++, which we will explore in the future.

We can also define our floats using scientific notation.

```
x = 1e3     # 1000.0
y = 2.5e-4  # 0.00025
```

## *Exercise 50* Identify the variable type

*Working Space*

Below are 5 variables defined in Python. Identify their type as `str`, `float`, `bool`, `list`, or `int` by filling in the table below.

```
a = 10
b = 3.5
c = "10"
d = True
e = [1, 2, 3]
```

| Variable | Value | Data Type |
|---|---|---|
| a | 10 | |
| b | 3.5 | |
| c | ”10” | |
| d | True | |
| e | [1, 2, 3] | |

*Answer on Page 1890*

### *Exercise 51* What does this do?

What is the output of the following code?

*Working Space*

```
1 print(type(3.0) == type(3))
```

*Answer on Page 1890*

## 18.4 Operations

Operations can be done on both variables and numbers alike. Most commonly they will be used for mathematical operations or simplifying operations of equivalence.

### 18.4.1 Arithmetic Operations

Math operations in Python are very similar to standard math operations. We need them for any type of data operation or math parsing.

```
1  a = 100
2  b = 3
3  c = 100.0
4  d = -100
5  print(a + b) # addition
6  print(a - b) # subtraction
7  print(a * b) # multiplication
8  print(a / b) # division (results in a float)
9  print(a % b) # modulus (remainder) operator - useful for division checks
10 print(a ** b) # exponentiation
11 print(a // b) # floor division
12 print(c // b) # floor division as a float
13 print(d // b) # negative flooring goes further DOWN
```

Output:

```
1 103
2 97
3 300
```

```
    33.333333333333336
    1
    1000000
    33
    33.0
    34
```

Note that division by `0` is not a valid operation, just as in algebra. This will cause an `ZeroDivisionError`.

### 18.4.2 Augmented Assignment Operators

You may want to do the following operation

```
x = 0
while True:
    print(x)
    x = x + 1
```

This would print the counting numbers, increasing by 1, forever. The `x = x + 1` rewrites the value of `x` equal to the current value of `x`, and adds one to it. There is a short hand for this kind of operation, called an *augmented assignment operator*, which combines assignment operators and operations. These are commonly used in loops and accumulation counters:

```
x = 10
x += 5   # same as x = x + 5
x -= 3   # same as x = x - 3
x *= 2   # same as x = 2 * x
x /= 4   # same as x = x / 4
```

Following order of operations, `x = 6` after all lines are executed.

### 18.4.3 Summary

- Math operations are done with `+`, `-`, `*`, and `/`
- Augmented Operations `+=`, `-=`, `*=`, `/=` are common for loops and counting operations.

### *Exercise 52* Tracing Chart

Let's create a tracing chart. What happens after each line of code is run?

*Working Space*

```
1 n = 5
2 m = "3"
3 n = n + 1
4 m = m + "1"
```

| Line Executed | n (value, type) | m (value, type) |
|---|---|---|
| After line 2 | | |
| After line 3 | | |
| After line 4 | | |

*Answer on Page 1890*

## 18.5 Input and Output

So far, we have only displayed information using the `print()` function. While output is useful, most programs also need a way to receive information from the user. In Python, this is done using the built-in `input()` function.

### 18.5.1 Getting User Input

The `input()` function pauses the program and waits for the user to type something into the console. Whatever the user types is then returned and can be stored in a variable.

```
1 name = input("Enter your name: ")
2 print("Hello,", name)
```

In this example:

- The text inside `input()` is displayed as a prompt in the console
- The program waits for the user to type a response and press Enter

- The entered text is stored in the variable `name`
- The `print()` function then outputs a greeting using that value

### 18.5.2 Input Always Returns a String

A very important rule in Python is that `input()` **always returns a string**, even if the user types a number. This means that mathematical operations cannot be performed on input values unless they are converted to a numeric type.

Consider the following incorrect example:

```
age = input("Enter your age: ")
print(age + 1)
```

This code will result in an error, because Python cannot add a number to a string.

To perform calculations, the input must be *cast* to a different datatype.

### 18.5.3 Casting Input Values

Casting is the process of converting one datatype into another. Python provides built-in functions such as `int()`, `float()`, and `str()` for this purpose.

```
age = int(input("Enter your age: "))
print(age + 1)
```

Here:

- `input()` returns a string
- `int()` converts that string into an integer
- The program can now perform arithmetic operations

### 18.5.4 Common Input Errors

If the user enters something that cannot be converted to the requested datatype, Python raises a runtime error called a `ValueError`. For example:

```
age = int(input("Enter your age: "))
```

If the user types `hello` instead of a number, Python will produce an error similar to the following:

```
ValueError: invalid literal for int() with base 10
```

This error occurs because the string `"hello"` cannot be converted into an integer. Handling these types of errors will be discussed later when we introduce `try-except` statements.

### 18.5.5 Summary

- `print()` is used to display output to the console
- `input()` is used to receive text input from the user
- `input()` always returns a string
- Casting is required to convert input into numeric types

## *Exercise 53* User Input Types

*Working Space*

```
x = input("Enter a number: ")
print(x + 5)
```

What will happen when this code is run and the user enters `10`? If there is an issue, explain what it is, why it occurs, and how to fix it.

*Answer on Page 1890*

## 18.6 Conditionals, Loops, and Match-Case

Most programs need to make decisions and repeat actions. Programmers do not want to write the same code multiple times, as this introduces *redundancy*. In Python, this is done using *conditionals* and *loops*. Conditionals allow the program to choose between different paths of execution, while loops allow code to be run multiple times.

### 18.6.1 Conditional Statements

Conditional statements execute code only if a given condition is true. Python uses the keywords `if`, `elif`, and `else` to define these conditions.

```
x = 10

if x > 0:
    print("x is positive")
elif x == 0:
    print("x is zero")
else:
    print("x is negative")
```

In this example:

- The expression after `if` is evaluated as a boolean
- If the condition is `True`, the indented block runs
- If it is `False`, Python checks the next condition. The `elif` block is the only way to check another branch after one is returned `False`. The `else` branch is a last resort, and only runs if all previous statements are false
- Only one branch of the conditional will execute

It is important to note that **indentation is required** in Python. All statements belonging to a conditional block must be indented at the same level.

### 18.6.2 Boolean Expressions

Conditions are built using comparison and logical operators. These expressions always evaluate to either `True` or `False`.

```
a = 5
b = 10

print(a < b)   # less than
print(a == b)  # equal to
print(a != b)  # not equal to
```

Logical operators can be used to combine conditions. Conditions can be *strung together* using `and`, `or`, and `not`

```
x = 7

if x > 0 and x < 10:
    print("x is between 1 and 9")
if x > 0 or x < -3:
    print("x is greater than 0 or less than -3")

y = False
print(not y)

```

### 18.6.3 Loops

Loops allow code to be repeated multiple times. Python provides two main types of loops: `for` loops and `while` loops.

To define a loop, Python uses the keywords `for` and `while`, followed by a condition or sequence to iterate over.

- the loop header, with correctly defined endpoints
- a colon :
- an indented block of code to iterate over

### 18.6.4 For Loops

A `for` loop is used to iterate over a sequence, such as a list or a range of numbers.

```
for i in range(5):
    print(i)
```

This code will print the numbers 0 through 4. The `range()` function generates a sequence of integers starting at 0 and stopping before the given value.

```
numbers = [10, 20, 30]

for n in numbers:
    print(n)
```

### 18.6.5 While Loops

A `while` loop repeats as long as a condition remains true.

```
x = 5

while x > 0:
    print(x)
    x -= 1
```

In this example:

- The condition is checked before each iteration
- The loop stops once the condition becomes `False`

Care must be taken to ensure that the condition eventually becomes false. Otherwise, the program will enter an infinite loop, one which never stops and the computer will run forever, risking hardware component issues.

### 18.6.6 Breaking and Continuing Loops

Python provides keywords to control loop execution.

```
for i in range(10):
    if i == 5:
        break
    print(i)
```

The `break` statement immediately exits the loop.

```
for i in range(5):
    if i == 2:
        continue
    print(i)
```

The `continue` statement skips the current iteration and moves to the next one.

### 18.6.7 Match-Case Statements

Python also provides the `match-case` statement, which allows a value to be compared against several possible patterns. This is similar to a switch statement in other programming languages.

```
command = input("Enter a command: ")

match command:
    case "start":
        print("Starting program")
    case "stop":
        print("Stopping program")
    case "pause":
        print("Pausing program")
    case _:
        print("Unknown command")
```

The underscore (`\_`) acts as a default case if no other patterns match.

### 18.6.8 Summary

- Conditionals allow programs to make decisions
- Boolean expressions evaluate to `True` or `False`
- Loops allow code to be repeated efficiently
- `for` loops iterate over sequences
- `while` loops repeat based on a condition
- `match-case` provides a structured way to handle multiple cases

## *Exercise 54* Predict the Output

Fill in the table with the outputs for the given inputs.

*Working Space*

```
x = int(input("Enter a number: "))

if x > 10:
    print("Large")
elif x == 10:
    print("Ten")
else:
    print("Small")
```

| Input | Output |
|---|---|
| 5 | |
| 10 | |
| 15 | |

*Answer on Page 1891*

## *Exercise 55* Fill in the Blanks: Loop Edition

Fill in the following code to print all even numbers from 0 to 20 (inclusive).

*Working Space*

```
for i in __________:
    print(__________)
```

*Answer on Page 1891*

## 18.7 Lists and Strings

We talked about creating strings, any set of characters between quotes. These can either be single quotes ' ' or double quotes " ".

Multiline strings must be surrounded by three quotations on each side of the text sequence.

```
a = """Lorem ipsum dolor sit amet,
consectetur adipiscing elit,
sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua."""
```

Strings are an example of a *sequence type*, meaning they store values in a specific order and allow individual elements to be accessed using an index.

### 18.7.1 Indexing Strings

Each character in a string has a position, called an *index*. Indexing in Python begins at 0, so the first character is at index 0. Think of it like asking the question for each letter: "How far from the start is this element?" The first element is zero positions away from the start, so its index is 0. The last element is always at index $n-1$, where $n$ is the length of string. In memory, all of these characters are stored consecutively.

```
word = "Python"

print(word[0])
print(word[1])
print(word[5])
```

Output:

```
P
y
n
```

Attempting to access an index that does not exist will result in a runtime error.

### 18.7.2 Length of a String

The number of characters in a string can be found using the built-in `len()` function.

```
word = "Python"
print(len(word))
```

Output:

```
6
```

### 18.7.3 Iterating Over Strings

Strings can be iterated over one character at a time using a `for` loop.

```
for char in "Hello":
    print(char)
```

This loop runs once for each character in the string.

### 18.7.4 String Methods

Strings come with built-in functions, called *methods*, that perform common operations. These methods are called using dot notation.

```
text = "  Python Programming  "

print(text.upper())
print(text.lower())
print(text.strip())
```

Common string methods include:

- `upper()` converts all characters to uppercase
- `lower()` converts all characters to lowercase
- `strip()` removes leading and trailing whitespace

### 18.7.5 Lists

A list is another sequence type used to store multiple values in a single variable. Lists are created using square brackets, with elements separated by commas.

```
numbers = [1, 2, 3, 4]
```

Lists can store values of different datatypes, including strings, numbers, booleans, and even other lists.

```
data = [10, 3.14, True, "Python"]
```

### 18.7.6 Indexing Lists

Like strings, lists are indexed starting at 0.

```
numbers = [10, 20, 30]

print(numbers[0])
print(numbers[2])
```

Output:

```
10
30
```

For nested lists, you search an indexe using a bracket for every nested list until you reach the element you are aiming to reach. Later, we will talk about `.json` files, which are files essentially comprised of strung-nested lists.

```
nested_numbers = [0, [1, 2, 3], [4, 5, 6], [7, 8, 9]]

print(numbers[0])
print(numbers[1][2])
print(numbers[2][1])
```

Output:

```
0
3
5
```

### Slicing Strings

You can also use *slicing* in Python to get all elements between two indices. This syntax is Python-specific, although there are equivalences in other programming languages. This syntax works for *both* lists and strings.

For any sequence, we can slice it or operate on it with the following index notation:

```
sequence[start : stop : step]
```

where `start` is the index to begin at (inclusive), `end` is the index to end at (exclusive), and `step`, an optional argument for how many items to skip.

```
text = "abcdef"

text[1:4]    # 'bcd'
text[:3]     # 'abc'
text[3:]     # 'def'
text[::2]    # 'ace'
text[::-1]   # 'fedcba'  (reverse string)
```

The above example slices a string in a few different ways. Let's look at list slicing:

```
nums = [0, 1, 2, 3, 4, 5]

nums[2:5]    # [2, 3, 4]
nums[:4]     # [0, 1, 2, 3]
nums[4:]     # [4, 5]
nums[::2]    # [0, 2, 4]
nums[::-1]   # [5, 4, 3, 2, 1, 0]
```

Negative indices start from the end, and work the way up the sequence:

```
text = "python"

text[-3:]    # 'hon'
text[:-2]    # 'pyth'
```

Note that setting a variable to a sliced string creates a new string object, while leaving the original unchanged.

## *Exercise 56* Mystery Sequence

What will the code `seq[:]` output, assuming `seq = [a, f, j, k]` is a valid list?

*Working Space*

*Answer on Page 1891*

## *Exercise 57* Computer... Indexing!

*Working Space*

You are given the following code:

```
word = "computer"
```

What is the output of each of the following expressions?

```
print(word[0])
print(word[2])
print(word[-1])
print(word[-3])
print(word[3:6])
```

*Answer on Page 1891*

### 18.7.7 Modifying Lists

Unlike strings, lists are *mutable*, meaning their contents can be changed after creation.

```
numbers = [1, 2, 3]
numbers[1] = 10
print(numbers)
```

Output:

```
[1, 10, 3]
```

### 18.7.8 List Length and Iteration

The number of elements in a list can be found using the `len()` function.

```
numbers = [1, 2, 3]
print(len(numbers))
```

Lists can be iterated over using a `for` loop.

```
numbers = [1, 2, 3]

for n in numbers:
    print(n)
```

### 18.7.9 Common List Methods

Lists include built-in methods that allow elements to be added or removed.

```
numbers = [1, 2, 3]
numbers.append(4)
numbers.remove(2)
print(numbers)
```

Common list methods include:

- `append()` - adds an element to the end of the list
- `remove()` - removes the first occurrence of a value
- `pop()` – removes and returns an element by index

### 18.7.10 Strings vs Lists

Although strings and lists are both sequence types, there is an important difference between them.

- Strings are immutable and cannot be modified after creation
- Lists are mutable and can be changed

### 18.7.11 Summary

- Strings and lists are sequence types
- Indexing starts at 0
- The `len()` function returns the size of a sequence
- Strings are immutable
- Lists are mutable and support modification

## *Exercise 58* What's in your backpack?

Let's say the following code is run.

*Working Space*

```
1 items = ["pen", "book", "eraser"]
2
3 items.append("ruler")
4 items.pop()
5 items.remove("book")
6 items.append("marker")
```

Fill in the following chart with the contents of the `items` list after each operation.

| Line Execution | `items` content |
|---|---|
| After Line 1 | `items = ["pen", "book", "eraser"]` |
| After Line 3 | |
| After Line 4 | |
| After Line 5 | |
| After Line 6 | |

*Answer on Page 1892*

## 18.8 Is there more?

Of course, there is always something to learn with Python. We are going to do a deeper dive into functions, classes, and a few useful libraries in the next workbook. For now, this should be enough to get you started with Python. We will revisit many of these concepts in later chapters as we build more complex programs.

# Chapter 19

# Sets and Logic

The use of math usually falls into two categories:

- *Developing mathematical tools that let us make better predictions.* This is how engineers and scientists use math. It is usually referred to as *applied math*.
- *Creating interesting statements and proving them to be true or false.* This is known as *pure math*.

Many mathematical ideas start out as pure math, and eventually become useful. For example, the field of number theory is devoted to proving things about prime numbers. The mathematicians who created number theory were certain that it could never be used for any practical purpose. After a century or two, number theory was used as the basis of most cryptography systems.

Conversely, some ideas start out as a "rule of thumb" that engineers use, and are eventually rigorously defined and proven.

This course tends to emphasize applied math, but you should know something about the tools of pure math.

You can think of all the mathematical proofs as a tree. Each proof proves some statement true. To do this, the proof uses logic and statements that were proven true by other truths. So, the tree is built from the bottom up. However, the tree has to have a bottom. At the very bottom of the tree are some statements that we just accept as true without proof. These are known as *axioms*.

All of modern mathematics can be built from:

- A short list of axioms.
- A few rules of logic.

There have been several efforts to codify a small but complete axiomatic system. The most popular one is known as *ZFC*. "Z" is for the Ernst Zermelo, who did most of the work. "F" is for Abraham Fraenkel, who tidied up a couple of things. "C" is for The Axiom of Choice. As a community, mathematicians debate whether the Axiom of Choice should be an axiom; we get a couple of strange results if we include it in the system. If we do not, there are a few obviously useful ideas that we can't prove true.

ZFC has 10 axioms. We simply accept these 10 statements as true, and all the proofs of modern mathematics can be extrapolated from them. The 10 axioms are all stated in terms of sets. A few of them will be discussed below.

## 19.1 Sets

A *set* is a collection. For example, you might talk about the set of odd numbers greater than 5, or the set of all protons in the universe.

We have a notation for sets. For example, here is how we define S to be the set containing 1, 2, and 3:

$$S = \{1, 2, 3\}$$

We say that 1, 2, and 3 are *elements* of the set S. (Sometimes we will also use the word "member")

If you want to say "2 is an element of the set S" in mathematical notation, it is done like this:

$$2 \in S$$

If you want to say "5 is *not* an element of the set S", it looks like this:

$$5 \notin S$$

We have notation for a few sets that we use all the time:

| Set | Symbol |
|---|---|
| The empty set | $\emptyset$ |
| Natural numbers | $\mathbb{N}$ |
| Integers | $\mathbb{Z}$ |
| Rational numbers | $\mathbb{Q}$ |
| Real numbers | $\mathbb{R}$ |
| Complex numbers | $\mathbb{C}$ |

The empty set is the set that contains nothing. It is also sometimes called *the null set*.

Often, when we define a set, we start with one of these big sets and say "The set I'm talking about is the members of the big set, but only the one for which this statement is

true". For example, if you wanted to talk about all the integers greater than or equal to -5, you could do it like this:

$$A = \{x \in \mathbb{Z} \mid x \geq -5\}$$

When you read this aloud, you say "A is the set of integers x where x is greater than or equal to negative 5."

### 19.1.1 And and Or

Sometimes you need the members to satisfy two conditions; for this, we use "and":

$$A = \{x \in \mathbb{Z} \mid x > -5 \text{ and } x < 100\}$$

This is the set of integers that are greater than -5 *and* less than 100. In this book, we usually just write "and", but if you do a large amount of set and logic work, you will use the symbol $\wedge$:

$$A = \{x \in \mathbb{Z} \mid (x > -5) \wedge (x < 100)\}$$

Sometimes, you want a set that satisfies at least one of two conditions. For this, you use "or":

$$A = \{x \in \mathbb{Z} \mid x < -5 \text{ or } x > 100\}$$

These are the number that are less than -5 or greater than 100. Once again, there is a symbol for this:

$$A = \{x \in \mathbb{Z} \mid (x < -5) \vee (x > 100)\}$$

### 19.1.2 How simple are sets?

Sets are so simple that some questions just don't make any sense:

- "What is the first item in the set?" makes no sense to a mathematician. Sets have no

order.

- "How many times does the number 6 appear in the set?" makes no sense. 6 is a member, or it is not.

### 19.1.3 Subsets

If every member of set A is also in set B, we say that "A is a subset of B."

For example, if $A = \{1, 4, 5\}$ and $B = \{1, 2, 3, 4, 5, 6\}$, then A is a subset of B. There is a symbol for this:

$$A \subseteq B$$

Remember the table of commonly used sets? We can arrange them as subsets of each other:

$$\emptyset \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

Note that subsets have the transitive property: $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ thus $\mathbb{N} \subseteq \mathbb{Q}$

Note that if A and B have the same elements, $A \subseteq B$ *and* $B \subseteq A$. In this case, we say that the two sets are equal.

We also have a symbol for "is not a subset of": $A \not\subset B$

### 19.1.4 Union and Intersection of Sets

If you have two sets A and B, you might want to say "Let C be the set containing element that are in *either* A or B." We say that C is the *union* of A and B. There is notation for this too:

$$C = A \cup B$$

For example, if $A = \{1, 3, 4, 9\}$ and $B = \{3, 4, 5, 6, 7, 8\}$, then $A \cup B = \{1, 3, 4, 5, 6, 7, 8, 9\}$.

You also want to say "Let C be the set containing elements that are in *both* A and B." We say that C is the *intersection* of A and B. There is notation for this too:

$$C = A \cap B$$

For example, if $A = \{1, 3, 4, 9\}$ and $B = \{3, 4, 5, 6, 7, 8\}$, then $A \cap B = \{3, 4\}$.

### 19.1.5 Venn Diagrams

When discussing sets, it is often helpful to have a Venn diagram to look at. Venn diagrams represent sets as circles. For example, the sets $A$ and $B$ above could look like this:

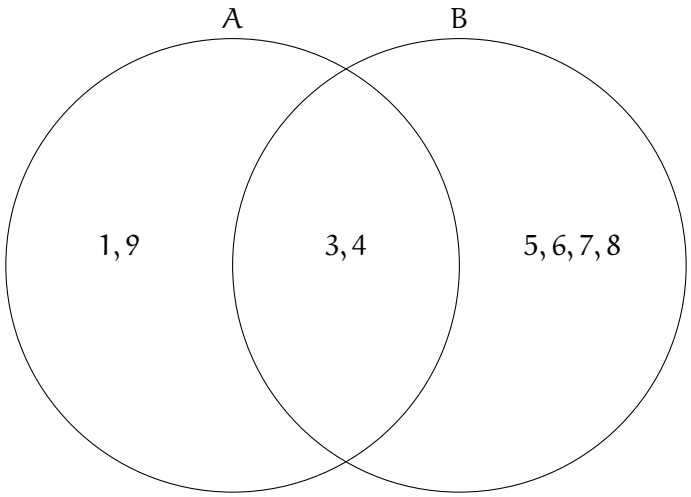



Figure 19.1: A Venn Diagram of the set of numbers $\{1, 9, 3, 4, 5, 6, 7, 8\}$

It makes it easy to see that $A$ and $B$ have a non-empty intersection, but they are not subsets of each other.

Often we won't even show the individual elements. For example, in the universe of all polygons, some rectangles are squares. Here's the Venn diagram:

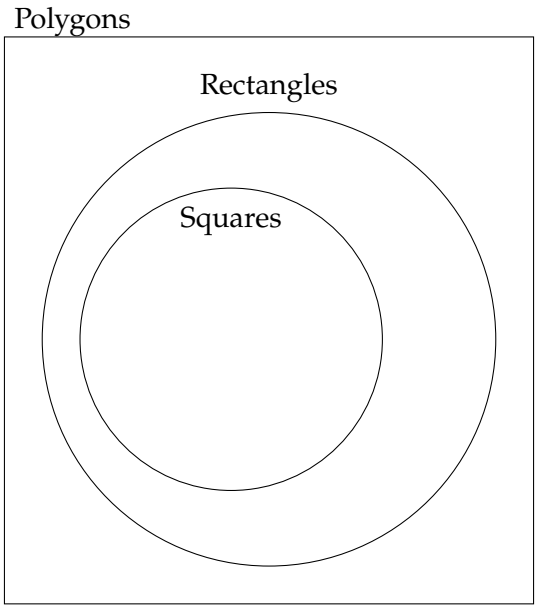



Figure 19.2: A Venn Diagram of polygons, rectangles, and squares.

As the combinations get more complex, we sometimes use shading to indicate what part we are talking about. For example, imagine we wanted all the rectangles with area greater than 5.0 that are not squares. The diagram might look like this:

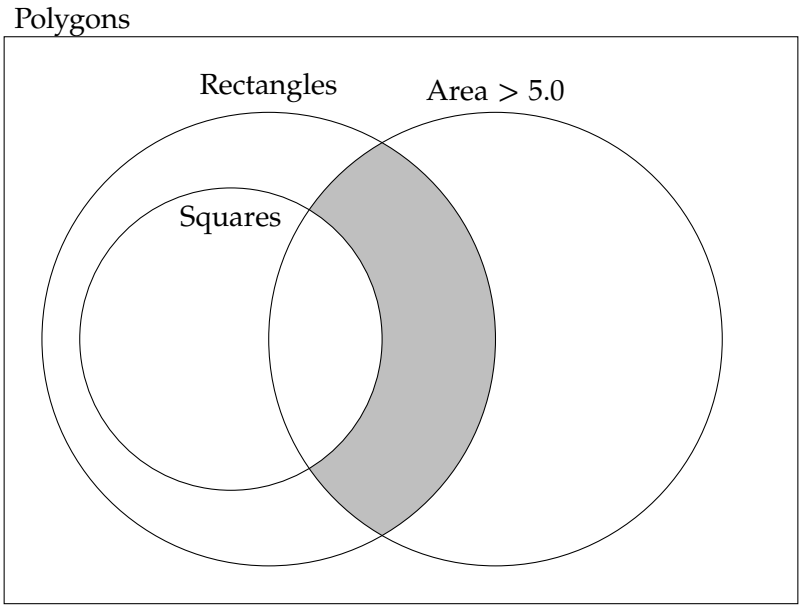



Figure 19.3: A Venn Diagram of the polygons, rectangle, and square number, and rectangles that have an area greater than 5.0 that are not squares.

## 19.2 Logic

We use a lot of logic in set theory. For example, the shaded region above represents all the polygons for which all the following are true:

- It is a rectangle.
- It is *not* a square.
- It has an area greater than 5.0.

## 19.3 Implies

In logic, we will often say "$a$ implies $b$". That means "If the statement $a$ is true, the statement $b$ is also true." For example: "$p$ is a square" implies "$p$ is a rectangle".

There is notation for this: an arrow in the direction of the implication.

$$p \text{ is a square} \implies p \text{ is a rectangle}$$

Notice that implication has a direction: "$p$ is a rectangle" does *not* imply "$p$ is a square".

Implications can be chained together: If $A \implies B$ and $B \implies C$, then $A \implies C$.

## 19.4 If and Only If

If the implication goes both ways, we use "if and only if". This means the two conditions are equivalent. For example: "$n$ is even if and only if there exists an integer $m$ such that $2m = n$".

There is a notation for this too:

$$p \text{ is even} \iff \text{there exists an integer } m \text{ such that } 2m = n$$

There is even notation for "there exists". It is a backwards capital E, represented by $\exists$:

$$p \text{ is even} \iff \exists\, m \in \mathbb{Z} \text{ such that } 2m = n$$

## 19.5 Not

The not operation flips the truth of an expression:

- If $a$ is true, not$(a)$ is false.
- If $a$ is false, not$(a)$ is true.

We sometimes talk about "notting" or "negating" a value. We won't use it much, but there is a symbol for this: $\neg$.

We might create a *logic table* for negation that shows all the possible values and their negation:

| A | $\neg$A |
|---|---|
| F | T |
| T | F |

This table says "If A is false, $\neg$A is true. If A is true, $\neg$A is false."

Most logic tables are for operations that take more than one input. For example, this logic table shows the values for and-ing and or-ing:

| A | B | A and B | A or B |
|---|---|---|---|
| F | F | F | F |
| F | T | F | T |
| T | F | F | T |
| T | T | T | T |

Notice that we have to enumerate all possible combinations of the inputs of A and B.

When a variable like A can only take two possible values, we say it is a *boolean* variable. (George Bool did important work in this area.)

### *Exercise 59* Logic Table

*Working Space*

Make a logic table that enumerates all possible combinations of boolean variables A and B and shows the value of the two following expressions:

- $\neg$ (A or B)
- ($\neg$A) and ($\neg$B)

*Answer on Page 1892*

## 19.6 Cardinality

Informally, the *cardinality* of a set is the number of elements it contains. So, $\{1, 3, 5\}$ has a cardinality of 3. The null set has a cardinality of zero.

Things get a little trickier if a set is infinite. We say two infinite sets A and B have the same cardinality if there is some mapping that pairs every member of A with a member of B and mapping that pairs every member of B with a member of A.

For example, the set of all natural numbers is $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$. The set of all even numbers is $\{2, 4, 6, 8, \ldots\}$. These have the same cardinality because we can pair each natural number $n$ with an even number $2n$.

## 19.7 Complement of a Set

Most sets exist in a particular universe, for example you might talk about the even numbers as a set in the integers. You can then talk about the set's *complement*: the set of everything else. For example, the complement of the even numbers (inside the integers) is the odd numbers.

If you have a set A, its complement is usually denoted by $A'$.

See the Venn Diagram shown in Figure 19.4.

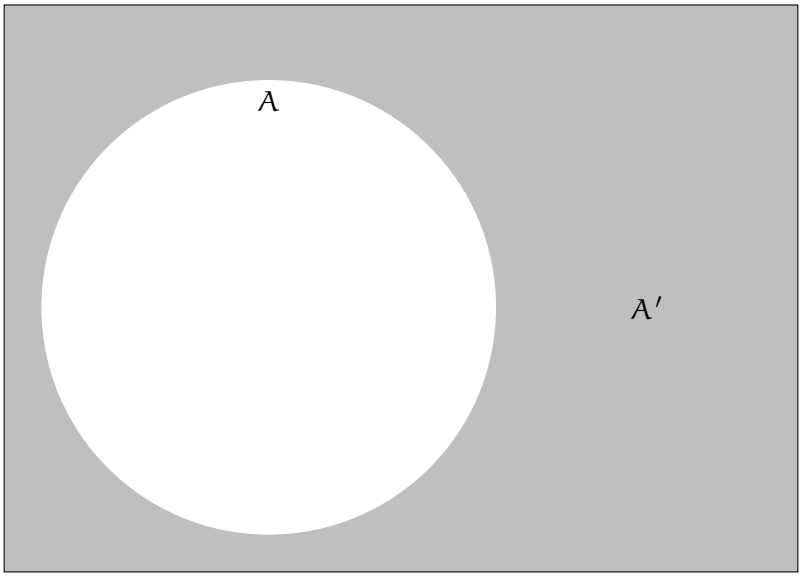



Figure 19.4: The complement of $A$ is $A'$.

Outside of logic, the concept of a set

## 19.8 Subtracting Sets

If you have sets $A = \{1, 2, 3, 4\}$ and $B = \{1, 4\}$, it makes sense to subtract $B$ from $A$ by removing 1 and 4 from $A$.

If $A$ and $B$ are sets, we define $A - B$ to be $A \cap B'$. Take a second to look at this diagram and convince yourself that the white region represents $A - B$ and that it is the same as $A \cap B'$.

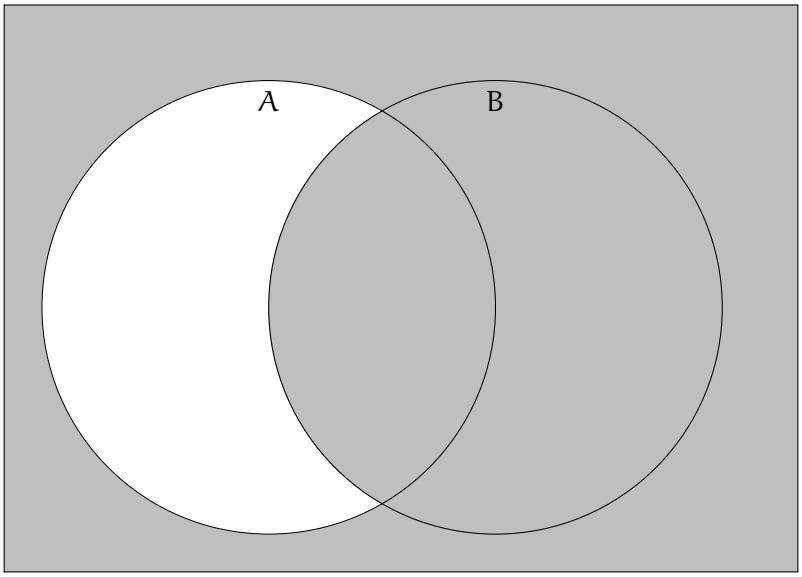



Figure 19.5: The subtraction of sets A and B.

## 19.9 Power Sets

It is not uncommon to have a set whose elements are also sets. For example, you might have the set that contains the following two sets: $\{1, 2, 3\}$ and $\{2, 3, 4\}$. You might write it like this: $\{\{1, 2, 3\}, \{2, 3, 4\}\}$. (Note that this set has a cardinality of 2 — it has two members that are sets.)

Given any set $A$, you can construct its *power set*, which is the set of all subsets of $A$. For example, if you have a set $\{1, 2, 3\}$, its power set is $\{\{1, 2, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1\}, \{2\}, \{3\}, \emptyset\}$.

If a set has $n$ elements, its power set has $2^n$ elements. Note that the last element must be the empty set, because there are no subsets of an empty set. Recall that order does not matter in sets, so $\{1, 2\}$ and $\{2, 1\}$ are considered the same.

## 19.10 Booleans in Python

In Python, we can have variables hold boolean values: `True` and `False`. We also have operators: `not`, `and`, and `or`.

For example, you could find out what the expression "`a and not b`" is if both variables are false like this:

```
1 a = False
2 b = False
```

```
result = a or not b
print(f"a={a}, b={b}, a or not b = {result}")
```

This would print out:

```
a=False, b=False, a or not b = True
```

What if you wanted to try all possible values for `a` and `b`? You could use `itertools`.

```
import itertools

all_combos = itertools.product([False, True], repeat=2)
for (a, b) in all_combos:
    result = a or not b
    print(f"a={a}, b={b}: a or not b = {result}")
```

Type it in and run it. You should get the whole logic table:

```
a=False, b=False: a or not b = True
a=False, b=True: a or not b = False
a=True, b=False: a or not b = True
a=True, b=True: a or not b = True
```

If you had three inputs into the expression, your truth table would have eight entries. For example, if you wanted to know the truth table for $a and not (b and c)$, here is the code:

```
all_combos = itertools.product([False, True], repeat=3)
for (a, b, c) in all_combos:
    result = a and not(b and c)
    print(f"a={a}, b={b}, c={c}: a and not (b and c) = {result}")
```

Type it in and run it. You should get:

```
a=False, b=False, c=False: a and not (b and c) = False
a=False, b=False, c=True: a and not (b and c) = False
a=False, b=True, c=False: a and not (b and c) = False
a=False, b=True, c=True: a and not (b and c) = False
a=True, b=False, c=False: a and not (b and c) = True
a=True, b=False, c=True: a and not (b and c) = True
a=True, b=True, c=False: a and not (b and c) = True
a=True, b=True, c=True: a and not (b and c) = False
```

## 19.11 The Contrapositive

Here is a statement with an implication: "If it has rained in the last hour, the grass is wet."

This is *not* equivalent to "If the grass is wet, it has rained in the last hour." (After all, the sprinkler may be running.)

However, it is exactly equivalent to its *contrapositive*: "If the grass is not wet, it has not rained in the last hour."

The rule can be written using symbols:

$$(A \implies B) \iff (\neg B \implies \neg A)$$

## 19.12 The Distributive Property of Logic

Many ideas from integer arithmetic have analogues in boolean arithmetic. For example, there is a distributive property for booleans. These two expressions are equivalent:

- A and (B or C)
- (A and B) or (A and C)

So are these:

- A or (B and C)
- (A or B) and (A or C)

## 19.13 Exclusive Or

The expression "a or b" is true in any of the following conditions:

- a is True and b is False.
- a is False and b is True.
- Both a and b are True.

Sometimes engineers need a way to say "Either a or b is true, but not both." For this, we use *exclusive OR* (or XOR). You may see XOR written symbolically as $\oplus$ or just used as XOR. indexXOR

Here, then, is the logic table for XOR

| A | B | XOR(a,b) |
|---|---|---|
| F | F | F |
| F | T | T |
| T | F | T |
| T | T | F |

In Python, Logical XOR is done using `!=`:

```
1 just_one = (a != b)
```

(Take 10 seconds to confirm that this is the same as the logic table above.)

# Chapter 20

# Proofs

## 20.1 Introduction

Let's talk about *proofs*. In mathematics, a proof shows that a new statement aligns with all true statements which have come before it. A proof is a sequence of statements that starts with some assumptions and ends with a new, unproven conclusion. If the conclusion is true, then we say the proof is *valid*.

In mathematics, a proof shows that a new statement aligns with all true statements which have come before it.

The general structure will follow this pattern: We want to prove that $P \implies Q$ (this is said statement $P$ *implies* statement $Q$).

We will assume some statement is true, and work to prove a different statement is true based on our assumptions and the rules of logic.

Here are some definitions to get us started:

- A *proposition* is a statement that is either true or false.
- A *theorem* is a proposition that has been proven to be true.
- A *lemma* is a *helper* proposition that is proven or known to be true and is used in the proof of a theorem.
- A *corollary* is a proposition that follows directly from a theorem.
- An *implication* is a theorem that relies on one proposition being equivalent to another.
- $P$ is the *hypothesis* or *premise*.
- $Q$ is the *conclusion*.
- The symbol $\forall$ represents the mathematical symbol for "for all", often used to iterate through all numbers of a subset.
- The symbol $\exists$ represents the mathematical symbol "there exists".
-

## 20.2 Trivial Proof

The easiest of proof is a **Trivial Proof**. If we know that Q is true always, we trivially know that P is true, and that P implies Q. This is most common when P and Q are unrelated statements with no affect on each other.

**Theorem 1.** *If* Q *is true, then* $P \implies Q$ *is true for any proposition* P.

*Proof.* Let P be the statement "The sky is blue" and Q be the statement "$2+2=4$". Since Q is always 1true, the implication $P \implies Q$ is true regardless of the truth value of P.

This follows directly from the definition of logical implication (or can be verified via a truth table), where an implication is true whenever the consequent is true. □

## 20.3 Direct Proof

A **Direct Proof** involves assuming the hypothesis (P) is true and then showing that the conclusion (Q) must also be true. This is done sequentially using logical deductions, equivalences, and previously established definitions.

To walk through an example, let's first define even and odd numbers.

**Even and Odd Numbers**

We say an integer $n$ is *even* if there exists an integer $k$ such that $n = 2k$.

We say an integer $n$ is *odd* if there exists an integer $k$ such that $n = 2k+1$.

**Theorem 2.** *The sum of two even integers is even.*

*Proof.* Let $a$ and $b$ be even integers. Then there exist integers $k$ and $m$ such that

$$a = 2k \quad \text{and} \quad b = 2m.$$

Consider their sum:

$$a + b = 2k + 2m = 2(k+m).$$

Since $k+m$ is an integer, it follows that $a+b$ is even. □

## 20.4 Proof By cases

A **proof by cases** is a technique where you split a problem into a finite number of distinct scenarios that cover all possibilities, then prove the statement in each scenario.

The structure is generally as follows:

1. Identify all possible cases (they must be exhaustive (cover all cases) and non-overlapping).
2. Prove the statement separately in each case. Usually, a direct proof is used in each case.
3. Conclude the statement holds in the general case.

Let's prove the following statement using a proof by cases:

**Theorem 3.** *For any integer $n$, $n^2$ is even or odd.*

*Proof.* Every integer is either even or odd. Therefore there are only two cases to consider:

**Case 1.** *$n$ is even. Then, there exists an integer $k$ such that $n = 2k$. Therefore,*

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

*which is even.*

**Case 2.** *$n$ is odd. Then, there exists an integer $k$ such that $n = 2k + 1$. Therefore,*

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

*which is odd.*

Since these cases cover all integers, the claim holds for all integers $n$. □

## *Exercise 60* Absolute Value Proof

Prove the following claim using a proof by cases:

*Working Space*

**Theorem 4.**

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

*Answer on Page 1892*

## 20.5 Disproof By Counter-example

A proof by counterexample,more often called a **disproof by counterexample** is used to show a statement is false by finding one case or instance (a *counterexample*) where it fails.

**Theorem 5.** *All even numbers are divisible by* 4.

*Proof.* Consider the integer 2. It, by definition, is even, as $\exists\, k = 1$ s.t $2 \cdot 1 = 2$, which is even.

However, we must check if 2 is divisible by 4. An integer is divisible by 4 if it can be written in the form $4m$ for some integer $m$. Suppose, for purposes of disproving, that $2 = 4m$ for some integer $m$. Then

$$m = \frac{2}{4} = \frac{1}{2},$$

which is not an integer. Hence, 2 is not divisible by 4.

Thus, we have found an even integer that is not divisible by 4, contradicting the claim that *all* even integers have this property. Therefore, the statement is false. □

## 20.6 Proof By Contraposition

Proof by contraposition is a method used to prove an implication of the form

$$P \implies Q.$$

Instead of proving this statement directly, we prove its **contrapositive**:

$$\neg Q \implies \neg P.$$

The key idea is that an implication and its contrapositive are logically equivalent. We assume Q is false ($\neg Q$ is true), and directly show that P is false ($\neg P$ is true). If the negation of Q is true and the negation of P is true, it is equivalent to P and Q being valid:

$$\neg Q \implies \neg P \iff P \implies Q$$

This may make more sense with an example:

**Theorem 6.** *Let* $n$ *be an integer. If* $3n + 2$ *is even, then* $n$ *is even.*

*Proof.* Using proof by contraposition, we will prove the above theorem. Assume $n$ is odd.

Then, $\exists\, k$ s.t $n = 2k + 1$. Then, through mathematical operations, Substitute into $3n + 2$:

$$\begin{aligned} 3n + 2 &= 3(2k + 1) + 2 \\ &= 6k + 3 + 2 \\ &= 6k + 5 \\ &= 2(3k + 2) + 1 \end{aligned}$$

which is odd.

Therefore, the contrapositive is true, and hence the original statement follows. □

## 20.7 Proof By Contradiction

Rationality states that you cannot believe two ideas which oppose each other at the same time. This is called a *contradiction*. For example, the statement "P is true and P is false" is a contradiction.

A proof by contradiction is a method of proving a statement by assuming the opposite of what you want to prove and then showing that this assumption leads to a contradiction or impossibility, represented by $\rightarrow\leftarrow$. This method is often used when direct proof is difficult or impossible. To prove the statement $P \implies Q$, we assume P is true and $\neg Q$ is true. We then show this leads to a contradiction ($\rightarrow\leftarrow$), which proves that if P is true, Q must also be true. One of these contradictions looks like:

- $1 = 0$ (or any similar false integer statement)
- $x \in A \wedge x \notin A$ (for set theory)
- $P \wedge \neg P$ (for logic)

We will use a lot of contradictory statements and proofs when we talk about thought experiments and paradoxes!

**Theorem 7.** *Let $n$ be an integer. If $3n + 2$ is even, then $n$ is even.*

*Proof.* Assuming $3n + 2$ is even. Further, for contradiction, assume $n$ is odd. Then, $\exists\, k$ s.t $n = 2k + 1$. Then, through mathematical operations,

$$\begin{aligned} 3n + 2 &= 3(2k + 1) + 2 \\ &= 6k + 3 + 2 \\ &= 6k + 5 \\ &= 2(3k + 2) + 1 \end{aligned}$$

which is odd. This contradicts our assumption that $3n + 2$ is even. Therefore our assump-

tion that $n$ is odd must be false, and $n$ is even. □

## 20.8 Proof By Induction

A *proof by induction* is a methodology of proof that involves a (usually) proving a recursively defined statement. We use what is called the Principle of Mathematical Induction to assist with these proofs. We start with a simple case of the problem. Often, if we can prove the simplest case, we can prove all other cases by showing the each case depends on the previous one.

Recursion is useful in both mathematics and computer science because it provides a way to define a problem as correct by breaking it into smaller, similar subproblems. In mathematics, recursive definitions (such as sequences, factorials, or tree structures) are able to use proof techniques like induction, as their formulas rely on the previous term.

In computer science, recursion simplifies the implementation of algorithms that operate on data that can be broken down into smaller versions of itself, such as trees (which can be broken into smaller trees), graphs (which can be broken into subgraphs), and divide-and-conquer algorithms. We are going to cover recursive algorithms in depth in the Advanced Python Chapter. In general, there are 5 steps to this proof structure:

1. Define a specific proposition in the form of $P(n)$ *works correctly for all valid inputs of size* $n$. This is often given to you.
2. Define a *base case*. Just as in recursive definitions, proofs by induction us a base case as an "exit condition", or a start point to build off of. The base case proves that the proposition works for the smallest input size, which is usually $n = 0$ or $n = 1$. We call this $P(0)$.
3. Define an *inductive hypothesis*. Here, we assume the proposition remains correct for size $k$, some integer $k \geq 0$. We don't actually prove anything yet!
4. *Inductive Step*. Using the inductive hypothesis, prove $P(k+1)$, which states that proposition is correct for size $k+1$. Note here, that most times, the inductive step reduces the problem to a smaller, already proved case size. We say that proving $P(k+1)$ is recursively defined by $P(k)$.
5. Conclude the proof by stating *By the Principle of Mathematical Induction, the proposition holds for all* $n \geq P(0)$, where $P(0)$ is the base case size. This step is optional but good practice.

Let's look at an example to clear things up!

**Theorem 8.** *For all integers* $n \geq 4$, $2^n < n!$. *Equivalently, for all integers* $n \geq 0$, $2^{n+4} < (n+4)!$.

*Proof.* We prove the statement by induction on $n$.

**Basis Step.** Let $n = 4$. This is our smallest input value within our subset. Then we have $2^4 = 16$ and $4! = 24$. $16 < 24$, so the proposition holds for the basis step.

**Inductive Hypothesis.** Suppose for some $k \in \mathbb{Z}_{\geq 4}$, $2^k < k!$.

**Inductive Step.** We want to show $P(k+1)$, such that $2^{k+1} < (k+1)$. Let $n = k+1$. Then we have

$$2^{k+1} = 2 \cdot 2^k$$

By the hypothesis,

$$2 \cdot 2^k < 2 \cdot k!$$

. Since $k \geq 4$, $k + 1 < 2$, making

$$2 \cdot k! < (k+1)k! = (k+1)!$$

. Therefore, by the PMI, the proposition holds for all $n \geq 4$. □

## 20.9 Summary

In this chapter we talked about mathematical proofs. A proof is a logical argument that starts at a proposition and leads to a conclusion $P \implies Q$. There are many proof methods, even some we did not talk about in this chapter, so make sure to watch addition videos for your own learning. The most important proofs techniques covered are:

- Direct Proofs, where we assume the starting statement $P$ is true, and use logical to derive to $Q$
- Contraposition, where we prove $\neg Q \implies \neg P$, which is equivalent to $P \implies Q$
- Contradiction, where we prove by deriving impossibility from an assumption that the proposition could be false. This will come back in our paradoxes chapter!
- Induction, where a base case is used as a starting step to reach all other steps. These will also come back in my computer science chapters!

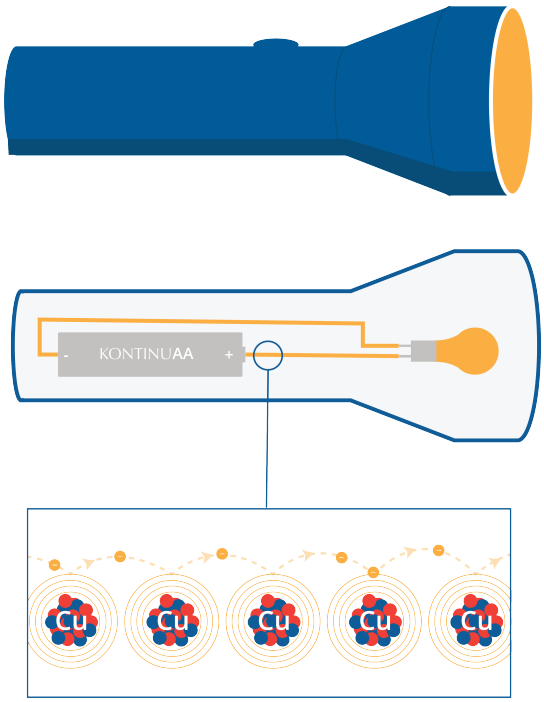
Overall, this chapter builds up many of the methodologies needed for proofs used to rigorously verify mathematical claims.

Chapter 21

# Introduction to Electricity

What happens when you turn on a flashlight? The battery in the flashlight acts as an electron pump, and the electrons flow through the wires to the lightbulb (or LED). As the electrons pass through the lightbulb, they excite the molecules within, which gives off light and heat. (LEDs also give off light and heat, but they give off much less heat.) The electrons then return to the battery to be pumped around again.

When electricity is flowing through a copper wire, the protons and neutrons of the copper stay put, while the electrons jump between the atoms on their way from the battery to the lightbulb and back again.




In some materials, like copper and iron, electrons are loosely bound to their nuclei, forming a sea of electrons, which allows energy to flow. These are good *electrical conductors*. In the presence of an electric field, which we will talk about in a future chapter, the electrons are free to move about, allowing *charge* to flow free. In other materials, like glass and

plastic, electrons don't leave their nuclei as easily. This makes them are terrible electrical conductors – we call these materials *electrical insulators*. For example, the plastic around a wire is used for electrical insulation.

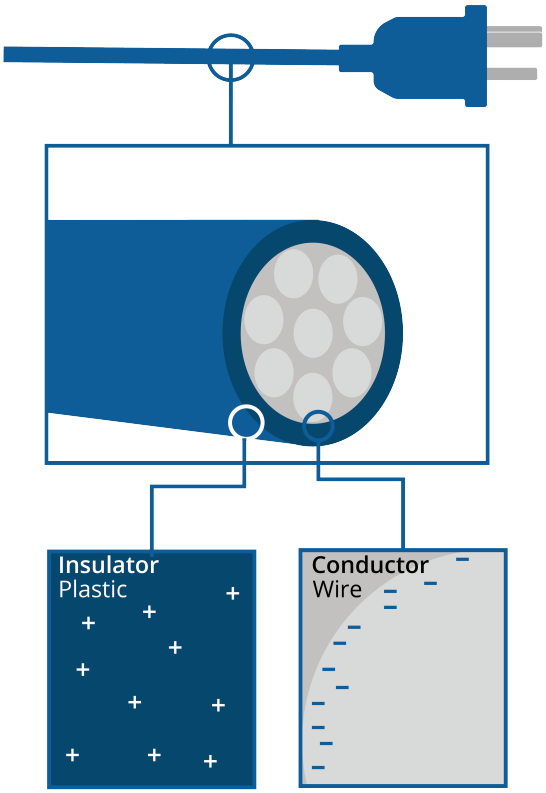



Figure 21.1: A cross section of an outlet plug.

## 21.1 Units

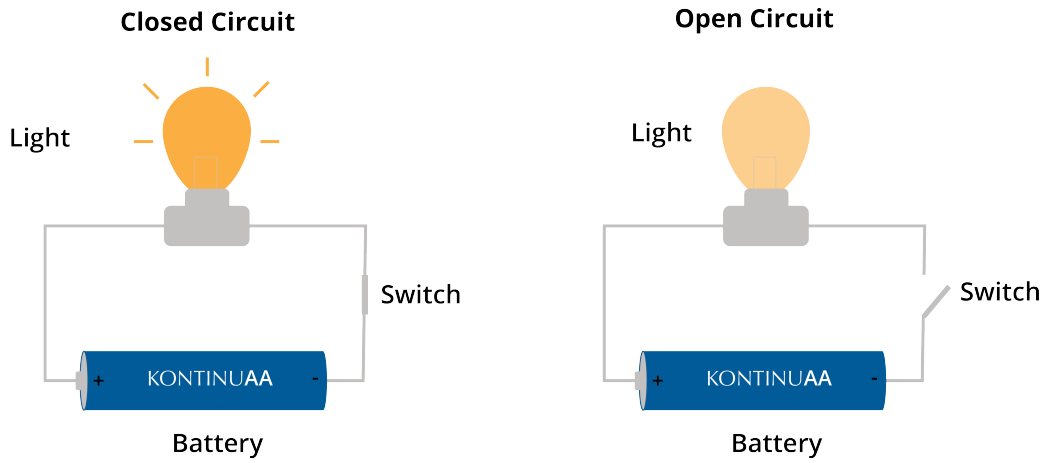
Electrons are very small, so to study them, scientists came up with a unit that represents *a lot* of electrons. 1 *coulomb* is about 6,241,509,074,460,762,608 electrons. When 5 coulombs enter one end of the wire every second (and simultaneously 5 coulombs exit the other end), we say "This wire is carrying 5 amperes of current."

(Truthfully, we usually shorten ampere to just "amp". This is sometimes a little awkward, because we also often shorten the word "amplifier" to "amp", but you should generally be able to tell which is which from the context.)

If you look at the circuit breakers or fuses for your home's electrical system, you'll see that each one is rated in amps. For example, maybe the circuit that supplies power to your kitchen has a 10 amp circuit breaker. If, for some reason, more than 10 amps tries to pass

through that wire, the circuit breaker will turn off the whole circuit.

When your flashlight is on, it pushes about 1 amp of current through the lightbulb (When it is off, there is no current in the lightbulb).




The lightbulb creates *resistance* that the current pushes through. Think of it like plumbing: The current is the amount of water passing through a pipe. The resistance is something that tries to stop the current – like a ball of hair, similar how different surfaces apply friction when pushing a box. The battery is what allows the current to push through the resistance; we call that pressure *voltage*. Especially in physics, voltage is often referred to as *electromagnetic potential*.

**Superconductor** is an extreme example of a conductor. Here, there is *absolutely no resistance* to the passing of charge. Materials that are conductors can become superconductors by decreasing their temperature to extremely low temperatures, even to absolute zero.

## 21.2 Capacitors

Two conductors separated by a distance that carry equal but opposite charges ($+Q$ and $-Q$) are a system that we refer to as **capacitors**. Most real-life capacitors (like the cylindrical electrolytic ones) are actually long strips of foil rolled up. We treat them as parallel plates, or metal plates with d between them.

Capacitors are similar to batteries, in the way that they store energy, however, capacitors store energy in the form of an electric field. As a result, capacitors store much less energy but can release it at a significantly higher rate. The measure of how much a capacitor

holds is called **capacitance**, represented with the symbol C and measured in Farads (F), where 1 F $= \frac{1 \text{ Coulomb}}{1 \text{ Volt}}$.[1]

Capacitance (C) is defined as the ratio of the magnitude of the charge (Q) on either conductor to the potential difference (V) between them:

$$C = \frac{Q}{V}$$

We will explore the underlying nature of charge in greater detail in the Charge chapter. For a parallel-plate capacitor with a vacuum between the plates ($\kappa = 1$), the capacitance is determined by its geometry:

$$C = \epsilon_0 \frac{A}{d}$$

Think of an medical defibrillator, a machine that delivers a specific amount of volts within an instantaneous moment of time. The goal of a defibrillator is not just to provide *charge*, but to provide Energy ($U_C$) to *reset* the heart's electrical rhythm. The energy stored is given by:

$$U_C = \frac{1}{2} C V^2$$

which comes from the area under a **Voltage** (V) vs **Charge** Q, and the fact that potential charge is $Q = CV$

Defibrillators often charge to several thousand volts. Because V is squared in the energy equation, increasing the voltage is the most efficient way to increase the *shock* power. To keep the device portable while maintaining high $U_C$ or Electric Potential Energy, manufacturers use advanced dielectrics ($\kappa$) to maximize C without making the internal plates massive. When the paddles touch the chest and the button is pressed, the capacitor discharges through the patient's body. The patient's chest acts as a resistor in an RC circuit. The current I follows

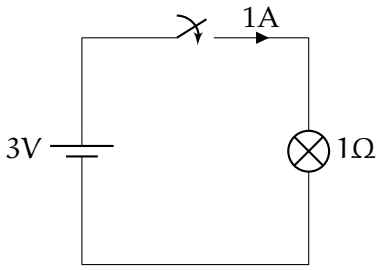
$$I(t) = \frac{V_0}{R} e^{-t/RC}$$

this is a great example of current following an example of exponential decay!

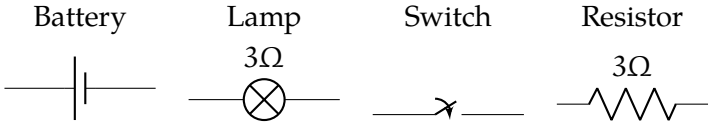
## 21.3 Circuit Diagrams

Here is a circuit diagram of your flashlight:

[1] 1 Farad is actually an enormous amount, enough to store power for the stereo system of an SUV! Smartphone circuits are 0.1010 μF, measured in microfarads, which are $10^{-6}$ F




The lines are wires. The symbols that we will use are:




The battery pushes the electrons from the positive end (the larger line) to the negative end (the smaller line), so the circuit must go around in a circle for the current to flow. This is why the current stops flowing when the switch breaks the circuit.

You can think of a switch as having zero resistance when it is closed and infinite resistance when it is open.

For our purposes, a lamp is just a resistor that gives off light.

## 21.4 Ohm's Law and Resistance

Current is a measure of how much *charge* per *time* is flowing through an area:

$$I = \frac{\Delta q}{\Delta t} \tag{21.1}$$

This is a measure of $1\frac{\text{coulomb}}{\text{second}}$, which we give the SI unit amperes or amps, represented with $A$.

Resistance is measured in *ohms*, and we use a Greek capital omega for that: $\Omega$

**Ohm's Law**

Whenever a voltage $V$ is pushing a current $I$ through a resistance of $R$ in given by some$\Omega$, the following is true:

$$V = IR$$

where $V$ is in volts, $I$ is in amps, and $R$ is in ohms.

Due to this relationship, we can find current given voltage and resistance using $I = \frac{V}{R}$. *Ohmic* materials maintain a constant resistance no matter how many volts are being put into it.

The resistance of a conductor depends on it's length, cross-sectional area, and resistivity. Due to Ohm's equation, we define resistance, measured in ohm's, as $1 \frac{\text{volt}}{\text{ampere}}$

## 21.5 Power and Watts

**Joule's Law**

When a current $I$ is passing through a resistance $R$, the power consumed is

$$W = I^2R$$

where $W$ is in watts, $I$ is in amps, and $R$ is in ohms.

Of course $V = IR$, so we can extend this to:

$$W = I^2R = IV = \frac{V^2}{R}$$

Your flashlight's batteries provide about 3 volts. How much battery power is the flashlight using when it is on? The power (in watts) produced by the battery is the product of the voltage (in volts) and the current (in amps). This means your flashlight is giving off $3volts \times 1amp = 3watts$ of power. Some of that power is given off as light, some as heat.

A watt is 1 joule of energy per second. We say that a watt is a measure of *power*.

When we talk about how much energy is stored in a battery, we use a unit like a kilowatt-hour. A kilowatt-hour is equivalent to 3.6 million joules.

### Portable Chargers

A portable charger labeled 165 W means that it's *maximum power output* is 165 watts. This is the maximum amount of voltage it can deliver to your device at any given moment.

Often, there is a noted voltage and current label on the back of the device.Power is the product of Voltage ($V$) and Current ($I$), expressed by the formula:

$$P = V \times I$$

In the case of a 165 W charger, it might output something like 20 V at 8.25 A ($20 \times 8.25 = 165$).

However, the 165 Watt charger will not always deliver 165 Watts to your phone. That is enough to power a gaming PC! Instead, protocols within modern chargers limit how much a device actually gets charged. Most smartphones only need 20 W. Chargers that are labeled fairly high in wattage often have multiple ports (this one has 4), and splits the charging load among all of them.

Now, if you see a measurement in mAh (milliampere-hours) or Wh (Watt-hours), that is the maximum *capacity* of the portable charger. This 165 W has a 25,000 mAh rating, meaning it stores approximately 92.5 Wh, or watt-hours. This comes from $\text{Wh} = \frac{\text{mAh} \times \text{V}}{1000}$, where most power banks deliver a nominal 3.7 V.[2]

## 21.6 Another great use of RMS

In many electrical problems, the voltage fluctuates a great deal. For example, the fluctuations in voltage makes the sound that comes out of an audio speaker.

You can use the root-mean-squared of the voltage to figure out the average power your speaker is consuming.

Let's say that the RMS of the voltage you are sending to the speaker is $V_{rms}$ and the resistance of the speaker is $R$ ohms. This means the power consumed by the speaker is:

$$P = \frac{V_{rms}^2}{R}$$

Similarly, if you know the RMS of the current you are pushing through the speaker is $I_{rms}$, then the power consumed by the speaker is:

$$P = I_{rms} R$$

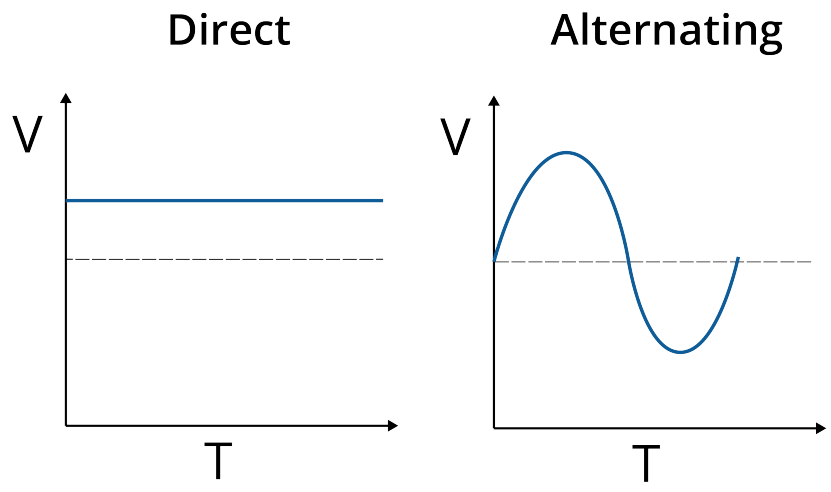
[2] Airlines often do not let you take anything over 100 Wh.

## 21.7 Electricity Dangers

Large amounts of electricity moving through your body can hurt or even kill you. You must be careful around electricity.

That said, your body is not a very good conductor, so low-voltage systems (like a flashlight) don't have enough voltage to move significant amounts of current through your body.

However, the electricity in a power outlet has much more voltage. The voltage in these outlets is fluctuating between positive and negative, so we call it *Alternating Current* or AC.




In most countries, the RMS of the voltage between 110 and 240 V. (The peak voltage is always $\sqrt{2}$ times the RMS value. In the US, for example, people say "Our outlets supply 120 V." They mean that the RMS of the voltage difference between the wire and the earth is 120V. The peak voltage is almost 170V.)

How much current can a human handle? Not much. You can barely feel 1 mA moving through your body, but at 16 mA, your muscles will clench and you won't be able to relax them — many people die from electrocution because they grab a wire which pushes enough current through their body to prevent them from letting go of the wire. At 20 mA, a human's respiratory muscles become paralyzed.

The fuse breaker in a house will often allow 20 A to flow through the circuit before it shuts off the power. Always be very sure to shut off the power before touching any of the

wiring in your house.

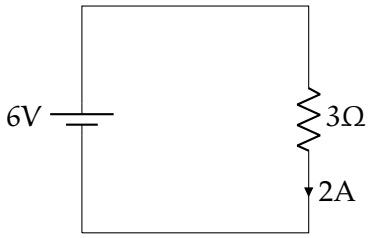
While water is actually a mediocre conductor, it can still deliver enough current to kill you. If you see a wire in a puddle, you should not touch the puddle. Interestingly, because of the salt, sea water is more than 100 times better at conducting electricity than the water you drink.

If you hold a wire in each hand, how many Ohms of resistance will your body have? Once it gets past your skin, you will look like a bag of salt water to the electricity. After the skin, your body will have a resistance of about 300Ω. However, the skin is a pretty good insulator. If you have dry, calloused hands, your skin may add 100,000Ω to the resistance.

CHAPTER 22

# DC Circuit Analysis

In the most basic circuit, you have only a battery and a resistor:

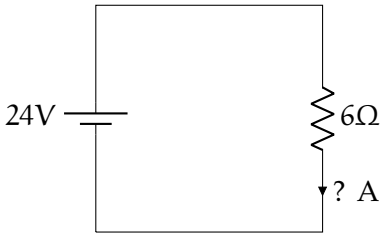



For this situation, you only need Ohm's Law: $V = IR$. In this case, $6V = 3\Omega \times 2A$.

### *Exercise 61* Ohm's Law

*Working Space*

How many amps are going around the circuit?

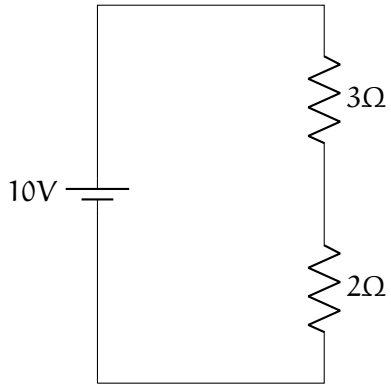



*Answer on Page 1893*

## 22.1 Resistors in Series

When you have two resistors wired together in a long line, we say they are "in series." If you have two resistors $R_1$ and $R_2$ wired in series, the total resistance is $R_1 + R_2$.

In this diagram, for example, the total resistance is $5\Omega$.




The current flowing through the circuit, then, is $10/5 = 2A$.

By Ohm's law, the voltage drop across the upper resistor is $IR = 2A \times 3\Omega = 6V$.

The voltage drop across the lower resistor is $IR = 2A \times 2\Omega = 4V$.

### 22.1.1 Kirchhoff's Voltage Law

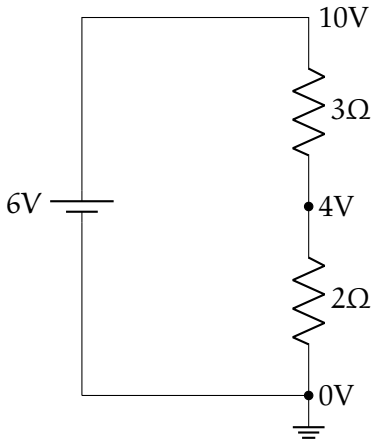
Notice that the battery pumps the voltage up to 10V, then the two resistors drop it by exactly 10V. This is known as "Kirchhoff's Voltage Law":

**Kirchhoff's Voltage Law**

As you make a loop around a circuit, the sum of the voltage increase must equal the sum of the voltage decrease. Mathematically, this is

$$\sum_{i=1}^{n} V_i = 0 \tag{22.1}$$

The negative end of the battery is connected to "ground" (it has zero voltage). We can then draw a diagram with the voltages (That symbol in the lower right represents a connection to ground).

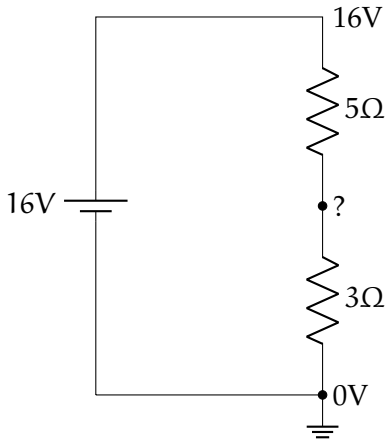



## *Exercise 62* Resistors In Series

*Working Space*

What is the current going around the circuit?

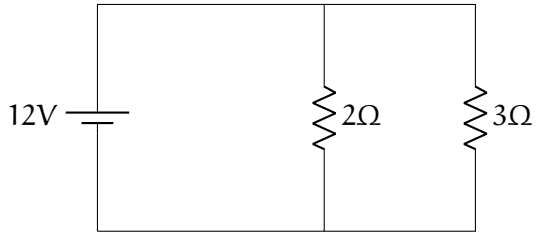
What is the voltage drop across each resistor?



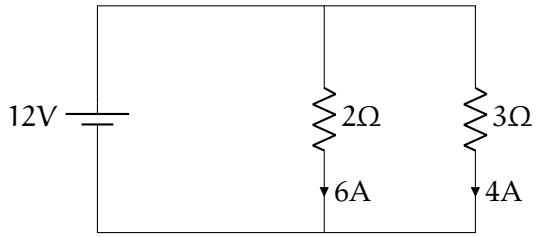

*Answer on Page 1893*

## 22.2 Resistors in Parallel

Observe the following circuit. Note that the current can go two different paths.




There is 12 volts pushing current through both resistors. So 6A will go through the 2Ω resistor and 4A will go through the 3Ω resistor. Note that even though there is a path of least resistance (2Ω), the current is still divided evenly among both branches.




Thus, a total of 10 A will be going through the battery.

Imagine you are a battery. You can't see that you have two resistors. What does it feel like to you? $\frac{V}{I} = R$, and $V = 12$ and $I = 10$. This means the effective resistance of the two resistors in parallel is $\frac{12}{10}$ or $\frac{6}{5}\Omega$.

**Resistance in Parallel**

If you have several resistances $R_1, R_2, \ldots, R_n$ wired in parallel, their effective resistance $R_t$ is given by

$$\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}$$

In our example:

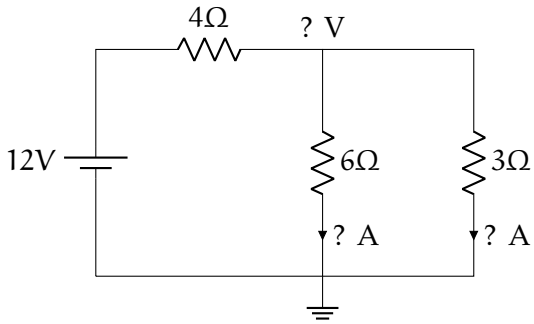
$$\frac{1}{R_t} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

Thus, $R_t = \frac{6}{5}\Omega$.

### *Exercise 63* Resistors In Parallel

*Working Space*

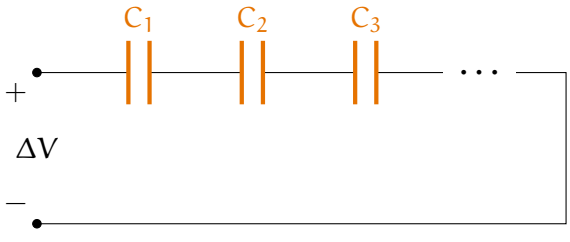
What is the current going through the battery? What is the drop over the 4Ω resistor? What is the current in each branch?




*Answer on Page 1893*

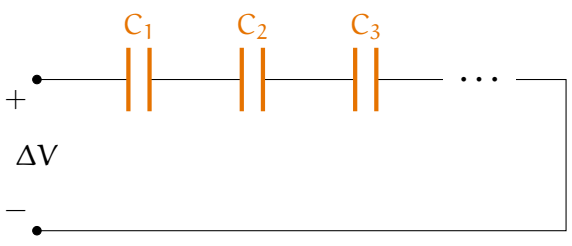
## 22.3 Capacitors in series and parallel

Recall that a capacitor is similar to a battery in that it stores voltage, but it is stored in the form of an electric field. Capacitors are represented with two differently colored lines. Here is an circuit diagram of capactors in **parallel** Capacitors are said to be in parallel if they share the same potential difference. You may see this represented mathematically:




$$C_p = \sum_i C_i \tag{22.2}$$

Capacitors are **series** if they all share the same charge magnitude. This would be



mathematically represented as

$$\frac{1}{C_S} = \sum_i \frac{1}{C_i} \tag{22.3}$$

A common mistake in calculating series capacitance is neglecting to take the reciprocal to find the equivalent capacitants after finding the reciprocal of each individual capacitor. If you have $n$ capacitors in series all with value $C$, the total capacitance is $C_S = \frac{C}{n}$.

## 22.4 Kirchhoff's Current Law

**Kirchhoff's Current Law** states that all currents coming into a junction (or node) must equal the currents leaving that junction.

Why? Because first of all, energy is always conserved, meaning it cannot be lost or destroyed within the equations in the first place.

Secondly, each circuit is conservative, meaning it cannot lose voltage, and thus, cannot lose current. As you go around a loop, the starting point must have the same chargeable amount as it did when you began, so any increases and decreases along the loop have to cancel out for a total change of zero.

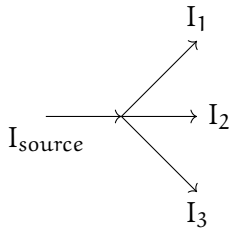
**Kirchhoff's Current Law**

The sum of current into a junction equals the sum of current out of the junction.

Mathematically, this can be expressed as

$$\sum_{i=1}^{n} I_i = 0 \tag{22.4}$$

where $n$ is the total number of branches with currents flowing towards or away from the node.




Kirchhoff's Current Law states that all currents going out of a junction must equal the current going into a junction. This is aligned with the Conservation of energy: **the total energy of an isolated system remains constant**. In this case,

$$I_{source} = I_1 + I_2 + I_3 \tag{22.5}$$

### *Exercise 64* Kirchhoff's Current Law

At a junction, 5 conductors meet at a single common junction. Given Branch Currents:

- $i_1 = 5$ A (entering the node)
- $i_2 = 3$ A (leaving the node)
- $i_3 = 2$ A (entering the node)
- $i_4 = 9$ A (leaving the node)

Find the direction of the unknown current $i_x$.

*Answer on Page 1894*

## 22.5 Summary

This chapter introduces the fundamentals of Direct Current (DC) circuits, focusing on the relationship between power sources and resistive elements.

- **Basic circuits**: The analysis begins with a minimal circuit configuration consisting of a voltage source (battery) and a load (resistor).
- **Voltage** (V): The electrical potential difference provided by the battery that drives the flow of charge.
- **Resistance** (R): The property of a material or component that opposes or *resists* the flow of electric current.
- **Direct Current** (DC): A type of electrical flow where the charge moves in a single, constant direction.
- **Kirchhoff's Laws**: Kirchhoff's Laws tell us that charge cannot be lost or pile up at a junction, and any supplied charged must be dropped to zero before the charge cycles around.

In a few workbooks, we will be doing a deep dive into charge, and how charge creates electric fields, electric flux, dipoles, and spherical electric potential.

Chapter 23

# Fertilizer

One of the biggest problems humans face is: how can we get enough food to feed everyone? In 1950, there were 2.5 billion people on the planet, and about 65% were malnourished. In 2019, there were 7.7 billion people on the planet, and only 15% are malnourished. How did crop yields increase so much? There were several factors: better crop varieties, reliable irrigation, increased mechanization, and affordable fertilizers.

When a plant grows, it takes molecules out of the soil and uses them to build proteins. It primarily needs the elements nitrogen ($N$), phosphorus ($P$), and potassium ($K$).

When you buy a bag of fertilizer at the store, it typically has three numbers on the front. For example, you might buy a bag of "24-22-4". This means that 24% of the mass of the bag is nitrogen, 22% is phosphorus, and 4% is potassium.

Potassium comes as potassium carbonate ($K_2CO_3$), potassium chloride ($KCl$), potassium sulfate ($K_2SO_4$), and potassium nitrate ($KNO_3$). Any blend of these chemicals is known as "potash". Potash is dug up out of mines.

Phosphorus is also mined, but is refined into phosphoric acid ($H_3PO_4$) before it is put into fertilizer.

Nitrogen is an especially interesting case for 2 reasons:

- Worldwide, farmers apply more nitrogen to their soil than potassium or phosphorous combined.
- 78% of the air we breathe is nitrogen in the form of $N_2$, but neither plants nor animals can utilize nitrogen in that form.

## 23.1 The Nitrogen Cycle

Converting the $N_2$ in the air into a form that a plant can use is known as *nitrogen fixation*. For billions of years, there were only two ways that nitrogen fixation occurred on earth:

- The energy from lightning causes $N_2$ and $H_2O$ to reconfigure as ammonia ($NH_3$) and nitrate ($NO_3$). This accounts for about 10% of all naturally occurring nitrogen fixation.
- Cyanobacteria are responsible for the rest. They convert $N_2$ into ammonia.

Let's say that you are eating soybeans. There is a cyanobacteria called *rhizobia* that has a symbiotic relationship with soybean plants. Rhizobia fixes nitrogen for the soybean plant. The soybean plant performs photosynthesis, then gives sugars to the rhizobia.

The proteins in the soybeans contain nitrogen from the rhizobia. When you eat them, you use some of the nitrogen to build new proteins. You probably don't use all the nitrogen, so your cells release ammonia into your blood.

Ammonia likes to react with things, so your liver combines the ammonia with carbon dioxide to make urea ($CO(NH_2)_2$). Your kidneys take the urea out of your blood and mix it with a bunch of water and salts in your bladder. When you urinate, the urea leaves your body.

If you urinate on the ground, the nearby plants can take the nitrogen out of the urea.

When you die, the nitrogen in your proteins will return to the soil as ammonia and nitrate.

For centuries, farms got their nitrogen from urine, feces, and rotting organic material. There were two challenges with this:

- Human pathogens had to be kept away from human food.
- There was simply not enough to support 7.7 billion people.

This meant we had to figure out how to do nitrogen fixation at an industrial level.

## 23.2 The Haber-Bosch Process

During World War I, two German scientists, Fritz Haber and Carl Bosch, figured out how to make ammonia from $N_2$ and $H_2$ using high temperatures and pressures. This is how nearly all nitrogen fertilizer is created today.

Where do we get the $H_2$? From methane ($CH_4$) in natural gas. Today, 3-5% of the world's natural gas production is consumed in the Haber-Bosch process.

The ammonia is converted into ammonium nitrate ($NH_4NO_3$) or urea before it is shipped to farms.

## 23.3 Other nutrients

Healthy plants require several other elements that are sometimes applied as fertilizer: calcium, magnesium, and sulfur.

Finally, tiny amounts of copper, iron, manganese, molybdenum, zinc, and boron are sometimes needed.

Chapter 24

# Concrete

To make concrete, you mix cement with water and an aggregate (sand or rock). The cement is usually only about 10 to 15 percent of the mixture. The cement reacts with the water, and the resulting solid binds the aggregate together. In 2019, the world consumed 4.5 billion tons of cement.

Concrete is hard and durable. The mortar between the pyramids at Giza is concrete — it is now 5000 years old. Today, we use concrete to build many structures including buildings, bridges, airport runways, and dams. Grout and mortar are related materials but different in that they are used as bonding or filler materials within construction.

There are many kinds of cement, but the most common is Portland cement. It is made by heating limestone (calcium carbonate) with clay (for silicon) in a kiln. Two things come out of the kiln: Carbon dioxide and a hard substance called "clinker". The clinker is ground up with some gypsum before it is sent to market.

The carbon dioxide is released into the atmosphere, making it an exothermic reaction. Cement manufacturing is responsible for about 8% of the world's $CO_2$ emissions; it is a major contributor to climate change.

Especially hard concrete, like that used in a nuclear power plant, can support 3,000 kg per centimeter without being crushed. However, if you pull on two ends of a piece of concrete, it comes apart relatively easily. We say that concrete can handle a lot of *compressive stress*, but not much *tensile* stress.

## 24.1 Steel reinforced concrete

Many places where we use concrete (like in a bridge), we need both compressive and tensile stress. Often, the top of a beam is undergoing compression and the bottom of the beam is undergoing tension.

FIXME Picture here

Steel has tremendous tensile strength, but not as much compressive strength as concrete. To get both tensile *and* compressive strength, we often bury steel bars or cables inside the concrete. This is known as *steel-reinforced concrete*. The concrete generally does a very good job protecting the steel, which keeps it from rusting.

You may have heard of *rebar* before. That is just short for "reinforcing bar". Typically, rebar

has bumps and ridges that keep the bar and the concrete from moving independently.

## 24.2 Recycling concrete

Many concrete structures only last about 100 years. When they are demolished, the concrete can be reused as aggregate in other projects. Often, the concrete bits are mixed with cement and made into concrete once more.

If the concrete to be reused is reinforced with steel, the steel has to be removed and recycled separately. The concrete is then crushed into small pieces.

CHAPTER 25

# Metals

Elements that transmit electricity well, even at low temperatures, are called *metals*. Many metals are likely familiar to you, such as aluminum, iron, copper, tin, gold, silver, and platinum. Aluminum and iron are particularly common; together they make up about 14% of the earth's crust.

An *alloy* is a mixture of elements that includes at least one metal. Brass, for example, is an alloy of copper and zinc. Bronze is an alloy of copper and tin.

## 25.1 Steel

One of the most common alloys is steel, which is an alloy of iron and carbon. In pure iron, the molecules slip past each other easily, so pure iron is relatively soft and easily deformed. The carbon in steel prevents that slipping, which is why steel is much, much harder than iron.

How much carbon does steel have? If you have less than 0.002% by weight, you end up with something very much like pure iron. As you increase the carbon, it gets harder and harder. Once it gets above about 2%, the result is very brittle.

If you add about 11% chromium to steel, you get *stainless steel*, which resists rusting.

## *Exercise 65* Tensile Strength

*Working Space*

The tensile strength of steel is usually between 400 MPa and 1200 MPa. A Mega Pascal (MPa) is the strength necessary to hold 1,000,000 newtons of force with a cable that has a 1 square meter cross section. Or, equivalently, to hold 1 newton of force with a cable that has a 1 square millimeter cross section.

If you have are buying a round cable that has a tensile strength of 700 Mpa and must hold a 100 kg man aloft, what is the diameter of the smallest cable you can use?

*Answer on Page 1894*

Here are some approximate tensile strengths of other materials:

| Material | Tensile strength (MPa) |
|---|---|
| Iron | 3 |
| Concrete | 4 |
| Rubber | 16 |
| Glass | 33 |
| Wood | 40 |
| Nylon | 100 |
| Human hair | 200 |
| Aluminum | 300 |
| Steel | 700 |
| Spider webs | 1000 |
| Carbon fiber | 4000 |

## 25.2 What metal for what task?

Copper is often used for electrical wires in your house and appliances. This is because it is very efficient at moving electricity (very little power is lost as heat). It is also very good

a transmitting heat, so you will often see copper pots and pans.

Aluminum is less dense than copper, and is still a relatively good conductor of electricity. This combination of lighter weight and conductivity is why the overhead wires in a power system are often made of aluminum.

Aluminum is not as strong as steel, but considerably lighter. It is often used structurally where weight is a concern, such as in skyscrapers, cars, airplanes, and ships.

Titanium is about as strong as steel, but it weights about half as much. Titanium is very difficult to work with, so it is used in places where weight and strength are very important and cost is not, such as in airplanes and bicycles. FIXME: We mention airplanes in both examples. Maybe we should clarify the role each one plays?

(Carbon fiber, which is light, strong, and very easy to work with, is replacing aluminum and titanium in many applications. 20 years ago, many expensive bicycles were made of titanium. These days the vast majority are made with carbon fiber.)

Zinc and tin are very resistant to corrosion, so they are often used as a coating to prevent steel from rusting. They are also used in many alloys for the same reason. In the United States, the penny is 97.5% zinc and only 2.5% copper.

Chapter 26

# Advanced Python Syntax and Libraries

## 26.1 Where did we leave off?

In the previous Introduction to Python chapter, We started by installing Python and setting up VSCode, then practiced using the console (terminal) to execute `.py` files. Recall that runs code **sequentially**, one line at a time, and that output appears in the console.

We can output to the screen with `print()`, including how Python formats output by default and how to customize it. Next, we introduced variables and common datatypes such as strings, integers, floats, booleans, lists, and dictionaries.

Mathematical operations, including arithmetic operators were demonstrated. We played with in `str()` and the method to get input from the console, `input()`. Finally, we worked with conditionals and loops to control program flow, then introduced strings and lists as sequence types, and worked with them. With these tools, you should now be able to write basic Python programs that read input, store data, make decisions, repeat actions, and produce useful output.

Let's now build on this foundation with some more advanced concepts, such as functions, classes, and libraries. We will also cover error handling, and experiment with basic graphing ability using the Matplotlib library.

## 26.2 Functions

As our programs get larger, we begin to repeat the same sets of steps again and again. Rather than copying and pasting code (which is hard to maintain and easy to break), we can group instructions into *functions*. A *function* is a named block of code that performs a task. Once a function is defined, it can be *called* (used) as many times as needed.

You have already used functions before, even if you did not know it! For example, `print()`, `input()`, `len()`, and `type()` are built-in Python functions.

### 26.2.1 Defining and Calling a Function

To define a function in Python, we use the `def` keyword, followed by:

- a function name
- parentheses `()`
- a colon `:`
- an indented block of code

```
def say_hello():
    print("Hello!")
    print("Welcome to Python.")
```

This code *defines* the function, but it does not run the code inside it yet. To execute the function, we must *call* it by using its name followed by parentheses:

```
say_hello()
```

When Python reaches a function call, it temporarily jumps into the function, runs the indented code, and then returns to the line after the call. Just like conditionals and loops, **indentation matters**.

### 26.2.2 Parameters and Arguments

Most functions are more useful when they can work with different values. A *parameter* is a variable name listed inside the parentheses of a function definition. An *argument* is the actual value you pass into the function when you call it.

```
def greet(name):
    print("Hello", name)
```

Here, `name` is a parameter. We can call the function with different arguments:

```
greet("Alex")
greet("Chicago")
```

### 26.2.3 Return Values

Some functions *return* a value. Returning a value allows the function to produce a result that can be stored in a variable or used in an expression.

```
def add(a, b):
    return a + b
```

Now we can use the returned value:

```
x = add(3, 4)
print(x)
```

Output:

```
7
```

We can alter the function to expect a certain parameter type, such as integers, and return a specific type as well:

```
def add(a: int, b: int) -> int:
    return a + b
```

It is important to note that Python won't raise a TypeError if you pass in the wrong type, but this is a helpful way to document your code for other programmers, which you may often work with, and yourself.

We can force the parameters to be a certain type, and raise an error if the wrong type is passed in:

```
def add(a: int, b: int) -> int:
    if not isinstance(a, int) or not isinstance(b, int):
        raise TypeError("Both arguments must be integers")
    return a + b
```

**Important:** `print()` displays information to the console, but it does not return a useful value. `return` sends a value back to where the function was called. Functions that do not have a `return` statement return `None` by default, and are often called *void* functions.

### 26.2.4 Scope

A variable created inside a function only exists inside that function. This idea is called *scope*. If you define a variable inside a function, you cannot use it outside of the function unless you return it.

```
def make_number():
    x = 10
    return x

y = make_number()
print(y)
```

The variable `x` exists only inside `make_number()`, but the value it returns is stored in `y` outside the function.

### 26.2.5 Summary

- Functions group code into reusable blocks
- Functions are defined with `def` and called using parentheses
- Parameters receive input values (arguments) when a function is called
- `return` sends a value back to the caller
- Variables inside a function have local scope

### *Exercise 66* Tracing a Function Call

Consider the following code:

```
def double(x):
    return x * 2

a = 3
b = double(a)
print(b)
```

What is printed? What is the value of `a` after the program runs?

*Working Space*

*Answer on Page 1894*

### *Exercise 67* Write a Function

Write a function called `is_even(n)` that returns `True` if `n` is even and `False` otherwise. Then show an example call to your function using `n = 7`.

*Working Space*

*Answer on Page 1895*

## 26.3 Classes and More Basic Object-Oriented Programming

So far, we have worked mostly with individual variables, lists, and functions. As programs grow larger, it becomes useful to group related data and behavior together. *Object-Oriented Programming* (OOP) is a programming style that organizes code around *objects* rather than individual functions.

In Python, objects are created from *classes*. A *class* is a blueprint for creating objects that contain both data (variables) and behavior (functions).

### 26.3.1 Defining a Class

A class is defined using the `class` keyword, followed by:

- the class name (by convention, written in `CamelCase`)
- a colon `:`
- an indented block of code

```
class Person:
    pass
```

This defines an empty class. It does not do anything yet, but it gives Python a new type called `Person`.

### 26.3.2 Creating Objects

An *object* is an instance of a class. Objects are created by calling the class name like a function.

```
p1 = Person()
p2 = Person()
```

Here, `p1` and `p2` are two separate objects, both created from the same class.

### 26.3.3 The `__init__` Method

Most classes need to store data. This is done using a special function called `__init__`. This function runs automatically when a new object is created.

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age
```

The parameter `self` refers to the current object being created. Each object gets its own copy of the variables defined using `self`.

```
p = Person("Alex", 20)
```

Now the object `p` has two attributes:

- `p.name`
- `p.age`

### 26.3.4 Accessing Object Attributes

Attributes are accessed using dot notation.

```
print(p.name)
print(p.age)
```

Output:

```
Alex
20
```

Each object stores its own values. Creating another object does not overwrite existing ones.

### 26.3.5 Methods

Functions defined inside a class are called *methods*. Methods describe behavior that belongs to the object.

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def greet(self):
        print("Hello, my name is", self.name)
```

Calling a method uses dot notation:

```
p = Person("Alex", 20)
p.greet()
```

Output:

```
Hello, my name is Alex
```

### 26.3.6 Printing Objects

By default, printing an object produces an unreadable result:

```
print(p)
```

Output (example):

```
<__main__.Person object at 0x7f9c1a3d>
```

This actually is the object's memory address in the computer's memory, but it may be a bit tedious want to print that and manually search ever byte. Instead, to control how an object is printed, we can define the special method `__str__`.

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def __str__(self):
        return f"{self.name} ({self.age} years old)"
```

Now printing the object gives a meaningful result:

```
p = Person("Alex", 20)
print(p)
```

Output:

```
Alex (20 years old)
```

### 26.3.7 Why Use Classes?

Classes allow us to:

- group related data and behavior together
- create many objects with the same structure
- write code that is easier to understand and maintain

### 26.3.8 Summary

- A class is a blueprint for creating objects
- Objects store data using attributes
- `__init__` initializes new objects
- Methods define behavior for objects
- `__str__` controls how objects are printed
- This was only a foundation of OOP; more advance concepts exist and are stronger in other programming languages such as Java and C++

## *Exercise 68* Tracing an Object

Consider the following code:

*Working Space*

```
class Counter:
    def __init__(self, value):
        self.value = value

    def increment(self):
        self.value += 1

c = Counter(5)
c.increment()
c.increment()
print(c.value)
```

What is printed?

*Answer on Page 1895*

## *Exercise 69* Custom Printing

Write a class called `Book` that stores a title and an author. Add a `__str__` method so that printing a book displays:

*Working Space*

```
Title by Author
```

*Answer on Page 1895*

## 26.4 Libraries

A *library* is a collection of pre-written code that provides additional functionality without requiring you to write everything from scratch. Python includes many built-in libraries that can be used to perform common tasks such as mathematical calculations, data han-

dling, and visualization. You'll often find that code you are seeking can be found through an open-source library which already exists.

To use a library, it must first be imported.

### 26.4.1 Importing Libraries

The simplest way to import a library is using the `import` keyword.

```
import math
print(math.sqrt(16))
```

In this example, the `math` library provides access to mathematical functions such as square roots.

It is also possible to import specific values or functions from a library.

```
from math import pi
print(pi)
```

This allows direct access to `pi` without referencing the library name.

### 26.4.2 Summary

- Libraries extend Python's functionality
- The `import` keyword is used to load libraries
- Specific components can be imported directly when needed

## 26.5 Matplotlib

Matplotlib is the most widely used plotting library in Python. It can produce simple charts quickly (line plots, scatter plots, bar charts, histograms) and also supports publication-quality figures with precise control over labels, legends, and layout. We will use it widely throughout this course to visualize data, create simulations for proving formulas, and experiment with datasets.

### 26.5.1 Installing and importing

If you are working locally, you can install Matplotlib with:

```
pip install matplotlib
```

In most scripts, you will import the plotting interface `pyplot`:

```
import matplotlib.pyplot as plt
```

### 26.5.2 A simple line graph

A line plot is ideal for showing how a value changes over time or across an ordered variable. Let's create `line.py`

```
import matplotlib.pyplot as plt # the matplotlib library, which we will call
↪  using plt

# two lists containing values
x = [0, 1, 2, 3, 4, 5]
y = [0, 1, 4, 9, 16, 25]

# a 2d plot, used for line graphs
plt.plot(x, y)
# add a title to our plot
plt.title("A Simple Line Plot")
# add axis labels
plt.xlabel("x")
plt.ylabel("y = x^2")
plt.show() # pops up a new window
```

Running `python3 line.py` will open a new *interactive window* in your computer, with the following image:

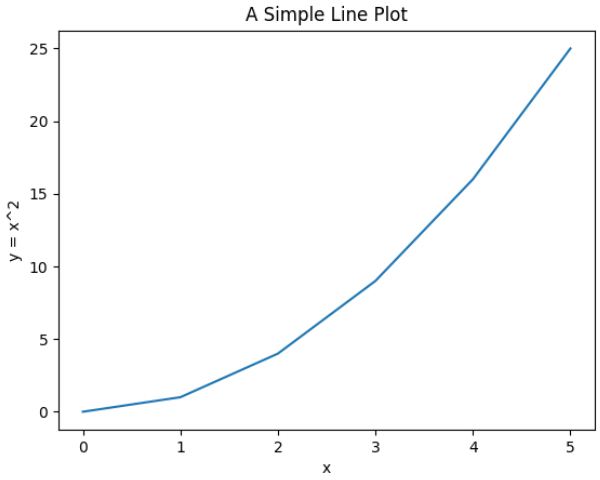



Figure 26.1: The output of line.py.

### 26.5.3 Labels, legends, and grids

A plot is more useful when it clearly communicates what each element represents. Let's add on to `line.py` by creating `lineTwoOutputs.py` and adding labels, a legend, and another set of data.

```
1  import matplotlib.pyplot as plt
2
3  x = [0, 1, 2, 3, 4, 5]
4  y1 = [0, 1, 4, 9, 16, 25]
5  y2 = [0, 1, 8, 27, 64, 125]
6
7  plt.plot(x, y1, label="x^2")
8  plt.plot(x, y2, label="x^3")
9
10 plt.title("Two Functions on One
   ↪  Axes")
11 plt.xlabel("x")
12 plt.ylabel("value")
13 plt.grid(True)
14 plt.legend()
15 plt.show()
```

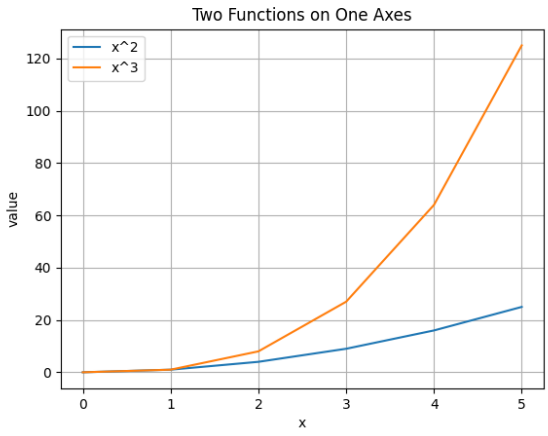



Figure 26.2: $y = x^2$ and $y = x^3$ on the same graphed.

### 26.5.4 Visualizing Relationships with scatter plots

Scatter plots are excellent for showing how two variables relate (for example, height and weight).

This very basic scatter plot shows the relationship between studying time and score.

```
import matplotlib.pyplot as plt

hours = [1, 2, 3, 4, 5, 6]
scores = [55, 60, 66, 72, 78, 85]

plt.scatter(hours, scores)
plt.title("Study Time vs. Score")
plt.xlabel("hours studied")
plt.ylabel("score")
plt.show()
```

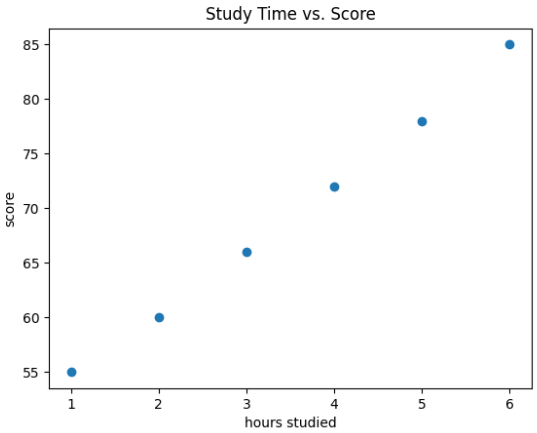



Figure 26.3: A scatterplot generated by matplotlib from the given data.

### 26.5.5 Histograms and Distributions

A histogram shows how values are distributed and how common different ranges are.

```python
import matplotlib.pyplot as plt
# a set of data, for example maybe these are all test scores
data = [4, 5, 5, 6, 7, 7, 7, 8, 8, 9, 10, 10, 10, 11, 12]

# creates a histogram plot
plt.hist(data, bins=5)
plt.title("Histogram of Values")
plt.xlabel("value")
plt.ylabel("frequency")
plt.show()
```

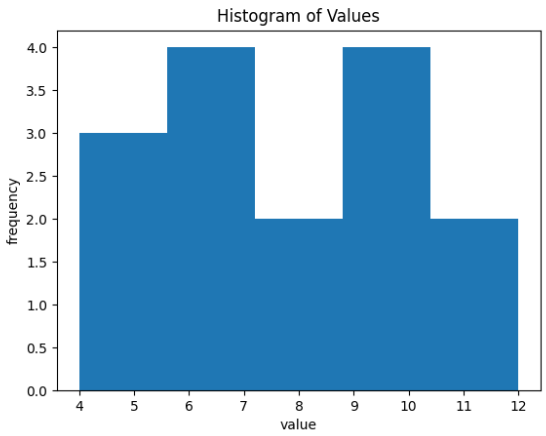



Figure 26.4: A histogram of a set of data.

### 26.5.6 The object-oriented approach (recommended)

Matplotlib has two main styles:

- **State-based** (using `plt.plot`, `plt.title`, ...): quick and convenient.
- **Object-oriented** (creating a `Figure` and `Axes`): more explicit and easier to scale.

For multi-plot layouts or different analysis of datasets, prefer the object-oriented approach:

```python
import matplotlib.pyplot as plt

x = [0, 1, 2, 3, 4, 5]
y1 = [0, 1, 4, 9, 16, 25]
y2 = [0, 1, 8, 27, 64, 125]

# two figures, side by side. axes becomes an indexable list
fig, axes = plt.subplots(nrows=1, ncols=2)

axes[0].plot(x, y1)
axes[0].set_title("x^2")
axes[0].set_xlabel("x")
axes[0].set_ylabel("value")

axes[1].plot(x, y2)
axes[1].set_title("x^3")
axes[1].set_xlabel("x")
axes[1].set_ylabel("value")

```

```
fig.suptitle("Two Subplots")
fig.tight_layout()
plt.show()
```

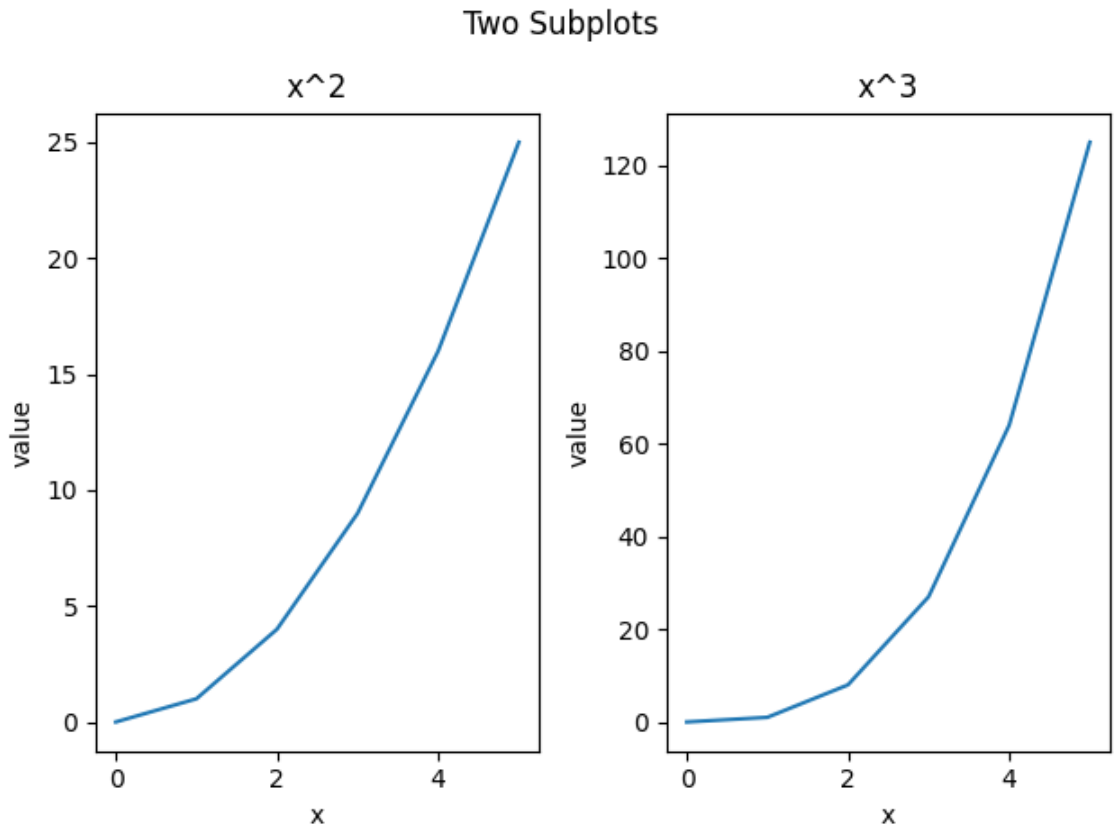



Figure 26.5: The two plots, generated side by side.

## 26.6 Errors

Even the most experienced programmers can make mistakes. If the code seems to run into issues, your code may crash or output an *error*. Python reports errors by raising a halt in the console output, which include a message explaining the problem and the exact line where it occurred. By learning how to read and interpret these messages, you'll be able to *debug*, or fix issues, in your programs more efficiently and write more reliable code.

You will encounter two main types of errors in your code: **Syntax Errors** and **Runtime Errors/Exceptions**.

- Syntax errors occur when your the formatting, or *syntax*. These errors can usually be found before your code is compiled.

- Runtime Errors, or *Exceptions*, occur when operations in your code cause an invalid calculation, or attempt to do an invalid action. These can range anywhere from basic misnamed variables to imported libraries crashing from incorrect data.

### 26.6.1 Syntax Errors

What do you notice is immediately wrong with this code?

```
    x = "Welcome Home"
    print(type(x)
```

If we run it, we get the output:

```
Traceback (most recent call last):
  File "<string>", line 1, in <module>
  File "/usr/lib/python3.12/py_compile.py", line 150, in compile
    raise py_exc
py_compile.PyCompileError:   File "./prog.py", line 2
    print(type(x)
          ^
SyntaxError: '(' was never closed
```

We notice that every opening parentheses, bracket, or brace must have a closing supplement. Without out, we run into Syntax Errors, letting us know our format is off.

Another type of Syntax Error can be found in incorrect indentation. Take for example the following code:

```
if True:
print("Hello!")
```

Output:

```
Traceback (most recent call last):
  File "<string>", line 1, in <module>
  File "/usr/lib/python3.12/py_compile.py", line 150, in compile
    raise py_exc
py_compile.PyCompileError: Sorry: IndentationError: expected an indented block
↪  after 'if' statement on line 1 (prog.py, line 2)
```

Here, there is no indentation (usually obtained by pressing the `Tab` key on your keyboard)

for lines under the conditional statement. This causes a Syntax Error to be raised.

### 26.6.2 Exceptions

Let's say I try to run the following code:

```
    print(x)
    x = "hello"
```

You may see the following output:

```
Traceback (most recent call last):
    File "./program.py", line 1, in <module>
NameError: name 'x' is not defined
```

You have encountered a *NameError*, because `x` was not assigned before it was attempted to be printed. This brings up an important note on Python code: **code is executed line-by-line, sequentially**. So even if you define `x` after you try and print it, Python will not understand what you are trying to print. Let's look at another example:

### 26.6.3 Try-Except

```
try:
    x = int(input("Enter a number: "))
    print(10 / x)
except ValueError:
    print("Please enter a valid integer.")
except ZeroDivisionError:
    print("Cannot divide by zero.")
```

Here, we attempt to divide 10 by some input number `x`. There are two `Exceptions` that could cause this to fail:

- The user inputs a string of characters, or anything that isn't an integer
- The user inputs 0, an invalid divisor.

We use a `Try-Except` block to check for different errors generated, similar to an if statement. The try block surrounds a block of code that may cause faulty runtime output or errors.

### *Exercise 70* Fix the Code

*Working Space*

```
items = ["pen", "book", "eraser",
↪  "ruler"]

index = input("Enter an index (0 to
↪  3): ")
print("You chose:", items[index])

items.remove("pencil")
print("Updated list:", items)
```

Name two errors that could occur when this code is run. For each error, explain why it occurs and how to fix it.

*Answer on Page 1895*

## 26.7 Recursive Functions

As we briefly mentioned in the proofs chapter, *recursion* is the process of having a given function call itself over an over. Imagine you are making a phone call, but you have the phone that you are speaking into as well as the phone you are calling. If you put them next to each other and say "Hello!", it will cause "Hello!" to be spoken by the phone speaker into the microphone phone infinitely, until you hang up on one of the phones.

A *recursive function* can be as simple as this

```
def foo():
    # recursive call
    foo()

foo()
```

The line that calls `foo()` within its self-definition is called a *recursive call*. However, doing this may cause you to get an error as follows

```
RecursionError: maximum recursion depth exceeded
```

What does this mean? Well, we did not create a way to get out of our recursive call once we start running. Your computer would call this function `foo()` forever until it either runs out of battery or the memory's call stack is hitting a maximum.[1] To prevent this, our recursive functions need to have an exit condition, called a *base case* or *base condition*. A base case is defined as a terminating condition that does not use a recursive call to provide an answer. This allows you to exit the function at some time, instead of continually calling a function forever. Let's look at the famous Fibonacci sequence.

The fibonacci sequence is a sequence of numbers where each term $F_n$ is defined by adding the previous two terms:

$$\text{fib}(n) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ \text{fib}(n-1) - \text{fib}(n-2) & n \geq 2 \text{ and } n \in \mathbb{Z} \end{cases}$$

In sequence notation we write the base cases as:

$$F_0 = 0, \quad F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2}$$

If we did not have a base case, how would we start defining the terms? Simply, we couldn't. There would be no way to define recursively-called previous terms explicitly as we wouldn't have a definite value for them.

### 26.7.1 Binary Search and Divide-and-Conquer Algorithms

If you look up information about recursive functions, you will find their most common use: *divide-and-conquer algorithms*. Divide-and-conquer algorithms are those that are run recursively to reduce problems into smaller versions of the same problem. As we talked about, problems involving trees, graphs, arrays are all best done with recursive functions as these are all structures that can be defined as smaller versions of the same structure.

A very common example of this is searching for a word in a dictionary. Let's say you are looking for the word *keyboardist*. For example purposes, let's say this specific dictionary is 2048 pages long (this will help our math). One way to find the word *keyboardist* would be

[1] In Java, you may get a `StackOverflowError`. In C or C++, you may get a message saying `Segmentation fault (core dumped)`. In programming languages much closer to memory, we say that the call stack reaches a maximum, or is *overflowed*. The call stack is one which is governed by the amount of RAM or random access memory. That's where the website Stack Overflow gets its name!

to search every individual page. This would be referred to as an *iterative* approach. This wouldn't take too long if we were searching for the word *apple*, but what about *zebra*? We would be searching almost all 2048 pages before reaching the word!

There is a much quicker and more efficient approach. Say we flip to approximately middle of the dictionary, page 1024. The first word is *muffin*. The word *keyboardist* must be to left of this of this page, since we know the dictionary is alphabetically sorted and $k$ comes before $m$. Let's split the 1024 remaining pages in half; we flip to page 512. The first word in page 512 is *committee*. We need to go back towards the original middle, since $c$ comes before $k$. So now we know *keyboardist* must lie somewhere between pages 512 and 1024.

We split that interval in half again and flip to page 768. Suppose the first word there is *gracious*. Since $g$ comes before $k$, we know our word must be farther to the right, so we now restrict our search to pages 768 through 1024. We continue in this way, each time cutting the remaining search region in half and deciding whether to move left or right depending on alphabetical order.

Eventually, we get to *keyboardist* on page 980. This method is much faster because every comparison eliminates about half of the remaining pages. Instead of checking pages one by one, we narrow the possibilities dramatically with each step. This process is a *divide-and-conquer* approach called *binary search*. Notice that at each step, we aren't actually doing one static action. We are reducing the possible page entries by half, and then recalling the same comparison on the function repeatedly until we reach our target. This is what makes our function a divide-and-conquer recursive algorithm.

Let's look at what a binary search for a list looks like in python:

```
1  def binary_search(arr, target, left, right):
2      if left > right:
3          return -1  # not found
4
5      mid = (left + right) // 2
6
7      if arr[mid] == target:
8          return mid
9      elif target < arr[mid]:
10         return binary_search(arr, target, left, mid - 1)
11     else:
12         return binary_search(arr, target, mid + 1, right)
```

Let's prove this algorithm is correct, just as we talked about in the proofs chapter. Since this algorithm is recursive and it functions by operating on smaller versions of itself, the best proof to use here is a *proof by induction*.

**Theorem 9.** *For any indices* `left` *and* `right`, *`binary_search(arr, target, left, right)` returns in index* `i` *with* `left` $\leq$ `i` $\leq$ `right` *and `arr[i] = target` if the target occurs within `arr[left...right]`, or returns* $-1$ *if the target is not found within the array.*

*Proof.* We prove the statement by induction on $n = \texttt{right} - \texttt{left} + 1$, the length of the interval to be searched.

**Basis Step.** If $n \leq 0$, then $\texttt{left} > \texttt{right}$. The function immediately returns $-1$ via line 3, which is correct. An empty interval contains no elements, so the target is not present.

**Inductive Hypothesis.** Assume the function works correctly for all intervals of length less than $n$. (Notice we are reducing the problem size).

**Inductive Step.** Now consider an interval of length $n > 0$, so $\texttt{left} \leq \texttt{right}$. The function computes $\texttt{mid} = \left\lfloor \frac{\texttt{left}+\texttt{right}}{2} \right\rfloor$. There are three cases:

- `arr[mid] == target`: If the target we are searching for is at the middle of the new computed index, we correctly return the that index on line 8.
- `target < arr[mid]`: Because the array is sorted, every element at positions $\texttt{left}, \texttt{left}+1, \ldots, \texttt{mid}$ is *at most* `arr[mid]`, so all of them are greater than the target. Therefore, if the target exists at all in `arr[left...right]`, it must lie in `arr[left...mid-1]`. The function then calls `binary_search(arr, target, left, mid - 1)`. The new interval has length strictly smaller than $n$, so by the inductive hypothesis, that recursive call gives the correct answer. Hence the original call is also correct.
- `target > arr[mid]`: Again, because the array is sorted, every element at positions $\texttt{mid}, \texttt{mid}+1, \ldots, \texttt{right}$ is *at least* arr[mid], so all of them are greater than the target. Therefore, if the target exists at all in `arr[left...right]`, it must lie in `arr[mid+1...right]`. The function then calls `binary_search(arr, target, mid + 1, right)`. The new interval, again, has length strictly smaller than $n$, so by the inductive hypothesis, that recursive call gives the correct answer. Therefore, the original call is also correct.

**Conclusion.** The proof covers all possible recursive calls and all of them accurately reduce the interval length in half. By induction on the interval length $n$, the function is correct for every valid call. □

## *Exercise 71* Recursive Factorial

*Working Space*

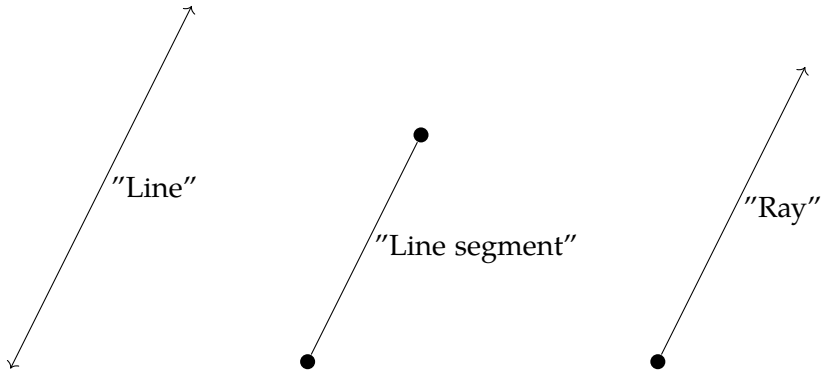
Write a pythonic function that recursively finds the factorial of a given number $n$. Assume $n \geq 0$.

*Answer on Page 1896*

# Chapter 27

# Angles

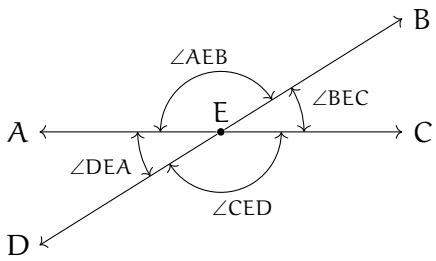
In the following recommended videos, the narrator talks about lines, line segments, and rays. When mathematicians talk about *lines*, they mean a straight line that goes forever in two directions. If you pick any two points on that line, the space between them is a *line segment*. If you take any line, pick a point on it, and discard all the points on one side of the point, that is a *ray*. All three have no width.




Watch the following videos from Khan Academy:

- Introduction to angles: https://youtu.be/H-de6Tkxej8
- Measuring angles in degrees: https://youtu.be/92aLiyeQj0w

When two lines cross, they form four angles:



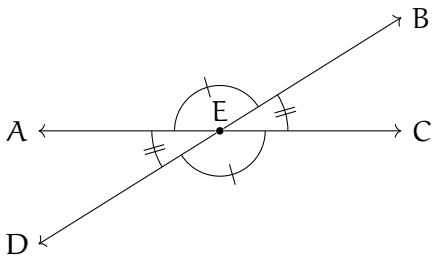

What do we know about those angles? (Note that m∠ means "the measure of angle")

- The sum of any two adjacent angles adds to be 180° So, for example, m∠AEB +

$m\angle BEC = 180°$. We use the phrase "adds to be 180°" so often that we have a special word for it: *supplementary*.

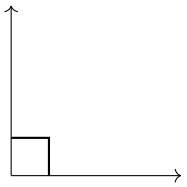
- The sum of all four angles is 360°.
- Angles opposite each other are equal. So, for example, $m\angle AEB = m\angle CED$.
- On any *horizontal* line segment with angles in between it, the sum of the angles must add up to 180°. For example, $\overline{AC}$ is a horizontal line, so angles between it must sum to 180°.

In a diagram, to indicate that two angles are equal we often put hash marks in the angle:




Here, the two angles with a single hash mark are equal, and the two angles with double hash marks are equal.

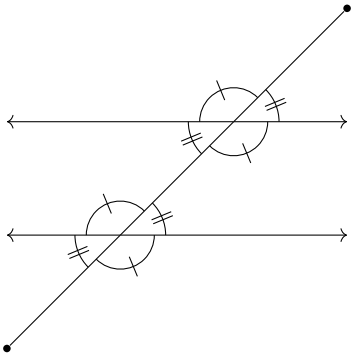
When two lines are perpendicular, the angle between them is 90°, and we say they meet at a *right angle*. When drawing diagrams, we indicate right angles with an elbow:



When an angle is less than 90°, it is said to be *acute*. When an angle is more than 90°, it is said to be *obtuse*.

Just as two angles which add up to 180° are referred to as supplementary angles, two (or more) angles which add up to 90° are *complementary* angles. For example, 2 45° angles are complementary, as well as 30° and 60°. If an angle is 65°, we say that its *complement* is $90° - 65° = 25°$.

If two lines are parallel (never intersecting lines with points all the same distance apart), line segments that intersect both lines form the same angles with each line:



## 27.1 Radians

As you've seen above, angles can be measured in degrees. Just like you can measure length in more than one unit (inches, meters, etc.), there is more than one unit to measure angles in. Angles can also be measured in *radians*. Radians are unitless (that is, you don't have to put a letter after the number) and there are $\pi$ radians across a straight line. On diagrams and equations, unknown angles in radians are usually represented with the greek letter $\theta$. This means $180^\circ$ is the same as $\pi$ radians. Radians come from comparing the length of an arc to the radius of a circle–which we will explore in a future chapter.

Generally, you can use the following formula for your conversions:

$$\text{angle in degrees} = \text{angle in radians} \cdot \frac{180^\circ}{\pi}$$

**Example**: An angle is measured to be $\frac{\pi}{2}$ radians. What is the angle in degrees?

**Solution**: Since we know that $\pi$ radians is the same as $180^\circ$, we can set up the unit conversion:

$$\frac{\pi}{2} \cdot \frac{180^\circ}{\pi} = 90^\circ$$

Therefore, a $\frac{\pi}{2}$ angle is $90^\circ$.

## Exercise 72

Convert the following angles from degrees to radians, or from radians to degrees.

1. $360°$
2. $\frac{\pi}{3}$
3. $225°$
4. $\frac{3\pi}{4}$
5. $30°$
6. $45°$

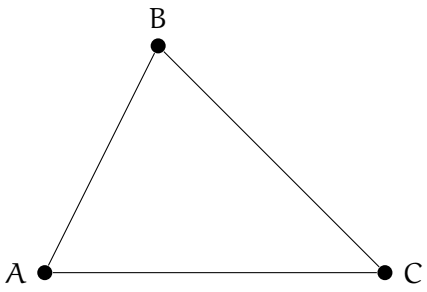
*Working Space*

*Answer on Page 1896*

CHAPTER 28

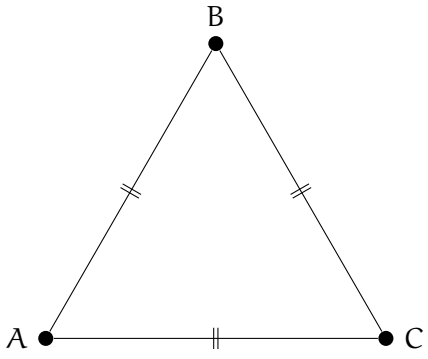
# Introduction to Triangles

Connecting any three points with three line segments will get you a triangle. Here is the triangle $ABC$, which was created by connecting three points $A$, $B$, and $C$:

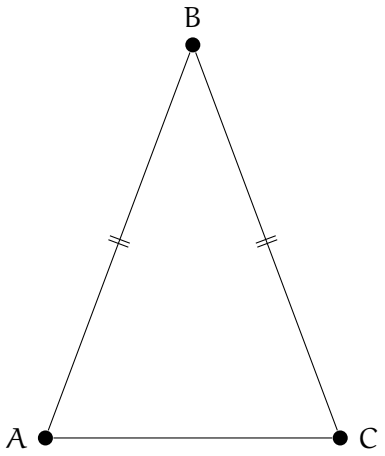



## 28.1 Equilateral, Isosceles, and Scalene Triangles

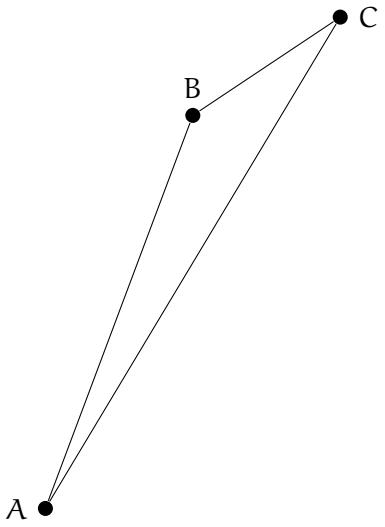
We talk a great deal about the length of the sides of triangles. If all three sides of the triangle are the same length, we say it is an *equilateral triangle*:




If only two sides of the triangle are the same length, we say it is an *isosceles triangle*:

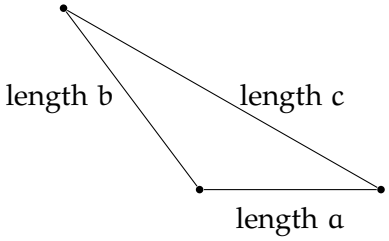



When all three sides of a triangle are different, the triangle is referred to as *scalene*. This consequently means all angle measures are different as well.



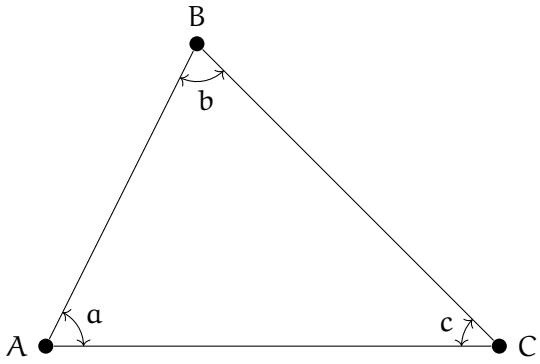

The shortest distance between two points is always the straight line between them. This means you can be certain that the length of one side will *always* be less than the sum of the lengths of the remaining two sides. This is known as the *triangle inequality*.

For example, in this diagram, $c$ must be less than $a + b$.

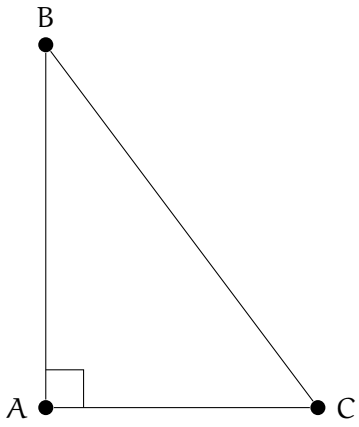



## 28.2 Interior Angles of a Triangle

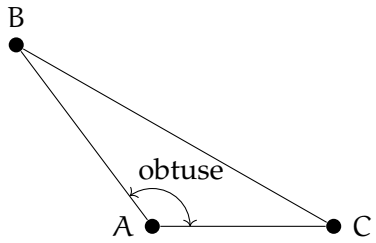
We also talk a lot about the interior angles of a triangle:




A triangle where one of the interior angles is a right angle is said to be a *right triangle*: The side opposite the right angle is the longest side, referred to as the hypotenuse.




If a triangle has an obtuse interior angle, it is said to be an *obtuse triangle*:




If all three interior angles of a triangle are less than 90°, it is said to be an *acute triangle*.

The measures of the interior angles of a triangle always add up to 180°. For example, if we know that a triangle has interior angles of 37° and 56°, we know that the third interior angle is 87°.

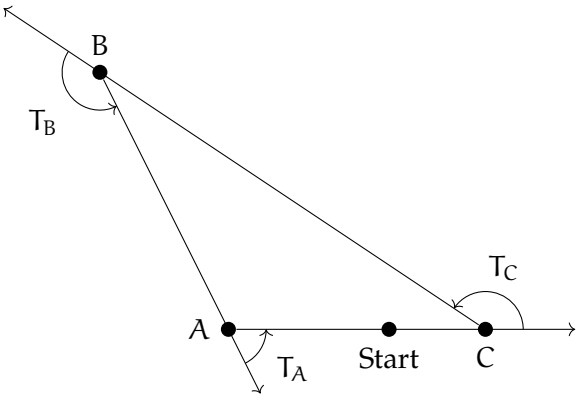
## *Exercise 73* Missing Angle

One interior angle of a triangle is 92°. The second angle is 42°. What is the measure of the third interior angle?

*Working Space*

*Answer on Page 1896*

How can you know that the sum of the interior angles is 180°? Imagine that you started on the edge of a triangle and walked all the way around to where you started. ( going counter-clockwise.) You would turn three times to the left:




After these three turns, you would be facing the same direction that you started in. Thus, $T_A + T_B + T_C = 360°$. The measures of the interior angles are a, b, and c. Notice that a and $T_A$ are supplementary. So we know that:

- $T_A = 180 - a$
- $T_B = 180 - b$
- $T_C = 180 - c$

So we can rewrite the equation above as

$$(180 - a) + (180 - b) + (180 - c) = 360°$$

Which simplifies to:

$$540° - (a + b + c) = 360°$$

Subtracting both sides from 540:

$$a + b + c = 180°$$

## *Exercise 74* Interior Angles of a Quadrilateral

Any four-sided polygon is a *quadrilateral*. Using the same "walk around the edge" logic, what is the sum of the interior angles of any quadrilateral?

*Working Space*

*Answer on Page 1896*

## 28.3 Is it an Equilateral, Isosceles, or Scalene Triangle with Python?

You have now seen three ways to classify a triangle by its sides, and one rule that decides whether three lengths make a triangle at all. A short Python function can apply both ideas to any triple of numbers you hand it.

```
1  def classify_triangle(a, b, c):
2      short1, short2, longest = sorted([a, b, c])
3      if short1 + short2 <= longest:
4          return "not a triangle"
5      if a == b == c:
6          return "equilateral"
7      if a == b or b == c or a == c:
8          return "isosceles"
9      return "scalene"
```

Sorting the three lengths puts the two shortest sides first, no matter what order they were given in. The triangle inequality check then compares those two shortest sides against the longest one, exactly as stated earlier in this chapter: the length of one side must always be less than the sum of the other two. Once that check passes, the function looks for three equal sides, then two, and falls back to scalene if neither pattern matches.

```
triangles = [(5, 5, 5), (5, 5, 8), (5, 12, 13), (2, 2, 5)]
for sides in triangles:
    print(sides, classify_triangle(*sides))
```

This prints:

```
(5, 5, 5) equilateral
(5, 5, 8) isosceles
(5, 12, 13) scalene
(2, 2, 5) not a triangle
```

Notice the last line. $2 + 2 = 4$, which is not greater than 5, so these three lengths cannot close into a triangle at all, and the function catches that before it ever checks for equal sides.

## *Exercise 75* Classifying Triangles

Classify each of these triples by hand, then check your answers by running them through `classify_triangle`: $(7, 7, 10)$, $(9, 40, 41)$, and $(3, 4, 9)$.

*Working Space*

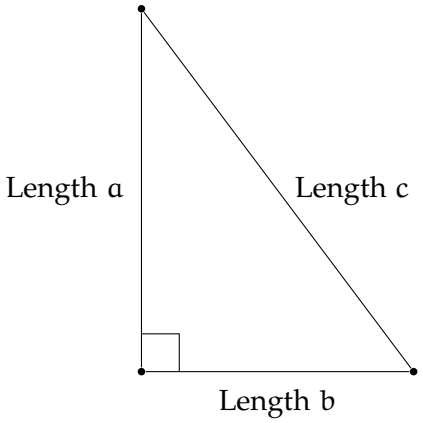
*Answer on Page 1896*

# Chapter 29

# Pythagorean Theorem

Watch's Khan Academy's Intro to the Pythagorean Theorem video at https://youtu.be/AA6RfgP-AHU.

If you have a right triangle, the edges that touch the right angle are called *the legs*. The third edge, which is always the longest and opposite the right angle, is known as *the hypotenuse*. The Pythagorean Theorem gives us the relationship between the length of the legs and the length of the hypotenuse.




The Pythagorean Theorem tells us that $a^2 + b^2 = c^2$, given that c is the hypotenuse.

For example, if one leg has a length of 3 and the other has a length of 4, then $a^2 + b^2 = 3^2 + 4^2 = 25$. Thus, $c^2$ must equal 25. This means you know the hypotenuse must be of length 5. This works for any right triangle

In reality, it rarely works out to be such a tidy number. For example, what is the length of the hypotenuse if the two legs are 3 and 6? $a^2 + b^2 = 3^2 + 6^2 = 45$. The length of the hypotenuse is the square root of that: $\sqrt{45} = \sqrt{9 \times 5} = 3\sqrt{5}$, which is approximately 6.708203932499369.

Common side lengths for these triangles are referred to as *Pythagorean triples*, meaning they evaluate to a whole number. Some common examples are $(3, 4, 5)$, $(5, 12, 13)$, and $(8, 15, 17)$. Multiples of right triangles are also triangles ie. $(3, 4, 5) \implies (6, 8, 10)$, which we will touch on next chapter.

There are also angle-based right triangles, consisting of ratios of the angles of the triangles. The most common ones are 45°-45°-90° and the 30°-60°-90°. We will discuss these further

in depth, but know for now that they are vital in trigonometry, and consist of Pythagorean triples as side lengths.

## *Exercise 76* Find the Missing Length

What is the missing measure? All missing values should be whole numbers, except d is an irrational number; answer should be a decimal approximation.

*Working Space*

Leg 1 = 6, Leg 2 = 8, Hypotenuse = a

Leg 1 = 5, Leg 2 = b, Hypotenuse = 13

Leg 1 = c, Leg 2 = 15, Hypotenuse = 17

Leg 1 = 3, Leg 2 = 3, Hypotenuse = d

*Answer on Page 1897*

A square's diagonal is a special case of the Pythagorean Theorem such that $c = \sqrt{a^2 + b^2} = \sqrt{2s^2}$.

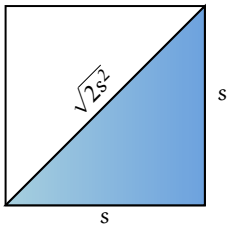
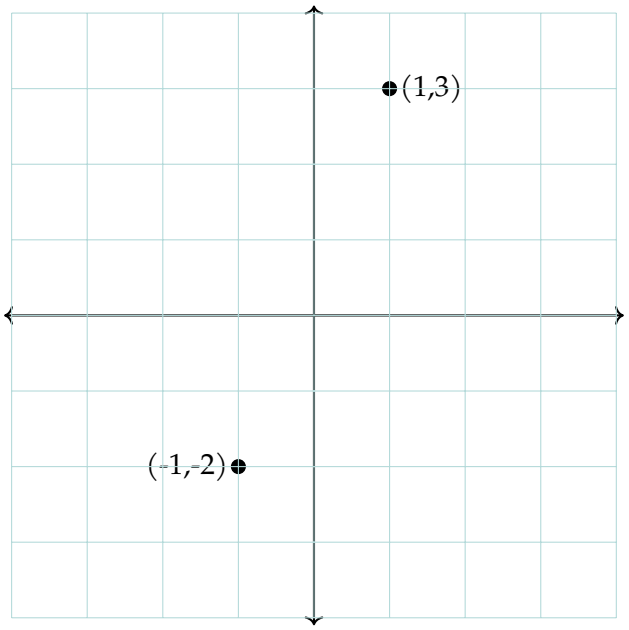


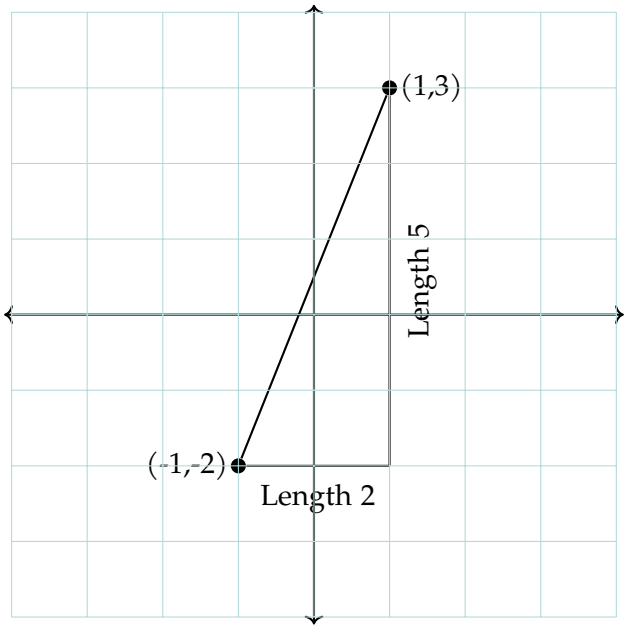

Figure 29.1: A special case of the Pythagorean Theorem where each side is side length s and the hypotenuse is $\sqrt{2s^2}$.

## 29.1 Distance between Points

What is the distance between these two points?



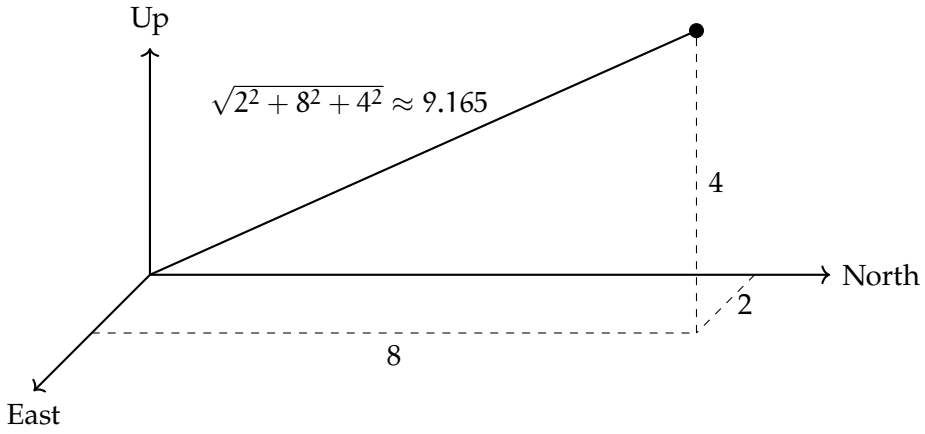

We can draw a right triangle and use the Pythagorean Theorem:




The distance between the two points is $\sqrt{2^2 + 5^2} = \sqrt{29} \approx 5.385165$. In other words, you square the change in $x$ and add it to the square of the change in $y$. The distance is the square root of that sum.

## 29.2 Distance in 3 Dimensions

What if the point is in three-dimensional space? For example, you move 2 meters East, 8 meters North, and 4 meters up in the air. How far are you from where you started? You just square each, sum them, and take the square root: $\sqrt{2^2 + 8^2 + 4^2} = \sqrt{84} = 2\sqrt{21} \approx 9.165$ meters.




This leads us to a formal definition of the distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Or in 3D space:

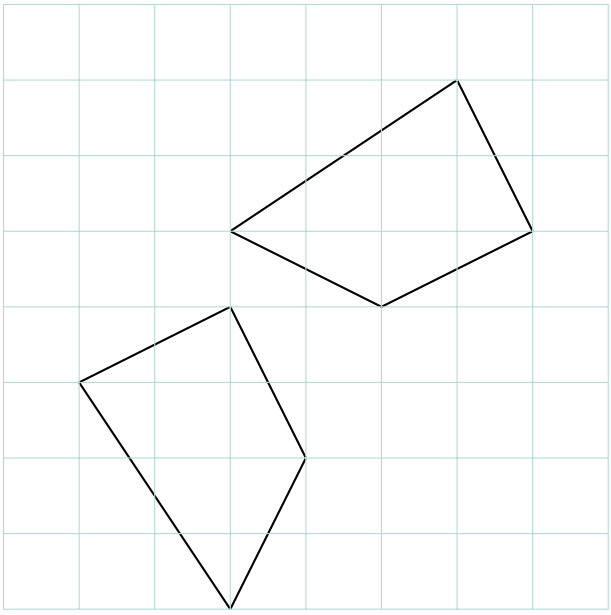
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

# Chapter 30

# Congruence

## 30.1 Rigid Transformations

Look at this picture of two geometric figures.



They are the same shape, right? If you cut one out with scissors, it would lay perfectly on top of the other. In geometry, we say they are *congruent*.

What is the official definition of "congruent"? Two geometric figures are congruent if you can transform one into the other using only rigid transformations. You might be wondering now, what are rigid transformations? A transformation is *Rigid* if it doesn't change the distances between the points or the measure of the angles between the lines they form. The following are all rigid transformations:

- Translations
- Rotations
- Reflections

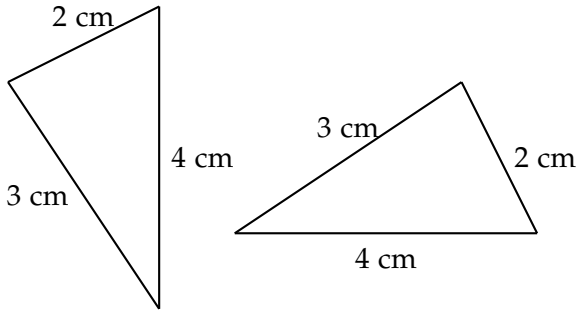
Once again, imagine cutting out one figure with scissors and trying to match it with the second figure; your actions are rigid transformations:

- Translations - Sliding the cutout left and right and up and down
- Rotations - Rotating the cutout clockwise and counterclockwise
- Reflection - Flipping the piece of paper over

A transformation is rigid if it is some combination of translations, rotations, or reflections.

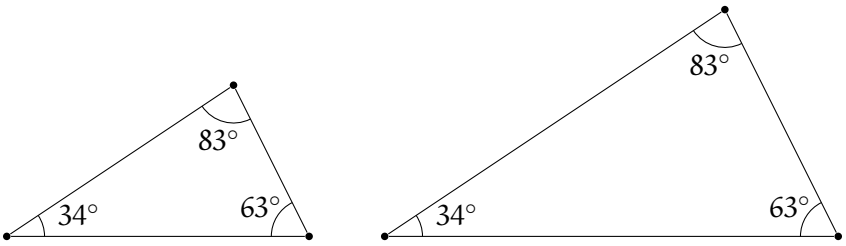
## 30.2 Triangle Congruency

If the sides of two triangles have the same length, the triangles must be congruent:




To be precise, the Side-Side-Side Congruency Test says that two triangles are congruent if three sides in one triangle are the same length as the corresponding sides in the other. We usually refer to this as the SSS test.

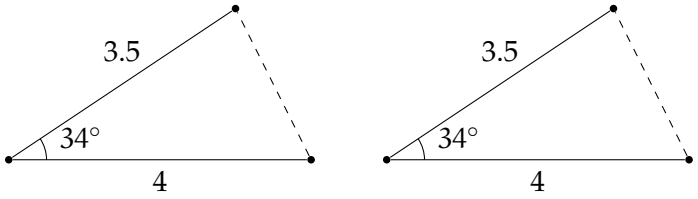
Note that two triangles with all three angles equal are not necessarily congruent. For example, here are two triangles with the same interior angles, but they are different sizes:




These triangles are not congruent, but they are *similar*. Meaning they have the same shape, but are not necessarily the same size.

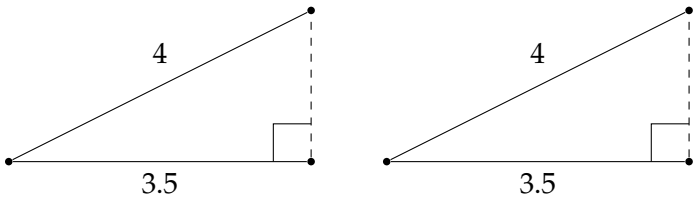
Therefore, if you know two angles of a triangle, you can calculate the third. This is why it makes sense to say "If two triangles have two angles that are equal, they are similar triangles." And if two similar triangles have one side that is equal in length, they must be the same size — so they are congruent. Thus, the Side-Angle-Angle Congruency Test says that two triangles are congruent if two angles and one side match.

What if you know that two triangles have two sides that are the same length, and that the angle between them is also equal?



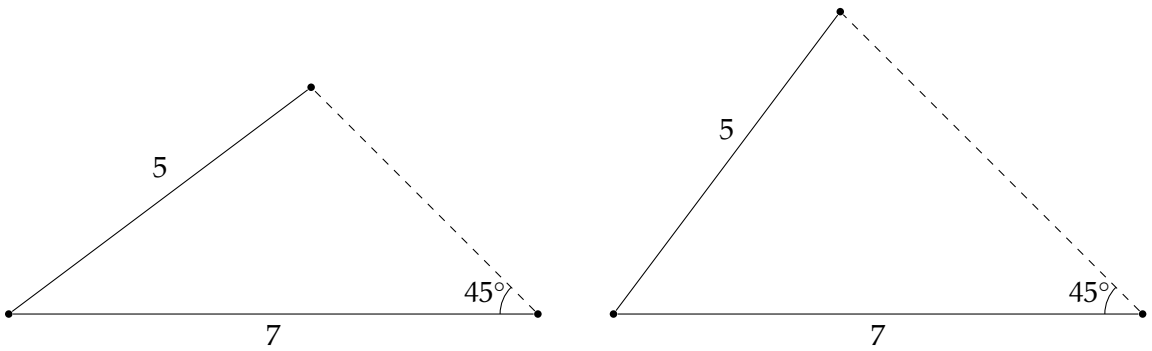

Yes, they must be congruent. This is the Side-Angle-Side Congruency Test.

What if the angle isn't the one between the two known sides? If it is a right angle, you can be certain the two triangles are congruent. (How do we know? Because the Pythagorean Theorem tells us that we can calculate the length of the third side. There is only one possibility, so all three sides must be the same length.)

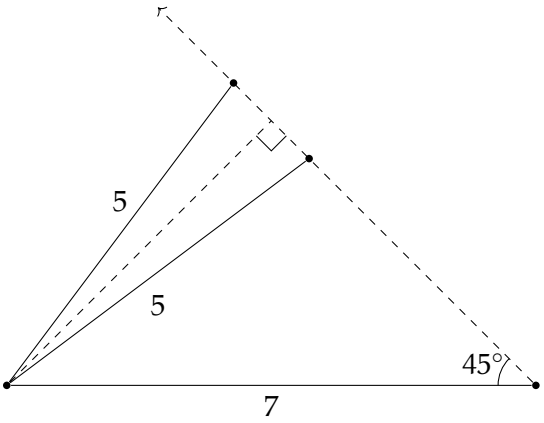



In this case, the third side of each triangle must be $\sqrt{4^2 - 3.5^2} \approx 1.9$.

What if the know angle is less than 90°? *The triangles are not necessarily congruent.* For example, let's say that there are two triangles with sides of length 5 and 7 and that the corresponding angle (at the end of the side of length 7) on each is 45°. Two different triangles satisfy this:




Let's look at this another way by laying one triangle on top of the other:




This is why there is *not* a general Side-Side-Angle Congruency Test.

Here, then, is the list of common congruency tests:

- Side-Side-Side: All three sides have the same measure
- Side-Angle-Angle: Two angles and one side have the same measure
- Side-Angle-Side: Two sides and the angle between them have the same measure
- Side-Side-Right: They are right triangles and have two sides the same measure

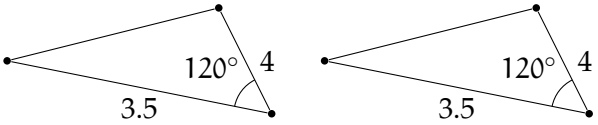
Once proven congruent, we can use the Congruent Parts of Congruent Triangles Theorem, or CPCTC. This concludes that their corresponding parts are also congruent. All three corresponding angles and all three corresponding sides must be the same (defined by congruence).

FIXME diagram here

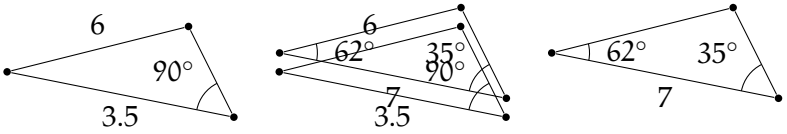
## *Exercise 77* Congruent Triangles

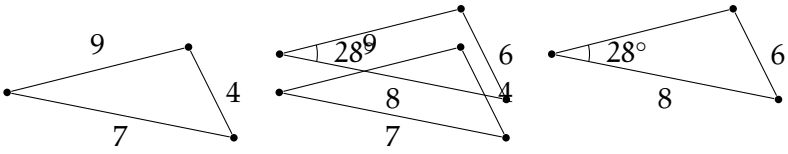
*Working Space*

Ted is terrible at drawing triangles; he always draws them exactly the same. Fortunately, he has marked these diagrams with the sides and angles that he measured. For each pair of triangles, write whether you know them to be congruent and which congruency test proves it. For example:




(These drawings are clearly not accurate, but you are told the measurements are.) The answer is "Congruent by the Side-Angle-Side test."



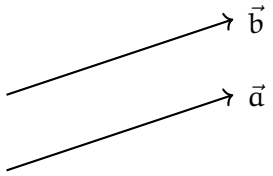

*Answer on Page 1897*

Chapter 31

# Parallel and Perpendicular

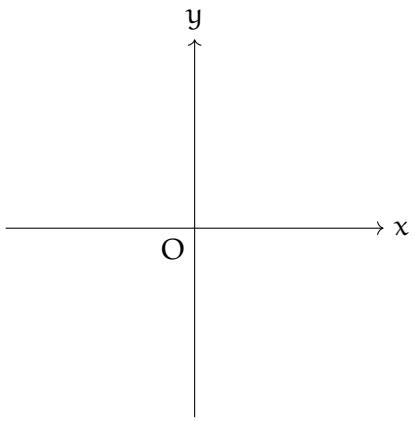
A vector is a line or ray with defined length (referred to as magnitude) and direction (given in degrees or radians). Velocity, force, and displacement are all examples of quantities that can be represented as vectors. Vectors are commonly drawn as arrows, where the length of the arrow corresponds to the magnitude and the direction of the arrow shows the direction of the vector.

Understanding how vectors relate to each other—such as being parallel or perpendicular—is fundamental in mathematics and physics. Two vectors are said to be parallel if they have the same or opposite direction. In simpler terms, if two vectors are pointing in the same direction (even if their magnitudes differ), they are considered parallel. For example, imagine you have a vector representing the direction and speed of a car moving north. If you have another vector representing the direction and speed of a different car also moving north, these vectors are parallel. We have already seen parallel lines when talking about parallel circuits, meaning they offer multiple paths of flow but don't intersect.




On the other hand, if two vectors point are in completely opposite directions, they are still considered parallel. For example, if one vector represents a car moving north and the other represents a car moving south, these vectors are parallel, but in opposite directions.

Perpendicular vectors, as the name suggests, are vectors that intersect each other at a right angle, forming a 90-degree angle. If we imagine a sheet of paper, drawing a horizontal vector and a vertical vector on that paper would create perpendicular vectors. In this case, the horizontal vector represents left-right direction, while the vertical vector represents up-down direction. Perpendicular vectors are often seen in geometric shapes, such as squares and rectangles, where their sides intersect at right angles. The coordinate planes are also perpendicular.




A fundamental property of perpendicular vectors is that their dot product is zero. The dot product is a mathematical operation that measures the extent to which two vectors align with each other. When two vectors are perpendicular, their dot product is always zero. This property provides a useful tool for determining whether two given vectors are perpendicular.

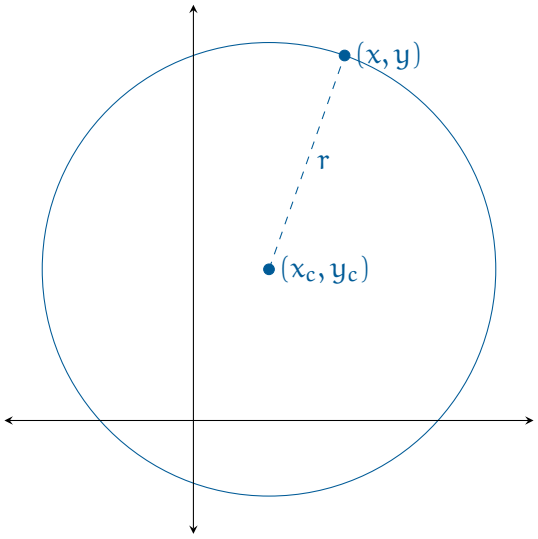
Understanding parallel and perpendicular vectors is essential in various areas of mathematics and physics. For example, in geometry, knowledge of perpendicular vectors helps us determine whether lines are perpendicular or parallel. In physics, vectors can represent forces, velocities, or displacements, and identifying parallel or perpendicular vectors aids in analyzing motion and forces acting on objects.

In summary, parallel vectors have the same or opposite direction, while perpendicular vectors intersect at a right angle. Recognizing these relationships between vectors enables us to solve problems involving geometry, physics, and many other fields. As you delve deeper into the exciting world of vectors, keep an eye out for parallel and perpendicular relationships, as they often hold valuable insights and solutions. We are going to get into graphing these two lines and their equations in the next chapter!

CHAPTER 32

# Circles

A circle is the set of points $(x, y)$ that are a particular distance $r$ from a particular point $(x_c, y_c)$. We say that $r$ is the *radius* and $(x_c, y_c)$ is the *center*.




**Area and Radius**

If the radius of a circle is $r$, the area of its interior $(a)$ is given by

$$a = \pi r^2$$

## *Exercise 78* Area of a Circle

*Working Space*

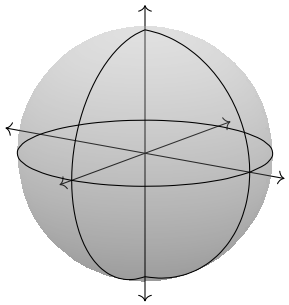
The paint you have says "One liter covers 6 square meters."

You are painting the top of a circular table with a radius of 3 meters.

How much paint will you need?

*Answer on Page 1897*

Note that a circle lives in a particular plane. In 3D, the points $(x, y, z)$ that are a particular distance $r$ from a particular point $(x_c, y_c, z_c)$ are a sphere:



The distance all the way across the middle of a circle (or a sphere) is its *diameter*. The diameter is always twice the radius.

For the rest of the chapter, we will be talking about circles, points, and lines *in a plane*.

**Circumference and Diameter**

The circumference $(c)$ of a circle is the distance around the circle. If the diameter is $d$,

$$c = \pi d$$

### *Exercise 79* Circumference

*Working Space*

Using a tape measure, you figure out that the circumference of a tree in your yard is 64 cm.

Assuming the trunk is basically circular, what is its diameter?

*Answer on Page 1898*

### *Exercise 80* Splitting a Pie

*Working Space*

A pie has a radius of 13 cm. 7 friends all want equal sized sectors. You have a tape measure to assist you.

How many centimeters will each outer crust be?

*Answer on Page 1898*

## 32.1 Arc Length

Previously, you learned that angles can be measured in degrees and radians. A circle is $360^o$ (see figure 32.1).

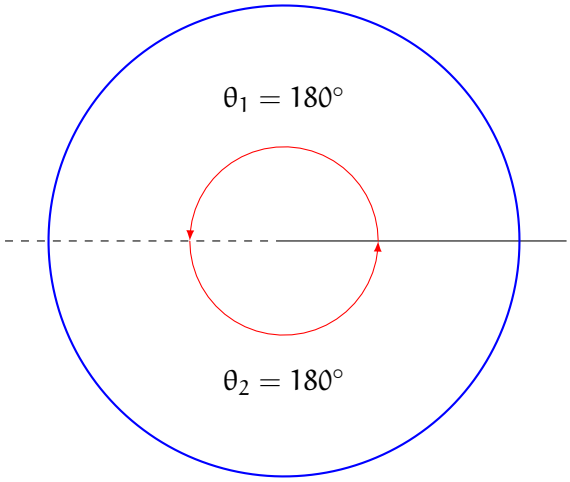



Figure 32.1: The total internal angle of a circle is $\theta_1 + \theta_2 = 360°$

This means a circle is also $2\pi$ radians:

$$360° \cdot \frac{\pi}{180°} = 2\pi$$

You may be wondering: why is it that there are $\pi$ radians in a 180° angle? A radian is defined such that one radian is the angle at the center of a circle which defines an arc of the circumference equal to the radius of the circle (see figure 32.2).

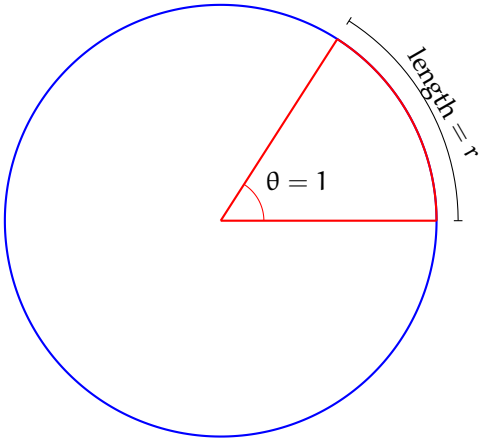
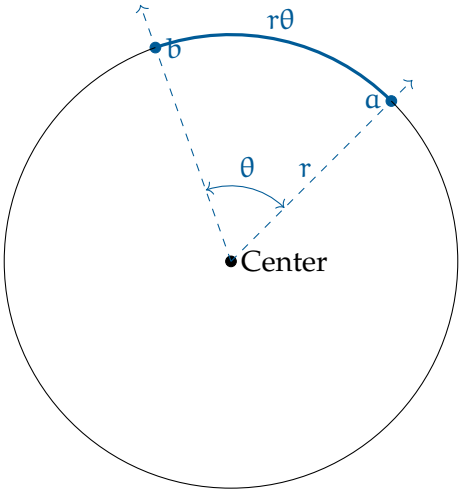



Figure 32.2: When the center angle is 1 radian, the length of the arc is equal to the radius of the circle

This makes it very straightforward to find the lengths of arcs if we know the center angle in radians. The arc length is just $r\theta$, where $\theta$ is the center angle in radians.

**Length of an Arc**

If you have two points $a$ and $b$ on a circle, the ray from the center through $a$ and the ray from the center through $b$ form an angle. If $\theta$ is the angle in radians and $r$ is the radius of the circle, the distance from $a$ to $b$ on the circle is $r\theta$.




This shows us why $\pi$ radians $= 180°$. Recall the formula for circumference: $c = \pi d$, where $d$ is the diameter of the circle. Since the diameter is twice the radius, we can also say that $c = 2\pi r$, where $r$ is the radius of the circle. The circumference of the circle is just an arc where the central angle is the entirety of the circle. Since we know that the length of an arc is $r\theta$, we can find the total internal angle of a circle in radians:

$$2\pi r = r\theta$$

$$\theta = 2\pi$$

This is how we know $360° = 2\pi$ radians.

## *Exercise 81* Angle of Rotation

A car tire has a radius of approximately 25 centimeters. If you roll your car forward 10 cm, by how many radians has your tire rotated?

*Working Space*

*Answer on Page 1898*

## *Exercise 82* Arc Length Ranking

Rank the following arc lengths from longest to shortest (the central angle that defines the arc and the radius of the circle are provided:

1. central angle of $\frac{\pi}{4}$ and a radius of 2 cm
2. central angle of $\pi$ and a radius of 1 cm
3. central angle of $\frac{\pi}{10}$ and a radius of 5 cm
4. central angle of $\frac{3\pi}{4}$ and a radius of 3 cm

*Working Space*

*Answer on Page 1898*

### *Exercise 83* Arc Length

*Working Space*

You have been asked to find the radius of a very large cylindrical tank. You have a tape measure, but it is only 15 meters long and doesn't reach all the way around the tank.

However, you have a compass. So you stick one end of the tape measure to the side of the tank and measure the orientation of the wall at that point. You then walk the 15 meters and measure the orientation of the wall there.

You find that 15 meters represents 72 degrees of arc.

What is the radius of the tank in meters?

*Answer on Page 1899*

## 32.2 Sector Area

We already know the area of a circle is given by $A = \pi r^2$. What about a piece of a circle? Let's start with a straightforward example:

**Example**: A pizza with a radius of 15 cm is divided into 6 equal pieces. What is the area of each piece?

**Solution**: First, we find the area of the entire pizza:

$$A = \pi r^2$$

$$A = \pi(18 \text{ cm})^2$$

$$A = 324\pi \text{ cm}^2 \approx 1018 \text{ cm}^2$$

Then, we divide by 6, since the pieces of equal sizes:

$$A_{\text{piece}} = \frac{A}{6} = \frac{324\pi \text{ cm}^2}{6} = 54\pi \text{ cm}^2$$

Let's use this to write a general formula for the area of a sector defined by a central angle $\theta$ (see figure 32.3). We know that when a circle is divided into 6 equal sectors, the central angle of each sector is $\theta = \frac{2\pi}{6} = \frac{\pi}{3}$. Additionally, we know the area of each sector is the total area divided by 6: $A_{\text{sector}} = \frac{\pi r^2}{6} = \frac{\pi}{6}r^2 = \frac{\theta}{2}r^2$.

**Area of a sector**

For a sector whose corner is at the center of a circle, the area is given by $A_{\text{sector}} = \frac{\theta}{2}r^2$, where $\theta$ is the central angle and $r$ is the radius.

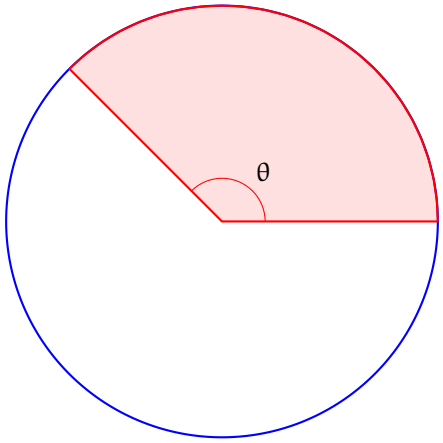



Figure 32.3: The area of a sector with central angle $\theta$ is $\frac{\theta}{2}r^2$

## *Exercise 84* Area of a sector

You are tasked with painting a large, circular logo on the side of a building. If a liter of paint covers 6 square meters and the logo is 5 meters wide, how many liters of red paint will you need to paint a sector whose central angle is $\frac{3\pi}{4}$ radians?

*Working Space*

*Answer on Page 1899*

## 32.3 Secants and Tangents

A line that is *tangent* to a circle touches it at exactly one point, as shown in figure 32.4.

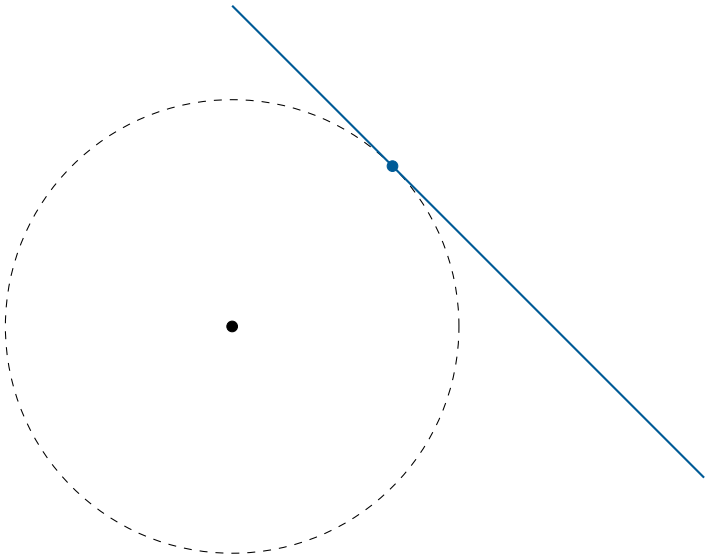


Figure 32.4: A line that is *tangent* to the circle.

The tangent line is always perpendicular to the radius to the point of tangency, as in figure 32.5.

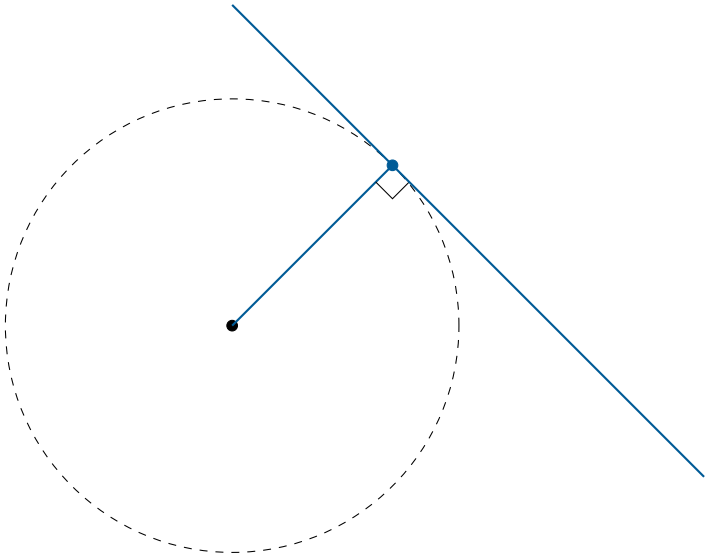


Figure 32.5: The radius and tangent line are perpendicular.

A secant line intersects a circle at *two points*, as shown in figure 32.6.

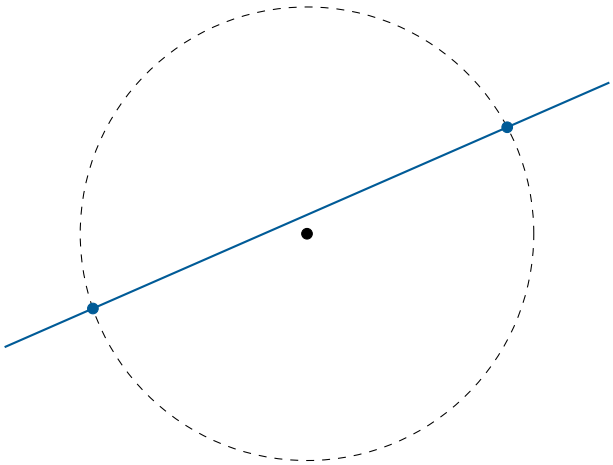


Figure 32.6: A line that is *secant* to the circle.

Notice that the secant line crosses the circle at two points, similar to how a chord intersects a circle. Every chord lies on a secant line but not every secant line contains a chord. These definitions are important, because secants and tangents will come up in calculus chapters in the future, so be able to recognize them. In calculus, we will use tangent lines to approximate the behavior of curves at instantaneous points, and secant lines to compute average rates of change.

## *Exercise 85* Painting a Comet

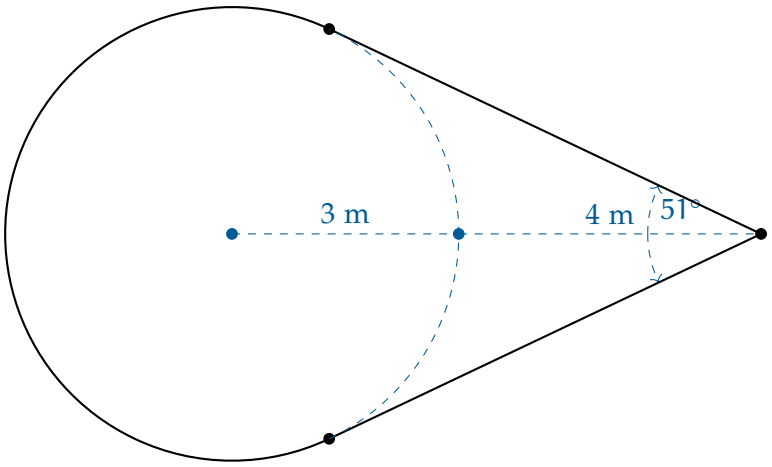
*Working Space*

You have been asked to paint a comet and its tail in yellow on the floor of a gymnasium.

A liter of yellow paint covers 6 square meters.

First you draw a circle with a radius of 3 meters. You then mark a point D on the floor 7 meters from the center of the circle. Then you draw two tangent lines that pass through D.

You use a protractor to measure the angle at which the tangent lines meet: about 51°




Before you paint the area contained by the circle and the two tangent lines, how much paint will you need?

*Answer on Page 1900*

Chapter 33

# Functions and Their Graphs

Functions are a major part of science, engineering, and math. You can think of a function as a machine: you put something into the machine, it processes it, and out comes something else: a product. Just as we often use the variable $x$ to stand in for a number, we often use the variable $f$ to stand in for a function. $f(x)$ is said as "f of x". However, this could be any letter or word that takes an input $x$, maybe even $\text{zebra}(x)$!

For example, we might ask, "Let the function $f$ be defined like this:

$$f(x) = -5x^2 + 12x + 2$$

What is the value of $f(3)$, said as "f of 3"?

You would run the number 3 through "the machine": $-5(3^2)+12(3)+2 = -7$. The answer would be "$f(3)$ is 7".

However, some functions are not defined for every possible input. For example:

$$f(x) = \frac{1}{x}$$

This is defined for any $x$ except 0, because you can't divide 1 by 0. The set of values that a function can process is called its *domain*, and resulting output values are called the range. It is important to note that functions have only one *output* for each *input*. However, multiple inputs can have the same output. A relationship where one input can result in the more than one output is not a function, but a relation. This can be proven by the vertical line test.

## *Exercise 86* Domain of a function

*Working Space*

Let the function $f$ be given by $f(x) = \sqrt{x-3}$. What is its domain?

*Answer on Page 1901*

## 33.1 Graphs of Functions

If you have a function, $f$, its graph is the set of pairs $(x, y)$ such that $y = f(x)$. We usually draw a picture of this set, called a *graph*. The graph not only includes the picture, but also the values of $x$ and $y$ used to create it.

Here is the graph of the function $f(x) = -5x^2 + 12x + 2$:

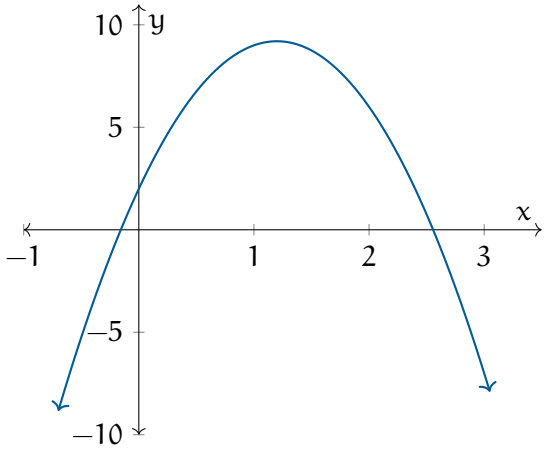



Figure 33.1: The function $f(x) = -5x^2 + 12x + 2$ *graphed* on an $xy$-plane from $x = -1$ to $x = 3$. The inputs $x$ are graphed along the $x$-axis, and the output of the function $f(x)$ is plotted along the $y$-axis.

(Note that this is just part of the graph; it goes infinitely in both directions. Remember your vectors!)

Here is the graph of the function $f(x) = \frac{1}{x}$:

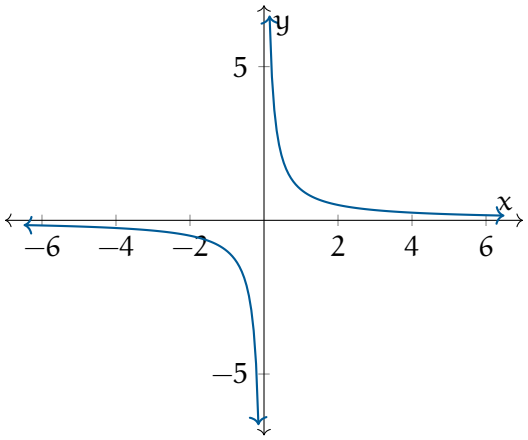



Figure 33.2: The function $f(x) = \frac{1}{x}$ graphed on a coordinate plane.

Notice the *asymptotes* at $y = 0$ and $x = 0$. Here, the function is undefined at $x = 0$ and $y = 0$, but is continually approaching it. It never actually reaches these values though! $f(0)$ is not defined as you cannot divide by zero (recall division rules) and as $x$ is increasing, $f(x)$ gets infinitesimally small but never equal to zero. We will talk more about these in the rational functions chapter.

To draw a graph, take each x value (usually in increments of 1) and determine its corresponding y value. Then, plot those points on a graph, using the origin as the starting point.

### *Exercise 87* Draw a graph

*Working Space*

Let the function $f$ be given by $f(x) = -3x + 3$. Sketch its graph.

*Answer on Page 1901*

## 33.2 Can this be expressed as a function?

Note that not all sets can be expressed as graphs of functions. For example, here is the set of points $(x, y)$ such that $x^2 + y^2 = 9$:

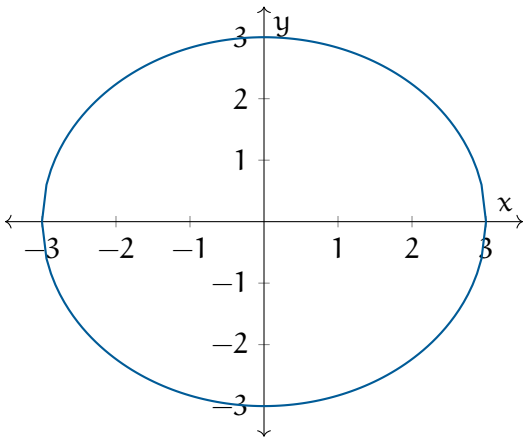



Figure 33.4: The graph of $x^2 + y^2 = 9$.

This cannot be the graph of a function, because what would $f(0)$ be? 3 or -3? This set fails what we call "the vertical line test": If any vertical line contains more than one point from the set, it isn't the graph of a function. For example, the vertical line $x = 2$ would cross the graph twice:

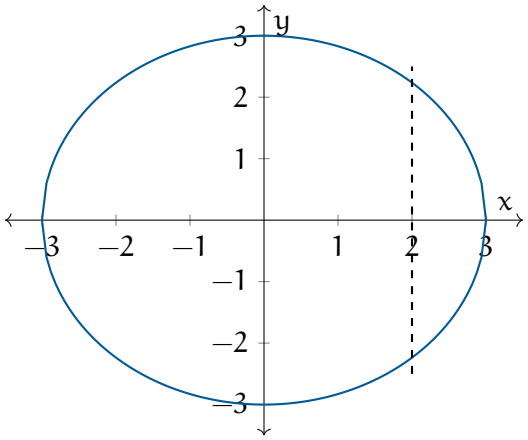



Figure 33.5: Vertical line test on $x^2 + y^2 = 9$.

## 33.3 Inverses

Some functions have inverse functions. If a function $f$ is a machine that turns number $x$ into $y$, the inverse (usually denoted $f^{-1}$) is the machine that turns $y$ back into $x$.

For example, let $f(x) = 5x + 1$. Its inverse is $f^{-1}(x) = (x - 1)/5$. (Spot check it: $f(3) = 16$ and $f^{-1}(16) = 3$)

Does the function $f(x) = x^3$ have an inverse? Yes, $f^{-1}(x) = \sqrt[3]{x}$. Let's plot the function (solid line) and its inverse (dashed):

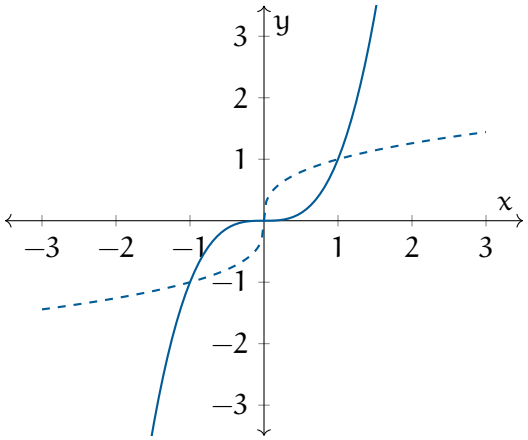



Figure 33.6: The function $f(x) = x^3$ and it's inverse.

The inverse is the same as the function, just with its axes swapped. The inverse function is *reflected* across the line $y = x$. This tells us how to solve for an inverse: We swap $x$ and $y$ and solve for $y$.

For example, if you are given the function $f(x) = 5x + 1$, its graph is all $(x, y)$ such that $y = 5x + 1$. The graph of its inverse is all $(x, y)$, such that $x = 5y + 1$. Solving for $y$ gives you $y = (x - 1)/5$. So we can say that $f^{-1}(x) = (x - 1)/5$, QED.

Not every function has an inverse. For example, $f(x) = x^2$. Note that $f(2) = f(-2) = 4$. What would $f^{-1}(4)$ be? 2 or -2? This implies the "horizontal line test": If any horizontal line contains more than one point of a function's graph, that function has no inverse. If a function passes the horizontal line test, it is called "one-to-one", meaning there is exactly one $x$ that gives each $y$.

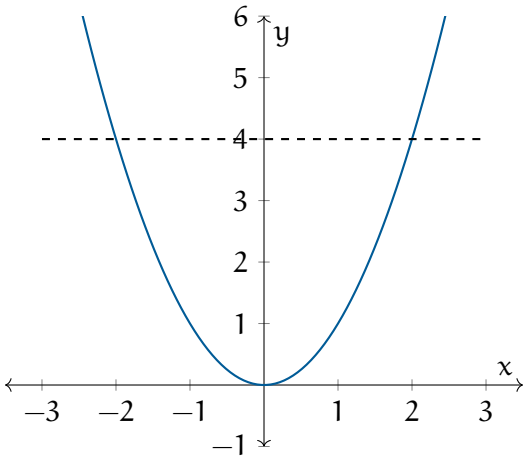



Figure 33.7: The horizontal line test run on $f(x) = x^2$.

In some problems, you need an inverse, but you don't need the whole domain, so you trim the domain to a set you can define an inverse on. This allow you to make claims such as "If we restrict the domain to the nonnegative numbers, the function $f(x) = x^2 - 5$ has an inverse: $f^{-1}(x) = \sqrt{x + 5}$.

This raises the question: What is the domain of the inverse function $f^{-1}$?

If we let $X$ be the domain of $f$, we can run every member of $X$ through "the machine" and gather them in a set on the other side. This set would be the *image* of the $f$ "machine". (This is the *range* of $f$.)

What is the image of $f(x) = x^2 - 5$? It is the set of all real numbers greater than or equal to -5. We write this:

$$\{x \in \mathbb{R} | x \geq -5\}$$

Now we can say: **The range (or image) of the function is the domain of the inverse**

**function.**

In inverse functions, the domain and range get swapped: the domain of the function is the range of the inverse function, and visa versa. In our example, we can use any number greater than or equal to -5 as input into the inverse function.

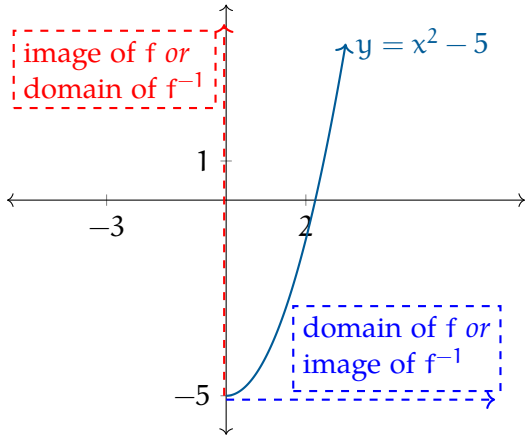



Figure 33.8: Domain and range of functions and their inverses get swapped.

## *Exercise 88* Find the inverse

*Working Space*

Let $f(x) = (x-3)^2 + 2$. Sketch the graph.

Using all the real numbers as a domain, does this function have an inverse?

How would you restrict the domain to make the function invertible?

What is the inverse of that restricted function?

What is the domain of the inverse?

*Answer on Page 1901*

### *Exercise 89*

A function is given by a table of values, a graph, or a written description. Determine whether it is one-to-one.

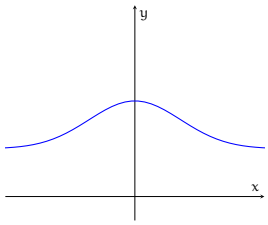
*Working Space*

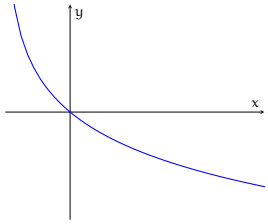
1.

| x | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| f(x) | 1.5 | 2.0 | 3.6 | 5.3 | 2.8 | 2.0 |

2.

| x | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| f(x) | 1.0 | 1.9 | 2.8 | 3.5 | 3.1 | 2.9 |

3. 


4. 


5. $f(t)$ is the height of a football $t$ seconds after kickoff

6. $v(t)$ is the velocity of a dropped object

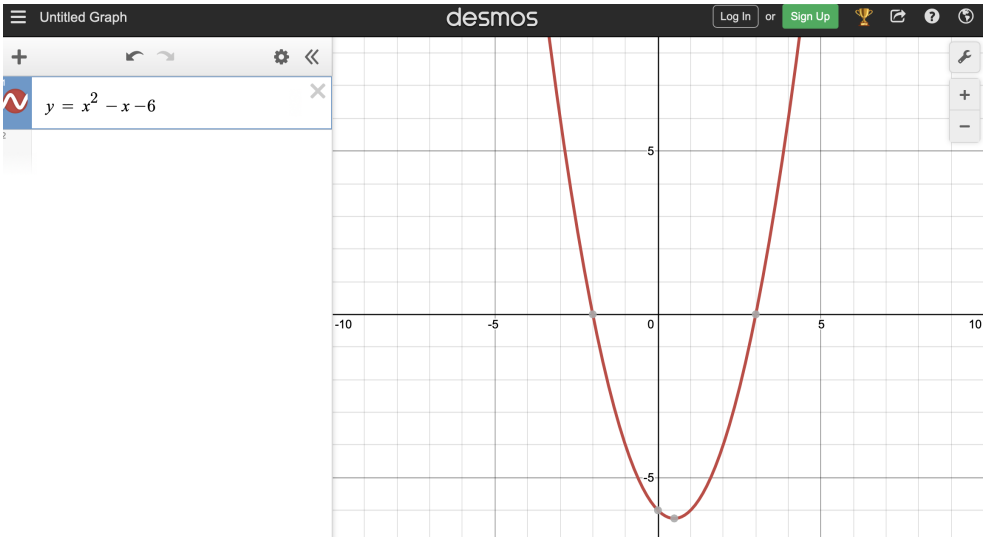
*Answer on Page 1902*

## 33.4 Graphing Calculators

One really easy way to understand your function better is to use a graphing calculator. Desmos is a great, free online graphing calculator.

In a web browser, go to Desmos: `https://www.desmos.com/calculator`

In the field on the left, enter the function $y = x^2 - x - 6$. (For the exponent, just prefix it with a caret symbol: "x^2".)




### Computing in Python

Python can draw the same graph. The listing below uses two libraries: NumPy, which builds lists of numbers called *arrays*, and Matplotlib, which plots them.

```
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return x**2 - x - 6

x = np.linspace(-5, 6, 200)
y = f(x)

plt.plot(x, y)
plt.axhline(0, color="black", linewidth=0.8)
plt.axvline(0, color="black", linewidth=0.8)
plt.grid(True)
plt.xlabel("x")
plt.ylabel("y")
plt.title("y = x^2 - x - 6")
plt.show()
```

The `def` line packages the rule $x^2 - x - 6$ into a Python function named `f`. Calling `f(3)` runs 3 through the machine, the same way $f(3)$ does on paper. The call `np.linspace(-5, 6, 200)` creates 200 evenly spaced $x$ values from $-5$ to 6, and passing that whole array to `f` returns the 200 matching $y$ values at once. Matplotlib then draws a point at each $(x, y)$ pair and joins neighboring points with short line segments. With 200 points, each

segment is short enough that the curve looks smooth. The calls `axhline` and `axvline` draw the x-axis and y-axis, and `plt.show()` opens the plot window.

## *Exercise 90* Reading values from f

Add this line after the plot code:

*Working Space*

```
print(f(0), f(3), f(-2))
```

What does it print? What do the three outputs tell you about the graph?

*Answer on Page 1903*

Desmos handles the asymptote of $\frac{1}{x}$ on its own. Matplotlib joins every pair of neighboring points, so the two branches of $\frac{1}{x}$ need separate arrays, one on each side of $x = 0$:

```
x_left = np.linspace(-6, -0.15, 100)
x_right = np.linspace(0.15, 6, 100)

plt.plot(x_left, 1 / x_left, color="C0")
plt.plot(x_right, 1 / x_right, color="C0")
plt.axhline(0, color="black", linewidth=0.8)
plt.axvline(0, color="black", linewidth=0.8)
plt.ylim(-7, 7)
plt.grid(True)
plt.title("y = 1/x")
plt.show()
```

Both calls use `color="C0"`, Matplotlib's default blue, so the two branches read as one function. The call `plt.ylim(-7, 7)` fixes the visible y range to match Figure 33.2.

## *Exercise 91* The phantom line

A student plots $\frac{1}{x}$ with a single array:

```
x = np.linspace(-6, 6, 100)
plt.plot(x, 1 / x)
plt.show()
```

The graph shows a vertical line running up the y-axis. The function has no value at $x = 0$. Where does the line come from?

*Working Space*

*Answer on Page 1903*

The inverse figure from earlier in this chapter takes three plot calls: the function, its inverse, and the line $y = x$ that reflects one onto the other.

```
def g(x):
    return x**3

def g_inv(x):
    return np.cbrt(x)

x = np.linspace(-3, 3, 200)

plt.plot(x, g(x), label="g(x) = x^3")
plt.plot(x, g_inv(x), linestyle="--", label="inverse")
plt.plot(x, x, color="gray", linewidth=0.8, label="y = x")
plt.xlim(-3.5, 3.5)
plt.ylim(-3.5, 3.5)
plt.gca().set_aspect("equal")
plt.grid(True)
plt.legend()
plt.show()
```

The function `np.cbrt` returns the cube root of every entry in an array. Setting the aspect to `"equal"` gives one unit the same length on both axes. Without it, the window stretches one axis and the reflection across $y = x$ looks lopsided. The `label` strings feed `plt.legend()`, which prints the key in the corner.

### *Exercise 92* The missing half

Replace `np.cbrt(x)` with `x**(1/3)` and run the code again. The dashed curve now appears only to the right of the y-axis, and Python prints the warning `invalid value encountered in power`. Explain what happened.

*Working Space*

*Answer on Page 1903*

## 33.5 Even and Odd Functions

### 33.5.1 Even Functions

An even function is symmetric about the y-axis. That means if you fold the graph along the y-axis, both sides will match perfectly. Note that the input value yields the same output regardless of whether the input value is positive or negative.

**Even Functions**

A function $f(x)$ is even if

$$f(-x) = f(x)$$

for all $x$ in its domain.

Examples:

- $f(x) = x^2$
- $f(x) = \cos(x)$
- any $f(x) = x^n$ where $n$ is an even number
- $f(x) = |x|$

On a graph, a function will be a mirror image across the y-axis.

### 33.5.2 Odd Functions

An odd function is symmetric about the origin. That means if you rotate the graph 180° about the origin, it lands on itself. Algebraically, when you input a value $k$, you get some output $n$; if you negate that input as $-k$, the output is also negated resulting in $-n$.

**Odd Functions**

A function $f(x)$ is odd if

$$f(-x) = -f(x)$$

for all $x$ in its domain.

Equivalently,

$$f(x) + f(-x) = 0$$

for all $x$ in its domain.

- $f(x) = x^3$
- $f(x) = \sin(x)$
- $f(x) = \tan(x)$
- any $f(x) = x^n$ where $n$ is an odd number

On a graph, rotating the graph 180° about the origin will yield the same graph.

### 33.5.3 Neither Even Nor Odd

Some functions are neither even nor odd.

For example, $f(x) = x^3 + 1$ does not satisfy either condition. **Computing in Python**

A plot can test for symmetry. Draw $h(x)$, $h(-x)$, and $-h(-x)$ on one set of axes. When the $h(-x)$ curve lies on top of $h(x)$, the function is even. A match between $-h(-x)$ and $h(x)$ means the function is odd.

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  def h(x):
5      return x**3 + 1
6
```

```
x = np.linspace(-2, 2, 200)

plt.plot(x, h(x), label="h(x)")
plt.plot(x, h(-x), linestyle="--", label="h(-x)")
plt.plot(x, -h(-x), linestyle=":", label="-h(-x)")
plt.grid(True)
plt.legend()
plt.show()
```

Writing `h(-x)` negates every entry of the array before it reaches `h`, and the extra minus sign in `-h(-x)` negates every output afterward. For $h(x) = x^3 + 1$, the dashed and dotted curves both separate from the solid one, so $x^3 + 1$ is neither even nor odd.

The function `np.allclose(a, b)` turns the visual check into a yes or no answer. It returns `True` when two arrays agree at every entry, allowing for tiny rounding errors:

```
print(np.allclose(h(-x), h(x)))   # even test
print(np.allclose(-h(-x), h(x)))  # odd test
```

Both lines print `False` for $h(x) = x^3 + 1$.

## *Exercise 93* One input is enough

Predict the output, then run it:

```
print(h(1), h(-1), -h(-1))
```

How do these three numbers show that `h` is neither even nor odd?

*Working Space*

*Answer on Page 1903*

### *Exercise 94* Swap the function

Change the body of `h` to `return np.abs(x)` and run the plot and both `allclose` tests. Then change it to `return x**3` and run everything again. For each version, which curve lands on top of the solid one, and what do the tests print?

*Working Space*

*Answer on Page 1903*

## 33.6 Piecewise Functions

Another type of function that will be integral to your mathematics career are *piecewise functions*. A piecewise function is one defined by two different equations. You chose which function to evaluate dependent on the domain. Let's look at a simple example:

$$f(x) = \begin{cases} \frac{1}{x} & x < 0 \\ x^2 & x \geq 0 \end{cases}$$

This piecewise function has two cases: when $x < 0$, we use the equation $\frac{1}{x}$; when $x \geq 0$, we use $x^2$. Notice that we don't consider $x = 0$ for $\frac{1}{x}$ since it would be undefined, but it is defined under $x^2$. The domain of $f(x)$ is defined for all values of $x$! Let's make a table of values:

| $x$ | $-2$ | $-1$ | $0$ | $0.1$ | $1$ | $2$ |
|---|---|---|---|---|---|---|
| $f(x)$ | $-\frac{1}{2}$ | $-1$ | $0$ | $0.01$ | $1$ | $4$ |

We can graph this like the following

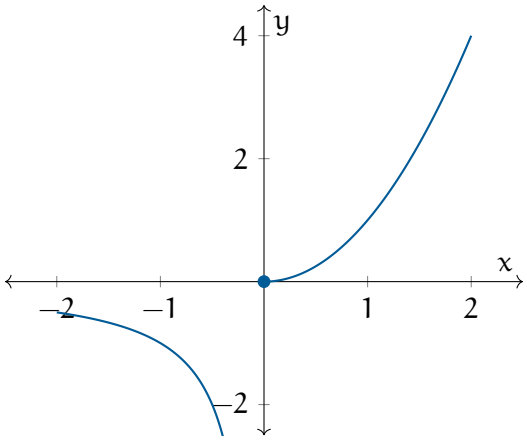



Figure 33.10: The piecewise function $f(x)$ graphed.

Notice the *closed circle*. This indicates that the value at the is included at a disjunction. An *open circle* would indicate that the value at that point is not included. Say we have the following function:

$$g(x) = \begin{cases} x^2 + 1 & x < 0 \\ x^2 & x \geq 0 \end{cases}$$

Graphing this, we obtain an open circle at $(0, 1)$ and a closed circle at $(0, 0)$.

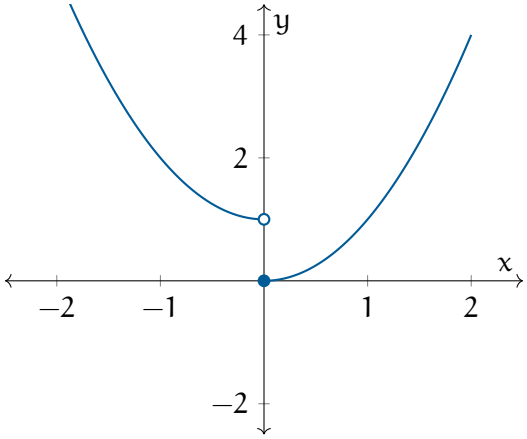



Figure 33.11: The piecewise function $g(x)$ graphed.

## *Exercise 95* Absolute Value

*Working Space*

Rewrite the absolute value function $|x|$ as a piecewise function.

*Answer on Page 1904*

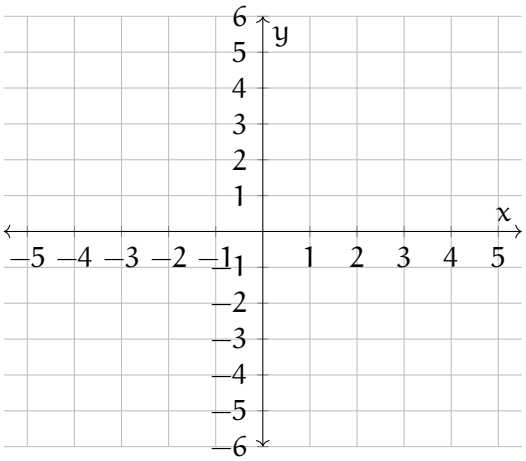
## *Exercise 96* Piecewise 1

*Working Space*

$$f(x) = \begin{cases} x+2 & \text{if } -5 \le x < -1 \\ x^2 & \text{if } -1 \le x \le 2 \\ 3-x & \text{if } 2 < x \le 5 \end{cases}$$

For the equation above, create a table for every integer from -5 to 5 (inclusive), and graph.

Here is a blank coordinate plane for plotting the piecewise function:




*Answer on Page 1904*

### 33.6.1 Floor and Ceiling Functions

Another common piecewise function you may see are the *floor function* and *ceiling function*. The floor function, which takes the any real number $x$, is defined as

$$\lfloor x \rfloor \text{ or } \text{floor}(x) = \begin{cases} n & \text{if } n \le x < n+1, \quad n \in \mathbb{Z} \end{cases}$$

This means that you take any number $x$ and find the closest integer less than or equal to it. For example, $\lfloor 0.7 \rfloor$ would be floored to 0.

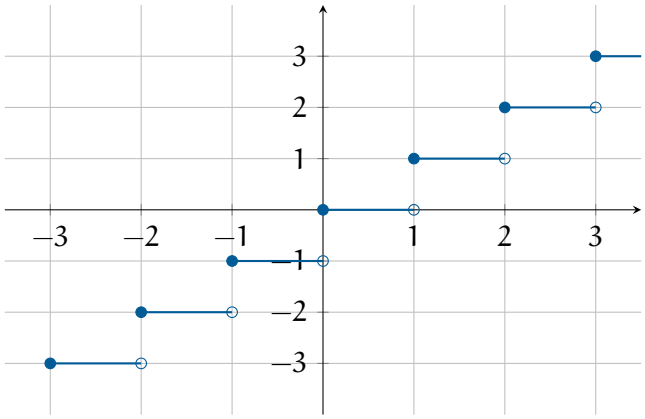



Figure 33.12: The flooring function, $f(x) = \lfloor x \rfloor$sdkblue,

Similarly, the ceiling function is defined as follows:

$$\lceil x \rceil \text{ or } \operatorname{ceil}(x) = \begin{cases} n & \text{if } n-1 < x \leq n, \quad n \in \mathbb{Z} \end{cases}$$

The ceiling function takes in any real numbers and returns the next greatest integer. For example, $\lceil 8.00001 \rceil$ returns 9, the next greatest integer. For any valid integer like $x = 7$, the $\lfloor 7 \rfloor$ and $\lceil 7 \rceil$ are both equal to 7.

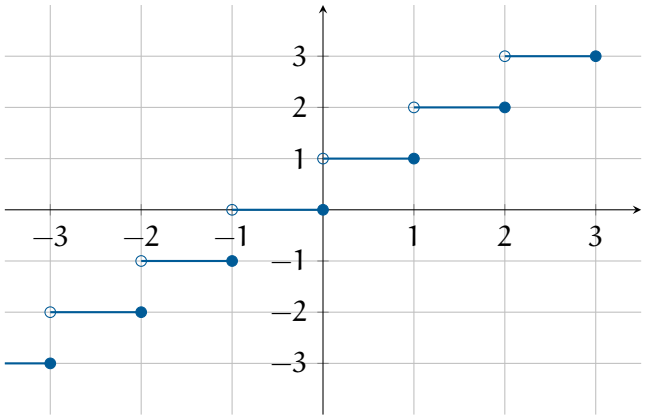



Figure 33.13: The ceiling function, $f(x) = \lceil x \rceil$.

## *Exercise 97* More TVs is Better

*Working Space*

A store sells televisions with pricing that depends on how many units a customer buys:

- For the first 5 TVs, the cost per TV is \$500
- For the next 10 TVs, the cost per TV is \$350
- Any additional TVs cost \$300

A hotel is buying TVs for each hotel room.

1. Construct a piecewise function $C(x)$ for the cost of TVs. Assume TVs must be a discrete integer (you cannot buy 8.5 TVs).
2. Find the cost of 12 TVs using your function.
3. When is average cost $<$ \$360?

*Answer on Page 1904*

### 33.6.2 Invalid Piecewise Functions

One thing to note when creating piecewise functions is to make sure your cases do not contradict each other, or in other words, create cases that define two different possibilities for $x$ at a given value. For example,

$$f(x) = \begin{cases} 10.2 & x = 4 \\ x^2 & x \geq 0 \end{cases}$$

This creates an impossible and thus invalid piecewise function, as at $x = 4$, $f(x)$ is both

defined as 10.2 and 16. This would be impossible ($\rightarrow\leftarrow$). In this case, either you define the value at DNE or undefined, or revise the piecewise function at that point. Most often, it is done as follows:

$$f(x) = \begin{cases} 10.2 & x = 4 \\ x^2 & x \geq 0,\ x \neq 4 \end{cases}$$

Then our value at $x = 4$ is equal to 10.2.

**Computing in Python**

Matplotlib draws a piecewise graph one piece at a time. Here is $g(x)$ from earlier in this section, with its open and closed circles:

```
import math
import numpy as np
import matplotlib.pyplot as plt

x_left = np.linspace(-2, 0, 100)
x_right = np.linspace(0, 2, 100)

plt.plot(x_left, x_left**2 + 1, color="C0")
plt.plot(x_right, x_right**2, color="C0")
plt.plot(0, 0, "o", color="C0")                              # closed circle
plt.plot(0, 1, "o", color="C0", markerfacecolor="white")  # open circle
plt.grid(True)
plt.title("piecewise g(x)")
plt.show()
```

Each piece gets its own array covering only its part of the domain. A single array across the whole domain would join the two pieces with a vertical segment at $x = 0$, the same phantom line that appears with $\frac{1}{x}$. The format string `"o"` draws one round marker, and `markerfacecolor="white"` hollows it out into an open circle.

Plotting shows the shape. Evaluating the function at a particular input takes `if` and `elif`, with one condition for each row of the cases. This is the function from Exercise 96:

```
def p(x):
    if -5 <= x < -1:
        return x + 2
    elif -1 <= x <= 2:
        return x**2
    elif 2 < x <= 5:
        return 3 - x
    return None

for x in range(-5, 6):
    print(x, p(x))
```

Python accepts chained comparisons such as `-5 <= x < -1`, written exactly in math notation. It checks the conditions from top to bottom and returns from the first one that is true. An input outside every case, such as `p(6)`, reaches the last line and returns `None`, Python's value for "nothing." That matches the function being undefined there. The call `range(-5, 6)` produces the integers from −5 up to 5; the stop value 6 is left out.

## *Exercise 98* Checking the table

Run the loop above. Compare its output with the table in the answer to Exercise 96. Which case does Python use at $x = -1$ and at $x = 2$?

*Working Space*

*Answer on Page 1905*

## *Exercise 99* The case that never runs

A student codes the revised function from the end of this section:

*Working Space*

```
def q(x):
    if x >= 0:
        return x**2
    elif x == 4:
        return 10.2
```

The call `q(4)` returns `16` instead of `10.2`. Find the error and fix it.

*Answer on Page 1905*

### *Exercise 100* Floor and ceiling in Python

Python's `math` module provides `math.floor` and `math.ceil`. Predict the output:

*Working Space*

```
print(math.floor(-2.5),
↪  math.ceil(-2.5),
↪  math.ceil(8.00001),
↪  math.floor(7))
```

*Answer on Page 1906*

## 33.7 Summary

This chapter talked about functions. Functions take in an input and spit out an output, like a magician putting a banana in a hat and releasing a white bunny. However, this is not magic, but math. Each input is given exactly one output. Not all functions have equations that are defined for all real numbers.

We can also graph functions according to their inputs (the $x$ values on the x-axis) and outputs (the returns of the function $f(x)$ plotted on the $y$ axis).

We established that relations must pass the vertical line test to be functions. Some functions have inverse functions, which reverse the action of the original function. A function has an inverse only if it is one-to-one, which can be checked with the horizontal line test. When a function and its inverse are graphed, they are reflections of each other across the line $y = x$ and the domain and range of a function switch roles in its inverse.

Functions can also be classified by symmetry (even or odd). Piecewise functions are by different formulas on different parts of the domain.

Chapter 34

# Transforming Functions

Recall how we could translate, mirror, or rotate shapes and they would still be congruent shapes? We can do the same with functions, but the functions are not always equal. Let's say we gave you the graph of a function $f$, like this:

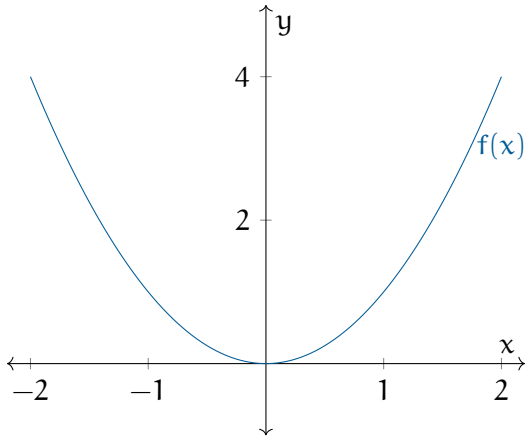



Figure 34.1: A graph of $x^2$.

We then tell you that the function is $g(x) = f(x) + 1.5$. Can you guess what the graph of $g$ would look like? It is the same graph, just translated up 1.5:

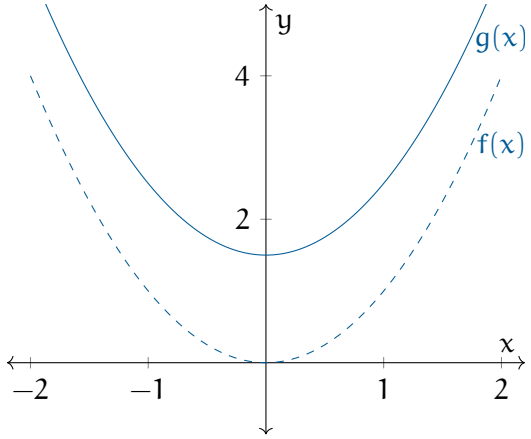



Figure 34.2: Shifted graph of $f(x) + 1.2$.

There are four kinds of transformations that we do all the time:

- Translation up and down in the direction of $y$ axis (the one you just saw)
- Translation left and right in the direction of the $x$ axis
- Scaling up and down along the $y$ axis
- Scaling up and down along the $x$ axis

Next, we will demonstrate each of the four using the graph of $\sin(x)$.

## 34.1 Translation up and down

When you add a positive constant to a function, you translate the whole graph up that much. A negative constant translates it down.

Here is the graph of $\sin(x) - 0.5$:

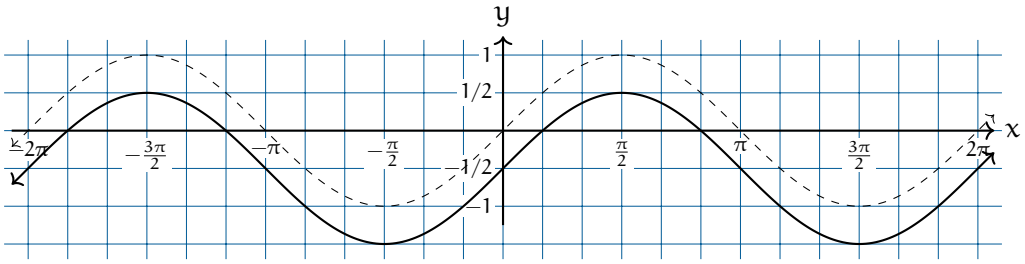



Figure 34.3: A graph of $\sin(x)$ dashed and $\sin(x) - 0.5$ in solid

## 34.2 Translation left and right

When you add a positive number to $x$ before running it through $f$, you translate the graph to the left by that amount. Adding a negative number translates the graph to the right.

Here is the graph of $\sin(x - \pi/6)$:

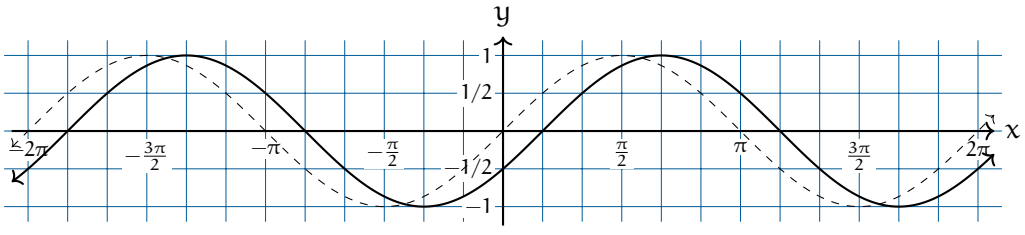



Figure 34.4: A graph of $\sin(x)$ in dashed lines and $\sin(x - \pi/6)$ in solid lines.

Notice the sign:

- Adding to x before processing with the function translates the graph to the *left*.
- Subtracting from x before processing with the function translates the graph to the *right*

## 34.3 Scaling up and down in the y direction

To scale the function up and down, you multiply the result of the function by a constant. If the constant is larger than 1, it stretches the function up and down.

Here is $y = 2\sin(x)$:

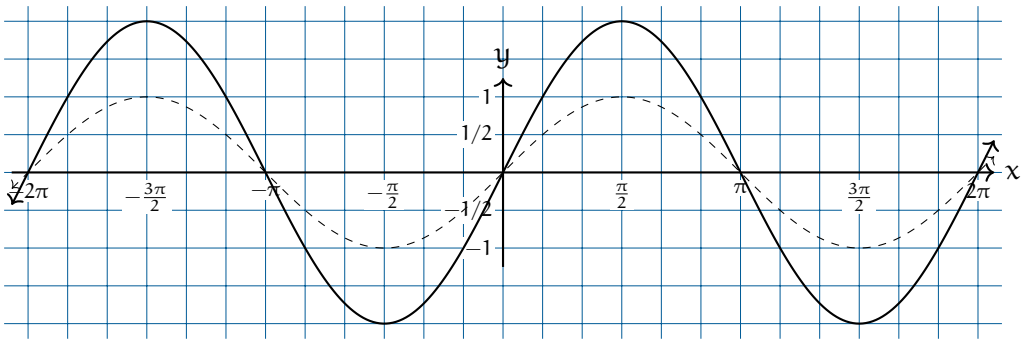



Figure 34.5: Scaling up and down in the y-direction requires a scalar outside of the function.

With a wave like this, we speak of its *amplitude*, which you can think of as its height. The baseline that this wave oscillates around is zero. The maximum distance that it gets from that baseline is its amplitude. Thus, the amplitude here has been increased from 1 to 2.

If you multiply by a negative number, the function gets flipped. Here is $y = -0.5\sin(x)$:

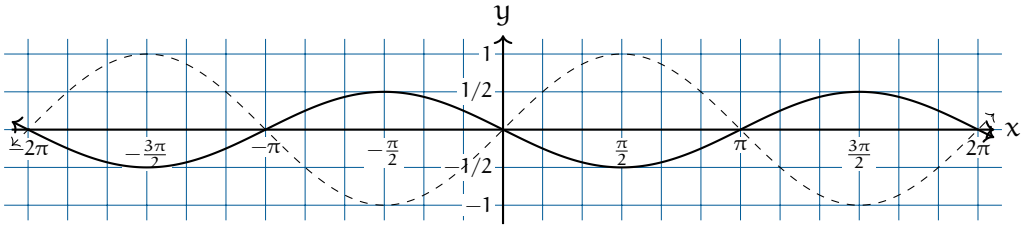



Figure 34.6: Scaling down and flipping or mirroring the function using $y = -0.5\sin(x)$.

Amplitude is never negative. Thus, the amplitude of this wave is 0.5.

## 34.4 Scaling up and down in the x direction

If you multiply x by a number larger than 1 before running it through the function, the graph gets compressed toward zero.

Here is $y = \sin(3x)$:

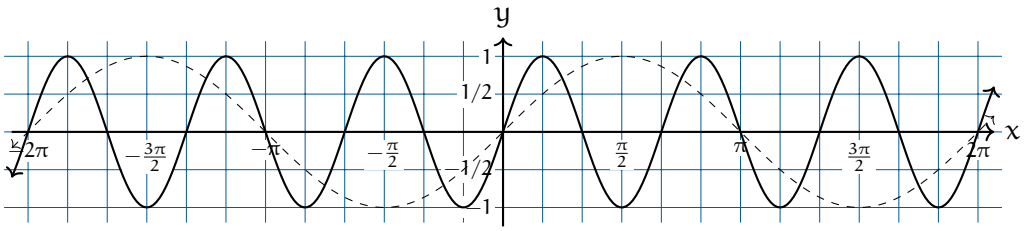



Figure 34.7: A solid $\sin(3x)$ compared to a solid $\sin(x)$ graph.

The distance between two peaks of a wave is known as its *wavelength*. The original wave had a wavelength of $2\pi$. The compressed wave has a wavelength of $2\pi/3$.

If you multiply x by a number smaller than 1, it will stretch the function out, away from the y axis.

If you multiply x by a negative number, it will flip the function around the y axis.

Here is $y = 2^{(-0.5x)}$. Notice that it has flipped around the y axis and is stretched out along the x axis.

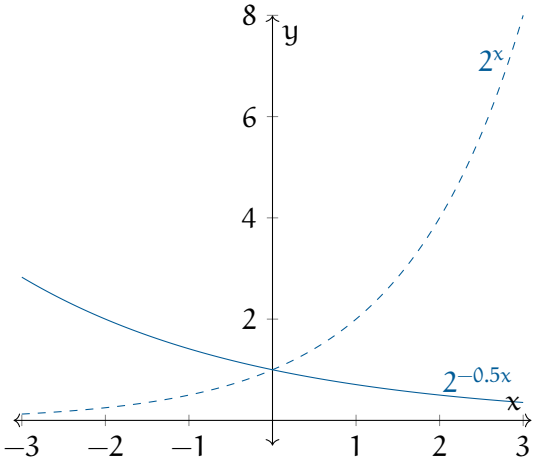



Figure 34.8: A base function $2^x$ in dashes compared to $2^{-0.5x}$ in solid.

| | |
|---|---|
| **Reflection over *x*-axis** | $(x, y) \to (x, -y)$ |
| **Reflection over *y*-axis** | $(x, y) \to (-x, y)$ |
| **Translation** | $(x, y) \to (x + a, y + b)$ |
| **Dilation** | $(x, y) \to (kx, ky)$ |
| **Rotation 90° counterclockwise** | $(x, y) \to (-y, x)$ |
| **Rotation 180°** | $(x, y) \to (-x, -y)$ |

Figure 34.9: A table of different transforms on a function

## 34.5 Order is important!

We can combine these transformations. This allows us, for example, to translate a function up 2, then scale along the $y$ axis by 3.

Here is $y = 2.0(\sin(x) + 1)$:

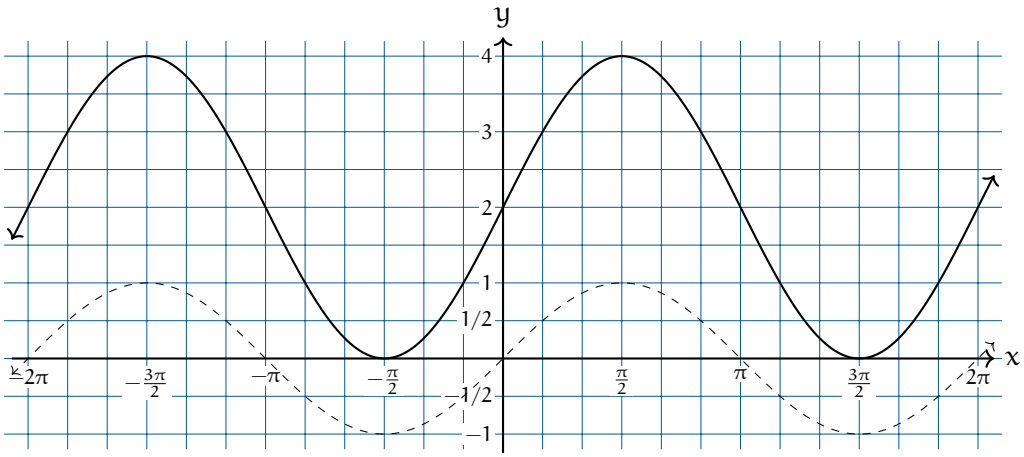



Figure 34.10: A graph of the base function $\sin(x)$ and the transformed function $2(\sin(x)+1)$

A function is often a series of steps. Here are the steps in $f(x) = 2(\sin(x) + 1)$:

1. Take the sine of $x$
2. Add 1 to that
3. Multiply that by 2

Note that this function can be distributed into $f(x) = 2\sin(x) + 2$ by bringing the 2 in both of the steps.

What if we change the order? Here are the steps in $g(x) = 2\sin(x) + 1$:

1. Take the sine of $x$
2. Multiply that by 2
3. Add 1 to that

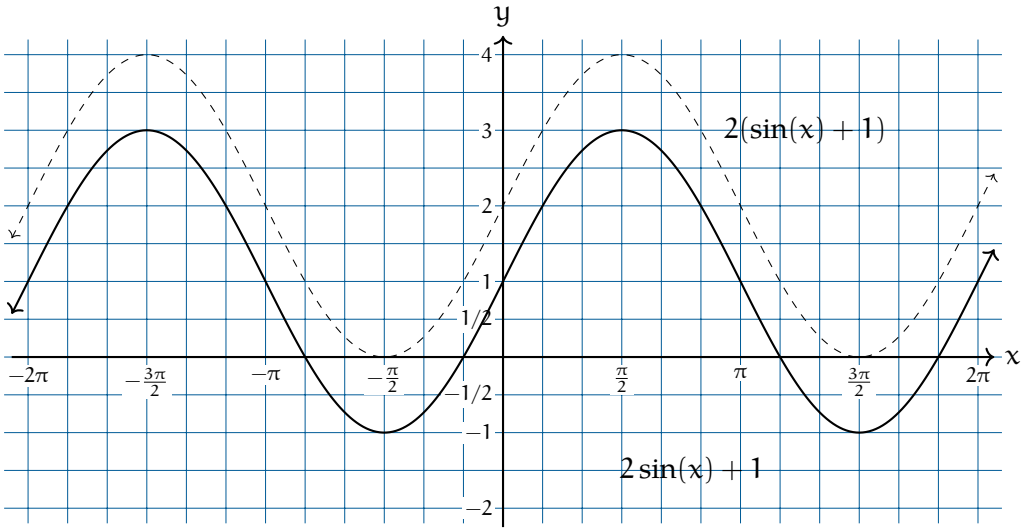


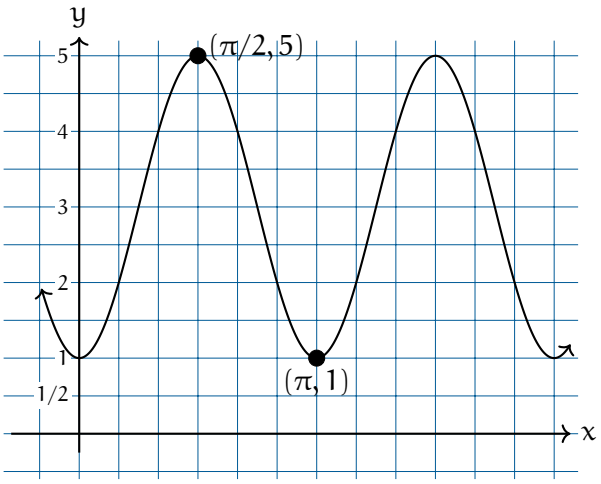

Figure 34.11: The functions $2\sin(x) + 1$ in solid compared to $2(\sin(x) + 1)$ in dashes.

The moral: You can do multiple transformations of your function, but the order in which you do them is important.

### *Exercise 101* Transforms

*Working Space*

Find a function that creates a sine wave such that the top of the first crest is at the point $(\frac{\pi}{2}, 5)$ and the bottom of the trough that follows is at $(\pi, 1)$.




*Answer on Page 1906*

## 34.6 Effects on even and odd Functions

Now that you've worked with transformations like shifts and scalings, you can use symmetry to quickly assess how those changes affect a function.

1. Vertical shifts (like $f(x)+c$) do not change whether a function is even or odd — but they break symmetry about the origin or y-axis. Example: $x^2+1$ is still even, but it's not symmetric about the origin.
2. Horizontal shifts (like $f(x-c)$) also typically destroy even/odd symmetry. Example: $f(x) = \sin(x)$ is odd. But $f(x) = \sin(x-\pi/2)$ is neither even nor odd.
3. Vertical scalings (like $a \cdot f(x)$) preserve symmetry type. Example: $x^2$ versus $3x^2$.
4. Horizontal scalings (like $f(bx)$) preserve symmetry as long as the center of the transformation stays aligned. Example: $f(x) = x^3$ is odd. $f(x) = (2x)^3$ is still odd.

Remember these key ideas:

1. Reflecting an even function over the y-axis? It remains unchanged.
2. Reflecting an odd function over the origin? Still unchanged.
3. But any shift left or right often breaks the symmetry, so check $f(-x)$ again if you've applied one.

Chapter 35

# Trigonometric Functions

As mentioned in an earlier chapter, in a right triangle where one angle is $\theta$, the sine of $\theta$ is the length of the side opposite $\theta$ divided by the length of the hypotenuse.

The sine function is defined for any real number. We treat that real number $\theta$ as an angle, we draw a ray from the origin out to the unit circle. The $y$ value of that point is the sine. For example, the $\sin(\frac{4\pi}{3})$ is $-\sqrt{3}/2$ (see figure 35.1).

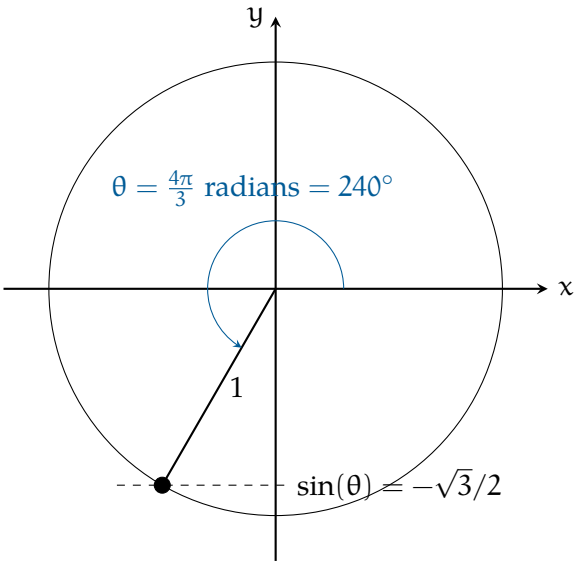



Figure 35.1: $\sin\frac{4\pi}{3} = \frac{-\sqrt{3}}{2}$

(Note that in this section, we will be using radians instead of degrees unless otherwise noted. While degrees are more familiar to most people, engineers and mathematicians nearly always use radians when solving problems. Your calculator should have a radians mode and a degrees mode; you want to be in radians mode.)

Similarly, we define cosine using the unit circle. To find the cosine of $\theta$, we draw a ray from the origin at the angle $\theta$. The $x$ component of the point where the ray intersects the unit circle is the cosine of $\theta$ (shown in figure 64.2).

From this description, it is easy to see why $\sin(\theta)^2 + \cos(\theta)^2 = 1$. They are the legs of a right triangle with a hypotenuse of length 1.

It should also be easy to see why $\sin(\theta) = \sin(\theta + 2\pi)$: Each time you go around the circle,

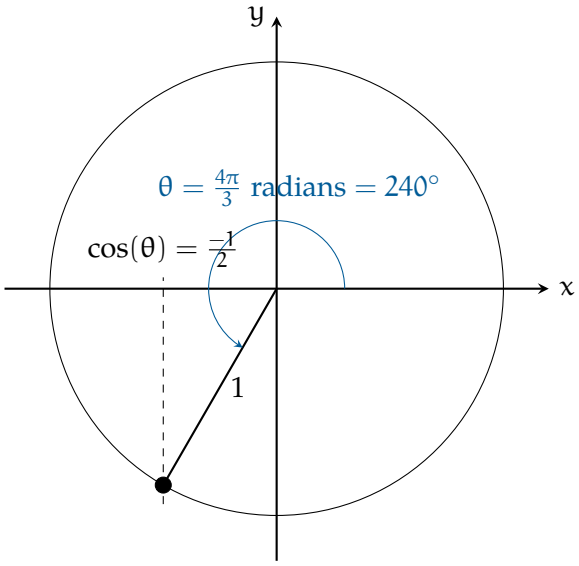



Figure 35.2: $\cos\frac{4\pi}{3} = \frac{-1}{2}$

you come back to where you started.

Can you see why $\cos(\theta) = \sin(\theta + \pi/2)$? Turn the picture sideways.

## 35.1 Graphs of sine and cosine

Here is a graph of $y = \sin(x)$:

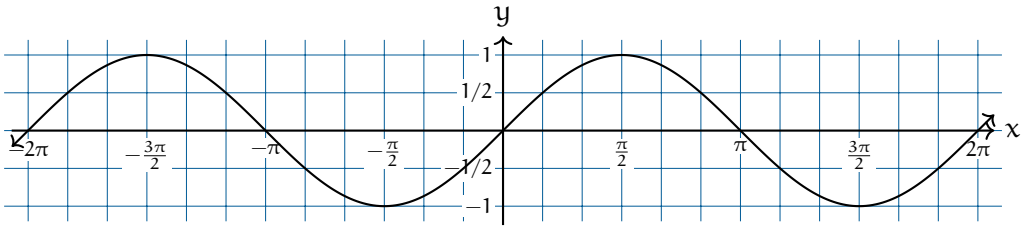



Figure 35.3: $y = \sin(x)$.

It looks like waves, right? It goes forever to the left and right. Remembering that $\cos(\theta) = \sin(\theta + \pi/2)$, we can guess what the graph of $y = \cos(x)$ looks like:

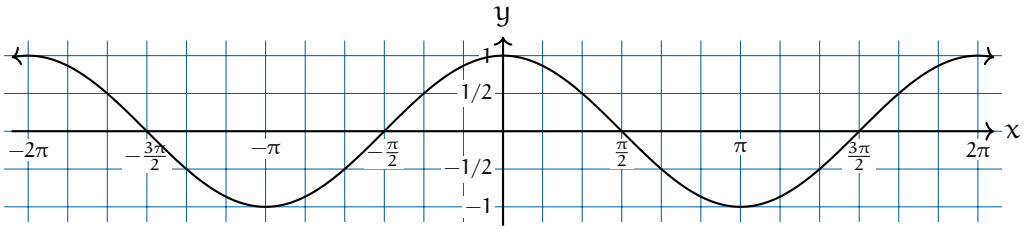


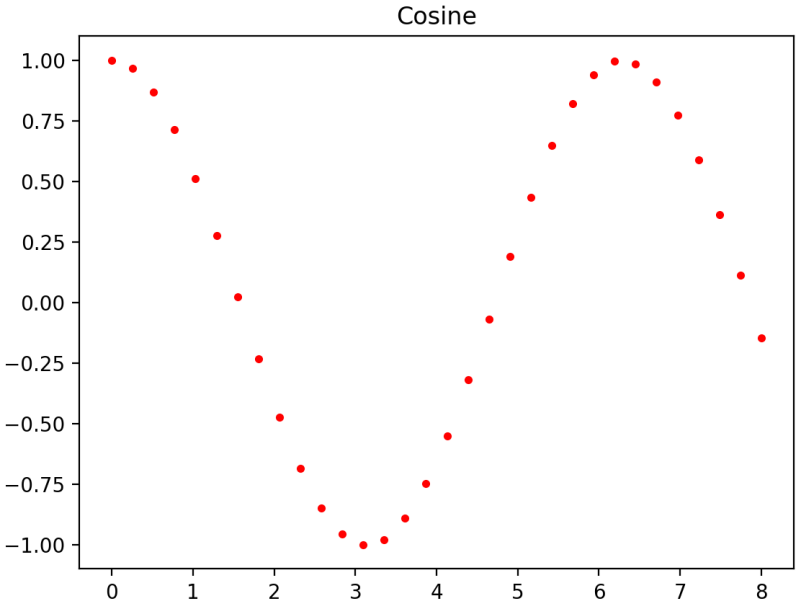

Figure 35.4: $y = \cos(x)$.

## 35.2 Plot cosine in Python

Create a file called `cos.py`:

```
import numpy as np
import matplotlib.pyplot as plt

until = 8.0

# Make a plot of cosine
thetas = np.linspace(0, until, 32)
cosines = []
for theta in thetas:
    cosines.append(np.cos(theta))

# Plot the data
fig, ax = plt.subplots()
ax.plot(thetas, cosines, 'r.', label="Cosine")
ax.set_title("Cosine")
plt.show()
```

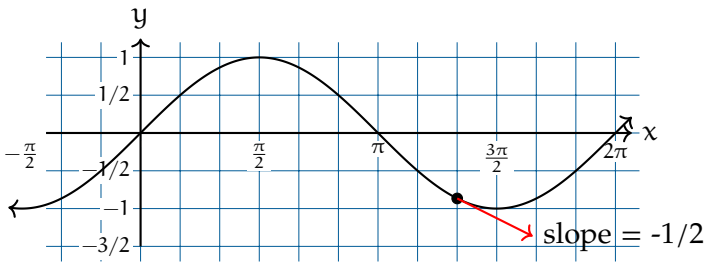
This will plot 32 points on the cosine wave between 0 and 8. When you run it, you should see something like this:




## 35.3 Derivatives of trigonometric functions

Here is a wonderful property of sine and cosine functions: At any point $\theta$, the slope of the sine graph at $\theta$ equals $\cos(\theta)$.

For example, we know that $\sin(4\pi/3) = -(1/2)\sqrt{3}$ and $\cos(4\pi/3) = -1/2$. If we drew a line tangent to the sine curve at this point, it would have a slope of -1/2:




We say "The derivative of the sine function is the cosine function."

Can you guess the derivative of the cosine function? For any $\theta$, the slope of the graph of the $\cos(\theta)$ is $-\sin(\theta)$.

## *Exercise 102* Derivatives of Trig Functions Practice 1

Use the limit definition of a derivative to show that $\frac{d}{dx}\cos x = -\sin x$

*Working Space*

*Answer on Page 1906*

The derivatives of all the trigonometric functions are presented below:

| | |
|---|---|
| $\frac{d}{dx}\sin x = \cos x$ | $\frac{d}{dx}\csc x = -\csc x \cdot \cot x$ |
| $\frac{d}{dx}\cos x = -\sin x$ | $\frac{d}{dx}\sec x = \sec x \cdot \tan x$ |
| $\frac{d}{dx}\tan x = \sec^2 x$ | $\frac{d}{dx}\cot x = -\csc^2 x$ |

**Example**: Find the derivative of $f(x)$ if $f(x) = x^2 \sin x$ **Solution**: Using the product rule, we find that:

$$\frac{d}{dx}f(x) = (x^2)\frac{d}{dx}(\sin x) + (\sin x)\frac{d}{dx}(x^2)$$

Taking the derivatives:

$$= x^2(\cos x) + 2x(\sin x)$$

## *Exercise 103* Derivatives of Trig Functions 2

Find the derivative of the following functions:

*Working Space*

1. $f(x) = \frac{\sec x}{1+\tan x}$
2. $y = \sec t \tan t$
3. $f(\theta) = \frac{\theta}{4-\tan \theta}$
4. $f(t) = 2\sec t - \csc t$
5. $f(\theta) = \frac{\sin \theta}{1+\cos \theta}$
6. $f(x) = \sin x \cos x$

*Answer on Page 1907*

## 35.4 A weight on a spring

Let's say you fill a rollerskate with heavy rocks and attach it to the wall with a stiff spring. If you push the skate toward the wall and release it, it will roll back and forth. Engineers would say "The skate will oscillate." Intuitively, you can probably guess:

- If the spring is stronger, the skate will oscillate more times per minute.
- If the rocks are lighter, the skate will oscillate more times per minute.

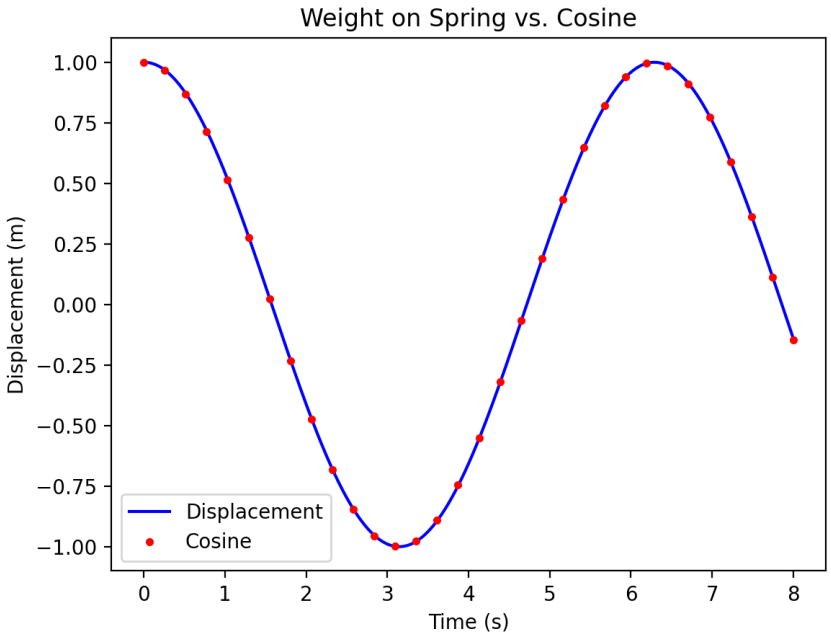
The force that the spring exerts on the skate is proportional to how far its length is from its relaxed length. When you buy a spring, the manufacturer advertises its "spring rate", which is in pounds per inch or newtons per meter. If a spring has a rate of 5 newtons per meter, that means that if you stretch or compress it 10 cm, it will push back with a force of 0.5 newtons. If you stretch or compress it 20 cm, it will push back with a force of 1 newton.

Let's write a simulation of the skate-on-a-spring. Duplicate `cos.py`, and name the new copy `spring.py`. Add code to implement the simulation:

```
import numpy as np
import matplotlib.pyplot as plt

until = 8.0

# Constants
mass = 100 # kg
spring_constant = -1 # newtons per meter displacement
time_step = 0.01 # s

# Initial state
displacement = 1.0 # height above equilibrium in meters
velocity = 0.0
time = 0.0 # seconds

# Lists to gather data
displacements = []
times = []

# Run it for a little while
while time <= until:
    # Record data
    displacements.append(displacement)
    times.append(time)

    # Calculate the next state
    time += time_step
    displacement += time_step * velocity
    force = spring_constant * displacement
```

```
    acceleration = force / mass
    velocity += acceleration

# Make a plot of cosine
thetas = np.linspace(0, until, 32)
cosines = []
for theta in thetas:
    cosines.append(np.cos(theta))

# Plot the data
fig, ax = plt.subplots()
ax.plot(times, displacements, 'b', label="Displacement")
ax.plot(thetas, cosines, 'r.', label="Cosine")

ax.set_title("Weight on Spring vs. Cosine")
ax.set_xlabel("Time (s)")
ax.set_ylabel("Displacement (m)")
ax.legend()
plt.show()
```

When you run it, you should get a plot of your spring and the cosine graph on the same plot.




The position of the skate is following a cosine curve. Why?

Because a sine or cosine waves happen whenever the acceleration of an object is proportional to -1 times its displacement. Or in symbols:

$$a \propto -p$$

where $a$ is acceleration and $p$ is the displacement from equilibrium.

Remember that if you take the derivative of the displacement, you get the velocity. And if you take the derivative of that, you get acceleration. So, the weight on the spring must follow a function $f$ such that

$$f(t) \propto -f''(t)$$

Remember that the derivative of the $\sin(\theta)$ is $\cos(\theta)$.

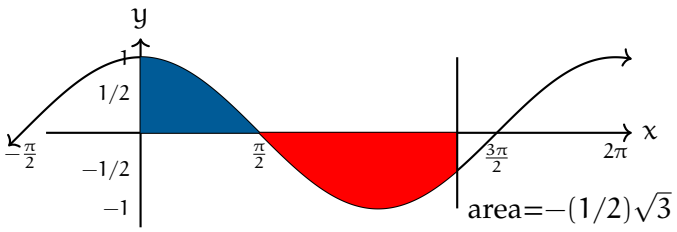
And the derivative of the $\cos(\theta)$ is $-\sin(\theta)$

These sorts of waves have an almost-magical power: Their acceleration is proportional to -1 times their displacement.

Thus, sine waves of various magnitudes and frequencies are ubiquitous in nature and technology.

## 35.5 Integral of sine and cosine

If we take the area between the graph and the $x$ axis of the cosine function (and if the function is below the $x$ axis, it counts as negative area), from 0 to $4\pi/3$, we find that it is equal to $-(1/2)\sqrt{3}$.




We say "The integral of the cosine function is the sine function."

### 35.5.1 Integrals of Trig Functions Practice

### Exercise 104

Evaluate the following integrals:

1. $\int \sec x \tan x \, dx$

*Working Space*

*Answer on Page 1907*

# Chapter 36

# Trigonometric Identities

## 36.1 The Unit Circle

There are some values of $\sin\theta$ and $\cos\theta$ that will be useful to know off the top of your head. The Unit Circle will help you in this memorization process (see figure 36.1). When a circle of radius 1 is centered at the origin, the Cartesian coordinates of any point on the circle correspond to the values of cosine and sine of the angle above the horizontal (how far you've rotated from the positive x-axis).

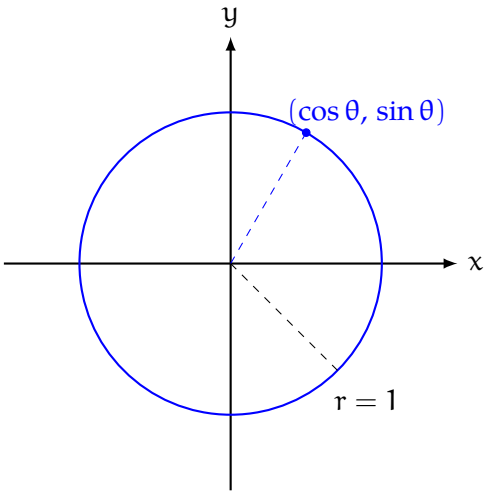



Figure 36.1: The Unit Circle is a circle with radius 1 centered at the origin

Let's take a closer look at a triangle in the first quadrant to see why this is true. Imagine some point on the circle, $(x_o, y_o)$. Drawing a line from that point back to the origin creates an angle $\theta$ between the imaginary line and the positive x-axis (see figure 36.2). Extending an imaginary vertical down to $(x_o, 0)$, then an imaginary horizontal from $(x_o, 0)$ to the origin, creates a right triangle. What can we say about the legs of the triangle?

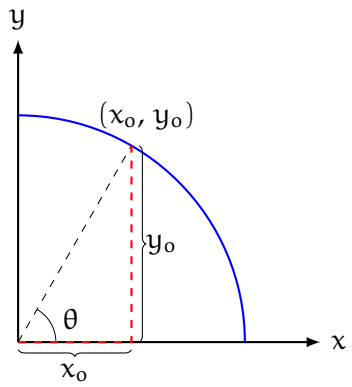



Figure 36.2: Drawing a line from any point on the circle to the origin creates an angle with the horizontal

Recall SOH-CAH-TOA from a previous chapter. This acronym tells us that, for a right triangle, the sine of an angle is given by the ratio of the length of the leg opposite the angle to the hypotenuse. In our case, then, $\sin\theta = \frac{y_o}{1} = y_o$. [Remember: We are dealing with the Unit Circle, which has a radius of one. Examining figure 36.2 shows you that the hypotenuse of the imaginary triangle is the same as the circle's radius.] This means that the y-coordinate of any point on the Unit Circle is the sine of the angle of rotation from the horizontal.

## Exercise 105

In a similar manner as we did with $\sin\theta$ above, prove the x-coordinate of any point on the unit circle is equal to $\cos\theta$, where $\theta$ is the angle of rotation from the horizontal.

*Working Space*

*Answer on Page 1908*

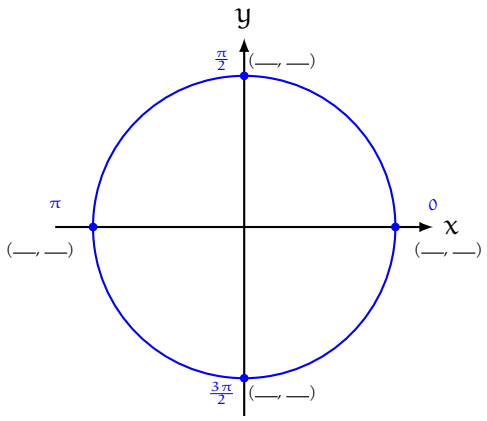
From these exercises, we can see that each $(x, y)$ coordinate on the circle is equal to $(\cos(\theta), \sin(\theta))$.

### *Exercise 106*

Fill in the unit circle with the coordinates for $\theta = 0$, $\pi/2$, $\pi$, and $3\pi/2$. Use this to determine:

1. $\sin \frac{\pi}{2}$
2. $\cos \frac{3\pi}{2}$
3. $\sin \pi$
4. $\cos -\pi$

*Working Space*




*Answer on Page 1908*

### 36.1.1 Exact Values of Key Angles

We will examine two triangles. First, a 30-60-90 triangle, then a 45-45-90 triangle. As shown in figure 36.3, you can get a 30-60-90 triangle with hypotenuse 1 by dividing an equilateral triangle in half. We will label the horizontal leg of the 30-60-90 triangle A and the vertical leg B.

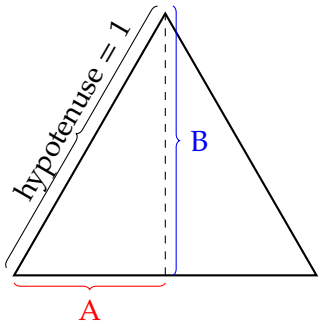



Figure 36.3: A 30-60-90 triangle is made by vertically bisecting an equilateral triangle

From the figure, we see that the length of A is half that of the hypotenuse, which in this case is $\frac{1}{2}$. This means the $\cos 60^\circ = \cos \frac{\pi}{3} = \frac{1}{2}$. To find the length of side B, we can use the Pythagorean theorem:

$$B^2 = C^2 - A^2, \text{ where C is the hypotenuse}$$

$$B^2 = 1^2 - \left(\frac{1}{2}\right)^2$$

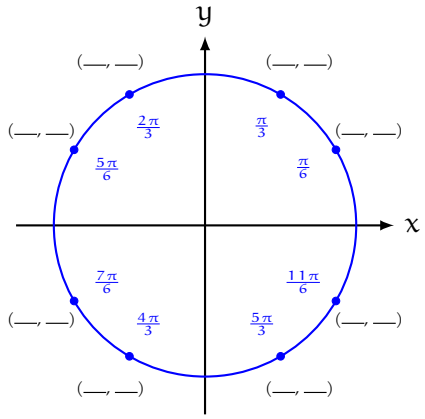
$$B^2 = \frac{3}{4}$$

$$B = \frac{\sqrt{3}}{2}$$

Therefore, $\sin 60^\circ = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$.

### Exercise 107

Use symmetry to complete the blank unit circle below. (Hint: We just showed that the $(x, y)$ coordinate for $\frac{\pi}{3}$ is $(1/2, \sqrt{3}/2)$).




*Working Space*

*Answer on Page 1909*

Now we will look at a 45-45-90 triangle (see figure 36.4), which will allow us to complete our Unit Circle. Recall that a 45-45-90 triangle is an isosceles triangle in addition to being a right triangle. This means both the legs are the same length. Using the Pythagorean theorem, we would say $A = B$. We also know that $C = 1$, since our triangle is inscribed in the unit circle. Let's find $A$:

$$A^2 + B^2 = C^2$$
$$A^2 + A^2 = 1^2$$
$$2A^2 = 1$$
$$A^2 = \frac{1}{2}$$
$$A = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Therefore, each leg has a length of $\sqrt{2}/2$, and the $(x, y)$ coordinates for $\theta = 45° = \pi/4$ are $(\sqrt{2}/2, \sqrt{2}/2)$.

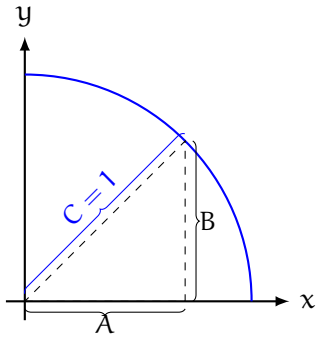
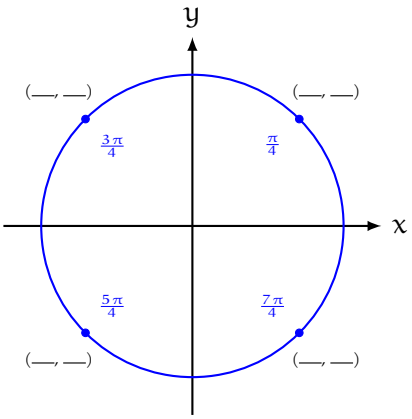



Figure 36.4: The two legs of a 45-45-90 triangle are the same length

## *Exercise 108*

Use symmetry to complete the blank unit circle below.




*Working Space*

*Answer on Page 1909*

### *Exercise 109*

Without a calculator and using only your completed unit circles, determine the value requested (angles are given in radians unless otherwise indicated).

1. $\cos \frac{3\pi}{2}$
2. $\sin \frac{\pi}{4}$
3. $\sin -\frac{\pi}{6}$
4. $\cos \frac{4\pi}{3}$
5. $\sin \frac{3\pi}{4}$
6. $\cos -\frac{\pi}{3}$
7. $\sin 45°$
8. $\sin 270°$
9. $\sin -60°$
10. $\sin 150°$

*Working Space*

*Answer on Page 1909*

## 36.2 Sum and Difference Formulas

Consider 4 points on the unit circle: B at $(1, 0)$, Q at some angle $\beta$, P at some angle $\alpha$, and A at angle $\alpha - \beta$ (see figure 36.5).

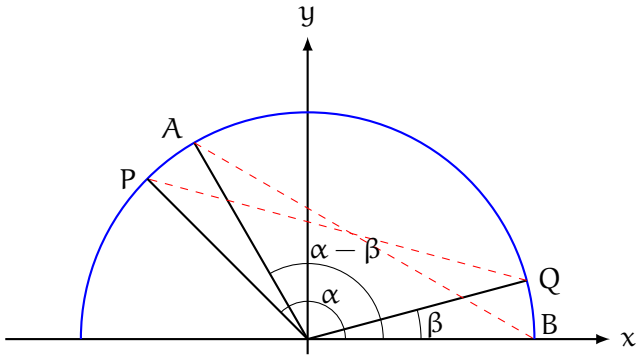



Figure 36.5: $\overline{AB} = \overline{PQ}$

The distance from P to Q is the same as the distance from A to B, since $\Delta POQ$ is a rotation of $\Delta AOB$. Because this is a Unit Circle, $P = (\cos\alpha, \sin\alpha)$, $Q = (\cos\beta, \sin\beta)$, and $A = (\cos\alpha - \beta, \sin\alpha - \beta)$. Let's use the distance formula to find the length of $\overline{PQ}$:

$$\overline{PQ} = \sqrt{(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2} =$$

$$\sqrt{\cos^2\alpha - 2\cos\alpha\cos\beta + \cos^2\beta + \sin^2\alpha - 2\sin\alpha\sin\beta + \sin^2\beta} =$$

$$\sqrt{(\cos^2\alpha + \sin^2\alpha) + (\cos^2\beta + \sin^2\beta) - 2\cos\alpha\cos\beta - 2\sin\alpha\sin\beta}$$

Recall that for any angle, $\theta$, $\sin^2\theta + \cos^2\theta = 1$. Substituting this identity, we see that:

$$\overline{PQ} = \sqrt{1 + 1 - 2\sin\alpha\sin\beta - 2\cos\alpha\cos\beta} = \sqrt{2 - 2\sin\alpha\sin\beta - 2\cos\alpha\cos\beta}$$

Let's leave this simplified equation for $\overline{PQ}$ alone for the moment and similarly find $\overline{AB}$:

$$\overline{AB} = \sqrt{[\cos(\alpha - \beta) - 1]^2 + [\sin(\alpha - \beta) - 0]^2} =$$

$$\sqrt{\cos^2(\alpha - \beta) - 2\cos(\alpha - \beta) + 1 + \sin^2(\alpha - \beta)} =$$

$$\sqrt{\cos^2(\alpha - \beta) + \sin^2(\alpha - \beta) + 1 - 2\cos(\alpha - \beta)}$$

$$= \sqrt{2 - 2\cos(\alpha - \beta)} = \overline{AB}$$

Recall that we've established $\overline{AB} = \overline{PQ}$. We can set the statements equal to each other:

$$\sqrt{2 - 2\sin\alpha\sin\beta - 2\cos\alpha\cos\beta} = \sqrt{2 - 2\cos(\alpha - \beta)}$$

Squaring both sides and subtracting 2, we find:

$$-2\sin\alpha\sin\beta - 2\cos\alpha\cos\beta = -2\cos(\alpha-\beta)$$

Finally, we can divide both sides by negative 2 to get the difference of angles formula for cosine:

$$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

There are similar formulas for the sine and cosine of the sum of two angles, and for the sine of the difference of two angles, which we won't derive here.

**Sum and Difference Formulas**

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$
$$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$
$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$$

### *Exercise 110*

Without a calculator, find the exact value requested:

*Working Space*

1. $\sin\frac{\pi}{12}$
2. $\cos\frac{7\pi}{12}$
3. $\tan\frac{13\pi}{12}$ (hint: $\tan\theta = \sin\theta/\cos\theta$)

*Answer on Page 1910*

## 36.3 Double and Half Angle Formulas

We can easily derive a formula for twice an angle by letting $\alpha = \beta$ for a sum formula.

**Example**: Derive a formula for $\cos 2\theta$ in terms of trigonometric functions of $\theta$.

**Solution**: Using the sum formula for cosine, we see that:

$$\cos 2\theta = \cos(\theta + \theta)$$

$$= \cos\theta\cos\theta - \sin\theta\sin\theta = \cos^2\theta - \sin^2\theta$$

Noting that $\sin^2\theta = 1 - \cos^2\theta$: $\boxed{\cos 2\theta = 2\cos^2\theta - 1}$ Alternatively, we could note that $\cos^2\theta = 1 - \sin^2\theta$: $\boxed{\cos 2\theta = 1 - 2\sin^2\theta}$

Or additionally, $\boxed{\cos 2\theta = \cos^2\theta - \sin^2\theta}$

## *Exercise 111*

Derive a formula for $\sin 2\theta$ in terms of trigonometric functions of $\theta$.

*Working Space*

*Answer on Page 1910*

We can use these double-angle formulas for find half-angle formulas. Consider the double-angle formula for cosine:

$$\cos 2\theta = 2\cos^2\theta - 1$$

Let $\theta = \alpha/2$, then:

$$\cos\alpha = 2\cos^2(\alpha/2) - 1$$

Rearranging to solve for $\cos(\alpha/2)$:

$$2\cos^2(\alpha/2) = \cos\alpha + 1$$

$$\cos^2(\alpha/2) = \frac{\cos\alpha + 1}{2}$$

$$\cos(\alpha/2) = \pm\sqrt{\frac{1+\cos\alpha}{2}}$$

### Exercise 112

Derive a formula for $\sin(\alpha/2)$.

*Working Space*

*Answer on Page 1910*

There are two identities that will be very useful for integrals in a future chapter:

**Squared Trigonometric Identities**

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$

These are just specific re-writings of the half-angle identities.

Chapter 37

# Volumes of Common Solids

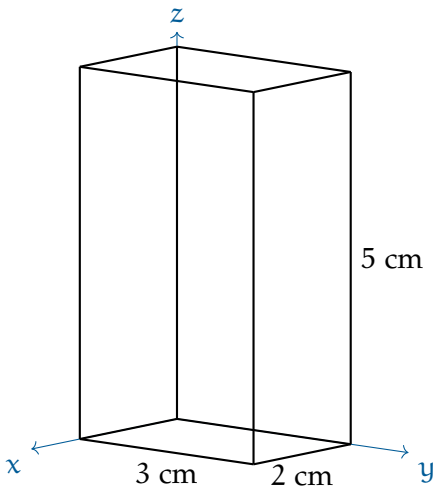
## 37.1 Rectangular Prism

The volume of a rectangular solid is the product of its three dimensions. If a block of ice is 5 cm tall, 3 cm wide, and 2 cm deep, its volume is $5 \times 3 \times 2 = 30$ cubic centimeters.

**Volume of a rectangular solid.**

A rectangular solid with height $h$, width $w$ and length/depth $l$ has volume:

$$V = lwh$$




A cubic centimeter is the same as a milliliter. A milliliter of ice weighs about 0.92 grams. This means the block of ice would have a mass of $30 \times 0.92 = 27.6$ grams.

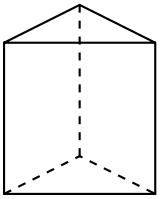
## 37.2 Triangular Prism

Triangular prisms are 3D versions of triangles (imagine stretching a triangle out of the page). It has 2 triangular faces and 3 rectangular faces.

**Volume of a triangular prism.**

Recall the area of a triangle is $V = \frac{1}{2}wh$ where w is the width or base and h is the height of the triangle. A triangular prism with height h, width $w$ and length/depth l has volume:

$$V = \frac{1}{2}lwh$$



## 37.3 Spheres

**Volume of a Sphere**

A sphere with a radius of r has a volume of

$$v = \frac{4}{3}\pi r^3$$

(For completeness, the surface area of that sphere would be

$$a = 4\pi r^2$$

Note that a circle of radius r is one quarter of this: $\pi r^2$.)

### *Exercise 113* Flying Sphere

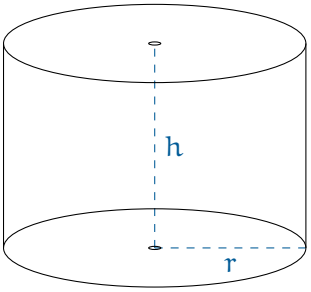
*Working Space*

An iron sphere is traveling at 5 m/s (and is not spinning). The sphere has a radius of 1.5 m. Iron has a density of 7,800 kg per cubic meter. How much kinetic energy does the sphere have?

*Answer on Page 1911*

## 37.4 Cylinders

The base and the top of a right cylinder are identical circles. The circles are on parallel planes. The sides are perpendicular to those planes.




**Volume of a cylinder**

The volume of the right cylinder of radius $r$ and height $h$ is given by:

$$v = \pi r^2 h$$

In other words, it is the area of the base times the height.

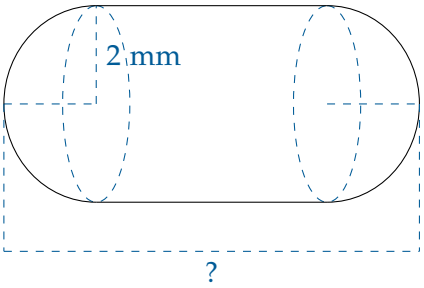
## *Exercise 114* Tablet

*Working Space*

A drug company needs to create a tablet with volume of 90 cubic millimeters.

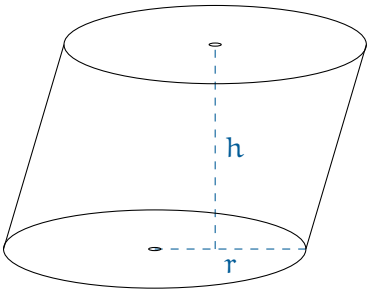
The tablet will be a cylinder with half spheres on each end. The radius will be 2mm.

How long do they need to make the tablet to be?




*Answer on Page 1911*

What if the base and top are identical, but the sides aren't perpendicular to the base? This is called *oblique cylinder*.

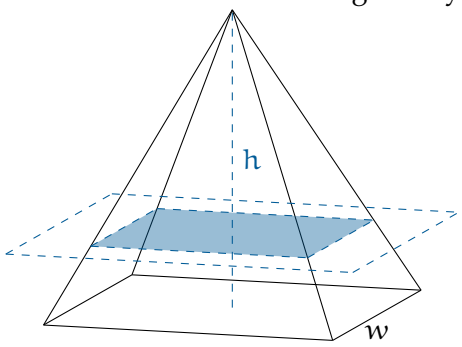



The volume is still the height times the area of the base. Note, however, that the height is measured perpendicular to the bottom and top.

Why is this the case?

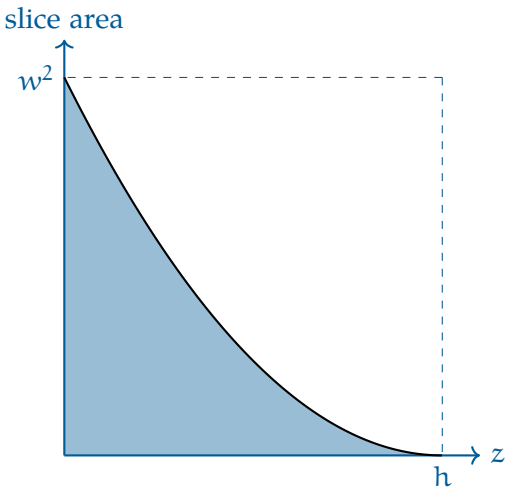
## 37.5 Pyramids

On a solid with a flat base, the line that we use to measure height is always perpendicular to the plane of the base. We can take slices through the solid that are parallel to that base plane. For example, if we have a pyramid with a square base, each slice will be a square — small squares near the top, larger squares near the bottom. The sides of the pyramids are all triangles, so these are referred to as Triangular Pyramids, just pyramids, or sometimes, Tetrahedrons.




We can figure out the area of the slice at every height $z$. For example, at $z = 0$, the slice would have area $w^2$. At $z = h$, the slice would have zero area. What about an arbitrary $z$ in between? The edge of the square would be $w(1 - \frac{z}{h})$. The area of the slice would be $w^2(1 - \frac{z}{h})^2$

The graph of this would look like this:




The volume is given by the area under the curve and above the axis. Once you learn integration, you will be extra good at finding the area under the curve. In this case, we will just tell you that the colored region in the picture is one third of the rectangle.

Thus, the area of a square-based pyramid is $\frac{1}{3}hw^2$.

In fact:

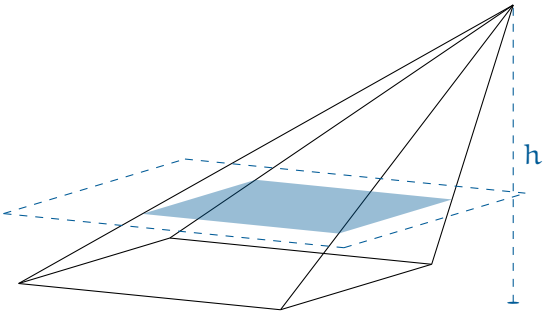
**Volume of a pyramid**

The volume of pyramid whose base has an area of $b$ and height $h$ is given by:

$$V = \frac{1}{3}hb$$

Regardless of the shape of the base.

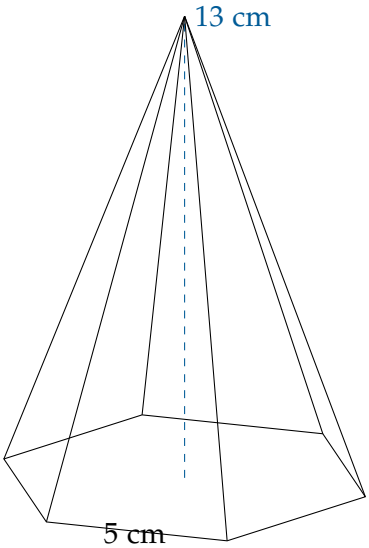
Note that this is true even for oblique pyramids:



## *Exercise 115* Hexagon-based Pyramid

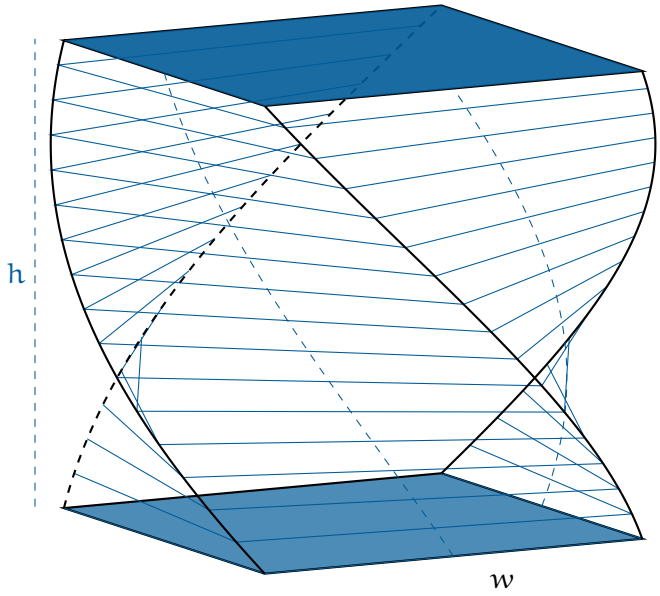
*Working Space*

There is a pyramid with a regular hexagon for a base. Each edge is 5 cm long. The pyramid is 13 cm tall. What is its volume?




*Answer on Page 1912*

Note that plotting the area of each slice and finding the area under the curve will let you find the area of many things. For example, let's say that you have a four-sided spiral, where each face has the same width $w$:




Every slice still has an area of $w^2$, which means this figure has a volume of $hw^2$.

## *Exercise 116* Volume of a building

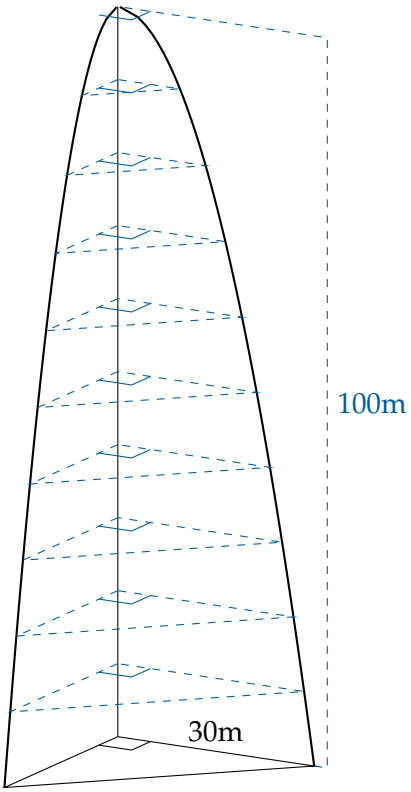
*Working Space*

An architect is designing a hotel with a right triangular base; the base is 30 meters on each leg. The building gets narrower as you get closer to the top, and finally shrinks to a point. The spine of the building is where the right angle is. That spine is straight and perpendicular to the ground.

Each floor has a right isosceles triangle as its floor plan. The length of each leg is given by this formula:

$$w = 30\sqrt{1 - \frac{z}{100}}$$

So, the width of the building is 30 meters at height $z = 0$. At 100 meters, the building comes to a point. It will look like this:




What is the volume of the building in cubic meters?

Chapter 38

# Conic Sections

In mathematics, conic sections (or simply conics) are curves obtained as the intersection of the surface of a cone with a plane. The three types of conic section are the hyperbola, the parabola, and the ellipse. The circle is a special case of the ellipse, though historically it was sometimes called a fourth type. All of the equations below can be graphed on programs like Desmos.

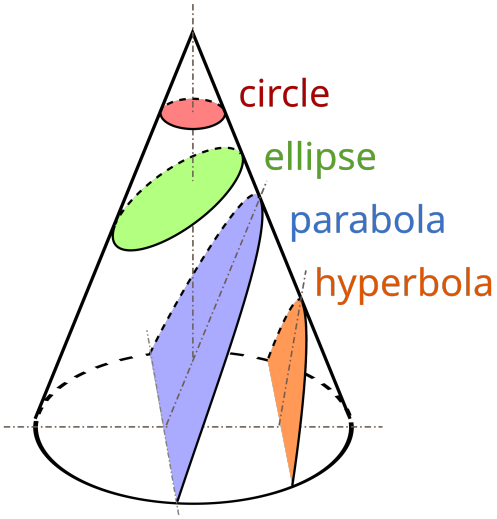



Figure 38.1: Visualization of conic sections.

Source: Wikimedia Commons, Public Domain: https://upload.wikimedia.org/wikipedia/commons/thumb/1/11/Conic_Sections.svg/1920px-Conic_Sections.svg.png

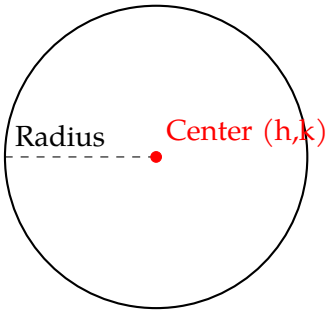
## 38.1 Definitions

Each type of conic sections can be defined as follows:

### 38.1.1 Circle

A circle is the set of all points in a plane that are at a given distance (the radius) from a given point (the center). The standard equation for a circle with center $(h, k)$ and radius $r$ is:

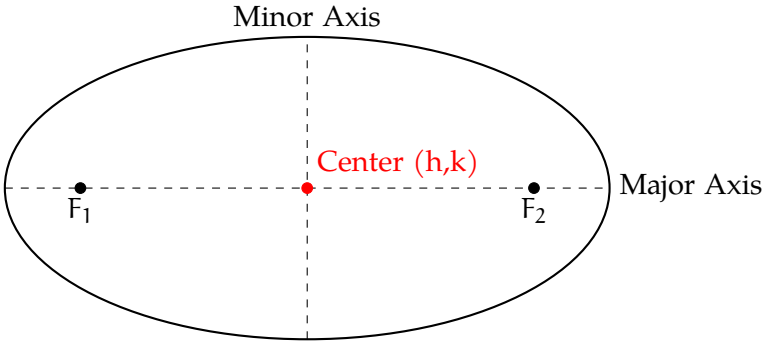
$$(x - h)^2 + (y - k)^2 = r^2 \tag{38.1}$$




### 38.1.2 Ellipse

An ellipse is the set of all points such that the sum of the distances from two fixed points (the foci) is constant. The standard equation for an ellipse centered at the origin with semi-major axis $a$ and semi-minor axis $b$ is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \tag{38.2}$$



### 38.1.3 Hyperbola

A hyperbola is the set of all points such that the absolute difference of the distances from two fixed points (the foci) is constant. A hyperbola is formed from slicing a *double-cone* — two cones placed tip-to-tip — parallel to or angled off of the central axes. The standard equation for a hyperbola centered at the origin is:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{38.3}$$

or

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1 \tag{38.4}$$

depending on the orientation of the hyperbola.

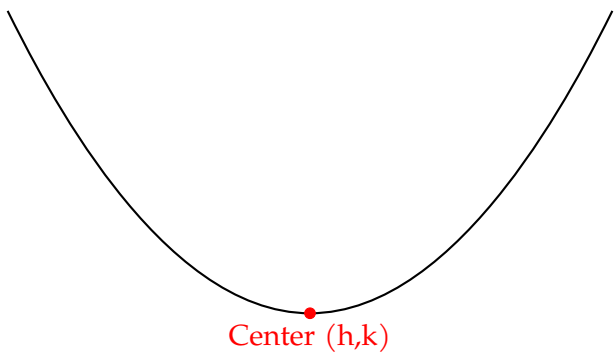
### 38.1.4 Parabola

A parabola is the set of all points that are equidistant from a fixed point (the focus) and a fixed line (the directrix). The standard equation for a parabola that opens upwards or downwards is:

$$y = a(x - h)^2 + k \tag{38.5}$$

and that opens leftwards or rightwards is:

$$x = a(y - k)^2 + h \tag{38.6}$$



where $(h, k)$ is the vertex of the parabola, and $a$ is a scalar.

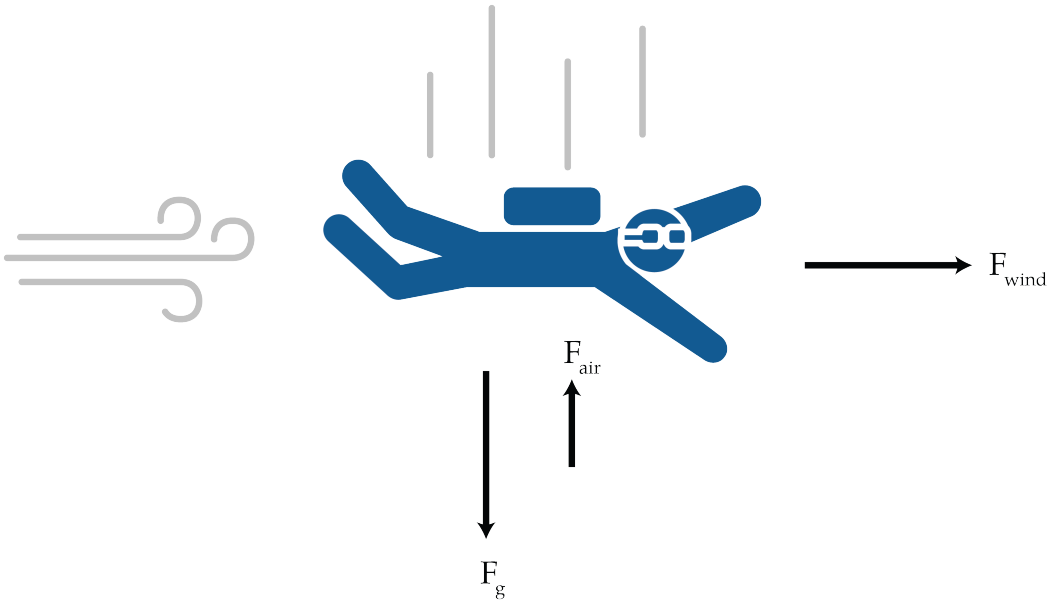
Note that only the parabola out of these four is a function, as it passes vertical line test. The other three cannot be expressed as functions, only equations.

Chapter 39

# Vectors

We have talked a some about forces, but in the calculations that we have done, we have only talked about the magnitude of a force. It is equally important to talk about its direction. To do the math on things with a magnitude and a direction (like forces), we need vectors.

For example, if you jump out of a plane (hopefully with a parachute), several forces with different magnitudes and directions will be acting upon you. Gravity will push you straight down. That force will be proportional to your weight. If there were a wind from the west, it would push you toward the east. That force will be proportional to the square of the speed of the wind and approximately proportional to your size. Once you are falling, there will be resistance from the air that you are pushing through — that force will point in the opposite direction from the direction you are moving and will be proportional to the square of your speed.




To figure out the net force (which will tell us how we will accelerate), we will need to add these forces together. To do this, we need to learn to do math with vectors.

## 39.1 Adding Vectors

A vector is typically represented as a list of numbers, with each number representing a particular dimension. For example, if you are creating a 3-dimensional vector representing a force, it will have three numbers representing the amount of force in each of the three axes. For example, if a force of one newton is in the direction of the x-axis, you might represent the vector as $v = [1, 0, 0]$. Another vector might be $u = [0.5, 0.9, 0.7]$ . You can see examples of 2-dimensional and 3-dimensional vectors in figures 39.1 and 131.1.

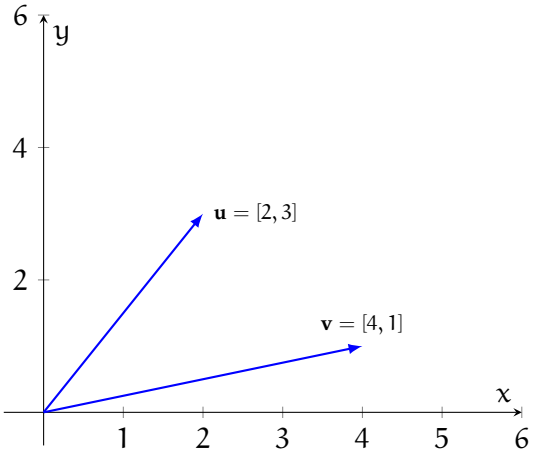



Figure 39.1: 2-dimensional vectors, **u** and **v**

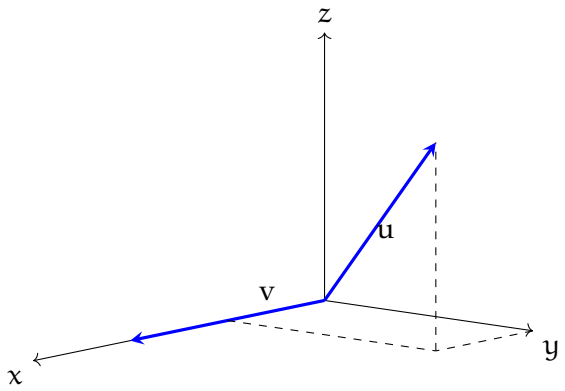



Figure 39.2: 3-dimensional vectors, **u** and **v**

Thinking visually, when we add to vectors, we put the starting point second vector at the ending point of the first vector. This is illustrated for 2-dimensional vectors in figure 39.3 and for 3-dimensional vectors in figure 39.4.

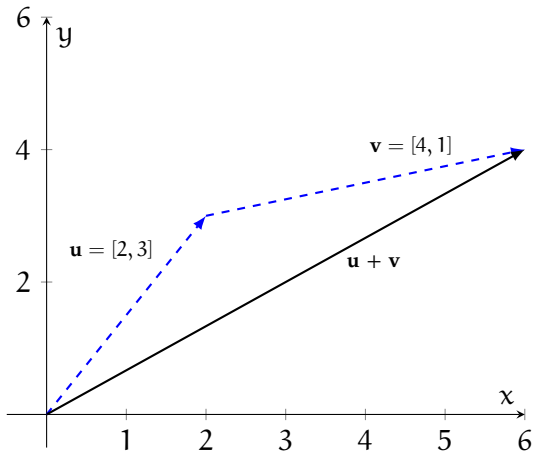



Figure 39.3: A visual representation of adding 2-dimensional vectors, **u** and **v**

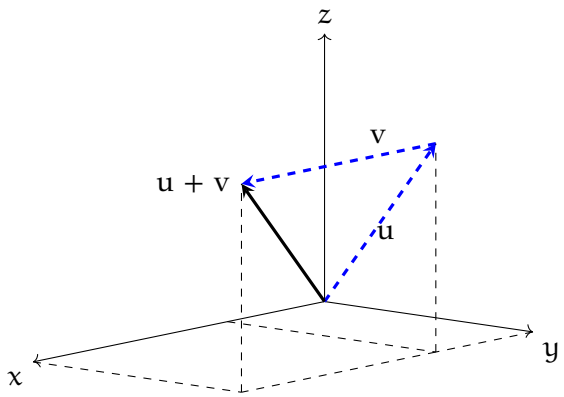



Figure 39.4: A visual representation of adding 3-dimensional vectors, **u** and **v**

If you know the vectors, you will just add them element-wise:

$$u + v = [0.5, 0.9, 0.7] + [1.0, 0.0, 0.0] = [1.5, 0.9.0.7]$$

These vectors have 3 components, so we say they are *3-dimensional*. Vectors can have any number of components. For example, the vector $[-12.2, 3, \pi, 10000]$ is 4-dimensional.

You can only add two vectors if they have the same dimension.

$$[12, -4] + [-1, 5] = [11, 1]$$

Addition is commutative; if you have two vectors $a$ and $b$, then $a + b$ is the same as $b + a$.

Addition is also associative: If you have three vectors $a$, $b$, and $c$, it doesn't matter which order you add them in. That is, $a + (b + c) = (a + b) + c$.

A 1-dimensional vector is just a number. We say it is a *scalar*, not a vector.

### *Exercise 117* Adding vectors

Add the following vectors:

- $[1, 2, 3] + [4, 5, 6]$
- $[-1, -2, -3, -4] + [4, 5, 6, 7]$
- $[\pi, 0, 0] + [0, \pi, 0] + [0, 0, \pi]$

*Working Space*

*Answer on Page 1913*

### *Exercise 118* Adding Forces

You are adrift in space, near two different stars. The gravity of one star is pulling you towards it with a force of $[4.2, 5.6, 9.0]$ newtons. The gravity of the other star is pulling you towards it with a force of $[-100.2, 30.2, -9.0]$ newtons. What is the net force?

*Working Space*

*Answer on Page 1914*

## 39.2 Multiplying a vector with a scalar

It is not uncommon to multiply a vector by a scalar. For example, a rocket engine might have a force vector $v$. If you fire 9 engines in the exact same direction, the resulting force vector would be $9v$.

Visually, when we multiply a vector $u$ by a scalar $a$, we get a new vector that goes in the

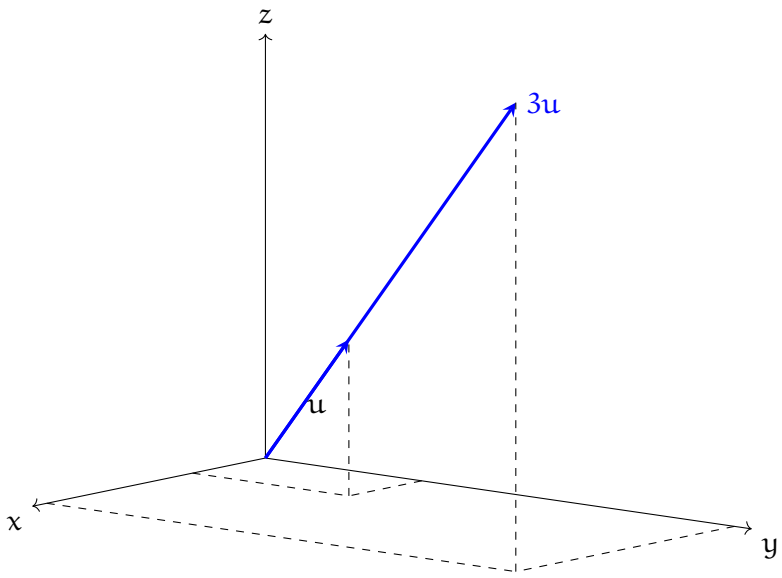



Figure 39.5: To multiply vectors, the vector gets stretched in the same direction a amount

same direction as u but has a magnitude a times as long as u. A visual is presented in figure 39.5.

When you multiply a vector by a scalar, you simply multiply each of the components by the scalar:

$$3 \times [0.5, 0.9, 0.7] = [1.5, 2.7, 3.6]$$

## *Exercise 119* Multiplying a vector and a scalar

Simplify the following expressions:

*Working Space*

- $2 \times [1, 2, 3]$
- $[-1, -2, -3, -4] \times -2$
- $\pi[\pi, 2\pi, 3\pi]$

*Answer on Page 1914*

Note that when you multiply a vector times a negative number, the new vector points in

the opposite direction (see figure 39.6).

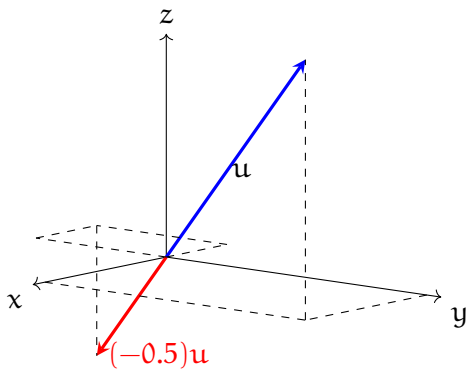



Figure 39.6: Multiplying a vector by a negative number reverses the direction of the vector.

## 39.3 Vector Subtraction

As you might guess, when you subtract one vector from another, you just do element-wise subtraction:

$$[4, 2, 0] - [3, -2, 9] = [1, 4, -9]$$

So, $u - v = u + (-1v)$.

Visually, you reverse the one that is being subtracted (see figure 39.7):

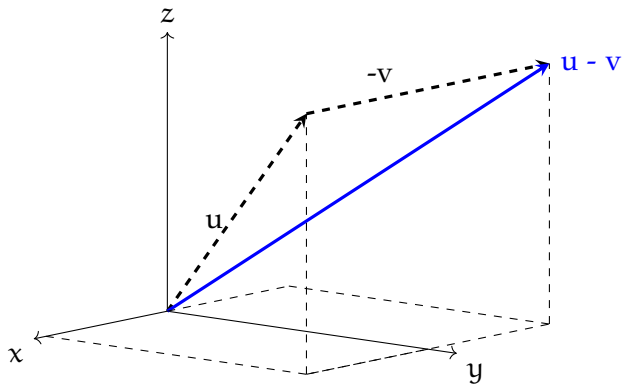



Figure 39.7: To subtract a vector, you reverse it, then add the reversed vector.

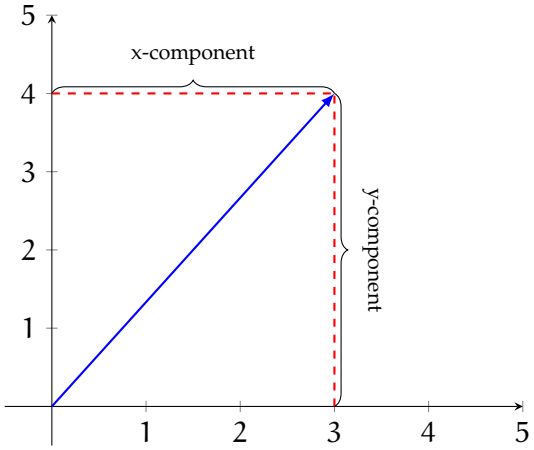



Figure 39.8: The magnitude of a vector can be thought of as the length of a hypotenuse of a right triangle.

## 39.4 Magnitude of a Vector

The *magnitude* of a vector is just its length. We write the magnitude of a vector $v$ as $|v|$.

We compute the magnitude using the pythagorean theorem. If $v = [3, 4, 5]$, then

$$|v| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50} \approx 7.07$$

(You might notice that the notation for the magnitude is exactly like the notation for absolute value. If you think of a scalar as a 1-dimensional vector, the absolute value and the magnitude are the same. For example, the absolute value of -5 is 5. If you take the magnitude of the one-dimensional vector $[-5]$,you get $\sqrt{25} = 5$.)

Where does this equation come from? Consider a 2-dimensional vector, $\mathbf{v} = [3, 4]$. This means the vector represents 3 units in the $x$-direction, and 4 units in the $y$-direction. We can then visualize a right triangle, with the vector being the hypotenuse and the legs being the $x$- and $y$-components of the vector (see figure 39.8). As you recall, the length of the hypotenuse of a right triangle is the square root of the sum of the squares of the legs. That is:

$$c = \sqrt{a^2 + b^2}$$

Where $c$ is the length of the hypotenuse and $a$ and $b$ are the lengths of the legs. See Figure **??**.

We won't prove it here, but this method holds for higher-dimension vectors as well.

**Magnitude of Vectors**

For an $n$-dimensional vector, $\mathbf{v} = [x_1, x_2, x_3, \cdots, x_n]$, the magnitude of the vector is given by:

$$|\mathbf{v}| = \sqrt{x_1^2 + x_2^2 + x_3^2 + \cdots + x_n^2}$$

Notice that if you scale up a vector, its magnitude scales by the same amount. For example:

$$|7[3, 4, 5]| = 7\sqrt{50} \approx 7 \times 7.07$$

Here is why that is true. Suppose we have a vector, $\mathbf{u} = [a, b, c]$. Then the magnitude of $\mathbf{u}$ is given by:

$$|\mathbf{u}| = \sqrt{a^2 + b^2 + c^2}$$

If we scale $\mathbf{u}$ to create $\mathbf{v}$ such that $\mathbf{v} = k\mathbf{u} = [ka, kb, kc]$, where $k$ is some constant. Then the magnitude of $\mathbf{v}$ is given by:

$$|\mathbf{v}| = \sqrt{(ka)^2 + (kb)^2 + (kc)^2}$$

We can expand and simplify this equation:

$$|\mathbf{v}| = \sqrt{k^2a^2 + k^2b^2 + k^2c^2}$$

$$|\mathbf{v}| = \sqrt{k^2\left(a^2 + b^2 + c^2\right)}$$

$$|\mathbf{v}| = \left(\sqrt{k^2}\right)\sqrt{a^2 + b^2 + c^2}$$

$$|\mathbf{v}| = |k|\sqrt{a^2 + b^2 + c^2} = |k|\,|\mathbf{u}|$$

So, if you scale a vector, the magnitude of the resulting vector is the absolute value of the scale factor times the magnitude of the original vector.

The rule then is: If you have any vector $v$ and any scalar $k$:

$$|kv| = |k||v|$$

### 39.4.1 Unit Vectors

A *unit vector* is a vector whose magnitude is 1. For any non-zero vector $\vec{\mathbf{v}}$, the unit vector $\vec{\mathbf{u}}$ pointing in the same direction is

$$\vec{\mathbf{u}} = \frac{\mathbf{v}}{|\mathbf{v}|}.$$

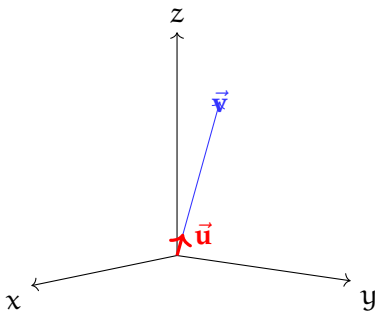
For example, if $\mathbf{v} = [3,4,5]$ then

$$|\mathbf{v}| = \sqrt{3^2+4^2+5^2} = \sqrt{50},$$

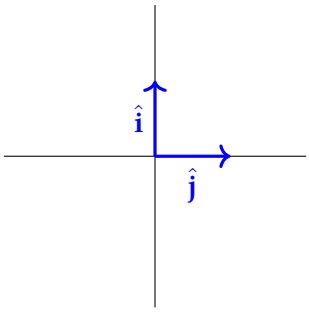
so

$$\vec{\mathbf{u}} = \frac{1}{\sqrt{50}}[3,4,5] = \left[\tfrac{3}{\sqrt{50}}, \tfrac{4}{\sqrt{50}}, \tfrac{5}{\sqrt{50}}\right].$$

$$\boxed{\hat{\mathbf{u}} = \frac{\mathbf{v}}{|\mathbf{v}|}}, \qquad |\vec{\mathbf{u}}| = 1.$$

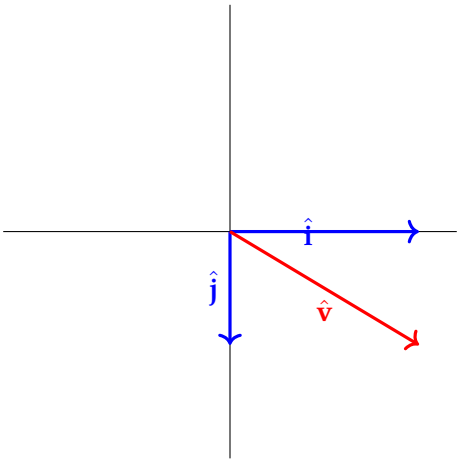



Unit vectors are often represented by placing a hat over the vector. In standard vector calculus, the variables $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{i}}$ for the x, y, and z variables respectively.




You may see vectors provided to you in the form $\mathbf{v} = 5\hat{\mathbf{i}} - 3\hat{\mathbf{j}}$. You may say this as '5 times the unit vector in the x-direction plus the -3 times the unit vector in the y-direction'. The

same may apply for 3D vectors as well.




## *Exercise 120* Magnitude of a Vector

Find the magnitude of the following vectors:

- $[1, 1, 1]$
- $[-5, -5, -5]$ (that is the same as $-5 \times [1, 1, 1]$)
- $[3, 4, -4] + [-2, -3, 5]$

*Working Space*

*Answer on Page 1914*

## 39.5 Vectors in Python

NumPy is a library that allows you to work with vectors in Python. You might need to install it on your computer. This is done with `pip`. `pip3` installs things specifically for Python 3.

```
1 pip3 install NumPy
```

We can think of a vector as a list of numbers. There are also grids of numbers known as *matrices*. NumPy deals with both in the same way, so it refers to both of them as arrays.

The study of vectors and matrices is known as *Linear Algebra*. Some of the functions we need are in a sublibrary of NumPy called `linalg`.

As a convention, everyone who uses NumPy, imports it as *np*.

Create a file called `first_vectors.py`:

```
import NumPy as np

# Create two vectors
v = np.array([2,3,4])
u = np.array([-1,-2,3])
print(f"u = {u}, v = {v}")

# Add them
w = v + u
print(f"u + v = {w}")

# Multiply by a scalar
w = v * 3
print(f"v * 3 = {w}")

# Get the magnitude
# Get the magnitude
mv = np.linalg.norm(v)
mu = np.linalg.norm(u)
print(f"|v| = {mv}, |u| = {mu}")
```

When you run it, you should see:

```
> python3 first_vectors.py
u = [-1 -2  3], v = [2 3 4]
u + v = [1 1 7]
v * 3 = [ 6  9 12]
|v| = 5.385164807134504, |u| = 3.7416573867739413
```

### 39.5.1 Formatting Floats

The numbers 5.385164807134504 and 3.7416573867739413 are pretty long. You probably want them rounded off after a couple of decimal places.

Numbers with decimal places are called *floats*. In the placeholder for your float, you can

specify how you want it formatted, including the number of decimal places.

Change the last line to look like this:

```
print(f"|v| = {mv:.2f}, |u| = {mu:.2f}")
```

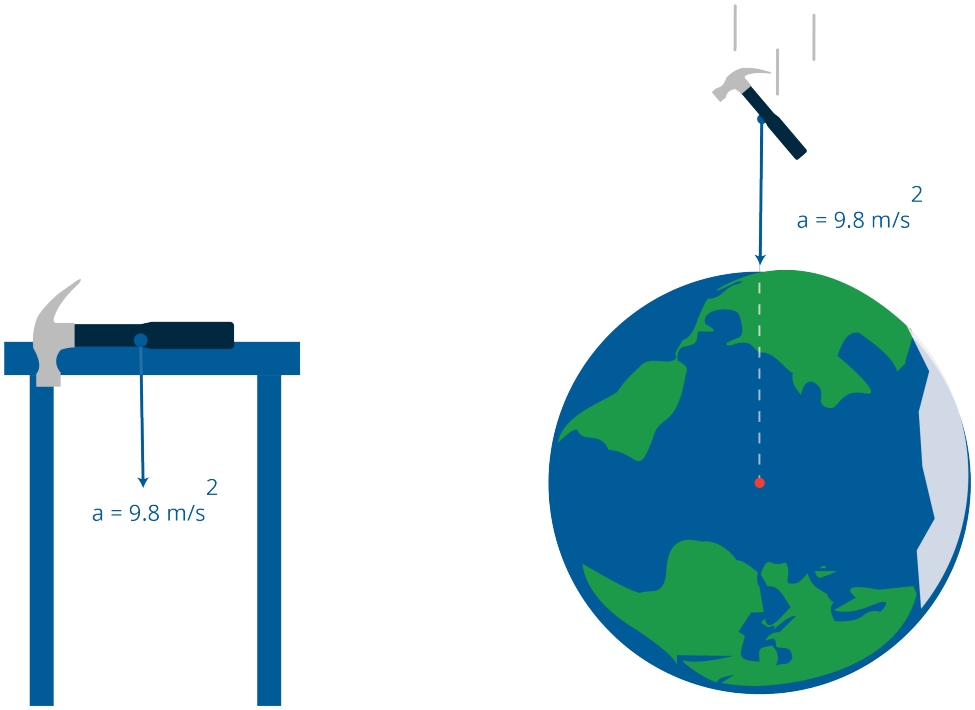
When you run the code, it will be neatly rounded off to two decimal places:

```
|v| = 5.39, |u| = 3.74
```

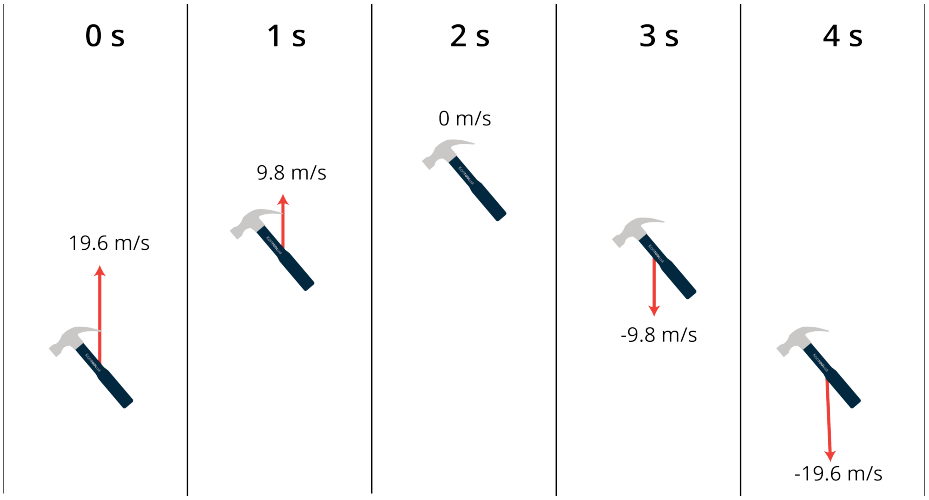
Chapter 40

# Falling Bodies

Gravity exists all around us. If you throw a hammer straight up in the air, from the moment it leaves your hand until it hits the ground, it is accelerating toward the center of the earth at a constant rate.




*Acceleration* can be defined as change in velocity. If the hammer leaves your hand with a velocity of 12 meters per second upward, one second later, it will be rising, and its velocity will have slowed to 2.2 meters per second. One second after that, the hammer will be falling at a rate of 7.6 meters per second. Every second the hammer's velocity is changing by 9.8 meters per second, and that change is always toward the center of the earth. When the hammer is going up, gravity is slowing it down by 9.8 meters per second, each second it is in the air. When the hammer is coming down, gravity is increasing the speed of its descent by 9.8 meters per second.




Acceleration due to gravity on earth is a constant negative 9.8 meters per second per second:

$$a = -9.8\frac{m}{s^2}$$

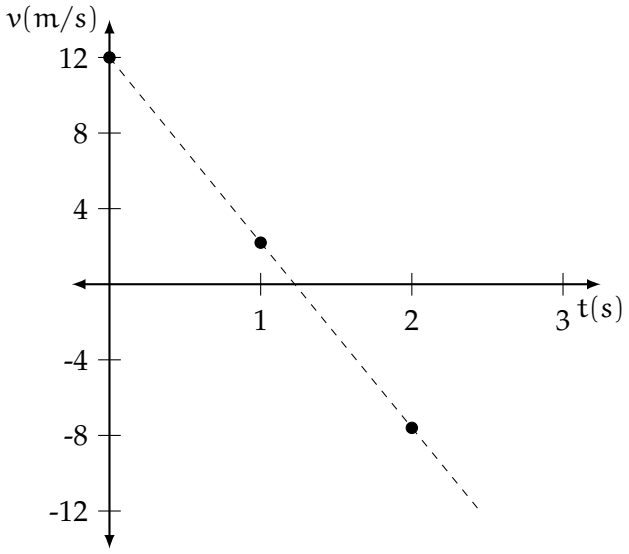
(Why is it negative? We are talking about height, which is generally considered to be increasing as you go away from the center of the earth. Since "up" is the positive direction, we take "down" as a negative direction. Therefore, since gravity pulls things "down", the acceleration due to gravity is considered negative.)

## 40.1 Calculating the Velocity

Given that the acceleration is constant, it makes sense that the velocity is a straight line. Assuming once again that the hammer leaves your hand at 12 meters per second, let's create a quick data table and graph of the hammer's velocity. Every second, we will subtract 9.8 m/s from the previous velocity.

| Time elapsed (s) | Velocity (m/s) |
|---|---|
| 0 | 12 |
| 1 | 2.2 |
| 2 | -7.6 |

Here is a plot of these data with a dashed fit line:




We can find an equation for the line: $v = 12 - 9.8t$. What is this saying? It says that at any time, t, after $t = 0s$, the velocity of the hammer is its starting velocity (12 m/s) plus the acceleration (-9.8 $m/s^2$) multiplied by the time elapsed (t). (Why did we say "plus" instead of "minus"? Well, since the acceleration *is negative*, $v_0 - 9.8t$ is the same as $v_0 + (-9.8t)$, and we are really adding the negative acceleration. When the acceleration is positive, we'll see a plus sign!)

## *Exercise 121* When is the apex of flight?

Given the hammer's velocity is calculated as $12 - 9.8t$, at what time (in seconds) does it stop rising and begin to fall?

*Working Space*

*Answer on Page 1914*

We can generalize this to other falling bodies. For any falling object with initial velocity $v_0$ and constant acceleration a, its velocity is given by:

$$v(t) = v_0 + a \cdot t$$

**Example**: The acceleration due to gravity on the Moon is approximately 1.63 $m/s^2$. If a hammer tossed upwards on the Moon takes 3.4 seconds to reach its apex, what was the hammer's initial velocity?

**Solution**: We know that $a = -1.63 m/s^2$ and $t = 3.4s$. Substituting and solving the velocity equation:

$$0\frac{m}{s} = v_0 + \left(-1.63\frac{m}{s^2}\right)(3.4s)$$
$$v - 0 = (1.63 \cdot 3.4)\,\frac{m}{s} \approx 5.5\frac{m}{s}$$

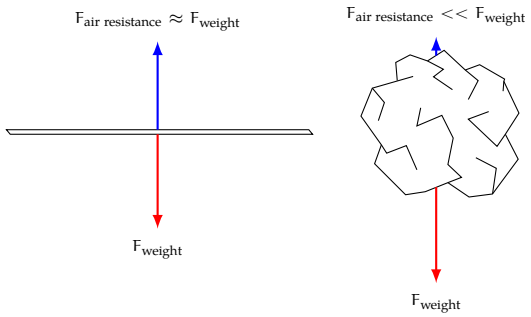
This hammer must have been tossed upwards with an initial velocity of approximately 5.5 m/s.

### 40.1.1 Air Resistance

At this point, we need to acknowledge air resistance. Gravity is not the only force on the hammer; as it travels through the air, friction with the air slows it down. This force is called *air resistance*, and for a large, fast-moving object (like an airplane) it is GIGANTIC force. For a dense object (like a hammer) moving at a slow speed (what you generate with your hand), air resistance doesn't significantly affect acceleration.

Consider two otherwise identical pieces of paper: one completely flat and unfolded, the other balled up as tightly as possible. If you dropped them both from the same height, which would hit the ground first? Intuitively, you know the balled-up paper would: but *why*? Since the papers have the same mass, you know that the force of gravity is the same on both papers! However, if you give this a try (go ahead, all you need to two sheets of printer paper and a safe place to drop them from), the acceleration of the two papers is different.

We can apply Newton's Second Law to explain our observations. First, let's draw diagrams to represent the forces acting on the papers:




For the crumpled paper, the decrease in surface area results in very little air resistance. As a result, the net force on the crumpled paper is approximately equal to the paper's weight.

$$F_{net}^{crumpled} = F_{air\ resistance} + F_{weight} \approx F_{weight}$$

$$F_{\text{net}}^{\text{crumpled}} = ma_{\text{crumpled}} \approx F_{\text{weight}} = mg$$
$$a_{\text{crumpled}} \approx g$$

Thus, you observe the crumpled paper to fall at nearly -9.8 $m/s^2$. The flat paper, on the other hand, has a much larger surface area, and therefore experiences significantly more air resistance. Thus, the observed acceleration of the flat paper is much slower:

$$F_{\text{net}}^{\text{flat}} = F_{\text{air resistance}} + F_{\text{weight}}$$
$$ma_{\text{flat}} = F_{\text{air resistance}} + mg$$
$$a_{\text{flat}} = \frac{F_{\text{air resistance}}}{m} + g$$

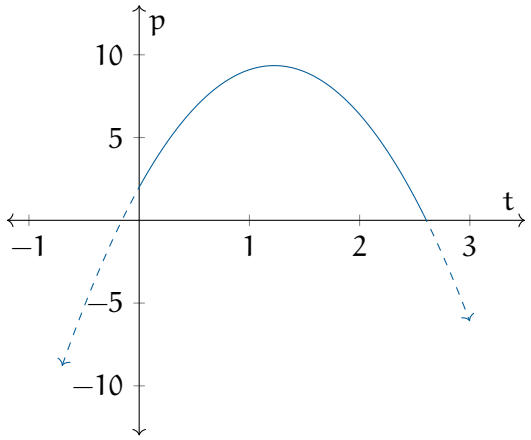
Since $g < 0$ and $F_{\text{air resistance}}$ is positive, the greater the air resistance the slower the paper falls.

## 40.2 Calculating Position

If you let go of the hammer when it is 2 meters above the ground, the height of the hammer is given by:

$$p = -\frac{9.8}{2}t^2 + 12t + 2$$

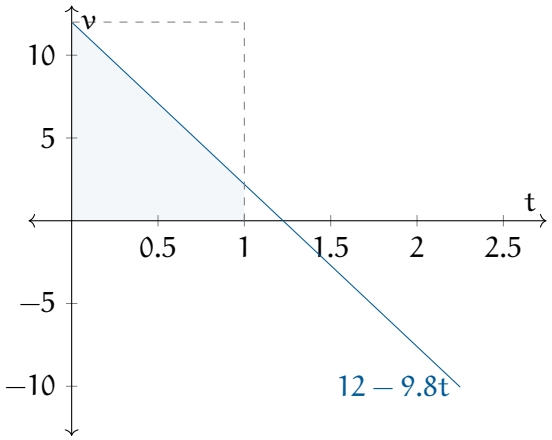
Here is a graph of this function:




How do we know? **The change in position between time 0 and any time t is equal to the area under the velocity graph between $x = 0$ and $x = t$.**

Let's use the velocity graph to figure out how much the position has changed in the first second of the hammer's flight. Here is the velocity graph with the area under the graph

for the first second filled in:



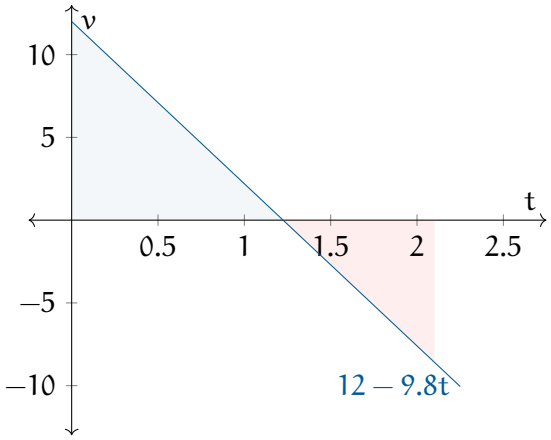

The blue filled region is the area of the dashed rectangle minus that empty triangle in its upper left. The height of the rectangle is twelve and its width is the amount of time the hammer has been in flight ($t$). The triangle is $t$ wide and $9.8t$ tall. Thus, the area of the blue region is given by $12t - \frac{1}{2}9.8t^2$.

That's the change in position. Where was it originally? 2 meters off the ground. This means the height is given by $p = 2 + 12t - \frac{1}{2}9.8t^2$. We usually write terms so that the exponent decreases, so:

$$p = -\frac{1}{2}9.8t^2 + 12t + 2$$

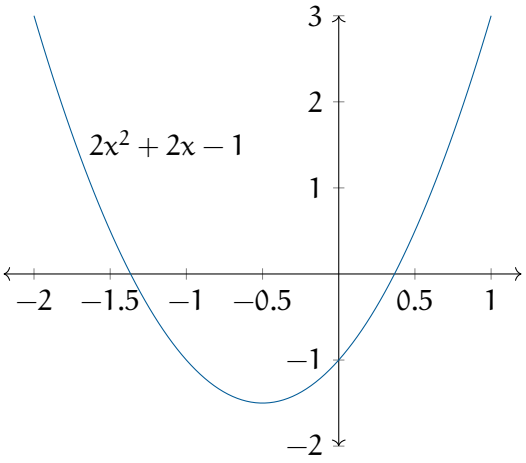
Finding the area under the curve like this is called *integration*. We say "To find a function that gives the change in position, we just integrate the velocity function." Much of the study of calculus is learning to integrate different sorts of functions.

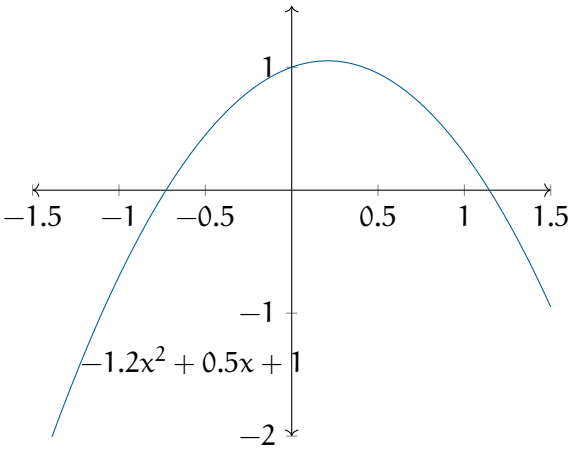
One important note about integration: Any time the curve drops under the $x$-axis, the area is considered negative. (Which makes sense, right? If the velocity is negative, the hammer's position is decreasing.)




## 40.3 Quadratic functions

Functions of the form $f(x) = ax^2 + bx + c$ are called *quadratic functions*. If $a > 0$, the ends go up. If $a < 0$, the ends go down.




The graph of a quadratic function is a *parabola*.

## 40.4 Simulating a falling body in Python

Now you are going to write some Python code that simulates the flying hammer. First, we are just going to print out the position, speed, and acceleration of the hammer for every 1/100th of a second after it leaves your hand. (Later, we will make a graph.)

Create a file called `falling.py` and type this into it:

```
# Acceleration on earth
acceleration = -9.8 # m/s/s

# Size of time step
time_step = 0.01 # seconds

# Initial values
speed = 12  # m/s upward
height = 2  # m above the ground
current_time = 0.0  # seconds after release

# Is the hammer still aloft?
while height > 0.0:

    # Show the values
    print(f"{current_time:.2f} s:")
    print(f"\tacceleration: {acceleration:.2f} m/s/s")
    print(f"\tspeed: {speed:.2f} m/s")
    print(f"\theight: {height:.2f} m")

    # Update height
    height = height + time_step * speed

    # Update speed
    speed = speed + time_step * acceleration

    # Update time
    current_time = current_time + time_step


print(f"Hit the ground: Complete")
```

When you run it, you will see something like this:

```
0.00 s:
        acceleration: -9.80 m/s/s
        speed: 12.00 m/s
        height: 2.00 m
0.01 s:
        acceleration: -9.80 m/s/s
        speed: 11.90 m/s
        height: 2.12 m
0.02 s:
        acceleration: -9.80 m/s/s
        speed: 11.80 m/s
        height: 2.24 m
0.03 s:
        acceleration: -9.80 m/s/s
        speed: 11.71 m/s
```

```
        height: 2.36 m
...
2.60 s:
        acceleration: -9.80 m/s/s
        speed: -13.48 m/s
        height: 0.20 m
2.61 s:
        acceleration: -9.80 m/s/s
        speed: -13.58 m/s
        height: 0.07 m
Hit the ground: Complete
```

Note that the acceleration isn't changing at all, but it is changing the speed, and the speed is changing the height.

We can see that the hammer in our simulation hits the ground just after 2.61 seconds.

### 40.4.1 Graphs and Lists

Now, we are going to graph the acceleration, speed, and height using a library called matplotlib. However, to make the graphs, we need to gather all the data into lists.

For example, we will have a list of speeds, and the first three entries will be 12.0, 11.9, and 11.8.

We create an empty list and assign it to a variable like this:

```
x = []
```

Next, we can add items like this:

```
x.append(3.14)
```

To get the first time back, we can ask for the object at index 0.

```
y = x[0]
```

Note that the list starts at 0. If you have 32 items in the list, the first item is at index 0; the last item is at index 31.

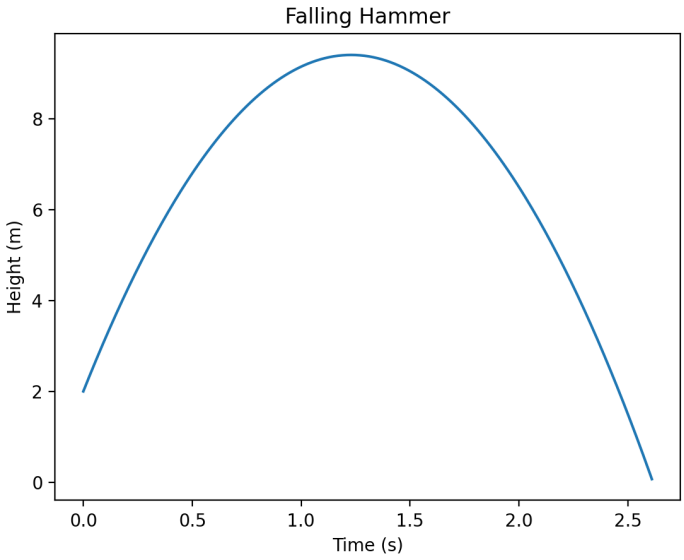
Duplicate the file `falling.py` and name the new copy `falling_graph.py`

We are going to make a plot of the height over time. At the start of the program, you will import the matplotlib library. At the end of the program, you will create a plot and show it to the user.

In `falling_graph.py`, add the bold code:

```
import matplotlib.pyplot as plt

# Acceleration on earth
acceleration = -9.8 # m/s/s

# Size of time step
time_step = 0.01 # seconds

# Initial values
speed = 12  # m/s upward
height = 2  # m above the ground
current_time = 0.0  # seconds after release

# Create empty lists
accelerations = []
speeds = []
heights = []
times = []

# Is the hammer still aloft?
while height > 0.0:

    # Add the data to the lists
    times.append(current_time)
    accelerations.append(acceleration)
    speeds.append(speed)
    heights.append(height)

    # Update height
    height = height + time_step * speed

    # Update speed
    speed = speed + time_step * acceleration

    # Update time
    current_time = current_time + time_step

# Make a plot
fig, ax = plt.subplots()
fig.suptitle("Falling Hammer")
ax.set_xlabel("Time (s)")
ax.set_ylabel("Height (m)")
ax.plot(times, heights)
plt.show()
```

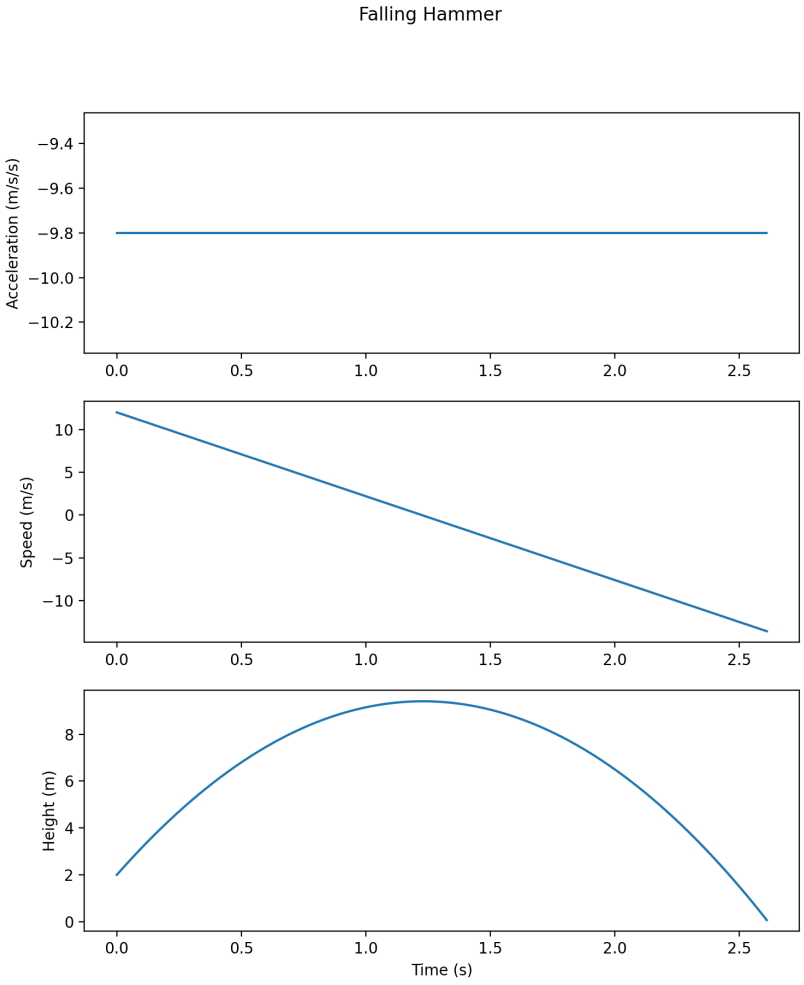
When you run the program, you should see a graph of the height over time.




It is more interesting if we can see all three: acceleration, speed, and height. So, let's make three stacked plots. Change the plotting code in `falling_graph.py` to:

```
# Make a plot with three subplots
fig, ax = plt.subplots(3,1)
fig.suptitle("Falling Hammer")

# The first subplot is acceleration
ax[0].set_ylabel("Acceleration (m/s/s)")
ax[0].plot(times, accelerations)

# Second subplot is speed
ax[1].set_ylabel("Speed (m/s)")
ax[1].plot(times, speeds)

# Third subplot is height
ax[2].set_xlabel("Time (s)")
ax[2].set_ylabel("Height (m)")
ax[2].plot(times, heights)
plt.show()
```

You will now get plots of all three variables:




This is what we expected, right? The acceleration is a constant negative number. The speed is a straight line with a negative slope. The height is a parabola. **The slope of the height graph is the speed, and the slope of the speed graph is acceleration**

A natural question at this point is "When exactly will the hammer hit the ground?" In other words, when does $\texttt{height} = 0$? The values of t where a function is zero are known as its *roots*. Height is given by a quadratic function. In the next chapter, you will get the trick for finding the roots of any quadratic function.

# Chapter 41

# Kinematics

How can we describe the motion of objects other than falling bodies? Kinematics is the description of motion.

**Kinematic Equations**

$$x_f = x_0 + v_0 t + \frac{1}{2} a t^2 \tag{41.1}$$

$$v_f = v_0 + a t \tag{41.2}$$

$$x_f = x_0 + \frac{1}{2} (v_f + v_0) t \tag{41.3}$$

$$v_f^2 = v_0^2 + 2a (x_f - x_0) \tag{41.4}$$

$$\Delta x - x_0 = v_{avg} t \tag{41.5}$$

Note that $v_0$ and $v_i$ are synonymous, some professors will use $v_0$ to mean "velocity not" or initial velocity.

Note that each equation is missing a certain variable (or two)

- (5.1) is missing $v_f$.
- (5.2) is missing $x_i$ or $x_f$
- (5.3) is missing $a$
- (5.4) is missing $t$
- (5.5) is missing $a$

It is important to note that these equations only work when the acceleration is held constant, which is referred to as *uniformly accelerated motion*.

**Example**: Terri and Jerry are running a race. Terri has a maximum acceleration of 3.4 $m/s^2$ and a top speed of 9.2 $m/s$. Jerry has a maximum acceleration of 3.7 $m/s^2$ and a top speed of 8.7 $m/s$. If the race is 200 $m$ long, who will win? Assume that each runner has maximum acceleration until they reach their top speed and that they will maintain that top speed once they reach it.

**Solution**: We want to know how long it takes each runner to complete the 200 $m$ race. For each runner, we will divide their run into two sections:

1. The time they are accelerating to their top speed
2. The time they are maintaining their top speed

We can use a table to track the results of our calculations:

| Runner | Terri | Jerry |
|---|---|---|
| Leg 1 (s) | | |
| Leg 2 (s) | | |
| Total (s) | | |

We'll begin with Terri's first leg. Taking the starting line as $x = 0$, we know that:

$$x_0 = 0 \text{ m}$$
$$v_0 = 0 \frac{\text{m}}{\text{s}}$$
$$a = 3.4 \frac{\text{m}}{\text{s}^2}$$

And since we want to know how long it takes Terri to reach her top speed, we also know that $v_f = 9.2$ m/s. Since we don't know how far Terri will run before she reaches her top speed (and we're not looking for that quantity), we need to select an equation that does not include $x$:

$$v_f = v_0 + at$$
$$9.2\frac{\text{m}}{\text{s}} = 0\frac{\text{m}}{\text{s}} + \left(3.4\frac{\text{m}}{\text{s}^2}\right) t$$
$$t = \frac{9.2\frac{\text{m}}{\text{s}}}{3.4\frac{\text{m}}{\text{s}^2}} \approx 2.7 \text{ s}$$

With a similar method, we can find how long it takes Jerry to reach his top speed:

$$t = \frac{v_f}{a} = \frac{8.7\frac{\text{m}}{\text{s}}}{3.7\frac{\text{m}}{\text{s}^2}} \approx 2.4 \text{ s}$$

Let's go ahead and record this in our table:

| Runner | Terri | Jerry |
|---|---|---|
| Leg 1 (s) | 2.7 | 2.4 |
| Leg 2 (s) | | |
| Total (s) | | |

Now that we know how much time it takes each runner to reach their top speed, we need to figure out how much time it takes them to complete the race from the point at which

each reaches their top speed. To do this, we will first have to find *where* each runner hits their top speed. (This is because we can't use $v_f = v_0 + at$ to find a time anymore, since from now on the runners' accelerations are zero, and all the other equations involve $x$.) For each runner, we know $a$, $v_0$, $v_f$, $x_0$, and $t$. You could choose any equation, but we will use this one:

$$x_f = x_0 + \frac{1}{2}(v_f + v_0)\,t$$

Since each runner begins on the starting line, $x_0 = 0$ m and $v_0 = 0$ m/s:

$$x_f = \frac{v_f \cdot t}{2}$$

For Terri:

$$x_f = \frac{\left(9.2\frac{m}{s}\right)(2.7\text{ s})}{2} \approx 12.4\text{ m}$$

For Jerry:

$$x_f = \frac{\left(8.7\frac{m}{s}\right)(2.4\text{ s})}{2} \approx 10.2\text{ m}$$

Now that we know where they reach their top speed, we can take that position as $x_0$ and find how long it takes each runner to reach the finish line at $x_f = 200$ m. Since $a = 0$ m/s$^2$, we can use:

$$x_f = x_0 + v_0 t$$

Rearranging to solve for $t$:

$$t = \frac{x_f - x_0}{v_0}$$

For Terri:

$$t = \frac{200\text{ m} - 12.4\text{ m}}{9.2\ \frac{m}{s}} \approx 20.4\text{ s}$$

And for Jerry:

$$t = \frac{200\text{ m} - 10.2\text{ m}}{8.7\ \frac{m}{s}} \approx 21.8\text{ s}$$

Completing our table, we see that Terri will win the race by finishing in the least amount of time:

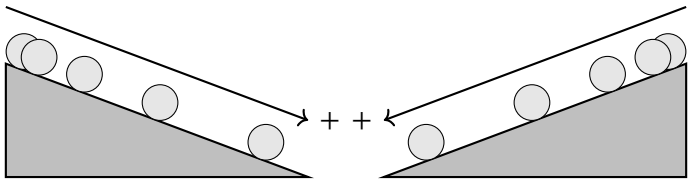
| Runner | Terri | Jerry |
|---|---|---|
| Leg 1 (s) | 2.7 | 2.4 |
| Leg 2 (s) | 20.4 | 21.8 |
| Total (s) | 23.1 | 24.2 |

## 41.1 Graphing Motion

### 41.1.1 Motion Diagrams

**Example**: Create a motion diagram of a ball rolling with a constant acceleration down an incline (the acceleration is in the same direction as motion).

**Solution**: The ball is accelerating down the ramp, so it will cover more distance each second. You could draw your ramp going left or right, as long as the distance covered each time interval increases as time passes.



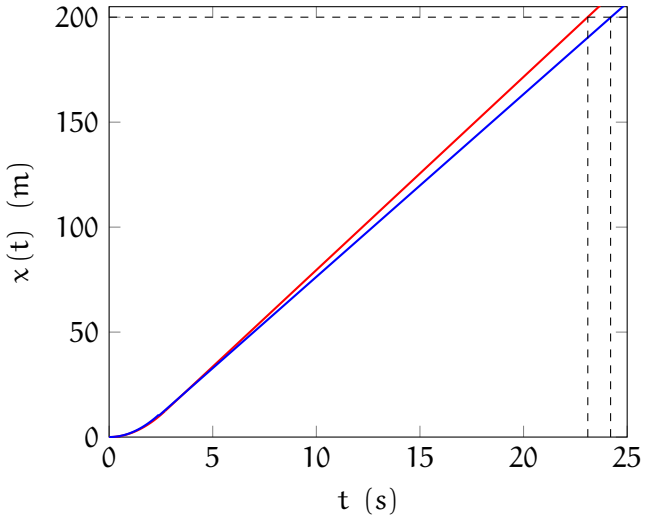
### 41.1.2 Position-Time and Velocity-Time Graphs

In the example problem above, graphs of each runner's motion would allow us to immediately see who would win. We can describe Terri's and Jerry's runs with piecewise functions:

$$x_{\text{Terri}}(t) = \begin{cases} \left(1.7\ \frac{\text{m}}{\text{s}^2}\right) t^2 & \text{if } 0 \leq t < 2.7\ \text{s} \\ 12.4\ \text{m} + \left(9.2\ \frac{\text{m}}{\text{s}}\right)(t - 2.7\ \text{s}) & \text{if } t \geq 2.7\ \text{s} \end{cases}$$

$$x_{\text{Jerry}}(t) = \begin{cases} \left(1.85\ \frac{\text{m}}{\text{s}^2}\right) t^2 & \text{if } 0 \leq t < 2.4\ \text{s} \\ 10.2\ \text{m} + \left(8.7\ \frac{\text{m}}{\text{s}}\right)(t - 2.4\ \text{s}) & \text{if } t \geq 2.4\ \text{s} \end{cases}$$

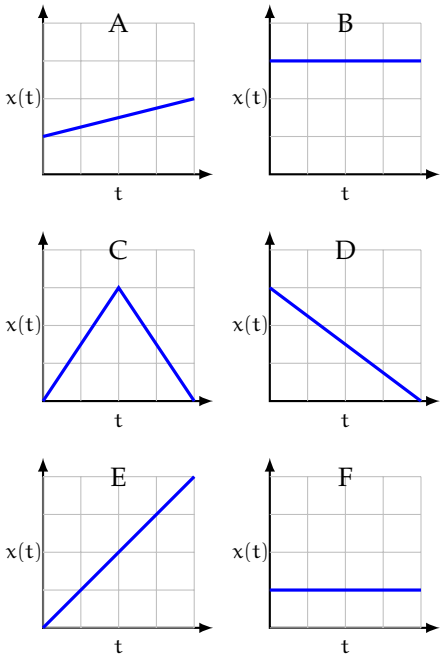
Graphing Terri in red and Jerry in blue:




Since the red function (Terri) crosses $x(t) = 200$ m first, we see that Terri will win.

## Exercise 122

The following graphs show the position of an object from t = 0 s to t = 10 s. The scales on the y-axes are all the same.




Rank the objects from least to greatest (all negative values are lower than all positive values) in terms of:

1. displacement from t = 0 s to t = 10 s.
2. instantaneous velocity at t = 7.5 s.
3. distance traveled from t = 0 s to t = 10 s

*Working Space*

*Answer on Page 1915*

## 41.2 Separation of Components

In the current chapter, we have learned how to use kinematics to describe one-dimensional motion. In the next chapter, you will learn to describe two-dimensional motion. It turns out that you can treat the different dimensions (horizontal and vertical motion) separately! Consider this scenario: A cannonball is shot from a cliff and the same instant an identical cannonball is dropped from the same cliff (see...). If the cannon is aimed horizontally, which cannonball will hit the ground first?

Go ahead and jot down what you think would happen: would the dropped ball hit first, the launched ball hit first, or would they both reach the ground below at the same time? Then, take a look at this video: https://www.youtube.com/watch?v=zMF4CD7i3hg. In it, two balls are released from the same height at the same time. One is dropped from rest while the other is launched horizontally (that is, its initial velocity is entirely in the $x$-direction). Based on the video, was your cannonball prediction correct? We'll learn to explain this phenomenon in the next chapter.

Chapter 42

# Projectile Motion

A projectile is an object that, once thrown or dropped, continues to move only under the influence of gravity. Throwing a baseball, shooting a cannon, and diving off a high diving board are all examples. NASA flight planners use projectile motion to plan flight paths for space vehicles, such as sending rovers to Mars. You've already learned how to describe and model one-dimensional projectile motion in the Falling Bodies chapter. Now, we consider projectiles that also have horizontal motion, and therefore are moving in two dimensions.

First, we will compare the motion of projectiles that are dropped versus horizontally launched from the same height. This will frame our discussion of the important concept of independence of motion: the vertical and horizontal motions of a projectile can be considered and described independent from each other. This will allow you to predict how far horizontally launched objects will travel before hitting the ground. Next, you'll learn to describe the motion of projectiles launched at an angle (like some heavy ground artillery). Finally, you'll use what you've learned to create a model of any projectile motion.

## 42.1 Comparing Projectiles

This video was mentioned at the end of the kinematics chapter: https://www.youtube.com/watch?v=zMF4CD7i3hg. From the video, we can see that the addition of horizontal motion does not effect how fast an object is acted upon by gravity. Both objects hit the ground at the same time, regardless of whether horizontal motion was added or not. This is because of a concept called *independence of motion*.

## 42.2 Independence of Motion

In projectile motion, such as a ball being thrown off of a cliff, the horizontal and vertical components of motion are independent of each other. This means that horizontal motion (and forces in the horizontal direction) do not affect vertical motion (and forces in the vertical direction), and vice versa. This is because gravity only acts in the vertical direction.[1]

If you recall the falling bodies chapter, you know that the vertical motion of an object

[1] This holds true for objects on sloped surfaces. Gravitational motion is just separated into components parallel and perpendicular to the slope. This will be covered in a future chapter.

| Horizontal Motion | Vertical Motion |
|---|---|
| $\Delta x = v_{0x}t$ | $\Delta y = v_{0y}t + \frac{1}{2}(-g)t^2$ |
| $v_x = v_{0x}$ (constant!) | $v_y = v_{0y} + (-g)t$ |
| $a_x = 0$ | $a_y = -g$ |

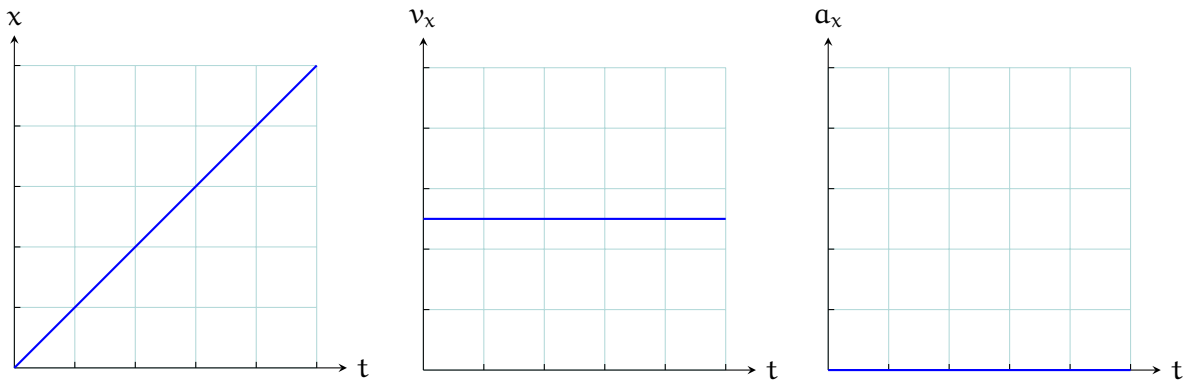
Figure 42.1: Projectile motion equations from kinematics equations.

in free fall is described by the equations of uniformly accelerated motion, with a constant acceleration of $9.8\frac{m}{s^2}$ downward (this may simplified to $10\frac{m}{s^2}$ for simplicity in many calculations).

In the horizontal direction, if we ignore air resistance (a common practice for elementary physics), there are no forces acting on the object. This means that the object will continue to move at a constant velocity in the horizontal direction. Note that from launch to landing, the time must be the same for both components. Take a minute to think about why this is if you are unsure.

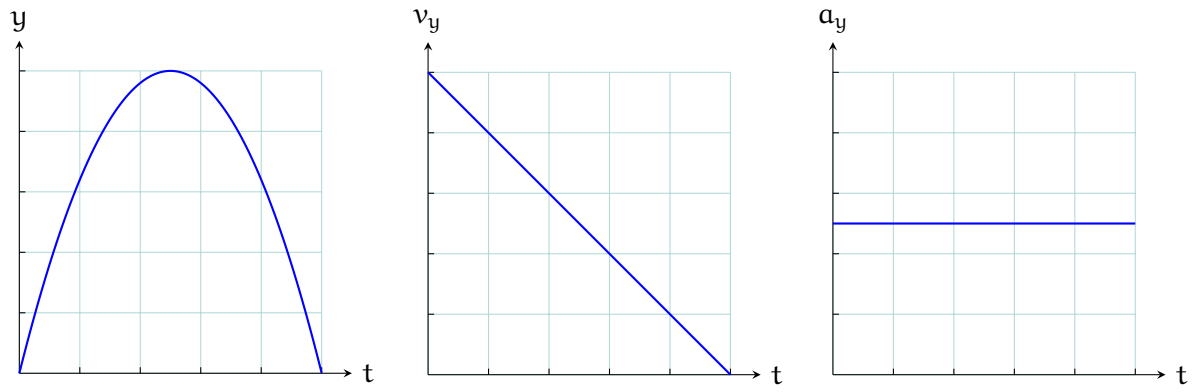
Because the horizontal and vertical motions are independent of each other, we can use different equations to describe the motion in each direction (See Figure 42.1). Note that the velocity in the horizontal direction is ***constant***, while the velocity in the vertical direction is ***not constant*** due to the acceleration of gravity ($9.8\frac{m}{s^2}$ downward).

Let's take a look at some simple graphs comparing x motions and y motions, shown in Figures 42.2 and 42.3. Note that these are a general representation of projectile motion, and the scales or values may not be accurate.




x-position vs. time x-velocity vs. time x-acceleration vs. time

Figure 42.2: Vertical motion graphs




y-position vs. time | y-velocity vs. time | y-acceleration vs. time

Figure 42.3: Vertical motion graphs

## 42.3 Horizontally-launched Projectiles

Imagine a ball is thrown horizontally off of a cliff. The ball has an initial horizontal velocity of 20 m/s, but no initial vertical velocity. The ball will continue to move horizontally at a constant velocity, while simultaneously accelerating downward due to gravity. The ball falls for 5 seconds before hitting the ground. We will use $-10\frac{m}{s^2}$ for the acceleration due to gravity to make calculations easier. See figure 42.4.

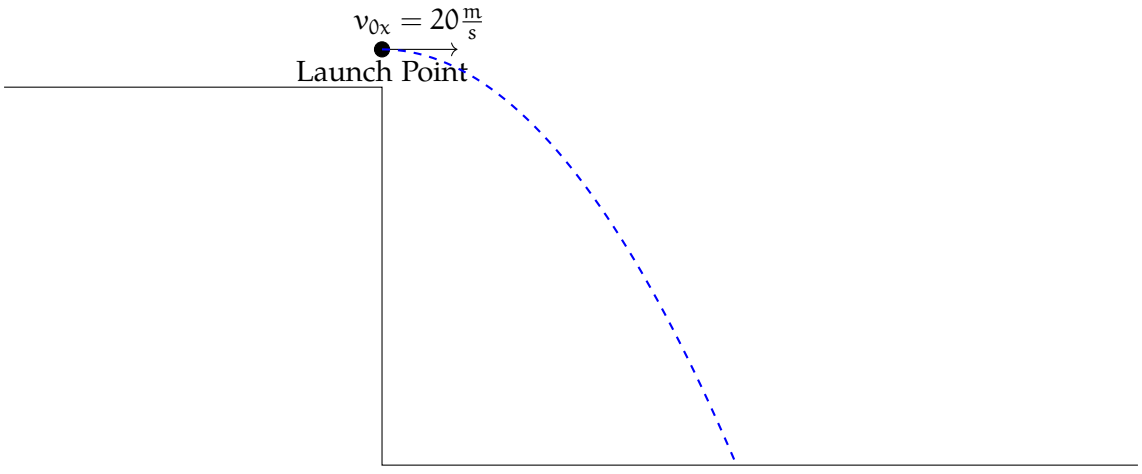



Figure 42.4: Diagram of a ball being thrown horizontally off of a cliff. Note: figure is not to scale.

Let's calculate the change in $y$.

The vertical motion can be described by the equation:

$$\Delta y = v_{0y}t + \frac{1}{2}(-g)t^2$$

Since the initial vertical velocity $v_{0y} = 0$, this simplifies to:

$$\Delta y = \frac{1}{2}(-g)t^2$$

Substituting in the values for $g$ and $t$:

$$\Delta y = \frac{1}{2}(-10\frac{m}{s^2})(5s)^2$$

Calculating this gives:

$$\Delta y = -125m$$

So the ball falls a vertical distance of 122.5 m from its origin before hitting the ground.

Now let's calculate the change in $x$. The horizontal motion can be described by the equation:

$$\Delta x = v_{0x}t$$

Substituting in the values for $v_{0x}$ and $t$:

$$\Delta x = (20\frac{m}{s})(5s) = 100m$$

## *Exercise 123* Horizontally-launched Projectile

A rock is thrown horizontally off of the edge of a cliff with an initial velocity of 8 m/s. If the rock falls for 3 seconds before hitting the ground, (a) how far from the base of the cliff does the rock land? (b) How high is the cliff?

Use $-10\frac{\text{m}}{\text{s}^2}$ for the acceleration due to gravity.

*Working Space*

*Answer on Page 1915*

### 42.3.1 Newton's Cannon and Escape Velocity

A real world application of horizontally launched projectiles is a thought experiment known as Newton's cannon (sometimes called Newton's cannonball). Newton theorized that a cannon on the top of a mountain above Earth's atmosphere could launch a cannonball at some high velocity could do one of three things:

- The cannonball would not have enough horizontal velocity, and fall back to the Earth.
- The cannonball would get shot at enough velocity to be in continuous orbit around the Earth. (This is how satellites stay in orbit around the Earth!).
- The cannonball would get shot at such a high velocity that it would escape the Earth's gravitational pull, and fly off into space. This is known as *escape velocity*.

Take a look at this simulation of Newton's cannon: https://physics.weber.edu/schroeder/software/NewtonsCannon.html. Then, watch this video: https://www.youtube.com/watch?v=ALRdYPMpqQs.

At anywhere below 7000 m/s, the cannonball falls back to Earth. In a range of 7,000 m/s to 8,000 m/s (the simulation doesn't go above 8,000 m/s), the cannonball enters an orbit around the Earth. For Earth, the escape velocity is $\approx$ 11,200 m/s.

For any given planet, the escape velocity can be calculated using the formula:

$$v_e = \sqrt{\frac{2GM}{d}}$$

where G is the gravitational constant, M is the mass of the planet, and d is the distance from the center of the planet to the object. We will cover this in more detail in the orbits chapter.

## 42.4 Projectiles launched at an Angle

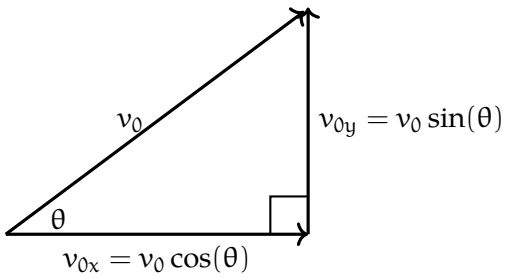
### 42.4.1 From the Ground

Now, let's imagine a ball is launched from the ground at an angle. The angle of launch influences both horizontal and vertical components of the initial velocity. We can use trigonometry to separate the initial velocity into its horizontal and vertical components.

$$v_{0x} = v_0 \cos(\theta) \qquad v_{0y} = v_0 \sin(\theta)$$

In the previous section, when an object is thrown directly horizontally, $\theta = 0°$, so $v_{0x} = v_0$. However, we now have $v_0$ being separated into two components, so the $v_{0x}$ will be less than $v_0$, as $\cos(\theta)$ and $\sin(\theta)$ are always less than or equal to 1.

The vertical component of $v_0$ is the initial velocity, $v_{0y}$. This component, unlike $v_{0x}$, is constantly changing due to gravity, at a rate of $-9.8\frac{m}{s^2}$. The peak of an object being launched at some angle is when $v_{0y}$ is equal to 0.

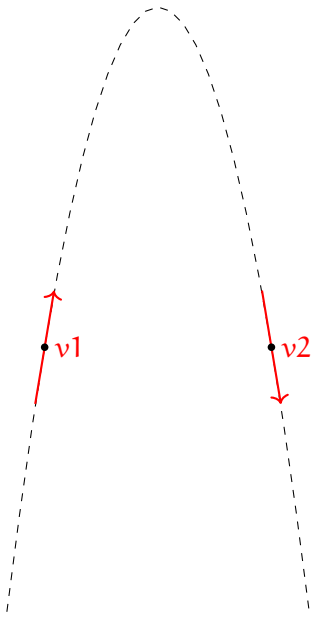



For example, take a watermelon launched at $20\frac{m}{s}$ at an angle of 30° above the horizontal. What would the horizontal and vertical components of the initial velocity be?

Well, we can use the equations above to find out:

$$v_{0x} = (20)\cos(30°) \approx 17.32 \text{ m/s}, \qquad v_{0y} = (20)\sin(30°) = 10 \text{ m/s}$$

It is important to note that the upwards velocity and downwards velocity are equal in magnitude at any given height (given the same vertical initial and final height), but *opposite in direction*, due to the symmetric nature of projectile motion.




$$\mathbf{v_1} = -\mathbf{v_2}$$

One last formula of note is the range formula, describing the horizontal range of any object launched at an angle on *level ground*:

**Range Formula**

$$R = \frac{v_0^2 \sin(2\theta)}{g} \tag{42.1}$$

where the object starts and lands at the same vertical height.

## *Exercise 124* Projectile Motion at an angle

A pumpkin is launched at 25 m/s at an angle of 40° above the horizontal. Find (a) the peak height of the pumpkin, and (b) the horizontal distance the pumpkin travels before hitting the ground. Use $-10\frac{\text{m}}{\text{s}^2}$ for the acceleration due to gravity. You may assume the ground is level the entire time.

*Working Space*

*Answer on Page 1916*

## *Exercise 125* Launch Angle for Maximum Range

(a) At what angle should you launch for an object to go the furthest given a maximum launch velocity? (b) How can you get the same horizontal distance with two different launch angles?

*Working Space*

*Answer on Page 1917*

### *Exercise 126* Projectile Motion at an angle

There is a target 100 meters away. I must shoot a bow at 14 meters per second. At what angle will I be able to hit my target? If I cannot hit the target, calculate the velocity needed to reach the target. Use $-10\frac{m}{s^2}$ for the acceleration due to gravity. You may assume the ground is level the entire time.

*Working Space*

*Answer on Page 1918*

#### 42.4.2 From a Height Above the Ground

Now we can imagine a ball being launched at some angle from a height above the ground. This is similar to the horizontally launched projectile, except now we have an initial vertical velocity component.

Let's say a ball is launched from a height of 50 m at an angle of 30° above the horizontal with an initial velocity of 20 m/s. We can find the horizontal and vertical components of the initial velocity as follows:

$$v_{0x} = (20)\cos(30^\circ) \approx 17.32 \text{ m/s}, \qquad v_{0y} = (20)\sin(30^\circ) = 10 \text{ m/s}.$$

The vertical motion can be described by the equation:

$$\Delta y = y_0 + v_{0y}t + \frac{1}{2}(-g)t^2 = 50 + 10t - 5t^2$$

Likewise, the horizontal motion of the projectile can be described by the equation:

$$\Delta x = v_{0x}t = 17.32t$$

To find the peak of the projectile, we can set $v_y = 0$ and solve for t:

$$\begin{aligned} v_y &= v_{0y} + (-g)t \\ 0 &= 10 - 5t^2 \\ t_{peak} &= \sqrt{2}\text{ s} \approx 1.414\text{ s} \end{aligned}$$

Substituting this time into the vertical and horizontal equations, we can find the peak height and horizontal distance at the peak:

$$\Delta y = 50 + 10(\sqrt{2}) - 5(\sqrt{2})^2 = 50 + 10\sqrt{2} - 10 = 40 + 10\sqrt{2} \approx 54.14\text{ m}$$

$$\Delta x = 17.32(\sqrt{2}) \approx 24.49\text{ m}$$

And to find the total time of flight, we can set $\Delta y = 0$ and solve for t:

$$\begin{aligned} 0 &= 50 + 10t - 5t^2 \\ 0 &= -5t^2 + 10t + 50 \\ 0 &= t^2 - 2t - 10 \end{aligned}$$

Using the quadratic formula, we find:

$$\begin{aligned} t &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-10)}}{2(1)} \\ &= \frac{2 \pm \sqrt{4 + 40}}{2} \\ &= \frac{2 \pm \sqrt{44}}{2} \\ &= 1 \pm \sqrt{11} \approx 4.32\text{ s (taking the positive root)} \end{aligned}$$

## 42.5 Simulating Projectile Motion

Chapter 43

# Free Body Diagrams

Now that you've mastered modeling *how things move*, you can learn to model *why things move*. Recall Newton's Second Law:

$$F = ma$$

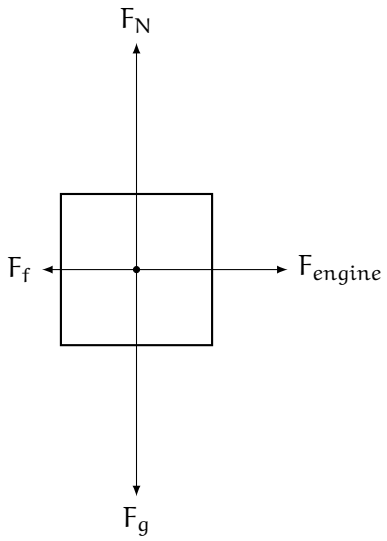
This is a simplification: what Newton's Second Law says in total is "the acceleration of a body is directly proportional to the *net force* acting on the body and inversely proportional to the mass of the body." Mathematically, that would be:

$$a = \frac{\Sigma F}{m}$$

Where $\Sigma F$ is the vector sum of all the individual forces acting on the body. This sum is also called the *net force*. To visualize the magnitude and direction of the net force, it can be very helpful to draw a free body diagram.

## 43.1 Interpreting Free Body Diagrams

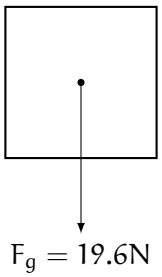
Before you learn to draw your own free body diagrams (FBDs), let's examine and analyze a few. Here is a FBD for an accelerating car:



We can consider the $x$ and $y$ axes separately. On the $y$-axis, we see that the weight ($F_g$) is the same magnitude as the normal force ($F_N$) because the arrows are the same length. This means in the $y$ direction, the forces are balanced, and thus we do not expect to see acceleration in the $y$ direction. This makes intuitive sense: if you are accelerating your car on a flat road, your car does not go up into the air. How about the $x$ direction? Here we see friction ($F_f$) pulling the car backwards while the engine is pushing the car forwards. In this case, the forces are *not* balanced. See how the $F_{engine}$ arrow is longer than the $F_f$ arrow? This means the forward force from the engine is *greater in magnitude* than the backward force of $F_f$. Therefore, the *net force* in the $x$ direction is to the right, and the car will accelerate to the right.

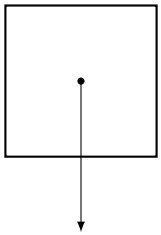
## 43.2 Drawing Free Body Diagrams

A free body diagram consists of the object, usually represented as a simple square, with various arrows representing the forces acting on the object. The direction and relative lengths of the arrows show the direction and relative strength of the forces. Here is a free body diagram for a 2-kg hammer in free fall through the air on Earth:



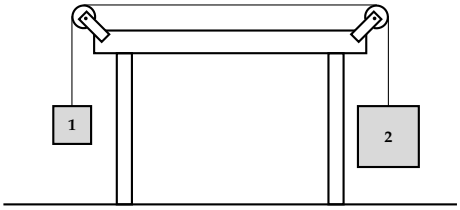

### 43.2.1 Objects At Rest

If an object is at rest, then its acceleration must be $0\frac{m}{s^2}$. By Newton's Second Law, the *net force* acting on that object must also be zero. Does this mean there are no forces acting on the object? NO! It means all the forces are *balanced*: for every upwards force, there is an equal downwards force, and so on. Consider the same 2-kg hammer, but this time it is sitting on a table. We know gravity must be acting on the hammer, so let's begin with that:


$F_g = 19.6N$

## *Exercise 127* Blocks on a Table

[This exercise was originally presented as a free-response question on the 2015 AP Physics 1 exam.] Two blocks are connected with a string and two pulleys over a table, as shown in the figure. Block 1 has a mass of 3 kg and Block 2 has a mass of 9 kg. The blocks are released from rest. Treat the string as massless and the pulleys as massless and frictionless when answering the questions below.

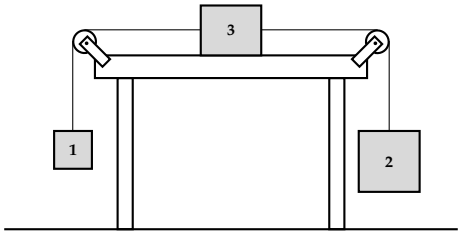



1. Draw free body diagrams for both blocks. Correctly show the relative magnitude of the forces using the relative lengths of the vectors.
2. Before doing any calculations, describe each block's acceleration as (a) positive or negative, (b) having a magnitude greater or less than $g$. Take the upward direction as positive.
3. Calculate the acceleration of each block. Take the upward direction as positive and express your answer in terms of $g$.

*Working Space*

*Answer on Page 1918*

Now here's an interesting question: what happens if we add another mass to the system? Imagine that a third block with a mass of 15 kg is laid on the table from the previous exercise and the string is attached to each side, as shown below.




We could redraw free body diagrams for all three blocks, noting that the tensions on either side of Block 3 will be different. Then, we would have another system of equations to solve and we could find the acceleration. Or, we could think about the entire system. Originally, the net force on the two-block system was 58.8 Newtons (the weight of the larger block minus the weight of the smaller block). And the mass of the entire two-block system was 12 kilograms. We can apply Newton's Second Law:

$$a = \frac{58.8\text{N}}{12\text{kg}} = 4.9\frac{\text{m}}{\text{s}^2} = \frac{1}{2}g$$

(Notice, this is the same answer we found in the previous exercise). What changes when we add a third, 15-kg block? The net force on the system doesn't change: the table pushes back up on the third block with a force equal to the block's weight. However, the total mass of the system has changed: it's increased. Now the total mass is 27 kg, and we can again apply Newton's Second Law:

$$a = \frac{58.8\text{N}}{27\text{kg}} \approx 2.18\frac{\text{m}}{\text{s}^2}$$

This answer makes sense: the mass of the system has increased while the net force acting on the system has stayed the same, resulting in a slower acceleration.

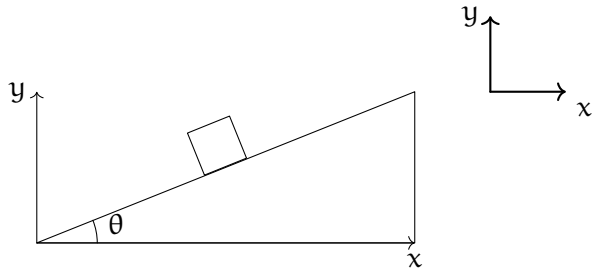
Now, what if instead of blocks resting on level tables, the blocks were on an incline? Let's take a look at this and more FBD's in the next chapter.

# Chapter 44

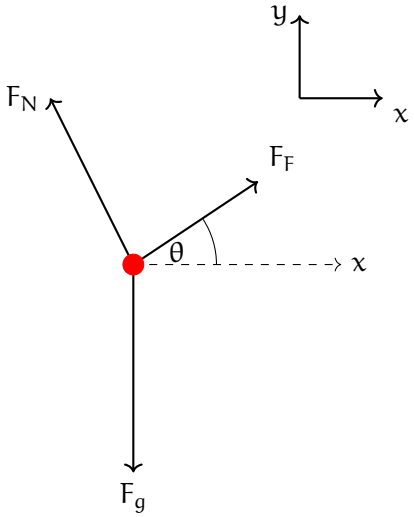
# Inclined planes

Take a look at this box resting on an incline plane due to static friction and gravity.

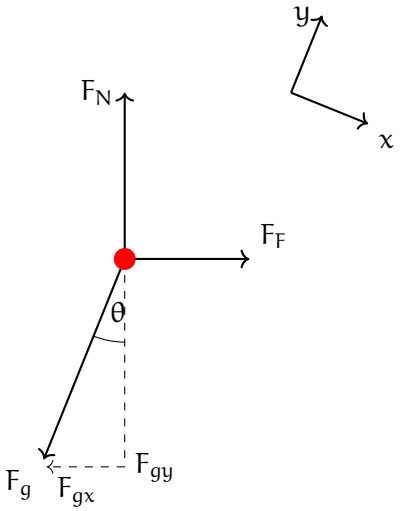
Figure 44.1: A box on an inclined plane with $a = 0$ due to static friction and gravity.




Now for objects on an incline, such as a box on a ramp, we will want to tip our coordinate system. Why you ask? Because we are not restricted to having our y-axis 90° vertically and our x-axis flat horizontally. In this case we can tip our axes so that the x-axis is parallel to the incline and the y-axis is perpendicular to the incline, aligning with the normal force. This will make our calculations much easier.




(a) Free body diagram in standard orientation.




(b) Axes rotated by θ.

Figure 44.2: Free body diagrams on an incline.

From this new perspective, we can see that the gravitational force $F_g$ can be broken down into two components: one parallel to the incline ($F_{gx}$)[1] and one perpendicular to the incline ($F_{gy}$). The normal force $F_N$ acts perpendicular to the surface of the incline, while the frictional force $F_F$ acts parallel to the surface, opposing motion. The coordinate planes in the corner are included to show the original coordinate system relative to our rotated FBD.[2]

We can set up the following equations based on Newton's second law:

$$\sum F_x = F_{gx} - F_F = ma_x = 0, \qquad \sum F_y = F_N - F_{gy} = ma_y = 0$$

Using the trigonometric components, we can express the forces in terms of the angle $\theta$:

$$\sum F_x = mg\sin(\theta) - F_F = ma_x = 0, \qquad \sum F_y = F_N - mg\cos(\theta) = ma_y = 0$$

Solving for the normal force and frictional force, we find:

$$F_N = mg\cos(\theta), \qquad F_F = mg\sin(\theta)$$

The following relationships are true for acceleration on an incline:

$$F_F = \mu F_N, \qquad a_x = g(\sin\theta - \mu\cos\theta)$$

Where $\mu$ is the coefficient of friction between the box and the incline.

Let's apply all of this to an example problem:

[1] Note that some professors will use down the incline as positive. This book will consistently use up the incline as positive as rotating coordinate plane aligns well with this.

[2] Note also that the dashed line components are not actual forces, they are drawn in for illustration of how forces at angles effect the standard forces.

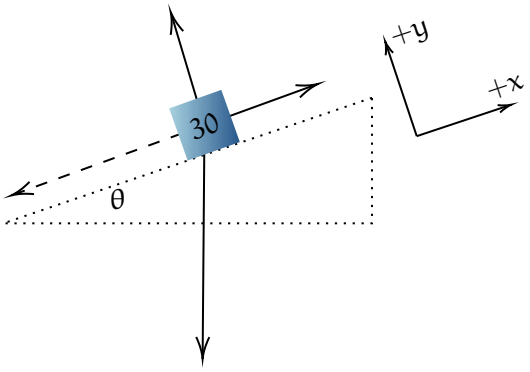
## *Exercise 128* Box sliding down an incline

A 30 kg block is sliding down an incline plane that makes a 30° with the horizontal. Given that the coefficient of kinetic friction is 0.3, find the following. Note that up the inclined plane is considered the positive x-direction.

(a) The normal force acting on the block.

(b) The frictional force acting on the block.

(c) The acceleration of the block down the incline.




*Working Space*

*Answer on Page 1919*

## *Exercise 129* Solving for angle $\theta$

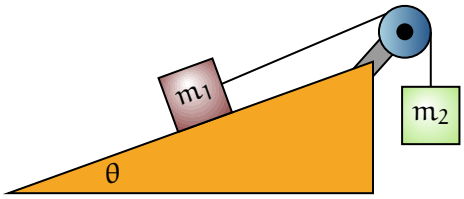
A 40 kg box is sitting on a rough inclined plane, held in place by static friction. The coefficient of static friction $\mu_s$ between the box and the incline is found to be 0.464. What is the maximum angle $\theta$ that the incline can be tilted to before the box starts to slide down?

*Working Space*

*Answer on Page 1920*

## *Exercise 130* Pulleys and Inclines

A block at rest of mass $m_1$ is resting on a rough inclined plane making an angle $\theta$ with the horizontal. The coefficient of kinetic friction between the block and the incline is $\mu_k$. The block is attached by a light string over a frictionless pulley to a hanging block of mass $m_2$. Once they are connected by a string, the hanging block begins to descend and the block of $m_1$ moves upwards along the incline. Draw the free body diagrams of each mass. Derive an equation for the acceleration of $m_2$, only in terms of $m_1, m_2, g, \theta,$ and $\mu_k$.



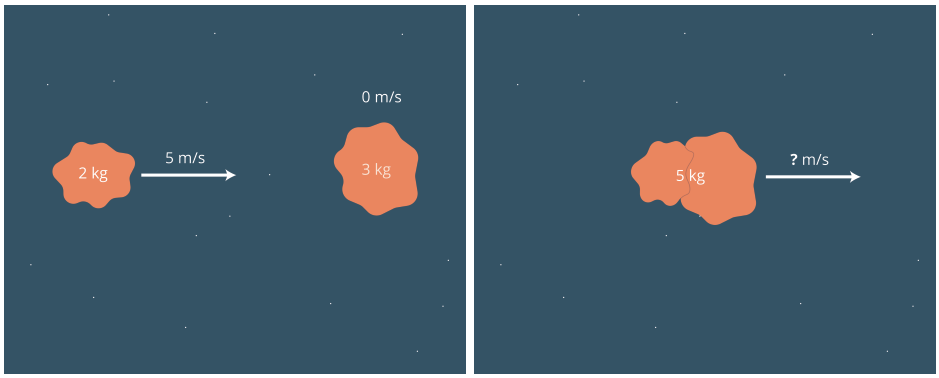

*Working Space*

*Answer on Page 1920*

# Chapter 45

# Momentum

Let's say a 2 kg block of putty is flying through space at 5 meters per second, and it collides with a larger 3 kg block of putty that is not moving at all. When the two blocks deform and stick to each other, how fast will the resultant big block be moving?




**Formula for Momentum**

Every object has *momentum*. The momentum is a vector quantity — it points in the direction that the object is moving and has a magnitude equal to its mass times its speed.

$$p = mv \tag{45.1}$$

Given a set of objects that are interacting, we can sum all their momentum vectors to get the total momenta. In such a set, the total momentum will stay constant.

$$p = p_1 + p_2 + \cdots + p_n = m_1v_1 + m_2v_2 + \cdots + m_nv_n \tag{45.2}$$

In our example, one object has a momentum vector of magnitude of 10 kg m/s, the other has a momentum of magnitude 0. Once they have merged, they have a combined mass of 5 kg. This means the velocity vector must have magnitude 2 m/s and pointing in the same direction that the first mass was moving.

**Conservation of Momentum**

**The total momentum of a system remains constant as long as no external forces act on it.** In a collision or interaction, the momentum before the event must equal the

momentum after the event:

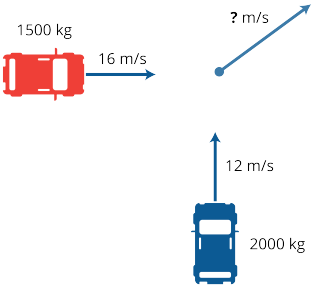
$$p_{\text{initial}} = p_{\text{final}} \tag{45.3}$$

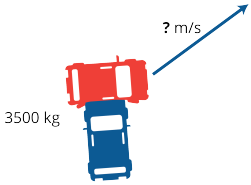
This applies whether the collision is elastic, inelastic, or perfectly inelastic. Even if kinetic energy is lost (as heat, sound, or deformation), momentum is still conserved because it depends only on mass and velocity, not on energy.

## *Exercise 131* Cars on Ice

A car weighing 1000 kg is going north at 12 m/s. Another car weighing 1500 kg is going east at 16 m/s. They both hit a patch of ice (with zero friction) and collide. Steel is bent, and the two objects become one. How what is the velocity vector (direction and magnitude) of the new object sliding across the ice?

*Working Space*




*Answer on Page 1921*

Note that kinetic energy ($1/2mv^2$) is *not* conserved here. Before the collision, the moving putty block has $(1/2)(2)(5^2) = 25$ joules of kinetic energy. Afterward, the big block has $(1/2)(5)(2^2) = 10$ joules of kinetic energy. What happened to the energy that was lost? It was used up deforming the putty.

What if the blocks were marble instead of putty? Then there would be very little deforming, so kinetic energy *and* momentum would be conserved. The two blocks would end up having different velocity vectors.

Let's assume for a moment that they strike each other straight on, so there is motion in only one direction, both before and after the collision. Can we solve for the speeds of the

first block ($v_1$) and the second block ($v_2$)?

We end up with two equations. Conservation of momentum says:

$$2v_1 + 3v_2 = 10$$

Conservation of kinetic energy says:

$$(1/2)(2)(v_1^2) + (1/2)(3)(v_2^2) = 25$$

Using the first equation, we can solve for $v_1$ in terms of $v_2$:

$$v_1 = \frac{10 - 3v_2}{2}$$

Substituting this into the second equation, we get:

$$\left(\frac{10 - 3v_2}{2}\right)^2 + \frac{3v_2^2}{2} = 25$$

Simplifying, we get:

$$v_2^2 - 4v_2 + 0 = 0$$

This quadratic has two solutions: $v_2 = 0$ and $v_2 = 4$. $v_2 = 0$ represents the situation before the collision. Substituting in $v_2 = 4$:

$$v_1 = \frac{10 - 3(4)}{2} = -1$$

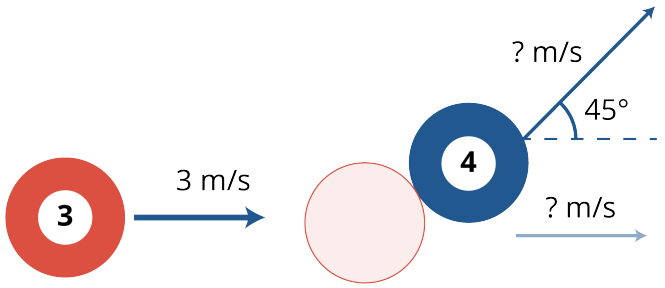
Thus, if the blocks are hard enough that kinetic energy is conserved, after the collision, the smaller block will be heading in the opposite direction at 1 m/s. The larger block will be moving at 4 m/s in the direction of the original motion.

## *Exercise 132* Billiard Balls

*Working Space*

A billiard ball weighing 0.4 kg and traveling at 3 m/s hits a billiard ball (same weight) at rest. It strikes obliquely (neither perpendicular nor parallel), so that the ball at rest starts to move at a 45 degree angle from the path of the ball that hit it.

Assuming all kinetic energy is conserved, what is the velocity vector of each ball after the collision?




*Answer on Page 1922*

## 45.1 Impulse

We can talk about a *change in Momentum* as what we refer to as *Impulse*. When an object has a change in momentum, it is said to have been given an Impulse. Since momentum is a vector quantity, impulse is as well.

**Formula for Impulse**

Impulse, **J** said to be the change in momentum, is given by:

$$\mathbf{J} = \Delta \mathrm{p} \tag{45.4}$$

Equivalently, it can be given as the following equation, in terms of force:

$$\mathbf{J} = F\Delta t \tag{45.5}$$

If the force varies with time, we use integration to find the impulse:

$$\mathbf{J} = \int_{t_0}^{t_1} F(t)dt \tag{45.6}$$

Both Equations (45.5) and (45.6) are referred to as the **Impulse-Momentum Theorem**.

By the impulse momentum theorem, a large force for a short time *or* a small force for a long time can produce the same impulse.

### 45.1.1 Golf Swings

The best example of impulse in action is a golf swing. Let's analyse this using equations:

The force of your swing will theoretically always be the same, as the maximum force you can apply is limited by human ability. You cannot change the human-provided force, however, you can increase the *contact time* of your club on the ball.

$$\mathbf{J} = F\Delta t$$

Since J is equivalent to $\Delta p$, $\Delta t$, the contact time of the swing, has a proportional relationship with the momentum of the ball, which starts of as 0. Thus, it can be said:

$$\Delta p \propto \Delta t$$

And, holding $F$, $m$, and assuming the golf ball is initially at rest, $p_i = 0$, we can say

$$v_f \propto \Delta t$$

So the longer the contact time of a golf swing (or any swing-based sport, really), the greater the velocity.

## 45.2 Collisions

When two (or more) objects collide, we can classify their collision as one of three main categories. These classifications tell us when we can apply the conservation of momentum, the conservation of kinetic energy, both, or neither. Momentum is only conserved when

the system is isolated, meaning no external forces act on the system.

### 45.2.1 Elastic Collision

Recall our billiard ball problem from Exercise 132. That problem provides both balls with velocity after the collision. In an ideal and more realistic billiards scenario, one ball transfers all of its velocity onto the other ball (in this case, the red ball stops and the blue continues with close to same velocity that the red ball had initially), as seen in Figure 45.1 This would be a *elastic collision*.

In an elastic collision, *both momentum and kinetic energy are conserved*. There is minimal to zero loss of energy in the collision. Although no collision is ever *truly* elastic (due to existing but minimal deformation of objects, sound, a transfer of heat through molecular changes, etc), we can think of a collision like this as perfectly elastic. The sound of billiard balls colliding is obsolete (especially compared to a car crash), so very little energy is lost to sound.

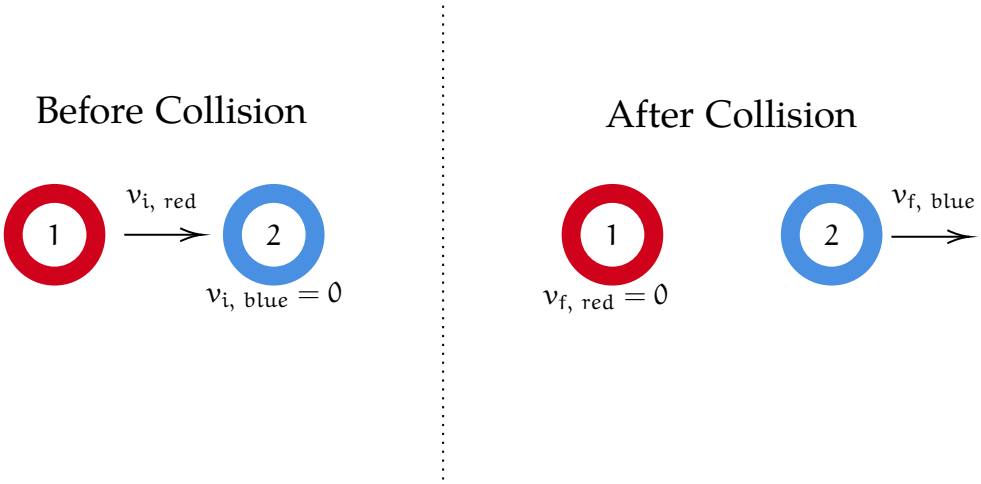



Figure 45.1: Example of an elastic biliard collision.

Another example to consider here is Newton's Cradle. There are two videos demonstrating momentum in Newton's Cradle on your digital resources.

In equation format, Elastic Collisions are represented by the sums of momentum and kinetic energy before and after the collision being equal:

$$\text{total } \mathbf{p}_{\text{before collision}} = \text{total } \mathbf{p}_{\text{after collision}} \implies m_1 v_{i,1} + m_2 v_{i,2} + \cdots = m_1 v_{f,1} + m_2 v_{f,2} + \cdots$$

*and*

$$\text{total } \mathbf{KE}_{\text{before collision}} = \text{total } \mathbf{KE}_{\text{after collision}}$$

### 45.2.2 Inelastic Collisions

Inelastic collisions, then, are ones in which the total kinetic energy after the collision is *different* than before the collision, in other words, there is a change in kinetic energy, usually lost due to friction, heat, or deformation.

However, momentum *is* conserved, meaning we can apply conservation of momentum principles.

$$\text{total } \mathbf{p}_{\text{before collision}} = \text{total } \mathbf{p}_{\text{after collision}} \implies m_1 v_{i,1} + m_2 v_{i,2} + \cdots = m_1 v_{f,1} + m_2 v_{f,2} + \cdots$$

A good example of an elastic collision is dropping a basketball on a floor from an initial height. It will never return to its initial height after "colliding" with the floor, as kinetic energy is 'lost' to sound and deformation of the ball. The missing height can be equated to the lost energy. See Figure 45.2

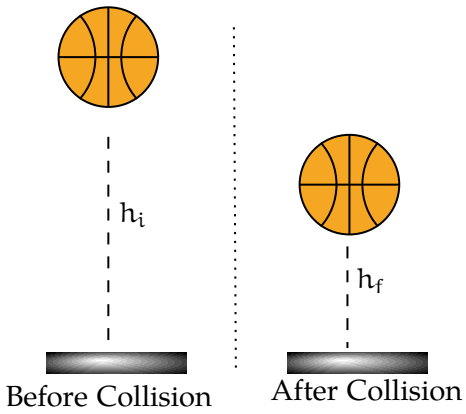



Figure 45.2: An example of a (partially) inelastic collision between the floor and a basketball.

### 45.2.3 Perfectly Inelastic

A collision is *perfectly inelastic* when the masses stick together after the collision.In this kind of collision, the *maximum* kinetic energy is lost (crumpling, bending, embedding of objects) and share the same final velocity.

The most common example of a perfectly inelastic collision is a car crash, especially high speed T-Bones. In this case, a car at a high speed collides with a car at a lower speed, resulting in a combined final speed as the cars interact on a molecular level and a large portion of energy is lost.

## *Exercise 133* Two Carts

Two carts on a frictionless track collide elastically.

Cart A has mass 2.0 kg and is moving to the right at 3.0 m/s.

Cart B has mass 1.0 kg and is initially at rest.

They collide head-on elastically.

Find the final velocities of both carts. Hint: You may have to do a few equation substitutions

*Working Space*

*Answer on Page 1923*

## *Exercise 134* Car Crash

Sometimes, car companies will intentionally crash cars in order to test the safety of their prototypes.

Two cars collide head-on on a straight, frictionless test track, where east is considered positive.

Car A has mass 1000 kg and is traveling *east* at 20 m/s. Car B has mass 1500 kg and is traveling *west* at 10 m/s.

During the collision, the cars do not stick together. After the collision, Car A is observed to move east at 5 m/s.

Find the final velocity of Car B. You are told kinetic energy is not conserved. Is this an elastic or inelastic collision?

*Working Space*

*Answer on Page 1925*

### *Exercise 135* Ready, Aim, Fire!

A $m_{bullet} = 10$ g bullet is fired horizontally at 500 m/s into a wooden block of mass $m_{block} = 2.00$ kg resting on a frictionless horizontal surface. The bullet embeds itself in the block, such that a loud bang was heard and the *block-bullet system* slides together after the collision.

(a) What is the velocity of the *block-bullet system* after the collision?

(b) How much kinetic energy was lost in the collision?

(c) A collision is considered *perfectly inelastic* if more than 95% of the initial energy is lost. Classify the *block-bullet system* collision by first finding the percent of energy lost.

*Working Space*

*Answer on Page 1925*

## 45.3 Momentum, Center of Mass, and Explosions

We have previously talked about finding the center of mass of an object (or objects). Now that we know about momentum, we can think about the center of mass in a different way.

Explosions typically involve an object breaking into pieces with internal forces (the forces of the explosion acting between fragments). The key is:

**Internal forces cannot change the motion of the center of mass.** This means that even during a violent explosion, the center of mass of the system continues moving exactly as it would if no explosion occurred. How does this work? It involves finding the sum of all momenta of the object.

*Before and after the explosion:* Total external force on the system is the only thing that can change center of mass (COM) motion.

Internal forces (the explosion pushing fragments apart) cancel out due to Newton's Third Law.

Therefore, momentum of the whole system stays the same (as long as there is no external *impulse*).

Let's say a projectile of mass M is moving, then explodes into two pieces with masses $m_1$ and $m_2$. Even if the fragments fly off in different directions with different speeds, the COM follows:

- the same trajectory,
- at the same velocity,
- as the original mass would have had if it had not exploded.

What can you do with this information?

- Find missing fragment velocities
- Find angles or directions after explosions
- Track motion of COM even when individual parts are complicated

A classic example of this is a mass following projectile motion. The mass splits into pieces at the peak of its motion. Even if pieces fly backwards, upwards, or along the same path, the COM stays on the original path.

If before the explosion the object moves with velocity $v$ then the COM velocity after explosion is still:

$$\vec{v}_{\text{COM}} = \vec{v}_{\text{before}}$$

After explosion:

$$M\vec{v}_{\text{COM}} = m_1\vec{v}_1 + m_2\vec{v}_2 + \dots$$

You use this to solve unknown fragment velocities or angles.

## *Exercise 136* Three-Fragment Projectile Explosion

*Working Space*

A projectile of mass 6.0 kg is launched from the ground and follows a parabolic arc. At the very top of its trajectory, its speed is measured to be 20 m/s, moving horizontally.

At that instant, it explodes into three fragments:

- Fragment A has mass 2.0 kg and flies off at 40° above the horizontal with a speed of 30 m/s.
- Fragment B has mass 1.0 kg and flies off at 60° below the horizontal with a speed of 25 m/s.
- Fragment C has mass 3.0 kg, and its direction is unknown, but it is known to move *somewhere* in the horizontal plane after the explosion.

Gravity acts only after the explosion. During the explosion itself, assume there are no external horizontal forces.

Find the **magnitude and direction** of the velocity of Fragment C.

*Answer on Page 1926*

This chapter has talked about Momentum, Impulse, Types of Collisions, and Explosions, all of which are intertwined topics in Newtonian Physics. Next, we will be talking about physics behind vehicles such as planes, quadcopters, helicopters, and boats.

CHAPTER 46

# Boats

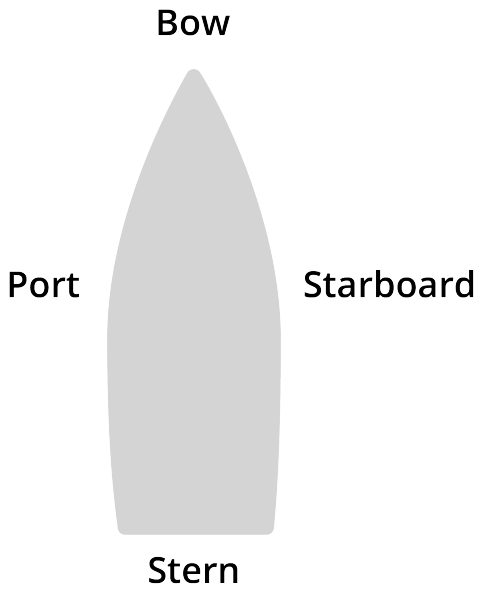
Engineers have been building boats for centuries. Through boat design, humanity learned the lessons that made airplanes and rockets possible. Learning about how boats work will give you a foundation for understanding more advanced concepts we will be covering later.

## 46.1 Basic Terminology

The front of a boat is called *the bow* (pronounced exactly the same as "bough"). The back of the boat is called *the stern*.

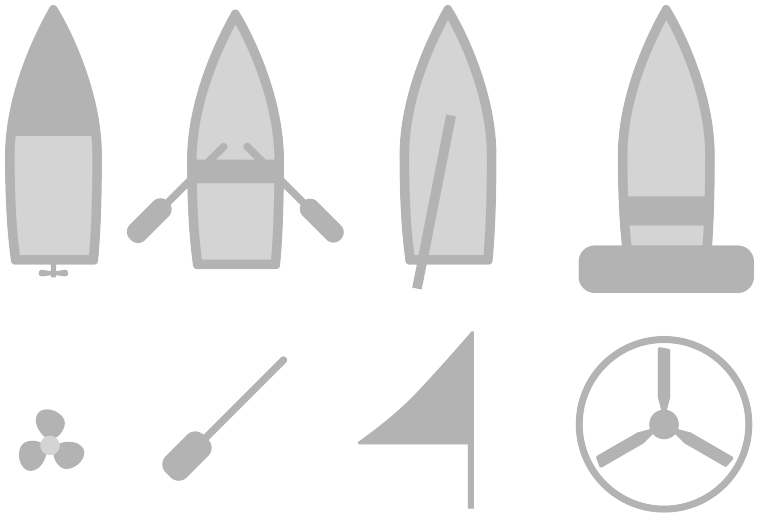
The underside of the boat is called *the hull*. The top of the boat is called *the deck*.

If you are standing at the stern and looking toward the bow, everything on your left is the *port* side. Everything on your right is the *starboard* side.




There are several different ways that boats are propelled:

- A motor turns a screw in the water, as in a motorboat. The screw is known as a *propeller*.
- A human pushes the water with a big spatula. If the spatula is attached to the boat with a pivot (as in a rowboat) it is an *oar*. If the spatula is not attached to the boat (as in a canoe), it is a *paddle*.
- A human pushes the ground beneath the water with a long pole. These boats are called *punts*, and are mainly used in more shallow waters where an oar or a paddle would be less convenient.
- The wind pushes the boat, as in a sailboat. The sails are held up by a *mast*.
- Some boats have a big fan that propel the boat. These are called *airboats*. Airboats are not the most efficient boats, but they can travel on waterways with water just a couple of inches deep.



In the terms of physics, each of these method provide a *thrust vector* which is applied to the boat at a particular place and in a particular direction.

### 46.1.1 Nautical Miles and Knots

A *nautical mile* is use to measure distances on a boat. The speed of a boat is usually measured in *knots*, which is just a quick way of saying "nautical miles per hour".

What is a nautical mile?

Latitude is the angle that a boat is above or below the equator. It is measured in degrees, minutes, and seconds. 60 minutes make one degree. 60 seconds make one minute. So, for example, if you are 12.45164° north of the equator, a sailor might say you were at Latitude 12°27′5"N. She might also say that you are a little more than 747 nautical miles from the equator. $(12 \times 60 + 27 = 747)$

A nautical mile is the distance of one minute of latitude. It is slightly longer than the normal (or *statute*) that you use on land: one nautical mile equals 1.151 statute miles or 1.852 km per hour.

## 46.2 Why Boats Float Upright

Early in this sequence, we discussed buoyancy as a quantity. The magnitude of the buoyant force is equivalent to the weight of the liquid displaced.

We can also talk about the direction of the buoyant force: buoyancy pushes in the opposite direction as gravity.

How do we design boats so that they don't flip over?

### 46.2.1 Center of Buoyancy

Let's say you have a rowboat. If you push down on point on the floor near the front, the front of the boat will go down and the back of the boat will rise. In other words, the boat will rotate in the direction you are pushing. Likewise, if you push on the floor near the back of the boat, the back will sink a little lower and the front will rise. However, there is a place, near the center of the boat, where if you push down, the boat will not rotate at all; it will simply sink a little lower in the water. That point is known as the *center of buoyancy*.

How can we calculate the center of buoyancy? Imagine the shape of the water that was displaced by the boat. Now imagine that shape filled with water. The center of mass of that water is the center of buoyancy of the boat.

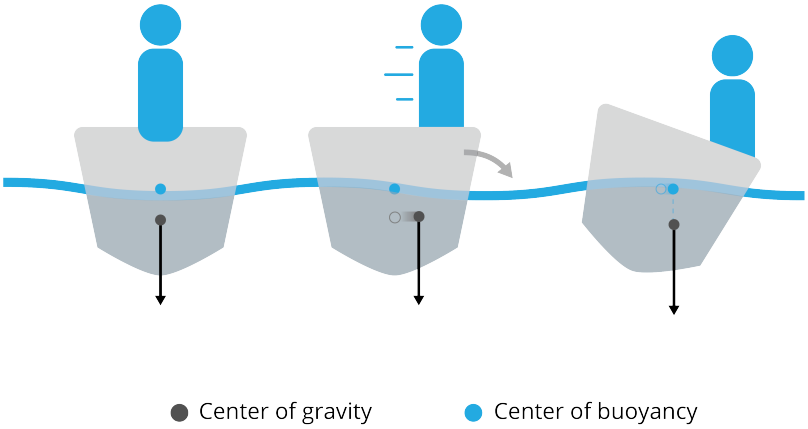
### 46.2.2 Center of Mass

Your boat and everything in it can be thought of as one object. That object has a center of mass. If you found the center of mass, you could balance the whole boat on it.

In a boat, if you move your body from the center of the boat to once side, you will have moved the center of mass. The boat will lean in that direction, which will change the center of buoyancy.

If you imagine a line is parallel to the force of gravity that passes through the center of mass of your boat, the boat will continue to increase its lean until the center of buoyancy is on that line.

If water comes over the sides of the boat before the center of gravity and center of buoyancy align, your boat will sink.



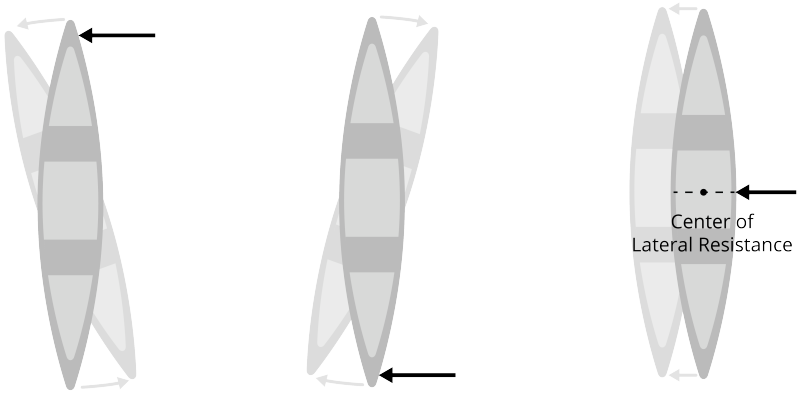

## 46.3 Center of Lateral Resistance

It isn't enough for a boat to float — for a boat to be useful, it must also be able to travel in a straight line.

Imagine that you are standing knee-deep in a lake next to a canoe. If you push the front of the canoe away from you, it will rotate — the back end will actually swing toward you. There is a point near the middle of the canoe where if you push it will not rotate in either direction — the boat will just slide sideways. This point is known as the *center of lateral resistance*.

The trick to making a boat travel in a straight line is to make sure that the line that contains the thrust vector passes through the center of lateral resistance.

An outboard motor allows you to direct the thrust vector. When the line of thrust passes the center of lateral resistance on the starboard side of the center of lateral resistance, the boat turns toward the port side.

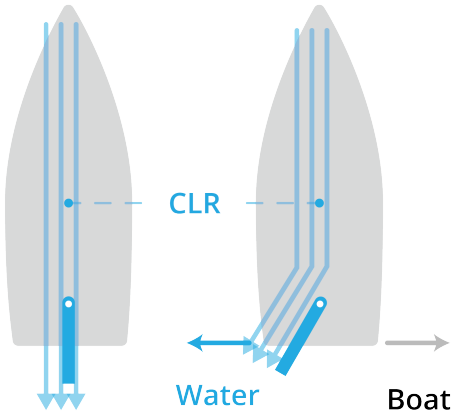



## 46.4 Steering with a Rudder

While outboard motors and airboats let you direct the thrust vector, most boats have a *rudder*. The rudder is a blade on a pivot near the back of the boat. The angle of the rudder can be adjusted so that water rushing past it gets pushed to one side or the other.

According to Newton's third law, when the water gets pushed to the left, the back of the boat gets pushed (with the same force) to the right. This causes the boat to rotate around its center of lateral resistance.

Note that a rudder only works when the boat is passing through the water.




## 46.5 Boat Length and Resistance

FIXME: Write about wave length, boat length, and Froude number.

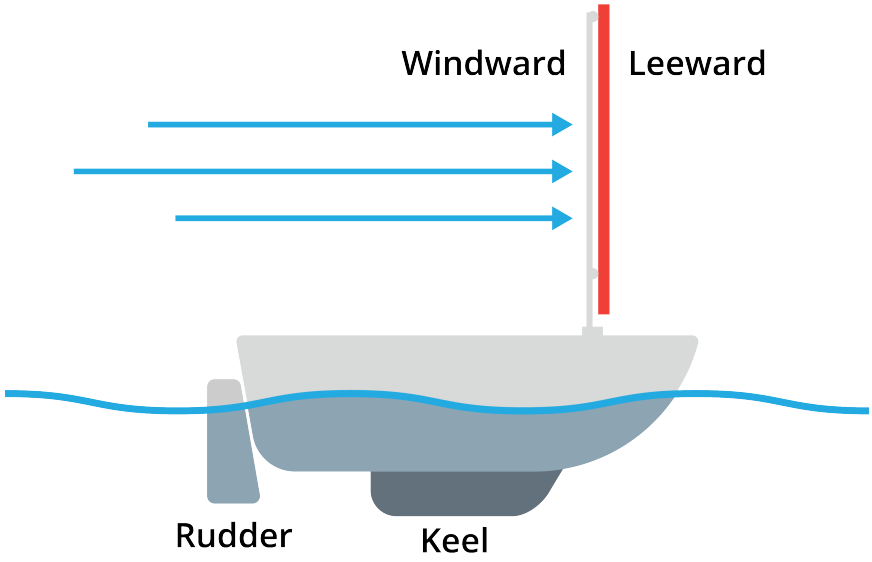
CHAPTER 47

# Sailboats

Imagine that you have a canoe, and you are about to paddle from one island to another that is directly east of where you are standing. There is also a steady wind coming from the west, and you have a big piece of plywood. You might be inspired to use it as a sail.

This situation is the most simple form of sailing: Wind comes from behind the boat and hits the sail which generates a force that pushes the boat in the direction of the wind.

The sail has two sides: The *windward* side is the one that is getting hit with the wind. The *leeward* side is the side away from the wind.




## 47.1 Magnitude of the Wind Force

The first natural question is: How much force will I have pushing my canoe through the water?

**Wind Force**

When the sail is perpendicular to the wind, the force of the wind on the sail in newtons ($F_w$) will be given by:

$$F_w = A\frac{dv^2}{2}$$

where $A$ is the area of the sail in square meters $d$ is the density of the gas in kg per cubic meter, and $v$ is the wind speed in meters per second.

For air at STP, $d$ is about 1.225 kg per cubic meter.

We call $\frac{dv^2}{2}$ the *wind pressure*. This is the amount of pressure that the windward side of plywood is experiencing that is above the pressure that the leeward side of the plywood is experiencing. (The leeward side might experience some turbulence, but the pressure it is experiencing is approximately 1 atmosphere.)

Let's say your canoe is standing still and the wind is 0.5 m/s. This means the wind pressure is

$$P = \frac{1.225(0.5^2)}{2} = 0.153125 \text{ newtons per square meter}$$

Let's say your plywood sail is 2 meters tall and 1.5 meters wide. What will be the force of the wind?

$$F_w = AP = (3)(0.153125) \approx 0.46 \text{ newtons}$$

This is a very intuitive idea: There is a difference between the pressure on the windward side and the pressure on the leeward side, and the plywood experiences a force that pushes the boat through the water.

## 47.2 Direction and Location of the Wind Force

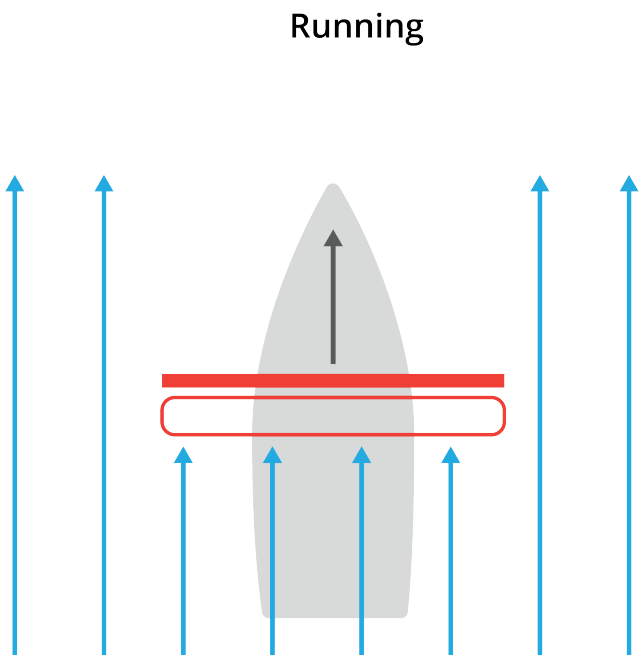
If there is low pressure on one side of the sail and high pressure on the other, the force vector will be perpendicular to the sail.

Where is this force vector applied? We can think of the force as being applied at the geometric center of the sail. This is called *the center of effort*. In this case, the center of effort is the exact center of the rectangular plywood.

The mast on a windsurfing board can be tilted from side to side. When the center of effort is over the center of lateral resistance, the board goes straight. To steer, the sailor moves the mast to one side of the center of lateral resistance, which rotates the board.

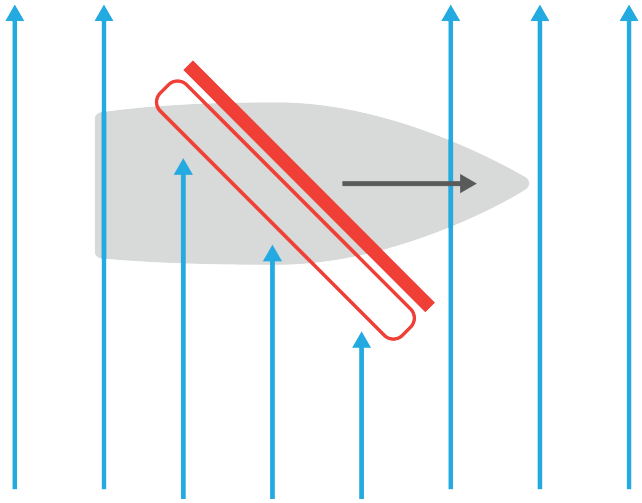
## 47.3 Beam Reach

When you are sailing in the same direction as the wind, sailors say you are *running*.




What if you want to go east and the wind (still 0.5 m per second) is now coming from the south? Sailing perpendicular to the wind is known as a *beam reach*.

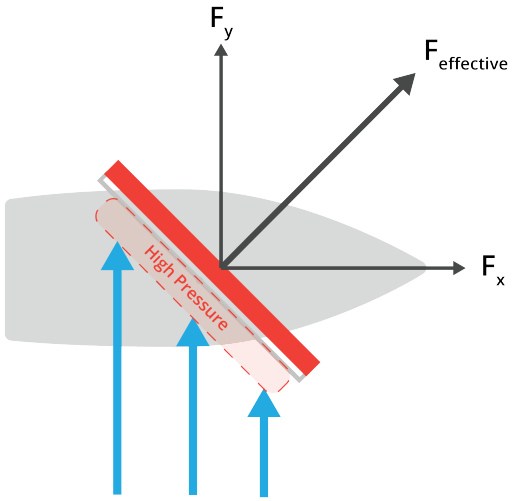
Beam Reach



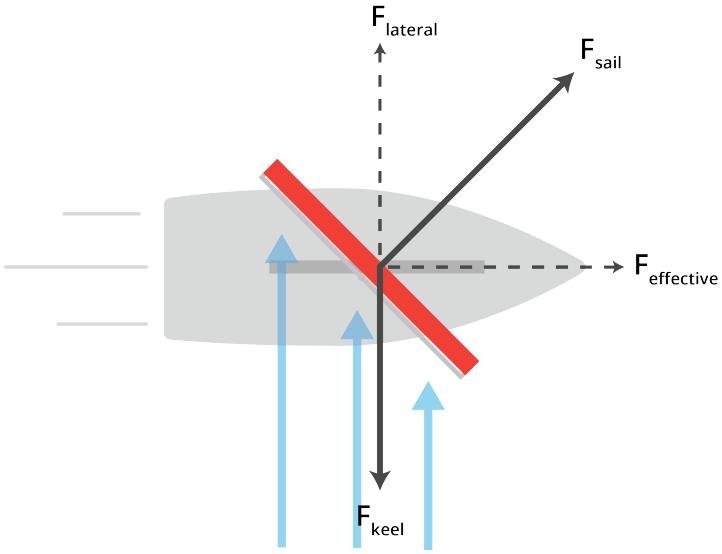
To do a beam reach, instead of mounting the plywood perpendicular to the boats direction of travel, you would mount it at a 45 degree angle. The wind pressure will build on the windward side of the plywood, and the plywood will experience a force pushing it at a 45° angle to the boat.

We can think of this force as having two components:

- One component pushes the boat forward (Yay!)
- One component pushes the boat sideways (Ugh!)




To minimize the effect of the sideways force, sailboats typically have a keel — a long fin on its underside that slows its sideways sliding.




Notice also that the "wind shadow" of the plywood is smaller when it is at a 45° angle to the wind. How much smaller? The effective area of your plywood has gone from 3 square meters to $\frac{3}{\sqrt{2}} \approx 2.12$ square meters. This means the force generated will be smaller, and some of it will be wasted pushing the boat sideways.

If we assume that the wind pressure is still 0.153125 newtons per square meter. The force on the plywood will be about

$$F_w = AP = \frac{3}{\sqrt{2}}(0.154125) \approx 0.325 \text{ newtons}$$

However, the direction of that force is not all in the direction you want to go, so the effective force is

$$F = F_w \frac{1}{\sqrt{2}} = \frac{3}{2}(0.154125) = 0.2311875 \text{ newtons}$$

Notice that we got twice as much effective force when we were running with the wind as when we are on a beam reach. However, any sailor will tell you that you can go much faster on a beam reach than you can running. Why?

## 47.4 Apparent Wind

When you are running, you can never go faster than the wind. As you go faster and faster, the wind that the boat experience decreases. For example, if you are going 0.2 m/s in a wind of 0.5 m/s, you (and your sail) will only experience wind at 0.3 m/s. We call the wind as experienced by the boat the *apparent wind*. The wind as observed by a stationary observer is called the *actual wind*.

If you are running with the wind, as you approach the speed of the wind, the force of that wind will decrease towards zero.

On a beam reach, as you go faster, the direction of the wind seems to change. If you are going 0.2 m/s east and the actual wind is 0.5 m/s from the south, the direction of the apparent wind will seem to come from about 22 degrees east of true south. The speed of the apparent wind will be about 0.54 m/s.

## 47.5 Close Reach

What if you want to go east, and the wind is coming from 40 degrees east of south? This would mean that you were sailing just 50 degrees away from straight into the wind. Is this possible?

If you put your sail at a 25 degree angle, you will still catch some wind and create some pressure on one side of the sail. Most of the resulting force would be trying to push your boat sideways, but some of it would be in the direction you were trying to travel.

Picking an angle for your sail that creates high pressure that makes a desirable force is known as the ”angle of attack”.

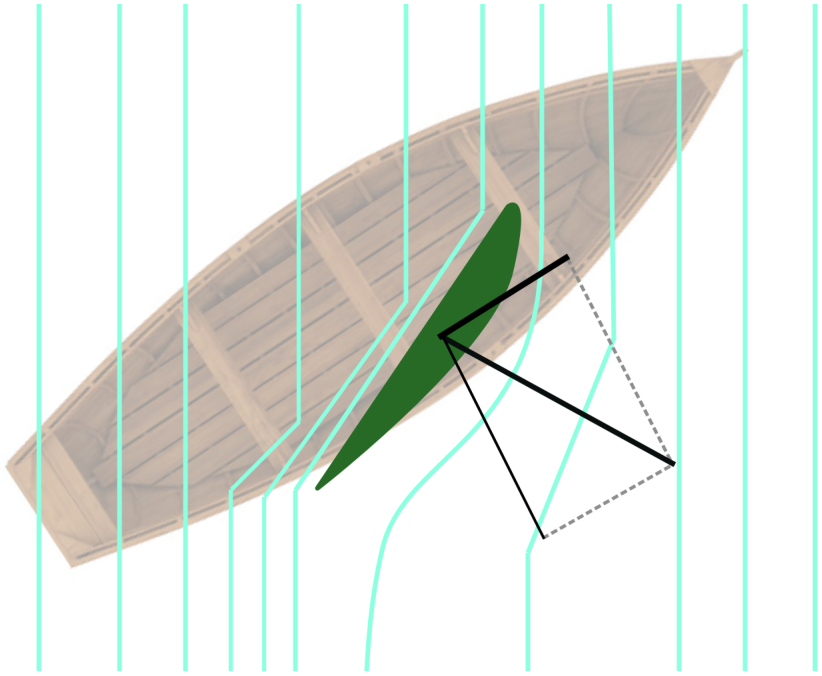
This is a result does not feel intuitive. A boat can sail into the wind!? The boat can’t sail directly into the wind – with each degree that the boat gets closer to straight into the wind, the force pushing it forward decreases and the force pushing it back increases. However, most boats can get within 45% if they have a well-shaped sail.

## 47.6 Shaping the Sail

Most of the power of the wind can be captured with a piece of flat plywood. The wind hits it and creates a high pressure on the windward side. What about the other side of the plywood?

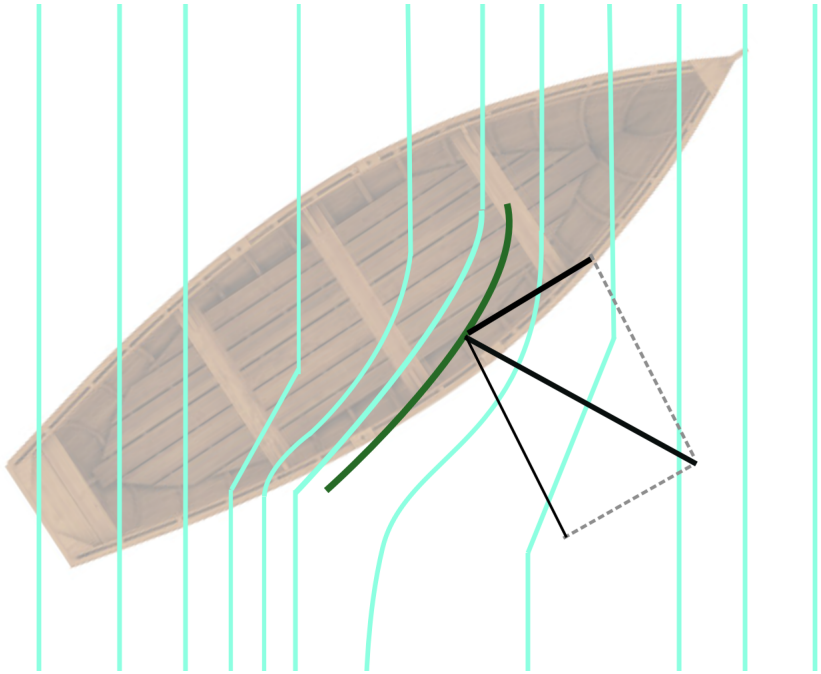
It turns out that if we can get the wind to travel smoothly over the back side of the plywood, the pressure on that side will be a little lower than if we had turbulence there. (We are not going to go too deeply into why. If you want to learn more, look up the Coanda effect.)

For example, if we were on a close reach, the very best sail we could have would gently pull the wind along its backside. It would look like this:



Of course, for the sail to work on either side of the boat, this asymmetrical design would not work. (Although, we should note that this design works great for airplane wings.)

When we make a sail out of cloth, we give it some curve known as *camber*. Slow winds require just a little camber; fast winds require more.



Some newer sailboats have wing sails that have two pieces that can be arranged to redirect the most air possible.

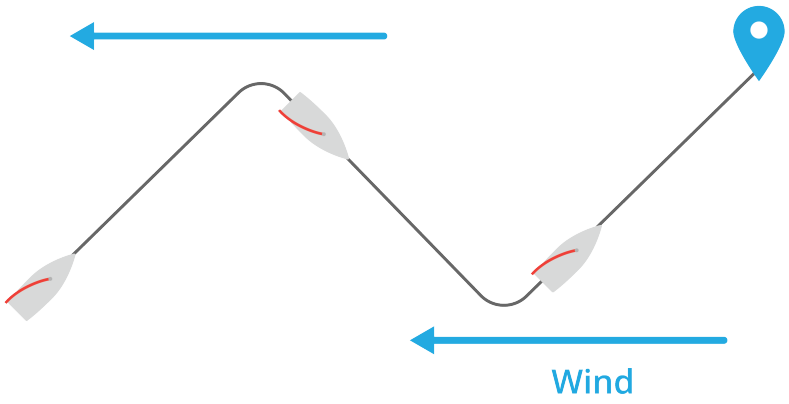
Note: when running with the wind, the turbulence on the leeward side of the sail is unavoidable. But when traveling perpendicular to the wind or on a close reach, the air should move smoothly over the leeward side of the sail. Many sailors have a piece of yarn taped on each side of the sail so they can see if the air is moving smoothly.

Many sailboats also have multiple sails. Besides the increase in the sail area, each sail also redirects the wind to pass smoothly over the leeward surface of the sail behind it.

## 47.7 Tacking into the Wind

What if the wind is coming from the east and you really need to go directly into the wind?

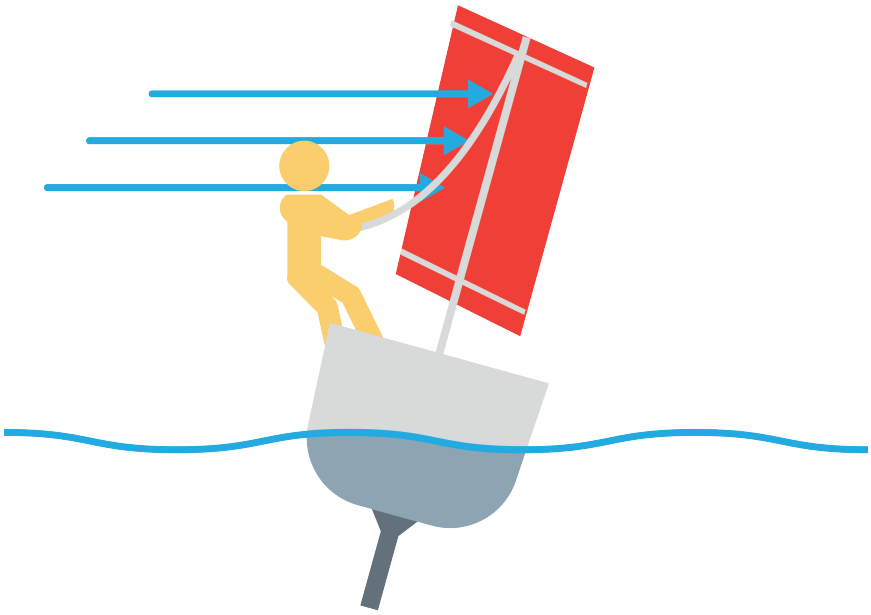
The boat will not travel straight into the wind, instead you will travel on a close reach with the wind coming from one side of the boat. You will then turn into the wind and continue turning until you are on a close reach with the wind coming from the other side. This is known as *tacking*.




## 47.8 Heeling

The center of effort is near the center of your sail, which is usually pretty high in the air. Yes, the force generated will push your boat, but it will also rotate your boat. Sailors call the resulting lean *heeling*. Heeling too far is problematic — the rudder gets pulled out of the water, which makes it hard to steer the boat.

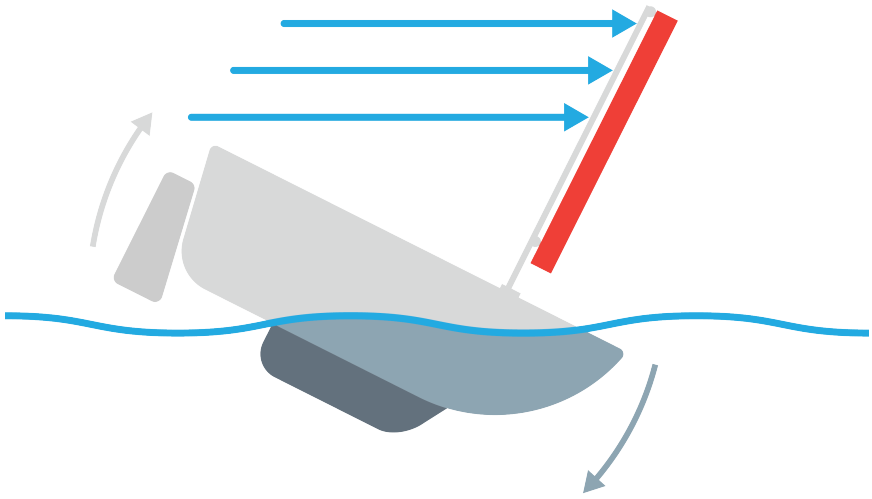
If a boat is heeling too far, sailors will move their weight to the windward side of the boat — some even wear harnesses that push their weight out beyond the edge of the hull. If that doesn't work, they will reduce the sideways force on the sail.



Can the boat flip over? (The word sailors use is *capsize*.) Most larger boats should not be able to capsize; as the sail gets pushed down toward the water, it loses power. So, putting some weight in the keel will ensure that the boat doesn't turn upside-down. Small boats (think 3 or 4 meters long) can capsize, but they are small enough that the sailor can usually get them upright again without assistance.

There are boats with multiple hulls. A catamaran has two identical hulls that are side-by-side. As a result, the catamaran will not heel much at all. However, if a big gust of wind comes up, one hull can be pulled completely out of the water. Catamarans can be capsized.

If a boat is running when a big gust comes up, that rotating force will try to push the bow underwater. If the boat is going very fast, this can result in a somersault as the front of the boat slows and dips suddenly and the back of the boat is pitched up over it.



CHAPTER 48

# Airplanes

Now that you understand how a sail works, you are ready to understand how an airplane works.

## 48.1 Drag and Thrust

As an airplane flies, the air around it creates a large force that is trying to slow the plane down. Pilots call this *drag*. To overcome drag, an airplane must have a source of *thrust*. This is usually done by pushing air toward the back of plane. You can think of drag as air resistance or air friction, and thrust as the applied force overcoming that friction.

There are three basic types of propulsion on airplanes:

- Propeller planes have fan blades that are spun by an internal combustion engine and push the air toward the back of the plane.
- Jet planes suck air into a tube, mix it with fuel, and ignite the mixture. There are fan blades inside the tube that ensure the power from the burn is efficiently converted into thrust.
- Turboprop planes use both propeller and jet technology — their propellers are turned by jet engines.

Drag is proportional to the square of the velocity. For example, if you double how fast you are flying, the drag force will increase by a factor of 4.

Passenger airplanes fly fast — they have no trouble reaching 1000 km per hour. A Boeing 747 burns about 4 liters of fuel per second; the flight from London to New York requires about 70,000 kg of fuel. When the plane takes off, the fuel for the journey often weighs twice as much as the passengers.

## 48.2 Lift

Unlike a hot air balloon, an airplane is heavier than the air it displaces, so the force of gravity will pull it from the sky unless there is a counteracting force. We call the counteracting force *lift*. Lift works just like sailing into the wind. By picking an angle of attack, we create high pressure under the wing. By creating a nice smooth path for the air

to travel over the top of the wing, we create low pressure over the wing. This difference creates the lift on the wings. Wings are shaped in this way to make it easier for the air to flow over the top of the wing, a shape called an *airfoil*. An airfoil is any shape that produces more lift than drag, such as a kite, a blade, an airplane wing, or sail.

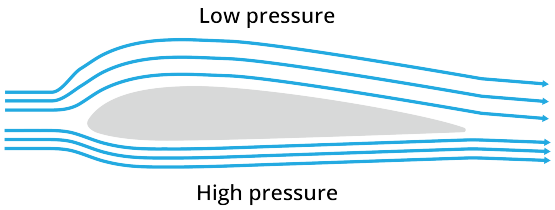



Figure 48.1: A diagram of the flow of air on a wing cross section.

(There are some very bad explanations of this idea that overstate the importance of the rounded top of the wing. Most airplanes can fly upside down — if the most important part were the shape of the top, this would be impossible. Instead, the most important part is the angle of attack.)

At high altitudes, the air is less dense, until space is reached where there is no air and it is an entire vacuum. If two identical planes are flying at the same speed, but at different altitudes:

- There is less drag on the higher airplane.
- The wings provide less lift on the higher airplane.
- The air around the higher plane has less oxygen per liter, which can affect how the fuel burns.

## 48.3 Control: Roll, Pitch, Yaw

The first control that a pilot has is the throttle. By increasing the throttle, the pilot increases the plane's thrust.

The pilot has a stick. Pulling back on the stick causes the *elevator* on the tail of the plane to go up. Air hitting the elevator pushes the tail down and the nose up.

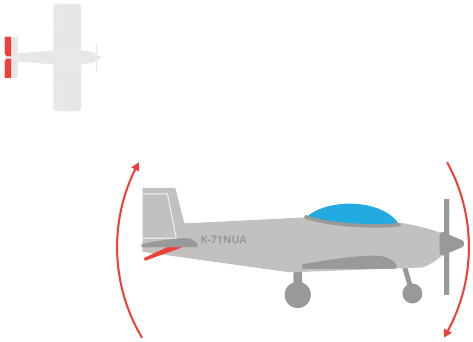



Figure 48.2: A diagram of a top-down and side view of an elevator and its pitch effect

Pushing the stick forward will push the tail up and the nose down. We say "The elevator controls the *pitch* of the airplane."

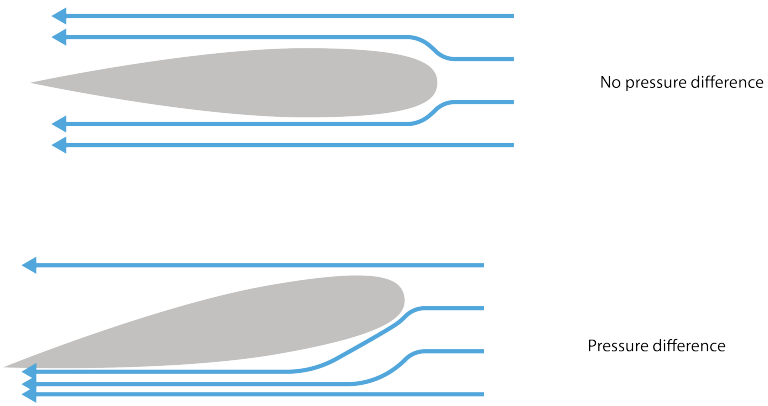



Figure 48.3: A view of how the wing of the airplane is effected by pressure.

The stick also goes left and right. Pushing the stick to the left, lifts the *aileron* on the left wing and lowers the aileron on the right wing. This pushes the left wing down and the right wing up. We say "The ailerons control the *roll* of the airplane".

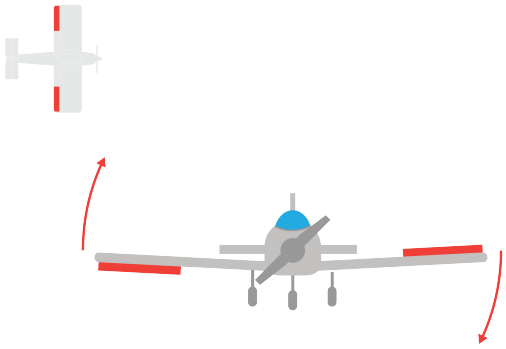


Figure 48.4: Aileron locations on the plane and their effect on roll.

The pilot controls the *rudder* on the tail with his feet. Pushing the right side down will push the rudder to the right. This will push the tail of the plane to the left and the nose to the right. We say "The rudder controls the *yaw* of the airplane."

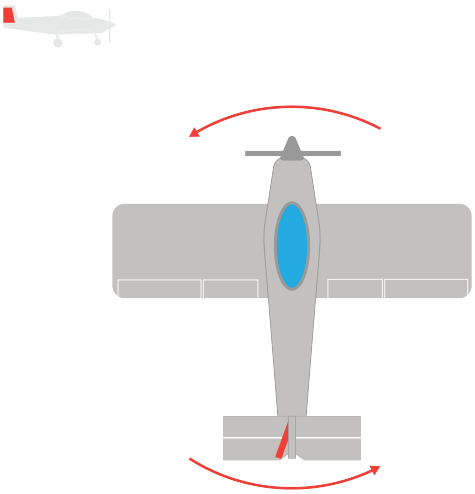


Figure 48.5: The rudder controls the yaw of the airplane, or top-down rotation

The flaps of a plane are surfaces on edges of the underside of the wings that are used to increase lift and drag during takeoff or landing.

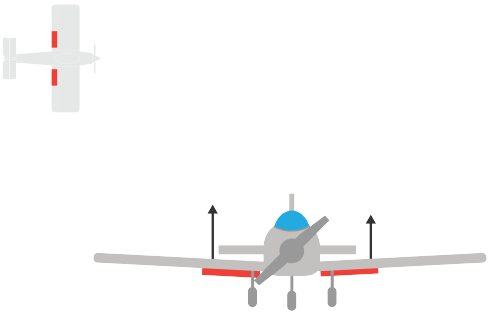


Figure 48.6: A diagram showing the location and effects of the flaps.

## 48.4 Thrust

There are several common ways that airplanes produce thrust.

The first is a propeller powered by a standard piston engine.

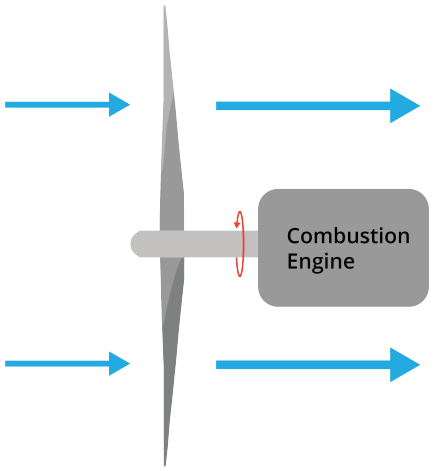



Figure 48.7: The propeller uses the air to push it forward (thrust). The wings of a propeller act similar to sails and wings of an airplane, separating airflow.

Another way to produce thrust is by using a *jet engine*. Jet engines take in air, increase its pressure and temperature, and shoots it out of the back of the engine.

Jet engines suck in air, compress it, add a fuel mixture, ignite the fuel, and shoot it out the back. Once the air exits the engine, it expands at a high speed, which further helps create thrust.

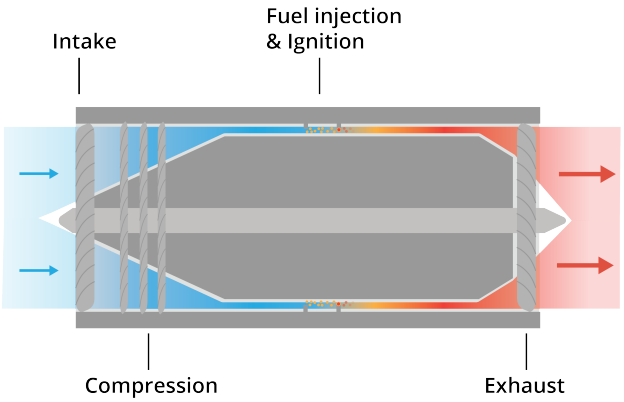



Figure 48.8: Cross section of a jet engine

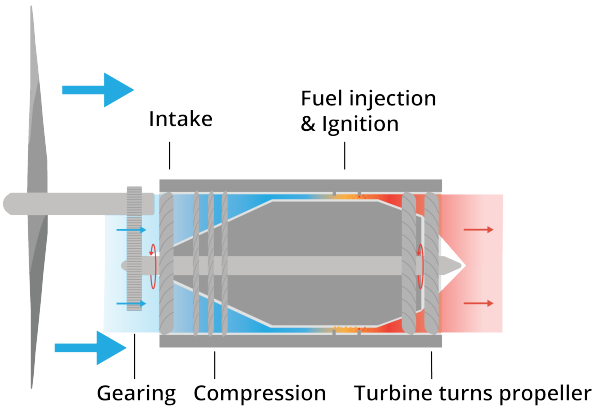



Figure 48.9: Cross section of a turboprop

## 48.5 Gliders

It is interesting to note that gliders fly without an engine. Gliding usually starts high in the air, where the glider has lots of potential energy. To stay aloft for a long time, glider pilots will look for places where air is rising, so they can ride those updrafts and regain that potential energy.

CHAPTER 49

# Data Tables and pandas

Much of the data that you will encounter in your career will come to you as a table. Some of these tables are spreadsheets, some are in relational databases, and some will come to you as CSV files.

Typically, each column will represent an attribute (like height or acreage) and each row will represent an entity (like a person or a farm). You might get a table like this:

| property_id | bedrooms | square_meters | estimated_value |
|---|---|---|---|
| 7927 | 3 | 921.4 | $ 294,393 |
| 9329 | 2 | 829.1 | $ 207,420 |

In most cases, one of the columns is guaranteed to be unique. We call this the *primary key*. In this table, `property_id` is the primary key; every property has one, and no two properties have the same `property_id`.

## 49.1 Data types

Each column in a table has a type, and these usually correspond pretty nicely with types in Python.

Here are some common datatypes:

| Type | Python type | Example |
|---|---|---|
| Integer | int | `910393` |
| Float | float | `-23.19` |
| String | string | `'Fred'` |
| Boolean | bool | `False` |
| Date | datetime.date | `2019-12-04` |
| Timestamps | datetime.datetime | `2022-06-10T14:05:22Z` |

Sometimes it is OK to have values missing. For example, if you had a table of data about employees, maybe one of the columns would be `retirement` — a date that tells you when the person retired. People who had not yet retired would have no value in this column. We would say that they have *null* for `retirement`.

Sometimes there are constraints on what values can appear in the column. For example, if the column were `height`, it would make no sense to have a negative value.

Sometimes a column can only be one of a few values. For example, if you ran a bike rental

shop, each bicycle's status would be "available", "rented", or "broken". Any other values in that column would not be allowed. We often call these columns *categorical*.

## 49.2 pandas

The Python community works with tables of data *very* often, so it created the pandas library for reading, writing, and manipulating tables of data.

When working with tables, you sometimes need to go through them row-by-row. However, for large datasets, this is very slow. pandas makes it easy (and very fast) to say things like "Delete every row that doesn't have a value for height" instead of requiring you to step through the whole table.

In pandas, there are two datatypes that you use a lot:

- a `Series` is a single column of data.
- a `DataFrame` is a table of data: it has a `Series` for each column.

In the digital resources, you will fined `bikes.csv`. If you look at it in a text editor, it will start like this:

```
bike_id,brand,size,purchase_price,purchase_date,status
5636248,GT,57,277.99,1986-09-07,available
4156134,Giant,56,201.52,2005-01-09,rented
7971254,Cannondale,54,292.25,1978-02-28,available
3600023,Canyon,57,197.62,2007-02-15,broken
```

The first line is a header; it tells you the name of each column. Next, the each bike is represented by values which are separated by commas. (Thus the name: CSV stands for "Comma-Separated Values".)

## 49.3 Reading a CSV with pandas

Let's make a program that reads `bikes.csv` into a pandas dataframe. Create a file called `report.py` in the same folder as `bikes.csv`.

First, we will read in the csv file. pandas has one series that acts as the primary key; it calls this one the index. When reading in the file, we will tell it to use the `bike_id` as the index series.

If you ask a dataframe for its shape, it returns a tuple containing the number of rows and

the number of columns. To confirm that we have actually read the data in, let's print those numbers. Add these lines to `report.py`:

```
import pandas as pd

# Read the CSV and create a dataframe
df = pd.read_csv('bikes.csv', index_col="bike_id")

# Show the shape of the dataframe
(row_count, col_count) = df.shape
print(f"*** Basics ***")
print(f"Bikes: {row_count:,}")
print(f"Columns: {col_count}")
```

Build it and run it. You should see something like this:

```
*** Basics ***
Bikes: 998
Columns: 5
```

Note that your table actually has six columns. The index series is not included in the shape.

## 49.4 Looking at a Series

Let's get the lowest, the highest, and the mean purchase price of the bikes. The purchase price is a series, and you can ask the dataframe for it. Add these lines to the end of your program:

```
# Purchase price stats
print("\n*** Purchase Price ***")
series = df["purchase_price"]
print(f"Lowest:{series.min()}")
print(f"Highest:{series.max()}")
print(f"Mean:{series.mean():.2f}")
```

Now when you run it, you will see a few additional lines:

```
*** Purchase Price ***
Lowest:107.37
Highest:377.7
Mean:249.01
```

What are all the brands of the bikes? Add a few more lines to your program that shows how many of each brand:

```
# Brand stats
print("\n*** Brands ***")
series = df["brand"]
series_counts = series.value_counts()
print(f"{series_counts}")
```

Now when you run it, your report will include the number of bikes for each brand from most to least common:

```
*** Brands ***
Canyon        192
BMC           173
Cannondale    170
Trek          166
GT            150
Giant         147
Name: brand, dtype: int64
```

`value_counts` returns a Series. To format this better, we need to learn about accessing individual rows in a series.

## 49.5 Rows and the index

In an array, you ask for data using the location (as an int) of the item you want. You can do this in pandas using `iloc`. Add this to the end of your program:

```
# First bike
print("\n*** First Bike ***")
row = df.iloc[0]
print(f"{row}")
```

When you run it, you will see the attributes of the first row of data:

```
*** First Bike ***
brand                     GT
size                      57
purchase_price        277.99
purchase_date     1986-09-07
```

```
status              available
Name: 5636248, dtype: object
```

Notice that the data coming back is actually another series.

The last line says that the name (the value for the index column) for this row is 5636248. In pandas, we usually use this to locate particular rows. For example, there is a row with `bike_id` equal to 2969341. Let's ask for one entry from the

```
print("\n*** Some Bike ***")
brand = df.loc[2969341]['brand']
print(f"brand = {brand}")
```

Now, you will see the information about that bike:

```
*** Some Bike ***
brand = Cannondale
```

pandas has a few different ways of getting to that value. All of these get you the same thing:

```
brand = df.loc[2969341]['brand'] # Get row, then get value
brand = df['brand'][2969341]     # Get column, then get value
brand = df.loc[2969341, 'brand'] # One call with both row and value
```

## 49.6 Changing data

One of your attributes needs cleaning up. Every bike should have a status, and it should be one of the following strings:"available", "rented", or "broken". Get counts for each unique value in status:

```
print("\n*** Status ***")
series = df["status"]
missing = series.isnull()
print(f"{missing.sum()} bikes have no status.")
series_counts = series.value_counts()
for value in series_counts.index:
    print(f"{series_counts.loc[value]} bikes are \"{value}\"")
```

This will show you:

```
*** Status ***
7 bikes have no status.
389 bikes are "rented"
304 bikes are "broken"
296 bikes are "available"
1 bikes are "Flat tire"
1 bikes are "Available"
```

Right away, we can see two easily fixable problems: Someone typed "Available" instead of "available". Right after you read the CSV in, fix this in the data frame:

```
mask = df['status'] == 'Available'
print(f"{mask}")
df.loc[mask, 'status'] = 'available'
```

When you run this, you will see that the mask is a series with `bike_id` as the index and `False` or `True` as the value, depending on whether the row's status was equal to "Available".

When you use `loc` with this sort of mask, you are saying "Give me all the rows for which the mask is True." So, the assignment only happens in the one problematic row.

Let's get rid of the mask variable and do the same for turning `Flat tire` into `Broken`:

```
df.loc[df['status'] == 'Available', 'status'] = 'available'
df.loc[df['status'] == 'Flat tire', 'status'] = 'broken'
```

Now those problems are gone:

```
7 bikes have no status.
389 bikes are "rented"
305 bikes are "broken"
297 bikes are "available"
```

What about the rows with no values for status? If we were pretty certain that the bikes were available, we could just set them to 'available':

```
missing_mask = df['status'].isnull()
df.loc[missing_mask, 'status'] = 'available'
```

Or maybe we would print out the IDs of the bikes so that we could go look for them:

```
missing_mask = df['status'].isnull()
missing_ids = list(df[missing_mask].index)
print(f"These bikes have no status:{missing_ids}")
```

However, let's just keep the rows where the status is not null:

```
missing_mask = df['status'].isnull()
df = df[~missing_mask]
```

At the end of your program, write out the improved CSV:

```
df.to_csv('bikes2.csv')
```

Run the program and open `bikes2.csv` in a text editor.

## 49.7 Derived columns

Let's say that you want to add a column with age of the bicycle in days:

```
bike_id,brand,size,purchase_price,purchase_date,status,age_in_days
5636248,GT,57,277.99,1986-09-07,available,13061
4156134,Giant,56,201.52,2005-01-09,rented,6362
7971254,Cannondale,54,292.25,1978-02-28,available,16174
```

Your first problem is that the `purchase_date` column looks like a date, but really it is a string. So, you need to convert it to a date. You can do this by applying a function to every item in the series:

```
df['purchase_date'] = df['purchase_date'].apply(lambda s:
↪  datetime.date.fromisoformat(s))
```

(With pandas, there is often more than one way to do things. pandas has a `to_datetime` function that converts every entry in a sequence to a `datetime` object. Here is another way to convert the string column in to a date column:

```
df['purchase_date'] = pd.to_datetime(df['purchase_date']).dt.date
```

You can look up `dt` and `date` if you are curious.)

Now, we can use the same trick to create a new column with the age in days:

```
today = datetime.date.today()
df['age_in_days'] = df['purchase_date'].apply(lambda d: (today - d).days)
```

When you run this, the new `bikes.csv` will have an `age_by_date` column.

Chapter 50

# Rotorcraft

## 50.1 Introduction

This chapter moves on from airplane movement to quadcopters! You may here rotor-winged aircraft, drones, or unmanned aerial vehicle, which all refer to similar types of the same structure.

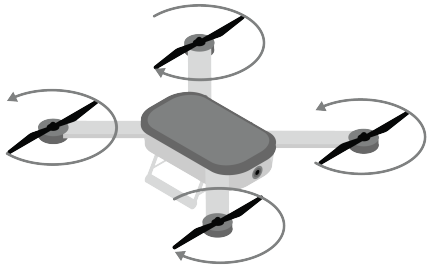


Figure 50.1: The general look of a quadcopter. Includes four rotors, a frame called a fuselage, and generally a head or designated front.

Before we dive in depth, let's introduce a few terms that will be repeated within this chapter.

**Quadcopter** Often defined as an aircraft with 4 rotors, synonymous with drone.

**Rotor** The rotor is the part of the aircraft which causes lift by spinning at a high velocity. It consists of a central *hub* with two or more *blades* attached. You may hear the term propeller like from our airplanes chapter, but the more correct term for quadcopters is *rotor*.

**UAS / UAV** Unmanned Aircraft System or Unmanned Aircraft Vehicle. Often used to

describe drones or semi-autonomous quadcopters. This includes the drones themselves, the software needed to avoid obstacles and track positioning, and equipment used by people to communicate with them.

**Fuselage** The main body of a quadcopter or aircraft; synonymous to frame.

**Torque** We have not talked about torque yet, but it is an important circular motion concept. Torque is the rotational equivalent of force, and is a measure of how much a force acting on an object causes that object to rotate. For now, understand that it is affects quadcopter movement and is a force that needs to be counteracted in order to maintain stability.

**Isometric** We also want to be clear about our use of views in our diagrams. An *isometric* perspective is one that visually represents all 3 views of a 3D object in a 2D space (a printed diagram). When we get to calculus, we will talk about *isometry* as transformation or mapping between two spaces that perfectly preserves distances.

## 50.2 VTOL

The strength of quadcopters lies in their propulsion system. As compared to airplanes, which take off by accelerating down *runways* and gaining enough lift to overcome its weight, quadcopters use spinning rotors to generate this lift. This applies Newton's Third Law — for every action (the downwards lift of the rotors), there is an equal and opposite reaction (the upwards lift of the quadcopter). This allows quadcopters to take off vertically, eliminating the need for a runway. This is a very special type of propulsion known as Vertical Take-Off and Landing (VTOL).

Specifically, the drone's rotors needs to generate enough lift to counteract its own weight and put it into motion in the air. Recall that there are *equal but opposite* forces in play here, so the lift created by the rotors can be found by

$$F_{\text{rotors}} = m_{\text{quadcopter}}\, g + F_{\text{vertical lift}}$$

This is a rudimentary equation that demonstrates the following: the amount of lift created by rotors is the excess force after the rotors are able to overcome the weight of the quadcopter.

Here is an isometric view of vertical flight of a quadcopter

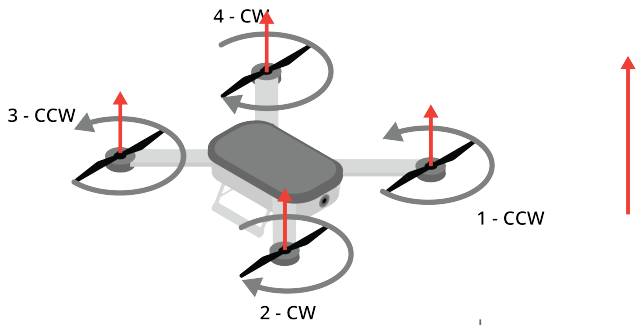



Figure 50.2: A quadcopter taking vertical flight. Notice how the red arrows, representing the strength of the upwards force, are all equivalent.

If we just want our quadcopter to *hover in place*, the forces vertically need to be equal. This means that the weight of the quadcopter will equal the lift generated by the rotors

$$F_{rotors} = m_{quadcopter}\ g \tag{50.1}$$

## 50.3 Rotor Structure

Now, the remaining question is how do rotors work?

Recall our airplanes chapter; the wings are what create lift, and this is heavily dependent on the wings shape. On quadcopters, each rotor is made up of two or more *blades* attached to a central hub, and these blades are specifically designed to push air *opposite* the direction the aircraft needs to go.

[PLACEHOLDER: ISOMETRIC VIEW OF BLADE WITH LABELING]

Each blade has a *root*, which connects to the hub, and a *tip* at its outer end. Along its length, the blade also has a *twist*, where the blade's angle gradually changes from root to tip, similar to the shape of a screw thread.

[PLACEHOLDER: SPLIT FIGURE OF BLADE TWIST COMPARED TO SCREW MECHANISM]

The direction the air is pushed depends on which way the quadcopter goes; reversing the rotation of a rotor reverses the direction of the air pushed, and thus the direction of the quadcopter.

Rotors are typically described by three numbers: their *diameter*, their *pitch*, and their *number of blades*. Typically, you will see something like *10x5 inch*, which refers to the diameter and pitch, and the visual will tell you the number of blades.

The diameter measures the tip to tip distance that a propeller can reach. If there is only 1 blade, this measurement is twice the hub-to-tip distance. Larger diameter rotors move a bigger column of air, but at slower speeds. Thrust scales roughly equal to the square of velocity, while power scales to the cube of velocity. Bigger props are meaningfully more efficient. This is the same reason helicopters use one large slow rotor while quadcopters use several small fast ones.

Two is the typical amount of blades per propeller, and ensure symmetry along the Y-Axis, as well as mass-production capability. Adding blades increases total thrust at a given diameter and RPM, but each additional blade partly flies through the turbulent wake left by the blade ahead of it (this is called *aerodynamic interference*), so you get diminishing thrust returns per blade.

The pitch is the distance the propeller would move after one 360° rotation. Similar to how a screw moves when twisted, this measurement predicts how the propeller would move in vertical space. Lower pitch means less torque needed and higher maximum RPM, but less thrust and vertical movement is produced per rotor. Higher pitch generates more thrust and RPM, but draws more torque from the motor.

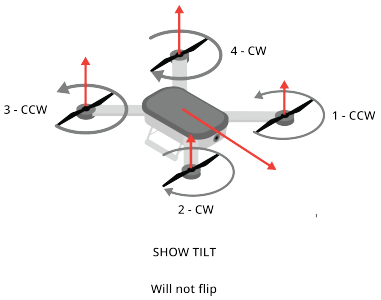
## 50.4 Movement, Spin Direction, and Torque

Generally, two rotors spin clockwise (CW) and two spin counter-clockwise (CCW). Rotors positioned diagonally from each other spin in the same direction. Rotors right next to each other spin in opposite directions. If all four rotors spun in the same direction, the body of the drone would violently spin in the opposite direction. By having two CW and two CCW motors, the forces cancel each other out. This balance allows the drone to hover completely still and turn smoothly post-VTOL. There is a guide in your digital resources that goes into blade selection based on your use case, very interesting!

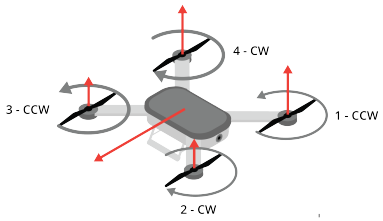
In theory, quadcopters that are symmetrical such as drones could be flown in any direction (this is called *headless flight*). This is easier for beginners as the direction away from the operator is forward, and the direction toward the operator is backwards, regardless of rotation.

Often a front is decided so that the fuselage body can we can determine roll, pitch, and yaw; this front is called the head. That way, its rotation and flight is based on its relative position in the world, not the operator's location. In the following discussion on forces

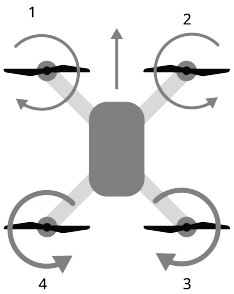
present on quadcopters, we will assume that there is no drag and our center of gravity is in the center of the fuselage, just as with all other ideal scenarios.



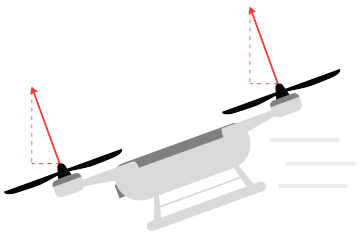

(a) `forwardIso.png`




(b) `lateralIso.png`




(a) `forwardTop.png`

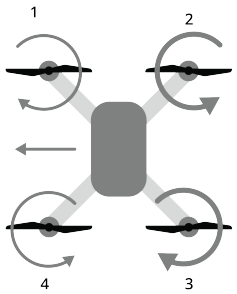


(b) `forwardSide.png`

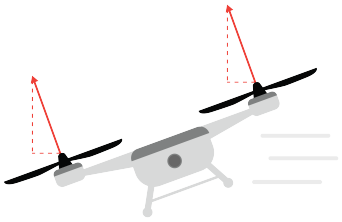
When flying a quadcopter forward, we apply *pitch* to the quadcopter (figure 50.3a). When doing this, we generate a greater amount of torque from the back two rotors (see 50.4a). This causes the copter to tilt forward with the back rotors higher than the front two rotors (see 50.4b), and direct its motion in the direction of the head. Backwards motion is very similar; the *front* rotors relative to the head spin at a greater rate, and as a result, tilt higher than the back motors, causing backwards motion.

As with airplanes, left and right movement, or *lateral movement*, is controlled by *roll*. Figure 50.3b shows a quadcopter rolling to the left. To produce this roll, the rotors to the *right* of the head spin faster than the rotors to the left, generating extra lift on the right side of the fuselage. This lifts the right side and tips the left side down, tilting the quadcopter's

thrust to the left and carrying it in that direction (see 50.5b). Rightward movement works the same way in reverse: the rotors to the *left* spin faster, lifting the left side and rolling the quadcopter to the right.

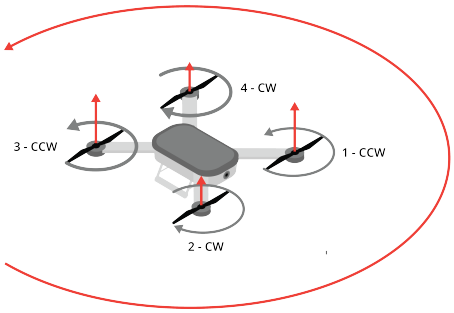



(a) `lateralTop.png`

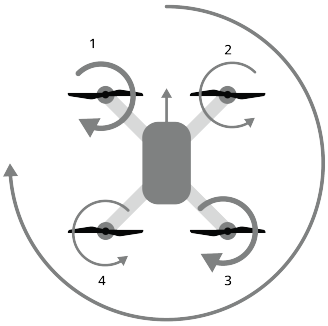


(b) `lateralSide.png`

Rotation of the ahead about the center of gravity is called *yaw*. Instead of two adjacent rotors having increased speeds, we adjust two diagonal rotors to have greater speeds then the other two. Increasing the clockwise rotors will cause a counterclockwise or left rotation. Rightward rotation is similar: increasing the counterclockwise rotors causes a clockwise




(a) `rotationIso.png`




(b) `rotationTop.png`

## 50.5 Helicopters

Instead of a fixed wing like on an airplane, or four small rotors on a quadcopter, a *helicopter* generates its lift from a single, much larger *main rotor*. The physics is the same as before: the spinning blades push air downward, and Newton's Third Law pushes the aircraft up in response. However, a single big rotor introduces two new problems a quadcopter doesn't have to solve.

First, on a quadcopter, half the rotors spin clockwise and half spin counter-clockwise, so their reaction torques cancel out. A helicopter's main rotor only spins one way, so its reaction torque has nothing to cancel it. Without a fix, the fuselage would spin in the opposite direction of the rotor. Most helicopters solve this with a small *tail rotor* that pushes sideways at a proportional speed to counteract the twist (see Figure 50.7).

Second, a quadcopter changes direction by speeding up or slowing down individual rotors. A helicopter's main rotor spins at a nearly constant speed instead; direction and lift are controlled by changing the *pitch* of the blades as they spin, using controls called *collective* and *cyclic*. We'll cover the tail rotor first, then come back to how collective and cyclic control the main rotor.

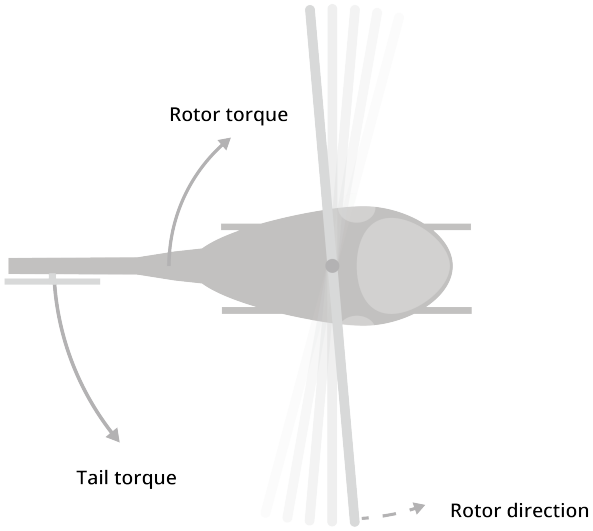
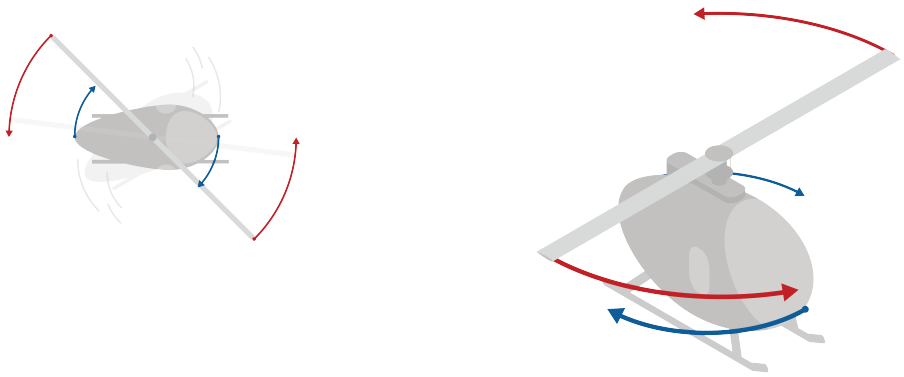



Figure 50.7: A helicopter uses one large rotor that rotates in one direction. A smaller tail rotor counteracts this torque.

### 50.5.1 Tail Rotor

On a helicopter, the tail rotor spins vertically, pushing air horizontally to counteract the torque of the main rotor. The pilot controls the tail rotor with pedals, which change the pitch of the tail rotor blades. This changes how much air is pushed sideways, and thus how much torque is counteracted.

Let's imagine a helicopter with a main rotor that spins clockwise and *no tail rotor*. Without a tail rotor, the fuselage would have a violent torque applied to it, and would spin counter-clockwise. There is nothing to counteract this torque, so the fuselage would spin faster and faster until the helicopter is destroyed or crashes.



(a) A helicopter without a tail rotor experiences a net torque that spins the fuselage in the opposite direction of the main rotor.

(b) The fuselage continues to rotate faster and faster unless a counter-torque is provided.

Figure 50.8: A helicopter becomes unbalanced without a tail rotor.

Because of this, we need a way to balance this torque. We add a rotor in the back of the helicopter that pushes air in the opposite direction the torque from the main rotor (see Figure 50.9).

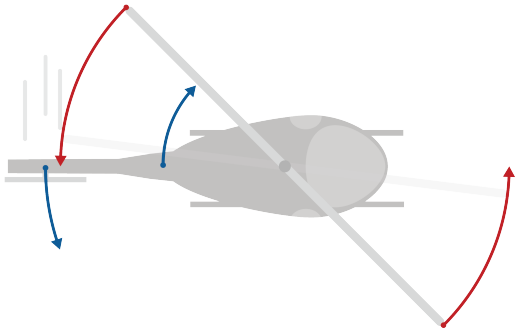


Figure 50.9: A tail rotor provides the counter-torque needed to keep the helicopter balanced and prevent violent movement.

If the pilot wants to yaw left, they increase the pitch of the tail rotor blades, pushing more air to the right and counteracting more of the main rotor's torque. To yaw right, they decrease the pitch of the tail rotor blades, pushing less air to the right and allowing more of the main rotor's torque to spin the fuselage clockwise (see Figure 50.10).

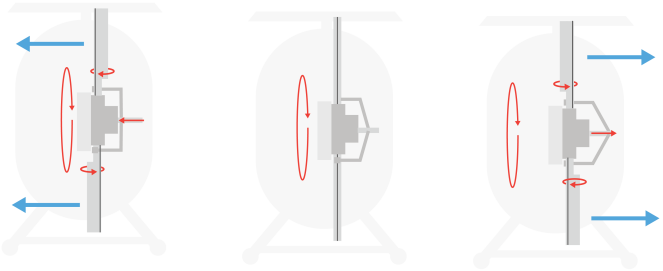


Figure 50.10: The tail rotor changes thrust to control the helicopter's yaw.

### 50.5.2 Main Rotor Movement

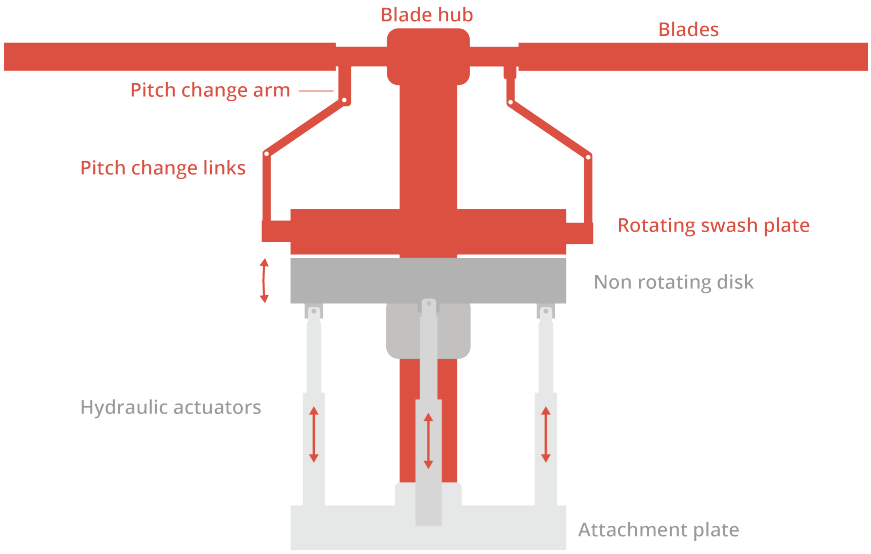



Figure 50.11: `mechanicalControl.png`

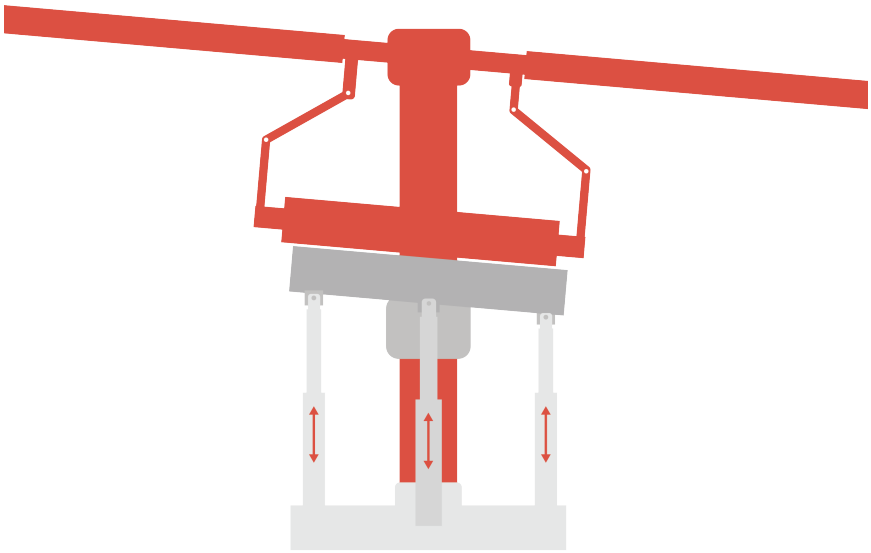


Figure 50.12: `mechanicalControlMovement.png`

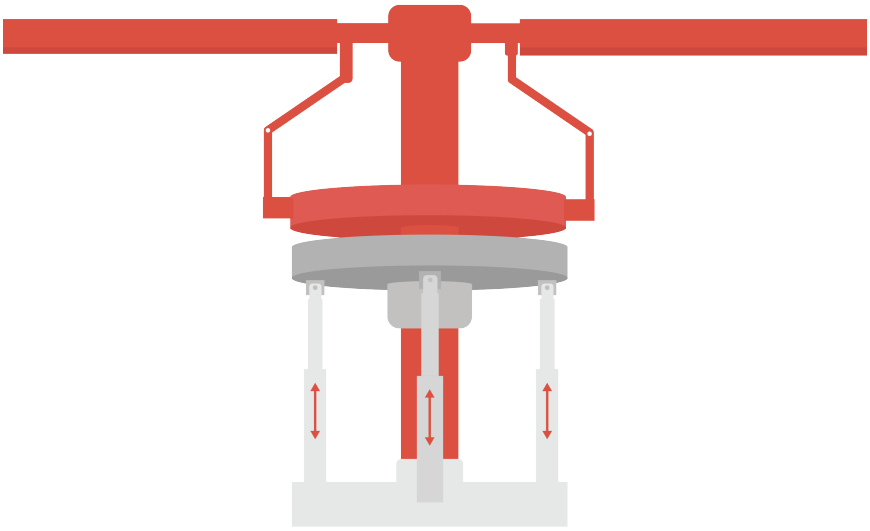


Figure 50.13: `mechanicalControlMovement2.png`

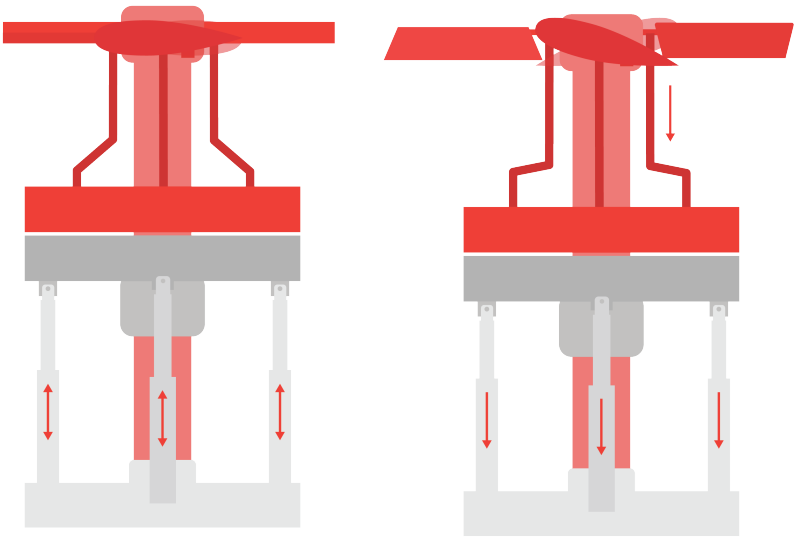
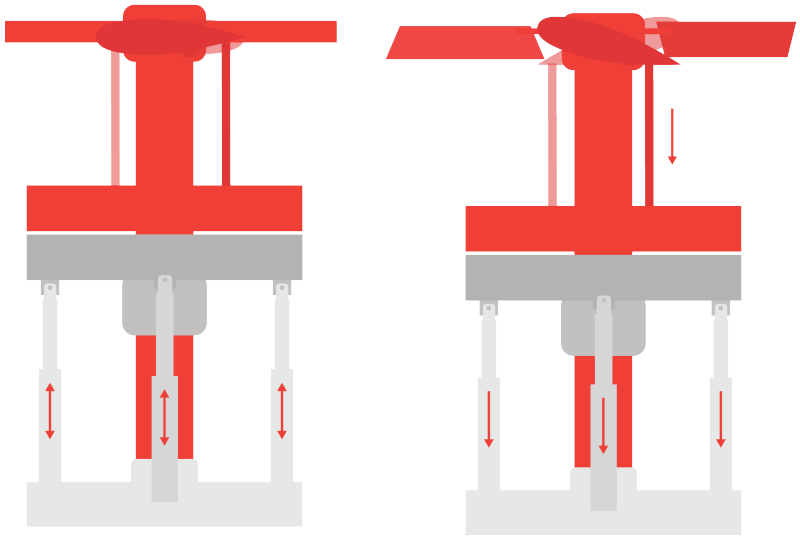



Figure 50.14: `collective.png`



Collective

Figure 50.15: `collective4Version.png`

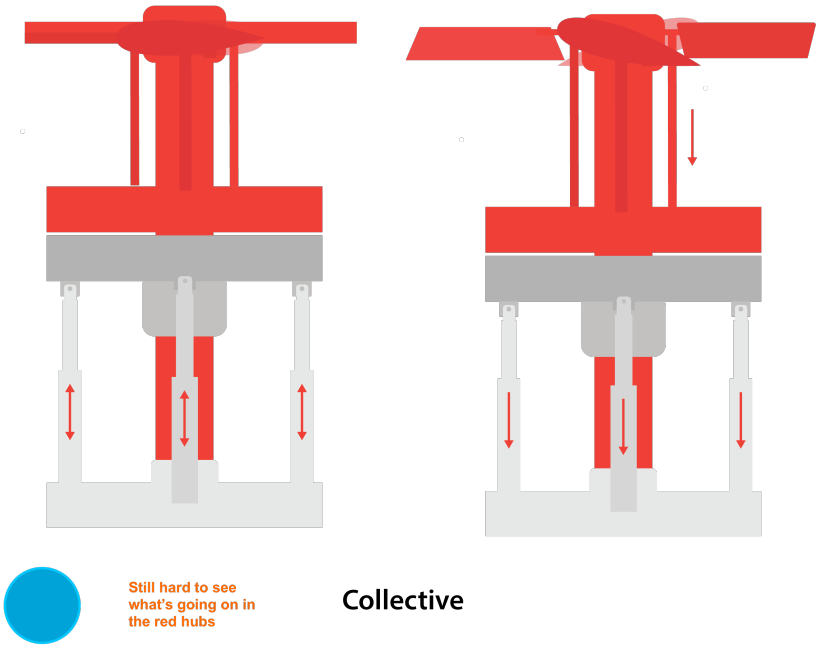
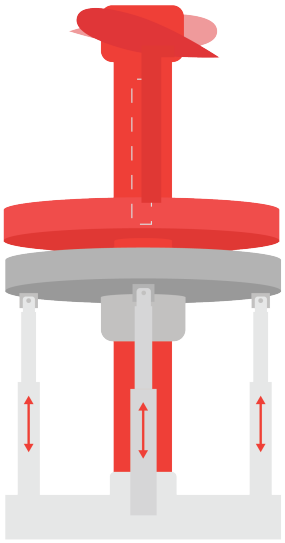



Figure 50.16: `collectiveQuad.png`



In reality, it's 90° away

Figure 50.17: `cyclic1.png`

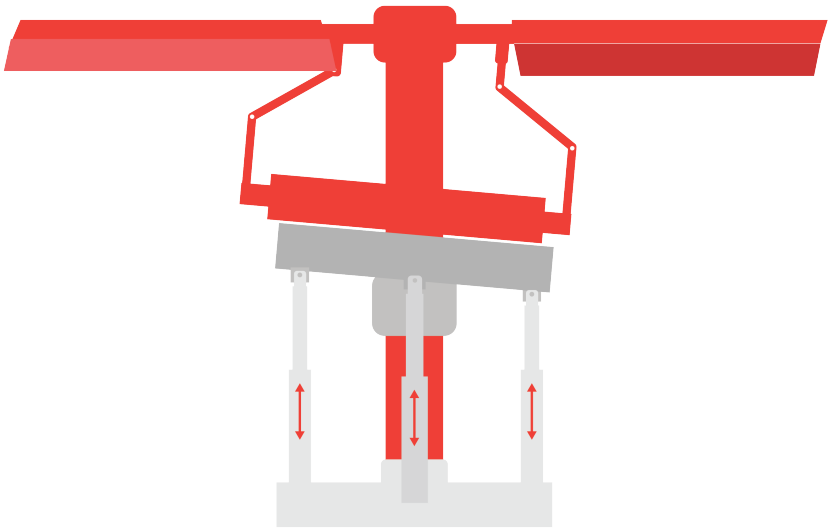


Figure 50.18: `cyclic2.png`

### 50.5.3 Isometric Copter Views

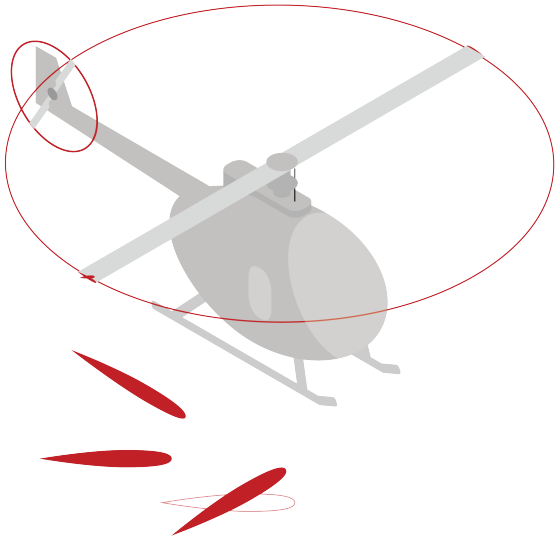


Figure 50.19: `iso-36.png`

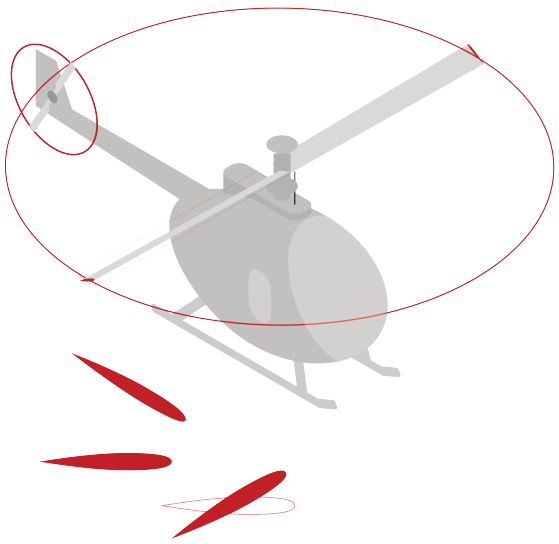


Figure 50.20: `iso-37.png`

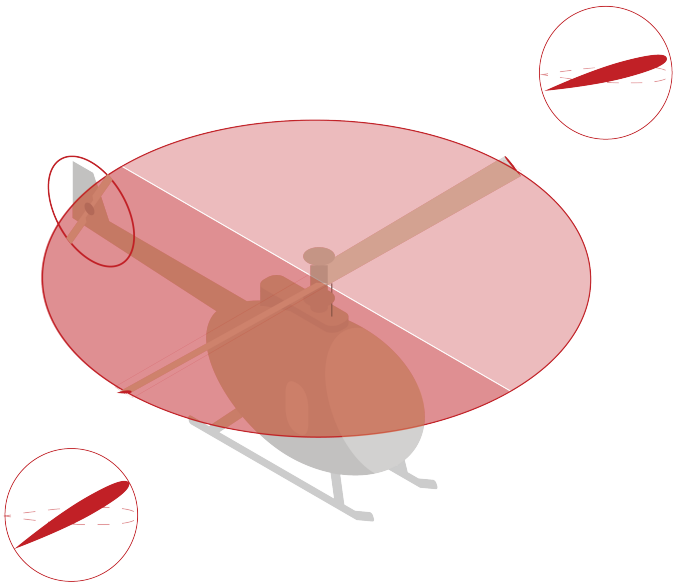


Figure 50.21: `iso-38.png`

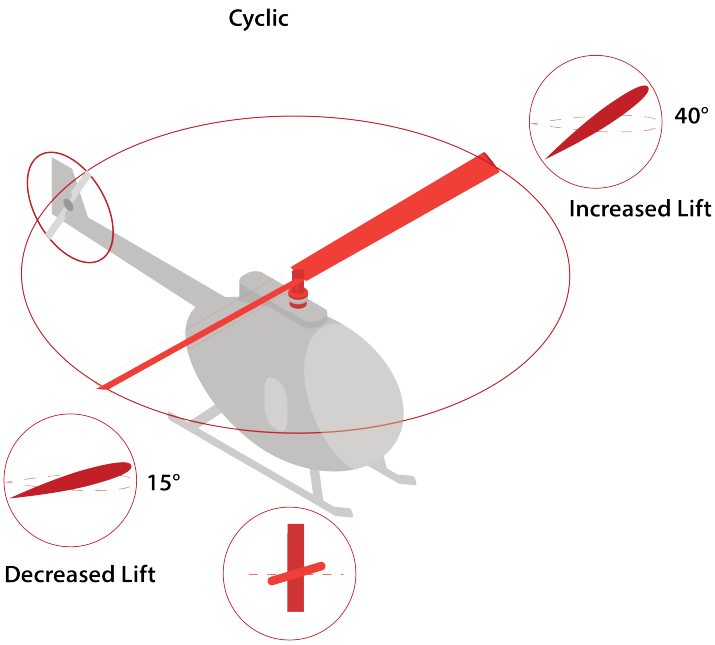



Figure 50.22: `iso-48.png`

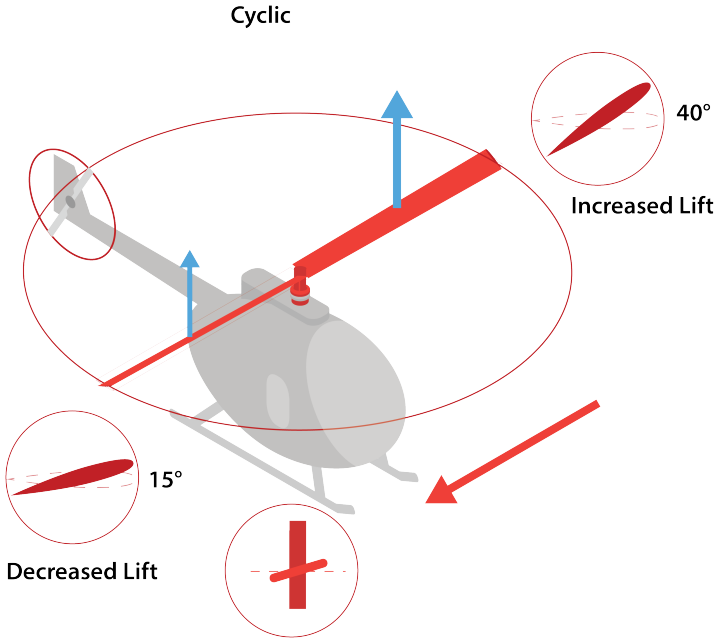



Figure 50.23: `iso-48 2.png`

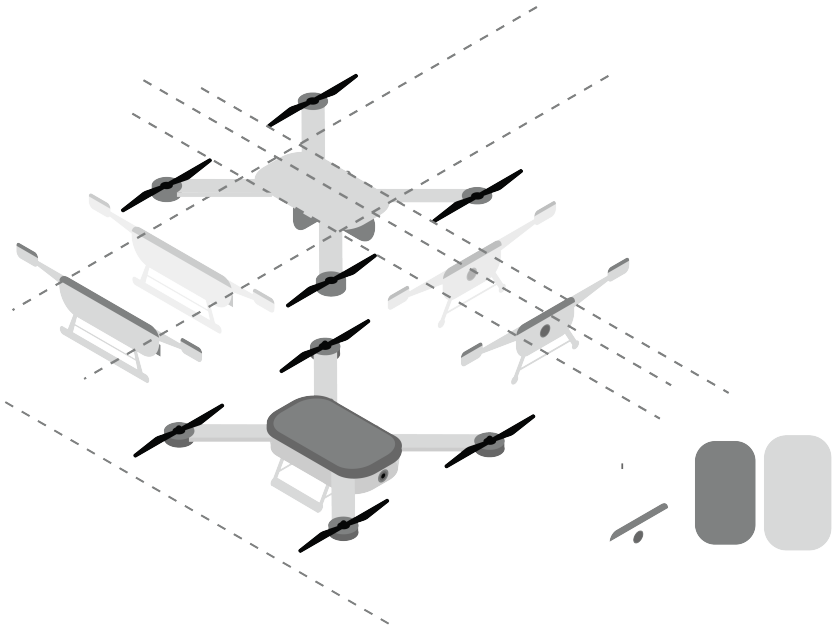


Figure 50.24: `iso.png`

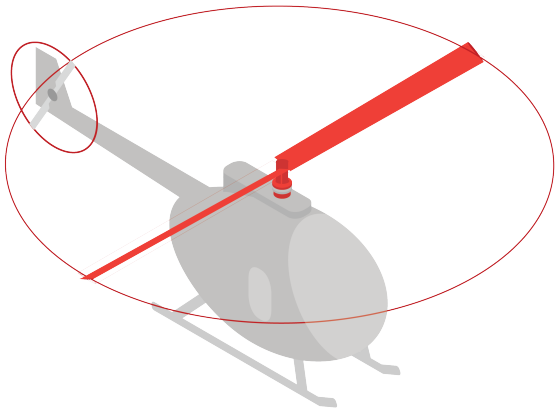


Figure 50.25: `iso46.png`

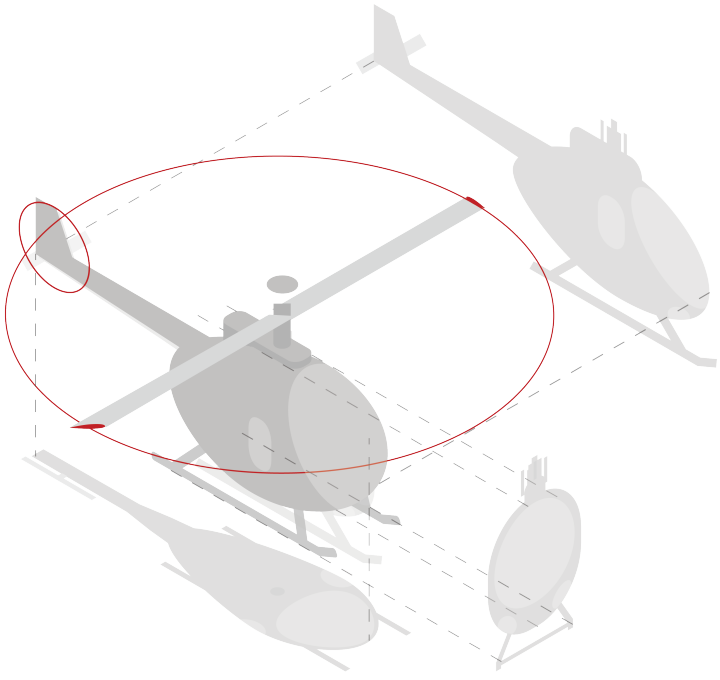


Figure 50.26: `isoprocess.png`

### 50.5.4 Copter Movement

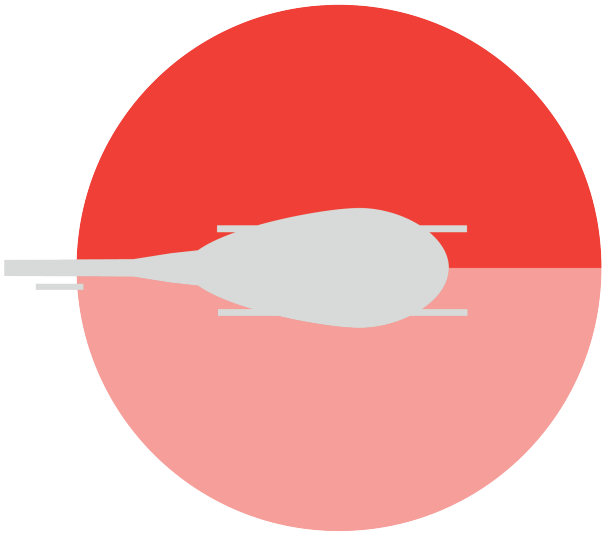


Figure 50.27: `helicopterAbove1.png`

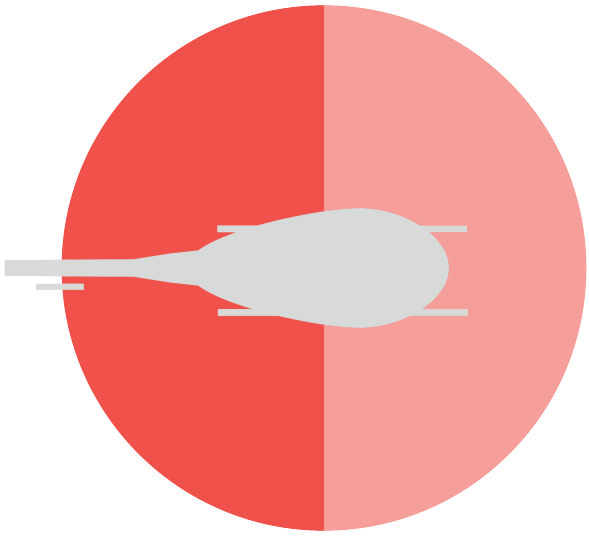


Figure 50.28: `helicopterAbove2.png`

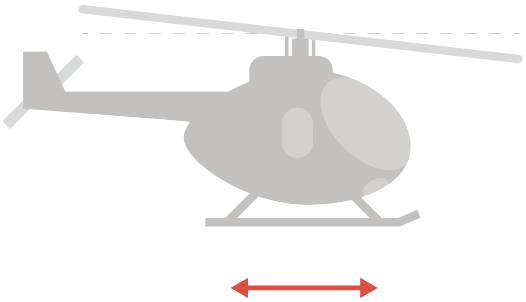


Figure 50.29: `helicopterMovement-10.png`

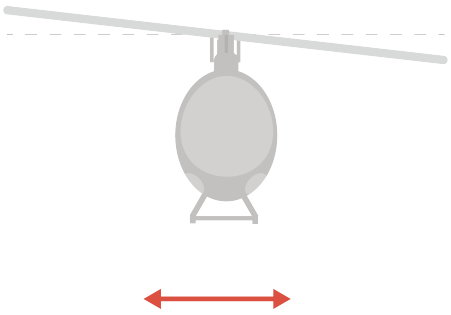


Figure 50.30: `helicopterMovement-11.png`

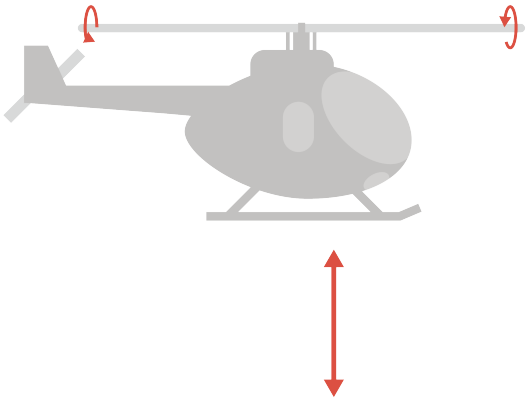


Figure 50.31: `helicopterMovement-12.png`

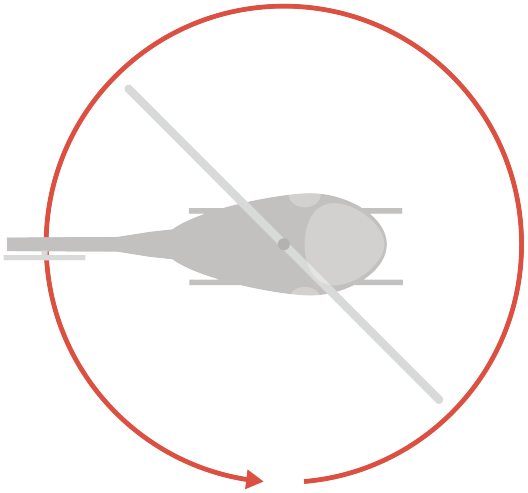


Figure 50.32: `helicopterMovement-13.png`

CHAPTER 51

# Atmospheric Pressure

The air you breathe is a blend of gases:

1. 78% nitrogen in the form of $N_2$
2. 21% oxygen in the form $O_2$
3. 1% other gases (mostly argon)

If you fill a balloon with helium (He), the helium will push against the interior of the balloon with a certain amount of pressure. The pressure is the same at every point in the interior of the balloon. Pressure, then, is force spread over some area. Force is commonly measured in newtons. Pressure is measured in *pascals*. A pascal is 1 newton per square meter.

We don't usually think about it, but the air outside the balloon is also pushing against the exterior of the balloon. We call this *barometric pressure* or *atmospheric pressure*, and it is caused by gravity pulling on the gas molecules above the balloon.

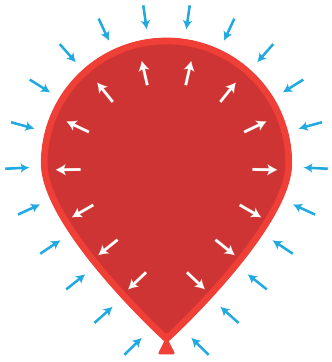


Figure 51.1: Air pushes both ways on a balloon.

Imagine a square meter on the ground at sea level. Now imagine the column of air above it — reaching all the way to the top of the atmosphere.

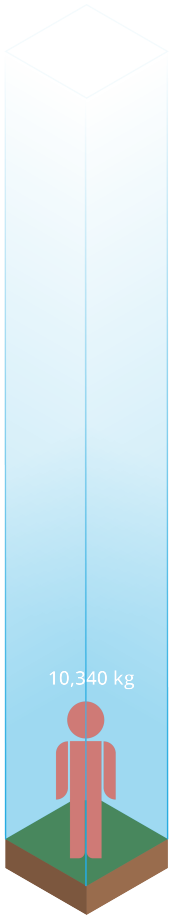



Figure 51.2: An air column surrounding a person experiences a large amount of pressure.

The air inside that column has a mass of about 10,340 kg. One kilogram on the earth experiences a gravitational force of 9.8 N. So the atmospheric pressure all around you is about 101,332 pascals. When dealing with such large numbers, we often use kilopascals. We'd say the barometric pressure at sea level is about 101.3 kPa.

That's a lot of pressure! Why doesn't your ribcage collapse crushing your lungs? The air *inside* your lungs is the same pressure as the air pushing on the outside of your rib cage.

This is why we go about our lives relatively oblivious to this huge force that is all around us — but you can see it sometimes. For example, if you suck the air out of a plastic bottle, the bottle will be crushed by the barometric pressure.

## 51.1 Altitude and Atmospheric Pressure

If you let go of the balloon, as it rises through this column there will be less and less air mass above it, and thus less and less atmospheric pressure on the outside of the balloon.

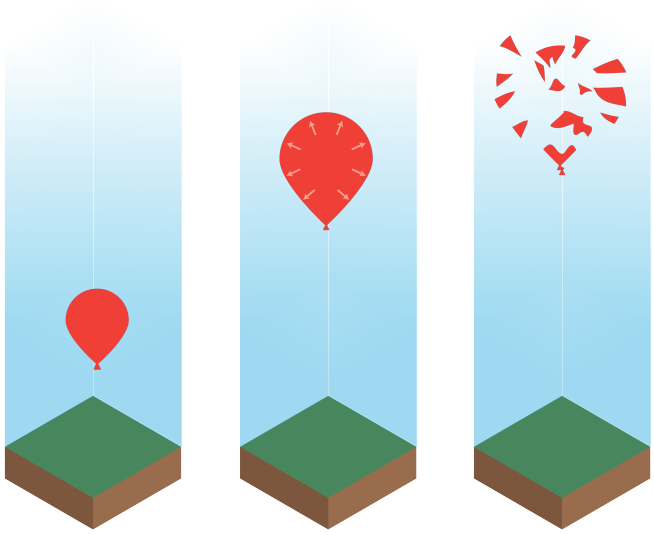


Figure 51.3: As a balloon rises, it reaches a point of less and less outer pressure.

What would be the atmospheric pressure at $h$ meters above sea level? Here is a handy formula for that:

$$p = 101,332 \times \left(1 - \left(2.25577 \times 10^{-5} \times h\right)\right)^{5.25588}$$

where $p$ is the atmospheric pressure in pascals.

### *Exercise 137* Atmospheric Pressure

*Working Space*

You are thinking about riding your bicycle to the top of Mount Everest. You are worried when the atmospheric pressure outside the tire drops, the tire will fail. (I have had a tire fail before; It is very, very loud.)

Calculate the atmospheric pressure at the top of Mount Everest (9,144 meters above sea level).

*Answer on Page 1927*

## 51.2 How a Drinking Straw Works

When you suck on a drinking straw, why does the beverage rise? It is actually pushed by atmospheric pressure.

Before you put your mouth on the straw, the atmospheric pressure is pressing on the entire surface of the liquid (even inside the straw) evenly. Gravity pulls on the liquid making the surface level.

When you suck some air out of the straw, the pressure on the surface inside the straw drops. The atmospheric pressure on the surface outside the straw pushes into the straw and the beverage rises.

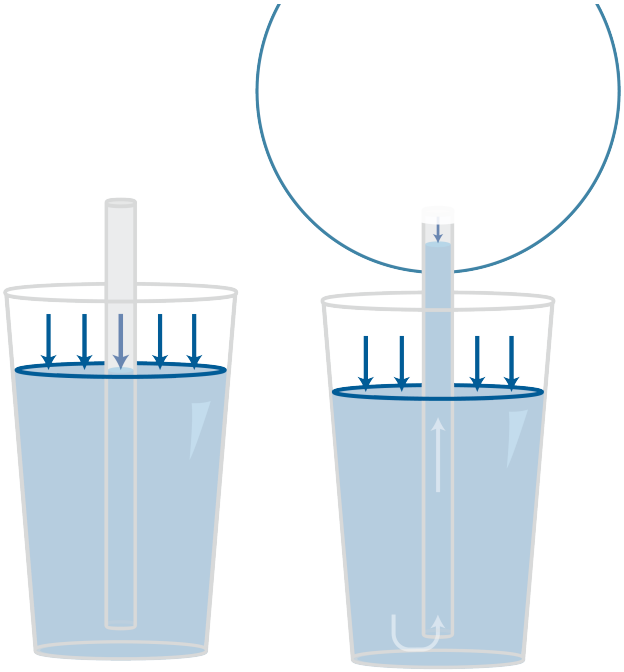


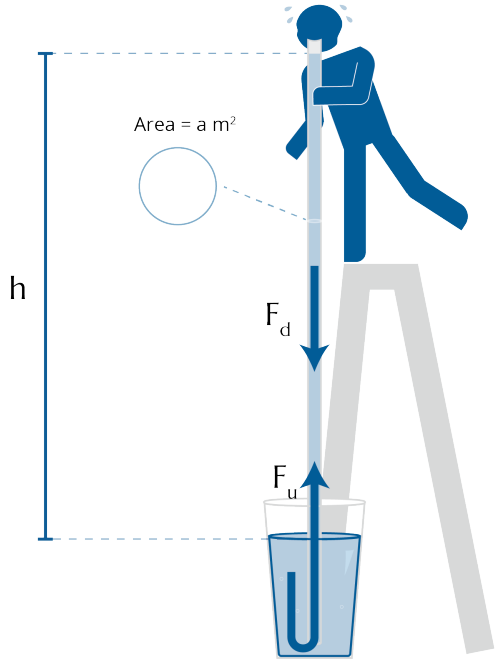
Figure 51.4: You use pressure to counteract gravity in a straw.

Of course, gravity is still trying to pull the liquid inside the straw back down. And for every inch that you lift the liquid in the straw, the force of gravity gets greater, demanding more suction.

### 51.2.1 The Longest Usable Straw

Assuming you are drinking water in a place with 100 kPa of atmospheric pressure, how high could you suck water with a perfect vacuum? That is, given a very, very long and very, very stiff drinking straw, if you created a pressure of 0 Pa inside, how far above the surface of the glass could you get the water?

Let's say a cross-section of the straw has an area of $a$ square meters, and the very top of the column of water is $h$ meters above the surface in the glass.




With how many newtons of force is the atmosphere pushing the water upward? 100 kPa = 100,000 newtons per square meter. So:

$$F_u = (100,000)a$$

With how many newtons of force is gravity pulling the water in the straw downward? The volume of the water is $ah$. A cubic meter of liquid water weights 1000 kg. The force of gravity is 9.8 Newtons per kg.

$$F_d = (ah)(1000)(9.8)$$

The water will stop rising when $F_u = F_d$. So, to find $h$, we substitute in:

$$(100,000)a = (ah)(1000)(9.8)$$

Notice that we can divide both sides by $a$ getting:

$$h = \frac{100,000}{9,800} = 10.2 \text{ meters}$$

A perfect vacuum would only be able to drag the water up 10.2 meters.

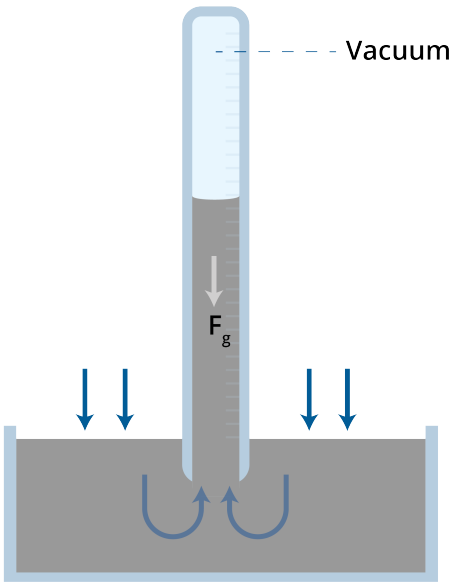
### 51.2.2 Millimeters Mercury

The density of mercury is 13,500 kg per cubic meter. How far up a straw would a perfect vacuum pull mercury?

$$h = \frac{100,000}{(9.8)(13,500)} \approx 0.756 \text{ meters}$$

When the atmospheric pressure is 100kPa, the mercury will rise 756 mm into a vacuum.

This is actually how scientists measure atmospheric pressure. They have a long glass tube filled with mercury. One end is closed off and pointed into the sky (exactly opposite the direction of gravity). The other end is placed into a dish of mercury. There are millimeter marks on the glass tube.




We use fluctuations in the atmospheric pressure to help us predict the weather. You might

hear a weather nerd with a barometer in their house say, "Wow, the barometer has gone from 752 to 761 millimeters mercury in the last hour. A high-pressure system is moving in."

## 51.3 How Siphon Works

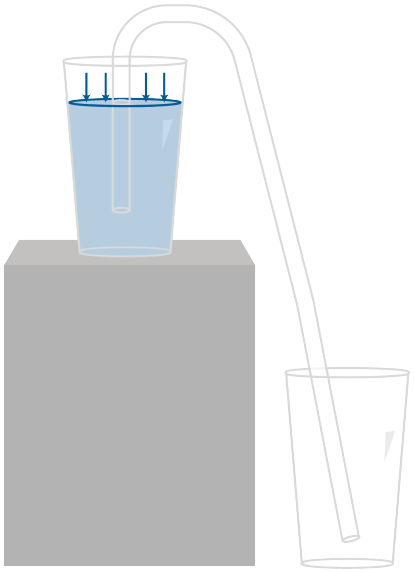
Let's say you had two cups on a table; one is filled with water, one is empty. And you connected them with an empty U -shaped straw. Water will not crawl up the empty straw — the pressure on each end of the straw is the same, and crawling the straw (against gravity) would require energy.

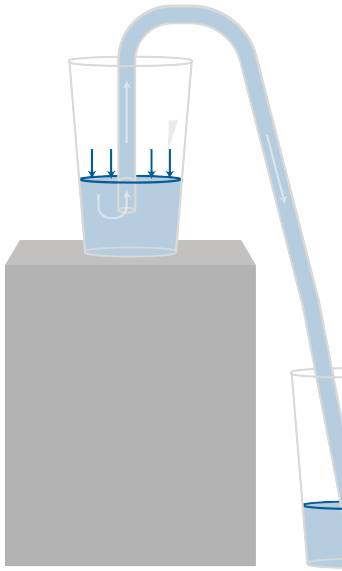
However, what if the straw were filled with water? Then the force of gravity is pulling the water on each side of the hump in different directions. However, the side going into the empty glass pulls a little harder. That is sufficient to create enough suction to pull the water in the other side up and over the hump.

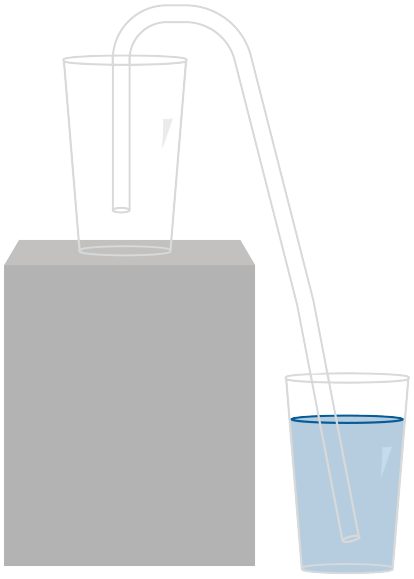
In turn, this will pull more water. Water will continue to flow from the full glass to the empty one until their surfaces are at the same level. At this point, the pull of gravity is the same on each side of the tube.

This is known as a *siphon*. Notice that atmospheric pressure makes the siphon possible. When the water on one side is pulled down by gravity, the atmospheric pressure pushes the other side up. If you were on a planet with plenty of gravity, but no atmosphere, a siphon wouldn't work.

A siphon is especially useful when you want to get liquid out of a container that is too big to pour. For example, if you wanted to take the gasoline out of a car, you could use any flexible tubing to make a siphon. You would put the hose in the gas tank, suck enough gas up into the hose to get the siphon going, then put the hose into your jug. (If you ever do this, be extra careful not to suck any of the gasoline into your mouth; ingesting even a little bit of gasoline can make you incredibly sick, or even kill you.)







There are two rules to siphons:

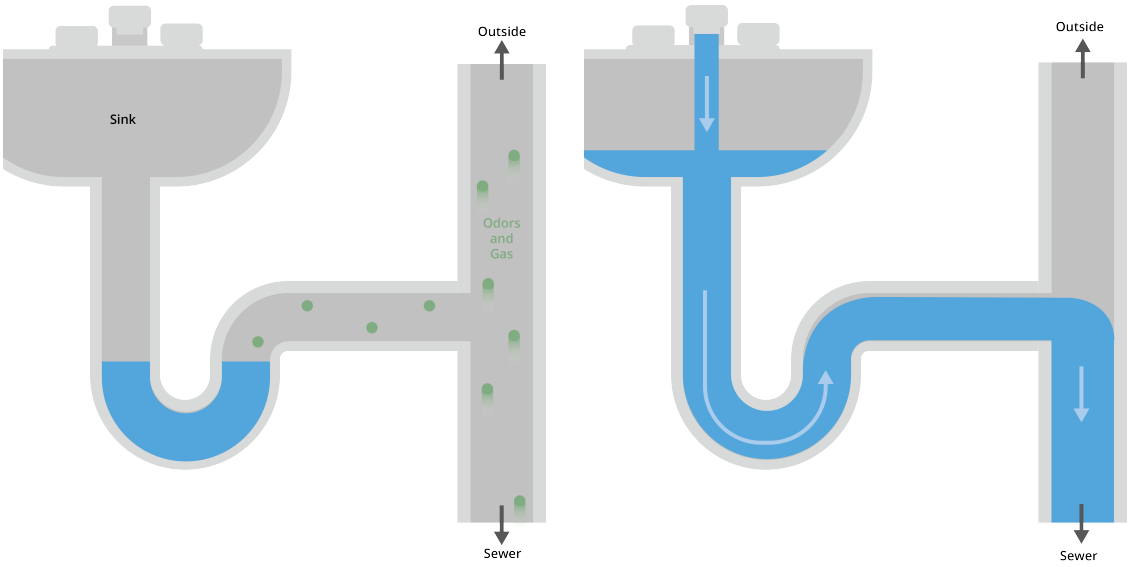
- The peak of the siphon needs to be low enough for atmospheric pressure to push the liquid that high. For water at sea-level, for example, the peak of the siphon can't be more than 10.2 meters above the surface of the source liquid.
- The tube has to carry the liquid to a lower level than the surface of the source liquid. If the destination end of the siphon is submerged, the surface of the liquid it is submerged in must be lower than the surface of the source liquid. If the destination

end of the siphon is not submerged, its opening must be lower than the surface of the source liquid.

As long as you follow these two rules, your siphons can be very creative. For example, every toilet has a siphon in it.

## 51.4 How a Toilet Works

Before we talk about toilets, you should know about P-traps. The drain from every sink, shower, and toilet in your house curves up and then down. This is known as a *P-trap*. The P-trap should always have some water in it. That keeps stinky (and flammable!) gases in the sewer from coming up and into your house.




If one of your fixtures (especially one that hasn't been used in a while) smells like raw sewage, run some water to ensure that the P-trap is full.

Now, back to the toilet! The drain in the bottom of the toilet is connected to a siphon into the sewer. The siphon is filled with air most of the time. However, when you flush the toilet, water rushes from the tank into the bowl, which also fills the siphon. Once the siphon is filled, it pulls water out of the toilet until air starts to enter the siphon. At that point, the water stops flowing.

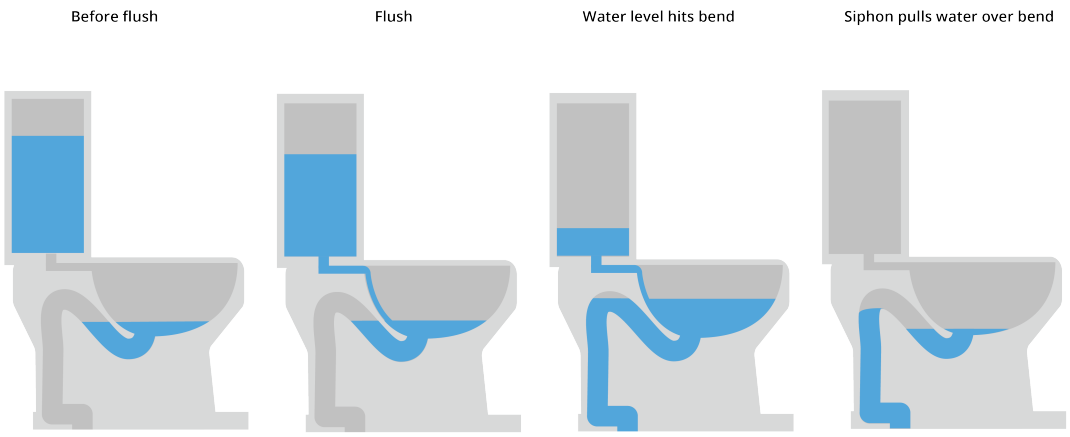



Figure 51.5: The steps a toilet takes when siphoning out waste.

The toilet tank is pretty simple: it has a float and valve that opens with the float is too low. So, anytime the water-level is too low, the value is open and slowly filling the tank. This means the tank is nearly always filled with a precise amount of water.

When you flush, a small door in the bottom of the tank is opened and the water rushes into the bowl. When the water is out, the door closes again so the tank can refill.

Chapter 52

# Solving Quadratics

A quadratic function has three terms: $ax^2+bx+c$. $a$, $b$, and $c$ are known as the *coefficients*. The coefficients can be any constant, except that $a$ can never be zero. (If $a$ is zero, it is a linear function, not a quadratic.)

When you have an equation with a quadratic function on one side and a zero on the other, you have a quadratic equation. For example:

$$72x^2 - 12x + 1.2 = 0$$

How can you find the values of $x$ that will make this equation true?

You can always reduce a quadratic equation so that the first coefficient is 1, so that your equation looks like this:

$$x^2 + bx + c = 0$$

For example, if you are asked to solve $4x^2 + 8x - 19 = -2x^2 - 7$

$$4x^2 + 8x - 19 = -2x^2 - 7$$

$$6x^2 + 8x - 12 = 0$$

$$x^2 + \frac{4}{3}x - 2 = 0$$

Here, $b = \frac{4}{3}$ and $c = -2$.

$x^2 + bx + c = 0$ when

$$x = -\frac{b}{2} \pm \frac{\sqrt{b^2 - 4c}}{2}$$

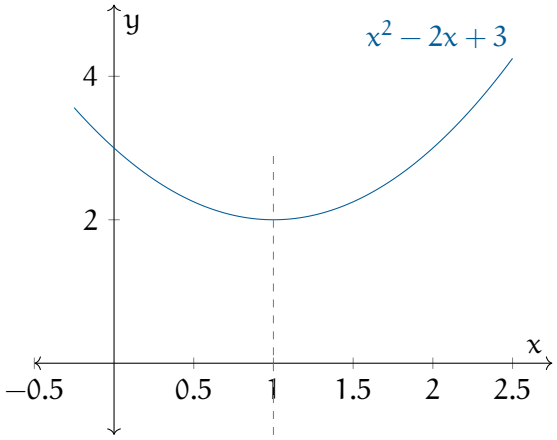
Only when $a = 1$

What does this mean?

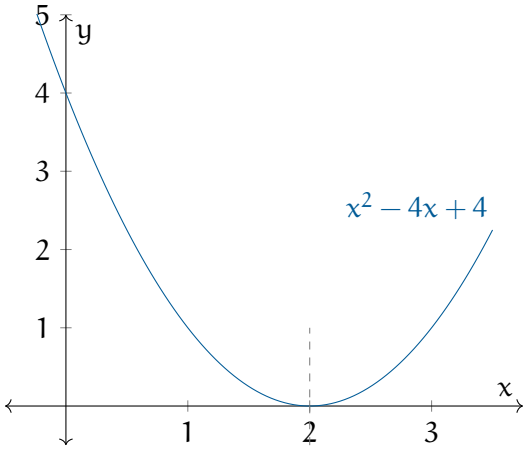
For any $b$ and $c$, the graph of $x^2 + bx + c$ is a parabola that goes up on each end. Its low point, the vertex, is at $x = -\frac{b}{2}$. For a quadratic with any value of $a$, the vertex is at

$x = -\frac{b}{2a}$, which is called the vertex formula.

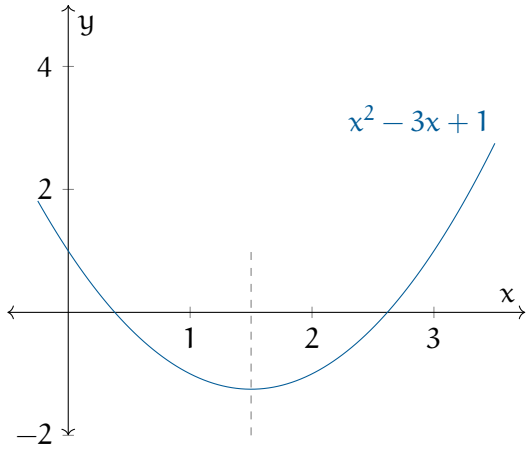
If there are no real roots ($b^2 - 4c < 0$), which means the parabola never gets low enough to cross the x-axis:




If there is one real root ($b^2 - 4c = 0$), it means that the parabola only touches the x-axis.




If there are two real roots ($b^2 - 4c > 0$), it means that the parabola crosses the x-axis twice as it dips below and then returns:




## *Exercise 138* Roots of a Quadratic

*Working Space*

In the last chapter, you found that the function for the height of your flying hammer is:

$$p = -\frac{1}{2}9.8t^2 + 12t + 2$$

At what time will the hammer hit the ground?

*Answer on Page 1927*

## 52.1 The Traditional Quadratic Formula

If the last explanation was a little tricky to understand, the quadratic formula is a nifty tool.

**The Quadratic Formula**

$ax^2 + bx + c = 0$ when

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This formula works for any quadratic equation, including ones where $a$ is not 1. Recall the equation from the start of this chapter, $6x^2 + 8x - 12 = 0$. Here $a = 6$, $b = 8$, and $c = -12$. Substitute these into the formula:

$$x = \frac{-8 \pm \sqrt{8^2 - 4(6)(-12)}}{2(6)} = \frac{-8 \pm \sqrt{64 + 288}}{12} = \frac{-8 \pm \sqrt{352}}{12}$$

Since $\sqrt{352} \approx 18.762$, the $+$ gives $x \approx \frac{10.762}{12} \approx 0.897$, and the $-$ gives $x \approx \frac{-26.762}{12} \approx -2.230$.

Watch the signs. Since $c = -12$, the term $-4ac$ becomes $-4(6)(-12) = +288$, so the number under the square root grows.

Earlier, you divided this equation by 6 to get $x^2 + \frac{4}{3}x - 2 = 0$. The formula for $a = 1$ gives the same roots: with $b = \frac{4}{3}$ and $c = -2$, $-\frac{b}{2} \pm \frac{\sqrt{b^2-4c}}{2}$ also comes out to $x \approx 0.897$ and $x \approx -2.230$. Use whichever form is easier for the numbers in front of you.

The formula also answers the question at the start of this chapter. For $72x^2 - 12x + 1.2 = 0$, the number under the square root is $(-12)^2 - 4(72)(1.2) = 144 - 345.6 = -201.6$. No real number is the square root of a negative number, so this equation has no real solutions.

## 52.2 Factoring Quadratics

The quadratic formula always works, but the arithmetic can get heavy, especially without a calculator. For some quadratics, factoring is faster.

Factoring rewrites a quadratic as a product of two simpler expressions. For example, $x^2 + 5x + 6$ can be written as $(x + 2)(x + 3)$. Expanding the product gives back the original: $x^2 + 3x + 2x + 6 = x^2 + 5x + 6$.

The factored form makes the roots easy to read because of the *zero product property*: if two numbers multiply to zero, at least one of them must be zero. So $(x + 2)(x + 3) = 0$ means $x + 2 = 0$ or $x + 3 = 0$, which gives $x = -2$ or $x = -3$. These are the same roots the quadratic formula gives, found without a square root.

Most quadratics you factor by hand have $a = 1$, in the form $x^2 + bx + c$. Certain quadratics with $a \neq 1$ also factor, using the AC method below.

### Types of Factoring Techniques

1. Factoring out the GCF (Greatest Common Factor)
   Always check for a common factor first:

$$2x^2 + 4x = 2x(x + 2)$$

2. Factoring Trinomials: $a = 1$
   For expressions like $x^2 + bx + c$, find two numbers that:
   - Multiply to $c$
   - Add to $b$

   $$x^2 + 5x + 6 = (x + 2)(x + 3)$$

3. Factoring Trinomials: $a \neq 1$
   Use the "AC method":
   (a) Multiply $a \times c$
   (b) Find two numbers that multiply to $ac$ and add to $b$
   (c) Split the middle term and factor by grouping

   $$6x^2 + 11x + 4 = 6x^2 + 3x + 8x + 4 = 3x(2x + 1) + 4(2x + 1) = (3x + 4)(2x + 1)$$

4. Special Cases:
   - Perfect Square Trinomials:
     $a^2 + 2ab + b^2 = (a + b)^2$
     $x^2 + 6x + 9 = (x + 3)^2$
   - Difference of Squares:
     $a^2 - b^2 = (a - b)(a + b)$
     $x^2 - 16 = (x - 4)(x + 4)$

**Tips**

- Always factor out the GCF first.
- Check your result by expanding (FOIL). The inverse of factoring is expanding using the FOIL method.
- Some quadratics cannot be factored using integers. For those, use the quadratic formula.

Chapter 53

# Complex Numbers

Complex numbers are an extension of real numbers, which in turn are an extension of rational numbers. In mathematics, the set of complex numbers is a number system that extends the real number line to a full two dimensions, using the imaginary unit, which is denoted by $i$, with the property that $i^2 = -1$.

An imaginary number is a number that is a multiple of the imaginary unit $i$. For example, $5i$ and $-3i$ are both imaginary numbers. While imaginary numbers are a subset of complex numbers (with real part equal to zero), not all complex numbers are imaginary. Complex numbers can have both a real and an imaginary part, while imaginary numbers specifically refer to those with no real part.

## 53.1 Definition

A complex number is a number of the form $a + bi$, where $a$ and $b$ are real numbers, and $i$ is the imaginary unit, with the property that $i^2 = -1$. The real part of the complex number is $a$, and the imaginary part is $bi$.

## 53.2 Why Are Complex Numbers Necessary?

Complex numbers are essential to many fields of science and engineering. Here are a few reasons why:

### 53.2.1 Roots of Negative Numbers

In the real number system, the square root of a negative number does not exist, because there is no real number that you can square to get a negative number. The introduction of the imaginary unit $i$, which has the property that $i^2 = -1$, allows us to take square roots of negative numbers and gives rise to complex numbers.

### 53.2.2 Polynomial Equations

The fundamental theorem of algebra states that every non-constant polynomial equation with complex coefficients has a complex root. This theorem guarantees that polynomial equations of degree $n$ always have $n$ roots in the complex plane.

### 53.2.3 Physics and Engineering

In physics and engineering, complex numbers are used to represent waveforms in control systems, in quantum mechanics, and many other areas. Their properties make many mathematical manipulations more convenient.

## 53.3 Adding Complex Numbers

The addition of complex numbers is straightforward. If we have two complex numbers $z_1 = a + bi$ and $z_2 = c + di$, their sum is defined as:

$$z_1 + z_2 = (a + c) + (b + d)i \tag{53.1}$$

In other words, you add the real parts to get the real part of the sum, and add the imaginary parts to get the imaginary part of the sum.

## 53.4 Multiplying Complex Numbers

The multiplication of complex numbers is a bit more involved. If we have two complex numbers $z_1 = a + bi$ and $z_2 = c + di$, their product is defined as:

$$z_1 \cdot z_2 = (a + bi) \cdot (c + di) = ac + adi + bci - bd = (ac - bd) + (ad + bc)i \tag{53.2}$$

Note the last term comes from $i^2 = -1$. You multiply the real parts and the imaginary parts just as you would in a binomial multiplication, and remember to replace $i^2$ with $-1$. See the Khan academy video in the digital resources for a more in-depth explanation.

CHAPTER 54

# Introduction to Sequences

A sequence is a list of numbers in a particular order. {1, 3, 5, 7, 9} is a sequence. So is $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \cdots\}$. There are many types of sequences. We will present two of the most common types in this chapter: arithmetic and geometric sequences.

Sequences are generally represented like this:

$$a_1, a_2, a_3, a_4, ..., a_n, ...$$

The first number, $a_1$, is called the *first term*, $a_2$ is the *second term*, and $a_n$ is the *nth term*. A sequence can be finite or infinite. If the sequence is infinite, we represent that with ellipses $(\cdots)$ at the end of the list, to indicate that there are more numbers.

We can also write formulas to represent a sequence. Take the first example, the finite sequence {1, 3, 5, 7, 9}. Notice that each term is two more than the previous term. We can define the sequence *recursively* by defining the $n^{th}$ term as a function of the $(n-1)^{th}$ term. In our example, we see that $a_n = a_{n-1} + 2$ with $a_1 = 1$ for $1 \leq n \leq 5$. This is called a recursive formula, because you have to already know the $(n-1)^{th}$ term to find the $n^{th}$ term.

Another way to write a formula for a sequence is to find a rule for the $n^{th}$ term. In our example sequence, the first term is 1 plus 0 times 2, the second term is 1 plus 1 times 2, the third term is 1 plus 2 times 2, and so one. Did you notice the pattern? The $n^{th}$ term is 1 plus (n-1) times 2. We can write this mathematically:

$$a_n = 1 + 2(n-1) \text{ for } 1 \leq n \leq 5$$

This is called the *explicit* formula because each term is explicitly defined. Notice that for the second way of writing a formula, we don't have to state what the first term is — the formula tells us.

## 54.1 Arithmetic sequences

Our first example sequence, {1, 3, 5, 7, 9} is a *finite*, *arithmetic* sequence. We know it is finite because there is a limited number of terms in the sequence (in this case, 5). How do we know it is arithmetic?

An arithmetic sequence is one where you add the same number every time to get the next term. Our example is an arithmetic sequence because you add 2 to get the next term every time. That number that you add is called the *common difference*, so we can say the sequence {1, 3, 5, 7, 9} has a common difference of 2. The common difference can be positive (in the case of an increasing arithmetic sequence) or negative (in the case of a decreasing arithmetic sequence). Formally, we can find the common difference of an arithmetic sequence by subtracting the $(n-1)^{th}$ term from the $n^{th}$ term:

$$d = a_n - a_{n-1}$$

### *Exercise 139*

Which of the following are arithmetic sequences? For the arithmetic sequences, find the common difference.

*Working Space*

1. $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, ...\}$
2. {5, 8, 11, 14, 17, ...}
3. {3, -1, -5, -9, ...}
4. {-1, 2, -3, 4, -5, 6, ...}

*Answer on Page 1927*

#### 54.1.1 Formulas for arithmetic sequences

If you are given an arithmetic sequence, you can write an explicit or recursive formula. You can think of the formula as a function where the domain (input) is restricted to integers greater than or equal to one. Let's write explicit and recursive formulas for the sequence {3, -1, -5, -9, ...}.

For either type of formula, we need to identify the common difference. Since each term is 4 less than the previous term, the common difference is -4 (see figure 54.1). This means the $n^{th}$ term is the $(n-1)^{th}$ term minus 4. The general form of a recursive formula is $a_n = a_{n-1} + d$, where d is the common difference. For our example, the common difference is -4, so we can write a recursive formula:

$$a_n = a_{n-1} - 4$$

However, this formula doesn't tell us what $a_1$ is! For recursive formulas, you have to specify the first term in the sequence. So, the *complete* recursive formula for the sequence is:

$$a_n = a_{n-1} - 4$$

$$a_1 = 3$$

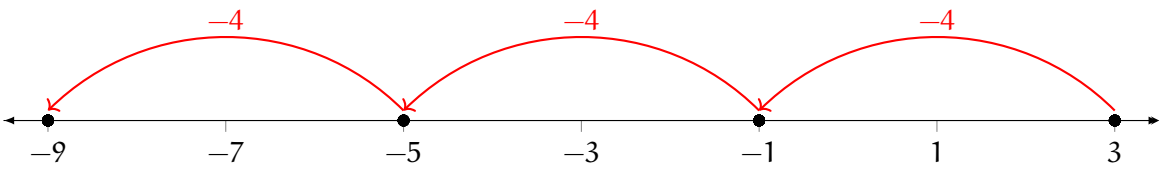



Figure 54.1: The common difference in the sequence {3, -1, -5, -9,...} is -4

Recursive formulas make it easy to see how each term is related to the next term. However, it is difficult to use recursive formulas to find a specific term. Say we wanted to know the $7^{th}$ term in the sequence. Well, from the formula, we know that:

$$a_7 = a_6 - 4$$

What is $a_6$? Again, we see that

$$a_6 = a_5 - 4$$

Now we have to find $a_5$! If we keep going, we see that:

$$a_5 = a_4 - 4$$

$$a_4 = a_3 - 4$$

$$a_3 = a_2 - 4$$

$$a_2 = a_1 - 4$$

Since we were told $a_1$, we can find $a_2$ and propagate our terms back up the chain to find $a_7$:

$$a_2 = 3 - 4 = -1$$

$$a_3 = a_2 - 4 = -1 - 4 = -5$$

$$a_4 = a_3 - 4 = -5 - 4 = -9$$

$$a_5 = a_4 - 4 = -9 - 4 = -13$$

$$a_6 = a_5 - 4 = -13 - 4 = -17$$

$$a_7 = a_6 - 4 = -17 - 4 = -21$$

Ultimately, we see that $a_7 = -21$. That was a lot of work! You can imagine that for higher $n$ terms, such as the $100^{th}$ or $1000^{th}$ term, this method becomes cumbersome. This is where the explicit formula is more useful.

The general form of an explicit formula for an arithmetic sequence is

$$a_n = a_1 + d \times (n - 1)$$

where $d$ is the common difference. For our example sequence, {3, -1, -5, -9, ...}, the common difference is $-4$. So the explicit formula is

$$a_n = 3 + (-4)(n - 1) = 3 - 4(n - 1)$$

You may be tempted to distribute and simplify, which is fine and yields an equivalent formula:

$$a_n = 7 - 4n$$

Now, to find the $7^{th}$ term, all we have to do is substitute $n = 7$:

$$a_7 = 3 - 4(7 - 1) = 3 - 4(6) = 3 - 24 = -21$$

We get the same answer with much less effort!

Another example of a recursive sequence is the Fibonacci sequence which we will cover in-depth later

## Exercise 140

An arithmetic sequence is defined by the recursive formula $a_n = a_{n-1} + 5$ with $a_1 = -4$. Write the first 5 terms of the sequence and determine an explicit formula for the same sequence.

*Working Space*

*Answer on Page 1928*

### *Exercise 141*

The first four terms of an arithmetic sequence are $\{\pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}\}$. What is the common difference? Write explicit and recursive formulas for the infinite sequence.

*Working Space*

*Answer on Page 1928*

## 54.2 Geometric sequences

Let's look at the other sequence given as an example at the beginning of the chapter: $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \cdots\}$. How is each term related to the previous term? Well, $\frac{1}{4}$ is half of $\frac{1}{2}$, and $\frac{1}{8}$ is half of $\frac{1}{4}$, so each term is the previous term multiplied by $\frac{1}{2}$. When each term in a sequence is a multiple of the previous term, this is a *geometric* sequence. The number we multiply by each time (in our example, this is $\frac{1}{2}$), which is called the *common ratio*. The common ratio can be positive or negative, but not zero.

An easy way to determine the common ratio $(r)$ is to divide the $n^{th}$ term by the $(n-1)^{th}$ term. In our example sequence, the first term is $\frac{1}{2}$ and the second is $\frac{1}{4}$.

$$r = \frac{a_2}{a_1} = \frac{1/4}{1/2} = \frac{1}{2}$$

which returns the common ratio we already identified, $r = \frac{1}{2}$.

If the common ratio is negative, then the sequence will "flip" back and forth from positive to negative. For example, suppose there is a geometric sequence such that $a_1 = 1$ and $r = -2$. Then the first 5 terms are {1, -2, 4, -8, 16}. Whenever you see a sequence going back and forth from positive to negative, that means the common ratio is negative.

For positive ratios $(r > 1)$, the sequence is increasing. And for negative ratios, $(r < 1)$, the sequence is decreasing.

### 54.2.1 Formulas for geometric sequences

Like arithmetic sequences, we can write recursive and explicit formulas. For geometric sequences, the recursive formula has the general form:

$$a_n = r(a_{n-1})$$

where r is the common ratio and $a_1$ is specified. For our example sequence, $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \cdots\}$, the recursive formula is:

$$a_n = \frac{1}{2}a_{n-1}$$

$$a_1 = \frac{1}{2}$$

In a geometric sequence, each term is the first term, $a_1$, multiplied by the common ratio, r, $n-1$ times. Therefore, the general form of an explicit formula for a geometric function is:

$$a_n = (a_1)r^{n-1}$$

Again, for our example sequence, $a_1 = \frac{1}{2}$ and $r = \frac{1}{2}$, so the explicit formula is:

$$a_n = (\frac{1}{2})(\frac{1}{2})^{(n-1)}$$

## *Exercise 142*

Which of the following are geometric sequences? For each geometric sequence, determine the common ratio.

*Working Space*

1. {2, -4, 6, -8,...}
2. {4, 2, 1, $\frac{1}{2}$, ...}
3. {-5, 25, -125, 525, ...}
4. {2, 0, -2, -4, ...}

*Answer on Page 1928*

### Exercise 143

A geometric sequence is defined by the recursive formula $a_n = a_{n-1} \times \frac{3}{2}$ with $a_1 = 1$. Write the first five terms of the sequence and determine an explicit formula for the same sequence.

*Working Space*

*Answer on Page 1928*

### Exercise 144

The first four terms of a geometric sequence are $\{-4, 2, -1, \frac{1}{2}\}$. What is the common ratio? Write recursive and explicit formulas for the infinite sequence.

*Working Space*

*Answer on Page 1928*

CHAPTER 55

# Exponents

Let's take quick look at exponents. Ancient scientists started coming up with a lot of formulas that involved multiplying the same number several times. For example, if they knew that a sphere was $r$ centimeters in radius, its volume in milliliters was

$$V = \frac{4}{3} \times \pi \times r \times r \times r$$

They did two things to make the notation less messy. First, they decided that if two numbers were written next to each other, the reader would assume that meant "multiply them". Second, they came up with the exponent, a little number that was lifted off the baseline of the text, that meant "multiply it by itself". For example $5^3$ was the same as $5 \times 5 \times 5$.

Now, the formula for the volume of a sphere is written

$$V = \frac{4}{3}\pi r^3$$

Tidy, right? In an exponent expression like this, we say that $r$ is *the base* and 3 is *the exponent*.

## 55.1 Identities for Exponents

What about exponents of exponents? What is $\left(5^3\right)^2$?

$$\left(5^3\right)^2 = (5 \times 5 \times 5)^2 = (5 \times 5 \times 5)(5 \times 5 \times 5) = 5^6$$

In general, for any $a$, $b$, and $c$:

$$\left(a^b\right)^c = a^{(bc)}$$

If you have $\left(5^3\right)\left(5^4\right)$, that is just $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$ or $5^7$

The general rule is that for any $a$, $b$, and $c$

$$\left(a^b\right)\left(a^c\right) = a^{(b+c)}$$

Mathematicians *love* this rule, so we keep extending the idea of exponents to keep this rule true. For example, at some point, someone asked "What about $5^0$?" According to the rule, $5^2$ must equal $5^{(2+0)}$ which must equal $\left(5^2\right)\left(5^0\right)$. Thus, $5^2$ must be 1. So mathematicians declared "Anything to the power of 0 is 1".

We don't typically assume that $0^0 = 1$. It is just too weird. So, we say that for any $a$ not equal to zero,

$$a^0 = 1$$

What about $5^{(-2)}$? By our beloved rule, we know that $\left(5^{-2}\right)\left(5^5\right)$ must be equal to $5^3$, right? So $5^{-2}$ must be equal to $\frac{1}{5^2}$.

We say that for any $a$ not equal to zero and any $b$,

$$a^{-b} = \frac{1}{a^b}$$

This makes dividing one exponential expression by another (with the same base) easy:

$$\frac{a^b}{a^c} = a^{(b-c)}$$

We often say "cancel out" for this. Here, we can "cancel out" $x^2$:

$$\frac{x^5}{x^2} = x^3$$

What about $5^{\frac{1}{3}}$? By the beloved rule, we know that $5^{\frac{1}{3}}5^{\frac{1}{3}}5^{\frac{1}{3}}$ must equal $5^1$. Thus, $5^{\frac{1}{3}} = \sqrt[3]{5}$.

We say that for any $a$ and $b$ not equal to zero and any $c$ greater than zero,

$$a^{\frac{b}{c}} = \left(\sqrt[c]{a}\right)^b = \left(\sqrt[c]{a^b}\right)$$

Before you go on to the exercises, note that the beloved rule demands a common base.

- We can combine these: $\left(5^2\right)\left(5^4\right) = 5^6$
- We cannot combine: $\left(5^2\right)\left(3^5\right)$

With that said, we note for any $a$,$b$, and $c$:

$$(ab)^c = \left(a^c\right)\left(b^c\right)$$

So, for example, if we were asked to simplify $\left(3^4\right)\left(6^2\right)$, we would note that $6 = 2 \times 3$, so

$$\left(3^4\right)\left(6^2\right) = \left(3^4\right)\left(3^2\right)\left(2^2\right) = \left(3^6\right)\left(2^2\right)$$

If these ideas are new to you (or maybe you are just having trouble remembering them), watch the Khan Academy's **Intro to rational exponents** video at https://youtu.be/lZfXc4nHooo.

Chapter 56

# Compound Interest

When you loan money to someone, you typically charge them some sort of interest. The most common loan of this sort is what the bank calls a "savings account". Any money you put in the account is loaned to the bank. The bank then lends it to someone else, who pays interest to the bank. The bank gives some of that interest to you. However, what if you leave the interest in your account? And you start making *interest on the interest*? This is known as *compound interest*.

## 56.1 An example with annual interest payments

Let's say that you put $1000 in a savings account that pays 6% interest every year. How much money would you have after 12 years? Let's make a spreadsheet.

| | A | B | C |
|---|---|---|---|
| 1 | **Interest Rate** | 6.00% | |
| 2 | | | |
| 3 | **After year:** | **Interest** | **Balance** |
| 4 | 0 | $0.00 | 1000 |
| 5 | 1 | $60.00 | $1,060.00 |

Create a new spreadsheet and edit the cells to look like this. All the cells in rows 1 - 4 are just values: just type in what you see.

The fifth row is all formulas:

| After year | Interest | Balance |
|---|---|---|
| = A4 + 1 | = B$1 * C4 | = C4 + B5 |

The interest rate field should be formatted as a percentage. One thing to know when dealing with percentages in the spreadsheet: If the field says "600%", its value is 6.

The cells in the Interest and Balance column should be formatted as currency.

You are about to make several copies of the cells in the fifth row, so make sure they look right.

Click on A5 and shift-click on C5 to select all three cells. Drag the lower-right corner down to fill the rows 6 - 15.

A5:C16 fx = A4 + 1

| | A | B | C |
|---|---|---|---|
| 1 | **Interest Rate** | 6.00% | |
| 2 | | | |
| 3 | **After year** | **Interest** | **Balance** |
| 4 | 0 | $0.00 | 1000 |
| 5 | 1 | $60.00 | $1,060.00 |
| 6 | 2 | $63.60 | $1,123.60 |
| 7 | 3 | $67.42 | $1,191.02 |
| 8 | 4 | $71.46 | $1,262.48 |
| 9 | 5 | $75.75 | $1,338.23 |
| 10 | 6 | $80.29 | $1,418.52 |
| 11 | 7 | $85.11 | $1,503.63 |
| 12 | 8 | $90.22 | $1,593.85 |
| 13 | 9 | $95.63 | $1,689.48 |
| 14 | 10 | $101.37 | $1,790.85 |
| 15 | 11 | $107.45 | $1,898.30 |
| 16 | 12 | $113.90 | $2,012.20 |
| 17 | | | |

Look at the numbers. The first interest payment is $60, but the last is $113.90. Your balance has more than doubled!

## 56.2 Exponential Growth

We figured this out numerically by repeatedly multiplying the balance by the interest rate. What if you wanted to know what the balance would be $n$ years after investing $P_0$ dollars with an annual interest rate of $r$? (Note that $r$ in our example would be 0.06, not 6.0.)

Each year, the balance is multiplied by $1+r$, so after one year, $P_0$ would become $P_0 \times (1+r)$. The next year you would multiply this number by $(1+r)$ again: $P_0 \times (1+r) \times (1+r)$. The next year? $P_0 \times (1+r) \times (1+r) \times (1+r)$ See the pattern? $P_n$ is this balance after $n$ years, then

$$P_n = P_0(1+r)^n \tag{56.1}$$

56.1 is referred to as the standard *compound interest formula*. Because $n$ is an exponent, we call this *exponential growth*. There are few things as terrifying to a scientist as the phrase "The population is undergoing exponential growth".

If the principal is not compounded annually, we can modify the equation to account for that by changing $n$.

$$n = mt$$

Where $t$ is the frequency of the compounding, called the period: (1: annually, 12: monthly, 52: weekly, 365: daily), and $m$ is the number of periods. This equation becomes

$$P_n = P_0(1 + \frac{r}{n})^{nt}$$

We will see these two terms pop up in other areas of the curriculum like circular motion and waves.

## 56.3 Sensitivity to interest rate

For most people, the first surprising thing about compound interest is how quickly your money grows after a few years. The second thing that is surprising is how much difference a small change in the percentage rate makes.

Let's add another set of columns that shows what happens to your money if you convince the bank to pay you 8% instead of 6%.

Copy everything from columns B and C:

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Interest Rate** | 6.00% | | 6.00% | |
| 2 | | | | | |
| 3 | **After year** | **Interest** | **Balance** | **Interest** | **Balance** |
| 4 | 0 | $0.00 | 1000 | $0.00 | 1000 |
| 5 | 1 | $60.00 | $1,060.00 | $60.00 | $1,060.00 |
| 6 | 2 | $63.60 | $1,123.60 | $63.60 | $1,123.60 |
| 7 | 3 | $67.42 | $1,191.02 | $67.42 | $1,191.02 |
| 8 | 4 | $71.46 | $1,262.48 | $71.46 | $1,262.48 |
| 9 | 5 | $75.75 | $1,338.23 | $75.75 | $1,338.23 |
| 10 | 6 | $80.29 | $1,418.52 | $80.29 | $1,418.52 |
| 11 | 7 | $85.11 | $1,503.63 | $85.11 | $1,503.63 |
| 12 | 8 | $90.22 | $1,593.85 | $90.22 | $1,593.85 |
| 13 | 9 | $95.63 | $1,689.48 | $95.63 | $1,689.48 |
| 14 | 10 | $101.37 | $1,790.85 | $101.37 | $1,790.85 |
| 15 | 11 | $107.45 | $1,898.30 | $107.45 | $1,898.30 |
| 16 | 12 | $113.90 | $2,012.20 | $113.90 | $2,012.20 |
| 17 | | | | | |

Now edit the second interest rate to be 8%:

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Interest Rate** | 6.00% | | 8.00% | |
| 2 | | | | | |
| 3 | **After year** | **Interest** | **Balance** | **Interest** | **Balance** |
| 4 | 0 | $0.00 | 1000 | $0.00 | 1000 |
| 5 | 1 | $60.00 | $1,060.00 | $80.00 | $1,080.00 |
| 6 | 2 | $63.60 | $1,123.60 | $86.40 | $1,166.40 |
| 7 | 3 | $67.42 | $1,191.02 | $93.31 | $1,259.71 |
| 8 | 4 | $71.46 | $1,262.48 | $100.78 | $1,360.49 |
| 9 | 5 | $75.75 | $1,338.23 | $108.84 | $1,469.33 |
| 10 | 6 | $80.29 | $1,418.52 | $117.55 | $1,586.87 |
| 11 | 7 | $85.11 | $1,503.63 | $126.95 | $1,713.82 |
| 12 | 8 | $90.22 | $1,593.85 | $137.11 | $1,850.93 |
| 13 | 9 | $95.63 | $1,689.48 | $148.07 | $1,999.00 |
| 14 | 10 | $101.37 | $1,790.85 | $159.92 | $2,158.92 |
| 15 | 11 | $107.45 | $1,898.30 | $172.71 | $2,331.64 |
| 16 | 12 | $113.90 | $2,012.20 | $186.53 | $2,518.17 |

## 56.4 Continuous Compound Interest

Let's say, hypothetically, that we compound interest on a given balance for an *indefinite* amount of time. Then, we take the formula

$$P_n = P_0(1 + \frac{r}{n})^{nt}$$

What if we divide our time period into infinite chunks? This would mean splitting our $n$ into much smaller pieces. We can use calculus notation to find the limit of this:

$$\lim_{n\to\infty} P_0(1 + \tfrac{r}{n})^{nt}$$

we can create a variable $x = \frac{n}{r}$. We do this so that we eliminate some other dummy variables. Now we have

$$\lim_{n\to\infty} P_0(1 + \tfrac{1}{x})^{xrt}$$

as we can eliminate $n$ with $n = xr$.

Now we can isolate parts of our compound interest equation into the limit, and extract variables out.

$$P_0 \left(\lim_{n\to\infty} (1 + \tfrac{1}{x})^{x}\right)^{rt}$$

Fun math fact: $\lim_{n\to\infty}(1 + \frac{1}{x})^x$ is equal to the mathematical constant $e$. We are going to dive into this in the *logs* chapter! For now, just understand that it is a constant in mathematics that describes many growth or decay values. With that, we can simplify the equation as the continuous compound interest equation (56.2)

$$P_0 e^{rt} \tag{56.2}$$

## 56.5 Practice Exercises

### *Exercise 145*

You deposit $1500 into a savings account that pays an annual interest rate of 5%, compounded annually. What is the balance after 8 years?

Note: you can also use excel or google sheets built in function `=FV()`.

*Working Space*

*Answer on Page 1928*

### *Exercise 146*

A bank offers two savings accounts, both with a stated annual interest rate of 6%. Account A compounds monthly ($t = 12$) and Account B compounds daily ($t = 365$). If you deposit $3000 into each account, which account has the higher balance after 5 years, and by how much?

Create two columns in Excel/Sheets showing the balance after each year for each account, using $3000 as the principal. Start with year 0 for the initial deposit, then fill in years 1 through 5 using the compound interest formula for monthly and daily compounding.

*Working Space*

*Answer on Page 1929*

CHAPTER 57

# Logarithms

After the world created exponents, it needed the opposite. We could talk about the quantity $? = 2^3$, that is, "What is the product of 2 multiplied by itself three times?" We needed some way to talk about $2^? = 8$, that is, "2 to the what is 8?" This is why we developed the logarithm.

Here is an example:

$$\log_2 8 = 3$$

In English, you would say, "The logarithm base 2 of 8 is 3."

The base (2, in this case) can be any positive number. The argument (8, in this case) can also be any positive number.

Try this one: What is the logarithm base 2 of 1/16?

You know that $2^{-4} = \frac{1}{16}$, so $\log_2 \frac{1}{16} = -4$.

## 57.1 Logarithms in Python

Most calculators have pretty limited logarithm capabilities, but python has a nice `log` function that lets you specify both the argument and the base. Start python, import the math module, and try taking a few logarithms:

```
>>> import math
>>> math.log(8,2)
3.0
>>> math.log(1/16, 2)
-4.0
```

Let's say that a friend offers you 5% interest per year on your investment for as long as you want. You wonder, "How many years before my investment is 100 times as large?" You can solve this problem with logarithms:

```
>>> math.log(100, 1.05)
94.3872656381287
```

If you leave your investment with your friend for 94.4 years, the investment will be worth 100 times what you put in.

## 57.2 Logarithm Identities

The logarithm is defined this way:

$$\log_b a = c \iff b^c = a$$

Notice that the logarithm of 1 is always zero, and $\log_b b = 1$.

The logarithm of a product:

$$\log_b ac = \log_b a + \log_b c$$

This follows from the fact that $b^{a+c} = b^a b^c$. What about a quotient?

$$\log_b \frac{a}{c} = \log_b a - \log_b c$$

Exponents?

$$\log_b (a^c) = c \log_b a$$

Notice that because logs and exponents are the opposite of each other, they can cancel each other out:

$$b^{\log_b a} = a$$

and

$$\log_b (b^a) = a$$

## 57.3 Changing Bases

We mentioned that most calculators have pretty limited logarithm capabilities. Most calculators don't allow you to specify what base you want to work with. All scientific calculators have a button for "log base 10". So, you need to know how to use that button to get logarithms for other bases. Here is the change-of-base identity:

$$\log_b a = \frac{\log_c a}{\log_c b}$$

For example, if you wanted to find $\log_2 8$, you would ask the calculator for $\log_{10} 8$, then divide that by $\log_{10} 2$. You should get 3.

## 57.4 Natural Logarithm

When you learn about circles, you are told that the circumference of a circle is about 3.141592653589793 times its diameter. Because we use this unwieldy number a lot, we give it a name: We say "The circumference of a circle is $\pi$ times its diameter."

There is a second unwieldy number that we will eventually use a great deal in solving problems. It is about 2.718281828459045 (but the digits actually go on forever, just like $\pi$). We call this number $e$. (We are not going to talk about why $e$ is special quite yet, but we will soon.)

Most calculators have a button labeled "ln". That is the *natural logarithm* button. It takes the log in base $e$.

Similarly, in python, if you don't specify a base, the logarithm is done in base $e$:

```
>>> math.log(10)
2.302585092994046
>>> math.log(math.e)
1.0
```

## 57.5 Logarithms in Spreadsheets

Spreadsheets have three log functions:

- `LOG` takes both the argument and the base. `LOG(8,2)` returns 3.
- `LOG10` takes just the argument and uses 10 as the base.

- `LN` takes just the argument and uses $e$ as the base.

Here is a plot from a spreadsheet of a graph of $y = \text{LOG}(x, 2)$.

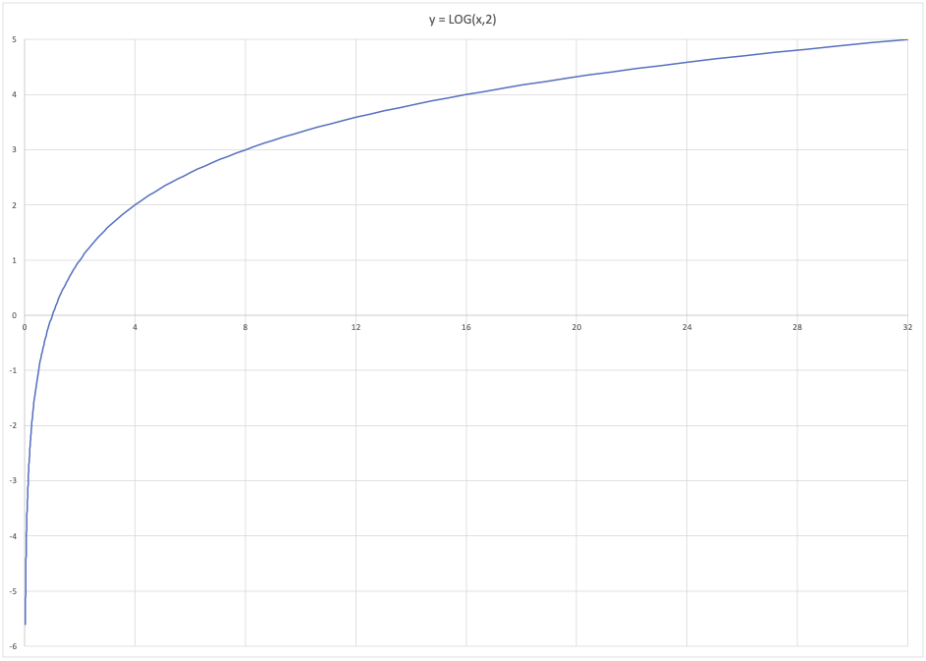



Figure 57.1: A graph of log base 2.

Spreadsheets also have the function `EXP(x)`, which returns $e^x$. For example, `EXP(2)` returns 7.38905609893065.

Chapter 58

# Exponential Decay

In a previous chapter, we saw that an investment of $P$ getting compound interest with an annual interest rate of $r$, grows exponentially. At the end of year $t$, your balance would be

$$P(1+r)^t$$

Because $r$ is positive, this number grows as time passes. You get a nice exponential growth curve that looks something like this:

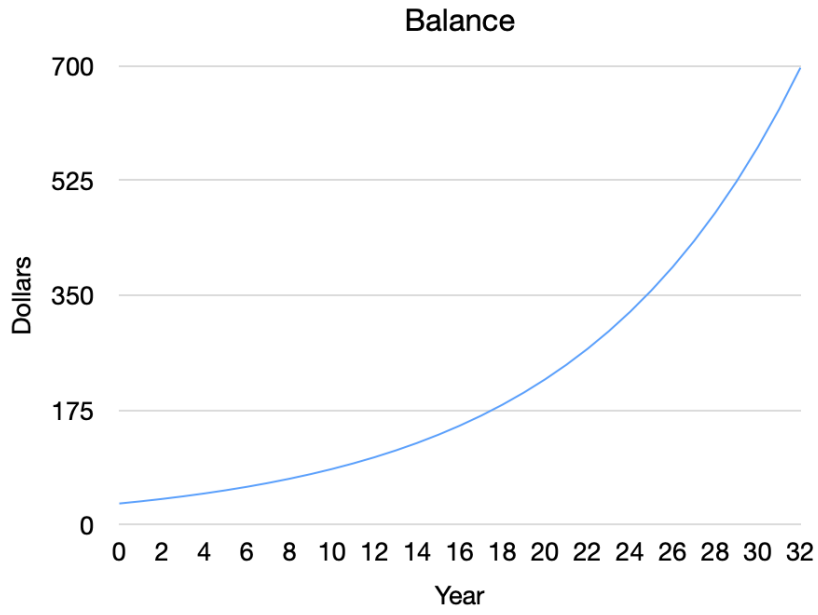



Figure 58.1: A diagram showing (exponential) compound interest.

This is \$30 invested with a 10% annual interest rate. So, the formula for the balance after $t$ years would be

$$(30)(1.1)^t$$

What if $r$ were negative? This would be *exponential decay*.

## 58.1 Radioactive Decay

Until around 1970, there were companies making watches whose faces and hands were coated with radioactive paint. The paint usually contained radium. When a radium atom decays, it gives off some energy, loses two protons and two neutrons, and becomes a different element (radon). Some of the energy given off is visible light. Thus, these watches glow in the dark.

How many of the radium atoms in the paint decay each century? About 4.24%.

Notice the quantity of atoms lost is proportional to the number of atoms you have. This is exponential decay. If we assume that we start with a million radium atoms, the number of atoms decreases over time like this:

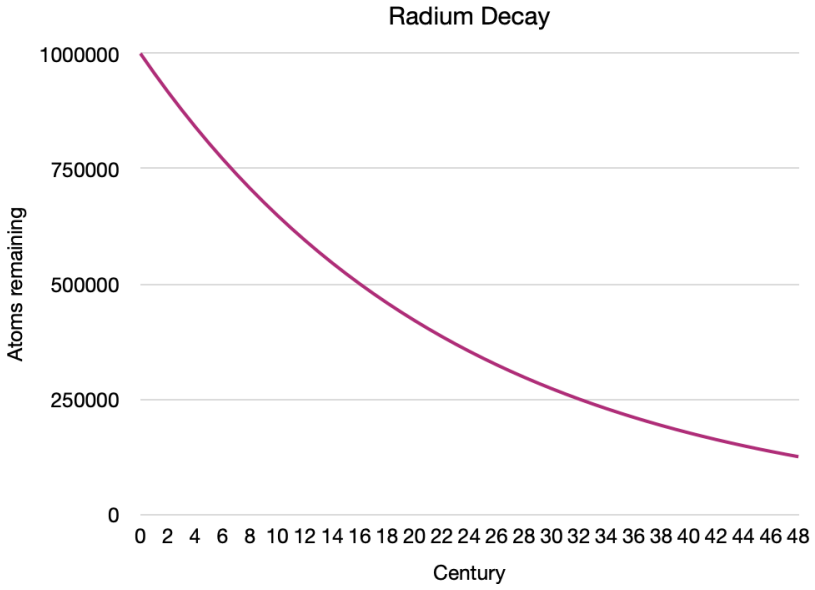



Figure 58.2: A diagram showing the decay and half life of radium.

- We start with 1,000,000 atoms.
- At 16 centuries, we have only 500,000 (half as many) left.
- 16 centuries after that, we have only 250,000 (half again) left.
- 16 centuries after that, we have only 125,000 (half again) left.

A nuclear chemist would say that radium has a *half-life* of 1,600 years; its lifespan decreases by half its original amount every 1,600 years. Note that this means that if you bought a watch with glowing hands in 1960, it will be glowing half as brightly in the year 3560.

How do we calculate the amount of radium left at the end of century t? If you start with P atoms, at the end of the t-th century, you will have

$$P\,(1 - 0.0424)^t$$

This is exponential decay.

## 58.2 Model Exponential Decay

Let's say you get hired to run a company with 480,000 employees. Each year, 1/8 of your employees leave the company for one reason or another (retirement, quitting, etc.). For some reason, you never hire any new employees.

Make a spreadsheet that indicates how many of the original 480,000 employees will still be around at the end of each year for the next 12. Next, make a bar graph from that data.

# Chapter 59

# Inverse Trigonometric Functions

Recall from the chapter on functions that an inverse of a function is a machine that turns $y$ back into $x$. The inverses of trigonometric functions are essential to solving certain integrals (you will learn in a future chapter why integrals are useful — for now, trust us that they are!). Let's begin by discussing the sin function and its inverse, $\sin^{-1}$, also called arcsin. (Note: this is not saying sin to the power of $-1$, it is a method of stating the inverse of the function).

Examine the graph of $\sin x$ in figure 59.1. See how the dashed horizontal line crosses the function at many points? This means the function $\sin x$ is not one-to-one. In other words, there is not a unique $x$-value for every $y$-value. This means that if we do not restrict the domain of $\arcsin x$, the result will not be a function (see figure 59.2). In figure 59.2, you can see that just reflecting the graph across $y = x$ fails the vertical line test: an $x$ value has more than one $y$ value.

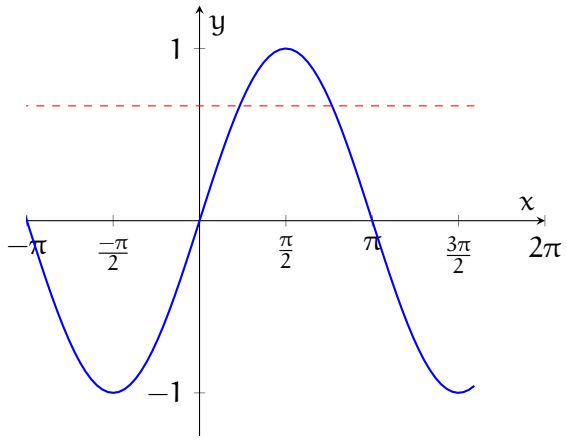



Figure 59.1: The horizontal line $y = \frac{2}{3}$ crosses $y = \sin x$ more than once

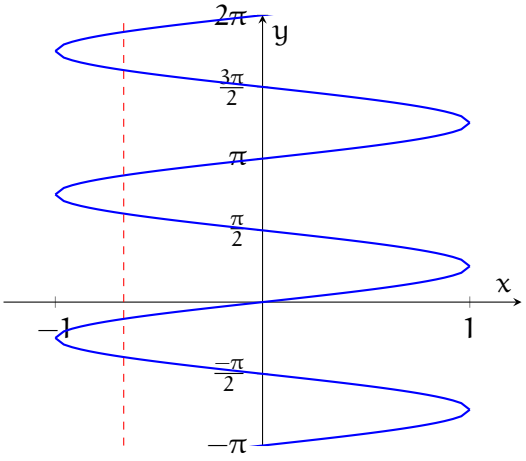



Figure 59.2: The inverse of an unrestricted sin function fails the vertical line test

## 59.1 Derivatives of Inverse Trigonometric Functions

| f | f′ |
|---|---|
| $\arcsin x$ | $\frac{1}{\sqrt{1-x^2}}$ |
| $\arccos x$ | $-\frac{1}{\sqrt{1-x^2}}$ |
| $\arctan x$ | $\frac{1}{1+x^2}$ |
| $\operatorname{arccsc} x$ | $-\frac{1}{x\sqrt{x^2-1}}$ |
| $\operatorname{arcsec} x$ | $\frac{1}{x\sqrt{x^2-1}}$ |
| $\operatorname{arccot} x$ | $-\frac{1}{1+x^2}$ |

## 59.2 Practice

### *Exercise 147*

Find the $f'$. Give your answer in a simplified form.

*Working Space*

1. $f(x) = \arctan x^2$
2. $f(x) = x\operatorname{arcsec}(x^3)$
3. $f(x) = \arcsin \frac{1}{x}$

*Answer on Page 1930*

Chapter 60

# Laws, Paradoxes, Thought Experiments, and the Illusions

A *paradox* is a statement that contradicts itself or defies intuition. Paradoxes often arise in mathematics, physics, and philosophy, challenging our realities and understanding of reality and logic.

We are going to talk about them here to introduce the idea of proofs by contradiction, which is a powerful tool in mathematics. A proof by contradiction assumes that the statement we want to prove is false and then shows that this assumption leads to a logical contradiction. We are also going to talk about a multitude of paradoxes in fields such as mathematics, physics, computer science, and architecture.

Some of these will also be *thought experiments*. Thought experiments involve considering an imaginary scenario that tests a theory or argument, and many times, one that would be rare or impossible to test. Scientists use this to refute past (incorrect) knowledge, establish new theories, or even provide scenarios about human nature[1]. Doing this allows engineers to isolate variables, look at edge cases, improve intution of what *should* happen, and consider how breakthroughs could occur. In fact, Isaac Newton used a thought experiment comparing a falling apple and the Moon to help develop his theory of universal gravitation (recall the Newton's cannon idea). In many of our physics scenarios, we imagine a *frictionless surface*. This will never truly exist, all materials involve some sort of friction, so we consider this a thought experiment.

Further, some statements which this chapter explores is laws of science and mathematics. Some of these laws are not intuitive,

If you recall, we talked in the proofs chapter about the different types of proofs, specifically proofs by contradiction. We will now talk about some examples of proofs by contradiction, and how they relate to paradoxes. We will use proofs by contradiction to prove many of our paradoxes.

The Barber's Paradox is a self-referential paradox that arises in set theory and logic.

The paradox is often stated as follows

**Theorem 10.** *In a town, there is a barber who shaves all and only those men who do not shave themselves. The question is: Does the barber shave himself?*

[1] https://en.wikipedia.org/wiki/Trolley_problem

Well there are two cases to consider:

- If the barber shaves himself, then according to the definition, he is now someone who *shaves himself*. The rule says the barber only shaves people who do not shave themselves. This is a contradiction because the barber cannot both shave himself and not shave himself at the same time.
- If the barber does not shave himself, then according to the definition, he is someone who does not shave himself. The rule says the barber shaves all those who do not shave themselves. This means the barber must shave himself, which again leads to a contradiction because he cannot both shave himself and not shave himself at the same time.

The reason this is a paradox is that no matter which path you take, you end up at a contradiction. The barber cannot exist under those rules. A paradox will often arise from statements that are self-referential or involve circular definitions, which is the case here. It is important to note that some paradoxes that we will talk about will have resolutions, but the Barber's Paradox is an example of a paradox that does not have a resolution.

In proofs by contraction we often intentionally create a contradiction to prove a theorem. In the Barber's Paradox, we assumed the two possible cases for the barber (shaving himself or not shaving himself) and showed that both lead to a contradiction.

## 60.1 Mathematical Paradoxes

### 60.1.1 Russell's Paradox

Russell's Paradox is a paradox within set theory. It arises from the statement that

**Theorem 11.** *Consider the set of all sets that do not contain themselves. Does this set contain itself?*

Let's consider some simple sets. The set of all apples *is not* an apple, it is a list of apples. Similarly, the set of all of ideas *is* an idea, it is a concept. For Russell's statement, let's create a master set R that contains only the sets that do not contain themselves. There are two cases:

- If R is not a member of itself, then its definition entails that it is a member of itself. $\rightarrow\leftarrow$
- If R is a member of itself, then it is not a member of itself, since it is the set of all sets that are not members of themselves. $\rightarrow\leftarrow$

Using symbols from our sets chapter,

$$\text{Let } R = \{x | x \notin x\}. \text{ Then } R \in R \iff R \notin R. \text{This is a } \rightarrow\leftarrow$$

### 60.1.2 Simpson's Paradox

Simpson's Paradox is a phenomenon in statistics where a trend or correlation that appears in several different groups of data can disappear or reverse when the groups are combined. Let's look at an example.

Imagine we are comparing two hospitals, Hospital A and Hospital B, for their success rates in completing an operational surgery. We have the following data:

| Hospital | Total Patients | Survived | Survival Rate |
|---|---|---|---|
| Hospital A | 1000 | 800 | 80% |
| Hospital B | 1000 | 890 | 89% |

From this, we see that the same number of patients were treated at both hospitals, but Hospital B has a higher survival rate. This would lead us to think that Hospital B is the better hospital, due to money, education of doctors, or whatever factor. However, let's look at the data when separated by those with *high-risk* surgeries versus those with *low-risk* surgeries.

Patients that had low-risk surgeries, or were in good health, are given by following data

| Hospital | Patients | Survived | Survival Rate |
|---|---|---|---|
| Hospital A | 100 | 98 | 98% |
| Hospital B | 900 | 850 | 94.4% |

And those with poor health and high-risk surgeries:

| Hospital | Patients | Survived | Survival Rate |
|---|---|---|---|
| Hospital A | 900 | 702 | 78% |
| Hospital B | 100 | 40 | 40% |

Looking at the data based on the health categorization, Hospital A is actually better in both categories. This "paradox" occurs due the unaccounted, **independent variable**: the incoming health of the patient. Hospital A took on the harder cases (90% of their patients are high-risk). Hospital B, although it initially was thought to be the better hospital, mainly took on those with good health (90% were low-risk).

For fun, let's plot this using pandas and matplotlib libraries in Python. The code is as follows:

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# aggregate the data
data = {
    'Condition': ['Good', 'Good', 'Poor', 'Poor', 'Total', 'Total'],
    'Hospital': ['A', 'B', 'A', 'B', 'A', 'B'],
    'Survival': [0.98, 0.94, 0.78, 0.40, 0.80, 0.89]
}
df = pd.DataFrame(data)

# split into subgroups and totals for showing different correlations
subgroups = df[df['Condition'] != 'Total']
totals = df[df['Condition'] == 'Total']


fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 5), sharey=True)
# regression lines and plots
ax1.plot([-0.2, 0.8], [0.98, 0.78], 'b--', marker='o', label='Trend A')
ax1.plot([0.2, 1.2], [0.94, 0.40], 'y--', marker='o', label='Trend B')
ax1.legend()
ax2.plot([-0.2, 0.2], [0.80, 0.89], 'k-', marker='o', label='Overall Trend')
ax2.legend()
sns.barplot(data=subgroups, x='Condition', y='Survival', hue='Hospital', ax=ax1)
ax1.set_title('Subgroups (A Wins)')

sns.barplot(data=totals, x='Condition', y='Survival', hue='Hospital', ax=ax2)
ax2.set_title('Total (B Wins)')

plt.tight_layout()
plt.show()
```

This creates two bar plots with opposing regression lines:

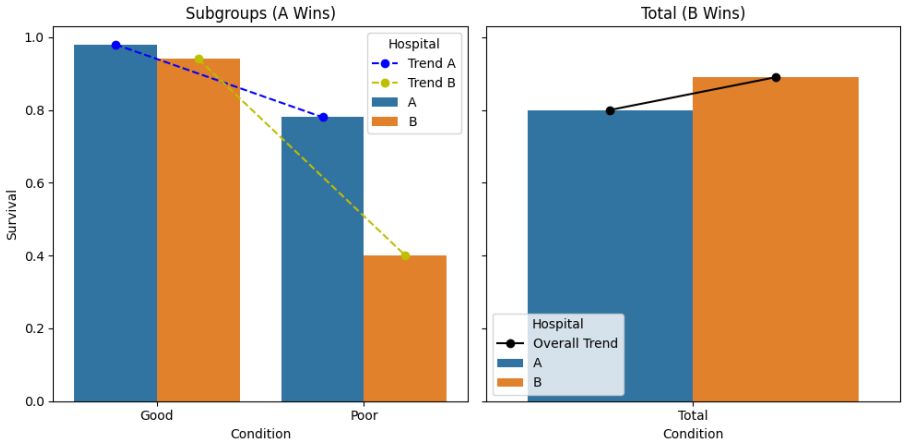



Figure 60.1: Simpson's Paradox: Subgroups vs. Total

### 60.1.3 Monty Hall Problem

Watch the following video the introduce the Monty Hall Problem.

The problem is as follows:

**Theorem 12.** *You are on a game show and are given the choice of three doors. Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to switch to door No. 2?" Is it to your advantage to switch your choice?*

ADD: an adobe illustration of the doors and the host opening one of them, and the probabilities of each door having the car after the host opens one.

Your choice of Door No. 1 has a 1/3 chance of being the car and a 2/3 chance of being a goat. When the host opens Door No. 3, which has a goat, the probability of each door changes! Even though there are two doors left, the probability is *50/50*. The probability of the car being behind Door No. 1 remains 1/3, while the probability of the car being behind Door No. 2 *increases to 2/3*. Therefore, it is to your advantage to switch to Door No. 2, as it has a higher probability of containing the car.

### 60.1.4 Birthday Problem

The birthday problem is a famous problem in probability and combinatorics. It goes as follows:

**Theorem 13.** *In a group of $n$ people, what is the probability that at least two people share the same birthday?*

The answer is often surprising to people because it is much higher than most people expect. Only 23 people are needed to guarantee a *majority*, where greater than 50% chance that at least two people share a birthday. Let's prove this.

Let $P(A)$ the probability that a group of $k$ people do *not* share a birthday. Further, let $P(B) = \neg P(A) = 1 - P(A)$ be the probability that at least two people share a birthday. We want to find the smallest $k$ such that $P(B) > 0.5$, as this constitutes a majority.

If there are only two people in a room(meaning $k = 2$), the probability that there does not exist a shared birthday is $P(A) = \frac{364}{365}$, meaning the probability that there is a shared birthday is $P(B) = 1 - P(A) = \frac{1}{365}$. If there are three people in a room, the probability that there does not exist a shared birthday is $P(A) = \frac{364}{365} \cdot \frac{363}{365}$, meaning the probability that there is a shared birthday is $P(B) = 1 - P(A) = 1 - \frac{364}{365} \cdot \frac{363}{365}$.

This pattern continues, and we can generalize the probability that there does not exist a shared birthday for $k$ people as follows:

$$P(A) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365 - k + 1}{365} = \prod_{i=0}^{k-1} \frac{365 - i}{365}.$$

Therefore, the probability that there is at least one shared birthday is:

$$P(B) = 1 - P(A) = 1 - \prod_{i=0}^{k-1} \frac{365 - i}{365}.$$

We now seek the smallest $k$ such that

$$1 - \prod_{i=0}^{k-1} \frac{365 - i}{365} > 0.5.$$

Equivalently,

$$\prod_{i=0}^{k-1} \frac{365 - i}{365} < 0.5.$$

Computing this product gives

$$P(A) \approx 0.5073 \quad \text{when } k = 22,$$

so

$$P(B) \approx 1 - 0.5073 = 0.4927 < 0.5.$$

But for $k = 23$,

$$P(A) \approx 0.4927,$$

so

$$P(B) \approx 1 - 0.4927 = 0.5073 > 0.5.$$

Thus, the smallest number of people needed so that the probability of at least two sharing a birthday exceeds 50% is 23 people. Much smaller than most people expect!

### 60.1.5 Zeno's Paradoxes

Zeno was a Greek philosopher who is famous for his paradoxes, who created challenging thought experiments and paradoxes that question our understand of motion, space, physics, and time. Zeno's paradoxes are often used to illustrate the concept of *infinity* and the nature of space and time.

#### The Dichotomy Paradox

His most famous paradoxes include is called the *Dichotomy Paradox*, which he stated as

> That which is in locomotion must arrive at the half-way stage before it arrives at the goal.

This paradox suggests that before an object can reach its destination, it must first reach the halfway point, and before it can reach the halfway point, it must reach the quarter point, and so on, leading to an infinite number of steps that must be completed before reaching the destination. This seems to imply that motion is impossible, as it would require completing an infinite number of tasks in a finite amount of time.

So then, how is motion possible?

The resolution to Zeno's paradoxes lies in the concept of *limits* and *infinite series*. In modern mathematics, we understand that an infinite number of steps can be completed in a finite amount of time if the steps get smaller and smaller. If we consider the remaining distance to be covered as 1 unit, the first step is $\frac{1}{2}$, the second step is $\frac{1}{4}$, the third step is $\frac{1}{8}$, and so on. The total distance covered after $n$ steps can be expressed as a *geometric series*:

$$S = \sum_{n=1}^{\infty} \frac{1}{2^n} = 1$$

For proof's sake, let

$$S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$

Then, multiplying both sides by $\frac{1}{2}$ gives

$$\frac{1}{2}S = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

$$S - \frac{1}{2}S = \frac{1}{2}$$

$$\frac{1}{2}S = \frac{1}{2} \quad \Longrightarrow \quad S = 1$$

## 60.2 Physics and Time Travel Paradoxes

A lot of what we have touched are counter-intuitive mathematical equations. Now we can look at some physics paradoxes and thought experiments that may mess with your brain a bit. A lot of these actually involve a concept called superposition.

### 60.2.1 Schrödinger's Cat

Erwin Schrödinger was a physicist with thoughts on quantum superposition, subatomic events, and early quantum theory. This thought experiment arose in a discussion with Albert Einstein, Niels Bohr, and Werner Heisenberg, all physicists from Europe working towards quantum development.

The thought experiment is as follows:

> Schrödinger imagined placing a cat in a sealed box with a device triggered by a quantum event (such as the decay of an atom, which depends on an electron-level process, or radioactive gunpowder, which Einstein preferred). The event has a 50% chance of happening and a 50% chance of staying dormant. If the event occurs, the cat dies; if not, it lives. However, you do not know whether the cat is alive or not until the box is opened. Schrödinger states that, the moment before the box is opened, the cat is equal parts dead and alive.

Why is this? Well, the cat, in Schrödinger's world, is theoretically governed by quantum physics and quantum position. **Quantum Superposition** states that an object, usually an electron, exists in a **superposition of states**, such that it occupies multiple positions or energy states until it is measured. Before measurement, the electron is not in one definite state; instead, it is described by a wavefunction that encodes multiple possibilities simultaneously. If electrons are fired at a barrier with two narrow slits and then detected on a screen behind it, they produce an interference pattern, one with alternating covered and uncovered regions. This is characteristic of wave behavior, showing that electrons are travelling in waves. Further, when we talk about nonpolar covalent bonds, the electrons are present in both atoms simultaneously.

We say that the cat is in a superposition of both equally possible dead or alive states. There is a great video on Schrödinger's Cat and its relation to quantum physics by TedEd in your digital resources, check it out!

#### Arrow Paradox

A very similar problem on superposition of an arrow comes from Zeno (who we talked about earlier) is the *Arrow Paradox*:

> If everything when it occupies an equal space is at rest at that instant of time, and if that which is in locomotion is always occupying such a space at any moment, the flying arrow is therefore motionless at that instant of time and at the next instant of time but if both instants of time are taken as the same instant or continuous instant of time then it is in motion.

In short, this is a fancy way of saying that if we look at an arrow in flight at any single instantaneous moment, it will appear to be at rest, and it occupies a specific position in space. In this tiny instant of time (recall $dt$ from calculus), the arrow is neither moving to where it is, nor to where it is not. This seems to suggest that the arrow is motionless at every instant of time, and therefore, it cannot be in motion.

However, the solution to this paradox lies in the concept of limits. Motion is not just about being at a specific position at an instant, but about the change in position over time. The arrow's motion can be understood as a continuous process, where its position changes smoothly over time, even though at any single instant it may appear to be at rest, just as single points on a curve can be thought of as having no length, but the curve itself can have a infinite length.

### 60.2.2 Time Travel Paradoxes

While we know (as of writing this chapter) that time travel is not real, it is fun to think about and arises in pop culture such as movies a lot! Time travel paradoxes are thought experiments which arise from the contradiction that arises within time travel or the foreknowledge of the future. Time travel raises logical problems because it allows events to influence their own causes.

The most well-known example is the **grandfather paradox**. Suppose a person travels back in time and prevents their grandfather from having children killing their grandfather. This would mean the time traveler is never born, as one of the time traveler's parents are never conceived. The time traveler, then, could not have gone back in time to interfere in the first place. The sequence of events becomes logically inconsistent: the same action both must and must not occur. This is a contradiction ($\rightarrow\leftarrow$), as we talked about, and many

people use this as an argument to prove that time travel cannot logically exist.

Another time travel paradox is preventing Abraham Lincoln's assassination. Imagine a time traveler goes back to 1865 and prevents Lincoln's assassination. At first glance, this seems like a straightforward intervention. However, if Lincoln survives, the historical conditions that motivated the time traveler to go back, such as learning about the assassination in the first place, no longer exist. This creates a contradiction: if Lincoln was never assassinated, why would the traveler have made the journey to stop it?

More generally, these are consistency paradoxes, where altering the past creates contradictions in the timeline. In physics, these scenarios contradict our existing knowledge, meaning either constraints exist to preserve consistency, or our understanding of time and causality is incomplete.

## 60.3 Architecture and Geometry Paradoxes

Architecture and geometric paradoxes come with their own set of paradoxes, often warming visual perception and challenging our understanding of space and geometry.

### 60.3.1 Penrose Steps

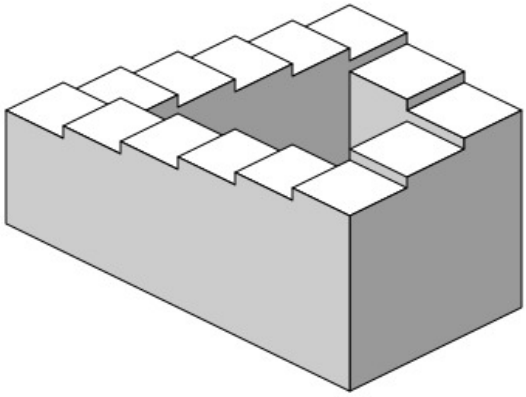


Figure 60.2: Penrose steps diagram from Wikipedia - By Sakurambo - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=794844

The penrose steps are an architectural paradox that creates the illusion of an infinite staircase.[2] The steps appear to be continuously ascending or descending, but in reality, they

[2]Christopher Nolan famously reconstructed this illusion in his film Inception (2010). If you are interested, the scene can be found here and behind the scenes here.

form a closed loop. This paradox is achieved through clever use of perspective and geometry, creating an optical illusion that tricks the viewer's perception of space.

### 60.3.2 Mobius Strip

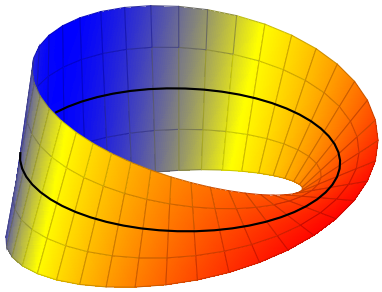


Figure 60.3: A parametric graph of a mobius strip.

The mobius strip is a piece of paper is a *surface* that is said to have one side and one boundary. It is formed most easily (you can do this at home!) by taking a strip of paper, giving it a half twist, and then taping the ends together. The resulting shape is a non-orientable surface, meaning that it has only one side and one edge. If you were to start drawing a line down the middle of the strip, you would find that you return to your starting point after traversing the entire length of the strip, without ever lifting your pen. There is no *inside* or *outside* to the mobius strip.

If you are curious, the mobius strip is defined by the following parametric equations:

$$\mathbf{r}(u, v) = \left((1 + v\cos\tfrac{u}{2})\cos u,\ (1 + v\cos\tfrac{u}{2})\sin u,\ v\sin\tfrac{u}{2}\right)$$

where $u$ ranges from $0$ to $2\pi$ and $v$ ranges from $-1$ to $1$. Do not worry, this is not something you need to know how to do at all! We will cover parametric in a future workbook.

### 60.3.3 Coastline paradox

The *coastline paradox* highlights an issue in measuring irregular geometric objects; their length is changes under changes in measurement scale. Unlike smooth curves in classical Euclidean geometry, natural coastlines exhibit intricate, self-similar structure across many scales. As the measuring unit becomes smaller, more of this fine structure is resolved, and the total measured length increases. This originated in the 1950's when a Lewis Fry Richardson was researching how border lengths could effect the likelihood of war, and there was a discrepancy between the Portuguese measurement of Spain's border compared to the Spanish. Simply, the shorter the unit (or ruler), the longer the measured border; the Spanish and Portuguese geographers were simply using different rulers. However, as

the "ruler length" approaches zero, the length grows infinity, rather than approaching a finite value!

ADD: Diagram from tikz of coastline

This phenomenon contradicts with calculus and its ideas on the definition of arc length. In calculus, the length of a curve $(y = f(x))$ over an interval is defined as the definite integral $\int f(x)\,dx$ between two points, which often leads to a finite value for smooth or simple functions. The coastline paradox demonstrates that this limiting process fails for highly irregular sets; the coastlines are not "smooth" enough. Consequently, arc length integrals cannot be utilized, and alternative methodologies are used to find the length of these areas, termed fractal surface perimeters.

## 60.4 Computer Science Paradoxes and Laws

### 60.4.1 Moravec's Paradox

*Hans Moravec*, a pioneer in robotics and Artificial Intelligence, wrote in 1988,

> it is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or [near] impossible to give them the skills of a one-year-old when it comes to perception and mobility

What he means by this statement, referred to as the *Moravec's Paradox* is that tasks that are *easy* for humans are *hard* for machines, and tasks that are *hard* for humans are relatively *easy* for machines.

A good example of this is chess. Chess robots have been being developed since the 1980's, and in 1997, a machine called Deep Blue by IBM defeated a world champion grandmaster in chess. Typically, chess has been a level of skill attributed to the most intelligent humans. However, contrary to what would arbitrarily think, chess is one of the easiest tasks for a computer to do.

Along with chess comes proving mathematical theorems, brute-force algorithms, multiplying large numbers, or even integration, all of which were generally thought of as difficult tasks to humans but for computers are the most simple. Why is this the case? Well, simply put, humans have skills that have naturally become second nature, or unconscious, to us, like movement, recognizing a friend's face, settings goals, etc. Humans naturally want to automate that which is not in tune with this unconscious behavior, like solving integrals, identifying the best path to hit all capitals of the United States, or even just take the best billiards shot.

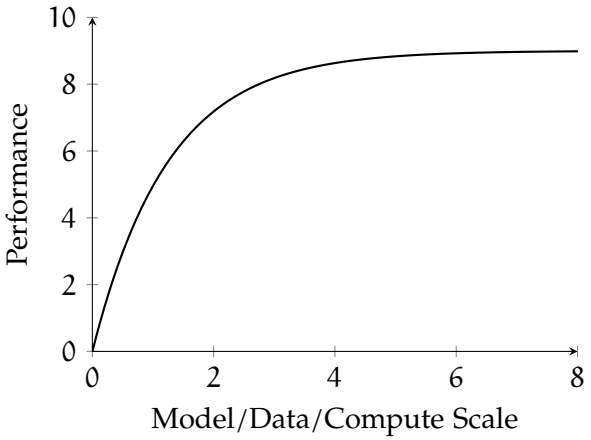
Intuitively, this means the best way for humans to learn is to practice cognitive skills like

reading carefully, practicing problems, or explaining ideas, in order to continue gaining mastery through repetition. Recognizing patterns or spotting errors takes practice and is harder to master for humans, but it takes human-centric skills to maintain and develop.

### 60.4.2 Neural Scaling Law

Imagine you are working for a factory that is producing pencils. With a single, static, automatic assembly line, you make pencils at a rate of 100 pencils per hour. By adding a second assembly line, you are now able to make pencils at 200 pencils per hour, double the rate it was previously. This shows that generally, production scales *linearly*.

However, not all systems scale this cleanly. Suppose instead that each additional assembly line becomes harder to integrate: workers need to coordinate more, machines interfere slightly, and space becomes constrained. Now, when you add a second line, production might increase to 180 pencils per hour instead of 200. Adding a third might bring you to 250, not 300. The total output still increases, but each new addition contributes less than the previous one. This is called *diminishing returns*.




Neural scaling laws behave more like this second scenario. Say you have a neural network model with a set of training params N, D, C, L, which stand for number of parameters, size of dataset, cost of computing, and a loss function, respectively. As you increase a model's size, the amount of data it sees, or the compute used to train it, performance improves steadily, not linearly. Early increases lead to large gains, while later increases produce smaller, more incremental improvements. Importantly, the improvement is still predictable and consistent, which is why researchers can estimate how much better a system will perform if they scale it up. Consequently, just overloading the model with a higher, almost infinite, magnitude of parameters will lead to a plateau of performance. The graph of performance vs parameters looks similar to an upside-down hockey stick, it scales really well for a while when parameters are being increased up to a point, but then it gradually decreases in rate of improvement of time.

CHAPTER 61

# Introduction to Data Visualization

It is difficult for the human mind to look at a list of numbers and identify the patterns in them, so we often use these numbers to make a picture. These pictures are called *graphs*, *charts*, or *plots*. Often, the right picture can make the meaning in the data obvious. **Data visualization** is the process of making pictures from numbers.

## 61.1 Types of Variables

Depending on the type of data and what you are trying to demonstrate about it, you will use different types of data visualizations. Before we get to the more complex types of data visualizations, let's look at the two types of variables we are usually looking to visualize: *categorical* and *numerical*.

*Categorical* or qualitative variables are those that can be put into categories. For example, the type of shoes a person wears (sneakers, sandals, etc.) is a categorical variable. The type of shoes sold is a categorical variable. The name of the salesperson (John, Jane, etc.) is also a categorical variable.

*Numerical* or quantitative variables are those that can be measured with numbers. For example, the number of boxes of cookies sold is a quantitative variable. The average temperature on a given day is also a quantitative variable. Quantitative variables can be further divided into *discrete* and *continuous* variables. Discrete variables are those that can only take on specific values, such as the number of pencils sold (you can't sell 3.5 pencils). Continuous variables can take on any value within a range, such as the average height of students in a class (which can be 5.5 feet, 1.65 m, etc.).

Further, we can also classify data into univariate, bivariate, and multivariate data. Univariate data has only one variable, such as the number of boxes of cookies sold. Bivariate data has two variables, such as the average temperature and total sales for a lemonade stand. Multivariate data has more than two variables, such as the average temperature, total sales, and type of ookie sold. We will only look at univariate and bivariate data!

## 61.2 How do we describe data?

There are different descriptive methods for representing data, and they can usually be grouped into three categories: *tabular*, *graphical*, and *numerical*.

*Tabular* methods include tables, which are a way of organizing data into rows and columns. They are useful for displaying raw data and making it easy to look up specific values. However, they can be difficult to interpret when dealing with large datasets or when trying to identify patterns.

*Graphical* methods include bar charts, line graphs, pie charts, and scatter plots. They are useful for visualizing data and making it easier to identify patterns and trends.

*Numerical* methods include measures of central tendency (mean, median, mode) and measures of variability (range, variance, standard deviation). They are useful for summarizing data and providing a numerical representation of the data's characteristics.

We will focus on graphical methods in this chapter, but we will also touch on tabular and numerical methods in later chapters.

## 61.3 Univariate Data

We will start with univariate data, which has only one variable. For example, the number of boxes of cookies sold by each salesperson is univariate data. Bar charts, line graphs, and pie charts are all good for visualizing univariate, *categorical* data. We will explore these types of graphs in the next sections.

### 61.3.1 Bar Chart

Bar charts are typically used to show the values of a categorical variable. Each bar represents a category, and the height of the bar represents the value of that category. Bar charts can have vertical or horizontal bars to represent this data.

Here is an example of a bar chart.

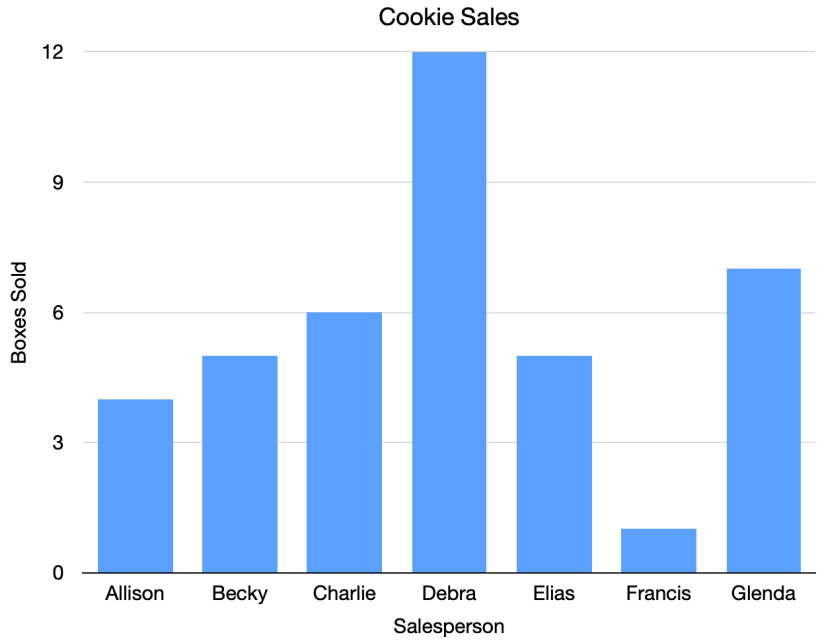



Figure 61.1: A bar chart showing cookie sales per person.

Each bar represents the cookie sales of one person. For example, Charlie has sold 6 boxes of cookies, so the bar goes over Charlie's name and reaches to the number 6.

Looking at this chart, you probably think, "Wow, Debra has sold a lot more cookies than anyone else, and Francis has sold a lot fewer."

The same data could be in a table like this:

| Salesperson | Boxes Sold |
|---|---|
| Allison | 4 |
| Becky | 5 |
| Charlie | 6 |
| Debra | 12 |
| Elias | 5 |
| Francis | 1 |
| Glenda | 7 |

A table (especially a large table) is often just a bunch of numbers. A chart helps our brains understand what the numbers mean.

Bar charts can also go horizontally.

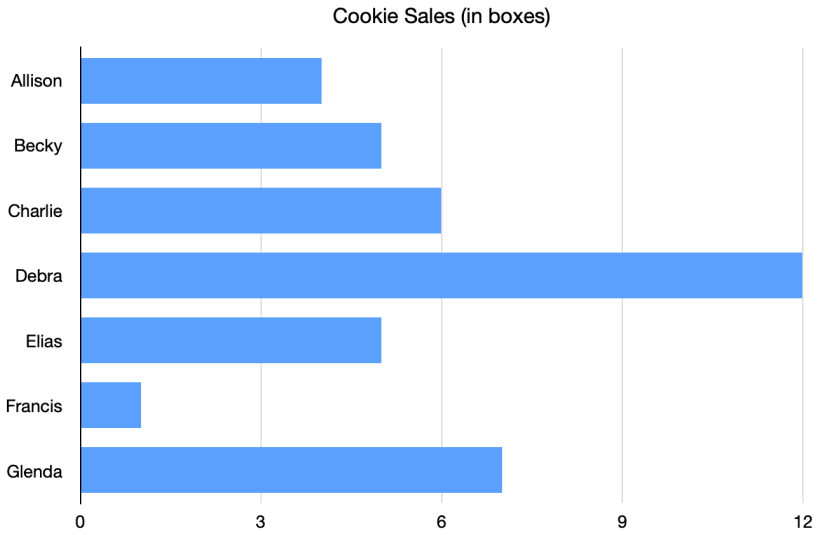



Figure 61.2: A horizontal bar chart showing the same cookie sales data.

Sometimes we use colors to explain what contributed to the number.

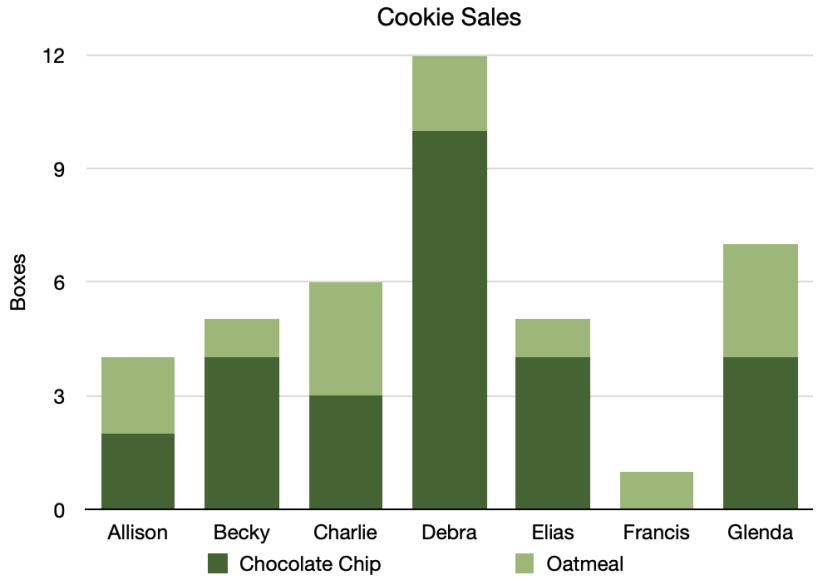



Figure 61.3: A bar chart with different colors showing types of cookies.

This tells us that Becky sold more boxes of chocolate chip cookies than boxes of oatmeal cookies.

**Make Bar Graph**

Go back to your compound interest spreadsheet and make a bar graph that shows both balances over time:

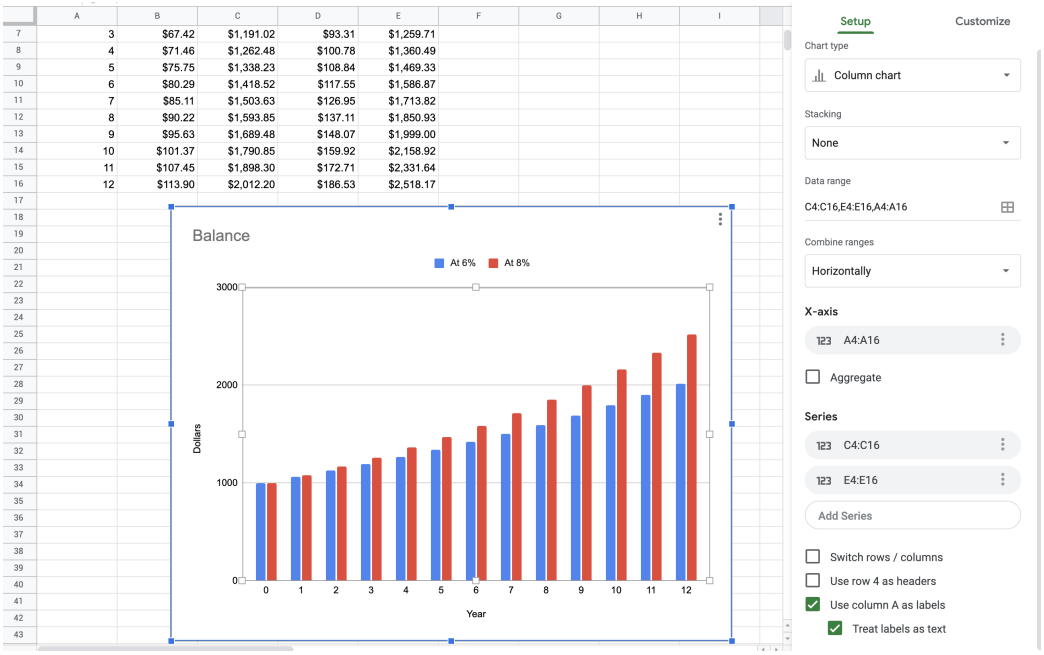



Figure 61.4: A bar graph made in google sheets showing interest.

The year column should be used as the x-axis. There are two series of data that come from C4:C16 and E4:E16. Tidy up the titles and legend as much as you like. Looking at the graph, you can see the balances start the same, but balance of the account with the larger interest rate quickly pulls away from the account with the smaller interest rate.

### 61.3.2 Line Graph

Here is a line graph:

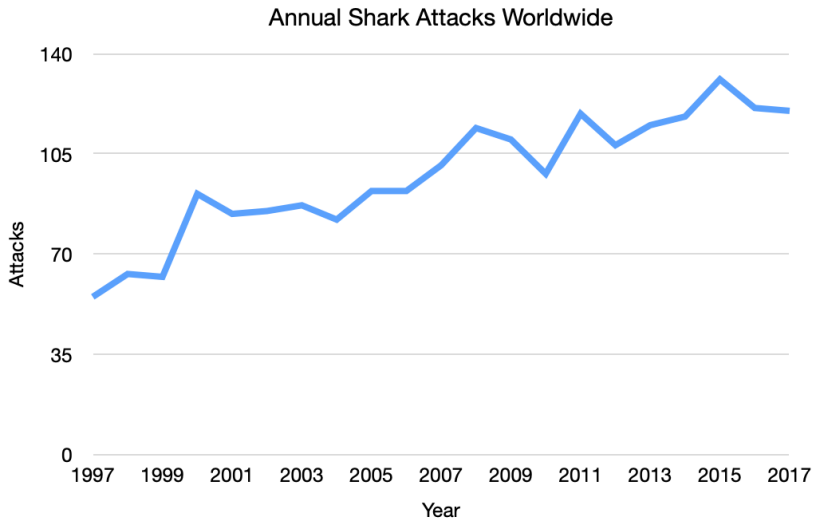



Figure 61.5: A line graph showing shark attacks per year over two decades.

These are often used to show trends over time. Here, for example, you can see that the number of shark attacks has been increasing over time.

You can have more than one line on a graph.

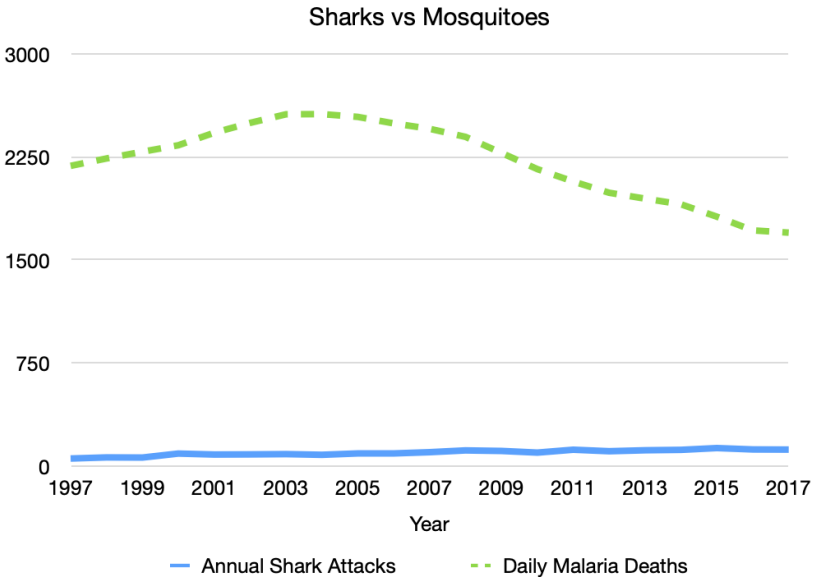



Figure 61.6: A line graph showing shark deaths versus mosquito deaths.

### 61.3.3 Pie Chart

You use a pie chart when you are looking at the comparative size of numbers. This is best for comparing percentages of a whole that sum to 100%. Here we can see that Nitrogen makes up 78% of the gases in the air.

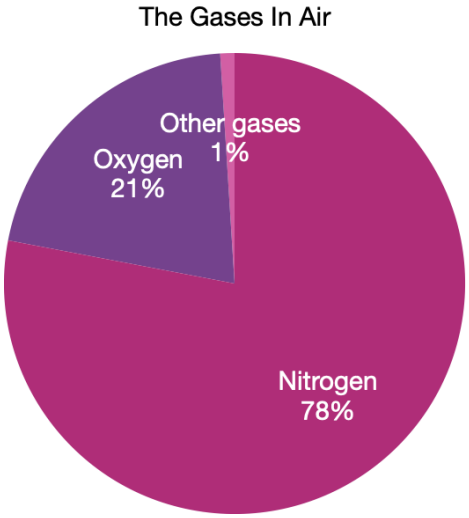



Figure 61.7: A pie chart of the various gases in the air.

#### When to use a Pie Chart Make a Pie Chart in Python

For *quantitative*, univariate data, we often use dotplots and stemplots to visualize smaller sets of data; and histograms, cumulative frequency plots, and boxplots to visualize larger sets of data. We will explore these types of graphs below.

### 61.3.4 Dotplot

A *dotplot* is one of the simplest ways to display single, quantitative variables. Each observation is represented by a dot placed above its value on a number line. If a value appears more than once, the dots stack vertically.

Dotplots are most effective for small to moderate data sets. If the data set is too large, a dotplot can become cluttered and hard to read. In that case, a histogram or a boxplot may be a better choice.

#### When to Use a Dotplot

- You have 1 variable.

- The variable is quantitative.
- You want to see the distribution and the individual data values.
- The data set is not too large.

**How to Make a Dotplot**

1. Draw a horizontal number line (x-axis) that covers the data range.
2. Choose a scale that fits the entire range of values.
3. Place one dot above the location of each observation.
4. If multiple observations have the same value, stack the dots vertically.

Here is an example of a dotplot.

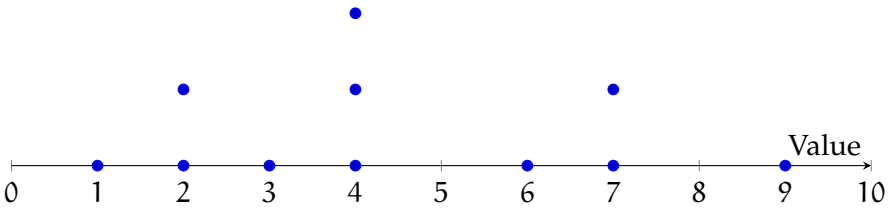



Figure 61.8: A dotplot showing a small sample distribution.

**How to Read a Dotplot**

Each dot shows the location of one data value. To find the value of a dot, look straight down to the number line.

A dotplot helps you describe the distribution by answering questions like:

- Spread: How spread out are the values (range and clustering)?
- Shape: Do the dots form a roughly symmetric shape, or is the distribution skewed?
- Center: Where is the distribution "balanced" (a typical value)?
- Outliers: Are there values far away from the rest?

Sometimes a dotplot shows a distribution with a long tail to one side. For example, salary data often has many values in the lower-to-middle range and a few extremely high values. This creates a right-skewed distribution.

Outliers can strongly affect numerical summaries, especially the mean.

**Connecting Outliers and the Mean (Seesaw Idea)** A useful way to think about the mean is as a balancing point. Imagine the dotplot as a seesaw: points far from the main cluster have more "pull." So a single extreme value can shift the mean noticeably, even if most data values stay the same.

**Comparative Dotplots**

A *comparative dotplot* uses the same numerical scale to compare two (or more) groups. This is useful when you want to compare distributions side-by-side.

When comparing groups, focus on differences in:

- center (typical value),
- spread,
- shape (skew/clusters),
- and outliers.

**Make a Dotplot in a Spreadsheet**

You can create a dotplot in Excel or Google Sheets using a scatter plot and a helper column.

1. Put your data values in column A (starting in A2). Sort them (recommended).
2. In cell B2, enter:
   `=COUNTIF($A$2:A2, A2)`
   Fill down. This creates stacking heights for repeated x-values.
3. Insert a Scatter chart using column A as x-values and column B as y-values.
4. Format the chart:
   - markers only (no lines),
   - hide the y-axis labels (y is only for stacking),
   - add titles and adjust the x-axis scale.

### 61.3.5 Stemplot

A *stemplot* (also called a *stem-and-leaf plot*) is an effective way to summarize **one quantitative variable** when the data set is not too large. Like a dotplot, it helps you see the overall distribution, but a stemplot has an extra advantage: it still shows the exact data values.

To make a stemplot, each observation is split into two parts:

- the stem, which is the left-most part of the number (leading digit(s))
- the leaf, which is the remaining part, usually the final digit

For example, 23 can be written as 2 | 3.

**How to Make a Stemplot**

1. Choose stems and leaves. Decide where to split each number into a stem and a leaf. There is no single rule for this. Your goal is to choose a reasonable number of stems.
2. Draw a vertical divider. Stems go on the left; leaves go on the right.
3. List all stems in order. Make sure the stems cover the full range of your data.
4. Write leaves for each observation. Place each leaf next to its correct stem.
5. Sort leaves. Put the leaves for each stem in increasing order.
6. Include a key. The key shows how to interpret the stems and leaves.

**Choosing the Number of Stems**

If you use *too few stems*, you can hide patterns because the data get lumped together. If you use *too many stems*, you can also hide patterns because the plot becomes too spread out.

A common strategy is to start with stems that represent groups of 10 (tens digits as stems), and then adjust if needed.

Sometimes a single stem represents too wide of an interval. In that case, you can split stems by repeating each stem twice:

- the first row holds leaves 0–4
- the second row holds leaves 5–9

For example, if your data run from 10 to 49, you can use stems

$$1, 1, 2, 2, 3, 3, 4, 4$$

where the first 1 represents 10–14 and the second 1 represents 15–19, and so on.

**Example**

Suppose these are housing costs (in dollars per month):

$$255, 259, 304, 306, 309, 317, 323, 327, 332, 334, 342, 349, 350, 351, 362, 367, 376, 384, 401, 403$$

If we use the hundreds digit as the stem and the last two digits as the leaf, we can split each stem into two rows: (top row for leaves 00–49, bottom row for leaves 50–99)

| Stem | Leaves |
|---|---|
| 2 | 55 59 |
| 2 | |
| 3 | 04 06 09 17 23 27 32 34 42 49 |
| 3 | 50 51 62 67 76 84 |
| 4 | 01 03 |
| 4 | |

**Key:** $3 \mid 04 = 304$ dollars per month

**How to Read a Stemplot**

A stemplot gives you several kinds of information at once:

- Exact values: Each leaf represents a single observation.
- Frequencies: The number of leaves in a row tells how many observations fall in that interval. For example, if a stem row has 9 leaves, that means 9 observations are in that range.
- Gaps: If a stem has no leaves, that means there are no observations in that interval.
- Shape and spread: You can see whether the values cluster in certain ranges and whether the distribution is skewed.
- Center: You can estimate a typical value by locating where the leaves cluster most.
- Outliers: Unusually high or low values may stand out because they appear far from the rest.

Just like with dotplots, you can imagine the distribution as something that could "balance" on a finger: the place it would balance is a rough sense of the center.

**Make a Stem Plot in a Spreadsheet Make a Stem Plot in Python**

### 61.3.6 Histogram

A *histogram* is one of the most popular ways to display single quantitative variables. It is especially useful for showing patterns in larger data sets, where dotplots or stemplots can become overwhelming. A histogram is like a stemplot turned on its side: instead of listing individual values, we group values into intervals called classes or bins, and draw a bar for each bin.

Histograms are best when you care about the overall shape of a distribution (clusters, skew, gaps, outliers), not the exact value of every data point.

A histogram can be drawn using frequency (counts), relative frequency (proportions), and percent frequency (percentages). These three versions have the same shape if they use the same bins. The difference is only the y-axis scale.

**How to Make a Histogram**

1. Create groups from continuous data. This process is called binning.
2. Draw the x-axis and y-axis, and scale them to fit the class intervals and the frequencies (or relative frequencies or percentages).
3. Draw one bar for each class. The height of each bar equals the frequency (or relative frequency or percent) for that class.
4. Draw the bars touching with no gaps (because the data values are continuous and the bins connect).

**How to Read a Histogram**

- Each bar represents a single class (bin). There is one bar per class.
- The bins are placed on the x-axis in increasing order, like a number line.
- The height of a bar tells how many observations fall in that class (for a frequency histogram).
- A class with no bar means no observations fall in that interval (a gap).

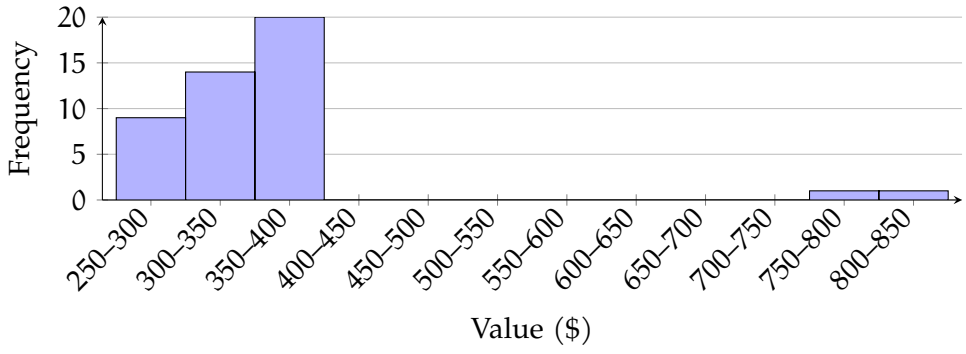



Figure 61.9: A histogram using 50-dollar bins illustrating the example counts (9, 14, 20, ... and 0,1,1 beyond $700).

**Example of Interpreting Bars**

Suppose a histogram uses the bins \$250–\$300, \$300–\$350, \$350–\$400, and so on.

- If the first bar (250–300) has height 9, then 9 observations are between \$250 and \$300.
- If the second and third bars (300–350 and 350–400) have heights 14 and 20, then $14 + 20 = 34$ observations are between \$300 and \$400.
- If there is no bar for 700–750, then there are 0 observations between \$700 and \$750.
- If the bars beyond 700 have heights 0, 1, and 1, then $0 + 1 + 1 = 2$ observations are at least \$700.

A histogram can help you to describe the distribution by focusing on shape, center, spread, gaps, and outliers.

### Making a Histogram in a Spreadsheet

You can make a histogram in Excel or Google Sheets.

1. Put the data in one column.
2. Choose bin widths (for example, bins of width 50: 250–300, 300–350, 350–400, etc.).
3. Insert a histogram chart (or use a column chart after building a frequency table).
4. Check the bin width and the endpoints. Changing the bins can change the appearance of the distribution.

### Making a Histogram in Python

Python can generate a histogram quickly and lets you change the bin width easily.

```
import matplotlib.pyplot as plt

# Replace this list with your data
data = [250, 275, 310, 320, 350, 365, 410, 415, 430, 500, 525, 610, 720, 840]

# Choose the number of bins (try changing this!)
plt.hist(data, bins=8)

plt.title("Histogram")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

Try changing the value of `bins`. Notice that the overall shape can look different depending

on how you group the data.

**How do we compare dotplots, stemplots, and histograms?**

For small data sets, dotplots and stemplots can be better because they show the exact values. In a histogram, you lose that detail because values are grouped into bins. In exchange, histograms make it easier to see the overall pattern in large data sets.

### 61.3.7 Cumulative Frequency Plot

A *cumulative frequency plot* (also called a *cumulative frequency chart*) shows the running total of how many data values are at or below each value (or at or below each class boundary). Instead of showing how many observations fall in each bin like a histogram, a cumulative frequency plot shows how the totals add up as you move from left to right.

Cumulative frequency plots are useful when you want to answer questions like:

- How many observations are less than or equal to a certain value?
- How many observations are greater than or equal to a certain value?
- What percent of the data fall below a certain cutoff?

**How to Make a Cumulative Frequency Plot**

1. Draw the x-axis and y-axis.
2. Set up bins for the data and identify the upper class boundaries (the right endpoints).
3. Compute the cumulative frequency at each upper class boundary.
   (This means: add the frequency of that class to the total from all previous bins.)
4. Plot a point at each upper class boundary with height equal to the cumulative frequency.
5. Connect the points with straight line segments.

**Interpreting the shape of your plot**

The steepness of the curve tells you where the data values are concentrated.

- If the curve rises quickly over a certain x-interval, many observations fall in that range.
- If the curve is almost flat, few observations fall in that range.

A cumulative frequency plot can also hint at skewness:

- **Right-skewed:** the curve increases quickly at smaller values, then levels off later.
- **Left-skewed:** the curve increases slowly at first, then rises sharply near the end.
- **Symmetric:** the curve often looks S-shaped.

### Example

Suppose $n = 80$ observations were grouped into bins, and the cumulative frequency at $x = 350$ is about 23. Then:

- About 23 observations are $\leq 350$.
- About $80 - 23 = 57$ observations are $\geq 350$.

The plot below shows a sample cumulative frequency curve. (These coordinates are an example format; you can replace them with your own cumulative totals.)

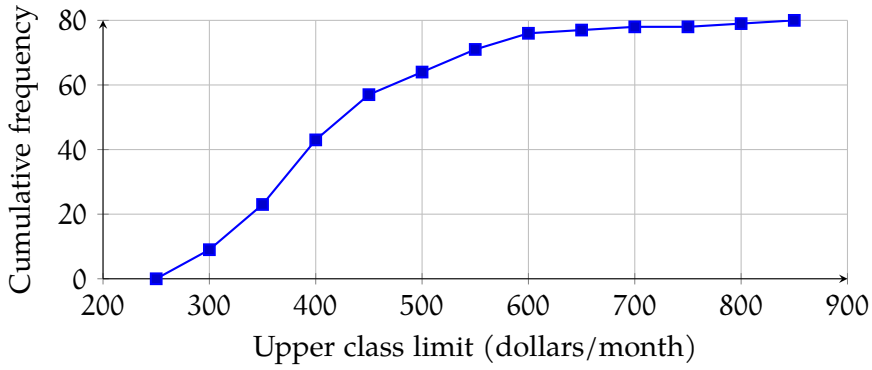



Figure 61.10: A cumulative frequency plot.

### Making a Cumulative Frequency Plot in Python

This example shows the basic idea: build bins, count how many values fall in each bin, then take a cumulative sum.

```
import numpy as np
import matplotlib.pyplot as plt

# Replace with your data
data = np.array([250, 275, 310, 320, 350, 365, 410, 415, 430, 500, 525, 610, 720,
↪  840])

# Choose bin edges (upper class boundaries are the right endpoints)
```

```
bins = np.arange(250, 851, 50)  # 250, 300, ..., 850

# Frequency in each bin
counts, edges = np.histogram(data, bins=bins)

# Cumulative frequency at each upper boundary
cumulative_counts = np.cumsum(counts)
upper_bounds = edges[1:]  # right endpoints

plt.plot(upper_bounds, cumulative_counts, marker='o')
plt.title("Cumulative Frequency Plot")
plt.xlabel("Upper class boundary ($)")
plt.ylabel("Cumulative frequency")
plt.grid(True)
plt.show()
```

If you change the `bin` width, the curve may look different, but it will always increase from left to right and end at $n$.

**Boxplot**

## 61.4 Bivariate Data

### 61.4.1 Scatter Plot

Sometimes, you have a large number of data points with two values, and you are looking for a relationship between them. For example, maybe you write down the average temperature and the total sales for your lemonade stand on the 15th of every month:

| Date | Avg. Temp. | Total Sales |
|---|---|---|
| 15 January 2022 | 2.6º C | $183.85 |
| 15 February 2022 | -4.2º C | $173.56 |
| 15 March 2022 | 13.3º C | $195.22 |
| 15 April 2022 | 26.2º C | $207.61 |
| 15 May 2022 | 27.5º C | $210.88 |
| 15 June 2022 | 31.3º C | $214.18 |
| 15 July 2022 | 33.5º C | $215.23 |
| 15 Aug 2022 | 41.7º C | $224.07 |
| 15 September 2022 | 20.7º C | $198.94 |
| 15 October 2022 | 17.2º C | $196.10 |
| 15 November 2022 | 1.7º C | $185.10 |
| 15 December 2022 | 0.2º C | $188.70 |

You may wonder, "Do I sell more lemonade on hotter days?"

To figure this out, you might create a scatter plot. For each day, you put a mark that represents that temperature and the sales that day:

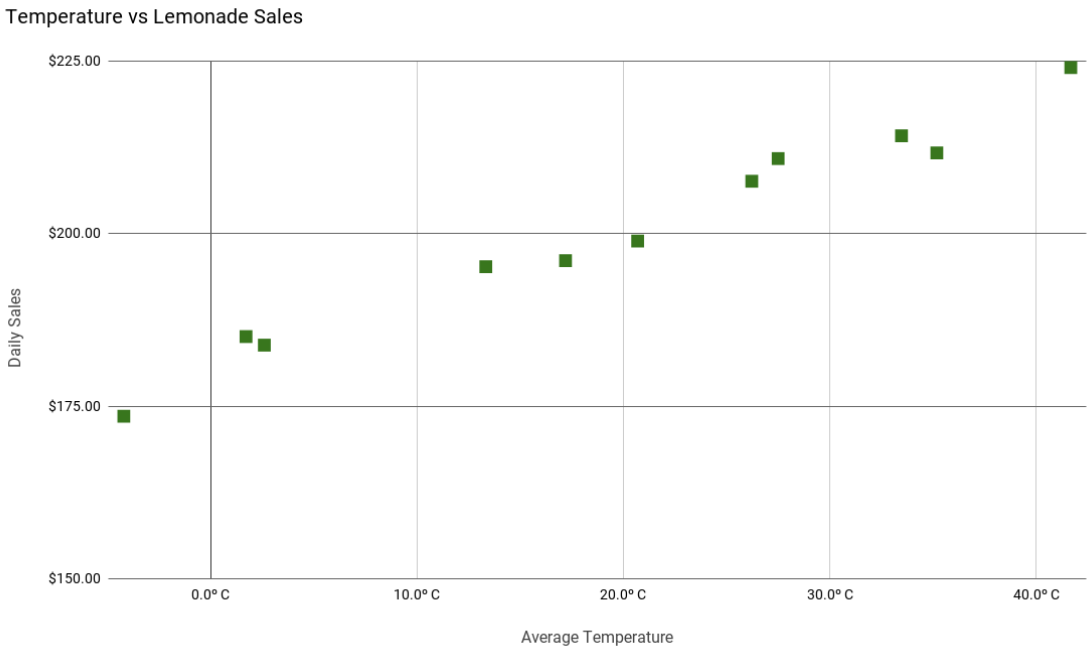



Figure 61.11: A scatterplot of temperature versus daily sales.

From this scatter plot, you can easily see that you do sell more lemonade as the temperature goes up. Drawing a *best-fit* line along the points will give you a *correlation coefficient*. A positive correlation coefficient will give you a positively proportional relationship, while a negative coefficient will give you a inversely proportional relationship.

### Making a Scatter Plot in Python

```
import matplotlib.pyplot as plt

# Avg. Temp (°C) and Total Sales ($)
temps = [2.6, -4.2, 13.3, 26.2, 27.5, 31.3, 33.5, 41.7, 20.7, 17.2, 1.7, 0.2]
sales = [183.85, 173.56, 195.22, 207.61, 210.88, 214.18, 215.23, 224.07, 198.94,
↪  196.10, 185.10, 188.70]

plt.scatter(temps, sales)
plt.title("Temperature vs. Lemonade Sales")
plt.xlabel("Average Temperature (°C)")
plt.ylabel("Total Sales ($)")
plt.grid(True)
plt.show()
```

CHAPTER 62

# Numerical Methods of Describing Data

In the Introduction to Data Visualization chapter, you learned to read a distribution from a graph. A histogram or boxplot can show you immediately whether data is skewed, where the center sits, and how spread out the values are. But a picture has limits. You cannot subtract two histograms. You cannot use a boxplot to predict a salary or compare a test score to a national average. For that, you need numbers.

This chapter develops the numerical tools that complement the graphical ones you already know. By the end, you will be able to describe any distribution precisely, locate any individual value within it, and avoid the most common errors people make when summarizing data.

## 62.1 Why Numerical Summaries Matter

Look back at the scatter plot from the previous chapter. To draw the line of best fit through that data, two quantities appeared in the formula: $S_x$ and $S_y$, the standard deviations of the two variables. The slope of the regression line depends directly on them. Before you can apply any of the methods in the next chapter, you need to know exactly what those quantities are and how to compute them. That is what this chapter is for.

There are three families of numerical measures, each answering a different question about a data set:

- *Measures of central tendency* ask: what is a typical value?
- *Measures of variation* ask: how spread out are the values?
- *Measures of position* ask: where does a particular value sit relative to the rest?

Each family has more than one tool, and choosing the right one depends on the shape of your data. That connection between shape and the choice of numerical summary is one of the central ideas of this chapter.

### 62.1.1 Summation Notation

Nearly every numerical measure involves adding things up. Before computing anything, we need a compact way to write sums.

Suppose you record the daily sales at a shop over $n$ days, calling them $X_1, X_2, \ldots, X_n$. Writing out every term quickly becomes unwieldy. The Greek capital letter $\Sigma$ (read as "sigma") gives us a cleaner way:

$$\sum_{i=1}^{n} X_i = X_1 + X_2 + \cdots + X_n$$

The expression below $\Sigma$ tells you where to start; the expression above tells you where to stop. When the range is clear from context, we often write $\sum X_i$ or simply $\sum X$.

Consider Ben, who scores the following on ten quizzes:

$$8,\ 9,\ 0,\ 10,\ 10,\ 8,\ 7,\ 9,\ 10,\ 5$$

Ben's total score is

$$\sum_{i=1}^{10} X_i = 8 + 9 + 0 + 10 + 10 + 8 + 7 + 9 + 10 + 5 = 76$$

This notation will appear in every formula in this chapter. When you see $\sum X_i$, read it as "add up all the values of $X$."

### *Exercise 148* Summation Practice

Sasha records the daily high temperatures (in degrees Celsius) over one week: 22, 19, 25, 21, 18, 23, 20.

1. Write the sum of all seven values using summation notation and compute it.
2. Compute $\sum_{i=1}^{7}(X_i - 20)$. What do you notice about how this compares to your answer in (a)?

*Working Space*

*Answer on Page 1930*

## 62.2 Measures of Center

You already know how to read the center of a distribution from a graph: it is roughly where the data balances. Now we want to pin that idea down to a single number. There are two main tools for this, and they do not always agree.

### 62.2.1 The Mean

The *mean* is the arithmetic average of a data set. The mean is the balancing point of the distribution: the point at which the data would balance if every value were a weight on a number line. This is the same seesaw idea from the graphical chapter, now expressed as a formula.

When the data represent an entire *population* of $N$ values, the mean is denoted by the Greek letter $\mu$ (read "mu"):

$$\mu = \frac{\sum_{i=1}^{N} X_i}{N}$$

When the data are a *sample* of $n$ values drawn from a larger population, the mean is denoted $\bar{X}$ (read "X-bar "):

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

The formula is identical in both cases. Only the notation changes to signal whether you are describing a full population or a sample.

Return to Ben's ten quiz scores from the previous section:

$$8,\ 9,\ 0,\ 10,\ 10,\ 8,\ 7,\ 9,\ 10,\ 5$$

$$\bar{X} = \frac{8+9+0+10+10+8+7+9+10+5}{10} = \frac{76}{10} = 7.6$$

Notice what that zero does. One failed quiz pulled the mean down from what would otherwise be a strong set of scores. This is the key property of the mean: it is not *resistant*. Every value in the data set contributes to it, so extreme values, whether unusually high or low, can pull it noticeably away from the center of the bulk of the data.

### 62.2.2 The Median

The *median* takes a different approach. Rather than averaging all the values, it finds the middle one. Half the data falls at or below the median; half falls at or above it.

To find the median $M$ of a data set with $n$ measurements:

1. Arrange all measurements in increasing order.
2. Compute $l = \frac{n+1}{2}$.
3. The median $M$ is the value of the $l$th measurement, counted from the smallest.

If $n$ is odd, $l$ is a whole number and the median is a value that actually appears in the data. If $n$ is even, $l$ falls halfway between two observations and the median is their average; it may not correspond to any value in the data.

Consider Daniel's car dealership, which tracks its daily sales. Over 14 days, the number

of cars sold was:

$$14,\ 9,\ 23,\ 7,\ 11,\ 23,\ 17,\ 11,\ 3,\ 24,\ 21,\ 2,\ 20,\ 20$$

Arranged in order:

$$2,\ 3,\ 7,\ 9,\ 11,\ 11,\ 14,\ 17,\ 20,\ 20,\ 21,\ 23,\ 23,\ 24$$

With $n = 14$:

$$l = \frac{14 + 1}{2} = 7.5$$

The median is the average of the 7th and 8th observations. Counting from the smallest, the 7th is 14 and the 8th is 17:

$$M = \frac{14 + 17}{2} = 15.5 \text{ cars}$$

Now notice what happens if the dealership's best day (24 cars) were replaced by an exceptional day of 80 cars. The mean would shift upward. The median would not move at all, as the 7th and 8th values in the sorted list are unchanged. This is what it means for the median to be resistant! That is, it does not react to extreme values the way the mean does.

### *Exercise 149* Computing Mean and Median

A small bookshop records the number of books sold each day over eight days:

12, 15, 9, 11, 14, 8, 13, 52

1. Compute the mean.
2. Compute the median.
3. The value 52 represents an unusually high day. It could be a sale event. Which measure is more affected by it? Explain why.
4. Remove the value 52 and recompute both the mean and the median using the remaining seven values. How much did each change?

*Working Space*

*Answer on Page 1930*

#### 62.2.3 Choosing Between Them

The mean and median each have their place, and the choice between them follows a clear rule. The right choice depends on the shape of the distribution, which you can now read directly from a histogram or boxplot.

For a *symmetric distribution*, the mean and median are close to each other. Either works, but the mean is generally preferred because it uses every observation and forms the foundation of most inferential statistics you will encounter later.

For a *skewed distribution* or any data set with outliers, the mean is pulled toward the tail while the median stays near the bulk of the data. In these cases, the median gives a more honest picture of a typical value.

You saw this in *Introduction to Data Visualization*. Most people earn within a moderate range, but a small number earn vastly more. The mean income for a country is often

much higher than what a typical person earns, precisely because it is dragged upward by those extreme high values. The median income is usually the figure reported when the goal is to describe a typical household.

The relationship between mean, median, and skew follows a consistent pattern:

- Symmetric: mean $\approx$ median
- Right-skewed: mean $>$ median
- Left-skewed: mean $<$ median

## *Exercise 150* Choosing a Measure of Center

For each situation, state whether you would report the mean or the median as the measure of center, and explain your reasoning.

1. The distribution of exam scores in a class of 30 students is roughly symmetric with no outliers.
2. The distribution of monthly salaries at a company is strongly right-skewed. The CEO earns considerably more than all other employees.
3. A histogram of the ages at which people in a city receive their first driving licence shows a large cluster between 16 and 19, and a long tail extending into the 30s and 40s.
4. After removing the CEO's salary from the company data in (b), the remaining distribution is roughly symmetric. Now which measure would you report, and does your answer change?

*Working Space*

*Answer on Page 1930*

## 62.3 Measures of Spread

Two data sets can share exactly the same mean and still tell completely different stories. What distinguishes them is spread: how far the values are scattered around the center. This section develops the numerical tools for measuring that scatter precisely.

### 62.3.1 Range and IQR

You already encountered these in the graphical chapter when building boxplots, so this is a brief review before introducing the more powerful measures.

The *range* is the simplest measure of spread:

$$R = \text{largest value} - \text{smallest value}$$

For the car dealership data, $R = 24 - 2 = 22$ cars. Range is easy to compute but unreliable, because it depends entirely on the two most extreme values and ignores everything in between. One unusual observation can make the range misleading.

The *interquartile range* (IQR) improves on this by focusing on the middle 50% of the data:

$$IQR = Q_3 - Q_1$$

For the dealership data, $Q_1 = 9$, $Q_3 = 21$, so $IQR = 12$ cars. Because IQR ignores the tails entirely, it is a resistant measure, so outliers do not affect it. When the median is the appropriate measure of center, IQR is the matching measure of spread.

### 62.3.2 Variance and Standard Deviation

Range and IQR both have a blind spot: they ignore most of the data. The *standard deviation* fixes this by taking every value into account. It measures how far, on average, each observation sits from the mean.

Before we can compute the standard deviation, we need the *variance*. The variance is the average of the squared distances from the mean. Squaring serves two purposes: it removes the sign so that values above and below the mean do not cancel each other out, and it penalizes large deviations more heavily than small ones.

The *population variance* is denoted $\sigma^2$:

$$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$$

The *sample variance* is denoted $s^2$:

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

We divide by $n - 1$ rather than $n$ when working with a sample. Dividing by $n$ would systematically underestimate the true population spread because a sample tends to cluster closer to its own mean than the full population does. Subtracting one corrects for that.

The standard deviation is simply the square root of the variance, which brings the units back to the same scale as the original data:

$$\sigma = \sqrt{\sigma^2} \qquad\qquad s = \sqrt{s^2}$$

A population standard deviation is denoted $\sigma$ ("sigma"); a sample standard deviation is denoted $s$.

### Computing Standard Deviation by Hand

Let us work through Ben's quiz score data step by step. The ten scores are $8, 9, 0, 10, 10, 8, 7, 9, 10, 5$ with $\bar{X} = 7.6$.

**Step 1.** Compute each deviation from the mean, $x_i - \bar{x}$:

| $x_i$ | $x_i - \bar{x}$ | $(x_i - \bar{x})^2$ |
|---|---|---|
| 8 | 0.4 | 0.16 |
| 9 | 1.4 | 1.96 |
| 0 | −7.6 | 57.76 |
| 10 | 2.4 | 5.76 |
| 10 | 2.4 | 5.76 |
| 8 | 0.4 | 0.16 |
| 7 | −0.6 | 0.36 |
| 9 | 1.4 | 1.96 |
| 10 | 2.4 | 5.76 |
| 5 | −2.6 | 6.76 |
| | | 86.40 |

**Step 2.** Divide the sum of squared deviations by $n - 1 = 9$ to get the sample variance:

$$s^2 = \frac{86.40}{9} = 9.60$$

**Step 3.** Take the square root:

$$s = \sqrt{9.60} \approx 3.10$$

Notice that the zero score contributes a squared deviation of 57.76, which is larger than the contributions of all nine other scores combined. One extreme value had an outsized effect on the standard deviation.

### Checking Your Work with Python

Hand computation is worth doing once, so the mechanics stay visible. After that, software does the arithmetic. Python's built-in `statistics` module computes the same sample variance and standard deviation directly:

```
import statistics

scores = [8, 9, 0, 10, 10, 8, 7, 9, 10, 5]

print("mean:    ", statistics.mean(scores))
print("variance:", statistics.variance(scores))   # divides by n - 1
print("stdev:   ", statistics.stdev(scores))       # divides by n - 1
```

This prints `mean: 7.6`, `variance: 9.6`, and `stdev: 3.0983866769659336`, matching the

by-hand result from Steps 1 through 3. `statistics.variance` and `statistics.stdev` use the sample formulas, dividing by $n - 1$, which is the same convention this chapter uses. The population versions, `statistics.pvariance` and `statistics.pstdev`, divide by $n$ instead.

## *Exercise 151* Computing Variance and Standard Deviation

*Working Space*

Daniel's car dealership records the number of cars sold each day over 14 days:

14, 9, 23, 7, 11, 23, 17, 11, 3, 24, 21, 2, 20, 20

The mean is $\bar{X} \approx 14.64$ cars.

1. The sum of squared deviations $\sum (x_i - \bar{x})^2$ for this data set is approximately 743.21. Use this to compute the sample variance $s^2$.
2. Compute the sample standard deviation $s$.
3. Standard deviation is measured in the same units as the original data. What are the units of $s$ here? What are the units of $s^2$?
4. A second dealership has a mean of 14.64 cars per day and $s = 2.1$ cars. Without any other information, which dealership has more consistent daily sales? Explain.

*Answer on Page 1931*

### 62.3.3 Interpreting Spread

Standard deviation is a unit of distance. Once you have computed $s$ for a data set, you can ask of any individual observation: how many standard deviations away from the mean is this value? That question will become the foundation of the next section on z-scores.

For now, focus on two properties that are always true.

First, a larger standard deviation means a wider distribution. Compare two students' quiz scores:

| | | |
|---|---|---|
| Susan: | 8, 9, 4, 8, 6, 8, 7, 9, 10, 5 | $s_S \approx 1.90$ |
| Joaquin: | 7, 8, 6, 7, 6, 6, 7, 5, 5, 7 | $s_J \approx 0.97$ |

Susan's scores vary more from quiz to quiz. Joaquin's are tightly packed. The standard deviations reflect exactly what you see in the data: $s_S > s_J$.

Second, standard deviation is not resistant. Because it is computed from the mean, and the mean is pulled by extreme values, any outlier also inflates the standard deviation. Recall Ben's quiz example: the single zero score contributed a squared deviation of 57.76 out of a total of 86.40. One observation accounted for two-thirds of the total spread.

This is why the choice of spread measure mirrors the choice of center measure:

- Symmetric data, no outliers: use mean and standard deviation
- Skewed data or data with outliers: use median and IQR

### *Exercise 152* Choosing a Measure of Spread

For each situation, state which measure of spread (standard deviation or IQR) is more appropriate, and explain why.

1. The distribution of heights of adult men in a country is roughly symmetric and bell-shaped with no outliers.
2. The distribution of house prices in a city is strongly right-skewed, with a small number of luxury properties selling for many times more than the typical home.
3. Reine removes all outliers from a right-skewed data set. The remaining distribution is approximately symmetric. Which measure of spread should she now report?

*Working Space*

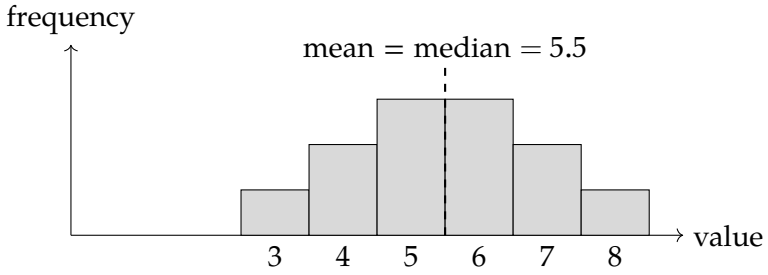
*Answer on Page 1931*

## 62.4 Connecting Shape to Numerical Measures

You already know how to read shape from a histogram or boxplot. This section translates what you see graphically into numerical consequences: specifically, what skew and outliers do to the measures you just computed.

### 62.4.1 Symmetric Data: Mean Approximately Equals Median

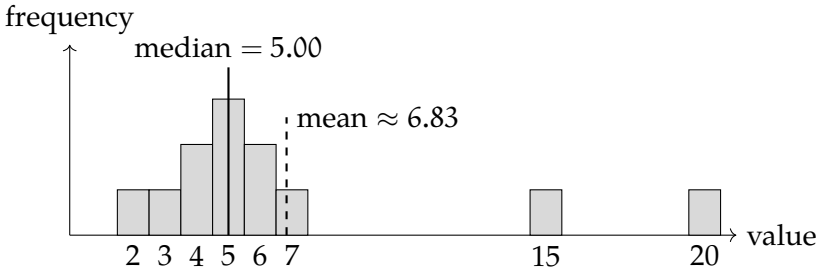
When a distribution is roughly symmetric, the mean and median land close to each other. Neither tail is pulling the mean away from the center of the data. This is the situation where mean and standard deviation are both appropriate, and where most inferential procedures work as intended.




A roughly symmetric data set: 3, 4, 4, 5, 5, 5, 6, 6, 6, 7, 7, 8. The mean and median land at exactly the same value.

### 62.4.2 Skewed Data: Mean Pulled Toward the Tail

When a distribution is skewed, the mean chases the tail while the median stays near the bulk of the data. The direction of the skew tells you exactly which way the mean is pulled:




The right-skewed data set below: two extreme values at 15 and 20 pull the mean well past the median.

- Right-skewed: mean $>$ median. The right tail pulls the mean upward.
- Left-skewed: mean $<$ median. The left tail pulls the mean downward.

This pattern follows directly from what you already know about the mean. Because the mean uses every value in its calculation, a cluster of extreme values in the right tail will drag it rightward. The median, which depends only on rank order, does not move.

Consider a small right-skewed data set:

$$2,\ 3,\ 4,\ 4,\ 5,\ 5,\ 5,\ 6,\ 6,\ 7,\ 15,\ 20$$

The mean is approximately 6.83 and the median is 5.00. The two extreme values at 15

and 20 pull the mean well above where most of the data sits. On a histogram, most bars would be clustered at the left with a long right tail. This shape corresponds to mean $>$ median.

### 62.4.3 Effects of Outliers

You have heard this term used all over our Statistics chapters! An *outlier* is a value that sits far from the rest of the data. You saw the numerical definition in the graphical chapter: any value more than $1.5 \times \text{IQR}$ below $Q_1$ or above $Q_3$ is flagged as an outlier.

What outliers do to each measure follows directly from whether that measure is resistant:

- Mean: pulled toward the outlier. One extreme value can move the mean substantially.
- Median: unaffected. Moving the most extreme value further into the tail does not change the middle rank.
- Standard deviation: inflated. Because it is computed from the mean, an outlier that moves the mean also increases the squared deviations.
- IQR: unaffected. It depends only on $Q_1$ and $Q_3$, which are determined by the middle of the data.

This is why mean and standard deviation always travel together, and median and IQR always travel together. They are basically matched pairs.

### 62.4.4 Changing Units on Summary Measures

One practical question comes up often: if every value in a data set is shifted or scaled, what happens to the summary measures? Suppose a teacher adds 20 points to every student's test score, or a government increases all property taxes by 2%. You do not need to recompute everything from scratch.

The rules depend on whether you are adding a constant or multiplying by one.

**Adding a constant $a$ to every value.** Shifting every observation by the same amount shifts the measures of center and position by the same amount. Measures of spread are unaffected, because the distances between values do not change.

**Multiplying every value by a constant $b$.** Scaling every observation multiplies all measures, such as center, spread, and position, by the same factor. The absolute value $|b|$ is used for spread measures, since spread is always non-negative.

These rules are summarized in the table below.

| **Measure** | $Y_i = X_i + a$ | $Y_i = b \cdot X_i$ |
|---|---|---|
| Mean | $\bar{Y} = \bar{X} + a$ | $\bar{Y} = b\bar{X}$ |
| Median | $\text{Median}(Y) = \text{Median}(X) + a$ | $\text{Median}(Y) = b \cdot \text{Median}(X)$ |
| Range | Unaffected | $\text{Range}(Y) = \lvert b \rvert \, \text{Range}(X)$ |
| Std dev | Unaffected | $s_Y = \lvert b \rvert \cdot s_X$ |
| $Q_1$, $Q_3$ | $Q_1 + a$, $Q_3 + a$ | $b \cdot Q_1$, $b \cdot Q_3$ |
| IQR | Unaffected | $\text{IQR}(Y) = \lvert b \rvert \cdot \text{IQR}(X)$ |

The key pattern: adding a constant moves everything but changes no distances; multiplying stretches or compresses everything including distances.

**Example.** A class takes a five-question test worth 20 points each. The summary statistics for their scores are:

| | |
|---|---|
| Mean | 62 |
| Median | 60 |
| Range | 45 |
| Standard deviation | 8 |
| $Q_1$ | 48 |
| $Q_3$ | 71 |
| IQR | 23 |

After grading, the teacher discovers a typographical error in question 2 that made it unanswerable. She adds 20 points to every student's score. The new summary statistics are:

| **Measure** | **Original** | **After adding 20** |
|---|---|---|
| Mean | 62 | $62 + 20 = 82$ |
| Median | 60 | $60 + 20 = 80$ |
| Range | 45 | 45 (unaffected) |
| Std dev | 8 | 8 (unaffected) |
| $Q_1$ | 48 | $48 + 20 = 68$ |
| $Q_3$ | 71 | $71 + 20 = 91$ |
| IQR | 23 | 23 (unaffected) |

Every student's score shifted by the same amount, so the spread of the distribution is unchanged. The center and quartiles each moved up by exactly 20.

### *Exercise 153* Changing Units

County residents vote to increase property taxes by 2% to fund local schools. The current summary statistics for property tax (in dollars per year) are:

| | |
|---|---|
| Mean | 12,000 |
| Median | 8,000 |
| Range | 30,000 |
| Standard deviation | 5,000 |
| $Q_1$ | 5,000 |
| $Q_3$ | 14,000 |
| IQR | 9,000 |

Each resident's new tax will be $1.02\times$ their current tax. Compute the new value of each summary measure.

*Working Space*

*Answer on Page 1931*

## 62.5 Z-scores

The previous section showed that adding a constant to every value shifts the mean but leaves the standard deviation unchanged. That means the distances between values, relative to the center, stay exactly the same. There is a natural way to express those distances as a single number: the *z-score*, also called a *standardized score*.

The z-score of a measurement tells you how far that value sits from the mean, expressed in units of standard deviations:

$$z = \frac{x - \bar{x}}{s}$$

For population data, replace $\bar{x}$ with $\mu$ and $s$ with $\sigma$. The interpretation is the same either way.

A positive z-score means the value is above the mean. A negative z-score means it is below. A z-score of zero means the value equals the mean exactly. The magnitude tells

you how many standard deviations separate the value from the center.

**Example.** A class takes a test with mean 74 and standard deviation 6. Two students receive the following scores:

$$\text{Amaka} \quad 88 \quad z = \frac{88-74}{6} = \frac{14}{6} \approx 2.33$$
$$\text{George} \quad 70 \quad z = \frac{70-74}{6} = \frac{-4}{6} \approx -0.67$$

Amaka scored about 2.33 standard deviations above the class average. George scored about 0.67 standard deviations below it. George is actually closer to the mean than Amaka: $|-0.67| < |2.33|$.

### Comparing Across Different Distributions

Z-scores become especially powerful when comparing measurements from distributions with different means and standard deviations. Without standardization, comparing values measured on different scales is meaningless. With z-scores, it becomes straightforward.

Suppose two cities report their daily high temperatures:

| City | Mean daily high | Standard deviation |
|---|---|---|
| North Bend | 80°F | 12°F |
| South Bend | 84°F | 4°F |

Both cities report a high of 95°F on the same day. Which city had the more unusually warm day?

The raw difference from the mean is larger for North Bend ($95-80=15$) than South Bend ($95-84=11$), but that ignores how variable the temperatures are in each city. Compute the z-scores:

$$\text{North Bend: } z = \frac{95-80}{12} = 1.25 \qquad \text{South Bend: } z = \frac{95-84}{4} = 2.75$$

South Bend's 95°F sits 2.75 standard deviations above its mean, which is far more unusual than North Bend's 1.25. The day was more remarkably warm in South Bend, even though it was not further above South Bend's mean in raw degrees.

## *Exercise 154* Computing and Interpreting Z-scores

Daniel's car dealership from earlier in this chapter had mean daily sales of $\bar{X} \approx 14.64$ cars and standard deviation $s \approx 7.56$ cars.

1. Compute the z-score for the best sales day (24 cars). Interpret it in context.
2. Compute the z-score for the worst sales day (2 cars). Interpret it in context.
3. A second dealership reports a best day of 31 cars, with mean 20 and standard deviation 4. Which dealership had the more unusually good best day? Show your work.
4. Daniel adds a fixed weekend bonus, increasing every daily total by 5 cars. Without recomputing, what is the z-score for the original best day of 24 cars under the new totals? Explain.

*Working Space*

*Answer on Page 1932*

Key Points:

- A z-score expresses a value as a number of standard deviations above or below the mean
- Positive z-scores are above the mean; negative z-scores are below it
- Z-scores allow meaningful comparison across data sets measured on different scales
- Adding a constant to every value does not change any z-score; z-scores are invariant under shifting

## 62.6 Percentiles and Quartiles

The mean, median, and standard deviation describe the whole distribution. Sometimes the question is narrower: where does one particular value sit relative to everyone else? Mary, who scores 1400 on the SAT, wants to know not just that the median was 1040, but what percentage of students she scored above. That is what percentiles and quartiles answer.

### Percentiles

The $k$*th percentile* $P_k$ is the value such that $k\%$ of observations fall at or below it and $(100-k)\%$ fall at or above it. For example, $P_{95}$ is the 95th percentile: a value at $P_{95}$ is higher than 95% of the data and lower than the remaining 5%.

Three percentiles you already know appear here under new names: $P_{25} = Q_1$, $P_{50} = Q_2 = M$ (the median), and $P_{75} = Q_3$.

To find $P_k$ for a data set of $n$ measurements:

1. Arrange all measurements in increasing order.
2. Compute $l = \dfrac{(n+1)k}{100}$.
3. $P_k$ is the value of the measurement in position $l$, counted from the smallest. If $l$ is not a whole number, interpolate between the two surrounding observations.

Return to the dealership data ($n = 14$, sorted):

$$2,\ 3,\ 7,\ 9,\ 11,\ 11,\ 14,\ 17,\ 20,\ 20,\ 21,\ 23,\ 23,\ 24$$

To find $P_{90}$:

$$l = \frac{(14+1)\times 90}{100} = 13.5$$

The 90th percentile falls halfway between the 13th and 14th observations. The 13th is 23 and the 14th is 24:

$$P_{90} = \frac{23+24}{2} = 23.5 \text{ cars}$$

On 90% of days, the dealership sold 23.5 or fewer cars.

To find $P_{10}$:

$$l = \frac{(14+1) \times 10}{100} = 1.5$$

Halfway between the 1st observation (2) and the 2nd (3):

$$P_{10} = \frac{2+3}{2} = 2.5 \text{ cars}$$

On 10% of days, the dealership sold 2.5 or fewer cars.

### Quartiles and the Five-Number Summary

*Quartiles* divide a sorted data set into four equal parts using the 25th, 50th, and 75th percentiles. You first encountered these when building boxplots in the *Introduction to Data Visualization* chapter. Here is the formal connection:

- $Q_1 = P_{25}$: 25% of values fall at or below $Q_1$
- $Q_2 = P_{50} = M$: the median
- $Q_3 = P_{75}$: 75% of values fall at or below $Q_3$

In practice, quartiles are computed by splitting the sorted data into a lower half and an upper half and taking the median of each. For even $n$, split the data at the midpoint with no overlap. For odd $n$, exclude the median from both halves.

For the dealership data, split at the midpoint ($n = 14$, so seven values on each side):

$$\text{Lower half: } 2,\ 3,\ 7,\ \mathbf{9},\ 11,\ 11,\ 14 \Rightarrow Q_1 = 9$$
$$\text{Upper half: } 17,\ 20,\ 20,\ \mathbf{21},\ 23,\ 23,\ 24 \Rightarrow Q_3 = 21$$

You may notice that the median-of-halves method gives $Q_1 = 9$, while the percentile formula gives $P_{25} = 8.5$. Both are correct, as different software and textbooks use slightly different conventions for quartile computation. Small disagreements between methods are normal and expected. On the AP exam, the median-of-halves method is standard.

Together with the minimum and maximum, the quartiles form the *five-number summary*:

the same five values you used to build a boxplot.

| Min | $Q_1$ | Median | $Q_3$ | Max |
|---|---|---|---|---|
| 2 | 9 | 15.5 | 21 | 24 |

The five-number summary and the boxplot are two representations of exactly the same information: one numerical, one graphical.

## *Exercise 155* Relative Standing

Use the dealership data:
2, 3, 7, 9, 11, 11, 14, 17, 20, 20, 21, 23, 23, 24
($n = 14$).

1. Compute $P_{50}$ using the formula $l = \frac{(n+1)k}{100}$. Does it match the median you computed earlier? What position in the sorted list does $l$ point to?
2. On what percentage of days did the dealership sell at most 21 cars?
3. A sales consultant tells Daniel that a day with 17 cars sold falls in the "third quarter" of the distribution. Verify this by identifying which quarter 17 belongs to.
4. Daniel wants to flag the bottom 10% of days as underperforming. Using $P_{10} = 2.5$ cars, how many of the 14 days fall in this category? List them.

*Working Space*

*Answer on Page 1932*

### 62.6.1 Computing with Python

You have now worked with all four families of numerical measures by hand: center, spread, shape, and standing. In practice, you will rarely compute them this way. Python's built-in `statistics` module handles all of them directly, using the dealership data set from throughout this chapter:

```
import statistics

sales = [14, 9, 23, 7, 11, 23, 17, 11, 3, 24, 21, 2, 20, 20]

print("mean:    ", statistics.mean(sales))
print("median:  ", statistics.median(sales))
print("variance:", round(statistics.variance(sales), 2))
print("stdev:   ", round(statistics.stdev(sales), 2))

q1, q2, q3 = statistics.quantiles(sales, n=4)
print("Q1, Q2, Q3:", q1, q2, q3)
```

This prints a mean of 14.642857142857142, a median of 15.5, a variance of 57.17, and a standard deviation of 7.56, matching this chapter's hand-computed values exactly.

Chapter 63

# Introduction to Graphs

## 63.1 Introduction

Some data is easier to work with if we imagine it as a set of connections. When we say *graph* in this chapter, we are referring to set of data defined by a collection of *nodes* (you may also hear vertex) and connected by *edges*. For example, on some social networks, each user can follow any number of other users. We can think of each user as a node, and the edge points from the user who follows to the user they follow:

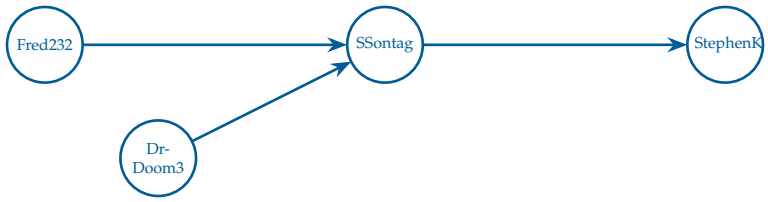



Figure 63.1: A graph showing four users and three "following" relationships.

Following is a *directed* relationship: Fred232 follows SSontag, but SSontag doesn't follow Fred232. So we would say that this is a *directed graph* with four nodes and three edges.

There are also undirected graphs. For example, you can imagine a graph that represents big data lines between cities. All the big data lines allow communications in both directions, like two-way roads:

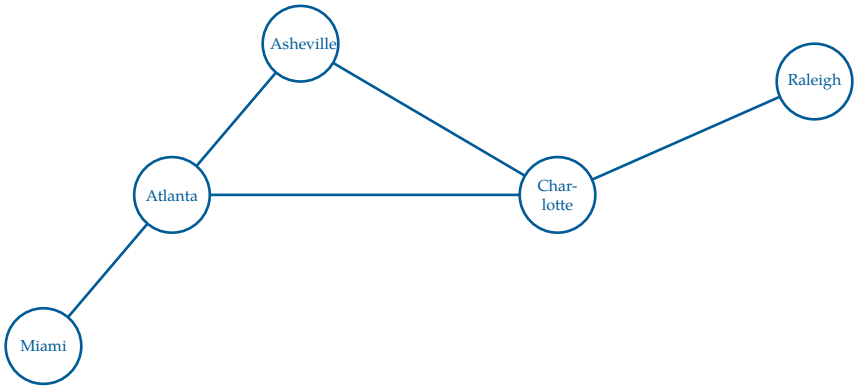



Figure 63.2: A set of cities, represented by nodes, on the east coast of the United States connected by roads, represented by edges.

The arrows are gone; if data can flow from Miami to Raleigh, then data can flow from Raleigh to Miama. A linked connection of edges between two or more nodes is called a *path*. Notice how Atlanta and Charlotte have 3 edges; we would say that Atlanta and Charlotte each have a *degree* of 3.

There is a whole branch of mathematics called *Graph Theory* that studies the properties of graphs. Here are two questions that mathematicians might ask about this graph:

- What is the smallest number of edges that we would need to follow to get from Miami to Raleigh?
- Does the graph have any paths where you could end up where you started?

That second question asks if there is a *cycle* in the graph. This graph has one cycle: Atlanta → Asheville → Charlotte → Atlanta.

Some graphs are *connected*: You can get from one node to any other node by following edges. Is this graph connected?

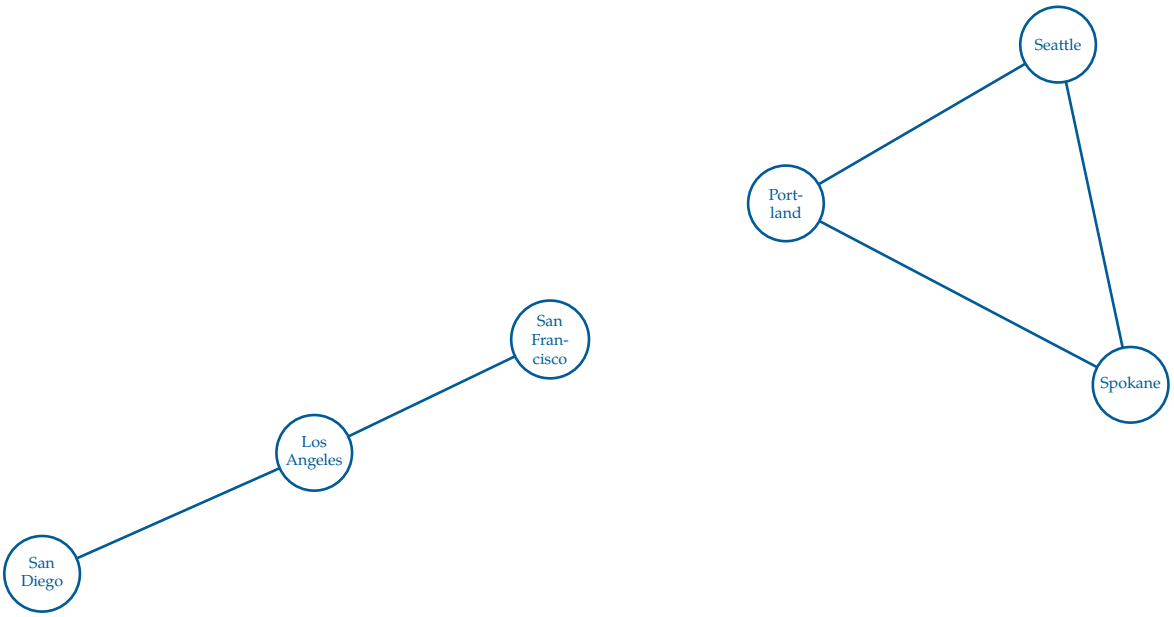



Figure 63.3: A set of cities (nodes) that are not all connected. Each set of *connected* cities forms a *connected component*.

This graph is *not* connected! You can't follow edges from San Diego to Seattle. However, you could identify two *connected components*, *subgraphs* of a graph that remain connected.

- San Diego, Los Angeles, San Francisco
- Portland, Seattle, Spokane

Connectivity is defined as, for all pairs of nodes in that graph, there exists a path between them. A subgraph is much simpler than a connected component, it's only requirement is to contain a subset of at least one node and, optionally, edges to go between nodes. Note that you cannot have a subgraph that contains a single node and an edge, because the edge would not be able to connect to anything else.

In graph data, the nodes and edges often have attributes. For example, a node representing a city might have a name and a population. An edge representing a data line might have a bandwidth (bits per second) and a latency (how many nanoseconds between when you put a bit into the pipe and when it comes out the other end). This is how we define a graph.

### *Exercise 156* Word Graphs

*Working Space*

Consider the concept of a graph as a set of nodes and edges. How would you create a graph where the nodes are words and the edges represent a one letter difference between two words?

For example, the word `cat` would be connected to `cot`, `cart`, and `cut`, but not to `dog`.

Define explicitly what the nodes would be in this graph, and what the edges would be. Draw a simple example of this graph with at least 5 nodes and 5 edges.

*Answer on Page 1933*

## 63.2 Finding Good Paths

For many problems, we are trying to find the best path from one node to another. If all the edges are the same, this usually means finding the path that requires walking the fewest edges.

Sometimes the edges have a cost attribute. For example, you might want to find the

cheapest way to ship a container from New York City to Long Beach, California. In this case the nodes are train depots. Each train line between the depots has a cost. What is the cheapest path?

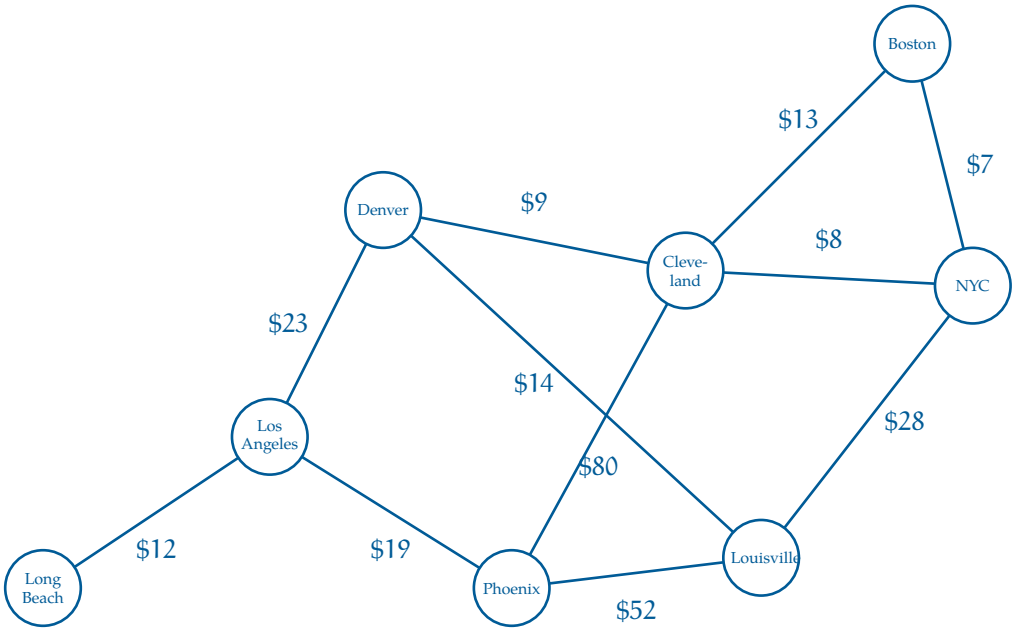



Figure 63.5: A graph of interconnected cities where each edge has a respective cost.

When edges have costs like this, we call the *weighted edges*, which forms a *weighted graph*. In weighted graphs, the *degree* of a node is split into the *in-degree* and *out-degree* of nodes. The in-degree identifies the incoming edges that originate from other nodes, and out-degree is the amount of edges going out of that node.

The graphs that you see here are really small, so finding efficient paths isn't difficult — you could just try all of them! However, in many computer programs, we are working with *millions* of nodes and edges. Efficient graph algorithms are *really* important.

## 63.3 Graphs in Python

There are many ways to represent graph data. There are database systems that are specifically designed to hold and analyze graph data. Not surprisingly, these are called *Graph Databases*.

You may also hear about *adjacency lists*, which store, for each node, the list of nodes it is connected to. Each direct connection from one node to another is called a *neighbor*. Simply, adjacency lists store lists of neighbors.

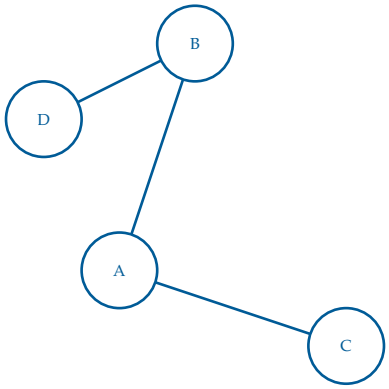



Figure 63.6: A simple (undirected unweighted) graph used to make an adjacency list.

As an adjacency list, this would look like:

- A: [B, C]
- B: [A, D]
- C: [A]
- D: [B]

## *Exercise 157* Directed Adjacency Lists

*Working Space*

Consider the concept of adjacency lists discussed above. How would the storage method change if the edges were to have weights?

*Answer on Page 1933*

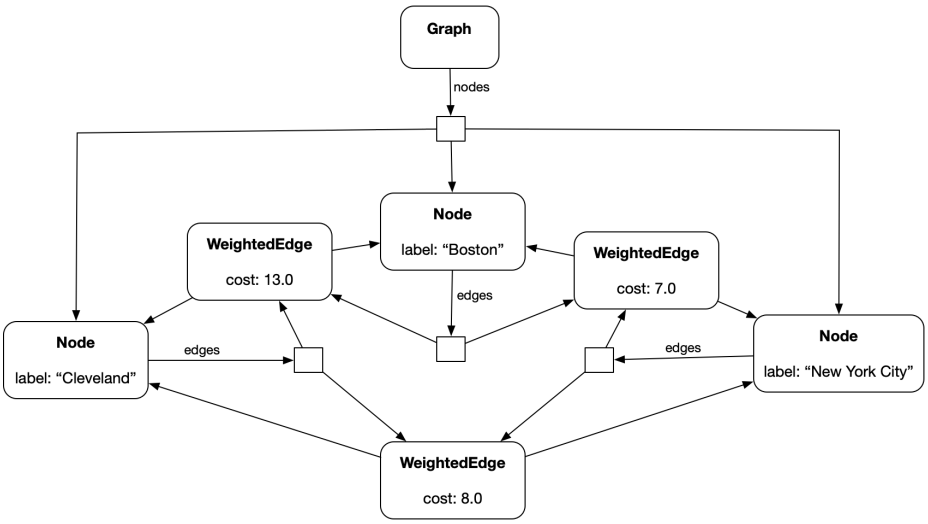
For simplicity, you are going to write Python classes that will let you represent an undirected graph with weighted edges, like the shipping problem above.

(Naturally, things would look a little different if the graph were directed or the edges were unweighted, but this is a good starting place.)

Create a file called `graph.py`. This will hold the code for your `Node` and `WeightedEdge` classes. We will also create a `Graph` class that will just hold onto the list of its nodes.

- A `Node` will have a label string and a list of edges that touch it.
- A `WeightedEdge` will have a cost and two nodes: `node_a` and `node_b`.
- A `Graph` will have a list of nodes.

Here is what the object diagram would look like if you had only three cities:




Put this code into `graph.py`

```
class Node:
  def __init__(self, label):
    self.label = label
    self.edges = []

  def __repr__(self):
    return f"(node:{self.label}, edges:{len(self.edges)})"

class WeightedEdge:
  def __init__(self, cost, node_a, node_b):
    self.cost = cost
    self.node_a = node_a
    node_a.edges.append(self)
    self.node_b = node_b
    node_b.edges.append(self)

```

```
def other_end(self, node_from):
    if self.node_a == node_from:
        return self.node_b
    if self.node_b == node_from:
        return self.node_a
    raise ValueError("node is not an endpoint of this edge")

class Graph:
  def __init__(self):
    self.nodes = []

  def add_node(self, new_node):
    self.nodes.append(new_node)

  def __repr__(self):
    return f"(Graph:{self.nodes})"
```

This is a graph class that handles undirected weighted graphs. Now, let's create some instances of `Node` and `WeightedEdge` and wire them together. Create another file in the same directory called `cities.py`. Put in this code:

```
import graph

# Create an empty graph
network = graph.Graph()

# Create city nodes and add to graph
long_beach = graph.Node("Long Beach")
network.add_node(long_beach)
los_angeles = graph.Node("Los Angeles")
network.add_node(los_angeles)
denver = graph.Node("Denver")
network.add_node(denver)
phoenix = graph.Node("Phoenix")
network.add_node(phoenix)
louisville = graph.Node("Louisville")
network.add_node(louisville)
cleveland = graph.Node("Cleveland")
network.add_node(cleveland)
boston = graph.Node("Boston")
network.add_node(boston)
nyc = graph.Node("New York City")
network.add_node(nyc)

# Create edges
graph.WeightedEdge(12, long_beach, los_angeles)
graph.WeightedEdge(23.0, los_angeles, denver)
graph.WeightedEdge(19, los_angeles, phoenix)
graph.WeightedEdge(52, phoenix, louisville)
graph.WeightedEdge(14, denver, louisville)
```

```
graph.WeightedEdge(80, phoenix, cleveland)
graph.WeightedEdge(9, denver, cleveland)
graph.WeightedEdge(8, cleveland, nyc)
graph.WeightedEdge(28, louisville, nyc)
graph.WeightedEdge(7, nyc, boston)
graph.WeightedEdge(13, cleveland, boston)

print(network)
```

Run it:

```
python3 cities.py
```

You should see some rather unexciting output:

```
(Graph:[(node:Long Beach, edges:1), (node:Los Angeles, edges:3), (node:Denver,
↪  edges:3),
(node:Phoenix, edges:3), (node:Louisville, edges:3), (node:Cleveland, edges:4),
(node:Boston, edges:2), (node:New York City, edges:3)])
```

But do not worry; we will make it more exciting in the next chapter!

Chapter 64

# The Dot Product

If you have two vectors $\mathbf{u} = [u_1, u_2, \ldots, u_n]$ and $\mathbf{v} = [v_1, v_2, \ldots, v_n]$, we define the *dot product* $\mathbf{u} \cdot \mathbf{v}$ as

$$\mathbf{u} \cdot \mathbf{v} = (u_1 \times v_1) + (u_2 \times v_2) + \cdots + (u_n \times v_n)$$

The output of the dot product is a *scalar* quantity.

For example,

$$[2, 4, -3] \cdot [5, -1, 1] = 2 \times 5 + 4 \times -1 + -3 \times 1 = 3$$

This may not seem like a very powerful idea, but dot products are *incredibly* useful. The enormous GPUs (Graphics Processing Units) that let video games render scenes so quickly? They primarily function by computing huge numbers of dot products at mind-boggling speeds.

### *Exercise 158* Basic dot products

Compute the dot product of each pair of vectors:

*Working Space*

- $[1, 2, 3]$, $[4, 5, -6]$
- $[\pi, 2\pi]$, $[2, -1]$
- $[0, 0, 0, 0]$, $[10, 10, 10, 10]$

*Answer on Page 1934*

## 64.1 Properties of the dot product

Sometimes we need an easy way to say "The vector of appropriate length is filled with zeros." We use the notation $\vec{0}$ to represent this. Then, for any vector $\mathbf{v}$, this is true:

$$\mathbf{v} \cdot \vec{0} = 0$$

The dot product is commutative:

$$\mathbf{v} \cdot \mathbf{u} = \mathbf{u} \cdot \mathbf{v}$$

The dot product of a vector with itself is its magnitude squared:

$$\mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2$$

If you have a scalar $a$, then:

$$(\mathbf{v}) \cdot (a\mathbf{u}) = a(\mathbf{v} \cdot \mathbf{u})$$

So, if $\mathbf{v}$ and $\mathbf{w}$ are vectors that go in the same direction,

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}||\mathbf{w}|$$

If $\mathbf{v}$ and $\mathbf{w}$ are vectors that go in opposite directions,

$$\mathbf{v} \cdot \mathbf{w} = -|\mathbf{v}||\mathbf{w}|$$

If $\mathbf{v}$ and $\mathbf{w}$ are vectors that are perpendicular to each other, their dot product is zero:

$$\mathbf{v} \cdot \mathbf{w} = 0$$

## 64.2 Cosines and dot products

Furthermore, dot products' interaction with cosine makes them even more useful is what makes them so useful: If you have two vectors $\mathbf{v}$ and $\mathbf{u}$,

$$\mathbf{v} \cdot \mathbf{u} = |\mathbf{v}||\mathbf{u}| \cos \theta$$

where $\theta$ is the angle between them.

So, for example, if two vectors $\mathbf{v}$ and $\mathbf{u}$ are perpendicular, the angle between them is $\pi/2$. The cosine of $\pi/2$ is 0. The dot product of any two perpendicular vectors is always 0. In fact, if the dot product of two non-zero vectors is 0, the vectors *must be* perpendicular (see figure 64.1 for an example of perpendicular 2-dimensional vectors).

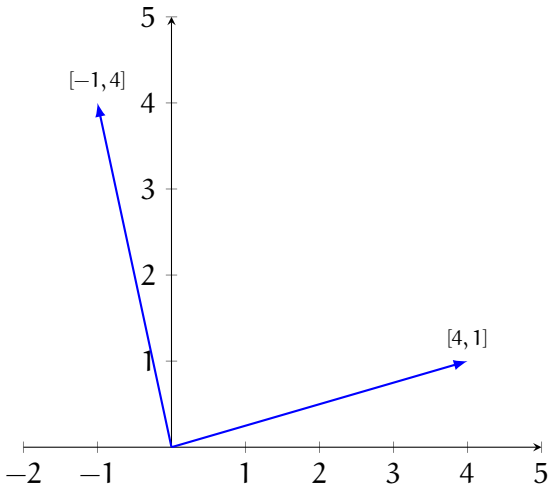



Figure 64.1: The dot product of any two perpendicular vectors is zero.

If you have two non-zero vectors $\mathbf{v}$ and $\mathbf{u}$, you can always compute the angle between them:

$$\theta = \arccos\left(\frac{\mathbf{v}\cdot\mathbf{u}}{|\mathbf{v}||\mathbf{u}|}\right)$$

Arccos is short for arccosine, or $\cos^{-}1$, and it is a function that is the inverse of cosine. Cosine takes an angle and gives back the scaled x-component of the angle. Arccosine takes the x-component of an angle and returns an angle with that x-component. However, there is a limit to what arccos can return. Let's look at cosine and its inverse, arccos (see figures 64.2 and 64.3).

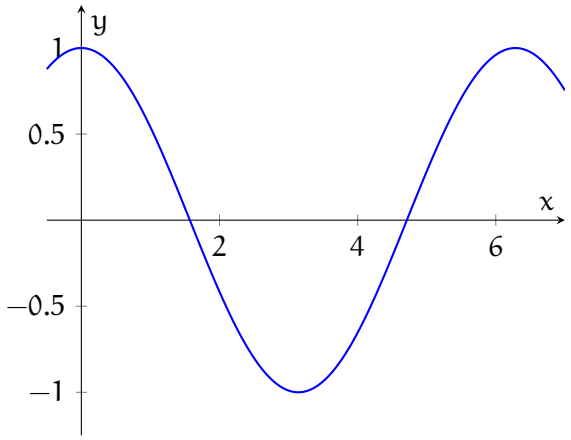



Figure 64.2: Cosine is a function: there is exactly one output for every input.

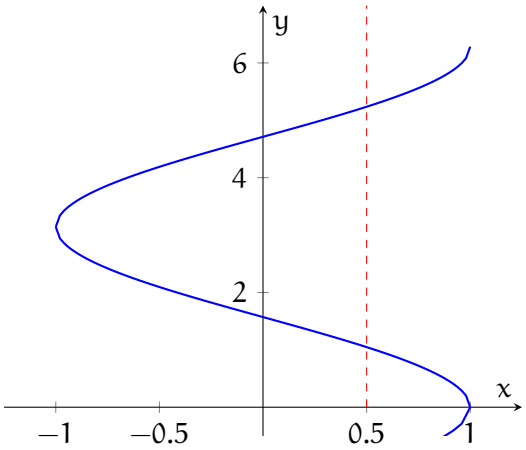



Figure 64.3: Arccos is not a function: there are many angles with the same x-component. Notice that one input value has many output values (see the red dashed line).

When you use a calculator to evaluate arccos, the calculator automatically restricts the results to between 0 and $\pi$. Let's look at an example of using the dot product to determine the angle between two vectors:

**Example**: What is the angle between $\mathbf{u} = \left[\sqrt{3}, 1\right]$ and $\mathbf{v} = [0, -1]$?

**Solution**: We know that $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}|\,|\mathbf{v}| \cos\theta$. Therefore, we also know that:

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}|\,|\mathbf{v}|}$$

First, let's compute the dot product:

$$\mathbf{u} \cdot \mathbf{v} = \sqrt{3} \cdot 0 + 1 \cdot -1 = -1$$

And therefore:

$$\cos\theta = \frac{-1}{|\mathbf{u}|\,|\mathbf{v}|}$$

Now, let's find the magnitudes of both vectors:

$$|\mathbf{u}| = \sqrt{\left(\sqrt{3}\right)^2 + (1)^2} = 2$$

$$|\mathbf{v}| = \sqrt{(0)^2 + (-1)^2} = 1$$

Substituting for the magnitudes, we find that:

$$\cos\theta = \frac{-1}{2 \cdot 1} = \frac{-1}{2}$$

To solve for $\theta$, we take the arccos of both sides:

$$\arccos(\cos\theta) = \theta = \arccos\frac{-1}{2}$$

What angles have a cosine of $-1/2$? We know that $2\pi/3$, $4\pi/3$, $8\pi/3$, etc., all have a cosine of $-1/2$. Because the range of arccos is restricted to between 0 and $\pi$, our result is:

$$\theta = \arccos\frac{-1}{2} = \frac{2\pi}{3}$$

.

Therefore, the angle between $\mathbf{u}$ and $\mathbf{v}$ is $2\pi/3$ (or $120°$).

### *Exercise 159* Using dot products

What is the angle between these each pair of vectors:

*Working Space*

- $[1, 0]$, $[0, 1]$
- $[3, 4]$, $[4, 3]$
- $[2, -1, 2]$, $[-1, 2, -2]$
- $[-5, 0, -1]$, $[2, 3, -4]$

*Answer on Page 1934*

## 64.3 Dot products in Python

NumPy will let you do dot products using the symbol @. Open `first_vectors.py` and add the following to the end of the script:

```
# Take the dot product
    d = v @ u
    print("v @ u =", d)

    # Get the angle between the vectors
    a = np.arccos(d / (mv * mu))
    print(f"The angle between u and v is {a * 180 / np.pi:.2f} degrees")
```

When you run it you should get:

```
v @ u = 4
The angle between u and v is 78.55 degrees
```

## 64.4 Work and Power

Earlier, we mentioned that mechanical work is the product of the force you apply to something and the amount it moves (formulaic representation: $W = \vec{F} \cdot \vec{d}$). For example,

if you push a train with a force of 10 newtons as it moves 5 meters, you have done 50 joules of work.

What if you try to push the train sideways? It moves down the track 5 meters, but you push it as if you were trying to derail — perpendicular to its the tracks and its motion. You have done no work, because the train didn't move at all in the direction you were pushing.

Now that you know about dot products: The *work* you do is the dot product of the force vector you apply and the displacement vector of the train. (The displacement vector is the vector that tells how the train moved while you pushed it.)

$$W = \vec{F} \cdot \vec{d} = |\vec{F}||\vec{d}| \cos\theta$$

$$P = \vec{F} \cdot \vec{v} = |\vec{F}||\vec{v}| \cos\theta$$

Similarly, we mentioned that *power* is the product of the force you apply and the velocity of the mass you are applying it to. It is actually the dot product of the force vector and the velocity vector.

For example, if you are pushing a sled with a force of 10 newtons and it is moving 2 meters per second, but your push is 20 degrees off, you aren't transferring 20 watts of power to the sled. You are transferring $10 \times 2 \times \cos(20 \text{ degrees}) \approx 18.8$ watts of power.

## *Exercise 160* Power at an angle

A person pushes a skateboard with a force of 40 N. The skateboard is moving forward at 3 m/s, but the person's push is angled 30° above the direction of motion. How much power is being transferred to the skateboard?

*Working Space*

*Answer on Page 1934*

Similarly, the same thing happens when pushing vertically, against forces such as gravity. If a cart moves a distance L up a ramp at angle $\theta$, its displacement is $\vec{d} = [L\cos\theta, L\sin\theta]$. Gravity is $\vec{F}_g = [0, -mg]$.

Therefore,

$$\vec{F}_g \cdot \vec{d} = [0, -mg] \cdot [L\cos\theta, L\sin\theta] = -mgL\sin\theta.$$

Since $L \sin \theta$ is the height of the ramp, gravity does $-mgh$ joules of work. This is where the get that a change in potential energy is a form of work being done.

## *Exercise 161* Work done on a Box

A force $\vec{F} = [12, 5]$ N pushes a box whose displacement is $\vec{d} = [3, 0]$ m. How much work is done?

*Working Space*

*Answer on Page 1935*

Chapter 65

# Manufacturing

If you try to think of any man-made object, whether it was made from woods, metals, or plastics, chances are it was produced through a manufacturing process.

Over time, these processes have been refined to be more efficient, cost-effective, and faster at producing the goods that we use on a daily basis.

New methods are also constantly being developed by engineers and scientists, and today the range of options available means that choosing the most appropriate manufacturing method involves finding the sweet spot between cost-effectiveness, yield, and time needed.

## 65.1 Woods and Metals Processes

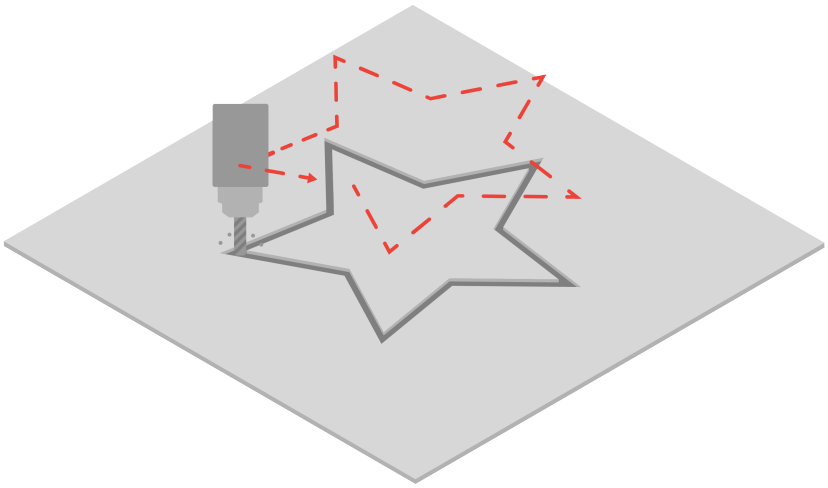
Manufacturing methods that are able to process woods and metals are typically the processes that are used to construct the vast majority of the built world around us.

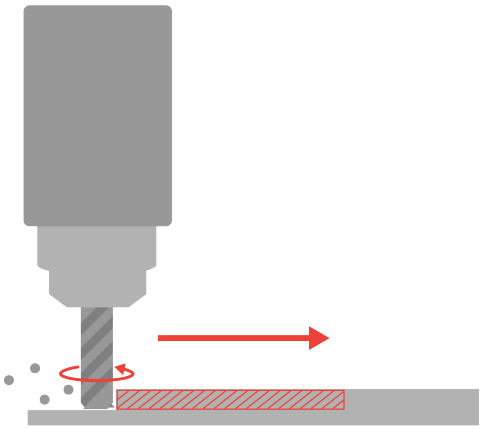
Infrastructure, transportation methods, and buildings would not exist without the advent of processes that allow us to accurately machine raw woods and metals into our desired forms.

The following sections will cover methods used to process both material types.

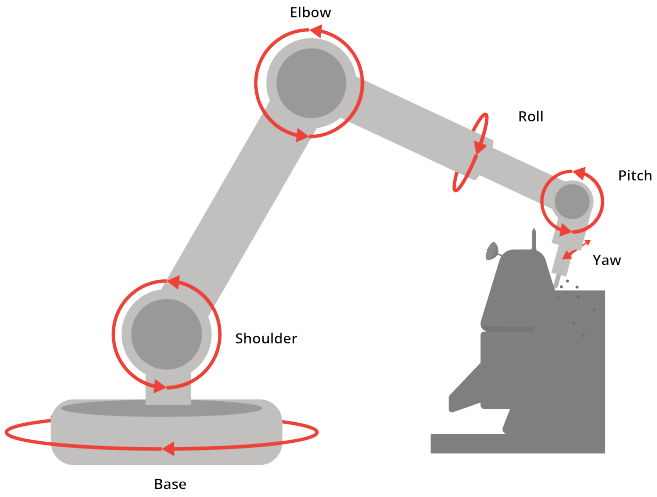
### 65.1.1 Milling

Mills are powerful tools that allow us to carve out complex shapes from blocks of raw material. A tool bit follows a path to remove the desired material, which makes it a *subtractive* manufacturing process. The tool bit rotates at a very high speed, which allows it to process harder materials such as woods and metals.





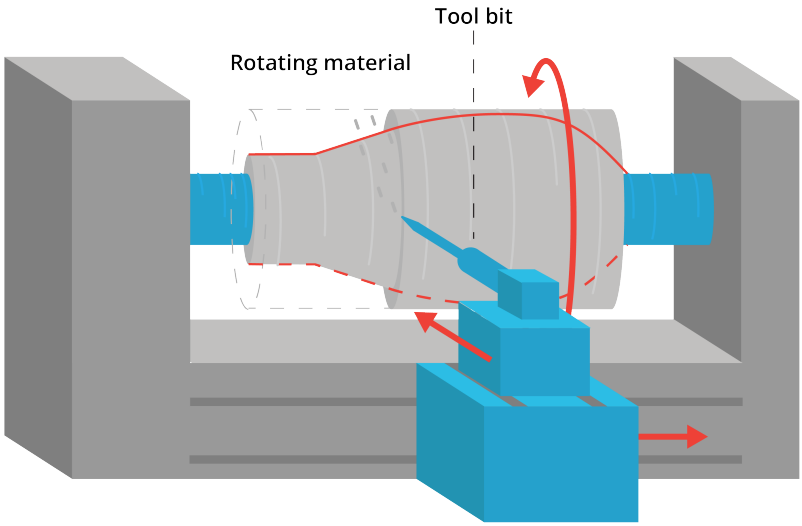
There are various types of mills, ranging from a 3-axis mill, which can cut out simple shapes in the X, Y, and Z axes, all the way to 6-axis mills, which can also rotate about those axes to create more complex curvatures.




In manufacturing settings where speed and repeatability is paramount, mills are often computer controlled. This functionality is referred to as *Computer Numerical Control*, or CNC. CNC mills are able to repeatedly follow a tool path, resulting in consistent and accurate parts.

### 65.1.2 Lathing

Lathes are tools that allow us to carve out complex cylindrical shapes from raw material. Like a mill, it also is a subtractive manufacturing process. However, lathes rotate the material itself at a high speed, rather than the tool bit. As the material rotates, the tool bit can be used to extract material layer by layer.




In the manufacturing industry, lathes are also often computer controlled. Alongside CNC mills, these CNC lathes are responsible for a majority of the objects that we interact with everyday.

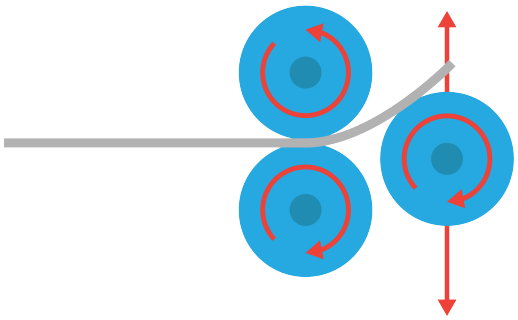
## 65.2 Metal-specific Processes

While mills and lathes are already able to cover the vast majority of manufacturing needs for woods and metals, there are certain processes that are specifically enabled by the unique properties of metals. More often than not, these processes leverage the malleability (ability to bend without breaking) of metals at room temperature or higher.

### 65.2.1 Sheet Metal Processes

While milling allows us to process blocks of metal to great effect, sometimes the application we need does not require material of such thickness.

This is where sheet metal comes in, as well as the methods we use to process it. One of the most common techniques in manufacturing is rolling, where a raw sheet is gradually rolled into the desired shape. This method is used to create many objects you may recognize, such as metal roofings, aircraft frames, and more.

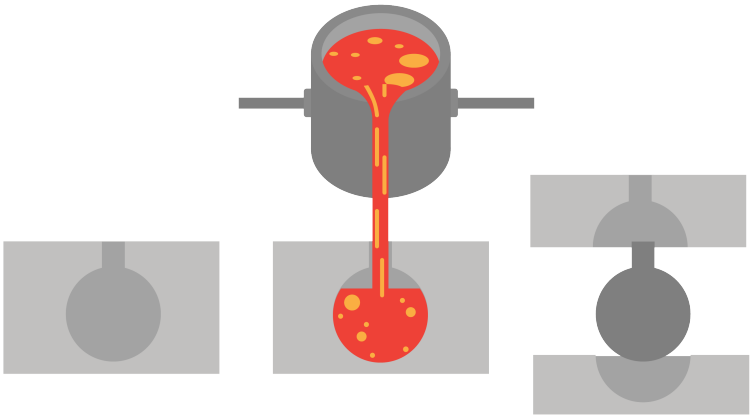


Another method is stamping, where a raw sheet is stamped into the desired shape. This allows us to create surfaces with complex geometries in an instant, and in large quantities. This method is often used for applications like the exterior panels of a car, where parts with compound curves are needed.

Lastly, there are also separate processes used to increase the strength of sheet metal parts. This falls under the category of sheet metal forming, and involves bending the edges of a part to form a reinforced flap. Almost all sheet metal parts are reinforced in this manner, as it is a relatively simple process and also helps to create a clean edge for a more finished look.

### 65.2.2 Casting

The last kind of metal-specific process we will cover is casting. Casting involves pouring molten metal into a mold, then letting the metal cool and set inside the mold. Smaller components with complex geometries and limited structural requirements (such as toys) are often cast, as it is an accurate and high-volume manufacturing method.



Casting also results in minimal material wastage, as it is not a subtractive manufacturing method where excess material is machined away, but rather only the specific amount of material required is poured in each time.

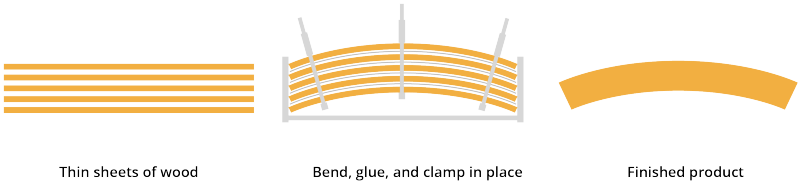
## 65.3 Wood-specific Processes

Similar to metals, there are also certain manufacturing processes that are enabled by the unique qualities of wood. These make use of the water content inherent in wood, and the flexibility it enables.

### 65.3.1 Bending

Turning raw wood into flat, workable pieces involves a variety of tools that you probably know of already, such as saws and drills. However, there are specific processes that help us create curved shapes with wood, in addition to the mill and the lathe mentioned earlier.

This is where bent lamination comes in. Bent lamination involves layering multiple thin veneers or strips of wood with adhesive, and clamping it to create the desired form while letting the glue dry. This method is often used for furniture production, enabling continuous curves in wood with tight radii.




Steam can also be used for bending, by helping soften the wood fibers to increase flexibility. Once the desired form is reached, the part can then cool down and harden. Unlike bent lamination, steam bending can be done without adhesives.

## 65.4 Plastic-specific Processes

Although plastics only came into prominence in the mid 20th century, they have changed manufacturing and, by extension, the world as we know it. Easily manufacturable, durable, and cost-effective, they have come to permeate almost everything we use on a daily basis.

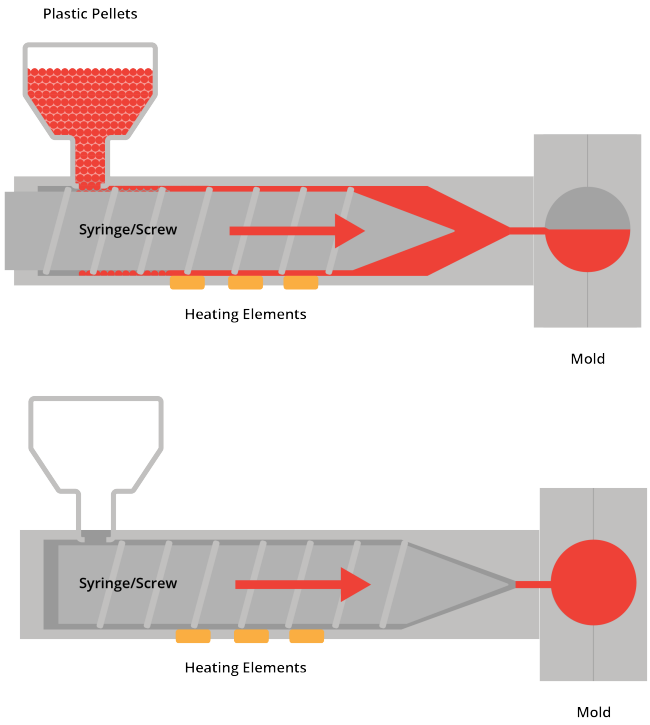
It must also be noted that these exact qualities have also resulted in the proliferation of plastics in our environment, and as such, usage of plastics should be well considered and limited. Alternative biodegradable materials are currently being trialed by scientists and would look to replicate many of the same qualities we expect from plastics, including its manufacturability.

### 65.4.1 Injection Molding

Injection molding is responsible for the vast majority of plastic products that you interact with on a daily basis. It is extremely quick, highly accurate, and has minimal material wastage, making it a popular and cost effective method of manufacturing plastic goods.

Similar to casting, injection molding involves injecting molten plastic into a mold, then allowing the part to cool and set inside the cavity.

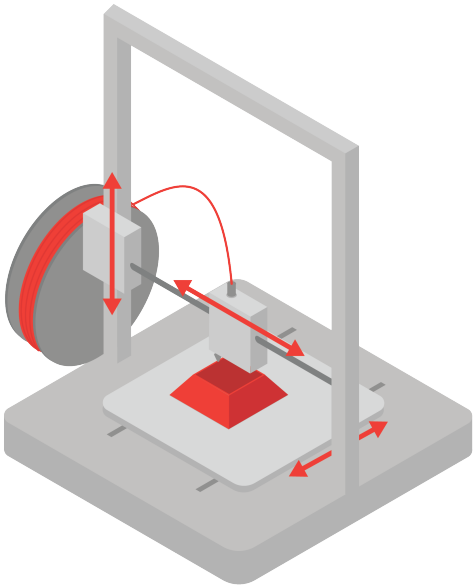
You can often tell when a part was produced through injection molding, with telltale signs such as the parting line. This is where the parts of the mold meet, forming a visible line on the surface of a part.




## 65.5 3D Printing

Injection molding has traditionally been the go-to technique for manufacturing plastic goods. However, new technologies result in the advent of new manufacturing methods. 3D printing is one such method, having come to prominence in the last few decades as a way to quickly prototype parts without having to create the molds needed for injection molding.

3D printing is an *additive* manufacturing process, where molten material is applied layer by layer to form the desired geometry. It allows for complex geometries, standardized batch production, and whilst the accuracy may currently lag behind traditional injection molding, it is also improving rapidly.

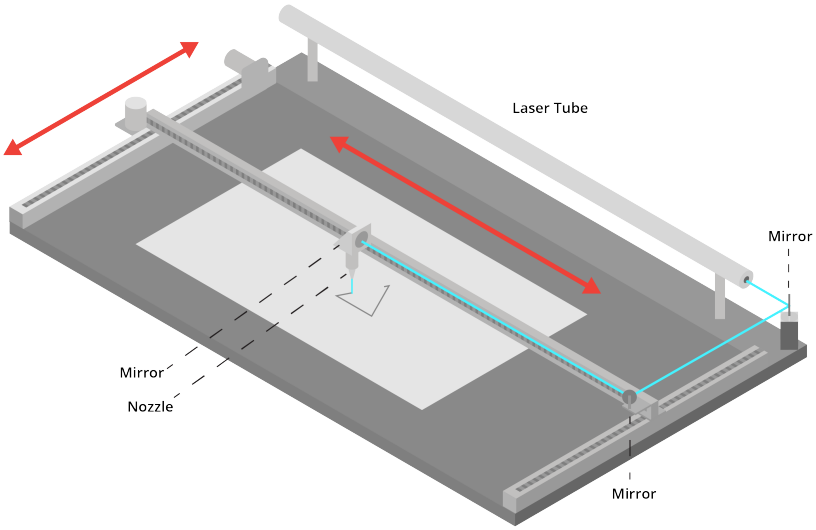


## 65.6 Laser Cutting and Water Jet

Similarly, another manufacturing method enabled by new technologies is laser cutting. Like 3D printing, it has come into prominence as a method to quickly prototype parts. However, it is not an additive manufacturing process.

Instead laser cutting uses a laser beam to cut and etch through sheets of plastic, however it can also be used for other materials such as fabrics and card stocks. Laser cutting is mostly limited by material thickness, and as such can only cut through thinner sheets of material.

Similarly, water jetting uses pressurized water to blast a stream of water through material (metal, wood, stone, or rubber usually). As the nozzle moves, the high pressure water traces a path throughout the material. Water jets are also limited to material thickness, as too thick of material may not easily be cut.


Laser Tube
Mirror
Mirror
Nozzle
Mirror

CHAPTER 66

# Introduction to Waves

When we talk about waves, we refer to so many phenomena in nature. Waves can be in the form of liquids, like when you throw a pebble into a pond and create a recurring effect of water. Waves can be ocean waves, controlled by the moon and the tides. Waves can be sound waves, bouncing back and forth travelling from a speaker. Waves can even my magnetic or electric, which we call electromagnetic waves, creating charge and magnetic fields. Light waves creates the large range of frequency of light we see with the human eye. All of these waves, however, have something in common: *they all share the same wave properties.*

We define waves as follows:

> A **wave** is a disturbance that transfers energy from one place to another without permanently transferring matter.

Another way to put it:

> A **wave** is a vibration of a system out of a restful state.

For example, when a water wave moves across a pond, the water itself does not travel across the pond. Instead, the water particles oscillate around their equilibrium positions while the wave carries energy outward.

In the case of a spring, which we will analyze deeper, energy (by the form of stretching or pushing) is transferred throughout the medium that is the spring.

## 66.1 Basic Properties of Waves

All waves can be described using several important properties. You will definitely see an overlap between trigonometric functions and wave properties. Often, we can describe waves that are generated with these trigonometric functions we have previously talked about.

There are essentially 5 terms we use to describe the properties of waves: **equilibrium**, **amplitude**, **wavelength**, **period**, and **frequency**.

### Equilibrium

An **equilibrium** state of a wave is what it would look no oscillation to be present. In other words, it is a state of rest for the wave. If you have a body of water that a pebble is thrown into, the equilibrium state is height or state of the water the instant before the pebble is thrown in. Same for a rope, a equilibrium state of a rope represents the rope before any motion is put into it.

### Amplitude

When a wave is created, it moves creates motion away from its *equilibrium* position.

The **amplitude** of a wave is the maximum displacement of the medium from its equilibrium position. In simpler terms, it measures how "large" the wave is.

The distance between the highest positive amplitude and the equilibrium position (called the *peak* or *crest*) and the lowest positive amplitude and the equilibrium position (called the *valley* or *trough*) are the same distance.

For a water wave, the amplitude is the height from the resting water level to the crest of the wave. For a sound wave, greater amplitude corresponds to louder sound. In light waves, larger amplitude corresponds to greater brightness.

### Wavelength

The **wavelength**, usually represented by the symbol $\lambda$ (lambda), is the distance between two identical points on consecutive waves, such as crest-to-crest or trough-to-trough.

Wavelength is usually measured in meters (or centimeters or micrometers).

### Frequency

The **frequency** of a wave is the number of complete waves that pass a point each second. Frequency is measured in hertz (Hz), where

$$1 \text{ Hz} = 1 \text{ wave per second.}$$

A wave with a higher frequency oscillates more rapidly. For sound waves, higher frequency corresponds to higher pitch. For light waves, frequency determines the color of

the light.

### Period

The **period** of a wave is the time required for one complete oscillation. The period and frequency are related by

where:

- T is the period (seconds),
- f is the frequency (hertz).

### Wave Speed

Waves travel at a certain speed depending on the medium through which they move. Wave speed is related to frequency and wavelength by

$$v = f\lambda$$

where:

- $v$ is wave speed,
- f is frequency,
- $\lambda$ is wavelength.

ADD: exercise in labelling

## 66.2 Two main types of waves

In waves around us, we can classify them under two categories: **longitudinal** and **transverse**. This classification depends on how particles are effected as the wave "passes".

### Longitudinal

For *longitudinal* waves, the direction of wave (or the disturbance in particles) is *parallel* to the direction of the vibration. The most common example of this a *sound wave*. A speaker playing a sound pushes one sound particle, pushing another, and then another, ultimately

continuing until the particles reach your ears. There is an amazing visual of this using a slinky in your digital resources!

### Transverse

In contrast, *transverse* waves move in a direction (or the disturbance in particles) is *perpendicular* to the wave's direction. Light and Electromagnetic waves and light waves are examples of transverse waves. The energy moves left to right (or horizontally) but the ends of the wave slide up and down vertically.

Again, see your digital resources for a visual of this using a slinky.

## 66.3 Compression and Rarefaction

Sound waves travel by creating disturbances in a medium such as air. When an object vibrates, it pushes nearby air particles together in some regions and spreads them farther apart in others. These alternating regions move outward through the medium as the sound wave travels.

Regions where particles are crowded closer together are called *compressions*. In these regions, the pressure and density of the air are higher. Regions where particles are spread farther apart are called *rarefactions*[1]. In these regions, the pressure and density are lower.

As the wave passes, the particles themselves do not travel long distances with the wave. Instead, they oscillate back and forth around their equilibrium positions while the disturbance moves through the medium.

## 66.4 Creating a Wave in Python

ADD ME

[1] Rarefaction regions and phases are commonly called decompression regions.

# Chapter 67

# Sound

When you set off a firecracker, it makes a sound.

Let's break that down a little more. Inside the cardboard wrapper of the firecracker, there is potassium nitrate ($KNO_3$), sulfur ($S$), and carbon($C$). These are all solids. When you trigger the chemical reactions with a little heat, these atoms rearrange themselves to be potassium carbonate ($K_2CO_3$), potassium sulfate ($K_2SO_4$), carbon dioxide ($CO_2$), and nitrogen ($N_2$). Note that the last two are gasses.

The molecules of a solid are much more tightly packed than the molecules of a gas. So after the chemical reaction, the molecules expand to fill a much bigger volume. The air molecules nearby get pushed away from the firecracker. They compress the molecules beyond them, and those compress the molecules beyond them.

This compression wave radiates out as a sphere; its radius growing at about 343 meters per second ("The speed of sound").

The energy of the explosion is distributed around the surface of this sphere. As the radius increases, the energy is spread more and more thinly around. This is why the firecracker seems louder when you are closer to it. (If you set off a firecracker in a sewer pipe, the sound will travel much, much farther.)

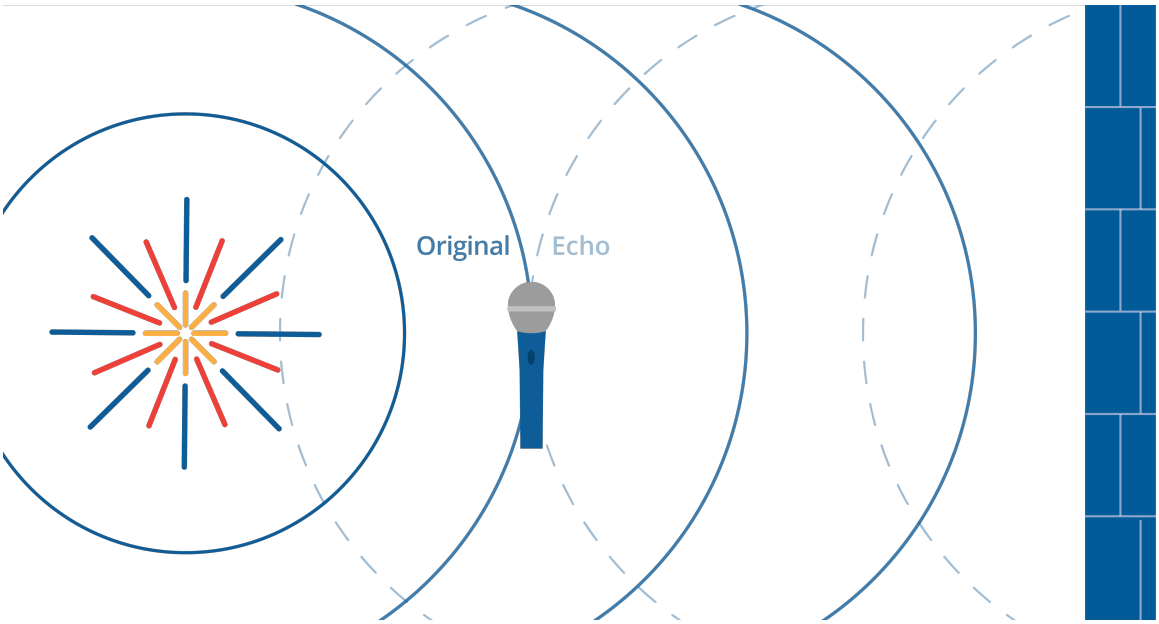



Figure 67.1: A firecracker exploding causes an initial sound wave and an echo.

This compression wave will bounce off of hard surfaces. If you set off a firecracker 50 meters from a big wall, you will hear the explosion twice. We call the second one an "echo".

The compression wave will be absorbed by soft surfaces. If you covered that wall with pillows, there would be almost no echo.

The study of how these compression waves move and bounce is called *acoustics*. Before you build a concert hall, you hire an acoustician to look at your plans and tell you how to make it sound better.

## 67.1 Pitch and frequency

The string on a guitar is very similar to the weighted spring example. The farther the string is displaced, the more force it feels pushing it back to equilibrium (remember the tension force?). Thus, it moves back and forth in a sine wave. (OK, it isn't a pure sine wave, but we will get to that later.)

The string is connected to the center of the boxy part of the guitar, which is pushed and pulled by the string. That creates compression waves in the air around it.

If you are in the room with the guitar, those compression waves enter your ear and push and pull your eardrum, which is attached to bones that move a fluid that tickles tiny hairs, called *cilia*, in your inner ear. This is how you hear.

We sometimes see plots of sound waveforms. The $x$-axis represents time. The $y$-axis represents the amount the air is compressed at the microphone that converted the air pressure into an electrical signal.

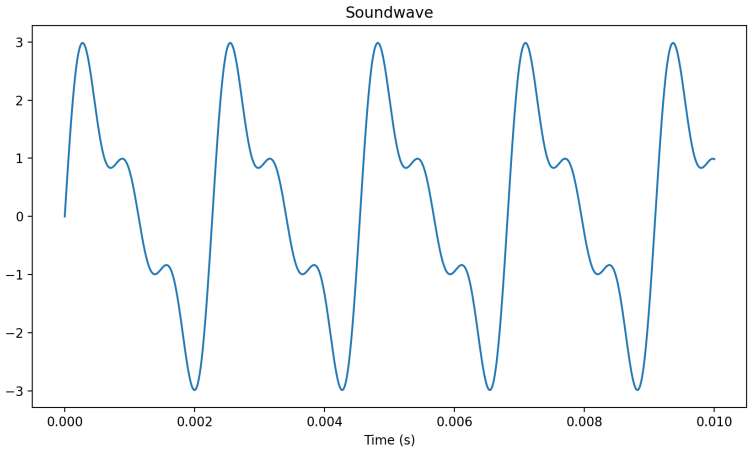



Figure 67.2: A soundwave graph over time.

If the guitar string is made tighter (by the tuning pegs) or shorter (by the guitarist's fingers on the strings), the string vibrates more times per second. We measure the number of waves per second and we call it the *frequency* of the tone. The unit for frequency is *Hertz*: cycles per second. The *period* is opposite of frequency: it is the time it takes for one cycle to complete. The unit for period is seconds per cycle.

Musicians have given the different frequencies names. If the guitarist plucks the lowest note on his guitar, it will vibrate at 82.4 Hertz. The guitarist will say "That pitch is low E." If the string is made half as long (by a finger on the 12th fret), the frequency will be twice as fast (164.8 Hertz), and the guitarist will say "That is E an octave up."

For any note, the note that has twice the frequency is one octave up. The note that has half the frequency is one octave down.

The octave is a very big jump in pitch, so musicians break it up into 12 smaller steps. If the guitarist shortens the E string by one fret, the frequency will be $82.4 \times 1.059463 \approx 87.3$ Hertz.

Shortening the string one fret always increases the frequency by a factor of 1.059463. Why?

Because $1.059463^{12} = 2$. That is, if you take 12 of these hops, you end up an octave higher. This, the smallest hop in western music, is referred to as a *half step*.

### *Exercise 162* Notes and frequencies

*Working Space*

The note A near the middle of the piano, is 440Hz. The note E is 7 half steps above A. What is its frequency?

*Answer on Page 1935*

## 67.2 Chords and harmonics

Of course, a guitarist seldom plays only one string at a time. Instead, they use the frets to pick a pitch for each string and strums all six strings.

Some combinations of frequencies sound better than others. We have already talked about the octave: If one string vibrates twice for each vibration of another, they sound sweet

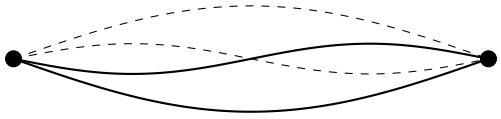
together.

Musicians speak of "the fifth". If one string vibrates three times and the other vibrates twice in the same amount of time, they sound sweet together.

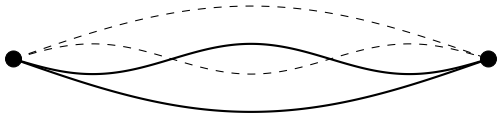
Likewise, if one string vibrates 4 times while the other vibrates 3 times, they sound sweet together. Musicians call this "the third."

Each of these different frequencies tickle different cilia in the inner ear, so you can hear all six notes at the same time when the guitarist strums their guitar.

When a string vibrates, it doesn't create a single sine wave. Yes, the string vibrates from end-to-end, and this generates a sine wave at what we call *the fundamental frequency*. However, there are also "standing waves" on the string. One of these standing waves is still at the centerpoint of the string, but everything to the left of the centerpoint is going up, while everything to the right is going down. This creates *an overtone* that is twice the frequency of the fundamental.



The next overtone has two still points — it divides the string into three parts. The outer parts are up, while the inner part is down. Its frequency is three times the fundamental frequency.



And so on. 4 times the fundamental, 5 times the fundamental, etc.

In general, tones with many overtones tend to sound bright. Tones with just the fundamental sound thin.

Humans can generally hear frequencies from 20Hz to 20,000Hz (or 20kHz). Young people tend to be able to hear very high sounds better than older people.

Dogs can generally hear sounds in the 65Hz to 45kHz range.

## 67.3 Making waves in Python

Let's make a sine wave and add some overtones to it. Create a file named `harmonics.py`.

```
import matplotlib.pyplot as plt
import math

# Constants: frequency and amplitude
fundamental_freq = 440.0 # A = 440 Hz
fundamental_amp = 2.0

# Up an octave
first_freq = fundamental_freq * 2.0 # Hz
first_amp = fundamental_amp * 0.5

# Up a fifth more
second_freq = fundamental_freq * 3.0 # Hz
second_amp = fundamental_amp * 0.4

# How much time to show
max_time = 0.0092 # seconds

# Calculate the values 10,000 times per second
time_step = 0.00001 # seconds

# Initialize
time = 0.0
times = []
totals = []
fundamentals = []
firsts = []
seconds = []

while time <= max_time:
    # Store the time
    times.append(time)

    # Compute value each harmonic
    fundamental = fundamental_amp * math.sin(2.0 * math.pi * fundamental_freq *
    ↪  time)
    first = first_amp * math.sin(2.0 * math.pi * first_freq * time)
    second = second_amp * math.sin(2.0 * math.pi * second_freq * time)

    # Sum them up
    total = fundamental + first + second

    # Store the values
    fundamentals.append(fundamental)
    firsts.append(first)
    seconds.append(second)
    totals.append(total)
```

```
47
48      # Increment time
49      time += time_step
50
51  # Plot the data
52  fig, ax = plt.subplots(2, 1)
53
54  # Show each component
55  ax[0].plot(times, fundamentals)
56  ax[0].plot(times, firsts)
57  ax[0].plot(times, seconds)
58  ax[0].legend()
59
60  # Show the totals
61  ax[1].plot(times, totals)
62  ax[1].set_xlabel("Time (s)")
63
64  plt.show()
```

When you run it, you should see a plot of all three sine waves and another plot of their sum:

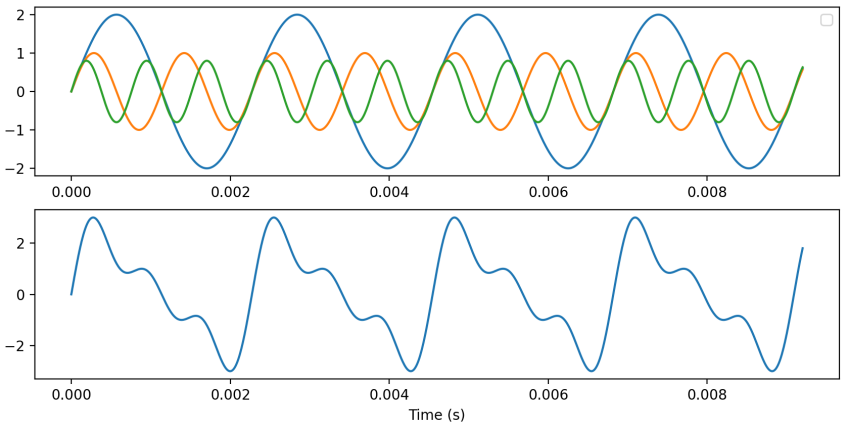



Figure 67.3: The output of harmonics.py

### 67.3.1 Making a sound file

The graph is pretty to look at, but make let's a file that we can listen to.

The WAV audio file format is supported on pretty much any device, and a library for writing WAV files comes with Python. Let's write some sine waves and some noise into a WAV file.

Create a file called `soundmaker.py`

```
import wave
import math
import random

# Constants
frame_rate = 16000 # samples per second
duration_per = 0.3 # seconds per sound
frequencies = [220, 440, 880, 392] # Hz
amplitudes = [20, 125]
baseline = 127 # Values will be between 0 and 255, so 127 is the baseline
samples_per = int(frame_rate * duration_per) # number of samples per sound

# Open a file
wave_writer = wave.open('sound.wav', 'wb')

# Not stereo, just one channel
wave_writer.setnchannels(1)

# 1 byte audio means everything is in the range 0 to 255
wave_writer.setsampwidth(1)

# Set the frame rate
wave_writer.setframerate(frame_rate)

# Loop over the amplitudes and frequencies
for amplitude in amplitudes:
    for frequency in frequencies:
        time = 0.0
        # Write a sine wave
        for sample in range(samples_per):
            s = baseline + int(amplitude * math.sin(2.0 * math.pi * frequency *
            ↪  time))
            wave_writer.writeframes(bytes([s]))
            time += 1.0 / frame_rate

        # Write some noise after each sine wave
        for sample in range(samples_per):
            s = baseline + random.randint(0, 15)
            wave_writer.writeframes(bytes([s]))

# Close the file
wave_writer.close()
```

When you run it, it should create a sound file with several tones of different frequencies and volumes. Each tone should be followed by some noise.

CHAPTER 68

# Applying Numerical Methods to Bivariate Data

## 68.1 Review of Scatter Plots

## 68.2 Correlation Coefficient

In the previous section, we used scatterplots to visually judge whether two variables are related. But how do we move beyond "this looks like a strong relationship" to something we can actually calculate and compare? That is exactly what the *correlation coefficient* gives us.

A correlation coefficient is a numerical measure used to judge the relation between two variables. The one we will use throughout this course is *Pearson's correlation coefficient*, also known simply as the correlation coefficient. It measures the strength and direction of the linear relation between two quantitative variables.

You will encounter two versions of this value depending on whether you are working with a full population or a sample drawn from one. When computed from a population, the correlation coefficient is written as $\rho$ (read as "rho"). When computed from a sample, it is written as $r$. In this course, we will almost always be working with sample data, so $r$ is the symbol you will use most.

No matter how strong or weak a relationship is, $r$ always falls within a fixed range:

$$-1 \leq r \leq +1$$

Think about what those endpoints mean. A value of $r = +1$ or $r = -1$ means the points in a scatterplot fall perfectly on a straight line. A value of $r = 0$ means there is no linear relationship between the two variables at all. Every other value of $r$ falls somewhere in between, and in the next subsection we will look at exactly how to interpret where $r$ lands.

## *Exercise 163* Boundaries of $r$

Answer the following questions about the range of the correlation coefficient.

1. A student computes $r = 1.3$ for a dataset. What does this tell you, without looking at the data?
2. Two researchers study the same population but one uses the full population and the other uses a random sample. What symbol does each researcher use for their correlation coefficient?
3. Can $r$ equal exactly $0.5$? Can it equal exactly $-1$? Explain what each of those values would mean about the data.

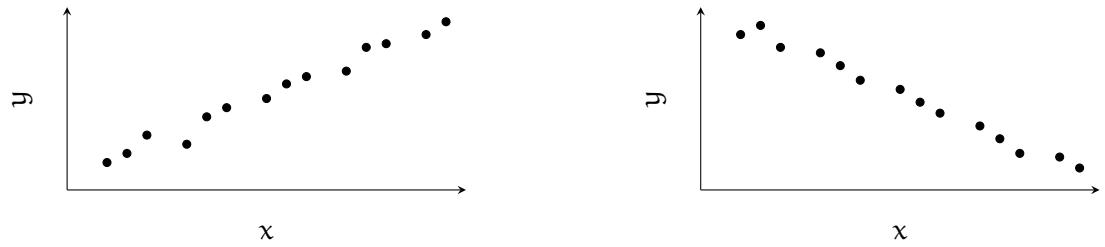
*Working Space*

*Answer on Page 1935*

**Direction**

Now that we know $r$ always falls between $-1$ and $+1$, the first thing to read from it is direction. The sign of $r$ tells you immediately whether the two variables move together or against each other.

A positive value of $r$ indicates that as $x$ increases, $y$ also tends to increase. Taller fathers tend to have taller sons, so the scatterplot shows an upward trend and $r$ is positive. A negative value of $r$ indicates that as $x$ increases, $y$ tends to decrease. As a phone gets older, its battery life tends to decrease, so the scatterplot shows a downward trend and $r$ is negative.




(a) Positive relation ($r > 0$) (b) Negative relation ($r < 0$)

Figure 68.1: Direction of a linear relation as shown in a scatterplot.

Notice that direction says nothing about strength. Both $r = +0.1$ and $r = +0.95$ are positive, but they describe very different relationships. That is what we look at next.

## *Exercise 164* Reading Direction from $r$

*Working Space*

For each pair of variables, predict whether $r$ would be positive, negative, or zero.

| $x$ | $y$ | Sign of $r$ |
|---|---|---|
| Hours studied | Exam score | |
| Temperature | Hot drinks sold | |
| Shoe size | Number of pets owned | |
| Age of a car | Resale value | |

*Answer on Page 1936*

### Strength

Direction tells us which way a relationship goes, but not how strong it is. For that, we look at the numeric value of $r$, ignoring its sign. The further $r$ is from zero, the stronger the linear relationship between the two variables. The closer $r$ is to zero, the weaker it is.

At the extremes, $r = +1$ or $r = -1$ indicates a *perfect correlation*, meaning every point in the scatterplot falls exactly on a straight line. In practice, real data almost never achieves this. On the other end, $r = 0$ means there is no linear relationship at all.

To compare the strength of two correlations, just compare the absolute values. If $r = -0.86$ for one pair of variables and $r = 0.75$ for another, the first pair has the stronger linear relationship because $0.86 > 0.75$.

Since interpreting strength can be subjective, statisticians often use cutoff values as a rough guide. The diagram below shows one commonly used set of cutoffs. Keep in mind that these are conventions, not rules. The key principle is always the same: the further from zero, the stronger the correlation.

| Very Strong | Strong | Weak | No Correlation | Weak | Strong | Very Strong |
|---|---|---|---|---|---|---|
| $-1.00$ to $-0.85$ | $-0.85$ to $-0.50$ | $-0.50$ to $-0.10$ | $-0.10$ to $+0.10$ | $+0.10$ to $+0.50$ | $+0.50$ to $+0.85$ | $+0.85$ to $+1.00$ |

Figure 68.2: One commonly used set of cutoff values for interpreting the strength of a correlation coefficient.

## *Exercise 165* Interpreting Strength

*Working Space*

The table below shows correlation coefficients computed from four different datasets. For each one, describe the direction and approximate strength of the relationship.

| r | Direction | Strength |
|---|---|---|
| $+0.92$ | | |
| $-0.34$ | | |
| $+0.07$ | | |
| $-0.88$ | | |

*Answer on Page 1936*

**Example**

Let us put direction and strength together using a real dataset. Consider a sample of 10 father-son pairs. The heights (in inches) are recorded below.

| Height of Father $x$ (inches) | Height of Son $y$ (inches) |
|---|---|
| 66 | 66 |
| 66 | 65 |
| 67 | 67 |
| 67 | 70 |
| 68 | 67 |
| 68 | 70 |
| 69 | 70 |
| 70 | 72 |
| 71 | 72 |
| 73 | 74 |

Before computing anything, we check whether a correlation coefficient is appropriate. Correlation only applies to linear relationships, so we first make a scatterplot to confirm the relation between the two variables is linear.

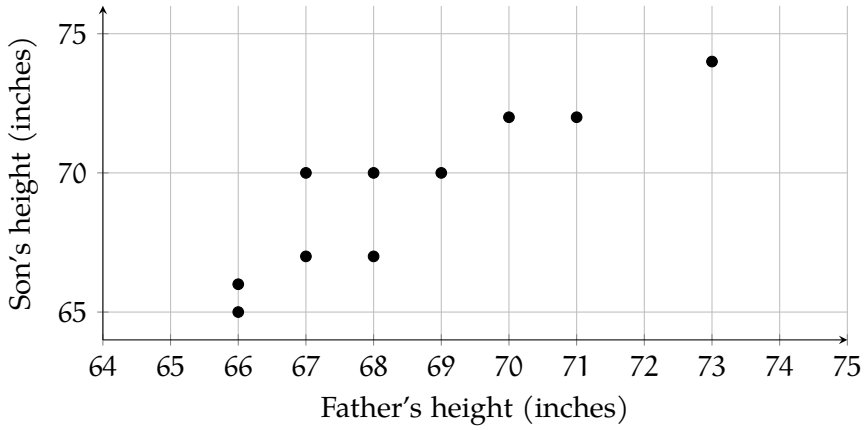



Figure 68.3: Scatterplot of father and son heights for a sample of 10 pairs.

The scatterplot shows a linear trend, so it is appropriate to compute $r$. In practice, correlation coefficients are computed using a calculator or statistical software rather than by hand. For this dataset, the computed value is approximately $r = 0.90$.

**Computing $r$ in Excel**

Open your data in Excel or Google Sheets with the $x$ values in column A and the $y$ values in column B. To compute $r$ directly, use the built-in correlation function:

```
=CORREL(A2:A11, B2:B11)
```

This returns the correlation coefficient between the two columns instantly. For the father-son data, this should return approximately $0.90$.

**Computing $r$ in Python**

```
import numpy as np

# Father and son heights in inches
x = [66, 66, 67, 67, 68, 68, 69, 70, 71, 73]
y = [66, 65, 67, 70, 67, 70, 70, 72, 72, 74]

r = np.corrcoef(x, y)[0, 1]
print(r)
```

`np.corrcoef()` returns a $2 \times 2$ matrix of correlation coefficients. The value at position `[0, 1]` is the correlation between `x` and `y`. For this dataset, the output should be approximately `0.90`.

Try replacing the data with your own values and observe how $r$ changes as the relationship between the two variables becomes stronger or weaker.

What does this tell us? The sign is positive, so taller fathers tend to have taller sons. The value is 0.90, which falls in the very strong range of our cutoff table. The points cluster tightly around a straight line, and the scatterplot confirms this.

Notice also what $r$ does not tell us. It describes the strength and direction of the linear relationship, but not the shape of any curve, and not whether one variable causes the other. A strong correlation between two variables does not mean one causes the other to change.

### *Exercise 166* Interpreting a Computed $r$

*Working Space*

*Answer on Page 1936*

## 68.3 Linear Regression

Suppose you wanted to predict a teacher's starting salary based on how many years of experience they have. You might expect that more experience leads to a higher salary, but how do you turn that intuition into a precise estimate? That is exactly what a *linear regression model* does. It gives you a straight-line equation that describes the relationship between two variables, so you can use the value of one to predict the value of the other.

**The Regression Model**

When one variable depends on or is explained by another, we call the variable being predicted the *dependent variable*. The variable doing the explaining is the *independent variable*. In our example, starting salary is the response variable and years of experience is the independent variable. It would make little sense to say that a teacher's salary determines how many years they have worked.

The linear relation between two variables is described by the *population regression equation*:

$$Y = \alpha + \beta X$$

where $Y$ is the response variable, $X$ is the independent variable, $\alpha$ is the $y$-*intercept* (the value of $Y$ when $X = 0$), and $\beta$ is the *slope* (the amount $Y$ changes for every one-unit increase in $X$). This equation describes the true relationship in the population, but in practice, we never know $\alpha$ and $\beta$ exactly. We estimate them from sample data.

Once we have estimates for $\alpha$ and $\beta$, we can write the *estimated regression line*:

$$\hat{y} = a + bx$$

The symbol $\hat{y}$ (read "$y$-hat") is the *predicted value* of $Y$ for a given value of $X$. Notice the difference in notation: $\alpha$ and $\beta$ are population parameters; $a$ and $b$ are their sample-based estimates.

Here is a quick reference for the symbols introduced so far in this chapter:

| Quantity | Population symbol | Sample symbol |
|---|---|---|
| Correlation | $\rho$ | $r$ |
| Intercept | $\alpha$ | $a$ |
| Slope | $\beta$ | $b$ |

Now, no data will ever fall perfectly on a straight line. A student who studies ten hours might score differently than the line predicts. The difference between what the model predicts and what is actually observed is called the *residual*, or *error*, and is denoted $e$:

$$e = y - \hat{y}$$

A positive residual means the observed value was higher than predicted; a negative residual means it was lower. From here on, salary will be expressed in thousands of dollars,

so a value of 38 means \$38,000; this is the same convention the data table will use later in this section. Going back to our teachers: if the model predicts a starting salary of 38 for a teacher with four years of experience, but that teacher actually earns 40.5, the residual is $40.5 - 38 = 2.5$, or \$2,500. This teacher earned \$2,500 more than the model expected.

## *Exercise 167* Computing a Residual

A regression model predicts that a teacher with seven years of experience will have a starting salary of 42.3 (in thousands of dollars). The teacher's actual starting salary is 40.1.

*Working Space*

1. Compute the residual.
2. Was the teacher's salary over-predicted or under-predicted? Explain.

*Answer on Page 1936*

The goal of linear regression is to find the line that makes these residuals as small as possible across all observations.

### The Least-Squares Regression Line

Every straight line you could draw through a scatterplot will produce a different set of residuals. Some lines will over-predict for certain point and under-predict for others. We want to know: which line does the best job overall? The answer is the *least-squares regression line*, also called the *line of best fit*. It is the line that minimizes the sum of the squared residuals, so, it makes $\sum e^2 = \sum (y - \hat{y})^2$ as small as possible.

Why should we square the residuals? First, squaring eliminates the sign, so positive and negative errors do not cancel each other out. Second, squaring penalizes large errors more heavily than small ones, which pushes the line toward a better overall fit. Now you know!

## *Exercise 168* Why Square the Residuals?

Suppose a regression line produces the following residuals for five data points: $3, -3, 2, -1, -1$.

1. Compute the sum of the residuals. What do you notice?
2. Compute the sum of the squared residuals.
3. Explain why simply minimizing the sum of the residuals would not be a useful goal.

*Working Space*

*Answer on Page 1936*

The least-squares regression line has two properties that always hold, regardless of the dataset.

First, it always passes through the point $(\bar{X}, \bar{Y})$ which represent the means (or average points) of the two variables. This point is sometimes called the *centroid* of the data.

Second, the slope $b$ of the line is always given by:

$$b = r\left(\frac{S_y}{S_x}\right)$$

where $r$ is the correlation coefficient, $S_y$ is the standard deviation of the $Y$-values, and $S_x$ is the standard deviation of the $X$-values. Notice that the slope depends directly on $r$: if there is no linear relationship ($r = 0$), the slope is zero, and the best prediction for any $X$ is simply $\bar{Y}$.

Once you have $b$, the $y$-intercept $a$ follows from the fact that the line passes through $(\bar{X}, \bar{Y})$:

$$a = \bar{Y} - b\bar{X}$$

These two formulas together give you the equation of the least-squares regression line:

$$\hat{y} = a + bx$$

## *Exercise 169* Computing the Line

For a dataset of teachers' years of experience (X) and starting salary in thousands of dollars (Y), the following summary statistics were computed: $\bar{X} = 6$, $\bar{Y} = 44$, $S_x = 3.2$, $S_y = 5.8$, $r = 0.91$.

1. Compute the slope $b$ of the least-squares regression line.
2. Compute the $y$-intercept $a$.
3. Write the equation of the least-squares regression line.
4. The line must pass through the centroid of the data. Verify this using your equation.

*Working Space*

*Answer on Page 1937*

### Interpreting Slope and Intercept

Having the equation of the least-squares regression line is only half the job. The other half is being able to explain what it means in plain language.

The slope $b$ tells you how much the predicted value of $Y$ changes for every one-unit increase in $X$. The key phrase you could see on an exam is **on average** because $b$ is an estimate based on sample data, not an exact rule.

For our teachers example, if $b = 1.65$, the correct interpretation is:

> For every additional year of experience, the predicted starting salary increases by \$1,650, on average.

Notice three things about that sentence. It names both variables. It gives the direction and amount of change. And it includes "on average." Leave out any one of those and the interpretation is incomplete.

## *Exercise 170* Interpreting the Slope

A least-squares regression line for predicting a teacher's starting salary (in thousands of dollars) from years of experience has slope $b = 1.49$.

1. Write a correct interpretation of the slope in context.
2. A student writes: "For every extra year of experience, salary goes up by \$1,490." What is wrong with this interpretation?

*Working Space*

*Answer on Page 1937*

The $y$-intercept $a$ is the predicted value of $Y$ when $X = 0$. Interpreting it requires care. Sometimes $X = 0$ is meaningful where a teacher with zero years of experience is simply a newly hired teacher. In that case the intercept has a natural interpretation: it is the predicted starting salary for someone entering the profession with no prior experience.

Other times $X = 0$ is outside the range of the data entirely, or physically impossible. In those situations, the intercept is mathematically necessary to define the line but has no useful real-world meaning. You should note that and move on.

## *Exercise 171* Interpreting the Intercept

Using the regression equation $\hat{y} = 34.10 + 1.649x$, where $x$ is years of experience and $\hat{y}$ is predicted starting salary in thousands of dollars:

1. Write a correct interpretation of the $y$-intercept in context.
2. Is this interpretation meaningful here? Explain.

*Working Space*

*Answer on Page 1937*

### The Coefficient of Determination

Once you have a regression line, a natural question is how well it actually predicts $Y$. The *coefficient of determination*, written $R^2$, answers this by measuring the proportion of the variation in $Y$ that is explained by its linear relationship with $X$.

For a single predictor, $R^2$ is simply the square of the correlation coefficient:

$$R^2 = r^2$$

Because $r$ always falls between $-1$ and $+1$, $R^2$ always falls between $0$ and $1$. An $R^2$ close to $1$ means the regression line accounts for nearly all of the variation in $Y$. An $R^2$ close to $0$ means the line explains almost none of it.

$R^2$ is usually reported as a percentage. For example, a correlation of $r = 0.80$ gives $R^2 = 0.64$, so you would say that 64% of the variation in $Y$ is explained by its linear relationship with $X$. The remaining $1 - R^2 = 36\%$ is left unexplained, due to other factors, natural variability, or measurement error.

**Worked Example**

A random sample of 10 teachers was selected from a large urban school district. Each teacher's years of experience at the time of hiring and their starting salary (in thousands of dollars) were recorded. The data is given in the table below.

| Experience (years) | Starting Salary ($1000s) |
|---|---|
| 3 | 36 |
| 8 | 43 |
| 1 | 32 |
| 0 | 30 |
| 5 | 38 |
| 10 | 47 |
| 2 | 33 |
| 7 | 42 |
| 4 | 37 |
| 6 | 41 |

Before doing any calculations, look at the data. As years of experience increase, starting salaries appear to rise steadily. There are no obvious outliers, and the values are spread across a reasonable range of experience levels. A scatterplot would confirm whether this relationship is linear and this is! There is a strong positive linear relationship between experience and starting salary.

From the data, we have $n = 10$ pairs of measurements. Using a calculator or software, we can obtain the following summary statistics:

$$\bar{X} = 4.6, \quad \bar{Y} = 37.9, \quad S_x = 3.169, \quad S_y = 5.013, \quad r = 0.9964$$

## *Exercise 172* Slope, Intercept, and the Regression Equation

Using the summary statistics above:

1. Compute the slope $b$ of the least-squares regression line. Interpret it in context.
2. Compute the $y$-intercept $a$. Interpret it in context.
3. Write the equation of the least-squares regression line.

*Working Space*

*Answer on Page 1937*

## *Exercise 173* Prediction, Residuals, and $R^2$

Using the regression equation $\hat{y} = 30.650 + 1.576x$:

1. Predict the starting salary for a teacher with 5 years of experience.
2. The teacher in the sample with 8 years of experience has an actual starting salary of \$43,000. Compute the residual and interpret it.
3. Compute the coefficient of determination $R^2$ and interpret it in context.
4. About what percentage of the variation in starting salaries is *not* explained by years of experience? Suggest one factor that might account for this.

*Working Space*

*Answer on Page 1938*

### Computing in Excel

You do not need to compute the slope, intercept, and $R^2$ by hand every time. Excel can produce all of these values at once, along with a scatterplot showing the line of best fit. Here is how.

First, enter your data into two columns, with experience values in column A and starting salary values in column B, with headers in row 1. Select both columns, then insert a scatterplot using **Insert** → **Chart** → **Scatter**. Once the chart appears, click on any data point, then select **Add Trendline**. In the trendline options panel on the right, choose **Linear** and check both **Display Equation on chart** and **Display R-squared value on chart**. Excel will draw the line of best fit and print the equation and $R^2$ directly on the chart.

To get the full regression output, use the **Data Analysis** Toolpak. Go to **Data** → **Data Analysis** → **Regression**. Set the Input Y Range to your salary column and the Input X Range to your experience column. Check **Labels** if your columns have headers. Click OK and Excel will produce a full regression table in a new sheet. The values you need are in the **Coefficients** column: the row labelled **Intercept** gives you $a$, and the row labelled **Experience** gives you $b$. The $R^2$ value appears near the top of the output under **R Square**.

### Computing in Python

Python can produce the same regression results as Excel, and a little more besides. The code below uses two libraries: `numpy` for the summary statistics and `scipy` for the regression, and `matplotlib` to draw the scatterplot with the line of best fit.

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

# Data
experience = [3, 8, 1, 0, 5, 10, 2, 7, 4, 6]
salary     = [36, 43, 32, 30, 38, 47, 33, 42, 37, 41]

# Least-squares regression
slope, intercept, r, p, se = stats.linregress(experience, salary)

print(f"Slope (b):     {slope:.3f}")
print(f"Intercept (a): {intercept:.3f}")
print(f"r:             {r:.4f}")
print(f"R-squared:     {r**2:.4f}")

# Scatterplot with regression line
x_line = np.linspace(0, 10, 100)
y_line = intercept + slope * x_line

plt.scatter(experience, salary, color="steelblue", label="Data")
plt.plot(x_line, y_line, color="firebrick",
```

```
        label=f"y = {intercept:.2f} + {slope:.2f}x")
plt.xlabel("Experience (years)")
plt.ylabel("Starting Salary ($1000s)")
plt.title("Experience vs. Starting Salary")
plt.legend()
plt.show()
```

Running this code produces the following output:

```
Slope (b):     1.673
Intercept (a): 30.203
r:             0.9957
R-squared:     0.9915
```

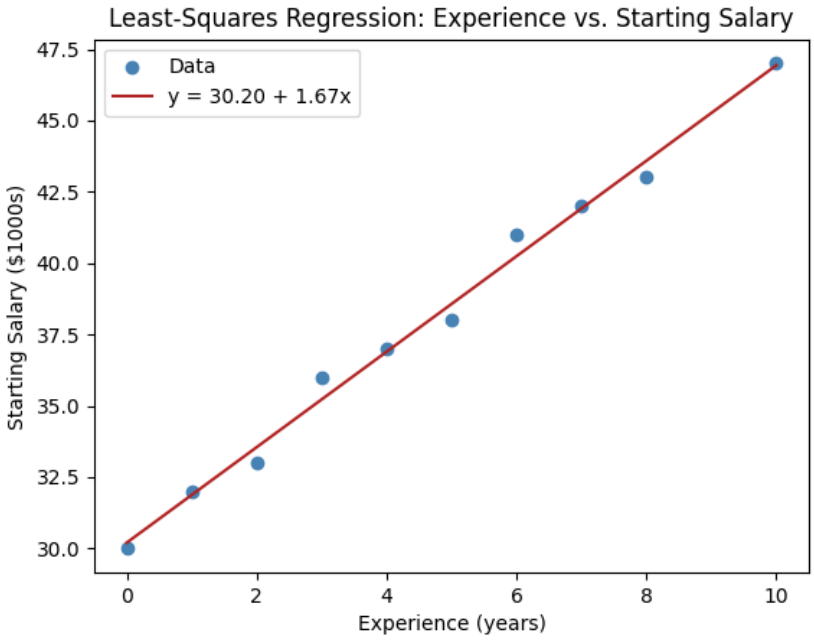



Figure 68.4: Experience vs. Starting Salary Graph generated from Python

These match the values we computed by hand exactly. The `stats.linregress` function does all the heavy lifting! It returns the slope, intercept, correlation coefficient r, a p-value, and the standard error of the slope in one line. For now, focus on the first four outputs. The p-value and standard error will become important when we study inference for regression later!

CHAPTER 69

# Data Analysis with jamovi

A student club surveyed 100 classmates about sleep, screen time, breakfast, and a short quiz. The club wants to know what the answers show. This chapter introduces Jamovi, which is a free software that does the data analysis calculations for you.

## 69.1 Getting Started with jamovi

*jamovi* is a free statistics program. You choose an analysis from a menu, tick a few boxes, and the results appear right away.

You can use jamovi in two ways:

- In a web browser, at `https://cloud.jamovi.org`, with a free Guest account. Nothing needs to be installed.
- On your own computer, by downloading it from `https://www.jamovi.org`. This version works without the internet.

The jamovi window has three parts:

- On the left, the *spreadsheet* holds the data. Each row is one person or thing, and each column is one variable.
- Across the top, the *analysis ribbon* lists the analyses you can run.
- The *results panel* on the right shows tables and graphs.

To try it, open an example dataset. Click the menu icon (three lines) in the top-left corner, then choose Open, Data Library, Examples, and Tooth Growth. On the Analyses tab, click Exploration, then Descriptives, and move `len` into the Variables box. A table of statistics appears without pressing any Run button.

### *Exercise 174* Which Part of the Window?

Name the part of the window you would use for each task: spreadsheet, analysis ribbon, or results panel.

*Working Space*

| Task | Part |
|---|---|
| Fix a number that was typed wrong | |
| Choose an analysis | |
| Read the mean | |

*Answer on Page 1938*

## 69.2 Setting Up Your Data

Find `student_survey.csv` in your digital resources. Click the menu icon, choose Open, then Browse, and select the file. In jamovi Cloud, you may need to upload it first.

| Column | What it records |
|---|---|
| `student_id` | A code number for each student |
| `travel` | How the student gets to school: Walk, Cycle, Bus, or Car |
| `breakfast` | Whether the student ate breakfast: Yes or No |
| `screen_time` | Hours on screens per day |
| `sleep_hours` | Hours of sleep per night |
| `sleep_quality` | The student's sleep rating: Poor, Fair, or Good |
| `quiz_score` | Score on a 20-point quiz |

jamovi gives each column a *measure type*, which tells it what kind of data the column holds:

- Nominal: categories with no order, such as travel method
- Ordinal: categories with an order, such as Poor, Fair, Good
- Continuous: measured numbers, such as hours of sleep
- ID: labels for each case, such as a student code

jamovi guesses these types when it opens a file, and it can guess wrong. To fix one, double-click the column name and pick the correct type.

Two cells in `sleep_hours` are empty because those students skipped the question. jamovi treats empty cells as *missing values* and leaves them out of calculations.

### *Exercise 175* Measure Types

Give the correct measure type for each column.

*Working Space*

| Column | Measure type |
|---|---|
| `student_id` | |
| `breakfast` | |
| `sleep_hours` | |
| `sleep_quality` | |

*Answer on Page 1939*

## 69.3 Describing Data

Click Exploration, then Descriptives. Move `quiz_score` and `sleep_hours` into the Variables box.

| | `quiz_score` | `sleep_hours` |
|---|---|---|
| N | 100 | 98 |
| Missing | 0 | 2 |
| Mean | 12.40 | 7.51 |
| Median | 12 | 7.60 |
| Standard deviation | 3.35 | 0.99 |
| Minimum | 4 | 4.5 |
| Maximum | 20 | 9.8 |

The average quiz score was 12.4 out of 20. Students slept 7.51 hours a night on average.

To compare groups, move a category into the Split by box. To draw a graph, open Plots and tick Histogram or Box plot. For a category such as `travel`, tick Frequency tables to

count how many students are in each group.

### *Exercise 176* Breakfast and Quiz Scores

Put `quiz_score` in the Variables box and `breakfast` in the Split by box.

1. What is the median quiz score for each group?
2. Which group did better?

*Working Space*

*Answer on Page 1939*

## 69.4 Comparing Two Groups

Students who ate breakfast averaged 13.28 on the quiz, and students who skipped it averaged 11.03. A t-test checks whether a gap this large could be due to chance.

Click T-Tests, then Independent Samples T-Test. Move `quiz_score` into Dependent Variables and `breakfast` into Grouping Variable. Under Tests, tick Welch's and untick Student's. Welch's version is the safer choice because it does not require the two groups to have the same spread.

jamovi reports $t = -3.40$ and $p = 0.001$. A small p-value, below 0.05, means the gap is unlikely to be due to chance alone.

The negative sign appears because jamovi subtracts the Yes group from the No group. The breakfast group still scored higher.

These data come from a survey, so they show a link between breakfast and quiz scores. They cannot show that breakfast causes higher scores.

### *Exercise 177* Reading the t-Test

*Working Space*

1. Is $p = 0.001$ below 0.05?
2. Write one sentence explaining what the test shows.

*Answer on Page 1939*

## 69.5 Looking at Relationships

Does more screen time go with less sleep? Click Exploration, then Scatterplot. Put `screen_time` on the X-Axis and `sleep_hours` on the Y-Axis.

The points slope downward: more screen time goes with less sleep.

Click Regression, then Correlation Matrix, and add both variables. The *correlation* $r$ runs from $-1$ to 1 and measures how closely the points follow a straight line. Here $r = -0.70$, a strong negative relationship.

For a line of best fit, open Regression, then Linear Regression. Put `sleep_hours` in Dependent Variable and `screen_time` in Covariates. jamovi gives the line

$$\widehat{\text{sleep}} = 9.55 - 0.44 \times \text{screen time}$$

Each extra hour of screen time goes with about 0.44 fewer hours of sleep.

### *Exercise 178* Making a Prediction

Use the line to predict the sleep of a student with 6 hours of screen time.

*Working Space*

*Answer on Page 1939*

## 69.6 Saving Your Work

Click the menu icon and choose Save As. jamovi saves an *.omv file*, which keeps your data and results together. In jamovi Cloud, download a copy to your own device before closing the tab.

To put a table or graph in a report, right-click it and choose Copy, then paste it into your document.

### *Exercise 179* Saving Choices

*Working Space*

1. Which file type keeps both your data and your results?
2. How do you move a table into a report?

*Answer on Page 1939*

With jamovi you can open data, set it up, describe it, test it, and save your results, all without writing code. Because it is free and runs in a browser, you can use it on almost any device.

CHAPTER 70

# Charge

If you rub a balloon against your hair, then place it next to a wall, it will stick. Why does this happen? This is exactly what will be explored in this chapter on **charge** and **electrostatics**. This is because the balloon has stripped some electrons from your hair, and now the ballon has slightly more electrons than protons. We say that it has gotten an *net electrical charge*, or an imbalance between protons and electrons. While any **friction** or contact between two objects can cause a difference in electric charge, *net charge cannot be created or destroyed*. In this case, the balloon has a negative electrical charge. Objects with slightly more protons than electrons have a positive charge. Consequently, your hair now has a positive charge. Note that the differences are equal but opposite.

An object with a negative charge and an object with a positive charge will be *attracted* to each other. Two objects with the same charge will be *repelled* by each other. This charge originates from a subatomic level, when protons (positively charged) and electrons (negatively charged) are present in an atom at *unequal amounts*.

This charge, referred to as *elementary charge* is represented by the base unit $e$ (not to be confused with the $e^x$ function in math disciplines). This value, known as the elementary charge ($e$), represents the smallest indivisible unit of electric charge. Much like counting physical coins, you cannot have a "fractional" coin. Similarly, an object's net charge must be an integer multiple of $e$. In physics, we describe this property by saying that charge is quantized ($Q = ne$ where $n$ is an integer).

In this way, we call elementary charge *quantized*. This was shown back in Because it is quantized, we represent the charge of a particle as $q$. The SI (International System of Units) measures charge in *coulombs*. The charge of a single proton is about $1.6 \times 10^{-19}$ coulombs.

Two objects with charges are bound to either attract (pull) or repel (push) on each other. This is a type of force referred to as *electric force*.

**Coulomb's Law**

If two objects with charge $q_1$ and $q_2$ (in coulombs) are $r$ meters from each other, the force of attraction or repulsion (electric force) is given by

$$F = k\frac{q_1 q_2}{r^2} \tag{70.1}$$

where $F$ is in newtons and $K$ is Coulomb's constant: about $8.988 \times 10^9$. You may use $9 \times 10^9\,N \cdot m^2/C$ in your calculations.

This $k$ value comes from $\frac{1}{4\pi\epsilon_0}$, a *relationship with the permittivity of free space*, $\epsilon_0$.[1] You may see this as $F_e$, $F_q$, or $F_c$, but they all refer to the electric force and the standard is $F_e$.

Note how similar this looks to Newton's Law of Universal Gravitation:

$$F_g = G\frac{m_1 m_2}{r^2}$$

Both laws follow an inverse-square relationship, meaning the force decreases with the square of the distance. However, gravity is always attractive, while electric force can be either attractive or repulsive depending on the signs of the charges ($q_1$ and $q_2$).

If this force is negative, it represents the idea that the two given particles are attracted to each other, as this would mean one of the particles must have a negative charge.

If this force is positive, it would mean there is a push or repelling force on each of the particles.

What if this force is zero? This would only result if one or both particles have zero charge. This would mean that they do not have an electric force, $F_E = 0$.

## *Exercise 180* Coulomb's Law

*Working Space*

Two balloons are charged with an identical quantity and type of charge: $-5 \times 10^{-9}$ coulombs. They are held apart at a separation distance of 12 cm. Determine the magnitude of the electrical force of repulsion between them.

*Answer on Page 1940*

[1] A note on the permittivity of free space: $\epsilon_0$ (or the electric constant) represents the capability of a vacuum to permit electric field lines. Its exact value in SI units is $\epsilon_0 \approx 8.854 \times 10^{-12}\ C^2/(N \cdot m^2)$. This tells us that the speed of light isn't just a number but rather it is strictly determined by how quickly the universe can allow electric and magnetic fields to change. Using $k$ allows for quick math, especially when fields become spherical and the $\pi$ factor is eliminated easily.

At this point, you might ask "If the wall has zero charge, why is the balloon attracted to it?" The answer: the electrons in the wall move away from the balloon, polarizing the atoms. The negative charge on the balloon pushes electrons away from itself, so the surface of the wall gets a mild positive charge. The negative charge on the balloon is attracted more to the positive charge on the surface of the wall than the negative charge on the inside, thus the balloon sticks. This is due to the idea of *conductors* and *insulators*.

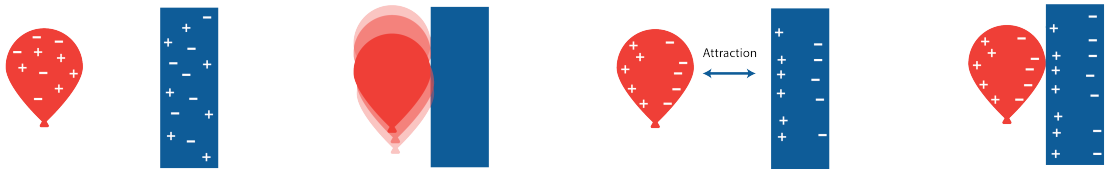



Figure 70.1: The negative balloon repels the wall's electrons, leaving a net positive surface. The balloon is then attracted to this induced positive charge.

Recall that a **conductor** is a type of material through which valence electrons (electrons in the outer shell) easily flows, and an **insulator** is a type of material through which charge does not easily flow, if at all. Metals, such as copper or zinc, are the best conductors because the electrons within the atoms are not tied to a nucleus, but rather form a sparse collection or sea in which the charge can flow through. Insulators like plastic, rubber, or glass are prevent charge flowing through as the electrons are tightly bound to their parent atoms or molecules.

### 70.0.1 Point Charge

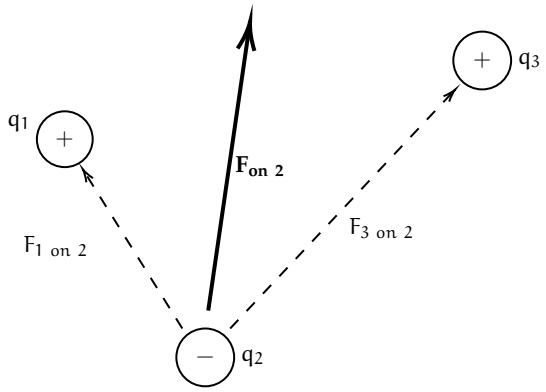
The wall and the balloon all contain a bunch of tiny particles which we can refer to as **point charges**. A point charge is an electric charge that is treated as a concentrated single point in space, with no physical size or volume. Similar to point masses, we use point charges to simulate effects of charge on other objects or points.

This charge acts parallel to the secant line that touches both point charges only once.

### 70.0.2 Superposition

What if there are more just one charges acting on a given point charge? Coulomb's law gives the force between two point charges, but in a system with three or more charges we apply the **principle of superposition**[2].

The principle of superposition states that the total electric force on a charge is the vector sum of the individual forces exerted on it by all other charges.[3]




## *Exercise 181* Finding net charge on a point charges

*Working Space*

This question is taken from the AP Physics C 19th edition practice workbook.

Consider four equal, positive point charges that are situated at the vertices of a square. Find the net electric force on a negative point charge placed at the square's center.

*Hint: draw a diagram to help you*

*Answer on Page 1940*

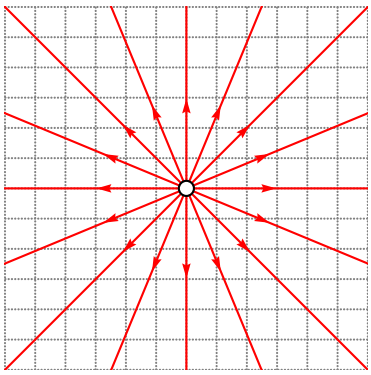
[2]Superposition is really a linear algebra topic, and not something that this textbook will cover, but it has so many real life applications and we recommend you do your own research! See the wiki page for more: https://en.wikipedia.org/wiki/Superposition_principle

[3]This is similar how we calculate tensions when there are multiple tensions (or just forces in general) on an object. The net force is the sum of all the forces after vector addition is applied.

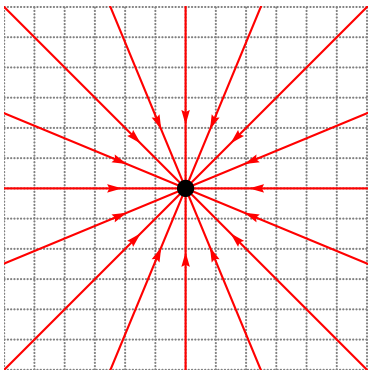
## 70.1 Electric Field

Recall that, when we talked about Gravitational Fields, we defined them as fields of infinitesimal points that each have a magnitude of force associated with them. This field revolves around the entire 2D circle or 3D sphere (generally point masses, charges, fields, etc are spherical).

Similar to how the gravitational field extends out from the center of a given point mass, an *electric field* of a point charge has (imaginary) vectors pointing out of it. However, the direction of the vectors[4] (and electric force) depends on the charge of the point charge. A negatively charged point mass has force lines pointing *towards* it, and a positively charged point mass has force lines pointing *away* from it (70.2a). In short, they always originate from positive charges and approach negative charges.



(a) A positively (open circle) charged electric field with corresponding electric field lines.



(b) A negatively (filled circle) charged electric field with corresponding electric field lines.

Figure 70.2: Two electric fields with different charges.

What are the *units* of an electric field? Well, by this equation, $\vec{\mathbf{F}}_e$ is a force which we know is in *Newtons* (N), and the test particle q is in Coulombs (C). The resulting units are $\mathrm{N\,C^{-1}}$, or Newton's per Coulomb. Note that this is equivalent to $1\,\mathrm{V\,m^{-1}}$, or one Volt per meter[5]. This is how electric field forces are converted to volts within batteries or circuits!

It is important to note that we can simplify this equation by substituting Coulomb's law:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_e}{q_t} = \frac{\frac{kq_1q_t}{r^2}}{q_t} = \frac{kQ}{r^2} \tag{70.2}$$

where Q is the electric field's source charge.

[4] You may hear the terms electric field lines and electric field vectors used interchangeably

[5] Equivalently, one $\mathrm{kV\,cm^{-1}}$, or kilovolt per centimeter is equal to $10^{-5} \times 1\mathrm{N\,C^{-1}}$. To reverse the operation, multiply the kilovolts per centimeter by $10^5$ to get the resulting Newtons per Coulomb

For positive charges Q, the force vectors get weaker the farther from the charge (visually, the force lines get shorter). In fact, it decreases by a factor of $\frac{1}{r^2}$. Think to yourself where this factor comes from. (Hint: refer back to Coulomb's law (70.1))

**Electric Field Force from a singular charge to a second charge**

A charge $q_1$ creates an electric field at a distance $r$ from it with strength:

$$E = \frac{kq_1}{r^2} \tag{70.3}$$

Introducing a new charge $q_2$, the force on it due to the electric field is:

$$F = q_2 \cdot \frac{kq_1}{r^2} \iff F = q_2 E \tag{70.4}$$

where $E$ is the electric field strength at the location of $q_2$. This is often referred to as the charge within a known field.

## *Exercise 182* Point Charges and Electric Fields

*This problem is adapted from an Example IB Exam.* Two isolated point charges, X of charge $+Q$ and Y of charge $+2Q$, are separated by distance $3d$. Point P is distance $d$ from X and distance $2d$ from Y.

What is the net electric field strength at P? Are the forces from each charge equal?

1. 0, equal forces
2. $\frac{2kQ}{d^2}$, equal forces
3. $\frac{kQ}{2d^2}$, equal forces
4. $\frac{2kQ}{1.5d^2}$, equal forces

*Working Space*

*Answer on Page 1940*

The density of the electric force lines at a given point determines the strength of the force at the point. Locations where field lines are more dense are much stronger than areas where the lines are sparse. This goes back to the idea of the force strength of an electric field.

If you recall the properties of conductors: *there can be no electrostatic field within the body of a conductor*. If an electric field were to exist inside the conductor, it would exert a force $F = qE$ on these free charges, causing them to accelerate. This internal motion continues until the charges redistribute themselves along the conductor's outer surface, creating an induced internal field that perfectly opposes and cancels the original field.

Electric field lines always

- Start at positive charges and terminate at negative ones. If it is a single charge, this is indicated by the direction of arrows off to infinity.
- The number of lines is proportional to the magnitude of charge.
- The line approaches a conductive surface perpendicularly.
- No two field lines ever cross in a charge-free region.

Keep these in mind for any sketches you may do.

## 70.2 Dipoles

An **electric dipole** is a set of two *point charges* that equal and opposite charges that are infinitesimally close together. Two point charges, one with charge $+q$ and the other one with charge $-q$ separated by a distance $d$, create an electric dipole.

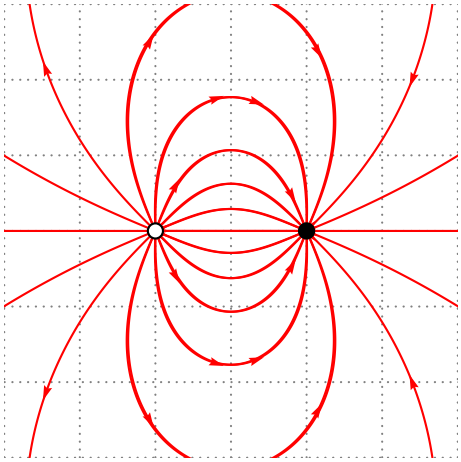


Figure 70.3: A basic dipole diagram.

Charge flows consistently from positive to negative, and has magnitude $\vec{\mathbf{p}} = q\vec{\mathbf{d}}$, where $\vec{\mathbf{p}}$ is the dipole moment with units $C \cdot m$, or Coulomb-meters.

## 70.3 Electric Flux and Gauss's law

**Electric Flux** is the number of field lines that pass through a given surface. Represented by $\Phi$, it is the measure of how much of an electric field itself passes through an area.

The electric flux depends on 3 factors:

- Electric Field Strength $(\mathbf{E})$
- Surface Area being measured, $(\mathbf{A})$. $\mathbf{A}$ must be perpendicular to the field lines.
- Angle at which the area is relative to the field lines.

If the electric field lines are perpendicular to the area $\mathbf{A}$, then the electric flux is

$$\Phi = \sum \mathbf{E} \cdot \mathbf{A} \tag{70.5}$$

Think of this as a paper directly perpendicular to a fan; the maximum amount of the paper is blown by the fan.

If the paper is at an angle to the fan, less flow from the fan affects the paper. If you tilt it until it's parallel to the wind, zero wind affects the paper. Similarly, the electric flux on a surface at a titled angle decreases how much flux is felt:

$$\Phi = \sum \mathbf{E} \cdot \mathbf{A} \cos(\theta) \tag{70.6}$$

In integral form (which we have not covered yet, so do not worry about using it), this is:

$$\Phi = \oint \mathbf{E} \cdot d\mathbf{A} \cos(\theta) \tag{70.7}$$

where the integral is on a closed area.

If you recall our discussion on resistance, flux and resistance are very similar measurements. In a wire, a larger cross-sectional area $(A)$ is like a highway for electrons. It allows more electrons to pass through easily, which decreases the resistance. If you maximize flux by having a huge area, you are essentially creating a path with very low resistance for a current.

### 70.3.1 Gauss's law

So far, we have imagined the electric field is only 2-dimensional plane (a circle). What if we consider them as a 3-dimensional boundary, similar to gravitation? This introduces the idea of *Gauss's Law for Electric Fields*:

**Gauss's Law**

Gauss's law states that the strength of an electric field for 3-Dimensional object can be calculated by:

$$\Phi = \oiint \vec{\mathbf{E}}\, d\vec{\mathbf{A}} \cos(\theta) = \frac{Q_{enc}}{\epsilon_0} \tag{70.8}$$

This is a formula for saying that every infinitesimal bit of a 2-Dimensional object is given a height and an electric strength, and this is equal to the enclosed area of the charge over the permittivity of free space constant.

For specifically spherically 3-Dimensional objects, inputting the formula for a sphere into A gives us:

$$E = \frac{q}{4\pi r^2 \epsilon_0} \tag{70.9}$$

where q is the source charge, r is the distance from that source charge, and $4\pi r^2$ is the surface area of the electric field. What this means is that electric field at all points r from the source charge are equal in magnitude. Pretty cool!

## 70.4 Continuous Charge Distributions

In the previous sections, we treated charges as discrete "dots" in space. However, in applications of charges such as a long charged wire, a flat metal plate, or a solid plastic sphere, charge is spread out over a macroscopic object. When we have a collection of so many point charges that they appear to form a smooth, "fluid-like" surface or volume, we call it a **continuous charge distribution**. This means that the charges, while not truly continuous, can be considered continuous as a long single stream.

To analyze these, we stop summing individual particles ($\sum$) and instead use calculus to integrate ($\int$) over the entire object. While we have not covered calculus yet, keep in mind that this is just a way of calculating the area or volume of a continuous solid.

### 70.4.1 Charge Densities

The first step in any continuous distribution problem is defining how the charge is spread. We use three Greek letters to represent the "concentration" of charge based on the object's

geometry:

| Type | Symbol | Definition | SI Units |
|---|---|---|---|
| Linear | $\lambda$ (lambda) | Charge per unit length | $C/m$ |
| Surface | $\sigma$ (sigma) | Charge per unit area | $C/m^2$ |
| Volume | $\rho$ (rho) | Charge per unit volume | $C/m^3$ |

Table 70.1: The three types of charge density used in electrostatics.

If the charge is distributed **uniformly**, these values stay constant. For example, for a rod of length L and total charge Q, the linear density is simply $\lambda = Q/L$. However, if these the charge is not constant per the unit of length, we must use calculus to find the total charge density. To find the total electric field $\vec{\mathbf{E}}$ at a point P near a charged object, we imagine breaking the object into many infinitesimal "pieces" of charge, $dq$. Each chunk acts like a tiny point charge and contributes a tiny bit of electric field, $d\vec{\mathbf{E}}$.

Recall our point charge formula:

$$d\vec{\mathbf{E}} = \frac{1}{4\pi\epsilon_0}\frac{dq}{r^2} \tag{70.10}$$

To find the total field, we integrate over the entire distribution:

$$\vec{\mathbf{E}} = \int d\vec{\mathbf{E}} = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r^2} \tag{70.11}$$

The key to solving these integrals is replacing $dq$ with the appropriate density:

- Lines: $dq = \lambda\, dl$
- Surfaces: $dq = \sigma\, dA$
- Volumes: $dq = \rho\, dV$

For now, you will not be quizzed on these integrals but know of their existence and that charge can be treated as a virtual stream along an object. This is why any part of the balloon from our first example will stick to any part of the wall.

### *Exercise 183* Belt being charged

*Working Space*

*This question is from an IB Electromagnetics problem.* A continuous conveyor belt of width 0.60 m travels at a constant speed of 2.5 $ms^{-1}$. The belt has a uniform distribution of charge of 5.0m$Cm^{-2}$ on the surface. As the belt passes a point A, all the charge is transferred into the wire with width along the horizontal of the belt.

What is the charge on the wire? *Hint: recall that current is* $\frac{\Delta Q}{\Delta t}$.

*Answer on Page 1941*

## 70.5 Electric Potential

Let's talk about electric potential and electric potential energy. Recall that objects can have *kinetic* and *potential* energy, that potential energy is gradually transported into kinetic energy.

Electric Potential describes the amount of work, $W_E$, done by an electric force. The change in **electric potential energy** is defined as the work done on that force:

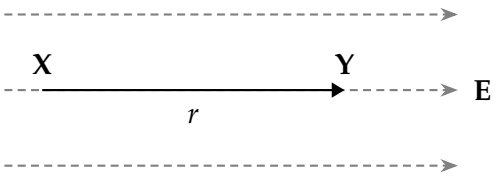
$$\Delta U_E = -W_E \tag{70.13}$$

Notice how this is similar to an object of mass $m$ undergoing work done by a gravitational field $\Delta U_G = -W_G$.

## *Exercise 184* Change in Electric Potential Energy

*Working Space*

A positive charge $+q$ moves a distance $r$ from position X to position Y in a **uniform** electric field $E$.




What is its change in electric potential energy? Assume moving in direction $r$ moves in the direction of $E$.

*Answer on Page 1941*

Electric Potential $(V)$ is the amount of work needed to move a test charge $q_t$ from a reference point to a specific point in a static electric field. It describes the potential for a charge to have energy at that location, regardless of whether a charge is actually there. Electric Potential Energy $(U_E)$ per unit charge where $(V = U_E/q)$.

Just as moving a mass up in height against gravity increases its gravitational potential, moving a positive charge "up" against an electric field increases its electric potential. In both cases, the field *tries* to push the object toward a state of lower potential energy. A positive charge naturally wants to move with the electric field toward lower potential; in this case, its electric potential energy is converted into kinetic energy. Conversely, to move a positive charge against an electric field, work must be done on the charge, causing its electric potential to increase.

Moving from two points not on the same electric field line introduces an angle component $\theta$:

The electric potential difference between two points A and B is defined by the line

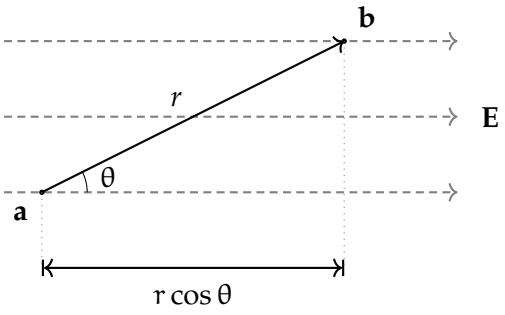



Figure 70.4: A gain of potential energy across an angle.

integral of the electric field:

$$\Delta V = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{l} \tag{70.14}$$

The negative sign indicates that the potential decreases in the direction of the electric field.

We can easily define two point charges $q_1$ and $q_2$ separated by distance $r$ to have potential energy

$$U = k\frac{q_1 q_2}{r} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r} \tag{70.15}$$

## *Exercise 185* Accelerating an Electron

*This is a sample problem from an IB Physics Exam.* An electron is accelerated from rest by an electric potential difference $V$. The electron reaches a speed of $9.4 \times 10^6$ m/s. Using the charge of an electron as $e = 1.6 \times 10^{-19}$ C and the mass of an electron as $m_e = 9.11 \times 10^{-31}$ kg, find the potential difference $V$ that accelerated the electron to this speed. You may use a calculator to simplify computations.

*Answer on Page 1942*

### 70.5.1 Point Charges, Uniform Fields, and Spherical Shells

In a **uniform electric field** (such as the space between two large parallel plates), the field strength $E$ is constant. The potential difference between two points separated by a distance $d$ along the field lines is:

$$\Delta V = -Ed \tag{70.16}$$

For a **point charge** Q, we typically set the reference point ($V = 0$) at infinity. The potential at a distance r from the charge is:

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{r} \tag{70.17}$$

Unlike the electric field, which is a vector and follows an $1/r^2$ factor, electric potential is a **scalar** and follows an $1/r$ factor.

### Surface Spherical Charge

For all spherical objects of radius R (surface spheres - assume hollow interior), it has potential inside and outside of the shell. For all points outside the shell, it is the same for any point charge Q, as seen in equation 70.17. For all points *inside* the shell, the potential is

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{R}$$

This means that moving a test charge q from the surface of the shell to the center of the shell, the potential does not change. The potential is constant within the shell.

When the sphere is *not hollow*, the interior charge changes to:

$$k = \frac{1}{4\pi\epsilon_0}\frac{Q}{2R}\left(3 - \frac{r^2}{R^2}\right)$$

### 70.5.2 Potential due to Configurations of Charge

Because potential is a scalar, finding the total potential from multiple charges is often simpler than finding the total electric field. We do not need to worry about components, instead, we simply sum the potentials:

$$V = \sum_i \frac{kq_i}{r_i} \quad \text{or for continuous distributions} \quad V = k\int \frac{dq}{r} \tag{70.18}$$

### 70.5.3 Conservation of Energy

When a charge q moves through a potential difference $\Delta V$, the change in its electric potential energy is $\Delta U_E = q\Delta V$. If no other forces are acting on the charge, the total mechanical energy is conserved:

$$K_a + U_{E,a} = K_b + U_{E,b} \implies \frac{1}{2}mv_a^2 + qV_a = \frac{1}{2}mv_b^2 + qV_b \tag{70.19}$$

### 70.5.4 Summary

| Particle | *wants* to move toward... | Potential Change ($\Delta V$) | Potential Energy Change ($U_E$) |
|---|---|---|---|
| Positive ($+q$) | Lower Potential | Negative (Decreases) | Negative (Decreases) |
| Negative ($-q$) | Higher Potential | Positive (Increases) | Negative (Decreases) |

## *Exercise 186* Electron between Field Plates

*This is a sample problem from an IB Physics Exam.* An electron that is travelling at a horizontal speed $v_x = 9.4 \times 10^6 \text{ms}^{-1}$ now enters a region of a *uniform electric field* between two parallel charged plates, separated by a distance of 4.0 cm. The electron is initially halfway between the plates and its initial velocity is parallel to the plates. The upper plate is positively charged and the lower plate is negatively charged. There is a potential difference between the plates of $V = 30$ V.

Find the horizontal distance travelled before hitting a plate. Which plate does the electron hit?

*Working Space*

*Answer on Page 1942*

## 70.6 Lightning

A cloud is a cluster of water droplets and ice particles. These droplets and ice particles are always moving up and down through the cloud. Recall that heat rises, so water is often trying to get higher in the cloud. In this process, electrons get stripped off and end up on the water droplets at the bottom of the cloud (water droplets collect at the bottom because they are denser). The air between the droplets is a pretty good insulator, which means the electrons are reluctant to jump anywhere. However, eventually, the charge gets so strong that even the insulating properties of the air is not enough to prevent the jump to earth, causing lightning. *Storms* are dense in clouds, so there is constant jumping of the electrons, which is why storms are prone to producing so much lightning. These electrons really want to jump to earth, and when given enough time and ability to bypass the air's

density. The electrons energy is visible as light and heat, and so we see a spark in the shape of a jagged line hitting a single point, the most conductive point on earth closest to the cloud. A great deal of lightning moves within a cloud or between clouds. However, sometimes it jumps to the earth. These bolts of lightning vary in the amount of electrons they carry, but the average is about 15 coulombs.

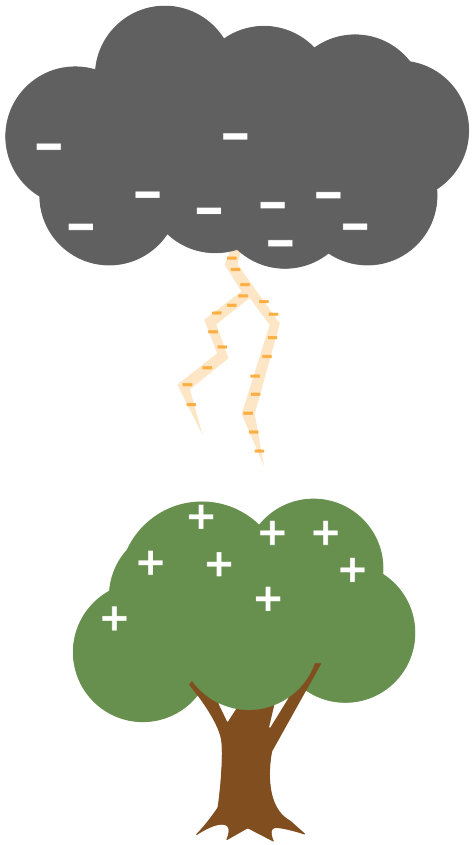


Figure 70.5: A difference of charge causes a spark of lightning to generate and transfer.

Thunder occurs because the electrons heat the air they pass through, causing the air to expand suddenly. The resulting shockwave is the sound we know as thunder.

## 70.7 But...

This idea that opposite charges attract creates some heavy questions that you do not yet have the tools to work with. So to these questions, the answer is basically "Don't ask that yet!"

However, you probably have these questions, so we will point you in the direction of the

answers.

The first is "In any atom bigger than hydrogen, there are multiple protons in the nucleus. Why don't the protons push each other out of the nucleus?"

We aren't ready to talk about it, but there is a force called *the strong nuclear force*, which pulls the protons and neutrons in the nucleus of the atom toward each other. At very, very small distances, it is strong enough to overpower the repulsive force due to the protons' charges.

Another question is "Why do the electrons whiz around in a cloud so far from the nucleus of the atom? Negatively charged electrons should cling to the protons in the center, right?"

We aren't ready to talk about this either, but quantum mechanics tells us that electrons like to live in a certain specific energy level. Hugging protons isn't one of those levels.

CHAPTER 71

# Alternating Current

We have discussed the voltage and current created by a battery. A battery pushes the electrons in one direction at a constant voltage; this is known as *Direct Current* or DC. A battery typically provides between 1.5 and 9 volts.

The electrical power that comes into your home on wires is different. If you plotted the voltage over time, it would look like this:

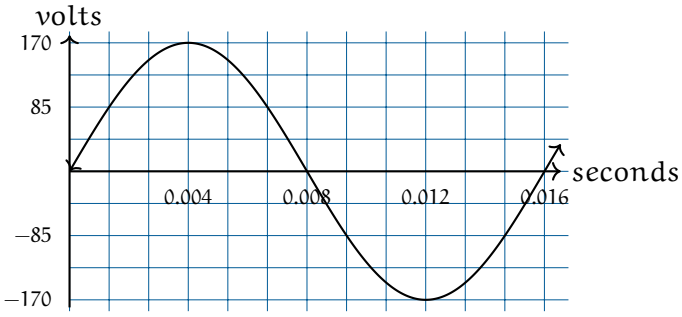



Figure 71.1: A graph of AC voltage over time.

The x axis here represents ground. When you insert a two-prong plug into an outlet, one is "hot" and the other is "ground". Ground represents 0 volts and should be the same voltage as the dirt under the building.

The voltage is a sine wave at 60Hz. Your voltage fluctuates between -170v and 170v. Think for a second what that means: The power company pushes electrons at 170v, then pulls electrons at 170v. It alternates back and forth this way 60 times per second.

## 71.1 Power of AC

Let's say you turn on your toaster, which has a resistance of 14.4 ohms. How much energy (in watts) does it change from electrical energy to heat? We know that $I = V/R$ and that watts of power are $IV$. So, given a voltage of $V$, the toaster is consuming $V^2/R$ watts.

However, $V$ is fluctuating. Let's plot the power the toaster is consuming:

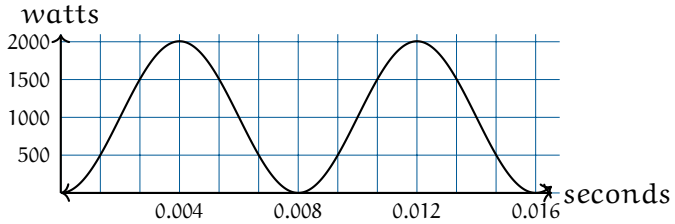



Figure 71.2: A plot of AC power over time.

Another sine wave! Here is a lesser-known trig identity: $(\sin(x))^2 = \frac{1}{2} - \frac{1}{2}\cos(2x)$

This is actually a cosine wave flipped upside down, scaled down by half the peak power, and translated up so that it is never negative. Note that it is also twice the frequency of the voltage sine wave.

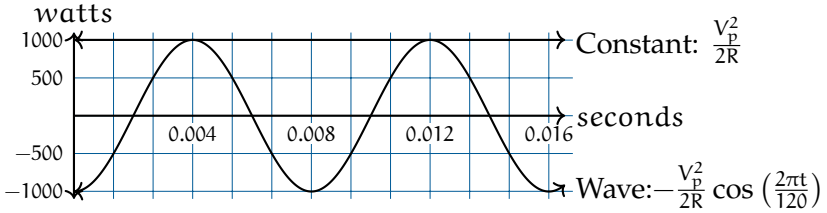
If we say the peak voltage is $V_p$ and the resistance of the toaster is $R$, the power is given by

$$\frac{V_p^2}{2R} - \frac{V_p^2}{2R}\cos\left(\frac{2\pi t}{120}\right)$$

As a toaster user and as someone who pays a power bill, you are mostly interested in the average power. To get the average power, you take the area under the power graph and divide it by the amount of time.

We can think of the area under the curve as two easy-to-integrate quantities summed:

- A constant function of $y = \frac{V_p^2}{2R}$
- A wave $y = -\frac{V_p^2}{2R}\cos\left(\frac{2\pi t}{120}\right)$




When we integrate that constant function, we get $\frac{tV_p^2}{2R}$.

When we integrate that wave for a complete cycle we get...zero! The positive side of the wave is canceled out by the negative side.

So, the average power is $\frac{V_p^2}{2R}$ watts.

Someone at some point said, "I'm used to power being $V^2/R$. Can we define a voltage measure for AC power such that this is always true?"

So we started using $V_{rms}$ which is just $\frac{V_p}{\sqrt{2}}$. If you look on the back of anything that plugs into a standard US power outlet, it will say something like "For 120v". What they mean is, "For 120v RMS, we expect the voltage to fluctuate back and forth from 170v to -170v."

Notice that this is the same Root-Mean-Squared that we defined earlier, but now we know that if $y = \sin(x)$, the RMS of $y$ is $1/\sqrt{2} \approx 0.707$.

For current, we do the same thing. If the current is AC, the power consumed by a resistor is $I_{RMS}^2 R$, where $I_{RMS}$ is the peak current divided by $\sqrt{2}$.

## 71.2 Power Line Losses

A wire has some resistance. Thinner wires tend to have more resistance than thicker ones. Aluminum wires tend to have more resistance than copper wires.

Let's say that the power that comes to your house has to travel 20 km from the generator in a cable that has about $1\Omega$ of resistance per km. Let's also say that your home is consuming 12 kilowatts of power. If that power is 120v RMS from the generator to your home, what percentage of the power is lost heating the power line?

10 amps RMS flow through your home. When that current goes through the wire, $I^2R = (100)(20) = 2000watts$ is lost to heat. This means the power company would need to supply 14 kilowatts of power, knowing that 2 kilowatts would be lost on the wires.

What if the power company moved the power at 120,000 volts RMS? Now only 0.01 amps RMS flow through your home. When that current goes through the wire $I^2R = (0.0001)(20) = .002$ watts of power are lost on the power lines.

This is much, much more efficient. The only problem is that 120,000 volts would be incredibly dangerous. So the power company moves power long distances at very high voltages, like 765 kV. Before the power is brought into your home, it is converted into a lower voltage using a *transformer*.

## 71.3 Transformers

A transformer is a device that converts electrical power from one voltage to another. A good transformer is more than 95% efficient. The details of magnetic fields, flux, and inductance are beyond the scope of this chapter, so we are going to give a relatively

simple (and admittedly incomplete) explanation for now.

A transformer is a ring with two sets of coils wrapped around it.

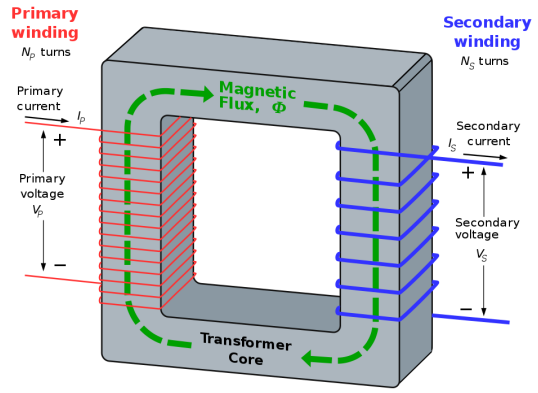



Figure 71.3: A diagram of a transformer from Wikipedia.

When alternating current is run through the primary winding, it creates magnetic flux in the ring. The magnetic flux induces current in the secondary winding. (This is called *induction*.)

If $V_P$ is the voltage across the primary winding and $V_S$ is the voltage across the secondary winding, they are related by the following equation:

$$\frac{V_P}{V_S} = \frac{N_P}{N_S}$$

where $N_P$ and $N_S$ are the number of turns in the primary and secondary windings.

There are usually at least two transformers between you and the very high voltage lines. There are transformers at the substation that make the voltage low enough to travel on regular utility poles. On the utility poles, you will see cans that contain smaller transformers. Those step the voltage down to the 120V RMS that your home uses.

## 71.4 Phase and 3-phase power

If two waves are "in sync", we say they have the same *phase*.

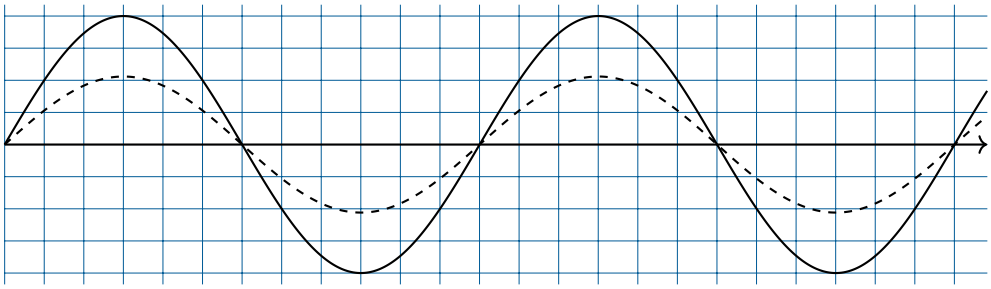


Figure 71.4: A diagram of two AC signals in sync.

If they are the same frequency, but are not in sync, we can talk about the difference in their phase.

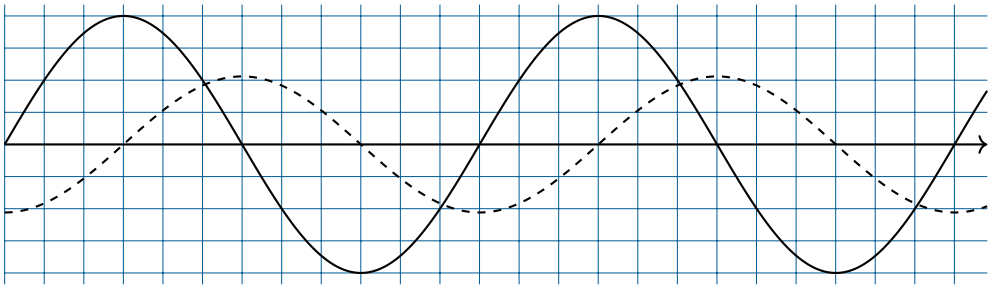


Figure 71.5: A diagram of two AC signals out of phase.

Here, we see that the smaller wave is lagging by $\pi/2$ or $90^\circ$.

In most power grids, there are usually three wires carrying the power. The voltage on each is $2\pi/3$ out of phase with the other two:

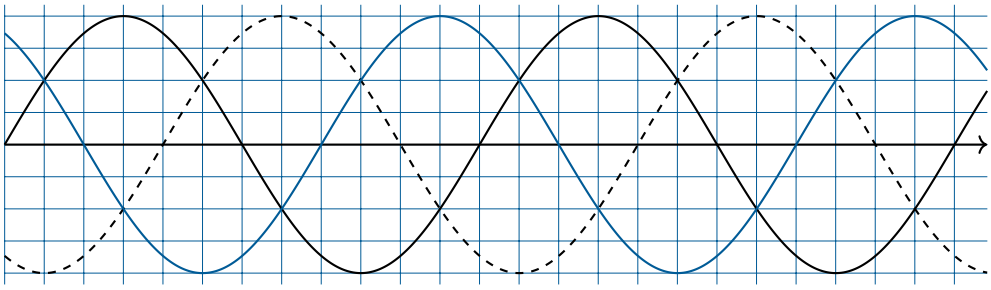


Figure 71.6: A diagram of three AC signals, each $2\pi/3$ radians out of phase.

This is nice in two ways:

- While the power in each wire is fluctuating, the total power is not fluctuating at all.

- While the power plant is pushing and pulling electrons on each wire, the total number of electrons leaving the load is zero.

(Both these assume that each wire is attached to a load with the same constant resistance.)

In big industrial factories, you will see all three wires enter the building. Large amounts of smooth power delivery means a great deal to an industrial user.

In residential settings, each home gets its power from one of the three wires. However, two wires typically carry power into the home. Each one carries 120V RMS, but they are out of phase by 180 degrees. Lights and small appliances are connected to one of the wires and ground, so they get 120V RMS. Large appliances, like air conditioners and washing machines, are connected across the two wires, so they get 240V RMS.

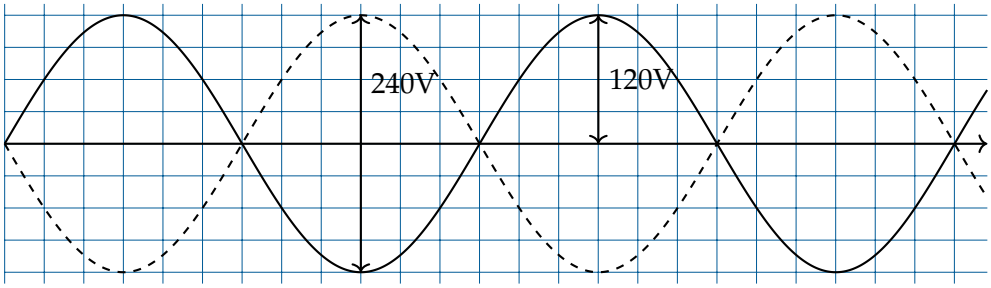



Figure 71.7: A diagram of AC out of phase in homes.

How do you get two circuits, 180 degrees out of phase, from one circuit? Using a center-tap transformer.

FIXME: Diagram here

Chapter 72

# Electromagnetic Induction

Electromagnetic induction is a fundamental concept in physics that describes how a changing magnetic field can produce an electric current. The phenomenon was discovered by Michael Faraday in 1831 and is the principle behind many electrical devices, including generators and transformers.

## 72.1 Faraday's Law of Induction

It is important to note that induction is the process of generating a current in a conductor by **changing** the magnetic field around it. In other words, if a conductor is moving through a **constant** magnetic field, no current will be induced. However, if the magnetic field is changing or if the conductor is moving through a gradient, a current will be induced.

The equation that describes electromagnetic induction is known as *Faraday's Law of Induction*:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

where $\mathcal{E}$ is the induced electromotive force (emf) in volts, and $\Phi_B$ is the magnetic flux through the circuit in webers (Wb). The negative sign indicates the direction of the induced emf, as described by Lenz's Law. Note that the induced emf is proportional to the **rate of change** of the magnetic flux. This means that a faster change in the magnetic field will result in a larger induced emf.

### 72.1.1 Magnetic Flux

Magnetic flux ($\Phi_B$) through a surface is defined as:

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

where $\vec{B}$ is the magnetic field and $d\vec{A}$ is an infinitesimal area vector perpendicular to the surface. You'll learn more about vector calculus in later workbooks.

## 72.2 Lenz's Law

Lenz's Law gives the direction of the induced current. It states that the induced current will flow in such a way that its magnetic field opposes the change in magnetic flux that produced it. This is reflected in the negative sign in Faraday's Law.

## 72.3 Induction in a Coil

When a coil of wire is placed in a region where the magnetic field changes over time, an emf is induced in the coil according to Faraday's Law. If the coil has N turns, the total induced emf is given by:

$$\mathcal{E} = -N\frac{d\Phi_B}{dt}$$

This means that the induced emf is proportional to both the number of turns in the coil and the rate of change of the magnetic flux through each turn.

If the magnetic field changes uniformly and the area of the coil is constant, the change in flux can be written as:

$$\Delta\Phi_B = \Delta B \times A$$

where $A$ is the area of the coil and $\Delta B$ is the change in magnetic field.

Therefore, the induced emf for a coil experiencing a uniform change in magnetic field is:

$$\mathcal{E} = -N\frac{A\Delta B}{\Delta t}$$

This principle is widely used in devices such as electric generators, where coils rotate in a magnetic field to produce electricity.

### *Exercise 187* Induced EMF in a Coil

A coil with 100 turns and area $0.2\,m^2$ experiences a change in magnetic field from 0 to 5 T over 0.1 s. Calculate the induced emf.

*Working Space*

*Answer on Page 1943*

## 72.4 Applications of Electromagnetic Induction

- **Electric Generators**: Convert mechanical energy into electrical energy using electromagnetic induction.
- **Transformers**: Change the voltage of alternating current (AC) electricity using induction between coils.
- **Induction Cooktops**: Use rapidly changing magnetic fields to heat cookware directly.

Chapter 73

# Electric Motor

Electric motors are devices that convert electrical energy into mechanical energy. They operate based on the interaction between magnetic fields and electric currents, producing rotational motion. As mentioned previously, when an electric current flows through a conductor, it generates a magnetic field around the conductor. If you place a magnet near this conductor, the magnetic field will interact with the magnet's field, causing the conductor to experience a force. This force can be harnessed to create motion, which is the fundamental principle behind electric motors.

## 73.1 Basic Electric Motor

The most common type of electric motor is the DC (Direct Current) brushed motor. It consists of a coil of wire (armature) that rotates within a magnetic field created by permanent magnets or electromagnets. The armature is connected to a commutator, which reverses the direction of the current in the coil as it rotates, ensuring continuous rotation in one direction. The name "brushed" refers to the use of brushes that make contact with the commutator to supply current to the armature.

TODO: Can there be a graphic of a basic electric motor? The generator image is similar, but the motor has a power source rather than a load.

Other types of electric motors include brushless DC motors, stepper motors, and AC (Alternating Current) motors. Each type has its own characteristics and applications, but they all operate on the same basic principles of electromagnetism.

## 73.2 Basic Electric Generator

Electric generators are the reverse of electric motors. They convert mechanical energy into electrical energy by using the principle of *electromagnetic induction*. When a conductor (such as a coil of wire) moves through a magnetic field, it induces an electric current in the conductor. This is covered in another chapter.

To construct a basic electric generator, all you need is an electric motor and a load.

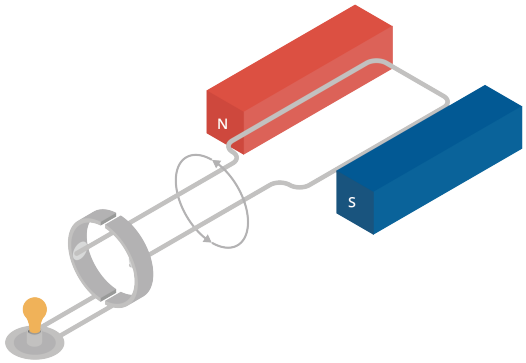



Figure 73.1: Basic Electric Generator Diagram

When you turn the motor with mechanical force, it will generate electricity that can be used to power a load. As the coil rotates within the magnetic field, the changing magnetic field induces a current in the coil, which can flow through the load, providing the necessary power.

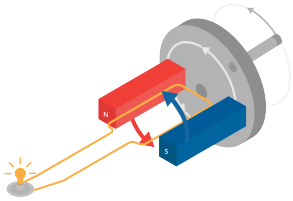



Figure 73.2: Basic Electric Generator in Motion with Handcrank

If you instead turn the motor with a wheel, you can generate electricity in any manner of ways. This simple principle is used all over the world to generate electricity. For example, in hydroelectric power plants, water is used to turn large turbines connected to generators, producing electricity on a massive scale. In wind power plants, wind turns the blades of a turbine, which is connected to a generator that produces electricity. In coal and natural gas power plants, steam is used to turn turbines connected to generators.

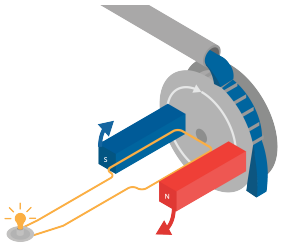



Figure 73.3: Basic Electric Generator in Motion with Wheel

# Chapter 74

# Drag

The very first computers were created to do calculations of how artillery would fly when shot at different angles. The calculations were similar to the ones you did earlier for the flying hammer, with two important differences:

- They were interested in two dimensions: the height and the distance across the ground.
- However, artillery flies a lot faster than a hammer, so they also had to worry about drag from the air.

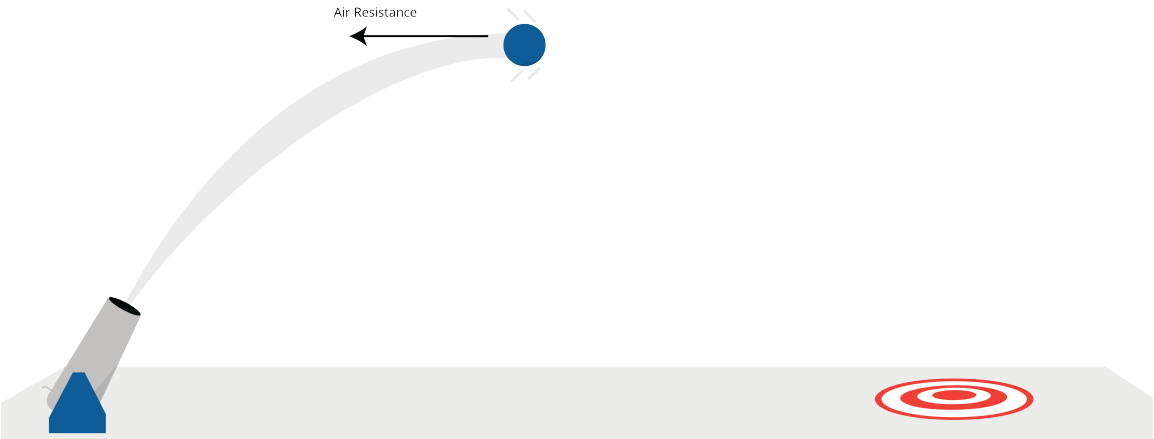



Figure 74.1: A cannonball shot has a parabolic trajectory.

## 74.1 Wind resistance

The first thing they did was put one of the shells in a wind tunnel. They measured how much force was created when they pushed 1 m/s of wind over the shell. Let's say it was 0.1 newtons.

One of the interesting things about the drag from the air (often called *wind resistance*) is that it increases with the *square* of the speed. Thus, if the wind pushing on the shell is 3

m/s, instead of 1 m/s, the resistance is $3^2 \times 0.1 = 0.9$ newtons.

(Why? Intuitively, three times as many air molecules are hitting the shell and each molecule is hitting it three times harder.)

So, if a shell is moving with the velocity vector $v$, the force vector of the drag points in the exact opposite direction. If $\mu$ is the force of wind resistance of the shell at 1 m/s, then the magnitude of the drag vector is $\mu|v|^2$ with $\mu$ being the wind resistance force.

## 74.2 Initial velocity and acceleration due to gravity

Let's say a shell is shot out of a tube at $s$ m/s, and the tube is tilted $\theta$ radians above level. The initial velocity will be given by the vector $[s\cos(\theta), s\sin(\theta)]$

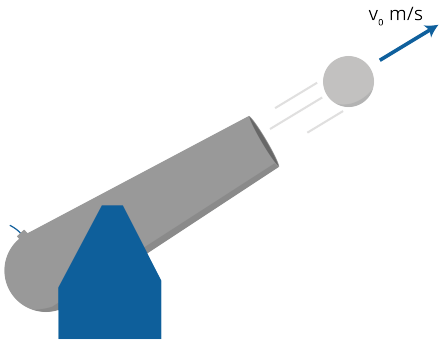



Figure 74.2: The initial velocity vector of the shell.

(The velocity of the shell is actually a 3-dimensional vector, but we are only going to worry about height and horizontal distance; we are assuming that the operator pointed it in the right direction.)

To figure out the path of the shell, we need to compute its acceleration. We remember that

$$F = ma$$

(Note that $F$ and $a$ are vectors.) Dividing both sides by $m$, we get:

$$a = \frac{F}{m}$$

Let's figure out the net force on the shell, so that we can calculate the acceleration vector.

If the shell has a mass of b, the force due to gravity will be in the downward direction, with a magnitude of 9.8b newtons.

To get the net force, we will need to add the force due to gravity with the force due to wind resistance.

## 74.3 Simulating artillery in Python

Create a file called `artillery.py`.

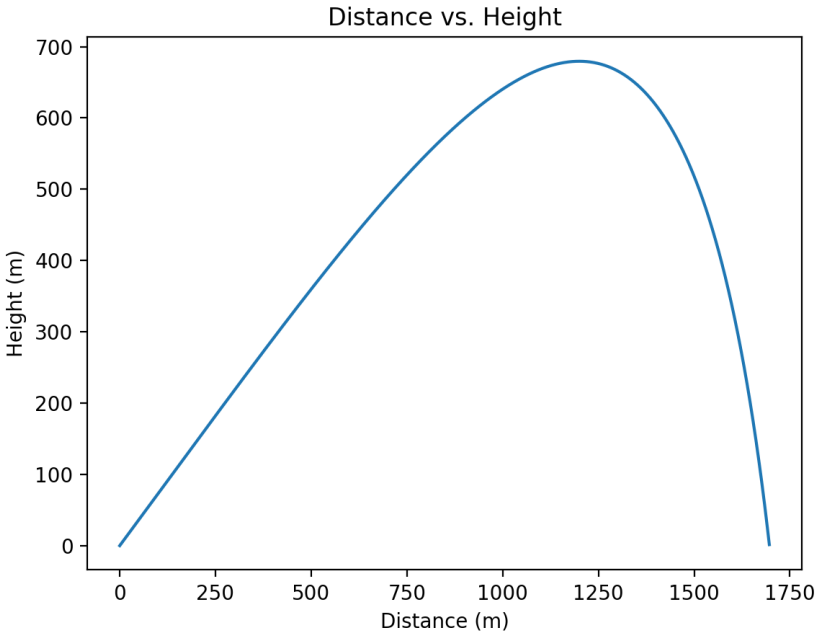
```
import numpy as np
    import matplotlib.pyplot as plt

    # Constants
    mass = 45 # kg
    start_speed = 300.0 # m/s
    theta = np.pi/5 # radians (36 degrees above level)
    time_step = 0.01 # s
    wind_resistance = 0.05 # newtons in 1 m/s wind
    force_of_gravity = np.array([0.0, -9.8 * mass]) # newtons

    # Initial state
    position = np.array([0.0, 0.0]) # [distance, height] in meters
    velocity = np.array([start_speed * np.cos(theta), start_speed *
    ↪  np.sin(theta)])
    time = 0.0 # seconds

    # Lists to gather data
    distances = []
    heights = []
    times = []

    # While shell is aloft
    while position[1] >= 0:
        # Record data
        distances.append(position[0])
        heights.append(position[1])
        times.append(time)

        # Calculate the next state
        time += time_step
        position += time_step * velocity

        # Calculate the net force vector
        force = force_of_gravity - wind_resistance * velocity**2

```

```
        # Calculate the current acceleration vector
        acceleration = force / mass

        # Update the velocity vector
        velocity += time_step * acceleration

    print(f"Hit the ground {position[0]:.2f} meters away at {time:.2f} seconds.")

    # Plot the data
    fig, ax = plt.subplots()
    ax.plot(distances, heights)
    ax.set_title("Distance vs. Height")
    ax.set_xlabel("Distance (m)")
    ax.set_ylabel("Height (m)")
    plt.show()
```

When you run it, you should get a message like:

```
Hit the ground 1696.70 meters away at 20.73 seconds.
```

You should also see a plot of the shell's path:



## 74.4 Terminal velocity

If you shot the shell very, very high in the sky, it would keep accelerating toward the ground until the force of gravity and the force of the wind resistance were equal. The speed at which this happens is called the *terminal velocity*. The terminal velocity of a falling human is about 53 m/s.

Note that kinematic equations do not apply to terminal velocity, because the acceleration is not constant. Instead, we can use the fact that at terminal velocity, the force of wind resistance equals the force of gravity.

### *Exercise 188* Terminal velocity

What is the terminal velocity of the shell described in our example?

*Working Space*

*Answer on Page 1943*

Chapter 75

# Electromagnetic Waves

Sound is a compression wave — to travel, it needs a medium to compress: air, water, etc. (Regardless of what you have seen in movies, sound does not travel through a vacuum!)

Light is an electromagnetic wave — it causes fluctuations in the electric and magnetic fields that are everywhere. It can cross a vacuum, as it does to reach us from the sun.

Electromagnetic waves travel at about 300 million meters per second in a vacuum. The waves travel slower through different types of matter. For example, an electromagnetic wave travels at 225 million meters per second in water.

Electromagnetic waves come in different frequencies. For example, the light coming out of a red laser pointer is usually about $4.75 \times 10^{14}$ Hz. The wifi data sent by your computer is carried on an electromagnetic wave too. It is usually close to $2.4 \times 10^{6}$ Hz or $5 \times 10^{6}$ Hz.

Because we know how fast the waves are moving, we sometimes talk about their wavelengths instead of their frequencies. Since we know its speed and its oscillation rate, we can calculate its wavelength. Since we want the distance per oscillation, we can divide the speed by the frequency. The light coming out of a laser pointer is $300 \times 10^{6} / 4.75 \times 10^{14} = 630 \times 10^{-9}$ m, or 630 nm.

### *Exercise 189* Wavelengths

*Working Space*

A green laser pointer emits light at $5.66 \times 10^{14}$ Hz. What is its wavelength in a vacuum?

*Answer on Page 1944*

This can be simplified as:

$$\lambda = \frac{c}{f}$$

where $c = 3 \times 10^8$ m/s (the speed of light in a vacuum)

We have given names to different ranges of the electromagnetic spectrum in Figure 75.1

| Name | Hertz | Meters |
|---|---|---|
| Gamma rays | $10^{20} - 10^{24}$ | $10^{-14} - 10^{-11}$ |
| X-rays | $10^{16} - 10^{20}$ | $10^{-11} - 10^{-8}$ |
| Ultraviolet | $10^{15} - 10^{16}$ | $10^{-8} - 10^{-7}$ |
| Blue | $\sim 6 \times 10^{14}$ | $\sim 5 \times 10^{-7}$ |
| Red | $\sim 4 \times 10^{14}$ | $\sim 7 \times 10^{-7}$ |
| Infrared | $10^{12} - 10^{14}$ | $10^{-6} - 10^{-3}$ |
| Microwaves | $10^{9} - 10^{12}$ | $10^{-3} - 10^{-1}$ |
| Radio waves | $10^{3} - 10^{9}$ | $10^{-1} - 10^{5}$ |

Figure 75.1: A table of different ranges of the electromagnetic spectrum.

(You may have heard of "cosmic rays" and wonder why they are not listed in this table. Cosmic rays are actually the nuclei of atoms that have been stripped of their electron cloud. These particles come flying out of the sun at very high speeds. They were originally thought to be electromagnetic waves, and were mistakenly named "rays".)

In general, the lower frequency the wave is, the better it passes through a mass. A radio wave, for example, can pass through the walls of your house, but visible light cannot. The people who designed the microwave oven chose the frequency of 2.45 GHz because the energy from those waves tended to get absorbed in the first few inches of food that it passed through.

## 75.1 The greenhouse effect

Humans have dug up a bunch of long carbon-based molecules (like oil and coal) and burned them, releasing large amounts of $CO_2$ into the atmosphere. It may not be obvious why that has made the planet warmer, but the answer is electromagnetic waves.

A warm object gives off infrared electromagnetic waves. That's why, for example, motion detectors in security systems are actually infrared detectors: even in a dark room, your body gives off a lot of infrared radiation.

You may have heard of "heat-seeking missiles." These are more accurately called "Infrared homing missiles" because they follow objects giving off infrared radiation – hot things like jet engines.

The sun beams a lot of energy to our planet in the form of electromagnetic radiation: visible light, infrared, ultraviolet. (How much? At the top of the atmosphere directly facing the sun, we get 1,360 watts of radiation per square meter. That is a lot of power!)

Some of that radiation just reflects back into space. 23% is reflected by the clouds and the atmosphere, while 7% makes it all the way to the surface of the planet and is reflected back into space.

The other 71% is absorbed. 48% is absorbed by the surface and 23% is absorbed by the atmosphere. All of that energy warms the planet and the atmosphere so that it gives off infrared radiation. The planet lives in equilibrium; the infrared radiation leaving our atmosphere is exactly the same amount of energy as that 71% of the radiation that it absorbs.

(If the planet absorbs more energy than it releases, the planet gets hotter. Hotter things release more infrared. When the planet is in equilibrium again, it stops getting hotter.)

So, what is the problem with $CO_2$ and other large molecules in the atmosphere? They absorb the infrared radiation instead of letting it escape into space. This means the planet must be hotter to maintain equilibrium.

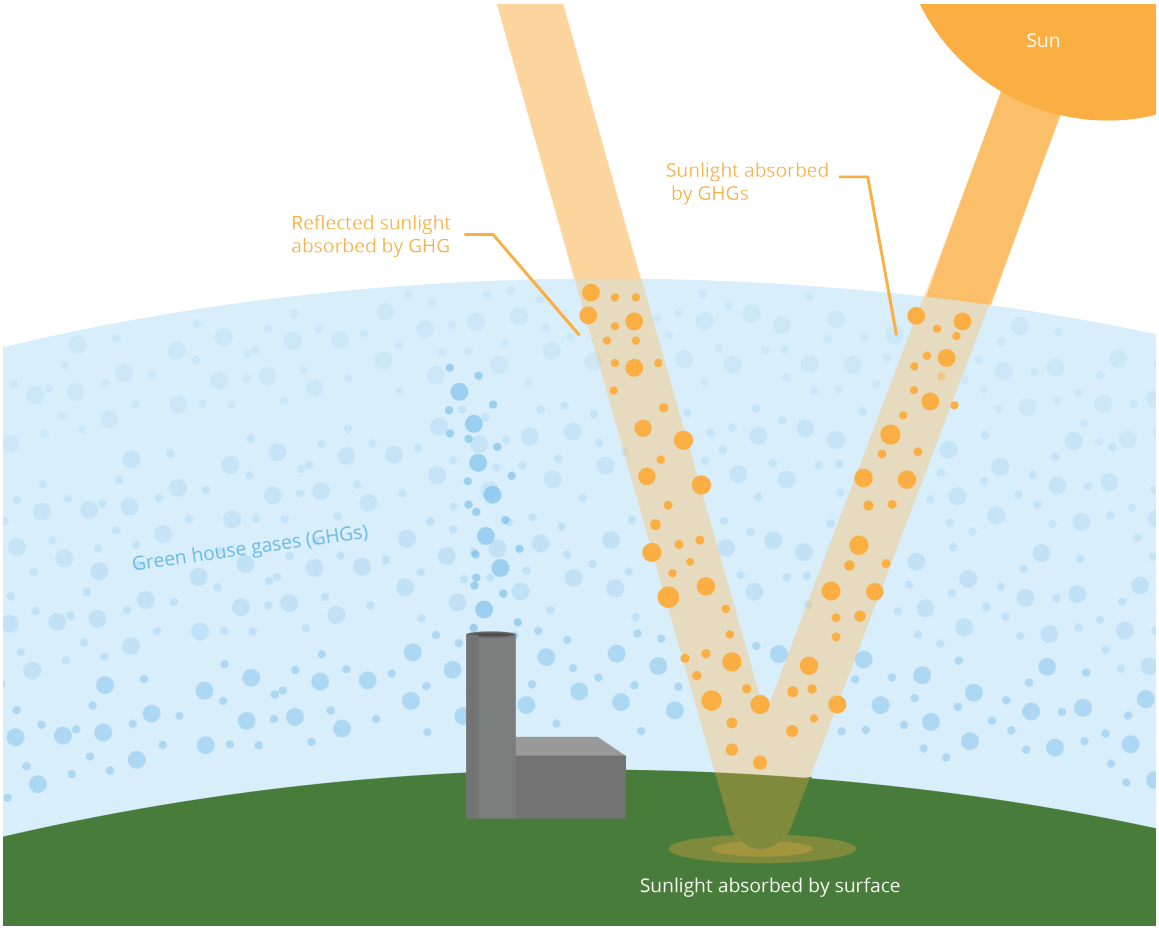



Figure 75.2: Greenhouse gas effects from the solar rays.

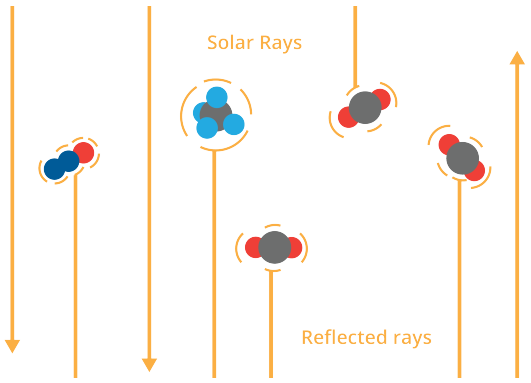



Figure 75.3: A zoomed in version of greenhouse gases.

The planet is getting hotter, and it is creating a multitude of problems:

- Weather patterns are changing, which leads to extreme floods and droughts.
- Ice and snow in places like Greenland are melting and flowing into the oceans. This is raising sea levels.
- Biomes with biodiversity are resilient. Rapidly changing climate is destroying biodiversity everywhere, which is making these ecosystems very fragile.
- In many places, permafrost, which has trapped large amounts of methane in the ground for millennia, is melting.

That last item is particularly scary, because methane is a large gas molecule — it absorbs even more infrared radiation than $CO_2$. As it escapes the permafrost, the problem will get worse.

Scientists are working on four kinds of solutions:

- **Stop increasing the amount of greenhouse gases in our atmosphere.** It is hoped that non-carbon based energy systems like solar, wind, hydroelectric, and nuclear could let us stop burning carbon. Given the methane already being released, it maybe too late for this solution to work on its own.
- **Take some of greenhouse gases out of our atmosphere and sequester them somewhere.** The trunk of a tree is largely made up of carbon molecules. When you grow a tree where there had not been one before, you are sequestering carbon inside the tree. There are also scientists that are trying to develop a process that pulls greenhouse gases out of the air and turn them into solids.
- **Decrease the amount of solar radiation that is absorbed by our planet and its atmosphere.** Clouds reflect a lot of radiation back into space. Could we increase the cloudiness of our atmosphere? Or maybe launch mirrors into orbit around our

planet?

- **Adapt to the changing climate.** These scientists are assuming that global warming will continue, and are working to minimize future human suffering. How will we relocate a billion people as the oceans claim their homes? When massive heat waves occur, how will we keep people from dying? As biodiversity decreases, how can we make sure that species that are important to human existence survive?

What are the greenhouse gases and how much does each contribute to keeping the heat from exiting to space? These numbers are still being debated, but this will give you a feel:

| | | |
|---|---|---|
| Water vapor | $H_2O$ | 36 - 72 % |
| Carbon dioxide | $CO_2$ | 9 - 26 % |
| Methane | $CH_4$ | 4 - 9 % |
| Ozone | $O_3$ | 3 - 7 % |

Notice that while we talk a lot about carbon dioxide, the most important greenhouse gas is actually water. Why don't we talk about it? Given the enormous surfaces of the oceans, it is difficult to imagine any way to permanently decrease the amount of water in the air. Additionally, a great deal of water in the air is in the form of clouds, which help reflect radiation before it is absorbed.

CHAPTER 76

# Reflection and Photons

What happens when light hits a mass?

## 76.1 Photons

In a previous chapter, we talked about light as a wave, and we mentioned that each color in the rainbow is a different wavelength. You can also think of light as particles of energy called *photons*. The term photons was coined by physicist Gilbert Newton Lewis, the same Lewis known for the Lewis dot structures we have talked about at the beginning of the sequence!

Every photon comes with an amount of energy that determines what color it is. When we are talking about light interacting with objects, your intuition will be right more often if you think of light as a beam of photons. These photons travel at the *vacuum speed of light*, more often called the ***vacuum speed of light***. This speed, in empty space or vacuums, is given by $2.9979 \times 10^8\,\mathrm{m/s}$. Photons also

- behave like particles and a wave.
- have zero mass
- has zero charge
- can be created and destroyed when radiation is present
- release electrons upon collisions.

**Formula for energy of a photon**

For any given photon, its energy, $E$, is given by

$$E = hf = \frac{hc}{\lambda}$$

where $c$ is the speed of light, $\lambda$ is the wavelength of the electromagnetic wave, and $h$ Planck's constant: $6.63 \times 10^{-34} m^2 kg/s$. Note that the angular frequency $f = \frac{c}{\lambda}$, and Planck's constant represents the quantum of action, establishing that energy is not continuous but exists in discrete packets called quanta. We will talk in a future

chapter about how charge and energy is quantized, or packeted in integerized values which we call discrete.

For example, a red laser light has a wavelength of about 630 nm. So, the energy in each photon is:

$$\frac{(300 \times 10^6)(6.63 \times 10^{-34})}{630 \times 10^{-9}} = 3.1 \times 10^{-19} \text{ joules}$$

For units sake, this is often abbreviated as $E = \frac{1240}{\lambda}$ where $\lambda$ is in nanometers and $E$ is in electronvolts (eV).

### 76.1.1 The Photoelectric Effect

In classical physics, scientists like Planck and Einstein thought that if you shined a light on a metal long enough or brightened it enough, it would eventually knock an electron loose, like a wave slowly pushing a boat to shore. Experiments showed that if the light was the wrong color, meaning too low of a frequency, nothing happened, no matter how bright or powerful the light was.

This showed that light is quantized, and in order to travel up or down the energy levels of an atom, it must absorb or emit a photon with the exact energy difference between those levels. This phenomenon is known as the *photoelectric effect*.

The effect is governed by the conservation of energy. The energy of an incoming photon ($E_{photon}$) must be greater than the bond holding the electron to the metal. This is described by

$$E_{photon} = \Phi + K_{max}$$

where

- $E_{photon} = hf$: The energy of the incoming photon, where $h$ is Planck's constant and $f$ is the frequency.
- $\Phi$ (*Work Function*): The minimum energy required to eject an electron from a specific metal. Each material has a different "cost" to let an electron go.
- $f_0$ (*Threshold Frequency*): The specific frequency where the photon has exactly enough energy to match the work function ($\Phi = hf_0$). If $f < f_0$, no electrons are ejected. If $f = f_0$, electrons are released but have zero kinetic energy ($K_{max} = 0$).
- $K_{max}$: The "leftover" kinetic energy of the electron once released from the metal.

This is how solar panels and digital camera sensors turn light into electricity. Solar panels work like a one-way road for light energy. First, sunlight hits the panel in the form of photons, which act like tiny kicks that knock electrons loose from their atoms. Because the panel has a built-in internal electric field, these loose electrons are forced to move in only one direction (back into the panel), creating a steady flow of Direct Current (DC) electricity that can power your home. We are going to look at camera sensors in depth in a future workbook.

### 76.1.2 Photons and Momentum

In classical mechanics, we define momentum as the product of mass and velocity ($p = mv$).

This leads to a common question: if a photon has zero rest mass, how can it possess momentum? Energy and momentum are linked, even for massless particles. For a photon, the momentum ($p$) is inversely proportional to its wavelength ($\lambda$):

$$p = \frac{h}{\lambda} \quad \text{or} \quad p = \frac{E}{c} \tag{76.1}$$

**Application:** For example, solar sails are spacecraft propulsion systems in space. These large, mirror-like sails in space can intercept millions of photons. As these photons reflect off the surface, they undergo a change in momentum ($\Delta p$), which, according to Newton's Second Law ($F = \frac{\Delta p}{\Delta t}$), exerts a constant force on the spacecraft, allowing for fuel-free propulsion.

#### Photonic Interactions

When a photon comes from the sun and hits an object, one of several things can happen:

- The energy of the photon is absorbed by the object. It makes the object a little warmer. If a large proportion of photons hitting the mass are absorbed like this, we say the object is "black".
- The photon bounces off the object. If the surface is very smooth, the photons bounce in a predictable manner, and we call this *reflection* and we say the object is "shiny".
- If the surface is rough and the photons are not absorbed, the photons are scattered in random directions. We call this *diffusion*. If most of the photons hitting an object are bounced in random directions, we say that the object is "white".
- The photon passes through the mass. If the mass has smooth surfaces and a consistent composition, the photons will pass through the mass in a predictable manner. We say that the mass is "transparent".

- If the photons pass through, but in an unpredictable, scattering manner, we say the mass is "translucent".

No object absorbs every photon, but chemists are always coming up with "blacker" materials. Vantablack, for example, is a super-black paint that absorbs 99.965% of all photons in the visible spectrum.

No object reflects every photon, but a mirror is pretty close. Let's talk about reflections in a mirror.

## 76.2 Reflection

When a beam of light hits a mirror, it bounces off the mirror at the same angle it approached from. That is, if it approaches nearly perpendicular to the mirror, it departs nearly perpendicular to the mirror. If it hits the mirror at a glancing angle, it departs at an angle close to the mirror's surface. A good video to see this in action can be found here: https://www.youtube.com/shorts/qA_VQZfTiUA

### 76.2.1 Law of Reflection

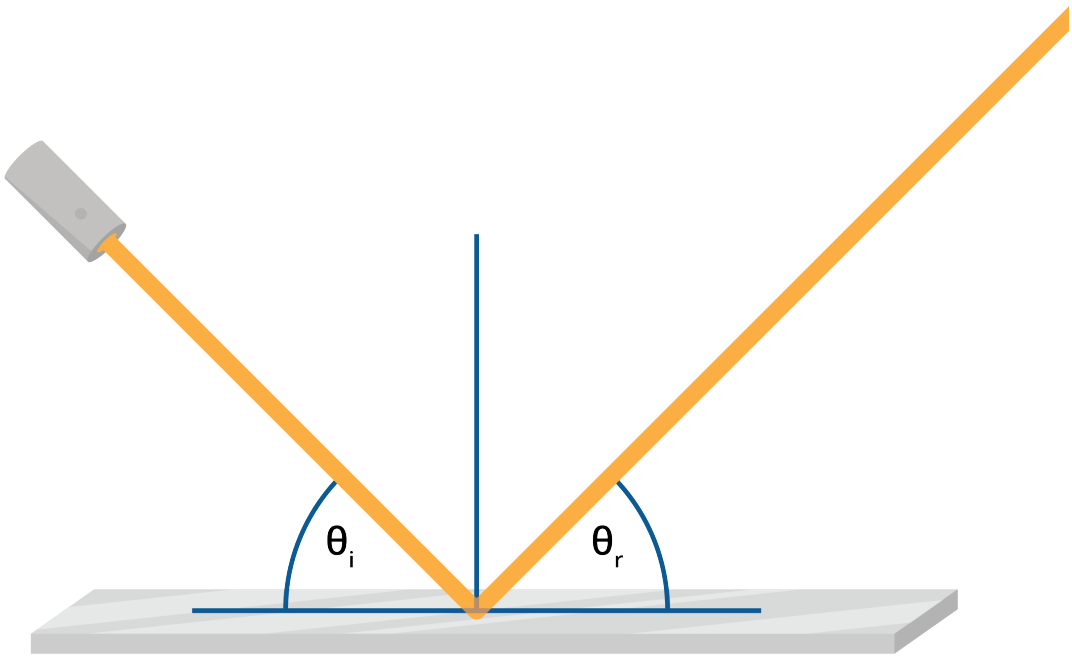


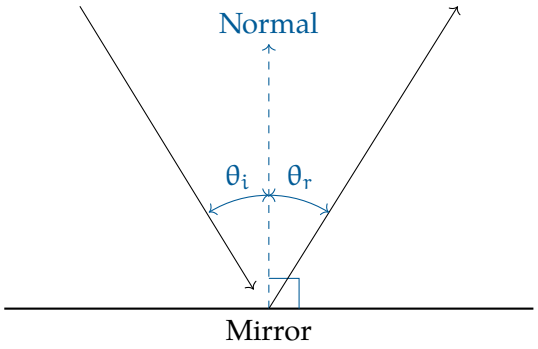

Figure 76.1: The angle of *incidence* is equal to the angle of *reflection*.

**Law of Reflection**

The angle of incidence, denoted as $\theta_i$, is equal to the angle of reflection, denoted as $\theta_r$. This law can be mathematically expressed as:

$$\theta_i = \theta_r$$

where $\theta_i$ is the angle between the incident light ray and the normal to the surface, and $\theta_r$ is the angle between the reflected light ray and the normal.




## *Exercise 190* Law of Reflection

*Working Space*

You are standing 4 meters from a mirror hung on a wall. The bottom of the mirror is the same height as your chin, so you can't see your whole body. You stick a piece of masking tape to your body.

You walk forward until you are only 3 meters from the mirror, then put a piece of masking tape on your body at the new cut-off point. Is the new masking tape higher or lower on your body?

*Answer on Page 1944*

### *Exercise 191* Photons and Color

*Working Space*

There are red photons.

Are there black photons?

Are there white photons?

Are there yellow photons?

*Answer on Page 1944*

## 76.3 Ellipses and Curved Mirrors

Flat mirrors are common and useful, but things get more interesting once you bend the mirrors. In this section, we are going to talk about a few different kinds of curved mirrors.

### 76.3.1 The Reflective Properties of Circles and Spheres

For example, if you were inside a circular room (a cylinder, actually), you could imagine standing in the center and pointing a flash light in any horizontal direction. The beam of light would bounce right back to you. See Figure 76.2

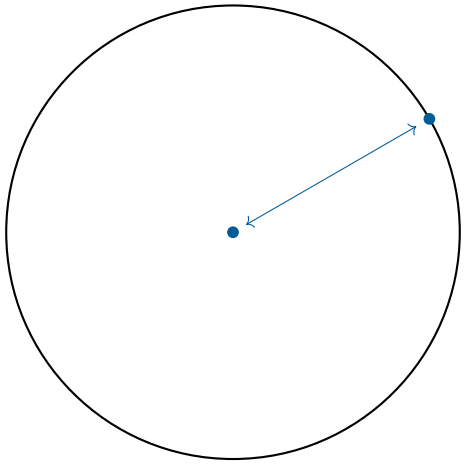


Figure 76.2: A circle would reflect back onto itself.

How do you know this? Because the tangent line is always perpendicular to the radius to the point of tangency:

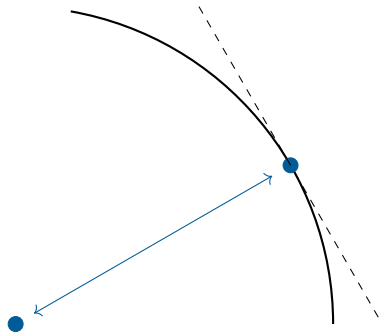


Figure 76.3: The reflection on a circle uses a radius which hits a tangent line as a perpendicular bisector.

You could create a spherical room with mirror walls. You would create a platform in the center where you could stand, and if you pointed your flashlight in any direction, its beam of light would shine back at you.

### 76.3.2 Ellipses and Ellipsoids

Intuitively, you know what an ellipse is: an oval. However, the ellipse is actually an oval with some special properties. This is a good time to talk about those properties.

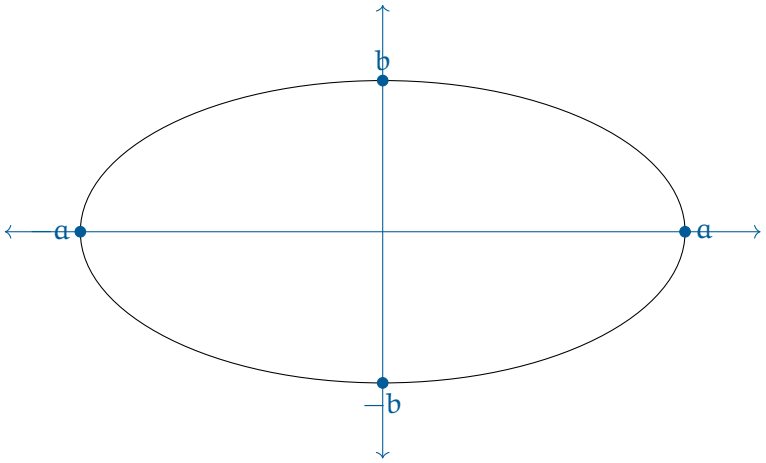
Mathematicians talk about a *standard* ellipse. A standard ellipse is centered on the origin $(0,0)$ and its long axis is parallel with the $x$-axis or the $y$-axis.

**Equation for a Standard Ellipse**

To be precise, a standard ellipse is the set of points $(x,y)$ that are solutions to the equation

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Note that $(a,0),(-a,0),(0,b),(0,-b)$ are all part of the set. The complete set looks like this:




The area contained inside this ellipse is given by

$$A = \pi a b$$

We can now talk about two special points: the *foci*. Each focal point is on the long axis of the ellipse. Let's assume for a moment that $a > b$. (Everything works the same if $b > a$, but it gets confusing if we try to deal with both cases simultaneously.)

If $p$ is a point on the ellipse, the distance from $p$ to focal point 1 plus the distance from $p$ to focal point 2 is always $2a$.

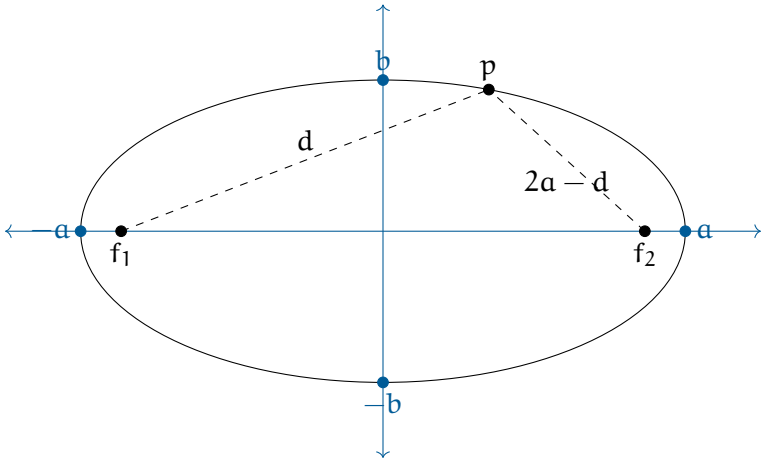



Figure 76.4: The both foci and point $p$ forms a triangle with legs $2a - d$ or $d$.

How do we find the foci? We know they are on the long axis and that they are symmetrical across the short axis. All we need to know is how far are they from the short axis.

**Distance from Center to the Foci**

If you have an ellipse with a long axis that extends $a$ from the center, and a short axis that extends $b$ from the center, the foci lie on the long axis and are $c$ from the center. Where

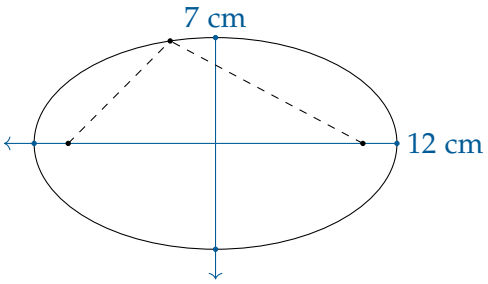
$$c = \sqrt{a^2 - b^2}$$

## *Exercise 192* Foci of an ellipse

*Working Space*

You need to draw an ellipse that is 12 cm long and 7 cm wide. You have a string, two pushpins, a ruler, and a pencil. Using the ruler, you draw two perpendicular axes.

You will stick one pin at each focal point. Each end of the string will be tied to a push pin. Using the pencil to keep the string taut, you will draw an ellipse.




How far from the short axis are the pushpins placed?

How long is the string between them?

*Answer on Page 1945*

**The Reflective Property of Ellipses**

Here is something else that is wonderful about an ellipse: Pick any point p on the ellipse. Draw a line from p to each focal point. Draw the line tangent to the ellipse at p. You will see that the angle between the tangent and the line to focal point 1 is equal to the angle between the tangent and the line to focal point 2. See Figure 76.5

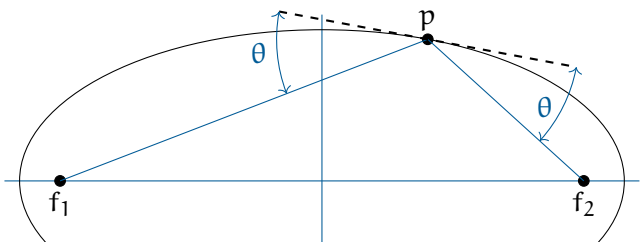



Figure 76.5: Reflective Property of Ellipses.

This is known as "The Reflective Property of Ellipses". Imagine you and your friend Fred are at an ellipse-shaped skating rink, and the edge of the rink is mirrored. You sit at one focal point and your friend sits at the other. If you point a flashlight at the mirror (in any direction!), the beam will bounce off the wall and head directly for Fred.

If Fred ducks out of the way, the beam will bounce again and head back to you.

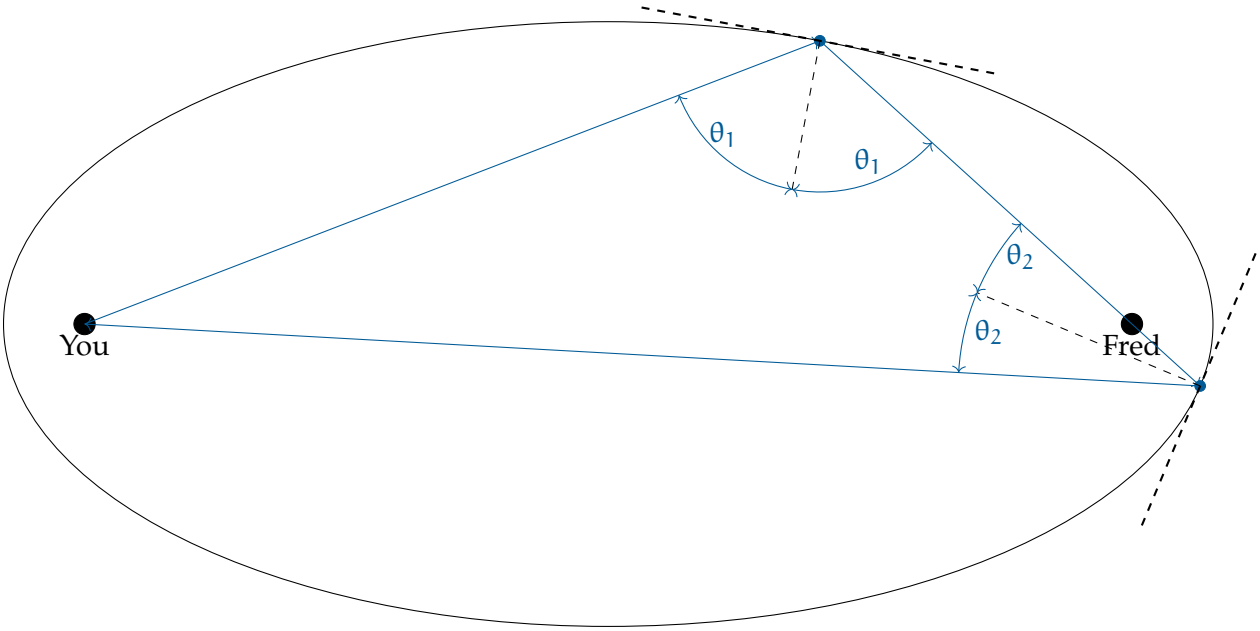



Figure 76.6: Your laser intersects Fred.

This will work for sound as well. If you whisper while on the focal point, Fred (at the other focal point) will hear you surprisingly well, because all the soundwaves that hit the wall will bounce (just like the light) straight at Fred.

### 76.3.3 Elliptical Orbits

One more fun fact about ellipses: We often imagine the planets traveling in circular orbits with the sun at the center — they actually travel in elliptical orbits, with the sun as one of the focal points.

The Earth is closest to the sun around January 3rd: 147 million km.

The Earth is farthest from the sun around July 3rd: 152 million km.

(Note that these dates are not the same as the solstices: The southern hemisphere is tilted the most toward the sun around December 21 and tilted most away around June 21.)

### 76.3.4 Ellipsoids

Just as we can pull the ideas of a circle into three dimensions to make a sphere, we can extend the ideas of the ellipse into three dimensions to talk about ellipsoids. Ellipsoids are like blimps.

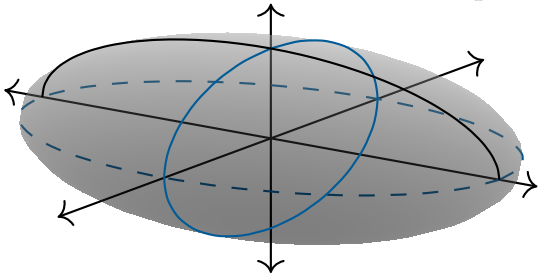
The standard ellipsoids are centered at the origin and aligned with the three axes.

**Equation for a Standard Ellipsoid**

To be precise, a standard ellipse is the set of points $(x, y, z)$ that satisfy the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Note that $(a, 0, 0), (0, b, 0), (0, 0, c)$ are all part of the set. The complete set looks like this:



The volume bounded by this ellipsoid is

$$V = \frac{4}{3}\pi abc$$

Of course, $a$, $b$, and $c$ can be any positive number, but in the real world, we find ourselves working regularly with ellipsoids where two of the numbers are the same.

### Oblate Spheroid

If two axes have the same length and one is shorter, you get something that looks like a sphere compressed in one direction — like a pumpkin. These are called *oblate spheroids*.

The Earth is actually an oblate spheroid; the axis that goes through the north and south pole is shorter than the axes that pass through the equator. How much shorter? Just a little, relatively speaking. The equator is 6,378 km from the center of the Earth; the north pole is 21 km closer.

### Prolate Spheroid

If two axes have the same length and one is longer, you get something that looks like a sphere stretched in one direction — like a rugby ball. It is called a *prolate spheroid*.

Like an ellipse, prolate spheroids have two focal points.

**Focal Points of a Prolate Spheroid**

If the long axis has a radial length of $a$ and the two shorter axes have radial length $b$, then the focal points are on the long axis. The distance from the center to the focal point is given by

$$c = \sqrt{a^2 - b^2}$$

For any point $p$ on the prolate spheroid, the sum of the distances from $p$ to the focus points will always be $2a$.

It has the reflective property: A photon shot in any direction from one focal point will bounce off the wall and head directly at the other.

## *Exercise 193* Volume of Ruby Ball

*Working Space*

Some jokesters once thought it would be fun to make something that looked like a rugby ball, but made out of lead.

A rugby ball is about 30 cm long and has a circumference of 60 cm at its midpoint. A cubic centimeter of lead has a mass of 11.34 grams.

How much would a solid (not hollow) lead ruby ball weigh?

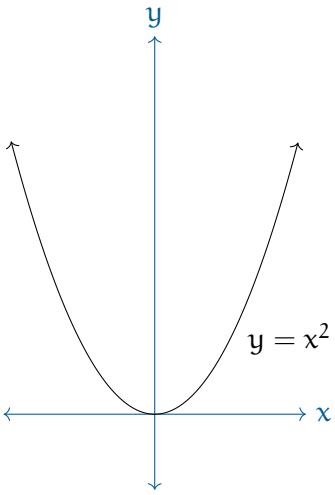
*Answer on Page 1945*

### 76.3.5 Parabolas and Parabolic Reflectors

You are familiar with quadratic functions:

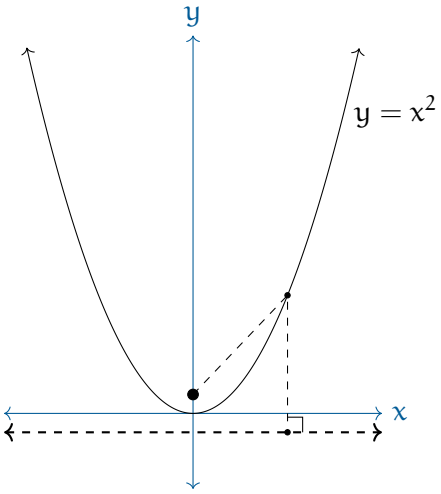
$$y = ax^2 + bx + c$$

If $a$ is not zero, the graph of a quadratic is a curved line called a *parabola*. The first parabola that most mathematicians think of is the graph of $y = x^2$:




Every parabola has a *focus* and a *directix*. The focus is a point on the parabola's axis of symmetry. The directrix is a line perpendicular to the axis of symmetry. Every point on the parabola is equal distance from the focus and the directrix.

For the graph of $y = x^2$, the focus is the point $(0, \frac{1}{4})$. The directrix is the line $y = -\frac{1}{4}$:




For example, the point $(1, 1)$ is on this parabola. It is 5/4 from the directorix. How far is it from the focus? 1 horizontally and 3/4 vertically.

$$\sqrt{1^2 + \left(\frac{3}{4}\right)^2} = \frac{5}{4}$$

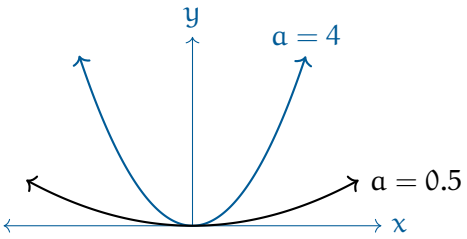
Thus, we have confirmed that $(1, 1)$ is equal distances from the focus and the directrix.

When we think about a parabola and its properties, we usually rotate and translate it to be symmetric around the $y$-axis, flip it so that it is low in the middle and rising on both sides, and push it up or down until the low point is on the $x$-axis.

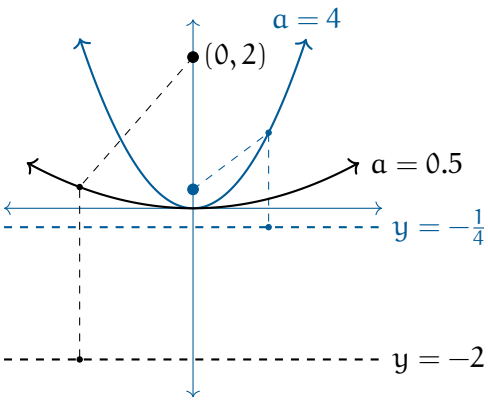
Then, they can all be written:

$$y = \frac{a}{4}x^2$$

where $a > 0$. If $a$ is small, the parabola opens wider.

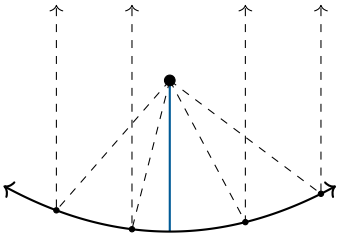



The focus is at $(0, \frac{1}{a})$ and the directorix is the line $y = -\frac{1}{a}$.



**Reflective Property of a Parabola**

Assume you have a parabola-shaped mirror. A beam of light shot from the focus will bounce off the mirror in the direction of the axis of symmetry:



This is why your flashlight has a parabolic mirror. The lightbulb is at the focus, so any photons that hit the mirror are redirected straight forward.

(Note that in the real world, we use parabolic dishes: a parabola rotated around its axis of symmetry.)

The reflection works exactly the same in reverse. There are solar cookers that are big parabolic mirrors. They let you put a pot on the focus point. You move the dish until its axis of symmetry is pointed at the sun.

You will also see a lot of antennas have parabolic dishes. Note that photons that come in parallel to the axis of symmetry are redirected to a single point: where the receiver is.

Figure 76.7: A real life parabolic dish as a satellite.

Sometimes in a science museum, you will see two parabolic dishes far apart and pointed at each other. One person speaks with their mouth at the focus of one. The other person listens with their ear at the focus of the other. Even though you are very far apart, it sounds like they are really, really close.

Figure 76.8: Two parabolic dishes send waves between each other.

This is because the speaker's parabolic wall focuses the sound energy in a nice beam the size of the wall pointed straight at the listener's parabolic wall. The listener's wall focuses

the energy of that beam at the listener's ear.

Chapter 77

# Refraction

## 77.1 Refraction

The refraction of light is a phenomenon where light changes its direction when it passes from one medium to another. The change in direction is due to a change in the speed of light as it moves from one medium to another. Do not confuse this with rarefaction, which sounds similar and is a similar wave concept for decompression!

## 77.2 Snell's Law

This phenomenon is explained by Snell's law, which states:

$$n_1 \cdot \sin(\theta_1) = n_2 \cdot \sin(\theta_2) \tag{77.1}$$

where:

- $n_1$ and $n_2$ are the indices of refraction for the first and second media, respectively. The index of refraction is the ratio of, c the speed of light in a vacuum to, $v$ the speed of light in the medium, $n = \frac{c}{v}$. It is a dimensionless quantity. $n \geq 1$ for all materials, such that $n_{\text{vacuum}} = 1$ and $n_{\text{air}} \approx 1$. See a list of all mediums here: https://en.wikipedia.org/wiki/List_of_refractive_indices
- $\theta_1$ and $\theta_2$ are the angles of incidence and refraction, respectively. These angles are measured from the normal (perpendicular line) to the surface at the point where light hits the boundary.

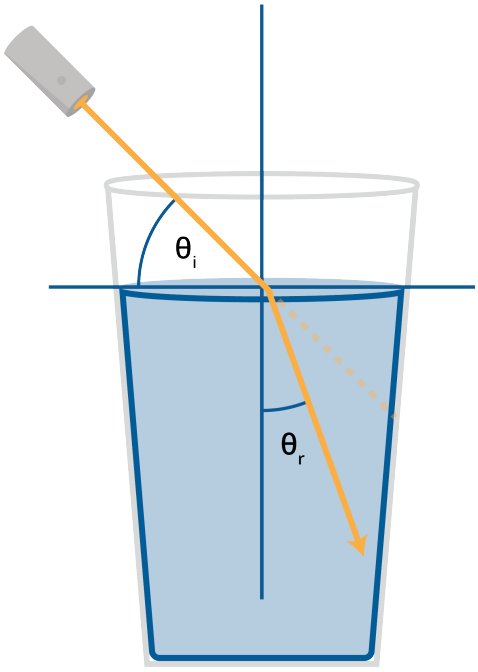



Figure 77.1: Refraction occurs when light changes the medium it is in.

The angle of incidence ($\theta_1$) is the angle between the incident ray and the normal to the interface at the point of incidence. Similarly, the angle of refraction ($\theta_2$) is the angle between the refracted ray and the normal.

When light travels from a medium with a lower refractive index to a medium with a higher refractive index, it bends towards the normal. Conversely, when light travels from a medium with a higher refractive index to one with a lower refractive index, it bends away from the normal.

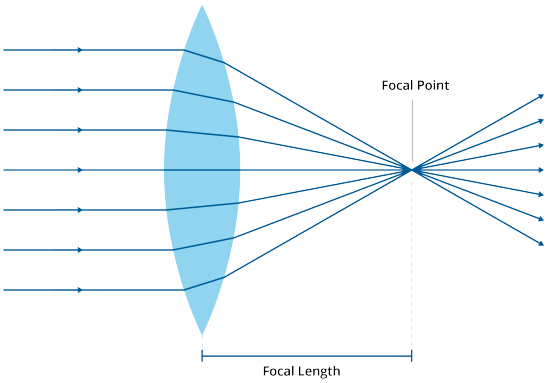
## 77.3 Diffraction

# Chapter 78

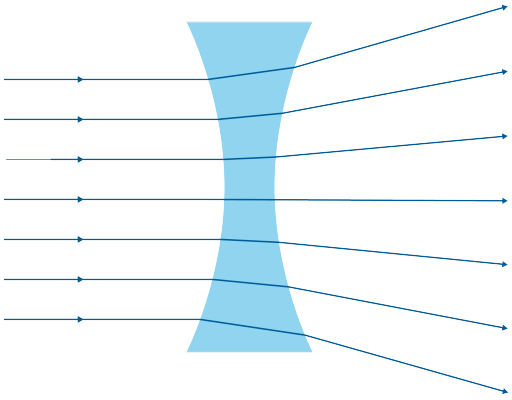
# Lenses

Lenses are optical devices with perfect or approximate axial symmetry that transmit and refract light, converging or diverging the beam. There are two main types of lenses, distinguished by their shape and the way they refract light:

- **Converging (or Convex) Lenses:** These are thicker at the center than at the edges. When parallel light rays enter a convex lens, they converge to a point called the focal point. Examples of converging lenses include magnifying glasses and camera lenses.




- **Diverging (or Concave) Lenses:** These are thinner at the center than at the edges. When parallel light rays enter a concave lens, they diverge or spread out. These lenses are often used in glasses to correct nearsightedness.



## 78.1 Focal Length

The focal length of a lens is the distance between the center of the lens and the focal point. It is determined by the lens shape and the refractive index of the lens material. For a converging lens, the focal length is positive; for a diverging lens, the focal length is negative.

## 78.2 Refractive Index

The refractive index of a material is a measure of how much the speed of light is reduced inside the material. The refractive index $n$ of a material is given by the ratio of the speed of light in a vacuum $c$ to the speed of light $v$ in the material:

$$n = \frac{c}{v}$$

The refractive index affects how much a light ray changes direction, or refracts, when entering the material at an angle. A higher refractive index indicates that light travels slower in that medium, and the light ray will bend more towards the normal.

Lenses work by refracting light at their two surfaces. By choosing the right lens shape and material, lenses can be designed to bring light to a focus, spread it out, or perform more complex transformations.

Chapter 79

# Cognitive Biases 1

In this section, we are going to take a look at research findings about *cognitive biases*. These are universal quirks found in the human thought process. Cognitive biases are a bit different from other kinds of biases, such as racial biases. Everyone, regardless of nationality, race, or gender is subject to these cognitive traps. You might be wondering, why do I need to learn about cognitive science in order to be an engineer? The most important tool we have as problem solvers is our own minds. We are going to be looking at ways that our minds can trip us up.

Our brains were designed over millions of years by the evolutionary process. The resulting mind is an amazing and powerful tool, but it is not flawless. The human brain has tendencies (or biases) that nudge us toward bad judgment and poor decisions.

When someone first gave you a hammer they handed it over with a warning: "Don't hit your thumb!" No matter how careful you are with the hammer, at some point you will still hit your thumb. It's the same with cognitive biases. In the course of life, all of us will fall prey to these cognitive biases. Knowing about them is the first step in protecting ourselves.

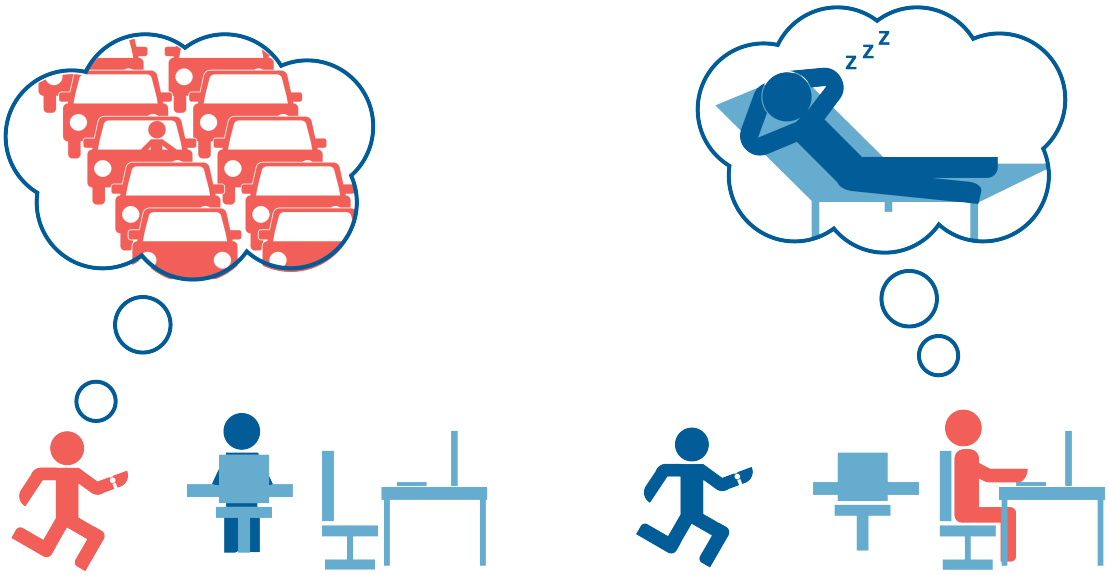
It would be irresponsible to teach you powerful ideas without also teaching you about the cognitive biases that follow them. There are about 50 that you should know about, but let's start with only a few.

## 79.1 Fundamental Attribution Error

You tend to attribute the mistakes of another person to their character, but attribute your own mistakes to the situation.

If someone asks you "Why were you late for work today?" you are likely to have an excuse, such as "I got stuck in a crazy traffic jam."

However, if you notice your coworker is late for work, you are likely to think "My coworker is lazy."




The solution? Cut people some slack. You probably don't know the whole story, so assume that their character is as strong as yours.

Or maybe you also need to hold yourself to a higher standard? Do you find yourself frequently rationalizing your bad judgment, lateness, or rudeness? This could be an opportunity for you to become a better person whose character is stronger regardless of the situation.

## 79.2 Self-Serving Bias

*Self-serving bias* is when you blame the situation for your failures, but attribute your successes to your strengths.

For example, when asked "Why did you lose the match?" you are likely to answer "The referee wasn't fair." When you are asked "Why did you win the match?" you are likely to answer "Because I have been training for weeks, and I was very focused."

This bias tends to make us feel better about ourselves, but it makes it difficult for us to be objective about our strengths and weaknesses.

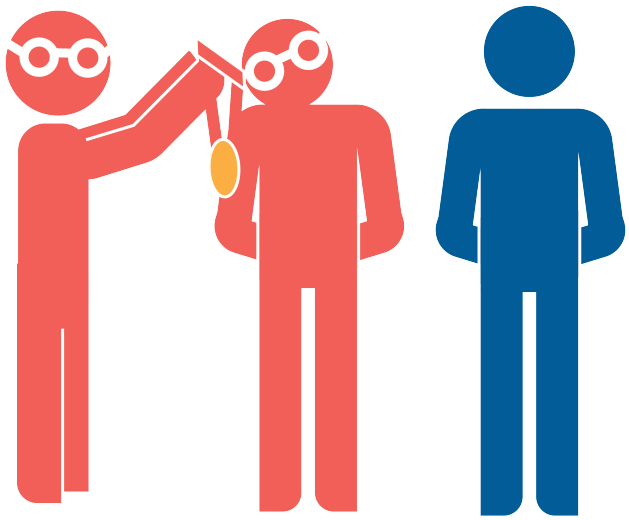
## 79.3 In-group favoritism

*In-group favoritism*: We tend to favor people who are in a group with us over people who are not in groups with us.

When asked "Who is the better goalie, Ted or John?"

If Ted is a Star Trek fan like you, you are likely to think he is also a good goalie.

As you might imagine, this unconscious tendency is the source of a lot of subtle discrimination based on race, gender, age, and religion. As we mentioned earlier, racial bias isn't a cognitive bias, but one can still feed into the other.



## 79.4 The Bandwagon Effect and Groupthink

*The bandwagon effect* is our tendency to believe the same things that the people around us believe. This is how fads spread so quickly: one person buys in, and then the people they know have a strong tendency to buy in as well.

*Groupthink* is similar: To create harmony with the people around us, we go along with things we disagree with.

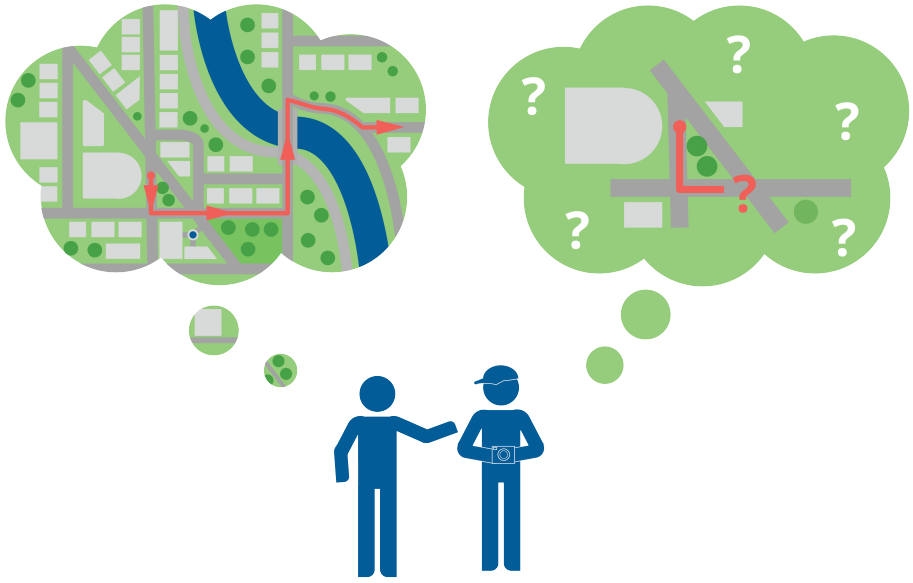
It takes a lot of perspective to recognize when those around us are wrong. And it takes even more courage to openly disagree with them.

## 79.5 The Curse of Knowledge

Once you know something, you tend to assume everyone else knows it too.

This is why teaching is sometimes difficult; a teacher will assume that everyone in the audience already knows the same things the teacher knows.

For example, imagine a local who has lived in a city for years giving directions to a tourist. The local has an in-depth understanding of the city, and gives overly quick and detailed instructions. The tourist politely smiles and nods, but stopped following after the local began listing unfamiliar street names and landmarks.



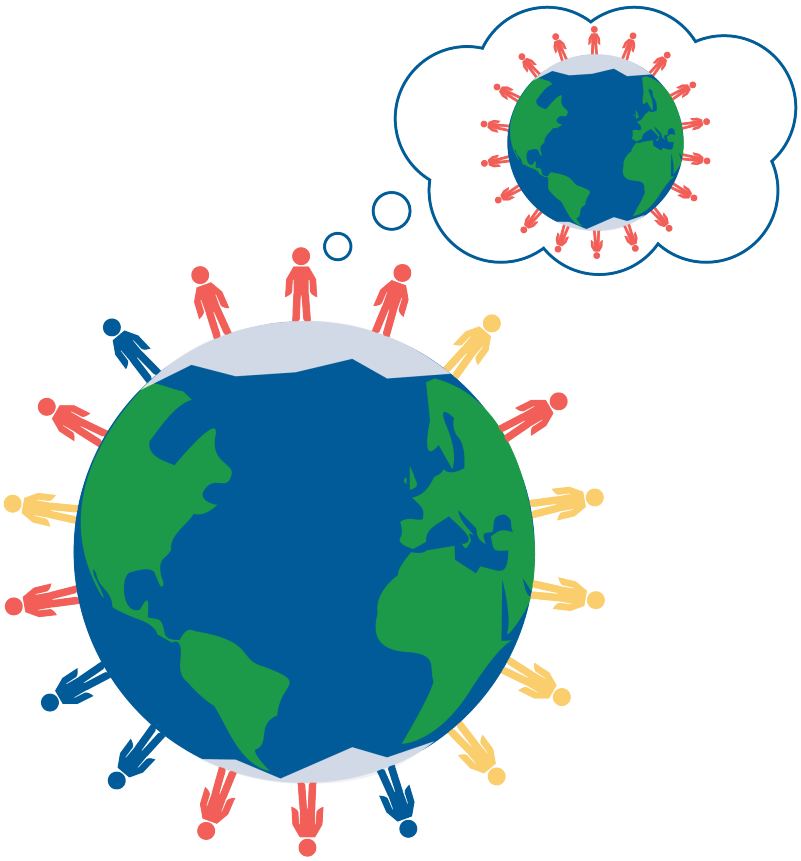
When we learn that a friend doesn't know something that we know, we are often very surprised. This surprise can sometimes manifest as hurtful behavior.

When we find a gap in our friend's knowledge, we try to remind ourselves that the friend certainly knows many things that we don't. We also try to imagine how it would feel if they teased us for our ignorance.

## 79.6 False Consensus

We tend to believe that more people agree with us than is actually the case. For example, if you are a member of a particular religion, you tend to overestimate the percentage of people in the world who are members of that religion.

When people vote in elections, they are often surprised when their preferred candidate loses. "Everyone, and I mean EVERYONE, voted for Smith!" they yell. "There must have been a mistake in counting the votes."

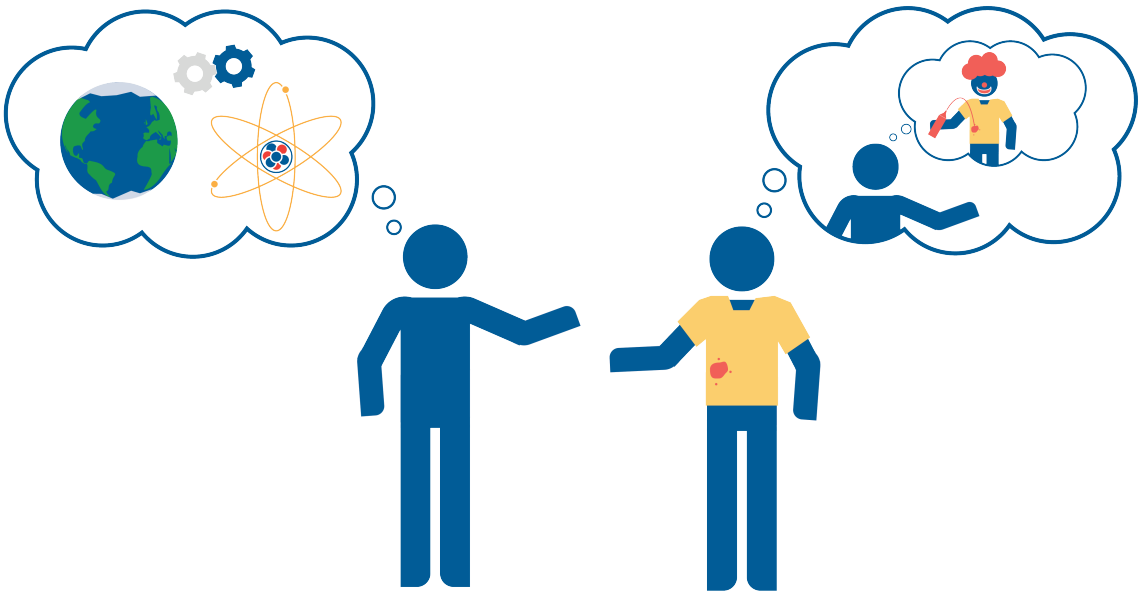


## 79.7 The Spotlight Effect

You tend to overestimate how much other people are paying attention to your behavior and appearance.

Think of six people that you talked to today. Can you even remember what shoes most of them were wearing? Do you care? Do you think any of them remember which shoes you wore today?

There is an old saying, "You would worry a lot less about how people think of you, if you realized how rarely they do."



## 79.8 The Dunning-Kruger Effect

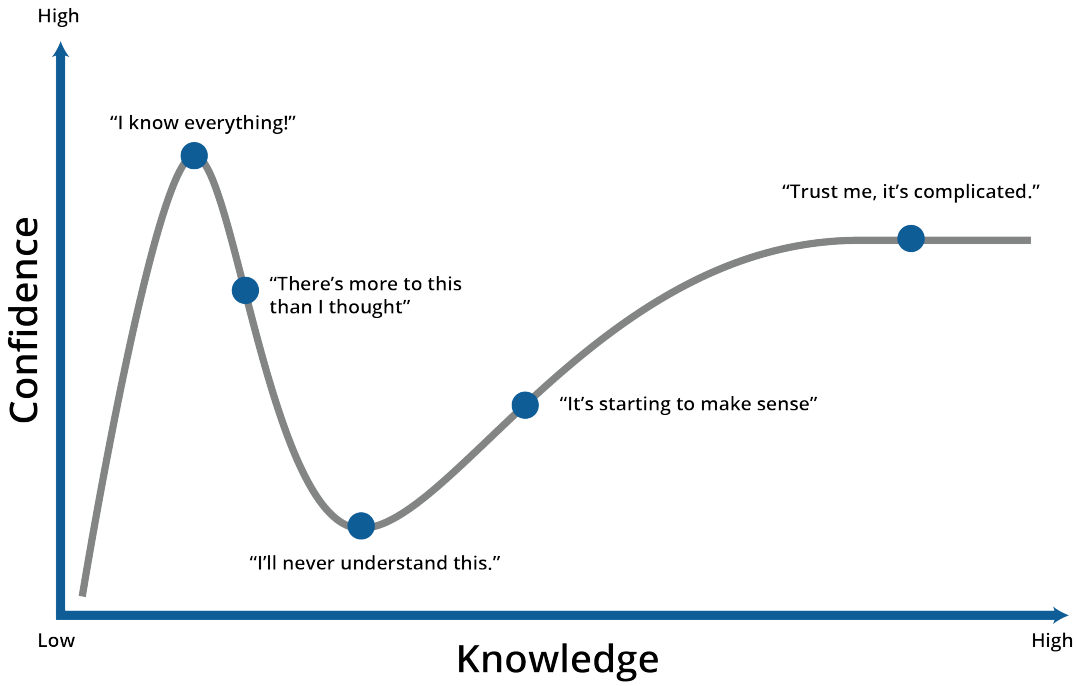
The less you know, the more confident you are.

When a person doesn't know all the nuance and context in which a question is asked, the question seems simple. Thus, the person tends to be confident in their answer. As they learn more about the complexity of the space in which the question lives, they often realize the answer is not nearly so obvious.

For example, a lot of people will confidently proclaim "Taxes are too high! We need to lower taxes." An economist who has studied government budgets, deficits, history, and monetary policy, might say something like "Maybe taxes *are* too high. Or maybe they are too low. Or maybe we are taxing the wrong things. It is a complex question."

When we are talking with people about a particular topic, we do our best to defer to the person in the conversation who we think has the most knowledge in the area. If we disagree with the person, we try to figure out why our opinions are different.

Similarly, you should assume that any opinion that is voiced in an internet discussion is, at best, wildly over-simplified. If you really care about the subject, read a book by a respected expert. Yes, a whole book — there are few interesting topics that can be legitimately explained in under 100 pages.

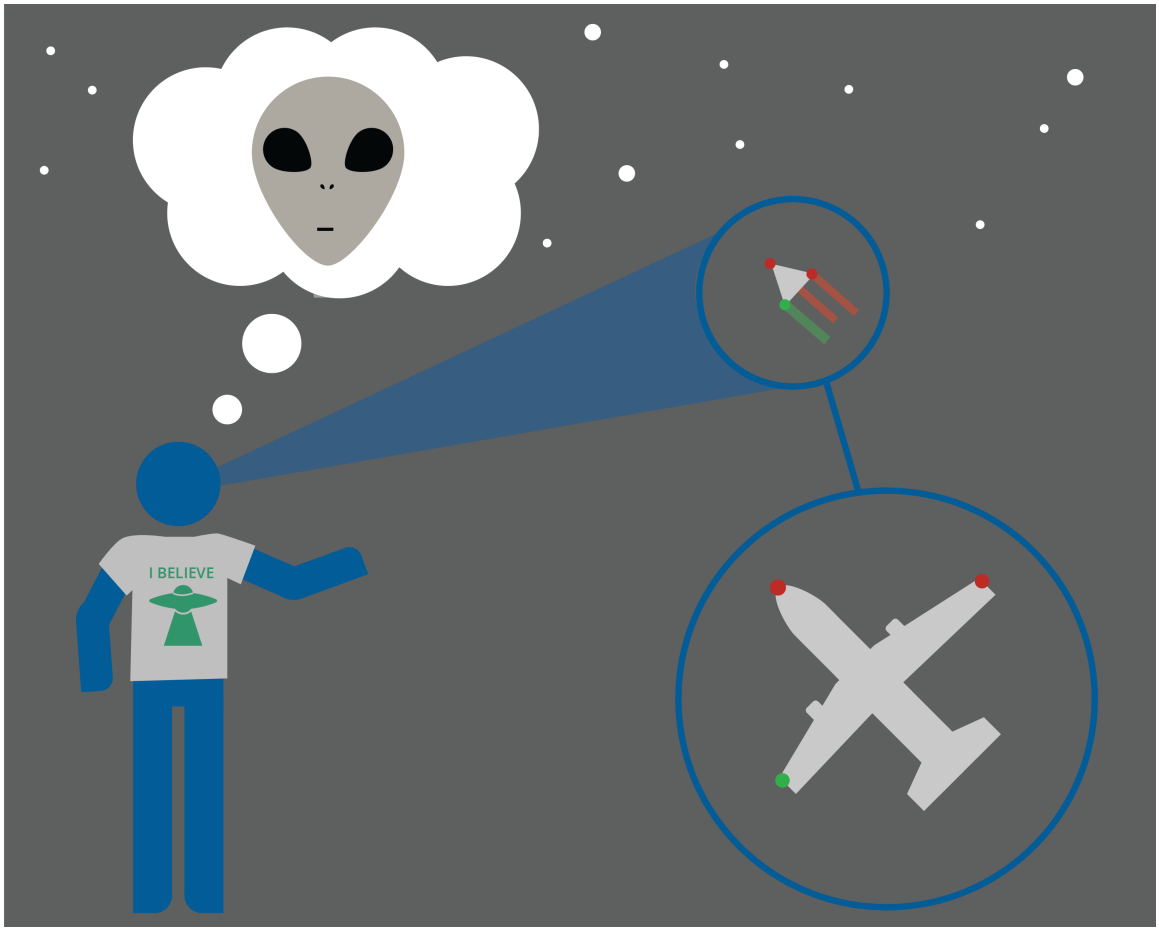



## 79.9 Confirmation Bias

You tend to find and remember information that supports beliefs you already have. You also tend to avoid and dismiss information that contradicts your beliefs.

If you believe that intelligent creatures have visited from other planets, you will tend to look for data to support your beliefs. When you find data that shows that it is just too far for any creature to travel, you will try to find a reason why the data is incorrect.

Confirmation bias is one reason why people don't change their beliefs more often.




Confirmation bias wrecks many, many studies. The person doing the study often has a hypothesis that they believe and very much want to prove true. It is very tempting to discard data that doesn't support the hypothesis. A person may even throw all the data away and experiment again and again until they get the result they want.

When you design an experiment, you must describe it explicitly before you start. You must tell someone: "If the hypothesis I love is incorrect, the results will look like this. If the hypothesis I love is correct, the results will look like that. And if the results look any other way, I have neither proved nor disproved the hypothesis."

Once the experiment is underway, you must not change the plan and you must not discard any data.

This is scientific integrity. You should demand it from yourself, and you should expect it from others.

**Watch a TED Talk and Learn More About Confirmation Bias**: What shapes our perceptions (and misperceptions) about science? In an eye-opening talk, meteorologist J. Marshall Shepherd explains how confirmation bias, the Dunning-Kruger effect and cog-

nitive dissonance impact what we think we know – and shares ideas for how we can replace them with something much more powerful: knowledge.

https://www.ted.com/talks/j_marshall_shepherd_3_kinds_of_bias_that_shape_your_worldview

## 79.10 Survivorship bias

You will pay more attention to those that survived a process than those who failed.

After looking at several old houses, you might say "In the 1880s, they built great houses." However, you haven't seen the houses that were built in the 1880s and didn't survive. Which houses tended to survive for a long time? Only the great houses – you are basing your opinion on a very skewed sample.

Chapter 80

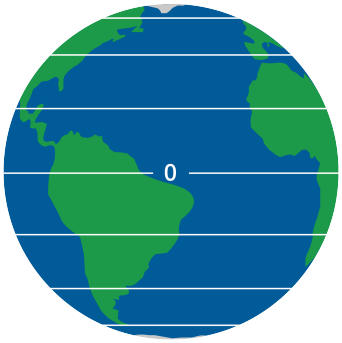
# Longitude and Latitude

## 80.1 Longitude and Latitude

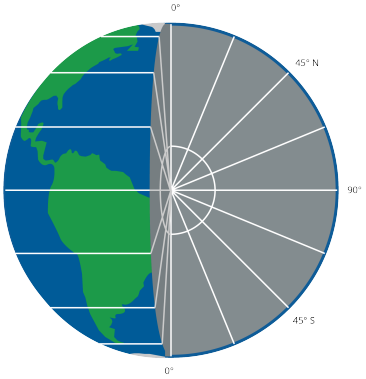
The Earth can be represented as a sphere, and the position of a point on its surface can be described using two coordinates: latitude and longitude.

Figure 80.1: A diagram of latitude and longitude.

Latitude is a measure of a point's distance north or south of the equator, expressed in degrees. It ranges from $-90°$ at the South Pole to $+90°$ at the North Pole, with $0°$ representing the Equator. (See figures 80.2a and 80.2b)



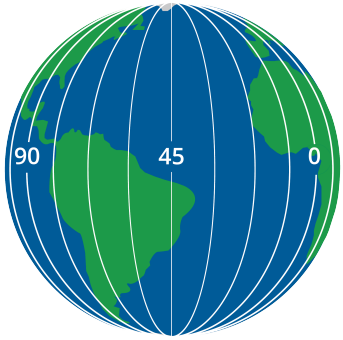

(a) Latitude drawn on the earth.



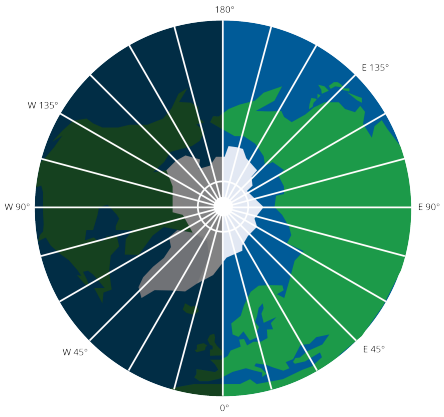

(b) A cross-section of the earth showing latitude.

Figure 80.2: Latitude illustrations

Longitude, on the other hand, measures a point's distance east or west of the Prime Meridian (which passes through Greenwich, England). It ranges from $-180°$ to $+180°$, with the Prime Meridian represented as 0°. (See figures 80.3a and 80.3b)




(a) Longitudinal lines drawn on the earth.




(b) A cross-section of the earth showing longitude.

Figure 80.3: Longitude illustrations.

The coordinates follow a system of latitude and then longitude (in the majority of contexts). Here is the Longitude and Latitude of four different large cities:

- **London, England**: 51° N 0° W
- **New York City, New York, USA**: 40° N 74° W

- **Sydney, New South Whales, Australia**: 33° S 150° E
- **Rio de Janeiro, Brazil**: 22° S 43° W

These coordinates may be broken down further into minute and second components as well, for further geometric accuracy. Each degree (°) is divided into 60 minutes (′). Each minute (′) is divided into 60 seconds (″). Just like in time. Note some do not

- **London, England**: 51° 30′ 26″ N 0° 7′ 39″ W
- **New York City, New York, USA**: 40° 42′ 46″ N 74° 0′ 22″ W
- **Sydney, New South Whales, Australia**: 33° 52′ 0″ S 150° 12′ 0″ E
- **Rio de Janeiro, Brazil**: 22° 54′ 0″ S 43° 12′ 0″ W

## 80.2 Nautical Mile

A nautical mile is a unit of measurement used primarily in aviation and maritime contexts. It is based on the circumference of the Earth, and is defined as one minute (1/60°) of latitude; in other words, **one minute of latitude is equal to 1 nautical mile.** This makes it directly related to the Earth's geometry, unlike a kilometer or a mile, which are arbitrary in nature. The exact value of a nautical mile can vary slightly depending on which type of latitude you use (e.g., geodetic, geocentric, etc.), but for practical purposes, it is often approximated as 1.852 kilometers or 1.15078 statute miles.

## 80.3 Haversine Formula

The *haversine formula* is an important equation in navigation for giving great-circle distances between two points on a sphere from their longitudes and latitudes. It is especially useful when it comes to calculating distances between points on the surface of the Earth, which we represent as a sphere for simplicity. See Figure 80.4

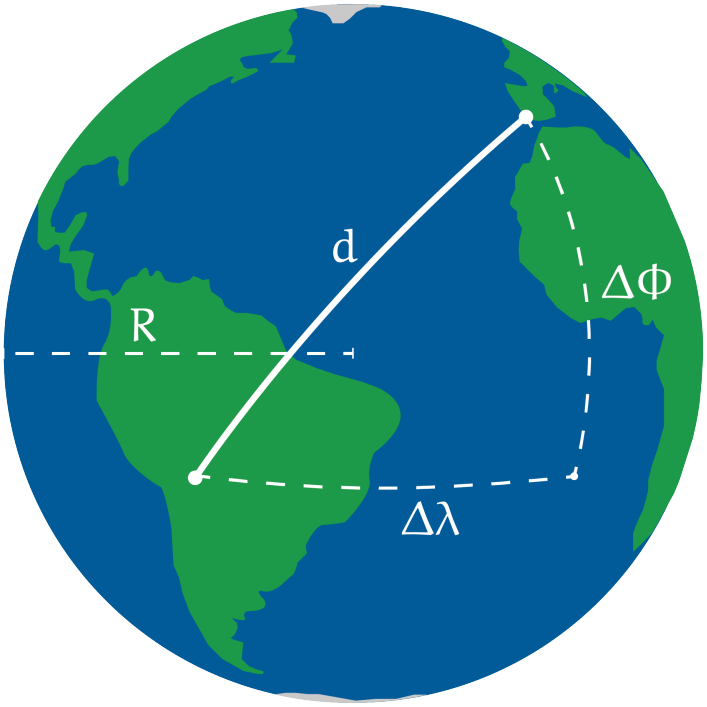



Figure 80.4: A diagram of the haversine formula.

In its basic form, the haversine formula is as follows:

The angle found using haversine $\text{haversine}(\theta) = \sin^2\left(\frac{\theta}{2}\right)$:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)$$

Compute $c$ which determines the angular distance using $\text{atan2}$, a computer science tangential function with 2 arguments equivalent to $2\arcsin(\sqrt{a})$[1]:

$$c = 2 \cdot \text{atan2}\left(\sqrt{a}, \sqrt{1-a}\right)$$

Find $d$, the true distance along the sphere:

[1]Note that this function is not available on calculators, but commonly used in computer science programs. You may use the singular argument equivalent in calculations.

$$d = R \cdot c$$

In one line, this is:

$$d = 2R \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)}\right)$$

Here, $\phi_1$ and $\phi_2$ represent the latitudes of the two points (in radians), $\Delta\phi$ and $\Delta\lambda$ represent the differences in latitude and longitude (also in radians), and $R$ is the radius of the Earth (using whatever unit of distance you desire). The result, $d$, is the distance between the two points along the surface of the sphere. Notice that $d$ is equal to the formula for the *arc length* of a sector. A sphere is just a bunch of circles with radius $r$, so it follows that we can use formulas for circles in our spherical calculations.

This is why planes fly arc-shaped paths rather than straight paths from points $A \rightarrow B$. Planes follow the shortest path between two points on a sphere, called a *great-circle route*. On a map, like plane flight-progress maps or the Mercator projection, the straight path looks very arc shaped, but this is just a geographical illusion. This straight distance can be calculated using the haversine formula, ultimately minimizing distance and fuel using the 'as-the-crow-flies' path.

CHAPTER 81

# Sampling Data

## 81.1 Planning a Study

Every day, decisions get made based on data. A city council votes on a new transit route based on ridership surveys. A school principal adjusts the lunch schedule based on student feedback. A pharmaceutical company decides whether to release a new drug based on clinical trial results. In every case, the quality of the decision depends entirely on the quality of the data behind it.

To collect data well, you need a good plan! That plan must specify what question you are trying to answer, how you will gather information, and who you will gather this information from.

### 81.1.1 Vocabulary

Five terms form the foundation of everything that follows. Learn to distinguish them from each other, because you will probably be tested on this.

***population*** The entire group of individuals or items that you want to learn about. The population is defined by the question you are asking.

***frame*** Also called a sampling frame. A list of all the members of the population from which a sample will be drawn. The frame is usually close to the population, but the two are not always identical. A school's enrollment database, a city's voter registration list, and a company's customer records are all examples of frames.

***sample*** The part of the population that is actually studied. You collect data from the sample and use it to draw conclusions about the population.

***sample survey*** The process of collecting information from a sample. The information gathered is then used to make inferences about population parameters.

***census*** The process of collecting information from every member of the population rather than just a sample. A census gives complete information, but it comes at a cost.

The relationship between these terms is important. The population is the full group you care about. The frame is your working list of that group. The sample is the slice of the frame you actually study. A census is what happens when the sample and the population are the same thing or equal in number.

## *Exercise 194* Identifying the Population and Sample

*Working Space*

A city's parks department wants to understand how often residents use public parks. Staff randomly select 200 households from the city's water billing records and conduct phone interviews.

1. What is the population?
2. What is the frame?
3. What is the sample?
4. Is this a census? Explain.

*Answer on Page 1945*

### 81.1.2 Why Not Always Use a Census?

A census tells you about everyone in the population. That sounds great, but it is not always realistic. If the group is small, like a football team, you can measure everyone.

Most of the time, a census is too difficult because:

- The population is too large. A company that wants to know what all households watch cannot contact every single one. - The measurement destroys the item. If testing a battery kills it, checking every battery would waste them all. - The information gets old too fast. If a company waits months to ask everyone, the results may no longer match what people think.

### *Exercise 195* Census or Sample?

*Working Space*

For each situation, decide whether a census is feasible or whether a sample should be used. Briefly explain your reasoning.

| Situation | Census or Sample? Explain |
|---|---|
| A school librarian wants to know which of her 8 student volunteers prefers morning or afternoon shifts. | |
| A battery manufacturer wants to test the lifespan of batteries by running them until they die. | |
| A government agency wants to estimate the average commute time of workers across a country of 40 million employed adults. | |
| A restaurant owner wants to know the dietary restrictions of the 22 guests attending a private event. | |

*Answer on Page 1946*

In most studies, sampling is the only practical option, and the quality of the conclusions you draw depends entirely on how well the sample was chosen.

## 81.2 Planning and Conducting Surveys

Not all samples are fair. How you pick people can make the group represent the population well, or it can leave out important parts.

### 81.2.1 Biased Sampling

A sample is *biased* if it favors some people or answers over others. Bias does not go away just because the sample is bigger.

These three common methods usually make biased samples.

*Judgmental sampling* is when one person decides who is included based on their own idea of what looks representative. A school inspector who visits only classrooms she already thinks are good is using judgmental sampling. The choice is based on their opinion.

A *convenience sample* uses whoever or whatever is easiest to reach. A journalist who interviews only people at the bus stop outside his office is using a convenience sample. Those people might not represent all commuters.

A *voluntary response sample* happens when people choose whether to answer. An online poll on a news site will attract mostly visitors with strong opinions. People who do not care much usually do not respond.

All three skip chance. Sampling methods based on random selection give everyone a fair shot and usually make more trustworthy results.

## 81.3 Simple Random Sampling

A *simple random sample* (SRS) is a sample chosen so every group of the same size has the same chance of being picked.

There are two ways to do it.

With *sampling with replacement*, you put someone back in the pool after choosing them. They could be picked again.

With *sampling without replacement*, you do not put them back. Each person can only be chosen once.

To get an SRS, you need a real random method.

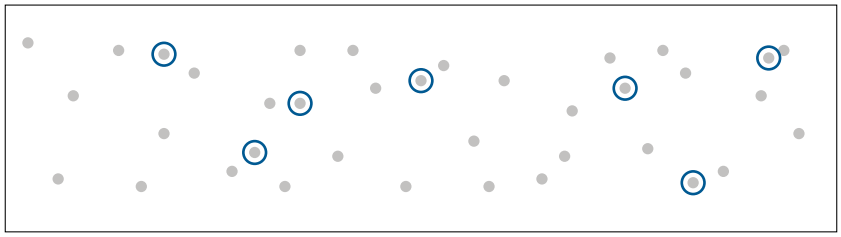


Figure 81.1: In simple random sampling, every individual in the population has an equal chance of being selected, and there is no grouping or pattern to who is chosen.

- Assign every member of the population a number. Use a random number table, entering at any random point and reading numbers sequentially. Select members whose numbers appear, skipping values outside your range and any repeats, until you have the sample size you need.

- Use a calculator. On a TI-84, `MATH` → `PRB` → `5:randInt(` generates random integers. The command `randInt(1,500,30)` produces 30 random integers between 1 and 500. Ignore repeats.
- Use a physical randomization device. Write each member's identifier on an identical slip of paper, mix thoroughly in a container, and draw without looking until the sample is complete.

The essential requirement in every case is that chance dictates who is selected, rather than convenience or judgment. This is what makes the sample fair and the results trustworthy.

## *Exercise 196* Selecting an SRS

A regional transit authority wants to survey 25 of its 300 bus drivers about workplace safety conditions.

1. Describe a complete procedure for selecting an SRS of 25 drivers using a random number table.
2. A supervisor suggests selecting the 25 drivers who are easiest to reach during the next shift change. What type of sample is this, and what bias might it introduce?
3. A colleague suggests selecting 5 drivers at random from each of the authority's 5 depots. Is this an SRS? Explain.

*Working Space*

*Answer on Page 1946*

**Sampling Real Data from the Web**

In this exercise you will load real world population data published by the World Bank, treat all 265 countries and territories as your population, draw a simple random sample, and compare your sample mean to the true population mean.

1. Import the necessary libraries.

```
import pandas as pd
```

2. Load the dataset directly from the web and filter to the year 2020.

```
url =
 ↪  "https://raw.githubusercontent.com/datasets/population/main/data/population.csv"
df = pd.read_csv(url)

pop2020 = df[df["Year"] == 2020][["Country Name",
 ↪  "Value"]].reset_index(drop=True)
print(pop2020.head())
print(f"Total entries: {len(pop2020)}")
```

3. Calculate the true population mean. This is the parameter you are trying to estimate.

```
true_mean = pop2020["Value"].mean()
print(f"True population mean: {true_mean:,.0f}")
```

4. Draw an SRS of 20 countries and compute the sample mean.

```
sample = pop2020.sample(n=20)
sample_mean = sample["Value"].mean()
print(f"Sample mean (n=20): {sample_mean:,.0f}")
```

5. Run step (d) five times and record the sample mean each time. How close is each to the true mean? Are they all the same?

6. Change the sample size to 80 and repeat step (e). How do the results compare? What does this tell you about the relationship between sample size and sampling error?

### 81.3.1 Other Methods of Random Sampling

Simple random sampling is not always the most practical or most efficient approach. Three other probabilistic methods appear regularly on the AP exam.

In *systematic sampling*, you list the population, choose a random start, and then pick every kth person. For example, to sample 40 students from 800, start at a random number between 1 and 20, then take every 20th student. This is simple when you have a full list.

In *stratified random sampling*, the population is split into groups called strata. Each group should be similar inside, like students in the same grade or patients in the same ward. Then you take a random sample from each group. This makes sure every part of the

population is represented. This method of random sampling is especially useful in medical studies, market research, political polling, educational testing, etc, where you want to make sure you have enough people from each group to compare them.

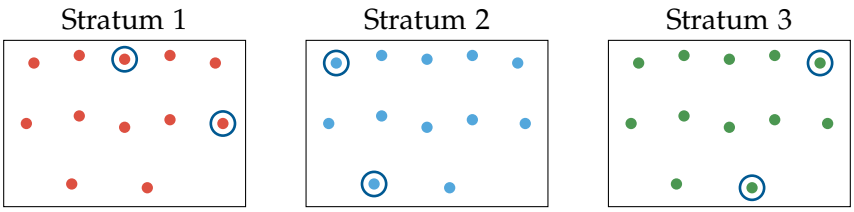



Figure 81.2: Each stratum is a single color, since members within a stratum share a characteristic. A small random sample (circled) is drawn from each stratum.

In *cluster sampling*, the population is split into groups called clusters. Each cluster should be like a mini-version of the whole population. Then you randomly pick some clusters and study everyone inside them. This works well when people are spread out over a wide area and it may be difficult or expensive to reach everyone. For example, a company that wants to survey its customers across the country might randomly select 10 cities and then survey all customers in those cities.

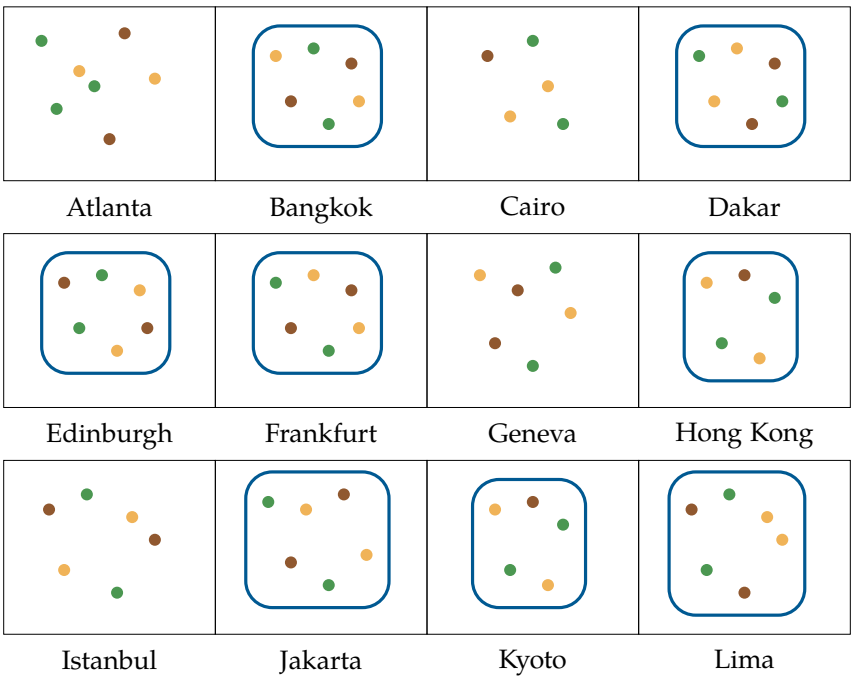



Figure 81.3: Each dot represents a customer and each color represents a different customer segment. Atlanta, Cairo, Geneva, and Istanbul were not selected. The other 8 cities were randomly selected as clusters, and every customer within each selected city becomes part of the sample.

**It is important to understand the difference between strata and clusters.** *Strata* are groups that are different from each other but similar within, and you sample from all of them. *Clusters* are groups that are similar to the whole population, and you sample only some of them but take everyone inside those.

In *quota sampling*, a non-probability sampling method where researchers select participants until predefined quotas for certain groups (ie: age, gender, ethnicity) are filled. The sample is structured to reflect specific population characteristics, but participants are not chosen randomly. For example, if a population is 60% female and 40% male, a researcher surveys people until they reach those proportions.

In *convenience sampling*, a non-probability sampling method where participants are selected because they are easy to access or available. For example, surveying students who happen to be in a campus cafeteria as you are eating lunch.

Bonus method: *Snowball sampling* is a non-probabilistic method used when the population of interest is difficult or impossible to reach through conventional sampling. The researcher identifies a small number of initial participants who meet the study criteria, and then asks those participants to refer others they know who also qualify. Each wave of referrals generates the next, like a snowball rolling downhill and growing as it goes. It is not a commonly tested method on the AP exam, but it is worth knowing as an example of a non-probabilistic sampling technique that can be useful in certain contexts, such as studying hidden or hard-to-reach populations (e.g., people experiencing homelessness, members of stigmatized groups, or individuals with rare diseases).

## *Exercise 197* Identifying Sampling Methods

*Working Space*

Identify the sampling method used in each scenario: SRS, systematic, stratified, or cluster.

| Scenario | Method |
|---|---|
| A supermarket chain selects 10 of its 80 stores at random and surveys every employee in those stores. | |
| A university randomly selects 50 students from each of its four faculties to participate in a wellbeing survey. | |
| A water utility selects a random starting meter on its numbered list and then tests every 25th meter for contamination. | |
| A film festival puts all 900 registered attendee names in a database and uses software to randomly select 90. | |

*Answer on Page 1946*

### Comparing Sampling Methods in Python

In this exercise you will apply three sampling methods to the same dataset and compare what each one produces. The dataset contains 30 students across three grade levels with their test scores.

**Setup: Create the dataset.**

```
import pandas as pd
import numpy as np

np.random.seed(42)

data = {
    "student_id": range(1, 31),
    "grade": ["9th"]*10 + ["10th"]*10 + ["11th"]*10,
    "score": (
        list(np.random.randint(60, 80, 10)) +
        list(np.random.randint(70, 90, 10)) +
        list(np.random.randint(80, 100, 10))
    )
```

```
}

df = pd.DataFrame(data)
print(df)
```

**Compute the true population mean.**

```
true_mean = df["score"].mean()
print(f"True mean score: {true_mean:.1f}")
```

1. **Simple Random Sample.** Draw an SRS of 9 students.

```
srs = df.sample(n=9)
print(f"SRS mean: {srs['score'].mean():.1f}")
print(srs[["student_id", "grade", "score"]])
```

   Run this three times. What grades tend to appear? Does the SRS always represent all three grades?

2. **Stratified Random Sample.** Draw 3 students from each grade level.

```
stratified = df.groupby("grade", group_keys=False).apply(
    lambda x: x.sample(n=3)
)
print(f"Stratified mean: {stratified['score'].mean():.1f}")
print(stratified[["student_id", "grade", "score"]])
```

   Run this three times. How does the sample mean compare to the true mean? Is it more or less stable than the SRS?

3. **Cluster Sample.** Randomly select one entire grade level as the cluster.

```
chosen_grade = pd.Series(df["grade"].unique()).sample(n=1).values[0]
cluster = df[df["grade"] == chosen_grade]
print(f"Cluster selected: {chosen_grade}")
print(f"Cluster mean: {cluster['score'].mean():.1f}")
```

   Run this three times. How much does the cluster mean vary? Why?

4. Fill in the table with your results.

| Method | Run 1 | Run 2 | Run 3 | Most stable? |
|---|---|---|---|---|
| SRS | | | | |
| Stratified | | | | |
| Cluster | | | | |
| True mean | 79.7 | | | |

5. Why does the stratified sample tend to produce a more stable estimate than the SRS? Why does the cluster sample vary the most?

## 81.4 Bias in Surveys

Even a good random sample can still give bad results if the survey is set up badly. Bias can happen at many steps, further down the line from when you pick people.

*Sampling error* is the normal difference between a sample and the population. Different samples can give different answers just by chance. This is not the same as bias. Sampling error gets smaller with a bigger sample.

The following types of bias are more serious because they do not shrink with larger samples. They push results systematically in a particular direction regardless of sample size.

*Response bias* occurs when respondents give answers that do not reflect their true beliefs or behaviors. This often happens when the topic is sensitive or when the presence of an interviewer creates social pressure. A survey on hand-washing habits conducted by a hygiene inspector at a food preparation facility will likely overstate how often workers wash their hands, because workers know what answer the inspector wants to hear.

*Nonresponse bias* occurs when individuals selected for the sample cannot be reached or choose not to participate, and those non-respondents differ systematically from those who do respond. A survey mailed to parents about school satisfaction will be completed at higher rates by engaged parents than by disengaged ones. The result overstates satisfaction among the parent community as a whole.

*Undercoverage bias* occurs when part of the population is excluded from the sampling frame entirely. A survey about internet access conducted through online banner advertisements cannot reach people who lack internet access. Those excluded are precisely the group most relevant to the question.

*Wording effect bias* occurs when the phrasing of a question influences the answer. Consider the difference between these two ways of asking essentially the same thing:

- "Do you support stricter emissions standards for vehicles?"
- "Do you support stricter emissions standards for vehicles, even if it means higher car prices and fewer model choices?"

The second version frames the question with costs attached, making a "no" response far more likely even among people who genuinely support the policy. Neutral, direct wording is essential for unbiased responses.

## *Exercise 198* Diagnosing Bias

Identify the most likely type of bias in each scenario and explain why it applies.

1. A streaming platform sends a satisfaction survey to all subscribers but receives responses from only 8% of them. The platform reports that 91% of subscribers are satisfied.
2. A survey on recycling habits is conducted by going door-to-door in affluent neighborhoods only, because those streets are easier and safer to walk.
3. An interviewer asks employees: "Our new open-plan office is designed to encourage collaboration. How much has it improved your productivity?"
4. A survey on alcohol consumption is conducted at a community health fair by a team wearing hospital uniforms. Respondents significantly underreport their weekly intake.

*Working Space*

*Answer on Page 1947*

CHAPTER 82

# Data tables in SQL

Most organizations keep their data as tables inside a relational database management system (compared to pandas, CSV, or spreadsheets). Developers talk to those systems using a language called SQL ("Structured Query Language").

Some relational database managers are pricey products you may have heard of before, such as Oracle or Microsoft SQL Server. Some are free, such as PostgreSQL or MySQL. These are server software that client programs talk to over a company's network.[1]

There is a library, called `sqlite`, that lets us create files that hold tables. We can use SQL to create, edit, and browse those tables. sqlite is free, fast, and very easy to install. We will use sqlite instead of a networked database management system.

If you look in your digital resources, you will find a file called `bikes.db`. We created this file using sqlite, and now you will use sqlite to access it.

In the terminal, get to the directory where `bikes.db` lives. To open the sqlite tool on that file:

```
> sqlite3 bikes.db
```

(If your system complains that there is no sqlite3 tool, you need to install sqlite. See this website: https://sqlite.org/)

Please follow along: type each command shown here into the terminal and see what happens.

We mostly run SQL commands in this tool, but there are a few non-SQL commands that all start with a period. To see the tables and their columns, you can run `.schema`:

```
sqlite> .schema
CREATE TABLE bike (bike_id int PRIMARY KEY, brand text, size int,
          purchase_price real, purchase_date date, status text);
```

[1] It should be noted that many notetaking and file storage applications allow you to run SQL-like queries to search your files to meet certain criteria. For example, Obsidian, the Markdown notetaking app, has a plugin called Dataview, which allows you to run searches for notes matching certain *metadata* attribute criteria. See more about this here: https://github.com/blacksmithgu/obsidian-dataview

That is the SQL command that we used to create the `bike` table. You can see all the columns and their types.

You want to see all the rows of data in that table?

```
sqlite> select * from bike;
4997391|GT|57|269.61|2009-05-03|rented
5429447|Cannondale|50|215.91|2002-02-17|broken
5019171|Trek|58|251.17|1985-07-11|rented
3000288|Cannondale|57|211.08|1993-01-05|broken
880965|GT|52|281.75|1995-08-02|available
...
```

You will see 1000 rows of data!

The SQL language is not case-sensitive, so you can also write it like this:

```
sqlite> SELECT * FROM BIKE;
```

Often, you will see SQL with just the SQL keywords in all caps:

```
sqlite> SELECT * FROM bike;
```

The semicolon is not part of SQL, but it tells sqlite that you are done writing a command and that it should be executed.

SQL lets you choose which columns you would like to see. The asterisk (FIXME) used above signifies all columns, and the `bike_id, brand` only gets the bike's id and brand from the dataframe:

```
sqlite> SELECT bike_id, brand FROM bike;
4997391|GT
5429447|Cannondale
5019171|Trek
3000288|Cannondale
...
```

Using WHERE, SQL lets you use conditions to decide which rows you would like to see, and can be combined with the common operators `AND`, `OR`, and `NOT`:

```
sqlite> SELECT * FROM bike WHERE purchase_date > '2009-01-01' AND brand = 'GT';
4997391|GT|57|269.61|2009-05-03|rented
326774|GT|56|165.0|2009-06-27|available
```

```
264933|GT|52|302.43|2009-07-09|available
5931243|GT|55|173.56|2009-11-26|rented
4819848|GT|51|221.71|2009-12-11|rented
9347713|GT|52|232.32|2009-06-13|available
3019205|GT|58|262.94|2009-08-22|available
```

Using DISTINCT, SQL lets you get just one copy of each value:

```
sqlite> SELECT DISTINCT status FROM bike;
rented
broken
available

Busted
Flat tire
good
out
Rented
```

You can also edit these rows. For example, if you wanted every status that is `Busted` to be changed to `broken`, you can use an `UPDATE` statement with a `SET`:

```
sqlite> UPDATE bike SET status='broken' WHERE status='Busted';
sqlite> SELECT DISTINCT status FROM bike;
rented
broken
available
Flat tire
good
out
Rented
```

You can insert new rows:

```
sqlite> INSERT INTO bike (bike_id, brand, size, purchase_price, purchase_date,
↪  status)
  ...> VALUES (1, 'GT', 53, 123.45, '2020-11-13', 'available');
sqlite> SELECT * FROM bike WHERE bike_id = 1;
1|GT|53|123.45|2020-11-13|available
```

Note that the bike_id here must be *unique*.

You can delete rows:

```
sqlite> DELETE FROM bike WHERE bike_id = 1;
sqlite> SELECT * FROM bike WHERE bike_id = 1;
```

To get out of sqlite, type `.exit`.

### *Exercise 199* SQL Query

Execute an SQL query that returns the `bike_id` (no other columns) of every Trek bike that cost more than $300.

*Working Space*

*Answer on Page 1947*

## 82.1 Using SQL from Python

The people behind sqlite created a library for Python that lets you execute SQL and fetch the results from inside a python program.

Let's create a simple program that fetches and displays the bike ID and purchase date of every Trek bike that cost more than $300.

Create a file called `report.py`:

```
import sqlite3 as db

con = db.connect('bikes.db')
cur = con.cursor()

cur.execute("SELECT bike_id, purchase_date FROM bike WHERE purchase_price > 330
↪  AND brand='Trek'")
rows = cur.fetchall()

today = datetime.date.today()
for row in rows:
  print(f"Bike {row[0]}, purchased {row[1]}")

con.close()
```

When you execute it, you should see:

```
> python3 report.py
Bike 4128046, purchased 2007-08-06
Bike 7117808, purchased 1995-03-12
Bike 7176903, purchased 1986-07-03
Bike 827899, purchased 2009-03-14
Bike 363983, purchased 1970-08-16
```

Chapter 83

# Diodes, Rectifiers, and Photovoltaics

Early in this course, we discussed some basic electronics components: resistors, capacitors, and inductors. In this chapter, we are going to introduce a semiconductor: *Diodes* ensure that current flows in only one direction.

You will learn a lot about semiconductors from studying diodes. We can then apply what you have learned to solar cells, transistors, and integrated circuits like CPUs.

We are going to focus on semiconductors made of silicon. You can make semiconductors out of other materials, but silicon has proved to be very versatile and cost-effective.

## 83.1 Silicon

A silicon atom has 14 protons and 14 electrons. The inner 10 electrons are arranged in two complete and stable shells. The outer four electrons are in an incomplete shell, which allows them to form bonds with other atoms. We call these *valence electrons* because they aren't tightly bound to their atoms. A complete outer shell would have eight electrons.

Atoms can share electrons to get complete shells. This is called *covalent bonding*. When atoms share electrons, they form a *covalent bond*. The electrons are shared between the atoms, and the atoms are held together by the shared electrons.

In a silicon crystal, each silicon atom shares its outer four electrons with four neighboring atoms. With its four electrons and an electron borrowed from each of it four neighbors, each silicon atom has eight electrons in its outer shell. This is a complete shell, and the atoms are stable. This is why glass is hard and stable. It also why electricity doesn't flow through glass – all the electrons are tightly bound to their atoms.

Phosphorus has five valence electrons. If you put a phosphorus atom into silicon crystal, it shares four of its electrons with its neighbors, but it has one electron left over after the complete shell of eight is built. As you add more phosphorus to a silicon crystal, it conducts more electricity. We call this *doping*.

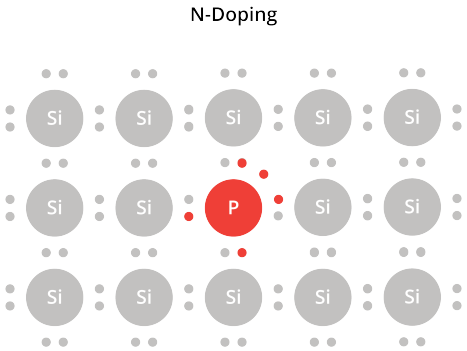



Figure 83.1: Phosphorus atom in a silicon crystal

Because this kind of doping frees up some electrons, which are negatively charged, we call this *n-doping*.

Boron has only three valence electrons. If you put a boron atom into silicon crystal, it shares three of its electrons with its neighbors, which leaves a ”hole” that would like to be filled by an electron. We call this *p-doping*.

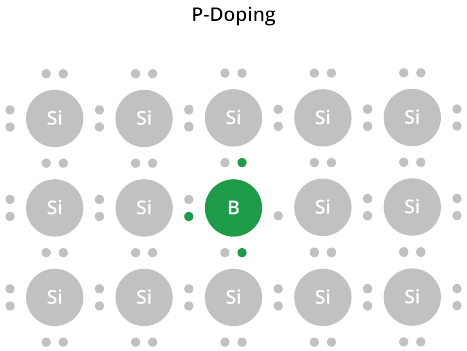



Figure 83.2: Boron atom in a silicon crystal

Both n-doped and p-doped silicon are conductors of electricity. In n-doped silicon, we say the loose electrons are the *carriers* of the current. In p-doped silicon, we say the holes are the carriers.

Note that doped silicon is still mostly silicon. In normal doping, maybe 1 in 10,000 atoms are replaced with boron or phosphorus.

## 83.2 Diodes

A diode is a semiconductor device that allows current to flow in one direction only.

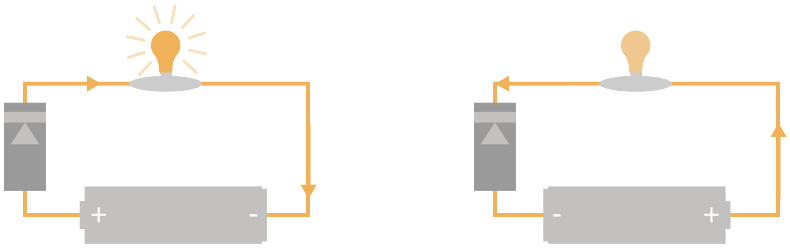



Figure 83.3: Current flows in one direction through the diode, not in the other direction.

Diodes can be made with silicon that has been n-doped and p-doped. A diode looks like this:

Figure 83.4: What Diodes Look Like

We say that current flows from the positive terminal to the negative terminal of a battery. However, remember that the electrons are actually moving from the negative terminal to the positive terminal. This is a common source of confusion in any discussion of diodes.

One end of the diode is called the *anode*. The anode is p-doped. The other end is called the *cathode*. The cathode is n-doped.

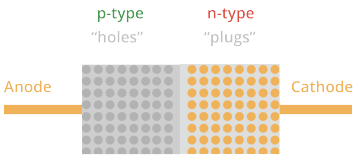



Figure 83.5: A p-n Diode

The region close to where the anode and cathode meet is called the *depletion region*. In this space, the spare electrons in the cathode (the n-doped material) migrate over to fill the holes in anode (the p-doped material). This creates an electric field: there are more electrons than protons on the anode side, and more protons than electrons on the cathode side. (How strong is this field? It is typically about 0.7 volts.)

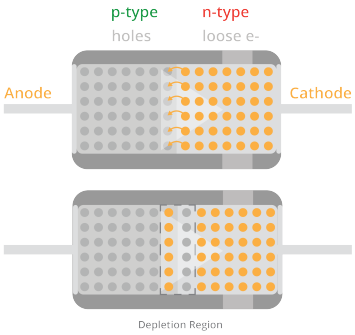



Figure 83.6: The depletion region in a diode

What happens if we overpower that 0.7 volts? What if we put 2 volts in the opposite direction? This shrinks the depletion region and the diode becomes a conductor. For example, if the negative end of a battery is connected to the cathode of a diode and the positive end is connected to the anode, current will flow through the diode.

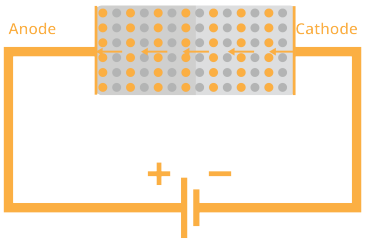



Figure 83.7: Electrons flow through the diode

If we reverse the battery, the depletion region expands, and the diode becomes an insulator.

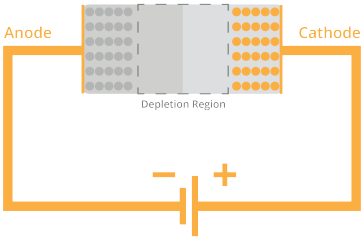



Figure 83.8: Reverse the electric field? No flow.

Here is the symbol for a diode:

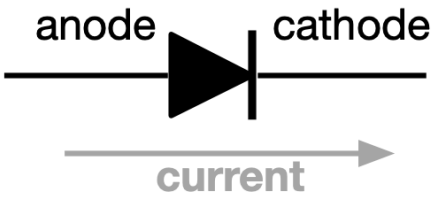



Figure 83.9: Diode Symbol

Current will flow in the direction of the arrow. It will not flow in the opposite direction.

### 83.2.1 Rectifiers

One of the really useful things we can do with a diode is to convert alternating current (AC) into direct current (DC). This is called rectification. Remember that AC pushes and pulls the electrons back and forth in the wire. Rectification ensures that the electrons only move in one direction.

A *half-wave rectifier* uses a single diode to ensure that the current flows in one direction through the load.

Figure 83.10: Half-wave Rectifier

Current flows in the direction of the arrow on the diode. If the voltage is reversed, the diode will block the current from moving.

A *full-wave rectifier* uses four diodes to swap the direction of the current through the load when necessary.

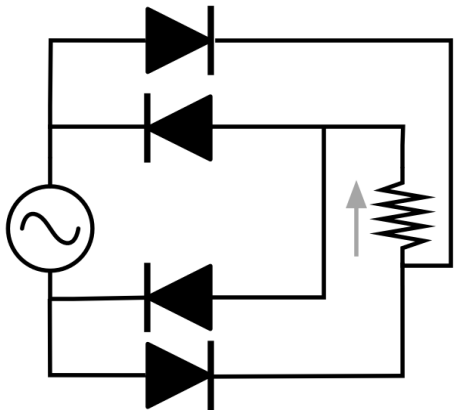


Figure 83.11: Full-wave Rectifier

### 83.2.2 Photovoltaics

Photovoltaics (or "Solar cells") are similar to diodes. They have a p-n junction with a 0.7 volt electrical field. The difference is that sunlight is allowed to shine on the cathode

side. When an atom is struck by a photon of light energy, the electron becomes excited. Sometimes is it excited enough to leap across that 0.7 volt barrier.

When it leaps across, there is an extra electron on the anode side of the barrier and an extra proton on the cathode side. The electron is pushed back to where it came from, but it can't go back over the barrier (unless more than 0.7 volts of charge have built up across the barrier). So, it goes the long way: through the load.

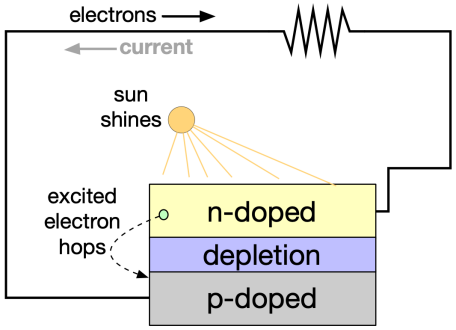



Figure 83.12: Solar Cell

CHAPTER 84

# How Cameras Work

Let's say it is a sunny day and you are standing in a field a few meters from a cow. You use the camera on your phone to take a picture of the cow. How does that whole process work?

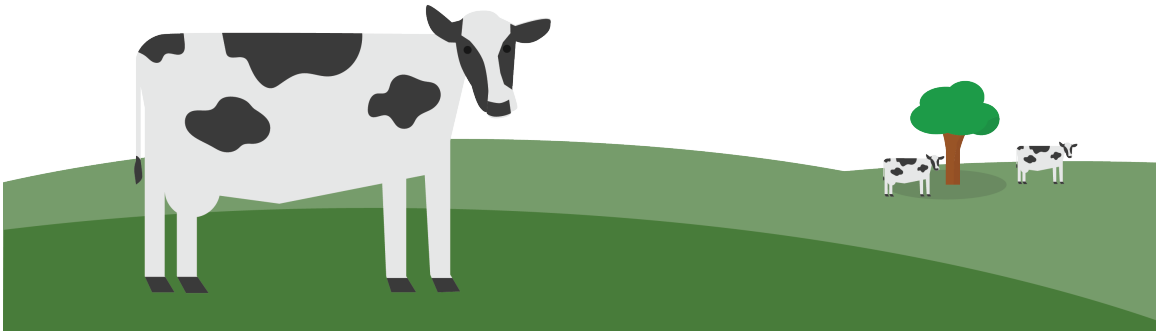


Figure 84.1: A cow in a field.

## 84.1 The Light That Shines On the Cow

The sun is a sphere of hot gas. About 70% of the gas is hydrogen. About 28% is helium. There's also a little carbon, nitrogen, and oxygen.

Gradually, the sun is converting hydrogen into helium through a process known as "nu-

clear fusion" (which we will be discussing more in a future chapter). A large amount of heat is created in this process. This heat makes the gases glow.

How does heat make things glow? The heat pushes the electrons into higher orbitals. When they come back down to a lower orbital, they release a photon of energy, which travels away from the atom as an electromagnetic wave. Recall that when we talked about reflections, we established that light is a constant stream of particles we call photons. These photons travel at the speed of light.

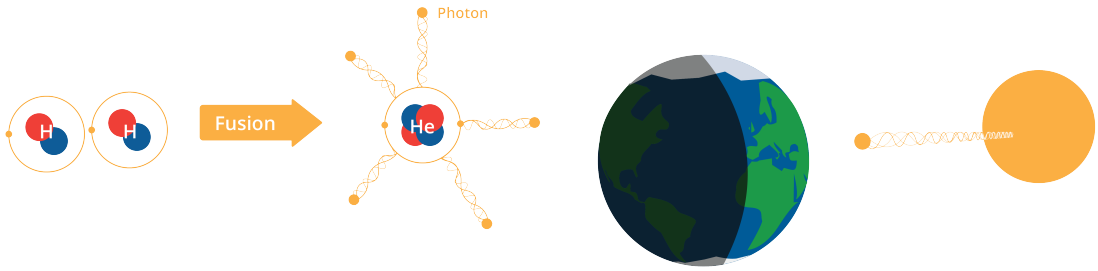



Figure 84.2: Photons are released when the sunlight hits the cow.

Heat is not the only way to push the electrons into a higher orbital. For example, a fluorescent lightbulb is filled with gas. When we pass electricity through the gas, its electrons are moved to a higher orbital. When they fall back to a lower orbital, light is created.

What is the frequency of the wave that the photon travels on? Depending on what orbital it falls from and how far it falls, the photon created has different amounts of energy. The amount of energy determines the frequency of the electromagnetic wave, and ultimately, its color!

In the sun, there are several kinds of molecules and each has a few different orbitals that the electrons can live in. Thus, the light coming from the sun is made up of electromagnetic waves of many different frequencies.

We can see some of these frequencies as different colors, but some are invisible to humans, such as ultraviolet and infrared.

## 84.2 Light Hits the Cow

When these photons from the sun hit the cow, the hide and hairs of the cow will absorb some of the photons. These photons will become heat and make the cow feel warm. Some of the photons will not be absorbed – they will leave the cow. When you say "I see the cow," what you are really saying is "I see some photons that were not absorbed by the cow."

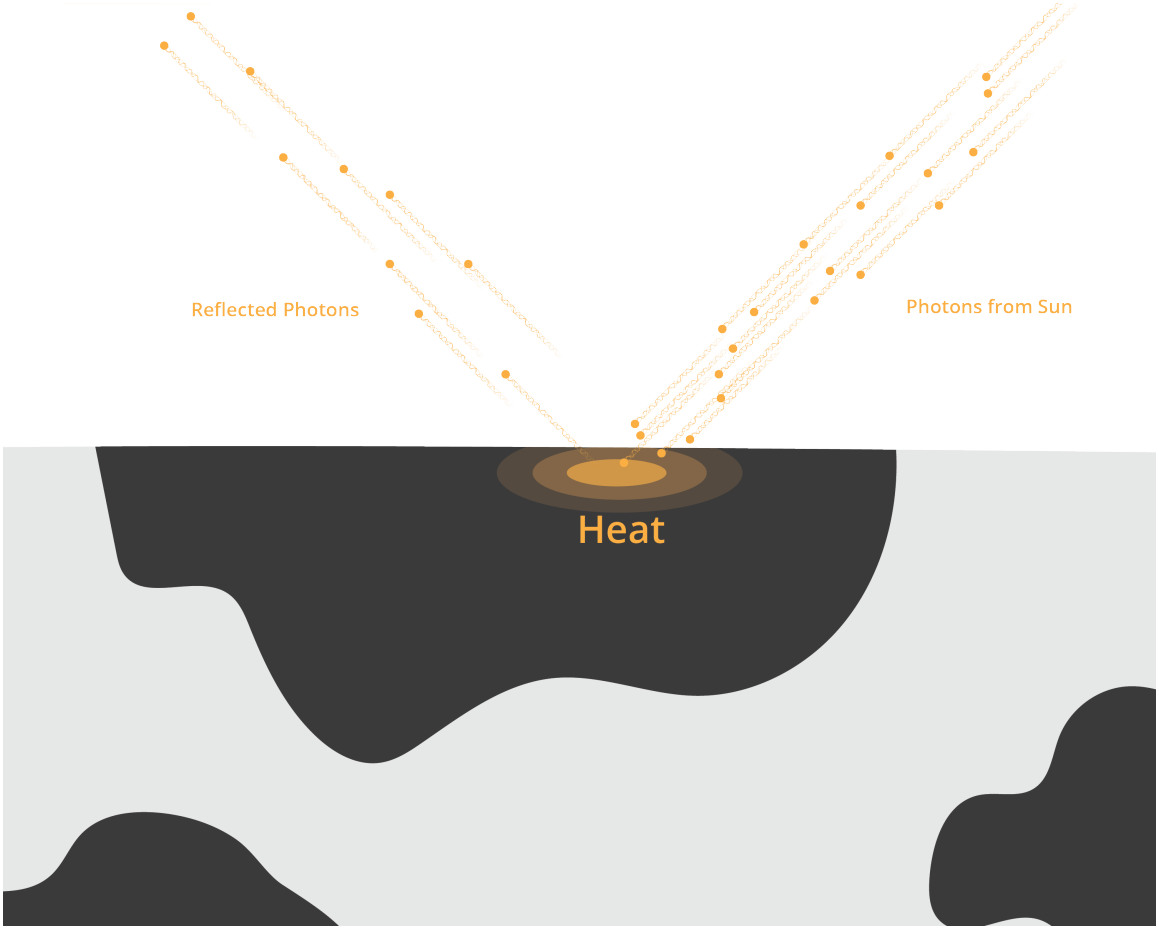



Figure 84.3: Photons hitting the cow create a heated spot, where some are absorbed.

Different materials absorb different amounts of each wavelength. A plant, for example, absorbs a large percentage of all blue and red photons that hit it, but it absorbs only a small percentage of the green photons that hit it. Thus, we say "That plant is green."

White things absorb very small percentages of photons of any visible wavelength. Black things absorb very *large* percentages of photons of any visible wavelength. That is why, on a hot summer day, a black car with black seats and interior will heat up on the inside much hotter than a white car.

Before we go on, let's review: The sun creates photons that travel as electromagnetic waves of assorted wavelengths to the cow. Many of those photons are absorbed, but some are not. Some of those photons that are not absorbed go into the lens of our camera.

## 84.3 Pinhole camera

The simplest cameras have no lenses. They are just a box. The box has a tiny hole that allows photons to enter. The side of the box opposite the hole is flat and covered with film or some other photo-sensitive material.

The photons entering the box continue in the same direction they were going when they passed through the hole. Thus, the photons that entered from high hit the back wall at a low point. The photons that came from the left hit the back wall on the right. This is how the image is projected onto the back wall, rotated 180 degrees; what was up is down, what was on the left is on the right.

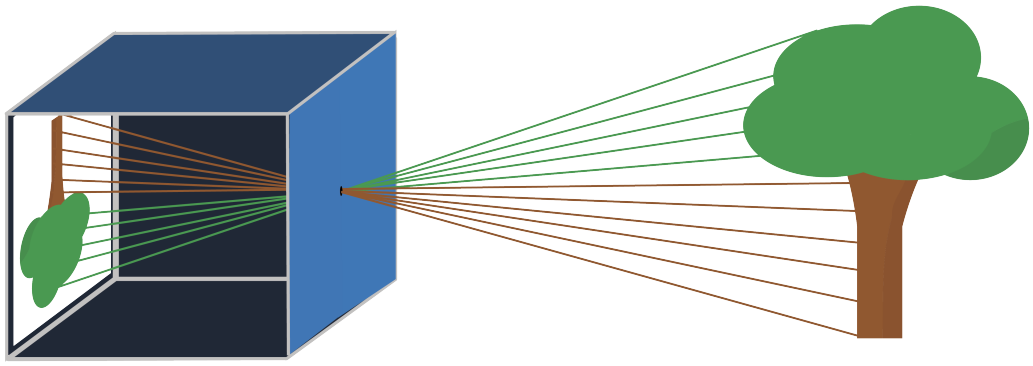


Figure 84.4: A pinhole camera flips the image it "sees" by 180°.

### *Exercise 200* Height of the image

Let's say that the pinhole is exactly the same height as the shoulder of the cow, and that the shoulder is directly above one hoof. This means the pinhole, the shoulder, and the hoof form a right triangle.

Now, let's say that the camera is being held perpendicular to the ground. The pinhole, the image of the shoulder, and the image of the hoof on the back wall of the camera now also form a right triangle.

These two triangles are similar.

The shoulder is 2 meters from the hoof. The cow is standing 3 meters from the camera. The distance from the pinhole to the back wall of the camera is 3 cm. How tall is the image of the cow on the back wall of the camera?

*Working Space*

*Answer on Page 1947*

## 84.4 Lenses

Now, a quick review: A photon leaves the sun in some random direction. It travels 150 million km from the sun and hits a cow. It is not absorbed by the cow, and heads off in a new direction. It passes through the pinhole and hits the back wall of the camera. That seems incredibly improbable, right?

It actually is relatively improbable, especially if there isn't a lot of light — like you are taking the picture at dusk. To increase the odds, we added a *lens* to the camera. If you focus a lens on a wall and you draw a dot on that wall, the lens is designed such that all the photons from the dot that hit the lens get redirected to the same spot on the back wall of the camera — regardless of which path it took to get to the lens.

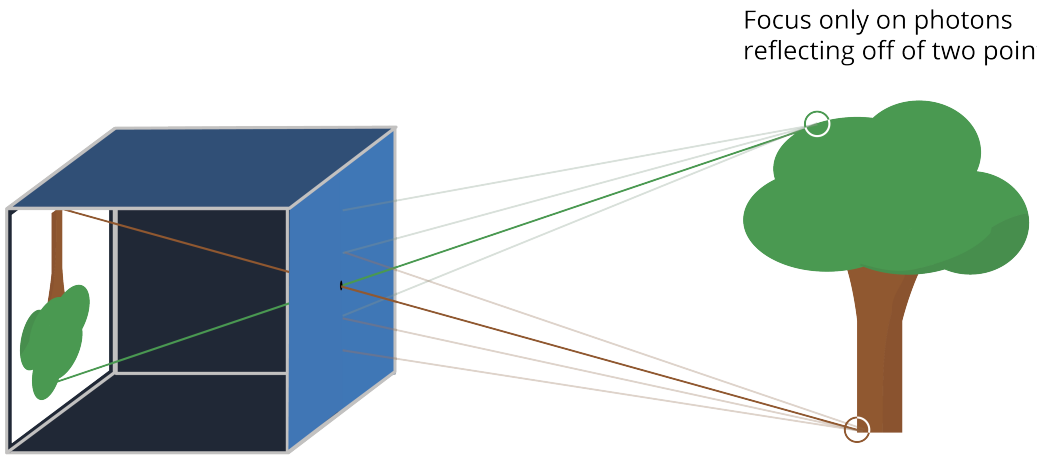



Figure 84.5: A pinhole with a lens allows you to choose the points it reflects.

Note that the image still gets flipped. There is a *focal point* that all the photons pass through.

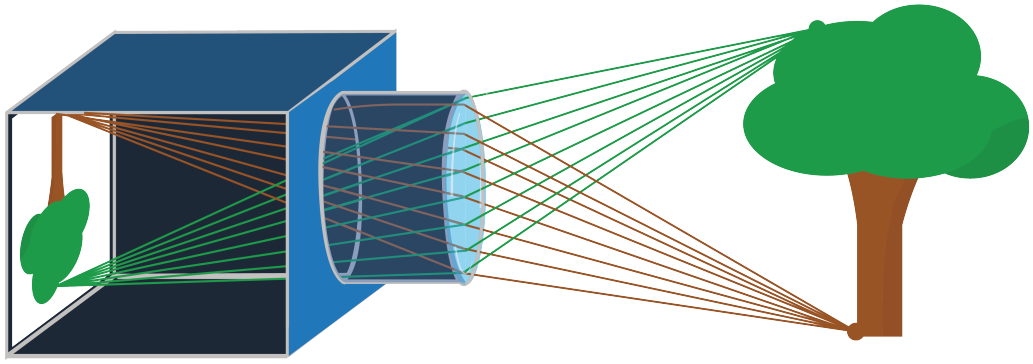


Figure 84.6: A lens still flips the image, but introduces a focal point.

The distance from the lens to its focal point is called the lens's *focal length*. Telephoto lenses, that let you take big pictures of things that are far away, have long focal lengths.

Wide-angle lenses have short focal lengths.

## 84.5 Sensors

The camera on your phone has a sensor on the back wall of the camera. The sensor is broken up into tiny rectangular regions called pixels. When you say a sensor is 6000 by 4000 pixels (most common ratio for photography), we are saying the sensor is a grid of 24,000,000 pixels: 6000 pixels wide and 4000 pixels tall.

In the sensor, each individual pixel is hit by an incoming photon. If the photon's energy (recall $E = hf$) is less than the Work Function $\Phi$ to dislodge an electron, the light simply passes through or reflects, and the pixel remains *dark*, or what we call *black*. If $hf \geq \Phi$, the photon is absorbed, and its energy is transferred to an electron, which *jumps* into the conduction band, appearing like *white* to us. Camera sensors then convert their pixels into a binary mapping, using voltage measurements.

Each pixel has three types of cavities that take in photos. One of the cavities measures the amount of short wavelength light, like blues and violets. One of the cavities measures the long wavelength light, like reds and oranges. One of the cavities measures the intensity of wavelengths in the middle, like greens. Thus, if your camera has a *resolution* of $6000 \times 4000$, the image is 24,000,000 pixels yields three numbers: intensity of long wavelength, mid wavelength, and short wavelength light, for a total of 72,000,000 numbers. We call these numbers "RGB" for Red, Green, and Blue. The RGB values range from $0 - 255$ for each channel. We will talk more about this when hexadecimal is introduced.

### *Exercise 201* Camera Pixels

A digital camera sensor is made of a material with a work function of $\Phi = 1.1$ electronvolts. If the sensor is exposed to monochromatic light with a wavelength of $\lambda = 1200$ nm, what does the resulting image look like? Why?

Hint: $E = \frac{1240}{\lambda}$ where $\lambda$ is in nanometers and $E$ is in electronvolts (eV).

*Working Space*

*Answer on Page 1947*

## 84.6 Aspect Ratio

When you are buying a monitor, TV, or even smartphone, you may hear the term **aspect ratio**. The aspect ratio is a set of two numbers representing the ratio between the width and the height. Conventionally, the width comes first, separated by a colon, and then the height:

$$\text{width} : \text{height}$$

Some of the most common aspect ratios are:

- $16 : 9$ (known as Widescreen)
- $17 : 9$
- $9 : 16$
- $4 : 3$ (known as Fullscreen)
- $4 : 5$
- $3 : 2$
- $1 : 1$ (perfect square, like on a smartwatch)

As we talked about above, the aspect ratio doesn't describe the actual size of the screen or image. Instead, it describes the shape. A movie theatre screen can have the same aspect ratio as your monitor, even if the screen is 360 inches and the monitor is 24 inches.

Aspect ratio describes the proportional relationship between width and height. Resolution describes the total number of pixels used to represent an image.

For example, both of the following resolutions have a $16 : 9$ aspect ratio:

$$1280 \times 720$$

$$1920 \times 1080$$

Mathematically, the aspect ratio is a dimensionless quantity. Since both width and height are measured in the same units (such as centimeters, inches, or pixels), the units cancel when forming the ratio. This means aspect ratio describes proportion rather than physical measurement. Most often, we determine the aspect ratio of a digital display by comparing the number of horizontal pixels to vertical pixels.

For example:

$$1920 \times 1080 \quad \Rightarrow \quad \frac{1920}{1080} = \frac{16}{9} = 16:9$$

Digital cameras and smartphone cameras capture images using a rectangular sensor. The shape of this sensor determines the default aspect ratio of the image. For example:

- Many digital cameras use a $3:2$ sensor.
- Micro Four Thirds cameras use $4:3$.
- Most video recording uses $16:9$.

Important: when you change the aspect ratio setting on a camera, the device does not change the sensor itself. Instead, it crops part of the image to match the selected ratio. Cropping reduces the number of recorded pixels, which may reduce file size. However, actual file size also depends on compression methods used by the device.

Human vision has a wider horizontal field of view than vertical field of view. Because of this, wider aspect ratios such as $16:9$ often feel more natural and immersive. This is one reason modern TVs and cinema formats favor wider screen shapes. However, with the rise of cellphones, we have switched to consistently using portrait or vertical mode, and video aspect ratios are more consistently becoming $9:16$. This allows for consistency in device quality, and commonly used for vertical videos, like in short-form content, compared to traditional widescreen videos, such as Youtube or traditional horizontal video capturing.

### 84.6.1 Printing

You can imagine this may cause a problem for printing out photos. Most printing paper is $8.5$ inches $\times$ $11$ inches or $8$ inches $\times$ $10$ inches. This causes an issue, as that fits a $8.5:11 \iff 17:22$ or $4:5$ aspect ratio, respectively. We combat this by using very standardized printing sizes (in inches):

- $6 \times 4 \to 3:2$,
- $7 \times 5 \to 7:5$,
- $10 \times 8 \to 5:4$,
- and, rarely, $8.5 \times 11 \to 1.29:1$

Going to any framing or printing store will give you a very standardized set of sizes, or some multiple of these.

Most smartphones use $4:3$ or $3:2$, which will work for $4 \times 6$ inches or $8 \times 6$ inches,

respectively.

# Chapter 85

# How Eyes Work

Dr. Craig Blackwell has made a great video on the mechanics of the eye. You should watch it: https://youtu.be/Z8asc2SfFHM

Mechanically, your eye works a lot like a camera. The eye is a sphere with two lenses on the front: The outer lens is called the *cornea*, while the second lens is simply called "the lens." Between the two lenses is an aperture that opens wide when there is very little light, and closes very small when there is bright light. The opening is called the *pupil* and the tissue that forms the pupil is called the *iris*. When people talk about the color of your eyes, they are talking about the color of your iris. The blackness at the center of your iris is your pupil.

There are two types of photoreceptor cells in your retina: rods and cones. The rods are more sensitive; in very dark conditions, most of our vision is provided by the rods. The cones are used when there is plenty of light, and they let us see colors.

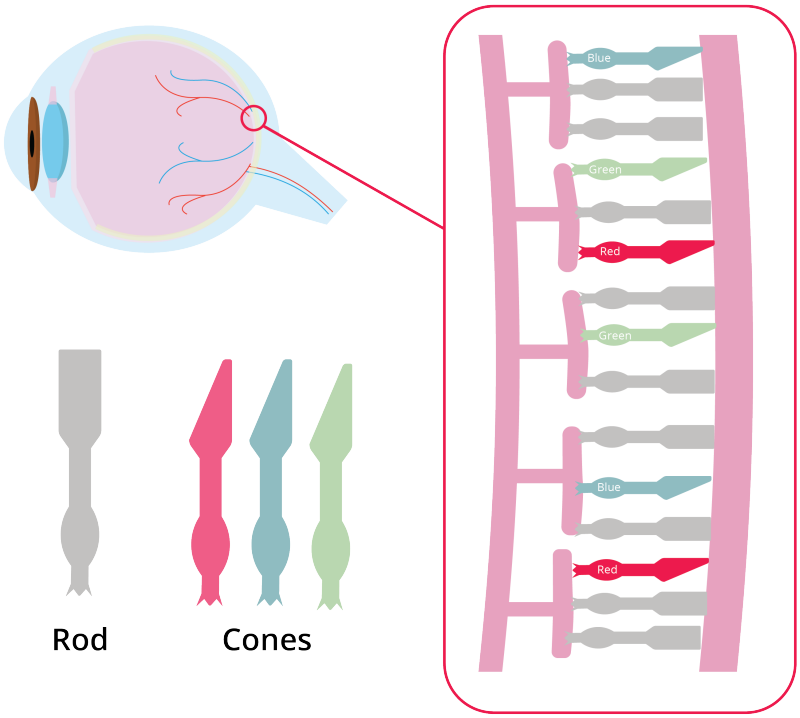



Figure 85.1: The cones and rods of the eye act as sensors and cognitive vision.

The white part around the outside of the eyeball? That is called the *sclera*.

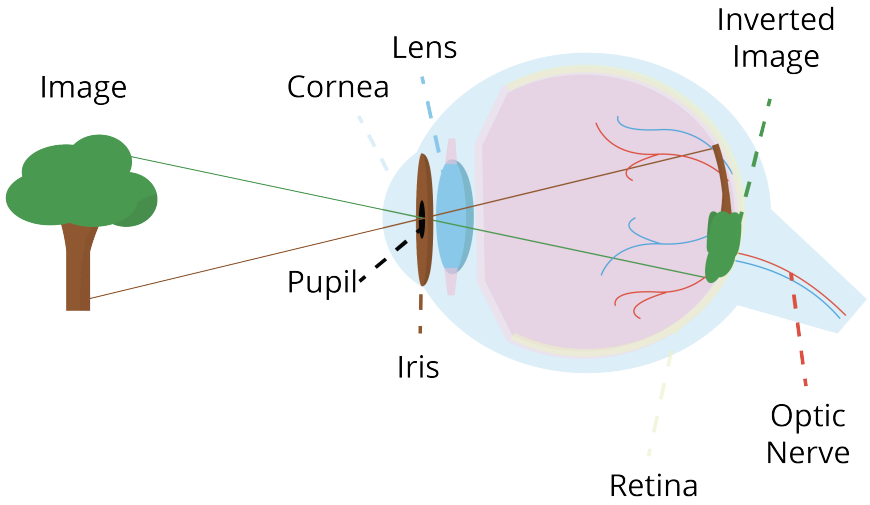



Figure 85.2: The eye works just like a camera's lens.

The walls of the eye are lined inside with the *retina*, which has sensors that pick up the light and send impulses down the optic nerve to your brain.

Just like a camera, the images are flipped when they get projected on the back of the eye (see Figure 85.2).

## 85.1 Eye problems

Now that you know the mechanics of the eye, let's go over a few things that commonly go wrong with the eye.

### 85.1.1 Glaucoma

The space between your cornea and lens is filled with a fluid called *aqueous humor*. To feed the cells of the cornea and lens, the aqueous humor carries oxygen and nutrients like blood would, but unlike blood, it is transparent so you can see. Aqueous humor is constantly being pumped into and out of that chamber. If aqueous humor has trouble

exiting, the pressure builds up and can damage the eye. This is known as *glaucoma*. See Figure 85.3.

### 85.1.2 Cataracts

The lens should be clear. As a person ages, the proteins in the lens break down and clump together, becoming opaque. This can also be accelerated by diabetes, too much exposure to sunlight, obesity, and high blood pressure. From the outside, the eye will look cloudy. This is called a *cataract*, and it makes it difficult for the person to see.

This problem can be corrected, however. The person's cloudy lens is removed and replaced with a clear, manufactured lens.

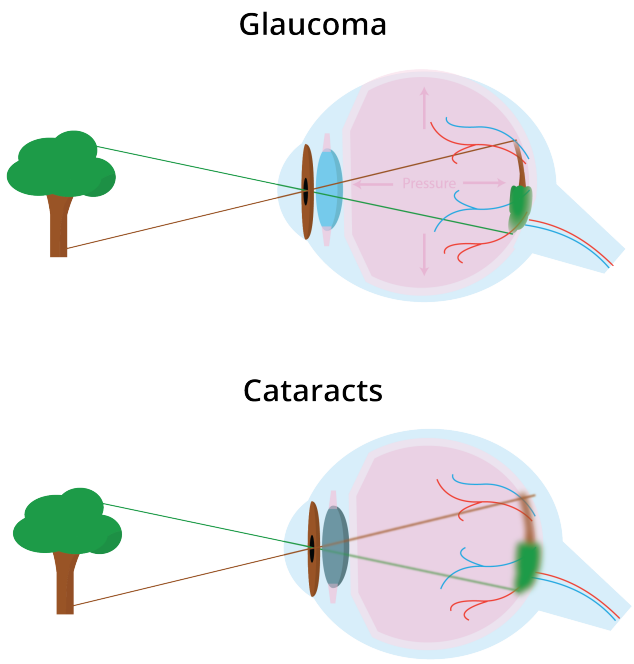



Figure 85.3: Cataracts and Glaucoma represented in the eye cross section.

### 85.1.3 Nearsightedness, farsightedness, and astigmatism

If you are in a dark room and a tiny LED is turned on, the photons from that LED can pass through your cornea in many different places. If your eye is focusing on that light correctly, all the photons should meet up at the same place on the retina.

FIXME: Diagram here

If the lenses are bending the light too much, the photons meet up before they hit the retina and get smeared a bit across it. To the person, the LED would appear blurry. The eye is said to be *nearsighted* or *myopic*. This signals that near objects appear correctly, but farther objects appear blurry.

If the lenses are not bending it enough, the photons would meet up behind the retina. Once again, they get smeared a bit across the retina and the LED looks blurry to the person. The eye is said to be *farsighted* or *hyperopic*. The objects which are distant can appear clearly, while closer objects are heavily blurred.

Your lenses are supposed to bend the photons the same amount vertically and horizontally. If one dimension is focused, but the other is myopic or hyperopic, the eye is said to have *astigmatism*.

Myopia, hyperopia, and astigmatism can be corrected with glasses or contact lenses. Doctors can also do surgical corrections, usually by changing the shape of the cornea.

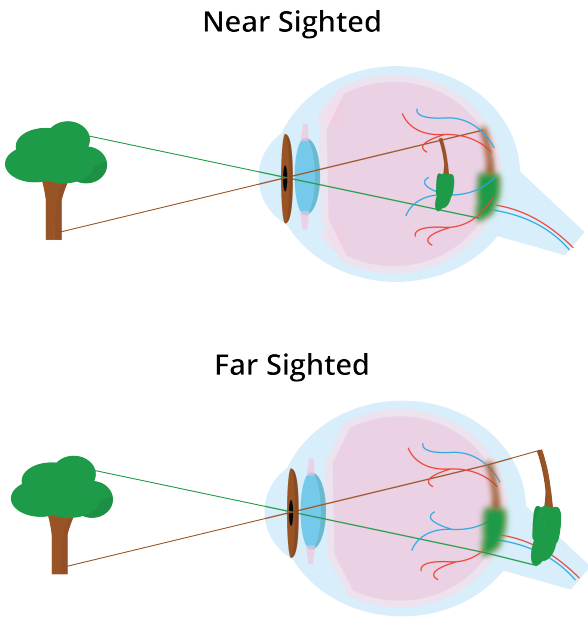



Figure 85.4: Nearsightedness versus farsightedness.

## 85.2 Seeing colors

TED-Ed has made a good video on how we see color. Watch it here: https://youtu.be/l8_fZPHasdo

When a rainbow forms, you are seeing different wavelengths separating from each other.

In the rainbow:

- Red is about 650 nm.
- Orange is about 600 nm.
- Yellow is about 580 nm.
- Green is about 550 nm.
- Cyan is about 500 nm.
- Blue is about 450 nm.
- Violet is about 400 nm.

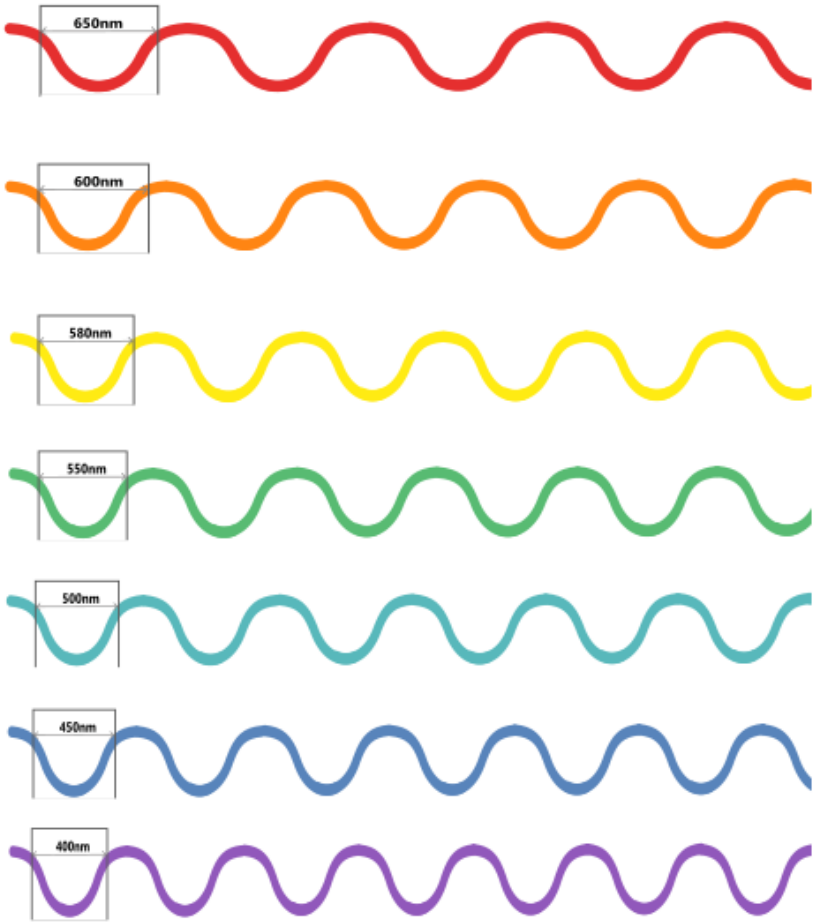



Figure 85.5: All colors have different wavelengths.

If you shine a light with a wavelength of 580 nm on a white piece of paper, you will see yellow.

However, if you shine two lights with wavelengths of 650 nm (red) and 550 nm (green), you will also see yellow.

Why? Our ears can hear two different frequencies at the same time. Why can't our eyes see two colors in the same place?

As mentioned above, the cone photoreceptors in our eyes let us see colors. There are three kinds of cones:

- Red: Cones that let us see the frequencies up to about 700nm.
- Green: Cones that are most sensitive to frequencies near 550nm.
- Blue: Cones that are most sensitive to frequencies near 450nm.

When a wavelength of 580 nm hits your retina, it excites the red and green receptors, and your brain interprets that mix as yellow.

Similarly, when light that contains both 650 nm and 550 nm waves hits your retina, it excites the red and green receptors, and your brain interprets that mix as yellow.

You can't tell the difference!

Now we know why the sensors on the camera are RGB. The camera is recording the scene as closely as necessary to fool your eye.

A TV or a color computer monitor only has three colors of pixels: red, green, and blue. By controlling the mix of them, it creates the sensation of thousands of colors to your eye.

## 85.3 Pigments

A color printer works in the opposite fashion. Instead of radiating colors, it puts pigments on the paper that absorb certain frequencies. A pigment that absorbs only frequencies near 650 nm (red) will appear to your eye as cyan. This makes sense, because the sensation of cyan is created when your blue and green receptors are activated.

Thus, pigment colors come in:

- Cyan: absorbs frequencies around red
- Magenta: absorbs frequencies around green

- Yellow: absorbs frequencies around blue

If you buy ink for a color printer, you know there is typically a fourth ink: black. If you put cyan, magenta, and yellow pigments on paper, the mix won't absorb all the visible spectrum in a consistent manner. Our eyes are pretty sensitive to this, so we would see brown. This is why we add black ink to get pretty grays and blacks.

We call this approach to color CMYK (as opposed to RGB). If an artist is creating an image to be viewed on a screen, they will typically make an RGB image. If they are creating an image to be printed using pigments, they typically create a CMYK image. (Most of us don't care so much — we just let the computer do conversions between the two color spaces for us.)

# Chapter 86

# Images in Python

An image is usually represented as a three-dimensional array of 8-bit integers. NumPy arrays are the most commonly used library for this sort of data structure.

If you have an RGB image that is 480 pixels tall and 640 pixels wide, you will need a $480 \times 640 \times 3$ NumPy array.

There is a separate library (imageio) that:

- Reads an image file (like JPEG files) and creates a NumPy array.
- Writes a NumPy array to a file in standard image formats

Let's create a simply python program that creates a file containing an all-black image that is 640 pixels wide and 480 pixels tall. Create a file called `create_image.py`:

```
import NumPy as np
import imageio
import sys

# Check command-line arguments
if len(sys.argv) < 2:
    print(f"Usage {sys.argv[0]} <outfile>")
    sys.exit(1)

# Constants
IMAGE_WIDTH = 640
IMAGE_HEIGHT = 480

# Create an array of zeros
image = np.zeros((IMAGE_HEIGHT, IMAGE_WIDTH, 3), dtype=np.uint8)

# Write the array to the file
imageio.imwrite(sys.argv[1], image)
```

To run this, you will need to supply the name of the file you are trying to create. The extension (like .png or .jpeg) will tell imageio what format you want written. Run it now:

```
python3 create_image.py blackness.png
```

Open the image to confirm that it is 640 pixels wide, 480 pixels tall, and completely black.

## 86.1 Adding color

Now, let's walk through the image, pixel-by-pixel, adding some red. We will gradually increase the red from 0 on the left to 255 on the right.

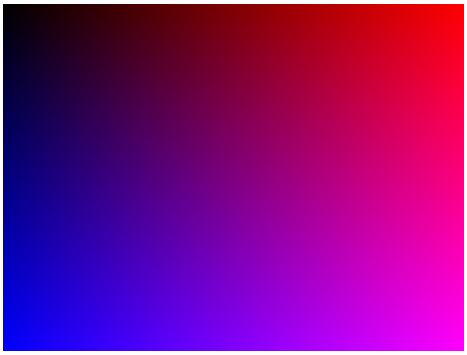
```
import NumPy as np
import imageio
import sys

# Check command-line arguments
if len(sys.argv) < 2:
    print(f"Usage {sys.argv[0]} <outfile>")
    sys.exit(1)

# Constants
IMAGE_WIDTH = 640
IMAGE_HEIGHT = 480

# Create an array of zeros
image = np.zeros((IMAGE_HEIGHT, IMAGE_WIDTH, 3), dtype=np.uint8)

for col in range(IMAGE_WIDTH):

    # Red goes from 0 to 255 (left to right)
    r = int(col * 255.0 / IMAGE_WIDTH)

    # Update all the pixels in that column
    for row in range(IMAGE_HEIGHT):
        # Set the red pixel
        image[row, col, 0] = r

# Write the array to the file
imageio.imwrite(sys.argv[1], image)
```

When you run the function to create a new image, it will be a fade from black to red as you move from left to right:

Now, inside the inner loop, update the blue channel so that it goes from zero at the top to 255 at the bottom:

```
    # Update all the pixels in that column
    for row in range(IMAGE_HEIGHT):

        # Update the red channel
        image[row,col,0] = r

        # Blue goes from 0 to 255 (top to bottom)
        b = int(row * 255.0 / IMAGE_HEIGHT)
        image[row,col,2] = b

imageio.imwrite(sys.argv[1], image)
```

When you run the program again, you will see the color fades from black to blue as you go down the left side. As you go down the right side, the color fades from red to magenta.



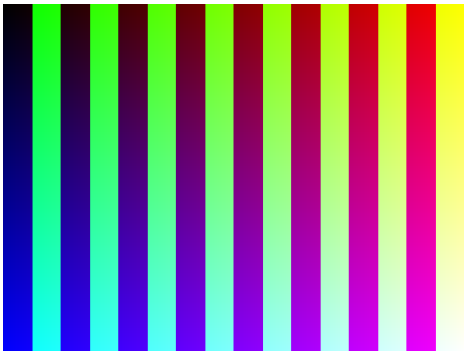
Notice that red and blue with no green looks magenta to your eye.

Next, let's add some stripes of green:

```
import NumPy as np
import imageio
```

```
import sys

# Check command line arguments
if len(sys.argv) < 2:
    print(f"Usage {sys.argv[0]} <outfile>")
    sys.exit(1)

# Constants
IMAGE_WIDTH = 640
IMAGE_HEIGHT = 480
STRIPE_WIDTH = 40
pattern_width = STRIPE_WIDTH * 2

# Create an image of all zeros
image = np.zeros((IMAGE_HEIGHT, IMAGE_WIDTH, 3), dtype=np.uint8)

# Step from left to right
for col in range(IMAGE_WIDTH):

    # Red goes from 0 to 255 (left to right)
    r = int(col * 255.0 / IMAGE_WIDTH)

    # Should I add green to this column?
    should_green = col % pattern_width > STRIPE_WIDTH

    # Update all the pixels in that column
    for row in range(IMAGE_HEIGHT):

        # Update the red channel
        image[row,col,0] = r

        # Should I add green to this pixel?
        if should_green:
            image[row,col,1] = 255

        # Blue goes from 0 to 255 (top to bottom)
        b = int(row * 255.0 / IMAGE_HEIGHT)
        image[row,col,2] = b

imageio.imwrite(sys.argv[1], image)
```

When you run this version, you will see the previous image in half the stripes. In the other half, you will see that green fades to cyan down the left side, and yellow fades to white down the right side.



## 86.2 Using an existing image

imageio can also be used to read in any common image file format. Let's read in an image and save each of the red, green, and blue channels out as its own image.

Create a new file called `separate_image.py`:

```
import imageio
import sys
import os

# Check command line arguments
if len(sys.argv) < 2:
    print(f"Usage {sys.argv[0]} <infile>")
    sys.exit(1)

# Read the image
path = sys.argv[1]
image = imageio.imread(path)

# What is the filename?
filename = os.path.basename(path)

# What is the shape of the array?
original_shape = image.shape

# Log it
print(f"Shape of {filename}:{original_shape}")

# Names of the colors for the filenames
colors = ['red','green','blue']

# Step through each of the colors
for i in range(3):

    # Create a new image
    newimage = np.zeros(original_shape, dtype=np.uint8)
```

```
31
32      # Copy one channel
33      newimage[:,:,i] = image[:,:,i]
34
35      # Save to a file
36      new_filename = f"{colors[i]}_{filename}"
37      print(f"Writing {new_filename}")
38      imageio.imwrite(new_filename, newimage)
```

Now, you can run the program with any common RGB image type:

```
1  python3 separate_image.py dog.jpg
```

This will create three images: `red_dog.jpg`, `green_dog.jpg`, and `blue_dog.jpg`.

CHAPTER 87

# Representing Natural Numbers

Natural numbers are positive whole numbers, such as 1, 2, 3, and so on. -5 is not a natural number. $\pi$ is not a natural number. $\frac{1}{2}$ is not a natural number.

## 87.1 Base-10 (Decimal)

You are used to seeing the natural numbers represented in a **base-10** *Hindu-Arabic* or the *Decimal* numeral system. That is, when you see 2531 you think "2 thousands, 5 hundreds, 3 tens, and 1 one." Rewritten, this is:

$$2 \times 10^3 + 5 \times 10^2 + 3 \times 10^1 + 1 \times 10^0$$

The *radix* is the amount of number of unique values a base can have before increasing the amount of digits. For example, values 0, 1, 2, $\cdots$, 8, 9 are all the values for base-10. Once 1 is added to 9, we need to increase the amount of digits in a number.

In any Hindu-Arabic system, the location of the digits is meaningful: 101 is different from 110; the position of a number indicates its magnitude. Here are those numbers in Roman numerals: CI and CX. Roman numerals didn't have a symbol for zero at all.

The Hindu-Arabic system gave us really straightforward algorithms for addition and multiplication. For addition, you memorized the following table:

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 2 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 3 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 4 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
| 5 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
| 6 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 7 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 8 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 9 | 9 | 10 | 11 | 12 | 13 | 14 | 15 16 | 17 | 18 | |

When you multiplied two number together, you simply multiplied each pair of digits. $254 \times 26$ might look like this:

| | | | | | |
|---|---|---|---|---|---|
| | | 2 | 5 | 4 | |
| | | × | 2 | 6 | |
| | | | 2 | 4 | $6 \times 4$ |
| | | 3 | 0 | | $6 \times 5$ |
| | 1 | 2 | | | $6 \times 2$ |
| | | | 8 | | $2 \times 4$ |
| | 1 | 0 | | | $2 \times 5$ |
| + | 4 | | | | $2 \times 2$ |
| 6 | 6 | 0 | 4 | | |

For multiplication, you memorized this table:

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 2 | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 |
| 3 | 0 | 3 | 6 | 9 | 12 | 15 | 18 | 21 | 24 | 27 |
| 4 | 0 | 4 | 8 | 12 | 16 | 20 | 24 | 28 | 32 | 36 |
| 5 | 0 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 |
| 6 | 0 | 6 | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 54 |
| 7 | 0 | 7 | 14 | 21 | 28 | 35 | 42 | 49 | 56 | 63 |
| 9 | 0 | 9 | 18 | 27 | 36 | 45 | 54 | 63 | 72 | 81 |

## 87.2 Base-2 (Binary)

Binary, on the other hand, has only 2 digits it can have: 0 or 1. This kind of number system is often used for electrical and computer systems because the values can only be **on** or **off**.

Each digit in a binary number represents a power of 2, starting from the rightmost digit (the least significant bit, or LSB). For example:

$$\cdots + 2^3 + 2^2 + 2^1 + 2^0$$

For example,

$$1011_2 = (1 \cdot 2^3) + (0 \cdot 2^2) + (1 \cdot 2^1) + (1 \cdot 2^0) = 8 + 0 + 2 + 1 = 11_{10}$$

[1]

[1]The subscripts represent the radix or base.

Because the radix is 2, the numbers increase differently:

$$
\begin{aligned}
&0 : 0 \\
&1 : 1 \\
&2 : 10 \\
&3 : 11 \\
&4 : 100 \rightarrow (2^2 \cdot 1) + 0 + 0 = 4 \\
&5 : 101 \\
&6 : 110 \\
&7 : 111 \\
&8 : 1000 \\
&\dots \\
&15 : 1111 \cdots
\end{aligned}
$$

### Conversion from Decimal to Binary

To convert $13_{10}$ to binary, repeatedly divide by 2 and record the remainders until the divisor reaches 0 or 1:

$13 \div 2 = 6$ remainder $16 \div 2 = 3$ remainder $03 \div 2 = 1$ remainder $11 \div 2 = 0$ remainder 1

Reading from bottom to top, $13_{10} = 1101_2$.

## 87.2.1 Bits and Bytes

A singular binary digit (0 or 1) is called a *bit*. A lightbulb, switch, or any sort of *Boolean* value (true or false) can be represented as a bit. A single binary digit is called a *bit*. Eight bits form a *byte*, which is a common unit of storage in computer systems:

$$1 \text{ byte} = 8 \text{ bits}$$

For example, the binary sequence 01001000 represents one byte of data.

We won't go into the depths of it here, but here are the basics of computer architecture. There are codes of 4-bit sequences that form up the *memory*. This can form the sections of it into different data types:

### Common Data Types in Memory

| Data Type | Size | Bit Length | Typical Use |
|---|---|---|---|
| Character | 1 byte | 8 bits | ASCII characters, small integers |
| Integer | 4 bytes | 32 bits | Whole numbers |
| Double | 8 bytes | 64 bits | Decimal (floating-point) values |

Strings are formed by listing characters in consecutive memory addresses such that they are marked by a starting memory address and ending memory address.

8-bit binary numbers can be used to signify positive or negative numbers. The most significant bit (MSB) will alter sign, causing the range to be $[-2^7,\ 2^7 - 1]$. If the byte is negative (in terms of bits), invert each bit and add 1 to the inverted result. The other way is to multiply the MSB by negative one, and add until the value is reached, as seen below.

$$-85_{10} = -1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 10101011$$

Binary can be easy converted to *Hexadecimal*.

## 87.3 Base-16 (Hexadecimal)

Hexadecimal (from the latin *hexa* and *deca* for 16) is a representation of number with a radix of 16. Digits are represented through numeric numbers 0-9 and *A-F*.

Counting to 16 in hex goes as follows:

1
2
3
...
9
A (decimal value 10)
B (decimal value 11)
C (decimal value 12)
D (decimal value 13)
E (decimal value 14)
F (decimal value 15)

Converting between binary and hexadecimal is easier because the hex radix ($16 = 2^4$) is a power of the binary radix ($2^1$).

It takes 4 binary digits to represent 1 hex digit: 15 = F

For example,

$$0x47 = 0100\ 0111_2$$

$$0xBD = 1011\ 1101_2$$

### 87.3.1 Hex Color Codes

Hex colors are represented by 6 hexadecimal digits, grouped as two each for red, green, and blue (RGB). Each pair ranges from $00$ (0 in decimal) to $FF$ (255 in decimal), allowing $256$ shades per color channel. For example:

$$\#FF0000 = \text{Red}, \quad \#00FF00 = \text{Green}, \quad \#0000FF = \text{Blue}$$

By mixing values, you can create millions of unique colors, e.g.:

$$\#FFA500 = \text{Orange}, \quad \#800080 = \text{Purple}, \quad \#FFFFFF = \text{White}, \quad \#000000 = \text{Black}$$

There are $256^3 = 16{,}777{,}216$ possible color combinations in the RGB hex system. The higher the red, green, or blue values are (closer to 255), the brighter the component becomes and the overall color shifts closer to white. Conversely, the lower the values are (closer to 0), the darker the component becomes and the overall color shifts closer to black.

Included in your digital resources are multiple hex videos, as well as an interactive hex color guessing game.

## 87.4 Base-8 (Octal)

Octal is another representation of radix 8. Digits range from 0 to 7, and the number of digits increase the threshold is passed.

Just as in the above systems, octal increases by the radix —– powers of 8. For example,

$$135_8 = (1 \times 8^2) + (3 \times 8^1) + (5 \times 8^0) = 64 + 24 + 5 = 93_{10}$$

### 87.4.1 Why Octal Matters

Octal was historically important in computing because early computers often worked with word sizes that were multiples of 3 bits (such as 12, 24, or 36). Since each octal digit corresponds exactly to 3 binary digits, it was a convenient shorthand for binary values before hexadecimal became more widespread. For example:

$$110101_2 = 65_8$$

Even though modern systems mostly use hexadecimal, octal is still used in some contexts, such as Unix file permissions (e.g., `chmod 755`).

CHAPTER 88

# Floating Point Arithmetic

## 88.1 Why Computers Can't Count Perfectly

Computers are known to be very precise. Modern processors can perform billions of calculations per second, so it's not intuitive to realize they have this persistent limitation: they cannot store most numbers exactly.

Let us think about what it would actually take to store a number like $\frac{1}{3}$. You probably know its decimal expansion:

$$\frac{1}{3} = 0.3333333\ldots$$

The digits go on forever. Now imagine you are a computer with a fixed amount of memory. You cannot store infinitely many digits. So you store 0.333 or 0.3333333, depending on how much space you have, and you accept that the result is not exactly right. That gap between the number you wanted and the number you actually stored is called *approximation error*, and it never fully goes away.

This is not just a problem for fractions with repeating decimals. Irrational numbers like $\sqrt{2} = 1.41421356\ldots$ and transcendental numbers like $\pi = 3.14159265\ldots$ also have infinite, non-repeating decimal expansions. Every one of them must be approximated when stored in a computer.

To handle this systematically, mathematicians and engineers developed the *floating-point system*. Before we look at how it works, it helps to understand the two core reasons why any finite system will always fall short of the real number line.

### 88.1.1 The Number Line is Continuous and Unbounded

Between any two real numbers, no matter how close together, there are infinitely many others. Pick any two decimals you like. For example, if you picked 1.00 and 1.01, you can always find a number between them: 1.005, 1.0051, 1.00500001, and so on without end. This is referred to as *continuity*. A computer's number system cannot be continuous. It stores numbers using a fixed string of digits, which means it can only represent finitely many values. Real numbers have to be rounded to the nearest representable value.

The real numbers stretch infinitely in both directions. There is no largest real number and no smallest positive real number. A computer's memory is finite, so its number system must have both a largest and a smallest representable value. Numbers too large to

store cause a problem called *overflow*. Numbers too small to distinguish from zero cause *underflow*.

### *Exercise 202* Overflow and Underflow

Suppose a floating-point system can only represent numbers between $10^{-4}$ and $10^{8}$.

1. Planck's constant is approximately $6.63 \times 10^{-34}$ J·s. Does it cause underflow, overflow, or neither in this system?
2. The speed of light is approximately $3.00 \times 10^{8}$ m/s. Does it cause underflow, overflow, or neither in this system?
3. Give one example of a number that would cause overflow.

*Working Space*

*Answer on Page 1948*

So what can we do?

We cannot fix these limitations, because they are baked into the nature of finite computation. What we can do is design the floating-point system carefully so that the errors it introduces are small, predictable, and manageable. That is exactly what engineers and mathematicians have spent decades working out, and it is what the rest of this chapter is about!

The key insight is that a floating-point number is not just a string of digits. A floating-point number has three parts: a *sign*, a *significand* (sometimes called the mantissa), and an *exponent*.

## 88.2 Binary and the Floating-Point System

In the previous section you saw that computers cannot store most numbers exactly. Now the natural question is: how do they store numbers at all? The answer starts at the hardware level, with the smallest possible piece of information a computer can hold.

### 88.2.1 Binary System

Every process inside a CPU comes down to the flow of electrical current through tiny switches called *transistors*. A transistor is either on (carrying current) or off (not carrying current). That gives us exactly two states, which we write as 0 and 1. This is why computers use *binary*: it is the only number system that hardware can physically implement with a simple on/off switch.

A single binary digit is called a *bit*. Eight bits grouped together form a *byte*. You have probably seen units like megabyte (MB) or gigabyte (GB) before. One megabyte is $10^6$ bytes, and one gigabyte is $10^9$ bytes. A modern smartphone holds tens of billions of bits.

## *Exercise 203* Bits and bytes

*Working Space*

1. A file is 4 megabytes. How many bytes is that? How many bits?
2. A hard drive stores 500 gigabytes. How many bits can it hold? Write your answer using scientific notation.
3. A single bit can be 0 or 1. Two bits together can be 00, 01, 10, or 11, which are four possible values. How many distinct values can 8 bits (one byte) represent? How about 64 bits?

*Answer on Page 1948*

### 88.2.2 Floating-Point Number

Recall that a floating-point number has three parts: a *sign*, a *significand* (sometimes called the mantissa), and an *exponent*.

Before we write any formulas, let us see why all three parts are necessary with a base-10 analogy.

Consider the number $-0.001234$. Written in scientific notation it is $-1.234 \times 10^{-3}$. Three pieces of information do all the work:

- The **sign** $(-)$ tells us the number is negative.
- The **significand** $(1.234)$ holds the meaningful digits.
- The **exponent** $(-3)$ tells us how large or small the number is by shifting the decimal point.

Now watch what the exponent does when we fix the significand at 0.1234 and vary it in base 10:

| Float | Value |
|---|---|
| $0.1234 \times 10^{-2}$ | 0.001234 |
| $0.1234 \times 10^{1}$ | 1.234 |
| $0.1234 \times 10^{4}$ | 1234 |
| $0.1234 \times 10^{8}$ | 12,340,000 |

The decimal point *floats*, which means it can sit anywhere, controlled entirely by the exponent. That is why these are called floating-point numbers. The significand always carries the same digits; only the scale changes.

## *Exercise 204* The floating point in action

The significand is fixed at 0.5050 and the base is 10. Fill in the value represented by each float.

| Float | Value |
|---|---|
| $0.5050 \times 10^{0}$ | |
| $0.5050 \times 10^{2}$ | |
| $0.5050 \times 10^{-1}$ | |
| $0.5050 \times 10^{5}$ | |

Now answer: if you want to represent 505,000,000 using the same significand, what exponent do you need?

*Working Space*

*Answer on Page 1948*

### 88.2.3 Floating-Point System

A floating-point system is defined by four numbers, written $(\beta, p, m, M)$. Every float in that system looks like this:

$$\pm\,(0.d_1\,d_2\,\cdots\,d_p)_\beta\ \times\ \beta^e$$

This looks intimidating at first, but each symbol has a simple job. Let us unpack them one at a time.

$\beta$ represents the *base*. This is the number that the exponent is a power of. In everyday life we use $\beta = 10$. Computers use $\beta = 2$ (binary).

$p$ represents the *precision*. This is how many digits the significand gets to use. The significand $0.d_1 d_2 \cdots d_p$ has exactly $p$ digits, each chosen from $\{0, 1, \ldots, \beta - 1\}$. More digits means more precision, that is, you can represent numbers more accurately. Fewer digits means faster arithmetic but more rounding error.

$e$ represents the *exponent*. This shifts the significand left or right. The exponent is not free to be any integer; it is capped between a minimum $m$ and a maximum $M$. Those bounds set the range of the system, i.e., how large and how small the representable numbers can be.

$m$ and $M$ represents the *exponent bounds*. The exponent must satisfy $m \le e \le M$. Together they control overflow and underflow. If a result needs an exponent below $m$, underflow occurs. If it needs an exponent above $M$, overflow occurs.

We say the system is *parametrized* by $(\beta, p, m, M)$. Changing any one of the four gives a new floating-point system with different capabilities and different limitations.

### *Exercise 205* Reading the parameters

A toy floating-point system has parameters $(\beta, p, m, M) = (10, 4, -3, 4)$.

*Working Space*

1. What base does this system use?
2. How many significant digits does it carry?
3. What are the smallest and largest allowed exponents?
4. Can this system represent 12,345? Try writing 12,345 in the form $0.d_1d_2d_3d_4 \times 10^e$ and check whether the exponent is within bounds.
5. Can it represent 0.00001? Show your reasoning.
6. Write two numbers that *can* be represented exactly, and one that causes overflow.

*Answer on Page 1948*

#### 88.2.4 Connecting to Binary Floating-Point

All of the above applies to a computer's binary floating-point system, with $\beta = 2$ instead of $\beta = 10$. The significand digits $d_1, d_2, \ldots, d_p$ are now bits: each one is either 0 or 1. The exponent is a power of 2.

So a binary float looks like:

$$\pm\,(0.d_1\,d_2\,\cdots\,d_p)_2 \;\times\; 2^e, \qquad d_i \in \{0, 1\}, \quad m \le e \le M$$

The specific choices of $p$, $m$, and $M$ define how much memory is used and how accurately numbers are stored. The international standard that fixes these choices for every modern computer is called *IEEE 754*, and it is the subject of the next section.

## 88.3 The IEEE 754 Standard

In 1985, after years of debate, the Institute of Electrical and Electronics Engineers published the *IEEE 754 standard*. It was accepted internationally in 1989, and it is still the standard used in virtually every computer, phone, and calculator you use today.

### 88.3.1 Floats

If we are going to store a float in a fixed number of bits, how do we split those bits among the three parts: sign, significand, and exponent?

For the sign, there are only two options, positive and negative, so a single bit handles it: 0 for positive, 1 for negative.

That leaves $N-1$ bits to split between the significand and the exponent. IEEE 754 defines two standard ways to do this split, called *single precision* and *double precision*.

| | Total bits | Significand bits ($p$) | Exponent bits |
|---|---|---|---|
| Single precision | 32 | 23 | 8 |
| Double precision | 64 | 52 | 11 |

In both cases the layout is the same. The sign occupies the first bit, the significand fills the next $p$ bits, and the exponent takes the remaining bits. Double precision uses twice as many bits in total, which gives it both greater precision and a wider range of representable exponents.

For single precision, the 8 exponent bits can represent integers up to $M = 127$, so the largest float in single precision is roughly $10^{38}$. Double precision is the standard used in modern computers.

### *Exercise 206* Bit Allocation

A floating-point system uses $N = 16$ bits total. The sign takes 1 bit and the significand takes 10 bits.

1. How many bits are left for the exponent?
2. If those exponent bits can represent integers from $-31$ to 32, what is the largest exponent M?
3. Would you expect this system to be more or less precise than single precision? Why?

*Working Space*

*Answer on Page 1949*

### 88.3.2 Machine Epsilon

The *machine epsilon* is commonly written as $u$. It measures the largest relative gap between a real number and its nearest float. In other words, it tells you the worst-case precision of your system.

For a system that rounds to the nearest float, machine epsilon is

$$u = \frac{1}{2}\beta^{1-p},$$

where $\beta$ is the base and $p$ is the precision. For IEEE 754 double precision, $\beta = 2$ and $p = 52$, giving $u \approx 2.22 \times 10^{-16}$. That means any real number is stored with a relative error no worse than about $2 \times 10^{-16}$ [correct punctuation] roughly 16 significant decimal digits of accuracy.

If you have access to Python, you can check your machine epsilon directly:

```
import sys
print(sys.float_info.epsilon)   # double precision machine epsilon
```

Every machine running IEEE 754 double precision should return the same value: `2.220446049250313e-16`.

## *Exercise 207* Machine Epsilon

A base-10 floating-point system has precision $p = 6$ and uses rounding.

1. Compute machine epsilon $u$ for this system.
2. A real number $x$ is stored as a float $\tilde{x}$. Using your answer above, what is the worst-case relative error $\frac{|\tilde{x}-x|}{|x|}$?
3. If $p$ were doubled to $p = 12$, how would $u$ change?

*Working Space*

*Answer on Page 1949*

**Good to know:** IEEE 754 has been updated since 1985. The current standard includes *quadruple precision* (128 bits) for applications that need the highest accuracy. This includes physics simulations and cryptographic computations. It can also define integer data types, `int8`, `int32`, and `int64`. These are used in situations where you only need whole numbers and want to avoid floating-point arithmetic entirely.

The existence of multiple precisions is a reminder that there is always a trade-off. More bits means more accuracy and a wider range, but also more memory and slower computation. Choosing the right precision for a given problem is the real engineering decision.

CHAPTER 89

# Introduction to Polynomials

Watch Khan Academy's **Polynomials intro** video at https://youtu.be/Vm7H0VTlIco

A *monomial* is the product of a number and a variable raised to a non-negative (but possibly zero) integer power. Here are some examples of monomials:

$3x^2$ $\pi x^2$ $7x$ $-\frac{2}{3}x^{12}$

$-2x^{15}$ $(3.33)x^{100}$ $3$ $0$

The exponent is called the *degree* of the monomial. For example, $3x^{17}$ has degree 17, $-7x$ has degree 1, and 3.2 has degree 0 (because you can think of it as $(3.2)x^0$).. The exponent cannot be $x$ (or any non-constant variable), as that becomes an exponential function rather than polynomial.[1]

The number in the product is called the *coefficient*. Example: $3x^{17}$ has a coefficient of 3, $-2x$ has a coefficient of -2, and $(3.4)x^{1000}$ has a coefficient of 3.4.

A *polynomial* is the sum of one or more monomials. Here are some polynomials:

$4x^2 + 9x + 3.9$ $\pi x^2 + \pi x + \pi$ $7x + 2$

$-2x^{10} + (3.4)x - 45x^{900} - 1$ $3.3$ $3x^{20}$

We say that each monomial is a *term* of the polynomial.

$x^{-5} + 12$ is *not* a polynomial because the first term has a negative exponent.

$x^2 - 32x^{\frac{1}{2}} + x$ is *not* a polynomial because the second term has a non-integer exponent.

$\frac{x+2}{x^2+x+5}$ is *not* a polynomial because it is not just a sum of monomials.

[1] A quadratic is a polynomial with a maximum degree of 2

## *Exercise 208* Identifying Polynomials

Circle only the polynomials.

*Working Space*

$-2x^3+\frac{1}{2}x+3.9(4.5)x^2+\pi x$ $\qquad 7$

$x^{\frac{2}{3}}$ $\qquad 3x^{20}+2x^{19}-5x^{18}$

$2x^{-10}+4x-1$

*Answer on Page 1949*

We typically write a polynomial starting at the term with the highest degree and proceed in decreasing order to the term with the lowest degree:

$$2x^9-3x^7+\frac{3}{4}x^3+x^2+\pi x-9.3$$

This is known as *the standard form*. The first term of the standard form is called *the leading term*, and we often call the coefficient of the leading term *the leading coefficient*. We sometimes speak of the degree of the polynomial, which is just the degree of the leading term.

## *Exercise 209* Standard of a Polynomial

Write $21x^2-x^3+\pi-1000x$ in standard form. What is the degree of this polynomial? What is its leading coefficient?

*Working Space*

*Answer on Page 1950*

## *Exercise 210* Evaluate a Polynomial

Let $y = x^3 - 3x^2 + 10x - 12$. What is $y$ when $x$ is 4?

*Working Space*

*Answer on Page 1950*

We would be remiss in our duties if we didn't mention one more thing about polynomials: Mathematicians have defined a polynomial to be a sum of a *finite* number of monomials.

It is certainly possible to have a sum of an infinite number of monomials like this:

$$1 + \frac{1}{2}x + \frac{1}{4}x^2 + \frac{1}{8}x^3 + \frac{1}{16}x^4 + \dots$$

This is an example of an *infinite series*, which we don't consider polynomials. Infinite series are interesting and useful, but we will not discuss them in detail until later in the course.

# Chapter 90

# Python Lists

Watch CS Dojo's **Introduction to Lists in Python** video at https://www.youtube.com/watch?v=tw7ror9x32s

To review, Python list is an indexed collection. The indices start at zero. You can create a list using square brackets.

You are now going to write a program that makes an array of strings. Type this code into a file called `faves.py`:

```
favorites = ["Raindrops", "Whiskers", "Kettles", "Mittens"]
favorites.append("Packages")
print("Here are all my favorites:", favorites)
print("My most favorite thing is", favorites[0])
print("My second most favorite is", favorites[1])
number_of_faves = len(favorites)
print("Number of things I like:", number_of_faves)

for i in range(number_of_faves):
    print(i, ": I like", favorites[i])
```

Run it:

```
$ python3 faves.py
Here are all my favorites: ['Raindrops', 'Whiskers', 'Kettles', 'Mittens',
↪  'Packages']
My most favorite thing is Raindrops
My second most favorite is Whiskers
Number of things I like: 5
0 : I like Raindrops
1 : I like Whiskers
2 : I like Kettles
3 : I like Mittens
4 : I like Packages
```

After you have run the code, study it until the output makes sense.

## *Exercise 211* Assign into list

Before you list the items, replace "Mittens" with "Gloves".

*Working Space*

*Answer on Page 1950*

## 90.1 Evaluating Polynomials in Python

First, before you go any further, you need to know that raising a number to a power is done with ** in Python. For example, to get $5^2$, you would write `5**2`.

Back to polynomials: if you had a polynomial like $2x^3 - 9x + 12$, you could write it like this: $12x^0 + (-9)x^1 + 0x^2 + 2x^3$. We could use this representation to keep a polynomial in a Python list. We would simply store all the coefficients in order:

```
1 pn1 = [12,-9,0,2]
```

In the list, the index of each coefficient would correspond to the degree of that monomial. For example, in the list, 2 is at index 3, so that entry represents $2x^3$.

In the last chapter, you evaluated the polynomial $x^3 - 3x^2 + 10x - 12$ at $x = 4$. Now you will write code that does that evaluation. Create a file called `polynomials.py` and type in the following:

```
1  def evaluate_polynomial(pn, x):
2      sum = 0.0
3      for degree in range(len(pn)):
4          coefficient = pn[degree]
5          term_value = coefficient * x ** degree
6          sum = sum + term_value
7      return sum
8
9  pn1 = [-12.0, 10.0, -3.0, 1.0]
10 y = evaluate_polynomial(pn1, 4.0)
11 print("Polynomial 1: When x is 4.0, y is", y)
```

Run it. It should evaluate to 44.0.

## 90.2 Walking the list backwards

Now you are going to make a function that makes a pretty string to represent your polynomial. Here is how it will be used:

```
def polynomial_to_string(pn):
    ...Your Code Here...

pn_test = [-12.0, 10.0, 0.0, 1.0]
print(polynomial_to_string(pn1))
```

This would output:

```
1.0x**3 + 10.0x + -12.0
```

This is not as simple as you might hope. In particular:

- You should skip the terms with a coefficient of zero
- The term of degree 1 has an x, but no exponent
- The term of degree 0 has neither an x nor an exponent
- Standard form demands that you list the terms in the reverse order from that of your coefficients list. You will need to walk the list from last to first.

Add this function to your `polynomials.py` file after your `evaluate_polynomial` function:

```
def polynomial_to_string(pn):

    # Make a list of the monomial strings
    monomial_strings = []

    # Start at the term with the largest degree
    degree = len(pn) - 1

    # Go through the list backwards stop after constant term
    while degree >= 0:
        coefficient = pn[degree]

        # Skip any term with a zero coefficient
        if coefficient != 0.0:

            # Describe the monomial
            if degree == 0:
                monomial_string = "{}".format(coefficient)
            elif degree == 1:
```

```
                monomial_string = "{}x".format(coefficient)
            else:
                monomial_string = "{}x^{}".format(coefficient, degree)

            # Add it to the list
            monomial_strings.append(monomial_string)

        # Move to the previous term
        degree = degree - 1

    # Deal with the zero polynomial
    if len(monomial_strings) == 0:
        monomial_strings.append("0.0")

    # Make a string that joins the terms with a plus sign
    return " + ".join(monomial_strings)
```

Note that in a list $n$ items, the indices go from 0 to $n-1$. When we are walking the list backwards, we start at `len(pn) - 1` and stop at zero.

Look over the code and google the functions you aren't familiar with. For example, if you want to know about the (join) function, google for "python join".

Now, change your code to use the new function:

```
pn1 = [-12.0, 10.0, -3.0, 1.0]
y = evaluate_polynomial(pn1, 4.0)
print("y =", polynomial\_to\_string(pn1))
print("    When x is 4.0, y is", y)
```

Run the program. Does the function work?

### *Exercise 212* Evaluate Polynomials

Using the function that you just wrote, add a few lines of code to `polynomials.py` to evaluate the following polynomials:

- Find $4x^4 - 7x^3 - 2x^2 + 5x + 2.5$ at $x = 8.5$. It should be 16481.875
- Find $5x^5 - 9$ at $x = 2.0$. It should be 151.0

*Working Space*

*Answer on Page 1950*

## 90.3 Plot the polynomial

We can evaluate a polynomial at many points and plot them on a graph. You are going to write the code to do this. Create a new file called `plot_polynomial.py`. Copy your `evaluate_polynomial` function into the new file.

Add a line at the beginning of the program that imports the plotting library matplotlib:

```
1 import matplotlib.pyplot as plt
```

After the `evaluate_polynomial` function:

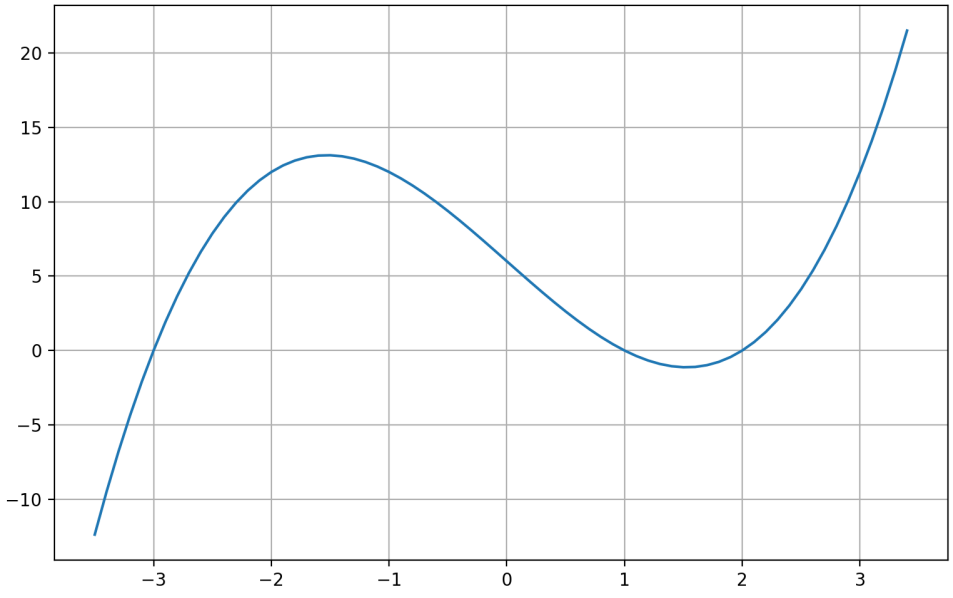
- Create a list with polynomial coefficients.
- Create two empty arrays, one for x values and one for y values.
- Fill the x array with values from -3.5 to 3.5. Evaluate the polynomial at each of these points; put those values in the y array.
- Plot them

Like this:

```
1 # x**3 - 7x + 6
2 pn = [6.0, -7.0, 0.0, 1.0]
3
```

```
# These lists will hold our x and y values
x_list = []
y_list = []

# Start at x=-3.5
current_x =-3.5

# End at x=3.5
while current_x <= 3.5:
    current_y = evaluate_polynomial(pn, current_x)

    # Add x and y to respective lists
    x_list.append(current_x)
    y_list.append(current_y)

    # Move x forward
    current_x += 0.1

# Plot the curve
plt.plot(x_list, y_list)
plt.grid(True)
plt.show()
```

You should get a beautiful plot like this:




If you received an error that the matplotlib was not found, use pip to install it:

```
$ pip3 install matplotlib
```

## *Exercise 213* Observations

Where does your polynomial cross the y-axis? Looking at the polynomial $x^3 - 7x+6$, could you have guessed that value?

*Working Space*

Where does your polynomial cross the x-axis? The places where a polynomial crosses the x-axis are called *its roots*. Later in the course, you will learn techniques for finding the roots of a polynomial.

*Answer on Page 1950*

Chapter 91

# Adding and Subtracting Polynomials

Watch Khan Academy's **Adding polynomials** video at https://youtu.be/ahdKdxsTj8E

When adding two monomials of the same degree, you sum their coefficients:

$$7x^3 + 4x^3 = 11x^3$$

Using this idea, when adding two polynomials, you convert them into one long polynomial, then simplify by combining terms with the same degree. For example:

$$\begin{aligned}(10x^3 - 2x + 13) + (-5x^2 + 7x - 12) \\ &= 10x^3 + (-2)x + 13 + (-5)x^2 + 7x + (-12) \\ &= 10x^3 + (-5)x^2 + (-2+7)x + (13-12) \\ &= 10x^3 - 5x^2 + 5x + 1\end{aligned}$$

## *Exercise 214* Adding Polynomials Practice

*Working Space*

Add the following polynomials:

1. $2x^3 - 5x^2 + 3x - 9$ and $x^3 - 2x^2 - 2x - 9$

2. $3x^5 - 5x^3 + 3x^2 - x - 3$ and $2x^4 - 2x^3 - 2x^2 + x - 9$

*Answer on Page 1951*

Notice that in the second question, the degree 1 term disappears completely: $(-x) + x = 0$

One more tricky thing can happen — sometimes the coefficients don't add nicely. For example:

$$\pi x^2 - 3x^2 = (\pi - 3)x^2$$

That is as far as you can simplify it.

## 91.1 Subtraction

Now, watch Khan Academy's **Subtracting polynomials** at https://youtu.be/5ZdxnFspyP8.

When subtracting one polynomial from the other, it is a lot like adding two polynomials. The difference: when making the two polynomials into one long polynomial, we multiply each monomial that is being subtracted by -1. For example:

$$\begin{aligned}(2x^2 - 3x + 9) - (5x^2 - 7x + 4) &= 2x^2 + (-3)x + 9 + (-5)x^2 + 7x + (-4) \\ &= (2-5)x^2 + (-3+7)x + (9-4) \\ &= -3x^2 + 4x + 5\end{aligned}$$

### *Exercise 215* Subtracting Polynomials Practice

Subtract the following polynomials:

*Working Space*

1. $(2x^3 - 5x^2 + 3x - 9) - (x^3 - 2x^2 - 2x - 9)$

2. $(3x^5 - 5x^3 + 3x^2 - x - 3) - (2x^4 - 2x^3 - 2x^2 + x - 9)$

*Answer on Page 1951*

## 91.2 Adding Polynomials in Python

As a reminder, in our Python code, we are representing a polynomial with a list of coefficients. The first coefficient is the constant term. The last coefficient is the leading coefficient. So, we can imagine $-5x^3 + 3x^2 - 4x + 9$ and $2x^3 + 4x^2 - 9$ would look like this: *FIXME: Diagram here*

To add the two polynomials then, we sum the coefficients for each degree. *FIXME: Diagram here*

Create a file called `add_polynomials.py`, and type in the following:

```
1  def add_polynomials(a, b):
2      degree_of_result = len(a)
3      result = []
4      for i in range(degree_of_result):
5          coefficient_a = a[i]
6          coefficient_b = b[i]
7          result.append(coefficient_a + coefficient_b)
8      return result
9
10 polynomial1 = [9.0, -4.0, 3.0, -5.0]
```

```
polynomial2 = [-9.0, 0.0, 4.0, 2.0]
polynomial3 = add_polynomials(polynomial1, polynomial2)

print('Sum =', polynomial3)
```

Run the program.

Unfortunately, this code only works if the polynomials are the same length. For example, try making `polynomial1` have a larger degree than `polynomial2`:

```
# x**4 - 5x**3 + 3x**2 - 4x + 9
polynomial1 = [9.0, -4.0, 3.0, -5.0, 1.0]

# 2x**3 + 4x**2 - 9
polynomial2 = [-9.0, 0.0, 4.0, 2.0]
polynomial3 = add_polynomials(polynomial1, polynomial2)
print('Sum =', polynomial3)
```

See the problem?

## *Exercise 216* Dealing with polynomials of different degrees

*Working Space*

Can you fix the function `add_polynomials` to handle polynomials of different degrees?

Here is a hint: In Python, there is a `max` function that returns the largest of the numbers it is passed.

```
biggest = max(5,7)
```

Here `biggest` would be set to 7.

Here is another hint: If you have an array `mylist`, `i`, a non-negative integer, is only a legit index if `i < len(mylist)`.

*Answer on Page 1951*

## 91.3 Scalar multiplication of polynomials

If you multiply a polynomial with a number, the distributive property applies:

$$(3.1)(2x^2 + 3x + 1) = (6.2)x^2 + (9.3)x + 3.1$$

(When we are talking about things that are more complicated than a number, we use the word *scalar* to mean "Just a number". This is the product of a scalar and a polynomial.)

In `add_polynomials.py`, add a function to that multiplies a scalar and a polynomial:

```
def scalar_polynomial_multiply(s, pn):
    result = []
    for coefficient in pn:
        result.append(s * coefficient)
    return result
```

Somewhere near the end of the program, test this function:

```
polynomial4 = scalar_polynomial_multiply(5.0, polynomial1)
print('Scalar product =', polynomial_to_string(polynomial4))
```

## *Exercise 217* Subtract polynomials in Python

*Working Space*

Now, implement a function that does subtraction using `scalar_polynomial_multiply` and `add_polynomials`.

It should look like this:

```
def subtract_polynomial(a, b):
    ...Your code here...

polynomial5 = [9.0, -4.0, 3.0,
 ↪  -5.0]
polynomial6 = [-9.0, 0.0, 4.0, 2.0,
 ↪  1.0]
polynomial7 =
 ↪  subtract_polynomial(polynomial5,
 ↪  polynomial6)
print('Difference =',
 ↪  polynomial_to_string(polynomial7))
```

*Answer on Page 1951*

CHAPTER 92

# Multiplying Polynomials

Watch Khan Academy's **Multiplying monomials** at https://www.youtube.com/watch?v=7EplN4Ch98Q.

To review, when you multiply two monomials, you take the product of their coefficients and the sum of their degrees:

$$(2x^6)(5x^3) = (2)(5)(x^6)(x^3) = 10x^9$$

If you have a product of more than two monomials, multiply *all* the coefficients and sum *all* the exponents:

$$(3x^2)(2x^3)(4x) = (3)(2)(4)(x^2)(x^3)(x^1) = 24x^6$$

## *Exercise 218* Multiplying monomials

Multiply these monomials:

*Working Space*

1. $(3x^2)(5x^3)$

2. $(2x)(4x^9)$

3. $(-5.5x^2)(2x^3)$

4. $(\pi)(-2x^5)$

5. $(2x)(3x^2)(5x^7)$

*Answer on Page 1951*

## 92.1 Multiplying a monomial and a polynomial

Watch Khan Academy's **Multiplying monomials by polynomials** at https://youtu.be/pD2-H15ucNE. When multiplying a monomial and a polynomial, you use the distributive property. After that, it is just multiplying several pairs of monomials:

$$(3x^2)(4x^3 - 2x^2 + 3x - 7)$$
$$= (3x^2)(4x^3) + (3x^2)(-2x^2) + (3x^2)(3x) + (3x^2)(-7)$$
$$= 12x^5 - 6x^4 + 9x^3 - 21x^2$$

## *Exercise 219* Multiplying a monomial and a polynomial

*Working Space*

Multiply these monomials:

1. $(3x^2)(5x^3 - 2x + 3)$

2. $(2x)(4x^9 - 1)$

3. $(-5.5x^2)(2x^3 + 4x^2 + 6)$

4. $(\pi)(-2x^5 + 3x^4 + x)$

5. $(2x)(3x^2)(5x^7 + 2x)$

*Answer on Page 1952*

## 92.2 Multiplying polynomials

Watch Khan Academy's **Multiplying binomials by polynomials** video at https://youtu.be/D6mivA_8L8U

When you are multiplying two polynomials, you will use the distributive property several times to make it one long polynomial. You will then combine the terms with the same degree. For example,

$$\begin{aligned}(2x^2-3)(5x^2+2x-7) &\\ &= (2x^2)(5x^2+2x-7)+(-3)(5x^2+2x-7)\\ &= (2x^2)(5x^2)+(2x^2)(2x)+(2x^2)(-7)+(-3)(5x^2)+(-3)(2x)+(-3)(-7)\\ &= 10^4+4x^3+-14x^2+-15x^2+-6x+21 = 10^4+4x^3+-29x^2+-6x+21\end{aligned}$$

One common form that you will see is multiplying two binomials together:

$$(2x+7)(5x+3) = (2x)(5x+3)+(7)(5x+3) = (2x)(5x)+(7)(5x)+(2x)(3)+(7)(3)$$

Notice the product has become the sum of four parts: the firsts, the inners, the outers, and the lasts. People sometimes use the mnemonic FOIL to remember this pattern, but there is a general rule that works for all product of polynomials, not just binomials: Every term in the first will be multiplied by every term in the second, and then just add them together.

So, for example, if you have a polynomial $s$ with three terms and you multiply it by a polynomial $t$ with five terms, you will get a sum of 15 terms — each term is a product of two monomials, one from $s$ and one from $t$. (Of course, several of those terms might have the same degree, so they will be combined together when you simplify. Thus you typically end up with a polynomial with less than 15 terms.)

Using this rule, here is how you would multiply $2x^2-3x+1$ and $5x^2+2x-7$:

$$\begin{aligned}(2x^2-3x+1)(5x^2+2x-7) &= \begin{array}{rcrcrc}(2x^2)(5x^2) & + & (2x^2)(2x) & + & (2x^2)(-7) & +\\ (-3x)(5x^2) & + & (-3x)(2x) & + & (-3x)(-7) & +\\ (1)(5x^2) & + & (1)(2x) & + & (1)(-7) & \end{array}\\ &= 10x^4+4x^3+(-14)x^2+(-15)x^3+(-6)x^2+21x+5x^2+2x+(-7)\\ &= 10x^4+(4-15)x^3+(-14-6+5)x^2+(21+2)x+(-7)\\ &= 10x^4-11x^3-15x^2+23x-7\end{aligned}$$

Note that the product (before combining terms with the same degree) has $3\times 3 = 9$ terms — every possible combination of a term from the first polynomial and a term from the second polynomial.

One common source of error is losing track of the negative signs. You will need to be really careful. We have found that it helps to use + between all terms, and use negative coefficients to express subtraction. For example, if the problem says $4x^2-5x-3$, you should work with that as $4x^2+(-5)x+(-3)$.

## *Exercise 220* Multiplying polynomials

Multiply the following pairs of polynomials:

1. $2x + 1$ and $3x - 2$

2. $-3x^2 + 5$ and $4x - 2$

3. $-2x - 1$ and $-3x - \pi$

4. $-2x^5 + 5x$ and $3x^5 + 2x$

*Working Space*

*Answer on Page 1952*

## *Exercise 221* Observations

Let's say you have two polynomials, $p_1$ and $p_2$. $p_1$ has degree 23. $p_2$ has degree 12. What is the degree of their product?

*Working Space*

*Answer on Page 1952*

Chapter 93

# Multiplying Polynomials in Python

At this point, you have created a nice toolbox of functions for dealing with lists of coefficients as polynomials. Create a file called `poly.py` and copy the following functions into it:

- `evaluate_polynomial`
- `polynomial_to_string`
- `add_polynomials`
- `scalar_polynomial_multiply`
- `subtract_polynomial`

Now, create another file in the same directory called `test.py`. Type this into that file:

```
import poly

polynomial_a = [9.0, -4.0, 3.0, -5.0]
print('Polynomial A =', poly.polynomial_to_string(polynomial_a))

polynomial_b = [-9.0, 0.0, 4.0, 2.0, 1.0]
print('Polynomial B =', poly.polynomial_to_string(polynomial_b))

# Evaluation
value_of_b = poly.evaluate_polynomial(polynomial_b, 3)
print('Polynomial B at 3 =', value_of_b)

# Adding
a_plus_b = poly.add_polynomials(polynomial_a, polynomial_b)
print('A + B =', poly.polynomial_to_string(a_plus_b))

# Scalar multiplication
b_scalar = poly.scalar_polynomial_multiply(-3.2, polynomial_b)
print('-3.2 * Polynomial B =', poly.polynomial_to_string(b_scalar))

# Subtraction
a_minus_b = poly.subtract_polynomial(polynomial_a, polynomial_b)
print('A - B =', poly.polynomial_to_string(a_minus_b))
```

When you run it, you should get the following:

```
Polynomial A = -5.0x^3 + 3.0x^2 + -4.0x + 9.0
Polynomial B = 1.0x^4 + 2.0x^3 + 4.0x^2 + -9.0
Polynomial B at 3 = 162.0
A + B = 1.0x^4 + -3.0x^3 + 7.0x^2 + -4.0x
-3.2 * Polynomial B = -3.2x^4 + -6.4x^3 + -12.8x^2 + 28.8
A - B = -1.0x^4 + -7.0x^3 + -1.0x^2 + -4.0x + 18.0
```

You are now ready to implement the multiplication of polynomials. The function will look like this:

```
def multiply_polynomials(a, b):
  ...Your code here...
```

It will return a list of coefficients.

In an exercise in the last chapter, you were asked " Let's say I have two polynomials, $p_1$ and $p_2$. $p_1$ has degree 23. $p_2$ has degree 12. What is the degree of their product?" The answer was $23 + 12 = 35$.

In our implementation, a polynomial of degree 23 is held in a list of length 24.

In Python, we will be trying to multiply a polynomial `a` and a polynomial `b` represented as lists. What is the degree of that product?

```
result_degree = (len(a) - 1) + (len(b) - 1)
```

Now, we need to create an array of zeros that is one longer than that. Here is a cute Python trick: If you have a list, you can replicate it using the * operator.

```
a = [5,7]
b = a * 4
print(b)
# [5, 7, 5, 7, 5, 7, 5, 7]
```

Here iss how you will get a list of zeros:

```
result = [0.0] * (result_degree + 1)
```

We will step through `a`, getting the index and value of each entry. You can do this in one

line using `enumerate`:

```
for a_degree, a_coefficient in enumerate(a):
```

For each of those, we will step through the entire b polynomial. As you multiply together each term, you will add it to the appropriate coefficient of the result.

Here is the whole function:

```
def multiply_polynomials(a, b): # What is the degree of the resulting
polynomial?  result_degree = (len(a) - 1) + (len(b) - 1)

    # Make a list of zeros to hold the coefficients result = [0.0] *
    (result_degree + 1)

    # Iterate over the indices and values of a for a_degree,
    a_coefficient in enumerate(a):

        # Iterate over the indices and values of b for b_degree,
        b_coefficient in enumerate(b):

            # Calculate the resulting monomial coefficient =
            a_coefficient * b_coefficient degree = a_degree + b_degree

            # Add it to the right bucket
            result[degree] = result[degree] + coefficient

    return result
```

Take a long look at that function. When you understand it, type it into `poly.py`.

In `test.py`, try out the new function:

```
# Multiplication
a_times_b = poly.multiply_polynomials(polynomial_a, polynomial_b)
print('A x B =', poly.polynomial_to_string(a_times_b))
```

This is an example of a *nested loop*. The outer loop steps through the polynomial a. For each step it takes, the inner loop steps through the entire polynomial b.

## 93.1 Something surprising about lists

You can imagine that you might want to create two very similar polynomials. Let's say polynomial c is $x^2 + 2x + 1$ and polynomial d is $x^2 - 2x + 1$. You might think you are very

clever to just alter that degree 1 coefficient like this:

```
c = [1.0, 2.0, 1.0]
d = c
d[1] = -2.0
```

If you printed out c, you would get $[1.0, -2.0, 1.0]$. Why? You assigned two variables (c and d) to the *the same list*. So, when you use one reference (d) to change the list, you see the change if you look at the list from either reference. *FIXME: Diagram of two references to the same list here.*

To create two separate lists, you would need to explicitly make a copy:

```
c = [1.0, 2.0, 1.0]
d = c.copy()
d[1] = -2.0
```

# Chapter 94

# Differentiating Polynomials

If you had a function that gave you the height of an object, it would be handy to be able to figure out a function that gave you the velocity at which it was rising or falling. The process of converting the position function into a velocity function is known as *differentiation* or *finding the derivative*. There are a bunch of rules for finding a derivative, but differentiating polynomials only requires three:

- The derivative of a sum is equal to the sum of the derivatives.
- The derivative of a constant is zero.
- The derivative of a nonconstant monomial $at^b$ ($a$ and $b$ are constant numbers, $t$ is time) is $abt^{b-1}$. This is referred to as the **power rule**.

**Power Rule**

For $f(x) = ax^n$, the derivative is $f'(x) = anx^{n-1}$.

Say, for example, that we tell you that the height (position) in meters of a quadcopter at second $t$ is given by $2t^3 - 5t^2 + 9t + 200$. You could tell us that its vertical velocity is $6t^2 - 10t + 9$.

We indicate the derivative of a function with an apostrophe (read as "prime") between the name of the function and the variable. For example, the derivative of $h(t)$ is $h'(t)$ (which is read out loud as "h prime of t").

## *Exercise 222* Differentiation of polynomials

Differentiate the following polynomials.

*Working Space*

1. $f(t) = 2t^3 - 3t^2 - 4t$
2. $g(t) = 2t^{-3/4}$
3. $F(r) = \frac{5}{r^3}$
4. $H(u) = (3u - 1)(u + 2)$

*Answer on Page 1952*

Notice that the degree of the derivative is one less than the degree of the original polynomial. (Unless, of course, the degree of the original is already zero.)

Now, if you know that a position is given by a polynomial, you can differentiate it to find the object's velocity at any time.

The same trick works for acceleration: Let's say you know a function that gives an object's velocity. To find its acceleration at any time, you take the derivative of the velocity function (the second derivative).

### *Exercise 223* Differentiation of polynomials in Python

Write a function that returns the derivative of a polynomial in `poly.py`. It should look like this:

```
def derivative_of_polynomial(pn):
  ...Your code here...
```

When you test it in `test.py`, it should look like this:

```
# 3x**3 + 2x + 5
p1 = [5.0, 2.0, 0.0, 3.0]
d1 =
↪  poly.derivative_of_polynomial(p1)
# d1 should be 9x**2 + 2
print("Derivative of",
↪  poly.polynomial_to_string(p1),"is",
↪  poly.polynomial_to_string(d1))

# Check constant polynomials
p2 = [-9.0]
d2 =
↪  poly.derivative_of_polynomial(p2)
# d2 should be 0.0
print("Derivative of",
↪  poly.polynomial_to_string(p2),"is",
↪  poly.polynomial_to_string(d2))
```

*Working Space*

*Answer on Page 1953*

## 94.1 Second order and higher derivatives

As seen from the example, with height, velocity, and acceleration, you can take the derivative of a derivative, which is called the second derivative and is indicated with two marks, like so:

$$\frac{d}{dx}f'(x) = f''(x)$$

When you have the height function (or position function, in the case of horizontal motion) of an object, the first derivative describes the velocity of the object, and the second deriva-

tive describes the acceleration. Suppose the motion of a particle is given by $s(t) = t^3 - 5t$, where $s$ is in meters and $t$ is in seconds. What is the acceleration when the velocity is 0? First, we find the velocity function, $s'(t)$, and the acceleration function, $s''(t)$:

$$s'(t) = 3t^2 - 5$$

$$s''(t) = 6t$$

To find where the velocity is 0, set $s'(t) = 0$:

$$3t^2 - 5 = 0$$

$$3t^2 = 5$$

$$t^2 = \frac{5}{3}$$

$$t = \sqrt{\frac{5}{3}} \approx 1.29s$$

(We ignore the other solution, $t = -\sqrt{\frac{5}{3}}$ because it is usual for time to start at zero.)

Next, we use $t \approx 1.29s$ in the acceleration function, $s''(t)$:

$$s''(\sqrt{\frac{5}{3}}) = 6\sqrt{\frac{5}{3}} \approx 7.75\frac{m}{s^2}$$

For higher order derivatives, you just keep taking the derivative! So a third derivative is found by taking the derivative of the second derivative, and so on.

## *Exercise 224* Using Derivatives to Describe Motion

The position of a particle is described by the equation $s(t) = t^4 - 2t^3 + t^2 - t$, where $s$ is in meters and $t$ is in seconds.

(a) Find the velocity and acceleration as functions of $t$.

(b) Find the velocity after 1.5 s.

(c) Find the acceleration after 1.5 s.

(d) Is the object speeding up or slowing down at $t = 1.5$? How do you know?

*Working Space*

*Answer on Page 1953*

# Chapter 95

# Python Classes

FIXME integrate this better with polynomials, but we need to cover the following

## 95.1 Object-Oriented Programming Introduction

Imagine you want to implement multiple dogs in python. It gets a bit complicated to do the following:

```
    dog1name = "Teddy"
    dog1age = 6
    dog1sound = "woof"

    dog2name = "Fluffers"
    dog2age = 2
    dog2sound = "bark"

    dog3name = "Bella"
    dog3age = 3
    dog3sound = "grr"

    print(f"{dog1name} is {dog1age} and says {dog1sound}!")
    print(f"{dog2name} is {dog2age} and says {dog2sound}!")
    print(f"{dog3name} is {dog3age} and says {dog3sound}!")


```

Instead, we can use classes, which are a way to create your own datatype. Classes can contain custom methods that either return, print, or calculate different values for you, and they can contain custom variables referred to as attributes.

```
class Dog:
    """A simple model of a dog."""
    # constructor - which creates a new model of a dog.
    def __init__(self, name: str, age: int, sound: str):
        self.name = name
        self.age = age
        self.sound = sound
    # method speak
    def speak(self) -> None:
```

```
        """Print a sentence describing the dog."""
        print(f"{self.name} is {self.age} and says {self.sound}!")

# create dog objects called "instances" with the given variables
dog1 = Dog("Teddy", 6, "woof")
dog2 = Dog("Fluffers", 2, "bark")
dog3 = Dog("Bella", 3, "grr")

# call the behavior on each object
dog1.speak()
dog2.speak()
dog3.speak()

```

Now let's talk about classes in the context of polynomials!

The built-in types, such as strings, have functions associated with them. So, for example, if you needed a string converted to uppercase, you would call its `upper()` function: -

```
my_string = "houston, we have a problem!"
louder_string = my_string.upper()
```

This would set `louder_string` to "HOUSTON, WE HAVE A PROBLEM!" When a function is associated with a datatype like this, it is called a *method*. A datatype with methods is known as a *class*. Creating a new version of a class in a variable is called an *instance*. For example, in the example, we would say "`my_string` is an instance of the class `str`. `str` has a method called `upper`"

The function `type` will tell you the type of any data:

```
print(type(my_string))
```

This will output

```
<class 'str'>
```

A class can also define operators. `+`, for example, is redefined by `str` to concatenate strings together:

```
long_string = "I saw " + "15 people"
```

## 95.2 Parent classes

Classes can have classes they inherit from (called "parent classes") or classes that inherit from them (called "child classes"). Parent classes (ie 'Animal') give a subclass (or child) different attributes or methods. Let's check out an example:

```
class Animal:
    """Generic animal base-class."""

    def __init__(self, name: str, age: int) -> None:
        self.name = name
        self.age = age

    def describe(self) -> None:
        """Print a basic description common to all animals."""
        print(f"{self.name} is {self.age} years old.")

class Dog(Animal):
    """Dog inherits name and age from Animal, adds its own sound."""

    def __init__(self, name: str, age: int, sound: str) -> None:
        super().__init__(name, age)      # initialize the Animal part
        self.sound = sound

    def speak(self) -> None:
        """Dog-specific implementation of speak()."""
        print(f"{self.name} is {self.age} and says {self.sound}!")


# demonstration
dogs = [
    Dog("Teddy", 6, "woof"),
    Dog("Fluffers", 2, "bark"),
    Dog("Bella", 3, "grr")
]

for dog in dogs:
    dog.describe()    # common behavior from Animal
    dog.speak()       # overridden behavior in Dog

```

Here, we have made a parent class for 'dog' called 'animal'.

## 95.3 Making a Polynomial class

You have created a bunch of useful python functions for dealing with polynomials. Notice how each one has the word "polynomial" in the function name like `derivative_of_polynomial`.

Wouldn't it be more elegant if you had a Polynomial class with a `derivative` method? Then you could use your polynomial like this:

```
a = Polynomial([9.0, 0.0, 2.3])
b = Polynomial([-2.0, 4.5, 0.0, 2.1])

print(a, "plus", b , "is", a+b)
print(a, "times", b , "is", a*b)
print(a, "times", 3 , "is", a*3)
print(a, "minus", b , "is", a-b)

c = b.derivative()

print("Derivative of", b ,"is", c)
```

And it would output:

```
2.30x^2 + 9.00 plus 2.10x^3 + 4.50x + -2.00 is 2.10x^3 + 2.30x^2 + 4.50x + 7.00
2.30x^2 + 9.00 times 2.10x^3 + 4.50x + -2.00 is 4.83x^5 + 29.25x^3 + -4.60x^2 +
↪  40.50x + -18.00
2.30x^2 + 9.00 times 3 is 6.90x^2 + 27.00
2.30x^2 + 9.00 minus 2.10x^3 + 4.50x + -2.00 is -2.10x^3 + 2.30x^2 + -4.50x +
↪  11.00
Derivative of 2.10x^3 + 4.50x + -2.00 is 6.30x^2 + 4.50
```

Create a file for your class definition called `Polynomial.py`. Enter the following:

```
class Polynomial:
    def __init__(self, coeffs):
        self.coefficients = coeffs.copy()

    def __repr__(self):
        # Make a list of the monomial strings
        monomial_strings = []

        # For standard form we start at the largest degree
        degree = len(self.coefficients) - 1

        # Go through the list backwards
        while degree >= 0:
            coefficient = self.coefficients[degree]

            if coefficient != 0.0:
                # Describe the monomial
                if degree == 0:
                    monomial_string = "{:.2f}".format(coefficient)
                elif degree == 1:
                    monomial_string = "{:.2f}x".format(coefficient)
```

```
                else:
                    monomial_string = "{:.2f}x^{}".format(coefficient, degree)

                # Add it to the list
                monomial_strings.append(monomial_string)

            # Move to the previous term
            degree = degree - 1

        # Deal with the zero polynomial
        if len(monomial_strings) == 0:
            monomial_strings.append("0.0")

        # Separate the terms with a plus sign
        return " + ".join(monomial_strings)

    def __call__(self, x):
        sum = 0.0
        for degree, coefficient in enumerate(self.coefficients):
            sum = sum + coefficient * x ** degree
        return sum

    def __add__(self, b):
        result_length = max(len(self.coefficients), len(b.coefficients))
        result = []
        for i in range(result_length):
            if i < len(self.coefficients):
                coefficient_a = self.coefficients[i]
            else:
                coefficient_a = 0.0

            if i < len(b.coefficients):
                coefficient_b = b.coefficients[i]
            else:
                coefficient_b = 0.0
            result.append(coefficient_a + coefficient_b)

        return Polynomial(result)

    def __mul__(self, other):

        # Not a polynomial?
        if not isinstance(other, Polynomial):
            # Try to make it a constant polynomial
            other = Polynomial([other])

        # What is the degree of the resulting polynomial?
        result_degree = (len(self.coefficients) - 1) + (len(other.coefficients) -
        ↪  1)

        # Make a list of zeros to hold the coefficients
        result = [0.0] * (result_degree + 1)
```

```
        # Iterate over the indices and values of a
        for a_degree, a_coefficient in enumerate(self.coefficients):

            # Iterate over the indices and values of b
            for b_degree, b_coefficient in enumerate(other.coefficients):

                # Calculate the resulting monomial
                coefficient = a_coefficient * b_coefficient
                degree = a_degree + b_degree

                # Add it to the right bucket
                result[degree] = result[degree] + coefficient

        return Polynomial(result)

    __rmul__ = __mul__

    def __sub__(self, other):
        return self + other * -1.0

    def derivative(self):

        # What is the degree of the resulting polynomial?
        original_degree = len(self.coefficients) - 1
        if original_degree > 0:
            degree_of_derivative = original_degree - 1
        else:
            degree_of_derivative = 0

        # We can ignore the constant term (skip the first coefficient)
        current_degree = 1
        result = []

        # Differentiate each monomial
        while current_degree < len(self.coefficients):
            coefficient = self.coefficients[current_degree]
            result.append(coefficient * current_degree)
            current_degree = current_degree + 1

        # No terms? Make it the zero polynomial
        if len(result) == 0:
            result.append(0.0)

        return Polynomial(result)
```

Create a second file called `test_polynomial.py` to test it:

```
from Polynomial import Polynomial

```

```
a = Polynomial([9.0, 0.0, 2.3])
b = Polynomial([-2.0, 4.5, 0.0, 2.1])

print(a, "plus", b , "is", a+b)
print(a, "times", b , "is", a*b)
print(a, "times", 3 , "is", a*3)
print(a, "minus", b , "is", a-b)

c = b.derivative()

print("Derivative of", b ,"is", c)

slope = c(3)
print("Value of the derivative at 3 is", slope)

```

Run the test code:

```
python3 test_polynomial.py
```

Chapter 96

# Common Polynomial Products

In math and physics, you will run into certain kinds of polynomials over and over again. In this chapter, we will cover some patterns that you will want to be able to recognize as you move through the sequence!

## 96.1 Difference of squares

To start off, watch ***Polynomial special products: difference of squares*** from Khan Academy at https://youtu.be/uNweU6I4Icw.

If you are asked what $(3x-7)(3x+7)$ is, you would use the distributive property to expand that to $(3x)(3x)+(3x)(7)+(-7)(3x)+(-7)(7)$. Two of the terms cancel each other, so this is $(3x)^2-(7)^2$. This would simplify to $9x^2-49$

You will see this pattern often. Anytime you see $(a+b)(a-b)$, you should immediately recognize it equals $a^2-b^2$. (Note that the order doesn't matter: $(a-b)(a+b)$ also equals $a^2-b^2$.)

Working the other way is important too. Any time you see $a^2-b^2$, that you should recognize that you can change that into the product $(a+b)(a-b)$. Making something into a product like this is known as *factoring*. You probably have done prime factorization of numbers like $42=2\times3\times7$. In the next couple of chapters, you will learn to factorize polynomials.

## *Exercise 225* Difference of Squares

Simplify the following products:

1. $(2x-3)(2x+3)$
2. $(7+5x^3)(7-5x^3)$
3. $(x-a)(x+a)$
4. $(3-\pi)(3+\pi)$
5. $(-4x^3+10)(-4x^3-10)$
6. $(x+\sqrt{7})(x-\sqrt{7})$ Factor the following polynomials:
7. $x^2-9$
8. $49-16x^6$
9. $\pi^2-25x^8$
10. $x^2-5$

*Working Space*

*Answer on Page 1954*

We are often interested in the roots of a polynomial. That is, we want to know "For what values of $x$ does the polynomial evaluate to zero?" For example, when you deal with falling bodies, the first question you might ask would be "How many seconds before the hammer hits the ground?" Once you have factored a polynomial into binomials, you can easily find the roots.

For example, what are the roots of $x^2-5$? You just factored it into $(x+\sqrt{5})(x-\sqrt{5})$ This product is zero if and only if one of the factors is zero. The first factor is only zero when $x$ is $-\sqrt{5}$. The second factor is zero only when $x$ is $\sqrt{5}$. Those are the only two roots of this polynomial.

Let's check that result. $\sqrt{5}$ is a little more than 2.2. Using your Python code, you can graph the polynomial:

```
import poly
import matplotlib.pyplot as plt

# x**2 - 5
pn = [-5.0, 0.0, 1.0]

# These lists will hold our x and y values
```

```
x_list = []
y_list = []

# Start at x=-3
current_x =-3.0

# End at x=3.0
while current_x < 3.0:
    current_y = poly.evaluate_polynomial(pn, current_x)

    # Add x and y to respective lists
    x_list.append(current_x)
    y_list.append(current_y)

    # Move x forward
    current_x += 0.1

# Plot the curve
plt.plot(x_list, y_list)
plt.grid(True)

#Additional graph customization (make axes visible, axes legend)
plt.axhline(0, color='black', linewidth=1)   # x-axis
plt.axvline(0, color='black', linewidth=1)   # y-axis
plt.title("Graph of y = x² - 5")
plt.xlabel("x")
plt.ylabel("y")

plt.show()
```

You should get a plot like Figure 96.1.

It does, indeed, seem to cross the x-axis near -2.2 and 2.2.

## 96.2 Powers of binomials

You can raise whole polynomials to exponents. Raising a polynomial to an exponent means taking the entire expression and multiplying it by itself repeatedly according to the value of the exponent. For example,

$$(3x^3+5)^2 = (3x^3+5)(3x^3+5)$$
$$= 9x^6 + 15x^3 + 15x^3 + 25 = 9x^6 + 30x^3 + 25$$

$(3x^3+5)^2$ is a binomial that has been raised to an exponent of 2. A polynomial with two terms is called a *binomial*. $5x^9 - 2x^4$, for example, is a binomial. In this section, we are going to develop some handy techniques for raising a binomial to some power.

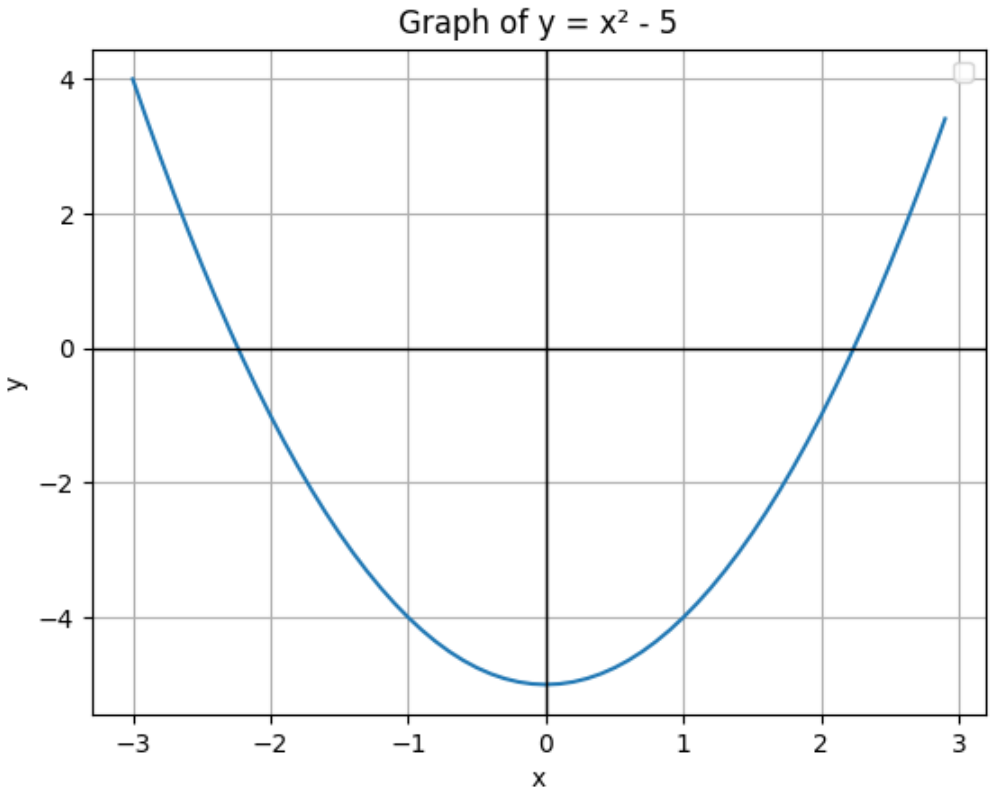



Figure 96.1: A graph of $x^2 - 5$ showing the roots on the x-axis.

Looking at the previous example, you can see that for any monomials $a$ and $b$, $(a + b)^2 = a^2 + 2ab + b^2$. This is referred to as a perfect square binomial. So, for example, $(7x^3 + \pi)^2 = 49x^6 + 14\pi x^3 + \pi^2$

## *Exercise 226* Squaring binomials

Simplify the following

1. $(x + 1)^2$
2. $(3x^5 + 5)^2$
3. $(x^3 - 1)^2$
4. $(x - \sqrt{7})^2$

*Working Space*

*Answer on Page 1954*

What about $(x+2)^3$? You can do it as two separate multiplications:

$$\begin{aligned}(x+2)^3 &= (x+2)(x+2)(x+2)\\ &= (x+2)(x^2+4x+4) = x^3+4x^2+4x+2x^2+8x+8\\ &= x^3+6x^2+12x+8\end{aligned}$$

In general, we can say that for any monomials $a$ and $b$, $(a+b)^3 = a^3+3a^2b+3ab^2+b^3$.

What about higher powers? $(a+b)^4$, for example? You could use the distributive property four times, but it starts to get pretty tedious.

Here is a trick. This is known as *Pascal's triangle*

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1 \\
1 & 5 & & 10 & & 10 & & 5 & 1 \\
1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \\
1 & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & 1 \\
 & & & & \cdots & & & & 
\end{array}$$

Each entry is the sum of the two above it.

The coefficients of each term are given by the entries in Pascal's triangle:

$$(a+b)^4 = 1a^4+4a^3b+6a^2b^2+4ab^3+1b^4$$

Notice how each term fits the form $ax^ky^m$. The coefficient $a$ in each term $ax^ky^m$ is known as the binomial coefficient or *binomial theorem*. The binomial coefficient is denoted as $\binom{n}{k}$ where $n$ is the exponent and $k$ is the term number (starting with 0). This is said "n choose k". For example, in $(a+b)^4$, the coefficient of the third term $(6a^2b^2)$ is $\binom{n}{k} = \binom{4}{2} = 6$. We will heavily discuss this in a future chapter on combinatorics.

## *Exercise 227* Using Pascal's Triangle

*Working Space*

1. What is $(x+\pi)^5$?

*Answer on Page 1955*

## *Exercise 228* Coding Pascal's Triangle

*Working Space*

Write a Python script that finds the row of Pascal's Triangle at a given index $n$.

*Hint*: it does not need to be recursive.

*Answer on Page 1955*

# Chapter 97

# Factoring Polynomials

We factor a polynomial into two or more polynomials of lower degree. For example, let's say that you wanted to factor $5x^3 - 45x$. You would note that you can factor out $5x$ from every term. Thus,

$$5x^3 - 45x = (5x)(x^2 - 9)$$

You might notice that the second factor looks like the difference of squares, so

$$5x^3 - 45x = (5x)(x + 3)(x - 3)$$

That is as far as we can factorize this polynomial.

Why do we care? The factors make it easy to find the roots of the polynomial. This polynomial evaluates to zero if and only if at least one of the factors is zero. Here, we see that

- The factor $(5x)$ is zero when $x$ is zero.
- The factor $(x + 3)$ is zero when $x$ is -3.
- The factor $(x - 3)$ is zero when $x$ is 3.

So, looking at the factorization, you can see that $5x^3 - 45x$ is zero when $x$ is 0, -3, or 3.

This is a graph of that polynomial with its roots circled:

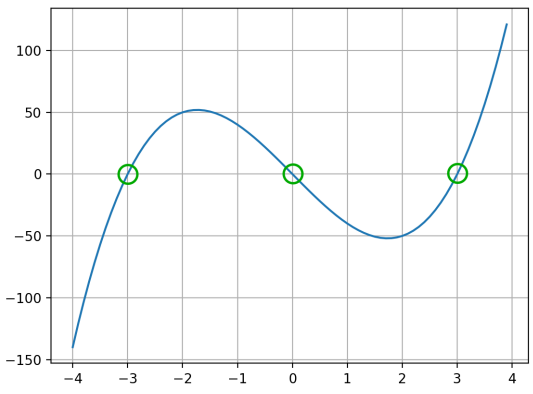



Figure 97.1: Factoring for roots makes it easier to find the roots rather than in expanded form.

## 97.1 How to factor polynomials

The first step when you are trying to factor a polynomial is to find the greatest common divisor for all the terms, then pull that out. In this case, the greatest common divisor will also be a monomial. Its degree is the least of the degrees of the terms, its coefficient will be the greatest common divisor of the coefficients of the terms.

For example, what can you pull out of this polynomial?

$$12x^{100} + 30x^{31} + 42x^{17}$$

The greatest common divisor of the coefficients (12, 30, and 42) is 6. The least of the degrees of terms (100, 31, and 17) is 17. So, you can pull out $6x^{17}$:

$$12x^{100} + 30x^{31} + 42x^{17} = (6x^{17})(2x^{83} + 5x^{14} + 7)$$

### *Exercise 229* Factoring out the GCD monomial

Factor out the greatest common divisor monomial from each polynomial.

*Working Space*

1. $18x^3 - 15x$
2. $4x^3 - 12x^2 + 16x$
3. $18x^{50} - 24x^{20} + 30x^8$

*Answer on Page 1955*

So, now you have the product of a monomial and a polynomial. If you are lucky, the polynomial part looks familiar, like the difference of squares or a row from Pascal's triangle.

Often, you are trying to factor a quadratic like $x^2 + 5x + 6$ in a pair of binomials. In this case, the result would be $(x + 3)(x + 2)$. Let's check that:

$$(x + 3)(x + 2) = (x)(x) + (3)(x) + (2)(x) + (3)(2) = x^2 + 5x + 6$$

Notice that 3 and 2 multiply to 6 and add to 5. If you were trying to factor $x^2 + 5x + 6$, you would ask yourself, "What are two numbers that when multiplied equal 6 and when added equal 5?" And you might guess wrong a couple of times. For example, you might say to yourself, "Well, 6 times 1 is 6. Maybe those work. But 6 and 1 add 7. So those don't work."

Solving these sorts of problems are like solving a Sudoku puzzle. You try things and realize they are wrong, so you backtrack and try something else.

The numbers are sometimes negative. For example, $x^2 + 3x - 10$ factors into $(x+5)(x-2)$.

## *Exercise 230* Factoring quadratics

Factor each quadratic into a pair of binomials.

*Working Space*

1. $x^2 + 14x + 33$
2. $x^2 - 16x + 60$
3. $x^2 + 3x - 54$
4. $x^2 - 3x - 10$

*Answer on Page 1955*

### Computing in Python

Python can check a factorization for you. Start by writing the original polynomial and its factored form as two Python functions:

```
def p(x):
    return 5*x**3 - 45*x

def f(x):
    return (5*x)*(x + 3)*(x - 3)
```

In Python, `**` means "raised to the power of," and every multiplication needs a `*`. You have to write `5*x`, since `5x` is an error.

First, confirm the roots we read off the factors:

```
for x in [0, -3, 3, 1]:
    print(x, p(x))
```

```
0 0
```

```
-3 0
3 0
1 -40
```

The values 0, -3, and 3 all give zero, so they are roots. The value 1 gives -40, so it is not a root.

Next, check that `p` and `f` are the same polynomial. If the factorization is correct, the two functions give the same output for every $x$:

```
for x in [0, 1, 2, 3]:
    print(x, p(x), f(x), p(x) == f(x))
```

```
0 0 0 True
1 -40 -40 True
2 -50 -50 True
3 0 0 True
```

Each line shows $x$, the value of the original, the value of the factored form, and whether they match. Every line says `True`.

How many values of $x$ do you need to check? Here is a useful fact: if two polynomials of degree at most $n$ give the same value at $n + 1$ different values of $x$, they are the same polynomial. Our polynomial has degree 3, so four matching values are enough.

Matching at the roots alone can hide a mistake. Suppose you forgot the 5 when you pulled out $5x$:

```
def g(x):
    return x*(x + 3)*(x - 3)

for x in [0, -3, 3, 1]:
    print(x, p(x), g(x), p(x) == g(x))
```

```
0 0 0 True
-3 0 0 True
3 0 0 True
1 -40 -8 False
```

The wrong factorization has the same roots, so it passes at 0, -3, and 3. The value $x = 1$ catches the mistake.

## *Exercise 231* Predict the output

Without running the code, predict what this prints. Which of the two values of x is a root?

```
def q(x):
    return x**2 + 3*x - 4

print(q(2))
print(q(-4))
```

*Working Space*

*Answer on Page 1955*

## *Exercise 232* Spot the sign error

A student claims that $x^2+3x-10=(x-5)(x+2)$ and runs this check:

```
def a(x):
    return x**2 + 3*x - 10

def b(x):
    return (x - 5)*(x + 2)

for x in [0, 1, 2]:
    print(x, a(x), b(x), a(x) ==
    ↪  b(x))
```

The output is:

```
0 -10 -10 True
1 -6 -12 False
2 0 -12 False
```

The first line says `True`. Why is the factorization still wrong? What is the correct factorization?

*Working Space*

*Answer on Page 1956*

## *Exercise 233* Spot the bug

A student wants to check that $x^2 + 5x + 6 = (x + 3)(x + 2)$, which we already know is correct, and writes:

```
def c(x):
    return x^2 + 5*x + 6

def d(x):
    return (x + 3)*(x + 2)

for x in [0, 1, 2]:
    print(x, c(x), d(x), c(x) ==
    ↪ d(x))
```

The output is:

```
0 8 6 False
1 12 12 True
2 16 20 False
```

The factorization is correct, so what went wrong?

*Working Space*

*Answer on Page 1956*

## *Exercise 234* Factor, check, and find the roots

Factor $2x^3 - 6x^2 - 8x$ completely. Write the original polynomial and your factored form as two Python functions, and check that they match at enough values of x. Then use your factors to list the roots, and confirm that each one gives zero.

*Working Space*

*Answer on Page 1956*

This last part is optional. It uses SymPy, a Python library for algebra. SymPy is a separate install, so you may need to run `python3 -m pip install sympy` first.

```
from sympy import symbols, factor, expand

x = symbols('x')
print(factor(5*x**3 - 45*x))
print(expand(5*x*(x + 3)*(x - 3)))
```

```
5*x*(x - 3)*(x + 3)
5*x**3 - 45*x
```

The line `x = symbols('x')` makes `x` a *symbolic variable*: a letter that stands for any number, which is how $x$ works in algebra. Earlier in this section, `x` always held one number at a time. Here it holds no number at all.

The function `factor` factors a polynomial, and `expand` multiplies the factors back out. SymPy lists $(x - 3)$ before $(x + 3)$; the order of the factors does not change the product.

SymPy also tells you when a polynomial cannot be factored using integers:

```
print(factor(x**2 + 1))
```

```
x**2 + 1
```

SymPy returns the polynomial unchanged, so $x^2+1$ has no factors with integer coefficients.

SymPy can factor every polynomial in this chapter in one line, which means it can also skip the thinking this chapter is teaching. Use it after you factor by hand, to check your answer.

If you get stuck on a problem, open `factor_hint.py`, type your polynomial on the line that starts with `polynomial =`, and run the file. It first shows a hint: the polynomial with its greatest common factor pulled out, found with SymPy's `factor_terms`. Try to finish the problem from there. When you press Enter, the file shows the full factorization.

## *Exercise 235* Predict what SymPy prints

Without running the code, predict the output.

```
from sympy import symbols, factor

x = symbols('x')
print(factor(x**2 - 9))
```

*Working Space*

*Answer on Page 1957*

## *Exercise 236* Factor by hand, then check with SymPy

Factor $3x^2+12x-15$ completely by hand. Then use `expand` to multiply your factors back out, and use `factor` to compare your answer with SymPy's.

*Working Space*

*Answer on Page 1957*

CHAPTER 98

# Practice with Polynomials

At this point, you know all the pieces necessary to solve problems involving polynomials. In this chapter, you are going to practice using all of these ideas together.

Watch Khan Academy's **Polynomial identities introduction here**: https://youtu.be/EvNKKyhLSpQ Also, watch the follow up here: https://youtu.be/-6qiO49Q180

*FIXME: Lots of practice problems here*

CHAPTER 99

# Graphing Polynomials

Polynomials are used to solve real-world problems! To do this, we need to know what these polynomials look like on a graph. You have many tools you can use to sketch out graphs. These graphs are simple visuals that help to evaluate several different points on the plot.

For instance, you can determine:

- where the graph crosses the y-axis, which you can find by evaluating the polynomial at $x = 0$.

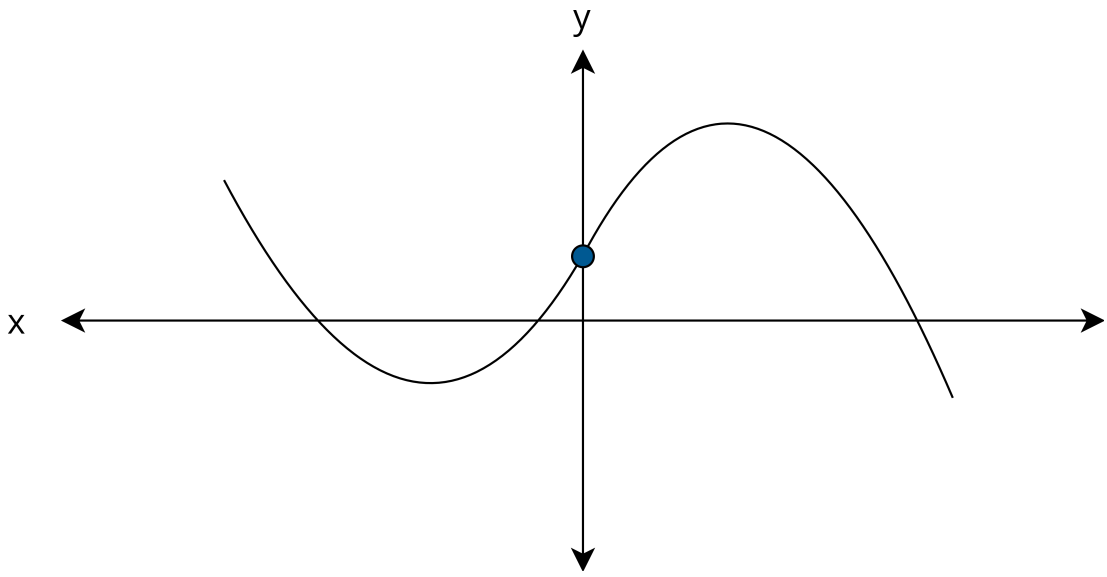



Figure 99.1: A graph showing the y-intercept of a polynomial

- where the graph crosses the x-axis, which you can find by solving for the roots of the polynomial.

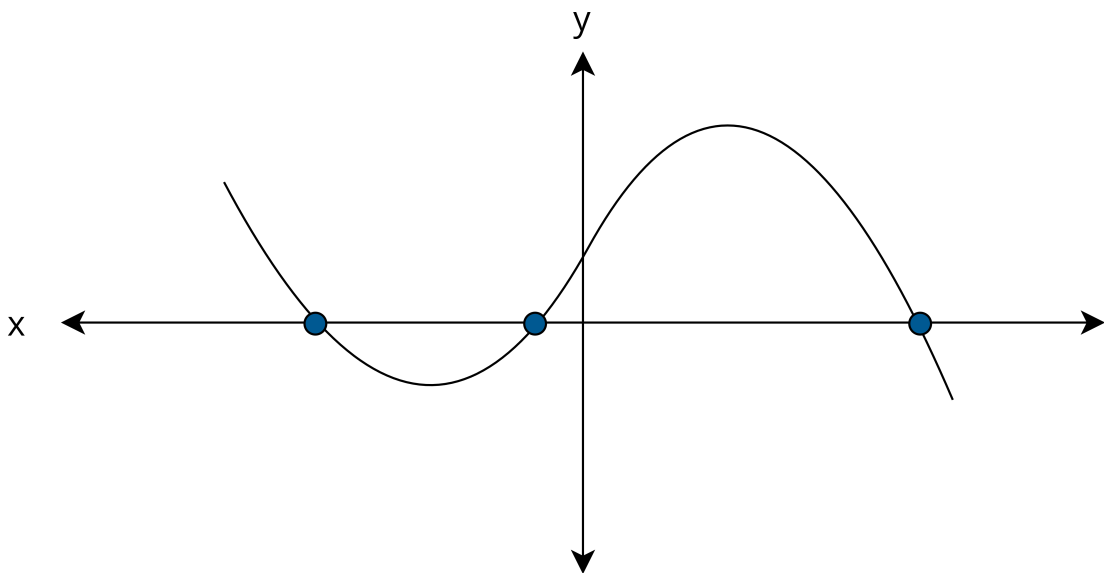



Figure 99.2: A graph showing the roots of a polynomial

- the level spots on the graph (often the top of a hump or the bottom of a dip). You can take the derivative of the polynomial (which is also a polynomial), and find the roots of that.

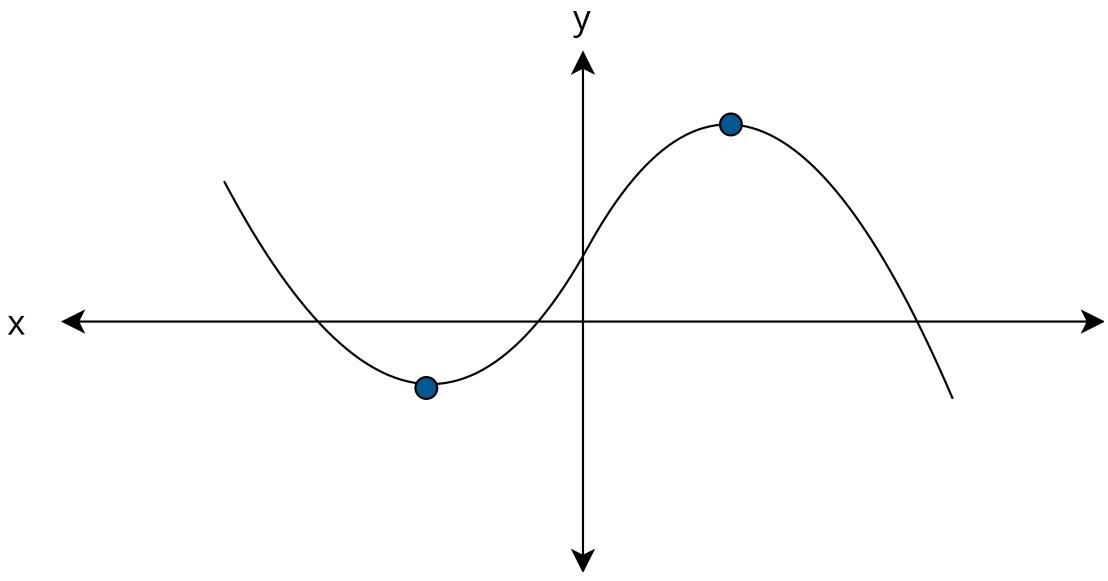



Figure 99.3: A graph showing "level spots" of a polynomial.

For example, if you wanted to graph the polynomial $f(x) = -x^3 - x^2 + 6x$, you might plug in a few values that are easy to compute:

- $f(-2) = -8$
- $f(-1) = -6$
- $f(0) = 0$
- $f(1) = 4$
- $f(2) = 0$

So, right away, we know two roots: $x = 0$ and $x = 2$. Are there others? We won't know until we factor the polynomial:

$$\begin{aligned} -x^3 - x^2 + 6x &= (-1x)(x^2 + x - 6) \\ &= (-1x)(x+3)(x-2) \end{aligned}$$

So, yes, there is a third root: $x = -3$

What about the level spots? $f'(x) = -3x^2 - 2x + 6$. Where is that zero?

$$\begin{aligned} -3x^2 - 2x + 6 &= 0 \\ x^2 + \frac{2}{3}x - 2 &= 0 \end{aligned}$$

We have a formula for quadratics like this:

$$\begin{aligned} x = -\frac{b}{2} \pm \frac{\sqrt{b^2 - 4c}}{2} &= -\frac{\frac{2}{3}}{2} \pm \frac{\sqrt{\left(\frac{2}{3}\right)^2 - 4(-2)}}{2} \\ &= -\frac{1}{3} \pm \frac{\sqrt{\frac{4}{9} + 8}}{2} \\ &= -\frac{1}{3} \pm \frac{\sqrt{\frac{85}{9}}}{2} \\ &= -\frac{1}{3} \pm \frac{\sqrt{85}}{6} \\ &\approx 1.20 \text{ and } -1.87 \end{aligned}$$

Now, you might plug those numbers in:

- $f(1.2) \approx 4.0$
- $f(-1.87) \approx -8.2$

## 99.1 Leading term in graphing

There is one more trick you need before you can draw a good graph of a polynomial. As you go father and farther to the left and right, where does the function go? In other words, does the graph go up on both ends (like a smile)? Or does it go down on both ends (like a frown)? Or does the negative end go down (frowny) while the positive end go up (smiley)? Or does the negative go up (smiley) and the positive end go down (frowny)?

Assuming the polynomial is not constant, there are only those four possibilities. It is determined entirely by the leading term of the polynomial. If the degree of the leading term is even, both ends go in the same direction (both are smiley or both are frowny). If the coefficient of the leading term is positive, the positive end is smiley.

The graph we are working on has a leading term of $-1x^3$. The degree is odd, thus the ends go in different directions. The coefficient is negative, so the positive end points down. Now, you can draw the graph, which should look something like this:

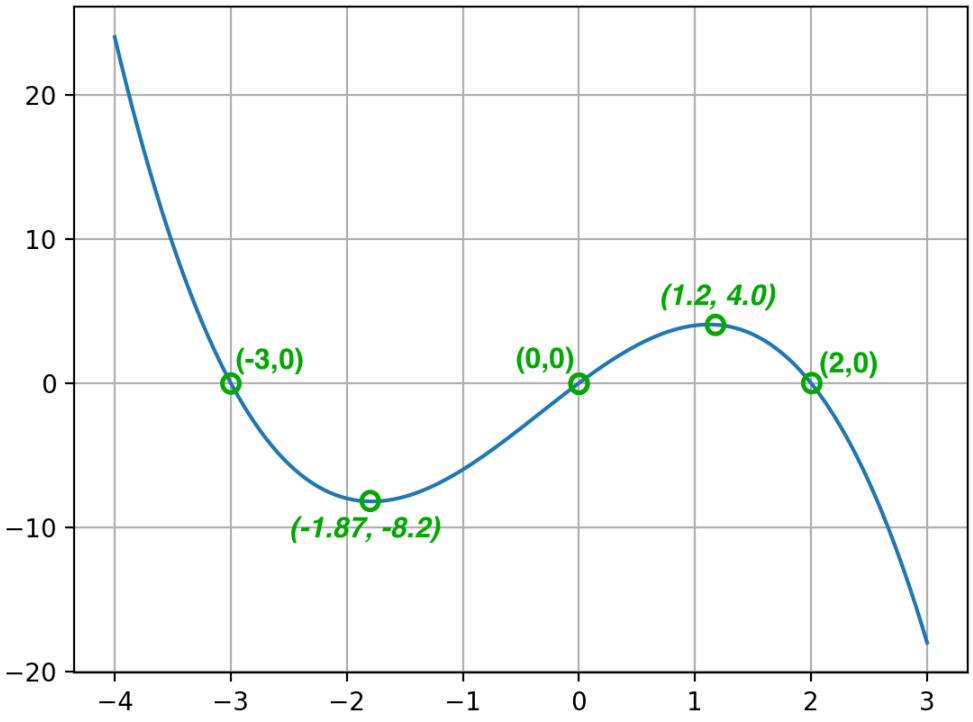



Figure 99.4: A graph with roots and significant points.

Chapter 100

# Interpolating with Polynomials

Let's say someone on a distant planet records video of a hammer being thrown up into the air. They send you three random frames of the hammer in flight. Each frame has a timestamp, and you can clearly see how high the hammer is in each one. Can you create a 2nd degree polynomial that explains the entire flight of the hammer?

In other words, you have three points $(t_0, h_0), (t_1, h_1), (t_2, h_2)$. Can you find $a, b, c$ such that the graph of $at^2 + bt + c = t$ passes through all three points?

The answer is yes. In fact, given any $n$ points[1], there is exactly one $n-1$ degree polynomial that passes through all the points.

There are a lot of variables floating around. Let's make it concrete: The photos are taken at $t = 2$ seconds, $t = 3$ seconds, and $t = 4$ seconds. In those photos, the height of the hammer is 5m, 7m, and 6m. So, we want our polynomial to pass through these points: (2, 5), (3, 7), (4,6).

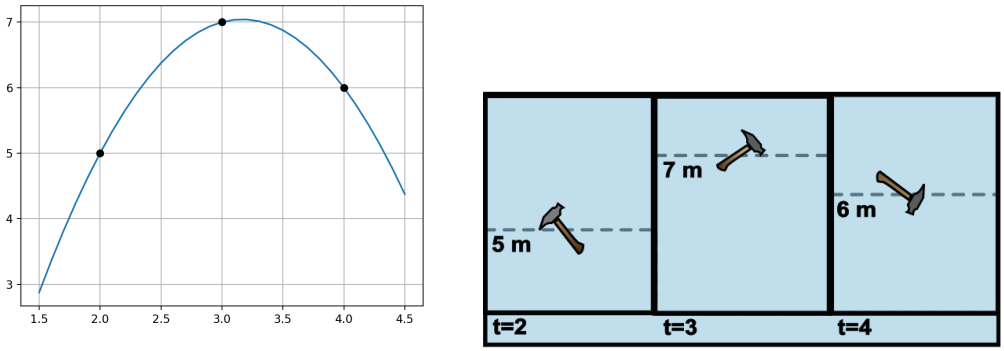



Figure 100.1: Left: the graph of the hammer going through the given points. Right: the three hammer frames.

How can you find that polynomial? Let's do it in small steps. Can you create a 2nd degree polynomial that is not zero at $t = 2$, but is zero at $t = 3$ and $t = 4$? Yes, you can: $(x-3)(x-4)$ has exactly two roots at $t = 3$ and $t = 4$. The value of this polynomial at $t = 2$ is $(2-3)(2-4) = 2$. We really want it to be 5m, so we can divide the whole polynomial by 2 and multiply it by 5.

[1] $n = 2$ will result in a line in slope intercept point, and $n = 1$ is a single point

Now we have the polynomial:

$$f_0(x) = \frac{5}{(2-3)(2-4)}(x-3)(x-4) = \frac{5}{2}x^2 - \frac{35}{2}x + 30$$

This is a second degree polynomial that is 5 at $t = 2$ and 0 at $t = 3$ and $t = 4$.

Now, we create a polynomial that is 7 at $t = 3$ and 0 at $t = 2$ and $t = 4$:

$$f_1(x) = \frac{7}{(3-2)(3-4)}(x-2)(x-4) = -7x^2 + 42x - 56$$

Finally, we create a polynomial that is 6 at $t = 4$ and zero at $t = 2$ and $t = 3$:

$$f_2(x) = \frac{6}{(4-2)(4-3)}(x-2)(x-3) = 3x^2 - 15x + 18$$

Adding these three polynomials together gives you a new polynomial that touches all three points:

$$f(x) = \frac{5}{2}x^2 - \frac{35}{2}x + 30 - 7x^2 + 42x - 56 + 3x^2 - 15x + 18 = -\frac{3}{2}x^2 + \frac{19}{2}x - 8$$

You can test this in Python with your `Polynomial` class. Create a file called `test_interpolation.py`. Add this code:

```
from Polynomial import Polynomial
import matplotlib.pyplot as plt

in_x = [2,3,4]
in_y = [5,7,6]

pn = Polynomial([-8, 19/2, -3/2])
print(pn)

# These lists will hold our x and y values
x_list = []
y_list = []

# Starting x
current_x = 1.5

while current_x <= 4.5:
    # Evaluate pn at current_x
    current_y = pn(current_x)

    # Add x and y to respective lists
    x_list.append(current_x)
```

```
    y_list.append(current_y)

    # Move x forward
    current_x += 0.05

# Plot the curve
plt.plot(x_list, y_list)

# Plot black circles on the given points
plt.plot(in_x, in_y, "ko")
plt.grid(True)
plt.show()
```

You should get a nice plot that shows the graph of the polynomial passing through those three points.

In general, then, if you provide any three points $(t_0, h_0), (t_1, h_1), (t_2, h_2)$, there is a second degree polynomial that pass through all three:

$$\frac{h_0}{(t_0 - t_1)(t_0 - t_2)}(x - t_1)(x - t_2) + \frac{h_1}{(t_1 - t_0)(t_1 - t_2)}(x - t_0)(x - t_2) + \frac{h_2}{(t_2 - t_0)(t_2 - t_1)}(x - t_0)(x - t_1)$$

What if you are given 9 points $((t_0, h_0), (t_1, h_1), \ldots, (t_8, h_8))$ and want to find a 8th degree polynomial that passes through all of them? Just what you would expect:

$$\frac{h_0}{(t_0 - t_1)(t_0 - t_2)\ldots(t_0 - t_8)}(x - t_1)(x - t_2)\ldots(x - t_8) + \ldots + \frac{h_8}{(t_8 - t_0)\ldots(t_8 - t_7)}(x - t_0)\ldots(x - t_7)$$

*FIXME: Do I need to define summation and prod here?*

The general solution is, given $n$ points, the $n - 1$ degree polynomial that goes through them is

$$y = \sum_{i=0}^{n} \left( \prod_{\substack{0 \le j \le n \\ j \ne i}} \frac{x - t_j}{t_i - t_j} \right) h_i$$

That would be tedious for a person to compute, but computers are perfect for this stuff. Let's create a method that creates instances of Polynomial using interpolation.[2]

[2]Note that other textbooks or resources may refer to this as *Lagrange Polynomial Interpolation*.

## 100.1 Interpolating polynomials in python

Your method will take two lists of numbers: one contains x-values and the other contains y-values. So comment out the line that creates the polynomial in `test_interpolation.py` and create it from two lists:

```
in_x = [2,3,4]
in_y = [5,7,6]
# pn = Polynomial([-8, 19/2, -3/2])
pn = Polynomial.from_points(in_x, in_y)
print(pn)
```

Add the following method to your Polynomial class in `Polynomial.py`

```
@classmethod
    def from_points(cls, x_values, y_values):
        coef_count = len(x_values)

        # Sums start with a zero polynomial
        sum_pn = Polynomial([0.0] * coef_count)
        for i in range(coef_count):

            # Products start with the constant 1 polynomial
            product_pn = Polynomial([1.0])
            for j in range(coef_count):

                # Must skip j=i
                if j != i:
                    # (1x - x_values[j]) has a root at x_values[j]
                    factor_pn = Polynomial([-1 * x_values[j], 1])
                    product_pn = product_pn * factor_pn

            # Scale so product_pn(x_values[i]) = y_values[i]
            scale_factor  = y_values[i] / product_pn(x_values[i])
            scaled_pn = scale_factor * product_pn

            # Add it to the sum
            sum_pn = sum_pn + scaled_pn

        return sum_pn
```

It should work exactly the same as before. You should get the same polynomial printed out as before as well as the same plot of the curve passing through the three points.

How about five points? Change `in_x` and `in_y` at the start of `test_interpolation.py`:

```
in_x = [1.7, 2, 2.7, 3.5, 4, 4.4]
```

```
in_y = [8, 12, 1, 4, -1, 6]
```

You should get a polynomial that passes through all five points:

$$11.21x^5 - 171.05x^4 + 1019.44x^3 - 2957.53x^2 + 4161.78x - 2258.75$$

It should look like this:

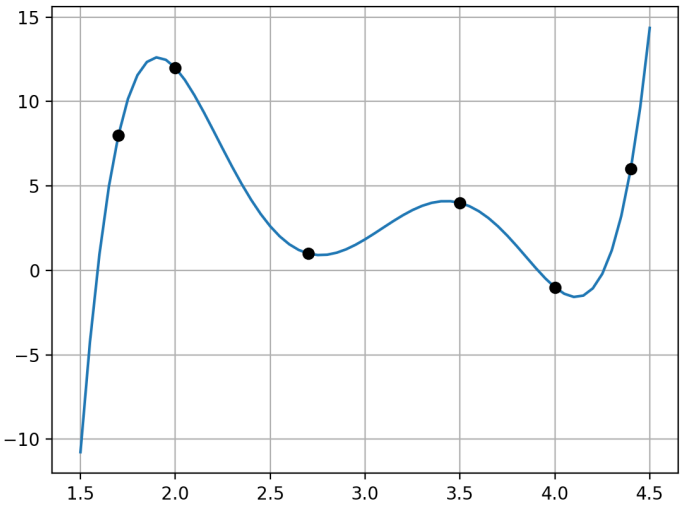



Figure 100.2: 5 point interpolation of points.

# Chapter 101

# Rational Functions

We have discussed addition, subtraction, and multiplication of polynomials. What about division?

A quotient of polynomials is called a rational expression. When the polynomials are factored and the stars align, we can simplify the rational expression to a single polynomial, just like we might reduce a fraction to lowest terms.

**Example**

$$\begin{aligned} \frac{(x+1)(x+5)}{x+5} &= (x+1)\cdot\frac{x+5}{x+5} \\ &= x+1 \qquad (x \neq 5) \end{aligned} \tag{101.1}$$

[1]

What if the polynomials are not factored? Factor them first.

**Example**

$$\frac{x^2+6x+5}{x+5} = \frac{(x+1)(x+5)}{x+5}$$

and simplify as in the previous example.

Now, let us consider a rational expression which can be simplified to a single polynomial — but in the denominator.

**Example**

$$\begin{aligned} \frac{x+5}{x^2+6x+5} &= \frac{x+5}{(x+1)(x+5)} \\ &= \frac{1}{x+1}\cdot\frac{x+5}{x+5} \\ &= \frac{1}{x+1} \end{aligned} \tag{101.2}$$

Consider this expression as a function: $f(x) = \frac{1}{x+1}$. As you might have guessed, this is called a rational function. We did not bother looking at the result of the previous example

[1] It is important to note that we need to restrict $x \neq 5$ since substituting $x = 5$ would produce division by zero, making the expression undefined. Canceling the $(x+5)$ factor is only valid when $x \neq 5$, so the simplified form $x+1$ and the original expression are equal everywhere *except* at $x = 5$.

as a function, because we already know that function type: it is a line with slope 1 and y-intercept 1. However, this rational function is another animal entirely. Let us examine our first rational function with a familiar concept: the y-intercept.

y-intercept: $f(0) = \frac{1}{0+1} = \frac{1}{1} = 1$. The graph contains the point $(0, 1)$.

Does f have an x-intercept? That would be an x-value where $f(x) = 0$. But a fraction equals 0 only when its numerator equals 0; since the numerator of this expression is always 1, f has no x-intercept.

Knowing the y-intercept, and that there is no x-intercept, is a comforting start. But things get weird when we consider a concept that has previously seemed quite simple: domain. Recall that the domain of a function is the set of all values which can be used as inputs. In this case, the domain includes all real numbers, with one exception. The number $-1$ is not a valid input because $f(-1) = \frac{1}{-1+1} = \frac{1}{0}$, which is undefined. So, we say that the domain is all real numbers except $-1$. This means the graph contains a point corresponding to every x-value except $-1$.

There is no point at $x = -1$, but there is a point at every other x-value, such as, for example, $-1.1$, or $-0.99999$. So, what is happening near $x = -1$?

| x | -1.1 | -1.01 | -1.001 | -0.999 | -0.99 | -0.9 |
|---|---|---|---|---|---|---|
| f(x) | -10 | -100 | -1000 | 1000 | 100 | 10 |

The function is going haywire. As we choose x-values closer and closer to $-1$, the resulting function values are larger and larger in magnitude. Also, they are negative on one side, but positive on the other. So, how does a graph go from y-values of $-10$, to $-100$, to $-1000$, all in a space of less than 0.1 on the x-axis? Then, from there, suddenly to big positive numbers on the other side of $x = -1$? All without ever crossing the x-axis (since there is no x-intercept)? Let's look at the graph.

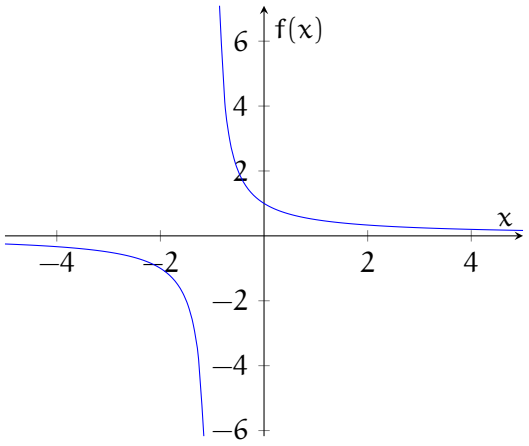



Figure 101.1: Graph of $f(x) = \frac{1}{x+1}$

We can see the y-intercept we found above. We can also see that the graph has no x-intercept, as expected. The phenomenon occurring at $x = -1$ is called a vertical asymptote. One other interesting feature of this graph is how it hugs the x-axis toward the left and right edges of the window. This makes the line $y = 0$ (the x-axis) a horizontal asymptote for this function. We can see why this is happening numerically by considering what happens for x-values far from 0. In this function, the result is a fraction with a numerator of 1 and a denominator that is large in size: a fraction that is close to 0.

| x | -1000 | -100 | -10 | 10 | 100 | 1000 |
|---|---|---|---|---|---|---|
| f(x) | -0.001 | -0.01 | -0.1 | 0.1 | 0.01 | 0.001 |

Let's examine another rational function. Begin by factoring to see if the function can be simplified.

$$g(x) = \frac{x^2 - 3x + 2}{x^2 - 4x + 3} = \frac{(x-1)(x-2)}{(x-1)(x-3)}$$

Consider the domain of g before continuing. Which values of x are valid inputs? Since substituting $x = 1$ or $x = 3$ would result in division by 0, these are not valid inputs. The domain of g is all real numbers except 1 and 3.

Now, for any x-value except 1, $\frac{x-1}{x-1} = 1$. This means that, for all x-values but 1, we can cancel those factors, leaving $g(x) = \frac{x-2}{x-3}$. (We will talk more about what is happening at $x = 1$ in a moment.)

This function has both x- and y-intercepts: y-intercept: $g(0) = \frac{0-2}{0-3} = \frac{2}{3}$. The graph contains the point $(0, \frac{2}{3})$. x-intercept: $g(x) = 0$ where the numerator equals 0 and the denominator does not equal 0. Since $x - 2 = 0$ when $x = 2$, the x-intercept is 2 and the graph contains the point $(2, 0)$.

The graph of g has a vertical asymptote at any x-value where substitution would result in dividing a nonzero number by zero. Thus, g has a vertical asymptote at $x = 3$.

Does g have a horizontal asymptote? Let us see what happens when we substitute x-values far from 0.

| x | -1000 | -100 | -10 | 10 | 100 | 1000 |
|---|---|---|---|---|---|---|
| g(x) | 0.999 | 0.990 | 0.923 | 1.143 | 1.010 | 1.001 |

As we move further away from the y-axis, the y-values become closer to 1. The horizontal asymptote describes the end behavior of the function, or what the graph looks like far from the y-axis. In this case, if we ignore the portion close to the y-axis, the graph begins to look like the line $y = 1$, making this the horizontal asymptote of g.

So, what is happening at $x = 1$? The value is not in the domain of the function, but there is no vertical asymptote there. That is because substituting any other value for x, even

values very close to 1, into $\frac{(x-1)(x-2)}{(x-1)(x-3)}$ gives the exact same number as substituting into $\frac{x-2}{x-3}$. So, there is a hole in the graph at $x = 1$, but nothing strange is happening on either side of 1. (Depending on the graphing software, the hole may not be visible.)

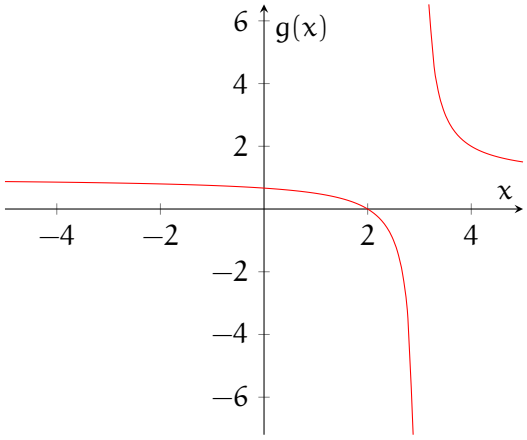



Figure 101.2: Graph of $g(x) = \frac{x^2-3x+2}{x^2-4x+3}$

## *Exercise 237* Rational Functions Practice 1

Determine the x- and y-intercepts and horizontal and vertical asymptotes of the rational function:

1. $\frac{2x+5}{x+4}$

*Working Space*

*Answer on Page 1957*

### Exercise 238

[This question was originally presented in the no-calculator section of the 2012 AP Calculus BC exam.] The line $y = 5$ is a horizontal asymptote of which of the following functions? Explain.

A. $y = \frac{\sin 5x}{x}$
B. $y = 5x$
C. $y = \frac{1}{x-5}$
D. $y = \frac{5x}{1-x}$
E. $y = \frac{20x^2-x}{1+4x^2}$

*Working Space*

*Answer on Page 1958*

In those examples, common factors cancel, leaving one polynomial. Of course, there is no guarantee that any two polynomials will have common factors, or even be factorable at all. Now, we consider an example that cannot be simplified. We will focus on just the asymptotes here.

$$h(x) = \frac{x^2}{x-1}$$

We see that the $x$-value 1 gives division of a non-zero number by zero, giving a vertical asymptote at $x = 1$. How about a horizontal asymptote? We examine values of $h$ for values of $x$ far from 0.

| $x$ | -1000 | -100 | -10 | 10 | 100 | 1000 |
|---|---|---|---|---|---|---|
| $h(x)$ | -999 | -99 | -9 | 11 | 101 | 1001 |

Rather than seeing function values leveling off as in the previous examples, we see function values that grow in size along with $x$. The function $h$ has no horizontal asymptote. Let's examine the graph:

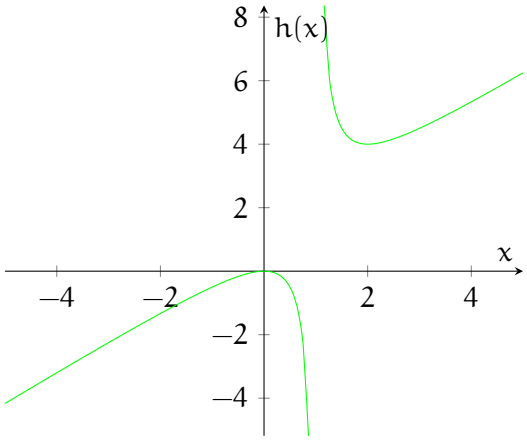



Figure 101.3: Graph of $h(x) = \frac{x^2}{x-1}$

This function exhibits a different type of end behavior: that of a line with slope 1. To see that, cover up the portion of the graph near the $y$-axis and focus on the left and right. The rather dull and time-consuming technique of polynomial long division can be used to rewrite the function as a quotient and a remainder. We encourage you to watch the Khan Academy video on the topic, but for now, let us instead use our knowledge of factoring techniques and a clever little trick.

$$\begin{aligned} h(x) &= \frac{x^2}{x-1} \\ &= \frac{x^2-1+1}{x-1} \\ &= \frac{x^2-1}{x-1} + \frac{1}{x-1} \\ &= \frac{(x-1)(x+1)}{x-1} + \frac{1}{x-1} \\ &= x+1+\frac{1}{x-1} \end{aligned} \tag{101.3}$$

We obtain a quotient of $x+1$ and a remainder of 1. It is the quotient that determines the end behavior of the graph. Why? Substituting $x$-values far from zero makes the remainder term very small, since it becomes a fraction with a large denominator but a numerator of only 1. So for $x$-values far from zero, the $y$-value is $x$ plus 1 plus a very small number (so small that we can justifiably ignore it). This means that far from the $y$-axis, the function acts like the quotient: the line $y = x+1$. We call this line an oblique asymptote. See below how the graph of $h(x)$ hugs that line.

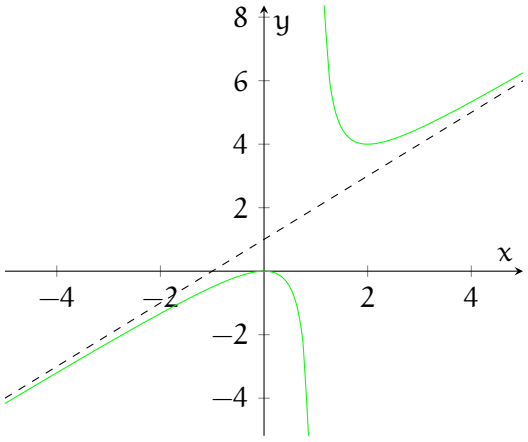



Figure 101.4: Graph of $h(x) = \frac{x^2}{x-1}$ and its oblique asymptote $y = x + 1$

## *Exercise 239* Rational Functions Practice 2

*Working Space*

Factor and simplify the rational function, then determine any holes and vertical and oblique asymptotes of the rational function.

1. $\frac{x^3+2x^2}{x^2+x}$

*Answer on Page 1958*

We have seen lines act as end behaviors. Are there other possibilities? Sure! Here is an example with parabolic end behavior.

$$k(x) = \frac{x^3}{x-2}$$

We use our add-subtract trick to reveal the quotient, which describes the end behavior.

$$\begin{aligned} h(x) &= \frac{x^3}{x-2} \\ &= \frac{x^3-8+8}{x-2} \\ &= \frac{x^3-8}{x-2} + \frac{8}{x-2} \\ &= \frac{(x-2)(x^2+2x+4)}{x-2} + \frac{8}{x-2} \\ &= x^2+2x+4+\frac{8}{x-2} \end{aligned} \tag{101.4}$$

The quotient, $x^2+2x+4$, should describe the end behavior. We confirm by graphing both $k$ and the quotient - the parabolic asymptote.

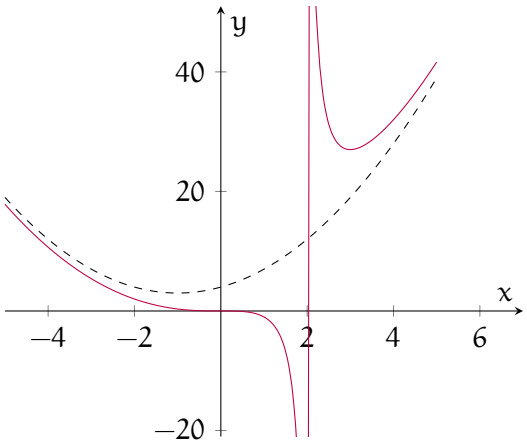



Figure 101.5: Graph of $k(x) = \frac{x^3}{x-2}$ and its parabolic asymptote $y = x^2+2x+4$

# Chapter 102

# Partial Fractions

How can you add fractions with different denominators, like $\frac{1}{x} + \frac{2}{x+3}$? You would need to make the denominators the same; after that, you could just add the numerators. You achieve this by multiplying the numerator and denominator of each fraction by the denominator of the other fraction:

$$\frac{1}{x} + \frac{2}{x+3} = \frac{1}{x}\left(\frac{x+3}{x+3}\right) + \frac{2}{x+3}\left(\frac{x}{x}\right)$$

Recall that when the numerator and denominator of a fraction are the same, the fraction is equal to one. So, we are not changing the *value* of each fraction, since we are just multiplying by one. Continuing, we can perform the multiplication and see that:

$$\frac{1}{x} + \frac{2}{x+3} = \frac{x+3}{x(x+3)} + \frac{2x}{x(x+3)}$$

$$= \frac{(x+3)+2x}{x(x+3)} = \frac{3x+3}{x^2+3x} = \frac{3(x+1)}{x^2+3x}$$

The inverse of this process is called **partial fraction decomposition** (or partial fraction expansion). This method has applications in many fields, but we will find it most useful as a tool to evaluate integrals in a later chapter.

Let $g(x)$ be a rational function such that

$$g(x) = \frac{P(x)}{Q(x)}$$

Where $P(x)$ and $Q(x)$ are polynomials. If $g(x)$ is proper (that is, the degree of $P$ is less than the degree of $Q(x)$ then we can express $g(x)$ as the sum of simpler rational fractions. If $g(x)$ is improper (that is, the degree of $P$ is greater than or equal to the degree of $Q$), then we must first perform long division to obtain a remainder, $R(x)$, where the degree of $R$ is less than the degree of $Q$:

$$g(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

### 102.0.1 Improper fractions

What is $\int \frac{x^3+x}{x-1}\,dx$. Using long division, we see that:

$$\frac{x^3+x}{x-1} = x^2+x+2+\frac{2}{x-1}$$

(see figure 102.1 for an explanation). Then we can also say that:

$$\frac{x^3+x}{x-1} = x^2+x+2+\frac{2}{x-1}$$

$$\begin{array}{r|l}
 & x^2+x+2 \\
\hline
x-1 & x^3+0x^2+x \\
 & \underline{-(x^3-x^2)} \\
 & \quad x^2+x \\
 & \quad \underline{-(x^2-x)} \\
 & \qquad 2x \\
 & \qquad \underline{-(2x-2)} \\
 & \qquad\quad 2
\end{array}$$

Figure 102.1: Evaluating $(x^3+x) \div (x-1)$ with the long division method

When you start with an improper fraction, use long division to reduce it to a term plus a proper fraction, then use the methods outlined below to further manipulate the proper fraction.

### Exercise 240

Use long division to reduce the following improper rational functions to a term plus a proper rational fraction.

1. $\frac{x^4+x^3+2x^2+2x-3}{x^2-3x+2}$
2. $\frac{2x^3+5}{x^3-3x^2+2x-4}$
3. $\frac{3x^4-2x^3-x^2+1}{x^3-3x}$

*Working Space*

*Answer on Page 1958*

#### 102.0.2 Proper fractions

When the order of the numerator is less than or equal to the denominator, there are three further possibilities.

**No repeated linear factors**

In the first case, the denominator, $Q(x)$ is composed of distinct linear factors. In this case, we can say that $Q(x) = (a_1x + b_1)(a_2x + b_2)\cdots(a_nx + b_n)$, where no factor is repeated (including constant multiples). Then, there exists $A, B, C, \cdots$, such that:

$$\frac{P(x)}{Q(x)} = \frac{A}{a_1x + b_1} + \frac{B}{a_2x + b_2} + \cdots$$

Let's see an example of this by decomposing $\frac{4x^2-7x-12}{x(x+2)(x-3)}$. We start by defining $A$, $B$, and $C$, such that:

$$\frac{4x^2 - 7x - 12}{x(x+2)(x-3)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-3}$$

Multiplying both sides by $x(x+2)(x-3)$ we get:

$$4x^2 - 7x - 12 = A(x+2)(x-3) + B(x)(x-3) + C(x)(x+2)$$

We have 3 unknowns and only one equation! Don't worry — remember this equation is

true for all $x$, so we can choose a convenient value of $x$ to isolate each unknown in turn. Starting, let $x = 0$. Then:

$$4(0)^2 - 7(0) - 12 = A(0+2)(0-3) + B(0)(x-3) + C(0)(x+2)$$

$$-12 = A(2)(-3) + 0 + 0$$

Notice that the $B$ and $C$ disappear, and we can solve for $A$:

$$A = \frac{-12}{-6} = 2$$

We can solve for $B$ by setting $x = -2$ and for $C$ by setting $x = 3$ (notice, we've used all three zeroes of the denominator polynomial):

$$4(-2)^2 - 7(-2) - 12 = A(-2+2)(-2-3) + B(-2)(-2-3) + C(-2)(-2+2)$$

$$4(4) + 14 - 12 = 0 + B(-2)(-5) + 0$$

$$16 + 2 = 10B$$

$$B = \frac{9}{5}$$

and

$$4(3)^2 - 7(3) - 12 = A(3+2)(3-3) + B(3)(3-3) + C(3)(3+2)$$

$$4(9) - 21 - 12 = 0 + 0 + C(3)(5)$$

$$36 - 33 = 15C$$

$$C = \frac{1}{5}$$

We can then decompose our original fraction:

$$\frac{4x^2 - 7x - 12}{x(x+2)(x-3)} = \frac{2}{x} + \frac{9}{5(x+2)} + \frac{1}{5(x-3)}$$

You can check your answer by cross-multiplying and adding. You should get the same rational function back.

**Repeated linear factors**

The second case is if $Q(x)$ has repeated factors (such as $x^2 + 8x + 16 = (x+4)^2$). Suppose the first linear factor, $(a_1x + b_1)$ is repeated $r$ times (that is, $Q(x)$ contains the factor $(a_1x + b_1)^r$). Then, instead of $\frac{A}{a_1x+b_1}$, we should write:

$$\frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \cdots + \frac{A_r}{(a_1x + b_1)^r}$$

Let's look at a concrete example to see how this works:

**Example**: Decompose $\frac{x^2+x+1}{(x+1)^2(x+2)}$

**Solution**: We start by defining:

$$\frac{x^2+x+1}{(x+1)^2(x+2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2}$$

Multiplying both sides by $(x+1)^2(x+2)$:

$$x^2+x+1 = A(x+1)(x+2) + B(x+2) + C(x+1)^2$$

Since there are only 2 roots to $(x+1)^2(x+2)$, we will use another method called "equating the coefficients" to find A, B, and C. We start by expanding the right side of the equation:

$$x^2+x+1 = A(x^2+3x+2) + B(x+2) + C(x^2+2x+1)$$

Distributing and combining, we find that:

$$x^2+x+1 = Ax^2+3Ax+2A+Bx+2B+Cx^2+2Cx+C$$

$$x^2+x+1 = (A+C)x^2+(3A+B+2C)x+(2A+2B+C)$$

For this equation to be true, we know that:

$$A+C=1$$

$$3A+B+2C=1$$

$$2A+2B+C=1$$

(That is, the coefficient for $x^2$ on the left, 1, must be equal to the coefficient for $x^2$ on the right, $(A+C)$, and so on.) We now have a system of 3 equations and 3 unknowns. When you solve for each, you should find that:

$$A=-2$$

$$B=1$$

$$C=3$$

Therefore,

$$\frac{x^2+x+1}{(x+1)^2(x+2)} = \frac{-2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2}$$

### Irreducible quadratic factors

Sometimes, we cannot express a polynomial as the product of two linear statements (that is, terms in the form $ax+b$). Take $x^2+1$, which cannot be expressed as the product of real, linear terms. What do you do if something like $x^2+1$ is in the denominator? In this

case, when we write an expression for $\frac{P(x)}{Q(x)}$, we include a term in the form:

$$\frac{Ax + B}{ax^2 + bx + c}$$

For example, we can write:

$$\frac{x}{(x-2)(x^2+1)(x^2+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{x^2+4}$$

**Example**: Decompose $\frac{2x^2-x+4}{x^3+4x}$

**Solution**: We begin by factoring the denominator:

$$x^3 + 4x = x(x^2+4)$$

Which cannot be factored further. Therefore, we define:

$$\frac{2x^2-x+4}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$$2x^2 - x + 4 = A(x^2+4) + (Bx+C)x$$
$$2x^2 - x + 4 = Ax^2 + 4A + Bx^2 + Cx$$

Which implies that:

$$2 = A + B$$
$$C = -1$$
$$4A = 4$$

Therefore, $A = 1$, $B = 1$, and $C = -1$ and we can say that:

$$\frac{2x^2-x+4}{x^3+4x} = \frac{1}{x} + \frac{x-1}{x^2+4}$$

**Repeated irreducible quadratic factors**

Lastly, the denominator might contain repeated irreducible quadratic factors. Similar to repeated linear factors, when setting up your partial fractions, instead of only writing

$$\frac{A}{ax^2 + bx + c}$$

For a quadratic factor that is repeated $r$ times, your equation should include:

$$\frac{A_1}{ax^2+bx+c} + \frac{A_2}{(ax^2+bx+c)^2} + \cdots + \frac{A_r}{(ax^2+bx+c)^r}$$

## Exercise 241

Decompose the following proper fractions

1. $\frac{x-4}{x^2+5x-6}$
2. $\frac{x^2+x+1}{(x^2+1)^2}$
3. $\frac{x^2+x+1}{(x+1)^2(x+2)}$

*Working Space*

*Answer on Page 1958*

# Chapter 103

# Limits

## 103.1 Introduction to Limits

To begin talking about limits, let's imagine a a scenario:

I flip a coin 10 times, and count the number of times it lands on heads. Let's say we get the following sequence

$$H, T, T, H, T, H, T, T, H, T$$

the fraction of heads to total flips is

$$\frac{\text{number of heads}}{\text{number of flips}} = \frac{4}{10} = 0.4$$

Now, if we were to flip the coin 100 times instead, and, we got 46 heads. Our sample proportion gets closer to 0.5, which is our *expected value* or *mean*:

$$\frac{\text{number of heads}}{\text{number of flips}} = \frac{46}{100} = 0.46$$

As we the number flips increases, the fraction of heads to total flips approaches the true probability of heads. Flipping the coin $10,000$ times would produce a result closer to the expected value, such as $5,230$ making a fraction of $0.523$. While all of the flips are individual events, the long running average stabilizes closer to the true probability.

This long-run stabilization is one version of the *Law of Large Numbers*, an important idea in statistics.

We will revisit this in our statistics and sequence converge chapters. For now, what does this tell us? The law of large numbers provides insight into the central idea of a limit: what value something approaches as an input or process continues.

## 103.2 Limits of Asymptotes

A function has a **vertical asymptote** at $x = a$ when the function values grow without bound as $x$ approaches $a$. A function's **end behavior** describes how the function values behave as $x \to \infty$ or $x \to -\infty$; if the function approaches a fixed value $L$, we say it has a **horizontal asymptote** at $y = L$.

The asymptotic behavior we see in rational functions suggests that we need to expand our

vocabulary of function characteristics. We examined vertical asymptotes and end behavior through graphs and tables, and discussed them in English. The language of limits enables us to discuss these attributes mathematically and with greater efficiency.

Let's revisit an example from the previous chapter. This function has a hole at $x = 1$, a vertical asymptote at $x = 3$, and a horizontal asymptote of $y = 1$.

$$f(x) = \frac{x^2 - 3x + 2}{x^2 - 4x + 3} = \frac{(x-1)(x-2)}{(x-1)(x-3)}$$

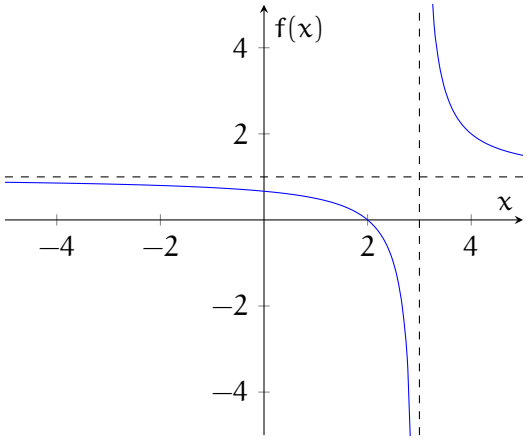



Figure 103.1: Graph of $f(x) = \frac{x^2-3x+2}{x^2-4x+3}$ with asymptotes

First, consider the vertical asymptote. We see that the graph goes down as it hugs the left side of the vertical asymptote, and goes up as it hugs the right side. We can describe these behaviors as the left- and right-hand limits, respectively. We say that the left-hand limit of $f$ at $x = 3$ is negative infinity. Another way of communicating this is to say that as $x$ approaches 3 from the left, the function approaches negative infinity. Symbolically, we summarize this as

$$\lim_{x \to 3^-} f(x) = -\infty$$

The little negative sign in $x \to 3^-$ indicates we are approaching $x = 3$ from the left (the negative side of the axis).

Similarly, the right-hand limit of $f$ at $x = 3$ is positive infinity. In other words, as $x$ approaches 3 from the right, the function approaches positive infinity. Symbolically, we write

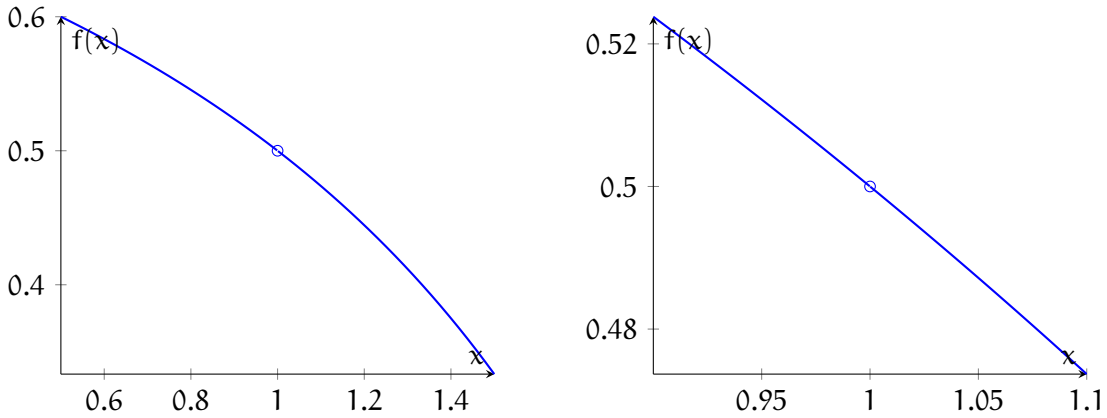
$$\lim_{x \to 3^+} f(x) = \infty$$

This time, the little $^+$ indicates we are approaching the x-value from the right (positive) side of the axis.

The limit of a function at a particular $x$-value is the $y$-value that the function approaches as it approaches the given $x$-value. In the previous example, we could only specify the left- and right-hand limits, because they were different. In cases where the left- and right-hand limits are equal, we can say that the function has a limit there. The hole in our function $f$ is one such value. We see that as we approach the hole from both the left and right, the function takes on values near $\frac{1}{2}$. This is more apparent numerically:

| $x$ | 0.9 | 0.99 | 0.999 | 1 | 1.001 | 1.01 | 1.1 |
|---|---|---|---|---|---|---|---|
| $f(x)$ | 0.5238 | 0.5025 | 0.5003 | undefined | 0.4998 | 0.4975 | 0.4737 |

We can also see this by zooming in on the graph (see Figure 103.2):




(a) Zoomed out near $x = 1$

(b) Zoomed in closer to $x = 1$

Figure 103.2: Two graphs of $f(x) = \frac{x^2-3x+2}{x^2-4x+3}$ zoomed in about $x = 1$

The left-hand and right-hand limits of $f$ at 1 are both $\frac{1}{2}$. Since they are equal, we can also say that the limit of $f$ at 1 is $\frac{1}{2}$. This allows us to efficiently discuss the behavior of $f$ at 1, even though the function is not defined there, as substituting 1 into the function gives division by zero.

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = \lim_{x\to 1} f(x) = \frac{1}{2}$$

We can also talk about limits at $x$-values where nothing weird is happening (that is, no hole or vertical asymptote). For example, as $x$ approaches 4 from the left and right, $y$ approaches 2.

| $x$ | 3.9 | 3.99 | 3.999 | 4 | 4.001 | 4.01 | 4.1 |
|---|---|---|---|---|---|---|---|
| $f(x)$ | 2.1111 | 2.0101 | 2.0010 | 2 | 1.9990 | 1.9901 | 1.9091 |

In this case, since nothing weird is happening, the limit is equal to the function value. This is an example of continuity, which we will discuss in more detail in the next chapter. By contrast, at the vertical asymptote $x = 1$, since the left- and right-hand limits are not equal, we say the function does not have a limit, or the limit does not exist.

Finally, let's consider the horizontal asymptote of $f$. The graph hugs the line $y = 1$ as $x$ goes far to the left and far to the right. We say that as $x$ approaches negative infinity, $f$ approaches 1; likewise, that as $x$ approaches positive infinity, $f$ approaches 1. We write these symbolically as $\lim_{x\to-\infty} f(x) = 1$ and $\lim_{x\to\infty} f(x) = 1$.

## *Exercise 242* Limits Practice 1

Determine the left- and right-hand limits of the function as $x$ approaches the given values. At $x$-values where the limit exists, determine it.

1. $p(x) = \frac{x+3}{x^2+9x+18}, x = -6, -5, -3, \infty$

*Working Space*

*Answer on Page 1958*

We have seen two weird behaviors of rational functions at certain $x$-values: holes and vertical asymptotes. Now, we will examine another type of weird behavior: jumps. This is a characteristic of some piecewise-defined functions. In piecewise-defined functions, the domain is divided into two or more pieces, and a different expression is used to give the y-value depending on which piece contains the $x$- value. One common piecewise-defined function is the floor function (shown in figure 103.3), sometimes denoted $\lfloor x \rfloor$. The standard floor function rounds any real number down to the nearest integer. So, for a price quoted in dollars and cents, the floor would just be the number of dollars.

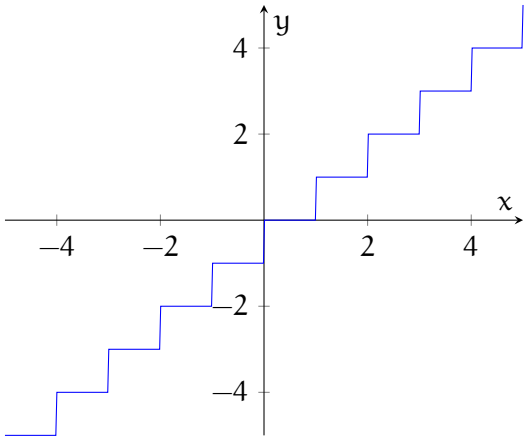



Figure 103.3: Graph of $y = \lfloor x \rfloor$

When $x$ is exactly 1, the function value is 1: the number of dollars in a price of \$1.00. When $x$ is any number greater than 1 but less than 2, the function value is still 1. Also, $\lfloor 1.01 \rfloor, \lfloor 1.5 \rfloor,$ and $\lfloor 1.99999 \rfloor$ are all 1. As we continue to look to the right, once $x$ equals exactly 2, $h$ jumps up to the value 2. So, $\lim_{x \to 2^-} \lfloor x \rfloor = 1$, while $\lim_{x \to 2^+} \lfloor x \rfloor = 2$.

Besides rational and piecewise defined functions, there are other functions with interesting limits. Consider the standard exponential function, $y = e^x$ (shown in figure 103.4).

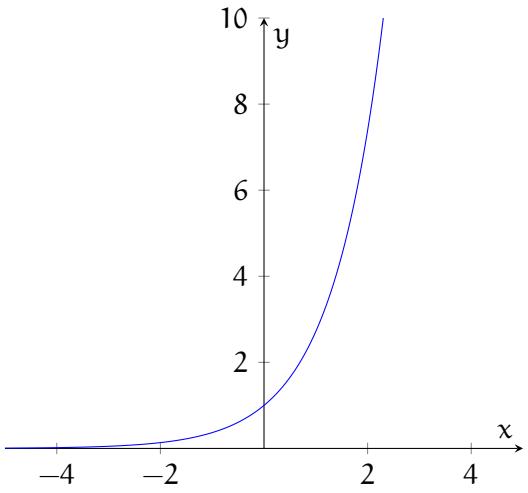



Figure 103.4: Graph of $y = e^x$

As $x$ increases, $y$ increases without bound; that is, $\lim_{x \to \infty} e^x = \infty$. However, looking far to the left, we see that $y$ hugs the $x$-axis. This is because raising $e$ to a large negative exponent is the same as 1 divided by $e$ raised to a large positive exponent; that is, 1 divided by a very large number, which yields a very small positive number. In limit notation, $\lim_{x \to -\infty} e^x = 0$. This example illustrates that horizontal asymptotes need not

model end behavior in both directions. Note that this reasoning holds for $y = b^x$ for any $b > 1$, so all such functions have the same horizontal asymptote, $y = 0$.

We know that the natural logarithm function, $y = \ln x$, is the inverse of $y = e^x$. Since inverse functions swap the role of $x$ and $y$, it stands to reason that a horizontal asymptote in one function corresponds with a vertical asymptote in the other function, and that is indeed the case (see figure 103.5).

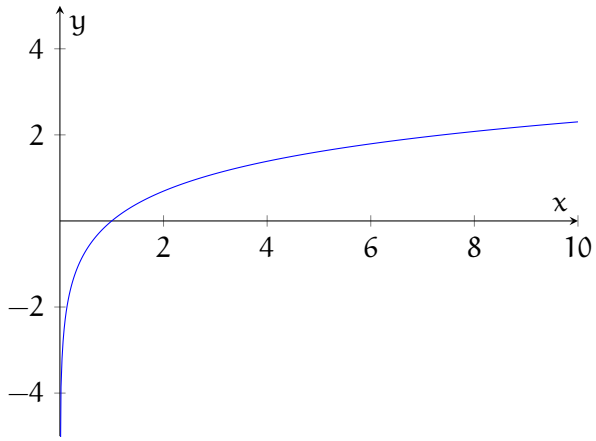



Figure 103.5: Graph of $y = \ln x$

An untransformed logarithm function is defined only for positive inputs. That is because it is not possible to find an exponent of a positive number that will yield a negative or zero result. What type of exponent on a positive number yields a number near zero? That would be a large-magnitude negative number. So, on the logarithm graph, large negative $y$-values correspond with $x$-values only slightly greater than zero. So, $\ln x$ (and $\log_2 x$, and indeed $\log_b x$ for any $b > 1$) approaches negative infinity as $x$ approaches 0 from the right. There is no left-hand limit at 0, however. In limit notation, $\lim_{x \to 0^+} \ln x = -\infty$.

## *Exercise 243* Limits Practice 2

State the asymptotes of the following transformed exponential and logarithmic functions. Give the limit statement which describes the behavior of the function along the asymptote.

*Working Space*

1. $y = 3^x + 1$
2. $y = \log_2(x - 4)$
3. $y = 2^{1-x}$
4. $y = \log_{10}(-2x)$

*Answer on Page 1959*

We next consider two functions that each have two horizontal asymptotes. These two seemingly obscure functions are quite important in data science.

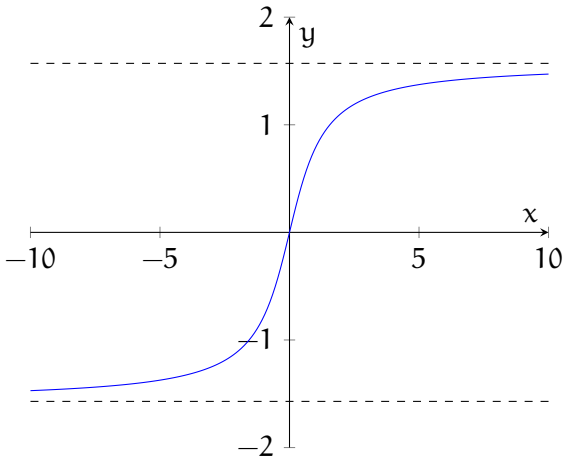



Figure 103.6: Graph of $y = \arctan x$

We know that the arctangent, or inverse tangent, function is the inverse of the piece of the tangent function which passes through the origin. The vertical asymptotes bounding this piece become horizontal asymptotes when the function is inverted.

Here are the equation and graph of the logistic function:

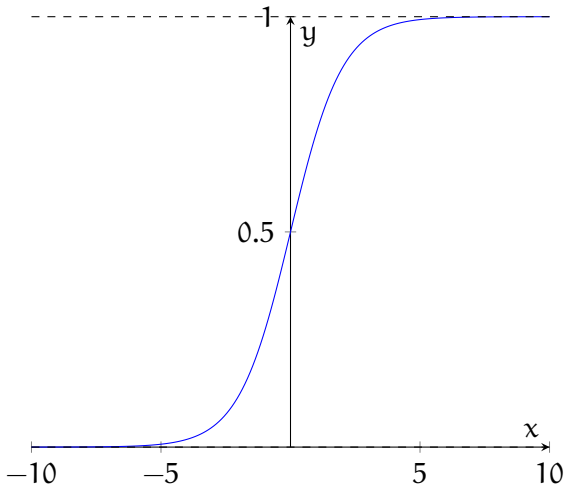



Figure 103.7: Graph of the logistic function, $y = \frac{1}{1+e^{-x}}$

For large magnitude negative values of x, the exponential term in the denominator becomes a very large positive value. The fraction thus becomes a positive number very close to zero. For large magnitude positive values of x, that exponential term becomes a very small positive number. Adding it to 1 yields a denominator just barely greater than 1. Dividing 1 by this number therefore yields a function value just barely less than 1. So, the logistic function yields values between 0 and 1, though never equaling either of these values exactly. It is precisely this characteristic which makes the logistic function so useful.

## *Exercise 244* Limits Practice 3

Using limit notation, state the limits as x approaches negative and positive infinity for the inverse tangent and logistic functions given above.

*Working Space*

*Answer on Page 1959*

As seen above, the limit of a function from the left may be different from the limit of the function from the right. Additionally, the actual *value* of the function may be different from the limit. Consider the piecewise function $h(x)$:

$$h(x) = \begin{cases} -x^2 + 3, & \text{if } x < 0 \\ 2, x = 0 \\ -x + 3, & \text{if } x > 0 \end{cases}$$

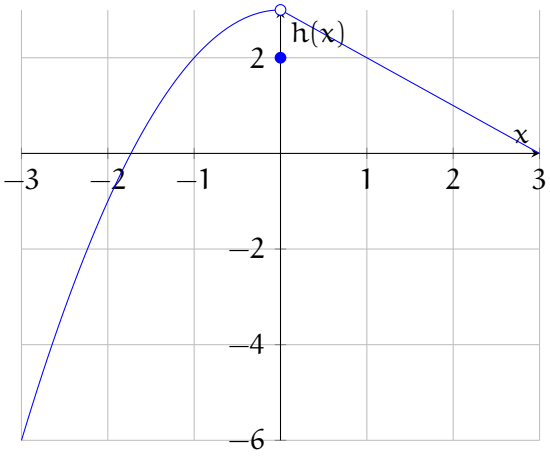



Figure 103.8: Graph of the piecewise function, $h(x)$

From examining the graph, we see that

$$\lim_{x \to 0_-} h(x) = \lim_{x \to 0_+} h(x) = 3$$

However, $h(0) = 2 \neq 3$. So, does this limit exist? It does! The limit of a function describes the *behavior* of the function around a particular value, not the value of the function itself. In order for a limit to exist, the limits from the left and right must be equal to each other, but not necessarily the actual value of the function.

Use the graph of $h(x)$ above and the graphs of $f(x)$ and $g(x)$ below to complete the following exercise.

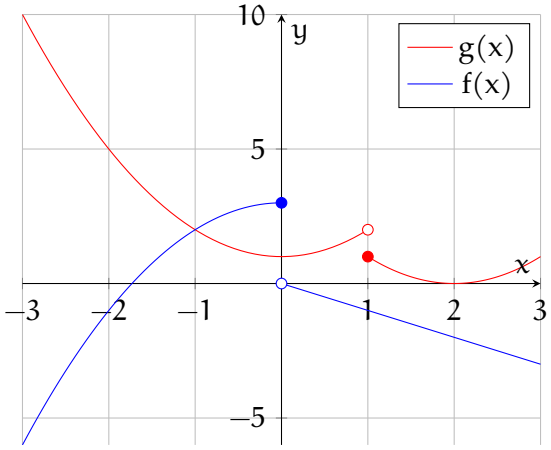



Figure 103.9: Piecewise functions $f(x)$ and $g(x)$

### *Exercise 245* Limits Practice 4

Working Space

Determine the limit from the left and the right for each function at the given value(s). State the limit at that value, if it exists.

1. $h(x), x = -1, 0, 1$
2. $f(x), x = -1, 0, 2$
3. $g(x), x = -2, 0, 1, 2$

*Answer on Page 1959*

## 103.3 Continuity

A note about continuity:

In order to be able to talk more about limits and know when we can apply certain rules and theorems, we first must discuss continuity. A function is continuous if there are no "jumps" or "gaps" in the graph of the function. For example, the function $f(x) = x^2$ is continuous for all real values of $x$. On the other hand, the function $g(x) = \tan(x)$ has many discontinuities, including at $x = \frac{\pi}{2}$. Let's examine the graph of each of these functions:

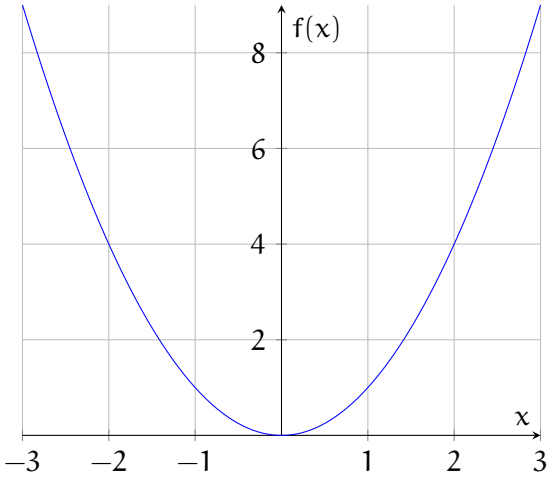



Figure 103.10: Graph of $f(x) = x^2$

If you wanted, you could trace your finger along the graph of $f(x)$ from $x = -3$ to $x = 3$ without ever picking up your finger. This means the function is continuous in the domain from $-3 \leq x \leq 3$. In this case, the domain of continuity *includes* the end points ($x = 3$ and $x = -3$). This is called a closed interval. In other cases, the function will be continuous right up to, but not including, the endpoints, as with the domains of continuity for our other example, $g(x) = \tan x$. This is called an open interval. Let's learn more about intervals of continuity by examining $g(x) = \tan x$.

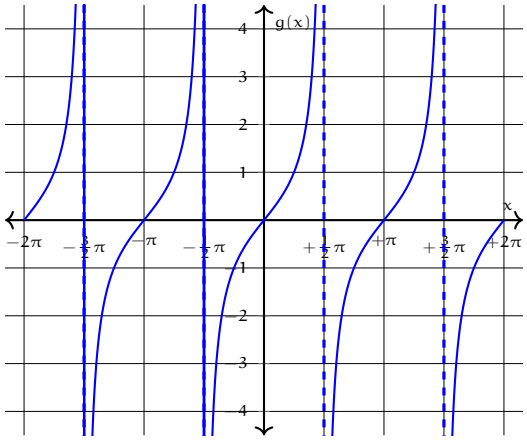



Figure 103.11: Graph of $g(x) = \tan x$

As you can see, if you trace your finger along the graph of the function starting at $x = 0$, you can continue without lifting your finger to $x = \frac{\pi}{2}$. As you approach $x = \frac{\pi}{2}$ from the left, the value of $g(x)$ approaches $\infty$. In order to continue tracing the function PAST $x = \frac{\pi}{2}$, you have to lift your finger and bring it down to $-\infty$. The function then continues continuously again until $x = \frac{3\pi}{2}$.

In the case of $g(x) = \tan x$, the function is continuous on *open intervals*, including the open interval $\frac{\pi}{2} < x < \frac{3\pi}{2}$.

There is a shorter way to represent open and closed domain intervals. We can represent that $f(x) = x^2$ is continuous on the closed interval $-3 \leq x \leq 3$ in the following way:

$$x \in [3, -3]$$

This reads as "x contained in the domain -3 to 3, inclusive". That is, all the values from -3 to 3, including the endpoints. The inclusion of the endpoints is implied by the use of *brackets*. For open intervals, we use parentheses to communicate that the interval goes up to, but does not include, the endpoints. For $g(x) = \tan x$, we can use parentheses:

$$x \in \left(\frac{-3\pi}{2}, \frac{-\pi}{2}\right)$$

because the $g(x) = \tan x$ is not continuous at $x = \frac{-3\pi}{2}$ or at $x = \frac{-\pi}{2}$.

Formally, a function $f(x)$ is continuous at $x = a$ if $\lim_{x\to a} f(x)$ exists *and* $\lim_{x\to a} f(x) = f(a)$. In other words, the limit is equal to the actual value of the function. Re-examine the graph of $h(x)$. We have already seen that $\lim_{x\to 0} h(x)$ exists and is equal to 3. However, $h(0) = 2 \neq \lim_{x\to 0} h(x)$. So $h(x)$ is not continuous at $x = 0$. Because $-x^2+3$ is evaluable all the way to $-\infty$ and $-x+3$ is evaluable all the way to $\infty$, the function $h(x)$ is continuous everywhere *except* $x = 0$. We can represent this mathematically by saying $h(x)$ is continuous on the domain $x \in (-\infty, 0) \cup (0, \infty)$. We use parentheses for $\pm\infty$ because we can never actually reach $\infty$. Additionally, the function is continuous up to, but not including 0, and the use of parentheses excludes $x = 0$ from the domain of continuity.

### 103.3.1 Continuity Practice

### Exercise 246

[This problem was originally presented as a calculator-allowed, multiple- choice question on the 2012 AP Calculus BC exam.] Suppose a function $f$ is continuous at $x = 3$. Classify the following statements as always true, sometimes true, or never true. Explain your answers.

1. $f(3) < \lim_{x\to 3} f(x)$
2. $\lim x \to 3^{+} f(x) \neq \lim_{x\to 3^{-}} f(x)$
3. $f(3) = \lim_{x\to 3^{+}} f(x) = \lim_{x\to 3^{-}} f(x)$
4. The derivative of $f$ at $x = 3$ exists.
5. The derivative of $f$ is positive for $x < 3$ and negative for $x > 3$.

Working Space

Answer on Page 1959

### Exercise 247 Limits Practice 5

State the location of discontinuities (if any) and explain why the function is discontinuous at that location:

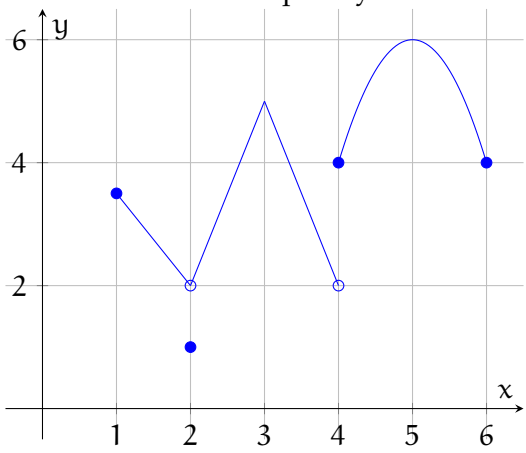
1. $f(x) = \frac{3x^2-8x-3}{x-3}$
2. $f(x) = \begin{cases} \frac{2}{x^4}, & \text{if } x <\neq 0 \\ 2, & \text{if } x = 0 \end{cases}$
3. $f(x) = \begin{cases} \frac{3x^2-8x-3}{x-3}, & \text{if } x \neq 3 \\ 1, & \text{if }, x = 3 \end{cases}$

Working Space

Answer on Page 1960

### *Exercise 248*

The graph of a function, $h(x)$, is shown. Classify each of the following statements as true or false and explain your answer.




1. $\lim_{x \to 2} h(x)$ exists
2. $\lim_{x \to 3} h(x)$ does not exist
3. $\lim_{x \to 4} h(x)$ exists
4. $h(x)$ is continuous at $x = 5$
5. $h(x)$ is not continuous at $x = 4$
6. $\lim_{x \to 2} h(x) = h(2)$

*Working Space*

*Answer on Page 1960*

## 103.4 Limits Rules

There are some mathematical properties of limits which allow us to determine the limit of complex functions without seeing a graph or using a calculator to generate a table.

The following laws are true given that $c$ is a constant, $\lim_{x \to a} f(x)$ exists, and $\lim_{x \to a} g(x)$ exists.

1. Sum Law $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$

2. Difference Law $\lim_{x\to a}[f(x)-g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$
3. Constant Multiple Law $\lim_{x\to a}[cf(x)] = c\cdot\lim_{x\to a} f(x)$
4. Product Law $\lim_{x\to a}[f(x)g(x)] = \lim_{x\to a} f(x)\cdot\lim_{x\to a} g(x)$
5. Quotient Law $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ given that $\lim_{x\to a} g(x) \neq 0$

These laws are fairly obvious — the limit of the sum of two functions is equal to the sum of the limits of each function individually. The only tricky one is the last: The limit of the quotient of two functions is equal to the quotient of the limits if and only if the limit of the function in the denominator does not equal zero. This makes sense, as we know dividing by zero yields an undefined result.

Let's practice applying these laws to evaluate the limits of the functions $f(x)$, shown in blue below, and $g(x)$, shown in red below:

$$f(x) = \begin{cases} -x^2+3, & \text{if } x \leq 0 \\ -x, & \text{if } x > 0 \end{cases}$$

$$g(x) = \begin{cases} x^2+1, & \text{if } x < 1 \\ (x-2)^2, & \text{if } x \geq 1 \end{cases}$$

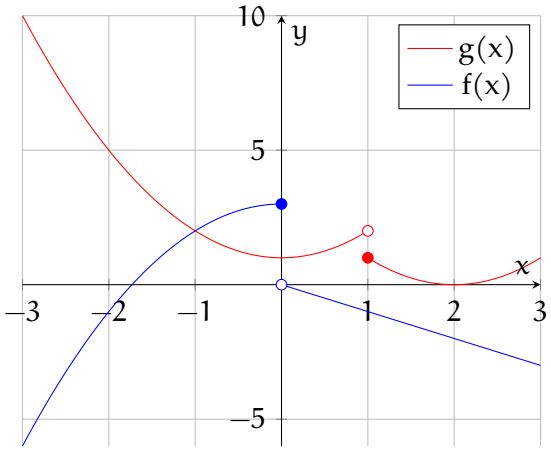



Figure 103.12: Graphs of the piecewise functions $f(x)$ and $g(x)$

We can use these laws to evaluate limits involving $f(x)$ and $g(x)$ (shown on the graph above). Here are some examples: Use the graphs of $f(x)$ and $g(x)$ given above to evaluate each limit, if it exists. If the limit does not exist, explain why. Two examples are given first:

**Example 1**: Evaluate $\lim_{x\to 0} f(x)\cdot g(x)$

**Solution 1**: From the Product Law, we know that:

$$\lim_{x \to 0} f(x) \cdot g(x) = \lim_{x \to 0} f(x) \cdot \lim_{x \to 0} g(x)$$

Looking at the graph, we can see that

$$\lim_{x \to 0} g(x) = 1$$

and there is a discontinuity in $f(x)$ at $x = 0$. Therefore,

$$\lim_{x \to 0} f(x) = \text{DNE}$$

Substituting this, we get:

$$\lim_{x \to 0} f(x) \cdot \lim_{x \to 0} g(x) = \text{DNE} \cdot 1 = \text{DNE}$$

Therefore, *the limit does not exist*.

**Example 2**: Evaluate $\lim_{x \to 2} f(x) - g(x)$

**Solution 2**: Applying the Difference Law, we see that:

$$\lim_{x \to 2} [f(x) - g(x)] = \lim_{x \to 2} f(x) - \lim_{x \to 2} g(x)$$

Examining the graph, we see that

$$\lim_{x \to 2} f(x) = -2$$

and

$$\lim_{x \to 2} g(x) = 0$$

Substituting these values, we get:

$$\lim_{x \to 2} f(x) - g(x) = -2 - 0 = -2$$

## *Exercise 249* Limits Practice 6

*Working Space*

1. $\lim_{x \to -3} \frac{f(x)}{g(x)}$
2. $\lim_{x \to 2} [f(x) + 5g(x)]$
3. $\lim_{x \to -1} \frac{3g(x)}{f(x)}$
4. $\lim_{x \to 0} f(x) \cdot 5g(x)$
5. $\lim_{x \to -1} f(x) - 3g(x)$

*Answer on Page 1960*

Recall that exponents represent repeated multiplication. Therefore, if we apply the Product Law multiple times, we obtain the Power Law for limits:

6. Power Law $\lim_{x \to \infty} [f(x)]^n = [\lim_{x \to \infty} f(x)]^n$ where n is a positive integer

There are two special limits that will be useful to us and are intuitively obvious, but we won't formally prove here.

7. $\lim_{x \to a} c = c$
8. $\lim_{x \to a} x = a$

Combining Law 8 with the Power Law, we find that:

9. $\lim_{x \to a} x^n = a^n$

And similarly, for square roots:

10. $\lim_{x\to a} \sqrt[n]{x} = \sqrt[n]{a}$ (if n is even, we assume $a > 0$)

Direct substitution property: If $f$ is a polynomial or rational function and $a$ is in the domain for $f$, then

$$\lim_{x\to a} f(x) = f(a)$$

Often, rational functions can be simplified. In an above example, we computed the limit by simplifying $f(x) = \frac{3x^2-8x-3}{x-3}$ to the simpler $g(x) = 3x+1$. This is a valid strategy because $\frac{3x^2-8x-3}{x-3} = 3x + 1$ when $x \neq 3$. Remember: a limit describes how a function behaves *as it approaches* $a$, not its value/behavior when $x$ *actually equals* $a$. This reveals the following useful rule:

If $f(x) = g(x)$ when $x \neq a$, then $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$, provided the limit exists.

## 103.5 Squeeze Theorem

The Squeeze Theorem states that if $f(x) \leq g(x) \leq h(x)$ when $x$ is near $a$ (except at $a$) and

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$$

then

$$\lim_{x\to a} g(x) = L$$

In other words, if $g(x)$ is between $f(x)$ and $h(x)$ near $a$, and $f$ and $h$ have the same limit, $L$, then the limit of $g$ must also be $L$.

**Example**: Let's examine the graph of $g(x) = x^2 \sin\frac{1}{x}$ and determine $\lim_{x\to 0} g(x)$:

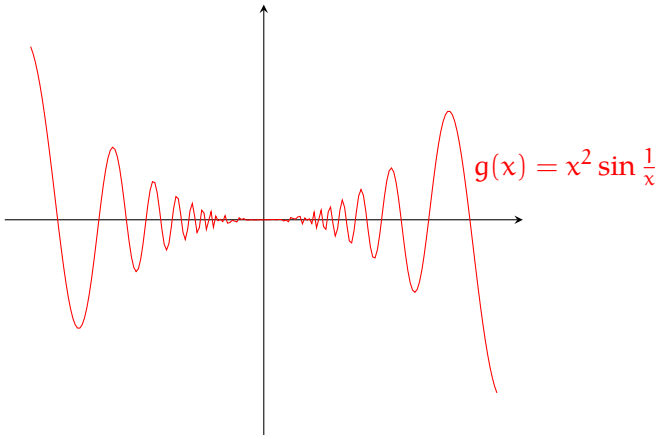



Figure 103.13: Graph of $g(x) = x^2 \sin\frac{1}{x}$

**Solution**: Because $\sin \frac{1}{x}$ is undefined at $x = 0$, we cannot compute the limit directly. However, from examining the graph, we can guess that $\lim_{x \to 0} g(x) = 0$. Feel free to confirm this with your calculator. We need to choose two functions: one that is larger than $g(x)$ near $x = 0$ and one that is smaller. Since $|\sin \frac{1}{x}| \leq 1$ (when $x \neq 0$), then

$$|x^2 \sin \frac{1}{x}| \leq x^2$$

and

$$-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

Let's confirm this by plotting $f(x) = -x^2$, $g(x)$, and $h(x) = x^2$ on the same graph:

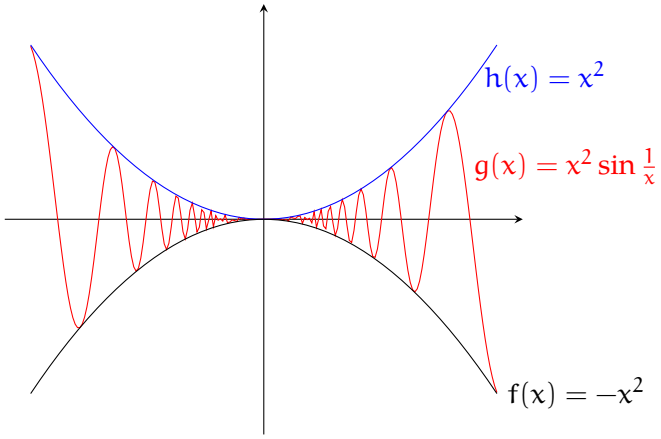



Figure 103.14: Squeeze Theorem example

As you can see, when $x$ is near 0, $f(x) \leq g(x) \leq h(x)$. Because $f(x)$ and $h(x)$ are both polynomials, their limits are straightforward:

$$\lim_{x \to 0} -x^2 = 0 \text{ and } \lim_{x \to 0} x^2 = 0$$

Then, by the Squeeze Theorem, we can say that:

$$\lim_{x \to 0} x^2 \sin \frac{1}{x} = 0$$

### 103.5.1 Squeeze Theorem Practice

## *Exercise 250* Squeeze Theorem 1

Use the Squeeze Theorem to show that $\lim_{x \to 0} \sqrt{x^3 + x^2} \cos \frac{1}{x} = 0$. Illustrate by graphing the functions you define as $f$, $g$, and $h$ on the same plot.

*Working Space*

*Answer on Page 1963*

### *Exercise 251* Squeeze Theorem 2

If $2x + 3 \leq f(x) \leq x^2 - 2x + 7$ for $x \geq 0$, find $\lim_{x \to 2} f(x)$

*Working Space*

*Answer on Page 1963*

### *Exercise 252* Squeeze Theorem 3

Prove that $\lim_{x \to 0^+} \sqrt{x} e^{\sin \frac{\pi}{x}} = 0$

*Working Space*

*Answer on Page 1963*

## 103.6 Intermediate Value Theorem

When considering functions that are continuous on a closed interval, the Intermediate Value Theorem can help us: Given a function, $f(x)$, that is continuous on the closed interval $[a, b]$ and $f(a) \neq f(b)$, there is at least one number $c$ such that $f(c) = N$, where $N$ is any number between $f(a)$ and $f(b)$. The theorem is illustrated in figures 103.15 and

103.16:

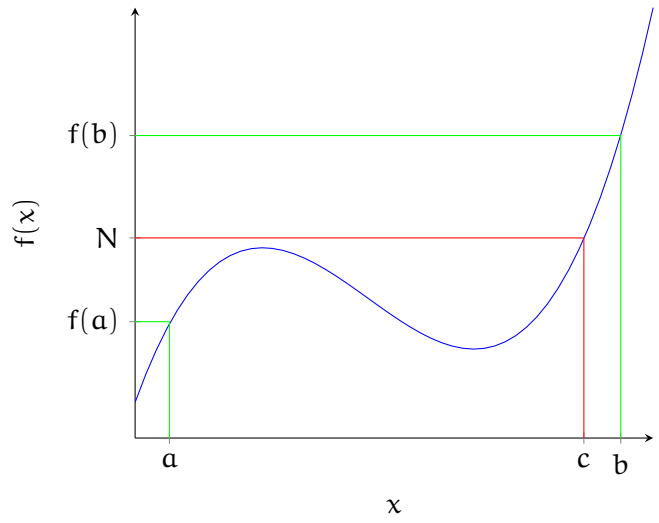



Figure 103.15: An example where one solution satisfies the Intermediate Value Theorem

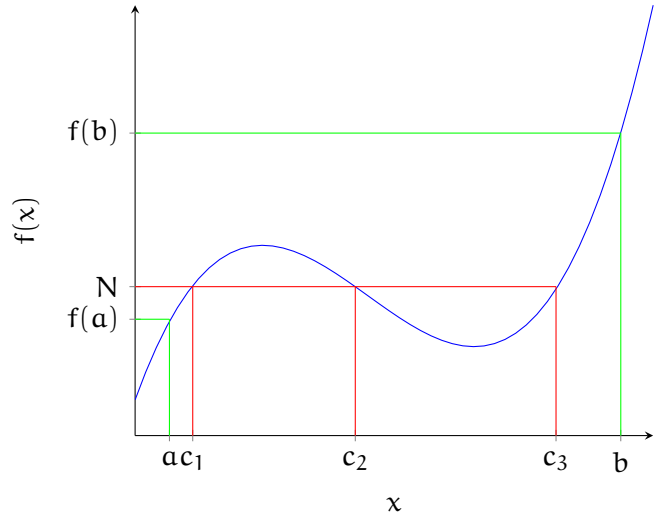



Figure 103.16: An example where more than one solution satisfies the Intermediate Value Theorem

Logically, we can think of the IVT this way: If a function is continuous on a closed interval, there are no gaps or breaks. If there are no gaps or breaks, then the function must pass through the line $y = N$, since it cannot jump over the line. Your graphing calculator uses IVT to find roots of functions:

**Example**: Show that there is at least one root of the equation $2x^3 - 6x^2 + 3x - 1 = 0$ between $x = 2$ and $x = 3$.

**Solution**: For IVT to apply, we must first check that the function is continuous on the

closed interval $x \in [2,3]$. We define $f(x) = 2x^2 - 6x^2 + 3x - 1$, which is continuous everywhere, because it is a polynomial function. *For more complex functions, always be sure to check the endpoints of an interval, since IVT only applies on closed intervals of continuity*. We will take $a = 2$, $b = 3$, and $N = 0$. We find the values of $f(x)$ at the endpoints:

$$f(2) = 2(2)^3 - 6(2)^2 + 3(2) - 1 = 16 - 24 + 6 - 1 = -3$$
$$f(3) = 2(3)^3 - 6(3)^2 + 3(3) - 1 = 54 - 54 + 9 - 1 = 8$$

Therefore, $f(2) < 0 < f(3)$ and according to IVT, there must exist some $c$ such that $f(c) = 0$ and there is a root to the equation in the interval $x \in (2,3)$.

## *Exercise 253* Intermediate Value Theorem Practice

Use the IVT to show there is a solution the given equation on the stated interval:

*Working Space*

1. $2x^4 + x - 12 = 0$ , $(1,2)$
2. $\ln(x) = 3x - 4\sqrt{x}$ , $(2,3)$
3. $2\sin x = 3x^2 - 2x$ , $(1,2)$

*Answer on Page 1964*

CHAPTER 104

# Parametric Functions

Throughout your study of functions, you have seen functions that look like this:

$$y = f(x)$$

Sometimes, to describe certain shapes or curves on a graph, we must use more than one function to do so. This means a shape, like a circle on the graph, can be represented using

$$y = f(t) \quad \text{and} \quad x = g(t).$$

In this case, there are two functions, $x(t)$ and $y(t)$, that depend on a third variable $t$, and together they help us visualize and define a curve.

The central word you should think about as you go through this chapter is *parameter*. A parameter is a variable that both $x$ and $y$ depend on. The most common example of a parametric function is the circle. To represent a circle on a graph, we can use the equations

$$x = \cos(\theta) \quad \text{and} \quad y = \sin(\theta).$$

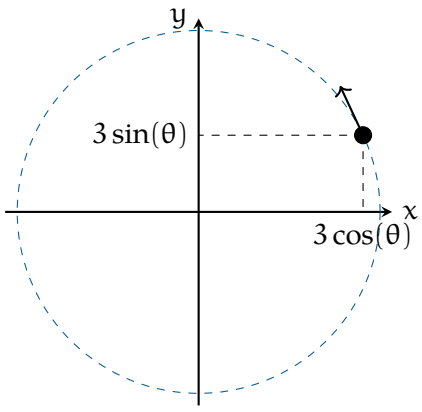



Figure 104.1: A diagram of a circle formed through a parametric functions.

In this case, the graph of the circle is determined by the value of the angle $\theta$, the parameter.

So, why parametric functions? When you look up the function that describes a circle, you might see something like

$$x^2 + y^2 = r^2.$$

The equation $x^2+y^2 = r^2$ does not describe $y$ as a single function of $x$. You would need two functions (one for the top half and one for the bottom half), and many curves cannot be written as a single function at all. Parametric equations solve this problem by allowing us to describe the entire curve with one pair of equations, using a parameter $t$. This parameter can be denoted by other letters, but we will use $t$ in this chapter.

Imagine a ball rolling across graph paper. At each moment, the ball is at a different location. If we label time with a variable $t$, then for every value of $t$ the ball has an $x$-coordinate and a $y$-coordinate. What if we had two equations, one that told us its $x$-position at time $t$ and one that told us its $y$-position? As $t$ changes, these two equations would trace the entire path of the ball. This is the basic idea behind parametric equations.

## 104.1 Conversion to Rectangular Form

In many cases, we can eliminate the parameter to convert a pair of parametric equations,

$$x = f(t), \quad y = g(t),$$

into a single equation relating $x$ and $y$.

In many cases, a parametric curve can be rewritten as a single equation involving only $x$ and $y$. This process is called eliminating the parameter, and the resulting equation is known as the rectangular form of the curve.

To eliminate the parameter, we solve one of the parametric equations for $t$ and substitute into the other equation.

**Example:**

Consider the parametric equations

$$x = 2t + 1, \quad y = 3t - 4.$$

**Solution:** Solve the equation for $x$ to isolate $t$:

$$t = \frac{x - 1}{2}.$$

Substitute into the equation for $y$:

$$y = 3\left(\frac{x - 1}{2}\right) - 4.$$

Simplifying,

$$y = \frac{3}{2}x - \frac{11}{2}.$$

**Example**

Consider the parametric equations

$$x = \cos t, \quad y = \sin t.$$

**Solution:** In this case, solving for $t$ directly is not convenient. Instead, we use the trigonometric identity

$$\cos^2 t + \sin^2 t = 1.$$

Substituting $x = \cos t$ and $y = \sin t$ gives

$$x^2 + y^2 = 1.$$

This rectangular equation describes a circle of radius 1 centered at the origin.

**Note**

Converting to rectangular form describes the shape of a curve, but it does not show the direction of motion or how many times the curve is traced. This information is contained in the parametric equations and the domain of the parameter!

## 104.2 Sketching Parametric Equations

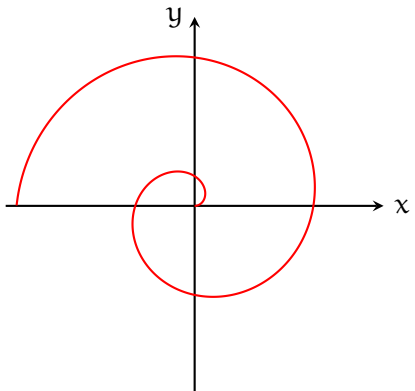
Parametric equations describe curves by expressing both $x$ and $y$ in terms of a third variable, usually $t$, called the parameter. When sketching a parametric curve, it is important to remember that the curve is traced as the parameter changes, rather than being drawn all at once as a function of $x$.

To sketch a parametric curve,

- choose several values of the parameter $t$,
- compute the corresponding $x$- and $y$-values,
- plot the resulting points in the $xy$-plane,
- and indicate the direction in which the curve is traced as $t$ increases.

Unlike rectangular graphs, parametric sketches also depend on the direction and speed of motion, which are determined by how $x(t)$ and $y(t)$ change as $t$ varies.

Here is an example of an Archimedean Spiral, a spiral where the radius $r$ is propor-



tional to the angle $\theta$. In parametric form, it is represented as

$$x(t) = t\cos(t), \quad y(t) = t\sin(t)$$

FIXME Sketched example

## Exercise 254

Working

Sketch the parametric equations

$$x = t + 1, \quad y = t^2.$$

In your sketch:

- create a table of values for $t$, $x$, and $y$,
- plot the corresponding points,
- and indicate the direction of motion as $t$ increases.

Answer on Page 1964

## 104.3 Using Python to Explore Parametric Curves

Python can be used to visualize parametric curves and explore how the point $(x(t), y(t))$ moves as $t$ changes. For example, we could write code that:

- plots points along a circle defined by $x = \cos(t)$ and $y = \sin(t)$,
- shows the distance from the origin,
- traces more complicated parametric curves like spirals or ellipses.

Below is a Python script that creates an animation for sketching parametric curves. There is a provision for changing the equations based on exercises and questions you come across.

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation

t = np.linspace(0, 3, 50)

x = 4 - 2*t
y = 3 + 6*t - 4*t**2

fig, ax = plt.subplots()
ax.set_aspect("equal", "box")

ax.axhline(0, color="black", linewidth=0.5)
ax.axvline(0, color="black", linewidth=0.5)

ax.set_xlabel("x(t)")
ax.set_ylabel("y(t)")
ax.set_title("Parametric Equation of Curve")

ax.set_xlim([x.min() - 1, x.max() + 1])
ax.set_ylim([y.min() - 1, y.max() + 1])

(line,) = ax.plot([], [], "b--")
(point,) = ax.plot([], [], "ro")

def update(frame):
    line.set_data(x[:frame + 1], y[:frame + 1])
    point.set_data([x[frame]], [y[frame]])
    return line, point

ani = FuncAnimation(fig, update, frames=len(t), interval=50, blit=True)
ani.save("parametric_animation.mp4")

plt.show()
```

## Exercise 255

*Working Space*

Consider the parametric equations

$$x = 4 - 2t, \quad y = 3 + 6t - 4t^2,$$

for $0 \leq t \leq 3$.

(a) Sketch the curve by creating a table of values for $t$, $x$, and $y$.

(b) Use the provided Python animation code to visualize the curve.

(c) Describe the direction in which the curve is traced as $t$ increases.

(d) Explain how the Python animation supports or clarifies your hand-drawn sketch.

*Answer on Page 1965*

## 104.4 Derivatives of Parametric Functions

Just like any other type of curve, you can be asked to determine the slope of the tangent to the point on a parametric curve. To do this, you should know how to find the derivative of the curve! You could do this by eliminating the variable t and solving for the derivative of the rectangular equation. However, there is an easier way!

The derivative of a parametric curve can be found by dividing $\frac{dy}{dt}$ by $\frac{dx}{dt}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

Derivatives of parametric functions are used to describe the slope of a curve at a given value of the parameter. This slope indicates how the curve is changing at that point. In some problems, the slope is used to find the equation of a tangent line to the curve. In other problems, the slope is used to analyze the behavior of the curve, such as determining where the curve has horizontal or vertical tangents.

### Example

Find $\frac{dy}{dx}$ for the parametric equations

$$x = t^2 + 1, \quad y = 3t - 2.$$

#### Solution

First compute the derivatives with respect to t:

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = 3.$$

Now divide:

$$\frac{dy}{dx} = \frac{3}{2t}.$$

As a check, eliminate the parameter. From $x = t^2 + 1$, we get

$$t = \sqrt{x - 1}.$$

Substitute into $y = 3t - 2$:

$$y = 3\sqrt{x - 1} - 2.$$

Differentiating gives

$$\frac{dy}{dx} = \frac{3}{2\sqrt{x-1}},$$

which matches the parametric result.

### Exercise 256

*Working Space*

Consider the parametric equations

$$x = t^3 - 3t, \quad y = t^2.$$

(a) Compute $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$.

(b) Find the values of $t$ for which the curve has horizontal tangents.

(c) Find the values of $t$ for which the curve has vertical tangents.

(d) Identify the points on the curve where these tangents occur.

*Answer on Page 1965*

## Exercise 257

*Working Space*

Consider the parametric equations

$$x = 2t + 1, \quad y = t^2.$$

(a) Find $\frac{dy}{dx}$ in terms of t.

(b) Evaluate $\frac{dy}{dx}$ at $t = 1$.

(c) Interpret your result as the slope of the tangent line at the corresponding point.

*Answer on Page 1966*

FIXME Expand on section

## 104.5 Length of Curve Traced by Parametric Equations

In the Python animation, a point moves along the curve as the parameter $t$ increases. This animation highlights an important idea: parametric equations do not only describe the shape of a curve, they describe motion along that curve.

At each moment in time, the point is located at $(x(t), y(t))$. As $t$ changes, the point moves from one position to the next. Sometimes this movement covers a small distance, and sometimes it covers a larger distance, even though the point remains on the same curve. This means the point does not necessarily move at a constant rate.

Speed measures how fast the point is moving along the curve at a given value of $t$. Once the speed is known, we can determine how far the point travels over a time interval. The length of the curve traced by the parametric equations is the total distance traveled by the moving point.

### Speed of a Parametric Curve

When a point moves along a parametric curve, its velocity is determined by the derivatives $x'(t)$ and $y'(t)$. The speed of the point is the magnitude of the velocity vector:

$$\text{Speed} = \sqrt{(x'(t))^2 + (y'(t))^2}.$$

Speed describes how fast the point is moving at a particular moment, but it does not tell us the total distance traveled along the curve.

### Arc Length Formula

To find the length of the curve traced by a parametric equation from $t = a$ to $t = b$, we integrate the speed over that interval.

If a curve is defined parametrically by $x = x(t)$ and $y = y(t)$, where $x'(t)$ and $y'(t)$ are continuous on $[a, b]$, then the length of the curve is given by

$$L = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\, dt.$$

### Example: Finding Speed

Given

$$x'(t) = 8t - t^2, \quad y'(t) = -t + \sqrt{t^{1.2} + 20},$$

the speed of the particle at time $t$ is

$$\text{Speed} = \sqrt{(8t - t^2)^2 + \left(-t + \sqrt{t^{1.2} + 20}\right)^2}.$$

Evaluating at $t = 2$ gives a speed of approximately 12.3 cm/s.

### Example: Length of a Curve

Consider the parametric equations

$$x = \cos t, \quad y = \sin t,$$

for $0 \leq t \leq 2\pi$.

First compute the derivatives:

$$x'(t) = -\sin t, \quad y'(t) = \cos t.$$

Substitute into the arc length formula:

$$L = \int_0^{2\pi} \sqrt{(-\sin t)^2 + (\cos t)^2}\, dt.$$

Using the identity $\sin^2 t + \cos^2 t = 1$, this simplifies to

$$L = \int_0^{2\pi} 1\, dt = 2\pi.$$

Thus, the length of the curve traced is $2\pi$, which matches the circumference of the unit circle.

### Note

Speed describes how fast a point moves along a parametric curve at a given moment. Arc length measures the total distance traveled along the curve over an interval of time. For

parametric curves, arc length is found by integrating the speed.

### Exercise 258

How can two sets of parametric equations represent the same graph, but different curves?

*Working Space*

*Answer on Page 1966*

Chapter 105

# Vector-valued Functions

In the last chapter, you calculated the flight of the shell. For any time $t$, you could find a vector $[\texttt{distance}, \texttt{height}]$. This can be thought of as a function $f$ that takes a number and returns a 2-dimensional vector. We call this a *vector-valued* function from $\mathbb{R} \to \mathbb{R}^2 \to \mathbb{R}^3$[1].

## 105.1 Vector-valued functions: position

We often make a vector-valued function by defining several real-valued functions. For example, if you threw a hammer with an initial upward speed of 12 m/s and a horizontal speed of 4 m/s along the $x$ axis from the point $(1, 6, 2)$, its position at time $t$ (during its flight) would be given by:

$$f(t) = [4t + 1,\ 6,\ -4.8t^2 + 12t + 2]$$

In other words, $x$ is increasing with $t$, $y$ is constant, and $z$ is a parabola.

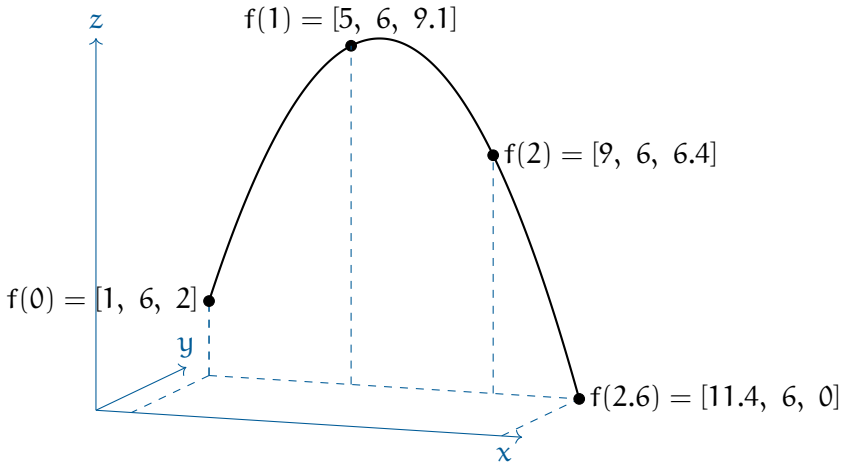



Figure 105.1: An example of a vector-valued function.

[1] the $\mathbb{R}$ symbol represents the set of all real numbers; the $\mathbb{R}^2$ symbol represents the set of all 2-dimensional vectors, and $\mathbb{R}^3$ represents the set of all 3-dimensional vectors

## 105.2 Finding the velocity vector

Now that we have its position vector, we can differentiate each component separately to get its velocity as a vector-valued function:

$$f'(t) = [4, 0, -9.8t + 12]$$

In other words, the velocity is constant along the $x$-axis, zero along the $y$-axis, and decreasing with time along the $z$ axis.

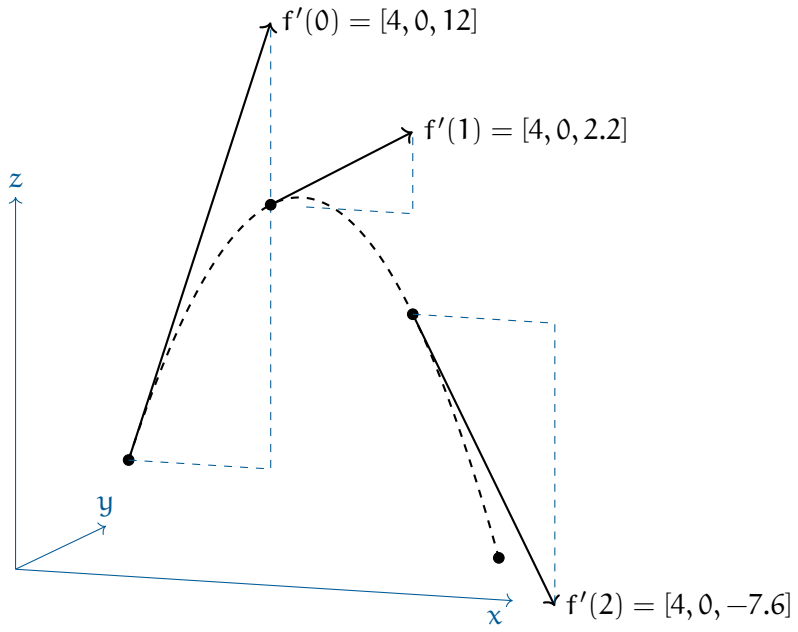



Figure 105.2: The derivatives of the position function (velocity) with respect to time.

## 105.3 Finding the acceleration vector

Now that we have its velocity, we can get its acceleration as a vector-valued function:

$$f''(t) = [0, 0, -9.8]$$

There is no acceleration along the $x$ or $y$ axes. It is accelerating down at a constant $9.8 m/s^2$.

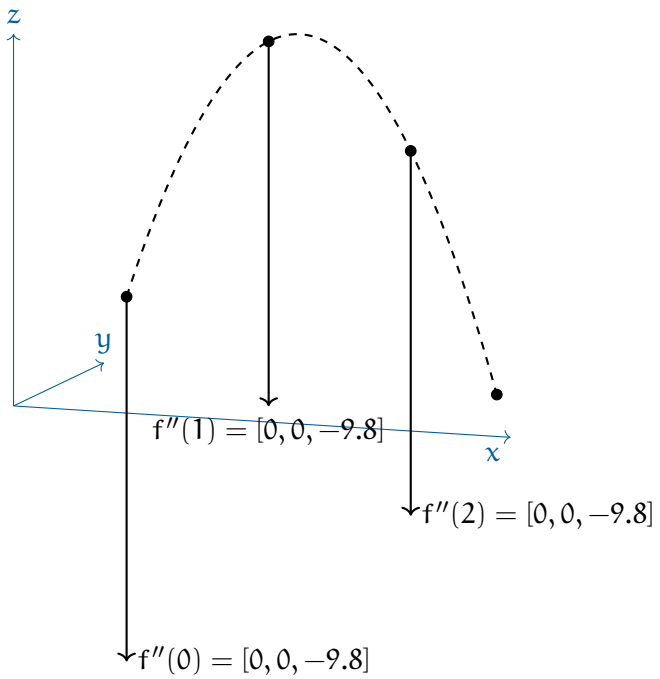



Figure 105.3: The acceleration vector is constant and points downward.

Chapter 106

# Circular Motion

We can recall the motion of 2D objects, such as a projectile launched or a car driving along a road. They have components of linear motion, such as position, velocity, acceleration, and force.

But that only works for objects moving in a straight line. What about objects moving in a circle?

## 106.1 Introduction to Uniform Circular Motion

Recall that in linear motion, You may be surprised to know that we can use very similar equations to describe circular motion, provided you use the correct variables.

The angular displacement, $\theta$, is the angle in *radians* that the object has traveled. The angular velocity, $\omega$, is the rate of change of the angular displacement, and the angular acceleration, $\alpha$, is the rate of change of the angular velocity.

| Linear motion | Rotational motion |
|---|---|
| $v = v_0 + at$ | $\omega = \omega_0 + \alpha t$ |
| $x = x_0 + v_0 t + \frac{1}{2}at^2$ | $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$ |
| $v^2 = v_0^2 + 2a(x - x_0)$ | $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$ |

Table 106.1: Kinematic equations for linear and rotational motion.

*Linear velocity*, $v$, is related to angular velocity by the equation:

$$v = r\omega$$

The *Period*, $T$, is the time it takes to complete one full rotation. The frequency, $f$, is the number of rotations per second. The two are related by:

$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$$

$$f = \frac{1}{T}$$

The *centripetal force*, usually labelled differently (such as tension or gravity), is given by

the equation:

$$F = mr\omega^2 = \frac{mv^2}{r}$$

The *angular velocity,* $\omega$ is rotations with respect to time. It can be defined in the following ways:

$$\omega_{inst} = \frac{d\theta}{dt}, \qquad \omega_{avg} = \frac{\Delta\theta}{\Delta t}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Angular acceleration:

$$\alpha = \frac{d\omega}{dt}, \qquad \alpha_{avg} = \frac{\Delta\omega}{\Delta t}$$

Note that all of the kinematic formulas still work as long as you stay in the same relative frame of reference (rotational or linear). You cannot "mix and match" the two sets of equations.

You may be asking, "If the object is moving at a constant speed, why is there acceleration?" Recall that acceleration is a vector quantity, meaning it has both magnitude and direction. Even though the speed (the magnitude of the velocity) is constant, the direction of the velocity vector is always changing. Since acceleration is the rate of change of velocity, any change in direction means there is acceleration. This acceleration is always directed toward the center of the circle, which is why it is called centripetal (center-seeking) acceleration.

Now that we've established the basic relationships for circular motion, let's apply them to a practical example.

## 106.2 The Flying Billiard Ball

Let's say you tie a 0.16 kg billiard ball to a long string and begin to swing it around in a circle above your head. The string, perpendicular, is 3 meters long, and the ball returns to where it started every 4 seconds. We will assume the ball moves at a constant speed. If you start your stopwatch as the ball crosses the x-axis, the coordinates of the position $(x, y, z)$ of the ball at any time $t$ given by:

$$p(t) = \left[3\cos\left(\frac{2\pi}{4}t\right), 3\sin\left(\frac{2\pi}{4}t\right), 2\right]$$

(This assumes that the ball would be going counter-clockwise if viewed from above. The

spot you are standing on is considered the origin $[0, 0, 0]$.)

Notice that the height is a constant — 2 meters in this case. That isn't very interesting, so we will talk just about the first two components. Figure 106.1 shows a visual of the situation:

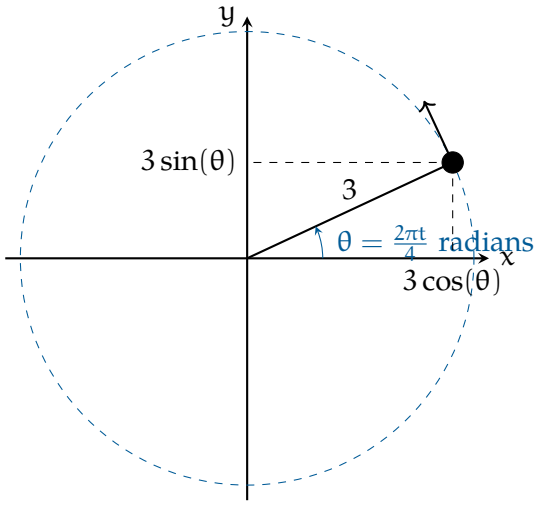



Figure 106.1: A diagram of our ball and string scenario.

In this case, the radius, r, is 3 meters. The period, T, is 4 seconds. In general, we say that circular motion is given by:

$$p(t) = \left[ r \cos \frac{2\pi t}{T}, r \cos \frac{2\pi t}{T} \right]$$

A common question is "How fast is it turning right now?" If you divide the $2\pi$ radians of a circle by the 4 seconds it takes, you get the answer "About 1.57 radians per second." This is known as *angular velocity* and we typically represent it with the lowercase Omega: $\omega$. (Yes, it looks a lot like a "w".) To be precise, in our example, the angular velocity is $\omega = \frac{\pi}{2}$. Note that in this scenario, the angular velocity and linear *speed* are constant. However, the *velocity* vector is not constant; as we will see in the next section, the direction of the velocity vector is always changing.

Notice that this is different from the question "How fast is it going? (referring to *linear velocity*)" This ball is traveling the circumference of $6\pi \approx 18.85$ meters every 4 seconds. This means the speed of the ball is about 4.71 meters per second.

It is very important to distinguish between angular velocity and linear velocity. Angular velocity is how fast the angle is changing and is referred to by $\omega$, while linear velocity, $v$, is how fast the object is moving along its path. While linear velocity has a constant speed,

it has always changing direction. The angular velocity is constant, but the linear velocity is not.

## 106.3 Velocity

The velocity of the ball is a vector, and we can find that vector by differentiating each component of the position vector.

For any constants $a$ and $b$:

| Expression $f(x)$ | Derivative $f'(x)$ |
|---|---|
| $a \sin bt$ | $ab \cos bt$ |
| $a \cos bt$ | $-ab \sin bt$ |

Thus, in our example, the velocity of the ball at any time $t$ is given by:

$$v(t) = \left[-\frac{3(2\pi)}{4} \sin \frac{2\pi t}{4}, \frac{3(2\pi)}{4} \cos \frac{2\pi t}{4}, 0\right]$$

Notice that the velocity vector is perpendicular to the position vector. It has a constant magnitude.

In general, an object traveling in a circle at a constant speed has the velocity vector:

$$v(t) = [-r\omega \sin \omega t, r\omega \cos \omega t]$$

where $t = 0$ is the time that it crosses the $x$ axis. If $\omega$ is negative, that means the motion would be clockwise when viewed from above.

The magnitude of the velocity vector is $r\omega$. See Figure 106.2 for a visual of the velocity vector. Note that the $z$-coordinate is constant, so its derivative is zero, and it is eliminated from our vector for simplification.

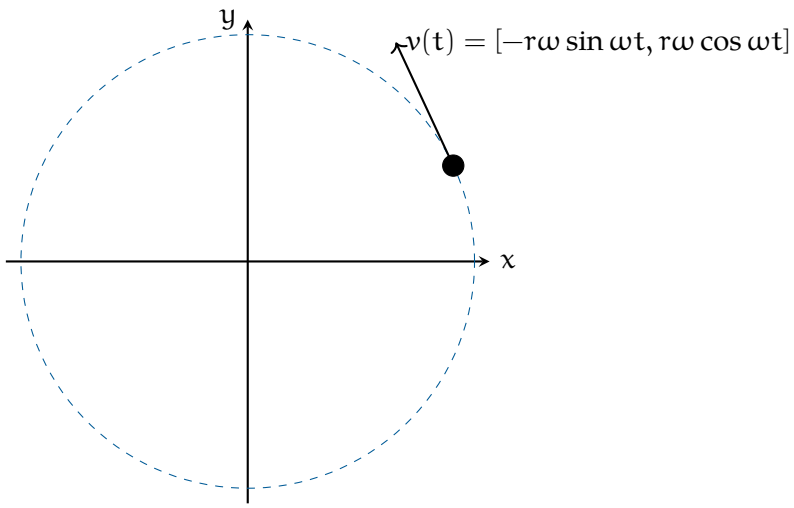



Figure 106.2: Velocity vector of the ball in circular motion.

### *Exercise 259* A Particle in Motion

[This question was originally presented as two multiple-choice problems on the 2012 AP Physics C exam.] The $x$ and $y$ coordinates of a particle as it moves in a circle are given by:

$$x = 5\cos(3t) y = 5\sin(3t)$$

What is the radius of the particle's circular path? What is the particle's *speed*? Based on your answers, how long does it take the particle to complete the circular path?

*Working Space*

*Answer on Page 1967*

## 106.4 Acceleration

We can get the (linear) acceleration by differentiating the components of the velocity vector.

$$a(t) = \left[-r\omega^2\cos\omega t, -r\omega^2\sin\omega t\right] = -r\omega^2\left[\cos\omega t, \sin\omega t\right]$$

Notice that the acceleration vector points **toward the center** of the circle it is traveling on. That is, when an object is traveling on a circle at a constant (uniform) speed (notice

that coefficients $r$ and $\omega$ are constant), its only acceleration is toward the center of the circle. The acceleration does not come from the motion itself, it comes from the constantly changing direction. This is known as *centripetal acceleration*.

The magnitude of the acceleration vector is $a_c = r\omega^2$, or more commonly, $a_c = \frac{v^2}{r}$. This gives us the common calculation for whatever force acts as the centripetal force: $F_c = ma_c = \frac{mv^2}{r}$.[1]

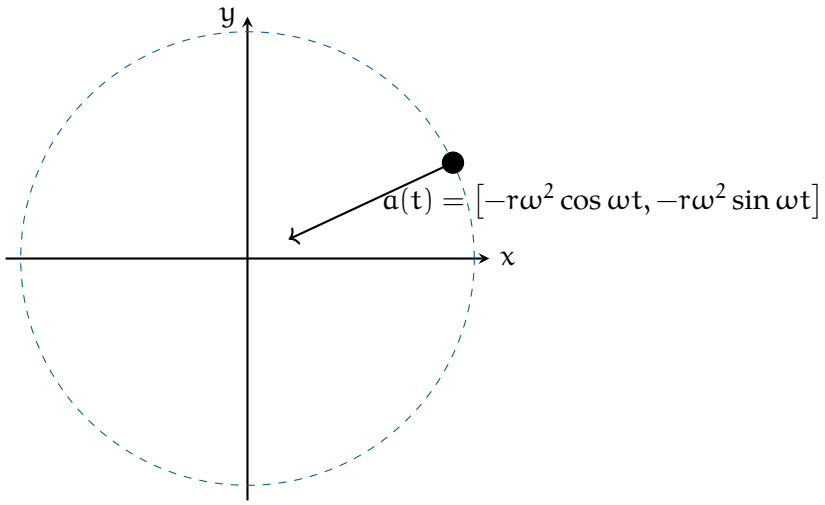
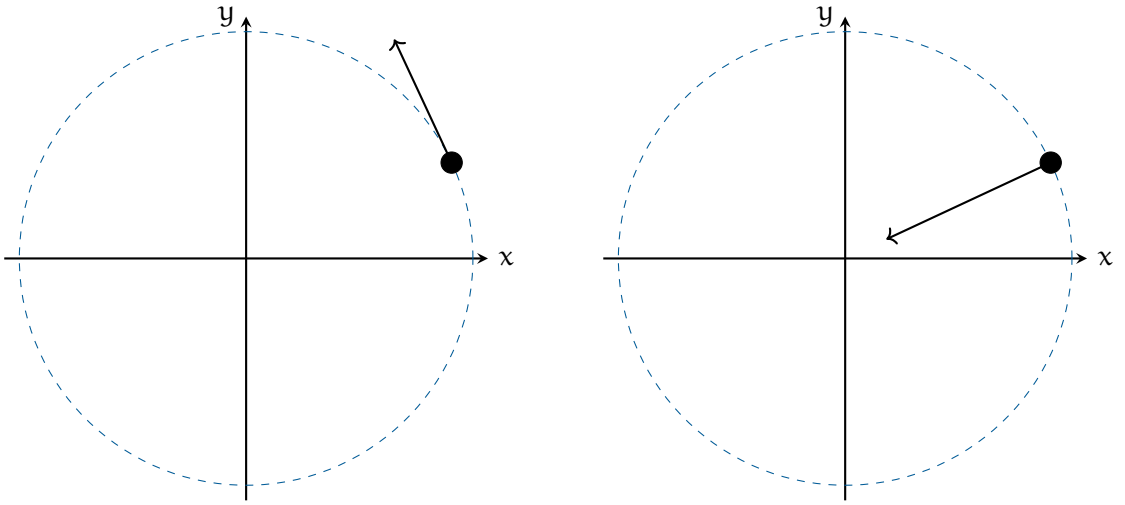



Figure 106.3: Acceleration vector of the particle in circular motion.

Let's compare the velocity and acceleration paths from figures 106.2 and 106.3. Notice that the velocity vector is always tangent to the circle, while the acceleration vector is always pointing toward the center of the circle.

[1] You will see centripetal acceleration noted as $a_c$ or $a_r$, standing for centripetal or radial, respectively. Note that both refer to the same acceleration.




(a) Velocity vector. (b) Acceleration vector.

Figure 106.4: Uniform circular motion: velocity (tangent) and acceleration (centripetal).

## 106.5 Centripetal force

What would happen if there were no force pulling the ball toward the center of the circle? It would fly off in a straight line, according to Newton's First Law.

The force you are exerting on the string is what causes it to accelerate toward the center of the circle. We call this the *centripetal force*. It makes sense that the centripetal force and centripetal acceleration are in the same direction, as they are proportional.

Recall that $F = ma$. The magnitude of the acceleration is $r\omega^2 = 3\left(\frac{2\pi}{4}\right)^2 \approx 7.4\text{m/s}^2$. The mass of the ball is 0.16 kg. So, the force pulling against your hand is about 1.18 newtons.

The general rule is that when something is traveling in a circle at a constant speed, the centripetal force needed to keep it traveling in a circle is:

$$F = mr\omega^2$$

If you know the radius $r$ and the speed $v$ of the object, here is the rule:

$$F = \frac{mv^2}{r}$$

We didn't introduce centripetal force as a type of force because it isn't its own force, like friction or gravity. Rather, calling a force "centripetal" tells us what the force is doing. A

centripetal force is any force that causes circular motion, a force which acts in the *radial* direction. For a satellite, the centripetal force is gravity. In the opening example with the billiard ball, the centripetal force is the tension in the string.

**Example**: A child is sitting on a merry-go-round as it spins. What provides the centripetal force?

**Solution**: In this case, the child is rotating horizontally, so the centripetal force must also be horizontal. Therefore, the centripetal force isn't the child's weight or the normal force. It is the *friction* between the child and the merry-go-round that keeps the child turning. Here is a free body diagram:

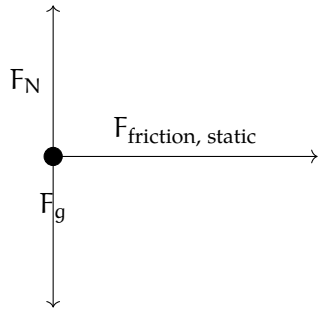



Figure 106.5: The static friction is what causes the centripetal force.

**Example**: If the child is 1.2 meters from the center of the merry-go-round and it spins at 0.25 rotations/second, what is the coefficient of friction between the child and the merry-go-round?

**Solution**: Using our FBD and applying Newton's Second Law, we see that:

$$F_N - F_g = 0 \rightarrow F_N = m_{child}g$$

$$F_f = mr\omega^2 = \mu F_N = \mu mg$$

Since the child isn't accelerating vertically, we know that the normal force equals the child's weight. The horizontal force, friction, must equal the mass of the child times the acceleration. Additionally, from the vertical component, we know the friction is equal to $\mu mg$, since the normal force is equal to the child's weight. Looking at the second equation, we see that:

$$mr\omega^2 = \mu mg$$

We were given $\omega$ in rotations per second, but we need radians per second:

$$\frac{0.25\text{rotation}}{1\text{second}} = \frac{1\text{rotation}}{4\text{seconds}} = \frac{2\pi\text{radians}}{4\text{seconds}} = \frac{\pi}{2}\frac{rad}{s}$$

We can divide $m$ from both sides of $mr\omega^2 = \mu mg$ (notice: we don't need to know the child's mass to determine the coefficient of friction!):

$$r\omega^2 = \mu g \rightarrow \mu = \frac{r\omega^2}{g} = \frac{(1.2m)\left(\frac{\pi}{2s}\right)^2}{9.8\frac{m}{s^2}} = 0.302$$

Therefore, the coefficient of friction between the child and the merry-go-round is 0.302.

### 106.5.1 Banked Turns

Have you ever been driving on the highway and taken a turn where the road is at an angle? Or maybe you've seen a sign like Figure 106.6 on the road as you take a turn.

Figure 106.6: A banked turn sign.

A banked curve is designed so that a car can safely turn without slipping, even in the rain, if the driver does not exceed the indicated speed. Engineers choose an angle such that the bank provides sufficient force to turn the car without friction. Let's look at a free body diagram for a car taking a banked turn (see figure 106.7).

In the past, we've split the vector for the force of gravity into components that are parallel and perpendicular to the ramp. This time, we will split the normal force into $x$ and $y$ components (see figure 106.8).

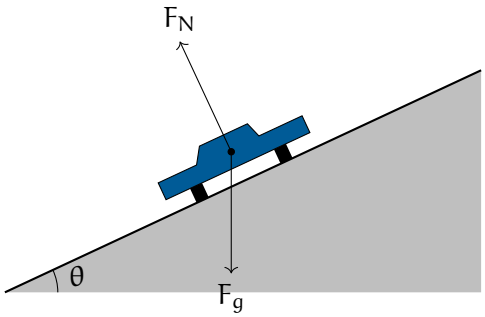



Figure 106.7: If there is no friction due to rain, then the only two forces acting on the car are gravity and the normal force.

Assuming the radius of the turn is 20 m, at what angle should the engineers build the banked turn? (Let's also assume the maximum speed is 25 mph, like the traffic sign above.) First, let's apply Newton's Second Law in the $x$ and $y$ directions:

$$(1)\ F_N \cos(\theta) - F_g = 0$$

$$(2)\ F_N \sin(\theta) = \frac{mv^2}{r}$$

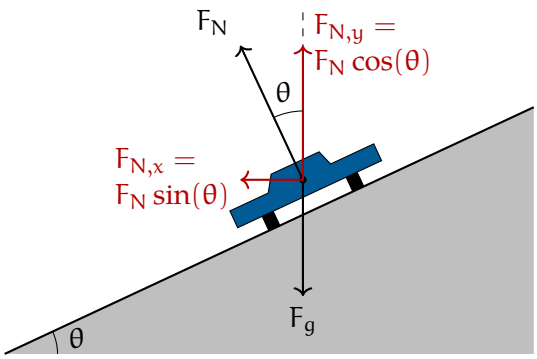



Figure 106.8: Geometrically, the x component of the normal force is given by $F_N \cos(\theta)$ and the y component by $F_N \sin(\theta)$.

We know that $F_g = mg$. From this and equation (1) we see that:

$$F_N = \frac{mg}{\cos(\theta)}$$

Substituting for $F_N$ into equation (2):

$$\left(\frac{mg}{\cos(\theta)}\right)\sin(\theta) = \frac{mv^2}{r}$$

The mass can be divided from both sides, and $\frac{\sin(\theta)}{\cos(\theta)} = \tan(\theta)$:

$$g\tan(\theta) = \frac{v^2}{r}$$

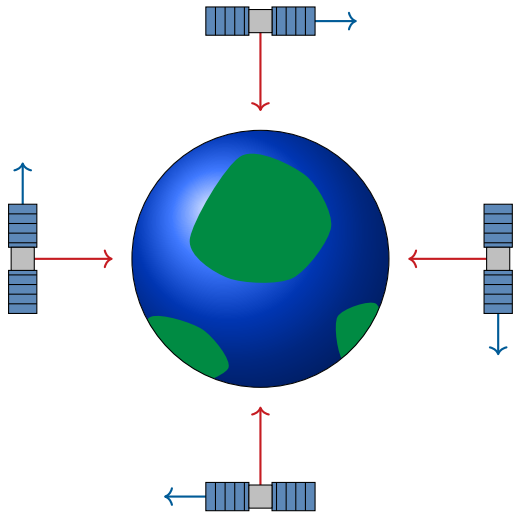
Solving for $\theta$ and substituting for the speed ($25\ \text{mph} \approx 11.18\frac{m}{s}$) and radius:

$$\theta = \arctan\left(\frac{v^2}{gr}\right) = \arctan\left(\frac{\left(11.18\frac{m}{s}\right)^2}{\left(9.8\frac{m}{s^2}\right)(20m)}\right)$$

$$\theta = \arctan(0.6377) \approx 32.53^\circ$$

## 106.6 Modeling Circular Motion

What causes circular motion is a constant force perpendicular to the motion. Consider a satellite circling the Earth. The only force acting on the satellite is gravity, yet the satellite does not fall to the Earth. Why? Let's look at the relative direction of motion and gravity for some different positions of the satellite (we'll assume this satellite is moving clockwise from our point of view):




No matter what position the satellite is in, the velocity and gravity vectors are perpendicular.

## *Exercise 260* Circular Motion

Just as your car rolls onto a circular track with a radius of 200 m, you realize your 0.4 kg cup of coffee is on the slippery dashboard of your car. While driving 120 km/hour, you hold the cup to keep it from sliding.

What is the maximum amount of force you would need to use? (The friction of the dashboard helps you, but the max is when the friction is zero.)

*Working Space*

*Answer on Page 1967*

## *Exercise 261* Twirling a Whistle

The lifeguard at a local pool is twirling their whistle horizontally. You wonder if the lifeguard could spin the whistle fast enough to break the string. The string the whistle is attached to can hold a maximum mass of 20 kg before breaking. If the lifeguard's whistle string is 0.35 m long and the average whistle has a mass of 165 grams, what is the maximum tangential speed the lifeguard can spin the whistle? How many rotations per second would the whistle be spinning at? Based on this, do you think the lifeguard is capable of spinning the whistle fast enough to break the string?

*Working Space*

*Answer on Page 1968*

## *Exercise 262* The Gravitron

The Gravitron is a carnival ride where riders "stick" to the wall of a spinning cylinder as the floor beneath them drops away. A video explanation is given here: https://www.youtube.com/watch?v=ifAY5tbYDmQ. Draw a free body diagram of a rider. If the coefficient of friction between a rider and the wall is 0.32 and the ride is 10 meters across, what angular velocity must the ride reach before the floor drops away?

*Working Space*

*Answer on Page 1969*

## 106.7 Flying Pigs and Tension

Take a look at the following two videos that involve flying pigs!

1. Flying Pig: Uniform Circular Motion and Centripetal Force
2. flying pig - circular motion [workthrough demo]

This involves a combination of circular motion and tension. Let's say we have a pig at making angle $\theta$ with the vertical and the string of length $L$, causing the pig to move in a circular path with radius $r$. We can draw the following two diagrams for our pig, as shown in Figure 106.9.

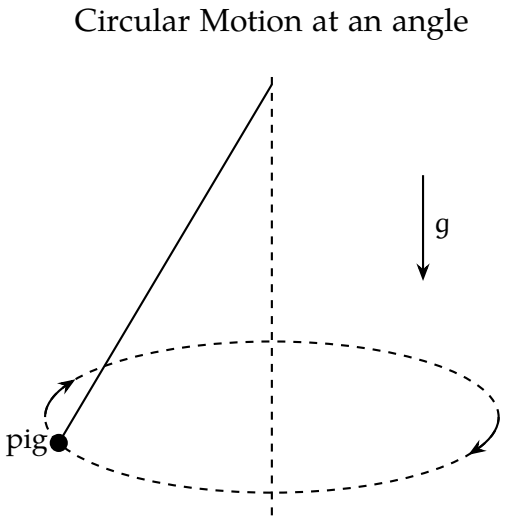



Figure 106.9: The uniform motion of a flying pig attached to a string.

If this is confusing, think of a tetherball from your local playground. It follows the same mechanics!

At any point, we can draw a free body diagram for the pig. Figure 106.10 shows the forces acting on the pig at all points.

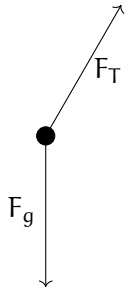



Figure 106.10: The FBD of the flying pig at any point.

To analyze the net forces, the tension force needs to be split into its $x$ and $y$ components. The $y$ component of the tension force must balance out the weight of the pig (as it is not moving vertically), while the $x$ component of the tension force provides the centripetal force to keep the pig moving in a circle.

$$\sum F_y = 0 \rightarrow F_{T_y} - F_g = 0 \rightarrow F_{T_y} = F_g = F_T \cos\theta = mg$$

The centripetal force is found by:

$$F_{T_x} = F_c = F_T \sin\theta = ma_r = \frac{mv^2}{r}$$

Dividing the second equation by the first equation gives:

$$\frac{F_T \sin(\theta)}{F_T \cos(\theta)} = \frac{\frac{mv^2}{r}}{mg} \rightarrow \tan(\theta) = \frac{v^2}{rg}$$

Solving for $\theta$ gives us $\theta = \arctan\left(\frac{v^2}{rg}\right)$. We can also solve for $v$:

$$v = \sqrt{rg\tan\theta} = \sqrt{Lg\sin\theta\tan\theta}$$

Since $r$ can be solved for $\theta$ using $r = L\sin\theta$.

Explicitly, we can express our tension force as $F_T = \frac{mg}{\cos(\theta)}$ (coming from our net force in the $y$-direction).

We can also explicitly solve for $\omega$:

$$\begin{aligned}\tan(\theta) &= \frac{v^2}{rg} \\ &= \frac{(r\omega)^2}{rg} = \frac{r\omega^2}{g} \\ \omega^2 &= \frac{g\tan(\theta)}{r} \\ \omega &= \sqrt{\frac{g\tan(\theta)}{r}}\end{aligned}$$

What does all of this tell us?

- The vertical component of tension balances the weight of the pig, so there is no vertical motion.
- The horizontal component of tension provides the centripetal force that keeps the pig moving in a circle.
- The faster the pig moves, the larger the angle $\theta$ and the greater the tension in the string.
- The tension is always greater than the pig's weight: $F_T = \frac{mg}{\cos\theta}$.
- The angular speed required for a given angle is $\omega = \sqrt{\frac{g\tan\theta}{r}}$. This increases with larger angles, and decreases with a shorter radius.

## *Exercise 263* Playground Tetherball

*Working Space*

A tetherball of mass .5kg is attached to a string of length 2.5m. When the ball is swung around in a horizontal circle, the string makes an angle of 38° with the vertical. If the ball is in uniform circular motion, what is the speed of the ball?

*Answer on Page 1970*

## 106.8 Non-uniform circular motion

We have talked enough about uniform circular motion, where the speed is constant. However, in many real-world scenarios, the speed is not constant. This is known as *non-uniform circular motion*. In this case, there are two components of acceleration: the centripetal acceleration $a_\perp$ (toward the center of the circle) and the tangential acceleration $a_\parallel$ (along the direction of motion). The tangential acceleration is responsible for changes in speed, while the centripetal acceleration is responsible for changes in direction. We will expand on this in the oscillations chapter, but this acceleration is caused by a *restoring force*, a force that tries to pull the object back toward equilibrium whenever it's displaced. We will introduce this concept here, but go into more detail in the oscillations chapter.

Let's look at a few examples of these concepts in action.

### 106.8.1 Roller Coaster Loop

Now we can talk about a roller coaster. A roller coaster loop is a great example of circular motion. As the roller coaster cart goes through the loop, it experiences both centripetal and tangential acceleration.

We will analyze the cart on a roller coaster at positions at the top, bottom, and sides of the loop, as shown in Figure 106.11

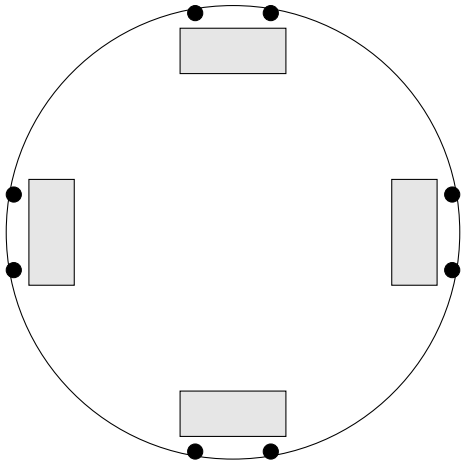


Figure 106.11: Carts at four positions around the loop with wheels on the track.

#### At the bottom

At the very bottom of the loop, the forces acting on the cart are gravity and the normal force from the track. Let's imagine the cart goes around the circular track infinitely without

"exiting" the loop. There must, then, be a force keeping the cart moving in a circle. This force is the centripetal force, which points toward the center of the loop. The only two forces acting on the cart are gravity (downward) and the normal force from the track (upward). We know that a centripetal force must satisfy the formula $F_{net} = F_N - F_g = \frac{mv^2}{r}$. Therefore, the normal force must be greater than the weight of the cart at the bottom of the loop: $F_N = \frac{mv^2}{r} + mg$. This is represented in Figure 106.12.

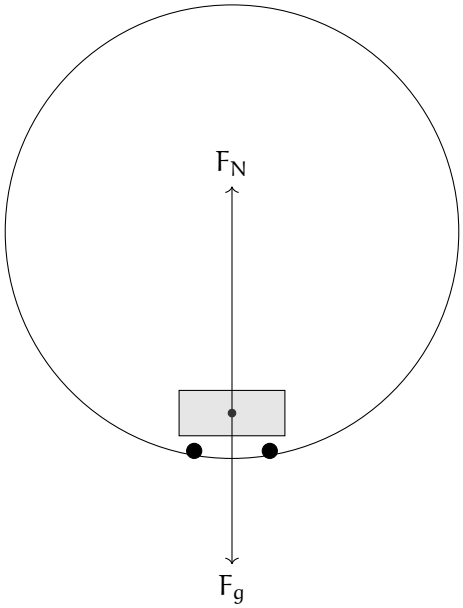



Figure 106.12: The FBD of the roller coaster at the bottom.

**At the top**

At the very top of the loop, the forces acting on the cart are gravity and the normal force from the track. Both forces point *downward* this time, toward the center of the loop. We can say that $F_{net} = F_N + F_g = \frac{mv^2}{r}$. Solving for the normal force gives: $F_N = \frac{mv^2}{r} - mg$.

Notice that the normal force is less than the weight of the cart at the top of the loop. This means that the centripetal acceleration would have to be greater than or equal $mg$. This is represented in Figure 106.13.

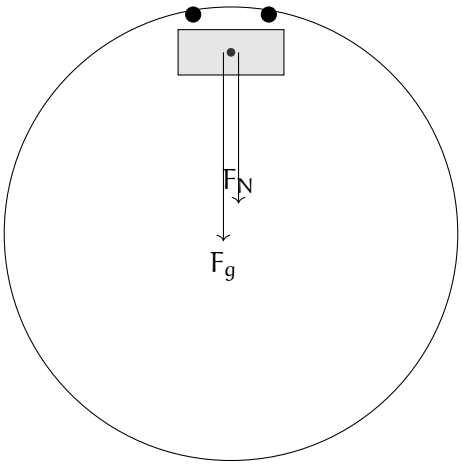



Figure 106.13: The FBD of the roller coaster at the top.

Note that for both the top and bottom of the loop, the minimum speed required to maintain contact with the track is given by $v = \sqrt{rg}$. If the cart goes any slower such that $v < \sqrt{rg}$, it will lose contact with the track, as there will not be enough normal force to provide the centripetal force needed to keep the cart moving in a circle. Think about where this observation comes from!

**At the sides**

Now at the sides of the loop (say $0$ and $\pi$ if we view the roller coaster as a unit circle), the net force is directed at an angle. The normal force from the track acts horizontally, pointing toward the center of the circle, while the gravitational force acts vertically downward. Since these two forces are perpendicular to each other, the net force is the vector sum of the normal and gravitational forces.

At the right side ($\theta = 0$), the normal force points leftward (toward the center), and gravity points downward, so the net force points diagonally down and to the left, toward the lower-left quadrant of the circle.

At the left side ($\theta = \pi$), the normal force points rightward (toward the center), and gravity still points downward, so the net force points diagonally down and to the right, toward the lower-left quadrant of the circle.

While the normal force provides the centripetal component of the net force, gravity introduces a tangential component, causing the cart to accelerate (speed up) along the track. Since the centripetal force is not directly towards the center of the circle, the cart must be speeding up. This is modeled in Figure 106.14.

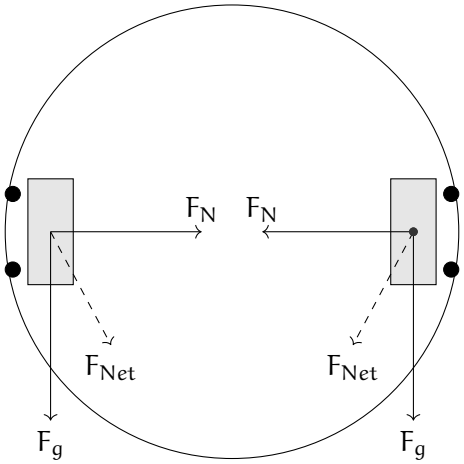



Figure 106.14: The FBD of the roller coaster at the sides.

## *Exercise 264* Roller Coaster Loop

This question is taken from the AP Physics C Review Book, 19th edition.

A roller-coaster car enters the circular loop portion of the ride. At the very top of the circle (where people are upside down), the car has a speed of 25 m/s, and the acceleration points straight down. If the diameter of the loop is 50 m and the total mass of the car plus passengers is 1,200 kg, find the magnitude of the normal force exerted by the track on the car at this point. Also find the normal force exerted by the track on the car when it is at the very bottom of the loop, given it is still traveling at 25 m/s. You may use $g = 10\ m/s^2$ for simpler calculations.

*Working Space*

*Answer on Page 1970*

Our next chapter will explore oscillations, restoring forces, simple harmonic motion, and more concepts of harmonic motion.

Chapter 107

# Oscillations

We just finished up a chapter about circular motion, where objects move around in a circle at either constant (uniform/harmonic) or changing (non-uniform) speeds.

In this chapter, we will study oscillatory motion, which is a type of motion where an object moves back and forth around an equilibrium point. Oscillations are common in many physical systems, such as springs, pendulums, and even electrical circuits. Oscillations are important not only because they appear everywhere in physical systems, but also because they provide a connection between simple motion and more complex behaviors. Small oscillations often behave in a very predictable way and can be described using *simple harmonic motion* (SHM), which is governed by restoring forces proportional to displacement.

We will also give an introduction to *differential equations* used to model oscillatory systems, often found in engineering and physics.

## 107.1 Sine and Cosine Graphs [Review]

Let's recall the two pillars of trigonometry and, consequently, simple harmonic motion: Sine and Cosine. Recall that these two functions are periodic, meaning they repeat their values in regular intervals or periods. The sine and cosine functions have a period of $2\pi$ radians or $360^\circ$, repeating their values every $2\pi$ radians.

As discussed in the circular motion chapter, the sine function represents the y-coordinate of a point on the unit circle as it moves around the circle, while the cosine function represents the x-coordinate. This is the fundamental behavior of the unit circle, where a point moves around the circle at a constant angular speed $\omega$. The position of the point at any time $t$ can be described using the following parametric equations:

$$x(t) = \cos(\omega t), \qquad y(t) = \sin(\omega t)$$

The general forms of the sine and cosine functions can be expressed as:

$$x(t) = A\cos(\omega t + \phi), \qquad y(t) = A\sin(\omega t + \phi)$$

where:

- $A$ is the amplitude, representing the maximum displacement from the equilibrium position.
- $\omega$ is the angular speed, which determines how quickly the oscillations occur.
- $\phi$ is the phase shift, which determines the initial position of the oscillation at $t = 0$.

Recall that the derivatives of these functions are also periodic, a key property that is useful for analyzing oscillatory motion:

$$\frac{d}{dt}\cos(\omega t) = -\omega \sin(\omega t)$$

$$\frac{d}{dt}\sin(\omega t) = \omega \cos(\omega t)$$

$$\frac{d^2}{dt^2}\cos(\omega t) = -\omega^2 \cos(\omega t)$$

In the motion we will talk about,

- $x(t)$ will be our position function
- $v(t)$ describes our velocity function, which can be the first derivative of our position
- $a(t)$ is our acceleration function, derived from $v(t)$

We notice that the second derivative produces a negative multiple of the original function. This can be simplified in terms of acceleration and position functions: $a(t) = -\omega^2 x(t)$.

## 107.2 Simple Harmonic Motion

Simple Harmonic Motion will reference many definitions we have seen before and will continue to see in this chapter. Here are some key terms to know:

**Cycle** A single complete execution of a periodically repeated motion. Another way to think about it is a particle, block, or object making a *round-trip*.

**Period** The time it takes to complete one cycle of motion. An identity that comes up often is $T = \frac{2\pi}{\omega}$.

**Frequency** The number of cycles completed per unit time. It is the reciprocal of the period, represented as **Hz**.[1]

[1] One **Hz** is equal to one cycle per second. For example, a frequency of 5 Hz means that 5 cycles are completed in one second. On a monitor, a refresh rate of 60 Hz means the screen updates 60 times per second. Higher refresh rates can lead to smoother motion perception.

**Equilibrium** The position at which the "default" position of an oscillating object is located. In terms of mechanics, this is the position where all the forces acting on the object are balanced or zero.

**Restoring Force** The force that acts to bring an oscillating object back to its equilibrium position. It is typically proportional to the displacement from equilibrium and acts in the opposite direction.

**Angular Speed** The rate at which an object moves around a circle, measured in radians per second.

So far, we have discussed oscillatory motion in general terms. Now, let's focus on a specific type of oscillatory motion known as Simple Harmonic Motion (SHM). SHM is characterized by the following properties:

- The motion occurs around an equilibrium position.
- The *restoring force* is directly proportional to the displacement from the equilibrium position and acts in the opposite direction.
- The motion is sinusoidal in nature, meaning it can be described using sine or cosine functions.
- The period and frequency of the motion are constant, regardless of the amplitude.
- The acceleration always points toward equilibrium and increases in size as the displacement grows.

So the motion continuously repeats itself in a regular pattern. It speeds up near equilibrium, and slows down to a stop at maximum displacement. Examples of these systems are

- A mass attached to a spring, either vertically or horizontally
- A simple pendulum
- Vibrations of a tuning fork or guitar string

We saw in our pendulums chapter how the system has both centripetal acceleration and acceleration due to gravity, causing oscillations back and forth with changing speed. Like all mechanical systems, an SHM system has both kinetic and potential energy. For general simple harmonic motion systems, we can describe its energy as follows:

- The total mechanical energy $E$ of the system is the sum of its kinetic energy $K$ and potential energy $U$:
$$E = K + U$$
- The kinetic energy $K$, system dependent, is highest when the object passes through

the equilibrium position.

- The potential energy $U$ depends on the specific system, such as spring potential energy or gravitational potential energy, but is highest at the maximum displacement from equilibrium.

## 107.3 Springs

Let's talk about springs! Springs are mechanical devices, usually made out of some rigid metal material, that store and release energy through deformation. When a spring is stretched or compressed from its equilibrium position, it exerts a restoring force that tries to bring it back to its original shape. The most common type of spring is a coil spring, which consists of a linear wire would into a helical shape.

### 107.3.1 Hooke's Law

In 1676, physicist Robert Hooke formulated a principle that describes the behavior of springs, now known as Hooke's Law. Hooke's Law states that the force exerted by an ideal spring is directly proportional to the displacement from its equilibrium position.

Mathematically, Hooke's Law can be expressed as:

$$F_s = -kx$$

where:

- $F_s$ is the restoring force exerted by the spring (in newtons, N)
- $k$ is the spring constant (in newtons per meter, N/m), which measures the stiffness of the spring. It is spring-specific
- $x$ is the displacement from the equilibrium position (in meters, m)
- The negative sign represents the fact that the force attempts to bring the spring back to its equilibrium position.

Since we know that force is related to mass and acceleration through Newton's Second Law, we can combine Hooke's Law with Newton's Second Law to analyze the motion of a mass attached to a spring. See Figure 107.1 for a diagram of a mass-spring system.

$$F_s = ma = -kx \tag{107.1}$$

$$ma = -kx \tag{107.2}$$

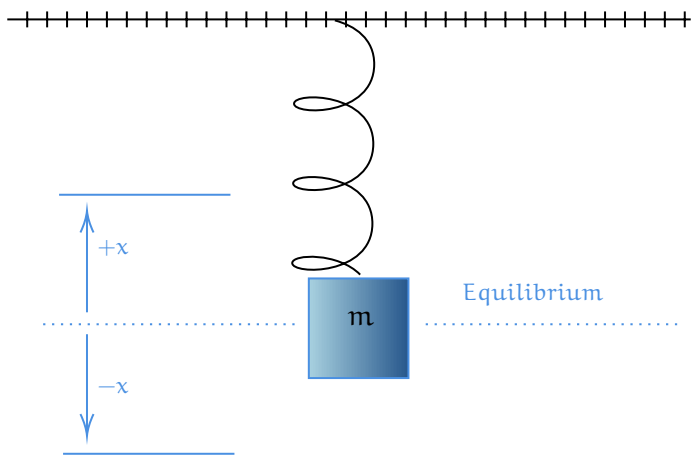



Figure 107.1: A diagram showing a block of mass $m$ displaced $+x$ and $-x$ from equilibrium.

But since above we found a relation between acceleration and position, we can write:

$$m\frac{d^2x}{dt^2} = -kx \tag{107.3}$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \tag{107.4}$$

We can see that this equation describes simple harmonic motion, where the acceleration is proportional to the negative of the displacement. In this form, we can identify the angular speed $\omega$ of the oscillation as:

$$\omega = \sqrt{\frac{k}{m}}$$

as the term in front of the $x$ in equation (107.4) is $\omega^2$. Thus, we can further define the period as

$$T = 2\pi\sqrt{\frac{m}{k}}$$

and the frequency as

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

In a later section of this chapter, we will solve this differential equation to find the position function $x(t)$ of general mass-spring systems.

## *Exercise 265* Mass-Spring Frequency Change

A block is attached to a spring and set into oscillatory motion, and its frequency is measured. If this block were removed and replace with a block of $1/2$ the mass, how would the frequency of oscillations compare to the original frequency?

1. The frequency would be halved.
2. The frequency would remain the same.
3. The frequency would increase by a factor of $\sqrt{2}$.
4. The frequency would decrease by a factor of $\sqrt{2}$.

*Working Space*

*Answer on Page 1971*

### 107.3.2 Spring Potential Energy

Let's recall that Work from a physics standpoint is defined as the transfer of energy through a certain distance via a force. When we alter the length of a spring through either compression or extension, we are performing work on the spring. This work is stored as potential energy within the spring, which can be released when the spring returns to its equilibrium position.

Let's find an equation for this change in potential energy, which gets stored up in the spring. Recall the two mathematical definitions for work:

- Work as the product of force and distance: $W = F \cdot d$
- Work as the integral of force over a distance: $W = \oint F \, dx$

We need to use the integral definition, since, by our definition of Hooke's Law, the force exerted by a spring varies with displacement, so every tiny bit of distance, or $dx$ is different. For any spring starting at equilibrium $(0)$, and moving to some final position $x_f$, we

can write:

$$\begin{aligned} W &= \int_{x_i}^{x_f} F_x \, dx \\ W_s &= \int_{x_i}^{x_f} F_s \, dx \\ &= \int_0^{x_f} -kx \, dx \\ &= \frac{-kx^2}{2}\Big|_0^{x_f} \\ &= \frac{-k(x_f)^2}{2} \end{aligned}$$

This work done on the spring is stored as potential energy, so we can say that the change in potential energy of the spring is equal to the negative of the work done by the spring force:

$$\Delta U_s = -W_s = \frac{1}{2}kx_f^2 \tag{107.5}$$

We can see this graphically in Figure 107.2, where the area under the force vs. displacement graph represents the work done on the spring, which is equal to the change in potential energy.

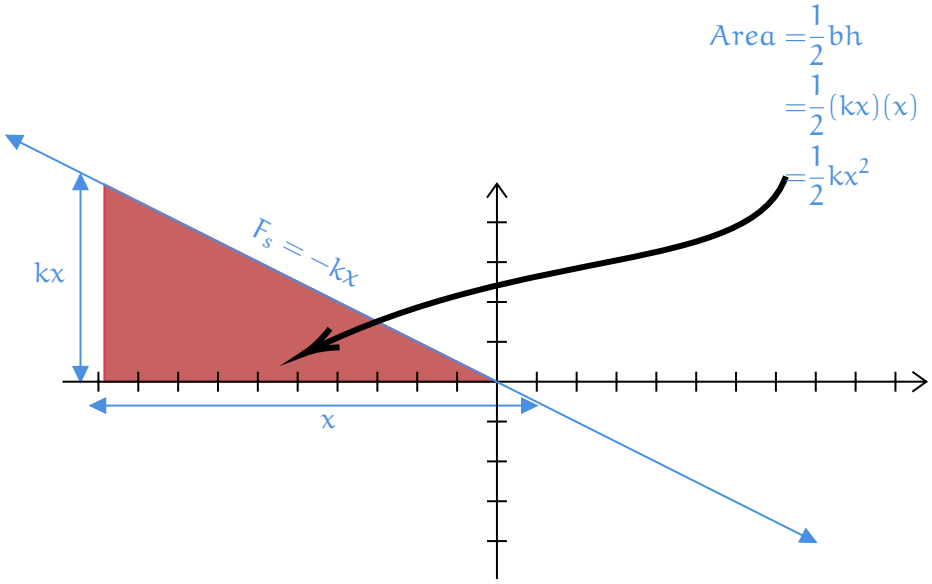



Figure 107.2: A graph of force versus displacement for a spring, equal to the work done and change in potential energy.

Notice that Equation (107.5) is similar in structure to the kinetic energy equation, $K = \frac{1}{2}mv^2$. This is not a coincidence, as both equations describe forms of energy in mechanical systems, and are derived from linear systems. When the effort (force) needed increases in

direct proportion to what you're trying to change, the energy required ends up depending on the square of that change. For a spring

- the farther you stretch or compress it, the more force it exerts to return to equilibrium (Hooke's Law)
- the more energy is stored in the spring (spring potential energy)

For kinetic energy

- the faster you try to move an object, the more force is needed to accelerate it (Newton's Second Law)
- the more energy the object has due to its motion (kinetic energy)

## *Exercise 266* Finding the equation for a mass-spring system

*Working Space*

This question is from an AP-style physics review. A block of mass 4 kg on a frictionless, horizontal table is attached to a spring with spring constant $k = 400\,\text{N/m}$ and undergoes simple harmonic oscillations about its equilibrium position and its amplitude $A = 6$ cm. If the block is at $x = 6$ cm at time $t = 0$ and has a velocity of 0 cm/s at that time, which of the following equations (with $x$ in cm and $t$ in seconds) describes the block's position as a function of time?

(A) $x(t) = 6\cos(10t)$

(B) $x(t) = 6\sin(10t)$

(C) $x(t) = 6\sin(10t + \frac{1}{2}\pi)$

(D) $x(t) = 6\sin(10\pi t + \frac{1}{2}\pi)$

(E) $x(t) = 6\sin(10\pi t - \frac{1}{2}\pi)$

*Answer on Page 1972*

### *Exercise 267* Energy Transfer in a Mass-Spring System

A pinball machine uses a spring to launch a ball of mass 0.2 kg. The spring has a spring constant of 500 N/m and is compressed by 0.24 m before release. Assuming energy is conserved and no frictional losses occur, what is the speed of the ball as it leaves the spring?

*Working Space*

*Answer on Page 1972*

## 107.4 Mass-Spring Systems and Linear Differential Equations

We have already seen that the motion of a mass attached to a spring can be model e d using different ial equations. Here, we will examine this connection more clo sel y.

### 107.4.1 Undamped Simple Harmonic Motion

Starting from Newton's Second Law and Hooke's Law, we derived the differential equation for simple harmonic motion:

$$m\frac{d^2x}{dt^2} + kx = 0. \tag{107.6}$$

The term$m\,\frac{d^2x}{dt^2}$comes directly from *Newton's Second Law*, representing the net force required to accelerate the mass. The term $kx$ arises from *Hooke's Law*, which states that the spring exerts a restoring force proportional to displacement and directed toward equilibrium. Equation (107.6) is an example of a **second-order linear differential equation with constant coefficients without damping** . It describes an ideal mass-spring system with no external influences: one that oscillates forever, undamped and unforced. The $ma(t)$ term is *Newton's Second Law* part of the equation, while the $kx(t)$ term comes from *Hooke's Law*. This is a second-order linear differential equation with constant coefficients. What if the spring system is be ing driven by a third force, like friction or damping? We can add an additional force term$F(t)$to the equation:$F_f(t) = -cv = -cx'$ for forcing function.

### 107.4.2 Introducing Damping

Real systems are often influenced by additional forces. One common example is a **damping force**, such as friction or air resistance, which opposes the motion of the mass. A typical damping force is proportional to the velocity and can be written as

$$F_d(t) = -c\,\frac{dx}{dt},$$

wherecis the damping coefficient. Including this force in Newton's Second Law modifies our differential equation. Instead of the undamped equation (107.6), we obtain:

$$ma(t) = -c\,v(t) - k\,x(t) \tag{107.7}$$

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0 \tag{107.8}$$

This is still a second-order linear differential equation with constant coeffici e nts, but its solutions behave very differently. Depending on the value of c, th e system may oscillate with decreasing amplitude, fail to oscillate at all, or re t urn to equilibrium as quickly as possible. Finding the roots of the characteristic equation associated with Equation (107.8) allows us to classify the system's behavior into three categories. Recall that the quadratic formula gives us the roots using our coefficients:

$$r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

| Characteristic Roots | Homogeneous Solution | Damping Type |
|---|---|---|
| $r_1, r_2 \in \mathbb{R}$, both negative | $y_h = C_1 e^{r_1 t} + C_2 e^{r_2 t}$ | Overdamped |
| $r$ real, repeated | $y_h = (C_1 + C_2 t)e^{rt}$ | Critically damped |
| $r = \alpha \pm i\beta$, $\alpha < 0$ | $y_h = e^{\alpha t}(C_1 \cos(\beta t) + C_2 \sin(\beta t))$ | Underdamped |
| $r = \pm i\beta$ | $y_h = C_1 \cos(\beta t) + C_2 \sin(\beta t)$ | Simple Harmonic Motion |

Table 107.1: Classification of solutions to the damped harmonic oscillator based on characteristic roots.

Without going into too much linear algebra, the function of a mass spring is giv e n by$y(t) = y_h(t) + y_p(t)$, where $y_h$ is the homogeneous solution (the p art we solved above by setting the differential equation to 0), and $y_p$ is the p articular solution, which depends on any external forcing functions. In this section, we focused on the homogeneous solution. If the equation (107.8) had a forcing function$F(t)$on the right side, we would need to find a particular solution$y_p$ to account for that.

## *Exercise 268* Linear Systems 1

*Working Space*

A 2 kg mass is attached to a spring with $k = 200\,\text{N/m}$ with no damping. Which differential equation describes the motion?

1. $x'' + 100x = 0$
2. $x'' + 10x = 0$
3. $x'' + 200x = 0$
4. $2x'' + 100x = 0$

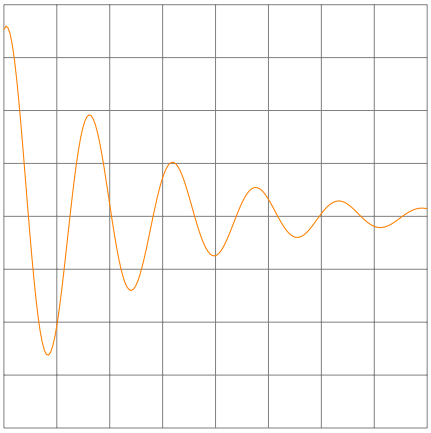
*Answer on Page 1973*

## *Exercise 269* Linear Systems 2

*Work*

Answer the following questions about the graph below, which shows the position of a mass on a spring as a function of time. The equation of motion for this system is given by:

$$x(t) = 0.15e^{-0.40t}\cos(4t + 0.30)$$



- Is this system undergoing simple harmonic motion, or is it damped? If it is damped, is it underdamped, overdamped, or critically damped?
- What would need to be changed in the equation of motion to make this system undergo simple harmonic motion instead?

*Answer on Page 1973*

### *Exercise 270* Linear Systems 3

*Working Space*

Given that a unit (1 kg) mass obeys:

$x'' + 16x = 0, \quad x(0) = 0.20, \quad v(0) = x'(0) = 0$

1. Solve for $x(t)$
2. Find the acceleration, $a(t)$, and the velocity, $v(t)$, as functions of time.

*Answer on Page 1973*

## 107.5 Pendulums

You may be surprised to know that pendulums act the exact same way as roller coasters from the circular chapter, in terms of forces. But, similar to springs, they oscillate back and forth unless acted upon by a damping force. A pendulum has an equilibrium position$\theta = 0$such that it is parallel to its gravitational component. A pendulum also has a maximum angle$\theta_{max}$from its equilibrium position. It will swing through a set arc in a repetitive motion, which is a type of motion we call simple harmonic motion, similar to springs (which we will also cover in the oscillations chapter). The tension force is always directed towards the pivot. This is diagramed below in Figure 107.3

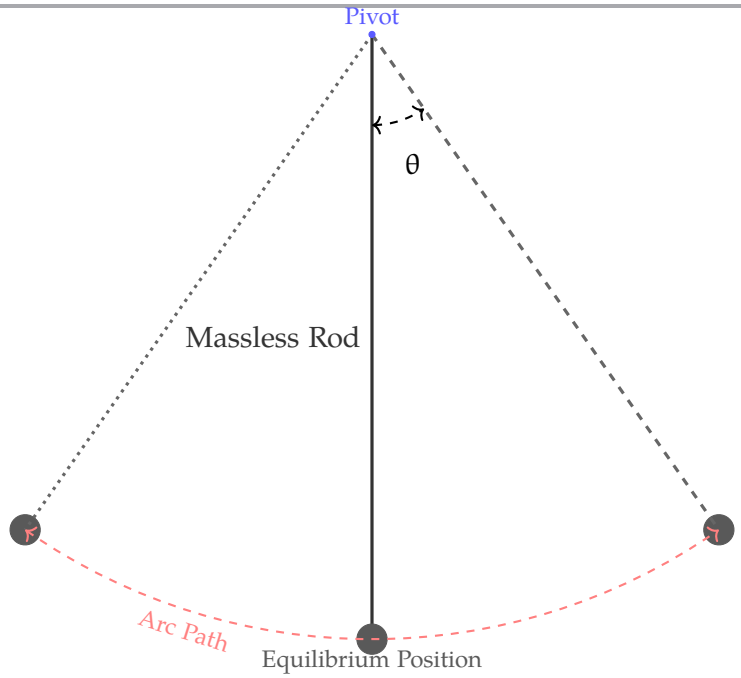



Figure 107.3: The anatomy of a pendulum.

When displaced by an angle$\theta$from the vertical equilibrium position ($\theta = 0$), gravity provides a *restoring force*that acts along the arc of motion. This restoring force is given by$F = -mg\sin\theta$, where the negative sign indicates that the force always points back toward the equilibrium position. The tension in the string keeps the bob moving along its curved path, while gravity continually tries to return it to rest. The net force shifts such that the acceleration arrow is *not*always pointing to the center. The net force is a combination of the tension (centripetal) force and the gravitational (tangential), so rather it points at different points along the arc of the circle, as it is always changing due to the height of the pendulum. This is demonstrated in Figure 107.4. Note the following differences in energy and forces at the equilibrium position versus the maximum angle positions, summarized in Table 107.2. Note here that the magnitude of the restoring

| **At** $\theta = 0$ | **At** $\theta = \theta_{max}$ |
|---|---|
| $F_{restoring} = 0$ | $F_{restoring} = max$ |
| $a_t$ (tangential acceleration) $= 0$ | $a_t = max$ |
| $PE = 0$ | $PE = mgh$ |
| $KE = max$ | $KE = 0$ |
| $V = max$ | $v = 0$ |

Table 107.2: Comparison of pendulum quantities at equilibrium and maximum displacement

force$-mg\sin(\theta)$, which is not *directly* proportional to the angular displacement$\theta$. However, for small angles (typically less than about 1 radian or 180°), we can use the small-angle approximation$\sin(\theta) \approx \theta$(in r adians), and state the restoring force as$F \approx -mg\theta$. This allows us to treat the pendulum's motion as simple harmonic motion for small oscil-

lations, where the restoring force is approximately proportional to the angular displacement.

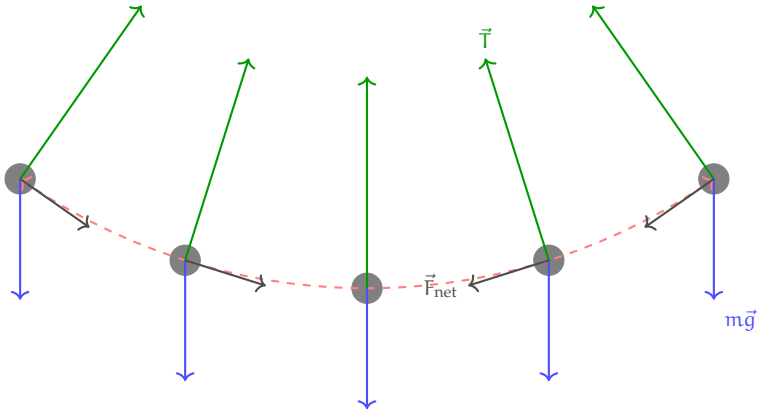



Figure 107.4: The net force at different locations of the arc.

## *Exercise 271* Pendulum Period Dependence

A simple pendulum consists of a mass $m$ attached to a massless rod $L$, swinging under the influence of gravity. Derive the equation for the period of a simple pendulum. Which of the following changes would affect the period $T$ of the pendulum's oscillation?

1. Increasing the mass $m$ of the pendulum bob.
2. Increasing the length $L$ of the string.
3. Increasing the amplitude (maximum angle) of the swing.
4. Decreasing the acceleration due to gravity $g$.

*Working Space*

*Answer on Page 1974*

This exercise showed you that the period of a simple pendulum depends on the length of the pendulum and the acceleration due to gravity, but not on the mass of the bob or the amplitude of the swing (for small angles). One derivation from the exercise above was the Equation **??**, which can be rearranged as $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$. Solving this differential equation[2] will yield the angular position function $\theta(t) = \theta_{max}\sin(\omega t + \phi_0)$ or equivalently, $\theta(t) = \theta_{max}\sin(\sqrt{\frac{g}{L}}t + \phi_0)$ of the pendulum over time, which describes its oscillatory motion. Note that this differential equation does not the depend on mass of the bob nor the amplitude of the swing $\theta_{max}$, a core factor of pendulum harmonic motion.

### *Exercise 272* Pendulum on the Moon

A simple pendulum has a period of 2 seconds when measured on Earth. If the same pendulum were taken to the Moon, where the acceleration due to gravity is approximately $\frac{1}{6}$, what would be its new period?

*Working Space*

a Approximately 0.8 seconds

b Approximately 2 seconds

c Approximately 4.9 seconds

d Approximately 6.8 seconds

*Answer on Page 1975*

## 107.6 Alternating Current

We have talked about Alternating Current (AC) in previous chapters, but now we will explore its connection to oscillations. AC is an electric current that periodically reverses direction, in contrast to Direct Current (DC), which flows in a single direction and provides constant voltage. The voltage and current in an AC circuit vary sinusoidally with time, making them ideal for modeling using oscillatory functions.

In Direct Current, $V = V_0$, which means the original voltage is equal to the voltage everywhere else.

[2] We will go more in depth about this in a future chapter.

In an AC circuit, the voltage at any time t behaves exactly like a mass on a spring, following a sinusoidal oscillation:

$$V(t) = V_{peak} \sin(\omega t + \phi) \tag{107.16}$$

where

- $V_{peak}$ is the maximum voltage reached in either direction
- $\omega$ is the angular frequency, related to standard frequency by $\omega = \frac{2\pi}{T} = 2\pi f$
- $\phi$ which is the phase constant, or the starting position of the wave

Note that, because of the trigonometric nature of sine, the wave fluctuates between $V_{peak}$ and $-V_{peak}$.

For sinusoidal voltages, the root mean squared (RMS) is derived as

$$\begin{aligned} V_{rms} &= \sqrt{\frac{1}{T}\int_0^T [V_{peak}\sin(\omega t + \phi)]^2 dt} \\ &= V_{peak}\sqrt{\frac{1}{2T}\int_0^T [1 - \cos(2\omega t + 2\phi)]dt} \\ &= V_{peak}\sqrt{\frac{1}{2T}\int_0^T dt} \\ &= \frac{V_{peak}}{\sqrt{2}} \end{aligned}$$

### 107.6.1 Impedance

In DC, we use Resistance (R). In AC oscillations, we use *Impedance*. Impedance is the "total opposition" to the oscillation and is a combination of standard resistance and Reactance (X).

Capacitors oppose changes in voltage, which we call Capacitive Reactance:

$$X_C = \frac{1}{\omega C}$$

Inductive Reactance comes from inductors which oppose changes in current (acting like "electrical inertia").

$$X_L = \omega L$$

Similarly, in AC circuits, the voltage and current waves do not always peak at the same time.

- In a purely inductive circuit, the current falls behind the voltage by $90^\circ$ $(\pi/2)$.
- In a purely capacitive circuit, the current leads the voltage by $90^\circ$ $(\pi/2)$.

### 107.6.2 Comparing Mass-Springs and AC

| Mechanical System (Mass-Spring) | Electrical System (AC Circuit) |
|---|---|
| Displacement (x) | Charge (q) |
| Velocity (v) | Current (i) |
| Mass (m) | Inductance (L) |
| Spring Constant (k) | Reciprocal Capacitance (1/C) |
| Damping/Friction (b) | Resistance (R) |

## 107.7 Summary

In this chapter, we have discussed oscillations via Simple Harmonic Motion, mass-springs, spring energy, pendulums, and AC currents.

- **Simple Harmonic Motion (SHM)**: A specific type of periodic motion where a restoring force (like a spring) constantly pulls an object back toward an equilibrium point. Key Parameters include:

  Oscillation above and below (or to and from) an equilibrium point.

  A restoring force, proportional to displacement.

  An angular velocity that is always changing, $\omega$

  Time for one full cycle $(T = 2\pi/\omega)$.

- **Hooke's Law**: For an ideal spring, $F_s = -kx$. This linear relationship leads to an angular frequency of $\omega = \sqrt{k/m}$.

- **Differential Equations**: Oscillations can be represented in the form $m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0$, where $m$ is the mass, $c$ is the damping constant, and $k$ is the spring constant.

  Underdamped: Oscillates with decaying amplitude.

  Critically Damped: Returns to equilibrium as quickly as possible without oscillating.

  Overdamped: Returns to equilibrium slowly without oscillating.

- **Energy Dynamics**: In a closed system, energy oscillates between Kinetic $(K)$ and Potential $(U)$. Total mechanical energy remains constant $(E = \frac{1}{2}kA^2)$, with $K$ peaking at equilibrium and $U$ peaking at maximum displacement.

- **Acceleration**: Acceleration is affected by both *centripetal acceleration* and *tangential acceleration*.

This chapter leads us into our next chapter on orbits, the oscillations of planetary motion.

CHAPTER 108

# Orbits

Gravity is the force that attracts two bodies toward each other. It is responsible for the behavior of orbital motion. Gravity is described by Newton's Law of Universal Gravitation, which states that every point mass attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The formula for gravitational force is:

$$F_g = \frac{Gm_1m_2}{r^2}$$

Where $F_g$ is the gravitational force between two objects, G is the gravitational constant, $m_1$ and $m_2$ are the masses of the objects, and $r$ is the distance between the centers of the two objects. G is a constant with a value of approximately $6.674 \times 10^{-11}$ N $m^2/kg^2$.

If we think about acceleration due to gravity where one mass is significantly larger than the other, we can rewrite the equation as:

$$A_g = \frac{Gm_1m_2}{r^2m_2} = \frac{Gm_1}{r^2}$$

Where $m_1$ is the mass of the more massive object and $m_2$ is the mass of the significantly less massive object. This cancelling of the $m_2$ mass is why the acceleration due to gravity is independent of the mass of the object in free fall.

A satellite stays in orbit around the planet because the pull of the planet's gravity causes it to accelerate toward the center of the planet.

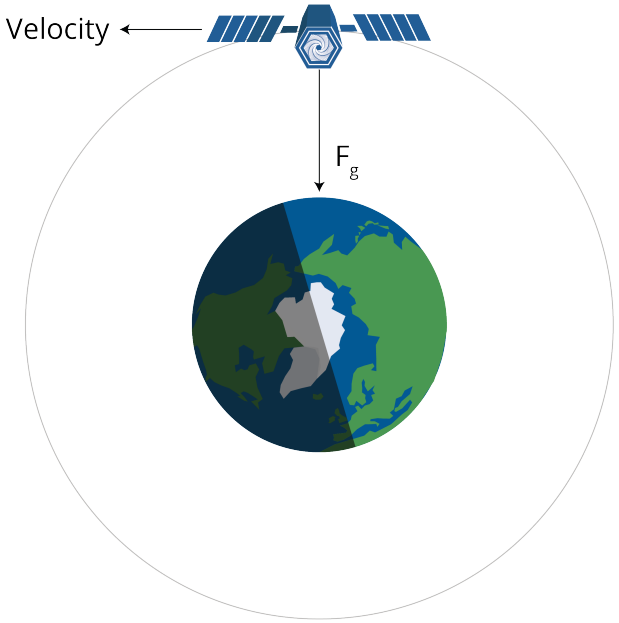



Figure 108.1: A satellite's centripetal force is gravity.

The satellite must be moving at a very particular speed to keep a constant distance from the planet — to travel in a circular orbit. If it is moving too slowly, it will get closer to the planet. If it is going too fast, it will get farther from the planet.

Figure 108.2: A diagram showing the required speeds for entering orbit.

The radius of a satellite in a low orbit is typically about 2 million meters above the ground. At that distance, the acceleration due to gravity is more like $6.8\text{m}/\text{s}^2$, instead of the $9.8\text{m}/\text{s}^2$ that we experience on the surface of the planet.

How fast does the satellite need to be moving in a circle with a radius of 8.37 million meters to have an acceleration of $6.8\text{m}/\text{s}^2$? Real fast.

Recall that the acceleration vector is

$$a = \frac{v^2}{r}$$

Thus the velocity $v$ needs to be:

$$v = \sqrt{ar} = \sqrt{6.8(8.37\text{x}10^6)} = 7,544 \text{ m/s}$$

(That's 16,875 miles per hour.)

When a satellite falls out of orbit, it enters the atmosphere at that 7,544 m/s. The air rushing by generates so much friction that the satellite gets very, very hot, and usually disintegrates.

## 108.1 Astronauts are not weightless

Some people see astronauts floating inside an orbiting spacecraft and think there is no gravity: that the astronauts are so far away that the gravity of the planet doesn't affect them. This is incorrect. The gravity might be slightly less (maybe 6 newtons per kg instead of 9.8 newtons per kg), but the weightlessness they experience is because they and the spacecraft are in free fall. They are moving so fast (in a direction perpendicular to gravity) that they don't collide with the planet.

### *Exercise 273* Mars Orbit

*Working Space*

The radius of Mars is 3.39 million meters. The atmosphere goes up another 11 km. Let's say you want to put a satellite in a circular orbit around Mars with a radius of 3.4 million meters.

The acceleration due to gravity on the surface of Mars is $3.721 \mathrm{m/s^2}$. We can safely assume that it is approximately the same 11 km above the surface.

How fast does the satellite need to be traveling in its orbit? How long will each orbit take?

*Answer on Page 1975*

## 108.2 Geosynchronous Orbits

The planet Earth rotates once a day. Satellites in low orbits circle the Earth many times a day. Satellites in very high orbits circle less than once per day. There is a radius at which a satellite orbits exactly once per day. Satellites at this radius are known as "geosynchronous" or "geostationary", because they are always directly over a place on the planet.

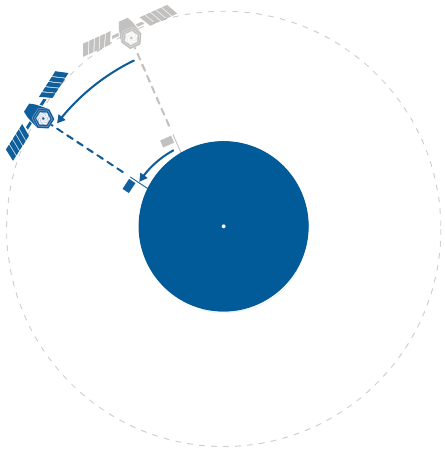


Figure 108.3: A satellite in geosynchronous orbit.

The radius of a circular geosynchronous orbit is 42.164 million meters. (About 36 km above the surface of the earth.)

A geosynchronous satellite travels at a speed of 3,070 m/s.

Geosynchronous satellites are used for the Global Positioning Satellite system, weather monitoring system, and communications system.

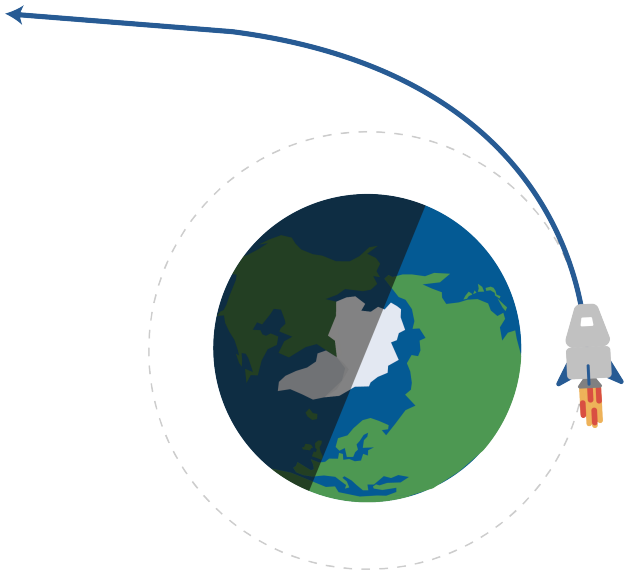


Figure 108.4: A satellite can reach a speed at which it "escapes" earth's orbit (centripetal force).

## 108.3 Kepler's Laws

There are three laws that describe the motion of planets and their elliptical properties.

1. Every planet moves along an ellipse with the sun at one focus.

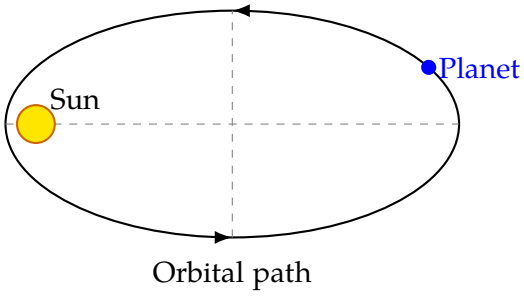



Figure 108.5: A diagram of Kepler's First Law.

2. As a planet moves around its orbit, it sweeps out equal areas in equal times. This means that a planet moves faster when it is closer to the sun and slower when it is farther from the sun. Note that this directly comes from the *Conservation of Angular Momentum*.

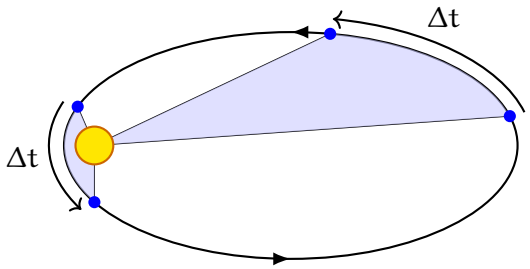



Figure 108.6: A diagram of Kepler's Second Law.

3. If $T$ is the period of a planet's orbit, and $a$ is the length of the semi-major axis of the ellipse, then $\frac{T^2}{a^3}$ is the same for all planets orbiting the same star.

## 108.4 Escape velocity

The escape velocity is the minimum speed needed for an object to "break free" from the gravitational attraction of a massive body or planet, without further propulsion. Gravity will never pull it back, because we assume no further forces act on the object. This is most commonly associated with a spaceship or satellite escaping from a planet's gravitational pull.

We need to understand that escape velocity relies on energy, rather than a force or acceleration. An object must have enough kinetic energy to overcome the gravitational potential energy pulling it back toward the planet. To escape from an orbit, the object's mechanical energy (the sum of its kinetic and potential energy) must be greater than or equal to zero.

We can derive its initial energy as: $E = K + U = \frac{1}{2}mv^2 - \frac{GMm}{R}$, where $m$ is the mass of the object, $M$ is the mass of the planet, $R$ is the distance from the center of the planet to the object, and $G$ is the gravitational constant.

After its escape, the object will be infinitely far away from the planet, and its potential energy will be zero. If we assume it just barely escapes, its kinetic energy will also be zero. Therefore, we set the initial energy equal to zero:

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0 \tag{108.1}$$

$$\frac{1}{2}mv^2 = \frac{GMm}{R} \tag{108.2}$$

$$v^2 = \frac{2GM}{R} \tag{108.3}$$

$$v_{esc} = \sqrt{\frac{2GM}{R}} \tag{108.4}$$

Notice that the mass of the object $m$ cancels out, meaning that the escape velocity is

independent of the mass of the escaping object. It only depends on the mass of the planet and the distance from its center.

Further, **notice that the escape velocity is $\sqrt{2}$ times the orbital velocity**, $v_{orb} = \sqrt{GM/R}$ at the same distance from the planet's center. This means that to escape from a circular orbit, an object must increase its speed by about 41.4%. For Earth, the escape velocity at the surface is approximately 11.2 km/s (about 25,000 mph). This means that any object, regardless of its mass, must reach this speed to escape Earth's gravitational pull without further propulsion.

## *Exercise 274* Escape Velocity from Mars

*Working Space*

The radius of Mars is $3.4 \times 10^6$ meters. The mass of Mars is $6.42 \times 10^{23}$ kg. A spaceship is sitting on the surface of Mars.

1. Find the escape velocity from the surface of Mars.
2. Suppose the spaceship is in a circular orbit 11 km above the surface of Mars. What is the escape velocity from that orbit?

*Answer on Page 1976*

Note that giving an object an extra height $h$ causes the escape velocity to decrease (as escape velocity is inversely proportional to the square root of the radius plus height). This is due to gravitation potential energy being less negative at larger radii. However, the effect is small compared to the overall radius of the planet. Problems may give you a radius plus height, or just the radius of the planet, or a total distance from the center, depending on the context, so read each problem carefully.

Chapter 109

# Circular Motion, continued

## 109.1 Review of Circular Motion Basics

We have previously discussed circular motion in the circular motion basics chapter, so let's do a quick review of the important concepts before we move on to rotational dynamics.

We have rotational kinematics, which describes the motion of objects rotating around a fixed axis. The key variables in rotational kinematics are angular displacement ($\theta$), angular velocity ($\omega$), and angular acceleration ($\alpha$). These variables are analogous to linear displacement, velocity, and acceleration in linear kinematics.

| Linear motion | Rotational motion |
|---|---|
| $v = v_0 + at$ | $\omega = \omega_0 + \alpha t$ |
| $x = x_0 + v_0 t + \frac{1}{2}at^2$ | $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$ |
| $v^2 = v_0^2 + 2a(x - x_0)$ | $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$ |

Table 109.1: Kinematic equations for linear and rotational motion.

When an object such as a wheel or a disk rotates about an axis, each point on the object moves along a circular path centered on that axis. The total distance traveled by a point during one complete rotation is the circumference of its circular path, which is given by $C = 2\pi r$. The rotation of the object is described by an angle $\theta$, usually measured in radians, where one full rotation corresponds to $2\pi$ radians.

## 109.2 Rigid Bodies

Although all points on a rotating disk pass through the same angular displacement $\theta$ in the same amount of time, they do not all travel the same distance. Points farther from the axis of rotation move along larger circles and therefore cover more distance during each rotation, even though the angular displacement is the same for all points. As a result, points farther from the axis have greater linear velocity and greater linear (tangential) acceleration than points closer to the center, as described by the relationships $v = r\omega$ and $a = r\alpha$. We call this points part of a *rigid body*.

A rigid body is an idealized object in which the distances between all points remain constant, even when forces are applied. All points along a radial line have the same angular displacement, angular velocity, and angular acceleration.

In other words, a rigid body does not deform: it does not stretch, compress, or bend. This assumption allows the object to be treated as a single system in which all parts move in a predictable way relative to one another.

In rotational motion, treating an object as a rigid body means that:

- All points in the object rotate together around a common axis.
- The angular velocity $(\omega)$ and angular acceleration $(\alpha)$ are the same for all points in the rigid body.
- The linear velocity $(v)$ and linear acceleration $(a)$ of points in the rigid body depend on their distance $(r)$ from the axis of rotation, following the relationships $v = r\omega$ and $a = r\alpha$.

This idea maintains for many of the problems and concepts we will discuss in this chapter.

## 109.3 Torque

Let's think about pushing a box along a frictionless surface. There is a linear relationship between how hard you push and how much the box accelerates. What if, instead, you apply the same force to a merry-go-round fixed at a central pivot? Where can you push the merry-go-round for the highest rotation? Obviously you cannot push the merry-go-round linearly (as it is fixed to the ground), so how far will the merry-go-round rotate? All of these questions can be answered by the concept of *torque*.

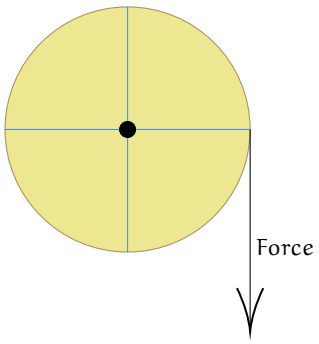



Figure 109.1: Merry-go-round with a singular applied force.

Torque is the concept of *rotational force*. Applying a torque to a surface, a *rigid body*, that has a fixed rotational pivot will cause the rigid body to *accelerate* around the pivot, ultimately producing an *angular acceleration*.

Imagine pushing open a door that is bound to a hinge on one side of the door. Pushing open the door at a point close to the hinge will require large amount of Torque, while

pushing closer to the door handle and farther away from the hinge is much easier to do, and requires less Torque.

Alternatively, think of a wrench rotating a large bolt. It will require more force from your arm closer towards the pivot, while pushing at the edge of the wrench will require less force to produce an equivalent torque. Take a look at this in action here.

**Torque**

Torque, represented through the greek letter $\tau$(tau), is defined as

$$\tau = rF \tag{109.1}$$

where $F$ is a force perpendicular to

$$\tau = rF\sin\theta \tag{109.2}$$

Notice that the torque around a pivot depends on the distance from the pivot, $r$, and the angle from the radius vector $\theta$. The $\sin\theta$ indicates that only the *perpendicular* part of the force impacts the torque. This is equivalent to the cross product of the two vectors:

$$\tau = r \times F \tag{109.3}$$

Torque has units of Newton-meters, but in this case, it is not an energy form, so cannot be equated to Joules. Torque is an *vector*, not a scalar.

Let's do an example proving this.

**Question**: A rod with a fixed end is being pulled by a tension force of 50 N at a 40° angle from the horizontal, at a distance of 2 m. Find the Torque on the rod.

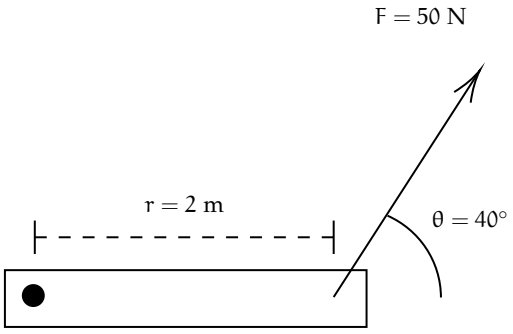



Figure 109.2: Main diagram.

**Answer**: Before just plugging in values, let's separate the force vector into components. The component that pulls perpendicular to the axis is $50\sin 40°$, while $50\cos 40°$ of the force is applied *pulling* the rod against the pivot, which produces no torque.

(a) $F\cos\theta$ (b) $F\sin\theta$

Figure 109.3: Components of the applied force.

Think back to the wrench example: you don't push or pull a wrench to loosen a bolt; you must apply a force *tangential* or *perpendicular* to the bolt to rotate it.

Since only the perpendicular component contributes to the torque, we calculate the torque to be $\tau = 50\sin 40^\circ\,\text{N}\,(2\text{m}) \approx 64.278\,\text{Newton-meters}$

## *Exercise 275* Door Torque

A door is connected to a wall by a hinge 0.5 m from the handle. A person opens the door with a tangential force of 38 N. What is the torque on the door?

*Working Space*

*Answer on Page 1976*

## *Exercise 276* Torque but with a twist!

A uniform horizontal beam is hinged at the left end and held in place by a force applied at its right end. The beam has length $L = 2.0$ m. A force of magnitude $F = 20$ N is applied at the right end of the beam, making an angle of 30° above the beam.

If the pivot point is at the left end of the beam,

1. Find the magnitude of the torque about the hinge.
2. State whether the torque tends to rotate the beam clockwise or counterclockwise.

*Working Space*

*Answer on Page 1976*

### 109.3.1 Torque directions

What happens when two torques are applied in *opposite*? Let's take a look at the merry-go-round again, but this time applied with two torques.

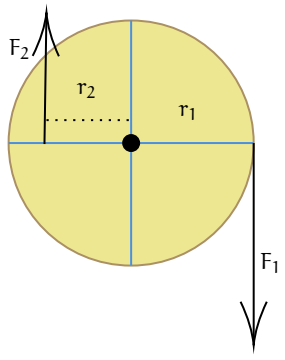



Figure 109.4: Merry-go-round with 2 forces in opposing directions.

Torques applied in opposition act *in opposite directions along the axis of rotation*. As torque is a vector defined by the *cross product equation* 109.3, so its direction is perpendicular to the plane formed by position vector and force which the plane live on. In problems, you will typically see the terms *out of* ($\odot$), or *into* ($\otimes$), the page, corresponding to the $+z$ or $-z$ directions, respectively. Remember these terms by the following visualizations: the out of symbol ($\odot$) looks like the tip of a feathered dart coming at you, while the into symbol ($\otimes$) looks like the feathers of a dart. Another way to think about this is using a drill and screw: rotating the screw counterclockwise causes it to advance toward you (out of the page), while rotating it clockwise causes it to retreat away from you (into the page).

To determine which direction a given torque points, one first identifies the sense of rotation the force would produce if acting alone: a force that tends to rotate the object counterclockwise in the plane produces a torque vector pointing out of the page, while a force that tends to rotate the object clockwise produces a torque vector pointing into the page. When two torques are opposite, one must therefore cause clockwise rotation and the other counterclockwise rotation, leading to torque vectors that are equal in line of action but opposite in direction.

The *net torque* is the sum of all torques present, similar to how net force is the sum of all forces. Generally, counterclockwise torques are positive, while clockwise torques are negative.[1] Using our merry go round example, we have $F_2$ is $\odot$while $F_1$ is $\otimes$.

$$\begin{aligned}\tau_{net} &= \sum \tau \\ &= \tau_{CCW} - \tau_{CW} \\ &= +(F_2)(r_2) - (F_1)(r_1)\end{aligned}$$

Because we know $\tau_1 > \tau_2$, the net torque is going to be negative, characterized with $\otimes$, and clockwise.

[1]Note that other videos or professors may do this opposite, using CW - CCW, but as long as you are consistent and clear, both are equally valid.

## *Exercise 277* Opposing Torques on a Beam

A uniform horizontal beam of length 4.0 m and mass 20 kg is supported by a pivot located 1.0 m from the left end. Two forces act on the beam, producing opposing torques about the pivot.

- A downward force of 50 N is applied at the left end of the beam.
- A downward force of unknown magnitude F is applied at the right end of the beam.
- The beam is in a state of static equilibrium.

Complete the following:

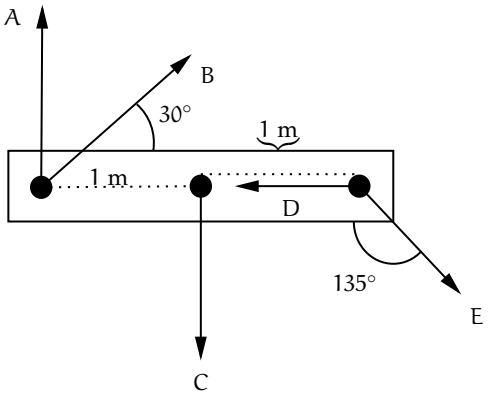
1. Find the Weight of the Beam
2. Write an equation for torque before solving for F
3. Solve for F, the unknown force acting on the right end

*Working Space*

*Answer on Page 1977*

## *Exercise 278* More Complicated Opposing Torque

Create an equation for equilibrium of the following forces. Note that you may pick your own pivot point.




*Working Space*

*Answer on Page 1977*

## 109.4 Moment of Inertia and Rotational $F = ma$

We know Newton's Second Law by heart now, $F_{net} = ma$. You probably could recite it in your sleep, but what about a rotational equivalent. Recall that from our first circular motion chapter, we have the equation $a = r\alpha$. Let's do some conversions to get this into an angular form:

$$\begin{aligned} F &= ma \\ F &= mr\alpha \\ rF &= mr^2\alpha \\ \tau &= mr^2\alpha \end{aligned} \tag{109.4}$$

From Equation 109.4, we can derive that the "mass equivalent" is $mr^2$, since $m$ is multiplied by $a$ in the linear $F = ma$, while $mr^2$ is multiplied by $\alpha$. Thus, we can state that for a point of distance $r$ from the pivot (or axis of rotation), that singular point has *rotation inertia*, also called *moment of inertia*, is defined as $mr^2$. For all point masses on an object, the total inertia is the sum of all point masses of the object.

**Moment of Inertia**

A mass's total moment of inertia is given by the sum of each mass multiplied the distance from the axis of rotation squared.

$$I = \sum mr^2 \tag{109.5}$$

for continuous, solid objects, this becomes

$$I = \int r^2 \, dm \tag{109.6}$$

From Equations 109.5 and 109.6, we see that mass farther from the axis matters much more than anything closer to the pivot. Or, for two objects, one solid (mass spread evenly throughout) and the other a hollow cylinder or hoop (mass confined to the edges), the hollow cylinder will have a greater moment of inertia.

By finding the net rotational inertia of an object, we can create an equation for the rotational equivalent of Newton's Second Law:

$$\sum \tau = \tau_{net} = I\alpha \tag{109.7}$$

### 109.4.1 Relating $\tau = I\alpha$ and other Torque Def'ns

Since we have already defined angular acceleration, and have just discussed moment of inertia, we can now relate our two torque Equations 109.2 and 109.7 by equating them, as they both equal net torque.

$$I\alpha = rF \sin\theta \tag{109.8}$$

This will be useful for solving rotational dynamics problems, and finding any of the variables $I$, $\alpha$, $r$, $F$, or $\theta$ when the others are known.

## *Exercise 279* Angular Acceleration of a Wheel

A wheel has $I = 0.80$ kg·m$^2$. A force $F = 25$ N is applied at radius $r = 0.40$ m at an angle $\theta = 60°$ to the radius.

Find $\alpha$.

*Working Space*

*Answer on Page 1978*

### *Exercise 280* Moment of Inertia from Multiple Torques

*Working Space*

A rigid wheel is mounted on a frictionless axle. Two forces act on the wheel:

- Force $F_1 = 30$ N is applied at radius $r_1 = 0.50$ m at an angle $\theta_1 = 90°$ to the radius, producing a counterclockwise torque.
- Force $F_2 = 20$ N is applied at radius $r_2 = 0.30$ m at an angle $\theta_2 = 45°$ to the radius, producing a clockwise torque.

The wheel undergoes an angular acceleration of $\alpha = 8.0\ \text{rad/s}^2$.

Find the moment of inertia I of the wheel.

*Answer on Page 1978*

## 109.5 Parallel-Axis Theorem

The **parallel-axis theorem** allows us to calculate the moment of inertia of a rigid body about any axis that is parallel to an axis passing through the center of mass. The theorem states that

$$I_s = I_{cm} + Md^2 \tag{109.9}$$

where $I_s$ is the moment of inertia about the axis of interest, $I_{cm}$ is the moment of inertia about a parallel axis through the center of mass, $M$ is the total mass of the object, and $d$ is the perpendicular distance between the two axes.

In many practical situations, such as a door rotating about its hinges or a rod rotating about one end, the axis of rotation is offset from the center of mass, making direct calculation more difficult. That's where the parallel axis theorem comes in!

### *Exercise 281* Parallel Axis Theorem

Prove that the Moment of Inertia of the end of a rod is $I_{end} = \frac{1}{3}ML^2$ given that the center of mass of the moment of inertia of a rod is $I_C M = \frac{1}{12}ML^2$.

*Working Space*

*Answer on Page 1979*

The parallel-axis theorem shows that the moment of inertia about an offset axis is always greater than the moment of inertia about the center-of-mass axis. The additional term $Md^2$ accounts for the translational motion of the object's center of mass relative to the new axis of rotation. As the distance between the axes increases, the moment of inertia increases accordingly.

If the mass of an object is not evenly distributed, the location of the center of mass shifts, but the parallel-axis theorem remains applicable. Once the center of mass is known, the theorem allows the moment of inertia about any parallel axis to be found by adding the contribution due to the displacement of the center of mass. In this way, the effects of mass distribution are fully captured by the center-of-mass term and the distance between the axes.

## 109.6 Rotational Kinetic Energy and Rolling Motion

Take a look at this video: Rotational Inertia: The Race Between a Ring and a Disc.

Let's analyze why the solid cylinder reaches the bottom first.

When two objects roll down an incline without slipping, their acceleration depends not only on gravity, but also on how their mass is distributed. This is because a rolling object must both translate and rotate, so gravitational potential energy is split between translational and rotational kinetic energy. The fraction that goes into rotation depends on the object's moment of inertia.

When an object is rolling, it has both translational kinetic energy and rotational kinetic energy. The total kinetic energy of a rolling object is the sum of these two forms of kinetic energy.

The total rotational kinetic energy of a rolling object comes from all of its individual point masses rotating around the axis. Each point mass has its own rotational kinetic energy

$K_i$, and the sum of all these individual kinetic energies gives the total rotational kinetic energy of the object:

$$\begin{aligned} KE_{\text{rotational}} &= \sum K_i \\ &= \sum \tfrac{1}{2} m_i r_i^2 \omega^2 \\ &= \tfrac{1}{2} \left( \sum m_i r_i^2 \right) \omega^2 \end{aligned}$$

This gives us the formula for rotational kinetic energy in terms of moment of inertia:

**Rotational Kinetic Energy**

The rotational kinetic energy of a rigid body rotating about a fixed axis is given by

$$KE_{\text{rotational}} = \tfrac{1}{2} I \omega^2 \tag{109.10}$$

where $I$ is the moment of inertia of the object about the axis of rotation, and $\omega$ is the angular velocity. This expression applies to rigid bodies rotating about a fixed axis.

Going back to our rolling objects, we can now see that the total kinetic energy of a rolling object is the sum of its translational and rotational kinetic energies:

**Total Kinetic Energy of a Rolling Object**

The total kinetic energy of a rolling object is given by

$$KE_{\text{total}} = KE_{\text{translational}} + KE_{\text{rotational}} = \tfrac{1}{2} m v^2 + \tfrac{1}{2} I \omega^2 \tag{109.11}$$

where $m$ is the mass of the object, $v$ is its linear velocity, $I$ is its moment of inertia, and $\omega$ is its angular velocity. For rolling without slipping, the translational and rotational motions are linked, but it is still useful to treat their kinetic energies separately.

The moment of inertia $I$ varies depending on how the mass is distributed in the object, and is a measure of how much the object resists rotational motion. For example, a solid disc has a different moment of inertia than a hollow disc of the same mass and radius. This difference in moment of inertia affects how much of the total kinetic energy is rotational versus translational.

The solid disc has most of its mass concentrated closer to the center, resulting in a lower moment of inertia. This means that less of its total kinetic energy is rotational, allowing more energy to be available for translational motion, which leads to a higher linear velocity down the incline.

**Relationship between translational velocity and moment**

For rolling objects of the same mass and radius starting from the same height, translational velocity is inversely related to moment of inertia. This relationship can be expressed as:

$$v \propto \frac{1}{I} \tag{109.12}$$

Explicitly, translational velocity is inversely related to moment of inertia. Objects with larger moments of inertia require more energy to rotate, leaving less energy available for translational motion.

Each object has a unique moment of inertia based on its mass distribution, which directly affects its rolling behavior. Here is a reference table with common moments of inertia for various shapes:

| **Object and Axis of Rotation** | **Moment of Inertia** |
|---|---|
| Solid cylinder or disk (symmetry axis) | $I = \frac{1}{2}MR^2$ |
| Solid cylinder or disk (central diameter) | $I = \frac{1}{4}MR^2 + \frac{1}{12}ML^2$ |
| Hoop (symmetry axis) | $I = MR^2$ |
| Hoop (diameter) | $I = \frac{1}{2}MR^2$ |
| Solid sphere (about diameter) | $I = \frac{2}{5}MR^2$ |
| Thin spherical shell (about diameter) | $I = \frac{2}{3}MR^2$ |
| Rod (about center, perpendicular to length) | $I = \frac{1}{12}ML^2$ |
| Rod (about one end, perpendicular to length) | $I = \frac{1}{3}ML^2$ |

Table 109.2: Common Moments of Inertia

A visualization of each of these can be found at Georgia State University's HyperPhysics website: HyperPhysics: Moments of Inertia.

## 109.7 Flywheel Energy Storage, Work, and Angular Momentum

A flywheel is a rigid body designed to store rotational energy. It typically consists of a heavy wheel or disk mounted on an axle, with much of its mass concentrated away from the axis of rotation. This mass distribution gives the flywheel a large moment of inertia, allowing it to resist changes in rotational speed. There is very little friction in the axle, so once the flywheel is spinning, it can maintain its angular velocity for a long time with minimal energy loss. We often eliminate friction in flywheel friction problems or

calculations.

Because of its large moment of inertia, a flywheel can smooth out variations in rotational motion. When energy is added, the flywheel stores it as rotational kinetic energy. When energy is removed, the flywheel releases this stored energy gradually, helping maintain a more uniform angular velocity. For this reason, flywheels are commonly used in engines, generators, and mechanical systems that require steady rotation.

This relies on the principle of *Conservation of Angular Momentum*, which states that in the absence of *external torques*, the total angular momentum of a system remains constant. For a flywheel, this means that if no external torque acts on it, its angular momentum will not change, allowing it to maintain its rotational speed even when external forces try to slow it down.

### 109.7.1 Angular Momentum

The connection between angular momentum and torque is direct: torque is the rate at which angular momentum changes. Applying a torque to a flywheel changes its angular momentum by increasing or decreasing its angular velocity. Because of the flywheel's large moment of inertia, a given torque produces only a gradual change in angular speed, contributing to the flywheel's stabilizing effect.

**Angular Momentum and the Conservation of Angular Momentum**

The angular momentum $L$ of a rotating rigid body is given by

$$L = I\omega \tag{109.13}$$

where $I$ is the moment of inertia of the body about the axis of rotation, and $\omega$ is its angular velocity. Angular momentum is a *vector* quantity, with both magnitude and direction.

Just as linear momentum can be written as $F = \frac{dp}{dt}$, torque can be expressed as the time rate of change of angular momentum:

$$\tau = \frac{dL}{dt} \tag{109.14}$$

We can also derive this as a Newtonian cross product between $r$ and $\mathbf{p}$, linear momentum. This works because angular momentum is the rotational equivalent of linear momentum, but depends on the choice of origin.

$$L = r \times p \tag{109.15}$$

Just like linear momentum, angular momentum is conserved in a system with no external torques. This is known as the *Conservation of Angular Momentum*:

$$L_{initial} = L_{final} \quad (\boldsymbol{\tau}_{net} = 0) \tag{109.16}$$

This equation shows that torque is responsible for changing angular momentum. Because a flywheel has a large moment of inertia, a given applied torque produces only a small angular acceleration, allowing the flywheel to respond smoothly to changes in applied forces.

### Angular Impulse

Just like in classical momentum, a change in angular momentum is called *angular impulse*. Angular Impulse is representative the cumulative effect of torque applied over given a time interval.

**Angular Impulse**

Angular impulse is defined by

$$\Delta \mathbf{L} = \int_{t_1}^{t_2} \boldsymbol{\tau}\, dt \tag{109.17}$$

Graphically, this is area under a *torque vs. time* graph.

Angular impulse has units Newton-meter-seconds, same as angular momentum:

$$N \cdot m \cdot s^{-1} = kg \cdot m^2 \cdot s^{-1}$$

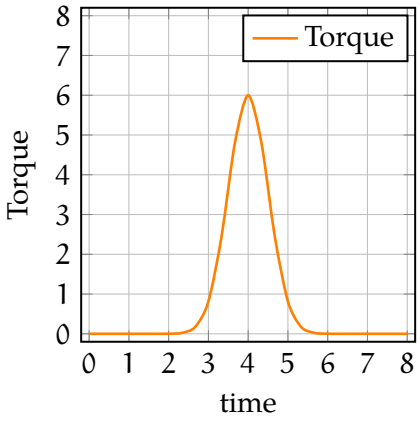
Angular impulse explains why sudden twists or torque have effects on rotational objects. A sudden hit to a door, instantaneous wrench forces, or off-center collisions cause a much better impulses. This impulse may be produced either by a large torque applied over a short time or by a smaller torque applied over a longer time. However, in many physical situations, applying a large torque over a near-instantaneous time interval is more effective and practical.

## *Exercise 282* Angular Impulse

*Working Space*

A kid rams a blue toy car into a red toy car, which is initially stationary. The collision causes the red toy car to start spinning in place.

Estimate, by counting the area under the curve, the angular impulse caused by a toy car crash.




*Answer on Page 1979*

### 109.7.2 Work done by Torque

Recall that work is a change in energy. Torque also does work when it causes rotation. For a constant torque acting on point mass through an angular displacement $\Delta\theta$ the work done is

$$W = Fr\theta = \tau\Delta\theta \tag{109.18}$$

This comes from the fact that linear work is defined as $W = F\Delta x$, and the arc length of a circle is defined as $s = r\theta$. Thus, we can substitute $r\theta$ for $\Delta x$ in the linear work equation to get the rotational work equation.

## 109.8 Summary

Here is a summary of linear quantities in physics, and their now introduced rotational components. You may need to refer to this often to so keep it handy.

Table 109.3: Correspondence Between Linear and Rotational Motion

| **Linear Motion** | **Rotational Motion** |
|---|---|
| Position $x$ | Angle $\theta$ |
| Velocity $v$ | Angular velocity $\omega$ |
| Acceleration $a$ | Angular acceleration $\alpha$ |
| Mass $m$ | Moment of inertia $I$ |
| Force $F$ | Torque $\tau$ |
| Momentum $p = mv$ | Angular momentum $L = I\omega$ |
| $F = \frac{dp}{dt}$ | $\tau = \frac{dL}{dt}$ |
| Work $W = F\Delta x$ | Work $W = \tau\Delta\theta$ |
| Kinetic Energy $\frac{1}{2}mv^2$ | Rotational KE $\frac{1}{2}I\omega^2$ |
| Power $P = Fv$ | Power $P = \tau\omega$ |

Chapter 110

# Rocketry

A rocket propels hot gases, which creates an equal and opposite reaction that pushes it forwards, even in a vacuum.

Even without anything to push against, the rocket can still move forward thanks to Newton's Third Law.

Imagine a spacecraft with a bowling ball attached to the back. If that spacecraft exerts a force to throw the bowling ball backwards, the ball will exert a force on the ship, moving it forwards.

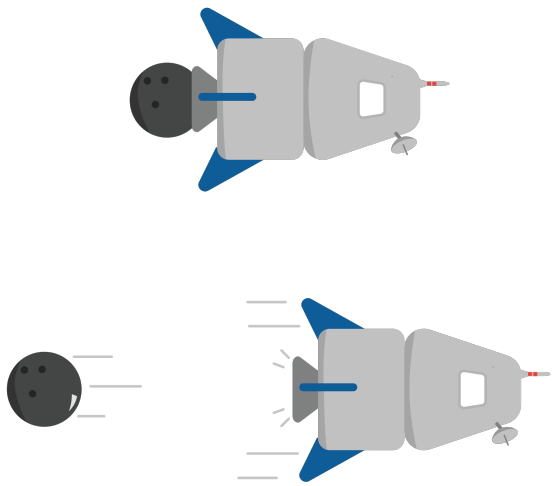


Figure 110.1: Newton's Third Law in action .

Instead of a bowling ball, real-life rockets usually "throw" particles of hot gas at very high speeds. Rockets carry their own oxidizer to provide oxygen to allow fuel to burn.

## 110.1 Types of rocket fuels

There are two main types of chemical rockets.

One type is a *Solid Fuel Rocket*, which ignites a solid fuel-oxidizer mix. Once the solid fuel is ignited, it can't be stopped until all of the fuel is exhausted.

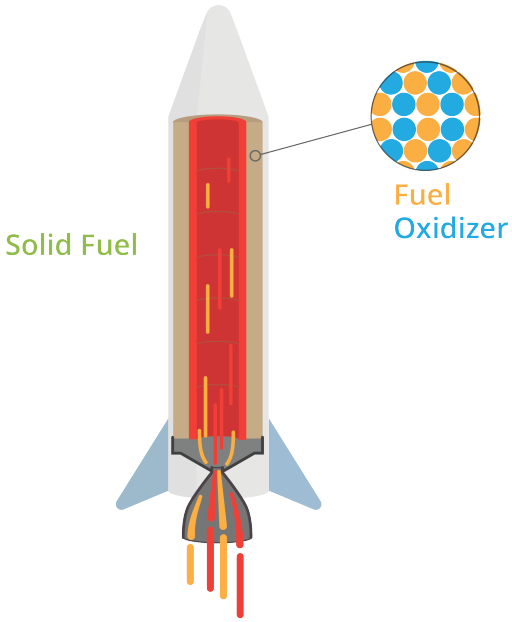



Figure 110.2: A solid fuel-oxidizer rocket.

The other main type of chemical rocket is called a *Liquid Fuel Rocket*. Liquid fuel rockets contain separate tanks for liquid fuel and liquid oxygen. Fuel pumps bring them both to a combustion chamber where they ignite and exit the rocket. Most liquid fuel engines can control their thrust.

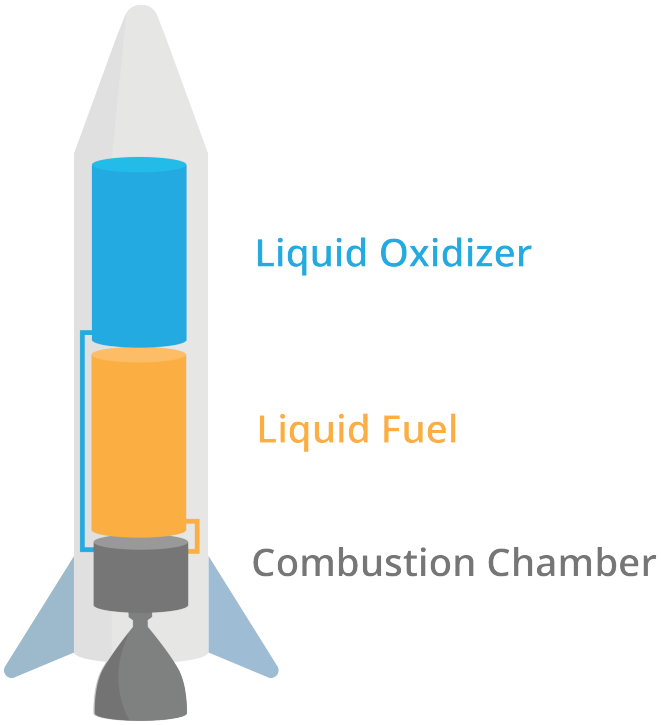



Figure 110.3: A liquid chemical phase.

## 110.2 Tyranny of the rocket equation

Chemical rockets can only burn the fuel that they bring with them. However, the more fuel you carry, the heavier the vehicle will be.

One way to help reduce this weight is by using *staging*.

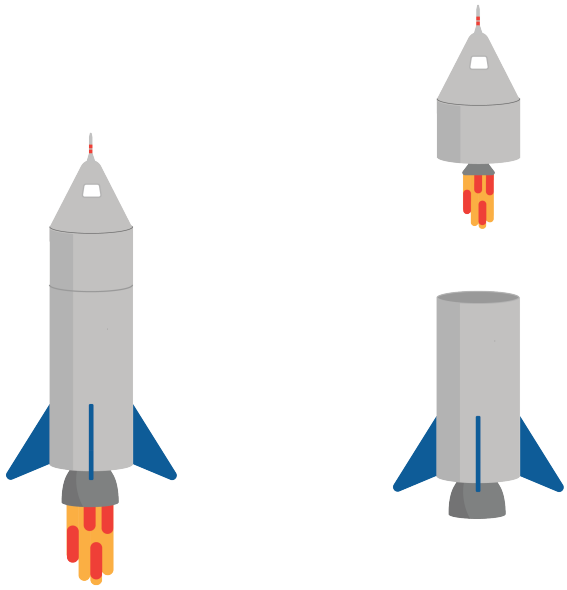


Figure 110.4: A dual stage rocket eliminates one of its stages throughout flight.

Staging allows rockets to drop unnecessary structural mass once they've used up a certain amount of fuel.

## 110.3 Control in atmosphere

There are several common ways that engineers have managed to control rockets' direction in the atmosphere. Usually, on-board sensors detect the orientation of the rocket, and can automatically adjust these controls to keep the rocket going the correct direction.

One method is using *movable fins*. The fins work similarly to control surfaces that we covered in the airplanes chapter.

Another method of control uses a *gimbaled engine*. A gimbaled engine is an engine that can pivot in a certain direction. By pivoting the engine, the rocket can change its direction of thrust, which changes the direction of the rocket. This is called *thrust vectoring*.

A more outdated method is using *vernier engines*, which are two smaller engines that control attitude. However, this adds a large amount of weight to the rocket, so they are less frequently used today.

### Pros and Cons of Rocket Control Systems in Atmosphere

- **Movable fins**
  - **Pros:** Simple, reliable, lightweight, effective at high speeds in atmosphere.
  - **Cons:** Ineffective in vacuum, limited control at low speeds or thin atmosphere.
- **Gimbaled engine (Thrust vectoring)**
  - **Pros:** Precise control, works in both atmosphere and vacuum, allows for rapid directional changes.
  - **Cons:** Mechanically complex, heavier, more expensive, potential failure points.
- **Vernier engines**
  - **Pros:** Provides fine attitude control, redundancy.
  - **Cons:** Adds significant weight, less efficient, rarely used in modern rockets.

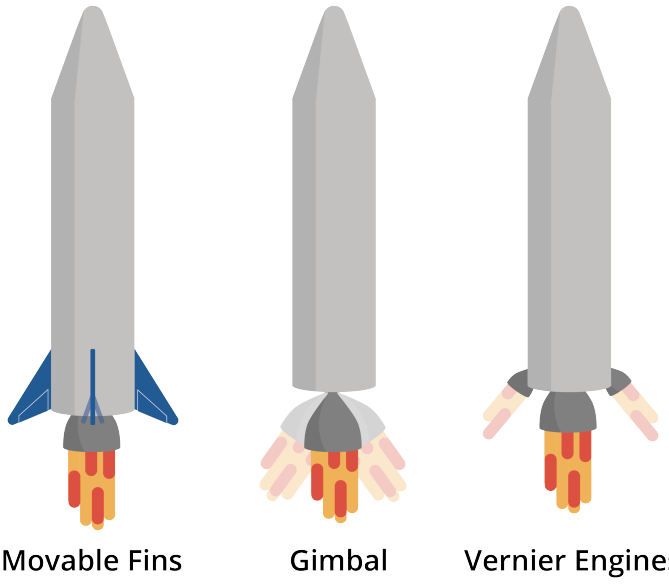



Figure 110.5: Three different methods of engines and their movement in outer space.

## 110.4 Control in space

The previous section describes ways that engineers control rockets in the atmosphere, but most rockets will end up in the vacuum of space. There are several common ways to adjust the orientation in space.

One method is using *RCS thrusters*. An RCS, or reaction control system, is a series of small thrusters that are used to change the direction and position of a spacecraft. RCS thrusters are usually small, and placed in blocks of 3-4 allowing for precise control in all three axes. They are often used for docking maneuvers, attitude control, and orbital adjustments.

Another method called *reaction wheels* uses angular momentum to rotate the spacecraft. By accelerating and decelerating wheels on three axes, the spacecraft can rotate in any direction.

A third common attitude control technology is *magnetorquer*. Magnetorquers use electromagnets and the earth's magnetic field to adjust the orientation of the spacecraft. Magnetorquers are common in small satellites, but they are not as common in larger spacecraft. To manipulate large spacecraft with magnetorquers effectively, you need either prohibitively high current or an external magnetic field more powerful than the Earth's magnetic field.

### Pros and Cons of Rocket Control Systems in Space

- **RCS Thrusters**
  - **Pros:** Precise control, enables translation and rotation, reliable, effective for docking and attitude adjustments.
  - **Cons:** Consumes propellant, limited by onboard fuel, adds weight and complexity.
- **Reaction Wheels**
  - **Pros:** No propellant required, precise attitude control, long operational life, quiet operation.
  - **Cons:** Limited torque, can saturate and require desaturation (often using RCS), mechanical failure risk.
- **Magnetorquers**
  - **Pros:** No propellant required, lightweight, simple design, effective for small satellites in low Earth orbit.
  - **Cons:** Only works near planetary magnetic fields, limited control authority, not suitable for deep space or large spacecraft.

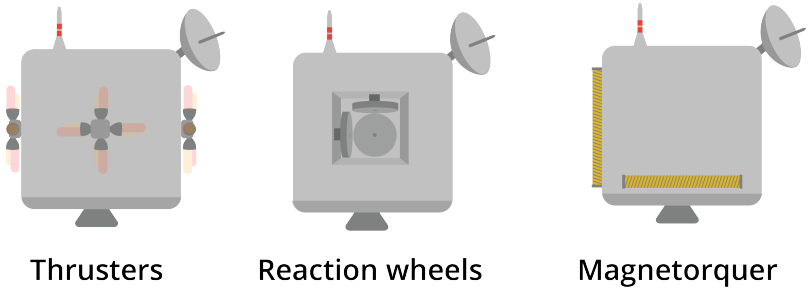



Figure 110.6: Control options for rockets in space.

## 110.5 Alternative propulsion

One type of alternate propulsion is called a *solar sail*. Solar sails have lightweight reflective surfaces that use photons in space to propel the spacecraft without on-board fuel.

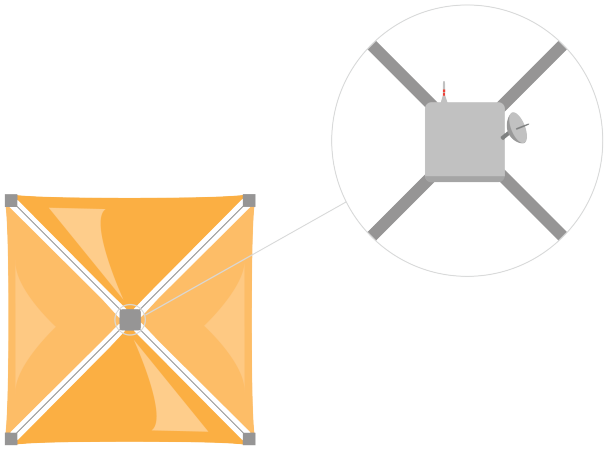


Figure 110.7: Solar sails propel the rocket with photon reflection.

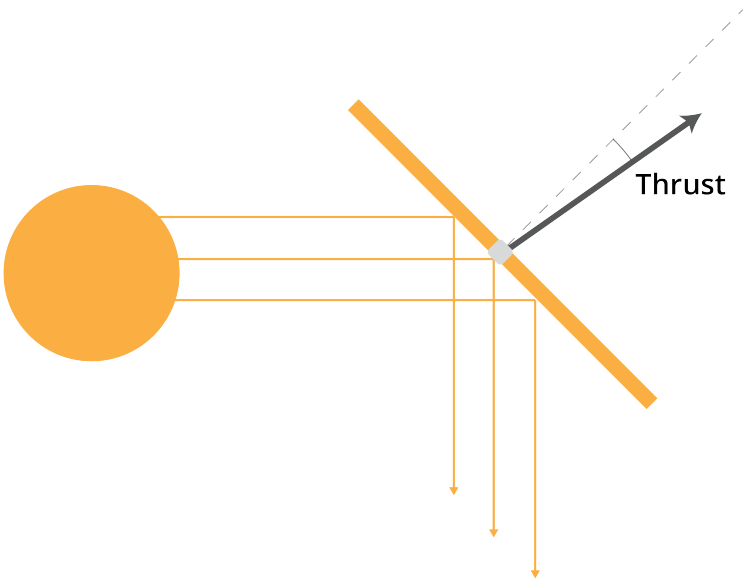



Figure 110.8: A zoomed in version of solar sails.

Photons reflect off of the surface of the sail. However, since the surface is not perfectly

reflective, some of those photons are absorbed, and they produce a horizontal equal and opposite reaction. That small force causes the net thrust to be slightly skewed away from a right angle to the sail.

*Ion propulsion* is a form of electric propulsion that accelerates ions to generate thrust. There are two main types: electrostatic and electromagnetic. Electrostatic ion propulsion uses electric fields to accelerate ions, typically xenon, through grids to produce thrust. Electrostatic thrusters use the Coulomb force to accelerate ions, whereas electromagnetic ion propulsion uses both electric and magnetic fields through the Lorentz force to accelerate ions. Both methods are highly efficient and suitable for long-duration space missions, but they produce low thrust compared to chemical rockets.

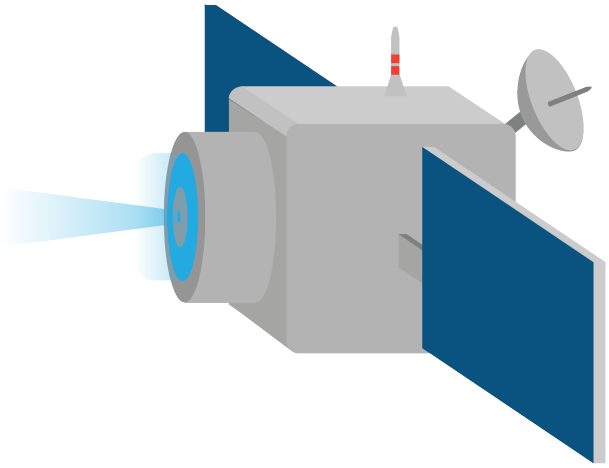


Figure 110.9: An ion thruster

Chapter 111

# Simulation with Vectors

In an earlier chapter, you wrote a Python program that simulated the flight of a hammer to predict its altitude. Your simulation dealt only with scalars. Now, you are ready to create simulations of positions, velocities, accelerations, and forces as vectors.

In this chapter, you are going to simulate two moons that, as they wandered through the vast universe, get caught in each other's gravity well. We will assume there are no other forces acting upon the moons.

## 111.1 Force, Acceleration, Velocity, and Position

We talked about the magnitude of a gravitational attraction between two masses:

$$F = G\frac{m_1 m_2}{r^2}$$

where $F$ is the magnitude of the force in newtons, $m_1$ and $m_2$ are the masses in kg, $r$ is the distance between them in meters, and $g$ is the universal gravitational constant: $6.67430 \times 10^{-11}$.

What is the direction? For the two moons, the force on moon 1 will pull toward moon 2. Likewise, the force on moon 2 will pull toward moon 1.

Of course, if something is big (like the sun), you need to be more specific: The force points directly at the center of mass of the object that is generating the force.

Each of the moons will start off with a velocity vector. That velocity vector will change over time as the moon is accelerated by the force of gravity. If you have a mass $m$ with an initial velocity vector of $\vec{v_0}$ that is being accelerated with a constant force vector $\vec{F}$, at time $t$, the new velocity vector will be:

$$\vec{v_t} = \vec{v_0} + \frac{t}{m}\vec{F}$$

If an object is at an initial position vector of $\vec{p_0}$ and moves with a constant velocity vector $\vec{v}$ for time $t$, the new position will be given by

$$\vec{p}_t = \vec{p}_0 + t\vec{v}$$

## 111.2 Simulations and Step Size

As two moons orbit each other, the force, acceleration, velocity, and position are changing smoothly and continuously. It is difficult to simulate truly continuous things on a digital computer.

However, think about how a movie shows you many frames each second. Each frame is a still picture of the state of the system. The more frames per second, the smoother it looks.

We do a similar trick in simulations. We say "We are going run our simulation in 2 hour steps. We will assume that the acceleration and velocity were constant for those two hours. We will update our position vectors accordingly, then we will recalculate our acceleration and velocity vectors."

Generally, as you make the step size smaller, your simulation will get more accurate and take longer to execute.

## 111.3 Make a Text-based Simulation

To start, you are going to write a Python program that simulates the moons and prints out their position for every time step. Later, we will add graphs and even animation.

We are going to assume the two moons are traveling the same plane so we can do all the math and graphing in 2 dimensions.

Each moon will be represented by a dictionary containing the state of the moon:

- Its mass in kilograms
- Its position — A 2-dimensional vector represent $x$ and $y$ coordinates of the center of the moon.
- Its velocity — A 2-dimensional vector
- Its radius — Each moon has a radius so we know when the centers of the two moons are so close to each other that they must have collided.
- Its color — We will use that when do the plots and animations. One moon will be red, the other blue.

There will then be a loop where we will update the positions of the moons and then

recalculate the acceleration and velocities.

How much time will be simulated? 100 days or until the moons collide, whichever comes first.

We will use numpy arrays to represent our vectors.

Create a file called `moons.py`, and type in this code:

```
import numpy as np

# Constants
G = 6.67430e-11               # Gravitational constant (Nm^2/kg^2)
SEC_PER_DAY = 24 * 60 * 60    # How many seconds in a day?
MAX_TIME = 100 * SEC_PER_DAY # 100 days
TIME_STEP = 2 * 60 * 60       # Update every two hours

# Create the initial state of Moon 1
m1 = {
    "mass": 6.0e22,  # kg
    "position": np.array([0.0, 200_000_000]),  # m
    "velocity": np.array([100.0, 25.0]),  # m/s
    "radius": 1_500_000.0,  # m
    "color": "red" # For plotting
}

# Create the initial state of Moon 2
m2 = {
    "mass": 11.0e22,  # kg
    "position": np.array([0.0, -150_000_000]),  # m
    "velocity": np.array([-45.0, 2.0]),  # m/s
    "radius": 2_000_000.0,  # m
    "color": "blue" # For plotting
}

# Lists to hold positions and time
position1_log = []
position2_log = []
time_log = []

# Start at time zero seconds
current_time = 0.0

# Loop until current time exceed Max Time
while current_time <= MAX_TIME:

    # Add time and positions to log
    time_log.append(current_time)
    position1_log.append(m1["position"])
    position2_log.append(m2["position"])
```

```
42
43      # Print the current time and positions
44      print(f"Day {current_time/SEC_PER_DAY:.2f}:")
45      print(f"\tMoon 1:({m1['position'][0]:,.1f},{m1['position'][1]:,.1f})")
46      print(f"\tMoon 2:({m2['position'][0]:,.1f},{m2['position'][1]:,.1f})")
47
48      # Update the positions based on the current velocities
49      m1["position"] = m1["position"] + m1["velocity"] * TIME_STEP
50      m2["position"] = m2["position"] + m2["velocity"] * TIME_STEP
51
52      # Find the vector from moon1 to moon2
53      delta = m2["position"] - m1["position"]
54
55      # What is the distance between the moons?
56      distance = np.linalg.norm(delta)
57
58      # Have the moons collided?
59      if distance < m1["radius"] + m2["radius"]:
60          print(f"*** Collided {current_time:.1f} seconds in!")
61          break
62
63      # What is a unit vector that points from moon1 toward moon2?
64      direction = delta / distance
65
66      # Calculate the magnitude of the gravitational attraction
67      magnitude = G * m1["mass"] * m2["mass"] / (distance**2)
68
69      # Acceleration vector of moon1 (a = f/m)
70      acceleration1 = direction * magnitude / m1["mass"]
71
72      # Acceleration vector of moon2
73      acceleration2 = (-1 * direction) * magnitude / m2["mass"]
74
75      # Update the velocity vectors
76      m1["velocity"] = m1["velocity"] + acceleration1 * TIME_STEP
77      m2["velocity"] = m2["velocity"] + acceleration2 * TIME_STEP
78
79      # Update the clock
80      current_time += TIME_STEP
81
82  print(f"Generated {len(position1_log)} data points.")
```

When your run the simulation, you will see the positions of the moons for 100 days:

```
1  > python3 moons.py
2  Day 0.00:
3          Moon 1:(0.0,200,000,000.0)
4          Moon 2:(0.0,-150,000,000.0)
5  Day 0.08:
6          Moon 1:(720,000.0,200,180,000.0)
```

```
        Moon 2:(-324,000.0,-149,985,600.0)
Day 0.17:
        Moon 1:(1,439,990.7,200,356,896.1)
        Moon 2:(-647,995.0,-149,969,507.0)
...
Day 100.00:
        Moon 1:(119,312,305.5,283,265,313.5)
        Moon 2:(17,393,287.9,-60,319,261.9)
Generated 1201 data points.
```

Look over the code. Make sure you understand what every line does.

## 111.4 Graph the Paths of the Moons

Now, you will use the matplotlib to graph the paths of the moons. Add this line to the beginning of `moons.py`.

```
import matplotlib.pyplot as plt
```

Add this code to the end of your `moons.py`:

```
# Convert lists to np.arrays
positions1 = np.array(position1_log)
positions2 = np.array(position2_log)

# Create a figure with a set of axes
fig, ax = plt.subplots(1, figsize=(7.2, 10))

# Label the axes
ax.set_xlabel("x (m)")
ax.set_ylabel("y (m)")
ax.set_aspect("equal", adjustable='box')

# Draw the path of the two moons
ax.plot(positions1[:, 0], positions1[:, 1], m1["color"], lw=0.7)
ax.plot(positions2[:, 0], positions2[:, 1], m2["color"], lw=0.7)

# Save out the figure
fig.savefig("plotmoons.png")
```

When you run it, your `plotmoons.png` should look like this:

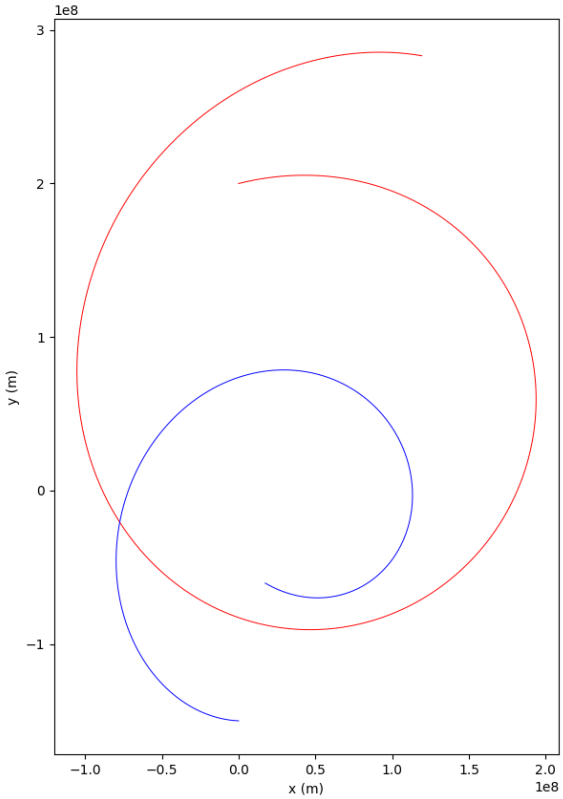



Figure 111.1: Output of moons_01.py.

It is nifty to see the paths, but we don't know where each moon was at a particular time. In fact, it is difficult to figure out which end of each curve was the beginning and which was the ending.

What if we added some lines and labels every 300 steps to put a sense of time into the plot? Add one more constant after the import statements:

```
PAIR_LINE_STEP = 300  # How time steps between pair lines
```

Immediately before you save the figure to the file, add the following code:

```
# Draw some pair lines that help the
# viewer understand time in the graph
i = 0
while i < len(positions1):

    # Where are the moons at the ith entry?
    a = positions1[i, :]
    b = positions2[i, :]
```

```
    ax.plot([a[0], b[0]], [a[1], b[1]], "--", c="gray", lw=0.6, marker=".")

    # What is the time at the ith entry?
    t = time_log[i]

    # Label the location of moon 1 with the day
    ax.text(a[0], a[1], f"{t/SEC_PER_DAY:.0f} days")
    i += PAIR_LINE_STEP
```

When you run it, your plot should look like this:

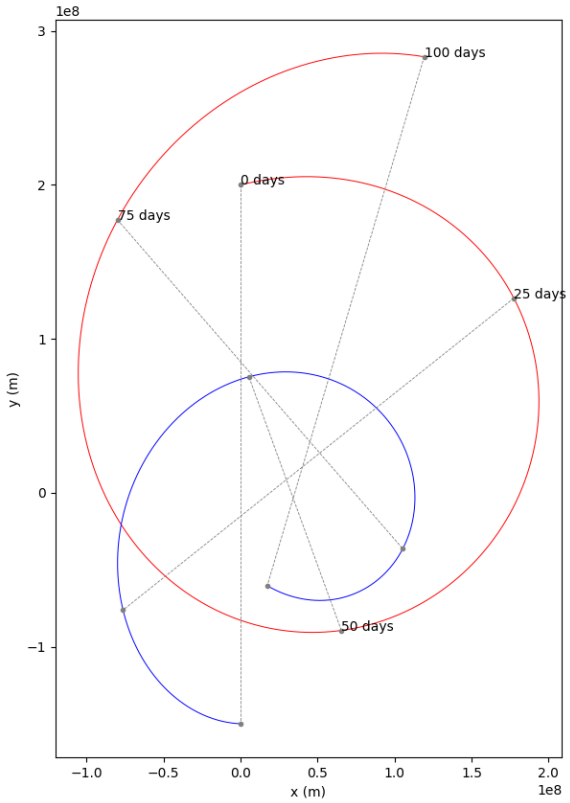



Figure 111.2: Plot moons_02.py.

Now you can get a feel for what happened. The moons were attracted to each other by gravity and started to circle each other. The heavier moon accelerates less quickly, so it makes a smaller loop.

Maybe we will get a better feel for what is happening if we look at more time. Let's increase it to 400 days. Change the relevant constant:

```
MAX_TIME = 400 * SEC_PER_DAY # 100 days
```

Now it should look like this:

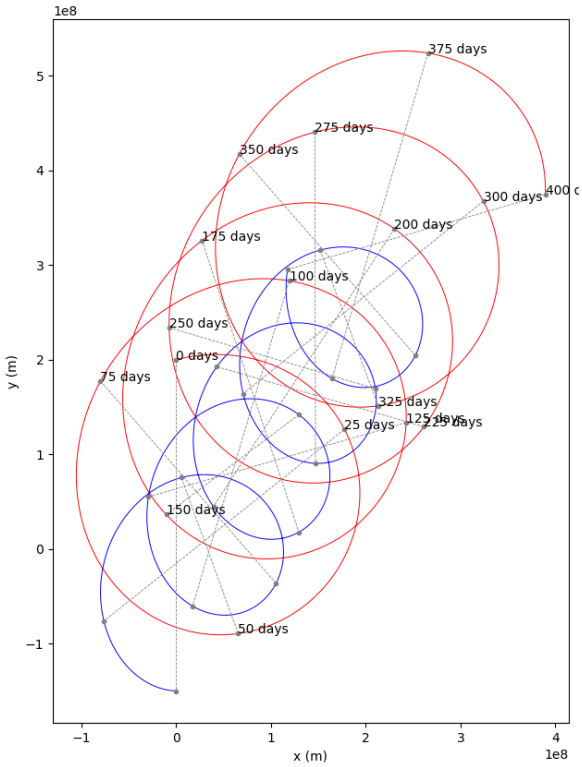



Figure 111.3: Output of plot moons_03.py.

Now you can see the pattern. They are rotating around each other and the pair is gradually migrating up and to the right.

## 111.5 Conservation of Momentum

Recall the idea of momentum being conserved in a system. In these programs, you are observing an extra important idea: *the momentum of a system will be conserved*. That is, absent forces from outside the system, the velocity of the center of mass will not change.

We can compute the initial center of mass and its velocity. In both cases, we just do a weighted average using the mass of the moon as the weight.

Immediately after you initialize the state of two moons, calculate the initial center of mass and its velocity:

```
# Calculate the initial position and velocity of the center of mass
tm = m1["mass"] + m2["mass"]  # Total mass
cm_position = (m1["mass"] * m1["position"] + m2["mass"] * m2["position"]) / tm
```

```
cm_velocity = (m1["mass"] * m1["velocity"] + m2["mass"] * m2["velocity"]) / tm
```

Let's record the center of mass for each time. Before the loop starts, create a list to hold them:

```
cm_log = []
```

Inside the loop (before any calculations), append the current center of mass position to the log:

```
    cm_log.append(cm_position)
```

Anywhere later in the loop (after you update the positions of the moon), update `cm_position`:

```
    # Update the center of mass
    cm_position = cm_position + cm_velocity * TIME_STEP
```

Now, let's look at the positions of the moons relative to the center of mass. Before you do any plotting, convert the list to a numpy array and subtract it from the positions:

```
cms = np.array(cm_log)

# Make positions relative to the center of mass
positions1 = positions1 - cms
positions2 = positions2 - cms
```

When you run it, you can really see what is happening:

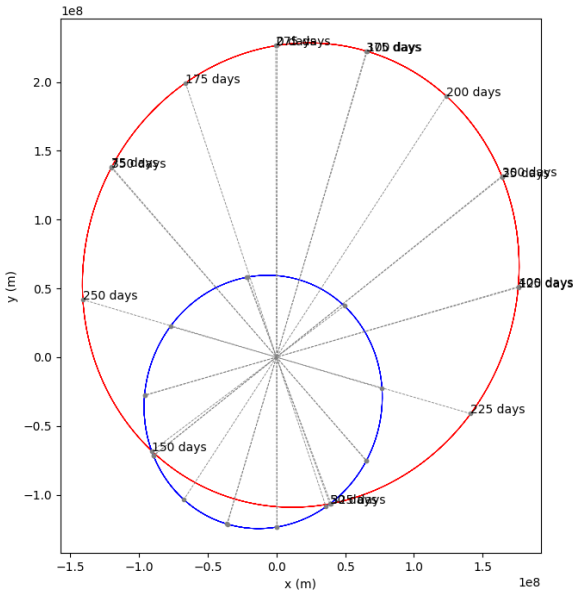



Figure 111.4: Output of plotmoons_04.py.

The moons are tracing elliptical paths. The center of mass is the focus point for both of them.

## 111.6 Animation

One of the features of matplotlib that not a lot of people understand is how to make animations with it. This seems like a really great opportunity to make an animation showing the position, velocity, acceleration of the moons. We will also show the center of mass.

The trick to animations is that you create a bunch "artist" objects. You create a function that updates the artists. matplotlib will call your functions, tell the artists to draw themselves, and make a movie out of that.

Make a copy of `moons.py` called `animate_moons.py`.

Edit it to look like this:

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  # Import animation support and artists
5  from matplotlib.animation import FuncAnimation
6  from matplotlib.patches import Circle, FancyArrow
```

```
from matplotlib.text import Text

# Constants
G = 6.67430e-11  # Gravitational constant (Nm^2/kg^2)
SEC_PER_DAY = 24 * 60 * 60  # How many seconds in a day?
MAX_TIME = 400 * SEC_PER_DAY  # 100 days
TIME_STEP = 12 * 60 * 60  # Update every 12 hours
FRAMECOUNT = MAX_TIME / TIME_STEP  # How many frames in animation
ANI_INTERVAL = 1000 / 50  # ms for each frame in animation

# The velocity and acceleration vectors are invisible
# unless we scale them up.  A lot.
VSCALE = 140000.0
ASCALE = VSCALE * 800000.0

# Create the initial state of Moon 1
m1 = {
    "mass": 6.0e22,  # kg
    "position": np.array([0.0, 200_000_000]),  # m
    "velocity": np.array([100.0, 25.0]),  # m/s
    "radius": 1_500_000.0,  # m
    "color": "red",  # For plotting
}

# Create the initial state of Moon 2
m2 = {
    "mass": 11.0e22,  # kg
    "position": np.array([0.0, -150_000_000]),  # m
    "velocity": np.array([-45.0, 2.0]),  # m/s
    "radius": 2_000_000.0,  # m
    "color": "blue",  # For plotting
}

# Calculate the initial position and velocity of the center of mass
tm = m1["mass"] + m2["mass"]  # Total mass
cm_position = (m1["mass"] * m1["position"] + m2["mass"] * m2["position"]) / tm
cm_velocity = (m1["mass"] * m1["velocity"] + m2["mass"] * m2["velocity"]) / tm

# Start at time zero seconds
current_time = 0.0

# Create the figure and axis
fig, ax = plt.subplots(1, figsize=(7.2, 10))

# Set up the axes
ax.set_xlabel("x (m)")
ax.set_xlim((-1.2e8, 4e8))
ax.set_ylabel("y (m)")
ax.set_ylim((-1.6e8, 5.5e8))
ax.set_aspect("equal", adjustable="box")
fig.tight_layout()

```

```
# Create artists that will be edited in animation
time_text = ax.add_artist(Text(0.03, 0.95, "", transform=ax.transAxes))
circle1 = ax.add_artist(Circle((0, 0), radius=m1["radius"], color=m1["color"]))
circle2 = ax.add_artist(Circle((0, 0), radius=m2["radius"], color=m2["color"]))
circle_cm = ax.add_artist(Circle((0, 0), radius=m2["radius"], color="purple"))
varrow1 = ax.add_artist(FancyArrow(0, 0, 0, 0, color="green",
↪  head_width=m1["radius"]))
varrow2 = ax.add_artist(FancyArrow(0, 0, 0, 0, color="green",
↪  head_width=m2["radius"]))
acc_arrow1 = ax.add_artist(
    FancyArrow(0, 0, 0, 0, color="purple", head_width=m1["radius"])
)
acc_arrow2 = ax.add_artist(
    FancyArrow(0, 0, 0, 0, color="purple", head_width=m2["radius"])
)


# This function will get called for every frame
def animate(frame):

    # Global variables needed in scope from the model
    global cm_position, cm_velocity, current_time, m1, m2

    # Global variables needed in scope from the artists
    global time_text, varrow1, varrow2, acc_arrow1, acc_arrow2, circle1, circle2,
    ↪  circle_cm

    print(f"Updating artists for day {current_time/SEC_PER_DAY:.1f}.")

    # Update the positions based on the current velocities
    m1["position"] = m1["position"] + m1["velocity"] * TIME_STEP
    m2["position"] = m2["position"] + m2["velocity"] * TIME_STEP

    # Update day label
    time_text.set_text(f"Day {current_time/SEC_PER_DAY:.0f}")

    # Update positions of circles
    circle1.set_center(m1["position"])
    circle2.set_center(m2["position"])

    # Update velocity arrows
    varrow1.set_data(
        x=m1["position"][0],
        y=m1["position"][1],
        dx=VSCALE * m1["velocity"][0],
        dy=VSCALE * m1["velocity"][1],
    )
    varrow2.set_data(
        x=m2["position"][0],
        y=m2["position"][1],
        dx=VSCALE * m2["velocity"][0],
        dy=VSCALE * m2["velocity"][1],
```

```
    )

    # Update the center of mass
    cm_position = cm_position + cm_velocity * TIME_STEP
    circle_cm.set_center(cm_position)

    # Find the vector from moon1 to moon2
    delta = m2["position"] - m1["position"]

    # What is the distance between the moons?
    distance = np.linalg.norm(delta)

    # Have the moons collided?
    if distance < m1["radius"] + m2["radius"]:
        print(f"*** Collided {current_time:.1f} seconds in!")

    # What is a unit vector that points from moon1 toward moon2?
    direction = delta / distance

    # Calculate the magnitude of the gravitational attraction
    magnitude = G * m1["mass"] * m2["mass"] / (distance**2)

    # Acceleration vector of moons (a = f/m)
    acceleration1 = direction * magnitude / m1["mass"]
    acceleration2 = (-1 * direction) * magnitude / m2["mass"]

    # Update the acceleration arrows
    acc_arrow1.set_data(
        x=m1["position"][0],
        y=m1["position"][1],
        dx=ASCALE * acceleration1[0],
        dy=ASCALE * acceleration1[1],
    )
    acc_arrow2.set_data(
        x=m2["position"][0],
        y=m2["position"][1],
        dx=ASCALE * acceleration2[0],
        dy=ASCALE * acceleration2[1],
    )

    # Update the velocity vectors
    m1["velocity"] = m1["velocity"] + acceleration1 * TIME_STEP
    m2["velocity"] = m2["velocity"] + acceleration2 * TIME_STEP

    # Update the clock
    current_time += TIME_STEP

    # Return the artists that need to be redrawn
    return (
        time_text,
        varrow1,
        varrow2,
```

```
        acc_arrow1,
        acc_arrow2,
        circle1,
        circle2,
        circle_cm,
    )


# Make the rendering happen
animation = FuncAnimation(
    fig,
    animate,
    np.arange(FRAMECOUNT),
    interval=ANI_INTERVAL
)

# Save the rendering to a video file
animation.save("moonmovie.mp4")
```

When you run this, it will take longer than the previous versions. You should have a video file that shows a simulation of the moons tracing their elliptical paths around their center of mass:

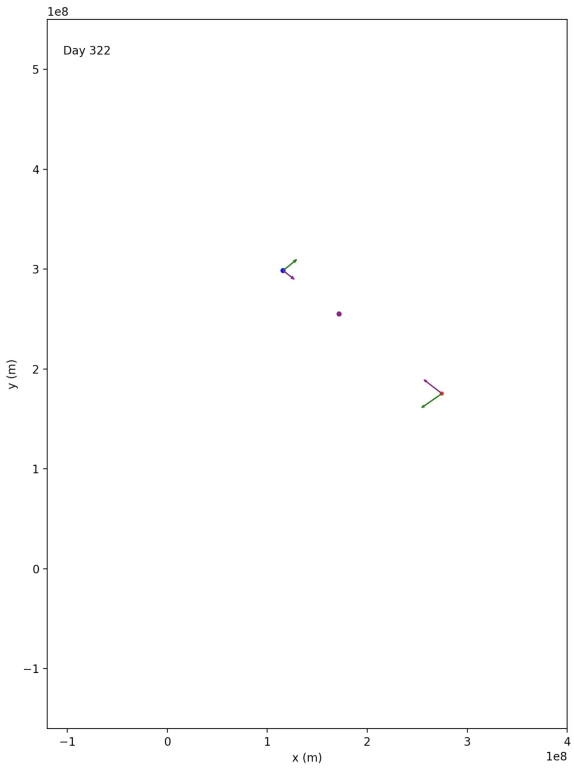



Figure 111.5: A still of the animation produced.

## 111.7 Challenge: The Three-Body Problem

It is time to stretch a little as a physicist and programmer: You are going to make a new version of `moons.py` that handles three moons instead of just two.

This is known as "The Three-Body Problem", and people have tried for centuries to come up with a way to figure out (from the initial conditions) where the three moons would be at time `t` without doing a simulation. And no one has.

For a lot of problems, the outcome is not very sensitive to the initial conditions. For example, consider the flight of a cannonball: If it leaves the muzzle of the cannon a little faster, it will go a little farther.

For the three-body problem, the outcome can be radically different even if the initial conditions are very similar.

(There is a whole field of mathematics studying systems that are very sensitive to initial conditions. It is known as *dynamical systems* or *chaos theory*.)

Copy `moons.py` to `3moons.py`. Here is a reasonable initial state for your third moon:

```
m3 = {
    "mass": 4.0e22,  # kg
    "position": np.array([50_000_000, 80_000_000]),  # m
    "velocity": np.array([-30.0, -35.0]),  # m/s
    "radius": 1_700_000.0,  # m
    "color": "green"
}
```

If we run that simulation for 100 days, we get a plot like this:

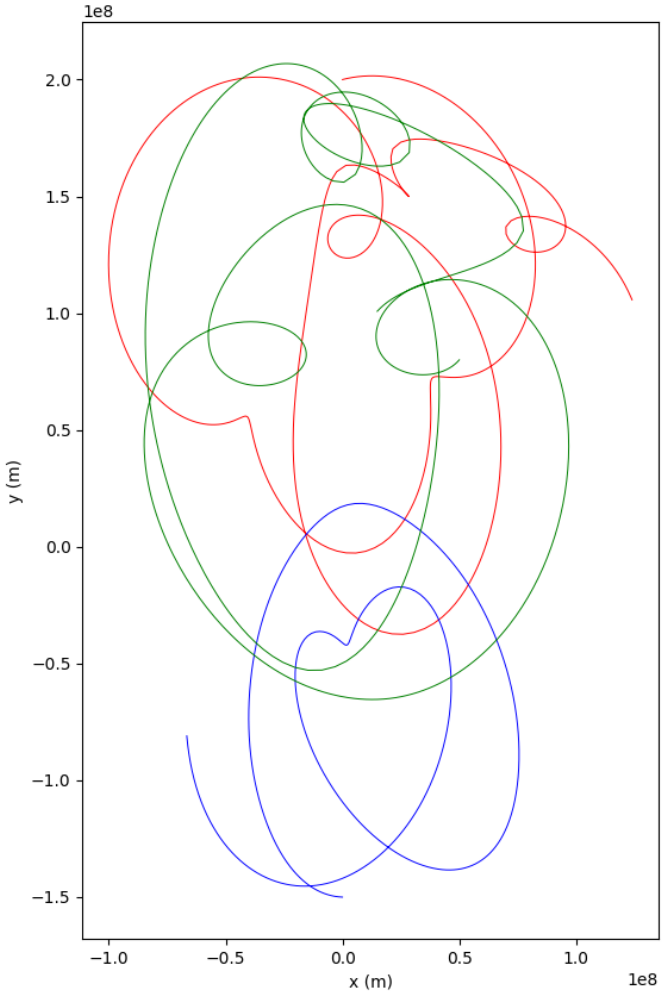



Figure 111.6: The Three Body Problem.

Visibly, you can see this is very different from the two-body problem that just traced ellipses around the center of mass.

Chapter 112

# Tides and Eclipses

Your life on earth involves many orbital paths:

- The Earth is spinning. If you are standing at the equator, you are traveling at 1,674 km per hour around the center of the planet. We are all spinning east, which is why the sun comes up in the east and sets in the west.
- The Earth is orbiting the sun. It takes 365.242 days for the Earth to go once around the sun. This is why different constellations appear at different times during the year — we only see the stars at night and the direction of night shifts as the Earth moves around the sun.
- The moon is orbiting the Earth. The moon travels once around the Earth once every 27.3 days.

You can see the effects of these orbits on our planet. Let's go over a few.

## 112.1 Leap Years

Note that it takes 365.242 days for the Earth to go around the sun. If we declared "The calendar will *always* be 365 days per year!" then the seasons would gradually shift by 0.242 days every year. After a century, they would have migrated 24 days.

So, we made a rule: "Every fourth year, we will add an extra day to the calendar!" The years 2021, 2022,and 2023 get no February 29th, but 2024 does.

That got us a calendar with an average 365.25 days per year, so the seasons would not have migrated as quickly, but they still would have migrated about three days every four hundred years.

So, we made another rule: 'There will be no February 29th in the three century years (multiples of 100) that are not multiples of 400.' So the year 1900 had no Feb 29, but the year 2000 had one. Now, the average number of days per year is 365.2425.

## 112.2 Phases of the Moon

The Earth, the moon, and the sun form a triangle. If you were standing on the moon, you could measure the angle between the light coming from the sun and the light going to the Earth. That angle would fluctuate between 0 degrees and 180 degrees.

- When the angle was close to 0, the people on Earth would see a full moon (fully illuminated disc).
- When the angle was close to 90 degrees, the people on Earth would see a half moon.
- When the angle was nearing 180 degrees, the people on Earth would see a slim crescent.
- When the angle was very close to 180 degrees, the moon would be dark. This is called a new moon (fully darked disc).

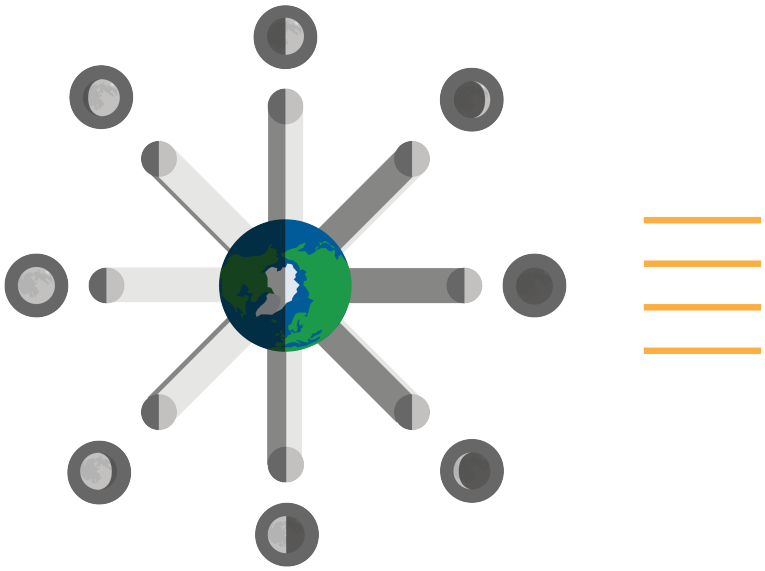


Figure 112.1: The moon revolves around the Earth in different phases every month.

Even though it takes 27.3 days for the moon to travel around the Earth once, it takes 29.5 days to get from one full moon to the next. Why? In the 27.3 days that it took the moon to travel around the Earth, the Earth has moved about 17 degrees around the sun. To get back into the same triangle configuration takes another 2.2 days.

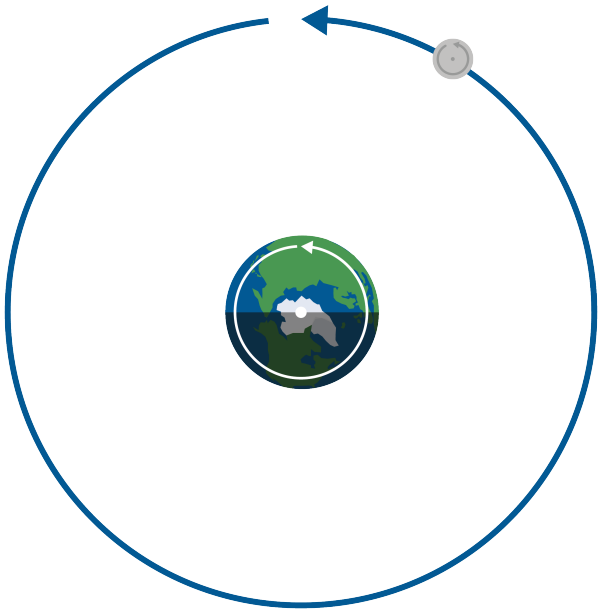


Figure 112.2: The moon and Earth rotate at the same time. Both revolve around its own axis at the same time.

Figure 112.3: A half moon circumstance in space.

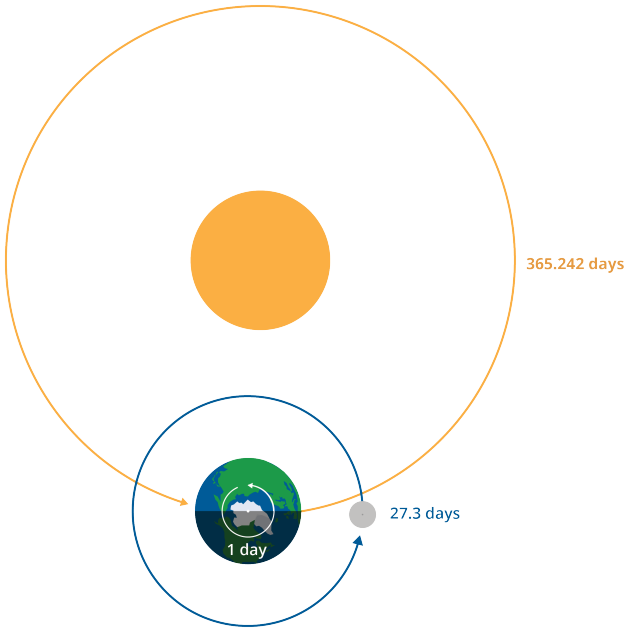



Figure 112.4: The earth, sun, and moon schedules of full rotations.

To explain why we often see a curve in the shadow of the moon, we can look at a ball that has one side painted yellow and the other red.

Figure 112.5: A rotating representation of the moon's phases.

As we rotate the ball, we can see that the straight color boundary between each hemisphere begins to look curved. The curve we see in the moon is due to this same basic principle of how the shading of spheres works.

FIXME: Add text about scale In all of these graphics, we have been using incorrect scale. Here is the true scale of the distance of the Earth and the moon with accurate radii:

Figure 112.6: The moon and Earth in their respective scale.

FIXME: Add text about tidal lock

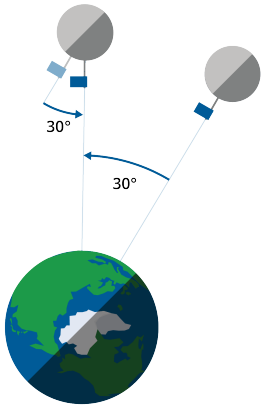



Figure 112.7: FIXME.

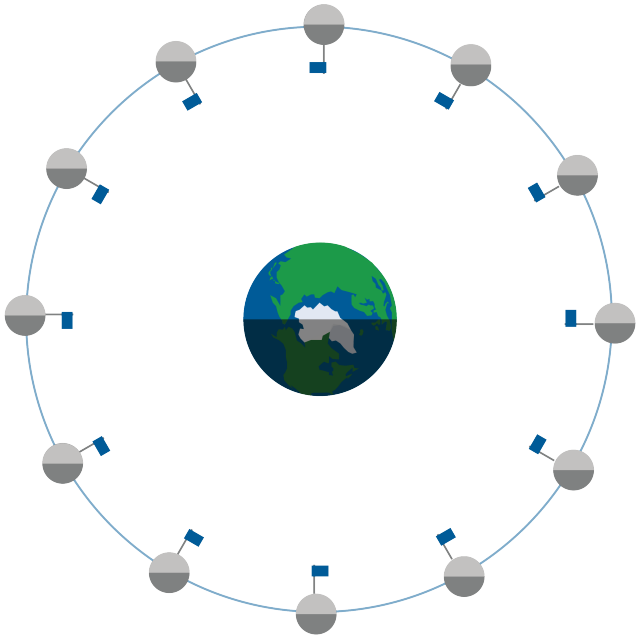


Figure 112.8: FIXME.

## 112.3 Eclipses

While the Earth orbits the sun and the moon orbits the Earth, the two orbits are *not in the same plane*. We call the plane that the Earth orbits the sun in the *ecliptic plane*. The plane of the moon's orbit is about 5° tilted from the ecliptic plane.

Note that the moon passes through the ecliptic plane only twice every 27.3 days. Imagine that the instant it passed through the ecliptic plane was also the precise instant of a full moon. The sun, the Earth, and the moon would be in a straight line! The earth would cast a shadow upon the moon — it would go from a bright full moon to a dark moon until the moon moved back out of the shadow of the earth. This is known as a *lunar eclipse*.

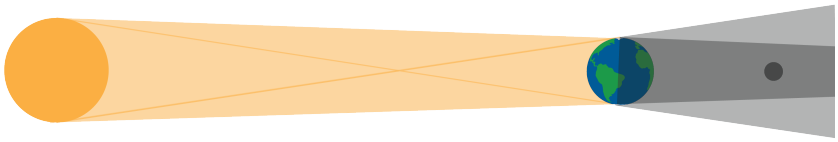


Figure 112.9: Lunar Eclipses happen when the Earth comes between the moon and the sun, causing the moon to be darkened by the Earth's shadow.

The diameter of the moon is a little more than a quarter the diameter of the Earth, so they don't have to be in perfect alignment for the moon to be darkened. Lunar eclipses actually happen once or twice per year.

Now, imagine that the instant the moon passed through the ecliptic plane was also the precise instant of a new moon. The sun, the moon, and the Earth would be in a straight line! The moon would cast a shadow upon some part of the Earth. To a person in that shadow, the sun disappear behind the moon. This is known as a *solar eclipse*.

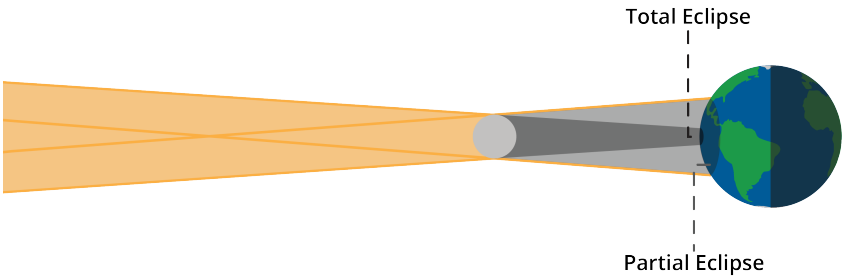



Figure 112.10: Partial and total solar eclipses.

The sun is pretty big, so if the moon blots out just part of it, we call it a *partial solar eclipse*, as seen in Figure 112.10. There are a few partial solar eclipses every year. Note that because the moon's shadow is too small to shade the whole Earth, only certain parts of the world will experience any solar eclipses.

Every 18 months or so, there is a total eclipse of the sun. Once again, only certain parts of the world experience it. You can expect to experience a total eclipse of the sun at your home about once every 375 years.

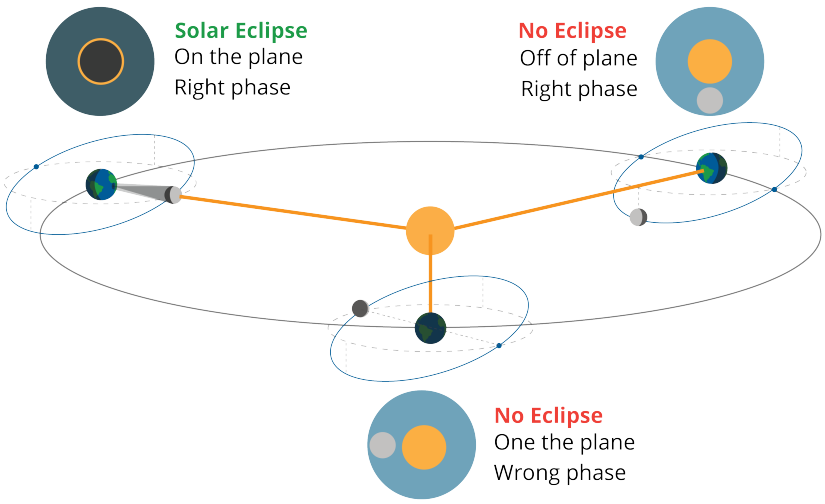



Figure 112.11: Different options for an eclipse.

## 112.4 The Far Side of the Moon

Like the earth, the moon spins on its axis. Due to Earth's gravity, the rotation of the moon slowed down until its spin matched the rate it orbits Earth. That is: we are always looking at the same side of the moon. Until we orbited the moon, we had no idea what the far side looked like.

Some people call it "The Dark Side of the Moon", but it gets just as much sunshine as the side that faces Earth. The name comes from the fact that we lose communication with spacecraft (like the Apollo missions to the moon) when they are on the far side of the moon. When we lose communications with a craft, we often say "It went dark".

## 112.5 Tides

When we say "The moon orbits the Earth", that is a bit of an oversimplification. The force of gravity that pulls the moon toward the Earth, also pulls the Earth toward the moon. The Earth is about 81 times heavier than the moon, so the moon moves more, but the moon definitely moves the Earth.

The center of the moon and the center of the Earth rotate around each other. The point they rotate around is inside the Earth, but it is closer to the surface of the earth than it is to the center of the Earth. This is referred to as a *barycenter*.

Orbits happen, remember, when the centripetal force is equal to gravitational force. So the centripetal force created by the Earth being swung by the moon is equal to the gravitational force that the moon exerts on all the mass on the moon.

However.

The parts of the Earth that are closer to the moon experience less centripetal force (away from the moon) and more gravitational force (toward the moon).

The parts of the Earth that are farther from the moon experience more centripetal force (away from the moon) and less gravitational force (away from the moon).

The effects are not big. For example, you won't notice that you can jump higher when the moon is overhead. You will lose only about 1/200,000 of your weight.

But the ocean is huge: 1/200,000 of its weight is a lot of force.

The water in the oceans bulges a little both toward the moon and away from it.

The Earth is still rotating. If you are at the beach as your longitude slides into one of these bulges, you say "Hey, the tide is rising!" The peak of these bulges is known as "*high tide*". Because there is a bulge on each side of the planet, high tide comes twice a day.

This is a *lunar tide* – because it is caused by the moon. There is a similar effect from the sun, but the sun is very, very far away: solar tidal forces are about half as powerful lunar tidal forces. When the sun and the moon work together, the tides are stronger. This is called a *spring tide*. Spring tides don't happen in the spring time; they happen close to full moons and new moons.

When the moon and the sun are working against each other, the tides are weaker. This is called a *neap tide*. Neap tides happen when you see a half moon in the sky.

### 112.5.1 Computing the Forces

We are enumerating several forces that shape the water on the planet. All these forces are pulling on your body too. In these exercises, you are going to calculate how each force would effect a 1 kg mass on the surface of the earth.

Here are some numbers you will need:

- The mass of the Earth: $5.97219 \times 10^{24}$ kg
- The mass of the sun: $1.9891 \times 10^{30}$ kg
- The mass of the moon: $7.347673 \times 10^{22}$ kg
- Radius of the Earth at the equator: $6,371$ km
- Average distance from the center of the Earth to the center of the sun: $149.6 \times 10^{6}$ km
- Average distance from the center of the moon to the center of the Earth: $384,467$ km.

## *Exercise 283* Life Among the Orbits 1: Earth Gravity

*Working Space*

If the Earth were still and alone in the universe, there would still be the force of gravity. We have said that a kilogram on the surface of the Earth is pulled toward the center of the earth with a force of 9.8 N.

**Confirm that the gravity of the Earth pulls a 1kg mass on the surface of the planet with a force of about 9.8 N.**

You will need the formula for gravitation:

$$F_g = \frac{g m_1 m_2}{r^2}$$

If we measure distance in km and mass in kg, the gravitation constant g is $6.67430 \times 10^{-17}$.

*Answer on Page 1979*

## *Exercise 284* Life Among the Orbits 2: Earth Centripetal Force

*Working Space*

What if we add the spinning of the Earth? The spinning would try to throw the kg into space. The formula for centripetal force is

$$F_c = \frac{mv^2}{r}$$

**Calculate the centripetal force on a 1 kg mass on the surface of the earth. It doesn't fly off into space, so the force due to gravity must be bigger. How many times bigger?**

Assume that the mass is on the equator, thus rotating around the Earth at 465 m/s.

**Does the centripetal force increase, decrease, or stay the same as you get closer to the north pole?**

*Answer on Page 1980*

## *Exercise 285* Life Among the Orbits 3: The Moon's Gravity

*Working Space*

Now we add the moon's gravitational force to our model.

When the moon is directly overhead, how strongly will it pull at the 1 kg mass on the equator?

When the moon is directly underfoot, how strongly will it pull at the 1 kg mass on the equator?

Is that a big difference?

*Answer on Page 1980*

## *Exercise 286* Life Among the Orbits 4: The Swing of the Moon

*Working Space*

Now we add the moon's motion. The moon and the Earth swing each other around. This creates a centripetal force. They both travel in nearly a circle centered at their center of mass.

**How far is the center of mass of the moon and the Earth from the center of the Earth?** (You can imagine a see-saw with the center of the Earth on one end and the center of the moon on the other. Where would the balance point be?)

**What point on the surface of the Earth is closest to the center of mass? How far is it?**

**What point on the surface of the Earth is farthest from the center of mass? How far is it?**

*Answer on Page 1981*

## *Exercise 287* Life Among the Orbits 5: Lunar Centripetal Force

*Working Space*

The moon swings us around that center of mass once every 27.3 days. (Forget about the spinning of the Earth for this part. ) What is the largest and smallest centripetal forces on the surface of the Earth created by this swinging

What is the largest centripetal force on a 1 kg mass with the moon directly underfoot? (You need an answer from the previous question: There is a point on the surface of the Earth that is 11,044,000 m from the center of gravity.)

What is the resulting centripetal force on a 1 kg mass with the moon directly overhead? (You will need the other answer from the previous exercise: That point is 1,698,000 m from the center of mass of the moon and the Earth.)

For this problem is probably easier to use this formula for centripetal force:

$$F_c = mr\omega^2$$

Where $m$ is mass in kg, $r$ is radius in m, and $\omega$ is the angular velocity in radians per second.

*Answer on Page 1981*

## *Exercise 288* Life Among the Orbits 6: Net Force

*Working Space*

Now add together the two forces at both the nearest point to the moon and the farthest.

*Answer on Page 1981*

### 112.5.2 Solar Tidal Forces

The sun has a much larger gravitational effect on the Earth than the moon does:

- When the sun is overhead, it will pull on a 1 kg mass with a force of about 0.00593 N.
- When the moon is overhead, it will pull on a 1 kg mass with a force of about 0.0000343 N.

Why are lunar tides about twice as powerful solar tides?

Tides occur because the pull of gravity and the pull of the centripetal force are out of balance somewhere on the planet. The sun is so far away that the effects of gravitational and centripetal forces are very close to equal everywhere on Earth.

Chapter 113

# Polar Coordinates

We have already seen how to plot a function with $(x, y)$ coordinates. For every $x$ that we put into a function, it returns a $y$. These pairs of coordinates tell us where on the $xy$-plane to graph the function. This coordinate system, where $x$ and $y$ are oriented horizontally and vertically, is called the *Cartesian* coordinate system. It can be used to describe 2D space, but it is not the only way.

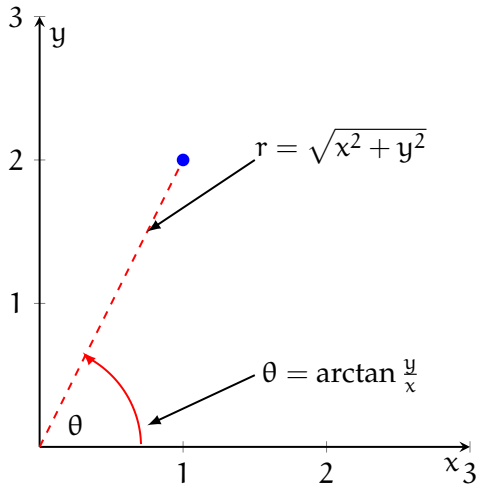



Figure 113.1: The point $(1, 2)$ is $\sqrt{5}$ units from the origin and approximately 1.107 radians counterclockwise from horizontal

Instead of thinking about the horizontal and vertical position, we could think about distance from the origin and rotation about the origin. Take the Cartesian coordinate point $(1, 2)$ (see figure 113.1). How far is $(1, 2)$ from the origin, $(0, 0)$? We can create a right triangle, where the legs are parallel to the $x$ and $y$ axes. This means the leg lengths are 1 and 2, and we can use the Pythagorean theorem to find the length of the hypotenuse (which is the distance from the origin to the point):

$$c^2 = a^2 + b^2$$

$$c^2 = 1^2 + 2^2 = 1 + 4 = 5$$

$$c = \sqrt{5}$$

Therefore, the Cartesian point $(1, 2)$ is $\sqrt{5}$ units from the origin. This is not enough to find our point: there are infinite points that are $\sqrt{5}$ from the origin (see 113.2). To identify a

particular point that is a distance of $\sqrt{5}$ from the origin, we also need an *angle of rotation*. By convention, angles are measured from the positive x- axis. This means points on the positive x-axis have an angle of $\theta = 0$, points on the positive y-axis have an angle of $\theta = \frac{\pi}{2}$, and so on.

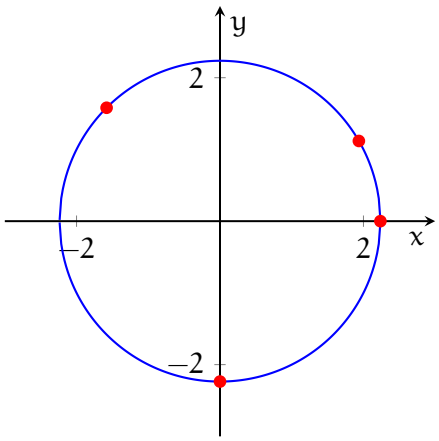



Figure 113.2: There are infinite points $\sqrt{5}$ from the origin, represented by the circle with a radius of $\sqrt{5}$ centered about the origin

We can use trigonometry to find the appropriate angle of rotation for our Cartesian point. There are many ways to do this, but using arctan is the most straightforward. Recall that:

$$\tan\theta = \frac{opposite}{adjacent}$$

That is, for a given angle in a right triangle, the tangent of that angle is given by the length of the opposite leg divided by the adjacent leg. In our case, the opposite leg is the vertical distance (y-value of the Cartesian point) and the adjacent leg is the horizontal distance (x-value of the Cartesian point), which means:

$$\tan\theta = \frac{2}{1}$$

$$\theta = \arctan 2 \approx 1.107 \text{ radians}$$

## 113.1 Plotting Polar Coordinate Points

How do we plot polar coordinate points? Begin by locating the angle given by the second coordinate (remember, the angle is measured counterclockwise from the horizontal). Your point will lie somewhere on this line. Next, move outwards along the angle by the radius given by the first coordinate.

**Example**: Plot the polar coordinate point $(2, \frac{\pi}{3})$.

**Solution**: Begin by locating $\theta = \frac{\pi}{3}$ (see figure 113.3)

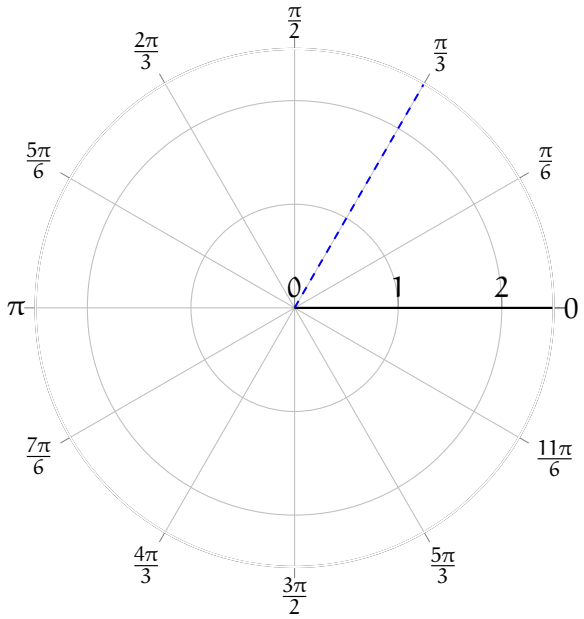



Figure 113.3: $\theta = \frac{\pi}{3}$

Then, move your finger or pencil along the line $\theta = \frac{\pi}{3}$ until you reach $r = 2$ (see figure 113.4).

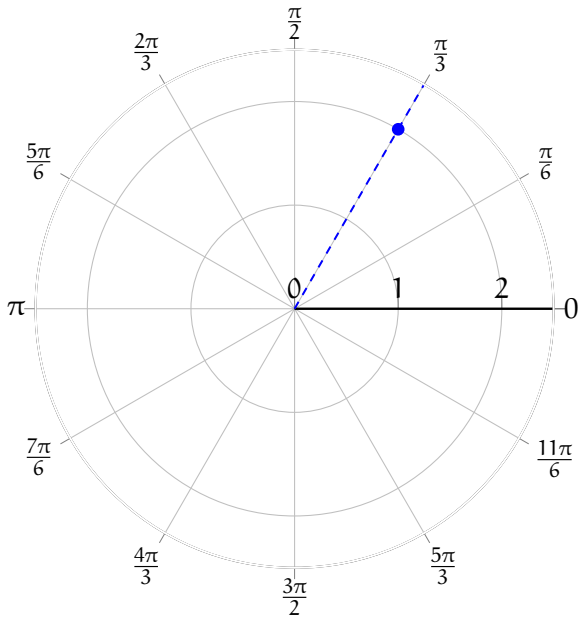


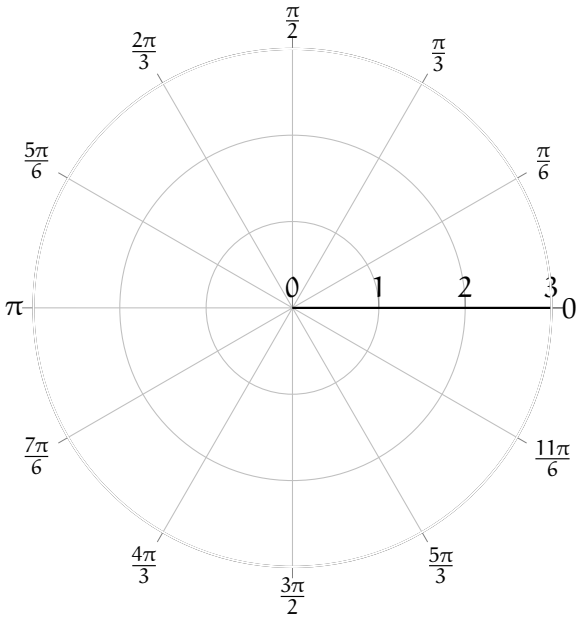

Figure 113.4: $(2, \frac{\pi}{3})$

### *Exercise 289*

Working Space

Plot the following polar coordinate points on the provided polar axis (hint: negative angles are taken counterclockwise):

1. $(1, \pi)$
2. $(1.5, \frac{\pi}{2})$
3. $(1.5, -\frac{\pi}{6})$
4. $(2, \frac{3\pi}{4})$




Answer on Page 1983

## 113.2 Equivalent Points

Unlike the Cartesian coordinate system, two different coordinates may lie at the same location. Consider the points $(1, \frac{\pi}{4})$ and $(-1, \frac{5\pi}{4})$ (see figure 113.5). When a radius is negative, you move *backwards* back over the origin, like a mirror image.

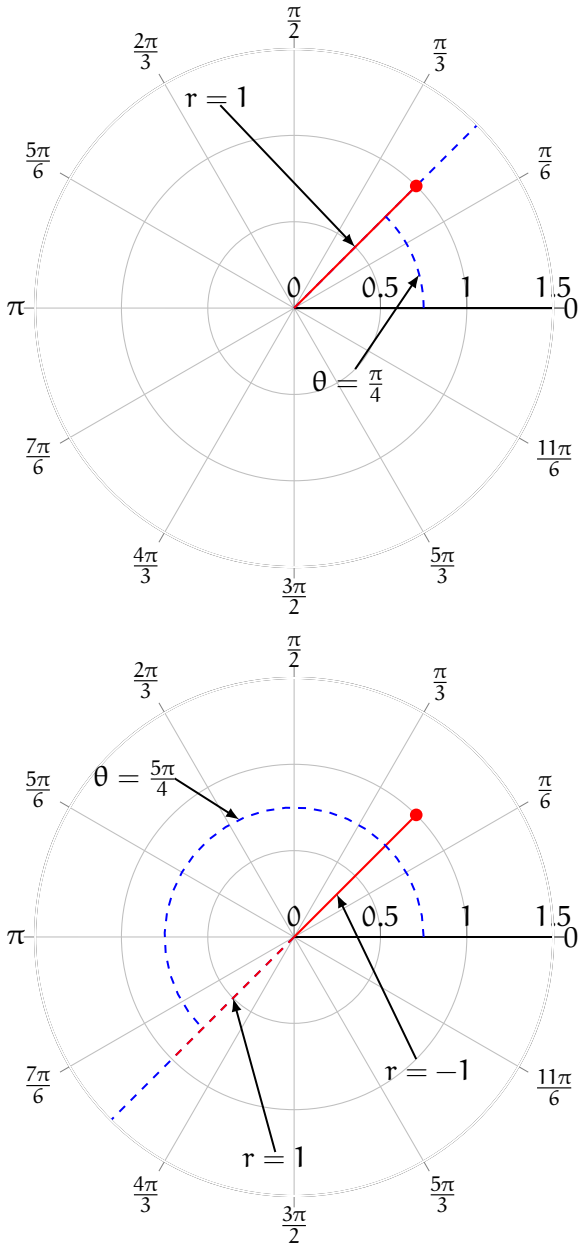



Figure 113.5: The polar coordinates points $(1, \frac{\pi}{4})$ and $(-1, \frac{5\pi}{4})$ are the same location on a polar axis

## 113.3 Changing coordinate systems

### 113.3.1 Cartesian to Polar

From the example above, you should see that a given Cartesian coordinate, $(x, y)$, can also be expressed as a polar coordinate, $(r, \theta)$, where $r$ is the distance from the origin and $\theta$ is the angle of rotation from the horizontal. (Note: Polar functions are generally given as $r$ defined in terms of $\theta$, which means the *dependent* variable is listed first in the coordinate pair, unlike Cartesian coordinates.) Additionally,

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \arctan \frac{y}{x}$$

**Example**: Express the Cartesian point $(-3, 4)$ in polar coordinates.

**Solution**: Taking $x = -3$ and $y = 4$, we find that:

$$r = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

We follow the convention of only taking the positive solution to the square root. Finding $\theta$:

$$\theta = \arctan \frac{4}{-3}$$

When you evaluate the arctan with a calculator, you are likely to get back $\theta = -0.928$. Recall that $\tan\theta = \tan\theta \pm n\pi$, where $n$ is an integer. We know our Cartesian point, $(-3, 4)$, is in the II quadrant, while the angle $-0.928$ radians would fall in the IV quadrant. So, clearly, $-0.928$ radians is not correct. Most calculators restrict the output of arctan to angles between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, because there are actually multiple angles where $\tan\theta = -\frac{4}{3}$. Since $\tan\theta = \tan\theta \pm n\pi$, we also know that:

$$\arctan -\frac{4}{3} = -0.928 \pm n\pi$$

Another possible $\theta$ is $-0.928 + \pi \approx 2.214$, which does fall in the appropriate quadrant. This means the polar coordinates $(5, 2.214)$ are the same as the Cartesian coordinates $(-3, 4)$. *Note:* It is standard practice to express angles in radians, and not degrees, when using polar coordinates.

### 113.3.2 Polar to Cartesian

We can also leverage our knowledge of right triangles to convert polar coordinates to Cartesian coordinates. Take the polar coordinate $(2, \frac{\pi}{4})$ (see figure 113.6). We can draw a right triangle with legs parallel to the $x$ and $y$ axes (not shown in the figure) and a hypotenuse that goes from the origin to the polar coordinate $(2, \frac{\pi}{4})$.

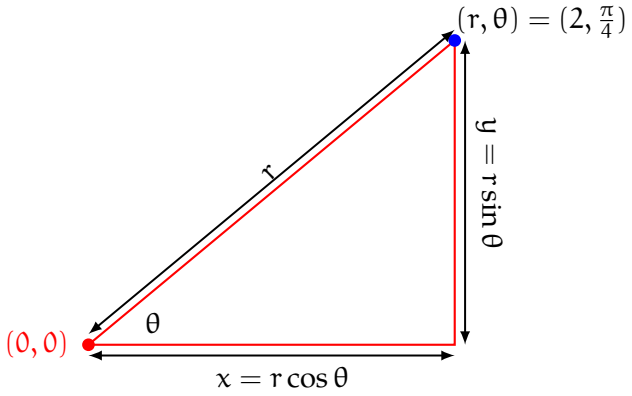



Figure 113.6: To convert from polar to Cartesian coordinates, use the identities $x = r \cos \theta$ and $y = r \sin \theta$

Recall from trigonometry that:

$$\sin \theta = \frac{\text{opposite leg}}{\text{hypotenuse}}$$

We know that the hypotenuse of this triangle has a length of $r$. The opposite leg is vertical and is the same length as the distance of the polar coordinate from the $x$-axis. Therefore, the length of the vertical leg represents the $y$ value of that same polar coordinate if it were expressed in Cartesian coordinates. So, we can say that:

$$\sin \theta = \frac{y}{r}$$

And therefore:

$$y = r \sin \theta$$

By a similar process, we also see that:

$$x = r \cos \theta$$

This is easy to visualize and understand for $0 \leq \theta \leq \frac{\pi}{2}$, but it also holds for other values of $\theta$.

**Example**: Express the polar coordinate $(\frac{3}{2}, \frac{2\pi}{3})$ in Cartesian coordinates.

**Solution**: From the polar coordinate, we see that $\theta = \frac{2\pi}{3}$ and $r = \frac{3}{2}$. Therefore:

$$x = r\cos\theta = \frac{3}{2} \cdot \cos\frac{2\pi}{3} = \frac{3}{2} \cdot -\frac{1}{2} = -\frac{3}{4}$$

$$y = r\sin\theta = \frac{3}{2} \cdot \sin\frac{2\pi}{3} = \frac{3}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$$

The Cartesian coordinate $(-\frac{3}{4}, \frac{3\sqrt{3}}{4})$ has the same location as the given polar coordinate.

## *Exercise 290*

Convert the following polar coordinates to Cartesian coordinates:

*Working Space*

1. $(2, \frac{3\pi}{2})$
2. $(\sqrt{2}, \frac{3\pi}{4})$
3. $(3, -\frac{\pi}{4})$
4. $(-3, -\frac{\pi}{3})$
5. $(2, -\frac{\pi}{2})$

*Answer on Page 1983*

### *Exercise 291*

Convert the following Cartesian coordinates to polar coordinates. Restrict $\theta$ to $0 \leq \theta < 2\pi$.

1. $(-4, 4)$
2. $(3, 3\sqrt{3})$
3. $(\sqrt{3}, -1)$
4. $(-6, 0)$
5. $(-2, -2)$

*Working Space*

*Answer on Page 1983*

## 113.4 Circles in Polar Coordinates

Many conic sections, including circles, are simpler to express as polar functions than as Cartesian functions. Consider a circle with a radius of 2 centered about the origin. The polar function for this is $r = 2$ for all $\theta$. Let's write a Cartesian function for the same circle.

We know that for every point on the circle, the distance to the origin is 2. This means that, by the Pythagorean theorem,

$$r^2 = x^2 + y^2$$

.

(see figure 113.7)

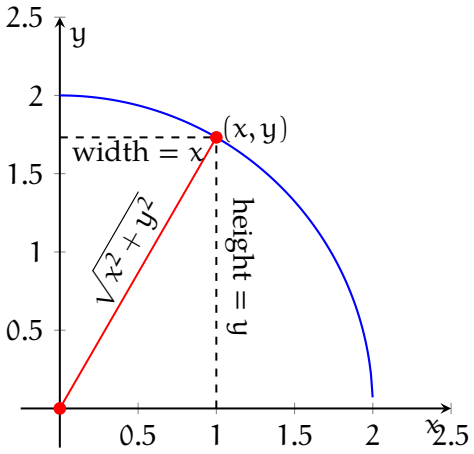



Figure 113.7: All $(x, y)$ pairs on the circle are the same distance from the origin

We can solve this equation for $y$, given that $r = 2$ (in this case):

$$y = \pm\sqrt{2^2 - x^2}$$

Notice that this is really two equations: $y = \sqrt{2^2 - x^2}$ and $y = -\sqrt{2^2 - x^2}$. This is more complex than the polar equation, $r = 2$.

As seen above, the equation of a circle with radius $R$ centered on the origin is simply $r = R$ in polar coordinates. What if we want a circle centered somewhere else? Polar coordinates are best when a circle is bisected by the $x$ or $y$ axis. Consider the polar equation $r = 3\sin\theta$. Let's use a table to find some points and plot the function:

| $\theta$ | $r = 3\sin\theta$ |
|---|---|
| $0$ | $0$ |
| $\frac{\pi}{6}$ | $\frac{3}{2}$ |
| $\frac{\pi}{4}$ | $\frac{3\sqrt{2}}{2}$ |
| $\frac{\pi}{3}$ | $\frac{3\sqrt{3}}{2}$ |
| $\frac{\pi}{2}$ | $3$ |
| $\frac{2\pi}{3}$ | $\frac{3\sqrt{3}}{2}$ |
| $\frac{3\pi}{4}$ | $\frac{3\sqrt{2}}{2}$ |
| $\frac{5\pi}{6}$ | $\frac{3}{2}$ |
| $\pi$ | $0$ |

Here is how those points look plotted (see figures 113.8 and 113.9):

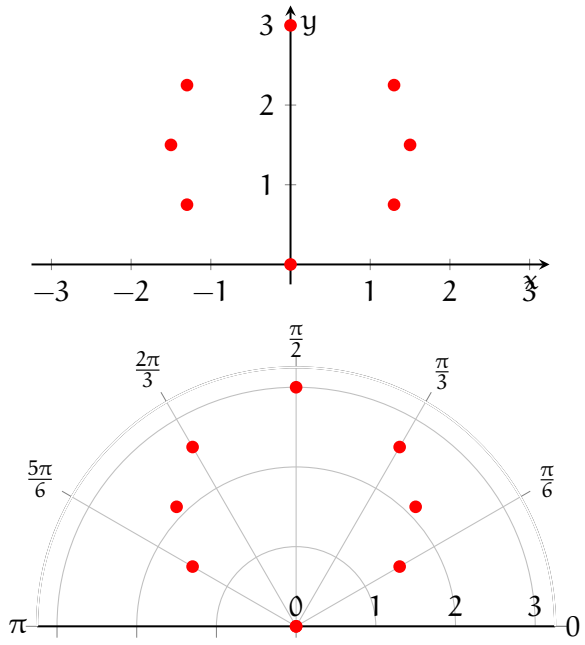



Figure 113.8: Several points for $r = 3\sin\theta$ plotted on Cartesian and polar coordinate systems

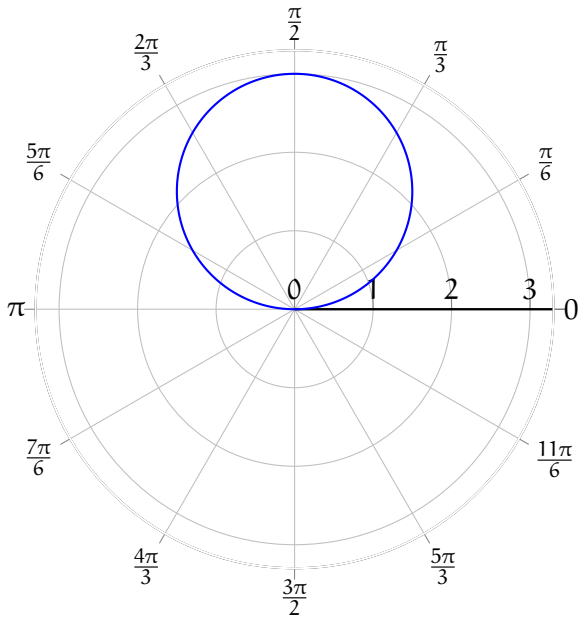
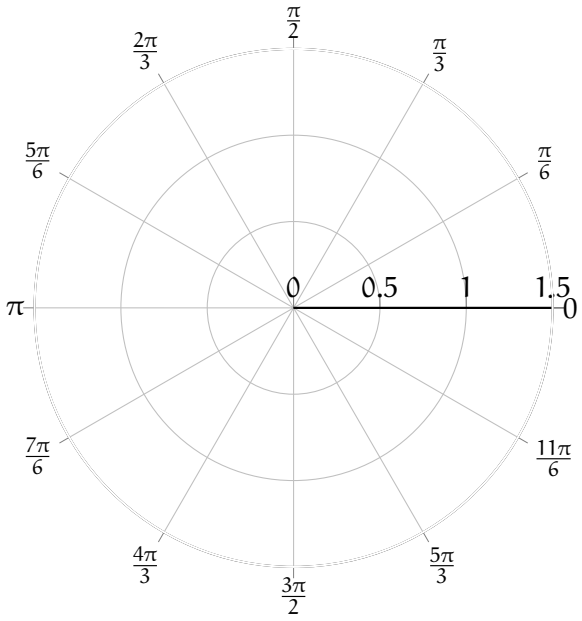



Figure 113.9: $r = 3\sin\theta$ plotted on a polar coordinate system

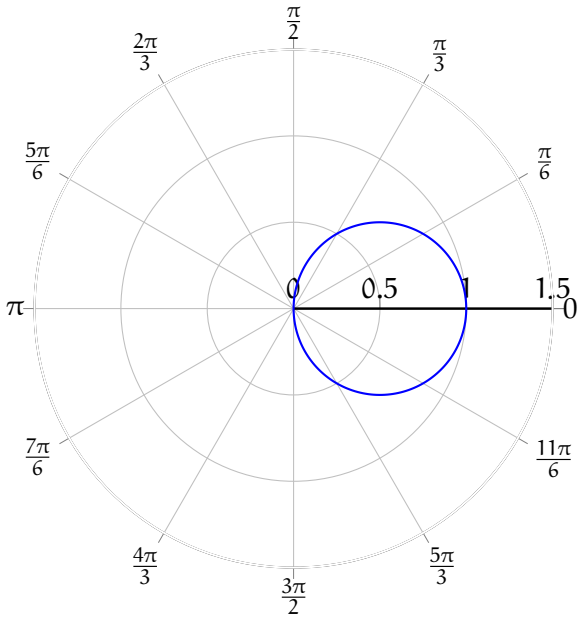
So, the polar equation $r = 3\sin\theta$ gives a circle with radius $\frac{3}{2}$ centered at $(0, \frac{3}{2})$.

**Example**: Describe the graph of $r = \cos\theta$. Feel free to make a rough plot on the blank

polar axis below:



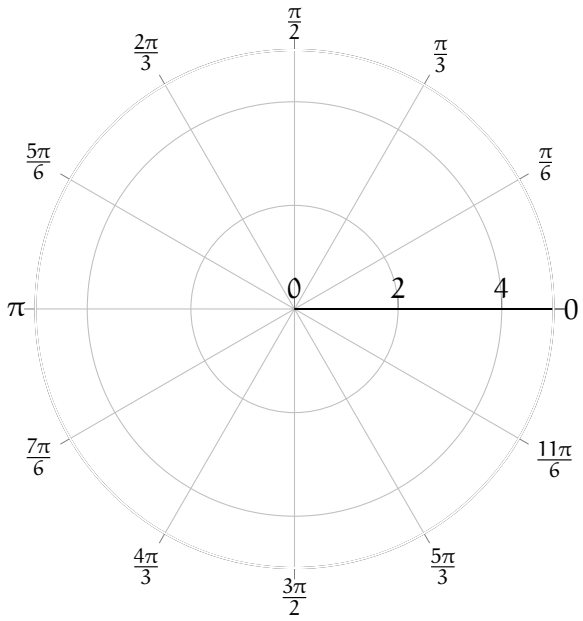

**Solution**: This plot will look like a circle of radius 0.5 centered at $(0.5, 0)$ (in polar coordinates).



### Exercise 292

Sketch the following polar functions on the provided polar axis for $0 \leq \theta < 2\pi$:

1. $r = 3$
2. $\theta = \pi$
3. $r = 2\cos\frac{\theta}{2}$
4. $r = -4\sin\theta$
5. $r = \theta$




Working Space

Answer on Page 1984

Chapter 114

# Differentiation

We have done some differentiation, but you haven't been given the real definition yet, because it is based on limits.

The idea is that we can find the slope between two points on the graph $a$ and $b$ like this:

$$m = \frac{f(b) - f(a)}{b - a}$$

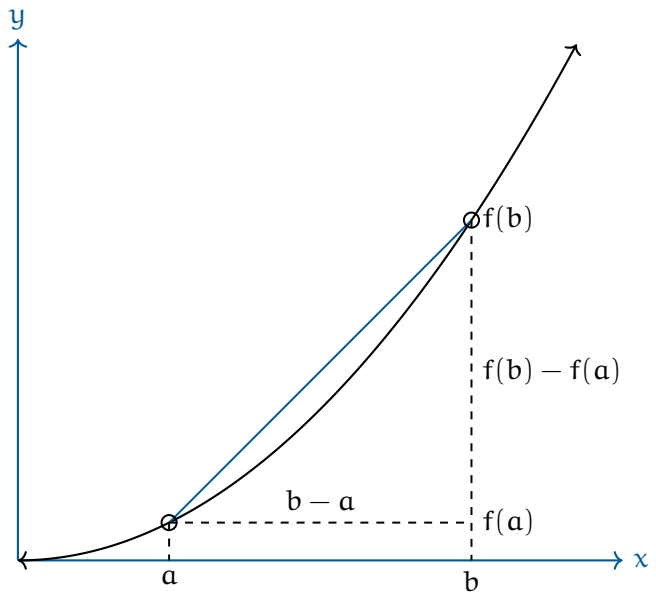



Figure 114.1: The slope of point $a$ using points $a$ and $b$.

If we want to find the slope at $a$, we take the limit of this as the $b$ goes to $a$:

$$f'(a) = \lim_{b \to b} \frac{f(b) - f(a)}{b - a}$$

This idea is usually expressed using $h$ as the difference between $b$ and $a$:

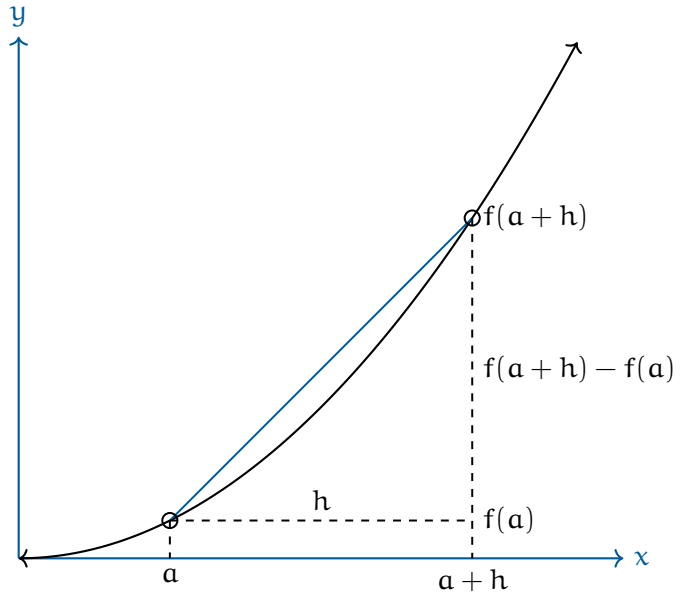



Figure 114.2: Limit of a point using h difference.

The formula then becomes:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

As $h$ decreases to very close to zero, almost infinitesimal, what we are finding is the sloe of the tangent line to point $a$.

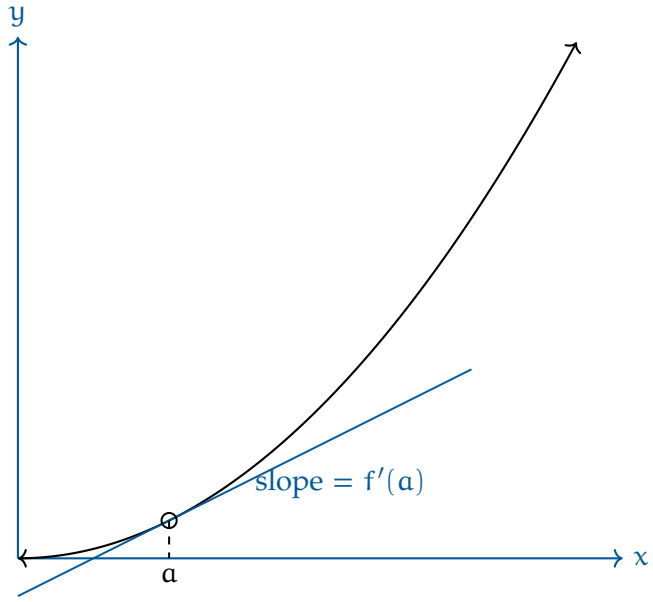



Figure 114.3: Slope at $a$ as $h$ approaches 0.

## 114.1 Differentiability

Warning: Not every function is differentiable everywhere. For example, if $f(x) = |x|$, you get a corner at zero.

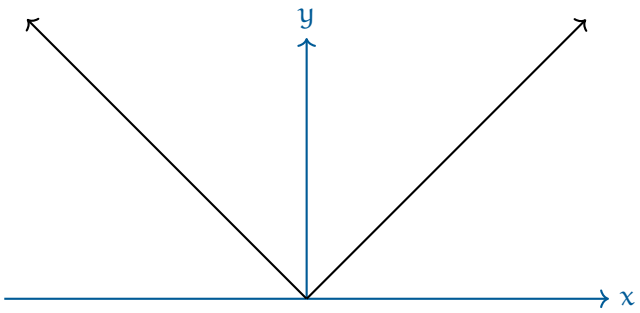



Figure 114.4: Absolute value function.

To the left of zero, the slope is -1. To the right of zero, the slope is 1. At zero? The derivative is not defined because the slopes are two different numbers.

If a function has a derivative everywhere, it is said to be *differentiable*. Generally, you can think of differentiable functions as smooth — their graphs have no corners or sharp turns. Another place where a function is not differentiable is at a vertical tangent, as the slope reaches $\pm\infty$.

It is important to note: if $f$ is differentiable at $a$, it *must* also be continuous at $a$.

Likewise, if $f$ is not continuous at $a$ it is not differentiable. An example of this is a jump discontinuity. FIXME diagram here of jump discontinuity.

### *Exercise 293*

[This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC exam.] Let $f$ be the function defined by $f(x) = \sqrt{|x-2|}$ for all $x$. Classify each of the following statements as true or false.

1. $f$ is continuous at $x = 2$.
2. $f$ is differentiable at $x = 2$.
3. $\lim_{x\to 2} f(x) = 0$.
4. $x = 2$ is a vertical asymptote of the graph of $f(x)$.

*Working Space*

*Answer on Page 1986*

## 114.2 Using the definition of derivative

Let's say that you want to know the slope of $f(x) = -3x^2$ at $x = 2$. Using the definition of the derivative, that would be:

$$f'(2) = \lim_{h\to 0} \frac{f(2+h) - f(2)}{h} = \lim_{h\to 0} \frac{-3(2+h)^2 - \left(-3(2)^2\right)}{h}$$
$$= \lim_{h\to 0} \frac{-12 - 12h + -3(h)^2 + 12}{h} = -12$$

# Chapter 115

# Derivatives

In calculus, the derivative of a function represents the rate at which the function is changing at a particular point. It is a fundamental concept that has vast applications in various fields, including physics.

## 115.1 Definition

The derivative of a function $f(x)$ at a point $x$ is defined as the limit:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{115.1}$$

provided this limit exists. In words, the derivative of $f$ at $x$ is the limit of the rate of change of $f$ at $x$ as the change in $x$ approaches zero. The derivative of a function is equal to the slope of the function. The derivative of a function, $f(x)$, is denoted as $f'(x)$ (read out loud as "f prime of x") or $df/dx$. The origin of this definition was shown in the previous chapter, Differentiation.

### 115.1.1 Estimating the Derivative

Consider the function $f(x) = x^2$. Suppose we want to write an equation for a line that is tangent to the curve at $x = 2$ (see figure 123.2). We already have a point that the line passes through: $(2, 4)$. To write an equation for the tangent line, we would need to know its slope, $m$.

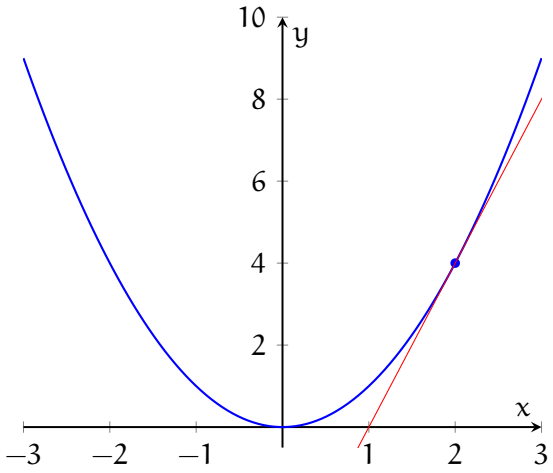



Figure 115.1: The red line is tangent to $f(x) = x^2$ at the point $(2, 4)$

We can estimate the slope by choosing points on either side of P, drawing a line through those points, and calculating the slope of that secant line (it is a secant line because it intersects the curve more than once). See figure 115.2 for a visualization.

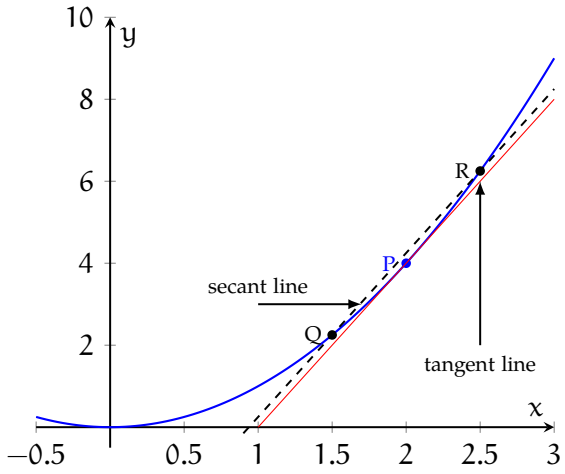



Figure 115.2: The slope of the secant line is approximately the slope of the tangent line

As the points Q and R get closer to P, the better the estimate becomes.

Much scientific data is not described as continuous functions, but rather as discrete data points. Consider the following data of a falling object:

| time (seconds) | height (m) |
|---|---|
| 0 | 50 |
| 0.5 | 48.775 |
| 1 | 45.1 |
| 1.5 | 38.975 |
| 2 | 30.4 |
| 2.5 | 19.375 |
| 3 | 5.9 |

A graph of the data is shown in figure 115.3.

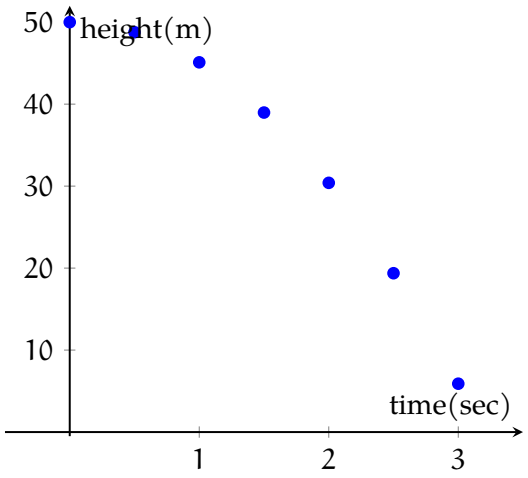



Figure 115.3: The height of a falling object over time

Suppose we wanted to estimate the velocity of the falling object at $t = 1.5s$. Recall that velocity is given by the change in position divided by the change in time. We can select data points on either side of $t = 1.5s$ and use them to find the average velocity from $t = 1s$ and $t = 2s$ (see figure 174.8):

$$v = \frac{h_2 - h_2}{t_2 - t_1} = \frac{30.4m - 45.1m}{2s - 1s} = -14.7\frac{m}{s}$$

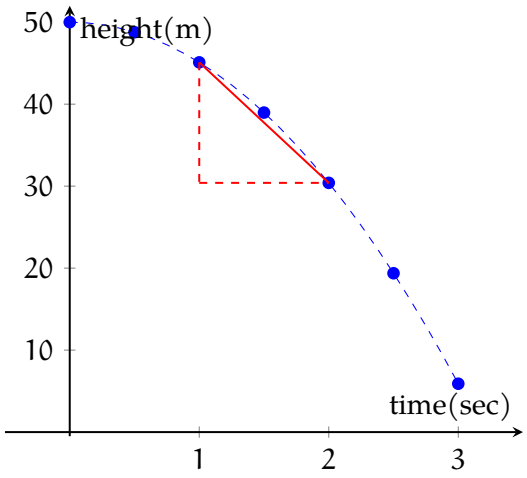



Figure 115.4: The slope of the line connecting the data points on either side of t = 1.5s is approximately the velocity of the falling object at t = s

**Example**: A 1000-gallon tank drains from the bottom in 30 minutes. The volume left in the tank is recorded every 5 minutes, as shown in the data table below. Use the data to estimate $V'(15)$ and $V'(25)$, including appropriate units. At which time is the tank draining faster?

| t (min) | V (gal) |
|---|---|
| 5 | 694 |
| 10 | 444 |
| 15 | 250 |
| 20 | 111 |
| 25 | 28 |
| 30 | 0 |

**Solution**: To estimate $V'(15)$, we find the slope of the line connecting the data points on either side of t = 15:

$$V'(15) \approx \frac{111\text{gal} - 444\text{textgal}}{20\text{min} - 10\text{min}}$$

$$V'(15) \approx \frac{-333\text{gal}}{10\text{min}}$$

$$V'(15) \approx -33.3\frac{\text{gal}}{\text{min}}$$

And we can use the data at t = 20 and t = 30 to estimate $V'(25)$:

$$V'(25) \approx \frac{0\text{gal} - 111\text{gal}}{30\text{min} - 20\text{min}}$$

$$V'(25) \approx \frac{-111\text{gal}}{10\text{min}}$$

$$V'(25) \approx -11.1\frac{\text{gal}}{\text{min}}$$

Both answers are negative because the tank is emptying, and the tank is draining faster at $t = 15$ than at $t = 25$.

### *Exercise 294*

[This question was originally presented as a free-response, calculator-allowed question on the 2012 AP Calculus BC Exam.] The temperature of water in a tub at time $t$ is modeled by a function, $W$, where $W(t)$ is measured in degrees Fahrenheit and $t$ is measured in minutes. Values of $W(t)$ at selected times for the first 20 minutes are given in the table. Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of the problem.

| $t$ (minutes) | $W(t)$ (degrees Fahrenheit) |
|---|---|
| 0 | 55.0 |
| 4 | 57.1 |
| 9 | 61.8 |
| 15 | 67.9 |
| 20 | 71.0 |

*Working Space*

*Answer on Page 1987*

## 115.2 The Derivative as a Function

We have seen how to estimate the value of a derivative at a specific point on a graph. Suppose we wanted to describe the slope of a graph everywhere. That is: can we find a function, $g(x)$ that describes the slope of another function, $f(x)$, over the domain of $f$? Using the definition of a derivative, we can. You have already seen an algorithm to find the derivatives of polynomial functions (see chapter Differentiating Polynomials FIXME

can this be linked). Recall that for a function, $f(x) = x^n$, the derivative is $f'(x) = nx^{n-1}$. Here is the proof:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^n - x^n}{h}$$

In order to expand the polynomial, $(x+h)^n$, we'll need to apply the Binomial Theorem, which tells us that:

$$(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \cdots + nab^{n-1} + b^n$$

Substituting this into our limit definition of a derivative, we see that:

$$f'(x) = \lim_{h \to 0} \frac{x^n + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \cdots nxh^{n-1} + h^n - x^n}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \cdots nxh^{n-1} + h^n}{h}$$

$$f'(x) = \lim_{h \to 0} nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}h + \cdots nxh^{n-2} + h^{n-1}$$

$$f'(x) = nx^{n-1}$$

**Example**: Use the limit definition of a derivative to find $f'(x)$ if $f(x) = 2x^3 - x^2$.

**Solution**: According to the limit definition, $f'$ is:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \frac{\left[2(x+h)^3 - (x+h)^2\right] - \left[2x^3 - x^2\right]}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{\left[2\left(x^3 + 3hx^2 + 3h^2x + h^3\right) - \left(x^2 + 2xh + h^2\right)\right] - 2x^3 + x^2}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{2x^3 - 2x^3 + 6hx^2 + 6h^2x + 6h^3 - x^2 + x^2 - 2xh - h^2}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{6hx^2 + 6h^2x + 6h^3 - 2xh - h^2}{h}$$

$$f'(x) = \lim_{h \to 0} 6x^2 + 6hx + 6h^2 - 2x - h = 6x^2 - 2x$$

Therefore, if $f(x) = 2x^3 - x^2$, then $f'(x) = 6x^2 - 2x$.

### *Exercise 295* Finding Functions for Derivatives

Use the limit definition of a derivative to find an equation for $f'(x)$.

*Working Space*

1. $f(x) = mx + b$
2. $f(x) = \sqrt{16 - x}$
3. $f(x) = \frac{x^2-1}{2x-3}$

*Answer on Page 1987*

## 115.3 Applications in Mathematics

### 115.3.1 l'Hôpital's Rule

Consider the function $h(x) = \frac{\ln x}{x-1}$ and suppose we are interested in the behavior of $h(x)$ around $x = 1$. As $x$ approaches 0 and the denominator $x - 1$ also approaches 0. This gives the an indeterminate result:

$$\lim_{x \to 1} \frac{\ln x}{x - 1} = \frac{0}{0}$$

This is exactly the situation where L'Hôpital's Rule applies.

In Chapter 3, we will learn about Quotient Rule, which when directly applied will give this indeterminate form! Keep an eye out for that.

Now, looking at the graph of $h(x)$ (see figure 115.5), we can guess that $\lim_{x \to 1} \frac{\ln x}{x-1} = 1$.

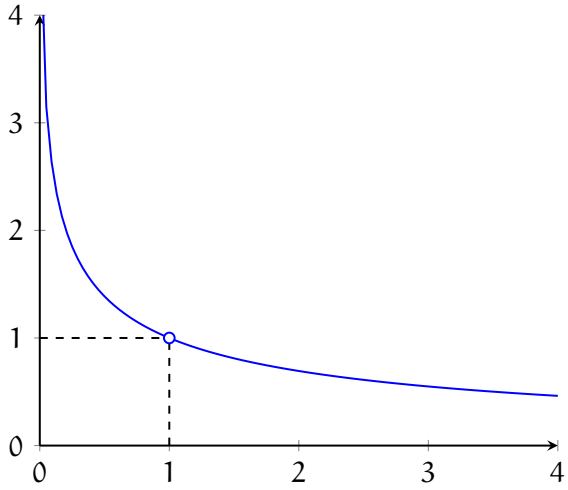



Figure 115.5: $h(x) = \frac{\ln x}{x-1}$

Let's examine the numerator and denominator separately: we'll define $f(x) = \ln x$ and $g(x) = x - 1$ (see figure 115.6).

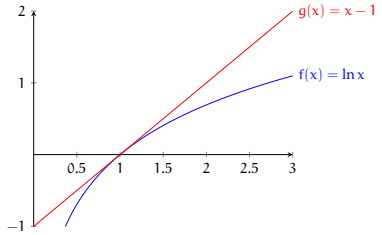



Figure 115.6: Examining each part of $\frac{\ln x}{x-1}$ separately

If we zoom in very far around $x = 1$, the graphs begin to look linear (see figure 115.7):

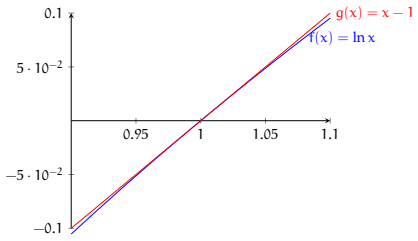



Figure 115.7: As we zoom in, the graph of $\ln x$ appears linear

We can approximate these graphs as linear functions with slopes $m_1$ and $m_2$, so that the blue curve is approximated as $y = m_1(x - 1)$ and the red curve is approximated as

$y = m_2(x-1)$. The ratio of the functions would then be

$$\frac{m_1(x-1)}{m_2(x-1)} = \frac{m_1}{m_2}$$

which is the same as the ratio of the derivatives of our linear approximations. This suggests l'Hopital's rule, that the limit of a ratio is the same as the limit of the ratio of the derivatives for certain indeterminate forms:

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

.

Let's apply l'Hopital's rule to our limit of $h(x)$:

$$\lim_{x\to 1} \frac{\ln x}{x-1} = \lim_{x\to 1} \frac{\frac{d}{dx}\ln x}{\frac{d}{dx}(x-1)} = \lim_{x\to 1} \frac{\frac{1}{x}}{1} = 1$$

Notice our result with l'Hôpital's rule matches our guess based on the graph of $h(x) = \frac{\ln x}{x-1}$.

L'Hospital's rule also applies to the indeterminate result $\frac{\pm\infty}{\pm\infty}$. For a limit of the form $\lim_{x\to a} \frac{f(x)}{g(x)}$, l'Hôpital's rule applies if:

1. the original limit is of the indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$
2. $f$ and $g$ are differentiable on an interval containing $a$ (but possibly not differentiable at $a$)
3. $g'(x) \neq 0$ on said interval

**Example**: Determine $\lim_{x\to\infty} \frac{e^x}{x^2}$.

**Solution**: We begin by evaluating the limit:

$$\lim_{x\to\infty} \frac{e^x}{x^2} = \frac{e^\infty}{\infty^2} = \frac{\infty}{\infty}$$

This is an indeterminate form that we can apply l'Hopital's rule to:

$$= \lim_{x\to\infty} \frac{\frac{d}{dx}e^x}{\frac{d}{dx}x^2} = \lim_{x\to\infty} \frac{e^x}{2x}$$

Evaluating this limit, we get another indeterminate form:

$$= \frac{e^{\infty}}{2 \cdot \infty} = \frac{\infty}{\infty}$$

Don't panic! We can apply l'Hôpital's rule again (in fact, we can apply l'Hôpital's rule as many times as needed to evaluate a limit, as long as we keep getting $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$):

$$= \lim_{x \to \infty} \frac{\frac{d}{dx} e^x}{\frac{d}{dx} 2x} = \lim_{x \to \infty} \frac{e^x}{2} = \frac{\infty}{2} = \infty$$

and therefore, $\lim_{x \to \infty} \frac{e^x}{x^2} = \infty$.

## Exercise 296

What is $\lim_{x \to 0} \frac{\tan x - x}{x^3}$?

*Working Space*

*Answer on Page 1988*

Note that this rule only applies if it is in *indeterminate form*. The most common indeterminate forms are

$$\frac{0}{0} \quad \text{and} \quad \frac{\infty}{\infty},$$

which you may end up with when evaluating quotients!

However, indeterminate forms can also appear in other ways. Some common examples include:

$$0 \cdot \infty, \quad \infty - \infty, \quad 0^0, \quad 1^{\infty}, \quad \infty^0.$$

These expressions are indeterminate because their values cannot be determined without evaluating further. In many cases, such limits can be rewritten into a quotient that results in either

$$\frac{0}{0} \quad \text{or} \quad \frac{\infty}{\infty}.$$

This allows l'Hôpital's Rule to be applied!

Example: Evaluate

$$\lim_{x \to 0^+} x \ln x.$$

Solution: As $x \to 0^+$, we have $x \to 0$ and $\ln x \to -\infty$, producing the indeterminate form

$$0 \cdot (-\infty).$$

Rewrite the expression as a quotient:

$$x \ln x = \frac{\ln x}{1/x}.$$

Now the limit has the form

$$\frac{-\infty}{\infty},$$

so l'Hôpital's Rule applies. Differentiate the numerator and denominator:

$$\lim_{x \to 0^+} \frac{\ln x}{1/x} = \lim_{x \to 0^+} \frac{1/x}{-1/x^2}.$$

Simplifying,

$$\lim_{x \to 0^+} (-x) = 0.$$

Therefore,

$$\lim_{x \to 0^+} x \ln x = 0.$$

## Exercise 297

Evaluate each of the following limits, using l'Hôpital's rule where needed.

*Working Space*

1. $\lim_{x \to 3} \frac{x-3}{x^2-9}$
2. $\lim_{x \to 1/2} \frac{6x^2+5x-4}{4x^2+16x-9}$
3. $\lim_{x \to 0^+} \frac{\ln x}{\sqrt{x}}$
4. $\lim_{x \to \pi} \frac{1+\cos x}{1-\cos x}$
5. $\lim_{x \to 1} \frac{x \sin x - 1}{2x^2-x-1}$

*Answer on Page 1989*

### 115.3.2 Mean Value Theorem

The Mean Value Theorem (MVT) states that on an interval $[a, b]$ where a continuous function $f$ is differentiable on an open interval $(a, b)$, there is at least one point where the tangent line to $f$ has the same slope as a line connecting the points $(a, f(a))$ and $(b, f(b))$. Consider the graph of $f(x) = x^2$ (see figure 115.8). The line connecting the points $(-1, 1)$ and $(2, 4)$ has a slope of $\frac{1}{2}$. MVT tells us there must be *at least one* point, $c$, on the interval $x \in (-1, 2)$ where $f'(c) = \frac{1}{2}$. We can find this point by setting $f'(x)$ equal to $\frac{1}{2}$:

$$2x = \frac{1}{2} \rightarrow x = \frac{1}{4}$$

Examining the figure 115.8, you can see that the tangent at $f(\frac{1}{4})$ (the black line) is parallel to the red line connecting $(-1, f(-1))$ and $(2, f(2))$.

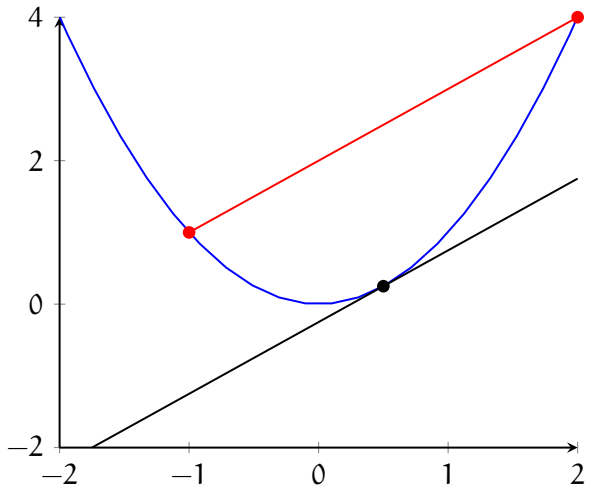



Figure 115.8: $f(x) = x^2$

Note that MVT doesn't tell us *where* $f'(x)$ is parallel to the line connecting $(a, f(a))$ and $(b, f(b))$, just that some value $c$ exists that satisfies the condition.

**Example**: Consider a hammer thrown upwards at 5 $\frac{m}{s^2}$ on Earth (where the acceleration due to gravity is approximately $-9.8\frac{m}{s^2}$).

**Solution**: We can use the MVT to show that there must be some point in the hammer's path upwards where the velocity of the hammer is exactly equal to its average velocity as it flies through the air.

The hammer's rise can be described with the function $y(t) = 5t - 4.9t^2$. The hammer

reaches its peak at approximately $t = 0.51$. So, we are looking for some value, $c$, such that

$$y'(c) = \frac{y(0.51) - y(0)}{0.51 - 0} = \frac{5(0.51) - 4.9(0.51^2)}{0.51} = \frac{1.2755}{0.51} = 2.5$$

Solving $y'(t) = 5 - 9.8t = 2.5$, we find that the $c$ that satisfies the MVT is approximately 0.255. This result is illustrated in figure 172.1:

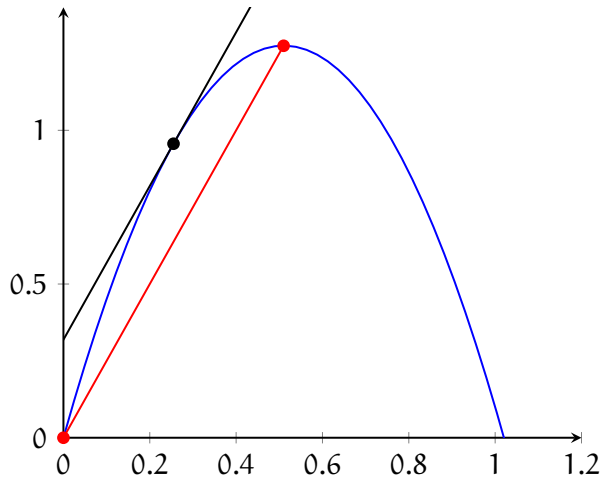



Figure 115.9: The height of a hammer tossed upwards at $5\frac{m}{s}$

**MVT Practice**

### *Exercise 298*

AT 3:30 PM, a car's speedometer reads $30\frac{mi}{hr}$. At 3:40 PM, it reads $50\frac{mi}{hr}$. Show that at some time between 3:30 and 3:340 PM, the car's acceleration is exactly $120\frac{mi}{hr^2}$.

*Working Space*

*Answer on Page 1989*

### *Exercise 299*

Find the number c that satisfies the MVT on the given interval.

(a) $f(x) = \sqrt{x}$, $[0, 4]$

(b) $f(x) = e^{-x}$, $[0, 2]$

(c) $f(x) = \ln x$, $[1, 4]$

*Working Space*

*Answer on Page 1990*

## 115.4 Applications in Physics

In physics, derivatives play a vital role in describing how quantities change with respect to one another.

### 115.4.1 Velocity and Acceleration

In kinematics, the derivative of the position function with respect to time gives the velocity function, and further taking the derivative of the velocity function gives the acceleration function. For example, if $s(t)$ represents the position of an object at time $t$, then the velocity $v(t)$ and acceleration $a(t)$ are given by:

$$v(t) = \frac{ds}{dt} \quad \text{and} \quad a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2} \tag{115.2}$$

**Practice**

A particle's motion is described by $s(t) = t^3 - 6t^2 + 6t$, where $t$ is measured in seconds and $s$ is measured in meters. Answer the following questions about the particle's motion:

### Exercise 300

Find the velocity at time t.

*Working Space*

*Answer on Page 1991*

### Exercise 301

What is the velocity after 2s? After 4s?

*Working Space*

*Answer on Page 1991*

### Exercise 302

When is the particle at rest?

*Working Space*

*Answer on Page 1991*

#### 115.4.2 Force and Momentum

In mechanics, the derivative of the momentum of an object with respect to time gives the net force acting on the object, as stated by Newton's second law of motion:

$$F = \frac{dp}{dt} \tag{115.3}$$

where F is the force, p is the momentum, and t is the time.

Chapter 116

# Rules for Finding Derivatives

Derivatives play a key role in calculus, providing us with a means of calculating rates of change and the slopes of curves. In this chapter, we present some common rules used to calculate derivatives.

## 116.1 Constant Rule

The derivative of a constant is zero. If $c$ is a constant and $x$ is a variable, then:

$$\frac{d}{dx}c = 0 \tag{116.1}$$

## 116.2 Power Rule

For any real number $n$, the derivative of $x^n$ is:

$$\frac{d}{dx}x^n = nx^{n-1} \tag{116.2}$$

## 116.3 Product Rule

The derivative of the product of two functions is:

$$\frac{d}{dx}(fg) = f'g + fg' \tag{116.3}$$

where $f'$ and $g'$ denote the derivatives of $f$ and $g$, respectively.

## 116.4 Quotient Rule

The derivative of the quotient of two functions is:

$$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2} \tag{116.4}$$

## 116.5 Chain Rule

The derivative of a composition of functions is:

$$\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x) \tag{116.5}$$

## 116.6 Practice

### *Exercise 303*

If $f$ is the function given, find $f'$.

*Working Space*

1. $f(x) = x \sin x$
2. $f(x) = (x^3 - \cos x)^5$
3. $f(x) = \sin^3 x$

*Answer on Page 1992*

### *Exercise 304*

Let $f(x) = 7x - 3 + \ln x$. Find $f'(x)$ and $f'(1)$

*Working Space*

*Answer on Page 1992*

### Exercise 305

[This question was originally presented as a multiple-choice, no-calculator question on the 2012 AP Calculus BC exam.] The position of a particle in the $xy$-plane is given by the parametric equations $x(t) = t^3 - 3t^2$ and $y(t) = 12t - 3t^2$. State a coordinate point $(x, y)$ at which the particle is at rest.

*Working Space*

*Answer on Page 1992*

### Exercise 306

Let $f(x) = \sqrt{x^2 - 4}$ and $g(x) = 3x - 2$. Find the derivative of $f(g(x))$ at $x = 3$.

*Working Space*

*Answer on Page 1992*

### Exercise 307

The particle's position on the x-axis is given by $x(t) = (t - a)(t - b)$, where $a$ and $b$ are constants and $a \neq b$. At what time(s) is the particle at rest?

*Working Space*

*Answer on Page 1992*

### Exercise 308

[This question was originally presented as a multiple-choice, no-calculator question on the 2012 AP Calculus BC exam.] Let $f(x) = \frac{x}{x+2}$. At what values of $x$ does $f$ have the property that the line tangent to $f$ has a slope of $\frac{1}{2}$?

*Working Space*

*Answer on Page 1993*

### Exercise 309

For $t \geq 0$, the position of a particle moving along the x-axis is given by $x(t) = \sin t - \cos t$. (a) When does the velocity first equal 0? (b) What is the acceleration at the time when the velocity first equals 0?

*Working Space*

*Answer on Page 1993*

### Exercise 310

The graph of $y = e^{(\tan x)} - 2$ crosses the x-axis at one point on the interval $[0, 1]$. What is the slope of the graph at this point?

*Working Space*

*Answer on Page 1994*

### Exercise 311

The function f is defined by $f(x) = \sqrt{25 - x^2}$ for $-5 \leq x \leq 5$.
(a) Find $f'(x)$.
(b) Write an equation for the line tangent to the graph at $x = -3$.

*Working Space*

*Answer on Page 1994*

### Exercise 312

For $0 \leq t \leq 12$, a particle moves along the x-axis. The velocity of the particle at a time t is given by $v(t) = \cos \frac{\pi}{6} t$. What is the acceleration of the particle at time $t = 4$?

*Working Space*

*Answer on Page 1995*

### Exercise 313

[This question was originally presented as a multiple-choice, calculator-allowed question on the 2012 AP Calculus BC exam.] Let f and g be the functions given by $f(x) = e^x$ and $g(x) = x^4$. On what intervals is the rate of change of $f(x)$ greater than the rate of change of $g(x)$?

*Working Space*

*Answer on Page 1995*

## 116.7 Conclusion

These rules form the basis for calculating derivatives in calculus. Many more complex rules and techniques are built upon these fundamental rules.

Chapter 117

# First and Second Derivatives and the Shape of a Function

## 117.1 Using first derivatives to describe a function

### 117.1.1 Critical Values

Let's re-examine our graph showing the height of a hammer tossed in the air:

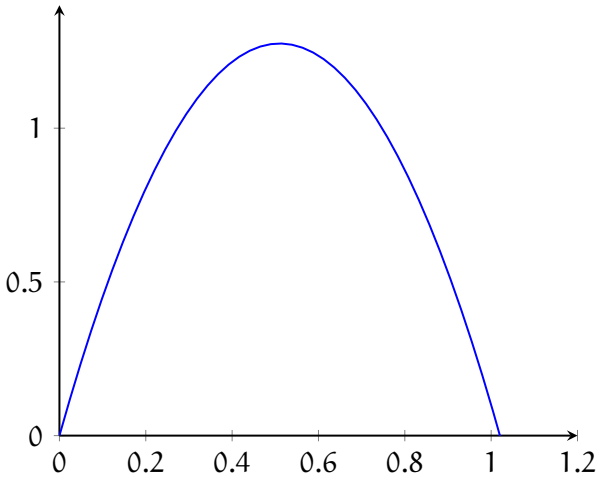



Figure 117.1: Height of a hammer over time

As you can see, the hammer reaches its peak around $t \approx 0.5s$ (see figure 117.1). Let's add tangent lines just before and after the peak of the hammer's path, so we can more easily examine how the slope of the graph changes:

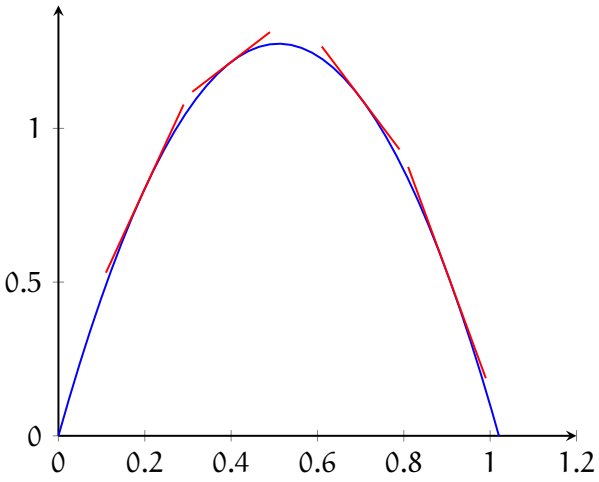



Figure 117.2: height of a hammer over time

In figure 117.2, we see that the slope changes from positive to negative as $t$ increases. That implies that $f'(t)$ also changes from positive to negative. In fact, at the highest point of the hammer's flight, the slope (and therefore $f'(t)$) is exactly zero! In general,

1. If $f'(x) > 0$ (positive slope) on an interval, then $f(x)$ is increasing on that interval.
2. If $f'(x) < 0$ (negative slope) on an interval, then $f(x)$ is decreasing on that interval.

**Example 1**: Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing.

**Solution**: We want to find the intervals where $f'(x) > 0$. First, we take the derivative to find $f'(x)$:

$$f'(x) = 12x^3 - 12x^2 - 24x$$

It will be easier to analyze the value of $f'(x)$ if we factor it so:

$$f'(x) = 12x(x-2)(x+1)$$

To determine where $f'(x) > 0$, we start by finding where $f'(x) = 0$ (in this case, this is true when $x = -1, 0, 2$). These values of $x$ are called *critical values*, and we will use them to divide $f'(x)$ into intervals. (Critical values are also called critical numbers, and we will use both in this text.) On each of these intervals, $f'(x)$ must be always positive or always negative. This is shown in the graph below:

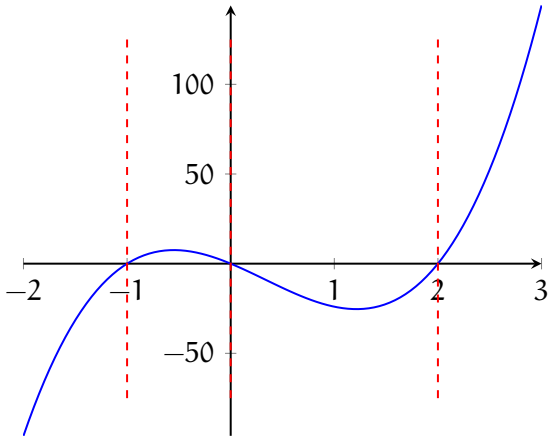



Figure 117.3: $f'(x)$ with critical values

As you can see in figure 117.3, $f'(x) > 0$ on two intervals: $x \in (-1, 0)$ and $x \in (2, \infty)$. These are open intervals because $f'(x) = 0$ at $x = -1$, $x = 0$, and $x = 2$. But what if we had a more complex function, or didn't have the resources to graph it?

We can use a table to help us analyze the value of $f'(x)$ (and therefore the behavior of $f(x)$). For each interval around the critical values, we can determine if $f'(x)$ is positive or negative by noting the value of the factors of $f'(x)$, which are $12x$, $x - 2$, and $x + 1$ in this case. For example, for $x < -1$, $12x < 0$, $(x - 2) < 0$, and $(x + 1) < 0$. Three negatives multiplied together is also negative. Therefore, for $x < -1$, $f'(x)$ is negative and $f(x)$ is decreasing. We can analyze all of the intervals similarly and log the results in a table:

| $x$ | $12x$ | $x - 2$ | $x + 1$ | $f'(x)$ | $f(x)$ |
|---|---|---|---|---|---|
| $x < -1$ | negative | negative | negative | negative | decreasing |
| $-1 < x < 0$ | negative | negative | positive | positive | increasing |
| $0 < x < 2$ | positive | negative | positive | negative | decreasing |
| $2 < x$ | positive | positive | positive | positive | increasing |

Notice the table method yields the same result as examining the graph: $f(x)$ is increasing for $x \in (0, -1)$ and $x \in (2, \infty)$, which can also be written as $x \in (-1, 0) \cup (2, \infty)$.

### *Exercise 314*

Let $g$ be the function given by $g(x) = x^2 e^{kx}$, where $k$ is a constant. For what value(s) of $k$ does $g$ have a critical value at $x = \frac{2}{3}$?

*Working Space*

*Answer on Page 1995*

### 117.1.2 Local Extrema

Examine the graphs of $x^2$, $\sin x$, and $y = \sqrt{4 - x^2}$ below. Each has a dot at a local extreme (either a local minimum or local maximum). Sketch what you think the tangent line to the graph would be at each local extreme. Use this to estimate the value of the derivative at that point.

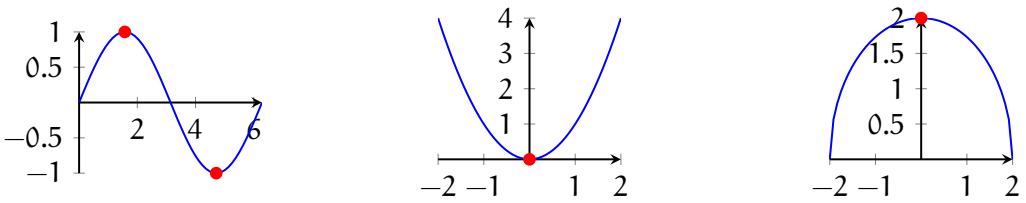



Figure 117.4: Three functions with highlighted points.

You should notice that all of the tangent lines are horizontal. Since the tangent lines at these local extrema have a slope of 0, that tells us $f'(x) = 0$ at these points as well. In fact, for *all* local minima and maxima, the value of the derivative is zero at that point. However, the converse statement is not necessarily true; just because the derivative is zero at some $x = c$, it does not mean there is a local extrema at $f(c)$. Consider $f(x) = x^3 + 3$, shown in figure 117.5:

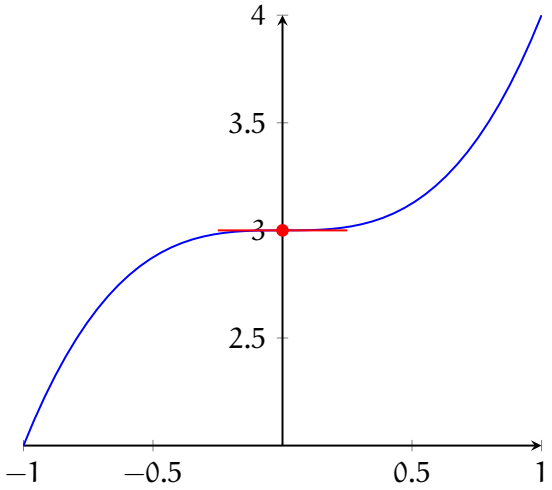



Figure 117.5: $f(x) = x^3 + 3$

At $x = 0$, $f'(x) = 0$, but there is not a local extreme. For a local extreme to exist, the graph of $f(x)$ must change from increasing to decreasing, or vice versa. Look closely at figure 117.5: the function is increasing for $x < 0$ and $x > 0$. Another way of saying this is to note that the graph of $f'(x)$ touches but does not cross the x-axis in this case:

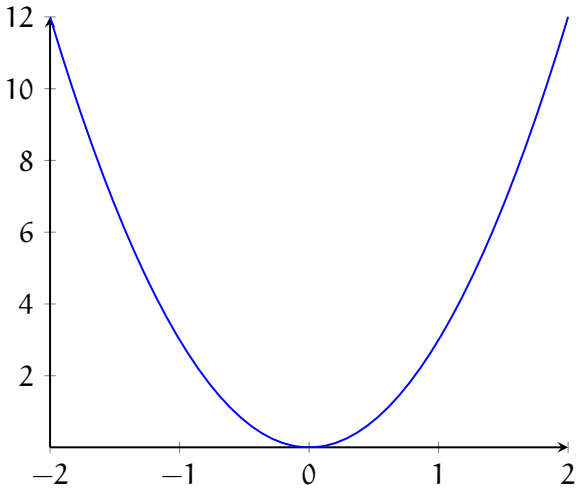



Figure 117.6: $f'(x) = 3x^2$

If $f(x)$ changes from increasing to decreasing, then $f'(x)$ is changing from positive to negative (i.e. crossing the x-axis). Look at the derivative of $f(x) = \sin x$, $f'(x) = \cos x$, presented in figure 117.7. The x-values where local extrema exist on $f(x)$ are marked in red (recall $\sin x = \pm 1$ when $x = \frac{n\pi}{2}$):

As you can see, local extrema are indicated when $f'(x)$ crosses the x-axis. If $f'(x)$ is negative to the left of $x = c$ and positive to the right, then $f(x)$ has a local minimum at $x = c$. On the other hand, if $f'(x)$ is positive to the left of $x = c$ and negative to the right, then $f(x)$ has a local maximum at $x = c$. Any value of $x = c$ where $f'(c) = 0$ is called a **critical number** or a **critical value**. Values where $f(c)$ does not exist are also a critical numbers.

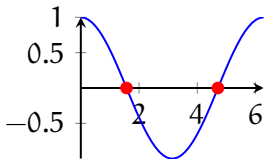


Figure 117.7: $f'(x) = \cos x$

### 117.1.3 Practice: Interval of Increasing and Decreasing, Local Extrema

#### *Exercise 315*

Let $f$ be the function given by $f(x) = 300x - x^3$. On which of the following intervals is $f$ increasing?

*Working Space*

*Answer on Page 1996*

#### *Exercise 316*

Find the intervals on which $f(x) = x^3 - 3x^2 - 9x + 4$ is increasing or decreasing. Then, find all local minimum and/or maximum values of $f(x)$.

*Working Space*

*Answer on Page 1996*

### 117.1.4 Global Extrema

Now that we've learned how to identify local minima and maxima, let's expand the discussion to include global extrema. A global extreme is an absolute minimum or maximum value of a function over a particular interval or the entire domain of the function. Let's examine the graph of $f(x) = x^4 - 5x^3 + 6x^2$ over the domain $x \in [-1, 4]$.

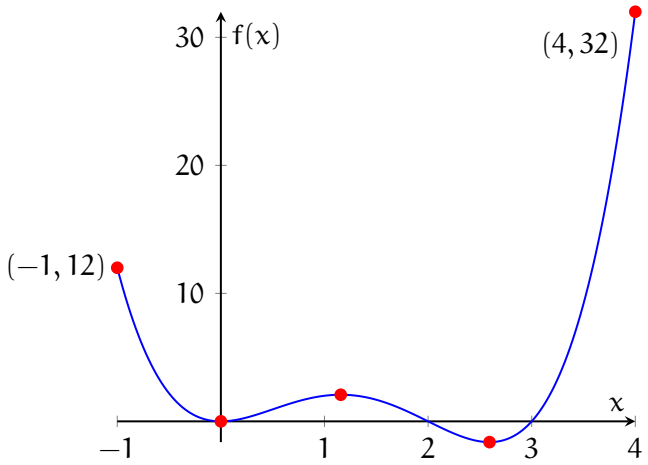



Figure 117.8: Graph of $f(x) = x^4 - 5x^3 + 6x^2$

As you can see in figure 117.8, $f(x)$ has two local minima and one local maximum. Additionally, the endpoints are labeled. To determine the *global* extrema, we need to examine the any local extrema (identified here graphically, but you can also identify them mathematically using that you learned in the "Local Extrema" subsection) **and** the endpoints of the domain (or the function's behavior at $\pm\infty$, if you are not restricted to a specific domain).

In the case of $f(x) = x^4 - 5x^3 + 6x^2$, for$x \in [-1, 4]$, the global maximum value is 32 at $x = 4$ and the global minimum is -1.623 at $x = 2.593$.

If a function is continuous on an interval, then there must exist a global maximum and global minimum on that interval. These global extrema may also be local extrema (as is the case for $f(2.593)$ in the example above) or not (as is the case for $f(4)$). Applying the Closed Interval Method is a straightforward way to identify global (absolute) extrema.

To find the global extrema of a continuous function, $f$, on a closed interval $[a, b]$:

1. Find the values of $f$ at the critical numbers of $f$ in $(a, b)$.
2. Find the values of $f$ at the endpoints of the interval.
3. The largest of the values from steps 1 and 2 is the absolute maximum; the smallest of the values is the absolute minimum.

Let's use the Closed Interval Method to determine the global extrema for the function $g(x) = x - 3\sin x$ on the interval $x \in [0, 2\pi]$.

To find the value of $g$ at any critical numbers, we must first identify the critical numbers. Recall that critical numbers are values where the first derivative of the function is 0 or

does not exist. To find critical numbers, we set $g'$ equal to 0:

$$g'(x) = 1 - 3\cos x = 0$$

$$3\cos x = 1$$

$$\cos x = \frac{1}{3}$$

$$x = 1.23, 5.052$$

Now, we substitute these critical numbers back into $g(x)$:

$$g(1.23) \approx -1.60$$

$$g(5.052) = 7.881$$

Now we need to check the endpoints:

$$g(0) = 0 - 3 * 0 = 0$$

$$g(2\pi) = 2\pi - 3 * 0 = 2\pi \approx 6.28$$

The results are presented in the table below:

| x | g(x) |
|---|---|
| 0 | 0 |
| 1.23 | -1.60 |
| 5.052 | 7.881 |
| 6.28 | 6.28 |

Therefore, for $g(x) = x - 3\sin x$ on the interval $x \in [0, 2\pi]$, the global maximum is $g(5.052) = 7.881$ and the global minimum is $g(1.23) = -1.60$.

### 117.1.5 Practice: Global Extrema

***Exercise 317***

Let f be the function defined by $f(x) = \frac{\ln x}{x}$. What is the absolute maximum value of f?

*Working Space*

*Answer on Page 1997*

### *Exercise 318*

Find the global minimum and maximum values on the stated interval.

*Working Space*

1. $f(x) = 12 + 4x - x^2$, $[0, 5]$
2. $f(t) = \frac{\sqrt{t}}{1+t^2}$, $[0, 2]$
3. $f(t) = 2\cos t + \sin 2t$, $[0, \frac{\pi}{2}]$
4. $f(x) = \ln x^2 + x + 1$, $[-1, 1]$

*Answer on Page 1997*

## 117.2 Sketching f from f'

Now that we know how the shape of $f$ is related to the value of $f'$, we can predict the shape of $f$ if we are given $f'$. Take the example $f'(x) = -(x-1)(x-5)$, shown in figure 117.9:

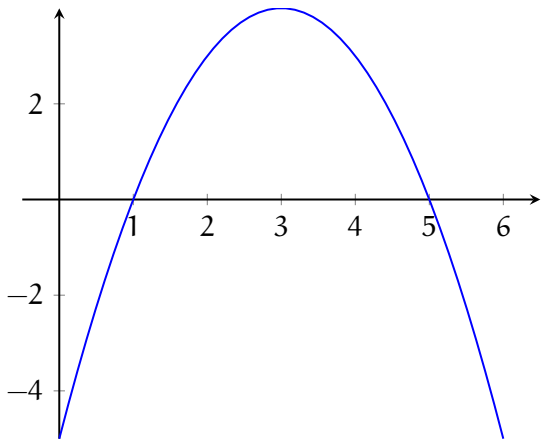



Figure 117.9: Graph of $f' = -(x-1)(x-5)$

Using the graph of $f'$, we can construct an approximate sketch of $f$. First, let's identify the critical numbers. Where does $f' = 0$? Take a second to examine the graph of $f'$ above and jot down what you think the critical numbers are.

You should recall that critical numbers are x-values where $f' = 0$. Examining the graph of $f'$, we see that $f' = 0$ at $x = 1$ and $x = 5$. We can now use a table to describe the behavior of $f$:

| $x$ | $x-1$ | $x-5$ | $f'$ | behavior of $f$ |
|---|---|---|---|---|
| $x < 1$ | negative | negative | negative | decreasing |
| $x = 1$ | zero | negative | zero | local minimum |
| $1 < x < 5$ | positive | negative | positive | increasing |
| $x = 5$ | positive | zero | zero | local maximum |
| $x > 5$ | positive | positive | negative | decreasing |

We can use this information to sketch a possible graph of $f$. We start by noting the location of local extrema:

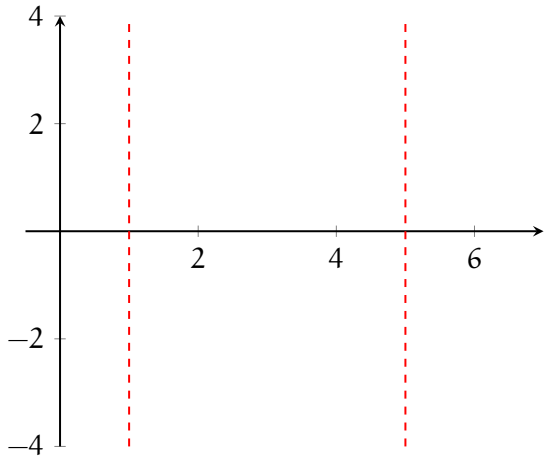



Figure 117.10: Possible graph of $f$

We know there is a local minimum at $x = 1$ and a local maximum at $x = 5$. We can add sketches around these values to indicate what we know about $f$:

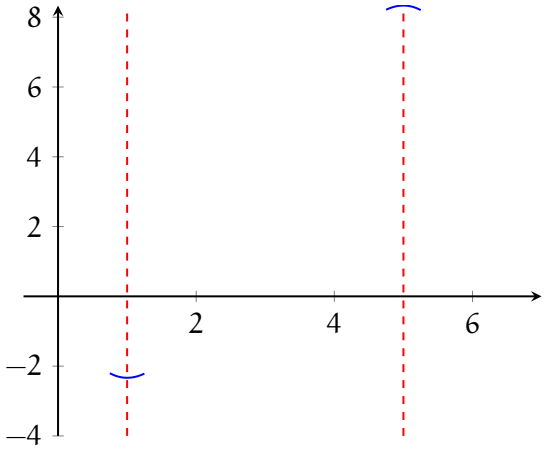



Figure 117.11: Possible graph of f

Lastly, we know f is increasing on $1 < x < 5$ and decreasing everywhere else, so we fill in the space between our local extrema:

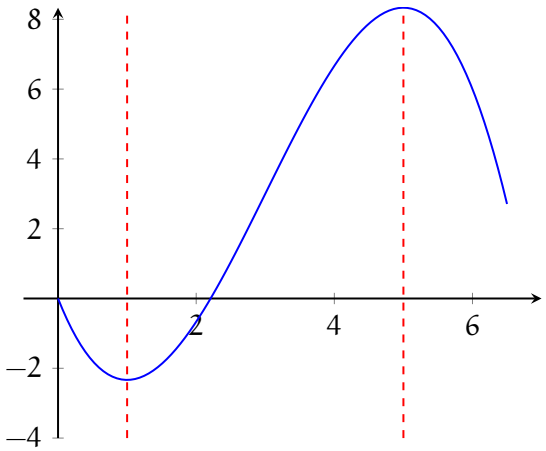



Figure 117.12: Possible graph of f

However, figure 117.12 is only a *possible* graph of f. Analyzing $f'$ reveals the shape of f, but not how high or low it is on the y-axis. Recall that the derivative of a constant is zero. Therefore, any $+c$ (where c is a constant) is lost when taking the derivative. So, there are many sketches of f that fulfill the behavior of f indicated by $f'$. You can see several of the possible sketches for f in figure 117.13.

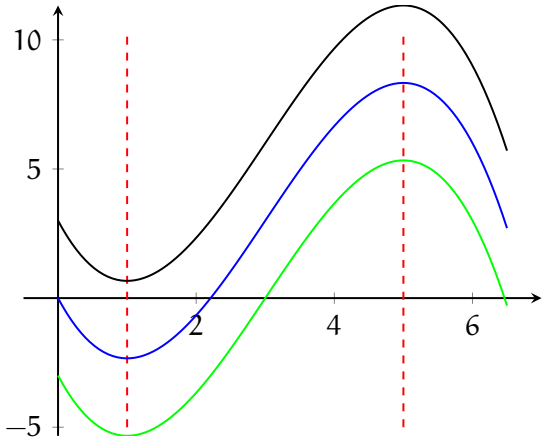



Figure 117.13: Possible graphs of f

### 117.2.1 Practice Sketching f from f′

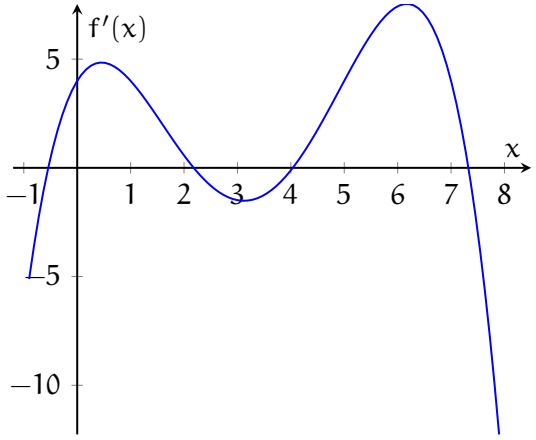



Figure 117.14: Graph of $f'(x)$

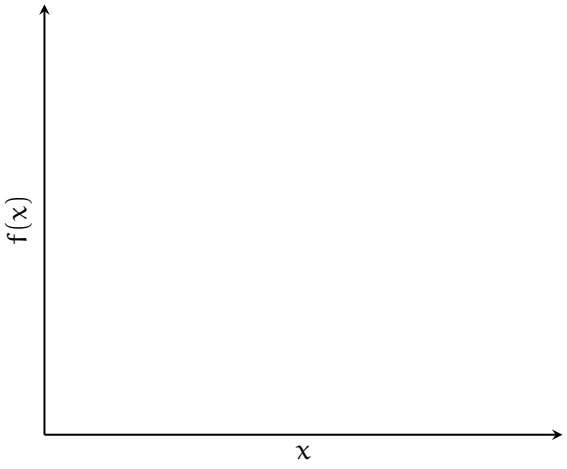
### *Exercise 319*

Use figure 117.14 to answer the following questions:

1. On what approximate intervals is f increasing or decreasing?
2. At what approximate values of x does f have a local maximum or minimum?
3. Sketch a possible graph of f in the space below:

*Working Space*

*Answer on Page 1999*

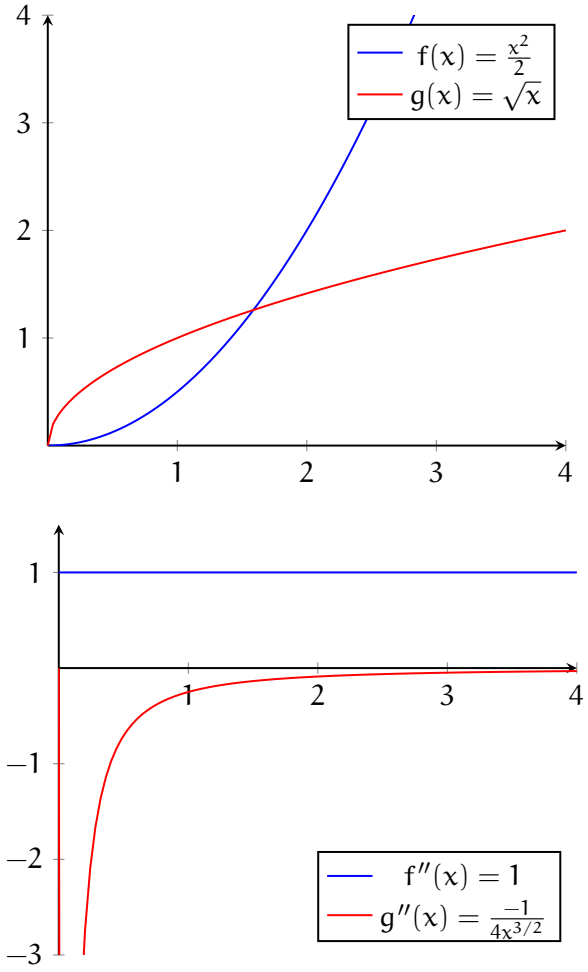



## 117.3 Using second derivatives to describe a function

### 117.3.1 Concavity

Let's examine two increasing functions, $f(x) = \frac{x^2}{2}$ and $g(x) = \sqrt{x}$:

Even though both of these functions are increasing, they have different shapes. $f(x)$ looks like a bowl. On the other hand, $g(x)$ looks like an upside-down bowl. These shapes are called *concave up* (in the case of $f(x)$) and *concave down* (in the case of $g(x)$). Both functions



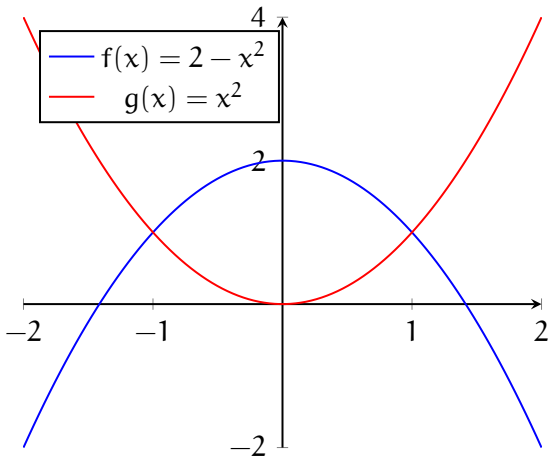

are increasing on the interval $x \in [0, 4]$, and therefore both $f'(x)$ and $g'(x)$ are positive on the stated interval. Let's look at their second derivatives, $f''(x)$ and $g''(x)$:

As you can see, $f''(x) > 0$ and $g''(x) < 0$. The second derivative tells us if a function is concave up or concave down. In general:

1. If $f''(x) > 0$ for all $x$ in a given interval, then the graph of $f$ is concave up on the interval.
2. If $f''(x) < 0$ for all $x$ in a given interval, then the graph of $f$ is concave down on the interval.

Additionally, the second derivative can help us determine if there is a local minimum or maximum at critical numbers. Look at the graphs of $f(x) = 2 - x^2$ and $g(x) = x^2$, which both have first derivatives equal to $0$ at $x = 0$:

When the graph is concave up, there is a local minimum where the first derivative equals




0. When the graph is concave down, there is a local maximum where the first derivative equals 0. This is summarized with the Second Derivative Test:

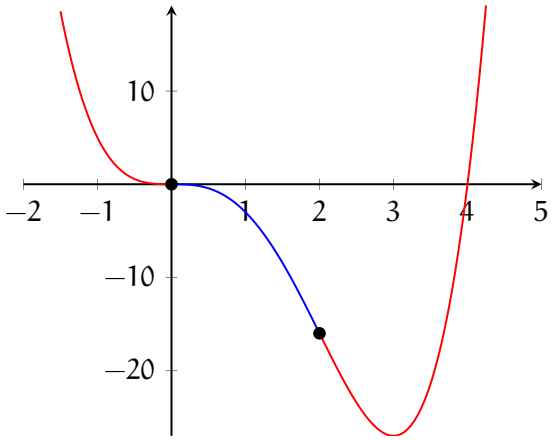
Suppose $f''$ is continuous near $c$. Then,

1. If $f'(c) = 0$ and $f''(c) > 0$, then $f$ has a local minimum at $c$.
2. If $f'(c) = 0$ and $f''(c) < 0$, then $f$ has a local maximum at $c$.

### 117.3.2 Inflection Points

If $f$ is concave up when $f'' > 0$ and concave down when $f'' < 0$, what about when $f'' = 0$? This is the value at which $f$ changes from concave up to concave down (or vice versa), which is called an *inflection point*. Similar to local extrema with $f'$, if there is an inflection point at $x = c$, then $f''(c) = 0$, but the converse is not necessarily true. To check if $x = c$ is an inflection point, then $f''$ should change signs on either side of $x = c$ ( either from positive to negative to from negative to positive).

Look at the graph of $f(x) = x^4 - 4x^3$. The concave up areas are shown in red, and the concave down in blue:




Let's examine $f''$ to confirm the inflection points are at $(0,0)$ and $(2,-16)$. First, we note that $f''(x) = 12x^2 - 24x$. Factoring, we see that $f''(x) = 12x(x-2)$, which has zeroes at $x = 0$ and $x = 2$. For $x < 0$, $f'' > 0$, and for $0 < x < 2$, $f'' < 0$; therefore, there is an inflection point in $f$ at $(0,0)$.

## Exercise 320

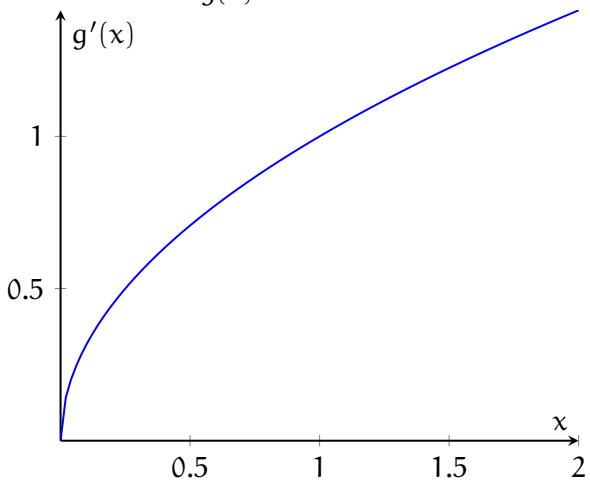
Prove that the other inflection point for $f(x) = x^4 - 4x^3$ is $(2,-16)$.

*Working Space*

*Answer on Page 2000*

### Exercise 321

The graph below shows $g'(x)$. Describe the behavior of $g(x)$ from $x = 0$ to $x = 2$.




*Working Space*

*Answer on Page 2000*

### Exercise 322

[This question was originally presented as a calculator-allowed, multiple-choice problem on the 2012 AP Calculus BC exam.] For $-1.5 < x < 1.5$, let $f$ be a function with first derivative given by $f'(x) = e^{(x^4-2x^2+1)} - 2$. State the interval(s) (to three decimal places) for which $f$ is concave down.

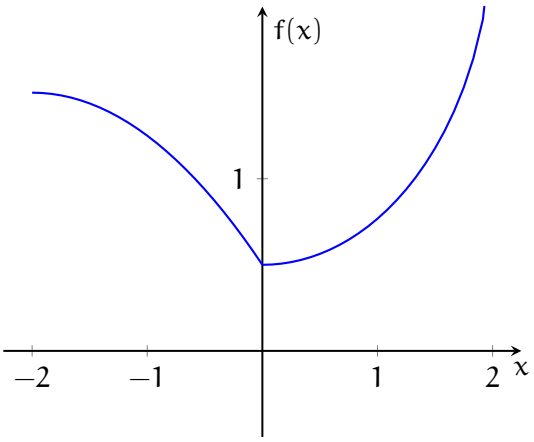
*Working Space*

*Answer on Page 2000*

## Exercise 323

[The following problem was originally presented as a calculator-allowed, multiple-choice question on the 2012 AP Calculus BC exam.] Consider the function, $f$, whose graph is shown below. Classify each of the following statements as true or false and explain.

1. $f' > 0$ for $x \in (-2, 0)$.
2. $f$ is differentiable at $x = 0$.
3. $f'' > 0$ for $x \in (0, 2)$
4. $f$ has a critical value at $x = 0$



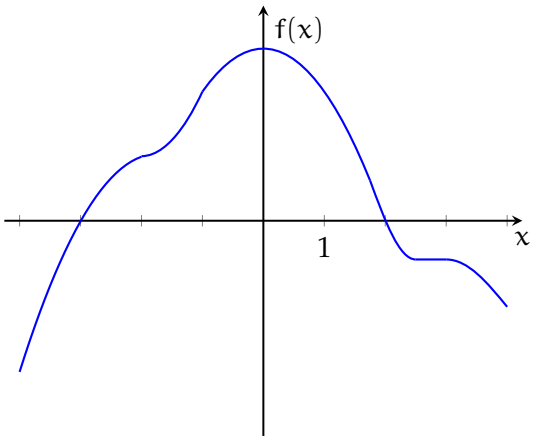

*Working Space*

*Answer on Page 2001*

### Exercise 324

[The following problem was originally presented as a calculator-allowed, multiple-choice question on the 2012 AP Calculus BC exam.] The graph of $f'$, the derivative of $f$, is shown below. Classify each of the following statements as true or false and explain your answer.

1. $f$ has a relative minimum at $x = -3$.
2. The graph of $f$ has a point of inflection at $x = -2$.
3. The graph of $f$ is concave down for $0 < x < 4$.




*Working Space*

*Answer on Page 2001*

### *Exercise 325*

[The following problem was originally presented as a calculator-allowed, multiple-choice question on the 2012 AP Calculus BC exam.] Let $f$ be a function that is twice differentiable on $-2 < x < 2$ and satisfies the conditions in the table below. If $f(x) = f(-x)$, what are the $x$-coordinates of the points of inflection of the graph of $f$ on $-2 < x < 2$?

| | $0 < x < 1$ | $1 < x < 2$ |
|---|---|---|
| $f(x)$ | Positive | Negative |
| $f'(x)$ | Negative | Negative |
| $f''(x)$ | Negative | Positive |

*Working Space*

*Answer on Page 2002*

Chapter 118

# Experimental Design

## 118.1 Planning and Conducting Experiments

Surveys tell you what people think or report. Well-designed experiments can show what causes what. The distinction determines what conclusions you are entitled to draw when the data comes in.

### 118.1.1 Experiments vs. Observational Studies

The defining question is whether the researcher assigns the conditions being studied or simply observes them as they exist.

In an *observational study*, the researcher collects data on *subjects*, the people or things being studied, without intervening. The variables of interest already exist in the subjects; the researcher records what is there. A sociologist who tracks the grades and sleep hours of university students over a semester, without changing anyone's sleep schedule, is conducting an observational study.

In an *experiment*, the researcher imposes conditions on subjects and then measures the effect or change. The researcher creates the differences rather than observing them.

Why does the distinction matter? Only a properly designed experiment can support a cause-and-effect conclusion. An observational study can show that two variables are associated, but a third variable you did not account for could explain the relationship entirely. This uninvited third variable is called a *confounding variable*: one whose effect on the response cannot be separated from the effect of the variable you are actually studying.

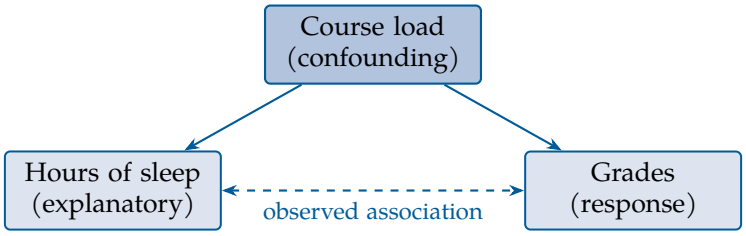



Figure 118.1: A confounding variable affects both the explanatory and response variables, so the observed association cannot be attributed to sleep alone.

In the observational sleep study, students who sleep more might also exercise more, eat

better, or have lower course loads. Any of these could be responsible for better grades. The data give no way to tell whether sleep or one of these other factors explains the difference.

There are situations where an experiment is unethical or impossible: you cannot randomly assign people to smoke for twenty years, or randomly select cities to experience floods. In these cases, observational data is all we have, and conclusions must be drawn with caution.

## *Exercise 326* Experiment or Observational Study?

*Working Space*

For each study, identify whether it is an experiment or an observational study. Then state whether a cause-and-effect conclusion is justified and why.

| Study | Type | Causal conclusion? |
|---|---|---|
| Researchers randomly assign 180 seedlings to three groups and water each group with a different fertilizer solution. After eight weeks they measure plant height. | | |
| A researcher records the screen time and attention span of 500 children aged 8 to 12, then looks for a relationship between the two. | | |
| A company randomly selects half its customer service team to complete a new communication training program, then compares customer satisfaction scores between the two groups. | | |
| A historian finds that cities with more libraries per capita had higher literacy rates in the early 20th century. | | |

*Answer on Page 2002*

### 118.1.2 Vocabulary for Experiments

Experiments have their own vocabulary. These terms build on each other, so it helps to learn them in order.

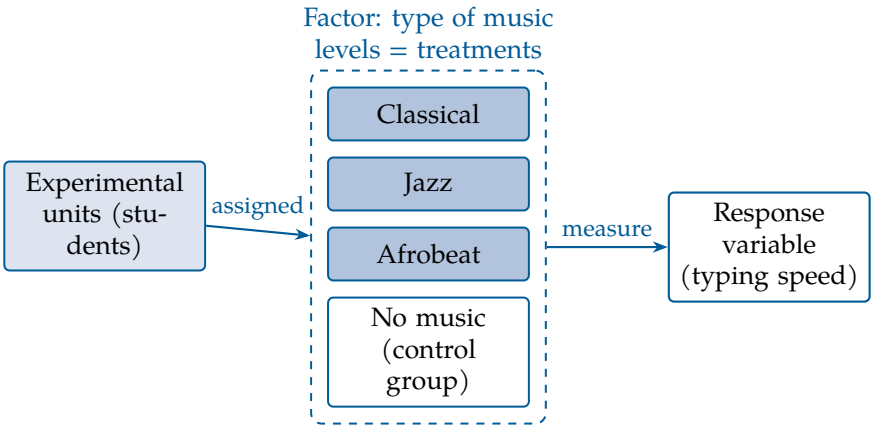



Figure 118.2: The parts of a one-factor experiment on background music and typing speed.

***Explanatory variable*** Also called the independent variable. The variable whose effect on the response is being studied. In a study of whether background music affects typing speed, the type of music is the explanatory variable.

***Response variable*** Also called the dependent variable. The outcome being measured. In the same study, typing speed is the response variable.

***Experimental unit*** The smallest unit to which a treatment is applied. In a study of plants, each plant is an experimental unit. In a study of people, each person is an experimental unit.

***Factor*** A variable whose effect on the response is of interest. Factors can be qualitative (type of music: classical, jazz, Afrobeat) or quantitative (temperature in degrees Celsius).

***Levels*** The specific values of a factor used in the experiment. If the factor is type of music with three options, there are three levels.

***Treatment*** A specific combination of factor levels applied to experimental units. With one factor and three levels, there are three treatments. With two factors at two levels each, there are $2 \times 2 = 4$ treatments.

***Control group*** A group that receives no treatment or a standard baseline condition. It provides a reference point for comparison with the treatment groups.

***Placebo*** A dummy treatment that resembles the real treatment but contains no active component. Used to isolate the genuine effect of the treatment from the *placebo effect*: the tendency for people to show change simply because they believe they are receiving treatment.

### 118.1.3 The Three Principles of Experimental Design

A well-designed experiment rests on three principles.

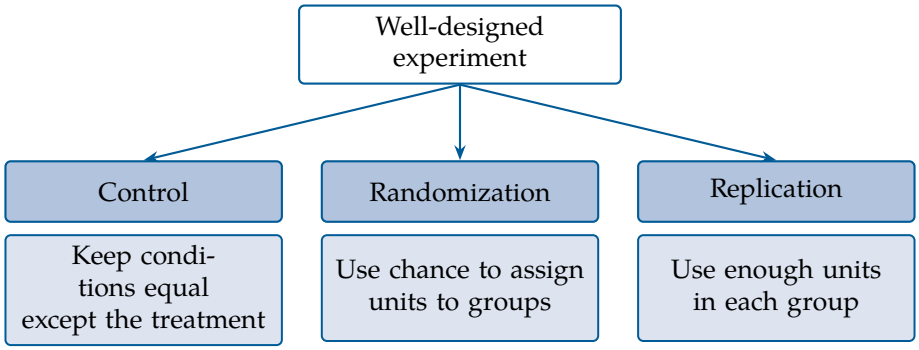



Figure 118.3: The three principles of experimental design.

*Control* means keeping everything the same for all groups except the treatment. It is not just having a control group. Control helps make sure the treatment is the only real difference.

*Randomization* means using chance to place experimental units into groups. It helps keep the groups fair and spreads unknown factors across all groups.

*Replication* means including enough experimental units in each group so that chance variation among individuals does not hide the treatment effect. If a new training method is tested on only two employees per group, a single unusually high or low performer could dominate the group's result. With 30 employees per group, individual extremes average out.

Randomization also protects against a *lurking variable*: a variable that affects the response but is not measured or included in the study. A lurking variable becomes a confounding variable when it is also linked to the explanatory variable, so that its effect cannot be separated from the effect of the treatment. Random assignment spreads lurking variables across all groups by chance, which keeps them from lining up with one treatment.

### *Exercise 327* Applying the Three Principles

A horticulture researcher wants to test whether exposing seedlings to classical music for two hours per day increases their growth rate compared to no music exposure. She has 60 seedlings of the same species, grown from the same seed batch, available for the study.

1. Describe how randomization should be applied.
2. What does control mean in this context? List at least three variables that should be held constant.
3. The researcher initially plans to use 5 seedlings per group. A colleague recommends 30 per group. Which principle does this change address, and why?
4. Should there be a control group? What would it look like?

*Working Space*

*Answer on Page 2002*

#### 118.1.4 Blinding

Subjects who know their treatment may behave or report differently because of that knowledge. The person measuring outcomes faces the same risk, since knowing a subject's group can sway their judgment, often without their awareness. Blinding prevents both kinds of bias.

In a *single-blind experiment*, either the subjects do not know which treatment they received, or the person measuring outcomes does not know, but not both.

In a *double-blind experiment*, neither the subjects nor the people measuring outcomes know the treatment assignments. Double-blind experiments are preferred whenever possible

because they remove bias from both sources at once.

Double-blinding is not always possible. If subjects can easily identify their treatment condition from its appearance or feel, blinding them is impossible. A study comparing two different workout splits cannot prevent participants from knowing which one they are following. In such cases, the best available design blinds the people measuring outcomes, even though the subjects know their treatment.

### 118.1.5 Random Selection vs. Random Assignment

Two design choices decide what you may conclude: (1) Were subjects randomly assigned to treatments? (2) Were they randomly selected from a larger population? These are separate decisions, and each one supports a different type of conclusion.

*Random assignment* of treatments is what allows you to conclude that a treatment caused an observed difference. When treatments are assigned by chance, any pre-existing differences among subjects are distributed randomly across groups. If one group then performs meaningfully better than another, the treatment is the only consistent explanation left. A difference large enough to be unlikely due to chance alone is called *statistically significant*. For example, suppose a noisy-room group scores 8 points lower than a quiet-room group, and random assignment alone would rarely produce a gap that large. The gap is then statistically significant. Statistically significant differences between treatment groups are evidence that the treatment caused the effect.

*Random selection* of subjects from a population is what allows you to generalize conclusions beyond the people in your study. If subjects were randomly drawn from a well-defined population, the results of the experiment can be extended to that population. If subjects were not randomly selected, the conclusions apply only to the subjects in the study, or to others who are very similar to them.

These two ideas operate independently. An experiment can have one, both, or neither. The table below summarizes what each combination allows.

| | **Random assignment: Yes** | **Random assignment: No** |
|---|---|---|
| **Random selection: Yes** | Can conclude causation. Can generalize to the population. | Cannot conclude causation. Can generalize to the population. |
| **Random selection: No** | Can conclude causation. Cannot generalize beyond the study subjects. | Cannot conclude causation. Cannot generalize beyond the study subjects. |

Remember: the two are separate ideas, and a causal claim about a broader population needs both.

## *Exercise 328* Random Assignment vs. Random Selection

*Working Space*

For each study, answer three questions: (1) Was there random assignment? (2) Was there random selection? (3) What conclusions are justified?

1. A researcher posts a sign-up sheet in a university cafeteria asking for volunteers to participate in a study on the effect of background noise on reading comprehension. Sixty students volunteer. The researcher randomly assigns 30 to read in a noisy room and 30 to read in a quiet room. Students in the noisy room score significantly lower on a comprehension test.
2. A public health agency randomly selects 400 adults from the voter registration rolls of a large city. Researchers observe whether each person exercises regularly and whether they report high stress levels. Those who exercise regularly report significantly lower stress.
3. A school district randomly selects 10 of its 60 elementary schools to pilot a new reading curriculum. All students in those 10 schools use the new curriculum; the remaining 50 schools continue with the standard curriculum. At year end, students in the pilot schools score significantly higher on a standardized reading test.
4. A fitness company recruits 80 volunteers through social media and randomly assigns 40 to a new training app and 40 to a standard printed workout plan. After 12 weeks, the app group shows significantly greater improvement in endurance.

*Answer on Page 2003*

## 118.2 Experimental Designs

There are three experimental designs you should be able to recognize, describe, and distinguish from one another. Each one handles *variability*, the natural differences among experimental units, in its own way.

### 118.2.1 Completely Randomized Design

The simplest structure is a *completely randomized design*. All available experimental units are assigned to treatments purely by chance, with no prior sorting or grouping. Randomization alone is responsible for balancing any existing differences among units across the treatment groups.

Suppose an agricultural research station wants to compare the yield of three varieties of wheat: a standard variety currently in use, and two new varieties developed in the lab. The station has 90 available plots of farmland. In a completely randomized design, the experiment proceeds as follows:

1. Measure the current condition of each plot (soil quality, drainage, sun exposure).
2. Number the plots 1 to 90 and use a random number generator to assign 30 plots to variety 1, 30 to variety 2, and 30 to variety 3.
3. Plant the assigned wheat variety in each plot and manage all plots identically in terms of water, fertilizer, and pest control.
4. At harvest, measure the yield from each plot.
5. Compare average yields across the three groups.

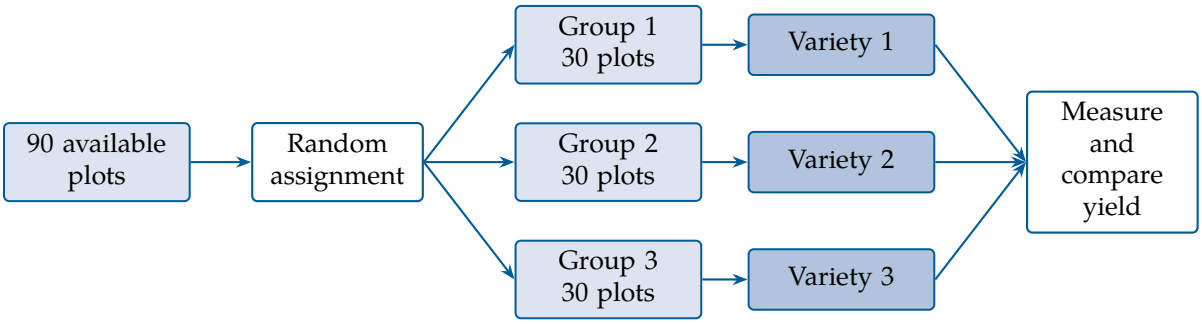



Figure 118.4: Schematic diagram of a completely randomized experiment.

This design works well when the experimental units are reasonably similar to begin with. If the 90 plots varied dramatically in soil quality, however, randomization might not perfectly balance soil quality across the three groups, particularly with a smaller experiment. This is the motivation for blocking.

## *Exercise 329* Designing a Completely Randomized Experiment

A sports science department wants to test whether three different warm-up routines (static stretching, dynamic stretching, or no warm-up) affect sprint times in recreational runners. Sixty volunteers have been recruited.

1. Identify the experimental units, the factor, the levels, and the treatments.
2. Describe a complete procedure for randomly assigning the 60 volunteers to the three groups.
3. At the end of the study, the dynamic stretching group has the fastest average sprint times. Can the researchers conclude that dynamic stretching caused the improvement? Explain.

*Working Space*

*Answer on Page 2003*

### 118.2.2 Randomized Block Design

When experimental units differ in some important, identifiable way before the experiment begins, a completely randomized design may leave a large amount of unexplained variability in the results. A more efficient approach is to group similar units into blocks before assigning treatments.

In a *randomized block design*, experimental units are first divided into groups of similar units called blocks. Within each block, all treatments are randomly assigned. Because units within a block are similar to each other, the comparison between treatments within a block is cleaner than a comparison that mixes unlike units together. The blocking factor is a variable known to affect the response that the researcher controls for by separating units before randomization.

Return to the wheat experiment. Suppose the research station has two soil zones: sandy

soil in the northern half of the land and clay soil in the southern half. Soil type is known to affect yield. If the three varieties are distributed randomly across all 90 plots without regard to soil type, differences in soil quality may obscure or exaggerate differences between varieties. The solution is to block on soil type.

1. Divide the 90 plots into two blocks: 45 sandy-soil plots and 45 clay-soil plots.
2. Within the sandy-soil block, randomly assign 15 plots to variety 1, 15 to variety 2, and 15 to variety 3.
3. Within the clay-soil block, randomly assign 15 plots to variety 1, 15 to variety 2, and 15 to variety 3.
4. Plant, manage, and harvest as before.
5. Compare yields within each block, then examine how results differ across soil types.

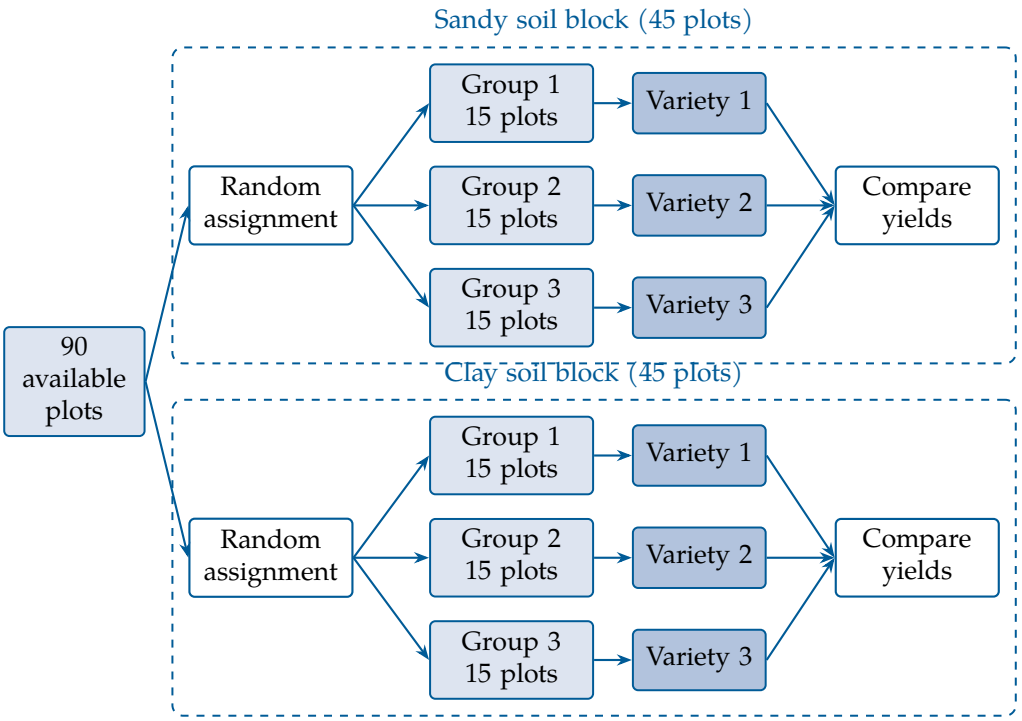



Figure 118.5: Schematic diagram of a randomized block design.

Notice that results must be compared within blocks. Comparing a variety 1 plot on sandy soil to a variety 2 plot on clay soil would mix up the variety effect with the soil effect. The whole point of blocking is to make comparisons fair by ensuring the only difference within each comparison is the treatment.

A good blocking variable is one you can observe before the experiment begins and one you expect to affect the response. In the wheat experiment, soil type is visible and known to matter. In a study of a new study-skills program, prior academic performance would be

a natural blocking variable. The rule: block on what you can see and control; randomize against what you cannot.

## *Exercise 330* Identifying the Block Design

For each study, identify the blocking factor and explain why blocking is preferable to a completely randomized design.

*Working Space*

1. A study tests two methods of teaching reading comprehension to primary school students. Researchers believe that students who already read above grade level will respond differently from those reading at or below grade level.

2. An engineering team tests four surface coatings on metal panels exposed to salt water. The available panels come from two different steel mills and are known to differ in base composition.

*Answer on Page 2004*

### 118.2.3 Matched-Pairs Design

A *matched-pairs design* is a special case of blocking in which each block contains exactly two experimental units and there are exactly two treatments to compare.

Pairs of units are matched on some relevant characteristic, and within each pair, one unit is randomly assigned to each treatment. Because the two units in each pair are as similar as possible, any difference in their responses can be attributed to the treatment rather than to pre-existing differences.

There are two common versions of this design.

In the first, two different subjects are matched. A study comparing two physiotherapy

protocols for knee rehabilitation might pair patients by age and injury severity. Within each pair, one patient is randomly assigned to protocol 1 and the other to protocol 2. The comparison is made within pairs, not across the whole sample.

In the second version, each subject serves as their own pair, receiving both treatments in a randomly determined order. This is sometimes called a *repeated measures design*. A washout period between treatments allows the first treatment's effect to clear before the second begins. Because each subject is compared only to themselves, all individual differences are perfectly controlled.

Consider a study testing whether caffeine affects short-term memory performance. Each participant completes a memory task twice: once after consuming a caffeinated drink, and once after consuming an identical-looking decaffeinated drink. The order is randomly determined for each participant. This is a matched-pairs design because each participant's caffeinated performance is paired with their own decaffeinated performance. Comparing one participant's caffeinated score to a different participant's decaffeinated score would not be effective.

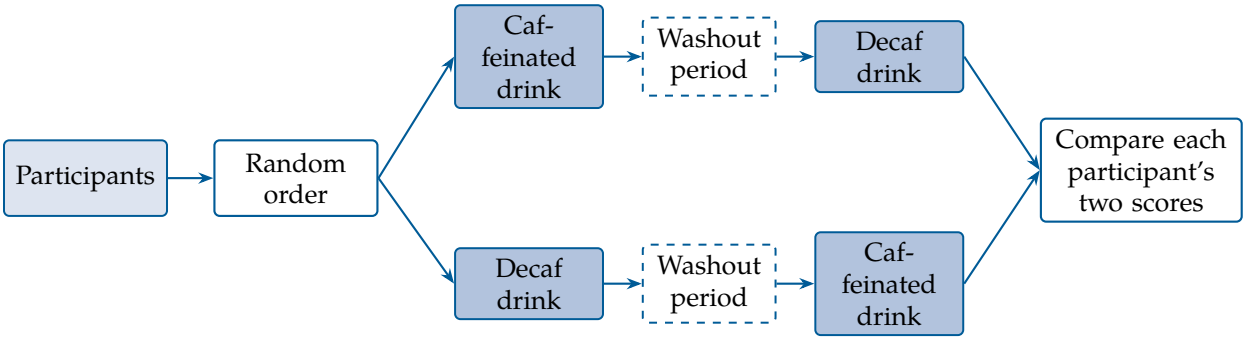



Figure 118.6: Schematic diagram of a matched-pairs design (repeated measures version).

## *Exercise 331* Matched Pairs or Randomized Block?

*Working Space*

Classify each study as a matched-pairs design or a randomized block design with more than two units per block. Explain your reasoning.

1. A study tests three different revision strategies (practice testing, summarizing, and re-reading) on university students. Students are divided into blocks by their first-year GPA: high (3.5 and above), mid (2.5 to 3.49), and low (below 2.5). Within each GPA block, students are randomly assigned to one of the three strategies.
2. A study compares the effectiveness of two soil treatment methods on crop yield. Farmers are paired by farm size and climate zone. Within each pair, one farm uses method 1 and the other uses method 2.
3. A study tests whether a standing desk reduces reported lower back discomfort. Each participant uses a standard seated desk for four weeks, then a standing desk for four weeks, with the order randomly determined for each person.

*Answer on Page 2004*

### 118.2.4 Choosing a Design

Knowing what each experimental design is called is not enough. The AP exam regularly asks you to justify why a particular design is appropriate for a given situation. That requires understanding what problem each design solves and when those problems arise.

The starting point is always the completely randomized design. It is the simplest structure, and when experimental units are reasonably similar to one another, it is the right choice. Random assignment spreads pre-existing differences among units across the treatment groups by chance. With enough units, those differences tend to balance out. If you have no strong reason to expect a particular characteristic of the units to consistently affect the response, use a completely randomized design.

Blocking makes sense when, before the experiment begins, you can identify a characteristic of the units that you expect to affect the response. Ignoring such a characteristic adds extra variability to your results. Units in the same treatment group will differ in ways unrelated to the treatment, and those differences make a real treatment effect harder to detect. Blocking solves this by grouping similar units first. Treatments are then compared within groups of units that are alike in the relevant way, which reduces variability and makes the experiment more sensitive.

To decide whether to block, ask yourself: is there a variable I can observe right now, before any treatment is applied, that I expect to affect how units respond? If yes, and if it is practical to measure and group by that variable, block on it.

Use matched pairs when there are exactly two treatments. Either pair up units that match on a relevant characteristic, or give both treatments to each unit. Because the two units in a pair are as similar as possible, the within-pair comparison is very clean. When each unit receives both treatments (with a washout period between them), individual differences drop out of the comparison entirely.

A design justification on the AP exam names the design, the variable it accounts for, and why that variable matters. Here is a model for the wheat experiment:

> Use a randomized block design, blocking on soil type. Soil type can be observed before planting, and it is expected to affect yield. Comparing the three varieties within each soil type removes soil-related variability from the comparison, so a difference in yield is easier to attribute to the variety.

The same three parts work for any design: name it, name the variable it handles, and explain how handling that variable makes the treatment comparison fairer.

## *Exercise 332* Justifying a Design Choice

*Working Space*

For each situation, recommend a design (completely randomized, randomized block, or matched pairs) and write a clear justification that explains why that design is more appropriate than the alternatives.

1. A software company wants to test whether a new user interface reduces the time it takes to complete a standard task, compared to the existing interface. It has 80 volunteers available. None of them have used either interface, and all have similar levels of computing experience.
2. A sleep research lab wants to compare the effect of two room temperatures (cool vs. warm) on sleep quality. They have 40 adult volunteers. Each person's natural sleep patterns vary night to night, but researchers expect consistent individual differences in baseline sleep quality across people.
3. Bluewave School District wants to compare three different formats for a professional development workshop (in-person, hybrid, and fully online) on teachers' reported confidence in teaching a new curriculum. The district's teaching staff includes both primary school teachers and secondary school teachers, and prior research suggests that these two groups respond very differently to professional development formats.
4. A skincare company wants to test whether a new moisturizer reduces skin dryness compared to a standard moisturizer. They recruit 50 volunteers and note that participants' baseline skin dryness varies widely.

*Answer on Page 2004*

Chapter 119

# Optimization

Optimization is a branch of mathematics that involves finding the best solution from all feasible solutions. In the field of operations research, optimization plays a crucial role. Whether it is minimizing costs, maximizing profits, or reducing the time taken to perform a task, optimization techniques are employed to make decisions effectively and efficiently.

In this chapter, the primary goal is to find the maximum or minimum value of a function, often referred to as the objective function, by finding the point on the graph where the derivative is equal to zero. This is usually referred to as the critical point!

At a critical point where the derivative is zero, the graph of the function has a horizontal tangent line. At such a point, the function may be reaching a peak or a valley.

Not every critical point corresponds to a maximum or minimum. Some critical points may instead be points of inflection. This is when the second derivative of the point is also equal to zero! You will see this in a few examples!

## 119.1 Extrema

There are two types of extrema: relative extrema and absolute extrema.

A relative maximum or minimum occurs when the function reaches a peak or valley compared to nearby values. This does not guarantee that the value is the highest or lowest value of the function overall.

An absolute maximum or minimum is the greatest or least value that the function takes on a given interval.

An absolute maximum or minimum can occur either at a critical point or at an endpoint of the interval.

For this reason, when solving optimization problems on a restricted domain, it is important to test both critical points and endpoints.

## 119.2 The Second Derivative Test

Once a critical point has been found, we often want to determine whether it corresponds to a maximum or a minimum. One method for doing this is the Second Derivative Test.

If a function has a critical point at $x = c$, then:

- if the second derivative is positive at $x = c$, the function is concave up and the point is a relative minimum;
- if the second derivative is negative at $x = c$, the function is concave down and the point is a relative maximum;
- if the second derivative is zero at $x = c$, the test is inconclusive.

If the second derivative is always positive or always negative on an interval, then any critical point in that interval must be an absolute minimum or absolute maximum, respectively.

## 119.3 Solving Optimization Problems

Optimization problems apply the ideas of derivatives, critical points, and extrema to real situations. Although the context of each problem may be different, the method used to solve them follows the same general pattern.

## 119.4 General Strategy for Solving Optimization Problems

Most calculus optimization problems follow these steps:

1. Identify the quantity to be optimized.
2. Write an equation for the objective function. Sometimes, you might be given an equation already. In that case, you will skip this step. For word problems, you will typically be required to write the equations yourself.
3. Use the given constraints to rewrite the function using one variable.
4. Determine the appropriate domain.
5. Find the derivative of the function.
6. Find critical points by setting the derivative equal to zero.
7. Interpret the result in the context of the problem.

When solving optimization problems, some common mistakes include forgetting to define variables, not using the constraints of the problem appropriately, misinterpreting or failing to interpret the final answer, and giving answers without appropriate units. For word problems, be very vigilant about how your answer relates back to the question given! It serves as a small check for accuracy.

### 119.4.1 Example

Consider the function

$$f(x) = x^2 - 4x + 1.$$

We want to find the maximum or minimum value of this function.

First, take the derivative of the function:

$$f'(x) = 2x - 4.$$

Next, set the derivative equal to zero to find the critical point:

$$2x - 4 = 0.$$

Solving for $x$ gives

$$x = 2.$$

Now determine whether this critical point corresponds to a maximum or a minimum by using the second derivative.

The second derivative is

$$f''(x) = 2.$$

Because the second derivative is positive, the function is concave up, and the critical point corresponds to a minimum.

To find the minimum value of the function, substitute $x = 2$ back into the original function:

$$f(2) = 2^2 - 4(2) + 1 = -3.$$

Therefore, the function has a minimum value of $-3$ at $x = 2$.

### *Exercise 333*

Consider the function

$$f(x) = x^2 - 6x + 5.$$

Find the maximum or minimum value of the function using calculus.

*Working Space*

*Answer on Page 2005*

Now, let's look at a word example:

**Example**

The cost, in dollars, of producing $x$ units of a product is given by the function

$$C(x) = 2x^2 - 24x + 100.$$

Find the number of units that should be produced in order to minimize the cost, and determine the minimum cost.

To minimize the cost, take the derivative of the cost function:

$$C'(x) = 4x - 24.$$

Set the derivative equal to zero to find the critical point:

$$4x - 24 = 0.$$

Solving for $x$ gives

$$x = 6.$$

Next, take the second derivative:

$$C''(x) = 4.$$

Since the second derivative is positive, the cost function is concave up, and the critical point corresponds to a minimum.

Substitute $x = 6$ back into the original cost function:

$$C(6) = 2(6)^2 - 24(6) + 100 = 28.$$

Therefore, the cost is minimized when 6 units are produced, and the minimum cost is \$28.

### *Exercise 334*

A rectangular enclosure is to be built using 60 units of fencing. Three sides of the enclosure require fencing, while the fourth side is along a wall and does not require fencing.

Let $x$ represent the length of the side parallel to the wall and $y$ represent the width of the enclosure.

Find the dimensions of the enclosure that minimize the amount of fencing used.

*Working Space*

*Answer on Page 2006*

## 119.5 Helpful Table for Optimization Problems

| **Condition at a point $x = c$** | **What it tells you** | **What kind of point it could be** |
|---|---|---|
| $f'(c) \neq 0$ | The graph has a non-horizontal tangent | Not a maximum or minimum |
| $f'(c) = 0$ | The graph has a horizontal tangent | Possible maximum, minimum, or neither |
| $f'(c)$ does not exist | The graph may have a corner, cusp, or vertical tangent | Possible maximum or minimum so run second derivative test to check! |
| $f'(c) = 0$ and $f''(c) > 0$ | Graph is concave up at c | Local minimum |
| $f'(c) = 0$ and $f''(c) < 0$ | Graph is concave down at c | Local maximum |
| $f'(c) = 0$ and $f''(c) = 0$ | Second derivative test fails | Could be max, min, or neither |

## 119.6 Using Python to Visualize Your Optimization Problem

You may be familiar with the use of Python to find the derivative of functions. We are going to use Python to supplement your knowledge of Optimization problems! Below is a script that can be used to visualize what your world problems may be asking you to do! This script will help you with finding the first derivative, performing the second derivative test, and graphs of your equations showing these points.

```
import sympy as sp
import numpy as np
import matplotlib.pyplot as plt

# 1) Define the variable
x = sp.symbols('x')

# 2) Define the objective function (STUDENTS EDIT THIS)
f = 20*x - x**2

# 3) Derivatives
f_prime = sp.diff(f, x)
f_double_prime = sp.diff(f_prime, x)

# 4) Solve f'(x) = 0
critical_point = sp.solve(f_prime, x)[0]

# 5) Classify using second derivative
second_derivative_value = f_double_prime.subs(x, critical_point)

if second_derivative_value > 0:
    classification = "minimum"
elif second_derivative_value < 0:
    classification = "maximum"
```

```
else:
    classification = "inconclusive"

print("f(x) =", f)
print("f'(x) =", f_prime)
print("f''(x) =", f_double_prime)
print("Critical point:", critical_point)
print("Classification:", classification)

# 6) Plot
f_num = sp.lambdify(x, f, "numpy")

X = np.linspace(0, 20, 400)
Y = f_num(X)

xc = float(critical_point)
yc = float(f_num(xc))

plt.plot(X, Y)
plt.scatter([xc], [yc])
plt.title("Optimization")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.grid(True)

plt.annotate(
    classification,
    (xc, yc),
    xytext=(xc + 1, yc),
    arrowprops=dict(arrowstyle="->")
)

plt.show()
```

## Exercise 335

The Python script below uses derivatives to locate and classify a critical point of a function.

Working Space

1. Run the script using the function

$$f(x) = 20x - x^2.$$

   Record the critical point and whether it is classified as a maximum or minimum.

2. Change the objective function in the script to

$$f(x) = x^2 - 6x + 5.$$

   Run the script again. How does the classification change? Explain why this happens using the second derivative test.

3. Change the objective function to

$$f(x) = x^3 - 3x.$$

   Run the script. What does the script report for the classification? Explain why the second derivative test is inconclusive in this case.

4. Finally, choose your own quadratic function and predict whether the critical point will be a maximum or minimum before running the script. Check your prediction using both calculus and the graph.

Answer on Page 2007

Chapter 120

# Implicit Differentiation

Implicit differentiation is a technique in calculus for finding the derivative of a relation defined implicitly (that is, a relation between variables $x$ and $y$ that is not explicitly solved for one variable in terms of the other).

## 120.1 Implicit Differentiation Procedure

Consider an equation that defines a relationship between $x$ and $y$:

$$F(x, y) = 0$$

To find the derivative of $y$ with respect to $x$, we differentiate both sides of this equation with respect to $x$, treating $y$ as an implicit function of $x$:[1]

$$\frac{d}{dx}F(x, y) = \frac{d}{dx}0$$

Applying the chain rule during the differentiation on the left side of the equation gives:

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0$$

Finally, we solve for $\frac{dy}{dx}$ to find the derivative of $y$ with respect to $x$:

$$\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}$$

This result is obtained using the implicit differentiation method.

[1] This $\frac{d}{dx}$ form of the derivative is the same as $y'$ said as taking the derivative of $y$ with respect to $x$

## 120.2 Example

Consider the equation of a circle with radius $r$:

$$x^2 + y^2 = r^2$$

First, we will find $\frac{dy}{dx}$ without implicit differentiation. Next, we will apply implicit differentiation to get the same result.

### 120.2.1 Without Implicit Differentiation

First, we need to rearrange the equation to solve for $y$:

$$y^2 = r^2 - x^2$$

$$y = \pm\sqrt{r^2 - x^2}$$

We take the derivative of $y$ by applying the Chain Rule:

$$\frac{dy}{dx} = \frac{1}{2 \pm \sqrt{r^2 - x^2}} \cdot (-2x) = \frac{-x}{\pm\sqrt{r^2 - x^2}}$$

Notice the denominator of this fraction is the same as the solution we found for $y$, $y = \pm\sqrt{r^2 - x^2}$. So, we can also represent this as:

$$\frac{dy}{dx} = \frac{-x}{y}$$

### 120.2.2 With Implicit Differentiation

With implicit differentiation, we assume $y$ is a function of $x$ and apply the Chain Rule.

$$\frac{d}{dx}[x^2 + y^2] = \frac{d}{dx}[r^2]$$

For $x^2$ and $r^2$, we take the derivative as we normally would.[2] For $y^2$, we apply the Chain Rule, as outlined above.[3]

$$2x + 2y\frac{dy}{dx} = 0$$

Solving for $\frac{dy}{dx}$, we find

$$\frac{dy}{dx} = \frac{-x}{y}$$

, which is the same result as we found without implicit differentiation.

## 120.3 Folium of Descartes

It was relatively easy to rearrange the equation for a circle to solve for $y$, but that is not always the case. To help you understand this better, we will consider a famous curve known as the *Folium of Descartes*, given by the equation,

$$x^3 + y^3 = 3xy.$$

The word *folium* comes from the Latin word for "leaf," describing the curve's distinctive looped shape. The curve is named after René Descartes, a French mathematician who originally presented the curve as a challenge to fellow mathematician Pierre de Fermat, asking him to find the tangent line to the curve. Fermat was able to solve the problem with ease!

The Folium of Descartes (seen in Figure 120.1) is historically significant because unlike simpler curves such as circles or parabolas, it exhibits more complex behavior, including a self-intersection at the origin (the curve crosses itself!). This complexity makes it difficult to describe the curve using a single equation of the form $y = f(x)$.

To explicitly solve the equation $x^3 + y^3 = 3xy$ for $y$ requires multiple expressions to fully capture the entire curve. As a result, the familiar techniques used for functions defined explicitly in terms of $x$ are no longer sufficient. Instead, to find the slope of the tangent line at a point on the folium, we must use *implicit differentiation*.

[2] The $\frac{d}{dx}$ part disappears when taking the derivative of $x$, as the derivative of $x$ with respect to $x$ is just regular differentiation.

[3] Applying the chain rule is only allowed as $y$ is not the variable we were taking the derivative with respect to.

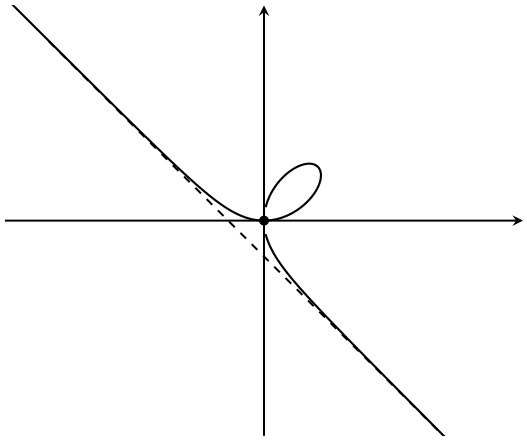


Figure 120.1: The Folium of Descartes $x^3 + y^3 = 3xy$ with asymptote $x + y + 1 = 0$.

### 120.3.1 Example: Tangent to Folium of Descartes

In this example, we will use implicit differentiation to easily find the tangent line at a point on the folium.

(a) Find $\frac{dy}{dx}$ if $x^3 + y^3 = 6xy$

(b) Find the tangent to the folium $x^3 + y^3 = 3xy$ at the point $(2, 2)$

(c) Is there any place in the first quadrant where the tangent line is horizontal? If so, state the point(s).

Solution:

(a) $\frac{d}{dx}[x^3 + y^3] = \frac{d}{dx}[3xy]$

$$3x^2 + 3y^2\frac{dy}{dx} = 3x\frac{dy}{dx} + 3y$$

$$x^2 + y^2\frac{dy}{dx} = x\frac{dy}{dx} + y$$

Rearranging to solve for $\frac{dy}{dx}$:

$$\frac{dy}{dx}(y^2 - x) = y - x^2$$

$$\frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$$

(b) We already have the coordinate point, $(2, 2)$, so to write an equation for the tangent

line, all we need is the slope. Substituting $x = 2$ and $y = 2$ into our result from part (a):

$$\frac{dy}{dx} = \frac{2 - 2^2}{2^2 - 2} = \frac{-2}{2} = -1$$

This is the slope, $m$. Using the point-slope form of a line, our tangent line is $y-2 = -(x-2)$.

(c) Recall that in the first quadrant, $x > 0$ and $y > 0$. We will set our solution for $\frac{dy}{dx}$ equal to 0:

$$\frac{y - x^2}{y + 2 - x} = 0$$

which implies that

$$y - x^2 = 0$$

Substituting $y = x^2$ into the original equation:

$$x^3 + (x^2)^3 = 3(x)(x^2)$$

$$x^3 + x^6 = 3x^3$$

Which simplifies to

$$x^6 = 2x^3$$

Since we have excluded $x = 0$ by restricting our search to the first quadrant, we can divide both sides by $x^3$:

$$x^3 = 2$$

$$x = \sqrt[3]{2} \approx 1.26$$

Substituting $x \approx 1.26$ into our equation for $y$:

$$y \approx 1.26^2 = 1.59$$

Therefore, the folium has a horizontal tangent line at the point $(1.26, 1.59)$.

## 120.4 Practice

## Exercise 336

[This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC Exam.] If $\arcsin x = \ln y$, what is $\frac{dy}{dx}$?

*Working Space*

*Answer on Page 2008*

## Exercise 337

[This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC Exam.] The points $(-1,-1)$ and $(1,-5)$ are on the graph of a function $y = f(x)$ that satisfies the differential equation $\frac{dy}{dx} = x^2+y$. Use implicit differentiation to find $\frac{d^2y}{dx^2}$. Determine if each point is a local minimum, local maximum, or inflection point by substituting the $x$ and $y$ values of the coordinates into $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

*Working Space*

*Answer on Page 2008*

CHAPTER 121

# Related Rates

In calculus, related rates problems involve finding a rate at which a quantity changes by relating that quantity to other quantities whose rates of change are known. The technique used to solve these problems is known as ”related rates” because one rate is related to another rate.

## 121.1 Steps to solve related rates problems

Step 1: Understand the problem. First, read the problem carefully. Understand what rates are given and what rate you need to find.

Step 2: Draw a diagram. For most problems, especially geometry problems, drawing a diagram can be very helpful.

Step 3: Write down what you know. Write down the rates that you know and the rate that you need to find.

Step 4: Write an equation. Write an equation that relates the quantities in the problem. This equation will be your main tool to solve the problem. Some cases may require an implicit equation to be constructed.

Step 5: Differentiate both sides of the equation. Now you can use calculus. Differentiate both sides of the equation with respect to time.

Step 6: Substitute the known rates and solve for the unknown. Now that you have an equation that relates the rates, substitute the known rates into the equation and solve for the unknown rate.

Step 7: Create a concluding statement/ Write a concluding statement that sums up what you have a calculated in a concise manner.

## 121.2 Example

Here is an example of a related rates problem:

*A balloon is rising at a constant rate of 5 m/s. A boy is cycling towards the balloon along a straight path at 15 m/s. If the balloon is 100 m above the ground, find the rate at which the distance from*

*the boy to the balloon is changing when the boy is 40 m from the point on the ground directly beneath the balloon.*

The problem can be modeled with a right triangle where the vertical side is the height of the balloon, the horizontal side is the distance of the boy from the point on the ground directly beneath the balloon, and the hypotenuse is the distance from the boy to the balloon.

Let $x$ be the distance of the boy from the point on the ground directly beneath the balloon, $y$ the height of the balloon above the ground, and $z$ the distance from the boy to the balloon. From the Pythagorean theorem, we have

$$z^2 = x^2 + y^2 \tag{121.1}$$

Differentiating both sides with respect to time $t$ gives

$$2z\frac{dz}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt} \tag{121.2}$$

Given that $\frac{dx}{dt} = -15$ m/s (the boy is moving towards the point beneath the balloon), $\frac{dy}{dt} = 5$ m/s (the balloon is rising), $x = 40$ m, $y = 100$ m, we can substitute these into the equation and solve for $\frac{dz}{dt}$.

All related rates problems are different, so it is important to continually do them so that you encounter many different examples. For more examples and practice, work through the problems included in your digital resources!

Chapter 122

# Multivariate Functions and Partial Derivatives

A real-valued multivariate function is a function that takes multiple real variables as input and produces a single real output. We generally denote such a function as $f : \mathbb{R}^n \to \mathbb{R}$, where $\mathbb{R}^n$ is the domain and $\mathbb{R}$ is the co-domain, (ie. $\mathbb{R}$ is the domain of one variable and $\mathbb{R}^2$ is the domain of a 2 variable function)

For example, consider a function $f$ that takes two variables, $x$ and $y$:

$$f(x, y) = x^2 + y^2$$

Here, $f : \mathbb{R}^2 \to \mathbb{R}$ takes an ordered pair $(x, y)$ from the 2-dimensional real coordinate space, squares each, and adds them to produce a real number.

In a similar way, a function $g : \mathbb{R}^3 \to \mathbb{R}$ could take three variables, $x$, $y$, and $z$, and might be defined as:

$$g(x, y, z) = x^2 + y^2 + z^2$$

FIXME graphic of this showing area splotch mapping to line segment

Here, the function squares each of the input variables, then adds them to produce a real number.

These functions are "real-valued" because their outputs are real numbers, and "multivariate" because they take multiple variables as inputs.

The concepts of limits, continuity, differentiability, and integrability can all be extended to multivariate functions, although they become more complex because we now have to consider different directions in which we approach a point, not just from the left or right, as in the univariate case. FIXME expand this?

For example, the partial derivative is the derivative of the function with respect to one variable, holding the others constant. It is one of the basic concepts in the calculus of multivariate functions.

For example, given the function $f(x, y) = x^2 + y^2$, the partial derivatives of $f$ are computed

as:

$$\frac{\partial f}{\partial x}(x, y) = 2x$$

$$\frac{\partial f}{\partial y}(x, y) = 2y$$

We will expand on these partial derivatives in the next chapter.

Chapter 123

# Partial Derivatives and Gradients

This chapter will introduce you to partial derivatives and gradients, equipping you with the tools to study functions of multiple variables. We will explore how these concepts provide valuable insights into optimization, vector calculus, and various fields of science and engineering.

Partial derivatives come into play when dealing with functions that depend on multiple variables. Unlike ordinary derivatives that consider changes along a single variable, partial derivatives focus on how a function changes concerning each individual variable while holding the others constant. In essence, partial derivatives measure the rate of change of a function with respect to one variable, while keeping the other variables fixed.

The notation for a partial derivative of a function $f(x, y, \ldots)$ with respect to a specific variable, say $x$, is denoted as $\frac{\partial f}{\partial x}$. Similarly, $\frac{\partial f}{\partial y}$ represents the partial derivative with respect to $y$, and so on. It is essential to remember that when taking partial derivatives, we treat the other variables as constants during the differentiation process.

The gradient is a vector that combines the partial derivatives of a function. It provides a concise representation of the direction and magnitude of the steepest ascent or descent of the function. The gradient vector points in the direction of the greatest rate of increase of the function. By understanding the gradient, we gain insights into optimizing functions and finding critical points where the function reaches maximum or minimum values.

Throughout this chapter, we will explore the following key topics related to partial derivatives and gradients:

- Calculating partial derivatives: We will delve into the techniques and rules for computing partial derivatives of various functions, including polynomials, exponential functions, and trigonometric functions. We will also explore higher-order partial derivatives and mixed partial derivatives.
- Interpreting partial derivatives: Understanding the geometric and physical interpretations of partial derivatives is essential. We will discuss the notion of tangent planes, directional derivatives, and the relationship between partial derivatives and local linearity.
- Gradient vectors and their properties: We will introduce this concept, including it connection to the direction of steepest ascent, its relationship with partial derivatives,

and how it relates to level curves and level surfaces.

- Applications of partial derivatives and gradients: We will explore various applications of these concepts, including optimization problems, constrained optimization, tangent planes, linear approximations, and their relevance in fields like physics, economics, and engineering.

By grasping the concepts of partial derivatives and gradients, you will unlock a powerful mathematical framework for analyzing and optimizing functions of multiple variables. These tools will equip you to tackle advanced calculus problems and gain deeper insights into the behavior of functions in diverse fields.

## 123.1 Calculating Partial Derivatives

For a function of two variables, $f(x, y)$, we can take the derivative with respect to $x$ or with respect to $y$. These are called the *partial derivatives* of $f$. Formally, the partial derivatives are defined as:

**Limit Definition of Partial Derivatives**

$$f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Let's consider a polynomial function of two variables: $f(x, y) = 3x^2 + y^3 + 4xy$. We will use the limit definition to find the partial derivative with respect to $x$, then compare this to what we already know about derivatives of single-variable functions. Recall that if we can describe a function as a sum of two other functions, the derivative of the original function is the same as the sum of the derivatives of the other functions. That is,

$$\text{if } f(x) = g(x) + h(x)$$

$$\text{then } f'(x) = g'(x) + h'(x)$$

Let's then define $r(x, y) = 3x^2$, $s(x, y) = y^3$, and $t(x, y) = 4xy$. And so $f(x, y) = r(x, y) + s(x, y) + t(x, y)$, which means $f_x(x, y) = r_x(x, y) + s_x(x, y) + t_x(x, y)$. Then,

$$f_x(x, y) = \lim_{h \to 0} \frac{r(x + h, y) - r(x, y)}{h} + \lim_{h \to 0} \frac{s(x + h, y) - s(x, y)}{h} + \lim_{h \to 0} \frac{t(x + h, y) - t(x, y)}{h}$$

$$= \lim_{h \to 0} \frac{3(x + h)^2 - 3x^2}{h} + \lim_{h \to 0} \frac{y^3 - y^3}{h} + \lim_{h \to 0} \frac{4(x + h)y - 4xy}{h}$$

$$= \lim_{h\to 0} \frac{3x^2 + 6xh + h^2 - 3x^2}{h} + 0 + \lim_{h\to 0} \frac{4xy + 4hy - 4xy}{h}$$

Notice that $s_x(x, y) = 0$. This term only had $y$, and its derivative with respect to $x$ is zero. Continuing,

$$f_x(x, y) = \lim_{h\to 0} \frac{6xh + h^2}{h} + \lim_{h\to 0} \frac{4hy}{h} = \lim_{h\to 0} 6x + h + \lim_{h\to 0} 4y$$
$$= 6x + 4y$$

As you can see, $r_x(x, y) = 6x$ and $t_x(x, y) = 4y$. Recall the polynomial rule for single derivatives. The derivative of $3x^2$ is $6x$, which is also what we see with the partial derivative in this case. What about the other term, $4xy$? Well, we know the derivative of $bx$, where $b$ is a constant, is $b$. The partial derivative of $4xy$ with respect to $x$ being $4y$ suggests the rule for determining partial derivatives:

**Rule for Finding Partial Derivatives of** $f(x, y)$

1. To find the partial derivative with respect to $x$, $f_x$, treat $y$ as a constant and differentiate with respect to $x$.
2. To find the partial derivative with respect to $y$, $f_y$, treat $x$ as a constant and differentiate with respect to $y$.

Let's check this by predicting $f_y$, then using the limit definition to confirm our prediction. Applying the polynomial rule, we predict that $f_y$ is:

$$f_y(x, y) = 3y^2 + 4x$$

Which we found by treating $x$ as a constant and taking the derivative of each term with respect to $y$. Let's see if we get the same result using the limit definition of the derivative with respect to $y$:

$$f_y(x, y) = \lim_{h\to 0} \frac{f(x, y + h) - f(x, y)}{h}$$
$$= \lim_{h\to 0} \frac{\left[3x^2 + (y + h)^3 + 4x\,(y + h)\right] - \left[3x^2 + y^3 + 4xy\right]}{h}$$
$$= \lim_{h\to 0} \frac{3x^2 + y^3 + 3y^2h + 3yh^2 + h^3 + 4xy + 4xh - 3x^2 - y^3 - 4xy}{h}$$
$$= \lim_{h\to 0} \frac{3y^2h + 3yh^2 + h^3 + 4xh}{h} = \lim_{h\to 0} 3y^2 + 3yh + h^2 + 4x = 3y^2 + 4x$$

Which is our expected result. In summary, you find the partial derivative with respect to a particular variable by treating all the other variables as constants and differentiating with respect to the particular variable, applying the rules of differentiation you've already learned.

### 123.1.1 Partial Derivative Notation

There are many ways to denote a partial derivative. We've already seen one way, $f_x$ and $f_y$. Another common notation uses a lowercase Greek letter delta, and a further uses capital D. They are shown below:

**Partial Derivative Notations**

$$f_x(x,y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x,y) = D_x f$$

$$f_y(x,y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x,y) = D_y f$$

## *Exercise 338* First Partial Derivatives

Find $f_x$ and $f_y$ for the following functions.

*Working Space*

1. $f(x,y) = 3x^4 + 4x^2y^3$
2. $f(x,y) = xe^{-y}$
3. $f(x,y) = \sqrt{3x + 4y^2}$
4. $f(x,y) = \sin x^2 y$
5. $f(x,y) = \ln(x^y)$

*Answer on Page 2008*

### 123.1.2 Partial Derivatives of Functions of More than Two Variables

The above method of determining partial derivatives applies to functions with three, four, or any number of variables.

**Example**: Find all the first derivatives of the function $f(x,y,z) = y\cos\left(x^2 + 3z\right)$.

**Solution**:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left[y\cos\left(x^2+3z\right)\right] = -y\sin\left(x^2+3z\right)\left(\frac{\partial}{\partial x}\left(x^2+3z\right)\right)$$

$$\frac{\partial f}{\partial x} = -2xy\sin\left(x^2+3z\right)$$

And

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left[y\cos\left(x^2+3z\right)\right]$$
$$\frac{\partial f}{\partial y} = \cos\left(x^2+3z\right)$$

And

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}\left[y\cos\left(x^2+3z\right)\right] = -y\sin\left(x^2+3z\right)\left(\frac{\partial}{\partial z}\left(x^2+3z\right)\right)$$
$$\frac{\partial f}{\partial z} = -3y\sin\left(x^2+3z\right)$$

### *Exercise 339* Partial Derivatives with 3 or More Variables

*Working Space*

Find all first partial derivatives of the following functions.

1. $f = \sin\left(x^2 - y^2\right) \cos\left(\sqrt{z}\right)$
2. $q = \sqrt[3]{t^3 + u^3 \sin(5v)}$
3. $w = x^z y^x$

*Answer on Page 2009*

#### 123.1.3 Higher Order Partial Derivatives

Just like with single-variable equations, we can take the partial derivative more than once. There are also several notations for second partial derivatives.

**Second Partial Derivative Notation**

$$(f_x)_x = f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2}$$

$$(f_x)_y = f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2}$$

Notice that for $\left(\partial^2 f/\partial y \partial x\right)$, we first take the derivative with respect to $x$, then with respect to $y$.

**Example**: Find all the second order partial derivatives of $f(x, y) = 2x^2 - x^3y^2 + y^3$.

**Solution**: We begin by finding $f_x$ and $f_y$:

$$f_x(x, y) = 4x - 3x^2y^2$$

$$f_y(x, y) = -2x^3y + 3y^2$$

We then take another partial derivative to find all the second order partial derivatives:

$$f_{xx}(x, y) = \frac{\partial}{\partial x} f_x(x, y) = \frac{\partial}{\partial x}\left(4x - 3x^2y^2\right) = 4 - 6xy^2$$

$$f_{xy}(x, y) = \frac{\partial}{\partial y} f_x(x, y) = \frac{\partial}{\partial y}\left(4x - 3x^2y^2\right) = -6x^2y$$

$$f_{yx}(x, y) = \frac{\partial}{\partial x} f_y(x, y) = \frac{\partial}{\partial x}\left(-2x^3y + 3y^2\right) = -6x^2y$$

$$f_{yy}(x, y) = \frac{\partial}{\partial y} f_y(x, y) = \frac{\partial}{\partial y}\left(-2x^3y + 3y^2\right) = -2x^3 + 6y$$

What do you notice about $f_{xy}$ and $f_{yx}$? They are the same! This is not a coincidence of the particular function used in the example. For most functions, $f_{xy} = f_{yx}$, as stated by Clairaut's theorem.

**Clairaut's Theorem**

If $f$ is defined on a disk $D$ and $f_{xy}$ and $f_{yx}$ are both continuous on $D$, then $f_{xy} = f_{yx}$ on $D$.

This is also true for third, fourth, and higher-order derivatives.

## *Exercise 340* Clairaut's Theorem

Show that Clairaut's theorem holds for the following functions (show that $f_{xy} = f_{yx}$).

*Working Space*

1. $f(x, y) = e^{2xy} \sin x$
2. $f(x, y) = \frac{x^2}{x+y}$
3. $f(x, y) = \ln(2x + 3y)$

*Answer on Page 2010*

## *Exercise 341* Second Order Partial Derivatives

Find all second order partial derivatives of the function.

*Working Space*

1. $f(x, y) = x^5y^2 - 3x^3y^2$
2. $v = \sin\left(p^3 + q^2\right)$
3. $T = e^{-3r}\cos\theta^2$

*Answer on Page 2011*

### 123.1.4 The Chain Rule

For single-variable functions, where $y = f(x)$ and $x = g(t)$, we have seen that:

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

Which is the Chain Rule for single-variable functions. For multi-variable functions, there are several versions of the Chain Rule, depending on how the variables and functions are defined. First, we consider the case where $z = f(x, y)$ and $x = g(t)$ and $y = h(t)$ (i.e. $f$ is a multi-variable function of $x$ and $y$, while $x$ and $y$ are single-variable functions of $t$).

This means that $z$ is an indirect function of $t$:

$$z = f(x, y) = f\left(g(t), h(t)\right)$$

Then the derivative of $z$ with respect to $t$ is given by:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

**Example**: If $z = xy^2 + 3x^4y$, where $x = 2\sin(t)$ and $y = \cos(3t)$, find $dz/dt$ when $t = \pi/2$.

**Solution**: First, we apply the Chain Rule to $z$:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial x}\frac{dy}{dt}$$

$$= \frac{\partial}{\partial x}\left[xy^2 + 3x^4y\right] \cdot \frac{d}{dt}\left[2\sin(t)\right] + \frac{\partial}{\partial y}\left[xy^2 + 3x^4y\right] \cdot \frac{d}{dt}\left[\cos(3t)\right]$$

$$= \left(y^2 + 12x^3y\right) \cdot \left(2\cos(t)\right) + \left(2xy + 3x^4\right) \cdot \left(-3\sin(3t)\right)$$

When $t = \pi/2$, $\cos(t) = 0$, $\sin(3t) = -1$, $x = 2$, and $y = 0$. Substituting:

$$\frac{dz}{dt} = (0 + 0) \cdot (0) + \left(0 + 3(2)^4\right) \cdot (-3 \cdot -1)$$

$$= 3(2)^4 \cdot 3 = 144$$

Another case is where $x$ and $y$ are also multi-variable functions. Consider $z = f(x, y)$, $x = g(s, t)$, and $y = h(s, t)$. This means $z$ is an indirect function of $s$ and $t$:

$$z = f(x, y) = f\left(g(s, t), h(s, t)\right)$$

In this case, there are two partial derivatives of $z$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

**Example**: Find $\partial z/\partial s$ and $\partial z/\partial t$ if $z = e^{2x}\cos y$, $x = s^2t$, and $y = st^2$.

**Solution**: First, let's find $\partial z/\partial s$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$= \frac{\partial}{\partial x}\left[e^{2x}\cos y\right] \cdot \frac{\partial}{\partial s}\left[s^2 t\right] + \frac{\partial}{\partial y}\left[e^{2x}\cos y\right] \cdot \frac{\partial}{\partial s}\left[st^2\right]$$

$$= \left(2e^{2x}\cos y\right) \cdot (2st) + \left(-e^{2x}\sin y\right) \cdot \left(t^2\right)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial s} = 4ste^{2s^2 t}\cos\left(st^2\right) - t^2 e^{2s^2 t}\sin\left(st^2\right)$$

And finding $\partial z/\partial t$:

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= \frac{\partial}{\partial x}\left[e^{2x}\cos y\right] \cdot \frac{\partial}{\partial t}\left[s^2 t\right] + \frac{\partial}{\partial y}\left[e^{2x}\cos y\right] \cdot \frac{\partial}{\partial t}\left[st^2\right]$$

$$= \left(2e^{2x}\cos y\right) \cdot \left(s^2\right) + \left(-e^{2x}\sin y\right) \cdot (2st)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial t} = 2s^2 e^{2s^2 t}\cos\left(st^2\right) - 2ste^{2s^2 t}\sin\left(st^2\right)$$

### *Exercise 342* The Chain Rule for Multivariable Functions

Find $dz/dt$ or $\partial z/\partial s$ and $\partial z/\partial t$.

*Working Space*

1. $z = \sin x \cos y$, $x = 3\sqrt{t}$, $y = 2/t$
2. $z = \sqrt{1 + xy}$, $x = \tan t$, $y = \arctan t$
3. $z = \arctan\left(x^2 + y^2\right)$, $x = t \ln s$, $y = se^t$
4. $z = \sqrt{x}e^{xy}$, $x = 1 + st$, $y = s^2 - t^2$

*Answer on Page 2011*

## 123.2 Interpreting Partial Derivatives

What is the meaning of a partial derivative? Recall that $z = f(x, y)$ plots a surface, $S$. Consider the function $z = \cos y - x^2$, shown in figure 200.4.

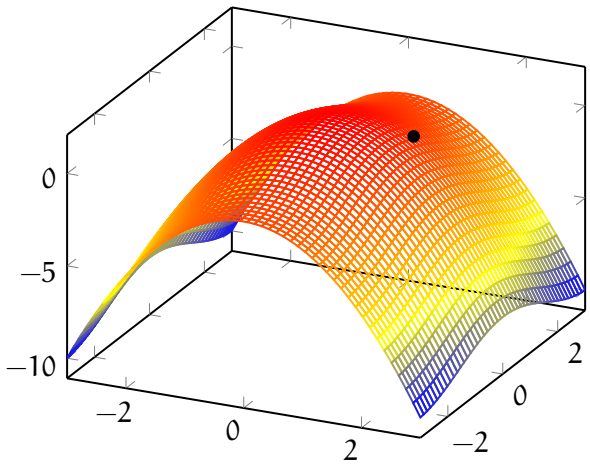



Figure 123.1: The surface $z = \cos y - x^2$

We can see that $f(1, \pi/3) = -1/2$; therefore, the point $(1, \pi/3, -1/2)$ lies on the surface $z = \cos y - x^2$ (the black dot shown in figure **??**). If we fix $y$ such that $y = \pi/3$, we are looking at the intersection between the surface and the plane $y = \pi/3$ (see figure **??**).

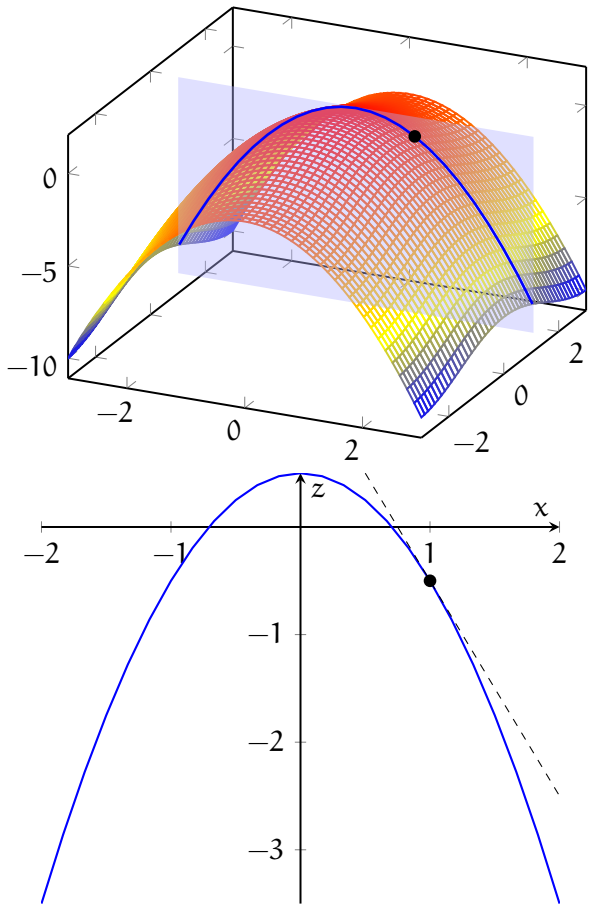



Figure 123.2: The intersection between the surface $z = \cos y - x^2$ and $y = \pi/3$ is the parabola $z(x) = 1/2 - x^2$

We can describe this intersection as $g(x) = f(x, \pi/3)$, so the slope of a tangent line to this intersection is given by $g'(x) = f_x(x, \pi/3)$. This means, geometrically, $f_x(1, \pi/3)$ is the slope of the line that lies tangent to $z = f(x, y)$ at the point $(1, \pi/3, -1/2)$ and in the plane $y = \pi/3$ (see figure 123.2). Alternatively, you could think of $f_x$ as the slope of the tangent line to the surface that is parallel to the x-axis.

Similarly, we can fix $x = 1$ and look at the intersection between the surface $z = \cos y - x^2$ and the plane $x = 1$ (see figure 123.3. Just like before, we can describe this intersection as $h(y) = f(1, y)$, which means the slope of a line tangent to the intersection is given by $h'(y) = f_y(1, y)$. Therefore, as with $f_x$, $f_y(a, b)$ gives the slope of a line tangent to the point $(a, b, f(a, b))$ and parallel to the y-axis.

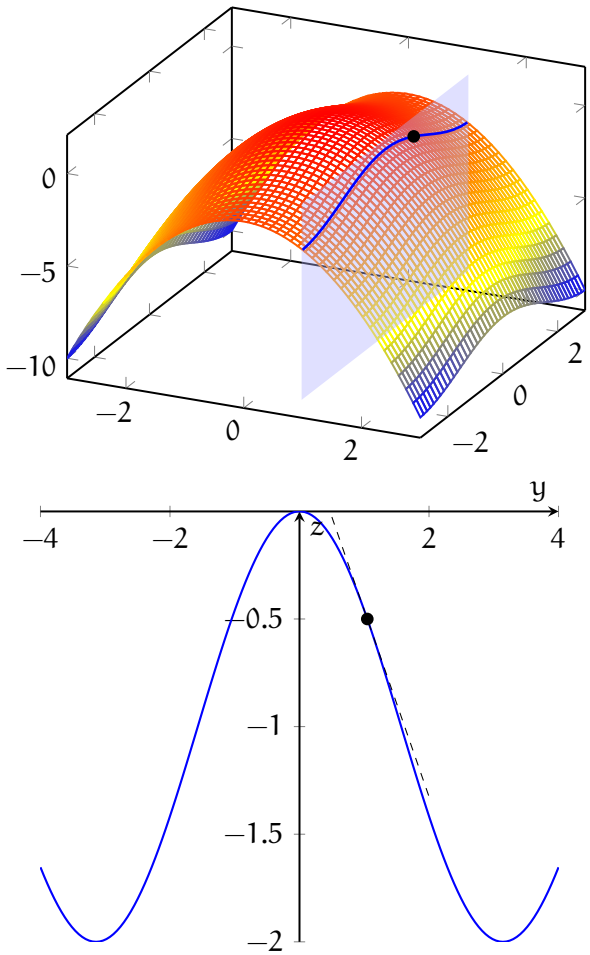



Figure 123.3: The intersection between the surface $z = \cos y - x^2$ and $x = 1$ is the trigonometric function $z = \cos y - 1$

**Example**: The density of bacterial growth at a point $(x, y)$ on a flat agar plate is given by $D = 45/\left(2 + x^2 + y^2\right)$. Find the rate of change of bacterial density at the point $(1, 3)$ (a) in the x-direction and (b) in the y-direction. Interpret the meaning of your results.

**Solution**: The rate of change of a two-variable function in the x-direction is given by the partial derivative with respect to x:

$$D_x = \frac{\partial}{\partial x}\frac{45}{2 + x^2 + y^2} = \frac{-45\left(\partial/\partial x\right)\left(2 + x^2 + y^2\right)}{\left(2 + x^2 + y^2\right)^2}$$

$$= \frac{-90x}{\left(2 + x^2 + y^2\right)^2}$$

The rate of change in the $x$-direction at $(x, y) = (1, 3)$ is given by:

$$D_x(1,3) = \frac{-90(1)}{(2+1^2+3^2)^2} = \frac{-90}{(12)^2} = \frac{-90}{144} = -\frac{5}{8}$$

This means that at $(1, 3)$, the density of bacteria is decreasing as you move away $x = 0$ along the line $y = 3$.

Similarly, the rate of change in the $y$-direction is given by the partial derivative with respect to $y$:

$$D_y = \frac{\partial}{\partial y}\frac{45}{2+x^2+y^2} = \frac{-45\,(\partial/\partial y)\left(2+x^2+y^2\right)}{(2+x^2+y^2)^2}$$
$$= \frac{-90y}{(2+x^2+y^2)^2}$$

The rate of change in the $y$-direction at $(x, y) = (1, 3)$ is given by:

$$D_y(1,3) = \frac{-90(3)}{(2+1^2+3^2)^2} = \frac{-270}{144} = -\frac{15}{8}$$

This means that at $(1, 3)$ the density of bacteria is decreasing faster along the $y$-direction than along the $x$-direction.

### *Exercise 343* Using partial derivatives to find tangent lines

*Working Space*

Find equations for tangent lines to the surface at the given xy-coordinate. In which direction is the function changing the fastest?

1. $z = x^2 e^{y/x}$, $(1, -1)$
2. $z = \cos x + y \sin y$, $(\pi, \pi/2)$
3. $z = x^2 y - 3xy^2$, $(3, 2)$

*Answer on Page 2014*

## 123.3 Gradient Vectors

The gradient vector is used to find the direction of the maximum rate of change of a surface (for example, the steepest part of a mountain). In order to understand the gradient, we must first discuss directional derivatives. Recall that the partial derivatives, $f_x$ and $f_y$, can be used to define a plane tangent to the surface $z = f(x, y)$ (see figure 131.3). Directional derivatives allow us to find the slope of the tangent plane in directions other than the x- and y-directions.

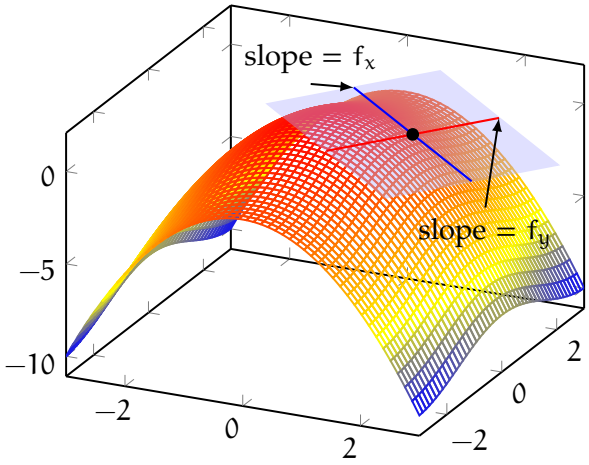



Figure 123.4: The directional derivatives, $f_x$ and $f_y$ define a tangent plane

### 123.3.1 Directional Derivatives

The contour map in figure 123.5 shows the elevation, $f(x, y)$ for a mountain. You already know that you can use the partial derivatives, $f_x$ and $f_y$ to find the rate of change in elevation going east-west or north-south. But what about other directions? Suppose the hiking path you're on goes north-east. How can you predict the steepness (i.e. the rate of elevation change) along this path? The directional derivative allows us to find the rate of change in any direction.

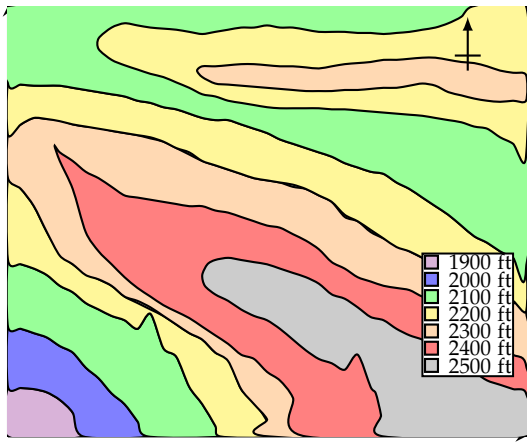



Figure 123.5: The contour plot shows the elevation of a mountain. $f_x$ gives the slope going east, while $f_y$ gives the slope going north

At some point, $(x_o, y_o)$, the partial derivatives $f_x(x_o, y_o)$ and $f_y(x_o, y_o)$ give the rate of change of elevation in the east-west and north-south directions, respectively (see figure 123.6).

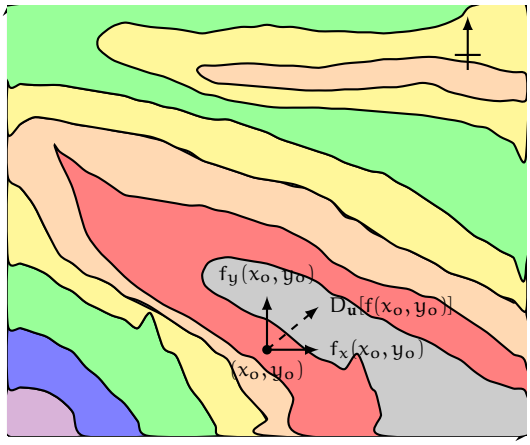



Figure 123.6: If $\mathbf{u}$ points north-east, then the directional derivative of $f(x, y)$ at $(x_o, y_o)$, $D_{\mathbf{u}}[f(x_o, y_o)]$, tells the rate of change going north-east

To find the rate of change at $(x_o, y_o)$, in the direction of some arbitrary unit vector, $\mathbf{u} = [a, b] = a\mathbf{i} + b\mathbf{j}$, we first note that the point $Q = (x_o, y_o, z_o)$, where $z_o = f(x_o, y_o)$, lies on the surface defined by $z = f(x, y)$. There is a vertical plane, P, that passes through Q and points in the direction of $\mathbf{u}$. This intersection defines curve C, which lies on the surface, and the slope of this curve at $Q = (x_o, y_o, z_o)$ is the directional derivative of H in the direction of $\mathbf{u}$ (see figure 123.7).

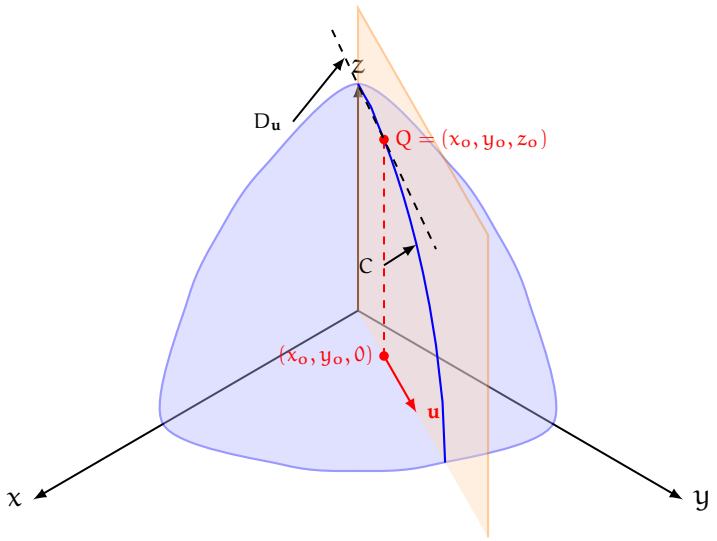



Figure 123.7: The slope of the curve formed between the plane parallel to $\mathbf{u}$ and the surface $z = f(x, y)$ is the directional derivative, $D_{\mathbf{u}}$

We can choose another point, $R = (x, y, z)$, the is $h$ units away from Q along $\mathbf{u}$ (see 123.8). Then the change in $x$ is $x - x_o = ha$ and the change is $y$ is $y - y_o = hb$. And the slope

from Q to R is given by:

$$\frac{\delta z}{h} = \frac{f(x,y) - f(x_o, y_o)}{h}$$

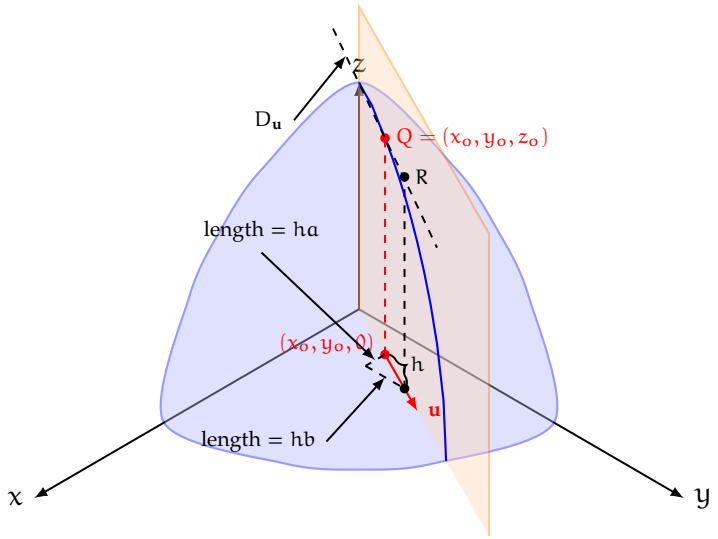



Figure 123.8: A second point, R, along **u** is h units away along **u**

We find the directional derivative by substituting for x and y and taking the limit as h goes to zero:

$$D_{\mathbf{u}} f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_o + ha, y_o + hb) - f(x_o, y_o)}{h}$$

How is this related to $f_x$ and $f_y$? Let's define $g(h)$ such that $g(h) = f(x_o + ha, y_o + hb)$. Then

$$g'(0) = \lim_{h \to 0} \frac{g(h) - g(0)}{h}$$

$$= \lim_{h \to 0} = \frac{f(x_o + ha, y_o + hb) - f(x_o, y_o)}{h} = D_{\mathbf{u}} f(x_o, y_o)$$

We can also apply the Chain Rule to $g(h)$:

$$g'(h) = \frac{\partial f}{\partial x}\frac{dx}{dh} + \frac{\partial f}{\partial y}\frac{dy}{dh} = f_x(x,y)a + f_y(x,y)b$$

Substituting $h = 0$, $x = x_o$, and $y = y_o$, we see that:

$$g'(0) = f_x(x_o, y_o)a + f_y(x_o, y_o)b$$

Which means that:

$$D_{\mathbf{u}}f(x_o, y_o) = f_x(x_o, y_o)a + f_y(x_o, y_o)b$$

So a directional derivative is:

**The Directional Derivative**

Let $f$ be a differentiable function and $\mathbf{u}$ be a unit vector, $\mathbf{u} = [a, b]$. Then the directional derivative in the direction of $\mathbf{u}$ is:

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b = \mathbf{u}_x \left[\frac{\partial}{\partial x}f(x, y)\right] + \mathbf{u}_y \left[\frac{\partial}{\partial y}f(x, y)\right] \tag{123.1}$$

Where $\mathbf{u}_x$ and $\mathbf{u}_y$ are the $x$- and $y$-components of $\mathbf{u}$, respectively.

**Example**: Find the directional derivative $D_{\mathbf{u}}f(x, y)$ if $f(x, y) = y^3 - 3xy + 4x^2$ and $\mathbf{u}$ is the unit vector given by the angle $\theta = \pi/3$. What is the rate of change in the direction of $\mathbf{u}$ at $(1, 2)$?

**Solution**: We can describe $\mathbf{u}$ thusly:

$$\mathbf{u} = \left[\cos\frac{\pi}{3}, \sin\frac{\pi}{3}\right] = \left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right]$$

And therefore:

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)\left(\frac{1}{2}\right) + f_y(x, y)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\partial}{\partial x}\left(y^3 - 3xy + 4x^2\right)\left(\frac{1}{2}\right) + \frac{\partial}{\partial y}\left(y^3 - 3xy + 4x^2\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{1}{2}\left(-3y + 8x\right) + \frac{\sqrt{3}}{2}\left(3y^2 - 3x\right)$$

$$= \frac{-3}{2}y + 4x + \frac{3\sqrt{3}}{2}y^2 - \frac{3\sqrt{3}}{2}x = \frac{3\sqrt{3}}{2}y^2 + \frac{8 - 3\sqrt{3}}{2}x - \frac{3}{2}y$$

And therefore $D_{\mathbf{u}}f(1, 2)$ is:

$$= \frac{3\sqrt{3}}{2}(2)^2 + \frac{8 - 3\sqrt{3}}{2}(1) - \frac{3}{2}(2) = 6\sqrt{3} + 4 - \frac{3\sqrt{3}}{2} - 3$$

$$= 1 + \frac{9\sqrt{3}}{2}$$

### 123.3.2 Unit Vectors in Two Dimensions

What if the given vector is not a unit vector? We can scale the given vector to find a unit vector in the same direction:

**Example**: Find the directional derivative of $f(x,y) = 3x\sqrt{y}$ at $(1,4)$ in the direction of $\mathbf{v} = [2,1]$.

**Solution**: First, we need to find a unit vector in the same direction as $\mathbf{v}$. There are several ways to do this. In two dimensions, a unit vector in the same direction as $\mathbf{v}$ can be found using trigonometry (see figure 123.9 for an illustration).

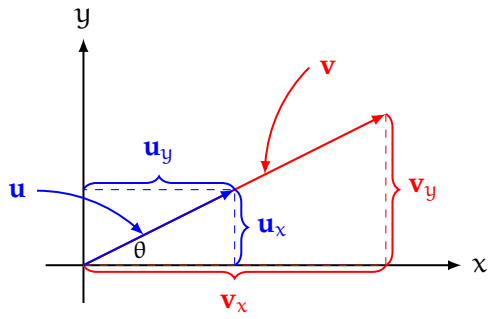



Figure 123.9: $\mathbf{u}$ is a unit vector in the same direction as $\mathbf{v}$

We know that $\theta = \arctan(\mathbf{v}_y/\mathbf{v}_x)$. Therefore, the x-component of the unit vector, $\mathbf{u}$, is given by:

$$\mathbf{u}_x = |\mathbf{u}|\cos\theta = \cos\left(\arctan\frac{\mathbf{v}_y}{\mathbf{v}_x}\right)$$

Similarly, we know that:

$$\mathbf{u}_y = |\mathbf{u}|\sin\theta = \sin\left(\arctan\frac{\mathbf{v}_y}{\mathbf{v}_x}\right)$$

(Recall that since $\mathbf{u}$ is a unit vector, $|\mathbf{u}| = 1$).

Let's use this method to find a unit vector, $\mathbf{u}$, in the same direction as $\mathbf{v} = [2,1]$:

$$\mathbf{u}_x = \cos\left(\arctan\frac{1}{2}\right) \approx \cos(0.464) = \frac{2}{\sqrt{5}}$$

$$\mathbf{u}_y = \sin\left(\arctan\frac{1}{2}\right) \approx \sin(0.464) = \frac{1}{\sqrt{5}}$$

Therefore, a unit vector in the same direction as $\mathbf{v}$ is $\mathbf{u} = \left[2/\sqrt{5}, 1/\sqrt{5}\right]$.

And we can find the directional derivative:

$$D_{\mathbf{u}}(x,y) = \mathbf{u}_x \left[\frac{\partial}{\partial x} f(x,y)\right] + \mathbf{u}_y \left[\frac{\partial}{\partial y} f(x,y)\right]$$

$$D_{\mathbf{u}}(x,y) = \left(\frac{2}{\sqrt{5}}\right)\left[\frac{\partial}{\partial x}\left(3x\sqrt{y}\right)\right] + \left(\frac{1}{\sqrt{5}}\right)\left[\frac{\partial}{\partial y}\left(3x\sqrt{y}\right)\right]$$

$$D_{\mathbf{u}}(x,y) = \left(\frac{2}{\sqrt{5}}\right)\left(3\sqrt{y}\right) + \left(\frac{1}{\sqrt{5}}\right)\left(\frac{3x}{2\sqrt{y}}\right)$$

$$D_{\mathbf{u}}(x,y) = \frac{12y + 3x}{2\sqrt{5y}}$$

To find the magnitude of the directional derivative at $(1,4)$, we substitute for $x$ and $y$:

$$D_{\mathbf{u}}(1,4) = \frac{12(4) + 3(1)}{2\sqrt{5(4)}} = \frac{51}{4\sqrt{5}} \approx 5.702$$

### 123.3.3 Unit Vectors in Higher Dimensions

The trigonometric explanation for finding unit vectors is more difficult to visualize in higher dimensions. However, there is another method that works well in 2, 3, and higher dimensions. Recall that the magnitude of a vector, $\mathbf{v} = [\mathbf{v}_x, \mathbf{v}_y]$ is given by $|\mathbf{v}| = \sqrt{(\mathbf{v}_x)^2 + (\mathbf{v}_y)^2}$. For a vector with $n$ dimensions, $\mathbf{v} = [\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n]$, the magnitude is given by $|\mathbf{v}| = \sqrt{(\mathbf{v}_1)^2 + (\mathbf{v}_2)^2 + \cdots + (\mathbf{v}_n)^2}$.

To find a unit vector, $\mathbf{u}$, in the same direction as $\mathbf{v}$, we can scale $\mathbf{v}$ up or down so that its magnitude is 1. We can do this by dividing by $\mathbf{v}$'s magnitude. Consider the two-dimensional vector used in the last example, $\mathbf{v} = [2,1]$. Its magnitude is:

$$|\mathbf{v}| = \sqrt{(2)^2 + (1)^2} = \sqrt{5}$$

Let's check if $\mathbf{v}/|\mathbf{v}|$ is a unit vector:

$$\frac{\mathbf{v}}{|\mathbf{v}|} = \left(\frac{1}{\sqrt{5}}\right)[2,1] = \left[\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right]$$

And the magnitude of this scaled vector is:

$$\left|\frac{\mathbf{v}}{|\mathbf{v}|}\right| = \sqrt{\left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2} = \sqrt{\frac{4}{5} + \frac{1}{5}} = \sqrt{1} = 1$$

Notice our unit vector is the same as we found using the trigonometric method above.

Another way to think of the question is: what factor, $k$, can we multiply $\mathbf{v}$ by to yield a vector with a magnitude of 1? Let's see this method for the 3-dimensional vector $\mathbf{v} = [3, 2, 1]$. We are looking for a $k$ such that:

$$|k\mathbf{v}| = 1$$

$$|k\mathbf{v}| = |[3k, 2k, 1k]| = \sqrt{(3k)^2 + (2k)^2 + (1k)^2}$$
$$= \sqrt{9k^2 + 4k^2 + k^2} = k\sqrt{14} = 1$$

Which implies that $k = 1/\sqrt{14}$, which is $1/|\mathbf{v}|$. And therefore a unit vector in the same direction as $\mathbf{v} = [3, 2, 1]$ is:

$$\mathbf{u} = \frac{1}{\sqrt{14}}[3, 2, 1] = \left[\frac{3}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}\right]$$

## *Exercise 344* Finding Directional Derivatives

Find the directional derivative of the function at the given point in the direction of the given vector.

1. $f(x,y) = e^{3x}\sin 2y$, $(0,\pi/6)$, $\mathbf{v} = [-3,4]$
2. $f(x,y) = x^2y + xy^3$, $(2,4)$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j}$
3. $f(x,y,z) = \ln\left(x^2 + 3y - z\right)$, $(2,2,1)$, $\mathbf{v} = [1,1,1]$

*Working Space*

*Answer on Page 2014*

### 123.3.4 Maximizing the Gradient

The directional derivative can be written as the dot product of two vectors:

$$D_u f(x,y) = a f_x(x,y) + b f_y(x,y) = [f_x(x,y), f_y(x,y)] \cdot \mathbf{u}$$

The first vector, $[f_x(x,y), f_y(x,y)]$, is called *the gradient of $f$*, and is noted as $\nabla f$.

**The Gradient**

For a two-variable function, $f(x,y)$, the gradient of $f$ is the vector:

$$\nabla f = [f_x(x,y), f_y(x,y)] = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}$$

Where $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors in the $x$- and $y$-directions, respectively.

Think back to the elevation example we opened the chapter with. What if we wanted to complete our ascent as quickly as possible? We would want to know the direction in which the elevation is changing the fastest. This occurs when the direction we are going is the same direction as the gradient vector, $\nabla f$.

Recall that the dot product is defined as:

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}|\,|\mathbf{v}| \cos\theta$$

Where $\theta$ is the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$. Applying this to the directional derivative, we see that:

$$D_u f = \nabla f \cdot \mathbf{u} = |\nabla f|\,|\mathbf{u}| \cos\theta = |\nabla f| \cos\theta$$

Which is at its maximum when $\nabla f$ and $\mathbf{u}$ point in the same direction (because $\cos(0) = 1$). Therefore, the gradient vector points in the direction of maximum change and the magnitude of that vector is the rate of maximum change.

**Example**: Find the maximum rate of change of $f(x,y) = 4y\sqrt{x}$ at $(4,1)$. In what direction does the maximum change occur?

**Solution**: We begin by finding $\nabla f$:

$$\nabla f = \left[\frac{\partial}{\partial x}\left(4y\sqrt{x}\right), \frac{\partial}{\partial y}\left(4y\sqrt{x}\right)\right]$$

$$\nabla f = \left[\frac{2y}{\sqrt{x}}, 4\sqrt{x}\right]$$

And thus,

$$\nabla f(4,1) = \left[\frac{2(1)}{\sqrt{4}}, 4\sqrt{4}\right] = [1,8]$$

Therefore, the maximum value of $\nabla f$ at $(4,1)$ is:

$$|\nabla f| = \sqrt{1^2 + 8^2} = \sqrt{65}$$

in the direction of the vector $[1,8]$.

## *Exercise 345* Using the Gradient to find Maximum Change

Suppose you are climbing a mountain whose elevation is described by $z = 3000 - 0.01x^2 - 0.02y^2$. Take the positive x-direction to be east and the positive y-direction to be north.

1. If you are at $(x, y) = (50, 50)$, what is your elevation?
2. If you walk south, will you ascend or descend?
3. If you walk northwest, will you ascend or descend? Will the rate of elevation change be greater or less than if you walked south?
4. In what direction should you walk for the steepest ascent? What will your ascension rate be?

*Working Space*

*Answer on Page 2017*

### 123.3.5 Conclusion

in summary, we have learned that the gradient of some multivariable function is the point of steepest increase of the function, represented as:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right] = \langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \rangle$$

The directional derivative is the gradient at some point $a$ dotted by the unit vector:

$$D_{\mathbf{u}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{u}$$

It is best to think of the gradient as an arrow pointing in the steepest uphill direction (vector), and the directional derivative as the slope of any specific uphill direction (scalar).

## 123.4 Applications of Partial Derivatives and Gradients

### 123.4.1 Laplace's Equation

A partial differential equation that has applications in fluid dynamics and electronics is Laplace's Equation. Solutions to Laplace's Equation are called *harmonic functions*.

**Laplace's Equation**

Consider a twice-differentiable function, $f$. In two dimensions, Laplace's Equation is given by:

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

And in three dimensions,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

Another way to represent Laplace's Equation is:

$$\delta f = \nabla^2 f = \nabla \cdot \nabla f = 0$$

Where $\nabla^2 = \delta$ is called the *Laplace operator*.

**Example**: Determine whether or not $f = x^2 + y^2$ is a solution to Laplace's Equation.

**Solution**: We are checking to see if $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$ for $f(x, y) = x^2 + y^2$. Finding $\partial^2 f / \partial x^2$:

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(x^2 + y^2\right)\right] \\ &= \frac{\partial}{\partial x}(2x) = 2 \end{aligned}$$

And finding $\partial^2 f / \partial y^2$:

$$\begin{aligned} \frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}\left(x^2 + y^2\right)\right] \\ &= \frac{\partial}{\partial y}(2x) = 2 \end{aligned}$$

Then $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 2 + 2 = 4 \neq 0$. Therefore, $f(x, y) = x^2 + y^2$ is not a solution to Laplace's Equation.

## *Exercise 346* Solutions to Laplace's Equation

Determine whether the function is a solution to Laplace's Equation.

*Working Space*

1. $f(x, y) = x^2 - y^2$
2. $f(x, y) = \sin x \cosh y + \cos x \sinh y$
3. $f(x, y) = e^{-x} \cos y - e^{-y} \cos x$

*Answer on Page 2018*

### 123.4.2 The Wave Equation

Another useful equation with partial derivatives is the Wave Equation:

$$\frac{\partial^2 f}{\partial t^2} = a^2 \frac{\partial^2 f}{\partial x^2}$$

Where $f$ is a function of $x$ and $t$ and $a$ is a constant. This equation describes waves, such as a vibrating string, light waves, or sound waves.

**Example**: Show that $f(x,t) = \sin(x - at)$ satisfies the Wave Equation.

**Solution**: First, we find the second partial derivatives:

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{\partial}{\partial t}(\sin(x - at))\right] = \frac{\partial}{\partial t}[-a\cos(x - at)] = -a^2 \sin(x - at)$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}(\sin(x - at))\right] = \frac{\partial}{\partial x}[\cos(x - at)] = -\sin(x - at)$$

And we see that:

$$a^2 \frac{\partial^2 f}{\partial x^2} = -a^2 \sin(x - at) = \frac{\partial^2 f}{\partial t^2}$$

Therefore, this function satisfies the Wave Equation.

### *Exercise 347* The Wave Equation

Show that the following functions satisfy the Wave Equation:

*Working Space*

1. $f(x,t) = \cos(kx)\cos(akt)$
2. $f(x,t) = \sin(x-at) + \ln(x+at)$
3. $f(x,t) = \frac{t}{a^2t^2-x^2}$

*Answer on Page 2018*

### 123.4.3 Cobb-Douglas Production Function

The Cobb-Douglas function describes the marginal utility of capital and labor as theorized by the economists Charles Cobb and Paul Douglas. Capital investments are things like new machinery, expanded factories, or raw materials. Labor investments involve hiring more workers or improving working conditions to improve work rates. We can describe total production, P, as a function of labor, L, and capital, K. Cobb and Douglas posit three conditions:

1. Without either labor or capital, production will cease.
2. The marginal utility of labor is proportional to the amount of production per unit of labor.
3. The marginal utility of capital is proportional to the amount of production per unit of capital.

The marginal utility of labor is given by the partial derivative, $\partial P/\partial L$ and the production per unit of labor is given by $P/L$. Therefore, statement 2 says that:

$$\frac{\partial P}{\partial L} = \alpha \frac{P}{L}$$

where $\alpha$ is some constant. Keeping $K$ constant at $K = K_0$, we have the differential equation:

$$\frac{dP}{dL} = \alpha \frac{P}{L}$$

Solving, we find that:

$$P(L, K_0) = C_1(K_0) L^{\alpha}$$

We make $C_1$ a function of $K_0$ because it could depend on $K_0$. In a similar manner to above, we can write statement 3 as a mathematical statement:

$$\frac{\partial P}{\partial K} = \beta \frac{P}{K}$$

where $\beta$ is also a constant. Keeping $L = L_0$ and solving, we see that:

$$P(L_0, K) = C_2(L_0) K^{\beta}$$

again, we assume $C_2$ is a function of the fixed labor, $L_0$. Combining these equations, we get:

$$P(L, K) = bL^{\alpha} K^{\beta}$$

where $b$ is a constant independent of capital and labor. Additionally, from statement 1, we know that $\alpha > 0$ and $\beta > 0$. What happens if both labor and capital are increased by a factor of $n$? Let's examine the effect on $P$:

$$P(nL, nK) = b(nL)^{\alpha}(nK)^{\beta}$$

$$P(nL, nK) = n^{\alpha+\beta} bL^{\alpha} K^{beta} = n^{\alpha+\beta} P(L, K)$$

Cobb and Douglas noted that if $\alpha + \beta = 1$, then $P(nL, nK) = nP(L, K)$, and therefore increasing labor and capital by a factor of $n$ increases production by a factor of $n$ as well. Therefore, the Cobb-Douglas equation assumes $\alpha + \beta = 1$ and can be written as:

$$P(L, K) = bL^{\alpha} K^{1-\alpha}$$

## *Exercise 348* Cobb-Douglas Production Model

Cobb and Douglas modeled production in the US from 1900 to 1922 with the equation $P(L, K) = 1.01L^{0.75}K^{0.75}$.

1. Express the marginal utility of labor as a function of L and K.
2. Express the marginal utility of capital as a function of L and K.
3. In 1916, $L = 382$ and $K = 126$ (compared to initial values of 100 in 1900). What is the marginal utility of labor in 1916? Of capital?
4. Based on your answer to the previous question, would you invest in capital or labor if you owned a factory in 1916? Why?

*Working Space*

*Answer on Page 2020*

Chapter 124

# Introduction to Linear Algebra

Welcome to the world of linear algebra, a branch of mathematics that relies on vectors, matrices, and linear transformations. You are familiar with most of these concepts, so in this workbook you will see how you can use them together to solve problems.

Let's review what you know.

- **Vectors**. In workbook 7, you saw how vectors can represent forces, as well as how to add and multiply them to figure out such things as rocket engine force and direction.
- **Matrices**. In workbook 3, you learned to use spreadsheets to solve problems numerically, such as how to figure out the number of barrels a cooper has to produce to achieve a certain take-home pay. Spreadsheets are essentially matrices — a row by column structure that contains values.
- **Linear transformations**. When you studied congruence in workbook 5, you were introduced to a few linear transformations, such as translation and reflection.

## 124.1 What's With the Linear?

You might be thinking, "Hey, haven't I been doing algebra already?"

You have! You have come a long way in your problem solving journey. You have used algebra to solve simple equations like $7x + 10 = 24$ and quadratic equations like $4x^2 + 9x + 31 = 0$. What distinguishes linear algebra is the focus on linear combinations. Any equation with a power greater than 1, such as a quadratic, is nonlinear. Want to see what we will be covering? Take a look at 3blue1brown's website with papers and videos on Linear Algebra: https://www.3blue1brown.com/topics/linear-algebra

We will first take a look at linear combinations

## 124.2 Linear Combinations

You won't see any **sin**, **cos**, or **tan** operations in this section. Linear operations do not use trigonometric functions; those operations are all nonlinear. A linear combination consists solely of addition and scalar multiplication. You will see that linear combinations allow you to solve many types of problems in science and engineering. Before we get deep into

the numbers, let's take a look at a few linear operations you can perform on images. This will give you an intuition for the underlying math. After that, we'll take a look at some numbers.

## 124.3 Image Operations

The simplest image, a bitmap, can be represented by a two-dimensional matrix of values — either 0 for black, or 1 for white. Grayscale images are also represented by a two-dimensional matrix of values, but the values typically range from 0 to 255. 0 is black, 255 is white, and the values in between represent shades of gray.

Color images are more complex. The simplest color image is a three-dimensional matrix of values. You can think of it as three 2D matrices, one to represent red values (R), another for green values (G), and the third for blue values (B). The combination of R, G, and B determines the color you see.

Working with images means working with millions of pixels. Fortunately, modern techniques make this a snap. Let's look at some common operations on an image of a rocket. Flipping is a linear operation.

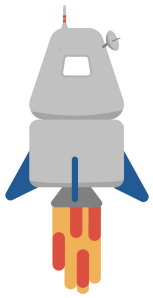


Figure 124.1: Rocket image.

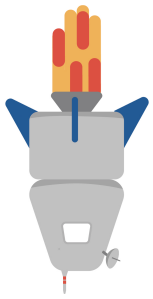


Figure 124.2: The rocket being flipped involves a binary linear equation.

Next, the image is rotated 90 degrees. This rotation is linear, but if you want to rotate it at an angle that isn't a multiple of 90, you would need trigonometry. This would be

treading into nonlinear territory, but that happens in the field of linear algebra. You will learn about nonlinear extensions later, which use trigonometric functions and imaginary numbers.

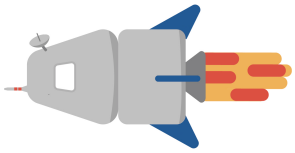


Figure 124.3: Rotating the rocket by 90°

Inversion is an interesting linear operation that involves redefining the red, green, and blue values, such that the new value is 1.0 minus the old value. The resulting black background gives the impression the rocket is in deep space, don't you think?

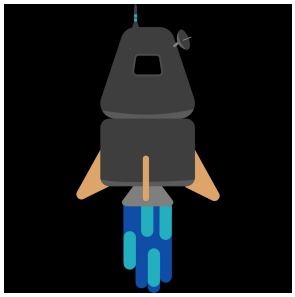


Figure 124.4: Inverting the rocket replaces each pixel with a "opposite" color.

It is possible to redefine the red, green, and blue values in many ways. Visit the NASA website and search for false color images. NASA and other scientists redefine colors to communicate such things as the amount of vegetation or water in an area, the temperatures of the sun's surface, and so on. Photographers often do this for artistic effect. For example, the image on the left was taken with an infrared camera. (This infrared is not the same as thermal kind you have likely seen before. This is the infrared that is emitted by living plants.) The image is further processed to swap channels. For example, the matrix representing red might be swapped with the matrix representing blue. The image on the right shows the image after swapping color values. All these swapping operations are linear.

Figure 124.5: Left: image with infrared camera. Right: image shown post swapping color values

## 124.4 The Numbers Behind Some Image Operations

You will see a few *matrices* in this section. Let's first look at how a spreadsheet can be represented as a matrix. Recall the barrel-making shop example. This is part of that spreadsheet.

| | A | B | C | D |
|---|---|---|---|---|
| 1 | **Barrels Produced** (per month) | 115 | 120 | 125 |
| 2 | **Materials cost** (per barrel) | $45.00 | $45.00 | $45.00 |
| 3 | **Sale price** (per barrel) | $100.00 | $100.00 | $100.00 |
| 4 | **Rent** (per month) | $2,000.00 | $2,000.00 | $2,000.00 |
| 5 | **Pretax Earnings** (per month) | $4,325.00 | $4,600.00 | $4,875.00 |
| 6 | **Taxes** (per month) | $865.00 | $920.00 | $975.00 |
| 7 | **Take home pay** (per month) | $3,460.00 | $3,680.00 | $3,900.00 |

Figure 124.6: Spreadsheet of the barrel example.

Represented as a matrix, it looks like the following. Note the differences: A matrix contains only values, no labels. This matrix uses floating point values, hence the inclusion of decimal points.

$$\begin{bmatrix} 115.0 & 120.0 & 125.0 \\ 45.0 & 45.0 & 45.0 \\ 100.0 & 100.0 & 100.0 \\ 2000.0 & 2000.0 & 2000.0 \\ 4325.0 & 4600.0 & 4875.0 \\ 865.0 & 920.0 & 975.0 \\ 3460.0 & 3680.0 & 3900.0 \end{bmatrix}$$

A matrix that represents an image contains only pixel values, whereas the barrel-making shop matrix represents seven kinds of variables: barrels produced, materials cost, sales price, rent, pretax earnings, taxes, and take home pay.

The simplest image to create is a bitmap, because that requires a matrix of zeros and ones. This is a matrix for a 10 pixel by 10 pixel image. Why use decimal points when this is obviously a matrix of integers? It turns out that when you use tools like Python to process matrices, you must be conscious of data types. Most of the Python methods we use for image operations expect floating points. A few expect integer types, but you'll see how to handle type conversion later, in the section on Python.

$$\begin{bmatrix} 0. & 0. & 0. & 0. & 1. & 0. & 0. & 0. & 1. & 0 \\ 0. & 0. & 0. & 0. & 1. & 1. & 1. & 1. & 1. & 1 \\ 0. & 0. & 0. & 0. & 1. & 0. & 0. & 0. & 1. & 0 \\ 0. & 0. & 0. & 0. & 1. & 0. & 0. & 0. & 1. & 0 \\ 0. & 0. & 0. & 0. & 1. & 0. & 0. & 0. & 1. & 0 \\ 0. & 0. & 0. & 0. & 1. & 0. & 0. & 0. & 1. & 0 \\ 0. & 0. & 0. & 0. & 1. & 0. & 0. & 0. & 1. & 0 \\ 0. & 0. & 0. & 0. & 1. & 1. & 1. & 1. & 1. & 1 \\ 0. & 0. & 0. & 0. & 1. & 0. & 0. & 0. & 1. & 0 \\ 0. & 0. & 0. & 0. & 1. & 0. & 0. & 0. & 1. & 0 \end{bmatrix}$$

When converted to an image, it is very tiny. This is an enlarged version, so you can see the pattern.

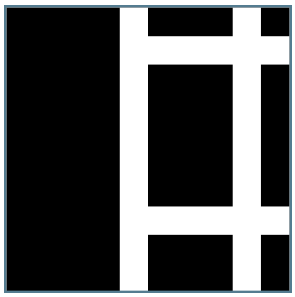


Figure 124.7: Resulting bitmap vectors as a black and white image.

We can create an inverse of this image by changing all the values in the matrix so that 0 becomes 1 and 1 becomes 0. (Technically this is not the way you would invert a matrix, as you will see in the Python section. For this example, you will get the same visual result, but when you start inverting matrices programmatically, you will learn the formal definition.)

$$\begin{bmatrix}
1. & 1. & 1. & 1. & 0. & 1. & 1. & 1. & 0. & 1 \\
1. & 1. & 1. & 1. & 0. & 0. & 0. & 0. & 0. & 0 \\
1. & 1. & 1. & 1. & 0. & 1. & 1. & 1. & 0. & 1 \\
1. & 1. & 1. & 1. & 0. & 1. & 1. & 1. & 0. & 1 \\
1. & 1. & 1. & 1. & 0. & 1. & 1. & 1. & 0. & 1 \\
1. & 1. & 1. & 1. & 0. & 1. & 1. & 1. & 0. & 1 \\
1. & 1. & 1. & 1. & 0. & 1. & 1. & 1. & 0. & 1 \\
1. & 1. & 1. & 1. & 0. & 0. & 0. & 0. & 0. & 0 \\
1. & 1. & 1. & 1. & 0. & 1. & 1. & 1. & 0. & 1 \\
1. & 1. & 1. & 1. & 0. & 1. & 1. & 1. & 0. & 1
\end{bmatrix}$$

When converted to an image and enlarged, it looks like this:

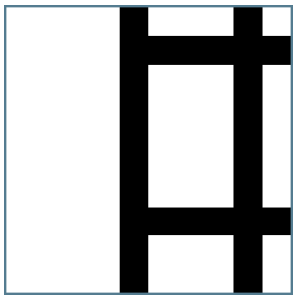


Figure 124.8: Inversion of the linear matrix.

Rotating the original matrix by 90 degrees gives this:

$$\begin{bmatrix}
0. & 1. & 0. & 0. & 0. & 0. & 0. & 1. & 0. & 0. \\
1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. \\
0. & 1. & 0. & 0. & 0. & 0. & 0. & 1. & 0. & 0. \\
0. & 1. & 0. & 0. & 0. & 0. & 0. & 1. & 0. & 0. \\
0. & 1. & 0. & 0. & 0. & 0. & 0. & 1. & 0. & 0. \\
1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. \\
0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. \\
0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. \\
0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. \\
0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0.
\end{bmatrix}$$

This is the resulting enlarged image:

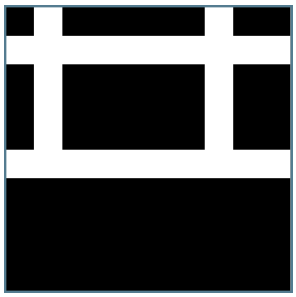


Figure 124.9: Rotating the image by 90deg.

You'll transpose many matrices in the upcoming pages. It requires swapping rows for columns.

$$\begin{bmatrix} 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. \\ 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. \\ 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. \\ 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. \\ 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. \\ 0. & 1. & 0. & 0. & 0. & 0. & 0. & 1. & 0. & 0. \\ 0. & 1. & 0. & 0. & 0. & 0. & 0. & 1. & 0. & 0. \\ 0. & 1. & 0. & 0. & 0. & 0. & 0. & 1. & 0. & 0. \\ 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. & 1. \\ 0. & 1. & 0. & 0. & 0. & 0. & 0. & 1. & 0. & 0. \end{bmatrix}$$

The resulting image looks like this:

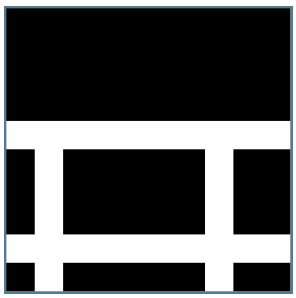


Figure 124.10: Transposed image reversing rows and columns.

What about adding images? That fits the definition of a linear combination. Recall that grayscale images have values from 0 to 255. To make things simple, let's define two matrices with values ranging from 0.0 to 1.0. When we want a grayscale image, it is easy to multiply the matrix by 255.

Let's call this matrix $f$.

$$\begin{bmatrix} 1.0 & 0.5 & 0.0 \\ 1.0 & 1.0 & 1.0 \\ 0.5 & 0.0 & 0.0 \end{bmatrix}$$

When multiplied by 255 and converted to a grayscale image:

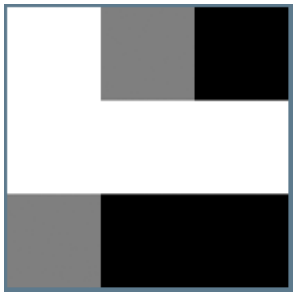


Figure 124.11: Bitmap of matrix $f$.

Let's call this matrix $g$:

$$\begin{bmatrix} 0.5 & 0.0 & 0.0 \\ 1.0 & 0.5 & 1.0 \\ 1.0 & 1.0 & 1.0 \end{bmatrix}$$

When multiplied by 255 and converted to a grayscale image:

Figure 124.12: Bitmap of matrix $g$.

When we add $f$ and $g$ we get $k$:

$$\begin{bmatrix} 1.5 & 0.5 & 0.0 \\ 2.0 & 1.5 & 2.0 \\ 1.5 & 1.0 & 1.0 \end{bmatrix}$$

However, the values in k exceed the range of 0.0 to 1.0, so we normalize by dividing the matrix by 2.0:

$$\begin{bmatrix} 0.75 & 0.25 & 0.00 \\ 1.00 & 0.75 & 1.00 \\ 0.75 & 0.50 & 0.50 \end{bmatrix}$$

When multiplied by 255 and converted to grayscale, we get:

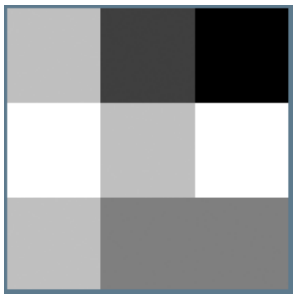


Figure 124.13: Adding f + g represented as a bitmap.

Let's go back to the first small grayscale image (from Figure 124.11):

Figure 124.14: Figure 11, repeated.

If you want to keep the pattern in the first column, you could multiply the matrix by a vector, [1.0 0.0 0.0]. The 1.0 will keep the values in the first column, but the 0.0 will knock out the other values because 0.0 times anything equals 0.0.

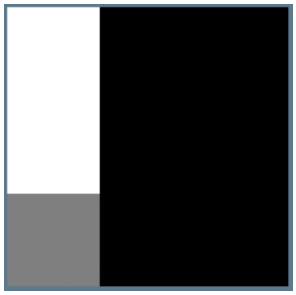


Figure 124.15: Multiplying channels by 0 results in them being represented as black bitmaps, while 1 keeps them the same.

What do you think will happen if you use the vector [0.0 1.0 0.0] or [0.0 0.0 1.0]? You'll get a chance later to use Python to perform image operations.

All the operations we performed on these images satisfy the requirement for linear combinations: preserving addition and scalar multiplication.

## 124.5 Applications of Linear Algebra

So far you've seen how linear operations on matrices can process images by:

- multiplying a matrix using a scalar (e.g., normalize, change the range)
- adding one matrix to another to get a composite image
- multiplying two matrices to perform a transform (e.g., flipping)
- multiplying a matrix with a vector (isolating a channel)

Many areas in engineering and science rely on the matrix operations defines by linear algebra. Besides image processing, linear algebra is used for:

- **Computer Graphics**. When you play a video game or watch the latest CG animation, matrix operations transform objects in the scene to make them appear as if moving, getting closer, and so on. You can represent the vertices of objects as vectors, and then apply a transformation matrix.
- **Data Analysis**. We live in an era in which it's easy to collect so much data that it's difficult to make sense of the data by just looking at it. You can represent the data in matrix form and then find a solution vector. For example, scientists use this technique to figure out the effectiveness of drug treatments on disease.
- **Economics**. Take a look at financial section of any news organization and you'll see headlines such as "Economic Data Points to Faster Growth" or "Is the Inflation

Battle Won?" Economists can use systems of linear equations to represent economic indicators, such as consumer consumption, government spending, investment rate, and gross national product. By using various methods that you'll learn about later, they can get a good idea of the state of the economy.

- **Engineering**. Engineers couldn't do without linear algebra. For example, the orbital dynamics of space travel relies on it. Engineers must predict and calculate the motion of planetary bodies, satellites, and spacecraft. By solving systems of linear equations engineers can make sure that a spacecraft travels to its destination without running into a satellite or space rock.

## 124.6 Let's Observe the Sun!

India recently sent the Aditya spacecraft on a mission to study the Sun. Without a thorough understanding of linear algebra (among other things), the engineers would not have accomplished the amazing feat of getting Aditya in a stable orbit around a Lagrange point.

In previous chapters you learned about gravity and its effects. A Lagrange point is a point in space between two bodies (e.g. Earth and Sun) where there is gravitational equilibrium. With the right trajectory, a spacecraft will orbit around a Lagrange point in a stable position that doesn't require much energy to maintain. That's called a Halo orbit. Because that there are no fueling stations in space, a Halo orbit will allow Aditya to maintain position for about 5 years. Pretty good mileage!

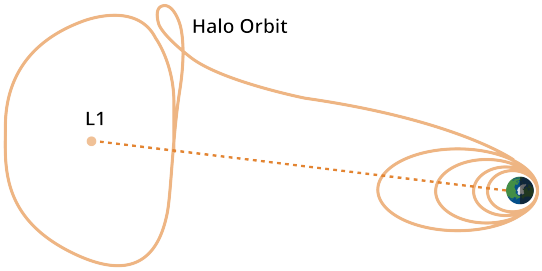



Figure 124.16: The Aditya Orbit.

Aditya's engineers had to calculate a looping maneuver that would precisely inject the Aditya spacecraft into the Halo orbit. They determined the angles and burn times for the thrust engine. If they were wrong in one direction, the spacecraft would fly off to the sun. The other direction would send the spacecraft back in the direction of Earth. Their success is due to a solid understanding of vectors and linear algebra.

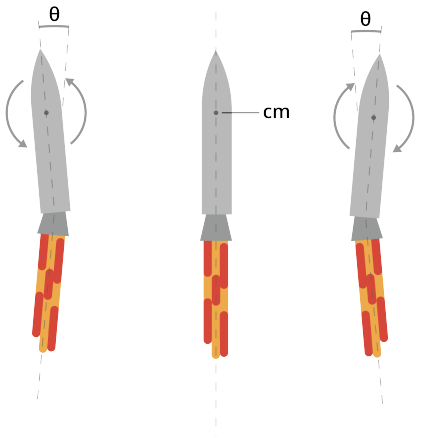



Figure 124.17: The gimbal of the rocket engine.

## 124.7 Images in Python

One of the wonderful things about python is the availability of libraries for specialized computation. The Python Imaging Library, PIL, is what you'll use to create images, read existing images from disk, and perform operation on images. To create and manipulate arrays, you will use NumPy.

Create a file called `image_creation.py` and enter this code:

```
# Import necessary modules
import numpy as np
import PIL
from PIL import Image
from PIL import ImageOps

# Create a 10 by 10 pixel bitmap Image.
# Using a decimal point ensure python see the values as floating point numbers
# Some image operations assume floats

bitmapArray = np.array([
[0., 0., 0., 0., 1., 0., 0., 0., 1., 0.],
[0., 0., 0., 0., 1., 1., 1., 1., 1., 1.],
[0., 0., 0., 0., 1., 0., 0., 0., 1., 0.],
[0., 0., 0., 0., 1., 0., 0., 0., 1., 0.],
[0., 0., 0., 0., 1., 0., 0., 0., 1., 0.],
[0., 0., 0., 0., 1., 0., 0., 0., 1., 0.],
[0., 0., 0., 0., 1., 0., 0., 0., 1., 0.],
[0., 0., 0., 0., 1., 1., 1., 1., 1., 1.],
[0., 0., 0., 0., 1., 0., 0., 0., 1., 0.],
[0., 0., 0., 0., 1., 0., 0., 0., 1., 0.]])

```

```
# Image.fromarray assumes a range of 0 to 255, so scale by 255

myImage = Image.fromarray(bitmapArray*255)
myImage.show()

# A window opens with an image so tiny you might think nothing is there
# Zoom in to see the pattern

# Transpose the array, create an image, and then show it.
# Note that you operate on the original array (not the image)
# Remember to zoom in to see the pattern

myImageTransposed = bitmapArray.transpose()
myImageTransposed.show()

# Invert the array. You'll use the NumPy invert method.
# The invert method assumes integer values. You need to convert the data type
# Numpy has a method for that

intBitmapArray = np.asarray(bitmapArray,dtype="int")
invertedArray = np.invert(intBitmapArray)

# Take a look at the array

invertedArray

# The values range from -2 to -1. Image values are positive.
# You need to change the range so the values are from 0 to 255
# Further you need to change back to floating point values because
# the PIL method requires them

invertedArray = (invertedArray + 2)*1.0
invertedImage = Image.fromarray(255*invertedArray)
invertedImage.show()

# Zoom in on the image and compare the pattern with the original

```

FIXME this needs improvement

## 124.8 Exercise

Create a python program that creates matrix f and matrix g from the previous section, and then performs all the operations shown in that section. If you are not sure how to accomplish something, consult the online documentation for the PIL and NumPy python libraries.

Chapter 125

# Vectors and Matrices

The last chapter provided an overview of linear algebra, using several image examples. In this chapter, we will focus primarily on vector-matrix multiplications. First, we will show how matrices can be used to represent a set of linear equations. Then, we will provide you with a general definition of vector-matrix multiplication, followed by a few examples. You will have an opportunity to solve a problem manually, then by using Python. In this chapter, we will use two-dimensional matrices for simplicity, but a matrix can have any number of dimensions.

## 125.1 Matrices

We've been looking at vectors. We've seen them in physics as a straight line comprised of $x$ and $y$ components, or represented as a column of numbers. For example, while we may write $\mathbf{v} = [1, 2, 3]$ in line, the vector is really:

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

A matrix can be made of many columns, like the $3 \times 3$ matrix shown below:

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ 3 & 9 & 27 \end{bmatrix}$$

We describe the size and shape of matrices by saying *an* $m \times n$ *matrix*, where $m$ is the number of *rows* and $n$ is the number of *columns*. A *vector* is simply a one-column matrix. For example, the vectors $\mathbf{v}$ above is of size$3 \times 1$. Matrices aren't restricted to 2 dimensions: a matrix can be 3, 4, or any number of dimensions. For example, a $3 \times 2 \times 4$ matrix would be made of 4 stacked $3 \times 2$ matrices. Visually, however, that is hard to represent, and even worse to compute.

A matrix is considered *square* when it has the same number of rows and columns. Most matrix operations have the restriction that the matrix must be square in order to perform operations, such as determinant calculation, finding an inverse, or eigenvector/eigenvalues, all of which you will learn in future chapters.3

## *Exercise 349* Matrix Dimensions 1

Write the dimensions of the following matrices:

*Working Space*

1. $\begin{bmatrix} -3 & 0 & 4 & -2 & -4 \\ -1 & 5 & 3 & 4 & -2 \\ -3 & 2 & 3 & -5 & 1 \end{bmatrix}$
2. $\begin{bmatrix} -3 & 1 \end{bmatrix}$
3. $\begin{bmatrix} -3 & 2 & -3 \\ 4 & 0 & -3 \\ -5 & -4 & 1 \\ 0 & -2 & 2 \end{bmatrix}$

*Answer on Page 2021*

## *Exercise 350* Matrix Dimensions 2

Create a matrix with the indicated dimensions.

*Working Space*

1. $1 \times 3$
2. $2 \times 4$
3. $4 \times 3$

*Answer on Page 2021*

### 125.1.1 Zero Matrices

Recall that we can represent a generic zero vector as $\mathbf{0}$ or $\vec{0}$ (you may see both), which indicates a vector of any number of dimensions filled with zeros. Just like vectors, there are *zero matrices*, which can be any number of dimensions, all filled with zeros. In two dimensions, zero matrices are denoted as $0_{m \times n}$, where the subscript is the dimension of

the matrix. The subscript can be expanded to denote any number of dimensions.

## 125.2 Operations of Matrices

### 125.2.1 Adding and Subtracting Matrices

Matrices that are the same dimension can be added and subtracted. Just like vectors, to add matrices you add the elements in the same position:

$$\begin{bmatrix} -2 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 2 \\ -1 & -4 \end{bmatrix} = \begin{bmatrix} -2+5 & -1+2 \\ 2+-1 & 4+-4 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$$

And to subtract matrices, you subtract the elements in the same position:

$$\begin{bmatrix} -2 & -1 \\ 2 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 2 \\ -1 & -4 \end{bmatrix} = \begin{bmatrix} -2-5 & -1-2 \\ 2-(-1) & 4-(-4) \end{bmatrix} = \begin{bmatrix} -7 & -3 \\ 3 & 8 \end{bmatrix}$$

Formally, for 2-dimensional matrices, we can say that:

**Adding and Subtracting Matrices**

For two $m \times n$ matrices, the sum of the matrices is the matrix of the sums of the elements in analogous positions:

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1n} \\ x_{21} & x_{22} & x_{23} & \cdots & \vdots \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ x_{m1} & x_{m2} & x_{m3} & \cdots & x_{mn} \end{bmatrix} + \begin{bmatrix} y_{11} & y_{12} & y_{13} & \cdots & y_{1n} \\ y_{21} & y_{22} & y_{23} & \cdots & \vdots \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ y_{m1} & y_{m2} & y_{m3} & \cdots & y_{mn} \end{bmatrix} =$$

$$\begin{bmatrix} x_{11}+y_{11} & x_{12}+y_{12} & x_{13}+y_{13} & \cdots & x_{1n}+y_{1n} \\ x_{21}+y_{21} & x_{22}+y_{22} & x_{23}+y_{23} & \cdots & \vdots \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ x_{m1}+y_{m1} & x_{m2}+y_{m2} & x_{m3}+y_{m3} & \cdots & x_{mn}+y_{mn} \end{bmatrix}$$

To subtract matrices, simply add the negative of the second matrix (that is, $\boldsymbol{A}$ - $\boldsymbol{B} = \boldsymbol{A}$ + - $\boldsymbol{B}$). Additionally, matrix addition is commutative ($\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A}$).

Matrices of different dimensions cannot be added or subtracted.

## *Exercise 351* Adding and Subtracting Matrices

Find $\boldsymbol{A} + \boldsymbol{B}$, $\boldsymbol{A}$ - $\boldsymbol{B}$, and $\boldsymbol{B}$ - $\boldsymbol{A}$.

*Working Space*

1. $\boldsymbol{A} = \begin{bmatrix} 0 & 4 & 0 & 5 \end{bmatrix}$ and $\boldsymbol{B} = \begin{bmatrix} -2 & 3 & -2 & 5 \end{bmatrix}$
2. $\boldsymbol{A} = \begin{bmatrix} 4 & -4 & -2 \\ 1 & -3 & 5 \\ -5 & 3 & 0 \end{bmatrix}$ and $\boldsymbol{B} = \begin{bmatrix} 5 & 0 & -1 \\ -5 & -3 & -2 \\ -5 & 3 & -4 \end{bmatrix}$.
3. $\boldsymbol{A} = \begin{bmatrix} -2 & -1 & -5 & -1 \\ 5 & -4 & 4 & 3 \end{bmatrix}$ and $\boldsymbol{B} = \begin{bmatrix} -5 & -2 & 3 & -5 \\ 0 & 5 & -4 & -3 \end{bmatrix}$.

*Answer on Page 2021*

### 125.2.2 Multiplying Matrices

Matrix multiplication has dimensional limits. We cannot multiply any two matrices; the first matrix must have the same number of columns as the second has number of rows. Let's examine the origin of the dimension limits on matrix multiplication. We begin with a review of the vector dot product.

Recall that in order to find the dot product of two vectors, they must be the same length (that is, the same number of dimensions). The result is always a scalar: one number. You can review finding the dot product of vectors and practice the dimension limits on the vector dot product in the next exercise.

## *Exercise 352* Vector Dot Product Review

Find all possible pairs of vectors that can be used to find a dot product, then find the dot products.

*Working Space*

1. $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
2. $\mathbf{b} = \begin{bmatrix} -3 \\ 3 \\ 5 \\ -5 \end{bmatrix}$
3. $\mathbf{c} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$
4. $\mathbf{d} = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$
5. $\mathbf{e} = \begin{bmatrix} 1 \\ -5 \\ 3 \\ 1 \end{bmatrix}$
6. $\mathbf{f} = \begin{bmatrix} 4 \\ 1 \\ -3 \end{bmatrix}$

*Answer on Page 2022*

To multiply two matrices, it is helpful to think of the rows of the first matrix and the columns of the second matrix as vectors. Let's see how this shakes out for two $2 \times 2$ matrices in Figure 125.1:

$$\begin{matrix} & & \mathbf{b}_1 & \mathbf{b}_2 \\ & & \downarrow & \downarrow \\ \mathbf{a}_1 \rightarrow & \begin{bmatrix} 5 & 4 \end{bmatrix} & \times & \begin{bmatrix} -1 & -2 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 \cdot \mathbf{b}_1 & \mathbf{a}_1 \cdot \mathbf{b}_2 \\ \mathbf{a}_2 \cdot \mathbf{b}_1 & \mathbf{a}_2 \cdot \mathbf{b}_2 \end{bmatrix} \\ \mathbf{a}_2 \rightarrow & \begin{bmatrix} -5 & 1 \end{bmatrix} & & \end{matrix}$$

$$\boldsymbol{A} \quad \times \quad \boldsymbol{B}$$

Figure 125.1: Each entry in $\boldsymbol{C}$, $c_{ij}$, is the dot product of the $i^{th}$ row of $\boldsymbol{A}$, $a_i$, and the $j^{th}$ column of $\boldsymbol{B}$, $b_j$.

Let's look at this more concretely. For two-dimensional matrices, it can be helpful to move your left index finger across the row and right index finger down the column, as shown in Figure 125.2.

$$\begin{bmatrix} 5 & 4 \\ -5 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & -2 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} 5\times\text{-}1+4\times\text{-}4 & \\ & \end{bmatrix}$$

$$\begin{bmatrix} 5 & 4 \\ -5 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & -2 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} -21 & 5\times\text{-}2+4\times\text{-}4 \\ & \end{bmatrix}$$

$$\begin{bmatrix} 5 & 4 \\ -5 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & -2 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} -21 & -26 \\ -5\times\text{-}1+1\times\text{-}5 & \end{bmatrix}$$

$$\begin{bmatrix} 5 & 4 \\ -5 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & -2 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} -21 & -26 \\ 0 & -5\times\text{-}2+1\times\text{-}4 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 4 \\ -5 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & -2 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} -21 & -26 \\ 0 & 6 \end{bmatrix}$$

Figure 125.2: You can use your fingers to trace across matrix $\boldsymbol{A}$ and down matrix $\boldsymbol{B}$ to find $\boldsymbol{A} \cdot \boldsymbol{B}$.

Since each entry in the product matrix is the dot product between a row of the first matrix and a column of the second matrix, the first matrix must have the same number of elements in each row as the second has in each column. Another way to say this is that the number of columns of the first matrix must match the number of rows in the second matrix.

**Matrix Multiplication**

For two-dimensional matrices, the inner dimensions must match in order to carry out matrix multiplication. That is, if we want to find $\boldsymbol{A} \times \boldsymbol{B}$, and $\boldsymbol{A}$ has dimensions $m \times n$, then $\boldsymbol{B}$ must have dimensions $n \times p$, where $m$, $n$, and $p$ are integers. The resulting matrix will have dimensions $m \times p$ ($m$ and $p$ may be equal or unequal). Note that we use $\times$ to differentiate from the dot product, represented by $\cdot$.

## *Exercise 353* Multiplying Matrices 1

Multiply the matrices.

*Working Space*

1. $\begin{bmatrix} -5 & -2 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} -3 & -1 & -5 \\ 3 & 0 & 3 \\ 4 & -1 & -4 \\ -1 & -4 & 2 \end{bmatrix}$

2. $\begin{bmatrix} 1 \\ 5 \\ -5 \\ 4 \\ 1 \end{bmatrix} \times \begin{bmatrix} 0 & 5 & 1 \end{bmatrix}$

3. $\begin{bmatrix} -1 & 4 & -4 \\ 5 & -3 & 5 \\ -1 & -4 & 4 \\ -4 & 1 & 4 \end{bmatrix} \times \begin{bmatrix} -3 & 5 & 1 \\ -3 & 0 & -3 \\ 0 & 3 & 0 \end{bmatrix}$

*Answer on Page 2022*

## *Exercise 354* Multiplying Matrices 2

Working Space

Find $A \times B$ and $B \times A$.

1. $A = \begin{bmatrix} -2 \\ 2 \\ 1 \\ -2 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 3 & -5 & -2 \end{bmatrix}$

2. $A = \begin{bmatrix} -4 & -2 \\ 2 & 5 \\ -3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -2 & -4 \\ 1 & -4 & 0 \end{bmatrix}$

3. $A = \begin{bmatrix} 2 & 0 & 1 & 4 \\ -4 & 0 & -5 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -3 \\ -4 & -1 \\ -2 & 3 \\ -5 & 1 \end{bmatrix}$

*Answer on Page 2022*

What have you noticed about the results of $A \times B$ as compared to $B \times A$? You should have noticed that the product matrices are *different dimensions*. This leads us to the next unusual property of matrix multiplication: it is *non-commutative*. That is, the *order* in which you multiply matrices affects the result. This is very different from scalar values!

As you saw in the second matrix multiplication exercise, $\boldsymbol{A}$ is a $2 \times 4$ matrix and $\boldsymbol{B}$ is a $4 \times 2$ matrix, then $\boldsymbol{AB}$ is a $2 \times 2$ matrix, while $\boldsymbol{BA}$ is a $4 \times 4$ matrix. It is obvious, then, that $A \cdot B \neq B \cdot A$. What if $\boldsymbol{A}$ and $\boldsymbol{B}$ are square matrices?

## *Exercise 355* Are Matrices Commutative?

Working Space

Find $A \cdot B$ and $B \cdot A$ if $A = \begin{bmatrix} -3 & 5 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 \\ 4 & -3 \end{bmatrix}$.

*Answer on Page 2023*

As you can see, even if $\boldsymbol{A}$ and $\boldsymbol{B}$ are square, matrix multiplication is still not commutative.

**Non-Commutation of Matrix Multiplication**

For two matrices $\boldsymbol{A}$ and $\boldsymbol{B}$, where neither is an identity matrix or a zero matrix:

$$A \times B \neq B \times A$$

### Properties of the Zero Matrix

Just like the number 0, the zero matrix, $\boldsymbol{0}$ has unique mathematical properties:

**Properties of the Zero Matrix**

For a matrix, $\boldsymbol{A}$, and a zero matrix, $\boldsymbol{0}$

1. $A + 0e = A$
2. $A + -A = 0e$
3. $0 \cdot A = 0e$

### The Identity Matrix

There is another special matrix, called the *identity matrix*, usually denoted with $\boldsymbol{I}$. An identity matrix is all zeroes except for a diagonal line of ones. A $3 \times 3$ identity matrix is shown below:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

All identity matrices are square (that is, they have the same number of rows as they do columns). The identity matrix has the special property that whenever a vector or matrix is multiplied by $\boldsymbol{I}$, it doesn't change. Let's look at some examples:

**Example**: If $\mathbf{x} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$, what is $I\mathbf{x}$? (Take $\boldsymbol{I}$ to be a $2 \times 2$ identity matrix.)

**Solution**:

$$I\mathbf{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \cdot (2) + 0 \cdot (-3) \\ 0 \cdot (2) + 1 \cdot (-3) \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

**Example**: If $B = \begin{bmatrix} -2 & 5 \\ 3 & -4 \end{bmatrix}$, what is $I \times B$?

**Solution**:

$$I \times B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -2 & 5 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} 1 \cdot (-2) + 0 \cdot (5) & 1 \cdot (5) + 0 \cdot (-4) \\ 0 \cdot (-2) + 1 \cdot (5) & 0 \cdot (5) + 1 \cdot (-4) \end{bmatrix} = \begin{bmatrix} -2 & 5 \\ 3 & -4 \end{bmatrix}$$

**Properties of the Identity Matrix**

An $n \times n$ identity matrix, $\boldsymbol{I}$, does not change any vectors or matrices it multiplies. That is:

1. $I \times \mathbf{x} = \mathbf{x}$
2. $I \times B = B$

where $\mathbf{x}$ is an $n \times 1$ vector and $\boldsymbol{B}$ is an $n \times p$ matrix ($p$ may be, but is not necessarily, equal to $n$).

### 125.2.3 Transposing a Matrix

The **transpose** of a matrix is an operation that flips a matrix over its diagonal. In other words, rows become columns and columns become rows. If a matrix $A$ has entries $a_{ij}$, then its transpose, written $A^\top$, sometimes $A^T$, has entries $a_{ji}$.

**Transpose of a Matrix**

If

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix},$$

then the **Transpose of A** is

$$A^\top = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}.$$

That is, entry $a_{ij}$ becomes $a_{ji}$.

**Example:** Let

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}.$$

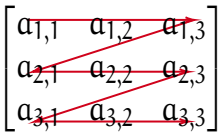
Then the transpose of $A$ is

$$A^\top = \begin{bmatrix} 2 & 0 \\ -1 & 3 \\ 4 & 5 \end{bmatrix}.$$

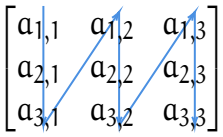
Notice how the first *row* of $A$ becomes the first *column* of $A^\top$, and so on.

### 125.2.4 Row-Major Order, Column-Major Order, and Computer Memory

Note that the longer the diagonal of $m \times n$, the longer this computation could take, especially by hand! For very large matrices, we love using computers to do the heavy lifting. Just as matrices are large sets of numbers, computer memory is essentially a large matrix, just stored in a very long string of characters. Remember how data is stored as a long, contiguous array? We can use this fact to simplify the expense of this operation. Instead of creating a whole new matrix, we can just swap the indices of the existing entries. This way, we don't have to move any data around in memory, just change how we access it. Many libraries and programming languages (Python, C++, Assembly, etc) default to **row-major order**, storing the matrix consecutively in rows in memory. Other languages (Fortran, MATLAB, R, etc) use **column-major order**, storing the matrix consecutively in columns in memory. When transposing a matrix stored in row-major order, we can just access the data as if it were stored in column-major order, and vice versa. This is as simple as swapping the bounds of the row/column operations. This trick saves us from having to move data around in memory, making the transpose operation much faster!




(a) Row-major ordering




(b) Column-major ordering

Figure 125.3: Row-major and column-major ordering of matrix elements.

### 125.2.5 Properties of Transposes

The transpose operation behaves nicely with matrix addition and multiplication:

(1)

(2) $(A^T)^T = A$

(3) $(A + B)^T = A^T + B^T$

(4) $(cA)^T = cA^T$ for any scalar $c$

(5) $(ABC)^T = C^T B^T A^T$ (order reverses)

Property 4 is especially important and often surprises students—the transpose of a product reverses the order.

Transposing a matrix is used throughout linear algebra, especially when:

- converting rows into column vectors (and vice versa),
- defining dot products using matrix multiplication,
- forming symmetric matrices ($A = A^T$),
- working with orthogonal matrices ($Q^T Q = I$),

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

More generally, a matrix with $m$ rows and $n$ columns is referred to as an $m \times n$ matrix, or simply an $m$-by-$n$ matrix; $m$ and $n$ are its dimensions.

The general form of a $2 \times 3$ matrix $A$ is:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix}$$

## 125.3 Types of Matrices

Matrices can be described by their shape:

**Row Matrix** A matrix of size $1 \times n$ with 1 row and $n$ columns.

**Column Matrix** A matrix of size $n \times 1$ with $n$ rows and 1 column, typically used to represent vectors.

**Square Matrix** A matrix of size $n \times n$, containing the same number of rows and columns

**Rectangular Matrix** A matrix of size $m \times n$, that has an unequal number of rows and columns.

They can also be described by their unique numerical properties. Special matrices that come in handy for certain types of computations. These are a few of the most common special matrices:

**Zero Matrix** A matrix that only contains entries that are zero.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The matrix above could also be

$$0_{2\times 2}$$

**Diagonal Matrix** A square matrix with nonzero entries along the diagonal, and zeroes everywhere else.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

**Identity Matrix** A diagonal matrix with entries of 1 along the diagonal.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

**Symmetric Matrix** A Symmetric matrix is a matrix, when transposed, becomes itself.

$$A = \begin{bmatrix} 20 & 40 & 60 \\ 40 & 50 & 80 \\ 60 & 80 & 100 \end{bmatrix}$$

**Triangular Matrix** This is a special square matrix that can be upper triangular or lower triangular. If upper, the main diagonal and all entries above it are nonzero while the lower entries are all zero. If lower, the main diagonal and all the entries below it are nonzero, while the upper entries are all zero. A matrix that is both upper and lower triangular becomes a diagonal matrix.

Upper Triangular Matrix

$$\begin{bmatrix} 2 & 3 & 1 \\ 0 & 5 & 4 \\ 0 & 0 & 6 \end{bmatrix}$$

Lower Triangular matrix

$$\begin{bmatrix} 7 & 0 & 0 \\ 2 & 5 & 0 \\ 4 & 3 & 9 \end{bmatrix}$$

### 125.3.1 Symmetric Matrices

If you want to find out if a square matrix is symmetric, you need to transpose it. If the transpose is equal to the original matrix, then the matrix is symmetric.

To transpose a matrix, flip it over its diagonal so that the rows and columns are switched, like this:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

After transposing:

$$A^T = \begin{bmatrix} a_{1,1} & a_{2,1} & a_{3,1} \\ a_{1,2} & a_{2,2} & a_{3,2} \\ a_{1,3} & a_{2,3} & a_{3,3} \end{bmatrix}$$

Note that $A^T$ means the transpose of A.

Let's see how this works for the following square matrix, A.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Switch the rows and columns to get the transpose:

$$A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Notice that $A = A^T$, so the matrix is symmetric.

What about matrix B?

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 7 & 8 & 9 \end{bmatrix}$$

Switch the rows and columns to get the transpose:

$$B = \begin{bmatrix} 1 & 2 & 7 \\ 2 & 4 & 8 \\ 3 & 5 & 9 \end{bmatrix}$$

Note that $B \neq B^T$, so B is not symmetrical.

## *Exercise 356* Matrix Transposition

Find the transpose of this matrix. Is it symmetric?

$$A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

*Working Space*

*Answer on Page 2023*

### 125.3.2 Can We Divide Matrices?

Matrices cannot be divided. Suppose we have a matrix, $\boldsymbol{A}$, a vector $\mathbf{x}$, and another vector $\mathbf{b}$ such that:

$$\boldsymbol{A} \cdot \mathbf{x} = \mathbf{b}$$

Now, if we know $\boldsymbol{A}$ and $\mathbf{x}$, it is easy to find $\mathbf{b}$. What if, on the other hand, we know $\boldsymbol{A}$ and $\mathbf{b}$ and want to find $\mathbf{x}$? We might be tempted to do something like this:

$$\mathbf{x} = \frac{\mathbf{b}}{\boldsymbol{A}}$$

While this would be correct if $\mathbf{x}$, $\mathbf{b}$, and $\boldsymbol{A}$ were scalars, but it is not for matrices. However, there is an analogy we can make. Instead of trying to divide by $\boldsymbol{A}$, we can multiply by its *inverse*:

**Inverse Matrices**

Given a matrix $\boldsymbol{A}$, and vectors $\mathbf{b}$ and $\mathbf{x}$, if

$$\boldsymbol{A} \times \mathbf{x} = \mathbf{b}$$

Then,

$$\mathbf{x} = A^{-1} \times \mathbf{b}$$

$A^{-1}$ is called the *inverse matrix*. *But,* the inverse *does not always exist*! We will explore inverse matrices and how to find them in the next chapter.

CHAPTER 126

# Linear Combinations of Vectors

In the introductory linear algebra chapter, you learned that vectors and matrices can be rotated, inverted, and added. In this chapter, we will explore linear combinations of vectors and the span of group of vectors. The **span** of a group of vectors is the set of vectors that can be made with linear combinations of the original group of vectors. We will offer mathematical and visual explanations later in the chapter. First, let's examine linear combinations.

A *linear combination* is simply the addition of vectors with leading scalar multipliers. In algebra, we can express polynomials similar to the form $5x + -4y$. In linear algebra, we do this with a vectors: $3\begin{bmatrix}2\\-1\end{bmatrix} + 2\begin{bmatrix}3\\5\end{bmatrix}$ is a linear combination of the vectors $\begin{bmatrix}2\\-1\end{bmatrix}$ and $\begin{bmatrix}3\\5\end{bmatrix}$. Another way to say this is:

**Linear Combination of Vectors**

A linear combination of a list of $n$ vectors, $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_n}$ takes the form:

$$a_1\mathbf{v_1} + a_2\mathbf{v_2} + \cdots + a_n\mathbf{v_n}$$

where $a_1, a_2, \cdots, a_n \in \mathbb{R}$. You may see the list of vectors as $\mathbf{x_1}, \mathbf{x_2}, \ldots \mathbf{x_n}$, both are acceptable variables.

**Example**: Find a linear combination of $\begin{bmatrix}2\\1\\-3\end{bmatrix}$ and $\begin{bmatrix}1\\-2\\4\end{bmatrix}$ that gives the vector $\begin{bmatrix}17\\-4\\2\end{bmatrix}$.

**Solution**: We are looking for $a_1$ and $a_2$ such that:

$$a_1\begin{bmatrix}2\\1\\-3\end{bmatrix} + a_2\begin{bmatrix}1\\-2\\4\end{bmatrix} = \begin{bmatrix}17\\-4\\2\end{bmatrix}$$

Looking at each dimension separately, we get the system of equations:

$$2a_1 + 1a_2 = 17$$

$$1a_1 - 2a_2 = -4$$

$$-3a_1 + 4a_2 = 2$$

If we can solve this system of equations, we will find $a_1$ and $a_2$. Let's multiply the first equation by 2 and add it to the second equation:

$$2(2a_1 + a_2) + (a_1 - 2a_2) = 2(17) + (-4)$$

$$4a_1 + 2a_2 + a_1 - 2a_2 = 34 - 4$$

$$5a_1 = 30$$

$$a_1 = 6$$

Now we can take $a_1$ and substitute it back into any equation in our system to find $a_2$. Let's use the third equation:

$$-3(6) + 4a_2 = 2$$

$$-18 + 4a_2 = 2$$

$$4a_2 = 20$$

$$a_2 = 5$$

Since we used all 3 equations, we know $a_1 = 6$ and $a_2 = 5$ are solutions to all 3 equations. If we had only used the first two equations to find $a_1$ and $a_2$, we would want to substitute our values back into the third equation to make sure our solution holds for that equation also.

Therefore,

$$6\begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} + 5\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ -4 \\ 2 \end{bmatrix}.$$

## *Exercise 357* Linear Combinations

Find a linear combination of the first two vectors that yields the third vector.

*Working Space*

1. $\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}-3\\1\end{bmatrix}, \begin{bmatrix}4\\5\end{bmatrix}$
2. $\begin{bmatrix}9\\4\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}-5\\3\end{bmatrix}$
3. $\begin{bmatrix}7\\-2\end{bmatrix}, \begin{bmatrix}-8\\4\end{bmatrix}, \begin{bmatrix}6\\-2\end{bmatrix}$

*Answer on Page 2023*

Sometimes, a set of vectors cannot be combined to make a specific vector. Take the pair of vectors we have looked at before: $\begin{bmatrix}2\\1\\-3\end{bmatrix}$ and $\begin{bmatrix}1\\-2\\4\end{bmatrix}$. Can we find a combination to make vector $\begin{bmatrix}17\\-4\\5\end{bmatrix}$? Let's try. We define $a_1$ and $a_2$ such that:

$$a_1\begin{bmatrix}2\\1\\-3\end{bmatrix} + a_2\begin{bmatrix}1\\-2\\4\end{bmatrix} = \begin{bmatrix}17\\-4\\5\end{bmatrix}$$

Which creates the system of equations:

$$2a_1 + a_2 = 17$$

$$a_1 - 2a_2 = -4$$
$$-3a_1 + 4a_2 = 5$$

We have two variables ($a_1$ and $a_2$) and three equations. Let's use the first two to find $a_1$ and $a_2$, then check our answers by substituting our solutions into the third equation. First, we'll multiply the second equation by $-2$ and add that to the first equation:

$$2a_1 + a_2 + (-2)(a_1 - 2a_2) = 17 + (-2)(-4)$$
$$2a_1 + a_2 - 2a_1 + 4a_2 = 17 + 8$$
$$5a_2 = 25$$
$$a_2 = 5$$

Substituting for $a_2$ back into the first equation and solving for $a_1$:

$$2a_1 + 5 = 17$$
$$2a_1 = 12$$
$$a_1 = 6$$

Now, let's check if $a_1 = 6$, $a_2 = 5$ is a solution to the third equation:

$$-3(6) + 4(5) = 5$$
$$-18 + 20 = 2 \neq 5$$

Therefore, there is no linear combination of the vectors $\begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$ that yields $\begin{bmatrix} 17 \\ -4 \\ 5 \end{bmatrix}$.

**Linear Combinations as systems.**

We are aiming to know if $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 = \mathbf{w}$. We can rewrite this as a matrix multiplication:

$$[v_1\ v_2] \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \mathbf{w}$$

Then, to check whether $w$ is a linear combination of $v_1$ and $v_2$, solve the system of equations. If a solution exists, then yes; if not, then no. Note that the $a_n$ values are only scalar numbers, while the $v_n$'s remain vectors.

We will talk more about systems of equations in Chapter 127.

## 126.1 Visualizing Linear Combinations

First, let's look at what vectors can be made from linear combinations of the 2-dimensional unit vectors $\mathbf{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Suppose we are looking for a linear combination of $\mathbf{i}$ and $\mathbf{j}$ to create the vector $\begin{bmatrix} 3 \\ -4 \end{bmatrix}$. We can find such a linear combination:

$$3\mathbf{i} + (-4)\,\mathbf{j} = 3\begin{bmatrix} 1 \\ 0 \end{bmatrix} - 4\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

In fact, with $\mathbf{i}$ and $\mathbf{j}$, we can create any 2-dimensional vector. To prove this, consider a generic vector, $\mathbf{z} = \begin{bmatrix} a \\ b \end{bmatrix}$, where $a, b \in \mathbb{R}$. We are looking for a linear combination of $\mathbf{i}$ and $\mathbf{j}$ such that:

$$c_1\mathbf{i} + c_2\mathbf{j} = \begin{bmatrix} a \\ b \end{bmatrix}$$

The above equation yields the system of equations:

$$c_1\,(1) + c_2\,(0) = a$$

$$c_1\,(0) + c_2\,(1) = b$$

And the solution to this system of equations is:

$$c_1 = a$$

$$c_2 = b$$

Therefore, using $\mathbf{i}$ and $\mathbf{j}$, we can construct any vector in $\mathbb{R}^2$ (that is, any vector in the $xy$-plane). What about combinations of other vectors?

Let's consider linear combinations of two vectors: $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. The vectors are shown in figure 126.1.

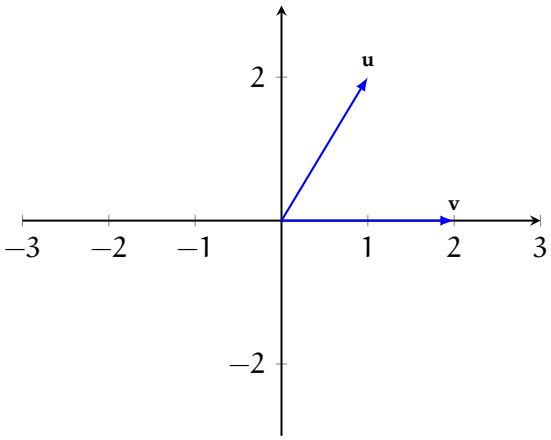
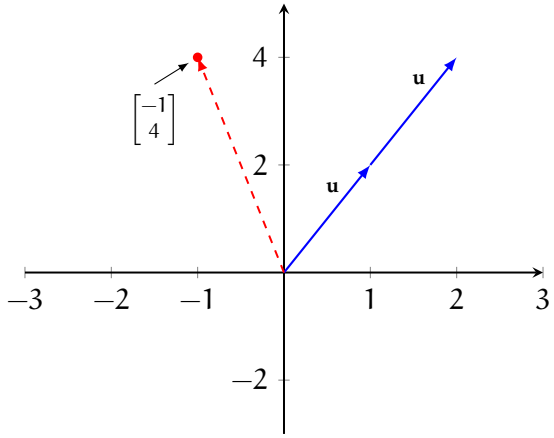
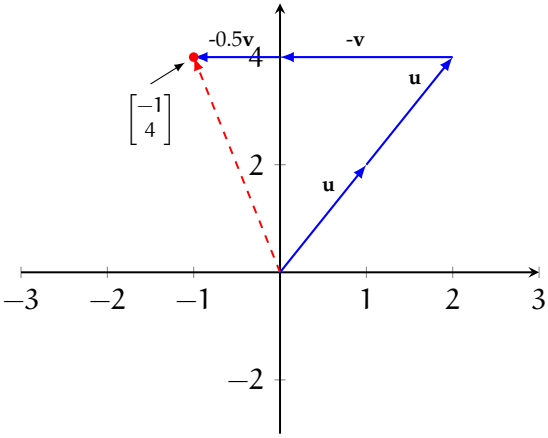



Figure 126.1: The vectors $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$.

Suppose we want to construct the vector $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Since only $\mathbf{u}$ has value in the $y$-dimension, we can start by adding $\mathbf{u}$ vectors to reach $y = 4$ (see figure 126.2a). Next, we can use $\mathbf{v}$ vectors to reach $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$ (see figure 126.2b).




(a) To create vector $\begin{bmatrix} 4 \\ -2 \end{bmatrix}$ with $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, we begin by adding two $\mathbf{u}$ vectors to reach a $y$-value of 4.



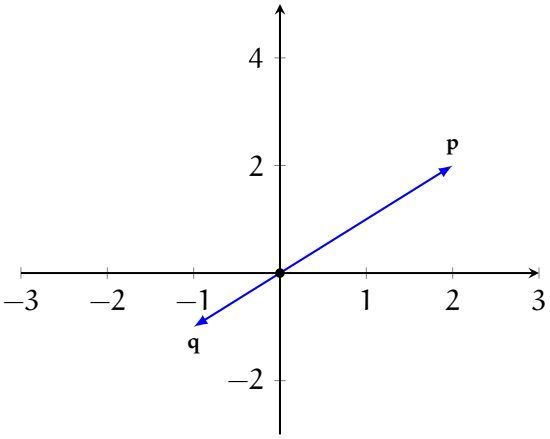

(b) If $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, then $2\mathbf{u} - 1.5\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$.

Figure 126.2: Using linear combinations of $\mathbf{u}$ and $\mathbf{v}$ to construct $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$.

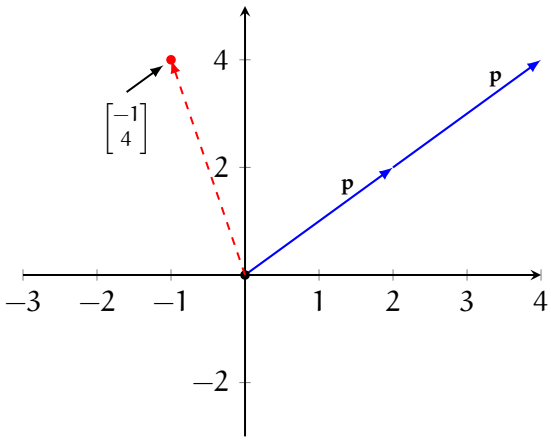
Using this method, we can imagine reaching any point in $\mathbb{R}^2$: we add or subtract as many $\mathbf{u}$ vectors as needed to reach the appropriate $y$-value, then add or subtract as many $\mathbf{v}$

vectors to reach the appropriate x-value. The vectors are *not multiples* of each other, so we can say that they span all of $\mathbb{R}^2$.

Let's look at another pair of vectors: $\mathbf{p} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ (see figure 126.3a). Again, let's try to use $\mathbf{p}$ and $\mathbf{q}$ to construct the vector $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$. We begin by using $\mathbf{p}$ to reach the $y$-value of 4 (see figure 126.3b).




(a) The vectors $\mathbf{p} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$.




(b) Adding two $\mathbf{p}$ vectors to reach $y = 4$.

Figure 126.3: Visualizing combinations of $\mathbf{p}$ and $\mathbf{q}$.

But now we run into a problem: no matter how many multiples of $\mathbf{q}$ vectors we add or subtract, we just move along the $\mathbf{p}$ vector and never reach our goal of $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$ (see figure 126.4). Notice that $\mathbf{p}$ and $\mathbf{q}$ lie on the same line, (for a better visualization, refer back to 126.3a). In other words, the vectors are *scalar multiples* of each other: $\begin{bmatrix} 2 \\ 2 \end{bmatrix} = -2 \cdot \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. When two vectors lie on the same line or a scalar multiples of each other, we call them *linearly dependent*.

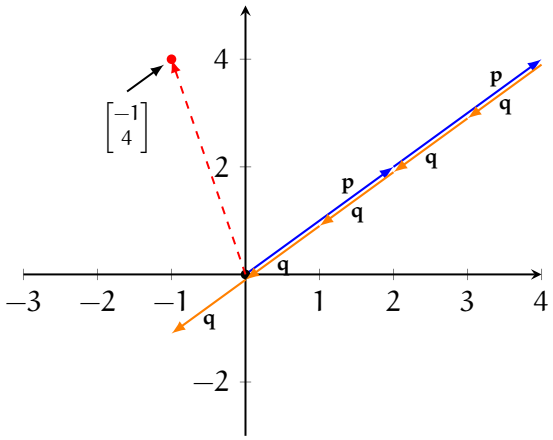



Figure 126.4: There is no linear combination of $\mathbf{p} = \begin{bmatrix}2\\2\end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix}-1\\-1\end{bmatrix}$ that yields the vector $\begin{bmatrix}-1\\4\end{bmatrix}$.

FIXME add summary

In the next chapter, we will be looking more at linear combinations, span, and independence in the form of systems of equations. Future chapters will look even further at these concepts, linear dependence and independence, and their applications to various subspaces.

Chapter 127

# Systems of Equations as Matrices

## 127.1 Systems of Equations, REF, and RREF

We have talked a lot about matrices and vectors. Now we can use matrices and vectors to solve systems of equations. A system of linear equations can be written compactly using matrices and vectors. This matrix representation allows us to use algebraic tools such as row operations and matrix multiplication to analyze and solve the system more efficiently. Let's look at a very simple example from a previous section.

$$\begin{cases} 1x_1 + 2x_2 = -1 \\ 2x_1 + 0x_2 = 4 \end{cases}$$

We can write this in the form of a matrix! This may come as a surprise to some of you. How can we do this? Let's rewrite the coefficients as a matrix and the $x_n$'s as column vector, and the constant vector, or right-hand-side vector, called $\vec{b}$. Here is the coefficient matrix which we will call $A$:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$$

Then we can write the column vector of $x$, which we will terminalize as $\vec{x}$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

And we can write the $\vec{b}$ as the following column vector:

$$\vec{b} = \begin{bmatrix} -1 \\ -4 \end{bmatrix}$$

Now, if we multiply $A$ and $\vec{x}$,

$$A\vec{x} = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1x_1 + 2x_2 \\ 2x_1 + 0x_2 \end{bmatrix}$$

We get exactly the left-hand sides of our original system of equations! Therefore, when we set this equal to the right-hand-side vector $\vec{b}$:

$$\vec{b} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

we get the fundamental system of equation formula represented as a matrix Equation

$$A\vec{x} = \vec{b} \tag{127.1}$$

This is a very important equation for linear algebra as a whole. Notice that the size of matrix has to respect the matrix multiplication rules:

$$\underset{m\times n}{A} \cdot \underset{n\times 1}{\vec{b}} = \underset{m\times 1}{\vec{p}}$$

## *Exercise 358* Matrix Equation

Rewrite the systems of equations in matrix form and identify $A$, $\vec{x}$, and $\vec{b}$.

$$\begin{cases} -4x_1 + 9x_2 - 8x_3 = 5 \\ -1x_1 + 0x_2 + 6x_3 = 7 \end{cases}$$

*Working Space*

*Answer on Page 2025*

## 127.2 Row-Echelon Form

This brings up the question, how can we solve for the $\vec{x}$? There are multiple ways, but most commonly we form an augmented matrix and solve for Row-Echelon Form.

### 127.2.1 The Augmented Matrix

What is the augmented matrix? It is a way of represented the A matrix and $\vec{b}$ vector. We write the A matrix, separated by a vertical line, and then writing the $\vec{b}$ vector:

$$[A \mid \vec{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

The left block contains all the coefficients of the system of equations, and the right block contains only the solutions. Then, to solve for $\vec{x}$, we perform *Elementary Row Operations* to find the *Row-Echelon Form*, mathematically stated as $\text{rref}(A|\vec{b})$. Let's continue with the matrix vector set we were experimenting with:

$$A\vec{x} = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1x_1 + 2x_2 \\ 2x_1 + 0x_2 \end{bmatrix}$$

This can be written as an augmented matrix:

$$\left[A|\vec{b}\right] = \left[\begin{array}{cc|c} 1 & 2 & -1 \\ 2 & 0 & 4 \end{array}\right]$$

### 127.2.2 Elementary Row Operations

The Row-Echelon Form of a matrix is an *equivalent* form of a matrix obtained through Gaussian and Gauss-Jordan elimination. There are lots of videos on this process online, so multiple videos included in your digital resources to walk you through the process. The rules of Gaussian Elimination are simple, only 3 different elementary row operations, or ERO's can be performed:

- Swapping two rows
- multiply a row by a non-zero scalar
- dd a multiple of one row to another row

Why do these operations preserve the given linear system? Swapping two rows only changes the order in which the equations are written and therefore has no effect on the solution set. Multiplying a row by a nonzero scalar corresponds to multiplying both sides of an equation by the same scalar, which does not alter its solutions. Finally, if two equations are satisfied by the same solution, then any linear combination of those

equations is also satisfied by that solution; in particular, adding a multiple of one equation to another preserves the solution set. Before and after each operation, the augmented matrix contains exactly the same information as the system of equations and therefore represents the same solution set.

With these ERO's, we are trying to achieve a matrix with the following properties:

- Leading entries (not necessarily 1) move to the right as you go down
- Rows of zeros, if exist, are at the bottom
- Each pivot has zeros in all positions below it in the same column.

Lets walk through the steps to simplify ref$(A|\vec{b})$:

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 2 & 0 & 4 \end{array}\right]$$
$$R_2 \to R_2 - 2R_1 \implies \left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & -4 & 6 \end{array}\right]$$

As you can see, we set Row 2 to be Row 2 minus 2 times Row 1. This gives us a zero in the first column of Row 2. Technically, this matrix is now in Row-Echelon Form (REF). However, we can continue applying elementary row operations to obtain an even more structured form called the *Reduced Row-Echelon Form*.

### 127.2.3 Reduced Row Echelon Form

A matrix is in *Reduced Row-Echelon Form* (RREF) if it satisfies all of the conditions for Row-Echelon Form, and in addition:

- Each leading entry is equal to 1
- Each leading entry is the only nonzero entry in its column

In other words, not only do pivots move to the right as we move down the rows, but each pivot column contains zeros both above and below the pivot, and each pivot is normalized to 1. These additional conditions make RREF especially useful, since the solutions to the system can often be read directly from the matrix.

Continuing our example, we start with the matrix in REF:

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & -4 & 6 \end{array}\right]$$

First, make the leading entry in Row 2 equal to 1 by dividing the row by $-4$:

$$R_2 \to -\frac{1}{4}R_2 \implies \left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & -\frac{3}{2} \end{array}\right]$$

Next, eliminate the entry above the pivot in column 2:

$$R_1 \to R_1 - 2R_2 \implies \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -\frac{3}{2} \end{array}\right]$$

The matrix is now in Reduced Row-Echelon Form.

### 127.2.4 Writing the Solution Vector

Recall that the augmented matrix represents a system of linear equations. After reducing the matrix to Reduced Row-Echelon Form, we obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -\frac{3}{2} \end{array}\right]$$

Each row now corresponds directly to an equation in the variables $x_1$ and $x_2$:

$$\begin{aligned} x_1 &= 2 \\ x_2 &= -\frac{3}{2} \end{aligned}$$

Thus, the solution to the system can be written as the vector

$$\vec{x} = \begin{bmatrix} 2 \\ -\frac{3}{2} \end{bmatrix}$$

Since every variable corresponds to a pivot column, the system has a *unique solution*.

We have successfully solved the system of equations using matrix methods:

$$\begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -\frac{3}{2} \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

This is a fundamental technique in linear algebra, and you will use it often in this module and beyond!

## 127.3 Larger RREF Example

Let's solve a larger system by finding $\vec{x}$ in the matrix equation

$$A\vec{x} = \vec{b}.$$

### 127.3.1 Write the System and Identify $A$, $\vec{x}$, and $\vec{b}$

Consider the following system of equations:

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 4 \\ 2x_1 + x_2 + 3x_3 + x_4 = 9 \\ x_1 + 3x_2 + 2x_3 + 2x_4 = 10 \\ 3x_1 + x_2 + 2x_3 + 4x_4 = 13 \end{cases}$$

The coefficient matrix, variable vector, and constant vector are:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 3 & 1 \\ 1 & 3 & 2 & 2 \\ 3 & 1 & 2 & 4 \end{bmatrix}, \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \qquad \vec{b} = \begin{bmatrix} 4 \\ 9 \\ 10 \\ 13 \end{bmatrix}.$$

So the matrix equation is:

$$A\vec{x} = \vec{b}.$$

### 127.3.2 Form the Augmented Matrix

$$\left[A|\vec{b}\right] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 2 & 1 & 3 & 1 & 9 \\ 1 & 3 & 2 & 2 & 10 \\ 3 & 1 & 2 & 4 & 13 \end{array}\right].$$

### 127.3.3 Row-Reduce to REF

Use the pivot in Row 1 to clear entries below it in column 1:

$$\begin{aligned} R_2 &\to R_2 - 2R_1 \\ R_3 &\to R_3 - R_1 \\ R_4 &\to R_4 - 3R_1 \end{aligned}$$

$$\Longrightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 2 & 1 & 1 & 6 \\ 0 & -2 & -1 & 1 & 1 \end{array}\right].$$

Now use the pivot in Row 2 (column 2) to clear entries below it in column 2:

$$\begin{aligned} R_3 &\to R_3 + 2R_2 \\ R_4 &\to R_4 - 2R_2 \end{aligned}$$

$$\Longrightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 3 & -1 & 8 \\ 0 & 0 & -3 & 3 & -1 \end{array}\right].$$

Next use the pivot in Row 3 (column 3) to clear the entry below it in column 3:

$$R_4 \to R_4 + R_3$$

$$\Longrightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 3 & -1 & 8 \\ 0 & 0 & 0 & 2 & 7 \end{array}\right].$$

At this point, the matrix is in Row-Echelon Form (REF).

### 127.3.4 Continue to RREF

Make the pivot in Row 4 equal to 1:

$$R_4 \to \frac{1}{2}R_4 \implies \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 3 & -1 & 8 \\ 0 & 0 & 0 & 1 & \frac{7}{2} \end{array}\right].$$

Clear the entries above the pivot in column 4:

$$\begin{aligned} R_1 &\to R_1 - R_4 \\ R_2 &\to R_2 + R_4 \\ R_3 &\to R_3 + R_4 \end{aligned}$$

$$\implies \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & \frac{1}{2} \\ 0 & -1 & 1 & 0 & \frac{9}{2} \\ 0 & 0 & 3 & 0 & \frac{23}{2} \\ 0 & 0 & 0 & 1 & \frac{7}{2} \end{array}\right].$$

Make the pivot in Row 3 equal to 1:

$$R_3 \to \frac{1}{3}R_3 \implies \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & \frac{1}{2} \\ 0 & -1 & 1 & 0 & \frac{9}{2} \\ 0 & 0 & 1 & 0 & \frac{23}{6} \\ 0 & 0 & 0 & 1 & \frac{7}{2} \end{array}\right].$$

Clear the entries above the pivot in column 3:

$$\begin{aligned} R_1 &\to R_1 - R_3 \\ R_2 &\to R_2 - R_3 \end{aligned}$$

$$\implies \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & -\frac{10}{3} \\ 0 & -1 & 0 & 0 & \frac{2}{3} \\ 0 & 0 & 1 & 0 & \frac{23}{6} \\ 0 & 0 & 0 & 1 & \frac{7}{2} \end{array}\right].$$

Make the pivot in Row 2 equal to 1:

$$R_2 \to -R_2 \implies \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & -\frac{10}{3} \\ 0 & 1 & 0 & 0 & -\frac{2}{3} \\ 0 & 0 & 1 & 0 & \frac{23}{6} \\ 0 & 0 & 0 & 1 & \frac{7}{2} \end{array}\right].$$

Clear the entry above the pivot in column 2:

$$R_1 \to R_1 - R_2$$

$$\implies \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -\frac{8}{3} \\ 0 & 1 & 0 & 0 & -\frac{2}{3} \\ 0 & 0 & 1 & 0 & \frac{23}{6} \\ 0 & 0 & 0 & 1 & \frac{7}{2} \end{array}\right].$$

This matrix is now in Reduced Row-Echelon Form (RREF).

### 127.3.5 Write the Solution Vector

Reading each row as an equation gives:

$$\begin{aligned} x_1 &= -\frac{8}{3} \\ x_2 &= -\frac{2}{3} \\ x_3 &= \frac{23}{6} \\ x_4 &= \frac{7}{2} \end{aligned}$$

Thus the solution vector is:

$$\vec{x} = \begin{bmatrix} -\frac{8}{3} \\ -\frac{2}{3} \\ \frac{23}{6} \\ \frac{7}{2} \end{bmatrix}.$$

You can verify the solution by confirming that $A\vec{x} = \vec{b}$ using matrix multiplication.

Note that solution sets must be one of the following:

- No Solution (ie. parallel lines),
- Exactly One Unique Solution (ie. lines or planes intersecting at one point),
- or Infinitely Many Solutions (ie. same line or plane is redundantly in the set)

## 127.4 Free Variables

In the reduced row-echelon form (RREF) of a matrix, pivots determine which variables are constrained by the system.

### 127.4.1 Zero rows and redundant equations

A row of all zeros in RREF represents a redundant equation. That row corresponds to the equation

$$0x_1 + 0x_2 + \cdots + 0x_n = 0,$$

which is always true and therefore imposes no restriction on the variables. Such a row arises when one of the original equations is a linear combination of the others.

Every row of zeros is a row without a pivot. Consequently, if a matrix has at least one zero row in RREF:

- The number of pivots is less than the number of rows,
- The rank of the matrix is less than the number of rows,
- The rows of the matrix are linearly dependent.

### 127.4.2 Free variables

Pivots are associated with *columns*, not rows. A variable is called a *free variable* if its column does not contain a pivot.

Thus:

- A *row* without a pivot (a zero row) indicates a redundant equation,
- A *column* without a pivot corresponds to a free variable.

Free variables can take arbitrary values and lead to infinitely many solutions when solving a homogeneous system.

### 127.4.3 Infinitely Many Solutions

**Example:** Consider the augmented matrix in RREF:

$$\begin{bmatrix} 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

Labeling the variable columns (and ignoring the constants c column), we write:

$$\begin{bmatrix} x_1 & x_2 & x_3 & c \\ 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

The pivot positions are in the columns corresponding to $x_1$ and $x_3$. Since the column corresponding to $x_2$ contains no pivot, $x_2$ is a free variable.

The system of equations represented by this matrix is:

$$\begin{aligned} x_1 &= 8, \\ x_3 &= -6. \end{aligned}$$

The variable $x_2$ does not appear in any equation and may take any real value. Since $x_1$ and $x_3$ are bound, we can say the solution is :

$$x_1 = 8, \qquad x_3 = -6, \qquad x_2 \in \mathbb{R}$$

written as a vector this is: $\vec{x} = \begin{bmatrix} 8 \\ x_2 \\ -6 \end{bmatrix}$ or in set builder notation: $\left\{ \begin{pmatrix} 8 \\ x_2 \\ -6 \end{pmatrix} : x_2 \in \mathbb{R} \right\}$

Thus, the presence of a column without a pivot leads to a free variable and infinitely many solutions.

### 127.4.4 Inconsistent sets

Let's look at another example: **Example** The augmented matrix

$$\begin{bmatrix} 1 & 2 & 4 & 12 \\ 12 & 24 & 48 & 9 \\ 3 & 6 & 12 & -4 \end{bmatrix}$$

simplifies to

$$\begin{bmatrix} 1 & 2 & 4 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

we can interpret this As

$$x_1 + 2x_2 + 4x_3 = 0,$$
$$0x_1 + 0x_2 + 0x_3 = 1.$$

Note that the equation $0 = 1$ is never true, so the system is *inconsistent*, and there is *no solution*.

The presence of an all zero row signifies an inconsistent system.

FIXME exercises

## 127.5 Basis, Dimension, Rank

Let's define a few terms. These definitions are very basic, and will come back in the future:

**Rank**

**Rank** is the number of pivots in the RREF state of a matrix of size $m \times n$. In other words, the number of nonzero rows in any row-echelon form of a matrix $A$ is denoted $\text{rank}(A)$. This has the constraints of:

- $\text{rank}(A) \leq m$
- $\text{rank}(A) \leq n$

**Dimension**

The **dimension** of the solution set of a system is the number of free variables. We calculate this with

$$\texttt{dimension} = \texttt{number of variables} - \texttt{rank}$$

or equivalently, for matrix $A$ of size $m \times n$

$$\texttt{dimension} = n - \text{rank}(A)$$

**Basis**

A **basis** is a minimal set of direction vectors that describes all solutions. Specifically, this consists of $n - \text{rank}(A)$ vectors, one for each free variable

In summary,

| | | |
|---|---|---|
| rank | = | number of pivots |
| number of free variables | = | $n$ − rank |
| dimension of solution set | = | $n$ − rank |
| number of basis vectors | = | $n$ − rank |

## Exercises

### *Exercise 359* Augmented Matrix 1

*Working Space*

Simplify the matrix:

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 1 & 1 & 0 & 2 \end{bmatrix}$$

to RREF state and determine the type of solution set.

*Answer on Page 2025*

## *Exercise 360* Augmented Matrices 2

Put the augmented matrix

$$\begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & -1 & 3 \\ 2 & 1 & 0 & 1 \end{bmatrix}$$

into RREF and determine its solution.

*Working Space*

*Answer on Page 2026*

## *Exercise 361* Augmented Matrices 3

Determine the solution for the following set:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 9 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

*Working Space*

*Answer on Page 2027*

## *Exercise 362* Rank 1

Find the rank of

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

*Working Space*

*Answer on Page 2027*

### *Exercise 363* Rank 2

Let $A$ be a $4 \times 6$ matrix with exactly 3 pivots in RREF.

- What is $\text{rank}(A)$?
- How many free variables are there?

*Working Space*

*Answer on Page 2027*

### *Exercise 364* Free Variables

The RREF of a system has the form:

$$\begin{bmatrix} 1 & 0 & -2 & 0 & 3 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1. Identify the pivot variables
2. Identify the free variables
3. State the dimension of the solution set

*Working Space*

*Answer on Page 2027*

## 127.6 Creating Matrices in Python

Recall that matrices can be represented by 2D arrays in Python. Let's explore this idea further with the module `numpy`. Create a file called `matrices_creation.py` and enter this code:

```
# import the python module that supports matrices
import numpy as np
# Use the function np.array to define a matrix that
# contains specific values that you supply.
A = np.array([[ 5, 1, 3],
              [ 1, -1, 8],
              [ 6, 2, 1]])
# The transpose function returns
A.transpose()
```

When you run it, you should see:

```
array([[ 5, 1, 6],
       [ 1, -1, 2],
       [ 3, 8, 1]])
```

As you can see, $A \neq A^T$, so A is not symmetric. Try another:

```
# create a matrix, B
B = np.array([[ 5, 1, 6],
              [ 1, -1, 2],
              [ 6, 2, 1]])
B.transpose()
```

When you run it, you should see:

```
array([[ 5, 1, 6],
       [ 1, -1, 2],
       [ 6, 2, 1]])
```

B is symmetric. You can actually transpose any matrix using this function, but a matrix cannot be symmetric unless it is square.

Try this code to see what happens when you transpose a rectangular matrix.

```
# create a matrix, J
J = np.array([[ 5, 1, 3, 0],
              [ 1, -1, 8, 11],
              [ 6, 2, 1,-7]])
J.transpose()
```

Note that transposing a rectangular matrix changes its dimension from 3 by 4 to 4 by 3.

You should see a transposed matrix, but it's not symmetric.

```
array([[ 5,  1,  6],
       [ 1, -1,  2],
       [ 3,  8,  1],
       [ 0, 11, -7]])
```

### 127.6.1 Creating Special Matrices in Python

Use the same file to add this code for creating a zero matrix.

```
# create an 8 by 10 Zero matrix.
 C = np.zeros((8, 10))
 C
```

When you run it, you should see:

```
array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

Add the following code to create an 8 by 8 Identity matrix.

```
# create an 8 by 8 Identity matrix
 D = np.eye(8)
 D
```

When you run it, you should see:

```
array([[1., 0., 0., 0., 0., 0., 0., 0.],
       [0., 1., 0., 0., 0., 0., 0., 0.],
       [0., 0., 1., 0., 0., 0., 0., 0.],
       [0., 0., 0., 1., 0., 0., 0., 0.],
       [0., 0., 0., 0., 1., 0., 0., 0.],
       [0., 0., 0., 0., 0., 1., 0., 0.],
       [0., 0., 0., 0., 0., 0., 1., 0.],
```

```
       [0., 0., 0., 0., 0., 0., 0., 1.]])
```

As you progress in your studies, you will learn the importance of diagonal matrices and of extracting the diagonal of a matrix. Let's see how to extract a diagonal, then create a diagonal matrix.

```
# create a  matrix
W = np.array([[1, 2, 3, 4],
              [5, 6, 7, 8],
              [-8, -7, -6, -5],
              [-4, -3, -2, -1]])
```

Extract the main diagonal using np.diag(<array>,<diagonal to extract>). Passing 0 as the second parameter specifies the main diagonal. A positive value extracts a diagonal from the upper part. A negative value extracts a diagonal from the lower part. Run this code then experiment passing other values to see what you get.

```
print(np.diag(W,0))
```

When you run it, you should see:

```
array([ 1,  6, -6, -1])
```

You can also use np.diag() to create a diagonal matrix from a 1D array. In this case, do not pass a second parameter.

```
Q = np.array([1, 2, 3])
DiagArray = np.diag(Q))
print(DiagArray)
```

When you run it you should see;

```
[[1 0 0]
 [0 2 0]
 [0 0 3]]
```

Python has functions for extracting upper and lower triangular matrices. Try these:

```
print(np.triu(W))
```

```
print(np.tril(W))
```

You should see:

```
[[ 1  2  3  4]
 [ 0  6  7  8]
 [ 0  0 -6 -5]
 [ 0  0  0 -1]]

[[ 1  0  0  0]
 [ 5  6  0  0]
 [-8 -7 -6  0]
 [-4 -3 -2 -1]]
```

## 127.7 Conclusion

In this module, we have explored the fundamental concepts of matrices, including their types, properties, and how they can be used to represent and solve systems of linear equations. We have also learned how to perform row operations to convert matrices into Row-Echelon Form (REF) and Reduced Row-Echelon Form (RREF), which are essential techniques in linear algebra. In the next chapter 130, we will talk deeper about the concepts of vector spaces and subspaces and how to find RREF, building on our understanding of matrices and their applications.

CHAPTER 128

# Applications of Matrices in System Form

In the chapter on linear combinations, we saw that we can linearly combine vectors to create other vectors. Consider 3 vectors:

$$\mathbf{x} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad \mathbf{z} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

We can write a linear combination of these vectors:

$$c\mathbf{x} + d\mathbf{y} + e\mathbf{z}$$

Which we can expand to show the vectors:

$$c \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + d \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} + e \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -c - d - 2e \\ 2c + 2d + e \\ d \end{bmatrix}$$

We can also represent this combination with a matrix where each column is one of the vectors:

$$\begin{bmatrix} -1 & -1 & -2 \\ 2 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} c \\ d \\ e \end{bmatrix} = \begin{bmatrix} -c - d - 2e \\ 2c + 2d + e \\ d \end{bmatrix}$$

## 128.1 Trail Mix for Mars

Let's look at an applied problem. Three astronauts, Pat, Kai, and River, are getting ready for a trip to Mars. NASA food service is preparing trail mix for the voyage, tailored to each astronaut's taste. The chef needs to submit a budget based on the cost of the trail mix for each astronaut. The mix is made up of raisins, almonds, and chocolate.

Pat prefers a raisins:almonds:chocolate ratio of 6:10:4, Kai likes 2:3:15, and River wants 14:1:5. The chef can buy a kg of raisins for $7.50, a kg of almonds for $14.75, and a kg of chocolate for $22.25. Assuming each astronaut will get 20 kg of trail mix, which astronaut

will cost more to feed?

First, set up a matrix to represent the raisins:almonds:chocolate ratios. (Conveniently, these ratios already add to 20.)

$$\text{MixRatios} = \begin{bmatrix} 6 & 10 & 4 \\ 2 & 3 & 15 \\ 14 & 1 & 5 \end{bmatrix}$$

Use a vector to represent the cost of each item:

$$\text{IngredientCost} = \begin{bmatrix} 7.50 \\ 14.75 \\ 22.25 \end{bmatrix}$$

To find the cost of trail mix for each astronaut, we simply find the dot product between the mix ratios and the ingredient costs to get:

Pat = \$281.50
Kai = \$615.50
River = \$231.00

## *Exercise 365* Vector Matrix Multiplication

*Working Space*

Multiply the array $A$ with the vector $v$. Compute this by hand, and make sure to show your computations.

$$A = \begin{bmatrix} 1 & -2 & 3 & 5 \\ -4 & 2 & 7 & 1 \\ 3 & 3 & -9 & 1 \end{bmatrix}$$

$$v = \begin{bmatrix} 2 \\ 2 \\ 6 \\ -1 \end{bmatrix}$$

*Answer on Page 2028*

## *Exercise 366* Using Vector Matrix Multiplication

A college professor offers three different methods of determining a student's final grade. In method A, the student's grade is 20% based on attendance, 50% homework, 15% midterm, and 15% final. This professor knows many students can learn the material without attending every class, so with method B the student's grade is 50% homework, 20% midterm, and 30% final. Last, the professor knows some students don't do the homework but still show they understand the material by doing well on the tests. With method C, a student's grade is 40% midterm and 60% final. The professor uses whatever method results in the highest grade to determine each student's final grade.

Suppose Suzy has attended 65% of classes, has an average homework grade of 30%, earned a 80% on the midterm, and earned a 75% on the final. What final grade will her professor post?

*Working Space*

*Answer on Page 2028*

### 128.1.1 Vector-Matrix Multiplication in Python

Most real-world problems use very large matrices, where it becomes impractical to do calculations by hand. As long as you understand how matrix-vector multiplication is performed, you will be equipped to use a computing language, like Python, to do the calculations for you.

Create a file called `vectors_matrices.py` and enter this code:

```
1 # import the python module that supports matrices
2 import numpy as np
```

```

# create an array
a = np.array([[5, 1, 3, -2],
              [1, -1, 8, 4],
              [6, 2, 1, 3]])

# create a vector
b = np.array([1, 2, 3, -8])

# calculate the dot product
print(a.dot(b))
```

When you run it, you should see:

```
[16, 6, 8]
```

## 128.2 Where to Learn More

Watch this video from Khan Academy about matrix-vector products: https://rb.gy/frga5

Chapter 129

# Linear Independence and Dependence

We have briefly mentioned the idea of linear independence and dependence in previous chapters. Let's explicitly define them using the concepts we have covered already.

## 129.1 Linear Independence

**Linear Independence**

**Linear independent** vectors are vectors that cannot be written as linear combinations (scalar multiples) of each other.

Mathematically, this means that

$$c_1v_1 + c_2v_2 \dots c_kv_k = 0$$

is only satisfied by

$$c_1 = c_2 = \cdots = c_k = 0$$

**Linear Dependence**

A set of vectors $\{v_1, \dots, v_k\}$ is **linearly dependent** if there exist scalars $c_1, \dots, c_k$, not all zero, such that

$$c_1v_1 + \cdots + c_kv_k = \mathbf{0}.$$

For two vectors, this reduces to one being a scalar multiple of the other.

When vectors are arranged as columns of a matrix, linear independence means that no column can be written as a linear combination of the others.

We say that each vector contributes a new *direction* not captured by any previous ones, meaning that no vector can be written as a linear combination of the others. Each vector adds something genuinely new to the set, so there is no redundancy.

## *Exercise 367* Testing Linear Independence

Determine whether each pair of vectors is linearly independent or linearly dependent.

*Working Space*

1. $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ -2 \end{bmatrix}$
2. $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -6 \\ -2 \end{bmatrix}$
3. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$
4. $\begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -2 \\ -3 \end{bmatrix}$

*Answer on Page 2029*

Any set of vectors that includes the zero vector is automatically linearly dependent, since

$$1 \cdot \vec{0} = \vec{0}$$

gives a nontrivial solution.

## *Exercise 368* Zero Vector and Dependence

Working Space

Determine whether each set of vectors is linearly independent.

1. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
2. $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$
3. $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

*Answer on Page 2029*

Geometrically, in $\mathbb{R}^2$: independent vectors are not collinear. In $\mathbb{R}^2$, the vectors are generally perpendicular.

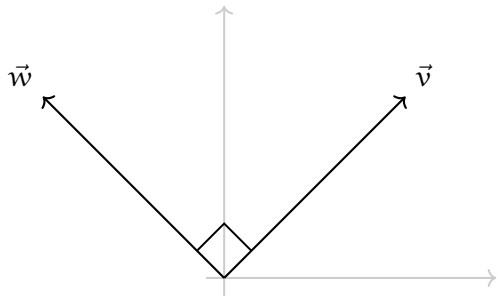



Figure 129.1: An example of independent vectors in $\mathbb{R}^2$.

In $\mathbb{R}^3$, two independent vectors lie in a plane, while three independent vectors do not lie in the same plane.

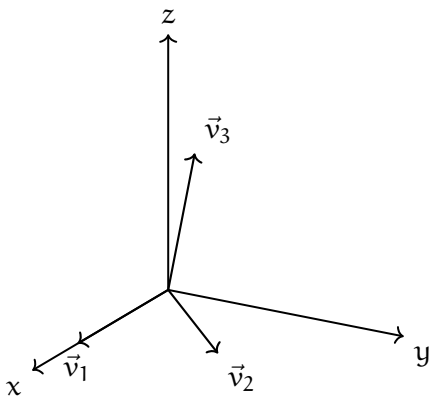



Figure 129.2: An example of linearly independent vectors in $\mathbb{R}^3$.

## *Exercise 369* Values Causing Dependence

*Working Space*

Find all values of k for which the vectors are linearly dependent.

1. $\begin{bmatrix} 1 \\ k \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$
2. $\begin{bmatrix} k \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2k \\ 2 \\ 4 \end{bmatrix}$

*Answer on Page 2029*

### 129.1.1 Connection to Span

The *span* of a set of vectors is the collection of all vectors that can be written as linear combinations of those vectors. In other words, the span consists of all vectors that are reachable using the given set.

We now use this idea to better understand linear independence.

Consider the vector space $\mathbb{R}^3$. Let

$$V = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Any vector in $\mathbb{R}^3$ can be written as a linear combination of these three vectors. Therefore,

$$\text{span}(V) = \mathbb{R}^3.$$

Moreover, no vector in $V$ can be written as a linear combination of the other two. Each vector contributes a new direction to the span. For this reason, the vectors in $V$ are *linearly independent*.

Equivalently, a set of vectors is linearly independent if every vector in their span has a *unique* representation as a linear combination of those vectors.

This example illustrates an important idea: a linearly independent set contains no redundant vectors, and when such a set spans a space, the set is called a basis for the space.

A set $B$ is a basis if its elements are linearly independent and every element of $V$ is a linear combination of elements of $B$. In other words, a basis is a linearly independent spanning set. A vector space can have several bases; however all the bases have the same number of elements, called the dimension of the vector space. We will dive into this concept more in the subspaces chapter, so don't worry if this is a bit of a jump!

## 129.2 Linearly Dependent Vectors: Geometrically

Two vectors are linearly dependent if one is a multiple of the other. Mathematically,

**Linearly dependent vectors in $\mathbb{R}^n$**

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are linearly dependent if there exists a scalar $a \in \mathbb{R}$ such that

$$\mathbf{v} = a\,\mathbf{u}.$$

Graphically, two linearly dependent vectors in $\mathbb{R}^2$ lie on the same line through the origin (or one of them is the zero vector).

If two vectors are linearly dependent, then linear combinations of them can only produce vectors lying on that same line. If they are *not* linearly dependent, they are called linearly *independent*, and their linear combinations can produce every vector in $\mathbb{R}^2$.

**Example**: Which of the following 3 vectors are linearly dependent, if any? $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -3 \\ 4 \\ -1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 6 \\ -8 \\ 2 \end{bmatrix}$.

**Solution**: Two vectors are linearly dependent if one is a scalar multiple of the other. Let's compare **u** and **v**. Since the first component of **u** is 1 and the first component of **v** is -3, let's multiply **u** by -3 to see if we get **v**:

$$-3\mathbf{u} = -3 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ -6 \\ -9 \end{bmatrix} \neq \mathbf{v}$$

Therefore, **u** and **v** are *not* linearly dependent. Now let's examine **v** and **w**. Again, we will use the first components: the first component of **w** is 6, so let's see if multiplying **v** by -2 yields **w**:

$$-2\mathbf{v} = -2 \begin{bmatrix} -3 \\ 4 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 2 \end{bmatrix} = \mathbf{w}$$

Therefore, **v** and **w** are linearly dependent. Since we already know that **u** and **v** are not linearly dependent, we also know that **u** and **w** are also not linearly dependent.

### *Exercise 370* Linear Dependence

*Working Space*

Identify which, if any, of the following vectors are linearly dependent:

1. $\mathbf{a} = \begin{bmatrix} -4 \\ 1 \\ 4 \end{bmatrix}$
2. $\mathbf{b} = \begin{bmatrix} -4 \\ 5 \\ -3 \end{bmatrix}$
3. $\mathbf{c} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$
4. $\mathbf{d} = \begin{bmatrix} 1 \\ -\frac{1}{4} \\ -1 \end{bmatrix}$
5. $\mathbf{e} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$
6. $\mathbf{f} = \begin{bmatrix} -6 \\ \frac{3}{2} \\ 6 \end{bmatrix}$

*Answer on Page 2030*

## 129.3 What's next?

In this chapter, we defined linear independence and linear dependence, which describe whether a set of vectors contains redundancy. Linearly independent vectors introduce new directions, while dependent vectors can be written as linear combinations of others.

These ideas lead naturally to the study of subspaces, which describe collections of vectors that are closed under addition and scalar multiplication. In the next section, we formalize how sets of vectors generate larger structures and how linear independence helps describe them efficiently.

Chapter 130

# Subspaces

Recall that, in Chapter 127, we established that all linear systems can be represented in matrix form as $A\vec{x} = \vec{b}$. In this chapter, we will explore the concept of subspaces, which are fundamental to understanding the structure of solutions to linear systems.

Quickly, let's review some vocabulary. The zero vector, denoted as $\vec{0}$, is the vector where all components are zero. It will be a column vector of appropriate size $n \times 1$.

If $A\vec{x} = \vec{0}$, where $\vec{b} = \vec{0}$, then the system is called **homogeneous**. If $A\vec{x} = \vec{b}$ where $\vec{b} \neq \vec{0}$, then the system is called **non-homogeneous**.

## 130.1 What is a Subspace?

A subspace $V$ of $\mathbb{R}^n$ is a subset $V$ of $\mathbb{R}^n$ satisfying the following properties:

- The zero vector $\vec{0}$ is in the subspace.
- If $\vec{u}$ and $\vec{v}$ are in the subspace, then their sum $\vec{u} + \vec{v}$ is also in the subspace.
- If $\vec{u}$ is in the subspace and $c$ is a scalar, then the scalar multiple $c\vec{u}$ is also in the subspace.

In short, a subspace is exactly the set of all linear combinations of some collection of vectors. Additionally, every subspace is a span.

**Subspace Span**

If $V$ is a subspace of $\mathbb{R}^n$, then there exists a set of vectors

$$\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\} \subseteq \mathbb{R}^n$$

such that

$$V = \text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$$

That is, every vector in $V$ can be written as a linear combination of vectors in $V$.

Because $V$ is closed under:

- vector addition, and
- scalar multiplication,

any linear combination of vectors in $V$ must also lie in $V$.

So if $\vec{u}, \vec{v} \in V$ and $a, b \in \mathbb{R}$, then

$$a\vec{u} + b\vec{v} \in V$$

The converse is also true: any span is a subspace.

$$\text{span}(V) \text{ is a subspace of } \mathbb{R}^n$$

For example, vectors $\vec{v_1} = [1, 0]$ and $\vec{v_2} = [0, 1]$ span all of $\mathbb{R}^2$, since they can be scaled to fit all of $\mathbb{R}^2$.

- the span $v_1$ is a line through $\vec{0}$
- the span $v_1, v_2$ is a line or plane through $\vec{0}$
- the span of $v_1, v_2, v_3$ is a line or plane through $\vec{0}$, or all of $\mathbb{R}^3$

We need to review and reinforce some vocabulary that will come up often during this section. Let's look at Basis and Dimension.

## 130.2 Basis and Dimension

A basis of a subspace $V$ is a set of linearly independent vectors that span $V$.

**Basis**

A set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\} \in \mathbb{R}^n$ is called a **basis** for $\mathbb{R}^n$ if

- the vectors span $\mathbb{R}^n$, and
- the vectors are linearly independent.

For example, a basis for $\mathbb{R}^3$ is given by the standard unit vectors:

$$V = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Note that the span of $V$ is all of $\mathbb{R}^3$, and the vectors in $V$ are linearly independent.

For a subspace $V$, there are often many possible bases. For example, another basis for $\mathbb{R}^3$ is given by the vectors:

$$W = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Put simply, a basis is a smallest possible set of vectors that can be used to build every vector in the space, with no redundancy. No vector in the basis can be written as a linear combination of the others.

A standard basis is a special type of basis that is often used for $\mathbb{R}^n$.

**Standard Basis**

The **standard basis** for $\mathbb{R}^n$ is the set of vectors

$$\{\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n\},$$

where $\vec{e}_i$ is the vector in $\mathbb{R}^n$ with a 1 in the $i$-th position and 0 in all other positions.

An example of this is the standard basis for $\mathbb{R}^2$:

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

Recall we may often simplify this to i hat and j hat notation:

$$\left\{\hat{i}, \hat{j}\right\} \qquad c_1\hat{i} + c_2\hat{j}$$

The standard basis consists of the vectors that point along the coordinate axes. Each standard basis vector measures exactly one coordinate.

The number of vectors in the basis is called the dimension of the subspace.

**Dimension**

The **dimension** of a vector space $V$ is the number of vectors in any basis for $V$. If $V$ has a basis consisting of $k$ vectors, then we say that $V$ has dimension $k$, and write

$$\dim(V) = k.$$

In particular, the dimension of $\mathbb{R}^n$ is $n$, since the standard basis contains exactly $n$ vectors.

**Properties of Dimension:**

- A basis of a subspace $\mathbb{R}^n$ contains exactly $n$ vectors.
- If $W$ is a linear subspace of $V$, then $\dim(W) \leq \dim(V)$.
- If $V$ is a finite-dimensional vector space and $W$ is a linear subspace of $V$ with $\dim(W) = \dim(V)$, then $W = V$.
- The space $\mathbb{R}^n$ has the standard basis $\{e_1, \ldots, e_n\}$, where $e_i$ is the $i$-th column of the corresponding identity matrix. Therefore, $\mathbb{R}^n$ has dimension $n$.

## 130.3 Nullspace

We now examine a fundamental example of a subspace: the **nullspace** of a matrix. Recall that a subspace is a subset of $\mathbb{R}^n$ that is closed under vector addition and scalar multiplication and contains the zero vector.

The nullspace of a matrix $A$, denoted $\text{Null}(A)$, is the set of all vectors $\vec{x}$ such that

$$A\vec{x} = \vec{0}.$$

That is, the nullspace is precisely the solution set of the homogeneous system $A\vec{x} = \vec{0}$.

**Nullspace**

The nullspace of a matrix $A$ is defined as

$$\text{Null}(A) = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\}.$$

The $\vec{x}$, then represents all vectors that get flattened to origin $(\vec{0})$.

Because the equation $A\vec{x} = \vec{0}$ is homogeneous, its solution set always contains the zero vector. Moreover, if $\vec{x}_1$ and $\vec{x}_2$ are solutions, then any linear combination of the form

$$a\vec{x}_1 + b\vec{x}_2$$

is also a solution. For this reason, the nullspace of a matrix is always a subspace of $\mathbb{R}^n$.

### 130.3.1 Linear Combinations and Span

Recall that a **linear combination** of vectors $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ is any vector of the form

$$a_1\vec{v}_1 + a_2\vec{v}_2 + \cdots + a_n\vec{v}_n,$$

where $a_1, a_2, \ldots, a_n \in \mathbb{R}$ are scalars.

The set of all possible linear combinations of a collection of vectors is called their **span**. If a subspace can be written as the span of one or more vectors, those vectors describe all possible directions within the subspace.

### 130.3.2 Example

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

To find the nullspace of $A$, we solve the homogeneous system $A\vec{x} = \vec{0}$:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This corresponds to the system of equations

$$x_1 + 2x_2 = 0, \qquad 2x_1 + 4x_2 = 0.$$

Solving for $x_1$ in terms of $x_2$ gives

$$x_1 = -2x_2.$$

Notice that $x_1$ can be written in terms of $x_2$. This implies *linear dependence*, the fact that the vectors can be written as combinations of each other. Thus, every vector in the nullspace can be written in the form

$$\vec{x} = x_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Therefore, the nullspace of $A$ is

$$\text{Null}(A) = \text{span}\left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}.$$

Geometrically, this nullspace is a line through the origin in $\mathbb{R}^2$, which is a one-dimensional subspace of $\mathbb{R}^2$. The nullspace contains all scalar multiples of the vector $\begin{bmatrix}-2\\1\end{bmatrix}$. This tells us that the solutions to the homogeneous system $A\vec{x} = \vec{0}$ form a line in $\mathbb{R}^2$. All vectors along this line produce the zero vector when substituted into the equation $A\vec{x} = \vec{0}$.

### 130.3.3 Example

Find a basis for the nullspace of the matrix

$$A = \begin{bmatrix}1 & 2 & -1 & 0\\2 & 4 & -2 & 0\\0 & 1 & 1 & 3\\0 & 2 & 2 & 6\end{bmatrix}.$$

First, find the RREF form of A. Notice immediately that $R_2 = 2R_1$ and $R_4 = 2R_3$ in $A$, so we will definitely have free rows. We get that

$$\text{rref}(A) = \begin{bmatrix}1 & 0 & -3 & -6\\0 & 1 & 1 & 3\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\end{bmatrix}$$

Our $\vec{x}$ vector has to be the same size as our column, so it must live in $\mathbb{R}^4$. This gives us $\vec{x} = \begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}$.

So finding $A\vec{x} = \vec{0}$ provides the systems of equations:

$$\begin{cases}x_1 - 3x_3 - 6x_4 = 0\\x_2 + x_3 + 3x_4 = 0\end{cases}$$

We can rewrite this as:

$$\begin{cases}x_1 = 3x_3 + 6x_4\\x_2 = -x_3 - 3x_4\end{cases} \implies x_3\begin{bmatrix}3\\-1\\1\\0\end{bmatrix} + x_4\begin{bmatrix}6\\-3\\0\\-1\end{bmatrix}$$

So a basis for the nullspace is formed by

$$\text{Null}(A) = \left\{ \begin{bmatrix} 3 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -3 \\ 0 \\ -1 \end{bmatrix} \right\}$$

## *Exercise 371* Finding a Basis for a Nullspace

*Working Space*

Consider the matrix

$$A = \begin{bmatrix} 1 & -1 & 1 & 3 & 2 \\ 2 & -1 & 1 & 5 & 1 \\ 3 & -1 & 1 & 7 & 0 \\ 0 & 1 & -1 & -1 & -3 \end{bmatrix}$$

When put in reduced row echelon form,

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 2 & -1 \\ 0 & 1 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

a Find a basis for the nullspace of $A$.

b Find a vector $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$ in the nullspace of $A$ such that $x_1 = -3$, $x_3 = 1$, and $x_4 = 2$.

*Answer on Page 2030*

**Key Idea: Nullspace and Linear Independence**

Solving for the *nullspace* of a matrix tells you whether a set of vectors is **linearly independent or linearly dependent**.

- If the nullspace contains *only* the zero vector, then the vectors are **linearly independent**.
- If the nullspace contains a *nonzero* vector, then the vectors are **linearly dependent**.

**Practical test (via RREF):**

- If every column has a *pivot*, the columns are linearly independent.
- If one or more columns lack a pivot, the columns are linearly dependent.

Equivalently, the equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution if and only if the columns of $A$ are linearly dependent.

## 130.4 Row Space

The row space of a matrix $A$, denoted $\text{Row}(A)$, is the subspace of $\mathbb{R}^n$ spanned by the row vectors of $A$. Each row vector can be viewed as a vector in $\mathbb{R}^n$, and the row space consists of all linear combinations of these row vectors.

**Row Space**

The row space of a matrix $A$ is defined as

$$\text{Row}(A) = \text{span}\{\text{row}_1, \text{row}_2, \ldots, \text{row}_m\},$$

where $\text{row}_i$ represents the $i$-th row of the matrix $A$. The row space consists of all directions in $\mathbb{R}^n$ that can be built from the rows of the matrix. Note that we are observing the matrix before it is transformed into RREF or altered in any way.

Another way to think about it is that the row space represents all possible linear combinations of the equations represented by the rows of the matrix $A$. Each row is "tested" on by $\vec{x}$ to produce a component of the output vector $A\vec{x}$. Equivalently, the row space is the set of all vectors that can be formed by adding and scaling the rows of $A$.

When we compute:

$$A\vec{x}$$

each row of $A$ gets dotted with $\vec{x}$ to produce a component of the output vector.

If each of the rows of $A$ are denoted as $\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_m$, then $A\vec{x}$,

$$\begin{bmatrix} \vec{r}_1 \cdot \vec{x} \\ \vec{r}_2 \cdot \vec{x} \\ \vdots \\ \vec{r}_m \cdot \vec{x} \end{bmatrix}$$

The row space, then, is a subspace of $\mathbb{R}^n$ that captures all possible linear combinations of the rows of $A$. The row space consists of all possible tests on $\vec{x}$ that can be performed by the rows of $A$.

$$A = \begin{bmatrix} \text{—}\vec{r}_1\text{—} \\ \text{—}\vec{r}_2\text{—} \\ \vdots \\ \text{—}\vec{r}_m\text{—} \end{bmatrix} \quad \text{with } \vec{r}_i \in \mathbb{R}^n$$

Why does the row space live in $\mathbb{R}^n$?

- $\vec{x}$ has $n$ components and lives in $\mathbb{R}^n$, and
- each row $\vec{r}_i$ has $n$ components and lives in $\mathbb{R}^n$

### 130.4.1 Example

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \end{bmatrix}$$

The row vectors of $A$ are

$$\vec{r}_1 = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}, \quad \vec{r}_2 = \begin{bmatrix} 0 & 1 & 3 \end{bmatrix}$$

dotting them with an $\vec{x}$ of appropriate size gives

$$A\vec{x} = \begin{bmatrix} \vec{r}_1 \cdot \vec{x} \\ \vec{r}_2 \cdot \vec{x} \end{bmatrix}$$

The row space of $A$ is the span of its row vectors:

$$\text{Row}(A) = \text{span}\left\{ \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 3 \end{bmatrix} \right.$$

## 130.5 Column Space

The column space of a matrix $A$, denoted $\text{Col}(A)$, is the subspace of $\mathbb{R}^m$ spanned by the column vectors of $A$. Each column of $A$ can be viewed as a vector in $\mathbb{R}^m$, and the column space consists of all linear combinations of these column vectors.

**Column Space**

The **column space** of a matrix $A$ is defined as

$$\text{Col}(A) = \text{span}\{\vec{c}_1, \vec{c}_2, \ldots, \vec{c}_n\},$$

where $\vec{c}_i$ represents the $i$-th column of the matrix $A$. The column space consists of all vectors in $\mathbb{R}^m$ that can be formed as linear combinations of the columns of $A$.

Equivalently, the column space can be described as the set of all possible outputs of the matrix:

$$\text{Col}(A) = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\}.$$

The columns of the *original* matrix $A$ that correspond to pivot columns in $\text{RREF}(A)$ span the column space of $A$. We will discuss this idea further in a future chapter, but it corresponds to the fact that the pivot columns are linearly independent and form a basis for the column space.

### 130.5.1 Example

Consider the matrix

$$A = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 2 & 7 & 3 & 9 \\ 1 & 5 & 3 & 1 \\ 1 & 2 & 0 & 8 \end{bmatrix}.$$

After applying elementary row operations, we find that the RREF of $A$ is

$$\text{RREF}(A) = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since there is a free row, there is a *dependent* row in the original matrix. The pivot columns are columns 1, 2, and 4. Thus, a basis for the column space of $A$ is given by the original

columns 1, 2, and 4:

$$\begin{bmatrix}1\\2\\1\\1\end{bmatrix}, \quad \begin{bmatrix}3\\7\\5\\2\end{bmatrix}, \quad \begin{bmatrix}4\\9\\1\\8\end{bmatrix}.$$

Thus the subspace spans $\mathbb{R}^4$ and consists of all linear combinations of these three vectors.

## *Exercise 372* Nullspace, Rowspace, and Columnspace

*Working Space*

Consider the matrix

$$A = \begin{bmatrix}1 & 2 & -1 & 3\\2 & 4 & -2 & 6\\1 & 1 & 0 & 2\\0 & 1 & 1 & 1\end{bmatrix}.$$

a Compute the reduced row echelon form of $A$.

b Identify the pivot columns and free columns of $A$. What does this tell you about the solutions?

c Find a basis for the **nullspace** $N(A)$. What is the dimension of $N(A)$?

d Find a basis for the **row space** $R(A)$. What is the dimension of $R(A)$?

e Find a basis for the **column space** $C(A)$. State clearly where the vectors for the basis come from.

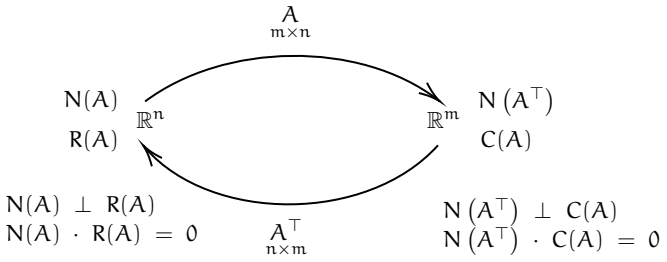
*Answer on Page 2031*

## 130.6 Summary

In this chapter, we have established a few different common subspaces: the Nullspace, Rowspace, and Columnspace. Each of these subspaces captures different structural information about a matrix and the system of equations it represents. The nullspace describes all input vectors that are sent to the zero vector, the column space contains all possible outputs of the matrix, and the row space encodes the independent constraints imposed by the system.

We have now set the stage for a new way of viewing matrices. Rather than thinking of a matrix solely as a collection of numbers or equations, we can begin to interpret a matrix as a transformation – a rule that takes input vectors and maps them to output vectors. In the next few chapters, we will build on our understanding of subspaces to study how matrices act on vectors and reshape space, providing a unifying viewpoint for many of the ideas introduced so far. We will look at the geometric implications of matrices and how they *transform* space. This is the basis of many computer graphic programs, like calculating lengths of shadows in VR games or simulations!

Before ending this chapter, take a look at this graphic. It shows the relationship between the subspaces we talked about, plus a new one: $\text{Null}(A^T)$, the transpose of the nullspace. If matrix $A$ has size $m \times n$, then $A^T$ has size $n \times m$. Note which subspaces live in $\mathbb{R}^n$ and which live in $\mathbb{R}^m$.




Instead of asking what vectors solve $A\vec{x} = \vec{0}$, we will begin asking:

*What does the matrix $A$ do to an arbitrary vector?*

The big take away: A matrix of size $m \times n$ can be viewed as a rule that takes vectors in $\mathbb{R}^n$ as inputs and produces vectors in $\mathbb{R}^m$as outputs.

# Chapter 131

# Span

## 131.1 Spans of Vectors

Knowing whether two vectors are linearly dependent or independent allows us to accurately describe the span of those two vectors (this expands to include any number of vectors). In the previous chapter, we saw that linear combinations of two linearly dependent vectors can only make vectors that lie on the same line as the two starting vectors. We saw this in 2D, but it also applies to 3D vectors. Consider the two vectors $\mathbf{u} = [2, 4, 3]$ and $\mathbf{v} = [4, 8, 6]$, shown in figure 131.1.

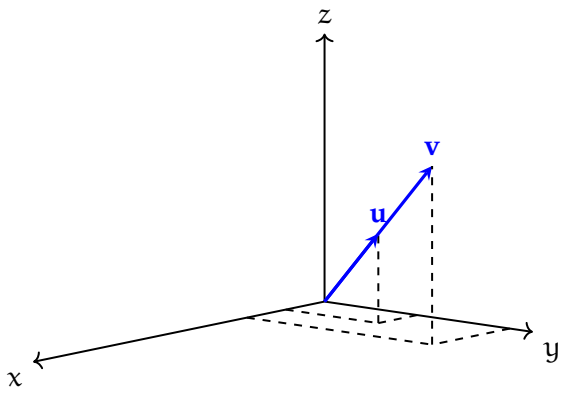



Figure 131.1: 3-dimensional vectors, $\mathbf{u}$ and $\mathbf{v}$

Notice that these two vectors are colinear (that is, they are on the same line), therefore they are linearly dependent and any combination of $\mathbf{u}$ and $\mathbf{v}$ will lie on the same line as $\mathbf{u}$ and $\mathbf{v}$. Therefore, we say the *the span of* $\mathbf{u}$ *and v is a line*. In fact, for any size list of linearly dependent vectors (whether it's one vector or one hundred), the span of that list is a line.

Now that you have a sense of what a span is, it is time for the formal mathematical definition. A vector span is the collection of vectors obtained by scaling and combining the original set of vectors in all possible proportions. Formally, if the set $S = \{\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_n}\}$ contains vectors from a vector space $V$, then the span of $S$ is given by:

$$\text{Span}(S) = \{a_1\mathbf{v_1} + a_2\mathbf{v_2} + \cdots + a_n\mathbf{v_n} : a_1, a_2, ..., a_n \in \mathbb{R}\} \tag{131.1}$$

This means that any vector in the Span$(S)$ can be written as a linear combination of the vectors in $S$.

### 131.1.1 Spans of Independent Vectors

What if our list of vectors aren't all linearly dependent on each other? We've seen in 2 dimensions that any two independent vectors can be linearly combined to create any vector in $\mathbb{R}^2$. So, the span is described as a *plane* (in fact, it is the entire $xy$-plane, which we also call $\mathbb{R}^2$). How does this expand to 3-dimensional vectors? Let's again consider two 3-dimensional vectors: $\mathbf{u} = [2, 4, 3]$ and $\mathbf{v} = [2, 1, 0]$, as shown in figure 131.2.

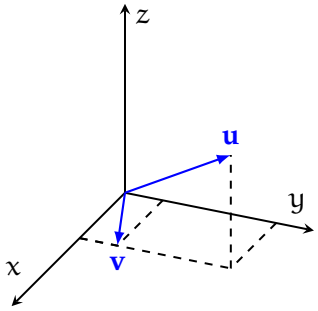



Figure 131.2: Linearly independent 3-dimensional vectors, $\mathbf{u}$ and $\mathbf{v}$

Just like in two dimensions, any two independent vectors in $\mathbb{R}^3$ define a plane (see figure 131.3). This also applies to higher dimensions: the span of any two linearly independent vectors is a plane.

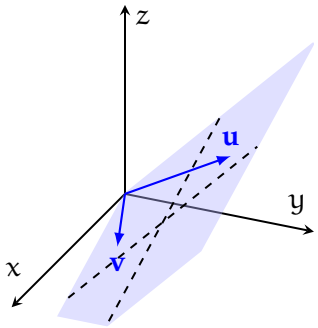



Figure 131.3: Linearly independent 3-dimensional vectors, $\mathbf{u}$ and $\mathbf{v}$, define a plane.

If we have 3 independent vectors, then we can define a *3-dimensional space*. To understand this, first imagine a plane formed by two independent 3-dimensional vectors like in figure 131.3). If a third independent vector is introduced, it must not lie on the plane: if it did, it would be a linear combination of the first two and therefore not independent. This third vector allows us to move off the plane, and therefore all three independent vectors span $\mathbb{R}^3$.

In review, 1 vector or set of dependent vectors span a *line*, 2 vectors or sets of dependent vectors span a *plane*, and 3 vectors or sets of dependent vectors span $\mathbb{R}^3$.

**Example**: Do the vectors $\mathbf{r} = [5, 4, -6]$, $\mathbf{s} = [0, -5, -10]$, and $\mathbf{t} = [0, 2, 4, ]$ span a line, plane, or $\mathbb{R}^3$?

**Solution**: We need to determine the number of *independent vectors*. First, we'll check if $\mathbf{r}$ and $\mathbf{s}$ are independent. They are independent if the only solution to the equation below is $a_1 = a_2 = 0$:

$$a_1 [5, 4, -6] + a_2 [0, -5, -10] = [0, 0, 0]$$

Which we can write as a system of equations:

$$5a_1 + 0a_2 = 0$$

$$4a_1 - 5a_2 = 0$$

$$-6a_2 - 10a_2 = 0$$

From the first equation, we see that $5a_1 = 0$ which implies that $a_1 = 0$. Substituting that into the second equation:

$$4(0) - 5a_2 = 0$$

$$-5a_2 = 0$$

$$a_2 = 0$$

Therefore, vectors $\mathbf{r}$ and $\mathbf{s}$ are independent. Now let's check $\mathbf{r}$ and $\mathbf{t}$:

$$a_1 [5, 4, -6] + a_2 [0, 2, 4, ] = [0, 0, 0]$$

Which we can re-write as a system of equations:

$$5a_1 + 0a_2 = 0$$

$$4a_1 + 2a_2 = 0$$

$$-6a_1 + 4a_2 = 0$$

Again, from the first equation, we see that $a_1 = 0$. Substituting into the second:

$$4(0) + 2a_2 = 0$$

$$2a_2 = 0$$

$$a_2 = 0$$

Therefore, $\mathbf{r}$ and $\mathbf{t}$ are also independent. Last, we'll check $\mathbf{s}$ and $\mathbf{t}$ for independence:

$$a_1 [0, -5, -10] + a_2 [0, 1, 2, ] = [0, 0, 0]$$

The system of equations:

$$0a_1 + 0a_2 = 0$$
$$-5a_1 + a_2 = 0$$
$$-10a_1 + 2a_2 = 0$$

The first equation doesn't tell us anything, since it would be true no matter what $a_1$ and $a_2$ are. We can solve the second equation for $a_2$ and substitute into the third equation:

$$a_2 = 5a_1$$
$$-10a_1 + 2(5a_1) = 0$$
$$-10a_1 + 10a_1 = 0$$

Which is also true for all $a_1$. In fact, there are many solutions to $a_1[0,-5,-10]+a_2[0,1,2,] = [0,0,0]$, $a_1 = 1$ and $a_2 = 5$ is an example. Therefore, $\mathbf{s}$ and $\mathbf{t}$ are *dependent*. So, we really have 2 independent vectors in the list, and therefore span$(\mathbf{r},\mathbf{s},\mathbf{t})$ is a plane.

### *Exercise 373* Determining Span

Geometrically describe (line, plane, or $\mathbb{R}^3$) the span of the list of vectors.

*Working Space*

1. $[1,2,4]$ and $[-2,-4,-8]$
2. $[2,0,0]$ and $[0,1,3]$
3. $[3,0,0]$ and $[0,3,3]$ and $[3,3,2]$

*Answer on Page 2032*

## 131.2 Where to Learn More

Watch this video on *Linear Combinations and Vector Spans from Khan Academy*: http://rb.gy/g1snk

The Wolfram Demonstrations website has a fun, interactive demo where you can enter values for 2D and 3D matrices and see how the area or volume changes. https://demonstrations.wolfram.com/DeterminantsSeenGeometrically/#more

If you are curious about the *Expansion of Minors*, see: https://mathworld.wolfram.com/DeterminantExpansionbyMinors.html

Chapter 132

# Determinants and Inverse Matrices

## 132.1 Determinants

Checking the independence of multitudes of vectors may take an immense amount of time. What if you had a list of 5, 10, or even 100 vectors? The determinant of a matrix is a scalar value that also indicates whether the columns of a matrix are linearly independent. So, if you put all your vectors together in a matrix and take the determinant of that matrix, the result will tell you if all the vectors are independent or not. For a 2D matrix, the determinant is the area of the parallelogram defined by the column vectors. For a 3D matrix, the determinant is the volume of the parallelepiped (a six-dimensional figure formed by six parallelograms, such as a cube).[1]

Let's plot the parallelogram for this matrix (see figure 132.1):

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

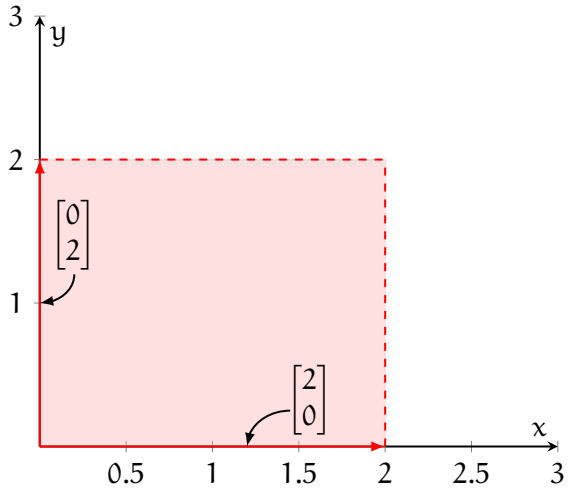



Figure 132.1: A parallelogram constructed from vectors $[2, 0]$ and $[0, 2]$

[1] Note that determinants can only be found for square, $n \times n$ matrices.

**2 by 2 Determinant**

The formal definition for calculating the determinant of a 2 by 2 matrix $A$ is:

$$\det(A) = (a \cdot d) - (b \cdot c)$$

where

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

For the matrix plotted above, the determinant is $(2 * 2) - (0 * 0)$. You can also see that 4.0 is the area, base $(2)$ times height $(2)$.

You can use the determinant to see what happens to a shape when it goes through a linear transformation. Let's scale the 2 by 2 matrix by 4:

$$\begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$$

Plot it (see figure 132.2):

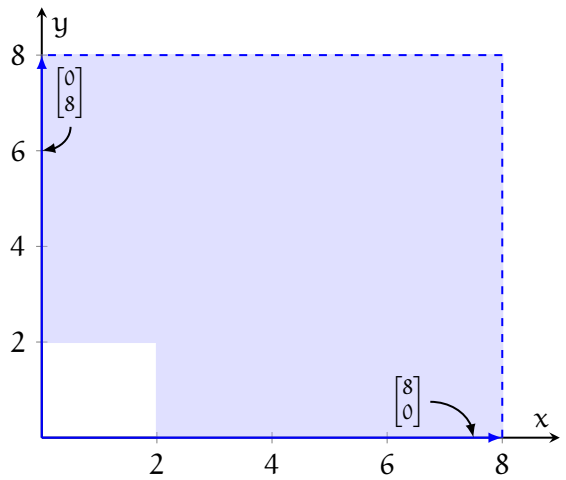



Figure 132.2: Scaling the matrix also scales the parallelogram.

Find the determinant using $(8 * 8) - (0 * 0) = 64$

You can see that scaling the matrix scaled the area by the scaling factor squared (see figure 132.3).

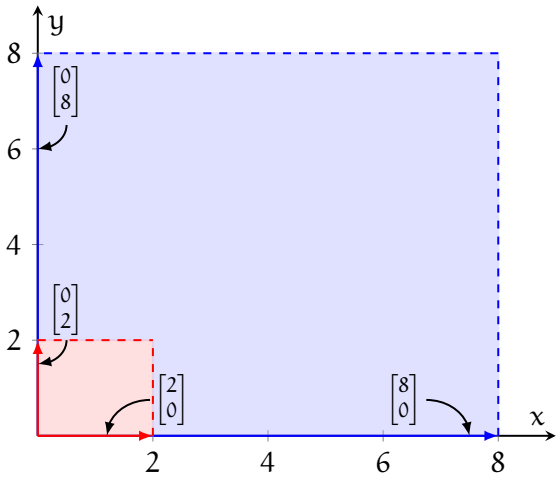



Figure 132.3: Scaling a matrix by a constant c increases the area of the parallelogram by a factor of $c^2$.

We can show why this is true mathematically. Suppose we have a 2 by 2 matrix A:

$$A = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$$

Then $\det(A) = wz - xy$. We can scale this matrix by a constant, c:

$$cA = c \cdot \begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} cw & cx \\ cy & cz \end{bmatrix}$$

And we can take the determinant:

$$\det(cA) = \det\left(\begin{bmatrix} cw & cx \\ cy & cz \end{bmatrix}\right) = cw(cz) - cx(cy) = c^2(wz - xy) = c^2 \cdot \det(A)$$

Therefore, scaling a 2 by 2 matrix by a factor changes the determinant by that factor squared. What about higher dimensions? If each side of a cube were scaled by a factor of c, then the volume of the cube would change by a factor of $c^3$ (feel free to confirm this yourself). And if a tesseract (a four-dimensional cube) had each side scaled by a factor of c, then the hypervolume (four-dimensional volume) would be scaled by a factor of $c^4$. Do you notice a pattern?

In fact, scaling an $n \times n$ matrix by a constant factor, c, changes the determinant of that $n \times n$ matrix by a factor of $c^n$.

What happens if the columns of a matrix are not independent? Let's plot this matrix (see

figure 132.4):

$$\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

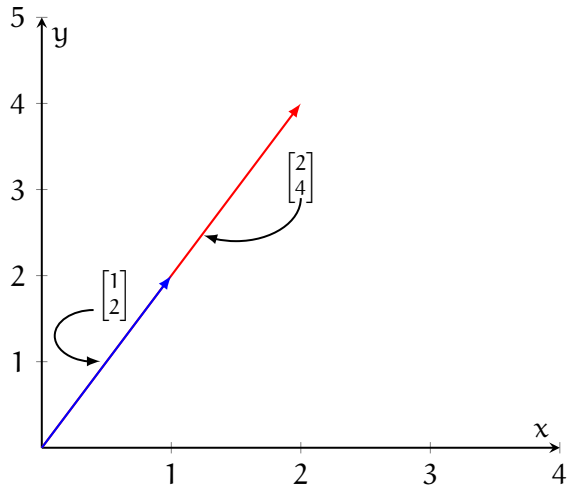



Figure 132.4: The vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ are co-linear, so there is no area between them and the determinant of $\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ is zero.

One vector overwrites the other. As you can see, the area is 0 because there is no space between the vectors. Therefore, the columns of the matrix are linearly dependent.

## *Exercise 374* Finding the Determinant

Plot the parallelogram represented by the columns of the matrix. What is the area of this parallelogram?

*Working Space*

1. $\begin{bmatrix} 1 & 4 \\ -3 & 1 \end{bmatrix}$
2. $\begin{bmatrix} 5 & -5 \\ 5 & -1 \end{bmatrix}$
3. $\begin{bmatrix} 0 & -5 \\ -2 & 0 \end{bmatrix}$

*Answer on Page 2032*

Calculating the determinant for a 2 by 2 matrix is easy. For a larger matrix, finding the determinant is more complex and requires breaking down the matrix into smaller matrices until you reach the 2x2 form. The process is called expansion by minors. For example,

**$3 \times 3$ Determinant**

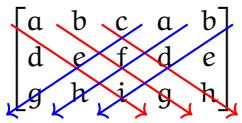
The determinant of a 3 by 3 matrix is found by

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \begin{bmatrix} d & e \\ g & h \end{bmatrix}$$

A trick for this is rewriting the matrix as a $3 \times 5$ augmented matrix with the first 2 columns after the third column:

$$B = \begin{bmatrix} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{bmatrix}$$

and solving the *down-right diagonals* minus the *down-left diagonals*:




Down-right diagonals (add) Down-left diagonals (subtract)

$$\det(B) = (a \cdot e \cdot i) + (b \cdot f \cdot g) + (c \cdot d \cdot h) - [(c \cdot e \cdot g) + (a \cdot f \cdot h) + (b \cdot d \cdot i)]$$

Note that this is the same multiplication as above, just formatted differently.

As you can see, this involves a recursive process of breaking a larger matrix into a smaller $2 \times 2$ matrix.

For our purposes, we simply want to first check to see if a matrix contains linearly independent rows and columns before using our Python code to solve.

## 132.2 Determinants in Python

Modify your code so that is uses the `np.linalg.det()` function. If the determinant is not zero, then you can call the `np.linalg.solve()` function. Your code should look like this:

```
# Are the rows and columns independent?
# Equivalently, is the determinant 0?
if (np.linalg.det(D) != 0):
    j = np.linalg.solve(D,e)
    print(j)
else:
    print("Rows and columns are dependent.")
```

How does this work below the hood? Let's also write a recursive python function that finds our determinant:

There are two base cases:

- The matrix is of size $1 \times 1$
- The matrix is of size $2 \times 2$

And further sizes can be simplified into one of the base cases by *cofactor expansion*. The

idea behind cofactor expansion is to break a big determinant into smaller ones until we reach cases we already know how to solve. Formally, this is written as

$$|A| = \sum_{j=1}^{n} (-1)^{(i+j)} a_{(ij)} M_{(ij)}$$

We do this as follows:

1. Pick the first row of the matrix.
2. For each entry in that row:
   - Remove the row and column containing that entry.
   - This creates a smaller matrix, called a *minor*, which are submatrices. Recall what we did above for our $3 \times 3$ determinant definition.
3. Compute the determinant of that smaller matrix.
4. Multiply the original entry (where the row and column lines originate from), the determinant of its minor, and an alternating sign $+, -, +, \ldots$.
5. Sum all of these results together.

This process repeats recursively: each smaller determinant is computed the same way, until we reach one of the base cases.

If we format our matrices as a nested array, we can use python's indexing to check and reduce the matrices. Take a look at this recursive determinant program:

```
def recursive_determinant(matrix):
    """
    Compute the determinant of a square matrix recursively.
    matrix: list of lists, forming a matrix of size n by n
    """
    n = len(matrix)

    # base case 1x1:
    if n == 1:
        return matrix[0][0]
    # base case 2x2
    if n == 2:
        return matrix[0][0] * matrix[1][1] - matrix[0][1] * matrix[1][0]

    det = 0

    for col in range(n):
        # build a minor matrix:
        minor = []
```

```
        # for every row in the matrix
        for row in matrix[1:]:
            # remove column col and store as new_row
            new_row = row[:col]+row[col+1:]
            # apprend new row to the minor matrix
            minor.append(new_row)

        sign = (-1) ** col #exponential for alternating sign
        det += sign * matrix[0][col] * recursive_determinant(minor)
    return det
```

We implemented our known base case, and recursively reduce our array until it fits a known base case. This is exactly how numpy's `np.linalg.det()` works.

## 132.3 Inverse Matrices

Now that we have talked about determinants, we can talk about Inverse Matrices. The idea of a matrix inverse is a natural generalization of the multiplicative inverse of a real number. For example, since

$$3 \cdot \frac{1}{3} = 1,$$

the number $\frac{1}{3}$ is called the multiplicative inverse of 3.

Similarly, for matrices, we define an inverse in terms of matrix multiplication and the identity matrix. If $A$ is an $n \times n$ matrix, its inverse (when it exists) is denoted by $A^{-1}$ and satisfies

$$AA^{-1} = A^{-1}A = I_n,$$

where $I_n$ is the $n \times n$ identity matrix.

**Inverse Matrix**

An inverse matrix is a square matrix that satisfies the following property:

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$$

where $A$ is the original matrix, $B$ is the inverse matrix, and $I_n$ is the identity matrix of size $n \times n$.

When such a matrix $B$ exists, it is denoted by $A^{-1}$, and is said to be *invertible*. Note that sometimes an inverse matrix may not exist.

It is important to note that not every matrix has an inverse. In particular, only *square*

*matrices* can be invertible, and even among square matrices, an inverse may fail to exist.

### 132.3.1 Existence of Square Matrices

#### Existence of the Inverse

A square matrix $A$ has an inverse if and only if its determinant is nonzero:

$$A^{-1} \text{ exists if and only if } \det(A) \neq 0.$$

If

$$\det(A) = 0,$$

then $A$ is called a **singular matrix**, and no inverse exists. Equivalently, an inverse matrix does not exist when the rows or columns of $A$ are *linearly dependent*. As discussed in the chapter on linear dependence, this occurs when one row (or column) is a scalar multiple of another, or when a row (or column) can be written as a linear combination of the others.

Linear dependence implies that the matrix does not contain enough independent information to reverse its action, making an inverse impossible.

### 132.3.2 Finding an Inverse Matrix

#### $2 \times 2$ Inverse

There are several methods for finding the inverse of a matrix. For small matrices, especially $2 \times 2$ matrices, there is a direct formula. For larger matrices, a more systematic method using row operations is required.

**Inverse of a $2 \times 2$ Matrix**

A trick for a $2 \times 2$ Matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Note that we multiply by the a new matrix has $a$ and $d$ swapped, and $b$ and $c$ switch signs. Finally, we divide by the determinant of the original $A$.

This formula is valid only when the determinant of $A$ is nonzero. Recall that the determinant of a $2 \times 2$ matrix is

$$\det(A) = ad - bc.$$

If $ad - bc = 0$, then $A$ is singular and has no inverse.

### *Exercise 375* $2 \times 2$ Practice

Find the inverse of the matrix of

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

*Working Space*

*Answer on Page 2034*

### $n \times n$ Inverse

For matrices of size $3 \times 3$ or greater, we create an augmented matrix.

Recall that we can create an augmented matrix by writing two matrices directly next to each other.

To find the inverse of a square matrix $A$:

1. Form the augmented matrix $[A \mid I_n]$, where $I_n$ is the identity matrix.
2. Use elementary row operations to transform the left side into $I_n$.
3. If this is possible, the right side of the augmented matrix becomes $A^{-1}$.

Symbolically, this process can be written as

$$[A \mid I_n] \longrightarrow [I_n \mid A^{-1}].$$

If row reduction does not result in the identity matrix on the left side, then $A$ does not have an inverse. In this case, the matrix is singular.

## *Exercise 376* A bigger inverse

*Working Space*

Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & 1 \\ 2 & 4 & -3 \end{bmatrix}$$

Check your answer using $AA^{-1} = I_3$

*Answer on Page 2034*

## *Exercise 377* Does a matrix exist?

*Working Space*

Given the matrix:

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 8 & 16 \\ 4 & 8 & 16 & 32 \\ 8 & 16 & 32 & 64 \end{bmatrix}$$

Does an inverse matrix exist? Explain why or why not

*Answer on Page 2036*

The augmented matrix row-reduction method not only provides a way to compute inverses, but also offers another criterion for invertibility: a matrix is invertible if and only if it can be row-reduced to the identity matrix.

### 132.3.3 Relation to $A\vec{x} = \vec{b}$

An inverse matrix is not only for square matrices, but also for systems of equations and our fundamental linear algebra equation $A\vec{x} = \vec{b}$. Let's look at an example.

Given

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

solve for $\vec{x}$ by first finding the inverse.

First, we can interpret our givens as a systems of matrices:

$$A\vec{x} = \vec{b} \quad \Longleftrightarrow \quad \begin{cases} 2x_1 + x_2 = 5 \\ x_1 + x_2 = 3 \end{cases}$$

Checking that an inverse matrix first exists, we have

$$\det(A) = 2(1) - 1(1) = 1 \neq 0$$

So we know an inverse matrix exists! To find $A^{-1}$

$$A^{-1} = 1 \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Here is the fun part! Let's do some computational equivalences:

$$\begin{aligned} A\vec{x} &= \vec{b} \\ A^{-1}(A\vec{x}) &= A^{-1}\vec{b} \\ I\vec{x} &= A^{-1}\vec{b} \\ \vec{x} &= A^{-1}\vec{b} \end{aligned} \tag{132.1}$$

Equation (132.1) show that, by only multiplying by the matrix equivalent of 1, the identity matrix I, we can state that $\vec{x} = A^{-1}\vec{b}$. Since we have both $A^{-1}$ and $\vec{b}$, we can find $\vec{x}$:

$$x = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Geometrically, this did a very simple operation. We used the fact that A vectors transforms $\vec{x}$ into $\vec{b}$. Then we noted that $A^{-1}$ undoes that transformation, recovering our original $\vec{x}$ by finding $A^{-1}\vec{b}$. This may be hard to visualize, so don't worry if it is hard to grasp immediately. We will review it again in a following chapter.

### 132.3.4 Matrix Inverse using NumPy

We can use our Python library NumPy to find inverses a lot easier using the `np.linalg.inv` function to find the inverse of a given array.

```
import numpy as np

A = np.array([[2, 1],
              [1, 1]])

A_inv = np.linalg.inv(A)
print(A_inv)
```

This outputs

```
[[ 1. -1.]
 [-1.  2.]]
```

If given a matrix that is *linearly dependent*, NumPy will raise an exception:

```
raise LinAlgError("Singular matrix")
numpy.linalg.LinAlgError: Singular matrix
```

letting us know that the inverse does not exist.

To improve our code, we check for the determinant of the matrix beforehand. In this code, we introduce a tolerance as computers handle numbers near zero as very small but non-zero digits.

```
import numpy as np

A = np.array([[2, 1],
              [4, 2]], dtype=float)

# BAD PRACTICE:
# A_inv = np.linalg.inv(A)
# print(A_inv)

detA = np.linalg.det(A)

tolerance = 1e-12

if abs(detA) < tolerance:
    print("Matrix is singular or nearly singular. No reliable inverse.")
```

```
else:
    A_inv = np.linalg.inv(A)
    print("Determinant:", detA) # you may see a large floating point number
    print("Inverse:\n", A_inv)
```

Output:

```
Matrix is singular or nearly singular. No reliable inverse.
```

## 132.4 Summary

In this chapter, we established the determinant of a matrix and inverse matrix.

The determinant of a matrix provides a powerful geometric and algebraic description of how a matrix acts on space.

In two dimensions, the determinant of a $2 \times 2$ matrix represents the signed area of the parallelogram formed by the images of the standard basis vectors. In higher dimensions, the determinant represents signed volume.

Further, if a matrix $A$ multiplies a region in space, then $|\det(A)|$ describes the factor by which areas or volumes are scaled, noting the sign and absolute value of the determinant as a scalar.

The inverse of a matrix *reverses the effect of the original matrix*. If $A$ is invertible, then multiplying by $A^{-1}$ restores any vector to its original position:

$$AA^{-1} = A^{-1}A = I.$$

Thus, invertibility, nonzero determinant, linear independence of rows and columns, and reversibility of action are all different ways of describing the same underlying property.

In the next chapter, we will interpret matrices as *functions* that transform vectors.

This idea will allow us to visualize matrix multiplication, understand invertibility deeper, and connect algebraic properties such as determinants to transformations of space.

The existence of an inverse matrix is closely related to whether a matrix represents a reversible transformation.

The next chapter will be very graph and program heavy, as we expand the geometric properties of determinants, inverses, and matrices.

Chapter 133

# Cross Product

## 133.1 The Cross Product

Similar to the dot product, we can also take the *cross product* of two vectors. While the dot product measures how closely the two vectors *align* and creates a *scalar*, the cross product measures how much two vectors *differ in direction*, producing a *vector* that is perpendicular to both vectors. The resulting vector's magnitude is equal to the area of the parallelogram spanned by the two input vectors. Before defining the cross product, recall the determinant of a $3 \times 3$ matrix. The determinant measures the *signed* volume scaling factor of the linear transformation represented by the matrix. If the sign is negative, orientation is reversed, as if flipping over a piece of paper.

The cross product formula often uses the notation of a $3 \times 3$ determinant as a compact way to organize its components. However, these were harder to solve without using the expansion by minors. It is important to understand that, in the case of the cross product, the "determinant" we write includes *unit vectors* in the first row; this is a memory device, not an actual determinant in the strict algebraic sense. However, understanding determinants is important before approaching the cross product.

Given two vectors in $\mathbb{R}^3$,

$$\vec{\mathbf{a}} = \langle a_x, a_y, a_z \rangle, \quad \vec{\mathbf{b}} = \langle b_x, b_y, b_z \rangle,$$

their *cross product* is defined as

$$\vec{\mathbf{a}} \times \vec{\mathbf{b}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

Expanding this yields:

$$\vec{\mathbf{a}} \times \vec{\mathbf{b}} = (a_y b_z - a_z b_y)\,\mathbf{i} - (a_x b_z - a_z b_x)\,\mathbf{j} + (a_x b_y - a_y b_x)\,\mathbf{k}.$$

where $\mathbf{i}, \mathbf{j}, \text{and} \mathbf{k}$ are unit vectors.

### 133.1.1 Geometric Interpretation

The cross product $\vec{\mathbf{a}} \times \vec{\mathbf{b}}$ has the following properties:

1. It is perpendicular to both $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$, making it *orthogonal* as well (orthogonality can be thought of as 3D perpendicularity).
2. Orthogonality implies $\vec{\mathbf{a}} \times \vec{\mathbf{b}} = \vec{\mathbf{c}}$, $\vec{\mathbf{a}} \cdot \vec{\mathbf{c}} = 0$ and $\vec{\mathbf{b}} \cdot \vec{\mathbf{c}} = 0$.
3. Its magnitude is
$$\|\vec{\mathbf{a}} \times \vec{\mathbf{b}}\| = \|\vec{\mathbf{a}}\| \, \|\vec{\mathbf{b}}\| \sin\theta,$$
where $\theta$ is the angle between $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$.
4. The magnitude equals the area of the parallelogram spanned by $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$.
5. Two parallel vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ result in the zero vector, $\vec{\mathbf{0}}$.

FIXME diagram here

## *Exercise 378* Using the cross product

Find the cross product of the vectors $\vec{v_1} = \langle 4, 5, 6 \rangle$ and $\vec{v_2} = \langle 3, 7, -8 \rangle$

*Working Space*

*Answer on Page 2036*

# Chapter 134

# Matrices as Transformations

Recall that a *function*, informally, is a rule that takes an input and produces an output.

For example, we know the common function $f(x) = x^3$ takes the input and cubes it, such that the result is $x \times x \times x$. Inputting $x = 3$ outputs $f(3) = 27$.

In Linear Algebra, we can think of matrices as a type of function. Recall our systems of matrices equation,

$$A\vec{x} = \vec{b}$$

We can rearrange this to look closer to function notation:

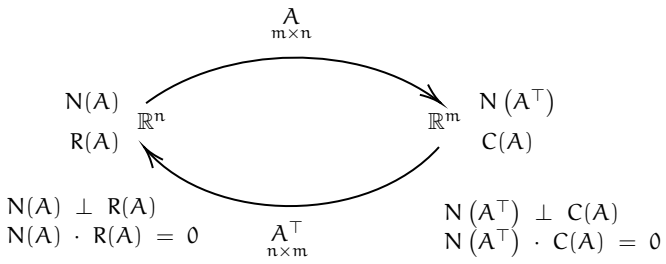
$$\vec{b} = A\vec{x}$$

We can change our input vector $\vec{x}$, which directly affects the output variable $\vec{b}$.

Recall our subspace diagram; the matrix $A$, which is size $m \times n$, takes the input vectors $\vec{x} \in \mathbb{R}^n$, and transforms them to the output $\vec{b} \in \mathbb{R}^m$.

The transformation $T$ is said to map from the reals of $n$ to the reals of $m$, such that: $T : \mathbb{R}^n \to \mathbb{R}^m$

- $\mathbb{R}^n$ is called the **domain**
- $\mathbb{R}^m$ is called the **codomain**




Let's call the transformation $T$, such that

$$T : \mathbb{R}^n \to \mathbb{R}^m$$

We can rewrite our transformation with matrix $A$

$$T(\vec{x}) = A\vec{x}$$

A matrix transformation is completely determined by where it sends the *standard basis vectors*. Each column of $A$ is the image of a basis vector under the transformation, and every output vector is a linear combination of these columns.

Consequently, the column space of $A$ represents the full set of possible outputs of the transformation.

### 134.0.1 The Identity Transformation

For this chapter, we will restrict our transformations to $\mathbb{R}^2 \to \mathbb{R}^2$, such that our matrix is $2 \times 2$. This will help calculations be simpler and also will allow us to make easy to follow transformation diagrams.

Let's take the simplest transformation; the *identity transformation*. Take the matrix:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Every vector in the subspace of $\mathbb{R}^n$ **stays where it is**.

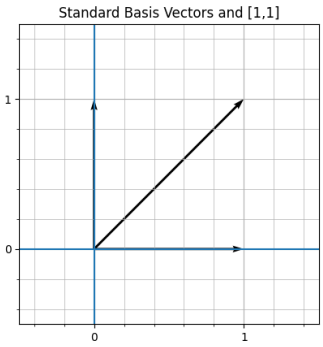
- $e_1$ stays $e_1$
- $e_2$ stays $e_2$

*Every vector stays where it originally started in this transformation.* In a way, no transformation is truly applied.

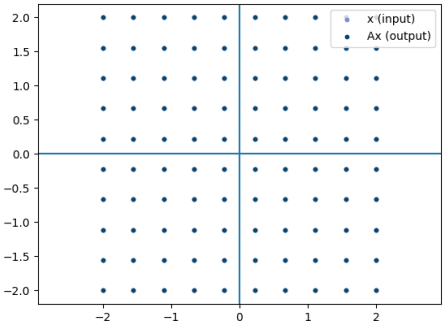
Under the identity transformation, nothing moves. We still get something very important out of this: the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ can be written as a combination of the basis vectors:

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1\vec{e}_1 + 1\vec{e}_2$$

When a linear transformation is applied, the way a vector is built from the basis vectors does not change, but *rather the direction of the basis vectors*.




(a) Identity vectors before transformation




(b) Identity vectors after transformation (no change)

Figure 134.1: An identity transformation acting on basis vectors.

A linear transformation acts on $\vec{x}$ by acting on each basis vector individually:

$$A\vec{x} = x_1 A\vec{e}_1 + x_2 A\vec{e}_2$$

such that any coefficient $x_1$ and $x_2$ are unchanged, while $e_1$ and $e_2$ are transformed.

Our chapter will also take a look at transforming a simple house. Here is the house before any transformations (or, with the identity transformation applied):

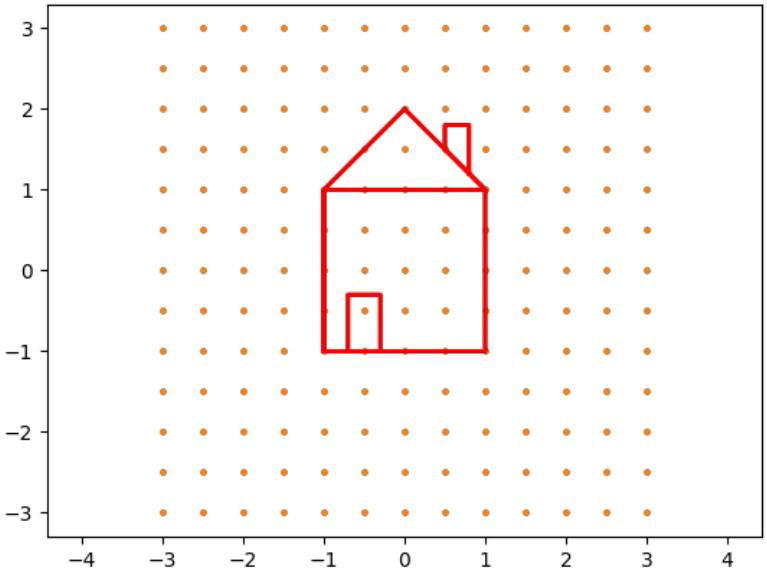



Figure 134.2: A simple house before any transformations.

Linear transformations preserve linear combinations while *altering the directions that define the coordinate system*.

In the next example, we will apply a non-identity matrix and observe how transforming the basis vectors reshapes the entire space. Each figure will show a grid of points in $\mathbb{R}^2$ alongside the resulting transformed points after applying the matrix transformation. We encourage you to play with the python script `transforms.py` to see how different matrices affect the transformation. Also, we will limit our view to a square region from $(-2, -2)$ to $(2, 2)$ and only use $2 \times 2$ matrices for simplicity.

### 134.0.2 Scaling and Shearing

## 134.1 Scaling

Now, let's try scaling up and down the basis vectors. Consider the matrix:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

This matrix will scale both the $e_1$ and $e_2$ vectors by a factor of 2, effectively doubling their lengths.

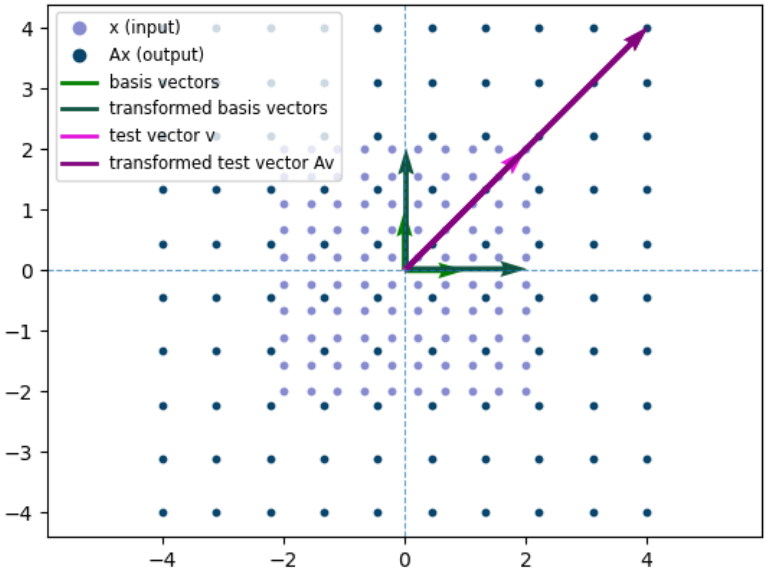



Figure 134.3: The effect of the scaling transformation $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ on a grid of points.

What happens to our house? Well, each part of the house is transformed linearly by 2, a linear *scaling*.

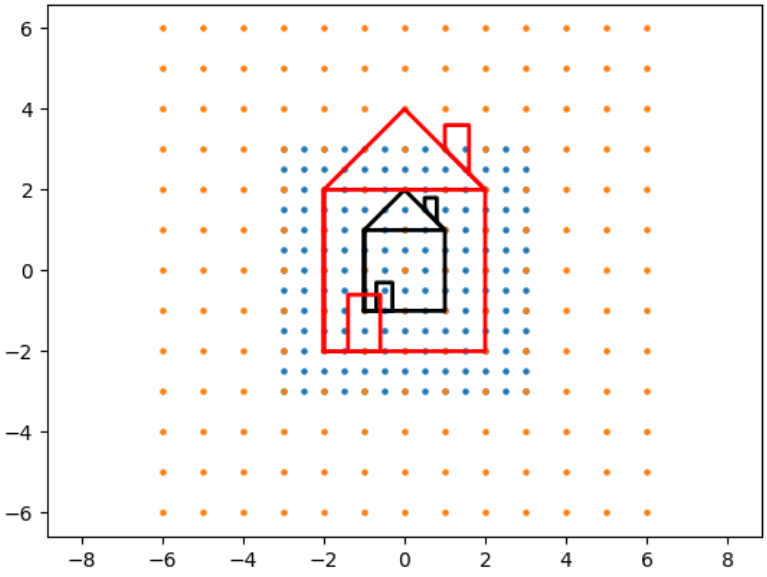



Figure 134.4: The effect of the scaling transformation $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ on a simple house.

This tells us that the transformation $A$ is a scaling transformation, which stretches the space by a factor of 2 in both directions. What do you think happens if we scale by a factor of $\frac{1}{2}$ instead? Try it out!

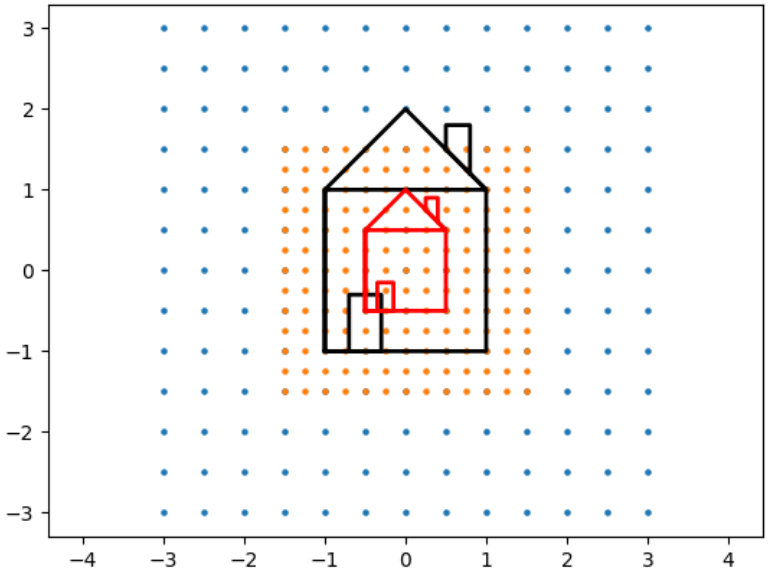



Figure 134.5: The effect of the scaling transformation $A = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$ on a simple house.

This, as expected, shrinks the house by a factor of $\frac{1}{2}$ in both directions.

Now, let's try scaling up *only one* of the basis vectors. Consider the matrix:

$$A_{\text{rightward shear}} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

This matrix will scale the $e_1$ vector by a factor of 1 (no change) and the $e_2$ vector by a factor of 2 (doubling its length).

Applying this transformation to the basis vectors produces this result:

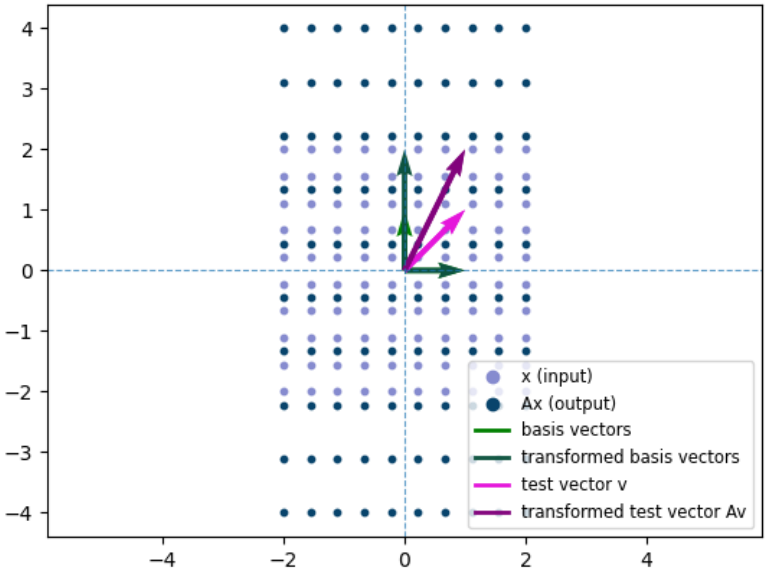



Figure 134.6: The effect of the scaling transformation $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ on the standard basis vectors.

Applying this transformation to our house:

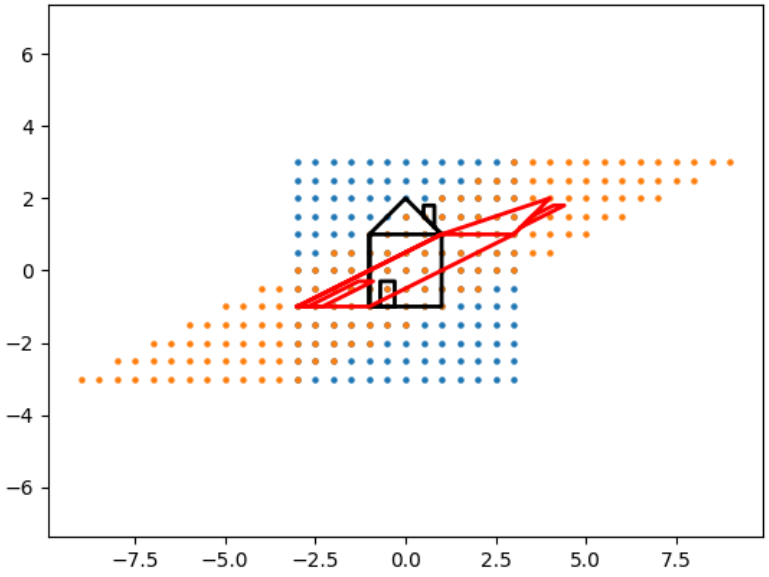



Figure 134.7: The effect of the shearing transformation $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ on a simple house.

## 134.2 Reflections

How would we do reflections across a given axis? Let's think about what happens to each point. Reflecting individual points is the same as reflecting the basis vectors, since every point is a linear combination of the basis vectors.

### 134.2.1 Reflection across the x-axis

For a reflection across the x-axis, the y-values of each point change sign, while the x-values stay the same.

$$(x, y) \mapsto (x, -y)$$

This means that $e_1$ stays $e_1$, while $e_2$ becomes $-e_2$. This allows us to rewrite a $2 \times 2$ matrix for this transformation:

$$A_{x-axis} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Below is a pink test vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and its transformation in purple, which is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

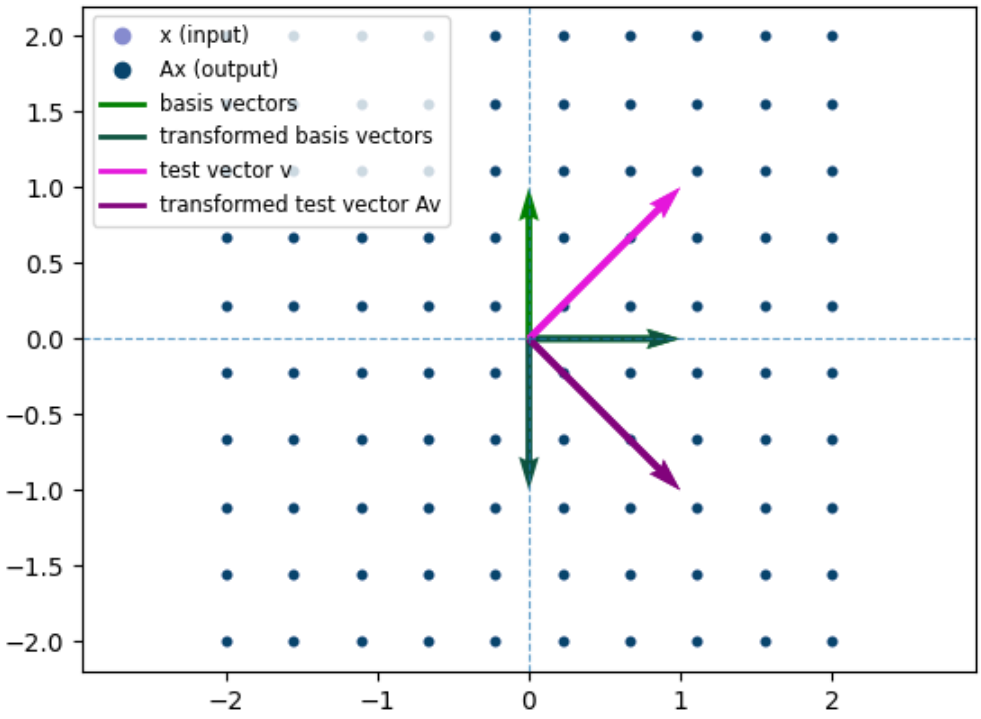



Figure 134.8: The effect of reflecting across the x-axis. Added is the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and its transformation in purple.

Also, we can reflect our house across the x-axis. We will see a difference in where the chimney, roof, and door are located.

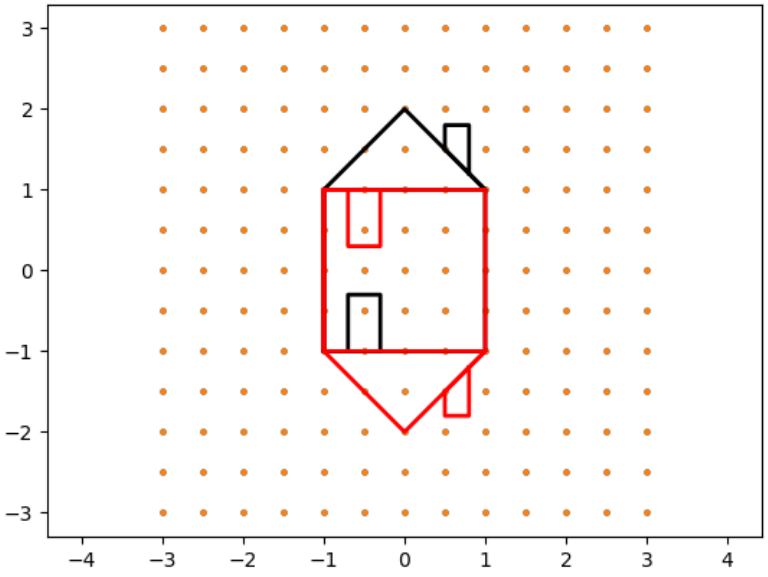



Figure 134.9: The effect of reflecting a house across the x-axis. Red house is the transformed house, while the black house is the original.

### 134.2.2 Reflection across the y-axis

What if we want to reflect across the y-axis instead? In this case, the x-values of each point change sign, while the y-values stay the same. As a function of points, this is

$$(x, y) \mapsto (-x, y)$$

Our basis vector $e_1$ becomes $-e_1$, while $e_2$ stays $e_2$. This allows us to rewrite a matrix for this transformation:

$$A_{y-axis} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

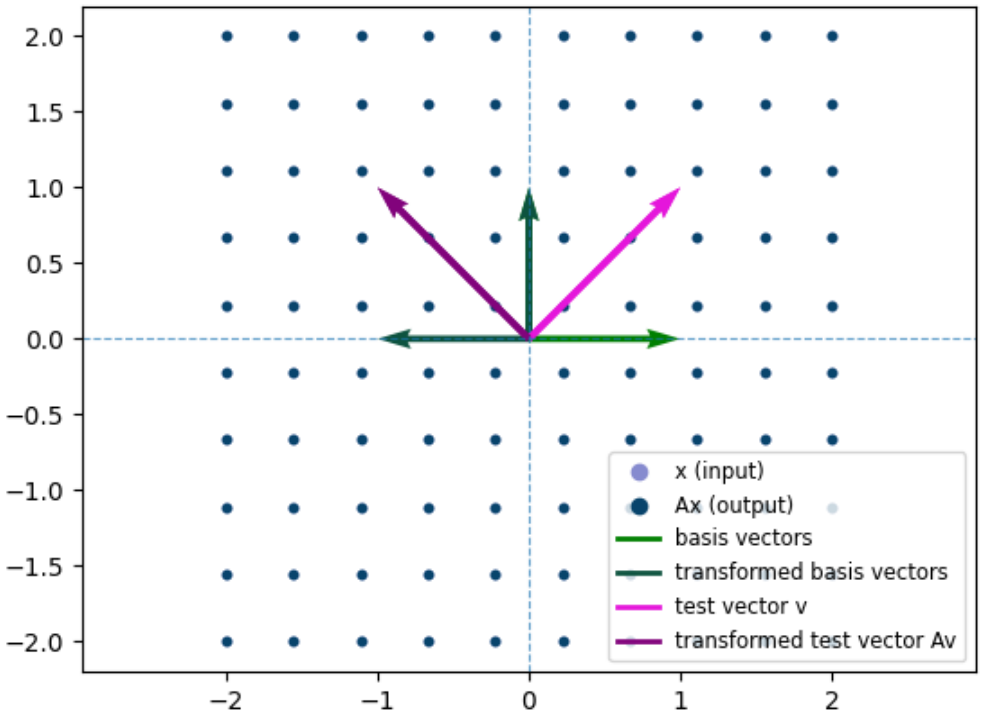



Figure 134.10: The effect of reflecting across the y-axis. Added is the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and its transformation in purple.

Notice that it is similar to the effect of a selfie, where the left and right sides are flipped. This may be more noticeable in the house transformation, where the door and chimney are on the opposite sides of the house.

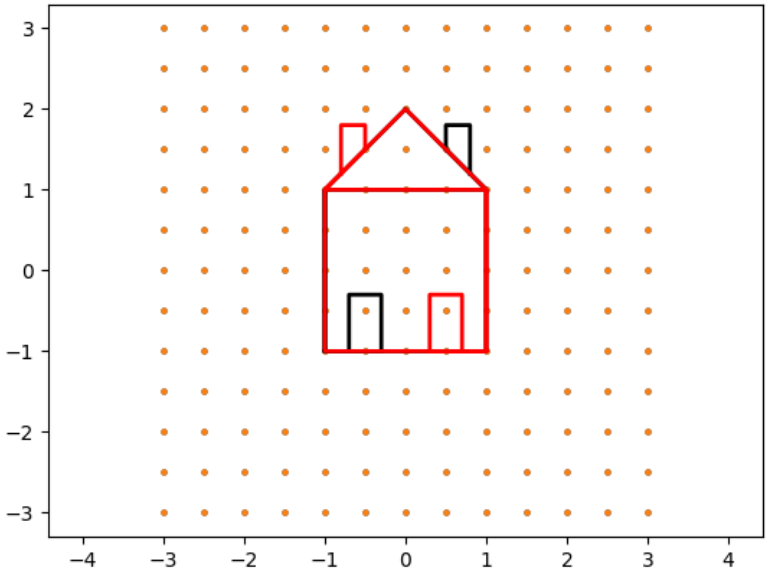



Figure 134.11: The effect of reflecting a house across the y-axis.

Note that the red house is still upright, but the left and right sides are flipped. This makes sense, since the y-values of the points are unchanged, while the x-values flip signs.

### 134.2.3 Reflection across the lines $y = x$ and $y = -x$

As we have seen, there are two standard lines important for inverse functions: $y = x$ and $y = -x$. We can also reflect across these lines.

The line $y = x$ is the line where the x and y values are equal. When we reflect across this line, the x and y values swap places. As a function of points, this is

$$(x, y) \mapsto (y, x)$$

Since each point swaps places, the basis vectors also swap places. $e_1$ becomes $e_2$, while $e_2$ becomes $e_1$. This allows us to rewrite a matrix for this transformation:

$$A_{y=x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Let's look at our house picture, reflected across the line $y = x$. Notice that the house is flipped across the line, such that the roof is now on the bottom and the door is on top.

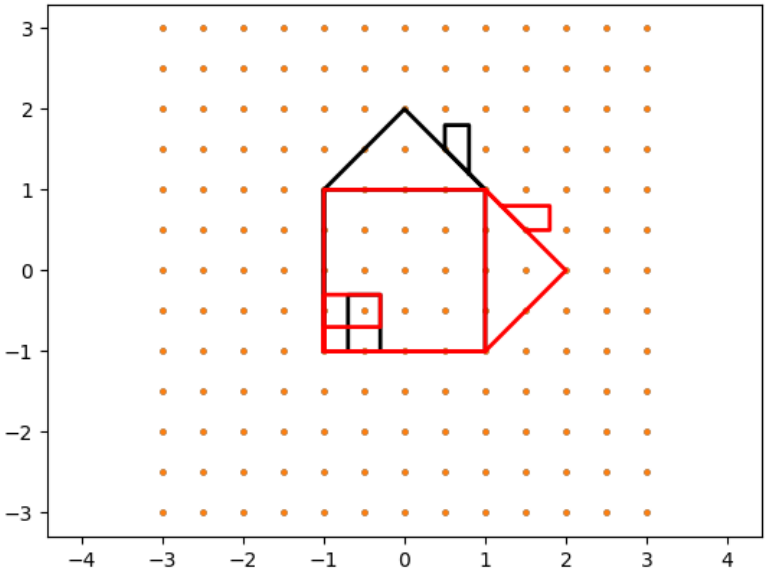



Figure 134.12: The effect of reflecting a house across the line $y = x$.

The line $y = -x$ is the line where the x and y values are equal but with opposite signs. When we reflect across this line, the x and y values swap places and flip signs. As a function of points, this is

$$(x, y) \mapsto (-y, -x)$$

Not only do the x and y values swap places, but they also flip signs. This means that $e_1$ becomes $-e_2$, while $e_2$ becomes $-e_1$. This allows us to rewrite a matrix for this transformation, where

$$A_{y=-x} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

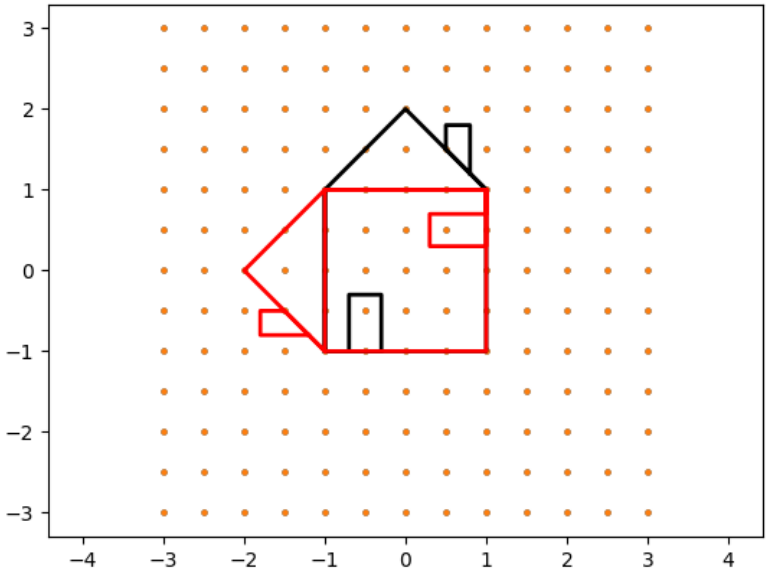
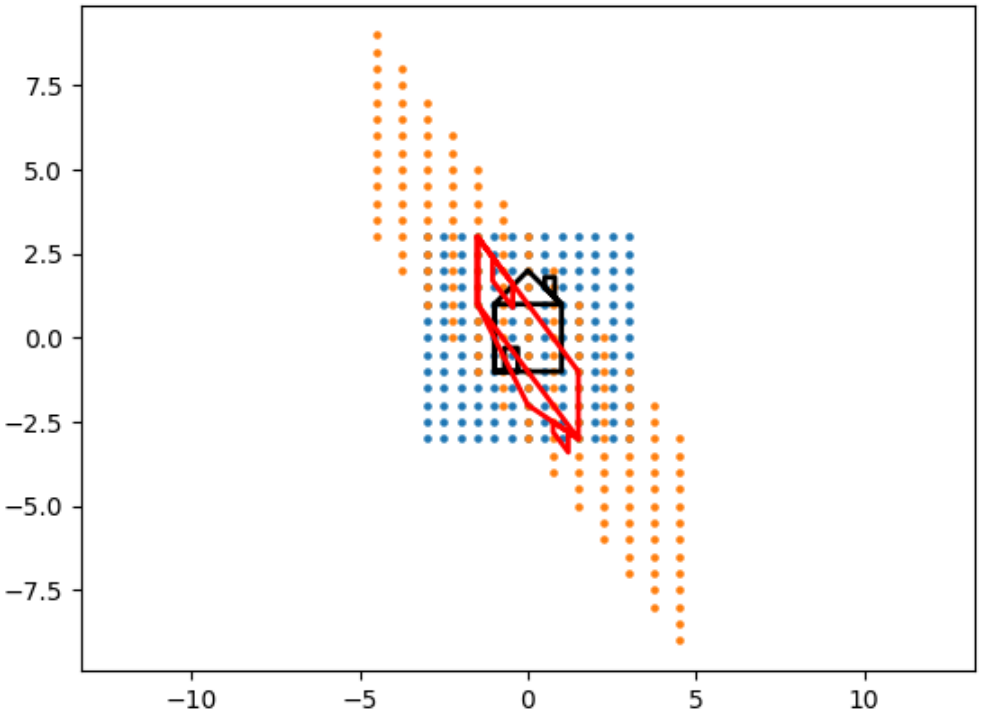



Figure 134.13: The effect of reflecting a house across the line $y = -x$.

### *Exercise 379* Guess the transformation matrix

Given the following transformation of the house, find the matrix that represents this transformation.

Hint: the grid is $3 \times 3$ unit direction with 13 points in each row/column.




*Work*

*Answer on Page 2036*

## 134.3 Rotations

Those past four transformations (134.9, 134.11, 134.12, 134.13) all had something similar

What if we want to rotate our house? We can also do this with a matrix transformation. It may look similar to the reflections, but it is not the same. Let's track our basis vectors to see where they end up:

$$\begin{bmatrix}1\\0\end{bmatrix} \mapsto \begin{bmatrix}0\\1\end{bmatrix}, \qquad \begin{bmatrix}0\\1\end{bmatrix} \mapsto \begin{bmatrix}-1\\0\end{bmatrix}$$

From this, we can write the matrix for a $90^\circ$ rotation clockwise:

$$A_{90^\circ}^{\text{cw}} = \begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix}$$

What about a $90^\circ$ rotation counterclockwise? We can track our basis vectors again:

$$\begin{bmatrix}1\\0\end{bmatrix} \mapsto \begin{bmatrix}0\\-1\end{bmatrix}, \qquad \begin{bmatrix}0\\1\end{bmatrix} \mapsto \begin{bmatrix}1\\0\end{bmatrix}$$

And so the matrix for a $90^\circ$ rotation counterclockwise is:

$$A_{90^\circ}^{\text{ccw}} = \begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}$$

However, we have learned trigonometric functions that result in these values. We know that at $90^\circ$, $\cos(90^\circ) = 0$ and $\sin(90^\circ) = 1$. So we could say:

$$A_{90^\circ}^{\text{cw}} = \begin{bmatrix}\cos(90^\circ) & \sin(90^\circ)\\-\sin(90^\circ) & \cos(90^\circ)\end{bmatrix}$$

Note that this aligns with our *polar* coordinate system where cosine is the x-value and sine is the y-value.

For a counterclockwise rotation of $90^\circ$, we have:

$$A_{90^\circ}^{\text{ccw}} = \begin{bmatrix}\cos(90^\circ) & -\sin(90^\circ)\\\sin(90^\circ) & \cos(90^\circ)\end{bmatrix}$$

But, we can generalize this to any angle $\theta$ instead of just $90^\circ$. For a clockwise rotation by $\theta$, we have:

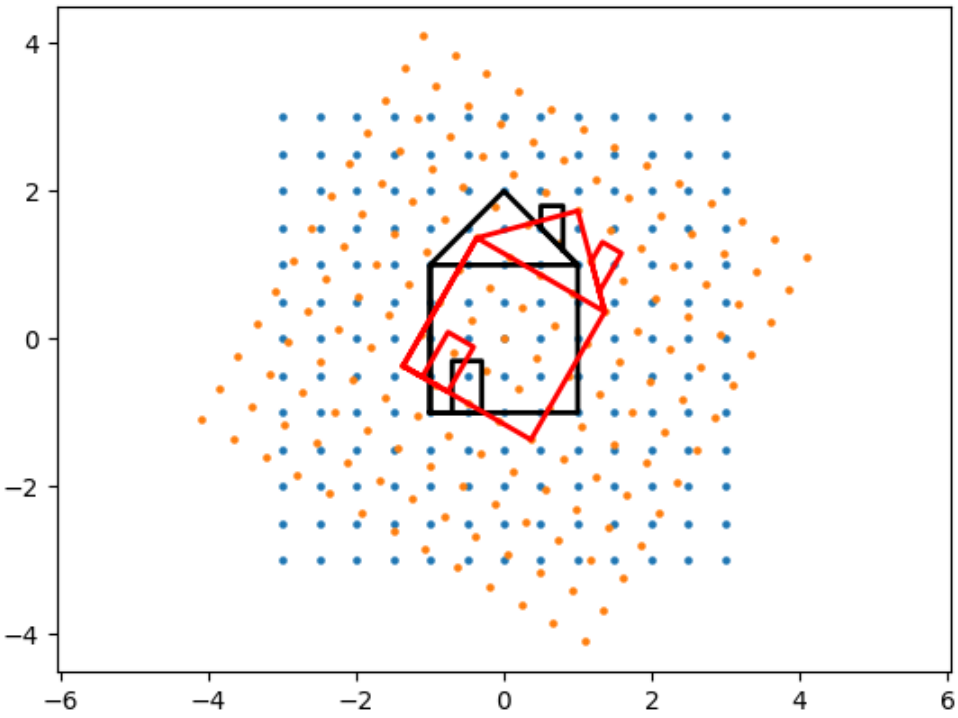
$$A_{\theta}^{\text{cw}} = \begin{bmatrix}\cos(\theta) & \sin(\theta)\\-\sin(\theta) & \cos(\theta)\end{bmatrix}$$

And for a counterclockwise rotation by $\theta$, we have:

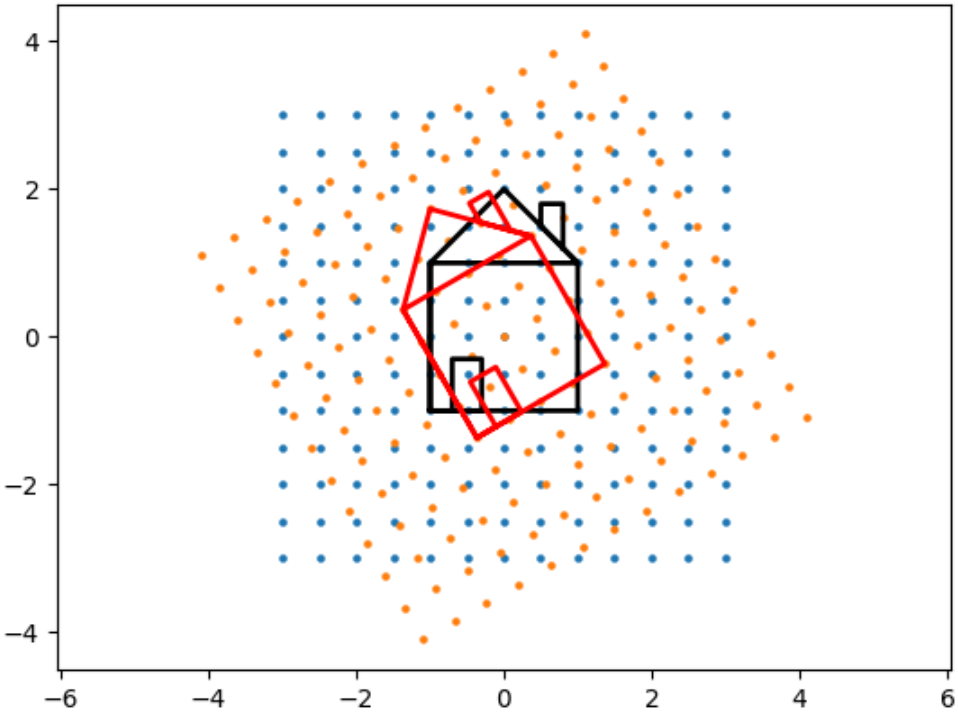
$$A_{\theta}^{\text{ccw}} = \begin{bmatrix}\cos(\theta) & -\sin(\theta)\\\sin(\theta) & \cos(\theta)\end{bmatrix}$$

Note how similar they are! The only difference is the sign of the sine values. Sine is an odd function, which means that $\sin(-\theta) = -\sin(\theta)$, so the sign of the sine values determines the direction of rotation.

Let's see what happens when we apply a $30^\circ$ clockwise and counterclockwise rotation to our house. We can calculate the matrix for this transformation using the cosine and sine values for $30^\circ$:




(a) 30 degree clockwise rotation




(b) 30 degree counterclockwise rotation

Figure 134.14: The effect of 30 degree rotations on our house

## 134.4 What if I want to undo my transform?

We have talked about how to apply a plethora transformations, but what if we want to undo a transformation? For example, if we have a house that is reflected across the rotated 30° clockwise (134.14a), how do we get it back to its original position?

This question brings us back to the application of those *inverse matrices* that we learned about in the previous chapters. If we have a transformation represented by a matrix $A$, then the inverse transformation is represented by the inverse matrix $A^{-1}$. Applying $A^{-1}$ to the transformed house will return it to its original position.

Additionally, given a transformed basis vector set $\vec{e_1}'$ and $\vec{e_2}'$, we can apply the inverse transformation to find the original basis vectors. This is because the inverse transformation will reverse the effects of the original transformation, allowing us to recover the original basis vectors from their transformed counterparts.

If you recall, for our $2 \times 2$ matrices, the inverse is simple! For a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Any larger matrices can get convoluted, so we use gaussian elimination to find the inverse or our numpy function `np.linalg.inv ()`.

The inverse transformation of our 30° clockwise rotation is going to be

$$\begin{aligned} A^{-1}_{30°\text{cw}} &= \frac{1}{\cos(30°)\cos(30°) - (-\sin(30°))\sin(30°)} \begin{bmatrix} \cos(30°) & \sin(30°) \\ \sin(30°) & \cos(30°) \end{bmatrix} \\ &= \frac{1}{\cos^2(30°) + \sin^2(30°)} \begin{bmatrix} \cos(30°) & \sin(30°) \\ \sin(30°) & \cos(30°) \end{bmatrix} \\ &= \frac{1}{1} \begin{bmatrix} \cos(30°) & \sin(30°) \\ \sin(30°) & \cos(30°) \end{bmatrix} \\ &= \begin{bmatrix} \cos(30°) & -\sin(30°) \\ \sin(30°) & \cos(30°) \end{bmatrix} \end{aligned}$$

Notice that this is the same as the matrix for a 30° counterclockwise rotation! This makes sense, since a 30° counterclockwise rotation would undo a 30° clockwise rotation.

Note that if the determinant is 0, then the associated linear transformation is not invertible. In this case, the transformation collapses space into a lower dimension (for example, a plane into a line or a line into a point), causing information to be lost. As a result, the transformation cannot be reversed, and therefore no inverse matrix exists.

Common examples of this are:

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (projection matrix (covered in a future chapter))

$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ (eliminates one direction)

$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ (duplicate / linearly dependent rows)

### *Exercise 380* Undoing a Shear

*Working Space*

A shear, by definition, leaves one axis fixed and slides points parallel to it. For example, an x-shear is given by the following relationship:

$$(x, y) \mapsto (x + ky, y)$$

1. Find the original matrix for an x-shear with a shear factor of $k$.
2. Find the inverse matrix for this transformation.

*Answer on Page 2037*

## 134.5 Summary and looking ahead

In this chapter, we have seen how matrices can be used to represent transformations in $\mathbb{R}^2$. We have explored various types of transformations, including scaling, shearing, reflections, and rotations. We have also discussed how to find the inverse of a transformation, which allows us to undo the effects of a transformation.

We want to look at these transformation as effects on directions. Specifically, are there any directions that this transformation does not mix with others?

- Reflections have the property that they flip one direction while keeping the other direction unchanged. For example, a reflection across the x-axis flips the y-direction

while keeping the x-direction unchanged. In this, some directions stay the same, while others are flipped. Certain directions are preserved, while others are not.

- Rotations, on the other hand, mix all directions together. There are no directions that remain unchanged under a rotation, except for the trivial case of a $0^\circ$ rotation (the identity transformation)
- Stretches and scalings also have the property that they stretch or shrink certain directions while keeping others unchanged. For example, a scaling transformation that stretches the x-direction while keeping the y-direction unchanged will preserve the y-direction while stretching the x-direction.

For some transformations, there exist special directions along which vectors are simply scaled. In studying linear transformations, it is natural to ask whether there exist directions that remain unchanged, except for cardinal scaling of its length. Before reading the next chapter, try to find a nonzero vector whose direction is *unchanged* by each transformation. Not every transformation will have such a vector, but some will.

Chapter 135

# Eigenvectors and Eigenvalues

Like many specialized disciplines, Linear Algebra uses many unfamiliar terms whose origins you might wonder about. Eigenvectors and eigenvalues are two of them. If you know German, you will recognize that eigen means inherent or a characteristic attribute. Named by the German mathematician David Hilbert, an eigenvector mathematically describes a characteristic feature of an object that remains unchanged after transformation. You can think of an eigenvector as the direction that doesn not change direction. An eigenvector characterizes a linear transformation, whereas its eigenvalue tells how much the vector is scaled. Eigenvalues can be negative or positive. A negative value indicates the direction of the eigenvector is reversed.

Eigenvalues and eigenvectors are a way to break down matrices, which can simplify many calculations and enable us to understand various properties of the matrix. They are widely used in physics and engineering for stability analysis, vibration analysis, and many other applications.

Let's look at a visual example.

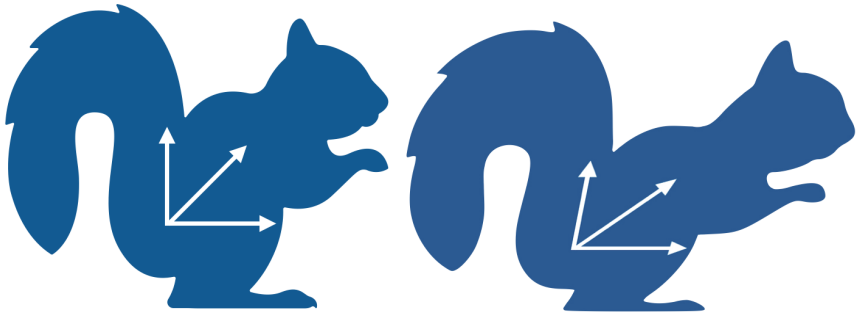


Figure 135.1: Standardized and skewed squirrel image.

You can see that the image on the right is a skewed version of the image on the left. Look closely at the vectors and you will notice that one of the vectors is pointing in the same direction in both images, while the direction of the other two vectors has changed. The eigenvector is the one at the bottom that points to 0 degrees (which you can think of due east) in both images. So, the characteristic attribute of both images is their horizontal direction.

When you overlay the vectors from one image over the other, you will notice that the

horizontal vector, while the same direction in both images, is a bit longer in the skewed version. The scale of the stretch is described by an eigenvalue.

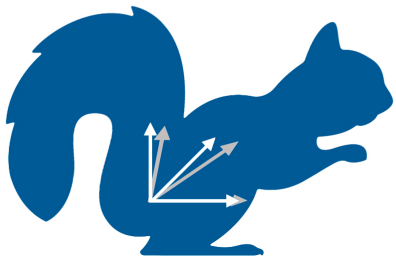


Figure 135.2: Standardized and skewed squirrel image overlaid on each other.

## 135.1 Definition

**Eigenvector and Eigen Equation**

Given a square matrix $A$, a non-zero vector $v$ is an eigenvector of $A$ if multiplying $A$ by $v$ results in a scalar multiple of $v$. In other words, the *eigenequation* is:

$$Av = \lambda v \tag{135.1}$$

where $\lambda$ is a scalar known as the eigenvalue corresponding to the eigenvector $v$.

An eigenvector is a nonzero vector that does not change direction when a linear transformation is applied.

It may stretch, shrink, or flip direction, but it *stays on the same line*.

## 135.2 Finding Eigenvalues and Eigenvectors

You find the eigenvalues of a matrix $A$ by solving the characteristic equation:

$$\det(A - \lambda I) = 0 \tag{135.2}$$

where $\det(.)$ denotes the determinant, $I$ is the *identity matrix* of the same size as $A$, and $\lambda$ is a scalar. In other words, we add $\lambda$ along the diagonal of a, and solve for its determinant.

$$A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{bmatrix}$$

Once your find the eigenvalues, you can find the corresponding eigenvectors by substituting each eigenvalue into the equation $Av = \lambda v$, and solving for $v$. $v$ is a vector which only gets *scaled*, with no change in direction.

For a non-zero **v** to satisfy this equation,

- 
- $\det(A - \lambda I) = 0$ must have a determinant of zero
- $A - \lambda I$ cannot have an inverse
- space is flattened on to a volume of 0.

FIXME visual representation graphically?

## 135.3 Example

For a $2 \times 2$ matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the characteristic equation is:

$$(a - \lambda)(d - \lambda) - bc = 0 \tag{135.3}$$

Solving this equation gives the eigenvalues. Substituting each eigenvalue back into the equation $Av = \lambda v$ gives the corresponding eigenvectors.

Let matrix $A =$

$$\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

The characteristic equation is:

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{bmatrix} = 0$$

$$(5 - \lambda)(2 - \lambda) - (4)(1) = 0$$

$$10 - 5\lambda - 2\lambda + \lambda 2 - 4 = 0$$

$$\lambda 2 - 7\lambda + 6 = 0$$

$$(\lambda - 6)(\lambda - 1) = 0$$

$$\lambda = 6, \lambda = 1$$

Now that you have the eigen values you can substitute these values into the equation:

$$|A - \lambda I| = 0$$

For $= 1$:

$$(A - \lambda I)v = O$$

$$\begin{bmatrix} 5-1 & 4 \\ 1 & 2-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Next, use elementary row transformation by multiplying row 2 by 4, then subtracting row 1.

$$\begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Now you can expand as an equation:

$$4x + 4y = 0$$

Assume $y = w$

$$4x = -4w$$

$$x = -w$$

The solution is:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -w \\ w \end{bmatrix} = w \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

So the eigenvector is:

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Now we need to substitute the other eigenvalue, 6, into the equation and follow the same procedure for finding the eigenvector.

$$\begin{bmatrix} 5-6 & 4 \\ 1 & 2-6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Next, use elementary row transformation by adding row 1 to row 2.

$$\begin{bmatrix} -1 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Expand as an equation:

$$-x + 4y = 0$$

$$\text{Assume} y = w$$

$$-x + 4w = 0$$

$$x = 4w$$

The solution is:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4w \\ w \end{bmatrix} = w \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

So the eigenvector is:

$$\begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

In conclusion, the eigenvectors of the given 2 x 2 matrix are:

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{and} \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

## 135.4 Eigenbasis and Eigenspace

Let $A$ be a square matrix and let $\lambda$ be an eigenvalue of $A$.

**Eigenspace**

The **eigenspace** corresponding to $\lambda$ is the set of all vectors $v$ such that

$$Av = \lambda v.$$

Equivalently, the eigenspace of $\lambda$ is the solution set of

$$(A - \lambda I)v = 0.$$

Note that there will likely be multiple lambdas for a given matrix, and therefore multiple eigenspaces for each matrix.

The eigenspace is a subspace consisting of all eigenvectors associated with $\lambda$, together including the zero vector, $\vec{\mathbf{0}}$.

**Eigenbasis**

An *eigenbasis* for a matrix $A$ is a basis of the vector space consisting entirely of eigenvectors of $A$.

$$\{v_1, v_2, \ldots, v_n\}$$

is an eigenbasis for $A$ if:

$$Av_i = \lambda_i v_i \quad \text{for each } i = 1, 2, \ldots, n$$

## *Exercise 381* A $2 \times 2$ matrix

*Working Space*

Find the eigenbasis of the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

*Answer on Page 2037*

## 135.5 Eigenvalues and Eigenvectors in Python

Create a file called `vectors_eigen.py` and enter this code:

```
# import numpy to perform operations on vector
import numpy as np
from numpy.linalg import eig

a = np.array([[2, 2, 4],
              [1, 3, 5],
              [2, 3, 4]])
eigenvalue,eigenvector = eig(a) #return a tuple of eigenvalues [l1, l2,..., ln],
↪  eigenvectors [ [...], [...], ...]

# The values are not in any particular order
print('Eigenvalues:', eigenvalue)

# The eig function returns the normalize vectors
print('Eigenvectors:', eigenvector)
```

## 135.6 Connection to transformations

In the previous chapter, we studied matrix transformations such as scaling, reflection, shear, and rotation. Each of these transformations changes vectors in space by stretching, shrinking, flipping, or rotating them. Most vectors change both their length and/or their direction under such transformations.

Eigenvectors arise naturally when a transformation leaves at least one direction unchanged. Different types of transformations exhibit different eigenvector behavior:

- **Scaling transformations** preserve all directions. Every nonzero vector is an eigenvector.
- **Reflections** preserve the direction of vectors along the axis of reflection and reverse vectors perpendicular to it. Vectors lying on the axis of reflection are unchanged by the transformation and therefore are eigenvectors with eigenvalue 1. Vectors perpendicular to the axis of reflection are flipped to point in the opposite direction, making them eigenvectors with eigenvalue $-1$.
- **Shear transformations** preserve exactly one direction, even though most vectors change direction.
- **Rotations in two dimensions** generally preserve no directions, and therefore have no real eigenvectors (except in the trivial cases of $\theta = \frac{\pi}{2}$ or $0$).

These observations explain why some matrices have many eigenvectors, some have only a few, and some have none at all. Eigenvectors exist precisely when a transformation has invariant directions.

Understanding eigenvectors as preserved directions allows us to connect algebraic computations to geometric behavior. When a matrix has enough independent eigenvectors, it becomes possible to describe the transformation entirely in terms of simple scaling along these special directions.

## 135.7 Summary

In summary, lets take a *transformation*:

Something that stretches, squishes, flips, or rotates space. This transformation could be represented by a matrix.

Most vectors in space will get changed in both direction and length when you apply the transformation.

But some special vectors don't change direction at all — they *only* get stretched or shrunk (and maybe flipped).

These special "unchanging-direction" vectors are called eigenvectors. The amount they get stretched (or shrunk) is their eigenvalue.

## 135.8 Where to Learn More

Watch this video from Khan Academy, *Introduction to Eigenvectors*: https://rb.gy/mse7i

# Chapter 136

# Projections

The word ”projection” has two main meanings in everyday life. One is a projection as a forecast or estimate of something in the future based on the current situation; another is the result of shining a light to cast a shadow or show a movie. Both these definitions apply to mathematical projection.

Projections are used in many fields, such as science, math, engineering, and finance. Here are a few examples:

- Investors evaluate risk and return of a portfolio by projecting an asset's return onto a reference portfolio.
- Astronomers analyze the motion of stellar objects by projecting the object's true motion onto the plane of the sky.
- Robotics engineers use projections to prevent robots from running into obstacles by projecting the robot's position onto the optimal path.

Mathematically, a projection describes the relationship of one vector to another in terms of direction and orthogonality. Given two vectors, $\mathbf{u}$ and $\mathbf{v}$, the projection of $\mathbf{u}$ onto $\mathbf{v}$ separates $\mathbf{u}$ into two components. The first component signifies how much $\mathbf{u}$ lies in the direction of $\mathbf{v}$. The second signifies the component of $\mathbf{u}$ that is orthogonal (perpendicular) to $\mathbf{v}$.

The figure 136.1 depicts a projection. The perpendicular line dropped from the end of $\mathbf{u}$ is the *orthogonal* component. The portion of $\mathbf{u}$ that lies in the direction of $\mathbf{v}$ is the blue segment.

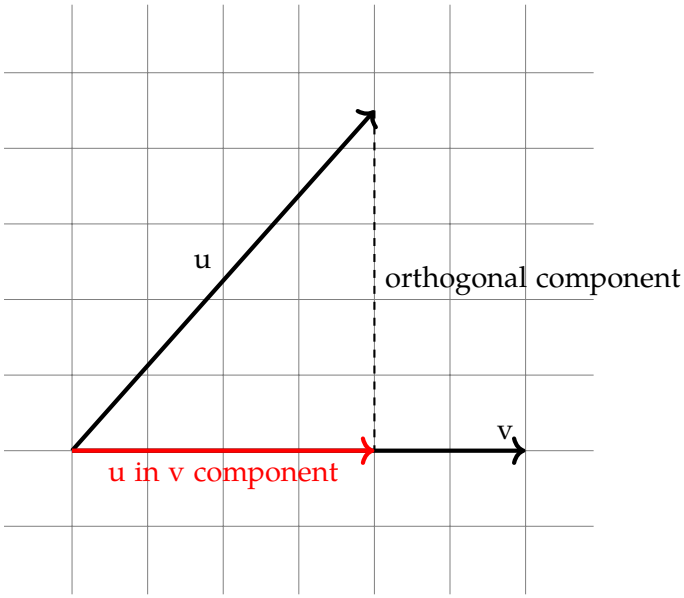



Figure 136.1: The projection of u onto v

You can also think of a projection as the shadow cast by one vector onto the other by an overhead light. See Figure 136.2

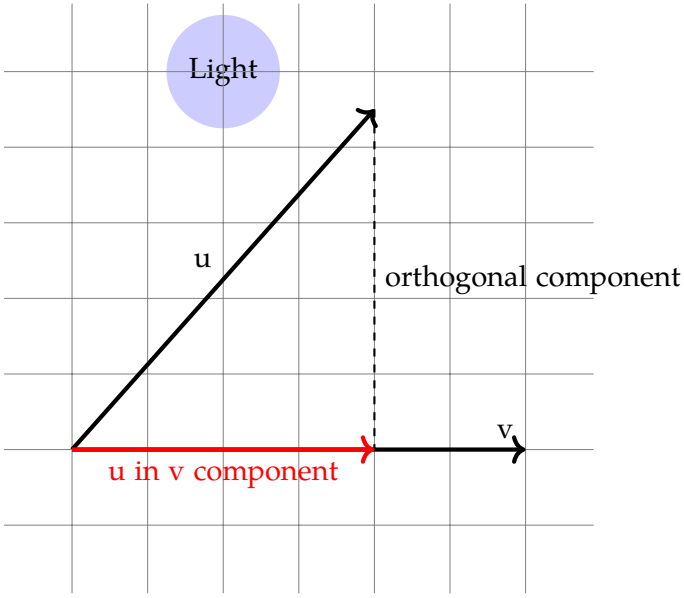



Figure 136.2: Projection of u onto v with a light included to simulate a shadow.

The projected vector can be in any direction and its length can extend beyond the vector onto which it is projecting. See Figure 136.3.

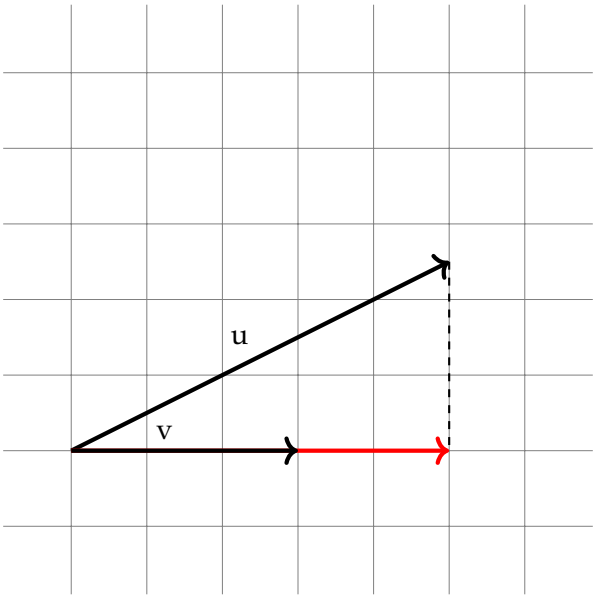



Figure 136.3: Projection vector extended beyond $v$.

To calculate the projection of $\mathbf{v}$ onto $\mathbf{u}$, use this formula:

$$\text{Projection of } \mathbf{u} \text{ onto } \mathbf{v}: \quad \text{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\,\mathbf{v} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\,\mathbf{v}$$

where $\mathbf{u} \cdot \mathbf{v}$ is the dot product of $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$. Note that either form of the equation works, and the $\mathbf{v}$ being multiplied by the dot product quotient cannot be cancelled because it is a vector, not a scalar.

Note that the denominator is the magnitude squared of vector $\mathbf{v}$.

$$\left(\sqrt{a_1^2 + a_2^2 + ... + a_n^2}\right)^2$$

You learned previously that this is the same as the dot product of a vector with itself.

$$v \cdot v$$

In the examples that follow, we will simplify to the dot product notation.

Let's look at a specific example:

$$u = (1, 4, 6)$$
$$v = (-2, 6, 2)$$

$$\mathbf{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{u \cdot v}{\| v \|^2} v$$

$$\mathbf{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{(1,4,6) \cdot (-2,6,2)}{(-2,6,2) \cdot (-2,6,2)} (-2,6,2)$$

$$\mathbf{proj}_{\mathbf{v}}(\mathbf{u}) = (\frac{34}{44} (-2,6,2)$$

$$\mathbf{proj}_{\mathbf{v}}(\mathbf{u}) = (-1.545, 4.64, 1.545)$$

As you work your way through this course, you will have a chance to apply the calculations you learn in this chapter to a variety of problems. Specifically, the next chapter shows how to transform a set of linearly independent vectors into a set of orthogonal ones. Projections are essential to that transformation.

## *Exercise 382* Projections

Find the projection of **a** on **b** where:

$$a = (1,3)$$

$$b = (-4,6)$$

*Working Space*

*Answer on Page 2038*

## 136.1 Projections in Python

Create a file called `projections.py` and enter this code:

```
import numpy as np

# define two vectors
a = np.array([1, 4, 6])
b = np.array([-2, 6, 2])

# use np.dot() to calculate the dot product
projection_a_on_b = (np.dot(a, b)/np.dot(b, b))*b

print("The projection of vector a on vector b is:", projection_a_on_b)
```

## 136.2 Where to Learn More

Watch this Introduction to Projections from Khan Academy from your digital resources: https://rb.gy/yf0i3

Chapter 137

# The Gram-Schmidt Process

The Gram-Schmidt Process is a method use to transform a set of linearly independent vectors to a set of orthogonal (perpendicular) vectors. The original vectors and the transformed vectors span the same subspace.

The process was named after two mathematicians: Jørgen Pedersen Gram, a Danish actuary mathematician, and Erhard Schmidt, a German mathematician. The men developed the orthogonalization process independently. Gram introduced the process in 1883, whereas Schmidt did his work in 1907. It was not named the Gram-Schmidt process until sometime later, after both mathematicians became well known in the mathematical community.

In the last chapter, you learned how to calculate a projection of one vector on another. Given two vectors, $u$ and $v$, the projection separates $u$ into the part that is orthogonal to $v$ and the part that has a dependency with $v$. The Gram-Schmidt Process builds on the notion of projections to iteratively strip away dependencies between vectors until the vectors that remain are orthogonal. If there happens to be a vector that is a linear combination of the others, that vector will be reduced to the zero vector. The vectors that remain define a new basis for space spanned by the original set of vectors.

Gram-Schmidt has many practical applications in science and engineering, such as:

1. In signal processing, it can represent an audio signal with fewer components making it easier to isolate and remove noise.
2. In statistics and data analysis, it can reduce the complexity of a dataset so that it is easier to see which aspects or features contribute to the analysis.

## 137.1 The Process

The Gram-Schmidt Process orthonormalizes a set of vectors in an inner product space, most commonly the Euclidean space $\mathbb{R}^n$. The process takes a finite, linearly independent set $S = \{v_1, v_2, \ldots, v_k\}$ for $k \leq n$, and generates an orthogonal set $S' = \{u_1, u_2, \ldots, u_k\}$ that spans the same k-dimensional subspace of $\mathbb{R}^n$ as S.

Let's look at how the process works. Given a set of vectors $S = \{v_1, v_2, \ldots, v_k\}$, the Gram-Schmidt Process is as follows:

1. Let $u_1 = v_1$.
2. For $j = 2, 3, \ldots, k$:
   (a) Let $w_j = v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, u_i \rangle}{\langle u_i, u_i \rangle} u_i$
   (b) Let $u_j = w_j$

Here, $\langle ., . \rangle$ denotes the inner product.

The set of vectors $S' = \{u_1, u_2, \ldots, u_k\}$ obtained from this process is orthogonal, but not necessarily orthonormal. To create an orthonormal set, you simply need to normalize each vector $u_i$ to unit length. That is, $u_i' = \frac{u_i}{\|u_i\|}$, where $\|.\|$ denotes the norm (or length) of a vector.

Among other things, making vectors orthonormal simplifies calculations makes it easier to define rotations and transformations, and provides a framework for calculations in fields such as quantum mechanics.

## 137.2 Example Calculation

Given a set of linearly independent vectors, we will use the Gram-Schmidt Process to find an orthogonal basis.

Let

$$W = Span(x_1, x_2, x_3)$$

where

$$x_1 = (1, 2, -2)$$
$$x_2 = (1, 0, -4)$$
$$x_3 = (5, 2, 0)$$

The three orthogonal vectors will define the same subspace as the original vectors.

The first vector of the orthogonal subspace is easy to define. We set it to be the same as $x_1$.

$$v_1 = x_1 = (1, 2, -2)$$

The second orthogonal vector is a projection of $x_2$ onto $v_1$. You learned projections in the last chapter, so this should be fairly straightforward.

$$v_2 = x_2 - \frac{x_2 v_1}{v_1 v_1} v_1$$

Substitute the values:

$$v_2 = (1, 0, -4) - \frac{(1, 0, -4)(1, 2, -2)}{(1, 2, -2)(1, 2, -2)} (1, 2, -2)$$

Calculate the coefficient for $v_1$:

$$v_2 = (1, 0, -4) - \frac{9}{9}(1, 2, -2)$$

Perform the subtraction:

$$v_2 = (0, -2, -2)$$

The third vector for the orthogonal subspace is a projection onto $v_1$ and $v_2$.

$$v_3 = x_3 - \frac{x_3 v_1}{v_1 v_1} v_1 - \frac{x_3 v_2}{v_2 v_2} v_2$$

Substitute the values:

$$v_3 = (5, 2, 0) - \frac{(5, 2, 0)(1, 2, -2)}{(1, 2, -2)(1, 2, -2)}(1, 2, -2) - \frac{(5, 2, 0)(1, 0, -4)}{(1, 0, -4)(1, 0, -4)}(1, 0, -4)$$

$$v_3 = (5, 2, 0) - (9/9)(1, 2, -2) - (-4/8)(1, 0, -4)$$

$$v_3 = (5, 2, 0) - (1, 2, -2) + (1/2)(1, 0, -4)$$

$$v_3 = (5, 2, 0) - (1, 2, -2) + (0, -1, -1)$$

$$v_3 = (4, -1, 1)$$

This set of vectors is orthogonal, so we need to normalize them so that the vectors are orthonormal. Recall that an orthonormal vector has a length of 1 and is computed using this formula:

$$\mathrm{normalizedVector} = \mathrm{vector}/\mathrm{np.sqrt}(\mathrm{np.sum}(\mathrm{vector} * *2))$$

Thus the normalized set of vectors is:

$$v_1 = (0.33, 0.67, -0.67)$$

$$v_2 = (0.0, -0.71, -0.71)$$

$$v_3 = (0.94, -0.24, 0.24)$$

## *Exercise 383* Gram-Schmidt Process

Use the Gram-Schmidt Process to find an orthogonal basis for the span defined by $x_1, x_2$ where:

$$x_1 = (1, 1, 1)$$

$$x_2 = (0, 1, 1)$$

*Working Space*

*Answer on Page 2038*

## 137.3 The Gram-Schmidt Process in Python

Create a file called `vectors_gram-schmidt.py` and enter this code:

```
# import numpy to perform operations on vector
import numpy as np

# Find an orthogonal basis for the span of these three vectors
x1 = np.array([1, 2, -2])
x2 = np.array([1, 0, -4])
x3 = np.array([5, 2, 0])

# v1 = x1
v1 = x1
print("v1 = ",v1)

# v2 = x2 - (the projection of x2 on v1)
v2 = x2 - (np.dot(x2,v1)/np.dot(v1,v1))*v1
print("v2 = ", v2)

# v3 = x3 - (the projection of x3 on v1) - (the projection of x3 on v3)
v3 = x3 - (np.dot(x3,v1)/np.dot(v1,v1))*v1 - (np.dot(x3,v2)/np.dot(v2,v2))*v2
print("v3 =", v3)

# Next, normalize each vector to get a set of vectors that is both orthogonal and
↪  orthonormal:
v1_norm = v1 / np.sqrt(np.sum(v1**2))
v2_norm = v2 / np.sqrt(np.sum(v2**2))
v3_norm = v3 / np.sqrt(np.sum(v3**2))
print("v1_norm = ", v1_norm)
print("v2_norm = ", v2_norm)
```

```
27 print("v3_norm = ", v3_norm)
```

## 137.4 Where to Learn More

Watch this video from Khan Academy about the Gram-Schmidt Process: https://www.youtube.com/watch?v=rHonltF77zI

Chapter 138

# Singular Value Decomposition

In the previous chapter, you learned how to calculate eigenvalues and eigenvectors. However, not every matrix has them. For those matrices, singular values and singular vectors are analogous features.

Singular Value Decomposition (SVD) is a matrix factorization technique that breaks down a matrix into three matrices that represent the structure and properties of the original matrix. The decomposed matrices make calculations easier and provide insight into the original matrix. Essentially, SVD can transform a high dimension, highly variable set of data into a set of uncorrelated data points that reveal subgroupings that you might not have noticed in the original data. SVD tells us that a linear transformation can be thought of as a rotation, scaling, and another rotation.

## 138.1 Definition

For any $m \times n$ matrix $A$, SVD decomposes the matrix into three matrices.

$$A = U\Sigma V^T \tag{138.1}$$

- $U$ is an orthogonal matrix whose size is $m \times m$. Its columns are the eigenvectors of $AA^T$. These are the left singular vectors of $A$. Because $U$ is orthogonal, $U^TU = I$.
- $V$ is an orthogonal matrix whose size is $n \times n$ matrix. Its columns are the eigenvectors of $A^TA$. These are the right singular vectors of $A$. Because $V$ is orthogonal, $V^TV = I$.
- $\Sigma$ is a diagonal matrix that is the same size as $A$. Its diagonal contains the singular values of $A$, arranged in descending order. These values are the square roots of the eigenvalues of both $A^TA$ and $AA^T$.

## 138.2 Applications of SVD

SVD has numerous applications:

- It is used in machine learning and data science to perform dimensionality reduction, particularly through a technique known as Principal Component Analysis (PCA).

- In numerical linear algebra, SVD is used to solve linear equations and compute matrix inverses in a more numerically stable way.
- It is used in image compression, where low-rank approximations of an image matrix provide a compressed version of the original image.

## 138.3 Calculating SVD Manually

You might be inclined to skip this example because the computations are lengthy. Why would anyone do this when they can use a computing language, like Python, to calculate the SVD with essentially one command? We show this so you can understand what goes on "under the hood" when you compute SVD programmatically.

After you read through this example, you will see how to use Python to compute SVD. Next, you will see an example of using SVD for image compression. Finally, you will be given an exercise to compute the SVD. For this, you will need to write your own Python script.

Let's find the SVD for matrix $A$. Recall that we want to find $U$. $\Sigma$, and $V^T$ such that:

$$A = U\Sigma V^T \tag{138.2}$$

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$U = AA^T$$

Calculating

$$AA^T$$

will give us a square matrix:

$$A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix}$$

Next, we will find the eigenvalues and eigenvectors of $A^T$. This is a chance to apply what you learned in the previous chapter. We know that:

$$Av = \lambda v \tag{138.3}$$

So:

$$\begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Rewrite as a set of equations:

$$11x_1 + x_2 = \lambda x_1$$

$$x_1 + 11x_2 = \lambda x_2$$

Then rearrange:

$$(11-\lambda)x_1 + x_2 = 0$$

$$x_1 + (11-\lambda)x_2 = 0$$

Solve for $\lambda$:

$$\begin{bmatrix} (11-\lambda), 1 \\ 1, (11-\lambda) \end{bmatrix} = 0$$

And as equations:

$$(11-\lambda)(11-\lambda) - 1 \cdot 1 = 0$$

$$(\lambda - 10)(\lambda - 12) = 0$$

These are the eigenvalues.

$$\lambda = 10$$

$$\lambda = 12$$

When substituted into the original equations, you get the eigenvectors. For

$$\lambda = 10$$

:

$$(11-10)x_1 + x_2 = 0$$

$$x_1 = -x_2$$

We will set

$$x_1$$

to 1 and get this eigenvector:

$$[1, -1]$$

For

$$\lambda = 10$$

:

$$(11-12)x_1 + x_2 = 0$$

$$x_1 = x_2$$

We will set

$$x_1$$

to 1 and get this eigenvector:

$$[1, 1]$$

The matrix is:

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Next, you need to apply the Gram-Schmidt process to the column vectors. After that, you will have $U$, the $m \times m$ matrix whose columns are eigenvectors of $AA^T$. These are the left singular vectors of $A$. After you apply Gram-Schmidt, you should end up with:

$$U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

The process for calculating $V$ is the same as the calculation for $U$, except:

$$V = A^T A$$

$$A^T A = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{bmatrix}$$

After applying the process we applied to solve for U, you get:

$$V = \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{5} & 1/\sqrt{30} \\ 2/\sqrt{6} & -1/\sqrt{5} & 2/\sqrt{30} \\ 1/\sqrt{6} & 0 & -5/\sqrt{30} \end{bmatrix}$$

However, you want $V_T$:

$$V_T = \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \\ 2/\sqrt{5} & -1/\sqrt{5} & 0 \\ 1/\sqrt{30} & 2/\sqrt{30} & -5/\sqrt{30} \end{bmatrix}$$

You have only to calculate $\Sigma$, a diagonal matrix that is the same size as $A$. The diagonal contains the singular values of $A$, arranged in descending order. These are the square roots of the eigenvalues of both $A^T A$ and $AA^T$.

Because the non-zero eigenvalues of U are the same as V, let's use the eigenvalues we calculate for U, 10 and 12. Note that $\Sigma$ will not be of the correct dimension to reconstruct the original matrix unless we add a column. By adding a zero column you'll be able to multiply between $U$ and $V$:

$$\Sigma = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{12} & 0 \end{bmatrix}$$

You can check your work by multiplying the decomposed matrices. This should return the original matrix.

$$A = U\Sigma V^T$$

$$= U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{12} & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \\ 2/\sqrt{5} & -1/\sqrt{5} & 0 \\ 1/\sqrt{30} & 2/\sqrt{30} & -5/\sqrt{30} \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{12}/\sqrt{2} & \sqrt{10}/\sqrt{2} & 0 \\ \sqrt{12}/\sqrt{2} & -\sqrt{10}/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \\ 2/\sqrt{5} & -1/\sqrt{5} & 0 \\ 1/\sqrt{30} & 2/\sqrt{30} & -5/\sqrt{30} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

## 138.4 Singular Value Decomposition with Python

Create a file called `vectors_decomposition.py` and enter this code:

```
# Singular-value decomposition
import numpy as np
from numpy import array
from scipy.linalg import svd
from numpy import diag
from numpy import dot
from numpy import zeros

# Define a matrix
A = array([[1, 2], [3, 4], [5, 6]])

print("Matrix (3x2) to be decomposed: ")
print(A)

# CalculateSVD
U, S, VT = svd(A)
print("Matrix (3x3) that represents the left singular values of A:")
print(U)
print("Singular values:")
print(S)
print("Matrix (2x2) that represents the right singular values of A:")
print(VT)

# Check if the decomposition by rebuilding the original matrix
# The singular values must be in an m x n matrix
# Create a zero matrix with the same dimension as A
Sigma = zeros((A.shape[0], A.shape[1]))
# Populate Sigma with n x n diagonal matrix
Sigma[:A.shape[1], :A.shape[1]] = diag(S)
# Reconstruct the original matrix
A_Rebuilt = U.dot(Sigma.dot(VT))
print("Original matrix:")
print(A_Rebuilt)
```

## 138.5 Sign Ambiguity

You might notice that at times, the absolute values in the $U$ and $V^T$ matrices are correct, but that the signs vary from what you see as the answer. For example, when you compare a manually calculated SVD with one done in Python the signs might not agree. Both decompositions of $A$ are valid. Both decompositions will satisfy:

$$A = U\Sigma V^T$$

Note that the $S$ diagonal values will always be positive.

The sign ambiguity has implications. For example, when using SVD to compress data, if some of the signs are flipped, the data can have artifacts. At this point in your education, you don't need to concern yourself with it, except when you are comparing SVD results for the same matrix.

### *Exercise 384* Single Value Decomposition

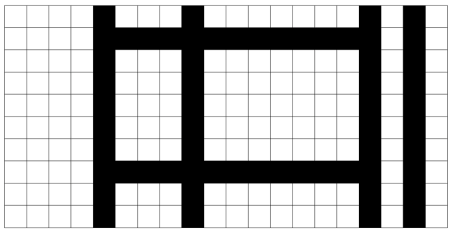
Modify your Python code to calculate SVD for the matrix in the worked out example. Did you arrive at the same answer? Keep in mind that Python will compute square roots and present fractions as decimal. Take a look at the signs for the values in the $U$ and $V^T$ matrices. Are they the same, or is this an example of sign ambiguity?

*Working Space*

*Answer on Page 2039*

## 138.6 SVD Applied to Image Compression

This image consists of a grid of 20 by 10 pixels, each of which is either black or white.



It is a simple image that has only two types of columns–ideal for data compression. A row is either the first pattern or the second.

We can represent the data as a 20 by 10 matrix whose 200 entries are either 0 for black or 1 for white.

$$\begin{bmatrix}
00001000100000001010\\
00001111111111111010\\
00001000100000001010\\
00001000100000001010\\
00001000100000001010\\
00001000100000001010\\
00001000100000001010\\
00001111111111111010\\
00001000100000001010\\
00001000100000001010
\end{bmatrix}$$

When you perform an SVD on this matrix, there are only two non-zero singular values, 6.79 and 3.72. (You are welcome perform the calculation in Python.) Thus, you can represent the matrix as:

$$A = U_1 S_1 V_1 + U_2 S_2 V_2$$

This means there are two $u$ vectors, each with 20 entries, two $v$ vectors each with 10 entries, and two singular values. Add those up: 2*20 + 2*10 + 2 = 62. This implies that the image can be represented by 62 values instead of 200. If you look back at the image, you can see that there are many dependent columns and very few independent ones.

This is a simple image and a small pixel matrix, but it should give you a sense of how SVD can decompose an image in a way that identifies how much of the image is redundant, and therefore can be compressed.

## 138.7 Where to Learn More

*We Recommend a Singular Value Decomposition*. This American Mathematical Society publication focuses the geometry of SVD. What we like about the article is that it shows both

graphically and numerically how SVD can be used for data compression on images and for noise reduction. The data compression example in your workbook is based on this article. https://www.ams.org/publicoutreach/feature-column/fcarc-svd

*Sign Ambiguity in Singular Value Decomposition* (*SVD*). This is a good article for those who want a deeper understanding of sign ambiguity. https://www.educative.io/blog/sign-ambiguity-in-singular-value-decomposition

*Singular Value Decomposition Tutorial*. This PDF starts by defining points, space, and vectors and works through all the concepts you need to tackle SVD. It is one of the few resources that has a completely worked out example of manually calculating SVD. The example in this chapter is from that tutorial. If you read the entire paper, you will find that it is a good review of the concepts you have studied in previous chapters. https://rb.gy/j6s0w

Chapter 139

# Tackling Difficult Problems: Positive Semidefinite Matrices

With all the computing power available today, you would think no problem would be too difficult to tackle. However, this is not the case. There is a category of problems called NP (nondeterministic polynomial time) whose solution is easy to verify, but whose computation is difficult to perform. This is because there is no straightforward algorithm and any brute-force method would take too much time. For these problems, all we can do is to develop an efficient algorithm that can find a solution in a reasonable amount of time. While the solution might not be the most optimal, the goal is to find the best solution possible in a short time.

As you learn more about optimization techniques, you will come across many efficiency algorithms that have been used throughout the years. In the 1990s, the field of optimization changed with the discovery that algorithms based on semidefinite positive matrices can achieve a higher efficiency than seen in the past. Today, there is an entire field of programming called Semidefinite Programming that is based on the use of semidefinite positive matrices.

First we will take a look at what some of the NP problems are. Next, we will describe the intuition behind semidefinite positive matrices. We will take a look at one NP problem, then introduce the Python module to use for solving these problems.

## 139.1 NP Problems

In the world of mathematics, easy problems are referred to as P, or polynomial time, problems. In simple terms, this means the problem can be solved quickly, and it is easy to verify that the solution is correct. Addition, subtraction, division, multiplication, square roots, and matrix-vector multiplication are just a few examples. NP problems, as stated in the introduction, can't be computed in a reasonable amount of time, but solutions are usually easy to verify. The game of Sudoku is one such example. It is easy to verify a correct solution, but writing a generalized algorithm to solve any Sudoku game is an NP problem.

NP problems show up in many other situations, such as:

- Designing robust communication networks

- Scheduling tasks without conflicts
- Managing supply chains
- Detecting patterns in biological networks
- Figuring out subgroups in social networks
- Predicting the structure of proteins

For each of these , think of large scale problems for which there are many variables. A cloud computing company that provides AI services to thousands of clients must be able to schedule tasks efficiently and in a timely manner, as well as manage the power needed for the computers and cooling the data center. Supply chain management is crucial to figuring out how to pick up, transport, and distribute goods to help provide disaster relief. Understanding protein structure is important for designing drugs that can tackle specific conditions. All we can do for each of these situations is to find an optimal solution, but we cannot find the definitive solution.

## 139.2 A Past Approach: Minimizing Errors in Neural Networks

When neural networks were first being developed to recognize things (faces, letters, music, and so on), the goal was to calculate weights between network nodes that would minimize the recognition error. The error space could be visualized as a surface of valleys and hills. The lowest point would have the least error. The idea behind the iterative weight calculations for training the network was to descend down the gradient until reaching the low point. Without getting into the mathematics, you can see by looking at this figure that there are two valleys. Some neural network training resulted in ending at a low point, but not the lowest point. Wouldn't it be great if you could formulate an optimization problem so that you would be guaranteed to land at the lowest point? That is where semidefinite programming can help.

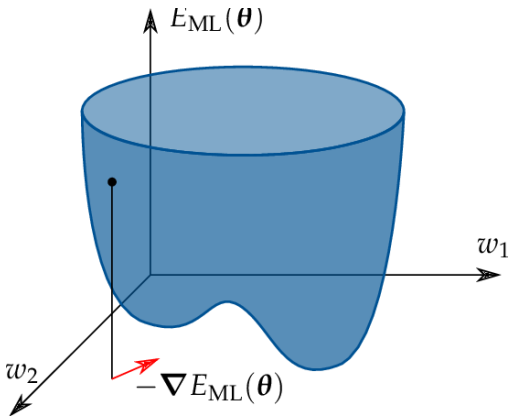


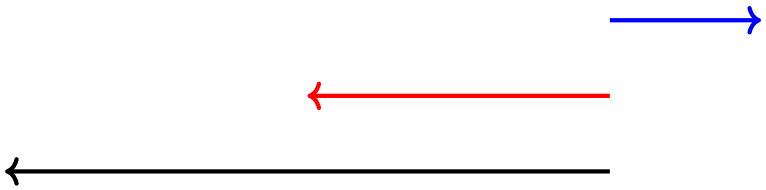

Figure 139.1: A neural network error surface with two valleys. Gradient descent can converge to a local minimum rather than the global minimum.

## 139.3 Positive Definite and Semidefinite Matrices

Unlike matrices that are defined by their content (such as identity matrix, zero matrix, and diagonal matrix), positive definite and semidefinite matrices are characterized by the result they produce. They have an analog in the scalar world, so let's first look at that. Let's take the scalars $a$ and $b$ and treat them as vectors. $ab$ is then the dot product. If $a > 0$, then $ab$ will take on the sign of $b$. If $a < 0$, then $ab$ will have the opposite sign of $b$. If you look at this in a graph, you can see that in the first case, $ab$ stays on the same side of the origin, but in the second case, $ab$ flips

For $a = [2], b = [4]$

For $a = [3], b = [-4]$. the result of multiplication flips a to the other side of the graph.



The notion of "staying on the same side" is positive definite. The notion of flipping is negative definite. Positive definite means that $x > 0$, so the result is a positive number. Positive semidefinite means that $x >= 0$, so the result is a non-negative number.

If you can formulate a problem as a positive semidefinite matrix, then you automatically constrain the result to the "same side." This constraint results in higher algorithmic efficiency.

Let's look at the formal definition:

A matrix is positive definite if, and only if:

$$x^T Ax > 0, x, x \neq 0$$

A matrix is positive semidefinite, if, and only if:

$$x^T Ax \geq 0$$

Further, a positive semidefinite matrix had an important property. Its eigenvalues are $\geq 0$. The eigenvalues of a positive definite matrix are $> 0$.

Take the triplet $(a, b, c)$ and the symmetric matrix:

$$\begin{bmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{bmatrix} \geq 0$$

For what values of $a, b, c$ is this matrix positive semidefinite? If you iterate through all possible combinations of a,b,c, then compute the eigenvalues for each matrix, you will find that some (like $0, 0, 0$) result in a positive semidefinite matrix and some (like $2, 2, 2$) are not positive semidefinite. If you plot the set of triplets that result in a semidefinite matrix, you will see an elliptope. This shape guarantees an optimal solution.

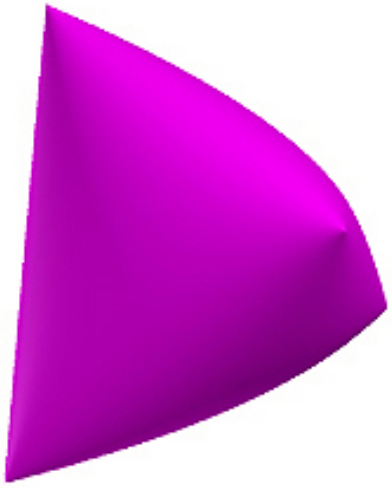


Figure 139.2: Elliptope visualized.

## 139.4 Identifying and Constructing a Positive Semidefinite Matrix

In the last section, you saw that being symmetric does not guarantee a positive semidefinite matrix. You also saw that a matrix of positive values does not guarantee a positive semidefinite matrix. The only way to check for positive semidefinite is to calculate the eigenvalues, which you learned in a previous workbook.

A surefire way to construct a positive semidefinite matrix is:

$$AA^T$$

### *Exercise 385* Figuring out if a matrix is positive semidefinite

Is this matrix positive definite? Show your work.

$$\begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix}$$

*Working Space*

*Answer on Page 2039*

### *Exercise 386* Creating a positive semidefinite matrix

Using any 3 by 3 matrix, create a positive semidefinite matrix.Then show it is positive semidefinite by calculating its eigenvalues. You can either compute this by hand or using Python. In either case, show your work.

*Working Space*

*Answer on Page 2039*

## 139.5 The Max Cut Problem

A famous NP problem is Max Cut. Given a graph of interconnected nodes, cut the graph to create two sets of nodes, such that the cut goes through as many edges as possible. (You can't cut an edge more than once.) Max Cut is important for binary classification in machine learning, circuit design, statistical physics, and more. There is no algorithm that will provide an exact solution. (If you could find one, you would be eligible to win a huge prize from the Clay Mathematics Institute!) Instead, you will see how to approximate a solution to this problem using a positive semidefinite matrix and a technique developed by mathematicians Michel Goemans and David Williamson.

You won't see all the complete details here, as this section is meant to be a quick introduction to how you can apply positive semidefinite matrices.

Take this simple graph of five nodes and six edges. Each node in the graph will take on

one of two values (1 or -1) to show which set they fall into after a cut is made. For any two connected nodes, $x_i \cdot x_j = 1$ if $x_i = x_j$ and $-1$ otherwise.

If you randomly assign 1 and $-1$ to the nodes, the chance of making the max cut is 0.5. By using semidefinite programming, you can achieve an algorithmic efficiency of 0.87.

The Goemans-Williamson technique can be used for any optimization problem where the variables take on the values of 1 and $-1$.

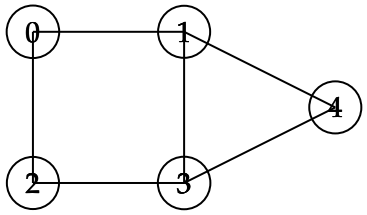



Figure 139.3: MaxCut algorithm visualized.

As a list of edges the graph is:

$$edges = [(0,1),(0,2),(1,3),(1,4),(2,3),(3,4)]$$

The optimization problem can be formulated as:

$$Max \sum_{edges(i,j)} \frac{1 - x_i x_j}{2}$$

for

$$x_i \in \{-1, 1\}$$

However, instead of allowing $x_i$ to be scalar, Goemans-Williamson defines $x_i$ as unit vectors.

$$x_i \in R^n, x_i = 1$$

and that makes the optimization equation:

$$Max \sum_{edges(i,j)} \frac{1 - x_i^T x_j}{2}$$

It is this "relaxation" that gets us to a semidefinite matrix, because we can now rewrite

the problem as a positive semidefinite matrix:

$$X = \left[x_i^T x_j\right]_{i,j}$$

Python has a module for solving optimization problems. Using this, you will get an optimum matrix, but to get the unit vectors, you will need to take the square root of the matrix.

$$X = [x_1 ... x_{1n}]\,[x_1 ... x_{1n}]^T$$

Next we need to go from unit vectors to scalars, using a process called rounding.

$$x^i \in R^n \rightarrow x^i \in \{-1, 1\}$$

Goemans-Williamson leveraged the fact that the end point of a unit vector is on a sphere. They generated a random plane to bisect the sphere. A vector on one side of the plane is assigned the value of 1 and a vector on the other side a value of $-1$.

You can then assign the scalar values to the nodes and make the cut accordingly. This particular cut will be 5, as shown.

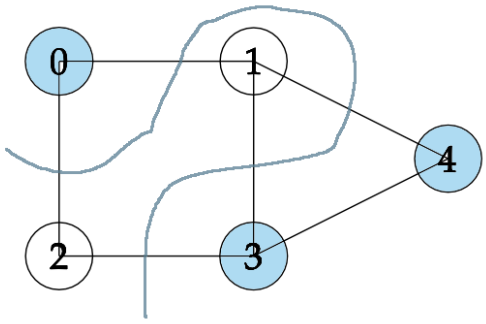



Figure 139.4: The result of the MaxCut.

## 139.6 The Max Cut Problem Solved in Python

```
# MaxCut Problem
import numpy as np
import scipy
from scipy.linalg import sqrtm
# cvxpy is a python module for solving optimization problems
import cvxpy as cp

# define the edges of the graph
edges = [(0,1),
         (0,2),
         (1,3),
         (1,4),
```

```
        (2,3),
        (3,4)]

#  Declare the matrix X to be positive semidefinite
X = cp.Variable((5,5),symmetric=True)
constraints = [X >> 0]

# Set diagonals to 1 to get unit vectors
constraints += [
    X[i,i] == 1 for i in range(5)
]

# Set the objective function
objective = sum(05.*(1 - X[i,j]) for (i,j)in edges)

# Set up the problem to maximize using the objective function and
# keeping within the set constraints
prob = cp.Problem(cp.Maximize(objective), constraints)

# Returns a positive semidefinite matrix
print(prob.solve())

# To get the vectors, take square root of the matrix
x = sqrtm(X.value)

# Generate a random hyperplane
u = np.random.randn(5) # normal to random hyperplane

# Pick values according to which side of the hyperplane the vectors are on
x = np.sign(x @ u)

```

Chapter 140

# Introduction to C

You already know how to write programs in Python. In the next few chapters, you are going to learn C. Your Python knowledge will make this easier, because C has the same fundamental ideas: variables, functions, loops, and conditionals. The syntax looks different, and there are a few genuinely new concepts, but you will recognize the shape of everything you write.

Why learn C? Two reasons. First, C is used wherever performance matters: operating systems, game engines, embedded systems, and scientific computing. Second, and more important for this course, C forces you to think about memory explicitly. Python quietly manages memory on your behalf; C makes you do it yourself. Once you understand how C uses memory, you will also understand what Python is doing underneath, and you will be ready to implement data structures from scratch.

## 140.1 Compiled vs. Interpreted

When you run a Python program, the Python interpreter reads your `.py` file and executes it line by line. There is no separate step between writing and running.

C works differently. Before a C program can run, a *compiler* must translate it into machine code, the raw instructions that your CPU executes directly. The compiler reads your `.c` file, checks it for errors, and produces an *executable* file. You then run that executable.

`hello.c` $\xrightarrow{\text{compiler}}$ `hello` (executable) $\xrightarrow{\text{run}}$ output

Figure 140.1: The compile-then-run cycle

Because the compiler sees your entire program before anything runs, it can catch many mistakes, like using a variable before declaring it or passing the wrong type to a function, that Python would only discover at runtime.

To compile a C file called `hello.c`, you run:

```
gcc hello.c -o hello
```

This `gcc` is the compiler that turns your C code into an executable. The `hello.c` is the input file, and `-o hello` tells the compiler to name the output executable `hello`. On

macOS and Linux systems, this may come pre installed. On Windows, however, this likely is not installed. You can find the installation instructions here.

The `-o hello` flag names the output executable `hello`. Then you run it:

```
./hello
```

## 140.2 Your First C Program

Create a file called `hello.c` and type the following:

```c
#include <stdio.h>

int main()
{
    printf("Hello, world!\n");
    return 0;
}
```

Compile and run it. You should see:

```
Hello, world!
```

Let's look at each piece.

`#include <stdio.h>` tells the compiler to include the standard input/output library, which provides `printf` for printing. Think of it like Python's `import` statement.

`int main()` is the function where your program starts. Every C program must have exactly one `main` function. The `int` means it returns an integer; `return 0` signals to the operating system that the program finished successfully.

`printf("Hello, world!\n");` prints a line of text. The `\n` is the newline character.

Notice the semicolons at the end of statements, and the curly braces `{ }` around the body of `main`. In Python, indentation defines blocks; in C, curly braces do.

## 140.3 Types and Variables

In Python, you write:

```
x = 5
y = 3.14
```

Python figures out the type of each variable automatically.

In C, you must declare the type of every variable when you create it:

```
int x = 5;
double y = 3.14;
```

This is called *static typing*. Once you declare a variable as `int`, it can only hold integers. The compiler enforces this and will refuse to compile code that mixes types incorrectly.

The common C types are:

**`int`** A whole number: `0`, `42`, `-7`. Typically 4 bytes.

**`double`** A floating-point number: `3.14`, `-0.001`. Typically 8 bytes. (There is also `float`, which uses 4 bytes and has less precision.)

**`char`** A single character: `'A'`, `'z'`, `'7'`. Use single quotes.

**`int`** Booleans in C are just integers: `0` means false, and any non-zero value means true. You can include `<stdbool.h>` to get the names `true` and `false`.

You can also declare a variable without immediately giving it a value:

```
int count;      /* declared but not initialized */
count = 10;     /* now it has a value */
```

Warning: an uninitialized variable in C contains whatever random bytes happen to be in that memory location. Unlike Python, C will not raise an error if you read an uninitialized variable; it will just give you garbage. Always initialize your variables.

Note that C uses `/* ... */` for comments, not `#`. You can also use `//` for single-line comments in modern C.

## 140.4 Printing with `printf`

`printf` uses format specifiers to print different types:

```c
int age = 17;
double height = 1.75;
printf("Age: %d, Height: %f\n", age, height);
```

Output:

```
Age: 17, Height: 1.750000
```

Common format specifiers:

| Specifier | Type |
|---|---|
| %d | int |
| %f | double or float |
| %c | char |
| %s | string (char pointer) |

## 140.5 Conditionals and Loops

The logic is the same as Python; only the syntax changes.

### 140.5.1 if / else

```c
int score = 85;

if (score >= 90) {
    printf("A\n");
} else if (score >= 80) {
    printf("B\n");
} else {
    printf("C or below\n");
}
```

The condition must be in parentheses `( )`, and each branch is surrounded by curly braces `{ }`.

### 140.5.2 for and while loops

```c
for (int i = 0; i < 5; i++) {
    printf("%d\n", i);
}
```

A C `for` loop has three parts separated by semicolons: the initializer (`int i = 0`), the condition (`i < 5`), and the update (`i++`). The `++` increments by 1, the same as `i = i + 1`.

```c
int i = 0;
while (i < 5) {
    printf("%d\n", i);
    i++;
}
```

## 140.6 Functions

In C, you must declare the return type and the type of each parameter:

```c
double square(double x)
{
    return x * x;
}

int main()
{
    printf("%f\n", square(4.0));   /* prints 16.000000 */
    printf("%f\n", square(2.5));   /* prints 6.250000  */
    return 0;
}
```

If a function returns nothing, its return type is `void`:

```c
void printGreeting(char *name)
{
    printf("Hello, %s!\n", name);
}
```

Functions must be defined *before* they are called, or you must provide a *forward declaration* (just the signature, ending in a semicolon) above `main`:

```c
double square(double x);   /* forward declaration */

int main()
{
    printf("%f\n", square(3.0));
    return 0;
}

double square(double x)    /* full definition can come later */
{
    return x * x;
}
```

## 140.7 The Stack: Blackboards for Functions

Every function, while it is running, needs somewhere to store its local variables. C gives each function a *frame*: a reserved chunk of memory that holds all of that function's local variables and parameters.

Think of a frame as a **blackboard**. When a function starts running, it gets a fresh blackboard. It can write anything it wants on that blackboard, create variables, change their values. When the function finishes, the blackboard is *erased and discarded*. Local variables do not survive after their function returns.

Where do these frames live? In a region of memory called the *stack*. The stack works exactly like a physical stack of objects: the most recently added item is always on top. When `main` calls a function, that function's frame is pushed onto the top of the stack. When the function returns, its frame is popped off, and `main`'s frame is on top again.

Here is a concrete example:

```c
#include <stdio.h>

int cookingMinutes(int weightPounds)
{
    int minutes = 15 + 15 * weightPounds;
    return minutes;
}

int main()
{
    int totalWeight = 10;
    int gibletsWeight = 1;
    int turkeyWeight = totalWeight - gibletsWeight;
    int result = cookingMinutes(turkeyWeight);
    printf("Cook for %d minutes.\n", result);
```

```
    return 0;
}
```

When `main` calls `cookingMinutes(9)`, the stack looks like this:

| **cookingMinutes**() | `weightPounds = 9`<br>`minutes = 150` |
|---|---|
| **main**() | `totalWeight = 10`<br>`gibletsWeight = 1`<br>`turkeyWeight = 9`<br>`result = ?` |

`main`'s blackboard sits below; `cookingMinutes`'s blackboard sits on top of it. The variable `minutes` inside `cookingMinutes` is completely separate from anything in `main`. Each function only sees its own blackboard.

When `cookingMinutes` returns 150, its frame is discarded. `main`'s frame resumes, and `result` is filled in with 150.

### 140.7.1 Parameters are copies

When you call `cookingMinutes(turkeyWeight)`, C copies the value of `turkeyWeight` into the new parameter `weightPounds` on `cookingMinutes`'s blackboard. Changing `weightPounds` inside the function would not change `turkeyWeight` in `main`. They are separate variables. This is called *pass-by-value*.

## 140.8 Addresses and Pointers

Every variable in your program lives at some address in memory. You can find out the address of a variable using the `&` operator, called the *address-of* operator:

```c
#include <stdio.h>

int main()
{
    int i = 17;
    printf("i stores its value at %p\n", (void *)&i);
    return 0;
}
```

Run this and you will see something like:

```
i stores its value at 0x7ffeea3c4738
```

The address is printed in hexadecimal. Your address will differ, because the operating system decides where in memory your variables live.

A *pointer* is a variable that holds an address. If you want to store the address of an `int`, you declare a variable of type `int *` ("pointer to int"):

```
int i = 17;
int *addressOfI = &i;
printf("i is at %p\n", (void *)addressOfI);
```

The `*` in the declaration is part of the type; it means "this variable holds an address, and the thing at that address is an `int`."

### 140.8.1 Reading and writing through a pointer

Once you have a pointer, you can use the `*` operator to read or write the value *at* the address the pointer holds. This is called *dereferencing* the pointer:

```
int i = 17;
int *p = &i;

printf("%d\n", *p);   /* prints 17 -- reads through p  */

*p = 89;              /* writes 89 to the address p holds */
printf("%d\n", i);    /* prints 89 -- i was changed!   */
```

To summarize:

- `&i` gives you the address of `i`.
- `int *p` declares a variable that can hold such an address.
- `*p` gives you the value stored at the address `p` holds.

### 140.8.2 NULL pointers

Sometimes you want a pointer that does not yet point to anything. Use `NULL`:

```
int *p = NULL;   /* p points to nothing */

if (p) {
    printf("%d\n", *p);     /* only runs if p is non-null */
} else {
    printf("p is null\n");
}
```

A null pointer is simply the address zero. **Never dereference a null pointer**, your program will crash. Always check before dereferencing a pointer that might be null.

## 140.9 Pass-by-Reference

Recall that when you call a function in C, arguments are *copied* into the function's frame. But what if you want a function to modify a variable in the calling function? You pass the *address* of that variable instead of its value. The function can then write to that address.

There is a standard C function called `modf()` that splits a floating-point number into its integer part and its fractional part, and needs to return *both*. Since a function can only return one value, it returns the fractional part and writes the integer part to an address you supply:

```
#include <stdio.h>
#include <math.h>

int main()
{
    double pi = 3.14;
    double integerPart;
    double fractionPart;

    /* Pass the address of integerPart so modf can write there */
    fractionPart = modf(pi, &integerPart);

    printf("Integer part:  %f\n", integerPart);
    printf("Fraction part: %f\n", fractionPart);
    return 0;
}
```

Output:

```
Integer part:  3.000000
Fraction part: 0.140000
```

Here is another way to think about it. Imagine you need to give a spy an assignment. You say: "I've left a length of steel pipe at the foot of the angel statue in the park. When you have the information, roll it up and leave it in the pipe. I'll pick it up Tuesday." In the spy business, this is called a *dead drop*.

Pass-by-reference works the same way. You are not giving the function the data directly, you are giving it a *location* where it can leave the result.

Here is our own pass-by-reference function that converts meters to feet and inches:

```c
#include <stdio.h>
#include <math.h>

void metersToFeetAndInches(double meters,
                           int *ftPtr,
                           double *inPtr)
{
    double rawFeet = meters * 3.281;
    int feet = (int)floor(rawFeet);
    double inches = (rawFeet - feet) * 12.0;

    if (ftPtr) *ftPtr = feet;
    if (inPtr) *inPtr = inches;
}

int main()
{
    int feet;
    double inches;
    metersToFeetAndInches(1.8, &feet, &inches);
    printf("%d ft %f in\n", feet, inches);
    return 0;
}
```

The caller declares the variables it wants filled in (`feet`, `inches`), passes their addresses to the function, and the function writes the results there.

## 140.10 Structs: Grouping Related Data

A *struct* lets you bundle several related values into a single named type. In Python you might use a dictionary for this; in C, a struct is the fundamental grouping mechanism.

```
struct Person {
    double heightInMeters;
    int weightInKilos;
};
```

This defines a new type called `struct Person`. You can then create variables of that type:

```
#include <stdio.h>

struct Person {
    double heightInMeters;
    int weightInKilos;
};

double bodyMassIndex(struct Person p)
{
    return p.weightInKilos / (p.heightInMeters * p.heightInMeters);
}

int main()
{
    struct Person mikey;
    mikey.heightInMeters = 1.70;
    mikey.weightInKilos  = 96;

    struct Person aaron;
    aaron.heightInMeters = 1.97;
    aaron.weightInKilos  = 84;

    printf("mikey BMI: %f\n", bodyMassIndex(mikey));
    printf("aaron BMI: %f\n", bodyMassIndex(aaron));
    return 0;
}
```

You access the members of a struct using a dot: `mikey.heightInMeters`.

When `main` is running, the stack looks like this:

| **main()** | `mikey.heightInMeters = 1.70`<br>`mikey.weightInKilos = 96`<br>`aaron.heightInMeters = 1.97`<br>`aaron.weightInKilos = 84` |
|---|---|

The two `Person` structs live right inside `main`'s frame on the stack.

## 140.11 The Heap

The stack is fast and automatic, but it has two constraints: stack frames are destroyed when functions return, and the sizes of all stack-allocated variables must be known at compile time.

The *heap* is a separate region of memory that has neither constraint. Memory on the heap persists until you explicitly release it, and can be any size determined at runtime.

In C, you claim heap memory with `malloc` and release it with `free`.

```c
#include <stdlib.h>

struct Person *mikey = malloc(sizeof(struct Person));
```

`malloc` asks the system for enough heap memory to hold a `Person` struct and returns the address of that memory. The variable `mikey` is a *pointer* to a `Person`, it lives on the stack, but the struct it points to lives on the heap.

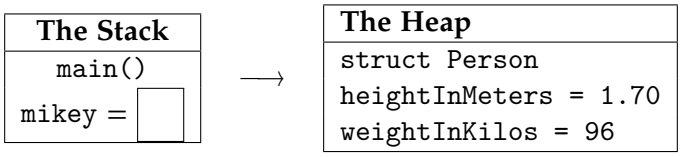



Figure 140.2: A pointer on the stack pointing to a struct on the heap

### 140.11.1 The arrow operator

When you have a pointer to a struct, you access its members using `->` instead of a dot:

```c
mikey->heightInMeters = 1.70;
mikey->weightInKilos  = 96;
```

The code `mikey->weightInKilos` means: "dereference the pointer `mikey` to get the struct, then access its `weightInKilos` member." It is shorthand for `(*mikey).weightInKilos`.

### 140.11.2 Releasing heap memory

When you are done with heap-allocated memory, you must release it with `free`:

```
free(mikey);
mikey = NULL;   /* good habit: prevents accidental reuse */
```

After `free`, the memory is returned to the heap for reuse. Setting the pointer to `NULL` prevents you from accidentally using it again.

If you forget to call `free`, the memory is never returned. This is called a *memory leak*. In a short-lived program it may not matter, but in a long-running program it can consume all available memory.

### 140.11.3 A complete heap example

```
#include <stdio.h>
#include <stdlib.h>

struct Person {
    double heightInMeters;
    int weightInKilos;
};

double bodyMassIndex(struct Person *p)
{
    return p->weightInKilos / (p->heightInMeters * p->heightInMeters);
}

int main()
{
    struct Person *mikey = malloc(sizeof(struct Person));
    mikey->heightInMeters = 1.70;
    mikey->weightInKilos  = 96;

    printf("mikey BMI: %f\n", bodyMassIndex(mikey));

    free(mikey);
    mikey = NULL;

```

```
    return 0;
}
```

`bodyMassIndex` now takes a `struct Person *` instead of a `struct Person` by value. This avoids copying the struct. Functions that work on heap-allocated data almost always take pointers.

### 140.11.4 Heap arrays

You can also allocate a contiguous block of memory for multiple values. Use `malloc` with a count:

```
/* Allocate space for 1000 doubles on the heap */
double *buffer = malloc(1000 * sizeof(double));

buffer[0] = 3.14;
buffer[1] = 2.72;

free(buffer);
buffer = NULL;
```

A pointer to the first element is all you need to work with the whole array. The heap memory is contiguous, so `buffer[0]`, `buffer[1]`, and so on are laid out one after another.

## 140.12 What Python Objects Actually Are

You now know everything you need to understand what a Python object is, under the hood.

When Python creates an object, it:

1. Allocates a struct on the heap large enough to hold the object's data (its instance variables).
2. Stores in that struct a pointer to the class's method table, so that method calls can find the right code.
3. Returns a pointer to that struct.

Your Python variable `obj` is, at the machine level, a pointer to a heap-allocated struct. When Python's garbage collector determines that nothing points to an object anymore, it calls the equivalent of `free` to release that memory.

The next chapter introduces C++ classes, and with this memory model in hand, you will implement linked lists, trees, and hash tables from scratch.

## 140.13 Exercises

1. Write a C function `celsiusToFahrenheit` that takes a `double` temperature in Celsius and returns the equivalent in Fahrenheit. Call it from `main` for the values 0, 20, 37, and 100, and print each result.

2. Write a function `swap` that takes two `int` pointers and swaps the values at the two addresses. Test it so that after calling `swap(&a, &b)`, the values of `a` and `b` have been exchanged.

3. Write a struct `Point` with `double` members `x` and `y`. Write a function `distance` that takes two `struct Point *` pointers and returns the Euclidean distance between them. Allocate two points on the heap, fill in their coordinates, print the distance, and then free the memory.

4. **Memory leak hunt.** The following program has a memory leak. Identify and fix it.

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int value;
};

void printValue(int x)
{
    struct Node *n = malloc(sizeof(struct Node));
    n->value = x;
    printf("%d\n", n->value);
}

int main()
{
    for (int i = 0; i < 5; i++) {
        printValue(i);
    }
    return 0;
}
```

5. A `struct tm` is defined in `<time.h>` and holds a broken-down calendar time. The function `time(NULL)` returns the number of seconds since midnight, January 1, 1970. The function `localtime_r(&seconds, &now)` fills a `struct tm` with the corresponding date and time. Here is a snippet to get you started:

```c
#include <stdio.h>
#include <time.h>
```

```c

int main()
{
    long secondsSince1970 = time(NULL);
    struct tm now;
    localtime_r(&secondsSince1970, &now);
    printf("The time is %d:%d:%d\n",
           now.tm_hour, now.tm_min, now.tm_sec);
    return 0;
}
```

Look up the other fields of `struct tm` and modify the program to print today's full date. Then modify it to print the date that falls exactly 4,000,000 seconds from now. (Hint: `tm_mon` is 0-indexed, so January is 0.)

Chapter 141

# Linked Lists and C++ transition

## 141.1 From C to C++

In this chapter we are going to transition from stand C programming to C++. Why C++? Because it is a more modern version of C++ that allows for a deeper insight into the structures that we put into the memory of the computer. C++ differs in that it allows for object-oriented programming, which is a programming paradigm that uses "objects" to represent data and methods. This allows for more complex data structures and algorithms to be implemented in a more organized and efficient manner. To do this, we will be utilizing *classes*, which we briefly covered in our Advanced Python chapter. In C++, classes are defined using the `struct` keyword.

There are two additional keyword differences you should be aware of regarding C vs C++. The primary one is the usage of the keyword *new*, which does not exist in C. In C, we have `malloc()`, which frees up a given amount of memory in bytes, and `sizeof`, which returns the size in bytes of a data structure.

```
1    // C : what you already know
2    struct Car *n = malloc(sizeof(struct Car));
```

C++ combines these two commands into one, called `new`.

```
1    // C++ : same thing, no sizeof, no cast
2    Car* n = new Car();
```

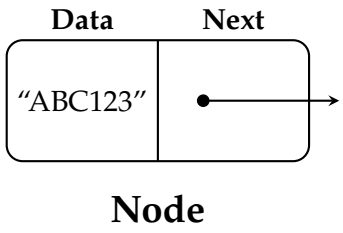
Here is a helpful video on some of the deeper differences that we will not be covering: https://www.youtube.com/watch?v=B_4x7WlQr7M

## 141.2 Linked Lists

A *linked list* is a linear data structure where each element is a separate object, called a *node*. Each node holds its own data and the address of the next node, called a *successor*, thus forming a chain-like structure. The first node in the list is called the *head*. The head is the only node you need to keep a direct reference to, since every other node can be reached by following successors from it. The last node, the one whose successor pointer is empty, is called the *tail*.

A simple node in a linked list can be represented in C++ as follows:

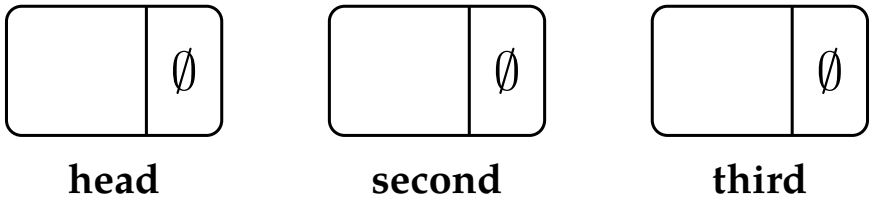
```
struct Node {
    int data;
    Node* next;
};
```



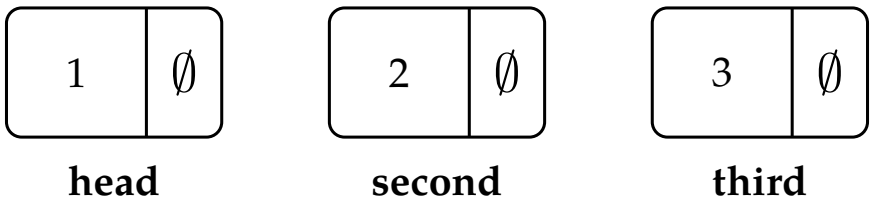

In this structure, 'data' is used to store the data and 'next' is a pointer that holds the address of the next Node in the list. When a pointer has nothing to point to, we mark its field with the empty-set symbol $\emptyset$ — the mathematical symbol for “nothing here.”

Here is a simple example of creating and linking nodes in a linked list:

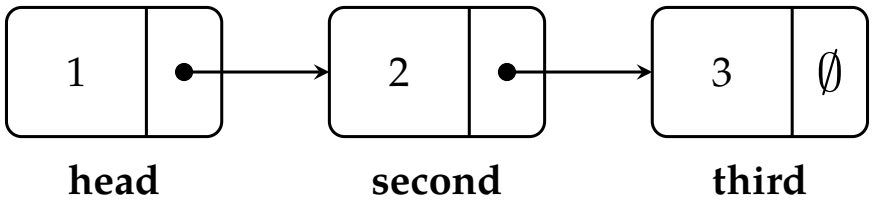
```
// Create nodes
Node* head = new Node();
Node* second = new Node();
Node* third = new Node();
```




```

// Assign data
head->data = 1;
second->data = 2;
third->data = 3;
```




```
// Link nodes
head->next = second;
second->next = third;
third->next = nullptr;  // The last node points to null, this is the TAIL
```



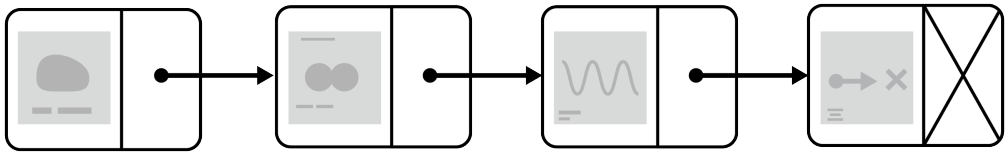

In this example, we first create three nodes using the ‘new‘ keyword, which dynamically allocates memory. We then assign data to the nodes and link them using the ’next’ pointer.

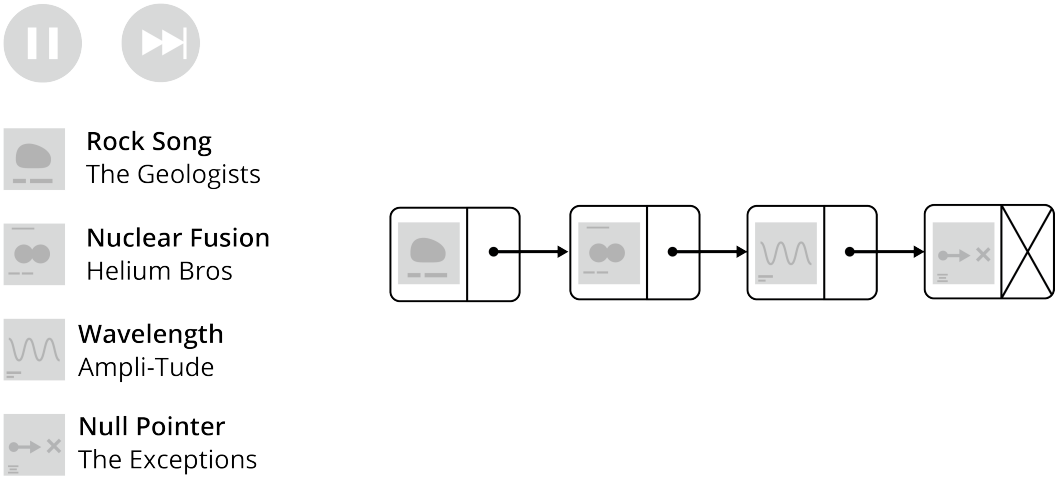
**Rock Song**
The Geologists

**Nuclear Fusion**
Helium Bros

**Wavelength**
Ampli-Tude

**Null Pointer**
The Exceptions

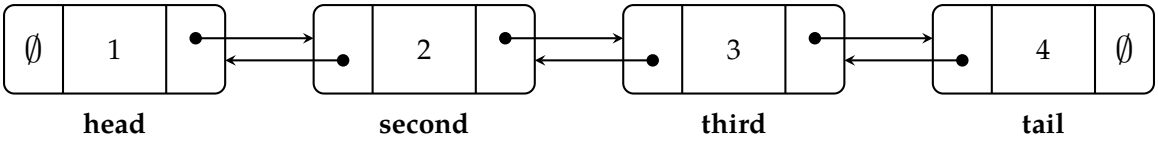



Rock Song
The Geologists
Nuclear Fusion
Helium Bros
Wavelength
Ampli-Tude
Null Pointer
The Exceptions

## 141.3 Doubly Linked Lists

Now that we have established the basics of linked lists, we can move on to a more complex structure: the *doubly linked list*. Doubly linked lists are strong because each node now has two pointers: the same next pointer, and now, the *predecessor*, which we will call `prev`.

```
struct DoublyNode {
    int data;
    DoublyNode* next;
    DoublyNode* prev;
};
```




Given this structure, we can traverse the entire list starting at the tail to the head or from the head to the tail.

Let's construct some code to analyze where the nodes would be in memory. We will create a doubly linked list with four nodes, and then we will print out the addresses of each node and their respective next and previous pointers.

Write the following code for the doubly linked list. There is a full file in your digital resources, here is a snippet of the code to get you started.

```

int main() {
    // Create nodes
    DoublyNode* head = new DoublyNode();
    DoublyNode* second = new DoublyNode();
    DoublyNode* third = new DoublyNode();
    DoublyNode* tail = new DoublyNode();

    // Assign data
    head->data = 1;
    second->data = 2;
    third->data = 3;
    tail->data = 4;

    // Link nodes forward
    head->next = second;
    second->next = third;
    third->next = tail;
```

```
    tail->next = nullptr;   // tail has no successor

    // Link nodes backward
    head->prev = nullptr;   // head has no predecessor
    second->prev = head;
    third->prev = second;
    tail->prev = third;

    // Print the address of each node, and the addresses stored in its
    // prev/next fields, so you can see both directions of the chain.
    std::cout << "head   is at: " << head   << "   prev: " << head->prev   << "
    ↪  (nullptr)" << "   next: " << head->next   << std::endl;
    std::cout << "second is at: " << second << "   prev: " << second->prev << "
    ↪  next: " << second->next << std::endl;
    std::cout << "third  is at: " << third  << "   prev: " << third->prev  << "
    ↪  next: " << third->next  << std::endl;
    std::cout << "tail   is at: " << tail   << "   prev: " << tail->prev   << "
    ↪  next: " << tail->next   << " (nullptr)" << std::endl;

    // Each DoublyNode takes up a fixed number of bytes in memory.
    std::cout << "\nsizeof(DoublyNode) = " << sizeof(DoublyNode) << " bytes" <<
    ↪  std::endl;

    // Clean up the memory
    delete head;
    delete second;
    delete third;
    delete tail;

    return 0;
}
```

When you run this, you'll get real addresses like these (yours specific addresses will differ!):

```
head   is at: 0x100c9dca0   prev: 0 (nullptr)   next: 0x100c9dbc0
second is at: 0x100c9dbc0   prev: 0x100c9dca0   next: 0x100c9dbe0
third  is at: 0x100c9dbe0   prev: 0x100c9dbc0   next: 0x100c9dc00
tail   is at: 0x100c9dc00   prev: 0x100c9dbe0   next: 0 (nullptr)

sizeof(DoublyNode) = 24 bytes
```

Computer memory is comprised of a very long string of bytes, and each byte has its own address. This line is a large, contiguous stream of ones and zeros when you get to the core of it. Often, programmers separate it into chunks of certain sizes so that we can make them look similar to grids of memory, like matrices.

Look closely at `second` and `third`: `0x100c9dbe0 - 0x100c9dbc0 = 0x20`, which is **32** in

decimal, not 24 bytes like `sizeof(DoublyNode)` had provided. The same is true between `third` and `tail`. Why, then, is the a difference between the 32 stored bytes and 24 bytes we would expect our memory to take up?

`DoublyNode` holds an `int` (4 bytes) plus two pointers (8 bytes each on a 64-bit machine), for 24 bytes total. But heap allocators don't hand out memory in exactly the size you ask for, they *round each allocation up* to a boundary of 16 bytes. Since 24 isn't a multiple of 16, the allocator rounds it up to the next one: 32. Your struct only uses 24 of those bytes; the other 8 are allocator *padding* that `sizeof()` never shows you. This is why the command new combines `malloc` and `sizeof`, and then proceeds to calculate the next multiple of 16 bytes to allocate.

Compare this to the `Node` struct from the previous section, which only has one pointer and one integer. It is 12 bytes, but memory allocation rounds it up to 16 bytes. This is why the addresses of the nodes in a linked list are not always exactly the size of the struct apart from each other. The addresses, too, may not always be consecutive.

Alternatively, comparing this to data structures in Python like Lists, we can see that Python's lists are more like arrays than linked lists. Python's lists are implemented as dynamic arrays, which means that they are contiguous blocks of memory that can grow and shrink in size. This allows for fast access to elements by index, but can be less efficient for inserting or deleting elements in the middle of the list, as it may require shifting elements around in memory. In contrast, linked lists allow for efficient insertion and deletion of elements at any position in the list, but require more memory overhead (like temporary variables) due to the additional pointers needed to link the nodes together.

# Chapter 142

# Bitmaps

Let's talk about another image format: the Bitmap. The Bitmap, denoted with file type `.bmp`, is an early file format for storing images on computers. Rows of different colors are stored as a grid of pixels. Each pixel is represented by a specific number of bits, which determines the color depth of the image. For example, a 24-bit bitmap can represent over 16 million colors, while an 8-bit bitmap can only represent 256 colors.

Bitmaps are uncompressed, meaning they can take up a lot of storage space compared to other image formats like JPEG or PNG. However, they are simple to read and write in code, making them useful for certain applications where image quality is more important than file size.

To understand how bitmaps work in practice, it helps to think of an image as nothing more than numbers stored in memory. Each pixel in a bitmap corresponds to a color value, and these values are usually written in hexadecimal (base-16) form. If you need a review, look back at Chapter 87

In a common 24-bit bitmap, each pixel is made up of three color components:

- Red
- Green
- Blue

The intensity of each color component is represented by 8 bits. In decimal, this means each component can have a value from `0` to 255, for a total of 256 possible values. In hexadecimal, this range is represented from `00` to `FF`.

Each pixel's color is then represented by a 6-digit hexadecimal number, where the first two digits represent the red component, the next two represent green, and the last two represent blue. A total of $16,777,216$ different colors can be represented $(256 \times 256 \times 256)$.

Together, the intensity of each component determines the final color of the pixel. For example:

- `0xFF0000` represents pure red
- `0x00FF00` represents pure green
- `0x0000FF` represents pure blue

- `0xFFFFFF` represents white
- `0x000000` represents black

Recall that the prefix `0x` indicates that the number is in hexadecimal format. When creating or manipulating bitmap images in code, you can directly set the color values of individual pixels using their hexadecimal representations. This allows for precise control over the image's appearance.

## 142.1 Bitmap Structure

When traversing a bitmap, it's important to understand its structure. A bitmap file typically consists of a header followed by the pixel data. The header contains metadata about the image, such as its width, height, and color depth. The pixel data is stored in a grid format, with each pixel represented by its color value.

In a 24-bit bitmap, each pixel is represented by 3 bytes (one byte for each color component). The pixel data is usually stored in a bottom-up order, meaning the first row of pixel data corresponds to the bottom row of the image.

### 142.1.1 Bitmap Headers

The bitmap header is typically constructed with a file section and info section. Following those comes the pixel data, represented as bytes.

```
| File Header (?) | Info Header (usually 40 bytes) | Pixel Data (rowSize*height)
↪  |
```

```

struct BITMAPFILEHEADER
{
    WORD bfType; //specifies the file type
    DWORD bfSize; //specifies the size in bytes of the bitmap file
    WORD bfReserved1; //reserved; must be 0
    WORD bfReserved2; //reserved; must be 0
    DWORD bfOffBits; //species the offset in bytes from the bitmapfileheader to
    ↪  the bitmap bits
};
struct BITMAPINFOHEADER
{
    DWORD biSize; //specifies the number of bytes required by the struct
    LONG biWidth; //specifies width in pixels
    LONG biHeight; //species height in pixels
    WORD biPlanes; //specifies the number of color planes, must be 1
```

```
    WORD biBitCount; //specifies the number of bit per pixel
    DWORD biCompression;//specifies the type of compression
    DWORD biSizeImage; //size of image in bytes
    LONG biXPelsPerMeter; //number of pixels per meter in x axis
    LONG biYPelsPerMeter; //number of pixels per meter in y axis
    DWORD biClrUsed; //number of colors used by th ebitmap
    DWORD biClrImportant; //number of colors that are important
};
```

You can refer to the BMP file format specification and the bitmap article for more information. Let's create variables to hold important fields:

- `int w = bih.biWidth;` - width of the image in pixels
- `int h = bih.biHeight;` - height of the image in pixels
- `int size = bih.biSizeImage;` - the total size of the pixel data in bytes
- Note that we create `bytesPerPixel = 3` - the number of bytes per pixel (3 for 24-bit RGB)

Let's also preemptively build a pixel structure that can contain the information of an individual value at coordinate $(x, y)$:

```
struct PIXEL
{
    // Each value is in range 0 to 255 represented as a byte
    BYTE b; // 1 byte
    BYTE g; // 1 byte
    BYTE r; // 1 byte
};
```

### 142.1.2 Padding

Remember that bytes are stored in a contiguous block of memory, as one long string of bytes. The computer does not inherently know where one row ends and the next begins (a concept called **row-major order**)[1]. We need to calculate the starting index of each pixel based on its row and column position. But, there is an issue: each row of pixel data in a bitmap must be aligned to a 4-byte boundary.

Padding exists in bitmap images to ensure that each row of pixel data is aligned to a 4-byte boundary in memory, which was a design choice made to improve performance and

[1]In row-major order, 2D arrays like bitmaps are stored sequentially in memory, row by row. To review this concept, refer back to the Vectors and Matrices Chapter, which explains how matrices can be analyzed in row-major order.

simplicity on early computer systems. Processors and hardware are more efficient when reading data that begins at predictable, aligned memory addresses, and forcing each row to occupy a size divisible by four bytes guarantees this alignment.

Because bitmap pixels do not always naturally fill a multiple of four bytes—especially in formats like 24-bit images where each pixel uses three bytes—extra, non-image bytes are added to the end of each row to reach the required alignment. These padding bytes do not represent color information and are ignored when displaying the image, but they ensure that each row starts at a consistent location in memory, making bitmap files easier and faster for software and hardware to process.

This means that if the width of the image (in bytes) is not a multiple of 4, we need to add padding bytes at the end of each row to ensure proper alignment. Padding, then, is extra bytes added to the end of each row of pixels so that the row size is a multiple of 4 bytes.

Let's look at an example. If we create a 4 pixel bitmap, it takes up

$$4 \text{ pixels} \times 3 \text{ bytes} = 12 \text{ bytes}$$

The memory looks something like

```
... | ?? | ?? | [B G R] [B G R] [B G R] [B G R] | ?? | ?? | ...
```

Since memory is contiguous, the question marks represent unrelated memory that does not belong to the bitmap's pixel data. Attempting to access memory outside the bounds of the image means the program is reading or writing data it does not own. Modern operating systems protect memory by dividing it into regions assigned to each program. If a program tries to access memory outside of its permitted region, the operating system immediately stops the program and reports an error known as a `Segmentation Fault`. This mechanism prevents programs from corrupting other data and helps ensure overall system stability.

12 bytes is evenly divisible by 4, so *no padding is needed*.

However, if we create a 3 pixel bitmap, it would take up:

$$3 \text{ pixels} \times 3 \text{ bytes} = 9 \text{ bytes}$$

9 is not evenly divisible by 4, so 3 bytes of memory must be added as padding for row-alignment.

```
... | ?? | ?? | [B G R] [B G R] [B G R] [ P ] [ P ] [ P ] | ?? | ?? | ...
```

Note that we read the pixels in `B G R` order in memory, but hex colors are read usually to

humans via `R G B` order. This gets into a concept called *little endian*. Put simply in a stack overflow post:

> RGB is a byte-order. But a deliberate implementation choice of most vanilla Graphics libraries is that they treat colours as unsigned 32-bit integers internally, with the three (or four, as alpha is typically included) components packed into the integer.
>
> On a little-endian machine (such as x86) the integer `0x01020304` will actually be stored in memory as `0x04030201`. And thus `0x00BBGGRR` will be stored as `0xRRGGBB00`!

So how can we create an equation for finding a pixel at the coordinates $(x, y)$ if memory is in a contiguous line?

The calculation `rawRowSize = width * bytesPerPixel` provides a raw row estimate, but if padding is included, we need to round up by 4. The multiples of 4 look like 0, 4, 8, 12, 16, 20, 24, 28, 32..

- If `rawRowSize` is already one of these, we want to keep it.
- If not, we want to round upwards to the next multiple.

One way to accomplish this rounding is to take advantage of integer division. By adding a small offset before dividing, we can ensure that any value which is not already divisible by 4 is pushed into the next group of four bytes. Specifically, adding 3 to the raw row size guarantees that the result will round upward when divided by 4 using integer arithmetic.

This gives us the following expression for the true number of bytes in a single row, including padding:

$$\texttt{rowSize} = \left\lfloor \frac{\texttt{rawRowSize} + 3}{4} \right\rfloor \times 4$$

In C++, this is commonly written as:

```
1 int rowSize = ((width * bytesPerPixel + 3) / 4) * 4;
```

This value represents the number of bytes that must be traversed in memory to move from the beginning of one row of pixels to the beginning of the next. Recall that our `bytesPerPixel` is 3, but can also be calculated by `int bytesPerPixel = bih.biBitCount / 8;`.

Once the padded row size is known, we can compute the location of any pixel at coordinates $(x, y)$. Because bitmap pixel data is stored row by row in a contiguous block of

memory, the offset to the start of row $y$ is given by

$$y \times rowSize$$

Within that row, each pixel occupies `bytesPerPixel` bytes, so the offset to column $x$ is:

$$x \times \texttt{bytesPerPixel}$$

Together, these two offsets combined yields the final memory index for pixel $(x, y)$:

$$\texttt{index}(x, y) = y \times \texttt{rowSize} + x \times \texttt{bytesPerPixel}$$

In code, this calculation appears as:

```
int offset = y * rowSize + x * bytesPerPixel;
```

This formula will come in handy later in our code!

Let's write the full code for bitmap analysis. Starting with store the bitmap that our program reads into a memory allocated array, we can load our bitmap into a file, and read all the file information from it. We also can then utilize our equations `rowSize` and `idx` to get the individual pixel at $(x, y) \in (w, h)$.

```
// program arguments: ./bitmap_reader inputname.bmp outputname.bmp
if (argc !=3) return 1;
string inputname = argv[1];
string outputname = argv[2];


FILE *inputfile = fopen(inputname.c_str(), "rb"); // read byte only mode

BITMAPFILEHEADER bfh;
BITMAPINFOHEADER bih;

fread(&bfh.bfType, 2, 1, inputfile);
fread(&bfh.bfSize, 4, 1, inputfile);
fread(&bfh.bfReserved1, 2, 1, inputfile);
fread(&bfh.bfReserved2, 2, 1, inputfile);
fread(&bfh.bfOffBits, 4, 1, inputfile);
fread(&bih, sizeof(BITMAPINFOHEADER), 1, inputfile);

int w = bih.biWidth;
int h = bih.biHeight;
int bytesPerPixel = bih.biBitCount / 8; // 3 BYTES, stored in BGR order
int rowSize = ((w * bytesPerPixel + 3) / 4) * 4;
int size = rowSize * abs(h);
```

```
bih.biSizeImage = size;
bfh.bfSize = bfh.bfOffBits + bih.biSizeImage;

fseek(inputfile, bfh.bfOffBits, SEEK_SET); // offset from header
BYTE* data = (BYTE *)malloc(size);
fread(data, size, 1, inputfile);
fclose(inputfile);

// ----- FORCE STANDARD BMP HEADER HERE -----
bih.biSize = 40;
bih.biSizeImage = size;
bfh.bfOffBits = 54;
bfh.bfSize = 54 + size;
```

From there, we can set up an output file:

```
FILE *outfile = fopen(outputname.c_str(), "wb"); // creates a new file in write
↪  bytes mode
BYTE* out = (BYTE *) malloc(size);

fwrite(&bfh.bfType,     2, 1, outfile);
fwrite(&bfh.bfSize,     4, 1, outfile);
fwrite(&bfh.bfReserved1,2, 1, outfile);
fwrite(&bfh.bfReserved2,2, 1, outfile);
fwrite(&bfh.bfOffBits,  4, 1, outfile);
fwrite(&bih, sizeof(BITMAPINFOHEADER), 1, outfile);
```

And then, we can loop through to the `w` and `h` to analyze each individual pixel. What we will do as a first test of pixel alteration is *swap red and blue values* in our output.

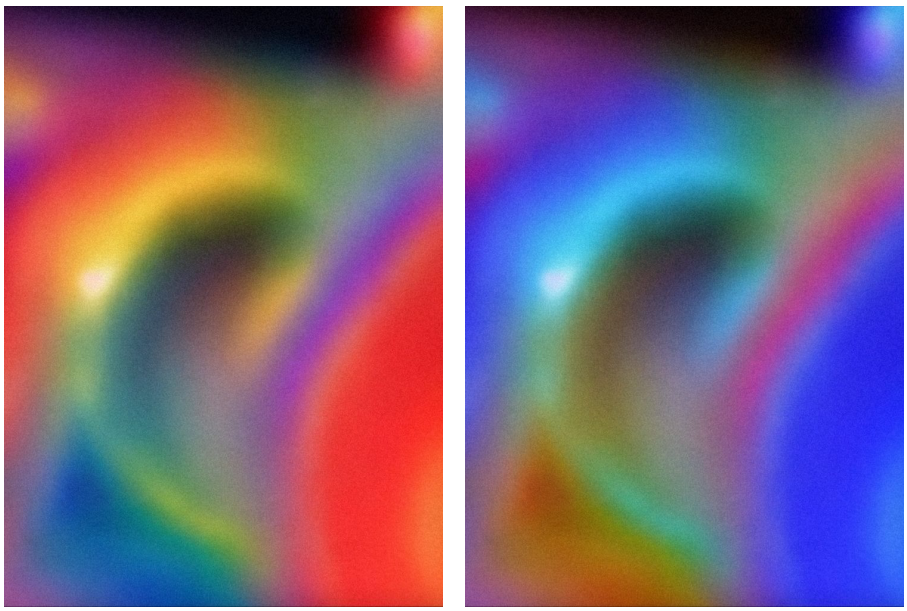
```
for (int x = 0; x < w; x++)
{
    for (int y = 0; y < h; y++)
    {
        int idx = y * rowSize + x * 3;

        PIXEL p;
        BYTE B = data[idx];
        BYTE G = data[idx + 1];
        BYTE R = data[idx + 2];
        p = { B, G, R };

        PIXEL o;
        o = { R, G, B }; // Swaps red and blue values
        out[idx + 0] = o.b;
        out[idx + 1] = o.g;
        out[idx + 2] = o.r;
    }
```

```
19 }
```

Note that we search our data array at `idx` to locate the blue component, and adding 1 and 2 respectively gets the next two bytes (green and red). This works because

- Memory is a linear array of bytes
- Pixels are stored contiguously
- The order is fixed by the BMP format

After compiling, running this using `./bitmap_reader cow.bmp output.bmp` and `./bitmap_reader gradient.bmp output.bmp` produces the following outputs (see Figures **??** and **??**):



(a) Input bitmap of a gradient. (b) Output bitmap.

Figure 142.1: Running `bitmap_reader.cpp` on a gradient, swapping red and blue channels.

(a) Input image of a highland cow. (b) Output bitmap.

Figure 142.2: Running `bitmap_reader.cpp` on a gradient, swapping red and blue channels.

The entire base code for swapping the red and blue channels is in your Digital Resources, as well as a basic copy paste input-output bitmap. A lot of this stays the same of other pixel-based modifications, such as the next Exercise.

### *Exercise 387* Eliminate the Green component

Now try it yourself:

Alter both `cow.bmp` and `gradient.bmp` such that the green channel is eliminated. Keep the red and blue channels the same (don't swap them).

*Working Space*

*Answer on Page 2040*

## 142.2 Creating a Bitmap with C++

What if we aren't given an input file? Can we create a bitmap?

Of course, let's create a simple 3x3 bitmap of the following colors:

```
[ Red ] [ White ] [ White ]
[ Black ] [ Blue ] [ White ]
[ Black ] [ Black ] [ Green ]
```

Although the image is conceptually two-dimensional, the bitmap file stores its pixel data as a one-dimensional sequence of bytes.

There are, however, a few constraints:

- 14-bit file header
- 40-byte info header
- pixel data with required row padding

```
| File Header (14) | Info Header (40) | Pixel Data |
```

**Calculations**:

- Each pixel $=3$ `bytes`
- Row $=3$ `pixels` $\times 3$ `bytes per pixel` $=9$ `pixels`
- $3$ `padding bytes` $\implies 12$ `bytes per row`
- $12$ `bytes` $\times 3$ `rows` $=36$ `bytes`

Here is the basic bitmap struct with filled in values:

```
#pragma pack(push, 1)
struct BITMAPFILEHEADER {
    uint16_t bfType = 0x4D42; // 'BM'
    uint32_t bfSize;
    uint16_t bfReserved1 = 0;
    uint16_t bfReserved2 = 0;
    uint32_t bfOffBits = 54; // 14 + 40
};

struct BITMAPINFOHEADER {
    uint32_t biSize = 40;
    int32_t  biWidth = 3;
    int32_t  biHeight = 3;
    uint16_t biPlanes = 1;
    uint16_t biBitCount = 24;
```

```
    uint32_t biCompression = 0; // BI_RGB
    uint32_t biSizeImage;
    int32_t  biXPelsPerMeter = 0;
    int32_t  biYPelsPerMeter = 0;
    uint32_t biClrUsed = 0;
    uint32_t biClrImportant = 0;
};
#pragma pack(pop)
```

These structures match the on-disk layout of a bitmap header exactly. The `#pragma pack` directive ensures that no padding bytes are inserted by the compiler.

Recall that we can reuse our calculations from the first example to find rowsize and other variables, but this time, we define them instead of fetching them from the input:

```
const int width = 3;
const int height = 3;
const int bytesPerPixel = 3;
const int rowSize = ((width * bytesPerPixel + 3) / 4) * 4; // 12
const int imageSize = rowSize * height; // 36
```

Now we can create a new file and write the file headers individually:

```
FILE* f = fopen("custom_made.bmp", "wb");

BITMAPFILEHEADER bfh;
BITMAPINFOHEADER bih;

bih.biSizeImage = imageSize;
bfh.bfSize = bfh.bfOffBits + imageSize;

fwrite(&bfh, sizeof(bfh), 1, f);
fwrite(&bih, sizeof(bih), 1, f);
```

At this point, the file contains only metadata. No pixel values have been written yet.

Recall that bitmap pixel data is stored bottom-up, meaning the first row written corresponds to the bottom row of the image.

```
Memory order:
Row 2 (bottom)
Row 1
Row 0 (top)
```

Let's write the third row first:

```
// ROW 3 (bottom)
// Black      Black          Green
row[0] = 0;   row[1] = 0;   row[2] = 0;      // Black
row[3] = 0;   row[4] = 0;   row[5] = 0;      // Black
row[6] = 0;   row[7] = 255; row[8] = 0;      // Green
fwrite(row, rowSize, 1, f);
```

Each group of three bytes represents one pixel in BGR order. Any remaining bytes in the row serve as padding and are ignored when the image is displayed.

Row 2:

```
// ROW 2
// Black     Blue           White
row[0] = 0;   row[1] = 0;   row[2] = 0;      // Black
row[3] = 255; row[4] = 0;   row[5] = 0;      // Blue
row[6] = 255; row[7] = 255; row[8] = 255;    // White
fwrite(row, rowSize, 1, f);
```

Row 1:

```
// ROW 1 (top)
// Red        White          White
row[0] = 0;   row[1] = 0;   row[2] = 255;    // Red
row[3] = 255; row[4] = 255; row[5] = 255;    // White
row[6] = 255; row[7] = 255; row[8] = 255;    // White
fwrite(row, rowSize, 1, f);
```

And that's it! The compiler automatically converts the 255s to `0xFF`. Close the file and run the program, and you get an extremely small output. Enlarge the output and you get:

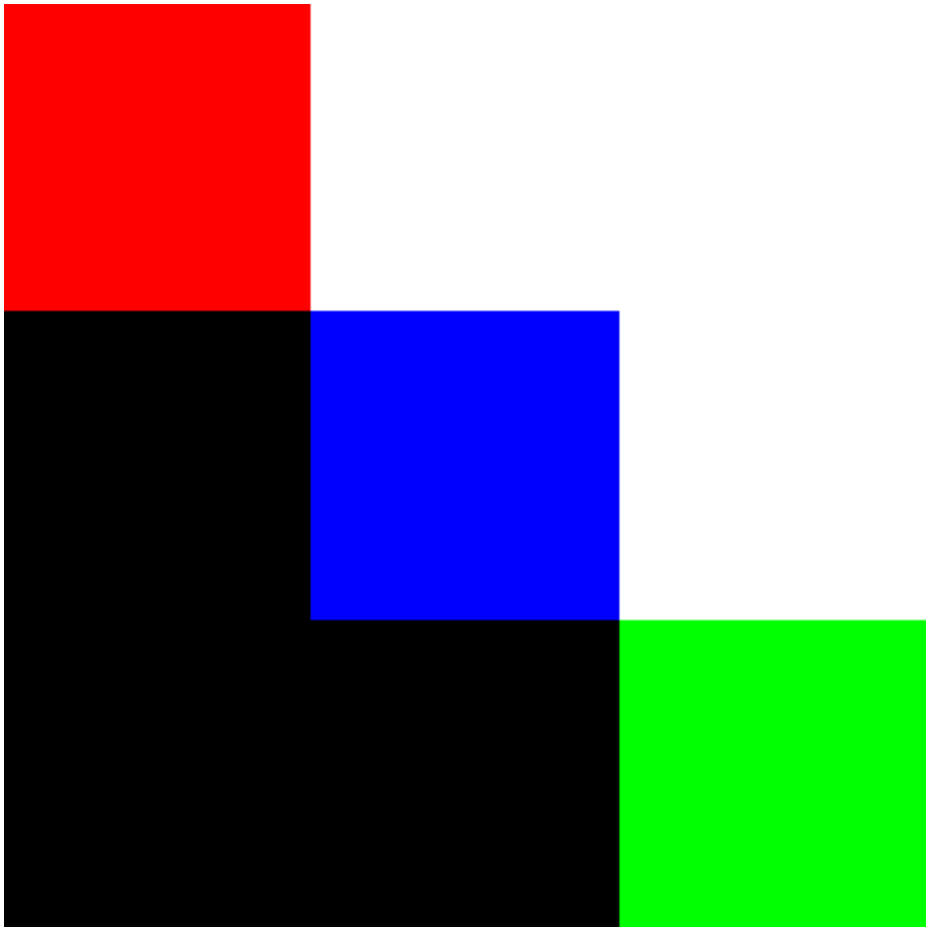


Figure 142.4: Your custom bitmap!

## *Exercise 388* 8 Pixel Colorful Bitmap

*Working Space*

Create a $4 \times 2$ bitmap with the following layout:

```
[ Yellow ] [ Magenta ] [ Cyan ] [ White ]
[ Black  ] [ Red     ] [ Green ] [ Blue  ]
```

Bitmap Requirements:

- Width: 4 pixels
- Height: 2 pixels
- Color depth: 24-bit (RGB, stored as BGR)
- Compression: None (BI_RGB)
- Row padding: Rows must be aligned to 4-byte boundaries

Answer the following questions before writing any code:

1. How many bytes does each pixel use?
2. How many bytes are required for one row before padding?
3. How many padding bytes are required per row?
4. What is the total size of the pixel data?
5. What is the total file size?

*Answer on Page 2041*

## 142.3 Connection to transformations

What if alter the pixel values in a more complex way? For example, instead of altering the colors themselves, we alter the pixel positions. This is what is called a transformation, and is a fundamental concept in computer graphics. A transformation is a mathematical operation that changes the position, size, or orientation of an object in a graphical space. In the context of bitmaps, transformations can be applied to the pixel data to achieve various effects, such as rotation, scaling, translation, and shearing.

What if you took a selfie, but then wanted to mirror it horizontally, to get a more representation of your face? This is a common transformation called a horizontal flip. To achieve this, you would iterate through each row of pixels in the bitmap and swap the pixels on the left side with the corresponding pixels on the right side. This can be done by calculating the index of each pixel and performing the necessary swaps. Note that nothing on the vertical direction of the image is changed, only the horizontal positions of the pixels are altered.

Since we are storing our pixel data as $(x, y)$ coordinates, we can easily apply transformations by manipulating these coordinates. For an image with width $w$ and height $h$, the horizontal flip transformation can be expressed mathematically as:

$$x' = w - 1 - x, \qquad y' = y$$

This operation does not change the pixel values themselves, but instead sets the new position of each pixel according to the transformation rule. The entire image transformation is therefore a systematic coordinate mapping applied across the matrix.

Without having to learn too much linear algebra, we can express this as a simple function: If we have a pixel coordinate expressed as a simple vector $\mathbf{p} = \begin{bmatrix} x \\ y \end{bmatrix}$, we can define a transformation function $T$ that takes this pixel coordinate and maps it to a new coordinate $\mathbf{p}'$:

$$\mathbf{p}' = T\mathbf{p}$$

In this case, the transformation function $T$ is a matrix that encodes the horizontal flip operation that obeys matrix multiplication. We have not talked about this yet, and it is a difficult concept to understand without having introduced linear algebra or matrices. So, we will not go into the details of how to construct this matrix, but just know that a horizontal flip can represented as:

$$T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

This matrix, when multiplied with the pixel coordinate vector $\mathbf{p}$, will yield the new coordinate $\mathbf{p}'$ that corresponds to the horizontally flipped position of the original pixel.

Specifically, the $-1$ in the first row of the matrix indicates that the x-coordinate is inverted (flipped), while the 1 in the second row indicates that the y-coordinate remains unchanged.

Let's do this with code! The only thing that changes in our for loop is the calculation of the output pixel's position. Instead of writing to the same index, we write to the flipped index:

```
for (int y = 0; y < h; y++)
{
    for (int x = 0; x < w; x++)
    {
        int newX = w - 1 - x; // Calculate the new x-coordinate for the
        ↪  horizontal flip
        int srcIdx = y * rowSize + x * 3;
        int dstIdx = y * rowSize + newX * 3;

        // read the pixel values from the source index
        BYTE B = data[srcIdx];
        BYTE G = data[srcIdx + 1];
        BYTE R = data[srcIdx + 2];

        //output the pixel values to the destination index
        out[dstIdx + 0] = B;
        out[dstIdx + 1] = G;
        out[dstIdx + 2] = R;
    }
}
```

Taking the input bitmap of the cow from 142.2a and running this code produces the horizontally flipped output in 142.6.

Figure 142.6: Output of a horizontally flipped cow bitmap after applying the new pixel position transformation.

Similarly, we can transform the bitmap in other ways, which we will learn more of how to do in the transformations chapter by experimenting with vectors. For now, remember that matrices are a powerful tool for representing transformations, and that by manipulating pixel coordinates, we can achieve a wide variety of effects on bitmap images.

## 142.4 Summary

In this chapter, we experimented with bitmaps. Specifically,

- loading a bitmap in C++

- creating and copying bitmap file and info headers
- creating a pixel structure to work with hexadecimal bytes and hexcodes
- swapping the channels of a bitmap
- creating a bitmap from scratch in C++

# Chapter 143

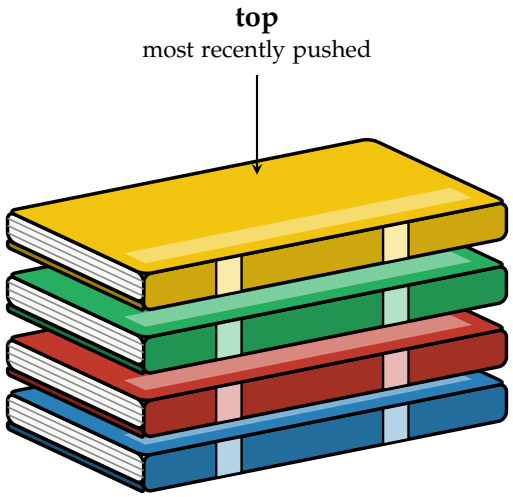
# Stacks

A *stack* is a linear data structure that only allows access at one end, called the *top*.

Data comes off in the reverse order it went on: the last thing pushed onto the stack is the first thing popped off. This ordering is called *LIFO*, for "last in, first out."

You have already seen a stack in a previous workbook, even if it wasn't called one: the call stack from the Introduction to C chapter behaves exactly this way, with each function call pushing a new frame on top and returning popping it back off.

We can think of this like a stack of books: you can only access the book on top, and you can only see the cover of the top book. If you want to get to a book underneath, you have to remove the books above it first.




Every stack, no matter how it is built underneath, supports the same handful of operations:

- `push(value)` put a new value on top of the stack.
- `pop()` remove and return the value on top of the stack.
- `peek()` look at the value on top of the stack, without removing it.
- `isEmpty()` true if the stack has nothing in it.

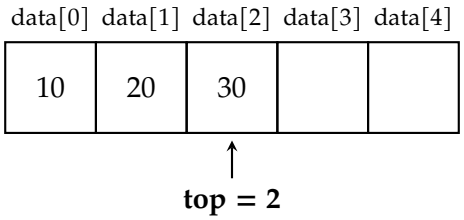
Notice that we can only use the value on top of the stack, no matter how large it is.

We are going to build a stack two different ways: once backed by a fixed-size array, and once backed by the linked-list nodes from the previous chapter. Both give you `push`/`pop`/`peek`/`isEmpty`, but they make very different tradeoffs underneath.

## 143.1 Array-Based Stack

The simplest way to build a stack is on top of a plain array. We keep a fixed-size buffer and a single integer, `top`, that tracks the index of the top element. An empty stack is represented by `top == -1`.

```
const int CAPACITY = 5;

struct ArrayStack {
    int data[CAPACITY];
    int top;   // index of the top element; -1 means empty
};
```




`push` and `pop` only ever touch the `top` index and whatever cell it points at, so both run in constant time, $O(1)$. The catch is `CAPACITY`: once the array is full, there is nowhere left to push, so a correct implementation has to check for and reject that case rather than writing past the end of the array.

```
bool isEmpty(const ArrayStack& s) {
    return s.top == -1;
}

bool isFull(const ArrayStack& s) {
    return s.top == CAPACITY - 1;
}

void push(ArrayStack& s, int value) {
    // prevent pushing onto a full stack
    if (isFull(s)) {
        std::cout << "push(" << value << ") failed: stack overflow" << std::endl;
        return;
    }
    s.top++;
```

```
    s.data[s.top] = value;
}

int pop(ArrayStack& s) {
    // prevent popping from an empty stack
    if (isEmpty(s)) {
        std::cout << "pop() failed: stack underflow" << std::endl;
        return -1;
    }
    int value = s.data[s.top];
    s.top--;
    return value;
}
```

Notice that `pop` does not actually erase `s.data[s.top]`, it just moves `top` down by one. The old value is still sitting in the array, but nothing can reach it anymore, since every operation only ever looks at indices `0` through `top`.

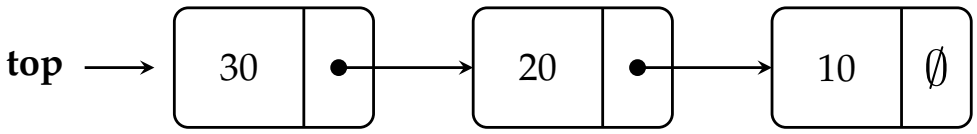
The full, runnable version of this file, including a `main()` that demonstrates both overflow and underflow, is in your digital resources. In this version, we added memory address output to demonstrate that shows how the array is a contiguous block.

## 143.2 Linked-List Stack

The array-based stack has one real weakness: `CAPACITY` is fixed at compile time. If you don't know in advance how much data you'll be pushing, you either waste memory reserving more than you need, or you risk overflow. A stack built on the `Node` type from the Linked Lists chapter has no limit, since it grows and shrinks on the heap as needed. The downside is that each element takes more memory, since it has to store a pointer to the next node.

```
struct Node {
    int data;
    Node* next;
};

struct LinkedStack {
    Node* top;   // nullptr means empty
};
```

Instead of an array index, `top` is now a pointer to the first node. Pushing means creating a new node whose `next` points at the old top, then moving `top` to the new node; popping is the reverse.




```
bool isEmpty(const LinkedStack& s) {
    return s.top == nullptr;
}

void push(LinkedStack& s, int value) {
    Node* n = new Node();
    n->data = value;
    n->next = s.top;    // the new node points at the old top
    s.top = n;          // the new node becomes the top
}

int pop(LinkedStack& s) {
    if (isEmpty(s)) {
        std::cout << "pop() failed: stack underflow" << std::endl;
        return -1;
    }
    Node* oldTop = s.top;
    int value = oldTop->data;
    s.top = oldTop->next;
    delete oldTop;
    return value;
}
```

Both `push` and `pop` still run in constant time (we will cover this in a future chapter), same as the array version, since neither one has to walk the list. But unlike the array version, `pop` here calls `delete` on the old top node; if it didn't, the node would sit on the heap forever.

The full version of this file is in your digital resource!

## 143.3 Array vs. Linked: Which One?

| | **Array-Based** | **Linked-Based** |
|---|---|---|
| Max size | Fixed at compile time | Limited only by available memory |
| Memory per element | Just the value | Value + one pointer |

If you know a safe upper bound on how much data you'll ever hold, the array version is the most efficient* choice. If you don't, the linked version is safer, since it will never overflow.

# Chapter 144

# Queues

Imagine you are at a concert. You are arriving at around 1pm, but the line has been growing since about 3am that same day (wow!). Naturally, by way of timing, the people who arrived earliest are going to be able to enter the venue first, while the people you arrive closest to the show time will enter the venue last.

Assuming supplements like VIP packages or fast-track lines are not of concern here, we can summarize this type of line in a similar phrase: **First In, First Out**. The first people to arrive in the line are the first people out of the line, and, subsequently, into the venue. We call this way of organizing data *FIFO*.

Similar to stacks, we only need a handful of operations to build a queue.

[placeholder: people in line image (maybe)]

Every stack, no matter how it is built underneath, supports the same handful of operations:

- `enqueue(value)` put a new value at the front of the queue. You may see this as `enq`
- `dequeue()` remove and return the value at the front of the queue. You may see this as `deq`
- `peek()` look at the value on front of the queue
- `isEmpty()` true if the queue has nothing in it.

Similarly, again, to stacks, we can implement this two ways: a fixed-array based method and a linked-list method. Both have benefits and tradeoffs.

## 144.1 Array-Based Queue

We can build a queue with an array, there is the same major constraint as with stacks: we need to know the size of the queue in advance.

```
1  const int CAPACITY = 5;
2
3  struct ArrayQueue {
4      int data[CAPACITY];
5      int front;   // index of the front element; -1 means empty
```

```
    int count = 0; // for ensuring we don't exceed CAPACITY

};

bool enqueue(Queue& q, int value) {
    if (isFull(q)) return false;
    int back = (q.front + q.count) % MAX;  // wraps around
    q.data[back] = value;
    q.count++;
    return true;
}

bool dequeue(Queue& q, int& out) {
    if (isEmpty(q)) return false;
    out = q.data[q.front];
    q.front = (q.front + 1) % MAX;         // wraps around
    q.count--;
    return true;
}

bool peek(const Queue& q, int& out) {
    if (isEmpty(q)) return false;
    out = q.data[q.front];
    return true;
}
```

TODO: Explain why we need to wrap around the array (circular array) and how it is done. Simplify above code.

## 144.2 Linked-List Queue

# Chapter 145

# Priority Queues

As mentioned in the last chapter, you are going to make a priority queue for use with Dijkstra's Algorithm. Using it will look like this algorithm called a *priority queue*:

```
import kpqueue

myqueue = pqueue.PriorityQueue()
myqueue.add(long_beach, 0) # Inserts first city and its cost
myqueue.add(san_diego, 14) # Puts San Diego after Long Beach
myqueue.add(los_angeles,12) # Inserts LA between Long Beach and San Diego
current_city = myqueue.pop() # Returns first city (Long Beach) and removes it
```

Now, if an item gets a new priority, we need to remove it and reinsert it in the new spot.

```
myqueue.add(city_a, 16) # Puts it last in the queue
myqueue.update(city_a, 16, 13) # Moves it to between LA and San Diego
```

## 145.1 A Naive Implementation of the Priority Queue

Create a file called `kpqueue.py`. Let's do a simple implementation that stores the priority and the data as tuple. And we will keep it sorted by the priority. If two tuples have the same priority, we will sort by the data.

Type this in to `kpqueue.py`:

```
class PriorityQueue:
    def __init__(self):
        self.list = []

    # Return and remove the first item
    def pop(self):
        if len(self.list) > 0:
            return self.list.pop(0)
        else:
            return None

    def __len__(self):
        return len(self.list)
```

```
14
15      def update(self, value, old_priority, new_priority):
16          old_pair = (old_priority, value)
17          self.list.remove(old_pair)
18          self.add(value, new_priority)
19
20      def add(self, value, priority):
21          pair = (priority, value)
22          # Add it at the end
23          self.list.append(pair)
24          # Resort the list
25          self.list.sort()
```

This will work fine, but it could be much more efficient:

- Every time we add a single element, we resort the whole list.
- The function `remove` is searching the list sequentially for the item to delete.

In a minute, we will revisit these inefficiencies and make them better.

## 145.2 Using the Priority Queue

We are going to change `graph.py` to use the priority queue. While we are doing so, why don't we also shrink the memory footprint of our program a bit.

Notice that as the algorithm is running, each node is in one of three states:

- Unseen: In the earlier implementation, these were the nodes with `math.inf` as their cost.
- Seen, but not finalized: These are "unvisited", but don't have `math.inf` as their cost.
- Finalized: These are the "visited" nodes — we know that their cost won't decrease any more.

We can shrink the memory foot print by not putting the unseen into the dist dictionary at all. And instead of a separate set for "unvisited", what if we moved finalized nodes and their distances into a separate dictionary?

Rewrite the `cost_from_node` function in `graph.py`:

```
1               # Visited nodes are already minimized, skip them
2               if v in finalized_dist:
3                   continue
```

```
                # What is the cost to this neighbor?
                alt = current_node_cost + edge.cost

                # Is this the first time I am seeing the node?
                if v not in seen_dist:

                    # Insert into the seen_dict, prev, and priority queue
                    seen_dist[v] = alt
                    prev[v] = current_node
                    pqueue.add(v, alt)

                else: # v has been seen. Is this a cheaper route?
                    old_dist = seen_dist[v]
                    if alt < old_dist:
                        # Update the seen_dict, prev, and priority queue
                        seen_dist[v] = alt
                        prev[v] = current_node
                        pqueue.update(v, old_dist, alt)

        return (finalized_dist, prev)
```

This should be have exactly the same result, except for the unreachable nodes. If you have a graph that is not connected, there will be nodes that cannot be reached from the origin. In the old version, these had a cost of `math.inf`. Now, they will just not be in the dictionary at all. So, change `cities.py` to deal with this:

```
if nyc in cost_from_long_beach:
    nyc_cost = cost_from_long_beach[nyc]
    print(f"\n*** Total cost from Long Beach to NYC: ${nyc_cost:.2f} ***")

    path_to_nyc = graph.shortest_path(prev, nyc)
    print(f"\n*** Cheapest path from Long Beach to NYC: {path_to_nyc} ***")
else:
    print("You can't get to NYC from Long Beach")
```

If you run `cities.py` now, it should behave exactly like the old version.

However, there is a bug. It will rear its head if two cities with the same cost are in the priority queue together. Change `cities.py` so that Denver and Phoenix have the same cost:

```
graph.WeightedEdge(12, long_beach, los_angeles)
graph.WeightedEdge(19, los_angeles, denver)
graph.WeightedEdge(19, los_angeles, phoenix)
```

Now try running it. You should get an error:

```
TypeError: '<' not supported between instances of 'Node' and 'Node'
```

What happened? The `loc_for_pair` method is comparing tuples made up of a `float` and a `Node`. The `float` comes first in the tuple, so that is compared first. However, if the two tuples have the same priority, it then compares nodes.

The error statement says, "Nodes don't have a less-than method; I don't know how to compare them."

Each `Node` lives at an address in memory. You can get that address as a number using the `id` function. The ID is unique and constant over the life of the object. It is a rather arbitrary ordering, but it will work for this problem. Add a method to your `Node` class:

```
    # Nodes will be ordered by their location in memory
    def __lt__(self, other):
        return id(self) < id(other)
```

Fixed.

Now, let's make the priority queue more efficient.

## 145.3 Binary Search

The phone company in every town used to print a thing called a phone book. The names and phone numbers were arranged alphabetically. As you might imagine, these books often had more than a thousand pages.

If you were looking for "John Jeffers", you would not start at the first page and read sequentially until you reached his name. You would open the book in the middle, and see a name like "Mac Miller", then think "Jeffers comes before Miller". You would then split the pages in your left hand in half and see a name like "Hester Hamburg" and think "Jeffers comes after Hamburg". You would then split the pages in your right hand, and continue on like this until you found the page with "John Jeffers" on it. That is binary search.

Binary Search is a search algorithm that finds the position of a target value within a sorted array. The binary search algorithm works by repeatedly dividing the search interval in half. If the target value is equal to the middle element of the array, the position is returned. If the target value is less than or greater than the middle element, the search continues in the lower or upper half of the array, respectively.

## 145.4 Algorithm

The binary search algorithm can be described as follows:

1. If the array is empty, the search is unsuccessful, so return "Not Found".
2. Otherwise, compare the target value to the middle element of the array.
3. If the target value matches the middle element, return the middle index.
4. If the target value is less than the middle element, repeat the search with the lower half of the array.
5. If the target value is greater than the middle element, repeat the search with the upper half of the array.
6. Repeat steps 2-5 until the target value is found or the array is exhausted.

```
class PriorityQueue:
    def __init__(self):
        self.list = []

    # Return and remove the first item
    def pop(self):
        if len(self.list) > 0:
            return self.list.pop(0)
        else:
            return None

    def __len__(self):
        return len(self.list)

    def add(self, value, priority):
        pair = (priority, value)
        i = self.loc_for_pair(pair)
        self.list.insert(i, pair)

    def update(self, value, old_priority, new_priority):
        old_pair = (old_priority, value)
        i = self.loc_for_pair(old_pair)
        del self.list[i]
        self.add(value, new_priority)

    def loc_for_pair(self, pair):
        # The range where it could be is [lower, upper)
        # Start with the whole list
        lower = 0
        upper = len(self.list)

        while upper > lower:
            next_split = (upper + lower) // 2
```

```
            v = self.list[next_split]
            if pair < v:  # pair is to the left
                upper = next_split
            elif pair > v:  # pair is to the right
                lower = next_split + 1
            else: # Found pair!
                return next_split
        return lower
```

If you try running it now, it should work perfectly.

You now have a graph class that will find the cheapest path quickly, even if it has thousands of nodes with thousands of edges.

Chapter 146

# Hash Tables

A hash table, also known as a hash map, is a data structure that implements an associative array abstract data type, a structure that can map keys to values. It uses a hash function to compute an index into an array of buckets or slots, from which the desired value can be found.

## 146.1 Structure of a Hash Table

A hash table is composed of an array (the 'table') and a hash function. The array has a predetermined size, and each location (or 'bucket') in the array can hold an item (or several items if collisions occur, as will be discussed later). The hash function is a function that takes a key as input and returns an integer, which is then used as an index into the array.

## 146.2 Inserting and Retrieving Data

When inserting a key-value pair into the hash table, the hash function is applied to the key to compute the index for the array. The corresponding value is then stored at that index.

When retrieving the value associated with a key, the hash function is applied to the key to compute the array index, and the value is retrieved from that index.

## 146.3 Handling Collisions

A collision occurs when two different keys hash to the same index. There are several methods for handling collisions:

- **Chaining (or Separate Chaining)**: In this method, each array element contains a linked list of all the key-value pairs that hash to the same index. When a collision occurs, a new key-value pair is added to the end of the list.
- **Open Addressing (or Linear Probing)**: In this method, if a collision occurs, we move to the next available slot in the array and store the key-value pair there. When looking up a key, we keep checking slots until we find the key or reach an empty slot.

## 146.4 Time Complexity

In an ideal scenario where hash collisions do not occur, hash tables achieve constant time complexity $O(1)$ for search, insert, and delete operations. However, due to hash collisions, the worst-case time complexity can become linear $O(n)$, where $n$ is the number of keys inserted into the table.

Using good hash functions and collision resolution strategies can minimize this issue and allow us to take advantage of the hash table's efficient average-case performance.

# Chapter 147

# Trees

Trees are one of the most versatile and widely used data structures in computer science. A *tree* is a hierarchical data structure consisting of nodes, where each node has a value and a set of references (*edges*) to its child nodes. The node at the top of the hierarchy is called the *root*, and nodes with the same parent are called *siblings*, as seen in Figure 147.1.

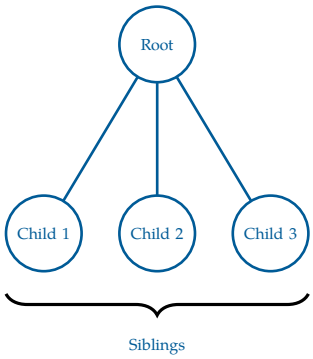



Figure 147.1: A tree with a root node and 3 children.

Trees are a subset of Graphs, a data structure comprised of nodes and edges. Graphs can be converted into trees by adding the following two constraints:

- The graph is connected, such that there exists a path between every pair of nodes
- The graph is acyclic; no cycles exist in the graph

With this comes the caveat that the number of edges in a tree is edges $= n - 1$, where $n$ is the number of nodes in the tree. This is because a tree with $n$ nodes must have exactly $n-1$ edges to maintain its connected and acyclic properties. If there were more than $n - 1$ edges, it would create a cycle, violating the acyclic property. If there were fewer than $n - 1$ edges, it would not be fully connected, violating the connected property.

Less formally, a *forest* is a collection of multiple trees (may be disconnected), and a *tree* is a forest with only one tree. Lots of agricultural terms in this chapter!

Trees have a unique structural property: between any two nodes, there exists exactly one simple path. This follows directly from the fact that trees are both connected and acyclic. As a result, there is only one way to move from a start node to an end node.

Although trees are often drawn with a root at the top, this is a matter of convention. Any node in a tree can be chosen as the root, and doing so induces a hierarchical structure in which edges are interpreted as parent-child relationships.

Once a root is selected, every node (except the root) has exactly one parent, and zero or more children. Because of this relationship, any randomly selected node can form a tree (we call this a *subtree*, consisting of the node and all its descendants), making the tree structure *recursive*. Moving from one node to the next is the recursive step that minimizes

the size.

Nodes with no children are called, as you'd expect, *leaves*, as they are the outermost node of a tree. The *depth* of a chosen node is the number of edges from the *root* to that node, and the *height* of a tree is the maximum depth among all nodes. Equivalent to the height of a tree is the maximum number of edges from the *root* to any leaf. You may see the *level* of a node referred to as the number of edges from the *root* to that node, similar to depth. By convention, the level and depth of the root is 0. See Figure 147.2.

Some more relevant terminology you may encounter in the context of trees:

- *Path*: A sequence of nodes where each adjacent pair is connected by an edge. This can be used to show the idea of how to get between two nodes in a tree, for example, between two cities in a flight path network.
- *Ancestor*: A node that lies on the path from the root to a given node. This is done by repeatedly traversing from a child node to a parent node until the selected node is reached.
- *Descendant*: A node that lies on the path from a given node to a leaf. This is done by repeatedly traversing from a parent node to a child node until a leaf is reached.

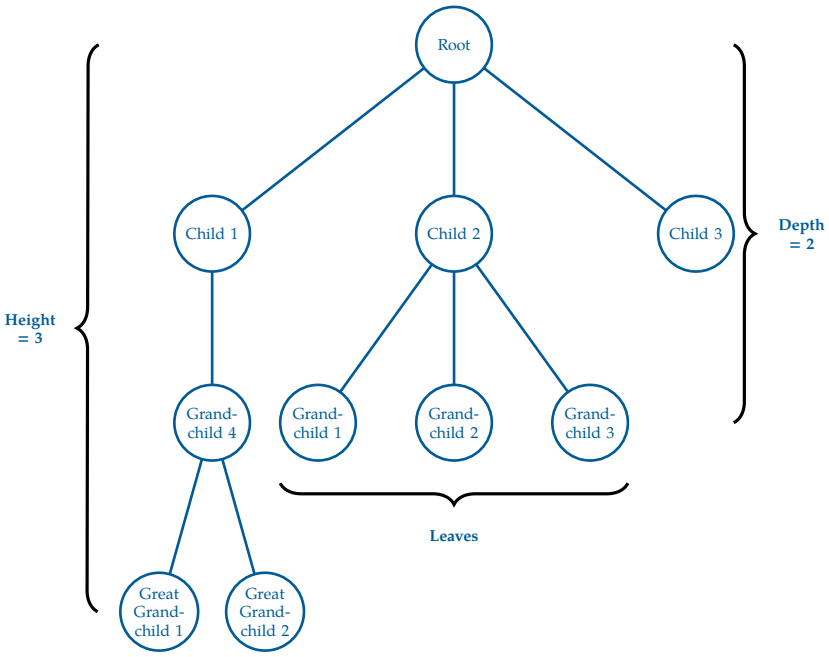



Figure 147.2: A tree with a root node and 3 children, a few grandchildren, and 2 great grandchildren.

FIXME tree proof

## 147.1 Binary Trees and Binary Search Trees

The most common type of tree is the *binary tree*, where each node has *at most* two children, referred to as the *left child* and the *right child*. Binary trees are widely used in various applications, including expression parsing, binary search trees, and heaps.

Let's put this practice. Recall our binary search implementation from the recursion chapter. Given a chosen word and a sorted list of words, such as a dictionary, we can find the page of the chosen word by repeatedly dividing the available pages in half and comparing

the elements on the page to the chosen word. This is much easier than, for example, going through every single page and checking if the word is on that page.

Say we wanted to represent our dictionary as a tree that we could recursively binary search through. For simplicity, we will limit our words to `mango`, `apple`, `pear`, `banana`, `anagram`, `peony` and `peach`. These 7 words form a binary tree of height 2, as seen in Figure 147.3. We select the word `mango` as the root as it is the middle word after sorting alphabetically.

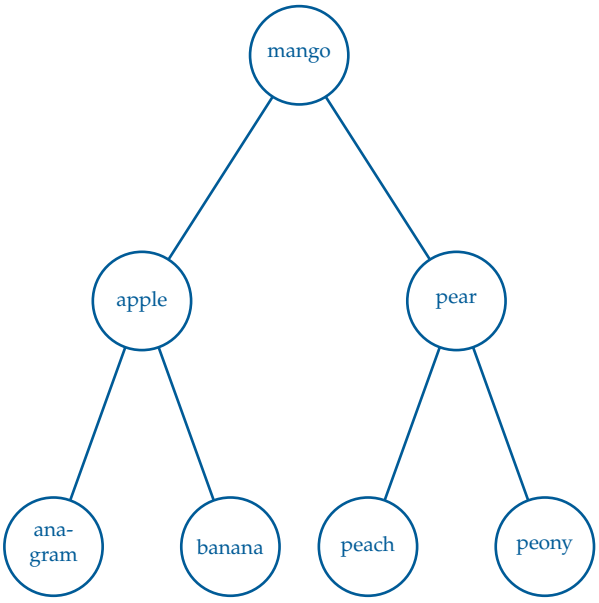


Figure 147.3: A binary search tree with 7 words.

Similar to binary search, we can find any word in the tree by starting at the root and comparing the target word to the current node's word. If the target word is less than the current node's word, we move to the left child; if it is greater, we move to the right child. We repeat this process until we find the target word or reach a leaf node, which tells us that the target word is not in the tree. This structure allows us to cut the available words in half with each step in the tree we go.

What if we want to insert into a binary search tree? We must complete four actions:

- Find the correct location for the new node.
- Create and insert the node at that location,
- Connect it to its respective parent(s).
- Ensure the BST properties are maintained.

Let's say, on our tree above, we want to add `orange` to the tree. Because of alphabetical sorting,

FIXME expand on addition, removal, and search of binary trees FIXME add exercise of a similar dictionary tree with more words, have student add and remove words, and search for words

## 147.2 File Structures and File System Hierarchies

Computers, since their first creation in the 1980s, have needed a way to store and organize data. Prior to computers, many systems of organization were used to store physical data, such as filing cabinets, libraries, and warehouses.

These systems often had a hierarchical structure, with a top-level category that branched out into subcategories, with these subcategories further branching out into more specific categories, and so on.

For example, a library might have two top-level categories: `fiction` and `non-fiction`. Within the category `non-fiction`, you may find `psychology`, `biographies`, `autobiographies`, `self-help`, `journalism` books, or even `nature guides`. Under the category `fiction`, you may find `romance`, `classics`, `crime`, `science fiction`, `thriller`, or `fantasy` novels. Of course, these sub-genres can have even further categorizations too.

A hospital may have patient records categorized by first letter of last name, and then sorted alphabetically from there.

With the development of the computer, sorting these types of hierarchical data became easy with the concept of directories. A *directory*, also called a *folder*, is a categorical way of organizing data (for example, by storing files or other subdirectories). We will stick to the more technical term of directories, however, any time you create a folder on your computer, you are making a *directory*. The name comes from how, in the twentieth-century, many phone books were referred to as telephone directories.

This structure naturally fits into the tree data structure:

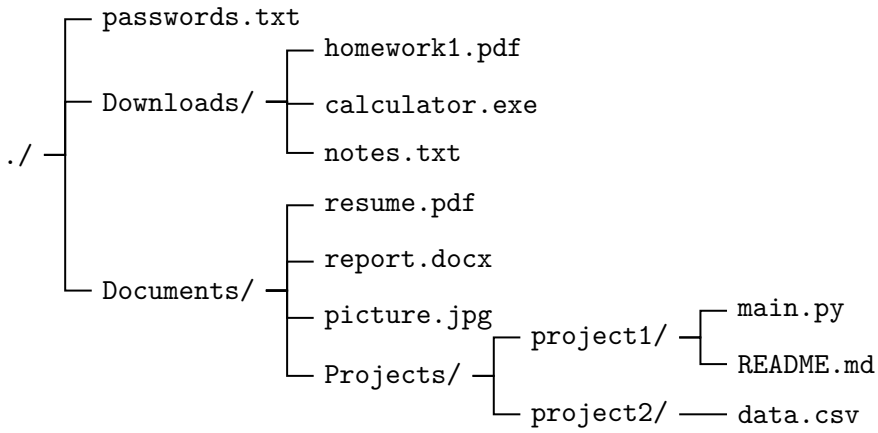
- The **root** of the tree is the top-level directory (`/` on Linux/Unix[1] systems or `C:\` on Windows).
- A **node** is naturally fits the idea of either *directories nodes* or *files nodes*.
- A **directory node** may have children, where its children are subdirectories or files.
- A **file node** is a **leaf**, meaning it has no children.

The **path** to a file describes the unique sequence of directories from the root to that file. Test again test again test again

### Example

A small file system might look like:

[1] There is a very helpful video in Linux File Systems in your digital resources. Check it out!




In tree terminology:

- `./` is the root node.
- `Downloads`, `Documents`, and its subdirectories are internal nodes (directories).
- All files, such as those ending in `.txt`, `.png`, `.exe`,`.csv`, `.docx`, `.py` files are leaf nodes.

Side note: please do not store your passwords in anywhere on your computer's memory!

## *Exercise 389* File Paths

*Working Space*

In the above File Hierarchy Tree, what is the path to `data.csv`? What is the depth of the file tree at that point?

*Answer on Page 2044*

## 147.3 Red-Black Trees and AVL Trees

[FIXME add content on red black trees and AVL trees]

## 147.4 B-Trees, Tries, and Suffix Trees

[FIXME add content on B-trees, tries, and suffix trees]

Chapter 148

# Tree Traversal

Trees can be "traversed" in a few different ways. When we say traversing a tree, we mean that we are visiting and extracting data from each node in a tree. We classify these *traversals* based on the order in which the nodes are visited.

This chapter will discuss *breadth-first search*, also known as *BFS*, which traverses in a *level-order*. Each level is traversed in sequential order.

Alternatively, there is *depth-first search* (*DFS*), which travels as far down a branch as possible before backtracking and checking other branches. Within DFS there are 3 common traversal methodologies: *in-order*, *pre-order* and *post-order*.

While the demonstrations for these traversals will be for trees, they work the same for any given graph, under one condition: using DFS or BFS on a *graph* requires you do have a preselected node a your *start node*.

## 148.1 BFS

Let's look at running BFS on a tree. Starting out with this tree:

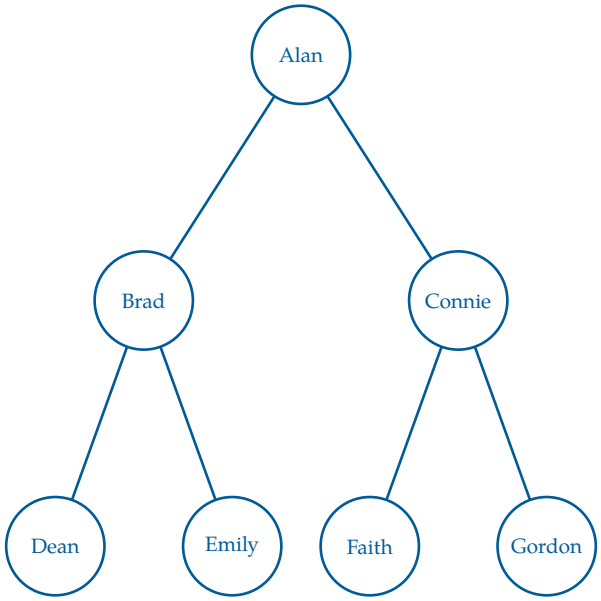



Figure 148.1: Our original tree containing a list of names to be traversed.

To perform BFS, we use a *queue*, which follows a first-in, first-out (*FIFO*) order. We begin by placing the root node into the queue. At each step, we remove the front node from the queue, visit it, and then add all of its children to the *back* of the queue.

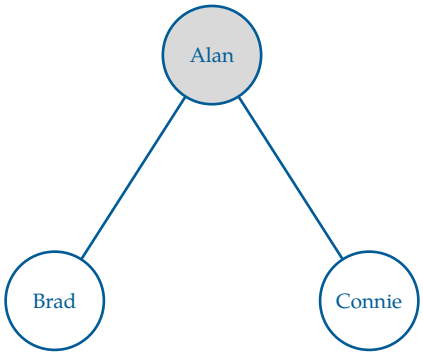



Figure 148.2: Step 1: Visit Alan, then add its children (Brad, Connie) to the queue.

Here, we visit the root node first, which is Alan. After visiting Alan, we add its children, Brad and Connie, to the queue. Since there are no other nodes to visit at this level, we move on to the next level of the tree.

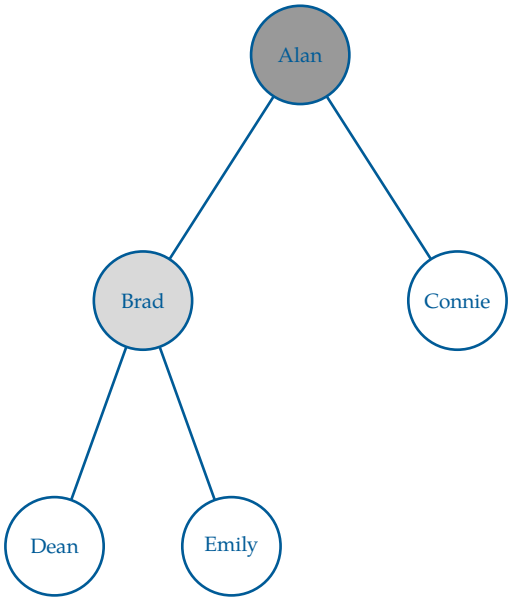



Figure 148.3: Step 2: Visit Brad, then add its children (Dean, Emily) to the queue.

We continue this process, visiting Connie next and adding its children, Faith and Gordon, to the queue. The queue now contains Connie, Dean, and Emily. Here is a step trace of the BFS traversal:

| Step | Queue | Output (visited) |
|---|---|---|
| Start | [Alan] | [] |
| 1 | [Brad, Connie] | [Alan] |
| 2 | [Connie, Dean, Emily] | [Alan, Brad] |
| 3 | [Dean, Emily, Faith, Gordon] | [Alan, Brad, Connie] |
| 4 | [Emily, Faith, Gordon] | [Alan, Brad, Connie, Dean] |

Continuing this process until the queue is empty gives the final traversal:

Alan, Brad, Connie, Dean, Emily, Faith, Gordon

This ordering reflects a level-by-level traversal of the tree, which is why BFS on a tree is also called level-order traversal.

To use BFS for *searching* rather than full traversal, we simply introduce a target value and stop the algorithm as soon as the target is found. Instead of visiting every node, we check each node when it is dequeued and terminate early if it matches the desired value.

For example, suppose we want to find "Faith" in the tree. BFS will examine nodes level by level: Alan, then Brad and Connie, then Dean, Emily, and Faith. As soon as Faith is encountered (we *dequeue* it from the queue), the search stops, without visiting any remaining nodes.

### 148.1.1 Pseudocode

Our steps can be reflected as pseudocode as follows:

**Algorithm 1** Breadth-First Search (BFS) - Iterative

```
procedure BFS(root, target)
    Q = empty queue
    ENQUEUE(Q, root)
    while Q is not empty do
        node = DEQUEUE(Q)
        if node = target then
            return node
        end if
        for each child of node do
            ENQUEUE(Q, child)
        end for
    end while
    return null
end procedure
```

This procedure is often used in game applications to find paths, such as maze solvers, chess algorithms, and tic-tac-toe solutions. It is also used in social network analysis to find the shortest path between two users, and in web crawling to explore the structure of websites.

## 148.2 DFS

In contrast, we can run a *depth-first search* on the original tree (see Figure 148.1). Depth-first search, as the name implies, goes as deep down the tree until reaching a leaf, and then backtracks to the next possible, unsearched path. DFS (implicitly) uses a stack to explore the graph or tree.

The steps are fairly simple:

1. Push the root node into the stack.
2. Pop a node from the stack, and mark it as visited.
3. Push the node's children onto the stack.
4. Repeat steps 2 and 3 until the stack is empty.

There are 3 orders in which we can traverse/search the deepest tree. We will look at `pre-order`, `post-order`, and `in-order`.

However, these orders essentially perform the same basic three tasks,

**V** Visit the selected node.

**L** Run DFS on the selected node's left subtree.

**R** Run DFS on the selected node's right subtree.

The recursion terminates when we reach a leaf. At this point, the algorithm returns to the previous call, which is how *backtracking* occurs. This backtracking is done via the stack we had mentioned previously.

### 148.2.1 Pre-order

In `pre-order` traversal, we visit the node *before* exploring its subtrees. The order is:

$$(V, L, R)$$

We visit the node *before* its children because the node's value is needed *prior* to exploring its subtrees. This is useful when the node represents a decision that affects which branch of the tree you go down, such as in tree copying or directory traversal like we talked about in the previous chapter.

Using our original tree, the pre-order results are:

Alan, Brad, Dean, Emily, Connie, Faith, Gordon

### 148.2.2 Post-order

In `post-order` traversal, we visit the node *after* exploring both subtrees. The order is:

$$(L, R, V)$$

We visit the node *after* its children because the node's computation depends on results from both subtrees. This is useful when combining subtree results or deleting trees, where children must be handled before the parent.

The post-order traversal of our original tree is

Dean, Emily, Brad, Faith, Gordon, Connie, Alan

### 148.2.3 In-order

In `in-order` traversal, we visit the node *between* exploring its subtrees. The order is:

$$(L, V, R)$$

We visit the node *between* its subtrees because the correct interpretation of the node depends on partially processed children. In-order only makes sense for binary trees, because it returns the *sorted order* of the binary search tree. It depends on left and right subchildren. This would be like the alphabetical order of our dictionary example from the previous chapter.

The in-order traversal of our original tree becomes

Dean, Brad, Emily, Alan, Faith, Connie, Gordon

### 148.2.4 Pseudocode

The previous sections (pre-order, in-order, and post-order) describe *when* a node is visited during a depth-first traversal.

The following algorithm instead describes the general *mechanism* of depth-first search. This implementation corresponds to a pre-order traversal. Other traversal orders (in-order and post-order) require additional state and cannot be obtained by simple reordering alone.

**Algorithm 2** Depth-First Search (DFS) - Iterative on a Tree

```
procedure DFS(root)
    S = empty stack
    Push(S, root)
    while S is not empty do
        node = Pop(S)
        visit(node)                                  ▷ V
        if node.left ≠ null then
            Push(S, node.left)                       ▷ L
        end if
        if node.right ≠ null then
            Push(S, node.right)                      ▷ R
        end if
    end while
end procedure
```

DFS is particularly useful for solving problems like connected-component detection in graphs and maze-solving.

## 148.3 Recursive vs Iterative Traversal

As we've established, trees are recursively defined structures, and as such, our algorithms for them can be defined as such too. This allows us to implement searches and traversals in two ways: *recursive* and *iterative*. Let's look at DFS recursively.

Importantly, they do not change the traversal order (pre-, in-, post-order); they only change how the traversal is implemented.

A *recursive traversal* relies on the program's *call stack* (recall from the Advanced Python Syntax chapter). Each time the algorithm explores a subtree, it makes a function call, and the system automatically remembers where to return when that call finishes. This matches the recursive definition of a tree, where each subtree is itself a tree. This allows us to get rid of the stack in DFS and solely rely on a computer's call stack.

### *Exercise 390* Recursive DFS

Take a look at the Iterative DFS code above.

How can we rewrite the pseudocode to be recursively define, taking advantage of a inherent recursive structure? Try pre-order recursively first. Assume the tree is a binary tree.

*Working Space*

*Answer on Page 2044*

Note that we can create recursive DFS just by reordering the three operations (`V`, `L`, `R`).

What about BFS? BFS is typically implemented iteratively because it relies on a queue. While recursive versions exist, they still depend on a queue and offer little conceptual advantage, so we omit them here.

#### 148.3.1 Pre-order and Post-order Numbering

There also exists pre-order and post-order numbering, regardless of which version of DFS is run. Pre-order and post-order numbering are techniques used during a depth-first search (DFS) traversal to record when each node is encountered and when it is fully processed. A *pre-order number* is assigned when the traversal first visits a node, before exploring any of its children. A *post-order number* is assigned after all of the node's descendants have been explored and the traversal is about to return to the node's parent. These numbers can be viewed as *timestamps* that describe the progress of the DFS.

Pre-order numbers capture the order in which nodes are discovered, while post-order numbers capture the order in which nodes are completed. This information is useful in many graph and tree algorithms, including dependency analysis, topological sorting (which we will talk about in the future), and determining ancestor-descendant relationships.

## *Exercise 391* DFS Pseudocode

*Working Space*

Alter the *recursive* DFS pseudocode to include an assignment of preorder and post order numbers for every node.

*Answer on Page 2045*

## *Exercise 392* Pre- and Post-order numbers

*Working Space*

Write the pre and post order numbers for the person graph we have been working with, Figure 148.1

*Answer on Page 2045*

Chapter 149

# Finding Shortest Paths

## 149.1 Introduction — BFS revisited

We have seen BFS as a way to search graphs (or trees). It is a pretty simple process:

- select a start node
- move one distance from the root, and visit all nodes ($v$) on that level
- when visiting, queue all child nodes ($v'$) of that node.
- keep going until either all nodes are marked *visited* or a provided *target node* is reached.

There is an issue here: what if the graph is weighted, and the paths are not equally weighted? For example, see this graph:

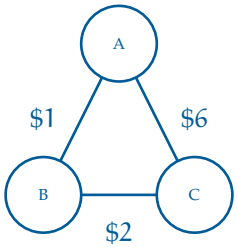



Figure 149.1: A graph of 3 nodes that would fail BFS.

BFS, here, would prefer $A \to C$, since it has an edge count of 1, whereas $A \to B \to C$ has an edge count of 2. However, the true shortest path, by weight, is $A \to B \to C$, with a weight of 3.

So BFS has a major limitation: it cannot handle weighted graphs. This is why we started it out on trees; typically trees do not have a set edge weight, so we were not limited by the order in which it expands.

To handle weighted graphs, we use a new algorithm that expands off of BFS, called *Dijkstra's Algorithm*. Dijkstra's Algorithm avoids the limitation of weights that BFS had by enqueueing the nodes in order of current shortest known distance.

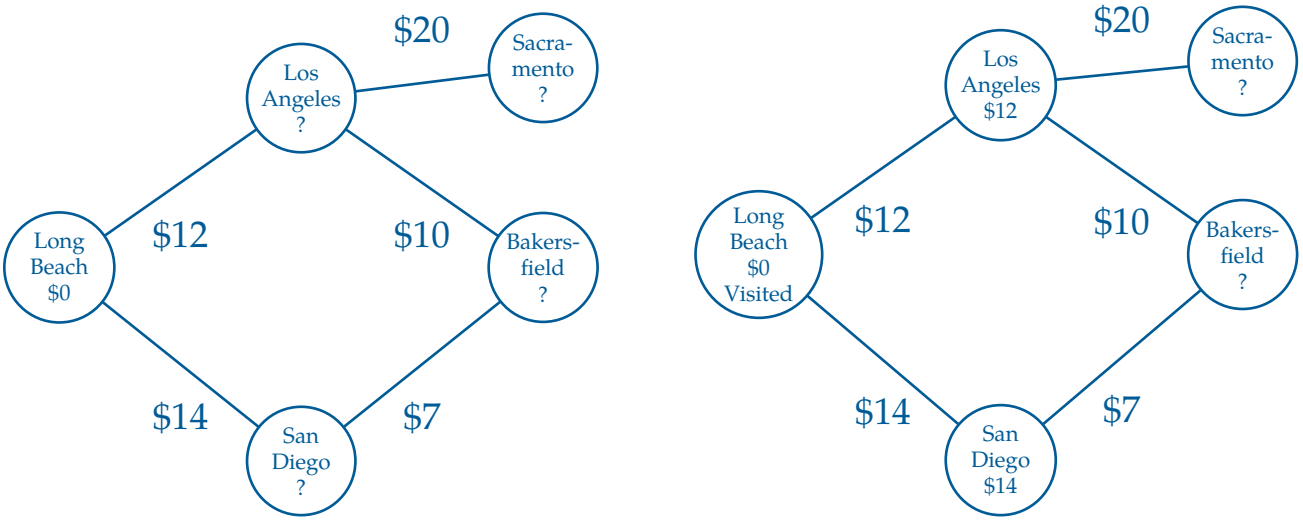
## 149.2 Dijkstra's Algorithm

Edsger W. Dijkstra (pronounced *DYKE-strah*) was a great Dutch computer scientist. He came up with an algorithm for finding the cheapest path through a graph with weighted edges. Today it is known as *Dijkstra's Algorithm*. It is used in a wide variety of common problems, and is also both simple and elegant.

### 149.2.1 Algorithm Description

You are going to mark each node with how much it would cost to get there from some chosen origin node. For example, if you are shipping a container from Long Beach, you will mark each city with the cost of getting the container to that city.

You start by marking the price for Long Beach to zero (the container is already there). You then mark each adjacent city with the cost on the edge (figure 149.2a). Next, you declare Long Beach to be "visited" (figure 149.2b).




(a) The initialized state of the graph, selecting Long Beach as our origin node.

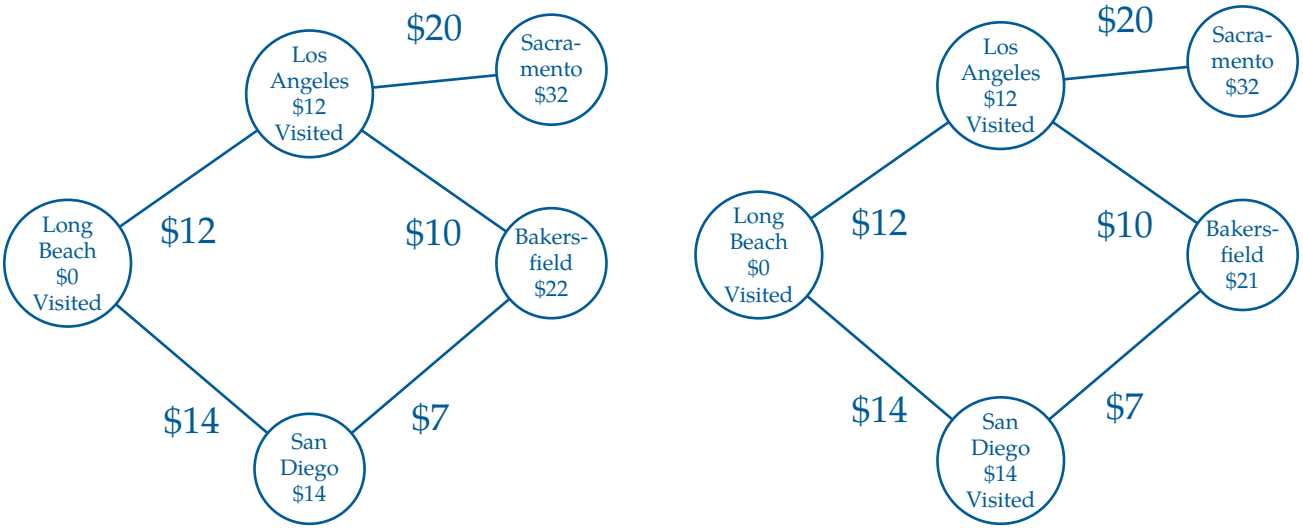
(b) Marking Long Beach as visited, we change the cost values of Los Angeles and San Diego to their respective edges.

Figure 149.2: A map of weighted cities.

After that, you find the cheapest of the unvisited nodes. In this case, Los Angeles is cheaper than San Diego, so that is the node you will visit next.

You mark all of the unvisited nodes adjacent to Los Angeles, with the price to ship it to Los Angeles plus the cost of shipping the container from Los Angeles to that city (figure

149.3a). Note that Bakersfield is marked with $22.




(a) After visiting Los Angeles.

(b) After visiting San Diego.

Figure 149.3: Visiting the next two minimum cost cities.

Now, the cheapest unvisited node is San Diego, so you mark its neighbors with the cost to ship to San Diego, plus the price to ship from San Diego to the neighbor (figure 149.3b).

Notice that Bakersfield is already labeled with $22 from a route through Los Angeles. But the price would be $21 if you shipped it to Bakersfield via San Diego. Because the new route is cheaper, you change the price to the lower value.

(What does it mean that a node is "visited"? If a node is marked visited, it is marked with a price that won't get any smaller.)

You continue visiting the cheapest unvisited node until all the nodes have been visited. Then you know every node has been marked with its lowest price.

In a big graph, each node may be marked several times in this process — each time with a lower price from a cheaper router.

Of course, once you have the price, you will ask, "What is the route that gets me that price?" So, we will also mark each node with the neighbor from which it would receive the shipment — the previous node. This is easy to do as we execute the algorithm.

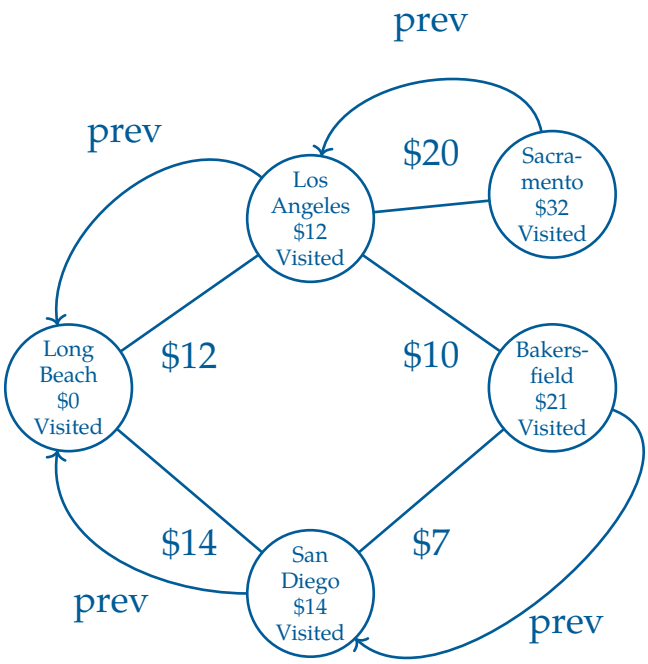



Figure 149.4: Marking each node with a previous pointer.

Now, to figure out the cheapest route from Long Beach to Bakersfield, we start at the destination and follow the `prev` pointer back through San Diego, then to Long Beach.

## *Exercise 393* Expanding on the Delivery Graph

Notice that the path between Los Angeles to Bakersfield is not used once we add in our previous labels. Why might this be?

*Working Space*

*Answer on Page 2045*

### 149.2.2 Implementation

We do not actually want to sully our graph objects with the three additional pieces of information we need:

- The current minimal cost from the origin node. This is usually called the `dist`, from "distance".
- The neighbor who gives us the current minimal cost. This is usually called `prev`, from "previous".
- Whether the city is visited or now.

So, we will keep them in collections external to the graph.

For example, to keep track of the `dist`, we will have a `dist` dictionary. Each node will be a key, the current minimal cost will be the value. If the node hasn't received even a first cost, we will put in infinity as the cost.

(After the algorithm is run, if the cost of a node is still infinity, that means that it cannot be reached from the origin node.)

We will also have a `prev` dictionary. The final node will be the key, and its previous neighbor will be the value.

Finally, the graph has a list of all the nodes, so we can just keep a set of the unvisited nodes.

Add a method to the `Graph` class that implements Dijkstra's algorithm:

```
    def cost_from_node(self, origin_node):
        # Cost of cheapest path from origin node discovered so far
        # Initially the origin is zero and all the other are infinity
        dist = {k: math.inf for k in self.nodes}
        dist[origin_node] = 0.0

        # The previous city on that cheapest path
        prev = {}

        # All the nodes start as unvisited
        unvisited = set(self.nodes)

        # While there are still unvisited nodes
        while unvisited:

            # Find unvisited node with lowest cost
            min_cost = math.inf
```

```
            for u in unvisited:
                if dist[u] < min_cost:
                    current_node = u
                    min_cost = dist[u]

            # If none are less than inf, we are done
            # This happens in graphs that are not connected
            if min_cost == math.inf:
                return (dist, prev)

            # Remove the lowest cost node from the unvisited list
            unvisited.remove(current_node)

            # Update all the unvisited neighbors
            for edge in current_node.edges:

                # What node is at the other end of this edge?
                v = edge.other_end(current_node)

                # Visited nodes are already minimized, skip them
                if v not in unvisited:
                    continue

                # Is this a shorter route?
                alt = dist[current_node] + edge.cost
                if alt < dist[v]:

                    # Update the distance and prev dicts
                    dist[v] = alt
                    prev[v] = current_node

        return (dist, prev)
```

Each time we visit a node, we have a decision to make:

$$\text{if } \text{dist}[u] + w(u, v) < \text{dist}[v] \text{ then update } \text{dist}[v]$$

This action, the core foundation of Dijkstra's, is done in lines 16-21. In Dijkstra's, the node with the minimum tentative distance can also be finalized, provided all edge weights are non-negative. Nodes left at infinity are not reachable from the chosen root node.

Append some code to your `cities.py` that tests this method:

```
(cost_from_long_beach, prev) = network.cost_from_node(long_beach)
print(f"\nMinimum costs from Long Beach = {cost_from_long_beach}")
print(f"\nLast city before = {prev}")

nyc_cost = cost_from_long_beach[nyc]

if nyc_cost < math.inf:
```

```
    print(f"\n*** Total cost from Long Beach to NYC: ${nyc_cost:.2f} ***")
else:
    print("You can't get to NYC from Long Beach")
```

When you run it, you should get a list of how much it costs to ship a container to each city from Long Beach:

```
Minimum costs from Long Beach = {(node:Long Beach, edges:1): 0.0,
(node:Los Angeles, edges:3): 12.0, (node:Denver, edges:3): 35.0,
(node:Phoenix, edges:3): 31.0, (node:Louisville, edges:3): 49.0,
(node:Cleveland, edges:4): 44.0, (node:Boston, edges:2): 57.0,
(node:New York City, edges:3): 52.0}
```

You will also get a collection of node pairs. What are these? For each node, you get the node that you would pass through on the cheapest route from Long Beach:

```
Last city before = {(node:Los Angeles, edges:3):(node:Long Beach, edges:1),
(node:Denver, edges:3):(node:Los Angeles, edges:3),
(node:Phoenix, edges:3):(node:Los Angeles, edges:3),
(node:Louisville, edges:3):(node:Denver, edges:3),
(node:Cleveland, edges:4):(node:Denver, edges:3),
(node:New York City, edges:3):(node:Cleveland, edges:4),
(node:Boston, edges:2): (node:Cleveland, edges:4)}
```

Your users will not want to read this; give them the shortest path as a list. Add a function to `graph.py` that turns the `prev` table into a path of nodes that lead from the origin to the destination:

```
def shortest_path(prev, destination):

    # Include the destination in the path
    path = [destination]
    current_node = destination

    # Keep stepping backward in the path
    while current_node in prev:

        # What node should come before the current node?
        previous_node = prev[current_node]

        # Insert it at the start of the list
        path.insert(0, previous_node)
        current_node = previous_node

    return path
```

Test that out:

```
if nyc_cost < math.inf:
    print(f"*** Total cost from Long Beach to NYC: ${nyc_cost:.2f} ***")

    path_to_nyc = graph.shortest_path(prev, nyc)
    print(f"*** Cheapest path from Long Beach to NYC: {path_to_nyc ***}")
else:
    print("You can't get to NYC from Long Beach")
```

This should look like this:

```
*** Cheapest path from Long Beach to NYC: [(node:Long Beach, edges:1),
(node:Los Angeles, edges:3), (node:Denver, edges:3), (node:Cleveland, edges:4),
(node:New York City, edges:3)] ***
\end{text}

% \subsection{Making it faster}

% On really big networks, doing a full Dijkstra's algorithm would take
% too long; luckily, there are many methods for getting similar results
% quickly.  When you ask for directions from Google Maps, it does not do
% a full Dijkstra's Algorithm for every possible route --- it would just
% take too long.

% However, there is a way to speed up this implementation. Look at this snippet:

% \begin{minted}{python}
%             # Find unvisited node with lowest cost
%             min_cost = math.inf
%             for u in unvisited:
%                 if dist[u] < min_cost:
%                     current_node = u
%                     min_cost = dist[u]
```

## 149.3 Bellman-Ford Algorithm

The Bellman-Ford Algorithm is another shortest path algorithm like Dijkstra's. However, unlike Dijkstra's Algorithm, Bellman-Ford can handle graphs with negative weight edges.

The algorithm works as follows:

1. Assign a tentative distance value for every node: set it to zero for our initial node and to infinity for all other nodes.
2. For each edge $(u, v)$ with weight $w$, if the current distance to $v$ is greater than the

distance to $u$ plus $w$, update the distance to $v$ to be the distance to $u$ plus $w$.

3. Repeat the previous step $|V| - 1$ times, where $|V|$ is the number of vertices in the graph.
4. After the above steps, if you can still find a shorter path, there exists a negative cycle.

If the graph does not contain a negative cycle reachable from the source, the shortest paths are well-defined, and Bellman-Ford will correctly calculate them. If a negative cycle is reachable, no solution exists, but Bellman-Ford will detect it.

Chapter 150

# Other Graph Algorithms and Proofs

This chapter is an overview of other graph algorithms, proofs, and concepts that are important to know. We will first cover adjacency matrices, then Prim's algorithm for minimum spanning trees, isomorphism, and topological sorting for directed acyclic graphs (DAGs). A lot of these are very advanced graph theory topics, so you won't be tested on too heavily of these topics. Many of these appear on AP/IB exams, or in advanced college level CS classes.

## 150.1 Adjacency Matrices

Recall that we can represent graphs as lists of data called adjacency lists. In code, this could look like the following:

- A: [B, C]
- B: [A, D]
- C: [A]
- D: [B]

This represents the following graph

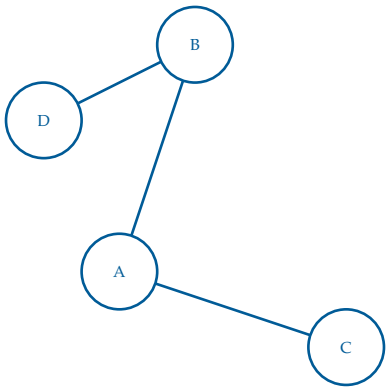



Figure 150.1: A simple (undirected unweighted) graph used to make an adjacency list and adjacency matrix.

We could also represent the same data as a *matrix*, which we have covered in linear algebra. The *adjacency matrix* is 2D array (as compared to an adjacency list, which consists of multiple 1D arrays) where `matrix[i][j]` indicates whether an edge exists from vertex $i$ to vertex $j$. Commonly, `matrix[i][j]` $= 1$ if and only if the edge from vertex $i$ to vertex $j$ exists, else `matrix[i][j]` $= 0$.[1] The above graph in Figure 150.1 can be represented as:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

For *undirected* graphs, the adjacency matrix is symmetric, and thusly $n \times n$, meaning that the transpose of matrix is the same matrix!

$$A = A^T$$

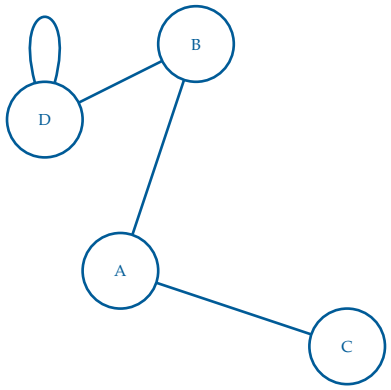



Figure 150.2: A simple graph with an added loop on vertex D.

By convention, a diagonal entry such as `matrix[k][k]`, where $(k \in \mathbb{Z})$, is non-zero only if there is a loop from a vertex to itself. The value of the entry represents the number of such loops. In this case, there are two edges from vertex (D) to itself in Figure 150.2, so the corresponding diagonal entry is (2).

For *directed* graphs, the adjacency matrix on appears a bit different. A *directed adjacency matrix* represents a directed graph, where edges have a direction (arrows). When $A_{ij} = 1$, then an edge exists from vertex $i$ to vertex $j$.

[1] If the edges are weighted, the matrix[i][j] is equal to the weight, or 0 if the edge does not exist.

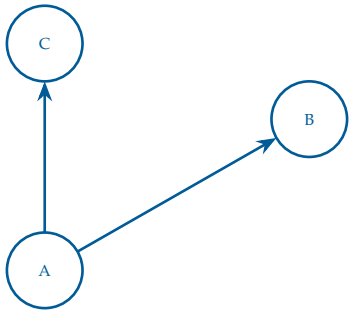



Figure 150.3: A simple directed graph used to make an adjacency matrix.

The resulting adjacency matrix is

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Notice how, now, $A \neq A^T$, because the edges are directional.

## *Exercise 394* Adjacency Matrix Exercise

*Working Space*

The following is an adjacency matrix for a weighted (undirected) graph.

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| A | 0 | 4 | 2 | 0 | 0 | 0 | 0 |
| B | 4 | 0 | 1 | 5 | 0 | 0 | 0 |
| C | 2 | 1 | 0 | 8 | 10 | 0 | 0 |
| D | 0 | 5 | 8 | 0 | 2 | 6 | 0 |
| E | 0 | 0 | 10 | 2 | 0 | 3 | 7 |
| F | 0 | 0 | 0 | 6 | 3 | 0 | 4 |
| G | 0 | 0 | 0 | 0 | 7 | 4 | 0 |

Draw the corresponding graph with weighted edges.

*Answer on Page 2046*

## 150.2 Prim's Algorithm and Kruskal's Algorithm for Minimum Spanning Trees

Prim's Algorithm is minimum spanning tree

FIXME https://github.com/KontinuaFoundation/sequence/issues/315

## 150.3 A* (A-star) Search Algorithm

The last shortest path we will discuss in this chapter is the *A-star Algorithm*, more commonly represented as *A* Algorithm*. In essence, it is an extension of Dijkstra's Algorithm in combination with a *priority-based heuristic function*. A heuristic is an introduced estimated cost from any node to the goal.

The key behind this implementation is the following function

$$f(n) = g(n) + h(n)$$

where

- $g(n)$: the cost of the path from the start node (usually $S$) to the current node $n$
- $h(n)$: the *heuristic* function that estimates the remaining cost from $n$ to the goal (usually $E$)
- $f(n) = g(n) + h(n)$: the estimated total cost of a path from $S$ to $E$ passing through node $n$

### 150.3.1 The Path Cost $g(n)$

$g(n)$ remains to be the same value that we used in Dijkstra's and other shortest path algorithms.

When moving from a node $u$ to an adjacent node $v$, the path cost is *updated* as

$$g(v) = g(u) + c(u, v)$$

where $c(u, v)$ is the cost of traversing the edge from $u$ to $v$. As we discussed in the chapter on Dijkstra's, we have a decision to make upon each visit to a node: Each time we visit a node, we have a decision to make:

$$\text{if } dist[u] + w(u, v) < dist[v] \text{ then update } dist[v]$$

### 150.3.2 The Heuristic Function $h(n)$

The heuristic $h(n)$ function depends on the problem domain. Its purpose is to estimate the remaining cost from node $n$ to the goal. While this is a difficult task, we can use basic algebra to make an informed guess.

For example, in a map-based pathfinding problem, $h(n)$ may be the straight-line distance, or *Euclidean distance* between node $n$ and the goal. Since the actual path cannot be shorter than the Euclidean distance, this provides a reasonable estimate of the remaining cost. If a nodes location is represented on a 2D plane (such as a map or graph) $(x_n, y_n)$ and the goal location as $(x_g, y_g)$, the Euclidean Heuristic

$$h(n) = \sqrt{(x_g - x_n)^2 + (y_g - y_n)^2}$$

Alternatively, many 2D games use a similar either *taxicab distance* (typically for 4-way movement),

$$h(n) = |x_g - x_n| + |y_g - y_n|$$

or *Chebyshev distance* (typically for 8-way movement in a game)

$$d = \max(|x_2 - x_1|,\ |y_2 - y_1|)$$

which takes the maximum of the horizontal and vertical movements.

Sometimes, graphs without direct spatial meaning become more difficult to work with because there is no obvious notion of distance between two nodes.

For general graphs, such as social networks, communication networks, or dependency graphs, the heuristic must be derived from problem-specific knowledge. For example, a social network application might use information such as mutual connections, community membership, or other metadata to estimate how *close* a user is to a target user.

In many cases, however, no suitable heuristic is known. In these situations, the heuristic is often defined as

$$h(n) = 0$$

for all nodes $n$. Substituting this into the A* evaluation function gives

$$f(n) = g(n)$$

which causes A* to reduce to to Dijkstra's algorithm. While this guarantees an optimal path, it loses the efficiency gains that a good heuristic can provide.

There is an important property that the Heuristic Function must satisfy:

$$h(n) \leq \text{true remaining cost}$$

When an admissible heuristic is used (such that $h(n) \leq$ true remaining cost), A* is guaranteed to find an optimal path if one exists.

### 150.3.3 The Total Estimated Distance $f(n)$ and Priority Queues

The purpose of the function

$$f(n) = g(n) + h(n)$$

is to assign a priority to each node during the search. Recall from our discussion of priority queues that each element is associated with a priority value, and the element with the highest (or lowest) priority is removed first.

In A*, the priority of a node is its estimated total distance $f(n)$. Nodes with smaller values of $f(n)$ are considered more promising because they represent paths that appear likely to reach the goal with a lower total cost.

As with Dijkstra's algorithm, A* maintains a priority queue containing nodes that have been discovered but not yet fully explored. However, instead of prioritizing nodes solely by their known distance from the start node, A* prioritizes nodes according to their estimated total distance

$$\text{priority}(n) = f(n)$$

At each step, the node with the smallest $f(n)$ value is removed from the priority queue and explored. When a neighboring node is discovered, its path cost $g(n)$ is updated, a new heuristic estimate $h(n)$ is calculated, and a new value of $f(n)$ is computed.

There are some good references about this in your digital resources, as well as a fun Minecraft-based explanation here.

## 150.4 Injection, Surjection, Bijection, and Isomorphism

Let's talk about Graph Isomorphism. To do this, we must first introduce the terms of *injection*, *surjection*, and *bijection*. These are types of functions in set theory as well as mathematics.

**Definition 1** (Injection)**.** *A function $f : A \to B$ is injective if different inputs always produce different outputs. Importantly, this means no two integers map to the same output.*

**Definition 2** (Surjection)**.** *A function $f : A \to B$ is surjective if every element of B is hit by at*

*least one element of A. In essence, all of the elements of B are covered.*

**Definition 3** (Bijection)**.** *A function* $f : A \to B$ *is bijective if it is both injective and surjective. Ultimately, the function creates no collisions and hits all functions of B.*

## *Exercise 395* Bijective, Surjective, Injective Exercise

*Working Space*

Determine whether each function is injective, surjective, bijective, or none of these.

1. $f : \{1, 2, 3\} \to \{a, b, c, d\}$ defined by

   $f(1) = a, \quad f(2) = b, \quad f(3) = c.$

2. $g : \{1, 2, 3, 4\} \to \{a, b, c\}$ defined by

   $g(1) = a, \quad g(2) = b, \quad g(3) = c, \quad g(4) = a.$

3. $h : \{1, 2, 3\} \to \{a, b, c\}$ defined by

   $h(1) = b, \quad h(2) = c, \quad h(3) = a.$

*Answer on Page 2046*

Now, let's talk about *Isomorphism*. A *graph isomorphism* is a bijection between the vertex sets of two graphs that preserves adjacency. Thus, an isomorphism not only matches vertices one-to-one, but also preserves the structure of the graph.

**Definition 4** (Graph Isomorphism)**.** *Let* $G = (V_G, E_G)$ *and* $H = (V_H, E_H)$ *be graphs. An isomorphism from* $G$ *to* $H$ *is a bijection*

$$f : V_G \to V_H$$

*such that for all vertices* $u, v \in V_G$*,*

$$\{u, v\} \in E_G \quad \iff \quad \{f(u), f(v)\} \in E_H.$$

*If such a function exists, we say that* $G$ *and* $H$ *are isomorphic and write*

$$G \cong H.$$

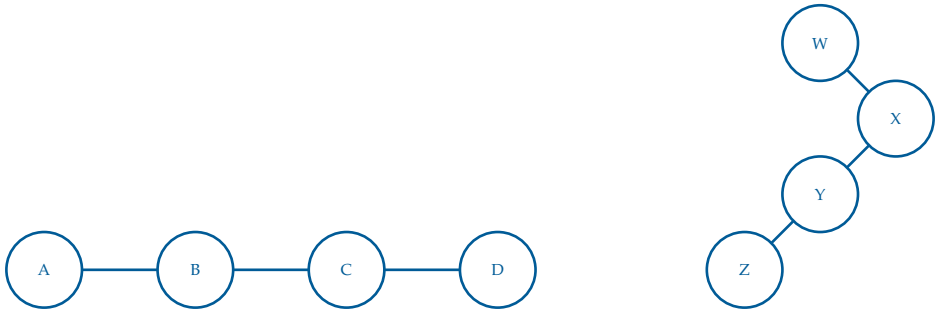



Figure 150.4: Two graphs that are isomorphic. Graph G (left) and Graph H (right).

We are able to say that these graphs (Figure 150.4) are isomorphic as one vertex from G can be paired with one vertex from H, while maintaining the same edge connections.

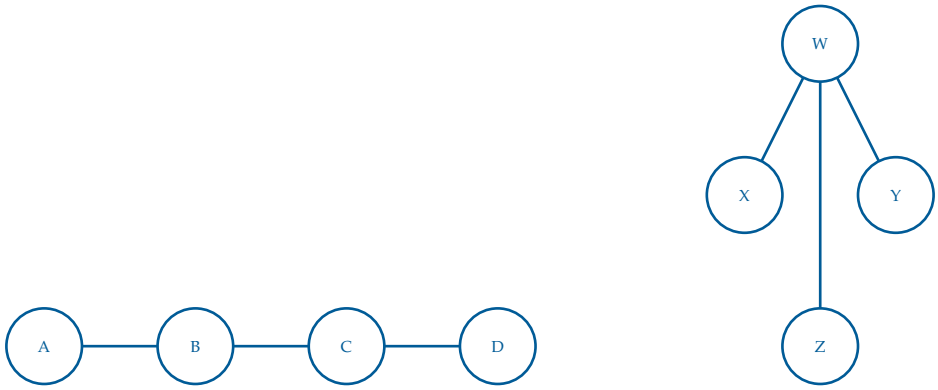



Figure 150.5: Graph X and Y; not isomorphic.

In Figure 150.5, we can see the vertices are not isomorphic without adding or subtracting edges. For example, Graph X does not have a vertex with 3 edges, unlike Graph Y.

If any of these differ, the graphs are not isomorphic:

- Number of vertices
- Number of edges
- Degree sequence (if degree sequences are intact, it does not guarantee isomorphism)
- Number of connected components
- Number of cycles of a given length

- Chromatic number

**Example.**

A common example that you will not be tested on but is important to know is the following. We may need to introduce a definition first.

**Definition 5** (Clique)**.** *A clique is a set of vertices in a graph such that every pair of distinct vertices is connected by an edge.*

**Definition 6** (Clique Subgraph)**.** *A* $k$*-clique is exactly a subgraph isomorphic to the complete graph* $K_k$*. A 3-clique would be a clique of size 3, called* $K_3$*.*

Now, given two graphs $G$ and $H$, how can we determine $G$ contain a subgraph isomorphic to $H$, a graph of smaller to equal size to $G$?

This may be easy if $G$ and $H$ are pretty small, but the math becomes infinitely harder as the number of vertices and edges go up.

We can simplify this by transforming the inputs of a clique problem to inputs of the subgraph isomorphism problem.

The clique problem gives us a graph $M$ and a threshold $k$. The output of this is whether $M$ contains a clique of size at least $k$.

To transform this into an instance of the subgraph isomorphism problem, construct a smaller subgraph $H = K_k$, the complete graph on $k$ vertices, and let $G = M$, the larger graph.

We now ask the subgraph isomorphism question:

> Does $G$ contain a subgraph isomorphic to $H$?

Since $H$ is the complete graph $K_k$, a subgraph of $G$ isomorphic to $H$ must contain $k$ vertices where every pair of vertices is connected by an edge. By definition, this is exactly a $k$-clique.

Therefore,

$$M \text{ contains a } k\text{-clique} \iff G \text{ contains a subgraph isomorphic to } K_k.$$

This shows that every instance of the Clique problem can be transformed into an instance of the Subgraph Isomorphism problem. We call this an *algorithmic reduction*.[2]

[2] Algorithmic Reductions are the basis of a set of problems called P, NP, and NP-Complete. See this video

## 150.5 Travelling Salesperson Problem and Nearest Neighbour

FIXME https://github.com/KontinuaFoundation/sequence/issues/312

## 150.6 Topological Sort and DAGs

Another common use of graph algorithms is sorting them in a meaningful order. Specifically, we can sort directed graphs that tells us exactly what order vertices need to be "visited" or operated on. We call this a *Topological Sorting* of the graph. A *topological order* is a linear ordering of the vertices in a graph such that for every directed edge $u \to v$ from vertex $u$ to vertex $v$, vertex $u$ comes before vertex $v$ in the ordering. The most common usage for this is scheduling jobs, queueing tasks with dependencies, or creating tentative project schedules.

To do this, we are going to use Python's `Graphlib`, and its `TopologicalSorter` module.

In the `graphlib` module, graphs are defined in the following format:

```
{
  node: {predecessors}
}
```

A basic example of this is the following

```
{
  "C": {"A", "B"}
}
```

Which says that A and B need to come before C, such that one possible ordering is

$$A \to B \to C$$

Now suppose we alter the graph format as follows:

```
{
  "C": {"A", "B"},
  "A": {"Z"},
  "B": {"Z"},
  "Z": {"C"}
}
```

for more.

Here,

- C depends on A and B
- A and B depend on Z
- Z depends on C

There is no way to determine a *starting point*. Now we have an ordering that contains a cycle. This implies that C must come before itself, which is impossible in a linear ordering. Therefore no topological ordering can exist.

**Definition 7** (Cycle)**.** *A cycle is a valid walk* $(v_1, v_2, \ldots, v_n)$ *where the only repeated vertex is* $v_n = v_1$*, meaning the last vertex equals the first vertex.*

So, we cannot do a topological sort on a graph that contains a cycle. This provides us with a theorem:

**Topological Sorting and DAGs**

A directed graph admits a topological ordering if and only if it is a *Directed Acyclic Graph* (DAG). That is, the directed graph may not contain any cycles (acyclic).

In fact, the python graphlib library will not produce a valid Topological Sort if a cycle exists. Running the following code

```
from graphlib import TopologicalSorter
graph = {
  "C": {"A", "B"},
  "A": {"Z"},
  "B": {"Z"},
  "Z": {"C"}
}
list(TopologicalSorter(graph).static_order())
```

will produce the following error

```
graphlib.CycleError: ('nodes are in a cycle', ['C', 'Z', 'A', 'C'])
```

The exception reports one cycle that prevents the ordering from being constructed. In this case, $C \to Z \to A \to C$ forms a directed cycle.

We can also introduce tasks that can be done in parallel. Create the following file `parallel_dag.py`

```
from graphlib import TopologicalSorter
tasks = {
    "Compile": set(),
    "Build Frontend": {"Compile"},
    "Build Backend": {"Compile"},
    "Test": {"Build Frontend", "Build Backend"},
    "Deploy": {"Test"},
}from graphlib import TopologicalSorter
tasks = {
    "Compile": set(),
    "Build Frontend": {"Compile"},
    "Build Backend": {"Compile"},
    "Test": {"Build Frontend", "Build Backend"},
    "Deploy": {"Test"},
}
from graphlib import TopologicalSorter

ts = TopologicalSorter(tasks)
ts.prepare()

while ts.is_active():
    ready = ts.get_ready()
    print(ready)

    for task in ready:
        ts.done(task)

```

The 5 nodes here are tasks for a basic software project. The ordering is as follows:

$$\text{Compile} \rightarrow \text{Build Frontend and Build Backend} \rightarrow \text{Test} \rightarrow \text{Deploy}$$

However notice that *Build Frontend* and *Build Backend* are not dependent on each other, so they could be done in *parallel*. The *Test* task cannot begin until both builds have finished. Running this code outputs the following

```
('Compile',)
('Build Frontend', 'Build Backend')
('Test',)
('Deploy',)
```

showing us that our sort outputs which ones can happen at the same time. In software development, many tasks have dependencies but can still be executed in parallel when those dependencies are satisfied. Topological sorting helps determine both the required execution order and which tasks may be performed concurrently (this is similar to running circuits in parallel!).

## 150.7 Proofs

This section will be a set of proofs or algorithm design problems for graph-related concepts that are important to know. At the basis level, these are just practice problems for graph understand, but can be applied in many real world scenarios.

### 150.7.1 Example 1: Bipartite

**Bipartite Graph**

A *Bipartite Graph* is a graph $G = (V, E)$ such that $V$ can be partitioned into two sets $(V = V_1 \cup V_2)$ and $(V_1 \cap V_2 = empty)$ such that there are no edges between vertices in the same set. FIXME Diagram

**Theorem 14.** *An undirected graph* $G$ *is bipartite if and only if it contains no cycles of odd length.*

### *Exercise 396* Algorithm for Determining Bipartite

*Working Space*

Design pseudocode for an algorithm that determines whether or not an undirected graph is bipartite. *Hint: use the theorem.*

*Hint 2: This function may be useful*

`FindOdd`$(G, e)$

**Require:** $G$, a graph on which DFS has been run; $e = (x, y)$, a back edge
**Ensure:** **true** if $e$ creates a cycle of odd length, **false** otherwise

```
1: if pre(x)−post(y) ≡ 0 (mod 2) then
2:     return true
3: else
4:     return false
5: end if
```

*Answer on Page 2047*

While you don't have to go through it on your own, here is the outline of how we can do a proof by contradiction for a graph, specifically proving theorem 14.

*Proof.* First, suppose G is bipartite. Then its vertices can be partitioned into two sets $V_1$ and $V_2$, with every edge going between $V_1$ and $V_2$. Any cycle in G must alternate between vertices in $V_1$ and vertices in $V_2$. Therefore, after an odd number of edges, the cycle would end in the opposite set from where it started, so it cannot return to its starting vertex. Hence every cycle has even length.

Conversely, suppose G contains no cycles of odd length. Choose any connected component of G, and pick a vertex $s$. Put each vertex at even distance from $s$ in $V_1$, and each vertex at odd distance from $s$ in $V_2$. We claim this gives a valid bipartition.

Assume for contradiction that there is an edge $(u, v)$ where $u$ and $v$ are in the same set. Then $u$ and $v$ have distances from $s$ with the same parity. Taking shortest paths from $s$ to $u$ and from $s$ to $v$, together with the edge $(u, v)$, creates a cycle of odd length. This contradicts the assumption that G has no odd cycles.

Thus no edge connects two vertices in the same set, so each connected component is bipartite. Applying this construction to every connected component of G, the whole graph G is bipartite.

Therefore, G is bipartite if and only if it contains no cycles of odd length. □

### 150.7.2 Example 2 and 3. Cycle Detection and Leaves

## *Exercise 397* Cycle Detection

*Working Space*

Design pseudocode for an algorithm which takes as an input an undirected graph, G, and an edge, $e = (u, v)$, and determines whether G has a cycle containing $e$.

*Answer on Page 2047*

## *Exercise 398* Leaf Removal

**Proof the following statement:** In any connected undirected graph, there is a vertex whose removal leaves the graph connected.

*Working Space*

*Answer on Page 2047*

Sometimes, we may need to create new graphs as counterexamples to disprove a statement. Consider the following:

**Theorem 15.** *An graph* G *is strongly connected if every vertex is reachable from every other vertex.*

Another way to say this is that a select vertex of the graph has a path to every other vertex on the graph. Often, this comes up in finding *subgraphs* of graphs that are strongly connected, called *strongly connected components*.

## *Exercise 399* Strongly Connected Graphs

Create an example of a strongly connected directed graph $G = (V, E)$ such that, for every $v \in V$, removing $v$ from G leaves a directed graph that is not strongly connected.

*Hint: the simplest answer is usually the best. Start with one vertex.*

*Working Space*

*Answer on Page 2048*

Chapter 151

# Making Web Requests with HTTP

The Hypertext Transfer Protocol (HTTP) is the protocol used for transmitting hypertext over the World Wide Web. It is the foundation of any data exchange on the web, and it is a protocol used for transmitting hypertext requests from clients (like a user's browser) to servers, which respond with the requested resources.

## 151.1 HTTP Requests

An HTTP request is made up of several components:

- **Method**: The HTTP method, like GET (retrieve data), POST (send data), PUT (update data), DELETE (remove data), and so on.
- **URL**: The URL of the resource to retrieve, send data to, update or delete.
- **Headers**: Additional information about the request or response, like the content type of the body.
- **Body**: The body of the request, used when sending data in POST or PUT requests.

One way to think about an HTTP request is to imagine ordering food at a restaurant. The method is the type of action you want to take, such as asking to see the menu, placing an order, changing an order, or canceling it. The URL is like telling the server exactly which restaurant counter to go to. The headers are like the extra details you give, such as saying you want vegetarian food or that you would like water with your meal. The body is the actual order information, such as the list of food you want to buy.

The method tells the server what kind of action the client wants to perform. For example, a GET request is used when the client wants to receive information from the server, such as loading a web page or asking for a list of items. A POST request is used when the client wants to send new information to the server, such as creating an account or submitting a form. A PUT request is used to update something that already exists, and a DELETE request is used to remove something.

The URL works like an address. It tells the request where to go. If the method explains what the client wants to do, the URL explains where that action should happen. For

example, if a web app wants a list of users, it might send a request to a URL that ends in `/users`. If it wants information for one specific user, it might send a request to a URL like `/users/5`.

Headers help the client and server understand each other better. They carry extra information along with the request or response, such as what type of data is being sent. This is similar to writing "Fragile" on a package or adding a note that says "Please handle with care." The message is still the same package, but the note helps explain how it should be handled.

The body is the part of the request that contains the actual data being sent. For example, when a user fills out a sign-up form, the name, email address, and password they enter may be placed in the body of the request. Not every request needs a body. GET requests usually do not send one, because they are mainly asking to receive information rather than send it.

## 151.2 Using HTTP with Web-Based APIs

Software developers often use HTTP to interact with web-based APIs. An Application Programming Interface (API) is a set of rules that allows programs to talk to each other. The developer creates the API on the server and allows the client to talk to it.

When a developer makes a request to an API endpoint, they're asking the server to either send them some data or receive some data from them. The response from the server will often be in a format like JSON or XML, which the developer can then use in their own application.

For example, a developer might make a GET request to `https://api.example.com/users` to retrieve a list of all users. The server would respond with a list of users in a format like JSON.

An API endpoint is a specific URL that a client can use to communicate with a server. Each endpoint usually has a clear purpose. One endpoint might return a list of books, while another might return information about a single book. In the same way that a restaurant has different counters for ordering food, paying, or picking up an order, an API can have different endpoints for different tasks.

When a client sends a request to an API, the server reads the request and decides how to respond. If the request is valid, the server sends back a response with the requested data or a message saying the action worked. If something is wrong, the server sends back a response explaining that there was a problem. This back-and-forth communication is one of the main ways web applications work behind the scenes.

For example, imagine a school website that stores a list of students. A client could send a GET request to an endpoint like `/students` to ask for the full list. It could send a

POST request to the same endpoint to add a new student. It could send a PUT request to `/students/3` to update the information for one student, or a DELETE request to `/students/3` to remove that student from the list. These requests follow the same HTTP ideas, even if the website or app itself looks very different on the screen.

Many modern apps work this way. A weather app may send a request to get the latest forecast. A music app may send a request to get a list of songs. A shopping app may send a request to place an order. Even though users may only see buttons, menus, and pictures, HTTP requests are often moving data in the background.

## 151.3 HTTP Responses

After a server receives an HTTP request, it sends back an HTTP response. The response is the server's answer to the client. Just as a request has several parts, a response also has its own important parts.

An HTTP response usually includes a status code, headers, and sometimes a body. The status code tells the client what happened. The headers give extra information about the response. The body contains the returned data, if there is any.

A simple way to think about a response is to imagine asking a librarian for a book. The librarian might say, "Here is the book you asked for," "That book is checked out," or "We do not have that book." These are different kinds of responses to the same kind of request.

Some common HTTP status codes are:

- 200: The request worked, and the server is sending back the requested data.
- 201: The request worked, and the server created something new.
- 400: The request had a problem, such as missing or incorrect information.
- 404: The server could not find what was requested.
- 500: Something went wrong on the server.

These codes help the client understand the result quickly. Instead of reading a long message first, the client can look at the status code and get a fast summary of what happened.

## 151.4 Data Formats in Responses

When a server sends data back to a client, that data needs to be organized in a way both sides can understand. Two common formats are JSON and XML. These formats are simply ways of arranging information so that programs can read it clearly and consistently.

JSON is one of the most common formats used with web APIs. It is popular because it is lightweight and easy for both humans and programs to read. A JSON response might include data written as key–value pairs. For example, a user might be described with data such as a name, age, and email address.

Here is a small example of JSON:

```
{
  "name": "Ava",
  "grade": 10,
  "club": "Robotics"
}
```

In this example, `"name"`, `"grade"`, and `"club"` are the labels, and `"Ava"`, `10`, and `"Robotics"` are the values. This is similar to filling out an information card. Each label tells what kind of information is being stored, and each value is the actual information.

XML is another format that may be used. It organizes information with opening and closing tags. While it is less common in many modern beginner examples, it is still useful to know that different APIs may return data in different formats.

# Chapter 152

# Using and Creating APIs

As a software engineer, you are likely familiar with building applications that interact with various external services and data sources. One of the most common methods for communication and integration is through HTTP APIs (Application Programming Interfaces). HTTP APIs provide a standardized way for applications to exchange data and functionality over the internet.

This chapter will introduce you to the world of HTTP APIs and explore how you can leverage them in your software development projects. We will cover the fundamental concepts, techniques, and best practices for effectively working with HTTP APIs.

An HTTP API allows two software systems to communicate and exchange information using the Hypertext Transfer Protocol (HTTP). It enables your application to make requests to an API server and receive responses in a structured format, such as JSON (JavaScript Object Notation) or XML (eXtensible Markup Language).

Using HTTP APIs offers a range of benefits for software engineers. It allows you to leverage external services and data sources, enabling your application to access functionality or retrieve valuable information from third-party systems. This opens up opportunities for integration with popular platforms, social media networks, payment gateways, geolocation services, and much more.

Throughout this chapter, we will explore various aspects of working with HTTP APIs, including:

- API endpoints and methods: Understanding how to interact with an API involves identifying the available endpoints and the supported methods, such as GET, POST, PUT, DELETE, and so on. We will discuss how to construct API requests and handle different response formats.
- Authentication and authorization: Many APIs require authentication to ensure secure access and protect sensitive data. We will delve into different authentication mechanisms, including API keys, tokens, OAuth, and other authentication protocols commonly used in API integrations.
- Request parameters and payloads: APIs often accept additional parameters or payloads to customize the request or send data for processing. We will explore how to pass query parameters, request headers, and request bodies when interacting with APIs.
- Error handling and status codes: Learning how to handle errors and interpret status

codes returned by APIs is crucial for building robust and resilient applications. We will discuss common status codes and best practices for handling various scenarios gracefully.

- Rate limiting and throttling: Many APIs impose restrictions on the number of requests you can make within a given timeframe to prevent abuse and ensure fair usage. We will cover techniques for handling rate limiting and implementing efficient strategies to manage API quotas.
- API documentation and testing: Proper documentation is essential for understanding an API's capabilities, endpoints, and expected behavior. We will explore how to read and interpret API documentation, as well as techniques for testing and validating API integrations.

By mastering the art of using HTTP APIs, you will expand your development toolkit and gain the ability to seamlessly integrate your applications with external services, leverage their functionalities, and build powerful, interconnected systems.

So, let's dive into the world of HTTP APIs and uncover the endless possibilities they offer for enhancing your software engineering projects.

## 152.1 Setting up an API

Create a file called `server.js`. This will be a basic webserver that we will use to test our an API. This is called a *RESTful API*. Write the following code to create a basic server for our API.

```
const express = require("express");

const app = express();
const PORT = 3000;

const student = {
  student_id: 42,
  name: "John Smith",
  major: "Computer Science",
  credits_completed: 88
};

app.get("/students/42", (req, res) => {
  res.json(student);
});

app.listen(PORT, () => {
  console.log(`API running at http://localhost:${PORT}`);
});
```

Then, run the following

```
npm init -y
npm install express
node server.js
```

This will create a new server, along with a ton of dependency files that javascript and node automatically install as part of your server.

### 152.1.1 What is an endpoint?

An *endpoint* is a URL exposed by the API. It usually is able to display a page or data based on the request of the endpoint.

We can navigate to the endpoint `GET /students/42`, where students is the resource that usually maps to a database, and 42 is the id of a student in the database. To retrieve this student from the database, we can navigate to

```
GET http://localhost:3000/students/42
```

which is an endpoint from our API. This will output

```
{
  "student_id": 42,
  "name": "John Smith",
  "major": "Computer Science",
  "credits_completed": 88
}
```

Note that this is done in json format, common for backend data parsing.

Each endpoint combines a resource path with an HTTP method to complete an action.

This leads us to the idea of *CRUD operations*. CRUD, which stands for CREATE, REMOVE, UPDATE, DELETE, are a form of outlining an API's routes. Calling a route that aims to alter should follow one of the CRUD operations. GET does not have an operation since it is *read-only*.

For a database representing students (assume more than just one), we would want to have routes that can complete each CRUD operation.

Before an API can receive data from a client, Express must parse JSON request bodies.

| HTTP Method | Purpose | CRUD Op | Example |
|---|---|---|---|
| GET | Fetch Data | N/A | Get student information |
| PUT | Update Data | UPDATE | Fully update student |
| PATCH | Update section of Data | UPDATE | Update a student's major |
| POST | Create Data | CREATE | Create a new student |
| DELETE | Remove Data | DELETE | Remove a student after graduation |

Add the following line after creating the application:

```
app.use(express.json());
```

This middleware automatically converts incoming JSON into a JavaScript object that can be accessed through `req.body`.

### 152.1.2 Reading all resources

Most APIs provide a route for retrieving an entire collection. Change the students array to look like the following.

```
const students = [
  {
    student_id: 42,
    name: "John Smith",
    major: "Computer Science",
    credits_completed: 88
  },
  {
    student_id: 43,
    name: "Jane Doe",
    major: "Mathematics",
    credits_completed: 61
  }
];

app.get("/students", (req, res) => {
  res.json(students);
});
```

The endpoint

```
GET /students
```

returns every student currently stored in the collection. Notice that we did not include an id. Now, there are more than one possible ids, so we need to make a general route that takes in an id of a student to fetch

```
app.get("/students/:id", (req, res) => {
  // use the id that is passed in the route
  const id = Number(req.params.id);
  // find a student from our students array
  const student = students.find((s) => s.student_id === id);

  // if the id is not found, we assume the student is not in the array
  if (!student) {
    return res.status(404).json({ error: "Student not found" });
  }

  res.json(student);
});
```

Start the server. Now, `http://localhost:3000/students` returns the entire students array, while `http://localhost:3000/students/43` only returns information about Jane Doe.

Routing to `http://localhost:3000/students/19` is invalid, so it returns

```
{"error":"Student not found"}
```

## 152.2 Requests with cURL

Curl, sometimes stylized as `cURL` or `curl`, is a command-line based resource used for sending API and HTTP requests for web server data.

The simplest curl command

```
curl http://example.com
```

which will return the `html` source code of the website `http://example.com`.

```
<!doctype html>
<html lang="en">

<head>
    <title>Example Domain</title>
    <link rel="icon" href="data:,">
    <meta name="viewport" content="width=device-width, initial-scale=1">
```

```
    <style>
        body {
            background: #eee;
            width: 60vw;
            margin: 15vh auto;
            font-family: system-ui, sans-serif
        }

        h1 {
            font-size: 1.5em
        }

        div {
            opacity: 0.8
        }

        a:link,
        a:visited {
            color: #348
        }
    </style>
</head>

<body>
    <div>
        <h1>Example Domain</h1>
        <p>This domain is for use in documentation examples without needing
        ↪  permission. Avoid use in operations.</p>
        <p><a href="https://iana.org/domains/example">Learn more</a></p>
    </div>
</body>

</html>
```

## 152.3 Updating and Deleting Resources with cURL

So far our API can read students and create new ones. To finish out the CRUD table from earlier, we need routes for `PUT`, `PATCH`, and `DELETE`, and we need to know how to trigger each of them with curl.

### 152.3.1 Fully replacing a resource with PUT

A `PUT` request means *replace this entire resource with what I'm sending you.* If you leave out a field, that field is expected to disappear or reset, because the whole object is being overwritten.

```
app.put("/students/:id", (req, res) => {
  const id = Number(req.params.id);
  const index = students.findIndex((s) => s.student_id === id);

  if (index === -1) {
    return res.status(404).json({ error: "Student not found" });
  }

  // PUT replaces the whole object, so we overwrite it completely
  students[index] = {
    student_id: id, ...req.body // this format keeps the same student_id and
    ↪  updates the rest of the fields with the new data
  };
  res.json(students[index]);
});
```

```
curl -X PUT http://localhost:3000/students/43 \
  -H "Content-Type: application/json" \
  -d '{
      "name": "Jane Doe",
      "major": "Physics",
      "credits_completed": 64
    }'
```

Notice we did not send `student_id` in the body – the route pulls it from the URL instead. Try leaving out `credits_completed` entirely and re-running the request. Because PUT replaces the whole object, that field will not be included in the response. This is the behavior that distinguishes PUT from PATCH.

### 152.3.2 Partially updating a resource with PATCH

A `PATCH` request means *only change the fields I'm sending you, and keep everything else the same.* This is almost always what you want when, say, updating a single field like a student's major.

```
app.patch("/students/:id", (req, res) => {
  const id = Number(req.params.id);
  const student = students.find((s) => s.student_id === id);

  if (!student) {
    return res.status(404).json({ error: "Student not found" });
  }

  // PATCH merges the incoming fields onto the existing object
  Object.assign(student, req.body);
```

```
  res.json(student);
});
```

```
curl -X PATCH http://localhost:3000/students/43 \
  -H "Content-Type: application/json" \
  -d '{ "major": "Physics" }'
```

Run a `GET` on `/students/43` afterward and confirm that `name` and `credits_completed` are untouched; only `major` changed. That single-field behavior is the whole point of PATCH.

### 152.3.3 Removing a resource with DELETE

A `DELETE` request removes a resource. Unlike POST, PUT, and PATCH, it typically does not need a request body; the URL alone tells the server what to remove.

```
app.delete("/students/:id", (req, res) => {
  const id = Number(req.params.id);
  const index = students.findIndex((s) => s.student_id === id);

  if (index === -1) {
    return res.status(404).json({ error: "Student not found" });
  }

  students.splice(index, 1);
  // status 204 means "success, but there is nothing to send back"
  res.status(204).send();
});
```

```
curl -X DELETE http://localhost:3000/students/44
```

Run `GET /students` again and Jerry Doe should be gone. If you run the same `DELETE` command a second time, you should get a `404`, since the student no longer exists to delete.

### 152.3.4 Headers with `-i` and `-v`

Every curl command so far has only shown you the body of the response. But remember: every HTTP exchange also includes headers and a status code, even when curl hides them by default. Two flags make that conversation visible.

`-i` includes the response headers above the body:

```
curl -i http://localhost:3000/students/43
```

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8
Content-Length: 78

{"student_id":43,"name":"Jane Doe","major":"Physics","credits_completed":64}
```

`-v` (verbose) goes further and shows the *request* curl sent, not just the response it received – useful when a request is failing and you aren't sure what curl is actually sending:

```
curl -v -X PATCH http://localhost:3000/students/43 \
  -H "Content-Type: application/json" \
  -d '{ "major": "Physics" }'
```

Reading a `-v` output line by line is one of the best habits a beginner can build: lines starting with `>` are what curl sent, lines starting with `<` are what the server sent back.

## 152.4 Authentication and API Keys

Every route we've built so far is completely open: anyone who knows the URL can read, create, update, or delete students. Real APIs usually can't work that way, since that data might be private, or changing it might cost money (imagine n API that sends text messages or charges a credit card). To restrict access, APIs commonly require an *API key*: a long, unique string that identifies who is making the request.

An API key is typically sent as a request header, most often `Authorization` or a custom header like `X-API-Key`:

```
curl http://localhost:3000/students \
  -H "Authorization: Bearer abc123yourkeyhere"
```

On the server, you check for that header before doing anything else:

```
const API_KEY = "abc123yourkeyhere";

function requireApiKey(req, res, next) {
  const authHeader = req.headers.authorization;
```

```
5
6   if (authHeader !== `Bearer ${API_KEY}`) {
7     return res.status(401).json({ error: "Unauthorized" });
8   }
9
10  next();
11 }
12
13 app.get("/students", requireApiKey, (req, res) => {
14   res.json(students);
15 });
```

If the header is missing or wrong, the client gets back a `401 Unauthorized` instead of the data.

A few rules are worth internalizing early, because they matter far beyond this textbook:

- **Never commit an API key to a public repository.** Once a key is pushed to code repository such as GitHub, assume it is compromised, even if you delete it in a later commit. If you do accidentally commit a key, revoke it immediately and generate a new one.
- **Never hardcode a real key directly in your source code.** The example does for simplicity. Real projects store keys in environment variables (often loaded from a `.env` file that is excluded from version control or public repositories) and read them with something like `process.env.API_KEY`.
- **Treat an API key like a password.** Anyone who has your key can act as you. They could be reading your data, spending your quota, or in some APIs, spending your money.
- **You can only access API's with their specific key.** If you have multiple API's, each one will have its own key. You cannot use the same key for different API's. For example, Spotify's public API has its own key, and a method for you to generate your own key. You cannot use a key from another API (such as OpenAI) to access Spotify's API. The strength here is that if one key is compromised, it does not compromise all of your API's. Secondly, this way, if a breach of data is attempted, Spotify knows who owns the key and can revoke it. Feel free to learn more about Spotify's API and how to generate your own key at https://developer.spotify.com/documentation/web-api/. Becoming increasingly more common is OpenAI's API, which has its own key and method for generating your own key. You can learn more about OpenAI's API at https://platform.openai.com/docs/guides/authentication.

This is the difference between *authentication* (proving *who* you are, usually with a key or token) and *authorization* (determining *what* you're allowed to do once you've proven who you are). A key can identify you correctly and still be denied access to a specific route if

you don't have permission for it.

## 152.5 Creating resources with POST

That's all great, but we can put cURL to work on much more laborious tasks. Going back to our created API example, we can post, update, and delete resources using cURL.

A POST request creates a new resource. Unlike GET requests, POST requests send data inside the request body.

```
app.post("/students", (req, res) => {
  const student = req.body;

  // basic validation: ensure a body is received
  if (!student) {
    return res.status(400).json({ error: "Invalid student data" });
  }

  students.push(student);

  // status 201 means that the student was pushed successfully
  res.status(201).json(student);
});
```

Notice the use of

```
req.body
```

which contains the JSON sent by the client.

HTTP requests are composed of headers and a body. Headers are metadata such as `Content-Type` (most commonly application/json) or `Authorization` that describe how the request should be handled. The body is the payload that carries actual data, like the student record being created by a POST request.

In Express, request headers are available on `req.headers`. For JSON payloads, middleware like `express.json()` parses the body so that the data is available on `req.body`.

A template for sending json in the body looks something like the following:

```
curl -X METHOD URL \
     -H "Header: Value" \
     -d 'JSON DATA'
```

To send a request body, include the body data in the HTTP request and set the appropriate `Content-Type` header. For example, with curl:

```
curl -X POST http://localhost:3000/students \
  -H "Content-Type: application/json" \
  -d '{
      "student_id": 44,
      "name": "Jerry Doe",
      "major": "Computer Science",
      "credits_completed": 23
    }'
```

This can be broken down as:

- `curl`: The command-line tool for making HTTP requests.
- `-X POST`: Use the POST HTTP method.
- `http://localhost:3000/students`: The endpoint being contacted. Must exist in our server routes.
- `-H "Content-Type: application/json"`: Tell the server the request body contains JSON.
- `-d '{ ... }'`: Send this JSON as the request body.

The request body in this example is the JSON string passed with `-d`. Express will parse that JSON into `req.body` when `express.json()` middleware is enabled.

Note that our server.js students array will not update with Jerry Doe, because the server only stores the students array upon initialization, but we can see our now three students by running a GET request on `/students`

```
[
  {
    "student_id": 42,
    "name": "John Smith",
    "major": "Computer Science",
    "credits_completed": 88
  },
  {
    "student_id": 43,
    "name": "Jane Doe",
    "major": "Mathematics",
    "credits_completed": 61
  },
  {
```

| cURL | Postman |
|---|---|
| The URL itself | The URL bar |
| `-X METHOD` | The method dropdown |
| `-H "Header: Value"` | The Headers tab |
| `-d '{...}'` | The Body tab (raw / JSON) |
| `-i` / `-v` output | The Response panel and Console |

```
    "student_id": 44,
    "name": "Jerry Doe",
    "major": "Computer Science",
    "credits_completed": 23
  }
]
```

## 152.6 Postman

Typing curl commands by hand is a great way to *learn* how HTTP requests actually work, because every piece of the request ( the method, the headers, the body) has to be spelled out yourself. But once you already understand what those pieces are, typing them out every single time gets tedious and error-prone, especially for requests with long JSON bodies or several headers.

*Postman* is a GUI application for building, sending, and saving HTTP requests. Everything you've been typing as curl flags has a matching box in Postman:

### 152.6.1 Building your first request in Postman

1. Open Postman (download available here ) and click **New** → **HTTP Request**.
2. Set the method dropdown (left of the URL bar) to `GET`, and enter `http://localhost:3000/students` in the URL bar.
3. Click **Send**. The response body appears in the panel below, along with the status code and response time: the same information `-i` shows you in curl, just laid out visually instead of printed as text.

Now recreate the POST request from earlier:

1. Change the method to `POST` and set the URL to `http://localhost:3000/students`.
2. Open the **Body** tab, select **raw**, and choose **JSON** from the format dropdown on the right. Postman will automatically set the `Content-Type: application/json` header for you. This is one thing it does that curl makes you do by hand.

3. Paste in a student object:

```
{
  "student_id": 45,
  "name": "Alex Rivera",
  "major": "Biology",
  "credits_completed": 12
}
```

4. Click **Send** and confirm you get a `201 Created` status back with the same object.

### 152.6.2 Why this matters once requests get complicated

A single GET request is barely worth using Postman for; a browser address bar does the job just as well. Postman starts to pay off once your API involves things that are painful to keep retyping in a terminal:

- **Collections.** Every request you build can be saved into a named *collection*, so your entire API (every route in this chapter, for example) can be saved, organized, and rerun with a click instead of retyped.
- **Environments.** A variable like `{{base_url}}` can be swapped between `http://localhost:3000` and a real deployed server without editing every saved request individually.
- **Authorization tab.** Instead of manually adding an `Authorization` header on every request, you can set your API key once per collection and Postman attaches it automatically.
- **Automated testing**: Postman can run a series of requests and check that the responses match expected values, so you can verify your API is working as intended after every change. This is great for testing before you have a front-end built, or for running regression tests after a code change.

Neither tool replaces the other. cURL is scriptable and works anywhere you have a terminal, including inside automated tests and shell scripts, which is why you'll keep seeing it later in this book. Postman is better suited to exploring an API interactively, organizing a large set of requests, and sharing that organized set with teammates. Most professional API developers end up using both, for different parts of their workflow.

## 152.7 Summary

This chapter touched on a lot of software and development concepts. Here are some of the key takeaways:

- API's are ways to connect and fetch data from a server. They are a way to connect two different applications together.
- API's are usually built with a RESTful architecture, which is a set of principles that follow HTTP protocol to perform CRUD operations on resources.
- **CURL** is a command line tool that allows you to send HTTP requests to an API. Another commonly used tool is **Postman**, which is a GUI application that allows you to send HTTP requests to an API.
- We created a basic API that allows us to perform CRUD operations on a student resource. We can create, read, update, and delete students from our API.

Chapter 153

# Data Compression and Decompression

Data compression and decompression are fundamental techniques used in modern computing, enabling efficient storage and transmission of data. The concept of entropy, borrowed from the field of information theory, plays a crucial role in determining the compression rate.

## 153.1 Data Compression and Decompression

Data compression is the process of reducing the amount of data needed to represent a particular set of information. The two main types of data compression are lossless and lossy. Lossless compression ensures that the original data can be perfectly reconstructed from the compressed data, whereas lossy compression allows some loss of data for more significant compression rates. Decompression is the reverse process of compression, reconstructing the original data from the compressed format.

## 153.2 Entropy

In information theory, entropy measures the unpredictability or randomness of information content. More specifically, it quantifies the expected value of the information contained in a message. Lower entropy implies less randomness and more repetitiveness, which in turn means the data can be compressed more.

## 153.3 Entropy and Compression

The role of entropy in data compression is fundamental. The entropy of a source of data is the minimum number of bits required, on average, to encode symbols drawn from the source. It serves as a lower bound on the best possible lossless compression rate.

For a source $X$ with probability distribution $p(x)$, the entropy $H(X)$ is defined as:

$$H(X) = -\sum_{x \in X} p(x) \log_2 p(x) \tag{153.1}$$

If the entropy of the data is high (i.e., the data is random and unpredictable), the potential for compression is low. On the other hand, if the entropy is low (the data is predictable), the data can be compressed to a smaller size.

Chapter 154

# Dealing with JSON and XML

# Chapter 155

# HTML

HTML, an abbreviation for Hypertext Markup Language, is the standard language for creating web pages and web applications. It is a cornerstone technology of the World Wide Web and forms the structure and layout of web content.

## 155.1 HTML Elements

An HTML document is composed of a series of elements, which are denoted by tags. Elements have an opening tag and a closing tag with content in between. Some elements, however, are self-closing and do not contain any content.

An element consists of three parts:

- **Opening tag**: The opening tag is the start of an element. The format of it is `<tagname>`.
- **Content**: Content is the data or information between the tags.
- **Closing tag**: The closing tag indicates the end of an element. The format of it is `</tagname>`.

For example, the paragraph tag `<p>` is used to denote a paragraph:

```
<p>This is a paragraph.</p>
```

This applies to a lot of HTML elements but not every one. There are HTML elements that are self-closing. A self-closing element does not require a closing tag. The image tag `<img>` is an example of a self-closing element.

```
<img src="image.jpg" alt="Description" />
```

## 155.2 Attributes

Attributes provide additional information about HTML elements. For some elements like `img`, they are required for the element to function correctly. They are always specified in the opening tag and consist of a name and value pair.

The syntax for attributes is:

```
<tagname attribute="value">Content</tagname>
```

There are so many attributes available, we cannot cover them all here. However, some common attributes that work on most elements include:

- **id**: The `id` attribute specifies a unique identifier for an element (should be unique within the entire page)
- **class**: The `class` attribute specifies one or more class names for styling with CSS
- **style**: The `style` attribute specifies inline CSS styling
- **title**: The `title` attribute specifies additional information displayed as a tooltip

Examples:

```
<p id="intro" class="highlight">
    Introduction paragraph
</p>

<div style="color: blue; font-size: 16px;">
    Styled text
</div>

<img src="photo.jpg" title="My Photo" alt="Me">
```

## 155.3 HTML Document Structure

A typical HTML document has a specific structure. Every HTML5 document begins with a DOCTYPE declaration, which tells the browser which version of HTML to expect.

The basic structure of an HTML document is:

```
<!DOCTYPE html>
<html>
<head>
    <title>Page Title</title>
</head>
<body>
    <!-- Content goes here -->
</body>
```

```
</html>
```

The main components are:

- **DOCTYPE declaration**: This informs the browser about the version of HTML. For HTML5, it is `<!DOCTYPE html>`.
- **html**: The `html` element is the root element that wraps all other content.
- **head**: This element contains metadata and information about the page, including:
  - `<title>`: This element sets the title of the page, which appears in the browser tab.
  - `<meta>`: This element includes metadata such as character encoding, and viewport settings.
- **body**: The `body` element contains all the visible content of the web page.

Here is a more complete example:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Website</title>
</head>
<body>
    <h1>Welcome</h1>
    <p>This is my website.</p>
</body>
</html>
```

## *Exercise 400* Building a Full HTML Document

Create the beginning of a web page for a chocolate chip cookie recipe. Add the `DOCTYPE` declaration, the `html` element, a `head` with a `title` that says "Chocolate Chip Cookies", and move the following content into the `body`.

```
<body>
  <h1 id="recipe-title">Chocolate
  ↪  Chip Cookies</h1>
  <p class="intro" title="A short
  ↪  recipe description">
    This recipe makes soft and
    ↪  sweet chocolate chip
    ↪  cookies.
  </p>
</body>
```

Write a full HTML5 page that wraps this content correctly.

*Working Space*

*Answer on Page 2048*

After this exercise, the page should look like the following:

**Chocolate Chip Cookies**

This recipe makes soft and sweet chocolate chip cookies.

Figure 155.1: HTML Structure Exercise Output

## 155.4 Web Development and HTML

The Web Development trio consists of three essential languages that work together to create modern web applications. These languages are HTML, CSS, and JavaScript.

- **HTML**: HTML provides the structure and content of web pages. It defines what elements exist on the page. Think of it as the skeleton and meat of a web page.
- **CSS**: Cascading Style Sheets (CSS) handles the presentation and styling. It controls how elements look, including colors, fonts, spacing, and layout. Think of it as the clothing and makeup that makes the page visually appealing.
- **JavaScript**: JavaScript adds interactivity and behavior. It enables dynamic content changes, form validation, animations, and user interactions. Think of it as the brain that allows the page to respond to actions like clicking a button or submitting a form.

## 155.5 Metadata

Metadata is information about the HTML document that is not directly displayed on the page. The information is provided through a series of `<meta>` tags with certain attributes and their corresponding values. These are placed in the `<head>` section and provides important information to browsers, search engines, and developers.

Common metadata elements include:

- `<meta charset="UTF-8">`: This line specifies the character encoding of the document. UTF-8 is the standard encoding for web pages.
- `<meta name="viewport" content="width=device-width, initial-scale=1.0">`: This line configures the viewport for responsive design. This tells mobile browsers how to scale the page.
- `<meta name="description" content="...">`: This line provides a brief description of the page content for search engines.
- `<meta name="keywords" content="...">`: This line lists relevant keywords for search engine optimization (SEO).
- `<meta name="author" content="...">`: This line specifies the author of the document.

Example of using multiple meta tags:

```
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <meta name="description" content="A tutorial about web development">
    <meta name="keywords" content="HTML, CSS, JavaScript">
    <meta name="author" content="John Doe">
    <title>Web Development Tutorial</title>
</head>
```

## 155.6 Div Elements

The `<div>` element is a generic container used to group content together. The name "div" stands for "division." `<div>` has no specific meaning to it. It is a neutral wrapper used for layout, styling, or organizing content with CSS and JavaScript.

The `<div>` element is one of the most commonly used HTML elements in existing applications. It is often used with `class` or `id` attributes to apply styling or add behavior.

Here's a basic example:

```
<div class="container">
    <h2>Section Title</h2>
    <p>This is some content inside a div.</p>
</div>
```

As mentioned, it is one of the most commonly used HTML elements. Here's an example with multiple divs for layout:

```
<div class="header">
    <h1>Website Title</h1>
</div>

<div class="content">
    <p>Main content goes here.</p>
</div>

<div class="footer">
    <p>&copy; 2024 My Website</p>
</div>
```

## *Exercise 401* Grouping Recipe Content with div

*Working Space*

Group the recipe title and introduction paragraph into a `div`. Add a "class" attribute to the `div` with the value "recipe-header".

```
<body>
  <h1 id="recipe-title">Chocolate
  ↪ Chip Cookies</h1>
  <p class="intro" title="A short
  ↪ recipe description">
    This recipe makes soft and
    ↪ sweet chocolate chip
    ↪ cookies.
  </p>
</body>
```

Add two more `p` elements for ingredients and steps in the body. Place the `p` elements into two separate `div` elements. The ingredients `div` should have a class attribute with a value of "ingredients-section". The steps `div` should have a class attribute with a value of "steps-section".

*Answer on Page 2048*

After this exercise, the page should now look like the following:

Figure 155.2: HTML Div Exercise Output

## 155.7 Semantic HTML

Unlike `<div>` elements, Semantic HTML elements are elements that clearly describe their meaning to both the browser and the developer. This improves code readability, helps with search engine optimization (SEO), and makes pages more accessible to assistive technologies like screen readers.

Semantic elements follow a logical structure that reflects the content of the page. Some common semantic elements include:

- `<header>`: The `header` element represents the top section of the page or a section, typically containing the logo, navigation, or introductory content.
- `<nav>`: The `nav` element contains navigation links to other pages or sections.
- `<main>`: The `main` element specifies the main content area of the page.
- `<article>`: The `article` element represents independent, self-contained content that could stand alone, such as a blog post or news article.
- `<section>`: The `section` element defines a generic section of content. This is preferred instead of using multiple divs for grouping related content.
- `<aside>`: The `aside` element contains supplementary content like sidebars, ads, or related links.
- `<footer>`: The `footer` element represents the bottom section of a page or section.

Here is an example of using semantic elements to structure a simple blog page:

```
<header>
    <h1>My Blog</h1>
    <nav>
        <a href="/">Home</a>
        <a href="/about">About</a>
        <a href="/contact">Contact</a>
    </nav>
</header>

<main>
    <article>
        <h2>Article Title</h2>
        <p>Article content here...</p>
    </article>

    <aside>
        <h3>Related Posts</h3>
        <ul>
            <li><a href="#post1">Post 1</a></li>
            <li><a href="#post2">Post 2</a></li>
        </ul>
    </aside>
</main>

<footer>
    <p>&copy; 2026 My Blog. All rights reserved.</p>
</footer>
```

## *Exercise 402* Using Semantic Elements

Replace general `div` containers with semantic elements where they make sense. Use `header` for the "recipe-header". Use `section` for the "ingredients-section" and "steps-section". Afterwards, put these inside a `main` element. Add a `footer` for the bottom note. There will be no content in the footer for now.

*Working Space*

```
<body>
  <div class="recipe-header">
    <h1 id="recipe-title">Chocolate Chip Cookies</h1>
    <p class="intro" title="A short recipe description">
      This recipe makes soft and sweet chocolate chip cookies.
    </p>
  </div>

  <div class="ingredients-section">
    <p>Ingredients</p>
  </div>

  <div class="steps-section">
    <p>Steps</p>
  </div>
</body>
```

*Answer on Page 2049*

After this, the web page should still look the same from the previous exercise.

## 155.8 Headings and Sections

HTML provides six levels of headings, ranging from `<h1>` to `<h6>`. Headings establish a hierarchical structure for page content. `<h1>` is the most important and typically used once per page for the main title, while `<h6>` is the least important.

```
<h1>Main Page Title</h1>
<h2>Major Section</h2>
<h3>Subsection</h3>
<h4>Sub-subsection</h4>
<h5>Minor heading</h5>
<h6>Least important heading</h6>
```

Headings are important for defining the structure of the content and improving accessibility. They allow users to quickly scan the page and understand the organization of information. Proper use of headings also benefits SEO by signaling the importance of content to search engines.

## *Exercise 403* Adding Section Headings

Replace the `p` elements in the ingredients and steps sections with `h2` tags.

*Working Space*

```
<main>
  <section
  ↪ class="ingredients-section">
    <p>Ingredients</p>
  </section>

  <section class="steps-section">
    <p>Steps</p>
  </section>
</main>
```

Update the main content to include headings both the Ingredients and Steps sections. Add an `id` attribute to each heading with the value "ingredients" and "steps" respectively.

*Answer on Page 2049*

After this exercise, the page should now look like the following:

# Chocolate Chip Cookies

This recipe makes soft and sweet chocolate chip cookies.

## Ingredients

## Steps

Figure 155.3: HTML Headings Exercise Output

## 155.9 Text Elements

HTML provides several elements for formatting text content. Here is a list of some of the most commonly used ones:

- `<p>`: The `p` element defines a paragraph. Browsers automatically add space before and after.
- `<br>`: The `br` element inserts a line break. This is a self-closing element.
- `<hr>`: The `hr` element displays a horizontal line. This is a self-closing element.
- `<strong>`: The `strong` element indicates strong importance. This is typically displayed in bold.
- `<em>`: The `em` element indicates emphasis. This is typically displayed in italics.
- `<b>`: The `b` element displays bold text without semantic importance.
- `<i>`: The `i` element displays italic text without semantic emphasis.
- `<mark>`: The `mark` element highlights text with a yellow background.
- `<code>`: The `code` element displays text in monospace font for code.

Examples:

```
<p>This is a <strong>important</strong> point.</p>

<p>This word is <em>emphasized</em>.</p>

```

```
<p><code>var x = 5;</code></p>

<p>This is <mark>highlighted</mark> text.</p>
```

## *Exercise 404* Using Text Elements

*Working Space*

Add a p element in the footer section of the cookie recipe page. The paragraph should say: "**Important Note:** Make sure not to overbake the cookies, or they will become hard."

```
<footer>
</footer>
```

Add a paragraph element inside of the `footer` element. Use the `strong` element to make "Important Note:" bold.

*Answer on Page 2049*

After this exercise, the page should now look like the following:

# Chocolate Chip Cookies

This recipe makes soft and sweet chocolate chip cookies.

## Ingredients

## Steps

**Important Note:** Make sure not to overbake the cookies, or they will become hard.

Figure 155.4: HTML Text Exercise Output

## 155.10 Links

Links are created using the `<a>` (anchor) element with the `href` attribute specifying the destination.

```
<a href="https://www.example.com">Visit Example</a>
```

Links can point to various destinations:

- **External websites**: `<a href="https://www.google.com">External Site</a>`. This will navigate the user to Google.
- **Other pages within the same site**: `<a href="about.html">About</a>`. This will navigate the user to the about.html page within the same website.
- **Email addresses**: `<a href="mailto:email@example.com">Email</a>`. This will open the user's email to send an email to the specified address.
- **Phone numbers**: `<a href="tel:+1234567890">Call Us</a>`. This will initiate a phone call on devices that support it. This can be useful for mobile users.
- **Specific locations on the same page**: `<a href="#section2">Jump to Section 2</a>` (where `<h2 id="section2">Section 2</h2>`). When the link is clicked, the page will scroll to the header with the id "section2".

Apart from the `href` attribute, here are some other useful attributes for links:

- `target="_blank"`: This attribute and value pair opens the link in a new tab.
- `target="_self"`: This attribute and value pair opens the link in the same tab. This is the normal behavior if the target attribute is not specified.
- `title="tooltip"`: This attribute and value pair displays text when the user hovers over the link like a tooltip.

## 155.11 Lists

HTML supports two main types of lists to organize content. These include unordered lists and ordered lists. Both types of lists use the `<li>` element to define individual list items.

### 155.11.1 Unordered Lists

Unordered lists display items with bullet points. As the name suggests, the order of items is not important like the items of a shopping list. They use the `<ul>` tag with list items defined by `<li>`:

```
<ul>
    <li>Apples</li>
    <li>Bananas</li>
    <li>Oranges</li>
</ul>
```

This would render an unordered list as:

- Apples
- Bananas
- Oranges

### 155.11.2 Ordered Lists

Ordered lists display items with numbers or other counters. As the name suggests, the order of items is important, like steps in a recipe. They use the `<ol>` tag:

```
<ol>
    <li>First step</li>
    <li>Second step</li>
    <li>Third step</li>
</ol>
```

The output of the above code would be:

1. First step
2. Second step
3. Third step

The `<ol>` tag supports the `type` attribute to customize or change the default numbering style. So instead of numbers, you can use letters or Roman numerals. Some of the common attributes include:

- `type="a"`: Lowercase letters (a, b, c...).
- `type="A"`: Uppercase letters (A, B, C...).
- `type="i"`: Lowercase Roman numerals (i, ii, iii...)
- `type="I"`: Uppercase Roman numerals (I, II, III...)

## *Exercise 405* Ingredients and Steps Lists

Use lists to show the cookie ingredients and the baking steps. The ingredients should be an unordered list, and the steps should be an ordered list.

*Working Space*

```
<section
↪  class="ingredients-section">
  <h2
  ↪  id="ingredients">Ingredients</h2>
  <!-- Add an unordered list of
  ↪  ingredients here -->
</section>

<section class="steps-section">
  <h2 id="steps">Steps</h2>
  <!-- Add an ordered list of steps
  ↪  here -->
</section>
```

Here is the list of ingredients:

2 1/4 cups flour. 1 teaspoon baking soda. 1 teaspoon salt. 1 cup butter, 3/4 cup sugar. 3/4 cup brown sugar, 2 eggs. 2 cups chocolate chips.

Here is the list of steps:

Preheat the oven to 375°F. Mix the dry ingredients in one bowl. Mix the butter, sugar, brown sugar, and eggs in another bowl. Stir the dry ingredients into the wet ingredients. Add the chocolate chips. Scoop the dough onto a baking sheet. Bake for 9 to 11 minutes.

*Answer on Page 2050*

After this exercise, the page should now look like the following:

# Chocolate Chip Cookies

This recipe makes soft and sweet chocolate chip cookies.

## Ingredients

- 2 1/4 cups flour
- 1 teaspoon baking soda
- 1 teaspoon salt
- 1 cup butter
- 3/4 cup sugar
- 3/4 cup brown sugar
- 2 eggs
- 2 cups chocolate chips

## Steps

1. Preheat the oven to 375°F.
2. Mix the dry ingredients in one bowl.
3. Mix the butter, sugar, brown sugar, and eggs in another bowl.
4. Stir the dry ingredients into the wet ingredients.
5. Add the chocolate chips.
6. Scoop the dough onto a baking sheet.
7. Bake for 9 to 11 minutes.

**Important Note:** Make sure not to overbake the cookies, or they will become hard.

Figure 155.5: HTML Lists Exercise Output

## 155.12 Navigation

The `<nav>` element is used to wrap navigational links within a navigation section. It's a semantic element that helps structure the page and signal to assistive technologies that this section contains navigation. This is commonly used for site menus, table of contents, or any set of links that help users navigate the website.

```
<nav>
    <ul>
        <li><a href="/">Home</a></li>
        <li><a href="/about">About</a></li>
        <li><a href="/services">Services</a></li>
        <li><a href="/contact">Contact</a></li>
    </ul>
</nav>
```

## *Exercise 406* Creating a Simple Navigation Bar

Add a navigation bar to the top of the cookie page within the `header` element. The navigation should link to the "Ingredients" and "Steps" sections.

*Working Space*

```
<header>
  <h1 id="recipe-title">Chocolate Chip Cookies</h1>
  <p class="intro" title="A short recipe description">
    This recipe makes soft and sweet chocolate chip cookies.
  </p>

  <!-- Add navigation here -->
</header>
```

Insert a `nav` element with links to each section. Each link should use the `href` attribute to point to the corresponding section's `id` (e.g., `href="#ingredients"` and `href="#steps"`) within the same page.

*Answer on Page 2050*

After this exercise, the page should now look like the following:

# Chocolate Chip Cookies

This recipe makes soft and sweet chocolate chip cookies.

- Ingredients
- Steps

## Ingredients

- 2 1/4 cups flour
- 1 teaspoon baking soda
- 1 teaspoon salt
- 1 cup butter
- 3/4 cup sugar
- 3/4 cup brown sugar
- 2 eggs
- 2 cups chocolate chips

## Steps

1. Preheat the oven to 375°F.
2. Mix the dry ingredients in one bowl.
3. Mix the butter, sugar, brown sugar, and eggs in another bowl.
4. Stir the dry ingredients into the wet ingredients.
5. Add the chocolate chips.
6. Scoop the dough onto a baking sheet.
7. Bake for 9 to 11 minutes.

**Important Note:** Make sure not to overbake the cookies, or they will become hard.

Figure 155.6: HTML Navigation Exercise Output

## 155.13 Tables

Tables organize data into rows and columns. They are created using the `<table>` tag. However, that is used as a container for table elements necessary to render the layout of the table. Those table elements include:

- `<tr>`: The `tr` element defines a table row
- `<th>`: The `th` element defines a table header cell. By default, it is displayed in bold and centered. It is used to indicate that the cell is a header for a column or row.
- `<td>`: The `td` element defines a table data cell. It is used for regular cells that contain data.
- `<thead>`: The `thead` element groups header rows.

- `<tbody>`: The `tbody` element groups body rows.
- `<tfoot>`: The `tfoot` element groups footer rows.
- `<caption>`: The `caption` element provides a title for the table.

Here's a basic table example:

```
<table>
    <caption>Monthly Sales</caption>
    <thead>
        <tr>
            <th>Month</th>
            <th>Sales</th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td>January</td>
            <td>$5000</td>
        </tr>
        <tr>
            <td>February</td>
            <td>$6000</td>
        </tr>
    </tbody>
    <tfoot>
        <tr>
            <th>Total</th>
            <td>$11000</td>
        </tr>
    </tfoot>
</table>
```

The output of the above code would be:

Table 155.1: Monthly Sales

| Month | Sales |
|---|---|
| January | $5000 |
| February | $6000 |
| **Total** | $11000 |

There are attributes that can be applied to table cells to make them span multiple rows or columns. These attributes are `rowspan` (row span) and `colspan` (column span). These attributes are used to merge cells together, which can be useful for creating more complex table layouts. `colspan` is used to make a cell span across multiple columns, while `rowspan` is used to make a cell span across multiple rows. The value of these attributes specifies the number of rows or columns the cell should span.

Here's an example of how to use them:

```
<table>
    <tr>
        <th colspan="2">Personal Information</th>
    </tr>
    <tr>
        <td>Name</td>
        <td>John Doe</td>
    </tr>
    <tr>
        <th>Age</th>
        <th rowspan="2">Skills</th>
    </tr>
    <tr>
        <td>30</td>
    </tr>
    <tr>
        <td colspan="2">HTML, CSS, JavaScript</td>
    </tr>
</table>
```

The output of the above code would be:

<table>
<tr><th colspan="2">Personal Information</th></tr>
<tr><td>Name</td><td>John Doe</td></tr>
<tr><td>Age</td><td>Skills<br>(spans 2 rows)</td></tr>
<tr><td colspan="2">HTML, CSS, JavaScript</td></tr>
</table>

## *Exercise 407* Recipe Facts Table

Create a new section for recipe facts. Inside the section, add a `h2` heading with the text "Recipe Facts".

*Working Space*

```
<section>
  <h2>Baking Tips</h2>
</section>
```

Add a table that shows basic facts about the chocolate chip cookie recipe, such as prep time, bake time, and servings. The table must have three rows and two columns. Add a `border` attribute to the `<table>` element with a value of "1" to make the table borders visible.

| | |
|---|---|
| Prep Time | 15 minutes |
| Bake Time | 10 minutes |
| Servings | 24 cookies |

*Answer on Page 2051*

After this exercise, the page should now look like the following:

# Chocolate Chip Cookies

This recipe makes soft and sweet chocolate chip cookies.

- Ingredients
- Steps

## Ingredients

- 2 1/4 cups flour
- 1 teaspoon baking soda
- 1 teaspoon salt
- 1 cup butter
- 3/4 cup sugar
- 3/4 cup brown sugar
- 2 eggs
- 2 cups chocolate chips

## Steps

1. Preheat the oven to 375°F.
2. Mix the dry ingredients in one bowl.
3. Mix the butter, sugar, brown sugar, and eggs in another bowl.
4. Stir the dry ingredients into the wet ingredients.
5. Add the chocolate chips.
6. Scoop the dough onto a baking sheet.
7. Bake for 9 to 11 minutes.

## Recipe Facts

| | |
|---|---|
| Prep Time | 15 minutes |
| Bake Time | 10 minutes |
| Servings | 24 cookies |

**Important Note:** Make sure not to overbake the cookies, or they will become hard.

Figure 155.7: HTML Tables Exercise Output

## 155.14 Forms

Forms allow users to input and submit data. The `<form>` element contains form controls and is submitted to a server to process the provided data. The most common form controls include text fields, checkboxes, radio buttons, and submit buttons. These form controls are created using input elements.

Here's an example of a basic form structure with various input elements:

```
<form>
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email">

    <input type="submit" value="Submit">
</form>
```

The `label` element is used to provide a label for form controls. The `for` attribute of the label should match the `id` of the corresponding input element to associate them together.

Input fields have various types, each serving a different purpose. The "type" attribute indicates the type of input field. Here's a list of the most common input types:

- `type="text"`: This is a single-line text input.
- `type="email"`: This is a single-line email input with validation that ensures the entered value is a valid email address.
- `type="password"`: This is a single-line password input (text is hidden).
- `type="number"`: This is a single-line numeric input.
- `type="checkbox"`: This is a checkbox for multiple selections.
- `type="radio"`: This is a radio button for single selection.
- `type="file"`: This is a file upload input.
- `type="date"`: This is a date picker input.
- `type="submit"`: This is a submit button.
- `type="reset"`: This is a reset button that will clear all form fields.

There are more form control elements besides the "input" element. Other form elements include:

- `<textarea>`: This is a multi-line text input.
- `<select>`: This is a dropdown menu. A dropdown is a list of options that can be expanded to show all available options.
- `<option>`: These are options within a dropdown.
- `<fieldset>`: This groups related form elements.

- `<legend>`: This is the title for a fieldset.

Here's an example that includes various form controls and uses fieldsets to group related information:

```
<form>
    <fieldset>
        <legend>Personal Information</legend>

        <label for="username">Username:</label>
        <input type="text" id="username" name="username">

        <label for="password">Password:</label>
        <input type="password" id="password" name="password">
    </fieldset>

    <fieldset>
        <legend>Preferences</legend>

        <label>
            <input type="checkbox" name="newsletter">
            Subscribe to newsletter
        </label>

        <label for="country">Country:</label>
        <select id="country" name="country">
            <option>Select a country</option>
            <option value="usa">USA</option>
            <option value="uk">UK</option>
            <option value="ca">Canada</option>
        </select>
    </fieldset>

    <button type="submit">Submit</button>
</form>
```

## 155.15 Images

Images are added using the `<img>` element, which is self-closing.

The basic syntax for the `<img>` element is:

```
<img src="image.jpg" alt="Description of image">
```

The most essential attributes for the `<img>` element are:

- `src`: The path to the image file. This is a required attribute that specifies the source of the image. The source path can route to an image file within the application or an external URL similar to a link. If the image cannot be found at the specified path, the browser will display a broken image icon. The `src` attribute is crucial for the image to load properly on the webpage. It is important to ensure that the path is correct and that the image file exists at that location.
- `alt`: This is alternative text displayed if the image cannot load or for screen readers. The `alt` attribute should provide a concise description of the image content.

Images come in various sizes. To control the display size of an image, you can use the `width` and `height` attributes.

- `width`: This specifies the width of the image in pixels or percentage.
- `height`: This specifies the height of the image in pixels or percentage.

Here's an example of using the `width` and `height` attributes to specify the dimensions of an image:

```
1 <img src="logo.png" alt="Company Logo" width="200" height="100">
```

For the next exercise, we will add an image to the cookie recipe page. Use the image below as a reference for the image you will add to the page.

Figure 155.8: Image of chocolate chip cookies

## *Exercise 408* The Recipe Image

Add a picture of the cookies to the recipe page add the bottom of the `header` element.

*Working Space*

```
<header>
  <h1 id="recipe-title">Chocolate Chip Cookies</h1>
  <p class="intro" title="A short recipe description">
    This recipe makes soft and sweet chocolate chip cookies.
  </p>

  <!-- nav is here. Just showing the header for context -->

  <!-- Add the image element here -->
</header>
```

Add an `<img>` element with the source set to "cookies.jpg". Make sure that the image file is in the same folder as your HTML file. The `alt` attribute should be set to "Fresh chocolate chip cookies on a plate". Set the width of the image to 500 pixels and the height to 300 pixels.

*Answer on Page 2051*

After this exercise, the page should now look like the following:

# Chocolate Chip Cookies

This recipe makes soft and sweet chocolate chip cookies.

- Ingredients
- Steps

## Ingredients

- 2 1/4 cups flour
- 1 teaspoon baking soda
- 1 teaspoon salt
- 1 cup butter
- 3/4 cup sugar
- 3/4 cup brown sugar
- 2 eggs
- 2 cups chocolate chips

## Steps

1. Preheat the oven to 375°F.
2. Mix the dry ingredients in one bowl.
3. Mix the butter, sugar, brown sugar, and eggs in another bowl.
4. Stir the dry ingredients into the wet ingredients.
5. Add the chocolate chips.
6. Scoop the dough onto a baking sheet.
7. Bake for 9 to 11 minutes.

## Recipe Facts

| | |
|---|---|
| Prep Time | 15 minutes |
| Bake Time | 10 minutes |
| Servings | 24 cookies |

**Important Note:** Make sure not to overbake the cookies, or they will become hard.

Figure 155.9: HTML Images Exercise Output

Chapter 156

# Introduction to Text

In computer systems, text is represented in files as a sequence of characters, each of which corresponds to a specific number, known as a character code. These character codes are then stored in the file as binary data.

## 156.1 Newlines and Carriage Returns

Two of the character codes that have special meanings are the newline (often represented as '\n') and the carriage return (often represented as '\r').

The newline character signifies the end of a line of text and the beginning of a new one. If you parse a text file with a Python script, it will see your 'enter' key press as a new line symbol. The carriage return character moves the cursor to the beginning of the line. The use of these characters can vary between operating systems. Unix-based systems (like Linux and MacOS) use the newline character to indicate the end of a line, while Windows systems use a combination of a carriage return and a newline ('\r\n').

## 156.2 ASCII

The American Standard Code for Information Interchange (ASCII) is one of the earliest character encodings. It uses 7 bits to represent each character, allowing it to define up to $2^7 = 128$ different characters. These include the English alphabet (in both lower and upper cases), digits, punctuation symbols, control characters (like newline and carriage return), and some other symbols.

## 156.3 UTF-8

UTF-8 (8-bit Unicode Transformation Format) is a variable-width character encoding that can represent every character in the Unicode standard, yet remains backward-compatible with ASCII. For the ASCII range (0-127), UTF-8 is identical to ASCII. However, it can use additional bytes (up to 4 bytes in total) to represent characters that are not included in ASCII, such as characters from other languages, emojis, and many other symbols. This has made UTF-8 a widely used encoding in many modern systems.

# Chapter 157

# Alphabets and Accents

Software developers often encounter text from diverse languages and cultures. As a developer, it is crucial to have a solid understanding of alphabets and accents to effectively handle and process multilingual text. Alphabets, the building blocks of written language, vary widely across different nations and regions. Meanwhile, accents, diacritical marks, and other phonetic notations play a crucial role in conveying the correct pronunciation and meaning of words.

This guide aims to provide software developers with a fundamental understanding of alphabets and accents to navigate the complexities of handling text from different nations. By familiarizing yourself with these concepts, you will be better equipped to develop robust applications, support multiple languages, and ensure accurate representation and interpretation of text data.

Alphabets are sets of letters or symbols used to represent the sounds of a language. While the Latin alphabet is widely used in many Western languages, numerous other alphabets exist, such as Cyrillic, Greek, Arabic, Devanagari, and Chinese characters. Each alphabet has its own unique set of letters, often organized in a specific order, and may include uppercase and lowercase variations.

Accents and diacritical marks are additional symbols added to letters to modify their pronunciation or provide additional phonetic information. Accents can appear above, below, or beside a letter, and they can change the sound, stress, or intonation of a word. For example, in French, the acute accent (’e) changes the pronunciation of the letter ”e” from /□/ to /e/.

When working with multilingual text, it is essential to consider various factors:

1. Character encoding: Different alphabets require specific character encodings to represent their letters digitally. Commonly used character encodings include ASCII, Unicode, and UTF-8. Understanding the appropriate encoding for each language is crucial to ensure proper text rendering and avoid data corruption.
2. Text input and validation: Building applications that handle user input requires robust text validation. Account for the diverse set of characters and possible accents that may appear in user-generated content. Implement proper validation and sanitization mechanisms to handle text input securely.
3. Sorting and collation: Sorting text from different languages involves considering the specific rules and conventions of each alphabet. Some languages may have unique

sorting orders, while others ignore accents or diacritics when determining the order of words. Take into account the appropriate sorting and collation algorithms to ensure consistent and accurate results.

4. Search and indexing: Efficient search and indexing systems must be capable of handling multilingual text. Consider appropriate text normalization techniques to account for different character representations (e.g., case-insensitive matching, ignoring accents), enabling users to find relevant content across languages and variations in spelling or diacritics.

By grasping the concepts of alphabets and accents, software developers can build robust, inclusive applications that handle multilingual text effectively. Understanding character encodings, implementing proper text validation, considering sorting and collation rules, and enabling efficient search capabilities are crucial steps toward supporting diverse linguistic communities and providing a seamless user experience across different languages.

Now, let's delve deeper into specific alphabets and accents commonly encountered in software development, exploring their unique characteristics and considerations for handling text from different nations.

# Chapter 158

# Making Plots with matplotlib

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. It is highly useful for presenting data in a more intuitive and easy-to-understand manner.

In order to use Matplotlib, you must first import it, typically using the following line of code:

```
import matplotlib.pyplot as plt
```

Let's create a simple line plot. Suppose we have a list of numbers and we want to visualize their distribution:

```
x = [1, 2, 3, 4, 5]
y = [1, 4, 9, 16, 25]

plt.plot(x, y)
plt.show()
```

Here, 'x' and 'y' are the coordinates of the points. The 'plt.plot' function plots y versus x as lines and/or markers. The 'plt.show' function then displays the figure.

Creating a bar plot follows a similar approach:

```
labels = ['A', 'B', 'C', 'D', 'E']
values = [5, 7, 9, 11, 13]

plt.bar(labels, values)
plt.show()
```

Here, 'labels' are the categories we are plotting, and 'values' are the respective sizes of those categories. The 'plt.bar' function creates a bar plot.

Matplotlib provides a variety of other plot types and customization options — everything from scatter plots and histograms to custom line styles and colors. Explore the official Matplotlib documentation to learn more about what this powerful library can offer.

Chapter 159

# Geocoding and Reverse Geocoding

Geocoding and reverse geocoding are essential processes in geographic information systems (GIS) that are used to convert between addresses and spatial data.

## 159.1 Geocoding

Geocoding is the process of converting addresses (like "1600 Amphitheatre Parkway, Mountain View, CA") into geographic coordinates (like latitude 37.423021 and longitude -122.083739), which you can use to place markers on a map, or position the map. The resulting latitude and longitude are often used as a key index in merging datasets based on location.

Here is an example of using Google's Geocoding service to get the longitude and latitude of the Dallas County Administration Building:

```

import requests
import json

# Encode the parameters
parameters = {"address": "411 Elm St, Dallas, TX 75202", "key": "YOUR_API_KEY"}
base_url = "https://maps.googleapis.com/maps/api/geocode/json?"

# Send the GET request
response = requests.get(base_url,  params=parameters)

# Convert the response to json
data = response.json()

# Extract the latitude and longitude
if len(data["results"]) > 0:
    latitude = data["results"][0]["geometry"]["location"]["lat"]
    longitude = data["results"][0]["geometry"]["location"]["lng"]
    print(latitude, longitude)
else:
    print(f"Could not find the latitude and longitude .")

```

## 159.2 Reverse Geocoding

Reverse geocoding, as the name implies, is the opposite process of geocoding. It involves converting geographic coordinates into a human-readable address. This can be useful in applications where you need to display an actual address to a user instead of latitude and longitude coordinates.

Here is an example of using Google's reverse geocoding API[1] to find the address at latitude = 33.9474096, longitude = -118.1179069

```
import requests
import json

api_key = "YOUR_API_KEY"
latitude = 33.9474096
longitude = -118.1179069

    # Encode the parameters
parameters = {"latlng": f"{latitude},{longitude}", "key": api_key}
base_url = "https://maps.googleapis.com/maps/api/geocode/json?"

# Send the GET request
response = requests.get(base_url, params=parameters)

# Convert the response to json
data = response.json()

# Extract the address
if len(data["results"]) > 0:
    address = data["results"][0]["formatted_address"]
    print(address)
else:
    print(f"Could not find the address")

```

[1]Note that the API key is something you obtain on your own. People do not share API keys for privacy and security reasons.

# Chapter 160

# Making a Map

Plotly is an open-source data visualization library for Python, R, and JavaScript. It allows for interactive plots, including geographical maps. In this brief example, we will learn how to create a simple annotated map using Plotly in Python.

To begin, you need to install Plotly. In Python, you can do this via pip:

```
pip install plotly
```

Once installed, you can create a map with annotations as follows:

```
import plotly.graph_objects as go

fig = go.Figure(data=go.Scattergeo(
    lon = [-75.789],
    lat = [45.4215],
    text = ['Ottawa'],
    mode = 'text',
))

fig.update_layout(
    title_text = 'Annotated Map with Plotly',
    showlegend = False,
    geo = dict(
        scope = 'world',
        projection_type = 'azimuthal equal area',
        showland = True,
        landcolor = 'rgb(243, 243, 243)',
        countrycolor = 'rgb(204, 204, 204)',
    ),
)

fig.show()


```

This code creates a world map and places a text annotation at the geographic coordinates for Ottawa. Then, the plotly is opened in your browser using a localhost IP address. The 'go.Scattergeo' function is used to define the geographical scatter plot (i.e., the annotation), while the 'update_layout' function is used to define the appearance and the

properties of the map itself.

In this example, you can replace the latitude, longitude, and text with the values corresponding to your desired location.

Chapter 161

# Introduction to Discrete Probability

Before we begin talking about discrete probability, let's address the word *discrete*. It means "individually separate and distinct" as in "You can think of light as a continuous wave or as a blast of discrete particles."

When we are talking about probabilities, some problems deal with discrete quantities like "What is the probability that I will throw these three dice and the numbers that roll face up sum to 9?" There are also problems that deal with continuous properties like "What is the probability that the next bird to fly over my house will weigh between 97.2 and 98.1 grams?" In this module, we are going to focus on the probability problems that deal with discrete quantities.

Watch Khan Academy's Introduction to Probability at https://youtu.be/uzkc-qNVoOk.

Let's say that you have a cloth sack filled with 100 marbles; 99 are red and 1 is white. If you reach in without looking and pull out one marble, you will probably pull out a red one. We say that "There is a 1 in 100 chance that you would pull out a white marble." Or we can use percentages and say "There is a 1% chance that you will pull out a white marble." Or we can use decimals and say "There is a 0.01 probability that you will pull out a white marble." In probability, we often talk about the probability of certain events. "Pulling out a white marble" is an event, and we can give it a symbol like $W$. In equations, we use p to mean "the probability of". Thus, we can say "There is a 0.01 probability that you will pull out a white marble," which becomes the equation

$$p(W) = 0.01$$

## 161.1 The Probability of All Possibilities is 1.0

We know that you are either going to pull out a red marble or a white marble, so the probability of a white marble being pulled and the probability of a red marble being pulled must add up to 100%. Therefore, the odds of pulling out a red marble must be 99%, or 0.99. If we let the event "Pull out a red marble" be given by the symbol R, we can say:

$$p(R) = 1.0 - P(W) = 1.0 - 0.01 = 0.99$$

Now, let's say that you take a marble from the bag, then toss a coin. What is the probability that you will pull a white marble, then get heads on the coin? It is the product of the two probabilities: $0.01 \times 0.5 = 0.005$, so one-half of a one percent chance. Do the probabilities still sum to 1?

- White and Heads = $0.01 \times 0.5 = 0.005$
- White and Tails = $0.01 \times 0.5 = 0.005$
- Red and Heads = $0.99 \times 0.5 = 0.495$
- Red and Tails = $0.99 \times 0.5 = 0.495$

$0.005 + 0.005 + 0.495 + 0.495 = 1.000$ Yes, the probabilities of all the possibilities still add to 1. This is only true because *the events do not effect each other*; the events are ***independent***.

## 161.2 Independence and Dependence

In the last section, you learned that the probability of two events ("Pulling a red marble from the bag" and "Getting tails in a coin toss") is the product of the probability of each event: $0.99 \times 0.5 = 0.495$.

This is true if the two events are *independent*; that is, the outcome of one doesn't change the probability of the other. The example we gave is independent: It doesn't matter what ball you pull from the bag, the outcome of the coin toss will always be 50-50.

What is an example of two events that are not independent? The probability that a person is a professional basketball player and the probability that someone wears a shoe that is size 13 or larger is *not* independent; rather it is dependent. After all, height is an advantage in basketball, and most tall people also have large feet. So, if you know someone is a basketball player, they likely wear large shoes.

## 161.3 Correlation does not Imply Causation

It is often said that "Correlation does not imply causation". What does this mean? Breaking it down:

- Correlation is a state in which two things increase or decrease together; two variables are *linearly* related
- Imply means suggests, as in *there is evidence for*
- Causation means there is a cause and effect relationship in place between the two variables.

So saying correlation does not imply causation simply means: just because two variables are linearly connected, it cannot be said one variable causes the other.

Here are some examples:

- Ice Cream Sales, Mosquito bites, and Sunburn cases all increase during the summertime. However, none of these are causally connected.
- Dinosaurs were not able to read books. However, this does not mean that the inability to read caused dinosaur extinction.
- There was a large increase in toilet paper sales and people working from home in the years 2020-2021. Does this mean that one causes the other? Not necessarily. In this case a third factor C, the pandemic, caused a large global quarantine, leading to increased toilet paper usage and the inability to work in an office.

### *Exercise 409* Rolling Dice

If you have three six-sided dice to roll, what is the probability that you will roll a five on all three dice?

*Working Space*

*Answer on Page 2052*

### *Exercise 410* Flipping Coins

If you have five coins to flip, what is the probability that at least one coin will come up heads?

*Working Space*

*Answer on Page 2052*

## 161.4 Why 7 is the most likely sum of two dice

If you roll two dice, the sum will be 2, 12, or any number in between. It is very tempting to assume that the likelihood of any of those numbers is the same. In fact, the probability

of a 2 is $\frac{1}{36} \approx 3\%$ and the probability of a 7 is $\frac{1}{6} \approx 17\%$. A 7 is six times more likely than a 12! Why?

When you roll the first die, there are six possibilities with equal probability. When you roll the second die, there are six possibilities with equal probability. So, there are a total of 36 possible events with equal probabilities: 1 then 1, 1 then 2, 2 then 1, 1 then 3, 3 then 1, and so on. Only one of these (1 then 1) adds to 2. However, six of these sum to 7: 1 then 6, 6 then 1, 2 then 5, 5 then 2, 3 then 4, and 4 then 3. So, a 7 is six times more likely than a 2.

Here is the complete table:

| Sum | | | | | | | Count | Probability |
|---|---|---|---|---|---|---|---|---|
| 2 | 1,1 | | | | | | 1 | 1/36 |
| 3 | 1,2 | 2,1 | | | | | 2 | 1/18 |
| 4 | 1,3 | 2,2 | 3,1 | | | | 3 | 1/12 |
| 5 | 1,4 | 2,3 | 3,2 | 4,1 | | | 4 | 1/9 |
| 6 | 1,5 | 2,4 | 3,3 | 4,2 | 5,1 | | 5 | 5/36 |
| 7 | 1,6 | 2,5 | 3,4 | 4,3 | 5,2 | 6,1 | 6 | 1/6 |
| 8 | | 2,6 | 3,5 | 4,4 | 5,3 | 6,2 | 5 | 5/36 |
| 9 | | | 3,6 | 4,5 | 5,4 | 6,3 | 4 | 1/9 |
| 10 | | | | 4,6 | 5,5 | 6,4 | 3 | 1/22 |
| 11 | | | | | 5,6 | 6,5 | 2 | 1/18 |
| 12 | | | | | | 6,6 | 1 | 1/36 |

If you don't believe these numbers, you could roll a pair of dice hundreds of times and make a histogram. However, it would be a tedious and time-consuming task — just the sort of thing that we make computers do for us.

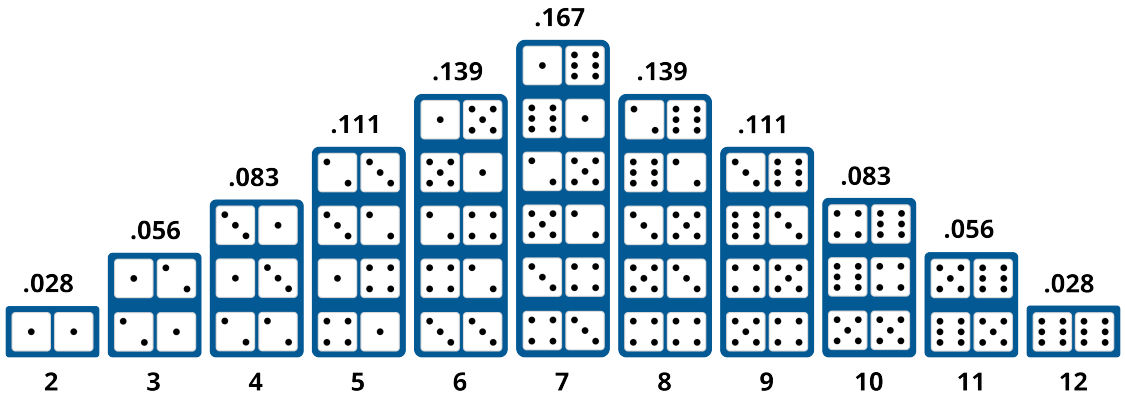



Figure 161.1: A histogram of probabilities of dice sums. This is referred to as a bell curve.

## 161.5 Random Numbers and Python

You are going to write a simulation of rolling dice in Python. To do this, you will need to generate a random sequence of numbers. The numbers will need to be in the range 1 to 6, and they will need to appear in the sequence with the same frequency. We say the sequence will follow *the uniform distribution*. In other words, the probability is uniformly distributed among the 6 possibilities.

Start python and try a few of the different ways to generate random numbers:

```
> python3
>>> import random
>>> random.random()  # Generates a random floating point number between 0 and 1
0.6840892758539989
>>> randrange(5)      # Generates an integer in the range 0 - 4
2
>>> x = ['Rock', 'Paper', 'Scissors']
>>> random.choice(x)   # Pick a random entry from the sequence
'Paper'
>>> x
['Rock', 'Paper', 'Scissors']
>>> random.shuffle(x)   # Shuffle the order of the sequence
>>> x
['Scissors', 'Paper', 'Rock']
>>> a = list(range(30))
>>> a
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,
  16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 29]
>>> random.sample(a, 10) # Return 10 randomly chosen items from the sequence
[8, 7, 20, 9, 25, 13, 23, 11, 14, 16]
```

Clearly, Python has many ways to do things that look random. However, here's the hidden truth: they aren't really random. The computer that you are using can't generate random data. Instead, it uses tricks to create data that look random; we call this *pseudorandom* data. Good pseudorandom algorithms are very important for cryptography and data security.

What if you want real random data? Some companies are using the decay of radioactive materials to generate real random data. You can pay to download it. For our purposes, Python's pseudorandom numbers are quite sufficient.

If we generate two random numbers in the range 1 through 6 and add them together, we will have simulated rolling a pair of dice. Like this:

```
>>> a = random.randrange(6) + 1
>>> b = random.randrange(6) + 1
>>> a + b
```

```
8
```

First, let's write a program that just rolls the dice 100 times and shows the result. Make a file `dice.py`:

```
import random

roll_count = 100

for i in range(roll_count):
    a = random.randrange(6) + 1
    b = random.randrange(6) + 1
    roll = a + b
    print(f"Toss {i}: {a} + {b} = {roll}")
```

When you run it, you should see something like:

```
> python3 dice.py
Toss 0: 6 + 6 = 12
Toss 1: 4 + 4 = 8
Toss 2: 4 + 2 = 6
Toss 3: 4 + 6 = 10
Toss 4: 4 + 4 = 8
...
Toss 98: 5 + 2 = 7
Toss 99: 5 + 2 = 7
```

Now we want to count occurrences of each possible outcome. Let's use an array of integers. We will start with an array of zeros. When we roll a 3, we will add 1 to item 3 in the array. (We can never roll a zero or a one, so those two entries will always be zero.)

```
import random

roll_count = 100

# Make an array containing 13 zeros
counts = [0] * 13

for i in range(roll_count):
    a = random.randrange(6) + 1
    b = random.randrange(6) + 1
    roll = a + b
    print(f"Toss {i}: {a} + {b} = {roll}")

    # Increment the count for roll
    counts[roll] += 1
```

```
16
17 print(f"Counts: {counts")}
```

When you run this, at the end you will see a count for each possible outcome:

```
1 ...
2 Toss 98: 3 + 2 = 5
3 Toss 99: 6 + 1 = 7
4 Counts: [0, 0, 2, 6, 16, 11, 13, 14, 11, 11, 6, 9, 1]
```

What was the count that we expected? For example, we expected to see a 2 about once every 36 rolls, right? It might be nice to compare our count to what we expected. Add a few more lines, and we are going to increase the number of rolls. You will probably want to delete the line that prints each roll separately:

```
1  import random
2
3  # Can't ever be 0 or 1
4  p = [0.0, 0.0, 1/36, 1/18, 1/12, 1/9, 5/36, 1/6, 5/36, 1/9, 1/12, 1/18, 1/36]
5  roll_count = 1000
6
7  # Make an array containing 13 zeros
8  counts = [0] * 13
9
10 for i in range(roll_count):
11     a = random.randrange(6) + 1
12     b = random.randrange(6) + 1
13     roll = a + b
14
15     # Increment the count for roll
16     counts[roll] += 1
17
18 for i in range(2,13):
19     print(f"{i appeared {counts[i]} times, expected {p[i] * roll_count:.1f}")}
```

Now you should see something like:

```
1 2 appeared 39 times, expected 27.8
2 3 appeared 55 times, expected 55.6
3 4 appeared 84 times, expected 83.3
4 5 appeared 110 times, expected 111.1
5 6 appeared 160 times, expected 138.9
6 7 appeared 176 times, expected 166.7
7 8 appeared 124 times, expected 138.9
8 9 appeared 93 times, expected 111.1
```

```
10 appeared 87 times, expected 83.3
11 appeared 49 times, expected 55.6
12 appeared 23 times, expected 27.8
```

Whenever you are dealing with random numbers, the outcome will seldom be *exactly* what you expected. In this case, however, you should see that your predictions are pretty close.

### 161.5.1 Making a bar graph

A bar graph is a nice way to look at quantities like this. Let's make a bar graph that shows the actual count and the expected count:

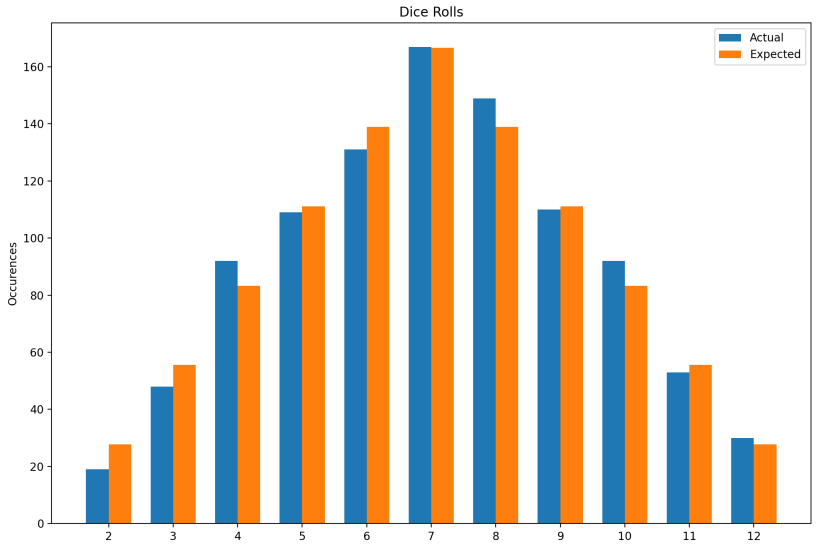



Figure 161.2: Dice roll histogram.

We need to describe the set of rectangles. To do this we will loop through each possible roll (2 - 12) and put data in four lists for each:

```
import random
import matplotlib.pyplot as plt

# Can't ever be 0 or 1
p = [0.0, 0.0, 1/36, 1/18, 1/12, 1/9, 5/36, 1/6, 5/36, 1/9, 1/12, 1/18, 1/36]
roll_count = 1000

```

```
# Make an array containing 13 zeros
counts = [0] * 13

for i in range(roll_count):
    a = random.randrange(6) + 1
    b = random.randrange(6) + 1
    roll = a + b

    # Increment the count for roll
    counts[roll] += 1

# Gather data for bar chart
bar_width = 0.35
expected = []
actual_starts = []
expected_starts = []
labels = []
actual = []
for i in range(2,13):
    expected.append(p[i] * roll_count)
    actual.append(counts[i])
    actual_starts.append(i - bar_width/2)
    expected_starts.append(i + bar_width/2)
    labels.append(i)

fig, ax = plt.subplots()

# Create the bars
ax.bar(actual_starts, actual, bar_width, label='Actual')
ax.bar(expected_starts, expected, bar_width, label='Expected')
ax.set_xticks(labels)

# Provide labels
ax.set_ylabel('Occurrences')
ax.set_title('Dice Rolls')
ax.legend()
plt.show()
```

# Chapter 162

# Beginning Combinatorics

Discrete probability problems often include some counting. For example, we figured out that there were 36 different ways the two dice could land, but all of them summed to some number 2 through 12. How many different ways could three eight-sided dice come up? We would need to count them, right? As the numbers get bigger, we will need some tricks so that we don't need to write them all down and count them one-by-one.

The branch of mathematics that focuses on tricks for counting is called *combinatorics*.

How can we be sure that there were 36 different configurations for the two 6-sided dice? The first die could have come up as any one of six numbers. For each of those, the second could have come up with any one of six numbers. Thus, the number of possibilities is $6 \times 6 = 36$.

How many different configurations for three 8-sided dice? $8 \times 8 \times 8 = 8^3 = 512$.

What about seven dice, each with 20 sides? There would be $20^7 = 1,280,000,000$ configurations. See, aren't you glad we don't need to write them all down?

Now, let's say that six people (Anne, Brock, Carl, Dev, Edgar, and Fred) are going to run a race. You have to make a plaque that says who won first place, who won second place, and who won third. If you want to get all the possible plaques created beforehand, and just pull the right one out as soon as the race ends, how many plaques would you need to get engraved?

Since you're making these plaques before the race even happens, you need one plaque for every possible outcome. So think of it as filling in three blanks: first place, second place, and third place. Any of the 6 people could fill the first blank. Whoever fills that blank can't also fill the second blank — a person can't come in both first *and* second — so only 5 people are left to fill the second blank. Likewise, only 4 people remain to fill the third blank, since two people have already been assigned to the first two spots. Multiplying these together, $6 \times 5 \times 4 = 120$, tells you exactly how many different first-second-third outcomes are possible — and thus how many plaques you'd need engraved.

What if the plaque includes all 6 places? Using the same reasoning, you'd fill in six blanks instead of three: $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$ plaques engraved. We use this process often enough that we gave it a name. We say "I need 6 factorial plaques engraved." When we write a *factorial*, we use an exclamation point:

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

We use the word "permutation" to mean a particular ordering. This rule says $n$ items can be ordered in $n!$ ways. Thus, mathematicians actually say "If you have a list of $n$ items then we can generate $n!$ different permutations of those items". We will discuss this further in the next chapter.

In Python, there is a `factorial` function in the math library:

```
> python3
>>> import math
>>> math.factorial(6)
720
```

Handy, right? Now you don't need to write a loop to calculate factorials.

Remember when we only wanted the first three names on the plaque? We can solve that problem using factorials:

$$6 \times 5 \times 4 = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = \frac{6!}{3!}$$

This formulation makes it easy to figure out on any calculator with a "!" button.

The rule on this is to fill $m$ positions from $n$ items. It can be done this many ways:

**Permutations**

For permutations where order is a factor, the formula for ${}^nP_m$ ($n$ objects pick $m$) is

$$\frac{n!}{(n-m)!}$$

## 162.1 Choose

Let's say that there are 12 kids in a classroom and you need a team of four to clean the top of the desks. How many different possible teams are there? You know that if you were giving out four different positions (Like the race gave out 1st, 2nd, and 3rd), the answer would be $12 \times 11 \times 10 \times 9$ or $12!/(12-4)!$.

However, once we pick the 4 people, we don't care what order they are in, right? In this problem, the team "Anne, Brad, Carl, and Don" is the same as the team "Carl, Don, Brad, and Anne".

Thus, the quantity $12!/(12-4)!$ is many times too large, because it counts each permutation separately. To get the right number, we just divide this by the number of possible permutations for a group of four people: $4!$

That gets us our answer: How many different teams of four can be chosen from 12 people?

$$\frac{12!}{(12-4)!4!} = 495$$

Generally, this formula, referred to as the *Binomial coefficient*. In combinatorics, we use this quantity a lot, so we have given it a name: *choose*

**Combinations**

For combinations, where order doesn't matter, the formula for ${}^{n}C_{k}$ ($n$ objects choose $k$) is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

We have also given it a notation. "12 choose 4" is written like this:

$$\binom{12}{4}$$

Python has the `math.comb` function:

```
> python3
>>> import math
>>> comb(12, 4)
495
```

We briefly saw this in the common polynomial products chapter. The binomial theorem says that the coefficients of the terms in the expansion of $(a+b)^n$ are given by the binomial coefficients $\binom{n}{k}$.

$$(a+b)^4 = 1a^4+4a^3b+6a^2b^2+4ab^3+1b^4 = \binom{4}{0}a^4+\binom{4}{1}a^3b+\binom{4}{2}a^2b^2+\binom{4}{3}ab^3+\binom{4}{4}b^4$$

## 162.2 Fractional Binomial Theorem

An issue arises when we have fractional exponents for this theorem. For example, what is $\sqrt{1+5x} = (1+5x)^{1/2}$? We can use the binomial theorem to expand this as follows:

$$(1+5x)^{1/2} = \binom{1/2}{0}1 + \binom{1/2}{1}5x + \binom{1/2}{2}(5x)^2 + \binom{1/2}{3}(5x)^3 + \dots$$

Pascal's triangle only works when the exponent $n$ is a non-negative integer. Here, the exponent is $\frac{1}{2}$, not a whole number. There's no "row $\frac{1}{2}$" of Pascal's triangle, so we need to extend the pattern.

This extension is called the ***general binomial theorem***. To make sense of $\binom{1/2}{k}$, we need a definition of the binomial coefficient that doesn't rely on factorials, since $(1/2)!$ doesn't make sense. Recall that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)(n-2)\cdots(n-k-1)}{k!}$$

Notice that the right-hand side only ever multiplies and divides — it never actually requires $n$ to be a whole number! This gives us a definition that works for *any* real number $n$, including fractions and negative numbers:

**Generalized Binomial Coefficient**

For any real number $n$ and non-negative integer $k$,

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$$

When $n$ is *not* a non-negative integer, like $n = \frac{1}{2}$, none of the factors $n, n-1, n-2, \dots$ is ever zero, so the expansion never terminates. Instead of a finite row of Pascal's triangle, we get an infinite series.

**General Binomial Theorem**

Recall that for $|x| < 1$, the geometric series gives

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

Since $\frac{1}{1-x} = (1-x)^{-1}$, this looks like a binomial expansion where the exponent $-1$ is not a positive integer. Using the generalized binomial coefficient,

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$$

For any real number $n$ and $|x| < 1$,

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

this is called the *binomial series*, and it generalizes the binomial theorem to arbitrary real exponents. Setting $n = -1$ recovers the geometric series above.

The condition $|x| < 1$ matters here for convergence. When $n$ was a positive integer, the expansion was just a polynomial, and polynomials are perfectly well-behaved for every value of $x$. But now that we have an infinite series, we need the terms to actually get smaller and smaller so the sum settles down to a finite value. If $|x| \geq 1$, the terms don't shrink, and the series doesn't converge to anything useful.

Let's compute the coefficients for $\binom{1/2}{k}$

$$\begin{aligned}
\binom{1/2}{0} &= 1 \\
\binom{1/2}{1} &= \frac{1}{2} \\
\binom{1/2}{2} &= \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!} = \frac{\frac{1}{2}\cdot\left(-\frac{1}{2}\right)}{2} = -\frac{1}{8} \\
\binom{1/2}{3} &= \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!} = \frac{3/8}{6} = \frac{1}{16}
\end{aligned}$$

So,

$$(1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots$$

Now we can go back to our original problem, $\sqrt{1+5x} = (1+5x)^{1/2}$. This is just $(1+x)^{1/2}$ with $x$ replaced by $5x$, so we substitute $5x$ everywhere:

$$\begin{aligned}
\sqrt{1+5x} &= 1 + \frac{1}{2}(5x) - \frac{1}{8}(5x)^2 + \frac{1}{16}(5x)^3 - \cdots \\
&= 1 + \frac{5}{2}x - \frac{25}{8}x^2 + \frac{125}{16}x^3 - \cdots
\end{aligned}$$

This expansion is only valid for $|5x| < 1$, i.e. $|x| < \frac{1}{5}$.

### *Exercise 411* Fractional Binomial Expansion

*Working Space*

1. Find the first four terms in the expansion of $(1+x)^{-1}$, in ascending powers of $x$. For what values of $x$ is this expansion valid?
2. Find the first three terms in the expansion of $\sqrt{1-2x}$, in ascending powers of $x$.

*Answer on Page 2052*

## 162.3 Applications in Probability

Suppose you flip a fair coin 5 times. What's the probability that you get exactly 3 heads?

Note that there are only two options: Heads (H) or Tails (T). Thus, there are $2^5 = 32$ possible sequences of flips. Each sequence is equally likely, so the probability of any particular sequence is $\frac{1}{32}$. Only a certain number of sequences will have exactly 3 heads.

One way to get 3 heads is HHHTT. There's also HTHHT, THHHT, and many other orderings also give you 3 heads. Before we can find the probability, we need to know *how many* different sequences of 5 coin flips contain exactly 3 heads. As we have discussed, we can use the $n$-choose-$k$ formula. There are

$$\binom{5}{3} = \frac{5!}{3!2!} = 10$$

such sequences. This is said "5 choose 3", or, more explicitly "get 3 successes in 5 trials".

Each individual sequence, like HHHTT, has probability $\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2$ of occurring, since each flip is independent. Since there are 10 equally likely sequences that give exactly 3 heads, we add up 10 copies of that probability:

$$\text{P(exactly 3 heads)} = \binom{5}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 = 10 \cdot \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$$

This pattern of counting the arrangements with $\binom{n}{k}$, then multiplying by the probability of one arrangement works whenever you have repeated *independent* trials with only two possible outcomes each time (success or failure).

### *Exercise 412* Free Throws

A free throw shooter has a past record of making 60% of her free throws. Over the next 6 free throws, what is the probability that she makes exactly 4 of them?

*Working Space*

*Answer on Page 2052*

This probability situation has been rightfully called the *binomial distribution*. This is a brief introduction to the binomial distribution, and we will discuss it in more detail in a later chapter.

Chapter 163

# Permutations and Sorting

In the previous chapter, we talked about permutations. If you have a list of four letters, like $[a, b, c, d]$, you can rearrange them in $4!$ ways:

| | | | | | |
|---|---|---|---|---|---|
| a,b,c,d | a,b,d,c | a, d, b, c | a, d, c, b | a, c, b, d | a, c, d, b |
| b,a,c,d | b,a,d,c | b, d, a, c | b, d, c, a | b, c, a, d | b, c, d, a |
| c,b,a,d | c,b,d,a | c, d, b, a | c, d, a, b | c, a, b, d | c, a, d, b |
| d,b,c,a | d,b,a,c | d, a, b, c | d, a, c, b | d, c, b, a | d, c, a, b |

You can make Python generate all the permutations for you:

```
from itertools import permutations
all_permutations = permutations(('a', 'b', 'c', 'd'))
for p in all_permutations:
    print(p)
```

## 163.1 Notation

How do we define or write down a single permutation? You could say something like "Swap the first and second items and swap the third and fourth items." However, that gets pretty difficult to read, so we usually write a permutation as two lines: the first line is before the permutation and the second line is after. Like this:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

We can also assign permutations to variables. For example, if we wanted the variable $A$ to represent "swapping the first and second item", we would write this:

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix}$$

And if we wanted $B$ to represent "swapping the third and fourth item", we would write:

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix}$$

Now, we can *compose* permutations together. For example, we might say:

$$B \circ A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

In other words, if we have the list $[a, b, c, d]$ and we apply permutation $A$, followed by permutation $B$, we get $[b, a, d, c]$.

**Important:** Note that permutations are applied from right to left. $B \circ A$ means "Applying $A$ and then $B$." Why does this matter? Permutations are not necessarily commutative. That is, if you have two permutations $S$ and $T$, $S \circ T$ is not always the same as $T \circ S$.

Also, note that "don't change anything" is a permutation. We call it *the identity permutation*. If you have four items, the identity permutation would be written:

$$I = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

(We use a capital "I" for the identity.)

### 163.1.1 Challenge

Find an example of two permutations $S$ and $T$, such that $S \circ T$ does not equal $T \circ S$.

## 163.2 Sorting in Python

One of the common forms of permutation in software is sorting. Sorting is putting data in a particular order. For example, in Python, if you had a list of numbers, you can sort it in ascending order like this:

```
my_grades = [92, 87, 76, 99, 91, 93]
grades_worst_to_best = sorted(my_grades)
```

Do you want to sort backwards?

```
my_grades = [92, 87, 76, 99, 91, 93]
grades_best_to_worst = sorted(my_grades, reverse=True)
```

Note that `sorted` makes a new list with the correct order. If you want to sort the array in place, you can use the `sort` method:

```
my_grades = [92, 87, 76, 99, 91, 93]
my_grades.sort(reverse=True)
```

## 163.3 Inverses

Think for a second about this permutation:

$$S = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

You could say this permutation shuffles a list. What is its inverse? That is, what is the permutation that unshuffles the items back to where they were originally?

$$S^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

In other words, the original moved an item in the first spot to the third spot; the inverse must move whatever was in the third spot back to the first spot.

(Notation note: In multiplication, $b \times b^{-1} = 1$, so we use "to the negative one" to indicate inverses in lots of places.)

Mechanically, how do you find the inverse? Flip the rows, then sort the columns using the top number:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \text{ flip } \rightarrow \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \text{ sort } \rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

Let's say you have two permutations A and B. Permuting by B then A would look like this:

$$C = A \circ B$$

If you know $A^{-1}$ and $B^{-1}$, what is $C^{-1}$? You would undo-A, then undo-B, so

$$C^{-1} = B^{-1} \circ A^{-1}$$

## 163.4 Cycles

Here is a permutation:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{pmatrix}$$

When this is applied, whatever is at 1 gets moved to 2, 2 gets moved to 4, and 4 gets moved to 1. That is a *cycle*: $1 \to 2 \to 4$, then it goes back to 1. It involves three locations, so we say it is a *3-cycle*.

There is another cycle in this permutation: $3 \to 5$, then it goes back to 3.

Because these cycles share no members, we say the cycles are *disjoint*.

Every permutation can be broken down into a collection of disjoint cycles.

$$T = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{pmatrix} = (1 \to 2 \to 4)(3 \to 5)$$

The first handy thing about this notation is that it makes it easy for us to describe the inverse. We just run the cycles backward:

$$T^{-1} = (4 \to 2 \to 1)(5 \to 3)$$

Starting with the list $[a, b, c, d, e]$, let's repeatedly apply the permutation T

| | |
|---|---|
| Initial | a, b, c, d, e |
| T applied | d, a, e, b, c |
| $T \circ T$ applied | b, d, c, a, e |
| $T \circ T \circ T$ applied | a, b, e, d, c |
| $T \circ T \circ T \circ T$ applied | d, a, c, b, e |
| $T \circ T \circ T \circ T \circ T$ applied | b, d, e, a, c |
| $T \circ T \circ T \circ T \circ T \circ T$ applied | a, b, c, d, e |

This permutation results in six combinations, then it loops back on itself. The number of

combinations is the least common multiple of all the cycles. In this case, there is a 3-cycle and a 2-cycle. The least common multiple of 2 and 3 is 6.

## 163.5 Further Learning

Check out this leetcode problem which combines permutations and programming! https://leetcode.com/problems/permutations/description/

Chapter 164

# Statistical Inferences for Proportions

## 164.1 Interval Estimation and the Confidence Interval

Suppose you had to guess your own weight, and then guess a classmate's weight. You would probably feel more confident about your own guess. You might say 140 pounds for both, but you would trust your own number a lot more than the one for your classmate. A point estimate only gives you a single number. It does not tell you how much to trust that number.

A point estimate is a single value that is calculated from a sample. It serves as our best guess for a population parameter. Although it can be useful, it is incomplete. Two point estimates can look identical on paper while carrying very different amounts of certainty behind them, like the way your weight guess and your classmate's guess might both be 140 pounds while being backed by very different amounts of information.

This is where the *confidence interval* comes in. A confidence interval gives a range of plausible values for the population parameter that is built around the point estimate. The *margin of error* is the "plus or minus" amount that sets the width of that range. It is driven by how confident we want to be and how much the estimate varies from sample to sample.

A wider interval is more likely to capture the true value, but it also tells you less. This tradeoff between confidence and precision runs through everything in this chapter.

## 164.2 Constructing a Confidence Interval for a Proportion

When we discussed confidence intervals, we saw that a wider interval gives us more certainty at the cost of being precise. That tradeoff is captured by a single number called the *critical value*, written $z^*$.

Let's say a city council wants to estimate the proportion of residents who support building a new network of bike lanes. Before collecting any data, they need to decide how confident they want to be in their eventual estimate, say 95%. That confidence level determines $z^*$.

A 95% confidence level leaves 5% split evenly between the two tails of the standard normal distribution, so 2.5% in each tail. We need the value $z^*$ such that the area to its right

is 0.025, which means the area to its left is 0.975. Using a standard normal table or a calculator's inverse normal function, $z^* \approx 1.96$.

If the city council instead wanted 99% confidence, would $z^*$ be larger or smaller than 1.96?

Once we have $z^*$, we still need to decide how precise we want our estimate to be. An interval can technically be true and still be worthless. If the council's confidence interval for bike lane support turned out to be "somewhere between 5% and 95%," that would be almost useless for making a decision, even if it is a perfectly valid 95% confidence interval. The width of a useful interval, called the margin of error, depends on how large a sample the council collects. Larger samples produce narrower, more informative intervals for the same confidence level.

This is why sample size planning matters just as much as the confidence level. A researcher who wants a useful interval has to think ahead about how much data collecting that precision will require.

**Computing in Excel**

**Computing in Python**

## 164.3 Inference: Tests of Significance

Building a confidence interval answers "what is the population proportion likely to be?" But sometimes the question you actually care about is different: has something changed? *Statistical inference* is the process of using information from a sample to answer a specific question or make a decision about a population parameter, and a *test of significance* is the tool built for exactly this kind of question.

Chapter 165

# Statistical Inferences for Means

In Statistical Inferences for Proportions, every question came down to a percentage: what proportion of trial users converted to paying customers, and whether that proportion changed after a redesign. Not every question reduces to a percentage.

Sometimes what you want to know is a genuine measurement, how long something takes, how much something weighs, how far something travels, on average.

A quantitative variable like this calls for the same two tools you already trust, a confidence interval to estimate a mean and a significance test to check a claim about one, but neither tool behaves the way it did for proportions. This is because sometimes you are missing crucial information: you almost never know the population's true standard deviation, only what your own sample can tell you. That gap forces a new distribution into the picture, which looks like the normal curve.

## 165.1 The t-Distribution and Conditions for Inference about a Mean

Suppose the streaming service now wants to know something different from Statistical Inferences for Proportions's conversion rate: how long, on average, a subscriber watches in a single sitting. A product analyst pulls a random sample of viewing sessions and calculates a sample mean watch time, $\bar{x}$. To turn that one number into a confidence interval or a significance test, you need the standard error of $\bar{x}$, and the standard error depends on the population standard deviation, $\sigma$. Here is the catch: nobody knows $\sigma$. The population's true spread in viewing habits is exactly the kind of thing you would need a census to pin down, and if you had a census, you would not need inference in the first place.

### 165.1.1 Meet the t-Distribution

The sample standard deviation $s$ estimates $\sigma$, but substituting $s$ changes $\dfrac{\bar{x}-\mu}{s/\sqrt{n}}$ from a normal distribution to a t-*distribution*.

The t-distribution is continuous, symmetric, and centered at 0, but shorter and thicker-tailed than the normal distribution. Its shape depends on *degrees of freedom*, abbreviated $df$. For a single mean, $df = n-1$. As $n$ grows, the t-distribution converges to the standard normal distribution.

The critical value $t^*$ replaces $z^*$ and is read from a t-table using both $df$ and the tail area.

For $n = 18$ ($df = 17$), a 90 percent confidence level gives $t^* = 1.740$, versus $z^* = 1.645$ for a known-$\sigma$ case. At $df = 100$, $t^* = 1.984$, much closer to $z^*$.

### *Exercise 413* Reading the t-Table

A logistics company samples $n = 25$ delivery times and wants to construct a 95 percent confidence interval for the true mean delivery time. State the degrees of freedom, and find $t^*$.

*Working Space*

*Answer on Page 2052*

#### 165.1.2 Conditions for Inference about a Mean

Two conditions must hold for a t-procedure to be valid.

Independence: the data come from a random sample or randomized experiment, and if sampling without replacement, the sample size is no more than 10 percent of the population.

Normal or large sample: the sampling distribution of $\bar{x}$ is approximately normal. Check this using a dotplot, stemplot, or normal probability plot for strong skew or outliers, or rely on the population being known normal, or $n \geq 30$.

Use $t$, not $z$, unless $\sigma$ is explicitly given, a case that rarely occurs in practice or on the AP exam.

- Independence: random sample or randomized experiment, and $n \leq 10\%$ of the population when sampling without replacement.
- Normal or large sample: no strong skew or outliers in the data, the population is known to be normal, or $n \geq 30$.
- Default to a t-procedure. Only switch to $z$ if $\sigma$ is explicitly given.

### *Exercise 414* Spot the Error

A student is asked to construct a confidence interval for a population mean using a sample of $n = 22$ measurements. The student writes, "Since $n = 22$ is less than 30, I cannot use a t-interval. I need to collect a bigger sample first." Explain what is wrong with this reasoning.

*Working Space*

*Answer on Page 2053*

#### 165.1.3 When the Data Comes in Pairs

Some data comes in pairs: two measurements per subject, such as watch time before and after a recommendation-algorithm update for the same subscribers. The two measurements within a pair are not independent.

Data like this is called *matched pairs*. Subtract to get one difference per subject, $d_i = x_i - y_i$, reducing the data to a single sample of differences. The t-distribution, degrees of freedom, and the independence and normality conditions all apply directly to that sample of differences. Independence now refers to the pairs being independent of one another, not the two measurements within a pair.

Matched pairs is not the same as comparing two independent means: two independent means needs two separate random samples, while matched pairs needs one sample of paired subjects.

### *Exercise 415* Paired or Independent?

For each scenario, state whether the data described would be analyzed as matched pairs or as two independent samples, and justify your answer in one sentence.

(a) Twenty subscribers each have their nightly watch time recorded for one week before a recommendation algorithm update and one week after.

(b) A researcher randomly splits 200 subscribers into two groups, shows one group only comedy titles and the other only drama titles for a month, and compares the two groups' average watch time.

*Working Space*

*Answer on Page 2053*

## 165.2 Constructing and Justifying a Confidence Interval for a Population Mean

You now have everything you need to actually build an interval. Every confidence interval takes the shape:

$$\text{point estimate} \pm \text{margin of error}$$

### 165.2.1 The Confidence Interval for a Population Mean

The point estimate for a population mean $\mu$ is the sample mean, $\bar{x}$. The margin of error is the critical value $t^*$ times the standard error of $\bar{x}$, which is $\frac{s}{\sqrt{n}}$. Putting the two together,

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

is a $(1-\alpha)100\%$ confidence interval for $\mu$, using $t^*$ from a t-distribution with $n-1$ degrees of freedom.

Back to the streaming service. The analyst samples $n = 24$ viewing sessions and finds a sample mean watch time of $\bar{x} = 47.8$ minutes with a sample standard deviation of $s = 9.6$ minutes. A dotplot of the sample shows no strong skew or outliers, and the sessions were pulled at random from the full session log, well under 10 percent of all sessions logged that month. Both conditions are satisfied, so a t-interval is appropriate.

For a 95 percent confidence interval, $df = 23$ and $t^* = 2.069$. The standard error is $\frac{9.6}{\sqrt{24}} \approx 1.960$, so the margin of error is $2.069 \times 1.960 \approx 4.05$ minutes. The interval is

$$47.8 \pm 4.05 \Rightarrow (43.75,\ 51.85)$$

so the analyst estimates the true mean session watch time to be between 43.75 and 51.85 minutes.

## *Exercise 416* Building a t-Interval

A city transportation office samples $n = 16$ commute times and finds $\bar{x} = 28.4$ minutes and $s = 6.2$ minutes. Conditions for inference are satisfied. Construct a 98 percent confidence interval for the true mean commute time.

*Working Space*

*Answer on Page 2053*

### 165.2.2 Interpreting and Justifying the Interval

An interpretation references two things: the confidence level and the population. For the streaming service example: the analyst is 95 percent confident that the true mean watch time per session, across all sessions, falls between 43.75 and 51.85 minutes.

Do not say there is a 95 percent probability that $\mu$ falls in this interval. Once calculated, $\mu$ either is or is not inside it. The 95 percent describes the method: if the sampling process were repeated many times, about 95 percent of the resulting intervals would capture the true $\mu$.

A confidence interval can justify a claim by checking whether the claimed value falls inside

or outside it. If the product team's target is an average watch time of at least 45 minutes, the interval $(43.75, 51.85)$ dips below 45, so it does not provide convincing evidence that the true mean is at least 45 minutes.

For a fixed confidence level, interval width shrinks as $n$ grows, roughly proportional to $\frac{1}{\sqrt{n}}$: quadrupling $n$ from 24 to 96 shrinks the margin of error from about 4.05 to about 1.95 minutes, since $\sqrt{4} = 2$. For a fixed sample, raising the confidence level widens the interval.

**Computing in Excel**

Excel does not have a single built-in function that hands back a finished confidence interval, but two functions cover everything once you have $n$, $\bar{x}$, and $s$.

If you have the raw session lengths in a column, say `A2:A25`, get the three summary values first:

- `=AVERAGE(A2:A25)` for $\bar{x}$
- `=STDEV.S(A2:A25)` for $s$
- `=COUNT(A2:A25)` for $n$

Suppose those land in cells `B1` ($n = 24$), `B2` ($\bar{x} = 47.8$), and `B3` ($s = 9.6$), with the confidence level in `B4` (`0.95`).

The most direct route is `CONFIDENCE.T`, which returns the margin of error by itself:

- `=CONFIDENCE.T(1-B4, B3, B1)` returns 4.054, the margin of error.

Then the interval bounds are just `=B2-C1` and `=B2+C1`, where `C1` holds that margin of error.

If you would rather see each piece of the formula spelled out, build it manually instead:

- `=B1-1` for $df$
- `=T.INV.2T(1-B4, C1)` for $t^*$, where `C1` is the $df$ cell (`T.INV.2T` takes the two-tailed probability, so use $1 - 0.95 = 0.05$ directly, not $0.025$)
- `=B3/SQRT(B1)` for the standard error
- `=C2*C3` for the margin of error, where `C2` and `C3` hold $t^*$ and the standard error

Both routes return the same margin of error, 4.054, and the same interval, $(43.75, 51.85)$. The manual version is worth building at least once, since it makes visible exactly where

the t-distribution and the sample size are doing their work; `CONFIDENCE.T` is what you would reach for afterward, once you trust the formula.

## *Exercise 417* A Matched-Pairs Interval

A call center tests a new training program on $n = 12$ employees, recording each employee's average call-handling time before and after training. The sample of differences (before minus after) has $\bar{d} = -3.2$ minutes and $s_d = 4.5$ minutes. Conditions for inference on the differences are satisfied. Construct a 90 percent confidence interval for $\mu_d$, and state what it suggests about the training program.

*Working Space*

*Answer on Page 2053*

## *Exercise 418* Spot the Error

A student reports the streaming-service interval $(43.75, 51.85)$ and writes, "There is a 95 percent probability that the true mean watch time is between 43.75 and 51.85 minutes." Explain what is wrong with this interpretation, and rewrite it correctly.

*Working Space*

*Answer on Page 2054*

## 165.3 Setting Up and Carrying Out a Test for a Population Mean

### 165.3.1 Setting Up the Test

A test for a population mean starts the same way a test for a proportion did: state hypotheses about the parameter, never about a sample statistic. The null hypothesis takes the form $H_0 : \mu = \mu_0$, where $\mu_0$ is whatever value is being challenged. The alternative hypothesis is $H_a : \mu < \mu_0$, $H_a : \mu > \mu_0$, or $H_a : \mu \neq \mu_0$, depending on which direction the question points.

Choosing the method and verifying conditions works exactly as it did in the last section: check independence, check that the population is normal or the sample is large enough and free of strong skew or outliers, and default to a t-test unless $\sigma$ is somehow known. If the data arrives as matched pairs, subtract first, then run everything that follows on the sample of differences, testing $H_0 : \mu_d = 0$ against whichever alternative fits the question.

### 165.3.2 Carrying Out the Test

The test statistic for a population mean is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

with $n - 1$ degrees of freedom. Find the p-value from the t-distribution, matching the tail(s) to $H_a$, and compare to $\alpha$. Reject $H_0$ if p-value $< \alpha$.

A streaming service wants to know whether average session watch time has increased above the old baseline of 42 minutes after a redesign. Sample: $n = 30$, $\bar{x} = 44.6$ minutes, $s = 8.2$ minutes. Conditions are satisfied.

$$H_0 : \mu = 42 \qquad H_a : \mu > 42$$

$\text{df} = 29$, standard error $= \dfrac{8.2}{\sqrt{30}} \approx 1.497$, so

$$t = \frac{44.6 - 42}{1.497} \approx 1.737$$

p-value $\approx 0.0465$. Since $0.0465 < 0.05$, reject $H_0$: sufficient evidence that mean session watch time increased above 42 minutes. A smaller p-value is stronger evidence against $H_0$ than one just under $\alpha$.

Matched-pairs example: call-handling time, $\bar{d} = -3.2$ minutes, $s_d = 4.5$ minutes, $n = 12$. Testing $H_0 : \mu_d = 0$ against $H_a : \mu_d < 0$ gives $\text{df} = 11$, $t \approx -2.463$, p-value $\approx 0.0157$.

At $\alpha = 0.05$, reject $H_0$: sufficient evidence the training program decreased average call-handling time. A confidence interval and a significance test on the same data will always agree.

## *Exercise 419* Setting Up and Testing a Claim

A bakery's recipe is supposed to produce loaves with a mean weight of 500 grams. A random sample of $n = 16$ loaves has $\bar{x} = 498.2$ grams and $s = 6.4$ grams, with no strong skew or outliers in the sample. At $\alpha = 0.05$, is there sufficient evidence that the true mean loaf weight differs from 500 grams?

*Working Space*

*Answer on Page 2054*

## *Exercise 420* A Matched-Pairs Test

A wellness program measures resting heart rate for $n = 15$ participants before and after an eight-week exercise regimen. The sample of differences (before minus after) has $\bar{d} = 4.8$ beats per minute and $s_d = 6.1$ beats per minute. Conditions for inference on the differences are satisfied. At $\alpha = 0.05$, is there sufficient evidence that the regimen decreased mean resting heart rate?

*Working Space*

*Answer on Page 2054*

### *Exercise 421* Spot the Error

A student sets up a test with the hypotheses $H_0 : \bar{x} = 500$ and $H_a : \bar{x} \neq 500$. Explain what is wrong with this setup.

*Working Space*

*Answer on Page 2054*

## 165.4 Confidence Intervals for a Difference in Two Means

Every example so far has looked at one population at a time. Just as often, the real question is a comparison: not "what is the mean watch time," but "is the mean watch time different between two groups."

### 165.4.1 Two Independent Samples

Comparing average watch time between free-tier and paid-tier subscribers, two disjoint groups, means the samples are independent. Let $\mu_1, \mu_2$ be the true means. The parameter of interest is $\mu_1 - \mu_2$, point estimate $\bar{x}_1 - \bar{x}_2$.

$$\text{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Degrees of freedom come from the Welch-Satterthwaite approximation: between $\min(n_1 - 1, n_2 - 1)$ and $n_1 + n_2 - 2$.

Conditions: both samples independently and randomly selected, each $n \leq 10\%$ of its population if sampling without replacement, and each sample free of strong skew/outliers or $n \geq 30$.

Sample: free-tier $n_1 = 20$, $\bar{x}_1 = 35.2$, $s_1 = 8.4$; paid-tier $n_2 = 18$, $\bar{x}_2 = 44.6$, $s_2 = 10.1$.

Conditions satisfied. For a 95 percent CI,

$$\text{SE} = \sqrt{\frac{8.4^2}{20} + \frac{10.1^2}{18}} \approx 3.032, \quad \text{df} \approx 33, \quad t^* \approx 2.035$$

Margin of error $\approx 6.17$ minutes, point estimate $= -9.4$ minutes:

$$-9.4 \pm 6.17 \Rightarrow (-15.57,\ -3.23)$$

95 percent confident the true difference (free-tier minus paid-tier) is between $-15.57$ and $-3.23$ minutes. Since the entire interval is negative, paid-tier subscribers watch more per session on average.

## *Exercise 422* Building a Two-Sample Interval

A tech reviewer compares battery life between two phone models. Model A: $n_1 = 15$, $\bar{x}_1 = 11.2$ hours, $s_1 = 1.8$ hours. Model B: $n_2 = 17$, $\bar{x}_2 = 9.8$ hours, $s_2 = 2.1$ hours. The two samples are independent, and conditions for inference are satisfied. Construct a 90 percent confidence interval for the true difference in mean battery life, Model A minus Model B. Technology gives df $\approx 29$.

*Working Space*

*Answer on Page 2054*

## *Exercise 423* Justifying a Claim

Using the battery-life interval from the previous exercise, does the data provide convincing evidence that Model A's true mean battery life exceeds Model B's by more than one hour? Explain using the interval.

*Working Space*

*Answer on Page 2055*

### *Exercise 424* Spot the Error

A student compares two independent samples with visibly different spreads, one sample's dotplot is tightly clustered, the other is spread out much wider, and writes, "Since both populations are approximately normal, I'll pool the two sample standard deviations into one estimate of $\sigma$ and use that in my confidence interval." Explain what is wrong with this reasoning for an AP Statistics response.

*Working Space*

*Answer on Page 2055*

## 165.5 Significance Tests for a Difference in Two Means, and Selecting the Right Procedure

### 165.5.1 Setting Up and Carrying Out the Test

TTesting a claim about $\mu_1 - \mu_2$ follows the same four-step process, usually $H_0 : \mu_1 - \mu_2 = 0$, with $H_a : \mu_1 - \mu_2 < 0$, $> 0$, or $\neq 0$.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

using the same Welch-approximated df and conditions from Section 4.

Free-tier versus paid-tier, same samples as Section 4: $n_1 = 20$, $\bar{x}_1 = 35.2$, $s_1 = 8.4$; $n_2 = 18$, $\bar{x}_2 = 44.6$, $s_2 = 10.1$. Testing whether paid-tier watches more on average:

$$H_0 : \mu_1 - \mu_2 = 0 \qquad H_a : \mu_1 - \mu_2 < 0$$

SE $= 3.032$ (from Section 4), so

$$t = \frac{35.2 - 44.6}{3.032} \approx -3.100$$

df $\approx 33$, one-sided p-value $\approx 0.00196$. Reject $H_0$: convincing evidence paid-tier subscribers

watch more per session on average. A test and a confidence interval on the same data will always agree.

### *Exercise 425* Running a Two-Sample Test

Two store locations track customer wait times. Store 1: $n_1 = 22$, $\bar{x}_1 = 8.3$ minutes, $s_1 = 2.1$ minutes. Store 2: $n_2 = 25$, $\bar{x}_2 = 9.6$ minutes, $s_2 = 2.8$ minutes. The samples are independent, and conditions for inference are satisfied. Test, at $\alpha = 0.05$, whether Store 1's mean wait time is shorter. Technology gives $df \approx 44$.

*Working Space*

*Answer on Page 2055*

#### 165.5.2 Selecting the Right Procedure

By this point you have four inference procedures for means and, from the last chapter, four more for proportions. The hardest part of a real problem is often not the arithmetic, it is recognizing which of those eight procedures actually fits. Figure 1 lays out the same three questions, in order, that will get you there every time.

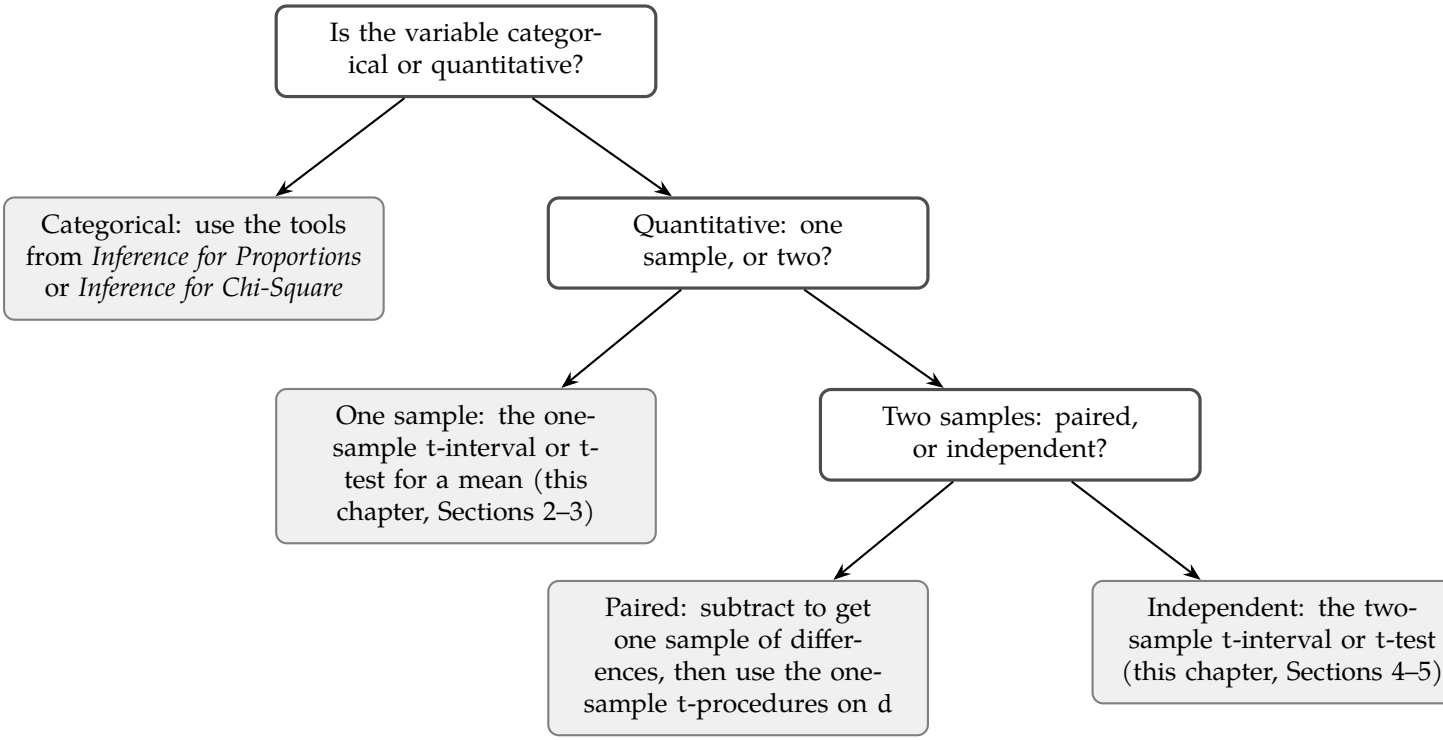



Figure 165.1: Selecting an inference procedure for a single quantitative or categorical variable

Categorical or quantitative is usually obvious from the variable itself. Paired or independent is worth slowing down for: does each measurement in one group have a natural partner in the other, or could the two groups' labels be shuffled without anything becoming meaningless? Same subscriber measured twice: shuffling is meaningless, so paired. Two disjoint groups: shuffling changes nothing, so independent.

### 165.5.3 Common Errors

Type I error, Type II error, and power all carry over from the last chapter without any changes to their definitions, just a change in what the parameter is. A Type I error here means rejecting $H_0 : \mu_1 = \mu_2$ when the two population means really are equal, concluding the streaming service's paid tier watches more when it actually does not. A Type II error means failing to reject $H_0$ when there really is a difference, missing a real effect. Power is still the probability of correctly detecting a real difference when one exists, and it still increases with a larger sample size, a larger true difference, or a higher significance level, exactly as it did for proportions.

## *Exercise 426* Naming the Error

In the store wait-time test from the previous exercise, the data led to rejecting $H_0$ in favor of Store 1 having a shorter mean wait time. Suppose, unknown to the researcher, the two stores' true mean wait times are actually equal. What type of error, if any, occurred? Explain.

*Working Space*

*Answer on Page 2055*

## *Exercise 427* Choosing the Procedure

For each scenario, state which of the four procedures from Figure 1 applies, quantitative one-sample, quantitative paired, quantitative independent-samples, or categorical.

*Working Space*

(a) A gym wants to know whether the average weight lifted by 30 randomly selected members changed after switching to a new equipment brand, measuring the same members' max lift before and after the switch.

(b) A researcher wants to compare the proportion of two different email subject lines that get opened.

(c) A nutritionist samples 40 adults at random and wants to estimate the true mean daily sugar intake in the population.

*Answer on Page 2056*

That decision flowchart will keep earning its keep. The next chapter turns to a different kind of categorical question, one with more than two possible outcomes at once, and the first thing you will need to decide, again, is which procedure actually fits the shape of the data in front of you.

CHAPTER 166

# Statistical Inferences for Slopes

## 166.1 The Regression Model and Why We Need Inference for the Slope

A sleep researcher pulls data from a sample of 8 teenagers: daily screen time, in hours, and how many hours they slept that night. She plots the points, fits a line, and finds a slope of $b_1 = -0.16$: for every extra hour of screen time, sleep drops by about ten minutes. Does this mean the effect is real across all teenagers, or could a sample this size show a slope like this by chance?

Recall from the Numerical Methods chapter that the least-squares line $\hat{y} = b_0 + b_1 x$ describes your sample: descriptive statistics, not inference.

A sample slope $b_1$ estimates an unknown population slope $\beta_1$, the same way $\hat{p}$ estimates $p$ and $\bar{x}$ estimates $\mu$. Statisticians describe the true relationship between two quantitative variables using a *population regression model*:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

$\beta_0$ is the population y-intercept, $\beta_1$ is the population slope, and $\varepsilon$ is random error: the part of $Y$ not explained by $X$. $\beta_0$ and $\beta_1$ are parameters, describing the entire population; $b_0$ and $b_1$ are statistics, calculated from your sample, used to estimate them.

The rest of this chapter reasons from $b_1$, the number your sample gives you, to $\beta_1$, the value you actually care about, using the same confidence interval and significance test tools you've already built for proportions and means.

### Estimating the Slope and Intercept

The formulas for $b_0$ and $b_1$:

$$b_1 = r\frac{s_y}{s_x} \qquad \text{and} \qquad b_0 = \bar{y} - b_1\bar{x}$$

where $r$ is the sample correlation coefficient and $s_x, s_y$ are the sample standard deviations of $X$ and $Y$. If you are working from raw sums instead of $r$, $s_x$, and $s_y$, an algebraically equivalent formula is

$$b_1 = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

Both formulas give the same $b_1$. You can use whichever matches the summary statistics you have.

The residual for any data point is still $e = y - \hat{y}$, where $\hat{y} = b_0 + b_1 x$ is the value your line predicts. The standard deviation of all the residuals in your sample, $s_e$, tells you the typical size of a prediction miss. You will use $s_e$ heavily, because it is the key in measuring how much $b_1$ itself might vary from sample to sample.

Following the sleep researcher's study, she samples 8 teenagers at random and records their daily screen time (X, in hours) and their sleep duration that night (Y, in hours):

| Screen time (hr) | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|
| Sleep duration (hr) | 7.85 | 8.05 | 7.40 | 7.55 | 6.90 | 7.35 | 6.70 | 6.95 |

For this sample, $\bar{x} = 5.5$, $\bar{y} = 7.34$, $s_x = 2.45$, $s_y = 0.47$, and $r = -0.854$. That gives

$$b_1 = (-0.854)\frac{0.47}{2.45} = -0.165 \qquad b_0 = 7.34 - (-0.165)(5.5) = 8.25$$

so $\hat{y} = 8.25 - 0.165x$. The negative slope matches what the researcher suspected: more screen time is associated with less sleep, at least in this sample. The question this chapter answers is whether that association is strong enough and consistent enough, to conclude something about $\beta_1$ for teenagers in general, or whether a slope this far from zero is a believable result of sampling variation alone.

- $Y = \beta_0 + \beta_1 X + \varepsilon$ is the population regression model; $\beta_0$ and $\beta_1$ are unknown parameters, and $\varepsilon$ is random error.
- $b_0$ and $b_1$ are sample statistics, calculated from your data, that estimate $\beta_0$ and $\beta_1$.
- $b_1 = r\dfrac{s_y}{s_x}$ and $b_0 = \bar{y} - b_1\bar{x}$, with an equivalent raw-sum formula available for $b_1$.
- The residual $e = y - \hat{y}$ and its standard deviation $s_e$ measure how far actual data points fall from the line.

## *Exercise 428* Identifying the Model

A horticulturist records the amount of fertilizer applied (in grams per plant) and the resulting plant height (in centimeters) for a random sample of tomato plants.

1. Identify $X$ and $Y$ in this study.
2. Write the population regression model for this relationship, using correct notation.
3. In context, what would it mean if $\beta_1 = 0$?
4. Explain why $\beta_1$ is not something the horticulturist can ever know exactly, even after collecting data from every plant she owns.

*Working Space*

*Answer on Page 2056*

**Computing in Excel**

Enter the sleep study data from earlier in this section into two columns: screen time in cells A2 through A9, and sleep duration in cells B2 through B9, with headers in row 1.

1. In an empty cell, enter `=SLOPE(B2:B9,A2:A9)`. Excel returns $-0.1649$, matching $b_1 = -0.165$ from the hand calculation (Excel simply keeps more decimal places).
2. In another cell, enter `=INTERCEPT(B2:B9,A2:A9)`. Excel returns $8.2506$, matching $b_0 = 8.25$.
3. To check the correlation behind the calculation, enter `=CORREL(A2:A9,B2:B9)`. Excel returns $-0.8536$, matching $r = -0.854$.
4. To find $s_x$ and $s_y$ directly instead of computing them by hand, use `=STDEV.S(A2:A9)` and `=STDEV.S(B2:B9)`. These return $2.4495$ and $0.4732$.

Watch the argument order: for both `SLOPE` and `INTERCEPT`, the known $y$-values are listed first, then the known $x$-values. Reversing the two ranges is a common mistake, and it may

hand you the wrong sign or a wrong intercept.

## 166.2 The Sampling Distribution of $b_1$ and the Confidence Interval for the Slope

Suppose the sleep researcher ran her study again with a brand new sample of 8 teenagers. A new sample of 8 teenagers would give a different fitted line and a different $b_1$, the same way $\hat{p}$ and $\bar{x}$ vary from sample to sample. That variation is the *sampling distribution* of $b_1$: the distribution of values you'd get from repeated random samples of the same size. Under the conditions for inference, this distribution is approximately normal, centered at $\beta_1$, with standard deviation $\sigma_{b_1}$.

$\sigma_{b_1}$ depends on $\sigma_\varepsilon$, the population standard deviation of the errors, which is unknown, so you estimate it with $s_{b_1}$, the *standard error of the slope*. Substituting $s_{b_1}$ for $\sigma_{b_1}$ turns

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

from normal into a t-distribution with $df = n-2$, the same trade as with means: estimated standard error in, precision out, t-distribution in place of normal.

Unlike proportions and means, you will almost never compute $s_{b_1}$ by hand: it depends on $s_e$, sample size, and the spread of your x-values, and doing that from scratch is tedious enough that no exam or workplace expects it. Instead, $s_{b_1}$ is read directly from a calculator's regression screen or software printout, in a column usually labeled "StDev" or "SE Coef." Recognizing and interpreting that column is the actual skill.

### Constructing the Confidence Interval

With $s_{b_1}$ in hand, a confidence interval for $\beta_1$ takes the same shape as every confidence interval you have built so far: statistic plus or minus a margin of error.

$$\text{Margin of Error} = t^* s_{b_1} \qquad \text{Confidence Interval: } b_1 \pm t^* s_{b_1}$$

where $t^*$ is the critical value from a t-distribution with $df = n-2$, chosen to match your desired confidence level.

Let's return to the sleep study. A statistical package fits the regression line to the 8 data points and reports the following output:

| Predictor | Coef | StDev | T | P |
|---|---|---|---|---|
| Constant | 8.2506 | 0.2448 | 33.70 | 0.000 |
| Screen time | −0.1649 | 0.0411 | −4.01 | 0.007 |

with $n = 8$, so $df = n - 2 = 6$. To build a 90% confidence interval for $\beta_1$, look up $t^* = t_{0.05}(6) = 1.943$. Then

$$\text{Margin of Error} = (1.943)(0.0411) = 0.0798$$

$$\text{Confidence Interval: } -0.1649 \pm 0.0798 = (-0.2447, -0.0850)$$

We are 90% confident that, for teenagers in general, each additional hour of daily screen time is associated with a change in average sleep duration of between $-0.245$ and $-0.085$ hours. Notice that this entire interval is negative. Since it does not contain 0, every plausible value of $\beta_1$ in this interval represents some negative relationship between screen time and sleep, which is a first hint that the effect the researcher found in her sample may reflect something real in the population, and not just sampling noise. Sections 3 and 4 make that hint precise, first by checking whether this procedure was even valid to use, and then by testing the claim directly.

- $b_1$ varies from sample to sample; its sampling distribution is centered at $\beta_1$ under the conditions for inference.
- The standard error $s_{b_1}$ is read directly from calculator or computer regression output; it is rarely computed by hand.
- $t = \dfrac{b_1 - \beta_1}{s_{b_1}}$ follows a t-distribution with $df = n - 2$.
- Confidence interval: $b_1 \pm t^* s_{b_1}$, with $t^*$ taken from the t-distribution at $df = n - 2$.
- If a confidence interval for $\beta_1$ does not contain 0, that is an early signal, not yet a conclusion, that the population slope may be nonzero.

## *Exercise 429* Reading Computer Output

A researcher studies the relationship between weekly study hours ($X$) and the improvement in exam score, in points ($Y$), for a random sample of $n = 12$ students. Regression output gives $b_1 = 2.35$ and $s_{b_1} = 0.68$.

1. Find the degrees of freedom for this problem.
2. Construct a 95% confidence interval for $\beta_1$. ($t^*_{0.025}(10) = 2.228$.)
3. Interpret the interval in context.

*Working Space*

*Answer on Page ??*

**Computing in Excel**

The regression output table above, including $s_{b_1}$, can be generated directly from the raw data rather than typed in by hand. Using the same screen time and sleep duration columns from Section 1 (A2:A9 and B2:B9):

1. Select a blank $2 \times 2$ range, say D2:E3. Type `=LINEST(B2:B9,A2:A9,TRUE,TRUE)` and confirm with Ctrl+Shift+Enter (in newer versions of Excel with dynamic arrays, a plain Enter is enough). This fills all four cells at once:

| | |
|---|---|
| $b_1 = -0.1649$ | $b_0 = 8.2506$ |
| $s_{b_1} = 0.0411$ | $s_{b_0} = 0.2448$ |

   The top row gives the coefficients, in reverse order from how you might expect (slope first, then intercept); the row directly below gives each one's standard error.

2. For the critical value, enter `=T.INV.2T(0.10,6)` in an empty cell. The first argument is 1 − confidence level, and the second is $df = n - 2$. Excel returns 1.9432, matching $t^*_{0.05}(6)$ from the table.
3. Margin of error: reference the two cells you already have, `=D3*[t* cell]`, giving 0.0798.
4. Confidence interval bounds: `=D2-[margin cell]` and `=D2+[margin cell]`, giving $-0.2447$ and $-0.0850$.

`LINEST` is the same idea as `SLOPE` and `INTERCEPT` from Section 1, but it returns the standard errors in the same step, so you never have to hunt for $s_{b_1}$ separately. The argument order is the same gotcha as before: known $y$'s first, known $x$'s second.

## 166.3 Conditions for Inference on the Slope

The confidence interval you just built in Section 2 assumed that it was valid to use a t-procedure at all. Every inference method in this book comes with conditions attached, and the slope is no exception.

You have already met four of these five conditions for proportions and for means.

- **Linearity**: the true relationship between $X$ and $Y$ is actually linear, not curved.
- **Independence**: individual observations do not influence one another; if sampling without replacement, the population is at least 10 times the sample size.
- **Normality**: for any fixed value of $X$, the residuals are approximately normally distributed.
- **Equal variance**: the spread of the residuals is roughly the same across all values of $X$.
- **Random**: the data come from a random sample or a randomized experiment.

These are remembered by the acronym LINER. Random and Independence are conditions you check from the design of the study itself, the same way you always have. Linearity, Normality, and Equal variance are different: you check all three using the same tool, a *residual plot*, which graphs each residual against its corresponding $x$-value.

### Checking the Conditions: The Sleep Study

Return to the sleep researcher's 8 teenagers. Before trusting the confidence interval from Section 2, check all five conditions.

**Random.** The researcher selected her 8 teenagers using a random sampling method, so this condition is satisfied by the design of the study.

**Independence.** Teenagers in general form a population far larger than 8, so sampling 8 of them without replacement does not meaningfully affect the independence of their screen time or sleep duration from one another. The 10% condition is easily satisfied.

**Linearity, Normality, and Equal variance.** These three come from the residuals themselves.

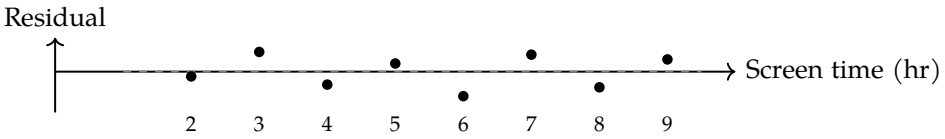



Figure 166.1: Residual plot for the sleep study regression.

The residuals in Figure 166.1 bounce above and below 0 with no visible curve, so linearity is reasonable: a straight line, rather than some curved model, is a sensible description of the relationship. The band of residuals also stays roughly the same width across the range of screen time values, from 2 hours to 9 hours, with no obvious funnel shape, so equal variance is reasonable as well.

Figure 166.2: Dotplot of the eight residuals from the sleep study regression.

The dotplot in Figure 166.2 shows no strong skew and no outliers among the eight residuals, so it is reasonable to proceed as though normality holds. With only $n = 8$ observations, though, this check is genuinely limited; a small sample simply does not give you enough information to rule out non-normality with much confidence. The honest conclusion is not "the residuals are normal," but "nothing in this small sample contradicts normality," which is a weaker, more defensible claim, and one that is worth stating explicitly on an exam or in a report.

With all five conditions reasonably satisfied, the confidence interval built in Section 2, and the hypothesis test coming in Section 4, both rest on solid ground.

- The five conditions for slope inference: Random, Linearity, Independence, Normality, Equal variance (LINER).
- Random and Independence come from the design of the study.
- Linearity, Normality, and Equal variance are checked using a residual plot (and, for normality specifically, a dotplot, boxplot, or normal probability plot of the residuals).
- A small sample size limits how confidently you can assess normality; conclusions should be stated as "nothing contradicts" rather than "confirms."

### *Exercise 430* Diagnosing a Residual Plot

A researcher studies the relationship between a car's weight and its fuel efficiency. The residual plot shows points that fan out widely for heavy cars but cluster tightly near 0 for light cars.

1. Which condition does this pattern call into question?
2. Explain, in your own words, why this pattern is a problem for the validity of a confidence interval or test built from this data.

*Working Space*

*Answer on Page 2056*

## 166.4 Testing a Claim About the Slope

The confidence interval in Section 2 hinted that the true slope $\beta_1$ is not 0, since $(-0.2447, -0.0850)$ never crosses it. A hint is not a conclusion. To make the claim precise, and to attach an actual probability to the strength of the evidence, you need a significance test.

### Step 1: State the Hypotheses

The sleep researcher's original question was directional: does more screen time predict *less* sleep, not just some unspecified difference. That points to a one-sided alternative.

$$H_0 : \beta_1 = 0 \qquad H_a : \beta_1 < 0$$

$H_0$ says screen time has no linear relationship with sleep duration in the population of teenagers; $H_a$ says the relationship is negative. Use $\alpha = 0.05$.

### Step 2: Name the Procedure and Verify Conditions

Use a one-sample t-test for the slope of a regression line, with $df = n - 2$. Section 3 already checked all five conditions, Random, Linearity, Independence, Normality, and Equal variance, against this exact data set, and found no serious violations. Since the same data set is being reused here, that check still applies; there is no need to redo it.

### Step 3: Calculate the Test Statistic and P-value

Recall the regression output from Section 2:

| Predictor | Coef | StDev | T | P |
|---|---|---|---|---|
| Constant | 8.2506 | 0.2448 | 33.70 | 0.000 |
| Screen time | −0.1649 | 0.0411 | −4.01 | 0.007 |

with $n = 8$, so $df = 6$. The test statistic is

$$t = \frac{b_1 - 0}{s_{b_1}} = \frac{-0.1649}{0.0411} = -4.01$$

matching the T column exactly, since that column is always $\frac{\text{Coef}}{\text{StDev}}$.

The P column, 0.007, is a two-sided p-value: it treats $H_a : \beta_1 \neq 0$ as the alternative, whether or not that is the alternative you actually care about. Since our alternative here is one-sided, that number needs one adjustment before you can use it.

**Converting a two-sided p-value to a one-sided p-value:**

1. Check whether the sign of t matches the direction claimed in $H_a$. Positive t matches $H_a : \beta_1 > 0$; negative t matches $H_a : \beta_1 < 0$.
2. If the sign matches, divide the reported two-sided p-value by 2.
3. If the sign does not match, the data point the opposite way from your claim. The correct one-sided p-value is actually greater than 0.5, no matter how small the two-sided number looks, and there is no evidence for $H_a$.

Here $t = -4.01$ is negative, matching the direction of $H_a : \beta_1 < 0$, so the sign check passes. The one-sided p-value is

$$\text{p-value} = \frac{0.007}{2} = 0.0035$$

### Step 4: State a Conclusion

Because $p = 0.0035 < \alpha = 0.05$, reject $H_0$. There is convincing statistical evidence that, in the population of teenagers this sample represents, more daily screen time is associated with a decrease in average nightly sleep duration. Because the screen time data were observed rather than randomly assigned, this conclusion is about association, not causation; some other factor could plausibly be driving both variables at once.

- Test statistic: $t = \dfrac{b_1 - 0}{s_{b_1}}$, with $df = n - 2$, using the value of $\beta_1$ specified in $H_0$ (usually 0).
- Standard computer output typically reports a two-sided p-value by default, regardless of which alternative you are actually testing.
- For a one-sided $H_a$, divide the reported two-sided p-value by 2 only if the sign of $t$ matches the direction of $H_a$; otherwise the true one-sided p-value exceeds 0.5.
- Reject $H_0$ when $p < \alpha$; state the conclusion in context, and be careful not to claim causation from an observational study.

#### Computing in Excel

Continuing from the `LINEST` output in Section 2 (cell D2 for $b_1$, cell D3 for $s_{b_1}$):

1. Test statistic: in an empty cell, enter `=D2/D3`. Excel returns $-4.0131$, matching the T column of the regression output.
2. One-sided p-value directly: enter `=T.DIST(D2/D3,6,TRUE)`. This gives the left-tail probability $P(T < t)$ for $df = 6$, which is exactly the one-sided p-value for $H_a : \beta_1 < 0$ when $t$ is negative. Excel returns 0.0035, with no separate halving step needed.
3. If you only have the two-sided p-value from a printed output table, use `=T.DIST.2T(ABS(D2/D3),6)` instead, which reproduces the halving rule from Step 3 directly. Both approaches agree here: 0.0035.

`T.DIST` assumes a lower-tail test by default; for $H_a : \beta_1 > 0$, use `=1-T.DIST(t,df,TRUE)` or the equivalent `=T.DIST.RT(t,df)` instead, since you need the upper tail.

#### Computing in Python

The full test, from raw data to conclusion, takes only a few lines with `scipy.stats`:

```
import numpy as np
from scipy import stats

```

```
x = np.array([2, 3, 4, 5, 6, 7, 8, 9])
y = np.array([7.85, 8.05, 7.40, 7.55, 6.90, 7.35, 6.70, 6.95])

result = stats.linregress(x, y)
df = len(x) - 2
t_stat = result.slope / result.stderr
p_one_sided = stats.t.cdf(t_stat, df)   # left tail, for Ha: beta1 < 0

print(f"b1 = {result.slope:.4f}, SE(b1) = {result.stderr:.4f}")
print(f"t = {t_stat:.4f}, df = {df}")
print(f"one-sided p-value = {p_one_sided:.4f}")
```

This prints `b1 = -0.1649, SE(b1) = 0.0411, t = -4.0131, df = 6`, and `one-sided p-value = 0.0035`, matching the by-hand and Excel results exactly. `stats.t.cdf(t_stat, df)` gives the lower-tail probability directly, which is the correct one-sided p-value whenever $H_a$ is "less than and t is negative; for a "greater than" alternative, use `1 - stats.t.cdf(t_stat, df)` instead.

Chapter 167

# Statistical Tables

This chapter collects the three tables printed on the AP Statistics formula sheet: standard normal probabilities, t critical values, and chi-square critical values. The layouts match the exam versions, so practice with these tables carries over directly to test day.

## 167.1 The Standard Normal Table

Table 167.1 and Table 167.2 give the *cumulative probability* $P(Z \le z)$, the area under the standard normal curve to the left of $z$. Each row gives $z$ to one decimal place, and each column adds the second decimal place. To find $P(Z \le 1.37)$, go to row 1.3 and column .07; the entry is 0.9147.

Three rules cover every question this table answers:

- Area to the left of $z$: read the table directly.
- Area to the right of $z$: subtract the table value from 1.
- Area between $z_1$ and $z_2$: subtract the smaller table value from the larger one.

To work backward from an area to a $z$-score, find the entry closest to the area in the body of the table, then read its row and column.

Table 167.1: Standard normal cumulative probabilities, $P(Z \leq z)$, negative $z$

| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| −3.4 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0002 |
| −3.3 | 0.0005 | 0.0005 | 0.0005 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0003 |
| −3.2 | 0.0007 | 0.0007 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0005 | 0.0005 | 0.0005 |
| −3.1 | 0.0010 | 0.0009 | 0.0009 | 0.0009 | 0.0008 | 0.0008 | 0.0008 | 0.0008 | 0.0007 | 0.0007 |
| −3.0 | 0.0013 | 0.0013 | 0.0013 | 0.0012 | 0.0012 | 0.0011 | 0.0011 | 0.0011 | 0.0010 | 0.0010 |
| −2.9 | 0.0019 | 0.0018 | 0.0018 | 0.0017 | 0.0016 | 0.0016 | 0.0015 | 0.0015 | 0.0014 | 0.0014 |
| −2.8 | 0.0026 | 0.0025 | 0.0024 | 0.0023 | 0.0023 | 0.0022 | 0.0021 | 0.0021 | 0.0020 | 0.0019 |
| −2.7 | 0.0035 | 0.0034 | 0.0033 | 0.0032 | 0.0031 | 0.0030 | 0.0029 | 0.0028 | 0.0027 | 0.0026 |
| −2.6 | 0.0047 | 0.0045 | 0.0044 | 0.0043 | 0.0041 | 0.0040 | 0.0039 | 0.0038 | 0.0037 | 0.0036 |
| −2.5 | 0.0062 | 0.0060 | 0.0059 | 0.0057 | 0.0055 | 0.0054 | 0.0052 | 0.0051 | 0.0049 | 0.0048 |
| −2.4 | 0.0082 | 0.0080 | 0.0078 | 0.0075 | 0.0073 | 0.0071 | 0.0069 | 0.0068 | 0.0066 | 0.0064 |
| −2.3 | 0.0107 | 0.0104 | 0.0102 | 0.0099 | 0.0096 | 0.0094 | 0.0091 | 0.0089 | 0.0087 | 0.0084 |
| −2.2 | 0.0139 | 0.0136 | 0.0132 | 0.0129 | 0.0125 | 0.0122 | 0.0119 | 0.0116 | 0.0113 | 0.0110 |
| −2.1 | 0.0179 | 0.0174 | 0.0170 | 0.0166 | 0.0162 | 0.0158 | 0.0154 | 0.0150 | 0.0146 | 0.0143 |
| −2.0 | 0.0228 | 0.0222 | 0.0217 | 0.0212 | 0.0207 | 0.0202 | 0.0197 | 0.0192 | 0.0188 | 0.0183 |
| −1.9 | 0.0287 | 0.0281 | 0.0274 | 0.0268 | 0.0262 | 0.0256 | 0.0250 | 0.0244 | 0.0239 | 0.0233 |
| −1.8 | 0.0359 | 0.0351 | 0.0344 | 0.0336 | 0.0329 | 0.0322 | 0.0314 | 0.0307 | 0.0301 | 0.0294 |
| −1.7 | 0.0446 | 0.0436 | 0.0427 | 0.0418 | 0.0409 | 0.0401 | 0.0392 | 0.0384 | 0.0375 | 0.0367 |
| −1.6 | 0.0548 | 0.0537 | 0.0526 | 0.0516 | 0.0505 | 0.0495 | 0.0485 | 0.0475 | 0.0465 | 0.0455 |
| −1.5 | 0.0668 | 0.0655 | 0.0643 | 0.0630 | 0.0618 | 0.0606 | 0.0594 | 0.0582 | 0.0571 | 0.0559 |
| −1.4 | 0.0808 | 0.0793 | 0.0778 | 0.0764 | 0.0749 | 0.0735 | 0.0721 | 0.0708 | 0.0694 | 0.0681 |
| −1.3 | 0.0968 | 0.0951 | 0.0934 | 0.0918 | 0.0901 | 0.0885 | 0.0869 | 0.0853 | 0.0838 | 0.0823 |
| −1.2 | 0.1151 | 0.1131 | 0.1112 | 0.1093 | 0.1075 | 0.1056 | 0.1038 | 0.1020 | 0.1003 | 0.0985 |
| −1.1 | 0.1357 | 0.1335 | 0.1314 | 0.1292 | 0.1271 | 0.1251 | 0.1230 | 0.1210 | 0.1190 | 0.1170 |
| −1.0 | 0.1587 | 0.1562 | 0.1539 | 0.1515 | 0.1492 | 0.1469 | 0.1446 | 0.1423 | 0.1401 | 0.1379 |
| −0.9 | 0.1841 | 0.1814 | 0.1788 | 0.1762 | 0.1736 | 0.1711 | 0.1685 | 0.1660 | 0.1635 | 0.1611 |
| −0.8 | 0.2119 | 0.2090 | 0.2061 | 0.2033 | 0.2005 | 0.1977 | 0.1949 | 0.1922 | 0.1894 | 0.1867 |
| −0.7 | 0.2420 | 0.2389 | 0.2358 | 0.2327 | 0.2296 | 0.2266 | 0.2236 | 0.2206 | 0.2177 | 0.2148 |
| −0.6 | 0.2743 | 0.2709 | 0.2676 | 0.2643 | 0.2611 | 0.2578 | 0.2546 | 0.2514 | 0.2483 | 0.2451 |
| −0.5 | 0.3085 | 0.3050 | 0.3015 | 0.2981 | 0.2946 | 0.2912 | 0.2877 | 0.2843 | 0.2810 | 0.2776 |
| −0.4 | 0.3446 | 0.3409 | 0.3372 | 0.3336 | 0.3300 | 0.3264 | 0.3228 | 0.3192 | 0.3156 | 0.3121 |
| −0.3 | 0.3821 | 0.3783 | 0.3745 | 0.3707 | 0.3669 | 0.3632 | 0.3594 | 0.3557 | 0.3520 | 0.3483 |
| −0.2 | 0.4207 | 0.4168 | 0.4129 | 0.4090 | 0.4052 | 0.4013 | 0.3974 | 0.3936 | 0.3897 | 0.3859 |
| −0.1 | 0.4602 | 0.4562 | 0.4522 | 0.4483 | 0.4443 | 0.4404 | 0.4364 | 0.4325 | 0.4286 | 0.4247 |
| −0.0 | 0.5000 | 0.4960 | 0.4920 | 0.4880 | 0.4840 | 0.4801 | 0.4761 | 0.4721 | 0.4681 | 0.4641 |

Table 167.2: Standard normal cumulative probabilities, $P(Z \leq z)$, positive $z$

| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| 2.1 | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| 2.2 | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| 2.3 | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| 2.4 | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| 2.5 | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| 2.6 | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| 2.7 | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| 2.8 | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| 2.9 | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| 3.0 | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| 3.1 | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| 3.2 | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| 3.3 | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| 3.4 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9998 |

### *Exercise 431* Areas under the standard normal curve

Use the standard normal table to find each probability.

*Working Space*

(a) $P(Z \leq 1.37)$

(b) $P(Z > -0.84)$

(c) $P(-1.25 < Z < 0.60)$

*Answer on Page 2057*

### *Exercise 432* From an area back to $z$

Find the $z$-score that has an area of 0.90 to its left.

*Working Space*

*Answer on Page 2057*

## 167.2 The t Table

Table 167.3 gives the critical value $t^*$ that has upper-tail probability $p$ for a $t$ distribution with $df$ degrees of freedom. The table is read two ways.

- For a confidence interval, use the confidence level C along the bottom row. A 95 percent interval with $df = 24$ uses the column above 95%, giving $t^* = 2.064$.
- For a significance test, locate the test statistic between two entries in its $df$ row. The tail probabilities at the top of those two columns bound the one-sided P-value. Double both bounds for a two-sided test.

When the exact $df$ is missing from the table, use the next smaller $df$ that appears. The smaller $df$ gives a slightly larger $t^*$, so the resulting interval is a little wider and the conclusion stays conservative. The last row, $df = \infty$, contains the standard normal critical values $z^*$.

Table 167.3: t distribution critical values, $t^*$ with upper-tail probability p

| | Tail probability p | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| df | .25 | .20 | .15 | .10 | .05 | .025 | .02 | .01 | .005 | .0025 | .001 | .0005 |
| 1 | 1.000 | 1.376 | 1.963 | 3.078 | 6.314 | 12.706 | 15.895 | 31.821 | 63.657 | 127.321 | 318.309 | 636.619 |
| 2 | 0.816 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303 | 4.849 | 6.965 | 9.925 | 14.089 | 22.327 | 31.599 |
| 3 | 0.765 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182 | 3.482 | 4.541 | 5.841 | 7.453 | 10.215 | 12.924 |
| 4 | 0.741 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776 | 2.999 | 3.747 | 4.604 | 5.598 | 7.173 | 8.610 |
| 5 | 0.727 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571 | 2.757 | 3.365 | 4.032 | 4.773 | 5.893 | 6.869 |
| 6 | 0.718 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447 | 2.612 | 3.143 | 3.707 | 4.317 | 5.208 | 5.959 |
| 7 | 0.711 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365 | 2.517 | 2.998 | 3.499 | 4.029 | 4.785 | 5.408 |
| 8 | 0.706 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306 | 2.449 | 2.896 | 3.355 | 3.833 | 4.501 | 5.041 |
| 9 | 0.703 | 0.883 | 1.100 | 1.383 | 1.833 | 2.262 | 2.398 | 2.821 | 3.250 | 3.690 | 4.297 | 4.781 |
| 10 | 0.700 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228 | 2.359 | 2.764 | 3.169 | 3.581 | 4.144 | 4.587 |
| 11 | 0.697 | 0.876 | 1.088 | 1.363 | 1.796 | 2.201 | 2.328 | 2.718 | 3.106 | 3.497 | 4.025 | 4.437 |
| 12 | 0.695 | 0.873 | 1.083 | 1.356 | 1.782 | 2.179 | 2.303 | 2.681 | 3.055 | 3.428 | 3.930 | 4.318 |
| 13 | 0.694 | 0.870 | 1.079 | 1.350 | 1.771 | 2.160 | 2.282 | 2.650 | 3.012 | 3.372 | 3.852 | 4.221 |
| 14 | 0.692 | 0.868 | 1.076 | 1.345 | 1.761 | 2.145 | 2.264 | 2.624 | 2.977 | 3.326 | 3.787 | 4.140 |
| 15 | 0.691 | 0.866 | 1.074 | 1.341 | 1.753 | 2.131 | 2.249 | 2.602 | 2.947 | 3.286 | 3.733 | 4.073 |
| 16 | 0.690 | 0.865 | 1.071 | 1.337 | 1.746 | 2.120 | 2.235 | 2.583 | 2.921 | 3.252 | 3.686 | 4.015 |
| 17 | 0.689 | 0.863 | 1.069 | 1.333 | 1.740 | 2.110 | 2.224 | 2.567 | 2.898 | 3.222 | 3.646 | 3.965 |
| 18 | 0.688 | 0.862 | 1.067 | 1.330 | 1.734 | 2.101 | 2.214 | 2.552 | 2.878 | 3.197 | 3.610 | 3.922 |
| 19 | 0.688 | 0.861 | 1.066 | 1.328 | 1.729 | 2.093 | 2.205 | 2.539 | 2.861 | 3.174 | 3.579 | 3.883 |
| 20 | 0.687 | 0.860 | 1.064 | 1.325 | 1.725 | 2.086 | 2.197 | 2.528 | 2.845 | 3.153 | 3.552 | 3.850 |
| 21 | 0.686 | 0.859 | 1.063 | 1.323 | 1.721 | 2.080 | 2.189 | 2.518 | 2.831 | 3.135 | 3.527 | 3.819 |
| 22 | 0.686 | 0.858 | 1.061 | 1.321 | 1.717 | 2.074 | 2.183 | 2.508 | 2.819 | 3.119 | 3.505 | 3.792 |
| 23 | 0.685 | 0.858 | 1.060 | 1.319 | 1.714 | 2.069 | 2.177 | 2.500 | 2.807 | 3.104 | 3.485 | 3.768 |
| 24 | 0.685 | 0.857 | 1.059 | 1.318 | 1.711 | 2.064 | 2.172 | 2.492 | 2.797 | 3.091 | 3.467 | 3.745 |
| 25 | 0.684 | 0.856 | 1.058 | 1.316 | 1.708 | 2.060 | 2.167 | 2.485 | 2.787 | 3.078 | 3.450 | 3.725 |
| 26 | 0.684 | 0.856 | 1.058 | 1.315 | 1.706 | 2.056 | 2.162 | 2.479 | 2.779 | 3.067 | 3.435 | 3.707 |
| 27 | 0.684 | 0.855 | 1.057 | 1.314 | 1.703 | 2.052 | 2.158 | 2.473 | 2.771 | 3.057 | 3.421 | 3.690 |
| 28 | 0.683 | 0.855 | 1.056 | 1.313 | 1.701 | 2.048 | 2.154 | 2.467 | 2.763 | 3.047 | 3.408 | 3.674 |
| 29 | 0.683 | 0.854 | 1.055 | 1.311 | 1.699 | 2.045 | 2.150 | 2.462 | 2.756 | 3.038 | 3.396 | 3.659 |
| 30 | 0.683 | 0.854 | 1.055 | 1.310 | 1.697 | 2.042 | 2.147 | 2.457 | 2.750 | 3.030 | 3.385 | 3.646 |
| 40 | 0.681 | 0.851 | 1.050 | 1.303 | 1.684 | 2.021 | 2.123 | 2.423 | 2.704 | 2.971 | 3.307 | 3.551 |
| 50 | 0.679 | 0.849 | 1.047 | 1.299 | 1.676 | 2.009 | 2.109 | 2.403 | 2.678 | 2.937 | 3.261 | 3.496 |
| 60 | 0.679 | 0.848 | 1.045 | 1.296 | 1.671 | 2.000 | 2.099 | 2.390 | 2.660 | 2.915 | 3.232 | 3.460 |
| 80 | 0.678 | 0.846 | 1.043 | 1.292 | 1.664 | 1.990 | 2.088 | 2.374 | 2.639 | 2.887 | 3.195 | 3.416 |
| 100 | 0.677 | 0.845 | 1.042 | 1.290 | 1.660 | 1.984 | 2.081 | 2.364 | 2.626 | 2.871 | 3.174 | 3.390 |
| 1000 | 0.675 | 0.842 | 1.037 | 1.282 | 1.646 | 1.962 | 2.056 | 2.330 | 2.581 | 2.813 | 3.098 | 3.300 |
| $\infty$ | 0.674 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960 | 2.054 | 2.326 | 2.576 | 2.807 | 3.090 | 3.291 |
| | 50% | 60% | 70% | 80% | 90% | 95% | 96% | 98% | 99% | 99.5% | 99.8% | 99.9% |
| | Confidence level C | | | | | | | | | | | |

### *Exercise 433* Critical values for confidence intervals

Find $t^*$ for each interval.

(a) A 95 percent confidence interval with $df = 24$.

(b) A 99 percent confidence interval with $df = 45$.

*Working Space*

*Answer on Page 2057*

### *Exercise 434* Bounding a P-value

A one-sided test with $df = 15$ produces $t = 2.30$. Use the t table to find bounds on the P-value.

*Working Space*

*Answer on Page 2057*

## 167.3 The Chi-Square Table

Table 167.4 gives the critical value $\chi^2$ that has upper-tail probability $p$ for a chi-square distribution with $df$ degrees of freedom. It extends the shorter table in the chi-square chapter to more tail areas and to $df = 100$. Chi-square tests use only the upper tail, so the P-value bounds come straight from the column headings with no doubling.

Table 167.4: Chi-square critical values, $\chi^2$ with upper-tail probability p

| | Tail probability p | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| df | .25 | .20 | .15 | .10 | .05 | .025 | .02 | .01 | .005 | .0025 | .001 | .0005 |
| 1 | 1.323 | 1.642 | 2.072 | 2.706 | 3.841 | 5.024 | 5.412 | 6.635 | 7.879 | 9.141 | 10.828 | 12.116 |
| 2 | 2.773 | 3.219 | 3.794 | 4.605 | 5.991 | 7.378 | 7.824 | 9.210 | 10.597 | 11.983 | 13.816 | 15.202 |
| 3 | 4.108 | 4.642 | 5.317 | 6.251 | 7.815 | 9.348 | 9.837 | 11.345 | 12.838 | 14.320 | 16.266 | 17.730 |
| 4 | 5.385 | 5.989 | 6.745 | 7.779 | 9.488 | 11.143 | 11.668 | 13.277 | 14.860 | 16.424 | 18.467 | 19.997 |
| 5 | 6.626 | 7.289 | 8.115 | 9.236 | 11.070 | 12.833 | 13.388 | 15.086 | 16.750 | 18.386 | 20.515 | 22.105 |
| 6 | 7.841 | 8.558 | 9.446 | 10.645 | 12.592 | 14.449 | 15.033 | 16.812 | 18.548 | 20.249 | 22.458 | 24.103 |
| 7 | 9.037 | 9.803 | 10.748 | 12.017 | 14.067 | 16.013 | 16.622 | 18.475 | 20.278 | 22.040 | 24.322 | 26.018 |
| 8 | 10.219 | 11.030 | 12.027 | 13.362 | 15.507 | 17.535 | 18.168 | 20.090 | 21.955 | 23.774 | 26.124 | 27.868 |
| 9 | 11.389 | 12.242 | 13.288 | 14.684 | 16.919 | 19.023 | 19.679 | 21.666 | 23.589 | 25.462 | 27.877 | 29.666 |
| 10 | 12.549 | 13.442 | 14.534 | 15.987 | 18.307 | 20.483 | 21.161 | 23.209 | 25.188 | 27.112 | 29.588 | 31.420 |
| 11 | 13.701 | 14.631 | 15.767 | 17.275 | 19.675 | 21.920 | 22.618 | 24.725 | 26.757 | 28.729 | 31.264 | 33.137 |
| 12 | 14.845 | 15.812 | 16.989 | 18.549 | 21.026 | 23.337 | 24.054 | 26.217 | 28.300 | 30.318 | 32.909 | 34.821 |
| 13 | 15.984 | 16.985 | 18.202 | 19.812 | 22.362 | 24.736 | 25.472 | 27.688 | 29.819 | 31.883 | 34.528 | 36.478 |
| 14 | 17.117 | 18.151 | 19.406 | 21.064 | 23.685 | 26.119 | 26.873 | 29.141 | 31.319 | 33.426 | 36.123 | 38.109 |
| 15 | 18.245 | 19.311 | 20.603 | 22.307 | 24.996 | 27.488 | 28.259 | 30.578 | 32.801 | 34.950 | 37.697 | 39.719 |
| 16 | 19.369 | 20.465 | 21.793 | 23.542 | 26.296 | 28.845 | 29.633 | 32.000 | 34.267 | 36.456 | 39.252 | 41.308 |
| 17 | 20.489 | 21.615 | 22.977 | 24.769 | 27.587 | 30.191 | 30.995 | 33.409 | 35.718 | 37.946 | 40.790 | 42.879 |
| 18 | 21.605 | 22.760 | 24.155 | 25.989 | 28.869 | 31.526 | 32.346 | 34.805 | 37.156 | 39.422 | 42.312 | 44.434 |
| 19 | 22.718 | 23.900 | 25.329 | 27.204 | 30.144 | 32.852 | 33.687 | 36.191 | 38.582 | 40.885 | 43.820 | 45.973 |
| 20 | 23.828 | 25.038 | 26.498 | 28.412 | 31.410 | 34.170 | 35.020 | 37.566 | 39.997 | 42.336 | 45.315 | 47.498 |
| 21 | 24.935 | 26.171 | 27.662 | 29.615 | 32.671 | 35.479 | 36.343 | 38.932 | 41.401 | 43.775 | 46.797 | 49.011 |
| 22 | 26.039 | 27.301 | 28.822 | 30.813 | 33.924 | 36.781 | 37.659 | 40.289 | 42.796 | 45.204 | 48.268 | 50.511 |
| 23 | 27.141 | 28.429 | 29.979 | 32.007 | 35.172 | 38.076 | 38.968 | 41.638 | 44.181 | 46.623 | 49.728 | 52.000 |
| 24 | 28.241 | 29.553 | 31.132 | 33.196 | 36.415 | 39.364 | 40.270 | 42.980 | 45.559 | 48.034 | 51.179 | 53.479 |
| 25 | 29.339 | 30.675 | 32.282 | 34.382 | 37.652 | 40.646 | 41.566 | 44.314 | 46.928 | 49.435 | 52.620 | 54.947 |
| 26 | 30.435 | 31.795 | 33.429 | 35.563 | 38.885 | 41.923 | 42.856 | 45.642 | 48.290 | 50.829 | 54.052 | 56.407 |
| 27 | 31.528 | 32.912 | 34.574 | 36.741 | 40.113 | 43.195 | 44.140 | 46.963 | 49.645 | 52.215 | 55.476 | 57.858 |
| 28 | 32.620 | 34.027 | 35.715 | 37.916 | 41.337 | 44.461 | 45.419 | 48.278 | 50.993 | 53.594 | 56.892 | 59.300 |
| 29 | 33.711 | 35.139 | 36.854 | 39.087 | 42.557 | 45.722 | 46.693 | 49.588 | 52.336 | 54.967 | 58.301 | 60.735 |
| 30 | 34.800 | 36.250 | 37.990 | 40.256 | 43.773 | 46.979 | 47.962 | 50.892 | 53.672 | 56.332 | 59.703 | 62.162 |
| 40 | 45.616 | 47.269 | 49.244 | 51.805 | 55.758 | 59.342 | 60.436 | 63.691 | 66.766 | 69.699 | 73.402 | 76.095 |
| 50 | 56.334 | 58.164 | 60.346 | 63.167 | 67.505 | 71.420 | 72.613 | 76.154 | 79.490 | 82.664 | 86.661 | 89.561 |
| 60 | 66.981 | 68.972 | 71.341 | 74.397 | 79.082 | 83.298 | 84.580 | 88.379 | 91.952 | 95.344 | 99.607 | 102.695 |
| 80 | 88.130 | 90.405 | 93.106 | 96.578 | 101.879 | 106.629 | 108.069 | 112.329 | 116.321 | 120.102 | 124.839 | 128.261 |
| 100 | 109.141 | 111.667 | 114.659 | 118.498 | 124.342 | 129.561 | 131.142 | 135.807 | 140.169 | 144.293 | 149.449 | 153.167 |

## *Exercise 435* Reading the chi-square table

*Working Space*

(a) Find the critical value for a chi-square test at $\alpha = 0.05$ with $\mathrm{df} = 12$.

(b) A goodness-of-fit test with $\mathrm{df} = 4$ produces $\chi^2 = 9.80$. Find bounds on the P-value.

*Answer on Page 2057*

# Chapter 168

# Conditional Probability

Let's say there is a virus going around, and there is a vaccine for it that requires two shots. You are working at a school, and you are wondering how effective the vaccines are. Some students are unvaccinated, some have had one shot, and some have had two shots. One day, you test all 644 students to see who has the virus. You end up with the following table:

| | $V_0$ | $V_1$ | $V_2$ |
|---|---|---|---|
| $T_+$ | 88 students | 36 students | 96 students |
| $T_-$ | 92 students | 76 students | 256 students |

Here is what each symbol means:

- $V_0$: student has had zero vaccination shots
- $V_1$: student has had one vaccination shot
- $V_2$: student has had both vaccination shots
- $T_+$: student tested positive for the virus
- $T_-$: student tested negative for the virus

So, for example, your data indicates that 76 students who had only one of the two shots tested negative for the virus.

Your principal has a few questions. The first is, "If I put five randomly chosen students in a study group together, what is the probability that one of them has the virus?"

The first thing you might do is make a new table that shows the probability of a randomly chosen student being in any particular group. You just divide each entry by 644 (the total number of students).

| | $V_0$ | $V_1$ | $V_2$ |
|---|---|---|---|
| $T_+$ | $p(V_0 \text{ AND } T_+) = 13.7\%$ | $p(V_1 \text{ AND } T_+) = 5.6\%$ | $p(V_2 \text{ AND } T_+) = 14.9\%$ |
| $T_-$ | $p(V_0 \text{ AND } T_-) = 14.3\%$ | $p(V_1 \text{ AND } T_-) = 11.8\%$ | $p(V_2 \text{ AND } T_-) = 39.8\%$ |

(In this table, we expressed the number as a percentage with a decimal point — you had to round off the numbers. If you wanted exact answers, you would have to keep each as a fraction: 36 students represents $\frac{9}{161}$ of the student body.)

## 168.1 Marginalization

Now we can sum across the columns and rows.

| | $V_0$ | $V_1$ | $V_2$ | sum |
|---|---|---|---|---|
| $T_+$ | 0.137 | 0.056 | 0.149 | $p(T_+) = 0.342$ |
| $T_-$ | 0.143 | 0.118 | 0.398 | $p(T_+) = 0.547$ |
| sum | $p(V_0) = 0.280$ | $p(V_1) = 0.174$ | $p(V_2) = 0.547$ | |

If a child is chosen randomly from the entire student body, there is a 34.2% that the student has tested positive for the virus, and there is 17.4% chance that the student has one shot of the vaccine.

This summing of the probabilities across one dimension is known as *marginalizing*. Marginalization is just summing across all the variables that you don't care about. If you don't care who has the virus, just the probability that a student has not received even one shot of the vaccine, you can simply marginalize all the vaccine statuses. To answer the principal's question, the easy thing to do is find the answer of the opposite: "if I put five randomly chosen students in a study group together, what is the probability that *none* of them has tested positive for the virus?"

The chance that a randomly chosen student doesn't have the virus ($p(T_-)$ is 54.7%. This means the chance that 5 randomly chosen students don't have the virus is $0.547 \times 0.547 \times 0.547 \times 0.547 \times 0.547 = 0.0489$ Thus, the probability of the opposite is $1.0 - 0.0489 = 0.951$

The answer, then, is "If you put 5 kids in a study group together, there is a 95.1 % probability that at least one of them has the virus."

## 168.2 Conditional Probability

Now the principal asks you, "What if I make a group of five kids who have had both shots of the vaccine? What are the odds that one of them has tested positive for the virus?"

This involves the idea of *Conditional probability*. You want to know the odds that a student doesn't have the virus, given that the student has had both shots of the vaccine.

There is a mathematical notation for this:

$$p(T_-|V_2)$$

That is the probability that a student who has had both vaccination shots will test negative for the virus.

How would you calculate this? You would count all the students who had a positive test *and* both vaccination shots, which you would divide by the total number of students who had both vaccination shots.

$$p(T_-|V_2) = \frac{256}{96+256} = \frac{8}{11} \approx 72.7\%$$

If we are working from the probabilities, you can get the same result this way. Divide the probability that a randomly chosen student had a positive test *and* both vaccination shots by the probability that a student had both vaccination shots:

$$p(T_-|V_2) = \frac{p(T_- \text{ AND } V_2)}{p(T_-)} = \frac{0.398}{0.547} \approx 72.7\%$$

Notice that this is different from $p(V_2|T_-)$, which is the probability that a student has had both vaccinations, given they tested negative for the virus.

Back to the principal's question: "If you have five students who have had both vaccinations, what is the probability that all of them tested negative for the virus?" The probability that one student is virus-free is $\frac{8}{11}$, so the probability that five students are virus-free is $\frac{8}{11}^5 \approx 0.203$. So, there is a 79.6% chance that at least one of the five has the virus.

## 168.3 Chain Rule for Probability

You just used this equality: For any events A and B

$$p(A|B) = \frac{p(A \text{ AND } B)}{p(B)}$$

This is more commonly written like this:

$$p(A \text{ AND } B) = p(A|B)p(B)$$

This is an abstract way of writing the idea, but the idea itself is pretty intuitive. The probability that you are going to buy a ticket and win the lottery is equal to the probability that you buy a ticket times the probability that you win, given that you have bought a ticket. (Here A is "win the lottery" and B is "buy a ticket".)

This is known as *The Chain Rule of Probability*. We can chain together as many events as we want: The probability that you are going to die in the car that you bought with your

winnings from the lottery ticket you bought is:

$$p(W \text{ AND } X \text{ AND } Y \text{ AND } Z) = p(W|X \text{ AND } Y \text{ AND } Z)p(X|Y \text{ AND } Z)p(Y|Z)p(Z)$$

where

- $W$ = Dying in car accident
- X = Buying a car with lottery winnings
- Y = Winning the lottery
- Z = Buying a lottery ticket

In English, the equation says:

"The probability that you will die in a car accident, buy a car with lottery winnings, win the lottery, and buy a lottery ticket is equal to the probability that you buy a lottery ticket times the probability that you win the lottery (given that you have bought a ticket) times the probability that buy a car with those lottery winnings (given that bought a ticket and won) times the probability that you crash that car (given that you have bought the car, won the lottery, and bought a ticket)."

CHAPTER 169

# Bayes' Theorem

Let's say that you are holding two bags of marbles. You know that one bag contains 60 white marbles and 40 red marbles, and you know that the other holds 10 white marbles and 90 red marbles. You don't know which is which, and you can't see the marbles.

Your friend says, "Guess which bag is mostly red marbles." You pick one.

"What is the probability that this is the bag that is mostly red marbles?" You think to yourself "There is a 50 percent chance that this bag is mostly red marbles, and there is also a 50 percent probability that it is the mostly-white-marbles bag."

You then pick one marble from the bag: it is red. Now you must update your beliefs. It is more likely that this is the mostly-red-marbles bag. What is the probability now?

Bayes Theorem gives you the rule for updating your beliefs based on new data.

## 169.1 Bayes Theorem

Let's say you have two events or conditions C and D. C is "The person has a cough" and D is "The person is waiting to see a doctor."

Using the chain rule of probability, we now have two ways to calculate $p(C \text{ AND } D)$:

$$p(C \text{ AND } D) = p(C|D)p(D)$$

(The probability the person is at the doctor multiplied by the probability they have a cough if they are at the doctor.)

or

$$p(C \text{ AND } D) = p(D|C)p(C)$$

(The probability the person has a cough multiplied by the probability they are at the doctor if they have a cough.) See Figure169.1

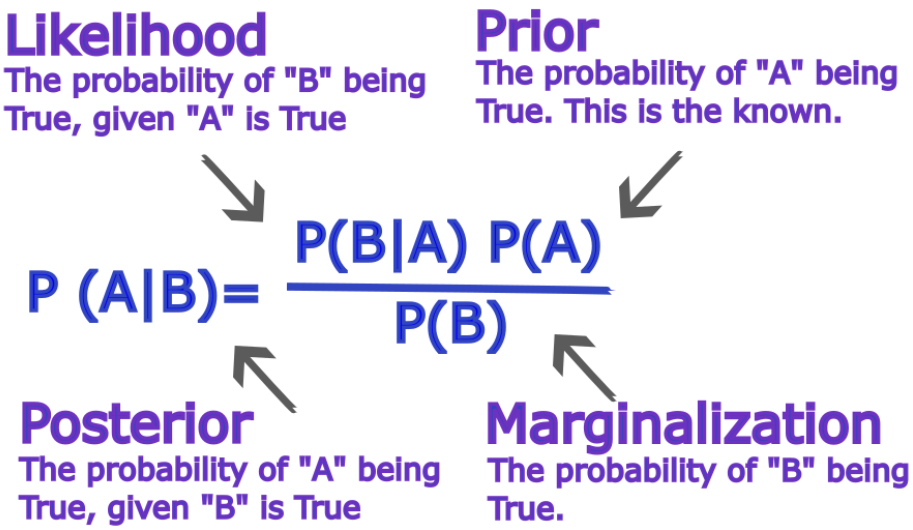



Figure 169.1: A labelled figure of Bayes's Theorem Equation.

Thus:

$$p(D|C) = \frac{p(C|D)p(D)}{P(C)}$$

Now, you can calculate $p(D|C)$ (in this case, the probability that you are waiting to see a doctor given that you have a cough.) if you know:

- $p(C|D)$ (The probability that you have a cough given that you are waiting to see a doctor)
- $p(D)$ (The probability that you are waiting for a doctor for any reason.)
- $p(C)$ (The probability that you have a cough anywhere)

Pretty much all modern statistical methods (including most artificial intelligence) are based on this formula, which is known as Bayes' Theorem. It was written down by Thomas Bayes before he died in 1761. It was then found and published after his death.

## 169.2 Using Bayes' Theorem

Back to the example at the beginning. To review:

- There are two bags that look exactly the same.

- Bag W has 60 white marbles and 40 red marbles.
- Bag R has 10 white marbles and 90 red marbles.
- You pull one marble from the selected bag: it is red.

What is the probability that the selected bag is Bag R? Intuitively, you know that the probability is now more than 0.5. What is the exact number?

In terms of conditional probability, we say we are looking for "the probability that the selected bag is Bag R, given that you drew a red marble", or $p(B_R|D_R)$, where $B_R$ is "the selected bag is Bag R" and $D_R$ is "you drew a red marble from the selected bag".

From Bayes' Theorem, we can write:

$$p(B_R|D_R) = \frac{P(D_R|B_R)P(B_R)}{P(D_R)}$$

$P(D_R|B_R)$ is just the probability of drawing a red marble given that the selected bag is Bag R. That is easy to calculate: There are 100 marbles in the bag, and 90 are red. Thus, $P(D_R|B_R) = 0.9$.

$P(B_R)$ is just the probability that you chose Bag R before you drew out a marble. Both bags look the same, so $P(B_R) = 0.5$. This is called *the prior*, because it represents what you thought the probability was before you got more information.

$P(D_R)$ is the probability of drawing a red marble. There was 0.5 probability that you put your hand into Bag W (in which 40 of the 100 marbles are red) and a 0.5 probability that you put your hand into Bag R (in which 90 of the 100 marbles are red). So

$$P(D_R) = 0.5\frac{40}{100} + 0.5\frac{90}{100} = 0.65$$

Putting it together:

$$p(B_R|D_R) = \frac{P(D_R|B_R)P(B_R)}{P(D_R)} = \frac{(0.9)(0.5)}{0.65} = \frac{9}{13} \approx 0.69$$

Thus, given that you have pulled a red marble, there is about a 69% chance that you have selected the bag with 90 red marbles.

## 169.3 Confidence

Bayes' Theorem, then, is about updating your beliefs based on evidence. Before you drew out the red marble, you selected one bag thinking it might contain 90 red marbles. How certain were you? 0.0 being complete disbelief and 1.0 entirely confidence, you were 0.5. After pulling out the red marble, you were about 0.69 confident that you had chosen the bag with 90 red marbles.

The question "How confident are you in your guess?" is very important in some situations. For example in medicine, diagnoses often lead to risky interventions. Few diagnoses come with 100% confidence. All doctors should know how to use Bayes' Theorem.

In a trial, a jury is asked to determine if the accused person is guilty of a crime. Few jurors are ever 100% certain. In some trials, Bayes' Theorem is an exceptionally important tool.

## 169.4 The Monty Hall Problem

The Monty Hall Problem is a popular example in discrete probability and probability theory that highlights an application of Bayes' theorem.

Here's the scenario[1]

- A contestant is presented with three doors. Behind one door is a brand new car; behind the other two doors are goats.
- The contestant selects one of the three doors, say Door A.
- The host, Monty Hall, who knows what is behind each door, then opens one of the remaining two doors, say Door B, revealing a goat.
- The contestant is then offered the choice to either stick with Door A or switch to the other remaining unopened Door C.

At first glance, many people assume that after Monty opens a door, there are two doors left and hence the probability of winning the car is $1/2$ regardless of whether one switches or stays. However, conditional probability and Bayes' Theorem reveal a different result.

Originally, the probability that you picked the right door, Door A, is $P(\text{A has the car}) = \frac{1}{3}$.

Then, Monty Hall reveals Door B to contain a goat behind it. This changes things, as Door A remains a probability of $\frac{1}{3}$, while Door C, the unopened door, now contains the summed probability of the other two doors, $\frac{2}{3}$.

Proving this with Bayes' Theorem:

[1] A few videos and a movie excerpt on this subject are linked in your digital resources.

$$\begin{aligned}
\mathrm{P}(\text{Car behind C} \mid \text{Monty opens B}) &= \frac{\mathrm{P}(\text{Monty opens B} \mid \text{Car behind C})\,\mathrm{P}(\text{Car behind C})}{\mathrm{P}(\text{Monty opens B})} \\
&= \frac{}{\mathrm{P}(\text{Monty opens B} \mid \text{Car behind A})\mathrm{P}(\text{Car behind A}) + \mathrm{P}(\text{Monty ope} \\
&= \frac{\frac{1}{3}}{(\frac{1}{2})(\frac{1}{3}) + 0 \cdot (\frac{1}{3}) + 1 \cdot (\frac{1}{3})} \\
&= \frac{\frac{1}{3}}{\frac{1}{6} + \frac{1}{3}} \\
&= \frac{\frac{1}{3}}{\frac{1}{2}} \\
&= \frac{2}{3}
\end{aligned}$$

Thus, it is always in your interest to switch. If you ever have the luxury to be on a game show, make sure to use Bayes' Theorem before making your guesses.

Chapter 170

# Introduction to Markov Chains

## 170.1 Markov Chains

Today was a `Sunny` day. However, I would like to know if it will be `Sunny` tomorrow. These are two categorical states of the weather.

How can I predict the weather? In this case, we are trying to determine values of *uncertainty*. We are uncertain of the weather the next day, or even, the weather in the next 5 days. How many days previously do I need to know the weather to make an accurate prediction?

This is the question that Russian mathematician Andrey Markov curious about:

> How much history of a state do you need to predict a current state?

His discovery was the following: you only need to know the current state to predict the *next state*. This is now known as the *Markov property*. We say a process is *Markovian* if, given the current state, knowledge of past states provides no additional information about the next state. This requires that the future state depends only on the present state and not on the sequence of events that preceded it. In other words, the process has no memory of past states. Generally, this chapter will deal with *discrete-time Markov chains*, where the process evolves in discrete time steps. We will discuss continuous-time Markov chains in a later chapter.

Suppose there are 3 days: Yesterday, Today, and Tomorrow. Information about Yesterday's weather does not provide any additional information about Tomorrow's weather once we know Today's weather.

$$\mathrm{P}(\text{Tomorrow} \mid \text{Today}, \text{Yesterday}) = \mathrm{P}(\text{Tomorrow} \mid \text{Today})$$

While weather, in reality, is not perfectly Markovian, this property allows us to model and predict it using a simplified framework. In a Markov chain, we can represent the states (`Sunny`, `Rainy`, `Cloudy`) and the probabilities of transitioning from one state to another.

Suppose we classify the weather as either `Sunny` or `Rainy`. We would call the *state space* $\mathcal{S} = \{\texttt{Sunny}, \texttt{Rainy}\}$. Notice the special S in the notation $\mathcal{S}$, which stands for *state space*.

Based on historical observations, we might estimate the following probabilities:

- If today is `Sunny`, tomorrow will be `Sunny` with probability 0.8 and `Rainy` with probability 0.2.
- If today is `Rainy`, tomorrow will be `Sunny` with probability 0.4 and `Rainy` with probability 0.6.

These probabilities completely describe our Markov chain. Once we know today's weather, we can predict the probabilities for tomorrow's weather without needing to know what happened last week.

The states and transitions probabilities can be represented using a *Markov chain*, also called a *state diagram*,(which looks similar to a graph!):

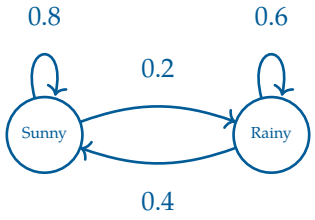



Figure 170.1: A Markov chain representing the weather with two states: `Sunny` and `Rainy`. The arrows indicate the probabilities of transitioning from one state to another.

Here, given the state of the weather today, we can predict the probabilities of the weather tomorrow by following the chain of arrows. There is a great visualization of Markov chains in your digital resources.

We can generalize the *Markov property* as the following equation:

$$P(X_{n+1} = j \mid X_0, X_1, \ldots, X_n) = P(X_{n+1} = j \mid X_n)$$

where $X_0, X_1, \ldots, X_n$ are states of a given process at times $0, 1, \ldots, n$ respectively. This means that the probability of transitioning to a next or future state $j$ depends only on the current state $X_n$ and not on any of the previous states.

As the number of states increases, the Markov chain can become more complex. However, the Markov property still holds, allowing us to model and analyze processes in the form of a matrix of transition probabilities. This matrix, known as the *transition matrix*, captures the probabilities of moving from one state to another in a compact form.

## *Exercise 436* Frank's Hot Dog Chain

*Working Space*

Frank is opening a new hot dog stand. Depending on how much they like his hot dogs, new customers have a chance of becoming a regular customer. If they do not like them, they may become an inactive customer.

A new customer has a:

- 40% chance of remaining new,
- 50% chance of becoming regular,
- 10% chance of becoming inactive.

A regular customer has a:

- 10% chance of reverting to new,
- 80% chance of staying regular,
- 10% chance of becoming inactive.

An inactive customer has a:

- 20% chance of returning as new,
- 30% chance of becoming regular,
- 50% chance of remaining inactive.

Assuming the $X_n$ are the states at of the $n$-th visit to the store (such that $X_0 \implies$ First Visit),

1. Identify the state space of this Markov chain.
2. Draw a Markov chain to represent this process.
3. Simplify this using the Markov property, and solve for the following probability:

   $$P(X_{n+1} = R \mid X_n = I, X_{n-1} = R, X_{n-2} = N)$$

*Answer on Page 2057*

## 170.2 Transition Matrices

A *transition matrix* (also sometimes called a *stochastic matrix*) records the probability of moving from one state to another in a Markov chain. Each row corresponds to the *current state*, and each column corresponds to the *next state*.

A transition matrix $P$ for a Markov chain with $n$ states can be represented as follows:

$$P = \begin{array}{c|cccc} & X_{t+1} & & & \\ X_t & 1 & 2 & \cdots & n \\ \hline 1 & p_{11} & p_{12} & \cdots & p_{1n} \\ 2 & p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n & p_{n1} & p_{n2} & \cdots & p_{nn} \end{array}$$

where the rows represent $X_t$ (current state at time $t$) and the columns represent $X_{t+1}$ (next state at time $t+1$).

The entry $P_{ij}$ represents the probability of transitioning from state $i$ at time $t$ to state $j$ at time $t+1$. In probability notation, we can express this as:

$$P_{ij} = P(X_{t+1} = j \mid X_t = i)$$

For the weather example, the transition matrix is

$$P = \begin{array}{c|cc} & \text{Sunny} & \text{Rainy} \\ \hline \text{Sunny} & 0.8 & 0.2 \\ \text{Rainy} & 0.4 & 0.6 \end{array}$$

The entry in row $i$ and column $j$ gives the probability of moving from state $i$ to state $j$ in a single time step.

## *Exercise 437* Frank's Hot Dog stand, returns

*Working Space*

Suppose Frank's hot dog stand is still running strong, but the customers are getting harder to please. The states still remain $\mathcal{S} = \{N, R, I\}$. The transition probabilities have changed, and the new transition matrix is given by:

$$P = \begin{bmatrix} 0.50 & 0.30 & 0.20 \\ 0.15 & 0.65 & 0.20 \\ 0.10 & 0.15 & 0.75 \end{bmatrix}$$

1. Find the probability that a regular customer becomes an inactive customer on their next visit.
2. Find the probability that an inactive customer becomes a regular customer after two visits.
3. Find the probability that a new customer becomes an inactive customer after three visits.

*Answer on Page 2058*

From this exercise, we saw how to calculate the probability of transitioning from one state to another after multiple steps. This can be done by multiplying the transition probabilities along the path of states and summing over all possible paths.

We can also do this process using matrix multiplication. Specifically, if we want to find the probability of transitioning from one state to another after $k$ steps, we can compute the $k$-th power of the transition matrix $P^k$. The entry in row $i$ and column $j$ of $P^k$ will give us the probability of transitioning from state $i$ to state $j$ in exactly $k$ steps.

The transition matrix $P$ gives probabilities for one state change. To find probabilities after two state changes, we compute $P^2$. This is done by multiplying the transition matrix by

itself:

$$P^2 = P \times P$$

For example, using the hot dog stand transition matrix, we would represent the probability of transitioning from a new customer to an regular customer after two steps as:

$$\sum_k P_{N,k} \cdot P_{k,R} = P^2_{N,R}$$

where $k$ ranges over all possible intermediate states (new, regular, inactive).

Therefore, if we want the probability of moving from state $i$ to state $j$ after two steps, we look at the $(i, j)$ entry of $P^2$, written $(P^2)_{ij}$.

Generally we can represent this as the following equation:

$$P^k_{ij} = P(X_{n+k} = j \mid X_n = i)$$

where $P^k_{ij}$ is the entry in row $i$ and column $j$ of the matrix $P^k$. This gives us the probability of transitioning from state $i$ to state $j$ in exactly $k$ steps.

$$\text{Probability Distribution after } k \text{ steps} = P^k$$

Note that this is only possible square matrices, which means the number of states must be the same for the current and next state. This is a common assumption in Markov chains, as it allows us to model processes where the set of possible states does not change over time.

## *Exercise 438* Chemical Change

A chemical changes colors approximately every 30 minutes. The state space of this chemical is $\mathcal{S} = \{\text{Green}, \text{Blue}\}$.

$$P = \begin{bmatrix} 0.2 & 0.8 \\ 0.1 & 0.9 \end{bmatrix}$$

Find the $P^3$, the transition matrix after 90 minutes. Then, find $P(X_{n+3} = \text{Green} \mid X_n = \text{Blue})$.

*Working Space*

*Answer on Page 2059*

## 170.3 Steady State Distribution

For many Markov chains, these probabilities eventually stabilize (recall **Law of Large Numbers**, $P^k$ as $k \to \infty$) as we request more and more transitions or future states (recall Law of Large Numbers). That is, after a large number of transitions, the proportion of the probability of each state approaches a fixed value. This long-run behavior is known as the *steady-state distribution* or *equilibrium distribution* of the Markov chain.

A steady-state distribution has a special property: applying the transition matrix does not change it. If $\mathbf{x}$ is a steady-state vector and $P$ is the transition matrix, then

$$P\mathbf{x} = \mathbf{x}.$$

In other words, once the chain reaches equilibrium, the probabilities remain unchanged from one time step to the next. This equation is also the eigenvector equation with eigenvalue 1. Therefore, steady-state distributions can be found by determining the eigenvector corresponding to the eigenvalue 1 and then normalizing its entries so that they sum to 1.

The following example illustrates how eigenvectors can be used to determine the long-run proportion of time a Markov chain spends in each state.

### *Exercise 439* Markov Chain in the form of Transition Matrix

*This question is taken from an IB practice exam.* A Markov Chain has two possible states, A and B. The transition matrix P is given by:

$$P = \begin{bmatrix} p & 1-q \\ 1-p & q \end{bmatrix}$$

Given one of the eigenvalues is 1.

1. Solve for the eigenvector corresponding to the eigenvalue 1.
2. Find an expression for the proportion of time the Markov chain spends in state A in the long run in terms of $p$ and $q$.
3. Hence or otherwise, find the proportion of time spent in state A when $p = 0.3$ and $q = 0.6$.

*Working Space*

*Answer on Page 2059*

## 170.4 Applications

### 170.4.1 PageRank

Imagine an individual browsing the internet. They start on a random webpage and click on links to navigate to other pages. The probability of moving from one webpage to another can be modeled as a Markov chain, where each webpage represents a state and the links between pages represent transitions with certain probabilities.

In this case, each page is a *state* and each link from one page to another represents a *transition* with a certain probability. The transition probabilities can be determined based on the structure of the web, such as the number of links on each page and the likelihood of a user clicking on those links. If a page has more links, the probability of transitioning to any particular linked page is lower. For example, if page $i$ has 3 *outgoing links*, the probability of transitioning to each linked page would be $\frac{1}{3}$. The more links to a given

page that exist (called *incoming links*), the more "important" that webpage is. Since these links are directed and each form a Markov Chain, recognize that this is *directed graph*.

Now, if we want to find the long-run proportion of time a user spends on each webpage, we can use the *steady-state distribution* of this Markov chain. The steady-state distribution will give us the PageRank of each webpage, which is a measure of its importance based on the structure of the web. A few issues arise in this model, such as the presence of *dead ends* (pages with no outgoing links) and *spider traps* (sets of pages that link to each other but not to the rest of the web). To find the ultimate page rank, we can use a modified version of the transition matrix that consists of a stabilized probability

$$\pi_{n+1} = \pi_n P$$

eventually this will converge to a steady-state distribution $\pi$ such that $\pi P = \pi$. This is the PageRank vector (similar to the eigenvector discussed above), which gives the relative importance of each webpage in the long run.

There may be an additional *damping factor* ($d$) that is introduced to compensate for the possibility that a user randomly stops following links and instead jumps to a different random page. This is often set to around $0.85$. The probability that they instead jump to any random page is $1-d$, usually $0.15$. In matrix terms, this leads to a modified transition model

$$G = dP + (1-d)\frac{1}{n}J,$$

where $J$ is the all-ones matrix and $n$ is the number of pages. This ensures the resulting chain is irreducible and aperiodic, so the steady-state distribution exists and is unique.

### 170.4.2 Hidden Markov Models and NLP

*Hidden Markov Models* (HMMs) are an expanded version of a Markov Process. In addition to having probabilities about transition from state to state, it also has influencing *observations*, called *emissions*. An emission is a *visible* output generated by those the states in play, often which are hidden or not visible.

Assume you have a patient in a hospital with two possible states: $\mathcal{S} = \{\text{Sick}, \text{Healthy}\}$. You cannot know the exact state at a current time with certainty. However, you have three observable factors: $\mathcal{O} = \text{no fever}, \text{fever}, \text{cough}$. You may also see the emission states symbolized with $\mathcal{E}$.

For each state change observation (for this example, lets say once a day), you need both the transition matrix and the *emission matrix*. We will define the transition matrix $T$ and emission matrix $E$ as follows:

$$S = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}, \qquad E = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.7 & 0.1 & 0.2 \end{bmatrix}$$

Note that the emission matrix is not necessarily square. In real life examples, there would be probably hundreds of possible states, each with tons of possible emissions.

The emission matrix describes the probability of observing each symptom given a particular hidden state. For example,

$$\mathrm{P}(\text{fever} \mid \text{Sick}) = 0.5,$$

while

$$\mathrm{P}(\text{no fever} \mid \text{Healthy}) = 0.7.$$

Suppose that over three consecutive days we observe the symptom sequence

$$\text{fever} \rightarrow \text{cough} \rightarrow \text{no fever}.$$

These observations alone do not reveal the patient's true health state. One possible hidden state sequence is

$$\text{Sick} \rightarrow \text{Sick} \rightarrow \text{Healthy},$$

while another is

$$\text{Healthy} \rightarrow \text{Sick} \rightarrow \text{Healthy}.$$

The goal of an HMM is to infer the most likely hidden state sequence that generated the observed emissions.

While we won't discuss them in this book, there are algorithms such as the Forward and Viterbi algorithms that can track and guess that most likely future state given the states available, or most likely emissions given a set of states. These use *dynamic programming*, a method that programming similar to excel processing.

### 170.4.3 Natural Language Processing

Another strong use of HMM is the basics of Natural Language Processing, and ultimately, predictive AI text, such as recommended words above your phone's keyboard.

Say we have a set of states represent parts of speech

$$\mathcal{S} = \{\text{Noun}, \text{Verb}, \text{Adjective}, \dots\}$$

These states are hidden because when you read a sentence, you only see the words, not their correct grammatical labels.

The observations are the words themselves:

$$\mathcal{O} = \{\text{dog}, \text{runs}, \text{fast}, \dots\}$$

Suppose we are given the sentence `dog runs fast`. The observations are known, but the hidden states are not. One possible hidden-state sequence is

$$\texttt{Noun} \rightarrow \texttt{Verb} \rightarrow \texttt{Adverb}$$

The transition matrix models how likely one grammatical tag is to follow another. For example,

$$\text{P}(\text{Verb} \mid \text{Noun}) = 0.6$$

might represent that verbs frequently follow nouns in the natural spoken language. The emission matrix models how likely a word is to be generated by a particular tag. For example,

$$\text{P}(\texttt{dog}|\text{Noun}) = 0.4, \text{P}(\texttt{runs}|\text{Verb}) = 0.5$$

## *Exercise 440* NLP Exercise

*Working Space*

Suppose an HMM uses the following part-of-speech tags:

$$\mathcal{S} = \{\text{Noun (N)}, \text{Verb (V)}, \text{Adjective (A)}\}.$$

From a training dataset of words, the following transition probabilities were estimated:

$$\mathrm{T} = \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.4 & 0.4 & 0.2 \\ 0.6 & 0.2 & 0.2 \end{bmatrix}$$

where the rows and columns correspond to the states

$$(\text{Noun}, \text{Verb}, \text{Adjective}).$$

Find the probability that an adjective occurs after a noun followed by a verb. That is, compute

$$\mathrm{P}(\mathrm{N} \to \mathrm{V} \to \mathrm{A}).$$

*Answer on Page 2060*

# Chapter 171

# Antiderivatives

In your study of calculus, you have learned about derivatives, which allow us to find the rate of change of a function at any given point. Derivatives are powerful tools that help us analyze the behavior of functions. Now, we will explore another concept called antiderivatives, which are closely related to derivatives.

An antiderivative, also known as an integral or primitive, is the reverse process of differentiation. It involves finding a function whose derivative is equal to a given function. In simple terms, if you have a function and you want to find another function that, when differentiated, gives you the original function back, you are looking for its antiderivative. Consider the graph of $f(x)$ below. We can sketch a possible antiderivative of $f$ by noting the slope of the antiderivative is equal to the value of $f$. We will refer to the antiderivative of $f$ as $F(x)$ (that is, $F'(x) = f(x)$).

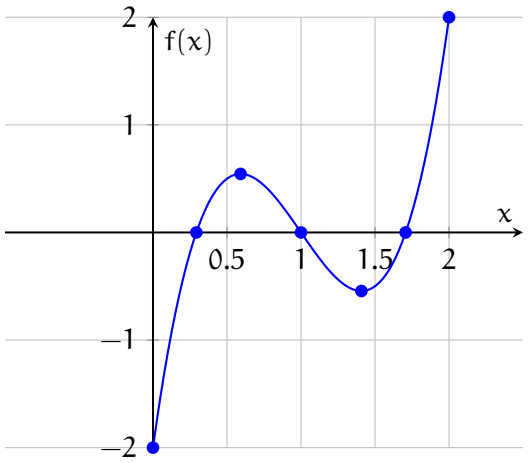



Figure 171.1: Plot of $f$ with select points

If we are given a coordinate for $F(x)$, then we can use the graph of $f(x)$ to sketch $F(x)$. Suppose we know that $F(0) = 1$. From the graph of $f$, we also know that

| $x$ | $f(x) =$ slope of $F(x)$ |
|---|---|
| 0 | -2 |
| $\approx 0.3$ | 0 |
| $\approx 0.6$ | $\approx 0.5$ |
| 1 | 0 |
| $\approx 1.4$ | $\approx -0.4$ |
| $\approx 1.7$ | 0 |
| 2 | 2 |

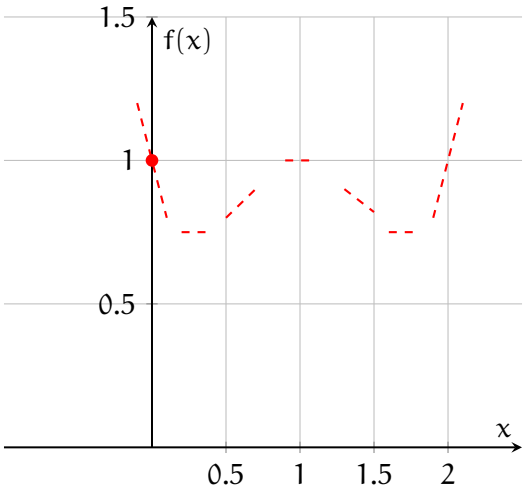



Figure 171.2: Beginning sketch of $F(x)$

We can then connect these slopes to have an approximate sketch of $F(x)$:

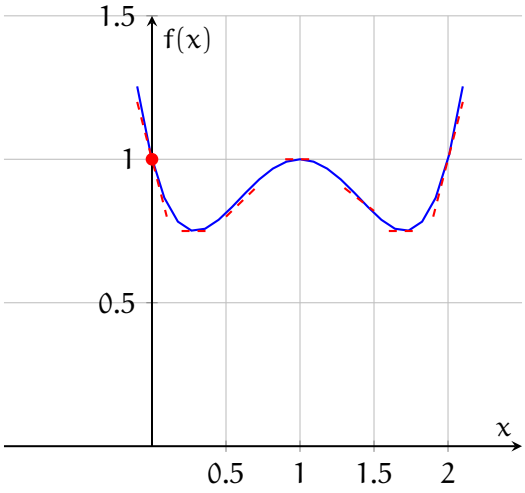



Figure 171.3: Sketch of $F(x)$

The symbol used to represent an antiderivative is $\int$. It is called the integral sign. For example, if $f(x)$ is a function, then the antiderivative of $f(x)$ with respect to $x$ is denoted as $\int f(x)\,dx$. The $dx$ at the end indicates that we are integrating with respect to $x$.

Another way to state this is that $F$ is the antiderivative of $f$ on an interval, $I$, if $F'(x) = f(x)$ over the interval. The relationship between $f$ and $F$ is discussed more in the chapter on the Fundamental Theorem of Calculus.

Finding antiderivatives requires using specific techniques and rules. Some common antiderivative rules include:

- The power rule: If $f(x) = x^n$, where $n$ is any real number except $-1$, then the antiderivative of $f(x)$ is given by $\int f(x)\,dx = \frac{1}{n+1}x^{n+1} + C$, where $C$ is the constant of integration.
- The constant rule: The antiderivative of a constant function is equal to the constant times $x$. For example, if $f(x) = 5$, then $\int f(x)\,dx = 5x + C$.
- The sum and difference rule: If $f(x)$ and $g(x)$ are functions, then $\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx$. Similarly, $\int (f(x) - g(x))\,dx = \int f(x)\,dx - \int g(x)\,dx$.

Antiderivatives have various applications in mathematics and science. They allow us to calculate the total accumulation of a quantity over a given interval, compute areas under curves, and solve differential equations, among other things.

## 171.1 General Antiderivatives

It is important to note that an antiderivative is not a unique function. Since the derivative of a constant is zero, any constant added to an antiderivative will still be an antiderivative of the original function. This is why we include the constant of integration, denoted by $C$, in the antiderivative expression.

Stated formally, if $F$ is an antiderivative of $f$ on interval $I$, then the most general antiderivative of $f$ on $I$ is $F(x) + C$, where $C$ is an arbitrary constant.

A concrete example of this is $f(x) = x^2$. Let us define $F(x)$ such that $F'(x) = f(x)$. That is, there is some function $F$ such that the derivative of $F$ is $x^2$. One possible solution for $F$ is $F(x) = \frac{1}{3}x^3$. You can check using the power rule that $\frac{d}{dx}F(x) = f(x)$. What if we added or subtracted a constant from $F$? Let's define $G(x) = \frac{1}{3}x^3 + 2$. Well, $G'(x) = f(x)$ also! The same applies for $H(x) = \frac{1}{3}x^3 - 7$. Several possible antiderivatives of $f(x) = x^2$ are shown in figure 171.4.

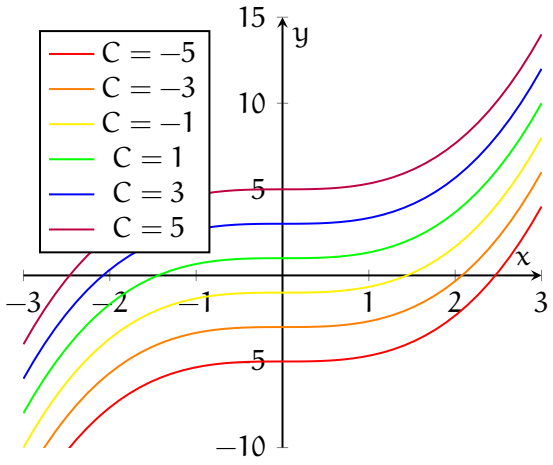



Figure 171.4: If $F'(x) = x^2$, then the general solution is $F(x) = \frac{1}{3}x^3 + C$

Since taking a derivative "erases" any constant, you must always add back in the unknown constant, C, when finding the general antiderivative.

## 171.2 Specific Antiderivatives

If you are given a condition, you can often solve for C and find a specific antiderivative. For example, suppose that in addition to knowing that $F'(x) = x^2$, we also know that $F(3) = 2$. We can use the fact that F passes through $(3, 2)$ to find the value of C:

$$F(x) = \frac{1}{3}x^3 + C$$

$$F(3) = \frac{1}{3}(3)^2 + C = 2$$

$$39 + C = 2$$

$$C = -7$$

Therefore, the specific solution to $F'(x) = x^2$ with the condition that $F(3) = 2$ is $F(x) = \frac{1}{3}x^3 - 7$.

## 171.3 Antiderivatives of Trig Functions

We already know that $\frac{d}{dx}\sin x = \cos x$. Taking $\sin x$ to be $F(x)$ and $\cos x$ to be $f(x)$, we see that $F'(x) = f(x)$ and therefore, $\sin x$ is the antiderivative of $\cos x$.

### *Exercise 441*

What is the antiderivative of $\sin x$? Explain your answer.

*Working Space*

*Answer on Page 2061*

You should have found that the antiderivative of $\sin x$ is $-\cos x$. Other general antiderivatives of trigonometric functions are presented in the table below.

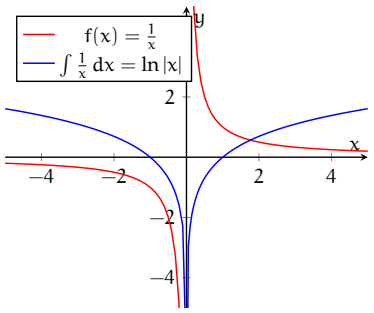
| Function | Antiderivative |
|---|---|
| $\cos x$ | $\sin x + C$ |
| $\sin x$ | $-\cos x + C$ |
| $\sec^2 x$ | $\tan x + C$ |
| $\sec x \tan x$ | $\sec x + C$ |
| $-\csc^2 x$ | $\cot x + C$ |
| $-\csc x \cot x$ | $\csc x + C$ |

Notice this is the flipped version of the derivatives of trigonometric functions presented in the Trigonometric Functions chapter. This hints at the relationship between derivatives and integrals: they are opposite processes.

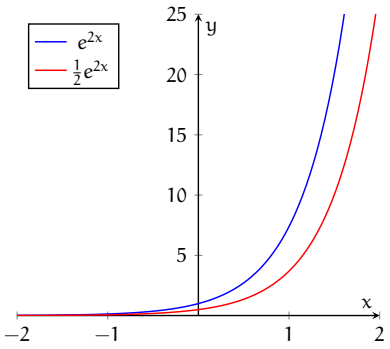
## 171.4 Other Important Antiderivatives

The power rule only applies when $n \neq -1$. So, what is the antiderivative of $f(x) = \frac{1}{x}$? Recall from the chapters on derivatives that $\frac{d}{dx} \ln x = \frac{1}{x}$ (see figure **??**). Therefore, the general antiderivative of $\frac{1}{x}$ is $\ln|x| + C$. We have to take the absolute value because of the domain restrictions of $\ln x$. Notice that for $x < 0$, the slope of $\ln|x|$ is negative and decreasing (becoming more negative), and the value of $\frac{1}{x}$ is also negative and decreasing. Similarly, for $x > 0$, the slope of $\ln|x|$ is positive and decreasing (becoming less positive) and the value of $\frac{1}{x}$ is also positive and decreasing.

Since the derivative of $e^x$ is $e^x$, it follows that the general antiderivative of $e^x$ is $e^x + C$. What if there is a multiplying factor in the exponent, such as $e^{kx}$? Recall that $\frac{d}{dx} e^{kx} = ke^{kx}$. It follows that $\frac{d}{dx} \frac{1}{k} e^{kx} = e^{kx}$. Therefore, the general antiderivative of $e^{kx}$ is $\frac{1}{k} e^{kx} + C$. (See figure **??** for an example where $k = 2$.)




(a) $\frac{1}{x}$ and its antiderivative $\ln|x|$




(b) $e^{2x}$ and its antiderivative $\frac{1}{2}e^{2x}$

Figure 171.5: Two examples of functions and their antiderivatives.

Often, the base of an exponential function is not $e$. We can also find the general antiderivative of $b^x$, where $b \neq e$. Recall that $\frac{d}{dx}b^x = \ln b\, b^x$. Therefore, $\frac{d}{dx}\frac{1}{\ln b}b^x = b^x$, and the general antiderivative of $b^x$ is $\frac{b^x}{\ln b}$.

## 171.5 Higher order antiderivatives

What if we are given the second order derivative, or a higher order? Take this example: $f''(x) = 2x + 3e^x$. The antiderivative of $f''$ is $f'$. Applying the power rule and knowing the antiderivative of $e^x$ is $e^x$, we find that $f'(x) = x^2 + 3e^x + C_1$. We designate the constant as $C_1$, because we will have to determine the antiderivative a second time and we don't want to confuse our constants with each other. To find $f$, we apply the power rule again, and we find that $f(x) = \frac{1}{3}x^3 + 3e^x + C_1x + C_2$. You can check if this is correct by taking the derivative of $f(x)$ twice, which should yield the $f''(x)$ originally given.

In summary, antiderivatives are the reverse process of differentiation. They help us find functions whose derivatives match a given function. Understanding antiderivatives is crucial for various advanced calculus concepts and real-world applications.

Now, let's explore different techniques and methods for finding antiderivatives and discover how they can be applied in solving problems.

## 171.6 Additional Practice

### Exercise 442

A particle moving in a straight line has an acceleration given by $a(t) = 6t+4$ (in units of $\frac{cm}{s^s}$). If its initial velocity is $-6\frac{cm}{s}$ and its initial position is $9cm$, what is the function $s(t)$ that describes the particle's position in $cm$?

*Working Space*

*Answer on Page 2061*

### Exercise 443

Let $f'(x) = 2\sin x$. If $f(\pi) = 1$. Write an expression for $f(x)$.

*Working Space*

*Answer on Page 2061*

### Exercise 444

Find the general antiderivatives of the following functions:

1. $f(x) = x^2 + 2x - 4$
2. $g(x) = \sqrt[3]{x^2} + x\sqrt{x}$
3. $h(x) = \frac{1}{5} - \frac{2}{x}$
4. $r(\theta) = 2\sin\theta - \sec^2\theta$

*Working Space*

*Answer on Page 2062*

### Exercise 445

Find the f that satisfies the given conditions:

1. $f'(\theta) = \sin\theta + \cos\theta$, $f(\pi) = 2$
2. $f''(x) = 12x^2 + 6x - 4$, $f(0) = 4$ and $f(1) = 1$

*Working Space*

*Answer on Page 2062*

Chapter 172

# Riemann Sums

## 172.1 The Meaning of the Area Under a Function

Let's look at the example of a hammer tossed in the air from a previous chapter. As you may recall, if a hammer is tossed up from the ground at 5 m/s, its velocity can be described as $v(t) = 5 - 9.8t$ (on Earth, where the acceleration due to gravity is approximately $-9.8\ \frac{m}{s^2}$). The velocity function of our hammer from when it is tossed ($t = 0$) to when it hits the ground ($t \approx 1.02$) is shown in figure 172.1.

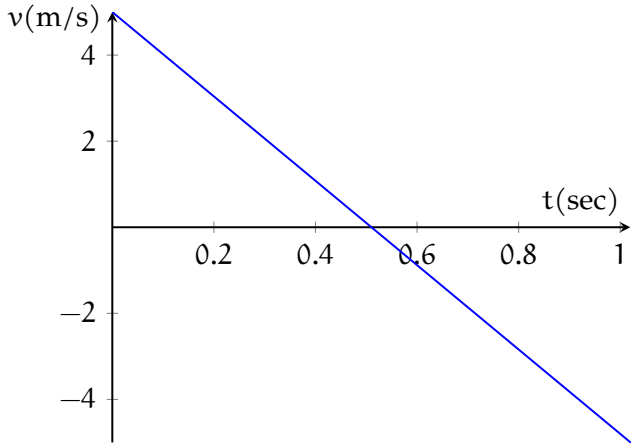



Figure 172.1: Velocity of a hammer thrown upwards at 5 m/s

Now, suppose we only have this velocity function, and we want to know how high above its initial position the hammer is tossed. Examine the graph: At approximately what time does the hammer reach its peak height? (Hint: what should the hammer's velocity be when it reaches its peak?). At the highest point of its flight, the hammer's velocity will be $0\ \frac{m}{s}$, which occurs at approximately $t = 0.5s$ (it's actually $t = 0.5102s$ but we don't need to be that precise for this example).

Now that we know *when* the hammer reaches its peak, how can we determine *how high* that peak is? Recall that velocity is the slope of the position-time graph. Since slope is change in position divided by change in time (in this case, as time is on the x-axis and position on the y-axis), then the slope must have units of [position]/[time] which could be $\frac{m}{s}$, $\frac{\text{miles}}{\text{hr}}$, and so on. These are units of velocity!

In figure 172.1, you can see that the units on the x-axis are seconds, and on the y-axis, the units are $\frac{m}{s}$. If we are looking for a *displacement* (that is, how far from its initial position

the hammer has traveled), we are looking for a solution with units of meters. To yield an answer with those units, we wouldn't use the slope of the graph; this would yield an answer with units $\frac{m}{s^s}$, the units for acceleration. Instead, we need to *multiply*! The area between the velocity function and the x-axis (see figure 172.2) can be found this way:

$$Area = \frac{1}{2}bh$$

where b is the base of the triangle and h is the height.

$$Area = \frac{1}{2}(0.5s)(5\frac{m}{s})$$

$$Area = 1.25m$$

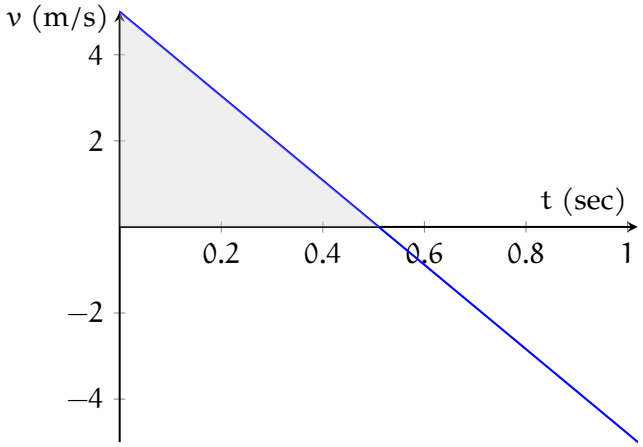


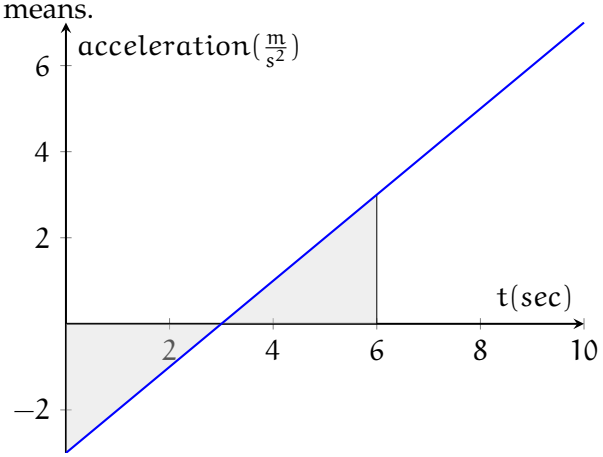

Figure 172.2: The area under $v(t)$ from $x = 0$ to $x = 0.5$ is equal to the displacement of the hammer

Notice that when multiplying the change in time (0.5 s) by the change in velocity (1.25 $\frac{m}{s}$), the seconds units cancel, yielding a result with units of meters. Therefore, the hammer reaches a peak height of $\approx 1.25$ m, which you can confirm by examining the graph originally presented for the hammer toss in the chapter on graph shape.

### 172.1.1 Determining the Meaning of the Area with Units

### *Exercise 446*

What units will the area shown in the graph have? Based on your answer, does the area represent a displacement, a net change in velocity, or a net change in acceleration? Calculate the shaded area [hint: areas below the x-axis are negative]. Write a sentence in plain English explaining what the are you calculated means.

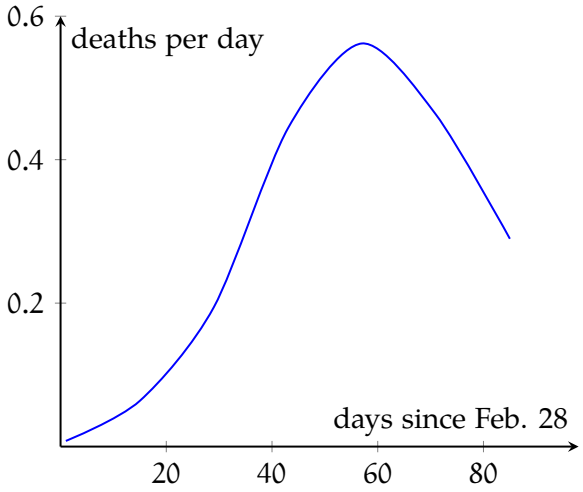



*Working Space*

*Answer on Page 2062*

## Exercise 447

The graph below shows historical data of the number of deaths due to SARS in Singapore over several months in 2003. What would the area under the curve represent?




*Working Space*

*Answer on Page 2063*

## Exercise 448

Oil leaked from a tank at a rate of $r(t)$ liters per hour. A site engineer recorded the leak rate over a period of 10 hours, shown in the table. Plot the data. How could you estimate the total volume of oil lost?

| t(h) | 0 | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|
| r(t)(L/h) | 8.7 | 7.6 | 6.8 | 6.2 | 5.7 | 5.3 |

*Working Space*

*Answer on Page 2063*

## 172.2 Estimating the area under functions

In the hammer example above, it was easy to determine the area under the function, since the area took the shape of a triangle. However, what about finding the area under a more complex function, such as $f(x) = \sin x + x$ (shown in figure 172.3)?

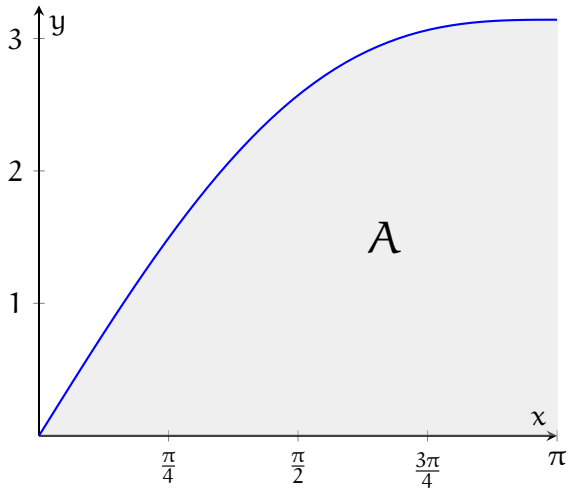



Figure 172.3: $f(x) = \sin x + x$

How can we determine the area under $f(x) = \sin x + x$ from $x = 0$ to $x = \pi$? We can *estimate* the area of that region by dividing the region into rectangles, finding the areas of the rectangles, and adding the areas. As an example, we will divide the region under $f(x) = \sin x + x$ into 4 intervals, shown in figure 172.4.

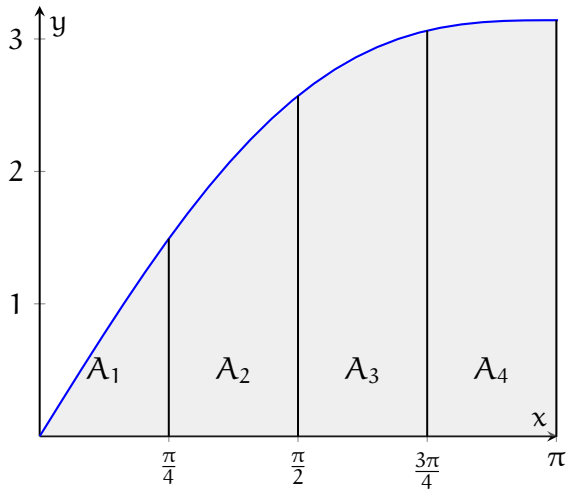



Figure 172.4: $f(x) = \sin x + x$ divided into 4 regions

As you can see in figure 172.4, each rectangle will have a width of $\frac{\pi}{4}$. But what about

the height? One way is to use the value of the function at the rightmost value of each rectangle, as shown in figure 172.5.

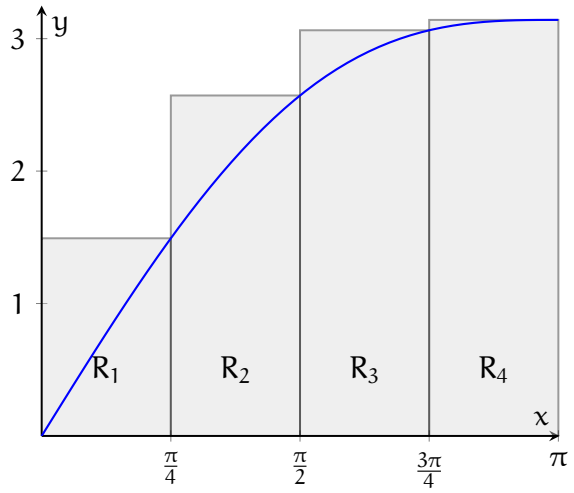



Figure 172.5: Four rectangle sections with heights determined by rightmost value of $f(x)$ on each interval

We can easily calculate the areas of each of these rectangles:

$$\frac{\pi}{4} \times f(\frac{\pi}{4}) + \frac{\pi}{4} \times f(\frac{\pi}{2}) + \frac{\pi}{4} \times f(\frac{3\pi}{4}) + \frac{\pi}{4} \times f(\pi)$$

$$\approx \frac{\pi}{4} \times (1.4925 + 2.5708 + 3.0633 + 3.1416) = 8.0646$$

Based on figure 172.5, will the calculated area be an overestimate or an underestimate? Each of the rectangles overshoots the function, so this will be an overestimate. What about using the leftmost value of $f(x)$ of each interval to determine the height of the rectangles? This is shown in figure 172.6.

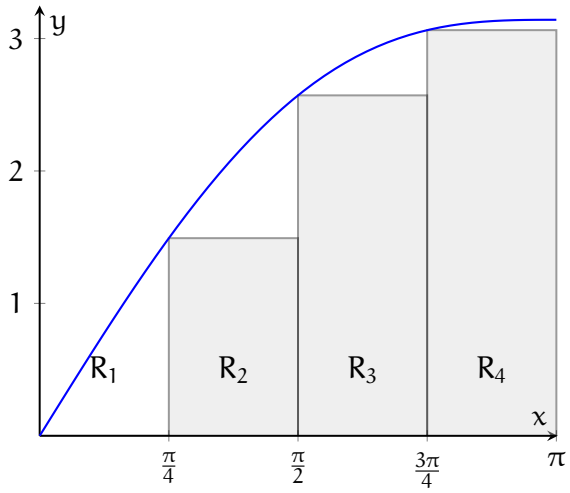


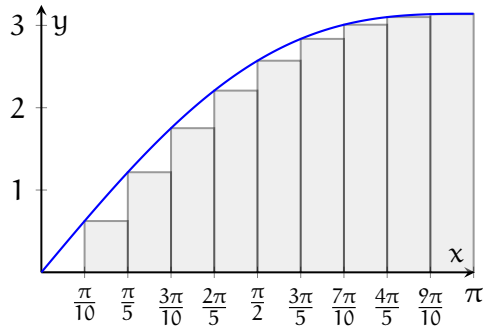

Figure 172.6: Four rectangle sections with heights determined by leftmost value of $f(x)$ on each interval

Notice that because $f(0) = 0$, the height of the first rectangle is zero, so we don't see it on the graph. To find the area of these rectangles:

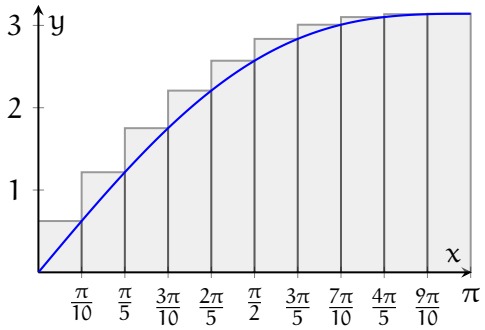
$$\frac{\pi}{4} \times f(0) + \frac{\pi}{4} \times f(\frac{\pi}{4}) + \frac{\pi}{4} \times f(\frac{\pi}{2}) + \frac{\pi}{4} \times f(\frac{3\pi}{4})$$

$$\approx \frac{\pi}{4} \times (0 + 1.4925 + 2.5708 + 3.0633) = 5.5972$$

This is an underestimate. Therefore, the true value of the area under $f(x) = \sin x + x$ is between 5.5972 and 8.0646. This is an awfully wide window! We can narrow our estimate by increasing the number of intervals. Graphs of $f(x)$ with 10 intervals are shown in figure 172.7.




(a) Left-endpoint Riemann sum




(b) Right-endpoint Riemann sum

Figure 172.7: Riemann sum approximations of $f(x) = \sin x + x$ on $[0, \pi]$ using 10 equal subintervals.

The total area for the left-determined rectangles is $\approx 6.4248$, and for the right-determined, it is $\approx 7.4118$. Therefore, we have narrowed the range for the true area under the curve to $6.4248 < A < 7.4118$. In general, as you increase the number of intervals, you get closer to the true area.

For a strictly increasing function, the right sum will be an overestimate and the left sum will be an underestimate of the true area under the curve. In the exercise below, you will examine a strictly decreasing function:

### Exercise 449

Estimate the area under the graph of $f(x) = \frac{1}{x}$ from $x = 1$ to $x = 2$ using four rectangles and right endpoints. Sketch the graph and the rectangles. Is your estimate an overestimate or an underestimate? Repeat using left endpoints.

*Working Space*

*Answer on Page 2063*

You should have found that for the strictly decreasing function $f(x) = \frac{1}{x}$, the right-

determined sum is an *underestimate*, while the left-determined sum is an *overestimate*.

## 172.3 The Riemann Sum

In the previous section, we estimated the area under functions by dividing the area into approximating rectangles. This method is called a *Riemann Sum*. We will use a general example to formally define the Riemann sum. Consider a generic function divided into strips of equal width (shown in figure 172.8). The width of each strip is

$$\Delta x = \frac{b - a}{n}$$

where $a$ is the left endpoint of the interval, $b$ is the right endpoint of the interval, and $n$ is the number of strips. So, the right endpoints of the sections are

$$x_1 = a + \Delta x$$

.

$$x_2 = a + 2\Delta x$$

$$\ldots \; x_n = a + n\Delta x$$

As above, we can use the value of the function to determine the height of a rectangle whose area approximates the area of the section. (E.g. for the $i^{th}$ strip, the width is $\Delta x$ and the height is $f(x_i)$, see figure 200.2). So, the total area approximated by the rectangles is

$$R_n = f(x_1)\Delta x + f(x_2)\Delta x + \ldots + f(x_n)\Delta x$$

This is the formal definition of the right Riemann sum. You can also take a left Riemann sum or a midpoint Riemann sum, as discussed below.

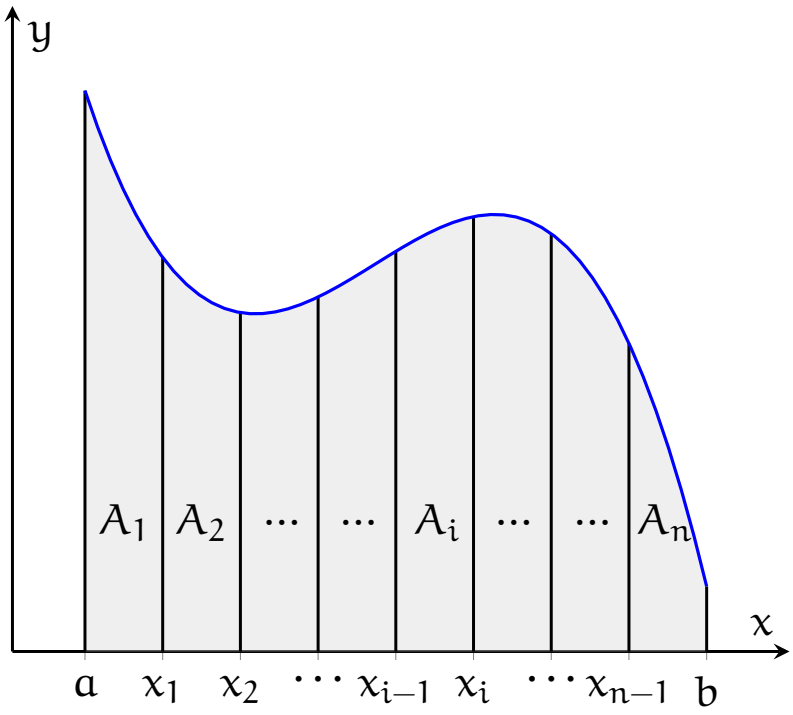



Figure 172.8: A representative function divided into $n$ strips of equal width

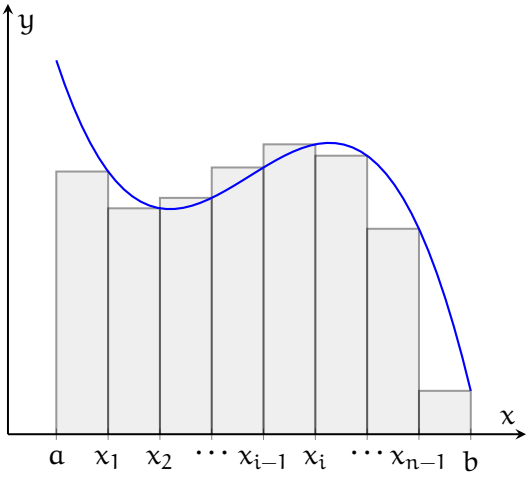



Figure 172.9: A representative function divided into $n$ rectangles of equal width, with rectangle height determined by the right endpoint of the subinterval

### 172.3.1 Right Riemann Sums

As seen above, a *right Riemann sum* uses the right-most value of $f(x)$ to determine the height of the rectangle (an example is shown in figure 172.10). We will refer to the right

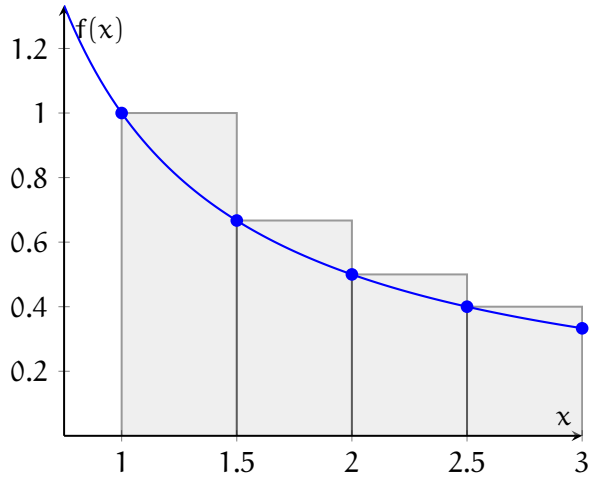



Figure 172.11: $L_4$ for $f(x) = \frac{1}{x}$

Riemann sum as $R_n$, where $n$ is the number of intervals.

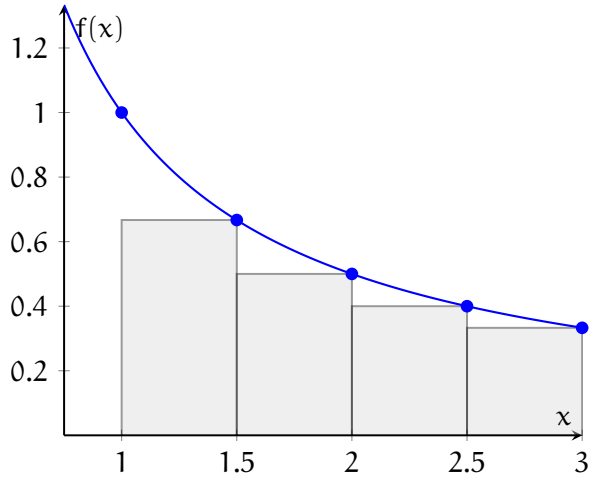



Figure 172.10: $R_4$ for $f(x) = \frac{1}{x}$

### 172.3.2 Left Riemann Sums

When taking a *left Riemann sum*, the height of the rectangle is determined by the value of the function at the lower (left-most) $x$-value. See figure 172.11. We will refer to left Riemann sums as $L_n$, where $n$ is the number of intervals. So, the total area approximated by a Left Riemann sum is

$$L_n = f(x_0)\Delta x + f(x_1)\Delta x + \ldots + f(x_{n-1})\Delta x$$

### 172.3.3 Midpoint Riemann Sums

A midpoint Riemann sum uses the value of $f(x)$ at the midpoint of the division to determine the height of the rectangle, as shown in figure 172.12. We will refer to the midpoint Riemann sum as $M_n$, where $n$ is the number of intervals. So, the total area approximated by the rectangles is

$$M_n = f(\frac{x_0 + x_1}{2})\Delta x + f(\frac{x_1 + x_2}{x})\Delta x + \ldots + f(\frac{x_{n-1} + x_n}{2})\Delta x$$

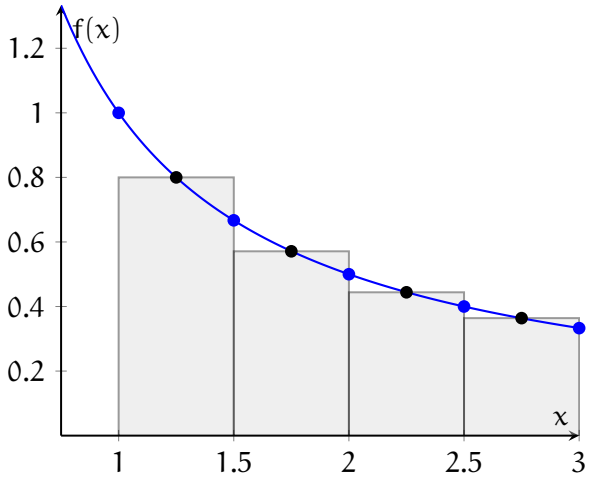



Figure 172.12: $M_4$ of $f(x) = \frac{1}{x}$

### 172.3.4 Trapezoidal Rule

We have established that the *Left Riemann Sums* are an *underestimate* and *Right Riemann Sums* are an *overestimate*, generally speaking for increasing functions. The *Midpoint Riemann Sum* generally gets more accurate, but still only uses one point of reference for the estimation. What if, instead, we used two? Specifically, we could use the two upper corners of the rectangle, which, if at different heights, form a trapezoid. Recall that the area of a trapezoid is $A = \frac{(a+b)}{2}h$. This allows us to use what is called the *Trapezoidal Rule*.

Let's look back at our $f(x) = \sin(x) + x$ example.

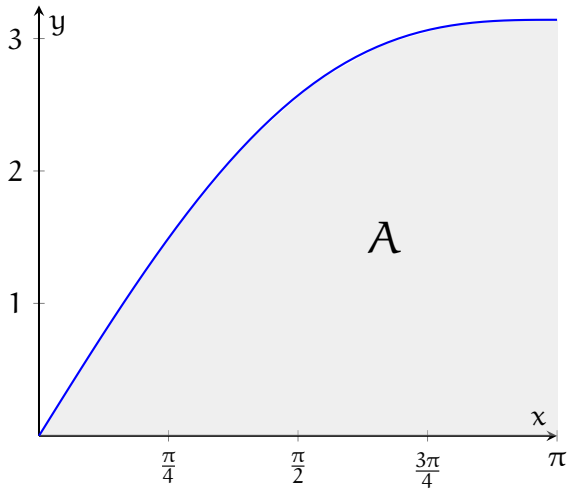



Figure 172.13: $f(x) = \sin x + x$

We have established that $f(x)$ is increasing on $[0, \pi]$. What we want to do is take the *average* of the left and right riemann sums. On each subinterval, the left Riemann sum uses a rectangle with height $f(x_i)$, while the right Riemann sum uses a rectangle with height $f(x_{i+1})$. Averaging these two rectangle areas gives

$$\frac{f(x_i) + f(x_{i+1})}{2}\Delta x$$

which is exactly the area of a trapezoid whose parallel sides have lengths $f(x_i)$ and $f(x_{i+1})$. Therefore, averaging the left and right sums is equivalent to approximating the region with trapezoids instead of rectangles.

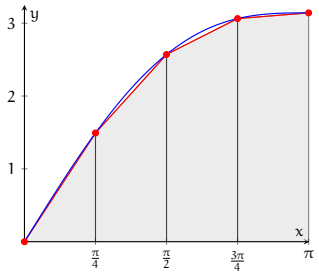
$$T_n = \frac{L_n + R_n}{2}$$

For our example, we found that

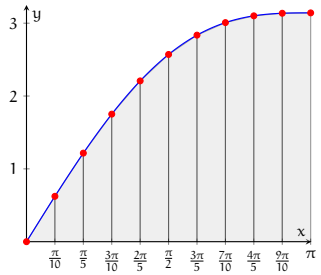
$$L_{10} \approx 6.4248, \qquad R_{10} \approx 7.4118$$

Finding our average gives us:

$$T_{10} = \frac{6.4248 + 7.4118}{2} = 6.9183$$




(a) $T_4$




(b) $T_{10}$

Figure 172.14: Trapezoidal approximations of $f(x) = \sin x + x$ on $[0, \pi]$. Increasing the number of subintervals improves the approximation.

Here we found that, gradually, the traprule estimates the area under under the curve to a closer accuracy.

### 172.3.5 Riemann sum sigma notation

As you may recall, mathematicians use sigma notation to concisely express sums, such as Riemann sums. We can rewrite the definition of a right Riemann sum in sigma notation:

$$\sum_{i=1}^{n} f(x_i)\Delta x$$

where $n$ is the number of subintervals. So, the actual area under the curve is the limit as $n$ approaches $\infty$ of the above sum. Let's apply this by writing a sum that represents the area, A, of the region that lies between the x-axis and the function $f(x) = e^{-x}$ from $x = 0$ to $x = 2$.

First, we find an expression for $\Delta x$:

$$\Delta x = \frac{2-0}{n} = \frac{2}{n}$$

Recall that $x_i = a + i\Delta x$. Since $a$ (the beginning of the interval) $= 0$, then the general expression for $x_i$ in this case is $0 + i \times \frac{2}{n} = \frac{2i}{n}$. Substituting our expressions for $\Delta x$ and $x_i$ into the sum formula, we see that:

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} e^{\frac{-2i}{n}} \frac{2}{n}$$

We can also interpret a sum as the area under a specific function. Take the expression:

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{\pi}{n} \sin \frac{i\pi}{n}$$

There are two expressions in the sum: $\frac{\pi}{n}$ and $\sin \frac{i\pi}{n}$. It makes sense that $\Delta x = \frac{\pi}{n}$ and $f(x_i) = \sin \frac{i\pi}{n}$. Because $\Delta x = \frac{b-a}{n} = \frac{\pi}{n}$, it follows that the interval of the area has a width of $\pi$. We will need to examine the other expression, $\sin \frac{i\pi}{n}$, to determine an exact window.

Since $f(x_i) = \sin \frac{i\pi}{n}$, it follows that the function we are looking for is a sine function. Further, the expression for $x_i = \frac{i\pi}{n}$. Recall that $x_i = a + i\Delta x$, where $a$ is the left-most boundary of the interval. Substituting what we have found already, we see that:

$$x_i = a + i\frac{\pi}{n} = \frac{i\pi}{n}$$

which implies that $a = 0$. Since we have established the interval is $\pi$ wide, we can infer that $b = \pi$. Therefore, the limit $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{\pi}{n} \sin \frac{i\pi}{n}$ is equal to the area under $f(x) = \sin x$ from $x = 0$ to $x = \pi$.

## Exercise 450

Use the formal definition of a Right Riemann sum to write a limit of a sum that is equal to the total area under the graph of $f$ on the specified interval. Do not evaluate the limit.

*Working Space*

1. $f(x) = \frac{2x}{x^2+1}$, $1 \le x \le 3$
2. $f(x) = x^2 + \sqrt{1+2x}$, $4 \le x \le 7$
3. $f(x) = \sqrt{\sin x}$, $0 \le x \le \pi$

*Answer on Page 2064*

| Event | Time (s) | Velocity (ft/s) |
|---|---|---|
| Launch | 0 | 0 |
| Begin roll maneuver | 10 | 185 |
| End roll maneuver | 15 | 319 |
| Throttle to 89% | 20 | 447 |
| Throttle to 67% | 32 | 742 |
| Throttle to 104% | 59 | 1325 |
| Maximum dynamic pressure | 62 | 1445 |
| Solid rocket booster separation | 125 | 4151 |

Figure 172.15: Speed of *Endeavour* from launch to booster separation

### *Exercise 451*

Use the formal definition of a Right Riemann sum to find a region on a graph whose are is equal to the given limit. Do not evaluate the limit.

1. $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{3}{n}\sqrt{1+\frac{3i}{n}}$
2. $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{\pi}{4n} \tan \frac{i\pi}{4n}$

*Working Space*

*Answer on Page 2064*

#### 172.3.6 Real-world Riemann Sums

Sometimes we are working from real data, and the intervals aren't evenly spaced. That's ok! We can still use Riemann sums to make an estimate. Consider the velocity data from the 1992 launch of the space shuttle *Endeavour*, shown in tabular form in figure 172.15:

We can use a Riemann sum to estimate how far the space shuttle traveled in the first 62 seconds of flight. First, let's visualize our data (see figure 172.16). There are 7 time intervals from the data, but we only need the first 6. We can find a reasonable range for the distance the space shuttle travels by finding the left and right Riemann sums. Remember: Because this data is strictly increasing, the left sum will be our lower bound and the right sum will be our upper bound.

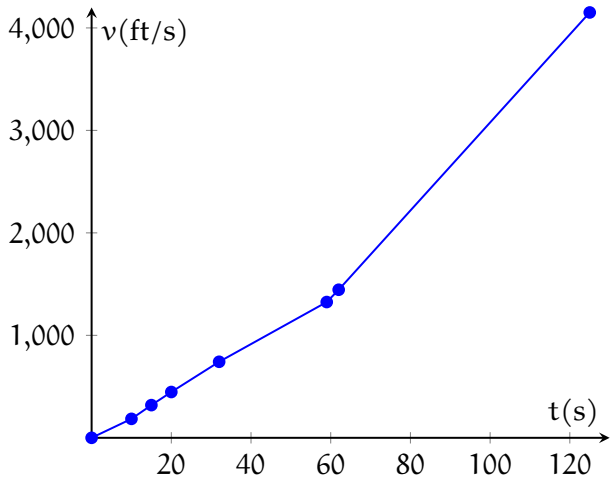



Figure 172.16: Plot of time, velocity data for the *Endeavour*

First, we'll find $L_6$. The width of the first interval is 10 seconds $(10-0=10)$ and the height of the rectangle will be $v(0)=0$. Calculations for the additional intervals are shown in the table:

| Interval | Width(s) | Height(ft/s) | Area(ft) |
|---|---|---|---|
| 1 | 10 | 0 | 0 |
| 2 | 5 | 185 | 925 |
| 3 | 5 | 319 | 1595 |
| 4 | 12 | 447 | 5364 |
| 5 | 27 | 742 | 20034 |
| 6 | 3 | 1325 | 3975 |

Adding the areas, we find the lower limit for the distance traveled is 31,893 feet. We can determine the upper bound, $R_6$, in a similar manner:

| Interval | Width(s) | Height(ft/s) | Area(ft) |
|---|---|---|---|
| 1 | 10 | 185 | 1850 |
| 2 | 5 | 319 | 1595 |
| 3 | 5 | 447 | 2235 |
| 4 | 12 | 742 | 8904 |
| 5 | 27 | 1325 | 35775 |
| 6 | 3 | 1445 | 4335 |

Adding the areas, we find the upper limit for the distance traveled is 54,694 feet. Therefore, the *Endeavour* traveled between 31,893 and 54,694 feet during the first 62 seconds of this flight.

## 172.4 Code for a Riemann Sum

You can create a program that automatically calculates a Riemann sum. Create a file called `riemann.py` and type the following into it:

```
import matplotlib.pyplot as plt
import sys
import math

from matplotlib.table import Rectangle

# Did the user supply two arguments?
if len(sys.argv) != 3:
    print(f"Usage: {sys.argv[0]} <stop> <divisions>")
    print(f"Numerically integrates 1/x from 1 to <stop>.")
    print(f"Calculates the value of 1/x at <divisions> spots in the range.")
    exit(1)

# Check to make sure the number of divisions is greater than zero?
divisions = int(sys.argv[2])
if divisions <= 0:
    print("ERROR: Divisions must be at least 1.")
    exit(1)

# Is the stopping point after 1.0?
stop = float(sys.argv[1])
if stop <= 1.0:
    print("ERROR: Stopping point must be greater than 1.0")
    exit(1)

start = 1.0
step_size = (stop - start)/divisions

print(f"Step size is {step_size:.5f}.")
x_values = []
y_values = []
sum = 0.0
for i in range(divisions):
    current_x = start + i * step_size
    current_y = 1.0/current_x
    area = current_y * step_size
    print(f"{i}: 1 / {current_x:.3f} = {current_y:4f}, area of rect = {area:8f}
    ↪ ")

    x_values.append(current_x)
    y_values.append(current_y)
    sum += area
    print(f"\tCumulative={sum:.3f},
    ↪ ln({current_x:.3f})={math.log(current_x):.3f}")

print(f"Numerical integration of 1/x from 1.0 to {stop:.4f} is {sum:.4f}")
```

```
print(f"The natural log of {stop:.4f} is {math.log(stop):.4f}")

# Create data for the smooth 1/x line
SMOOTH_DIVISIONS = 200
smooth_start = start - 0.15
smooth_stop = stop + 1.0
smooth_step = (smooth_stop - smooth_start)/SMOOTH_DIVISIONS
smooth_x_values = []
smooth_y_values = []
for i in range(SMOOTH_DIVISIONS):
    current_x = smooth_start + i * smooth_step
    current_y = 1.0/current_x
    smooth_x_values.append(current_x)
    smooth_y_values.append(current_y)

# Put it on a plot
fig, ax = plt.subplots()
ax.set_xlim((smooth_x_values[0], smooth_x_values[-1]))
ax.set_ylim((0, smooth_y_values[0]))
ax.set_title("Riemann Sums for 1/x")

# Make the Riemann rects
for i in range(divisions):
    current_x = x_values[i]
    next_x = current_x + step_size
    current_y = y_values[i]
    rect = Rectangle((current_x, 0), step_size, current_y, edgecolor="green",
    ↪  facecolor="lightgreen")
    ax.add_patch(rect)

# Make the true 1/x curve
ax.plot(smooth_x_values, smooth_y_values, c="k", label="1/x")

# Show the user
plt.show()

```

This program will calculate and display a graph of the left Riemann sum of $\frac{1}{x}$ from 1 to the provided stop value with the indicated number of subintervals. When you run it, you will see a graph in a new window and something like this in the terminal:

```
Step size is 0.40000.
0: 1 / 1.000 = 1.000000, area of rect = 0.400000
        Cumulative=0.400, ln(1.000)=0.000
1: 1 / 1.400 = 0.714286, area of rect = 0.285714
        Cumulative=0.686, ln(1.400)=0.336
2: 1 / 1.800 = 0.555556, area of rect = 0.222222
        Cumulative=0.908, ln(1.800)=0.588
3: 1 / 2.200 = 0.454545, area of rect = 0.181818
        Cumulative=1.090, ln(2.200)=0.788
```

```
4: 1 / 2.600 = 0.384615, area of rect = 0.153846
        Cumulative=1.244, ln(2.600)=0.956
5: 1 / 3.000 = 0.333333, area of rect = 0.133333
        Cumulative=1.377, ln(3.000)=1.099
6: 1 / 3.400 = 0.294118, area of rect = 0.117647
        Cumulative=1.495, ln(3.400)=1.224
7: 1 / 3.800 = 0.263158, area of rect = 0.105263
        Cumulative=1.600, ln(3.800)=1.335
8: 1 / 4.200 = 0.238095, area of rect = 0.095238
        Cumulative=1.695, ln(4.200)=1.435
9: 1 / 4.600 = 0.217391, area of rect = 0.086957
        Cumulative=1.782, ln(4.600)=1.526
Numerical integration of 1/x from 1.0 to 5.0000 is 1.7820
The natural log of 5.0000 is 1.6094
```

### *Exercise 452*

Use the Python program you created to find $L_{10}$, $L_{50}$, $L_{100}$, $L_{500}$, $L_{1000}$, and $L_{5000}$ for the function $\frac{1}{x}$ from $x = 1$ to $x = 5$. What do you notice about the results?

*Working Space*

*Answer on Page 2065*

## 172.5 Riemann Sum Practice

### Exercise 453

Working Space

| t (hours) | 4 | 7 | 12 | 15 |
|---|---|---|---|---|
| R(t) (L/hr) | 6.5 | 6.2 | 5.9 | 5.6 |

A tank contains 50 liters of water after 4 hours of filling. Water is being added to the tank at rate $R(t)$. The value of $R(t)$ at select times is shown in the table. Using a right Riemann sum, estimate the amount of water in the tank after 15 hours of filling.

Answer on Page 2065

### Exercise 454

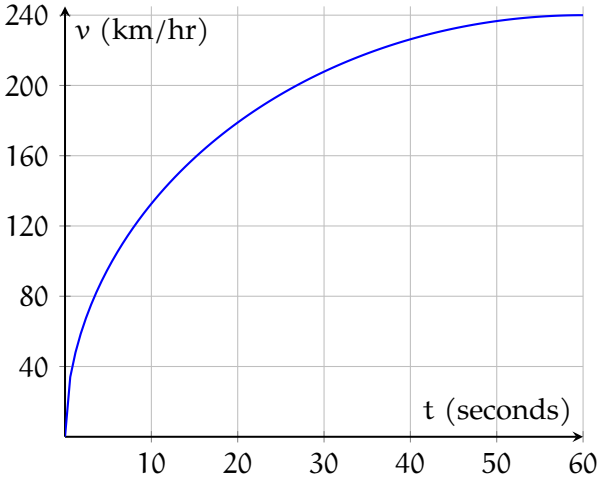
Working Space

Let $f(x) = x - 2\ln x$. Estimate the area under $f$ from $x = 1$ to $x = 5$ using four rectangles and the value of $f$ at the midpoint of each interval. Sketch the curve and your approximating rectangles.

Answer on Page 2065

## Exercise 455

A graph of a car's velocity over a period of 60 seconds is shown. Estimate the distance traveled during this period.




*Working Space*

*Answer on Page 2066*

# Chapter 173

# Stemming and Lemmatization

Stemming and lemmatization are two fundamental techniques in natural language processing that are used to prepare text data. They help in reducing inflectional forms of a word to a common base form.

### 173.0.1 Stemming

Stemming is the process of reducing a word to its word stem (its basic form). For example, a stemming algorithm would reduce the words ”jumping”, ”jumped”, and ”jumps” to the stem ”jump”.

It’s important to note that stemming may not always lead to actual words. For example, the stem of the word ”running” could be ”run” depending on the stemming algorithm used.

Stemming is generally simpler and faster than lemmatization, but it is also less precise.

### 173.0.2 Lemmatization

Lemmatization, on the other hand, reduces words to their base or root form, which is linguistically correct. For example, ”running” and ”runs” are both changed to ”run”.

Lemmatization uses a more complex approach to achieve this. It considers the morphological analysis of the words and requires detailed dictionaries, which the algorithm can look through to link the form back to its lemma.

### 173.0.3 Applications

To summarize, both stemming and lemmatization help in text normalization and preprocessing, but while stemming can be faster and simpler, lemmatization is more accurate, as it uses more informed analysis to create groups of words with similar meanings based on the context. These are both used to improve natural language processing (NLP) in software such as search engines, machine learning models, algorithmic searches, and even baseline programs for autofill and text prediction.

Chapter 174

# Definite Integrals

Integrals are a fundamental concept in calculus. They are used to calculate areas, volumes, and many other things. A definite integral calculates the net area between the function and the x-axis over a given interval.

Recall that you can use a Riemann sum to estimate the area under a function, and that as we increase the number of subintervals, the estimated area approaches the actual area. In sigma notation we can express a Riemann sum as

$$\sum_{i=1}^{n} f(x_i)\Delta x \tag{174.1}$$

## 174.1 Definition

The definite integral of a function $f(x)$ over an interval $[a, b]$ is defined as the limit of a Riemann sum as $n$ approaches $\infty$:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x \tag{174.2}$$

where $x_i^*$ is a sample point in the $i^{th}$ subinterval of a partition of $[a, b]$, $\Delta x = \frac{b-a}{n}$ is the width of each subinterval, and the limit is taken as the number of subintervals $n$ approaches infinity.

### *Exercise 456*

Express

$$\lim_{n\to\infty} \sum_{i=1}^{n} (x_i^3 + x_i \sin x_i)\Delta x$$

as an integral on the interval $[0, \pi]$. Do not solve.

*Working Space*

*Answer on Page 2067*

## 174.2 Positive and Negative Areas

What if the function dips below the x-axis? We consider that area negative. In other words, it represents a *decrease* as opposed to an increase. Consider an oscillating object where $v(t) = \sin \pi x$ (figure 174.1). From $t = 0$ to $t = 1$, the velocity is positive, which means the object is moving *away from* the starting position. This is a positive displacement. From $t = 1$ to $t = 2$, the velocity is negative. What does this tell you about the direction the object is moving and its displacement during this time period? A negative velocity means the object is moving *back towards* the starting position.

In general, areas above the x-axis are positive, while areas below the x-axis are negative.

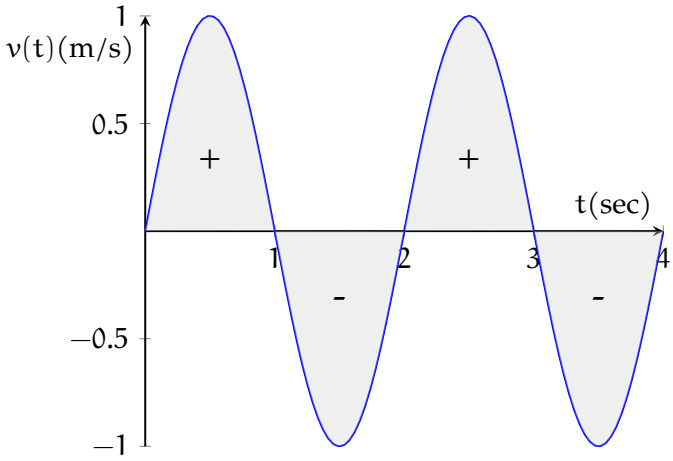



Figure 174.1: velocity of an oscillating object

## 174.3 Properties of Integrals

There are several important properties of integrals that will help us evaluate more complex integrals in the future. The following examples apply when $f(x)$ is continuous or has a finite number of jump discontinuities on the interval $a \leq x \leq b$.

### 174.3.1 What happens when $a = b$?

What if the endpoints of the integral are the same? Let's consider $\int_a^b x^2\,dx$, and take the limit as $b \to a$ (shown in figure 174.2). As you can see, as $b$ approaches $a$, the calculated area decreases. Intuitively, we can guess that when $b = a$, then the width of the area $(\Delta x)$ is zero, and therefore the area is also zero. Let's prove this formally.

Recall that $\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$, where $\Delta x = \frac{b-a}{n}$ and $x_i = a + i\Delta x$. To evaluate the integral when $b = a$, we will take the limit of the limit:

$$\lim_{b\to a} \lim_{n\to\infty} f(x_i)\frac{b-a}{n}$$

This can be rewritten as

$$\lim_{b\to a}(b-a) \lim_{n\to\infty} \frac{f(x_i)}{n}$$

We know that $\lim_{b\to a}(b-a) = (a-a) = 0$, and therefore

$$\int_a^a f(x)\,dx = 0 \cdot \lim_{n\to\infty} \frac{f(x_i)}{n} = 0$$

This is true for any function.

Note that when $a = b$, the integral just becomes a line. From geometry we also know that a line has no area, as the width or $dx$ is 0.

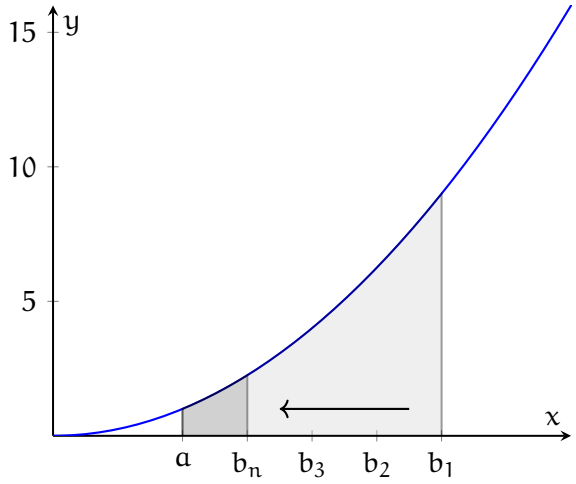



Figure 174.2: As b gets closer to a, the area represented by the integral decreases

### 174.3.2 The integral of a constant

When the function we are integrating is a constant (that is, it takes the form $f(x) = C$), the area is simply $(b - a) \cdot C$. This is shown graphically in figure 174.3.

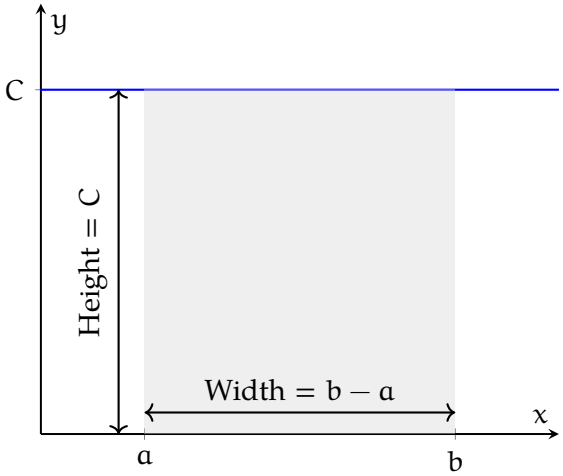



Figure 174.3: $\int_a^b f(x)\,dx = (b - a) \cdot C$

Since $f(x) = C$ is a horizontal line, the area under $f(x)$ is simply a rectangle. As you can see in figure 174.3, the width of the rectangle is $b - a$ and the height is C. To find the area of a rectangle, we multiply the width by the height, and therefore $\int_a^b C\,dx = (b - a) \cdot C$.

### 174.3.3 The integral of a function multiplied by a constant

How is $\int_a^b f(x)\,dx$ related to $\int_a^b C\cdot f(x)\,dx$? Intuitively, we know that multiplying a function by a constant, C, vertically stretches the graph by a factor of C. In turn, the area under the curve increases by a factor of C. Imagine a simple shape, like a triangle. If we keep the base of the triangle the same (analogous to the integral being over the same interval) and make the triangle three times taller (analogous to multiplying the function we're integrating by a factor of $C = 3$), then we would expect the total area of the triangle to be 3 times greater. Therefore, $\int_a^b C \cdot f(x)\,dx = C\int_a^b f(x)\,dx$.

### 174.3.4 Integrals of sums and differences of functions

If a function can be described as a sum of two other functions, then the integral of the original function is the same as the sum of the integrals of the two other functions. Concretely, we say $\int_a^b f(x)+g(x)\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$. Figure 174.4 shows $f(x) = x+2$, $g(x) = 4x^3 - 12x^2 + 10x$, and $f(x) + g(x)$. As you can see, the area under $f(x) + g(x)$ is equal to the area under $f(x)$ (the red area) plus the area under $g(x)$ (the diagonal lined area).

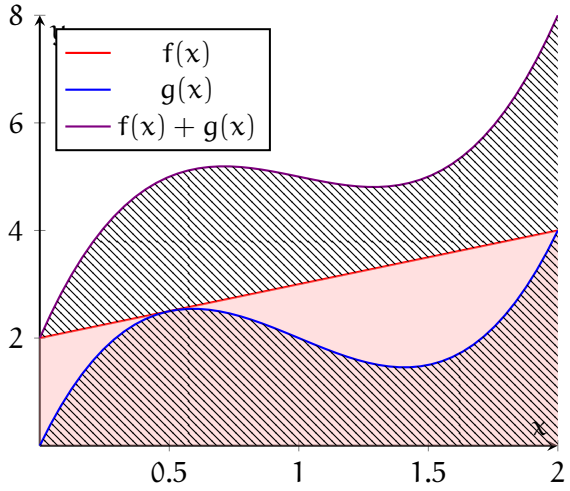



Figure 174.4: The integral of $f(x)+g(x)$ is equal to the integral of $f(x)$ plus the integral of $g(x)$

Mathematically, we can prove this by recalling that the limit of a sum is the sum of the limits:

$$\int_a^b f(x)+g(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n}[f(x_i)+g(x_i)]\Delta x$$

$$= \lim_{n\to\infty} [\sum_{i=1}^{n} f(x_i)\Delta x + \sum_{i=1}^{n} g(x_i)\Delta x]$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x + \lim_{n\to\infty} \sum_{i=1}^{n} g(x_i)\Delta x$$

$$= \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

Similar to the addition property, the integral of the difference between two function is equal to the difference of the integrals of two functions.

$$\int_a^b [f(x) - g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

. This is more difficult to visualize than addition, but we can easily prove it by applying the constant multiple and addition properties. Let's define $f(x) - g(x) = f(x) + (-g(x))$:

$$\int_a^b f(x) - g(x)\,dx = \int_a^b f(x) + (-g(x))\,dx$$

By the addition property,

$$= \int_a^b f(x)\,dx + \int_a^b -g(x)\,dx = \int_a^b f(x)\,dx + \int_a^b (-1)\cdot g(x)\,dx$$

By the constant multiple property:

$$= \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

### 174.3.5 Integrals of adjacent areas

If $c$ is some $x$-value between $a$ and $b$, then $\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$. This is shown graphically in figure 174.5. The total area from $x = a$ to $x = b$ is equal to the red area (the integral from $a$ to $c$) plus the blue area (the integral from $c$ to $b$).

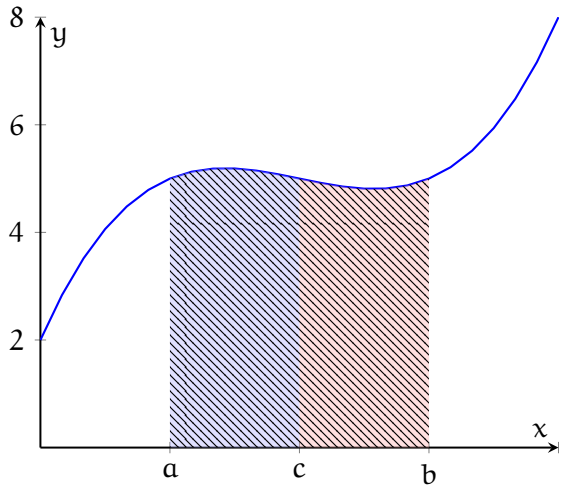



Figure 174.5: $\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$

### 174.3.6 Estimating the value of an integral

Suppose we need to know the area under a complex function. We can estimate a range for the value of the integral if we can bookend the function over the interval we are interested. Suppose there is some value $m$ such that $f(x) \geq m$, and some other value $M$ such that $f(x) \leq M$ on the interval we are interested in (see figure 174.6). The total area under $f(x)$ is the light blue plus the darker blue. The total area under $y = M$ is the darker blue, plus the light blue, plus the white area. The darker blue area under the curve has total area $m \cdot (b - a)$ and the rectangle under $y = M$ has total area $M \cdot (b - a)$ (since these are both integrals of a constant, which we learned about above). The actual area under our function is more than just the dark blue area, but less than the total area under $y = M$. Therefore, $m(b - a) \leq \int_a^b f(x)\,dx \leq M(b - a)$ if $m \leq f(x)$ and $M \geq f(x)$ on the interval $x \in [a, b]$.

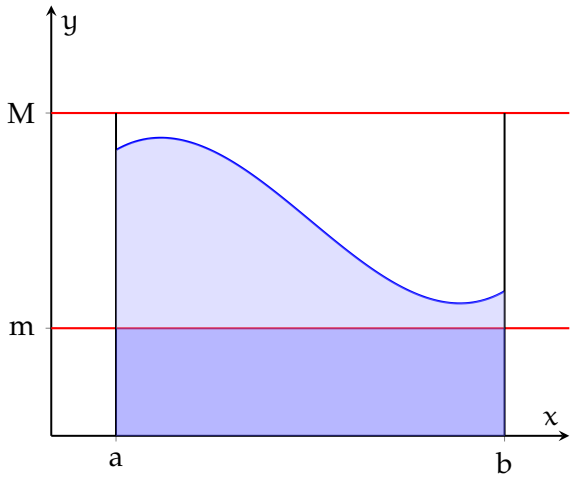



Figure 174.6: $m \leq f(x) \leq M$

### 174.3.7 Other Properties of Integrals

If $f(x) \geq 0$ over the for $a \leq x \leq b$, then $\int_a^b f(x)\,dx > 0$. We can make an intuitive, geometric argument to support this claim. Recall that areas above the x-axis are considered positive. If $f(x) \geq 0$, then all the area of the integral lies above the x-axisj therefore, the total area must be positive.

Similarly, if $f(x) \geq g(x)$ on the interval $a \leq x \leq b$, then $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$ (see figure 174.7). The entire area under $g(x)$ is contained in the area under $f(x)$. Therefore, $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$

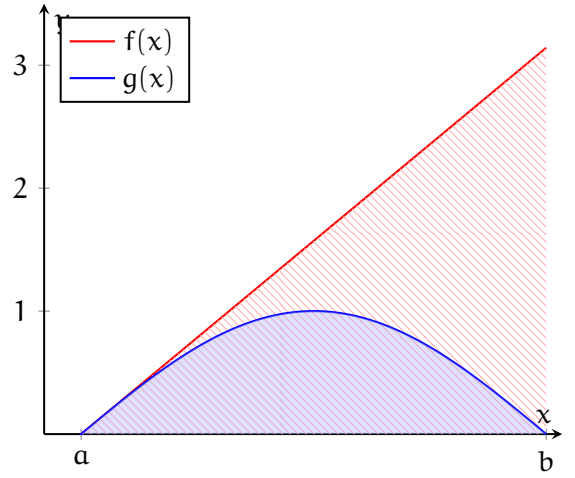



Figure 174.7: $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$

Lastly, we see what happens when we switch $a$ and $b$. While it is unusual to integrate from right to left (that is, in a case where $a > b$), this property will be useful. Recall that

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\frac{(b-a)}{n}$$

What is $\int_b^a f(x)\,dx$? Substituting, we see that

$$\int_b^a f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\frac{(a-b)}{n}$$

Noting that $(a-b) = -(b-a)$ we see:

$$\begin{aligned}\int_b^a f(x)\,dx &= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\frac{-(b-a)}{n} \\ &= (-1)\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\frac{(b-a)}{n} \\ &= (-1)\int_a^b f(x)\,dx\end{aligned}$$

Therefore, it is true that $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$.

## 174.4 Applications in Physics

We have already seen that the area under a velocity function is displacement, and the area under an acceleration function is change in velocity (Riemann Sums). We can use integrals to determine the change in position of an object over a given time frame. If we *also* know the object's starting position, then we can state the object's ending position. Consider the graph of an object's velocity in figure 174.8:

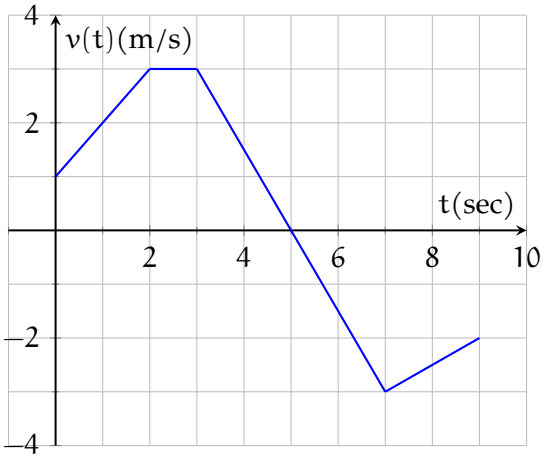



Figure 174.8: Velocity of an object from $t = 0$ to $t = 9$

We can determine the net displacement of the object from $t = 0$ to $t = 9$ by evaluating $\int_0^9 v(t)\,dt$. Since the definite integral is equal to the area under the curve, we need to find the total area. As the function consists of straight lines, we will leave the explicit calculation of the area as an exercise for the student. You should find that the total positive area (above the x-axis) is 10 meters, and the total negative area (below the x-axis) is 8 meters. Therefore, the object's displacement over the specified time interval is $10 - 8 = 2$ meters.

When you push on something to move it, you are applying a force over a distance (assuming you are strong enough to move it!). The integral of force as a function of distance is the *work* done on that object. Work is the change in kinetic energy (KE) of an object. Mathematically, this is

$$\int_a^b F(x)\,dx = \Delta KE = \frac{1}{2}m(v_f^2 - v_i^2)$$

.

If you integrate the force as a function of time, that is *impulse*. Impulse is the change in momentum (p) of the object. Mathematically, this is

$$\int_a^b F(t)\,dt = \Delta p = m(v_f - v_i)$$

.

Example problem: You push a 3 kg box with force $F(x) = 0.5x$, where $x$ is measured in meters and $F$ is measured in Newtons. If the box was initially at rest, what is its speed when it reaches the 2 meter mark? (Hint: $KE = \frac{1}{2}mv^2$.)

Solution: Change in kinetic energy is the area under a force-distance curve. We can plot the force applied to the box from d=0 to d=2 (see figure 174.9):

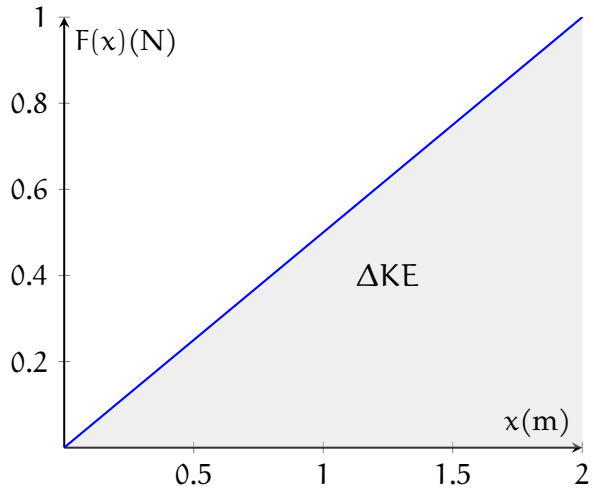



Figure 174.9: Force applied to a box over a distance; the shaded area represents the change in kinetic energy.

Given that the box's initial velocity is $0\frac{m}{s}$, we know that the initial kinetic energy (KE) is 0J. This implies that $KE_f = \Delta KE$. We can find $\Delta KE$ from the shaded area:

$$\Delta KE = \frac{1}{2}(2m)(1N) = 1J = KE_f$$

Solving for the final velocity:

$$KE_f = 1J = \frac{1}{2}(3kg)(v^2)$$

$$2J = (3kg)(v^2)$$

$$\frac{2}{3}\frac{m^2}{s^2} = v^2$$

$$v = \sqrt{\frac{2}{3}\frac{m^2}{s^2}} \approx 0.816\frac{m}{s}$$

## 174.5 Practice Exercises

### Exercise 457

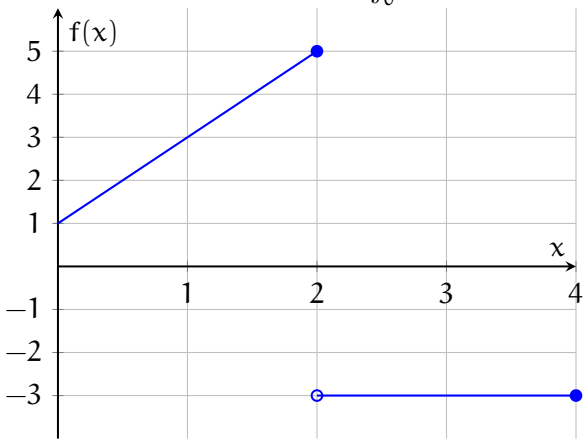
Given that $\int_0^1 x^2\,dx = \frac{1}{3}$, use the properties of integrals to evaluate $\int_0^1 (5-6x^2)\,dx$.

*Working Space*

*Answer on Page 2067*

### Exercise 458

[This question was originally presented as a multiple-choice problem on the 2012 AP Calculus BC exam.] The graph of f is shown. What is the value of $\int_0^4 f(x)\,dx$?

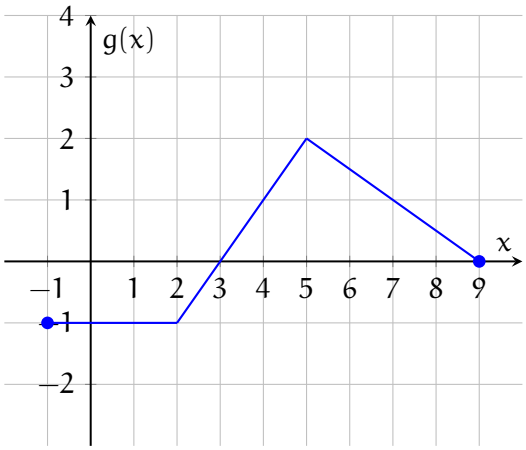



*Working Space*

*Answer on Page 2067*

### Exercise 459

[This question was originally presented as a multiple-choice problem on the 2012 AP Calculus BC exam.] The graph of the piecewise function $g(x)$ is shown. What is the value of $\int_{-1}^{9} 3g(x) + 2\,dx$?




*Working Space*

*Answer on Page 2068*

## Exercise 460

[This question was originally presented as a calculator-allowed, multiple- choice question on the 2012 AP Calculus BC exam.] If $f'(x) > 0$ for all real numbers and $\int_4^7 f(x)\,dx = 0$, which of the following could be a table of values for the function f?

(A)

| x | f(x) |
|---|---|
| 4 | -4 |
| 5 | -3 |
| 7 | 0 |

(B)

| x | f(x) |
|---|---|
| 4 | -4 |
| 5 | -2 |
| 7 | 5 |

(C)

| x | f(x) |
|---|---|
| 4 | -4 |
| 5 | 6 |
| 7 | 3 |

(D)

| x | f(x) |
|---|---|
| 4 | 0 |
| 5 | 0 |
| 7 | 0 |

(E)

| x | f(x) |
|---|---|
| 4 | 0 |
| 5 | 4 |
| 7 | 6 |

*Working Space*

*Answer on Page 2068*

Chapter 175

# The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus (FTC) is a theorem that connects the concept of differentiating a function with the concept of integrating a function. This theorem is divided into two parts.

## 175.1 First Part

The first part of the Fundamental Theorem of Calculus states that if $f$ is a continuous real-valued function defined on a closed interval $[a, b]$ and $F$ is the function defined, for all $x$ in $[a, b]$, by:

$$F(x) = \int_a^x f(t)\,dt \tag{175.1}$$

Then, $F$ is uniformly continuous and differentiable on the open interval $(a, b)$, and $F'(x) = f(x)$ for all $x$ in $(a, b)$. (That is $F(x)$ is the antiderivative of $f(x)$.)

## 175.2 Second Part

The second part of the Fundamental Theorem of Calculus states that if $f$ is a real-valued function defined on a closed interval $[a, b]$ that admits an antiderivative $F$ on $[a, b]$, and $f$ is integrable on $[a, b]$ (it need not be continuous), then

$$\int_a^b f(t)\,dt = F(b) - F(a). \tag{175.2}$$

We will also use shorthand as follows:

$$\int_a^b f(t)\,dt = F(t)|_a^b \tag{175.3}$$

Which means "F(t) evaluated from $t = a$ to $t = b$".

## 175.3 FTC and Definite Integrals

Let $f$ be a function that is continuous on the interval $x \in [a, b]$ and $g(x)$ is given by:

$$g(x) = \int_a^x f(t)\,dt$$

So, $g$ is continuous on $[a, b]$ and differentiable on $(a, b)$. Additionally,

$$g'(x) = f(x)$$

**Proof**: Let $x$ and $x + h$ be in $(a, b)$. Then,

$$g(x + h) - g(x) = \int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt$$

Recall from the chapter on definite integrals that we can split the first integral, rewriting it as:

$$g(x + h) - g(x) = \left[\int_a^x f(t)\,dt + \int_x^{x+h} f(t)\,dt\right] - \int_a^x f(t)\,dt$$

$$g(x + h) - g(x) = \int_x^{x+h} f(t)\,dt$$

And for $h \neq 0$:

$$\frac{g(x + h) - g(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt$$

Since $f$ is continuous, there is some $u$ in $(a, b)$, such that $f(u) = m$, where $m$ is the minimum value of $f$ on the interval $(a, b)$. Similarly, there is also some $v$, such that $f(v) = M$, where $M$ is the maximum value (see figure ??). We can then state the true inequality that:

$$mh \leq \int_x^{x+h} f(t)\,dt \leq Mh$$

Therefore, (assuming $h > 0$):

$$f(u) \leq \frac{1}{h}\int_x^{x+h} f(t)\,dt \leq f(v)$$

Substituting the equation above for the integral, we see that:

$$f(u) \leq \frac{g(x+h) - g(x)}{h} \leq f(v)$$

If we let $h$ approach zero, then the window that $u$ and $v$ are in collapses and $u$ and $v$ both approach $x$. Therefore,

$$\lim_{h \to 0} f(u) = \lim_{u \to x} f(u) = f(x)$$

Recall also that

$$\lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = g'(x)$$

Taking the limit as $h \to 0$ of the whole, inequality becomes the Squeeze Theorem:

$$\lim_{h \to 0} f(u) \leq \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} \leq \lim_{h \to 0} f(v)$$

$$f(x) \leq g'(x) \leq f(x)$$

Therefore, if $g(x) = \int_a^x f(t)\,dt$, then $g'(x) = f(x)$. Notice it doesn't matter what $a$ is!

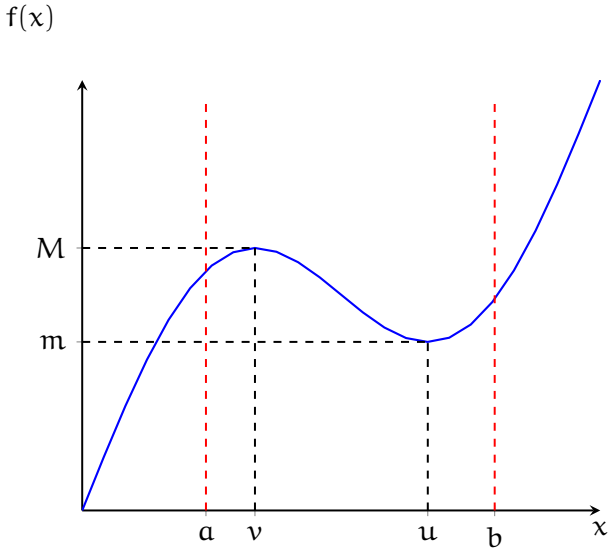



Figure 175.1: $f(v) = M$, the maximum value, and $f(u) = m$, the minimum value on the interval $x \in [a, b]$

### *Exercise 461*

[This question was originally presented as a no-calculator, multiple-choice problem on the 2012 AP Calculus BC Exam.] Let $g$ be a continuously differentiable function with $g(1) = 6$ and $g'(1) = 3$. What is the value of $\lim_{x\to 1} \frac{\int_1^x g(t)\,dt}{g(x)-6}$?
(A) 0
(B) $\frac{1}{2}$
(C) 1
(D) 2
(E) The limit does not exist

*Working Space*

*Answer on Page 2068*

## 175.4 The Meaning of the FTC

What the Fundamental Theorem of Calculus is really saying is that differentiation and integration are opposite processes. Mathematically, we can say

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

This may seem clunky, but many useful functions are defined this way. Consider the Fresnel function: $S(x) = \int_0^x \sin\frac{\pi t^2}{2}\,dt$. Originally used in optics, this equation is also used by civil engineers to design road and railway curves. According to FTC, then, $S'(x) = \sin\frac{\pi t^2}{2}$.

We can also apply the Chain Rule when taking derivatives of integrals. Let $f(x) =$

$\int_1^{x^4} \sec t \, dt$. What is $f'(x)$? First, let us define $u = x^4$. By the Chain Rule,

$$\frac{d}{dx}\int_0^{x^4} \sec t \, dt = \frac{d}{dx}\int_0^{u} \sec t \, dt$$

$$= \frac{d}{du}\left[\int_0^{u} \sec t \, dt\right]\frac{du}{dx}$$

$$= \sec u \frac{du}{dx}$$

Noting that $\frac{du}{dx} = \frac{d}{dx}x^4 = 4x^3$,

$$f'(x) = \sec x^4 (4x^3)$$

### 175.4.1 FTC Practice

***Exercise 462***

Use the Fundamental Theorem of Calculus to find the derivative of the function.

*Working Space*

1. $g(x) = \int_0^x \sqrt{t + t^3} \, dt$
2. $F(x) = \int_x^0 \sqrt{1 + \sec t} \, dt$
3. $h(x) = \int_1^{e^x} \ln t \, dt$
4. $y = \int_{\sqrt{x}}^{\frac{\pi}{4}} \theta \tan \theta \, d\theta$

*Answer on Page 2069*

## 175.5 Using Antiderivatives to Evaluate Definite Integrals

In everyday English, the FTC states that the integral from $a$ to $b$ of a function is the antiderivative of that function evaluated from $a$ to $b$. In the previous chapter, the integrals presented were of linear functions where the area under the curve could be equally calculated by hand. The FTC connects integrals to antiderivatives, allowing us to evaluate more complex integrals. Consider the following example:

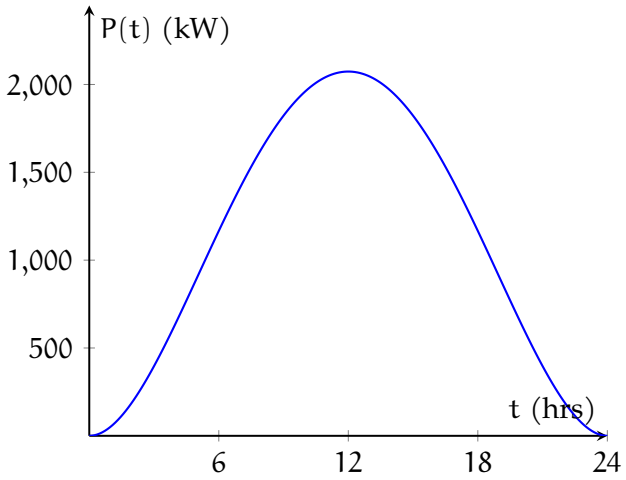



Figure 175.2: Power consumption of a household in a day

The power consumption of a household can be modeled as $P(t) = \frac{1}{10}t^2(t-24)^2$ from $t = 0$ to $t = 24$, where $P$ is measured in watts and $t$ is measured in hours ($t = 0$ is midnight). The total energy the household uses is given by $\int_0^{24} P(t)\,dt$. As you can see from the graph (see figure 175.2), we cannot simply use our geometry skills to determine the area under the curve.

To determine the total energy use, we need to evaluate $\int_0^{24} \frac{1}{10}t^2(t-24)^2\,dt$. First, we expand the polynomial:

$$E_{tot} = \frac{1}{10}\int_0^{24} t^2(t^2 - 48t + 576)\,dt = \frac{1}{10}\int_0^{24} t^4 - 48t^3 + 576t^2\,dt$$
$$= \frac{1}{10}\int_0^{24} t^4\,dt - \frac{24}{5}\int_0^{24} t^3\,dt + \frac{288}{5}\int_0^{24} t^2\,dt$$

Using the power rule to determine the antiderivatives of $t^4$, $t^3$, and $t^2$, we see:

$$= \frac{1}{10}\left[\frac{1}{5}t^5\right]_0^{24} - \frac{24}{5}\left[\frac{1}{4}t^4\right]_0^{24} + \frac{288}{5}\left[\frac{1}{3}t^3\right]_0^{24} = 26542.1Whr = 26.5421kWhr$$

### 175.5.1 Definite Integrals Practice

### Exercise 463

Evaluate the following integral:

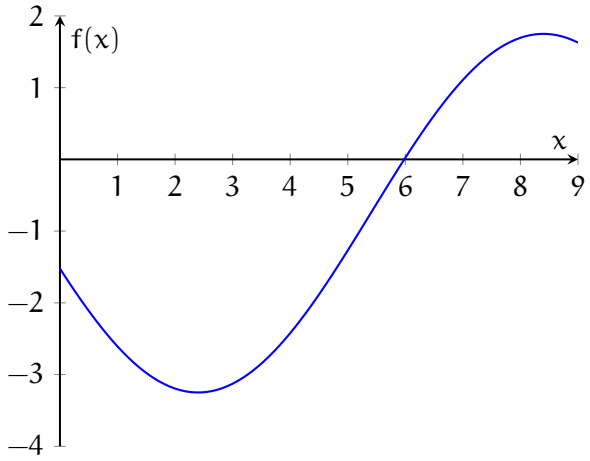
1. $\int_1^4 t^{-3/2}\,dt$

*Working Space*

*Answer on Page 2069*

### Exercise 464

[This question was originally presented as a multiple-choice, no-calculator problem on the 2012 Calculus BC exam.] The graph of a differentiable function $f$ is shown in the graph. $h(x) = \int_0^x f(t)\,dt$. Rank the relative values of $h(6)$, $h'(6)$, and $h''(6)$ from lowest to highest.

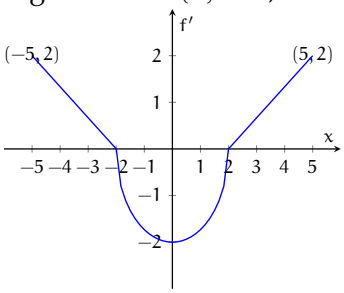



*Working Space*

*Answer on Page 2069*

### Exercise 465

The graph of $f'$ is shown in the graph and consists of a semi-circle and two line segments. If $f(2) = 1$, then what is $f(-5)$?




Working Space

Answer on Page 2069

### Exercise 466

[This question was originally presented as a calculator-allowed, multiple- choice question on the 2012 AP Calculus BC exam.] A particle moves along a line so that its acceleration for $t \geq 0$ is given by $a(t) = \frac{t+3}{\sqrt{t^3+1}}$. If the particle's velocity at $t = 0$ is 5, what is the velocity of the particle at $t = 3$?

Working Space

Answer on Page 2069

## 175.6 Average Value of a Function

The average value of a function, $f$, on an interval $[a, b]$ is given by:

**Average Value of a Function**

$$f_{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

**Example**: Find the average value of $f(x) = 3 + x^2$ on the interval $[-2, 1]$.

**Solution**: Taking $a = -2$ and $b = 1$, we have:

$$f_{avg} = \frac{1}{1-(-2)}\int_{-2}^{1}\left[3+x^2\right]\,dx$$

$$f_{avg} = \frac{1}{3}\int_{-2}^{1}\left[3+x^2\right]\,dx$$

$$f_{avg} = \frac{1}{3}\left[3x + \frac{1}{3}x^3\right]_{x=-2}^{x=1}$$

$$f_{avg} = \frac{1}{3}\left[\left(3\cdot 1 + \frac{1}{3}\cdot 1^3\right) - \left(3\cdot(-2) + \frac{1}{3}\cdot(-2)^3\right)\right]$$

$$f_{avg} = \frac{1}{3}\left[\left(3+\frac{1}{3}\right) - \left(-6-\frac{8}{3}\right)\right]$$

$$f_{avg} = \frac{1}{3}\left[\frac{10}{3}+6+\frac{8}{3}\right] = \frac{1}{3}(12) = 4$$

Therefore, the average value of $f(x) = 3 + x^2$ on the interval $[-2, 1]$ is 4.

## *Exercise 467*

[This question was originally presented as a multiple-choice, calculator- allowed problem on the 2012 AP Calculus BC Exam.] What is the average value of $y = \sqrt{\cos x}$ on the interval $0 \le x \le \frac{\pi}{2}$?

*Working Space*

*Answer on Page 2070*

Chapter 176

# Arc Lengths

## 176.1 Determining the Arc Length of a Curve

Another application of integrals is finding the length of a curve. In real life, we could do this by laying a piece of string up against the curve, then straightening out the string and measuring its length with a ruler (you may have done this in elementary school when you were first learning about the relationship between the radius and circumference of a circle).

Archimedes estimated the circumference of a circle by inscribing a circle with polygons of increasing numbers of sides. (Archimedes' proof that $\pi$ is between $3\frac{10}{71}$ and $3\frac{1}{7}$ is more complicated, but we will not dive into that here.) As we increase the number of sides of the inscribed polygon, the perimeter of the polygon (the sum of the lengths of the sides) gets closer to the circumference of the circle (see figure 176.1).

Now, it's easy to find the length of a polygon: just add up the length of the line segments! Using this, we can find the length of a curve by approximating it as many short lines and adding up the lengths of those lines.

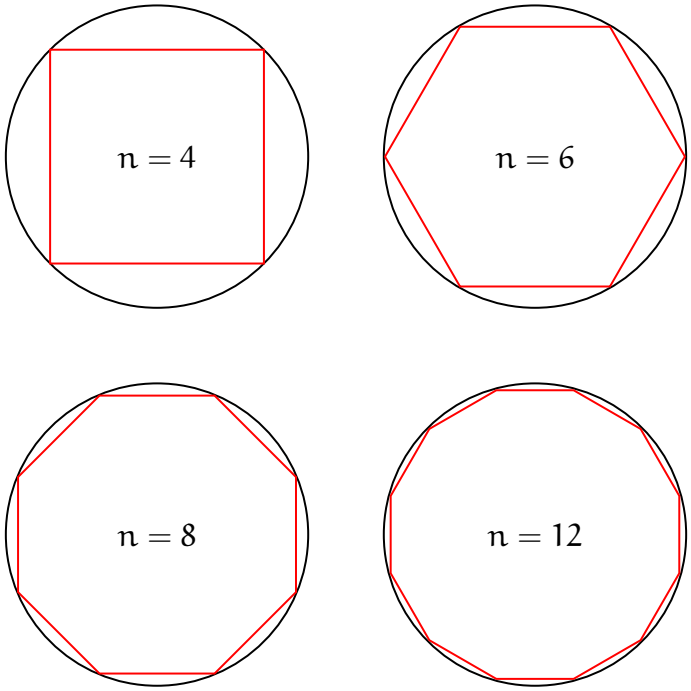



Figure 176.1: As $n$ increases, the perimeter of the inscribed polygon approaches the circumference of the circle

We can choose $n$ points along the graph of $f(x)$ and connect each point with a straight line (this is shown in figure 176.2). If we add up the length of the lines, we get an estimate of the length of the curve. We represent the length of the line between the $i^{th}$ point, $P_i$ and the previous point, $P_{i-1}$ as $|P_{i-1}P_i|$ (recall that the absolute value sign can be used to signify the length or **magnitude** of something). Therefore, the sum of the lengths of the lines approximating the curve is:

$$\sum_{i=1}^{n} |P_{i-1}P_i|$$

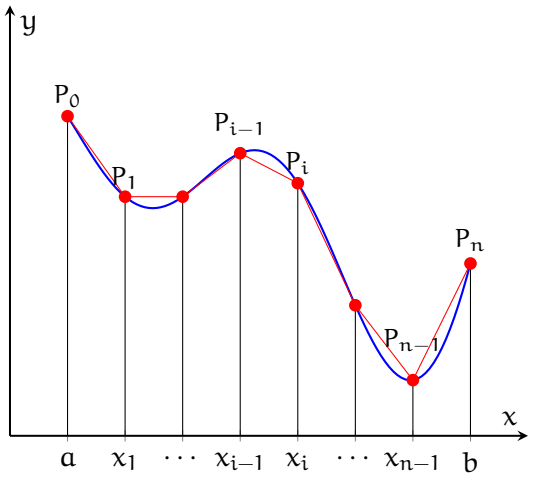



Figure 176.2: Polygon approximation of $f(x)$

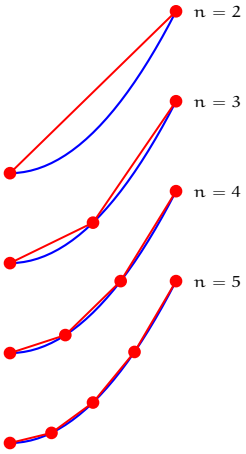



Figure 176.3: As the number of points increases, the total length of the lines segments approaches the true length of the curve

The more points we choose, the closer the lines lay to the actual curve (see figure 176.3), and the closer our estimate is to the true length. So, to find the true length, we will want to take $n$ to $\infty$. Therefore, the actual curve length is the limit as $n \to \infty$ of that sum:

$$L = \lim_{n\to\infty} \sum_{i=1}^{n} |P_{i-1}P_i|$$

The length of each segment can be found using the Pythagorean theorem. Recall that the distance between two points on the $xy$-plane is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. (For a reminder of why this is, see figure 176.4.) The coordinates of $P_{i-1}$ are $(x_{i-1}, f(x_{i-1}))$ and the coordinates of $P_i$ are $(x_i, f(x_i))$. Substituting this into the above sum, we see that the total length

of the segments is

$$\sum_{i=1}^{n} \sqrt{(x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2}$$

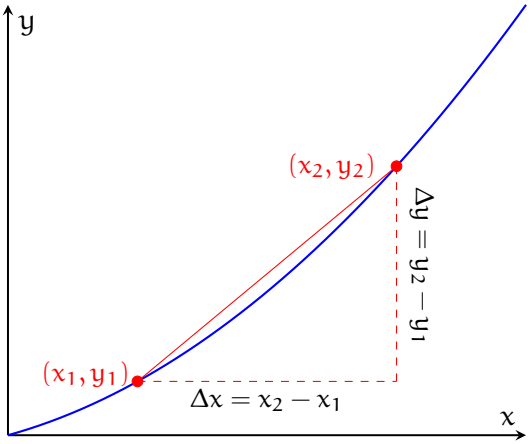



Figure 176.4: The distance between two points on the $xy$-plane

Recall the Mean Value Theorem, which states that there is some $x_i^*$ such that $f(x_i) - f(x_{i-1}) = f'(x_i^*)(x_i - x_{i-1})$. Substituting this into the above sum, we get:

$$\sum_{i=1}^{n} \sqrt{(x_i - x_{i-1})^2 + (f'(x_i^*)(x_i - x_{i-1}))^2}$$

Recall from the chapter on Riemann sums and the integral that we defined $\Delta x = x_i - x_{i-1}$ and we can further rewrite the sum as

$$\sum_{i=1}^{n} \sqrt{(\Delta x)^2 + [f'(x_i^*)\Delta x]^2} = \sum_{i=1}^{n} \sqrt{1 + [f'(x_i^*)]^2}\sqrt{(\Delta x)^2} = \sum_{i=1}^{n} \sqrt{1 + [f'(x_i^*)]^2}\Delta x$$

Putting this all together, we see that the actual length of the curve is defined as

$$L = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{1 + [f'(x_i^*)]^2}\Delta x$$

.

This is the definition of the integral of $\sqrt{1 + [f'(x)]^2}$ and therefore the length of some function $f(x)$ on the interval $a < x < b$ is $\int_a^b \sqrt{1 + [f'(x)]^2}\, dx$. In another notation, this is equivalent to $\int_a^b \sqrt{1 + (\frac{dy}{dx})^2}\, dx$.

## 176.2 Arc Length of Vector-valued Functions

Suppose you have a vector-valued function, $f(t) = [x(t), y(t)]$. A common example might be an artillery shell shot at an angle. For a shell shot with an initial velocity $v_o$ at angle $\theta$ from the ground, its position can be described with the vector-valued function $f(t) = [v_o \cos\theta(t), v_o \sin\theta(t) - 4.9t^2]$. (A concrete example where $v_o = 12\frac{m}{s}$ and $\theta = 30^o$ is shown in figure 176.5.)

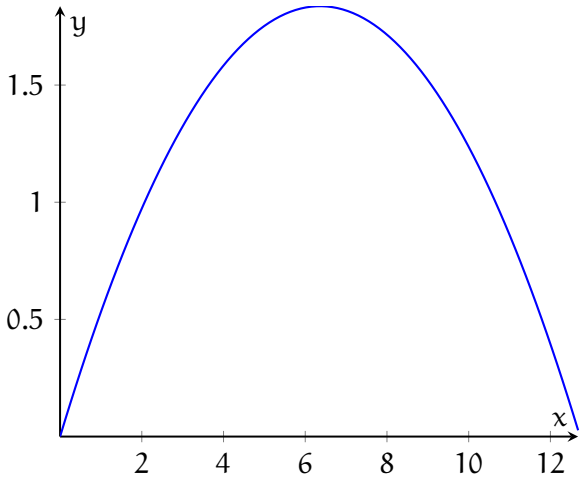



Figure 176.5: The path of an artillery shell with shot with initial velocity $v_o$ at angle $\theta$

How can we find the length of the flight path of the artillery shell? We can reinterpret the length integral for a vector-valued function, $f(t) = [x(t), y(t)]$.

$$L = \int_a^b \sqrt{1 + (\frac{dy}{dx})^2}\, dx = \int_a^b \sqrt{1 + \frac{(dy/dt)^2}{(dx/dt)^2}}\frac{dx}{dt}\, dt$$

Moving the $\frac{dx}{dt}$ under the square root, we see that

$$L = \int_a^b \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2}\, dt$$

which is equivalent to

$$L = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\, dx$$

[1]

[1]Note that this is similar to distance formula, just applied at every $dx$ along the curve. Zooming in on a small piece of the curve, it looks like a straight line, and seems as if the distance formula is being applied.

## 176.3 Applications in Physics

When we take the integral of a velocity function, we get the *displacement*. For example, if you drove to school and home again, your displacement would be zero. However, the *distance* you traveled is not zero! We can use the arc length formula to find the total distance traveled. (Remember, that if $x(t)$ is the object's position, then its velocity is given by $x'(t)$.)

Suppose a block is attached to a spring on a frictionless horizontal surface. You pull on the block, initiating harmonic motion described by $v(t) = (-0.16)\sin 9t$, where $v$ is in $\frac{m}{s}$ and $t$ is in $sec$. (Note: We are working in radians, not degrees.) What is the block's displacement from $t = 0$ to $t = 3$? What is the total distance the block moves from $t = 0$ to $t = 3$?

To find the displacement, we integrate the velocity function over the specified interval:

$$\int_0^3 (-0.16)\sin 9t\,dt = \frac{-0.16}{9}(-\cos 9t)\Big|_0^3$$

$$= \frac{0.16}{9}[\cos 27 - \cos 0] = \frac{(0.16)(-0.29-1)}{9} = -0.0229m$$

The position function and displacement are shown in figure 176.6. To find the total distance traveled, we need to find the length of the curve. Before we do so, take a minute to mentally predict: Will the distance be more or less than the displacement?

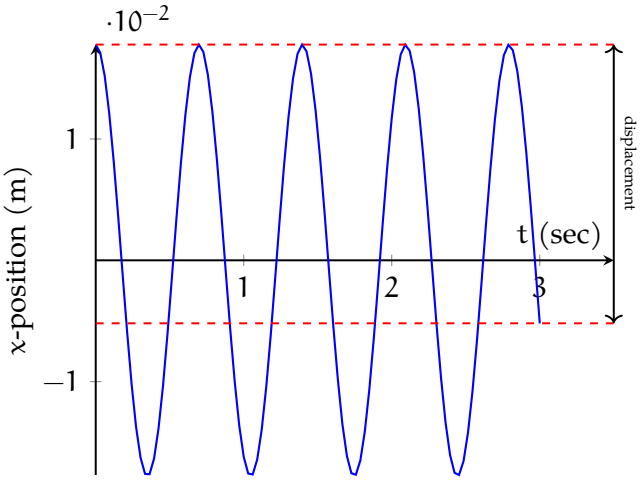



Figure 176.6: The position of the block with displacement shown. The distance traveled is the total length of the curve.

Recalling that the distance traveled by an object is $\int_a^b \sqrt{1+[x'(t)]^2}\,dt = \int_a^b \sqrt{1+[v(t)]^2}\,dt$,

we can write an integral to determine the total distance traveled by the block:

$$\int_0^3 \sqrt{1 + [-0.16 \sin 9t]^2}\, dt$$

.

Unfortunately, we do not know an antiderivative for this integral, and u-substitution won't help us. However, for definite integrals, calculators such as a TI-89 or Wolfram Alpha can easily use Riemann sums to determine the value of the integral to a high precision. Using such a tool, we find that the total distance traveled by the block is $\approx 3.019$ meters. Did you predict that the distance would be greater than the displacement?

## 176.4 Practice

### *Exercise 468*

Write an integral that gives the length of the requested curve.

*Working Space*

1. $y = \ln x$ from $x = 1$ to $x = 3$
2. $y = \sin x$ from $x = 0$ to $x = \pi$
3. $y = \frac{x^3}{3} + \frac{1}{4x}$ from $x = 1$ to $x = 4$
4. $y = \ln(\cos x)$ from $x = 0$ to $x = \frac{\pi}{3}$

*Answer on Page 2070*

### Exercise 469

The arc length function for a curve $f(x)$, where $f$ is an increasing function, is given by $s(x) = \int_0^x \sqrt{3t+5}\,dt$. If $f$ has $y$-intercept 2, find an equation for $f$. What point on $f$ is three units from the $y$-intercept? Give your answer to the thousandths place.

*Working Space*

*Answer on Page 2070*

### Exercise 470

[This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC exam.] Write an integral that gives the length of the curve $y = \ln x$ from $x = 1$ to $x = 2$.

*Working Space*

*Answer on Page 2071*

### Exercise 471

An out-of-control rocket ship is spiraling out of control through space. Its velocity can be described with the vector-valued function $v(t) = [-1412\sin t, 1412\cos t, t]$ where $v$ is in $\frac{m}{s}$ and $t$ is in sec. How far does the ship travel in the first 60 seconds? In the second 60 seconds? [Hint: in three dimensions, the length of a vector-valued function is $\int_a^b \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2 + (\frac{dz}{dt})^2}\,dt$]

*Working Space*

*Answer on Page 2071*

Chapter 177

# Continuous Probability Distributions

When we talked about the probability distribution of the sum of two dice, we assigned a probability to each of the 11 possibilities:

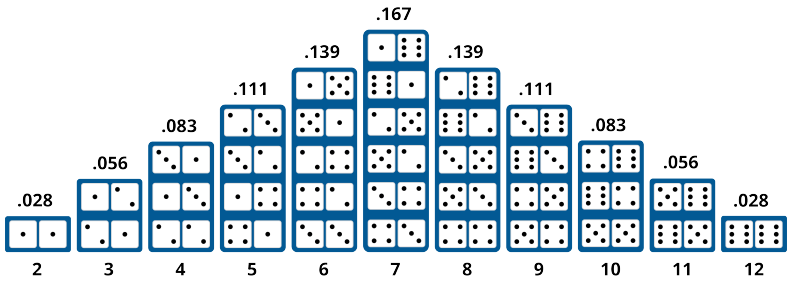



Figure 177.1: Rolling two dice probability distribution.

The probabilities all added up to be 1.0. That is a way of saying "100% of the times you throw the dice, the sum will be an integer between 2 and 12."

Now we need to talk about probabilities of properties that are continuous, not discrete. For example, we might want to ask the question, "If I randomly pick a cow from all the cows in the world, what is the probability that it will weigh less than 597.34 kg?" What does a probability distribution for a continuous variable look like?

## 177.1 Cumulative Distribution Function

Imagine that you live in ancient times. You buy, sell, and ship cows. A lot of people come into your office and brag about their cows: "Bessie is heavier than 99% of the cows in world!" So, you need to develop some statistics on cow weights.

You have ten cows. You weight them:

| Cow | Mass in kg |
|---|---|
| Cow 1 | 580.22 |
| Cow 2 | 540.07 |
| Cow 3 | 538.20 |
| Cow 4 | 512.39 |
| Cow 5 | 589.75 |
| Cow 6 | 456.91 |
| Cow 7 | 583.09 |
| Cow 8 | 493.56 |
| Cow 9 | 489.97 |
| Cow 10 | 704.15 |

If someone comes into your shop and says, "My Bessie is an astonishing 530 kg!" it would be cool to have a list on the wall that would let you yell back, "Half my cows are heavier than that, Silly!" So, you sort the cows by weight. For each weight, you say how many of your cows are lighter than that:

| Proportion of cows lighter | Mass in kg |
|---|---|
| 0.00 | 456.91 |
| 0.10 | 489.97 |
| 0.20 | 493.56 |
| 0.30 | 512.39 |
| 0.40 | 538.20 |
| 0.50 | 540.07 |
| 0.60 | 580.22 |
| 0.70 | 583.09 |
| 0.80 | 589.75 |
| 0.90 | 704.15 |

In fact, for easy reference, you make a plot:

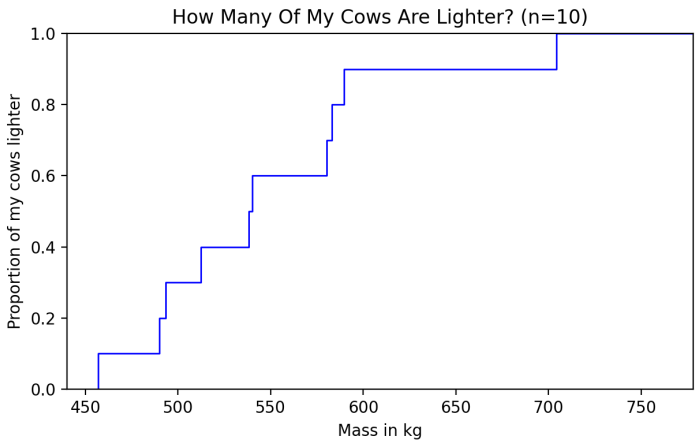



Figure 177.2: 10 cow cumulative distribution function.

Now, for any weight, you can quickly look up what proportion of your cows are lighter (and, if you subtract that from 1, what proportion of your cows are heavier).

See how jagged that graph is? That is because you only have the data for 10 cows. However, as the years pass and you weight thousands of cows, the plot will become smoother. Because it always accumulates more cows as you move from left to right, this is known as a *cumulative distribution function*, or CDF:

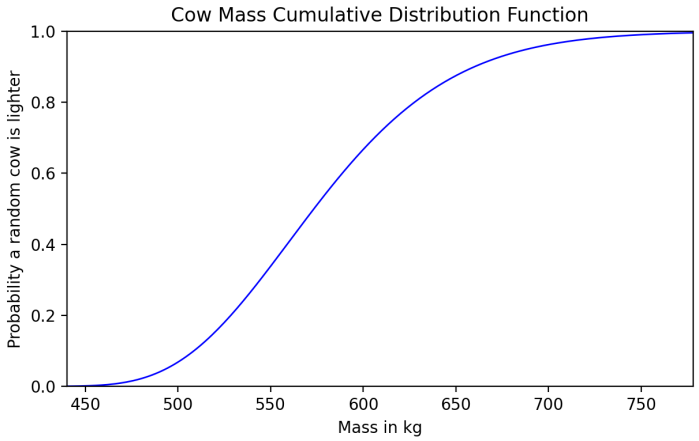



Figure 177.3: Cumulative distribution function for cow weights.

A cumulative distribution function always starts at 0 and ends at 1. On that journey, it never decreases. Reading the graph tells you what proportion of the population is *less than* any given value. If your data is normally distributed, the CDF will look like an S-curve, being symmetric closer to the probability 0.5.

Let's say you want to know what proportion of cows weigh between 620 kg and 635 kg. Using the CDF, you could figure out that 77% of all cows weigh less than 620 kg and 83% of all cows weigh less than 635 kg. Thus 6% of all cows must weigh more than 620 kg and less than 635 kg.

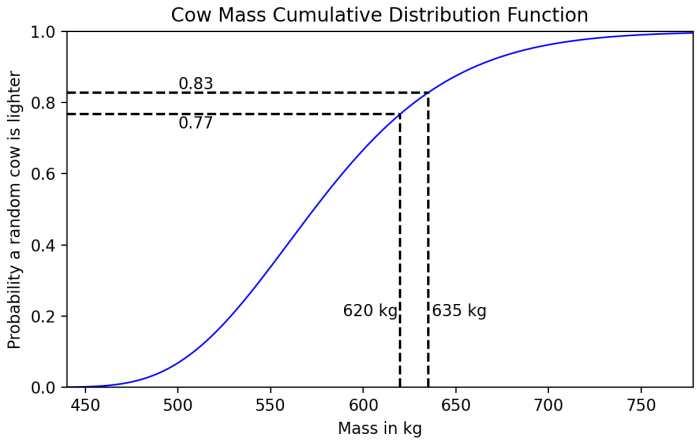



Figure 177.4: Cumulative distribution function with bounds from 620 kg to 635 kg.

## 177.2 Probability Density Function

The cumulative density function is handy, but some of its information can be hard to see. For example, how would you answer the question, "What is the most common weight of a cow?" You would squint at the CDF and try to determine where it was steepest.

To make these sorts of questions easier to answer, we take the derivative of the CDF to get the *probability density function* (or PDF). For the cows, it would look like this:

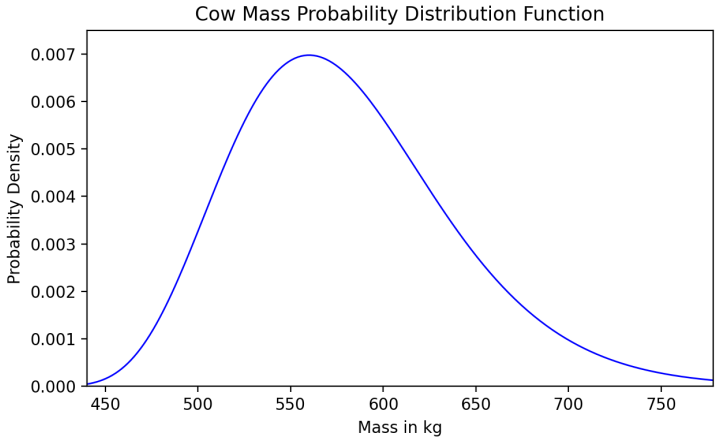



Figure 177.5: Probability density function for cow weights.

Now you can easily see that the CDF was steepest at about 560 kg. We call the highest point on the PDF the *most likely estimator*. For example, you might say "560 kg is the most likely estimator of cow mass." Sometimes we just say "the MLE".

Note that the MLE is often different from the mean or the median. In this case, for example, the distribution is skewed right — there are more cows that are heavier than the MLE than there are cows that are lighter than the MLE. The MLE would be less than the mean or the median.

Once again, let's say you want to know what proportion of cows weight between 620 kg and 635 kg. This is more difficult with a PDF than it is with a CDF. With a PDF, you have to find the area under the curve between $x = 620$ and $x = 635$.

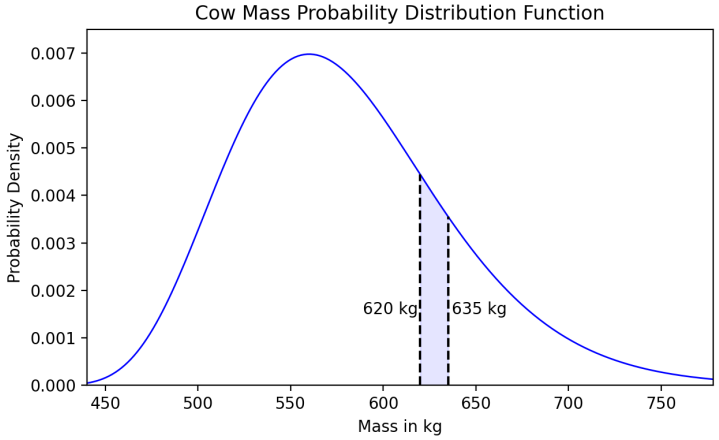



Figure 177.6: Probability density function with the integral from 620 kg to 635 kg.

This is why it is called a "probability density" — to get a true probability you need to multiply the density by the width of the region.

What is the area under the entire curve? If you integrated it, you would get the CDF. The CDF goes from 0 to 1.0. The area under a PDF is *always* 1.0.

## 177.3 The Continuous Uniform Distribution

The most simple continuous distribution is the uniform distribution between two numbers $a$ and $b$. The CDF is a straight line from zero at $a$ to 1 at $b$. For example, here is the CDF for the uniform distribution between 6 and 9:

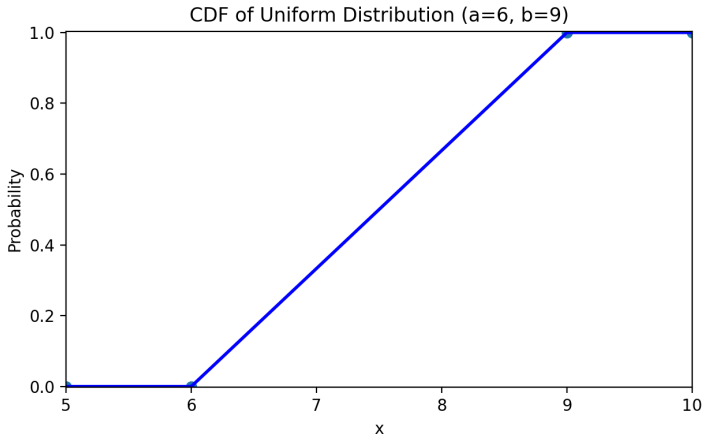



Figure 177.7: Cumulative distribution function for a uniform distribution between 6 and 9.

That line goes from 0 to 1 over a distance of 3, so its slope is $\frac{1}{3}$ between 6 and 9 and zero everywhere else. Thus, the PDF (its derivative) looks like this:

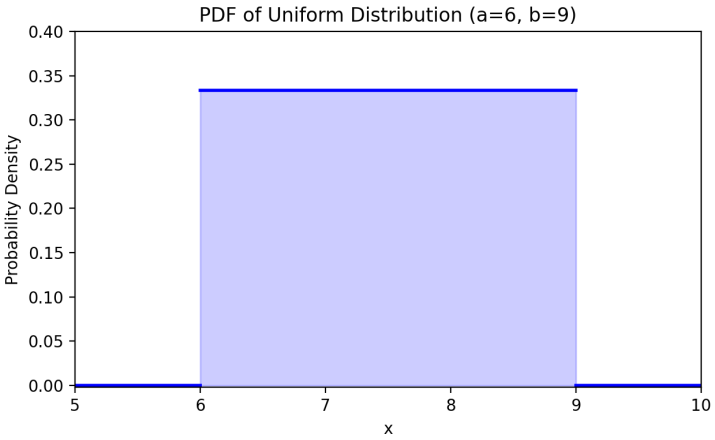



Figure 177.8: Probability density function for a uniform distribution from the continuous uniform distribution.

So, we can write the probability distribution of a continuous uniform distribution between $a$ and $b$ as:

$$p(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b. \end{cases}$$

Notice that if $a$ and $b$ are less than 1 apart, the value of $p(x)$ will be greater than 1. This is an important difference between a probability and a probability density:

- A probability will always be in the interval $[0, 1]$.
- A probability density will never be less than 0, but can be much larger than 1.

That said, the probability density will always integrate to 1.

Here is the PDF for a uniform distribution between 0.6 and 0.9:

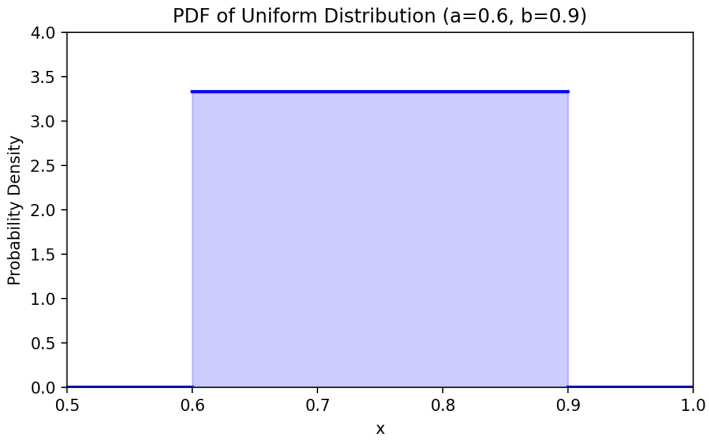



Figure 177.9: Uniform probability density function.

The mean and median of a uniform distribution between $a$ and $b$ is its midpoint: $\frac{a+b}{2}$.

The variance $(\sigma^2)$ is $\frac{(b-a)^2}{12}$.

## 177.4 Continuous Distributions In Python

The SciPy library has functions that let a programmer work with a large collection of different probability distributions.

For example, if you wanted to work with a continuous uniform distribution between 6 and 9, you would import the relevant functions like this:

```
1 from scipy.stats import uniform
```

Now, if you wanted a numpy array containing a sample of 300 numbers generated randomly from that distribution:

```
1 samples = uniform.rvs(loc=6,  scale=3,  size=300)
```

The `loc` argument is $a$. The `scale` argument is $b - a$.

If you wanted to know the value of the probability density function at 8 and 10, you could use the `pdf` function:

```
x_values = np.array([8, 10])
p_values  = uniform.pdf(x_values,  loc=6,  scale=3)
```

Now, `p_values` contains 0.33333 and 0.0.

To get the value of the cumulative distribution function at those points, you would use the `cdf` function:

```
cdf_values = uniform.cdf(x_values,  loc=6,  scale=3)
```

Now, `cdf_values` contains 0.666667 and 1.0.

The inverse of the CDF is very useful. It answers questions like "How heavy does a cow have to be to be in top 1%?"

```
bottom_top_percentiles = np.array([0.01,  0.99])
boundaries = uniform.ppf(bottom_top_percentiles,  loc=6,  scale=3)
```

Now, `boundaries` contains 6.03 and 8.97.

The SciPy library supplies these functions (`rvs`, `pdf`, `cdf`, and `ppf`) for over a hundred common continuous probability distributions.

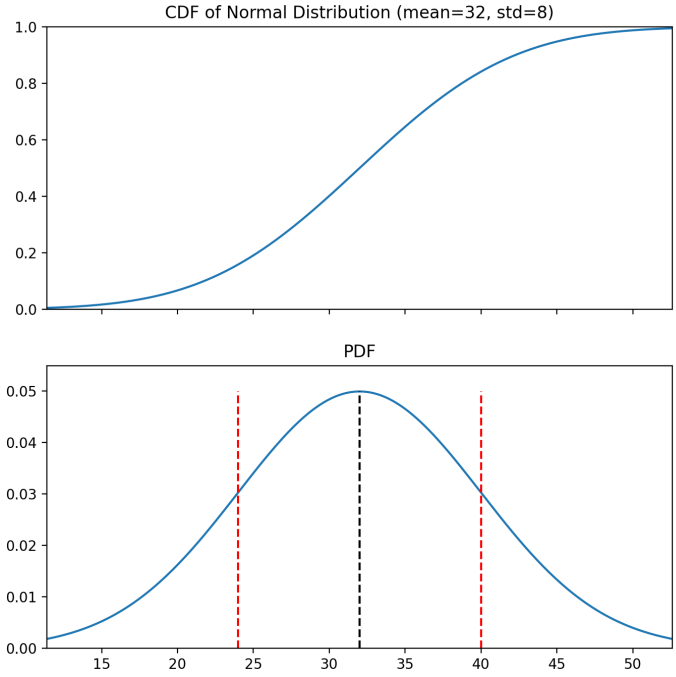
The common "bell curve" shaped distribution is called a Gaussian or Normal distribution. It is described by its mean (the midpoint of the bell) and its standard deviation. For the normal distribution, the standard deviation is the distance you have to go from the mean to reach 68% of the population. We will talk a lot more about the normal distribution in other chapters, but let's take this opportunity to plot the CDF and PDF of a normal distribution with a mean of 32 and a standard deviation of 8.

Create a file called `plot_norm.py` and add the following lines:

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

# Constants
```

```
MEAN = 32
STD = 8

# Plotting from the 0.5 percentile to the 99.5 percentile
x_min = norm.ppf(0.005, loc=MEAN, scale=STD)
x_max = norm.ppf(0.995, loc=MEAN, scale=STD)

# Make 200 points between x_min and x_max
x_values = np.linspace(x_min, x_max, 200)

# Get CDF for each x value
cdf_values = norm.cdf(x_values, loc=MEAN, scale=STD)

# Get PDF for each x value
pdf_values = norm.pdf(x_values, loc=MEAN, scale=STD)

# What is the highest density?
max_density = norm.pdf(MEAN, loc=MEAN, scale=STD)

# Make a figure with two axes
fig, axs = plt.subplots(nrows=2, sharex=True, figsize=(8, 8), dpi=200)
axs[0].set_xlim(left=x_min, right=x_max)

# Draw the CDF on the first axis
axs[0].set_title("CDF of Normal Distribution (mean=32, std=8)")
axs[0].set_ylim(bottom=0.0, top=1.0)
axs[0].plot(x_values, cdf_values)

# Draw the PDF on the second axis
axs[1].set_title("PDF")
axs[1].set_ylim(bottom=0.0, top=max_density * 1.1)
axs[1].plot(x_values, pdf_values)

# Add lines for mean,  mean-std, and mean+std
axs[1].vlines(MEAN - STD, 0, max_density, "r", linestyle="dashed")
axs[1].vlines(MEAN + STD, 0, max_density, "r", linestyle="dashed")

# Save out the figure
fig.savefig("norm_32_8.png")
```

The resulting plot should look like this:




What do those vertical lines mean? An ornithologist might tell you, "The wingspan of adult robins are normally distributed with a mean of 32 cm and a standard deviation of 8 cm." This means 68% of the population of adult robins would have wingspans between the two red lines.

## *Exercise 472* SciPy Stats

*Working Space*

Globally, the height of adult women is approximately normally distributed. The mean is 164.7 cm. The standard deviation is 7.1 cm.

Use Python and SciPy stats to answer these questions:

- To be in the tallest decile (the top 10%) of adult women, how tall does one need to be?
- What percentage of adult women are between 160 cm and 165 cm?

(In case you are wondering: For men the mean is 178.4 cm and the standard deviation is 7.6 cm.)

*Answer on Page 2071*

Chapter 178

# The Physics of Gases

Now, let's say you start to heat the helium inside the balloon. As the temperature goes up, the molecules inside will start to move faster.

Remember that the kinetic energy of an object with mass $m$ and velocity $v$ is given by

$$k = \frac{1}{2}mv^2 \tag{178.1}$$

So, you could say "As the temperature of the gas increases, the kinetic energy of the molecules increases." But a physicist would say "The temperature of the gas is how we measure its kinetic energy."

### 178.0.1 A Statistical Look At Temperature

If you say "This jar of argon gas is 25 degrees Celsius," you have told me about the *average* kinetic energy of the molecules in the jar. However, some molecules are moving very slowly. Others are moving really, really fast.

We could plot the probability distribution of the speeds of the molecules. For argon at 25 degrees Celsius, it would look like this:

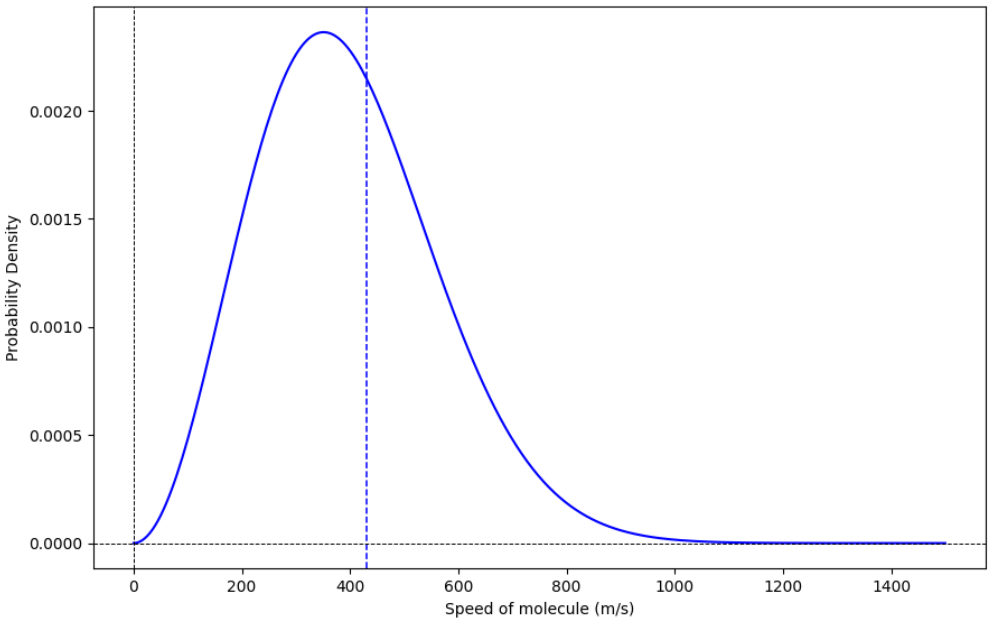



Figure 178.1: Probability distribution of molecular speeds for argon gas at 25 degrees Celsius.

The temperature, remember, is determined by the average kinetic energy of the molecules. Some molecules are moving slowly, and have less kinetic energy than the average. Some molecules are moving very quickly, and have more kinetic energy. The dotted line is the divider between the two groups: molecules moving at speeds to the left of the line have less kinetic energy than average; those on the right have more kinetic energy than average.

Where is that line? That is the RMS of the speeds of the molecules. That is, if we measured all the speeds of all the molecules $s_1, s_2, s_3, \ldots, s_n$, that line would be given by the root of the mean of the squares:

$$v_{\text{rms}} = \sqrt{\frac{1}{n}\left(s_1^2 + s_2^2 + s_3^2 + \ldots s_n^2\right)}$$

If you have the same gas at a lower temperature, the distribution shifts toward zero: the peak becomes taller and narrower, and the tail of very fast molecules becomes much shorter. In other words, fewer molecules have high speeds, and more molecules cluster near lower speeds. Let's look at an example.

Here is probability distribution of molecular speeds for argon gas at 25 degrees and -100 degrees Celsius.

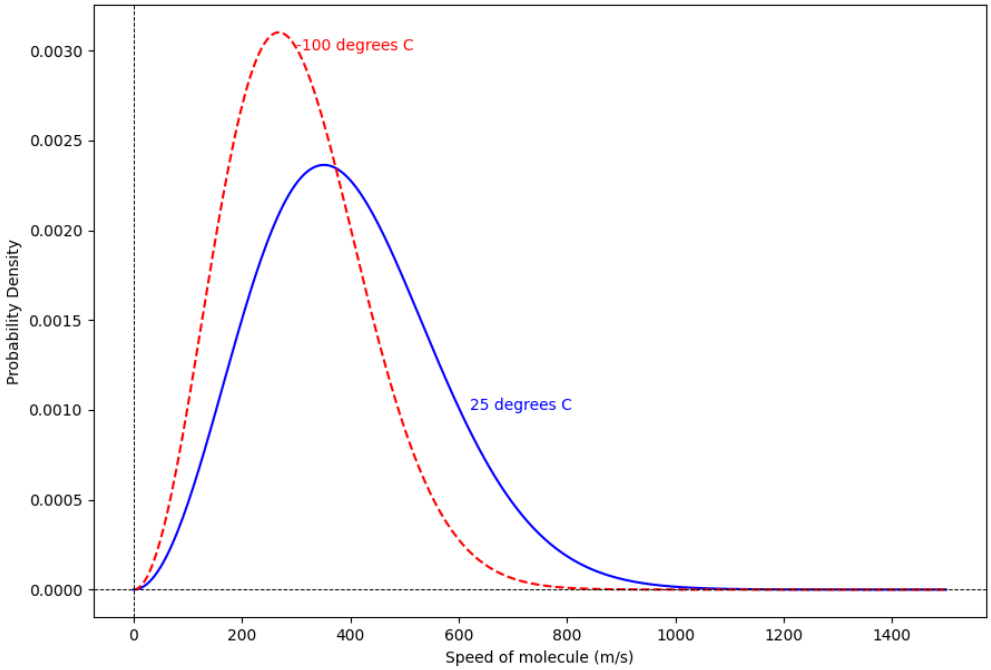



Figure 178.2: Probability distribution of molecular speeds for argon gas at 25 degrees and -100 degrees Celsius.

### 178.0.2 Absolute Zero and Degrees Kelvin

If you keep lowering the temperature, eventually, all the molecules stop moving. This is known as *absolute zero* — you can't make anything colder than absolute zero. Absolute zero is $-273.15°$ Celsius or $-459.67°$ Fahrenheit.

In addition to Celsius and Fahrenheit, there is a third temperature system: Kelvin. Kelvin has the same scale as Celsius, but it starts at absolute zero (such that every value differs by 273.15). So, 0 degrees Celsius is 273.15 degrees Kelvin, and 100 degrees Celsius is 373.15 degrees Kelvin.

Any time you are working with the physics of temperature, you will use Kelvin.

Sometimes, when reading about gases, you will see "STP" which stands for "Standard Temperature and Pressure." STP is defined to be $0°$ Celsius and 100 kPa.

## 178.1 Temperature and Volume

Let's say it's dawn, and a person is lying on a half-full air mattress in a field. The weight of the person will keep the pressure of the air inside constant (or pretty close).

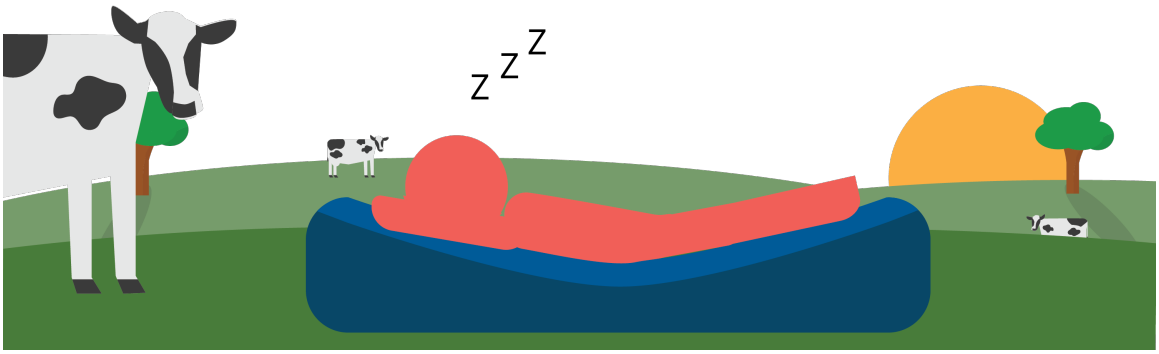



Figure 178.3: A man sleeping on an air mattress as the sun rises.

The molecules in the mattress are not entering or leaving that mattress. However, as the sun rises, the air inside will get warmer and expand. The person will be gently lifted by the expanding air. You might wonder: how much will the air expand?

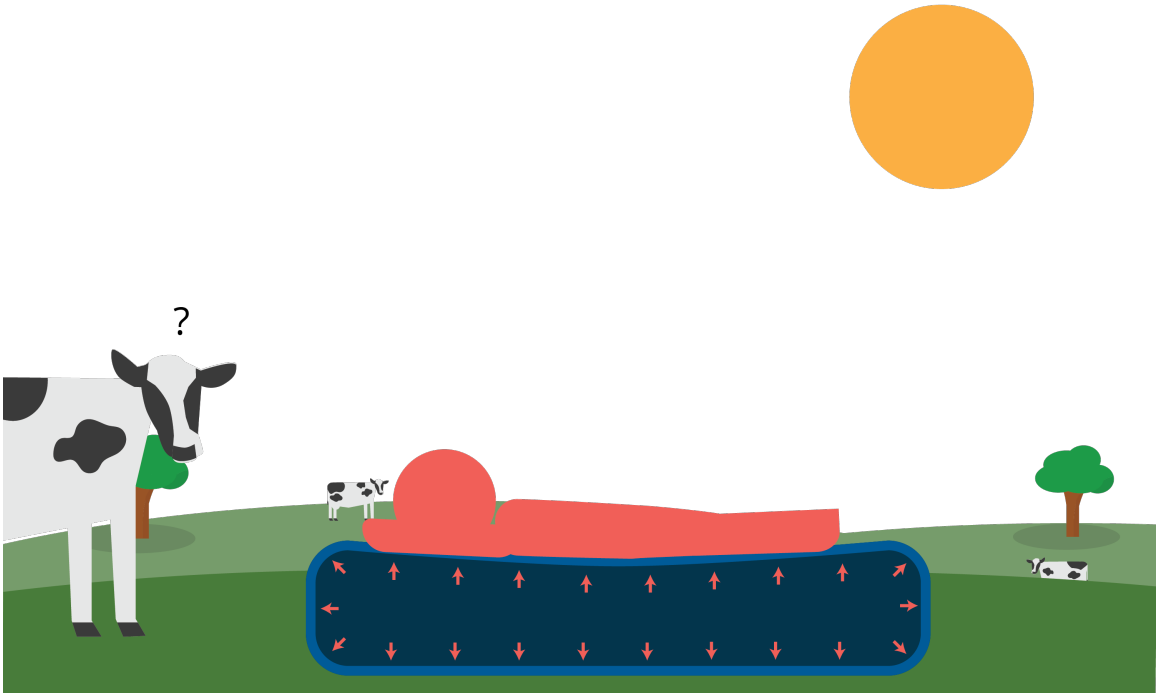



Figure 178.4: The air mattress inflates as the sun reaches its peak height in the sky.

If you have constant pressure and a constant number of molecules, the volume of the gas is proportional to the temperature in Kelvin:

$$V \propto T$$

This is known as Charles's Law.

### *Exercise 473* Temperature and Volume

*Working Space*

At dawn, the air inside mattress at dawn has a volume of 1000 liters and a temperature of 12 degrees Celsius.

At noon, that same air has a temperature of 28 degrees Celsius. The pressure on the gas has not changed at all.

What is the volume of the gas at noon?

*Answer on Page 2072*

Note: Volume and temperature are only proportional as long as the substance is a gas. We will talk about liquids and solids soon.

## 178.2 Pressure and Volume

As you increase the pressure on a gas, the molecules will get pushed closer together, and the volume will decrease.

For example, if you put the cap on an empty plastic bottle and squeeze it. As you put the gas inside the bottle under pressure, its volume will decrease.

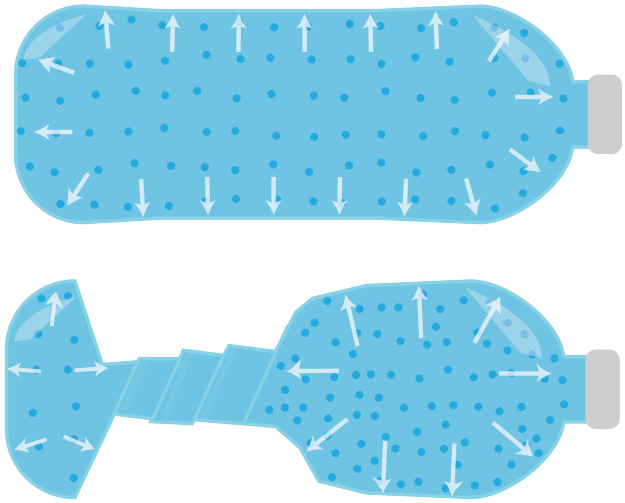


Figure 178.5: The water bottle shrinks as it is put under pressure.

If you keep the number of molecules and the temperature constant, the pressure of the gas and its volume are inversely proportional:

$$P \propto \frac{1}{V}$$

This is known as Boyle's Law. "But," you say with disbelief, "if I increase the pressure on my empty water bottle from 5 kPa to 10 kPa, the volume inside won't decrease by half!"

Don't forget that the air inside the bottle is under 101 kPa of atmospheric pressure before you even start to squeeze it.

### *Exercise 474* Temperature and Volume

*Working Space*

At an altitude where the atmospheric pressure is 100 kPa, you seal air in a 1 liter water bottle.

Squeezing the water bottle, you raise the internal pressure by 20 kPa. What is the volume inside the bottle now?

*Answer on Page 2072*

## 178.3 The Ideal Gas Law

You are gradually getting an intuition for the relationship between the number of molecules, the volume, the pressure, and the temperature of a gas. We can actually bring these together in one handy equation.

**Ideal Gas Law**

$$PV = nRT$$

where:

- $P$ is the pressure in pascals
- $V$ is the volume in cubic meters
- $n$ is the number of molecules in moles
- $R$ is the molar gas constant: 8.31446
- $T$ is the temperature in Kelvin

(You can remember this as the "Pivnert.")

From the name, you might predict the following: The Ideal Gas Law is not 100% accurate. However, for most purposes, it works remarkably well.

Notice that the ideal gas law says nothing about what kind of gas it is; it works regardless.

### *Exercise 475* Ideal Gas Law

*Working Space*

You have a cylinder containing $O_2$. The chamber inside has a radius of 12 cm and a length of 50 cm. The temperature inside the cylinder is 20 degrees Celsius. The pressure inside the tank is 600 kPa.

How many moles of $O_2$ are inside?

*Answer on Page 2073*

## 178.4 Molecules Like To Stay Close to Each Other

When two molecules get close to each other a few things can happen:

- They can under go a chemical reaction: electrons are exchanged or shared and a different molecule or molecules come into existence. This is the realm of chemistry, and we won't go into it in this course.
- One or both of them have so much kinetic energy that they just pass each other or bounce off each other. This is what happens in a gas.
- The two molecules can "stick" together. This is what happens in a liquid or a solid.

Why do they stick together if they aren't combined in a chemical reaction?

First, they don't get *too close*. If they get too close, their electron clouds repel each other with a strong force. This is what happens in a gas when two molecules bounce off of each other.

However, if the molecules are quite close to each other, there are forces that will attract them toward each other. These intermolecular forces are beyond the scope of this course, but they called Van der Waals forces and hydrogen bonds. The strength of these forces vary based on the two molecules involved.

Which is why some of the matter around you is in gas form (molecules that don't stick

together at the temperature and pressure you are living in because they have weak attractive forces) and some is non-gas (gangs of molecules with stronger attraction that makes them clump together as a liquid or a solid at that same temperature and pressure).

However, what if we change the temperature and pressure? We can change if and how the molecules clump together. This is known a *phase change*; We will cover it soon.

CHAPTER 179

# Kinetic Energy and Temperature of a Gas

As mentioned in the previous chapter, for a particular gas, the temperature (in Kelvin) is proportional to the average kinetic energy of the individual molecules.

Perhaps you want to warm 3 moles of helium gas (trapped in a metal cylinder) from 10 degrees Celsius to 30 degrees Celsius. How would you compute exactly how many Joules of energy this would require?

The amount of energy necessary to raise one mole of a molecule by one degree is known as *molar heat capacity*. (The molar heat capacity of liquid water, for example, is 75.38 J per mole-degree.)

With gases, there are actually two different possible situations:

1. Constant volume: As you heat the gas, the pressure and the temperature increase. This molar heat capacity is usually denoted as $C_{V,m}$.
2. Constant pressure: As you heat the gas, the temperature and the volume increase. This molar heat capacity is usually denoted as $C_{P,m}$.

All gases made up of one atom (Helium, for example, is a monoatomic gas) have the same values for $C_{V,m}$ and $C_{P,m}$:

$$C_{V,m} = \frac{3}{2}R \approx 12.47 \text{ Joules per mole-degree}$$

$$C_{P,m} = \frac{5}{2}R \approx 20.8 \text{ Joules per mole-degree}$$

(Remember from last chapter that $R$ is the ideal gas constant $\approx 8.31446$ Joules per mole-degree.)

## *Exercise 476* Warming Helium

*Working Space*

You have 3 moles of helium.

1. How many Joules would be required to warm 3 moles of helium gas by 20 degrees Celsius at constant volume?
2. How many Joules would be required to warm 3 moles of helium gas by 20 degrees Celsius at constant pressure?

*Answer on Page 2073*

## 179.1 Molecule Shape and Molar Heat Capacity

We told you that gases made up of one atom have the same values for $C_{V,m}$ and $C_{P,m}$:

$C_{V,m} = \frac{3}{2}R \approx 12.47$ Joules per mole-degree

$C_{P,m} = \frac{5}{2}R \approx 20.8$ Joules per mole-degree

For any molecule, it is generally true that

$$C_{P,m} \approx C_{V,m} + R$$

It is also true that for any molecule, there is some integer $d$ such that

$$C_{V,m} \approx \frac{d}{2}R$$

For example, for all monoatomic gases, $d = 3$. For diatomic gases (like $N_2$ and $O_2$, $d$ is 5.

$d$ is known as the *degree of freedom* of the molecule. When you study chemistry, they will teach you to predict $d$ based on the shape of the molecule.

Here are the relevant numbers for some gases you are likely to work with:

| Gas | type | $C_{V,m}$ | $C_{P,m}$ | $d$ |
|---|---|---|---|---|
| $He$ | monoatomic | 12.4717 | 20.7862 | 3 |
| $Ar$ | monoatomic | 12.4717 | 20.7862 | 3 |
| $O_2$ | diatomic | 21.0 | 29.38 | 5 |
| $N_2$ | diatomic | 20.8 | 29.12 | 5 |
| $HO_2$ (water vapor) | 3 atoms | 28.03 | 37.47 | 7 |
| $CO_2$ | 3 atoms | 28.46 | 36.94 | 7 |

## 179.2 Kinetic Energy and Temperature

For a sample of a gas, we can calculate its kinetic energy based on its molar heat capacity, the number of molecules, and the temperature:

$$E_K = C_{V,m} n T$$

where

- $E_K$ is the kinetic energy in Joules
- $C_{V,m}$ is the molar heat capacity of the gas at constant volume
- $n$ is the number of molecules in moles
- $T$ is the temperature in Kelvin (or Celsius as long as $T \neq 0$)

### *Exercise 477* Warming Helium Revisited

*Working Space*

How much kinetic energy does 3 moles of helium have at 10 degrees Celsius?

How much kinetic energy does 3 moles of helium have at 30 degrees Celsius?

What is the difference?

*Answer on Page 2073*

## 179.3 Why is $C_{V,m}$ different from $C_{P,m}$?

What if, instead of keeping the volume constant while we heat the molecules in the helium tank, we keep the pressure constant and let the gas expand? The change in kinetic energy is the same: 748 Joules.

However, we know that the molar heat capacity if we keep pressure constant is $\frac{5}{2}R$, so heating will require $\frac{5}{2}R(3)(20) = 1247$ Joules.

What happened to the 499 missing Joules!? Thermodynamics tells us energy is neither created nor destroyed, so it must have gone somewhere.

That energy was used pushing against the pressure as the gas expanded. For example, maybe the sample was in a balloon in space — the extra energy stretched the surface of the balloon.

The 499 Joules were converted into potential energy.

## 179.4 Work of Creating Volume Against Constant Pressure

Let's imagine that you had a total vacuum (zero pressure) with a piston. As you pulled the piston out, you would be pulling against the atmospheric pressure. How much energy would that require?

If you increased the volume of the vacuum by $V$ against a pressure of $P$, you would do

$VP$ work.

Let's check to make sure the 499 Joules mentioned above makes sense with this in mind.

No initial pressure was given in the problem, so let's just make one up and see how things work out: 100 kPa. Using the ideal gas law, the initial volume would be:

$$V_1 = \frac{nRT}{P} = \frac{(3)(8.31446)(283.15)}{100,000} = 0.07063 \text{ cubic meters}$$

The volume after we heated the gas and let it expand against 100 kPa would be:

$$V_2 = \frac{nRT}{P} = \frac{(3)(8.31446)(303.15)}{100,000} = 0.07562 \text{ cubic meters}$$

So, the volume increased by $0.07562 - 0.07063 = 0.00499$ cubic meters. Multiplying that by 100,000 pa, we get 499 Joules, as we expected!

## 179.5 Why does a gas get hotter when you compress it?

Now, imagine that there is gas inside the piston and you push on the piston to compress that air. The work that you do is converted into kinetic energy, and that kinetic energy raises the temperature of the gas.

For example, if you had two moles of argon gas in the piston, and you pushed the piston 0.1 meters with an average force of 50 newtons, you will have done 5 Joules of work.

How much would 5 Joules raise the temperature of 2 mole of a monoatomic gas?

$$\Delta T = \frac{5}{(2)(C_{V,m})} = 0.2^\circ \text{ Kelvin}$$

It works both ways: Compression makes a gas hotter; decompression makes a gas colder. You can sometimes experience the heat of compression when you pump up a bicycle tire — as you pump the tire will get warmer.

If you compress or decompress a gas without letting any heat enter or depart, we say the compression or decompression was *adiabatic*. In order to solve any interesting problems about heating/cooling due to compression/decompression, you will need to assume the process was adiabatic.

When a spacecraft enters the atmosphere, it has to deal with a lot heat. Some people

assume that heat is due to friction of the air rubbing against the spacecraft at over 7,000 meters per second. Actually, most of the heat is due to the compression of the air as it gets pushed out of the way of the spacecraft.

## 179.6 How much hotter?

Let's say you have an accordion-like container filled with helium at 100 kPa (about 1 atmosphere) and 300 degrees Kelvin. It holds 2 cubic meters. You then you put it in a vice and quickly compress it down to 0.5 cubic meters. Assuming it was adiabatic, how hot would the gas inside be after the compression?

Here is the challenging part: As you crush the container, the temperature and the pressure in the container are both increasing. So, as you go, it gets harder and harder to crush. Each milliliter of volume that you eliminate requires a little more work than the milliliter before.

Let's simulate the process in Python, after which we will give you the formula.

In the simulation, you will start with an initial volume of 2 cubic meters and crush it down to 0.5 cubic meters in 40 steps. At each step you will recalculate the temperature and pressure.

Next, you will plot the results. Make a file called `gas_crunch.py`:

```
import numpy as np
import matplotlib.pyplot as plt

V_initial = 2.0 # cubic meters
V_final = 0.5 # cubic meters
step_count = 40 # steps

T_initial = 300.0 # kelvin
P_initial = 100000 # pascals

# Constants
R = 8.314462618 # ideal gas constant
C_v = 3.0 * R / 2.0 # molar heat capacity (constant volume)

# Compute the number of moles
n = P_initial * V_initial/(R * T_initial)
print(f"The container holds {n:.2f} moles of helium")

# How much volume do we need to eliminate in each step?
# (in cubic meters)
step_size = (V_initial - V_final) / step_count

# For recording the state for each step
```

```
data_log = np.zeros((step_count, 3))

# Variables to update in the loop
V_current = V_initial
T_current = T_initial
P_current = P_initial

for i in range(step_count):
  # Record the current state
  data_log[i,:] = [T_current, V_current, P_current/1000.0]

  # Find how much energy to make the step at the current pressure
  E_step = step_size * P_current

  # Find how big the change in temperature will be from that energy
  delta_T = E_step / (n * C_v)

  # Update the current temperature, volume, and pressure
  T_current += delta_T
  V_current -= step_size
  P_current = n * R * T_current / V_current

print(f"Iterative:{T_current:0.3f} K, {V_current:0.3f} m3,
↪  {P_current/1000.0:0.3f} kPa")

fig, axs = plt.subplots(3,1,sharex=True, figsize=(8, 6))
axs[0].set_xlim((0,step_count))
axs[0].plot(data_log[:,0], 'k', lw=0.2)
axs[0].plot(data_log[:,0], 'r.')
axs[0].set_ylabel("Temperature (K)")

axs[1].plot(data_log[:,1],'k', lw=0.2)
axs[1].plot(data_log[:,1], 'r.')
axs[1].set_ylabel("Volume (cubic m)")

axs[2].plot(data_log[:,2], 'k', lw=0.2)
axs[2].plot(data_log[:,2], 'r.')
axs[2].set_ylabel("Pressure (kPa)")

axs[2].set_xlabel("Step")

fig.savefig('tvpplot.png')
```

When you run this, you will see how many moles of gas there are and reasonable estimates of the temperature, volume, and pressure:

```
> python3 gas_crunch.py
The container holds 80.18 moles of helium
Iterative:733.499 K, 0.500 m3, 977.999 kPa
```

And a good plot of the intermediate values:

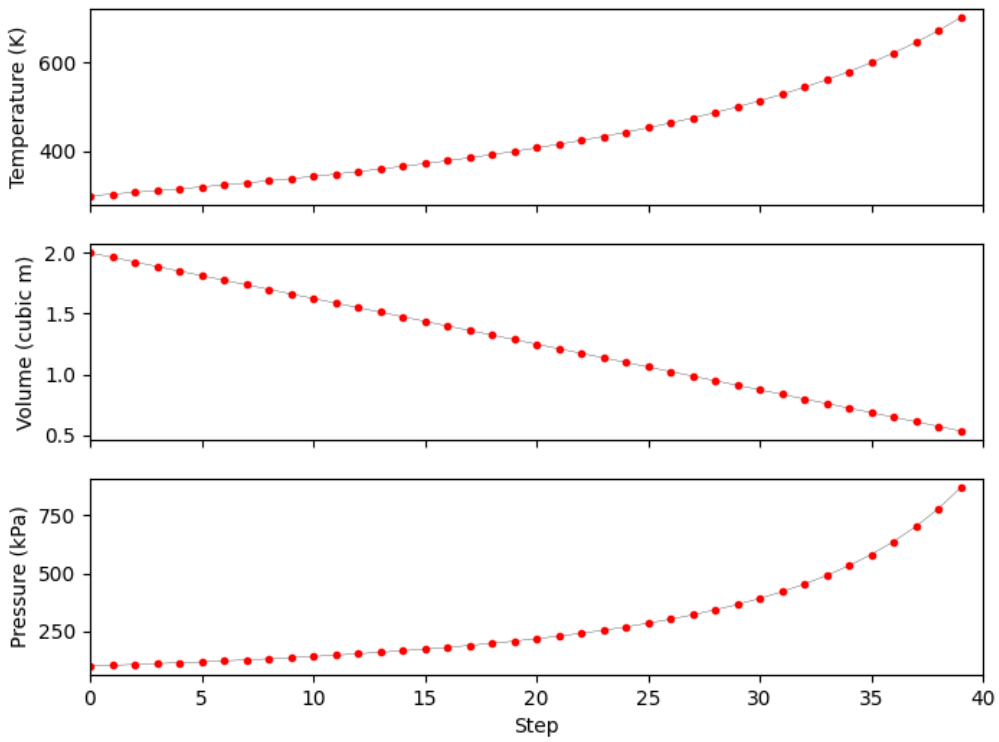



Figure 179.1: Chunk plots of temperatures every 40 steps.

However, we will get better estimates if we break it up into 400 steps instead of 40. Change the line that defines the number of steps:

```
step_count = 400 # steps
```

Now the predicted temperature and pressure should be something like 753.603° K and 1004.803 kPa. (This is much closer to the correct result: :755.953° K and 1007.937 kPa.)

What if you break it into 400,000 steps? Now, the result should be really, really close to correct. And the plot is quite accurate:

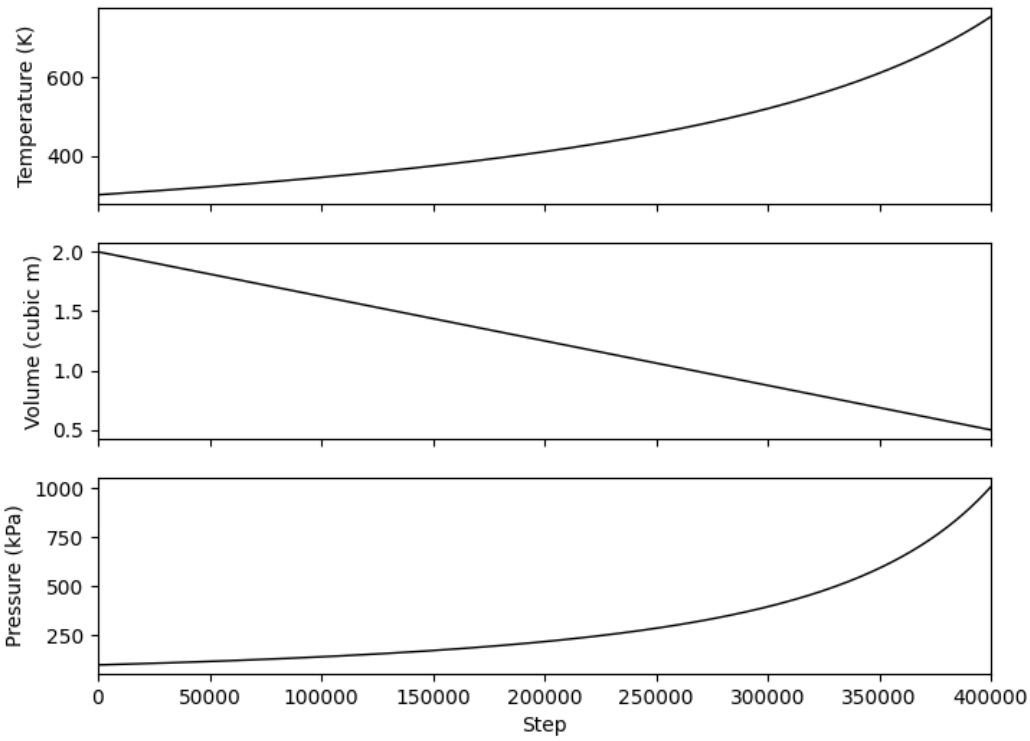



Figure 179.2: Chunk plots of temperatures every 400 steps.

(You can comment out the lines that make the red dots on the graphs. No one wants to see 400,000 red dots.)

It is inefficient to have to do long simulations to guess the final temperature and pressure. Fortunately, there are two handy rules you can use to skip this:

**Adiabatic Compression and Decompression**

Let

$$\gamma = \frac{C_{P,m}}{C_{V,m}}$$

In an adiabatic compression or decompression, $P$ and $V$ change, but

$$P\left(V^{\gamma}\right)$$

stays constant.

Also

$$T\left(V^{(\gamma-1)}\right)$$

stays constant

For a monoatomic gas:

$$\gamma = \frac{C_{P,m}}{C_{V,m}} = \frac{5}{3}$$

$$\gamma - 1 = \frac{2}{3}$$

Before the compression:

$$T\left(V^{(\gamma-1)}\right) = 300\left(2^{0.6667}\right) = 476.22$$

After the compression it has to be the same:

$$T\left(V^{(\gamma-1)}\right) = T\left(0.5^{0.6667}\right) = 476.22$$

Thus

$$T = 755.95^{\circ} \text{ Kelvin}$$

We can then use the ideal gas law to solve for the final pressure:

$$P = \frac{nRT}{V} = \frac{(80.18)(8.31446)(755.95)}{0.5} = 1007937 \text{ pascals}$$

That is hot! As you let it cool back down to 300 degrees Kelvin, how much heat would be released?

$$E = C_{V,m} n \Delta T = (12.47)(80.2)(755.95 - 300) \approx 456 \text{ kJ}$$

## 179.7 How an Air Conditioner Works

Once again, imagine the accordion-like container filled with helium. Let's say you walked it outside and compressed it from 2 cubic meters to 0.5 cubic meters in a vise. The container would get to 755.95 degrees Kelvin. You keep it compressed in the vise, but let it cool down outside. When it gets back to 300 degrees Kelvin, you walk it back inside.

Now, without letting any molecules in or out of the container, you release the vise. The gas is decompressed and gets very cold. How cold? Cold enough to accept about 456 kJ of kinetic energy from your house. That is, it would absorb heat from your house until the gas inside was the same temperature as your house.

So, you walk outside with your accordion and your vise, and repeat:

1. Compress the gas outside.
2. Let the hot gas cool down outside.
3. Walk the room-temperature compressed gas inside.
4. Decompress the gas inside.
5. Let the cold gas warm up inside.

You could keep your house cool on a hot day this way. This is not unlike how an air conditioner works.

There is a hose filled with refrigerant that does a loop:

- Outside, the refrigerant is compressed and allowed to cool to the outside temperature. (Usually there is a big fan blowing on a coil of refrigerant to speed the process.) Inside, the refrigerant is decompressed and allowed to warm to the inside temperature. (Usually there is a big fan blowing the air of the home past a coil of refrigerant to speed the process.)

In each pass of the loop, the refrigerant absorbs some of the kinetic energy from inside the house, and releases it on the outside.

This same mechanism can be used to heat your house. (Units that both heat and cool are known as *heat pumps*.) The heat pump does the process backwards: The hot compressed refrigerant cools down inside. The cold decompressed refrigerant warms up outside.

# Chapter 180

# Phases of Matter

You have experienced $H_2O$ in three phases of matter:

- Ice is $H_2O$ in the solid phase. At standard pressure, when the temperature of $H_2O$ is below $0°$ C, it is a solid.
- Water is $H_2O$ in the liquid phase. At standard pressure, when the temperature of $H_2O$ is between $0°$ C and $100°$ C, it is a liquid.
- Water vapor (or steam) is the gas phase. At standard pressure, when the temperature of $H_20$ is above $100°$ C, it is a gas.

Let's look at some of the properties of the three phases:

| Gas | Liquid | Solid |
|---|---|---|
| Assumes the volume and shape of its container | Assumes the shape, but not the volume, of its container | Retains its shape and volume |
| Compressible | Not compressible | Not compressible |

## 180.1 Thinking Microscopically About Phase

As mentioned in an early chapter, there are intermolecular forces that attract molecules to each other. A pair of molecules will have very strong intermolecular forces or very weak intermolecular forces, depending on what atoms they are made of.

For example, two helium molecules are very weakly attracted to each other due to weak intermolecular forces. Two molecules of $NaCl$ (table salt) will experience very strong intermolecular attraction.

In a gas, the molecules have lots of room to roam and lots of kinetic energy: The intermolecular attraction has very little effect.

In a liquid, the molecules are sticking close together, but are still moving around, sort of like bees in a hive.

In a solid, the molecules are not changing their configuration, and the kinetic energy they have is just expressed as vibrations within that configuration. You can imagine them like eggs vibrating in a carton.

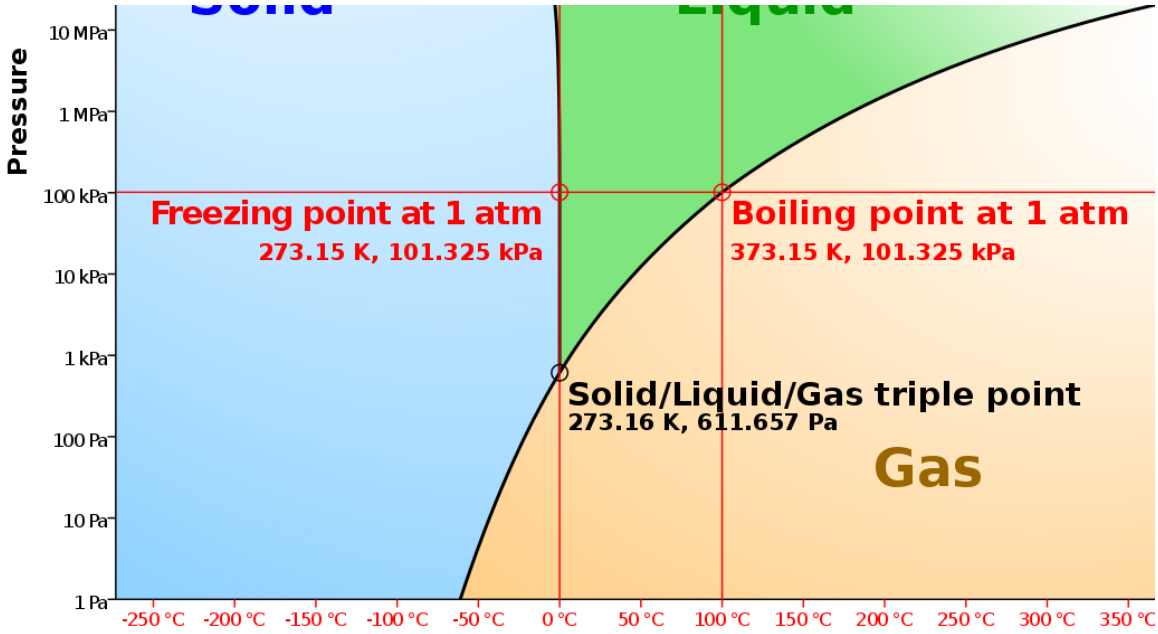
As you would expect, molecules with strong intermolecular attraction require more kinetic energy to change phases. For example, helium is a liquid below $-269^\circ$ C. NaCl, on the other hand, is a liquid between $801^\circ$ and $1,413^\circ$ C.

The temperatures I just gave you are at standard pressure (100 kPa or 1 atm). Pressure also has a role in phase change. In low pressure environments, it is much easier for the molecules to make the jump to being a gas.

For example, if you climb a mountain until the atmospheric pressure is 70 kPa, your water will boil at about $90^\circ$ C.

If you rise in a balloon until the atmospheric pressure is 500 Pa, if your water is colder than $-2^\circ$ C, it will be ice. If it is warmer it will vaporize. There is no liquid water at 500 Pa!

For any molecule, we could observe its phase at a wide range of temperatures and pressures. This would let us create a phase diagram. Here is the phase diagram for $H_2O$:




(FIXME: This diagram needs to be recreated prettier.)

## 180.2 Phase Changes and Energy

The molar heat capacity of ice is about 37.7 J/mol-K. That is, it takes about 37.7 Joules of energy to raise the temperature of one mole of ice by one degree kelvin.

The molar heat capacity of liquid water is about 75.4 J/mol-K. For water vapor, it is about

36.6 J/mol-K.

Imagine you have mole of ice at 173° K, and you are going to gradually add kinetic energy into it until you have steam at 473° K. You might guess (incorrectly) that the temperature vs. energy applied would look like this:

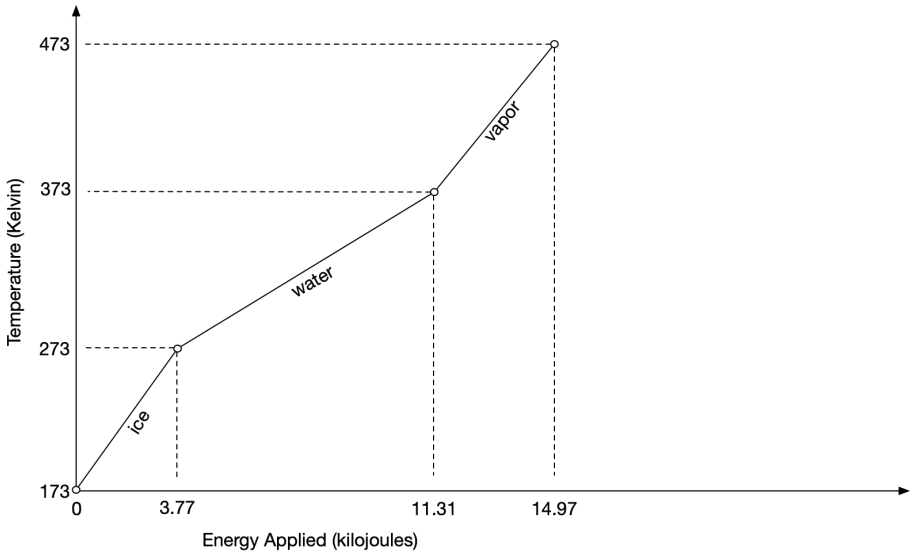



Figure 180.1: Assumption of temperature vs. energy applied.

However, once molecules are nestled into their solid state (like eggs in cartons), it takes extra energy to make them move like a liquid. How much more energy? For water, it is 6.01 kilojoules per mole. This is known as *the latent heat of melting* or *the heat of fusion*.

Similarly, the transition from liquid to gas takes energy. At standard pressure, converting a mole of liquid water to vapor requires 40.7 kilojoules per mole. This is known as *the latent heat of vaporization*. So, the graph would actually look like this:

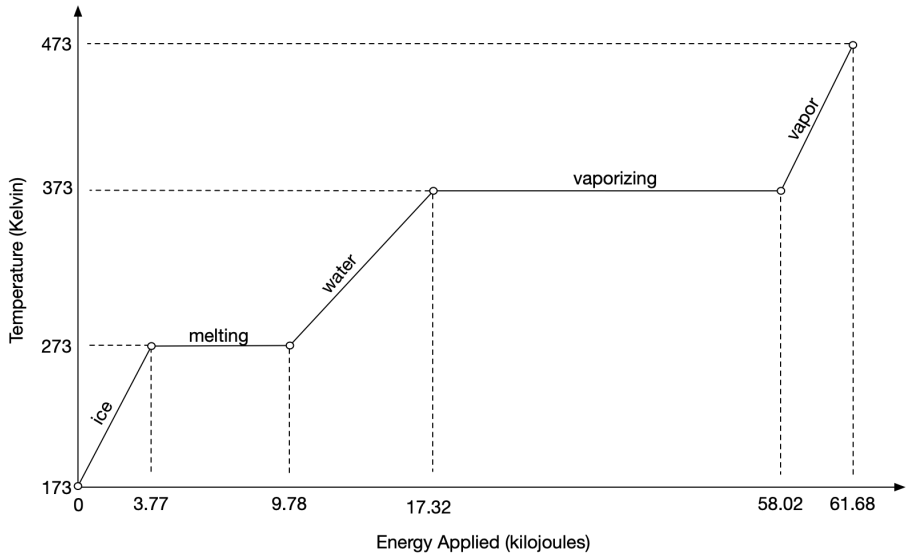



Figure 180.2: Actual temperature vs. energy applied.

Note that just as melting and vaporizing require energy, going the other way (freezing and condensing, respectively) give off energy. Thus, we can store energy using the phase change.

## 180.3 How a Rice Cooker Works

A rice cooker is a bowl with a lid and an electric heating element. You put rice and water into the bowl and turn on the heating element. The heating element pushes kinetic energy into the water, which gets warmer and eventually starts to boil.

How does the rice cooker know when to turn off (or at least down to a low-heat "keep warm" mode) before the rice starts to burn?

As long as there is a little liquid water in the bottom of the bowl, the rice won't burn, so the question is really "How does it know when there is no more liquid water in the bottom of the bowl?"

There is a mechanism (and there have been a few different versions of this mechanism) that monitors the temperature of the surface of the bowl. As long as there is liquid water in the bowl, it *cannot* go above 100° C! When all the water has been absorbed by the rice or turned to steam, the temperature rises above 100° C, and the mechanism cuts off the heat.

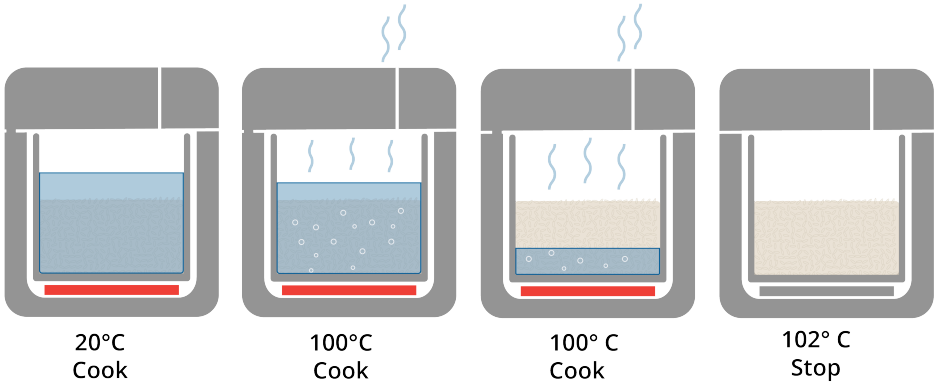



Figure 180.3: The various phases of a rice cooker.

## *Exercise 478* Using Water For Thermal Energy Storage

Tom is building a passive solar house; the front of his house is a greenhouse. He also likes to eat dinner in the greenhouse. He will have barrels (painted black) that hold 159 liters of water. His plan is to let the sun heat the barrels to 33° C by the time the sun goes down. (Any warmer and it would be unpleasant to eat dinner near them.)

At night, he will circulate air past the barrels and through his house. He is OK with the house and the barrels dropping to 17° C.

Looking at the insulation on his house and the expected nighttime temperatures, Tom estimates that he needs to store 300,000 kJ of energy in the barrels.

A mole of water is about 0.018 liters.

The molar heat capacity of water is about 75.38 J/mole-K.

How many barrels of water does Tom need to install in his greenhouse?

*Working Space*

*Answer on Page 2074*

### *Exercise 479* Using Mirabilite For Thermal Energy Storage

*Working Space*

Water barrels are going to take up too much of Tom's greenhouse!

There is a substance known as mirabilite, or Glauber's salt, or sodium sulfate decahydrate. It is relatively cheap to produce, and it has a melting point of 32.4° C.

The molar heat capacity of mirabilite is 550 J/mole-K.

The latent heat of melting mirabilite is 82 kJ per mole.

Mirabilite comes in a powder form. Assume that a mole of mirabilite occupies about 0.22 liters.

If Tom fills his barrels with mirabilite, how many barrels will he need?

*Answer on Page 2074*

## 180.4 Thinking Statistically About Phase Change

We like to say simple things like "At 100° C, water changes to vapor." However, remember what temperature is: Temperature tells you how much average kinetic energy the water molecules have. The key word here is *average*; Some molecules are going faster than average and some are going slower.

A puddle in the street on a warm night will evaporate. It isn't 100° C. Why would the puddle turn to vapor?

While the *average* molecule in the puddle doesn't have enough energy to escape the intermolecular forces, some of the molecules do. When a molecule on the surface has enough velocity (toward the sky!) to escape the intermolecular forces, it becomes vapor and leaves

the puddle.

What happens to the temperature of the puddle during this sort of evaporation? Temperature is proportional to the average kinetic energy of the molecules. If a bunch of molecules with a lot of kinetic energy leave, the average kinetic energy (of the molecules that remain) will decrease.

The most obvious example of this process is sweating: When your body is in danger of getting too hot, sweat comes out of your pores and covers your body. The fastest moving molecules escape your body, taking the excess kinetic energy with them.

### 180.4.1 Evaporative Cooling Systems

An evaporative cooling system (also called a "Swamp Cooler") uses this idea to cool air. You can imagine a fan drawing warm air through a duct from the outside. Before the air is released inside, it passes very close to a cloth that is soaked with water. The warm air molecules (which has a lot of kinetic energy) slam into the water molecules, some of which get enough kinetic energy to become vapor. The cool air and the water vapor then enter the room.

"Wait, wait, wait," you say, "The heat hasn't gone away. It is just transferred into the water molecules."

Remember that it takes 40.7 kilojoules to change a mole of liquid water at 100° C to water vapor at 100°. Escaping those intermolecular bonds takes a great deal of energy!

Thus, if a mole of water evaporates, it is because it has absorbed 40.7 kj of heat you can feel (*sensible heat*) and used it to liberate the molecules from their intermolecular bonds. For convenience, physicists call this *latent heat*.

### 180.4.2 Humidity and Condensation

When a puddle is evaporating on a warm day, there might be some water vapor already in the air. Even as the water molecules in the puddle are evaporating, some water molecules in the air are crashing into the puddle and become liquid again. (We say they *condensed*, thus the word *condensation* to describe the water that accumulates on a cold glass on a warm day.)

When there is a lot of water vapor in the air, the puddle will evaporate more slowly. In fact, if there is enough water vapor in the air, the puddle won't evaporate at all. At this point, we say "The relative humidity is 100%." That is, relative to the amount of water the air will hold, it already has 100% that amount.

Neither sweating nor evaporative cooling systems work well when the relative humidity is high.

As the temperature goes up, the air can hold more water. We usually notice it when it goes the other way: the air cools and has more water vapor than it can hold. Some of the water vapor condenses into water droplets. If the droplets land on something, we call it ”dew”. If it is high in the sky, we call the droplets ”a cloud”. If it is near the ground, we call it ”fog”.

Chapter 181

# The Piston Engine

Most cars, airplanes, and chainsaws get their power from burning hydrocarbons in a combustion chamber. We say they have *internal combustion engines*. There are many types of internal combustion engines: jet engines, rotary engines, diesel engines, and more. In this chapter, we are going to explain how one type, piston engines, work. Most cars have piston engines.

Most piston engines burn gasoline, which is a blend of liquid hydrocarbons. Hydrocarbons are molecules made of hydrogen and carbon (and maybe a little oxygen). In the presence of oxygen and heat, hydrocarbons burn — the carbon combines with oxygen to become $CO_2$, and the hydrogen combines with oxygen to become $H_2O$. In the process, heat is released, which causes the gases in the cylinder to create a lot of pressure on the piston.

## 181.1 Parts of the Engine

The engine block is a big hunk of metal. There are cylindrical holes bored into the engine block. A piston can slide up and down the cylinder. There are two valves in the wall of the cylinder:

- Before the burn, one valve opens to let ethanol and air into the cylinder.
- After the burn, the other valve opens to let the exhaust out.

There is also a spark plug, which creates the spark that triggers the burn.

As you give the engine more gas, the cylinder does more frequent burns. When the engine is just idling, the cylinder fires about 9 times per second. When you depress the gas pedal all the way down, it is more like 40 times per second.

The cylinder has a rod that connects it to the crank shaft. As the pistons move back and forth, the crank shaft turns around and around. Sometimes a piston is pushing the crankshaft, and sometimes the crank shaft is pushing or pulling the piston. All the cylinders share one crank shaft.

How many cylinders does a car have? Nearly all car models have between 3 and 8 cylinders. The opening of the valve and the firing of the spark plugs are timed so that cylinders all do their burns at different times. This makes the total power delivered to the crank

shaft smoother.

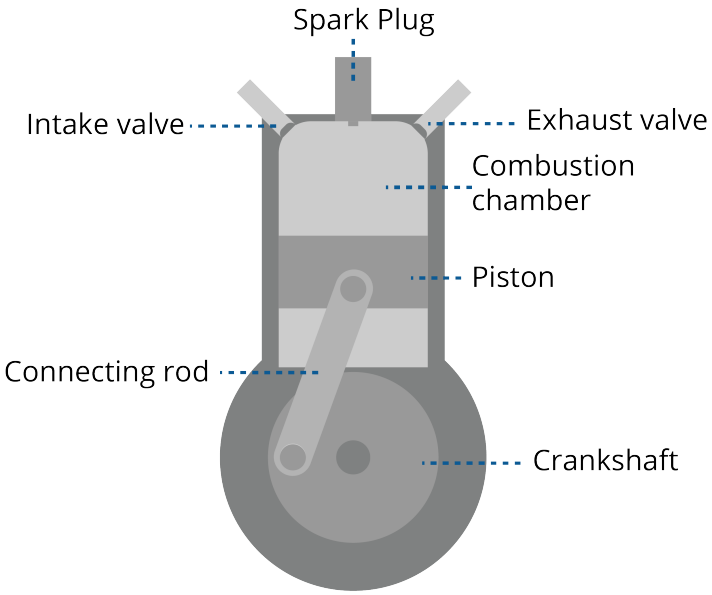



Figure 181.1: A cross-section of an 8 cylinder engine.

## 181.2 The Four-Stroke Process

Cars have four-stroke engines — this means for every two rotations of the crank shaft, each cylinder fires once. Smaller engines, like those in chainsaws, are often two-stroke engines — every cylinder fires every time the crank shaft rotates. For now, let's focus on four-stroke engines.

Here is the cycle of a single cylinder:

- As the drive shaft turns, it pulls the piston down. The intake valve opens and lets the gas/air mixture into the combustion chamber. See Figure 181.2.

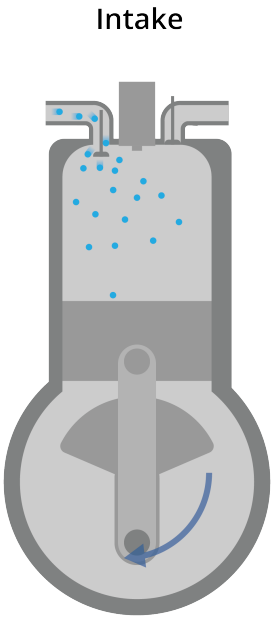



Figure 181.2: The intake stroke allows gas and oxygen into the combustion chamber.

- As the piston reaches the bottom of the stroke, the intake valve closes.
- Now, the crank shaft starts to push the piston up, compressing the gas and oxygen. See Figure 181.3.

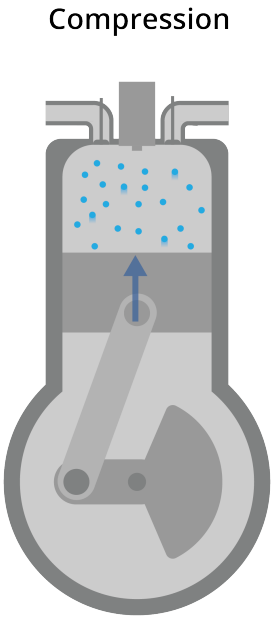



Figure 181.3: The compression stroke compresses the gas and oxygen.

- As the piston reaches the top of its stroke, the spark plug creates a spark. The fuel and oxygen burn quickly. The cool liquid fuel becomes hot carbon dioxide and water vapor.

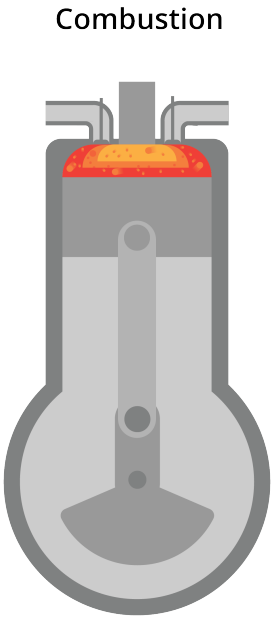



Figure 181.4: On the combustion stroke, the gas and oxygen merge into a hot ignited fuel.

- There is now very high pressure inside the cylinder. It pushes hard on the piston, which pushes the crank shaft.
- When the piston reaches the bottom of this stroke, the exhaust valve opens.
- As the crank shaft pushes the piston up, the carbon dioxide and water vapor is pushed out.

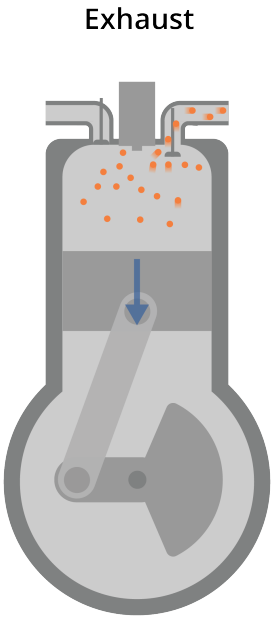



Figure 181.5: On the exhaust stroke, the burnt gases are expelled from the cylinder.

- When the piston reaches the top of this stroke, the exhaust valve is closed.

## 181.3 Dealing with Heat

Burning fuel inside a block of metal generates a large amount of heat. If there is too much heat, parts of the engine will start to melt. So, modern car engines are liquid cooled — there are arteries in the engine block carrying a liquid (called "coolant"). The hot coolant is pumped through the radiator (where the air passing through takes away the heat), then back into the engine.

Note that the heat that is carried away by the coolant is wasted energy. In fact, of the total energy created in burning the fuel, most car engines only transfer about 30% to turning the crank shaft. About 35% of the heat goes out with the exhaust. About 30% is carried away by the coolant. The remaining waste (usually about 5%) is lost to friction.

## 181.4 Dealing with Friction

From the description, it is clear that there is a great deal of metal sliding against metal, which would grind the engine up quickly if there were no lubrication. In a modern car, the moving parts in the engine are constantly bathed in oil. There is an oil pump that causes it to get sprayed on the crankshaft, the connecting rod, and in the cylinder under

the piston (that is, not on the combustion side).

The oil eventually falls through the oil into a pan at the bottom of the engine. The oil pump sucks the oil up, pushes it through a filter (so bits of metal are not pumped back into the engine), and is then sprayed on the moving parts again.

## 181.5 Challenges

With a piston engine, there are a lot of things that can go wrong. Let's enumerate a few:

- *The seal around the piston leaks*. Mechanics say "We aren't getting any compression." The cylinder doesn't get much power to the drive train.
- *The valves open or the spark plug fires at the wrong time.* This is known as a timing problem.
- *The spark plug doesn't make a spark*. The spark plug has two prongs of metal and electrons jump from one to the other. For a good spark, the prongs need to be a very precise distance apart. Sometimes you need to bend one of the prongs to get the right gap. This is known as *gapping*.
- *The mix of fuel and oxygen is wrong*. If there is too much fuel and not enough oxygen (so not all the fuel burns), we say the mix is too rich. If there is not enough fuel (so the pressure created by the burn is as high as possible), we say the mix is too lean.

## 181.6 How We Measure Engines

If you look up the specs on an engine, you will see the following:

- The number of cylinders
- The cylinder bore, which is the diameter of the cylinder
- The piston stroke, which is the distance the piston travels in the cylinder
- The compression ratio, which is the ratio between the maximum volume of the combustion change and the minimum volume of the combustion chamber
- What fuel it runs on

The difference between the minimum and maximum value of the cylinder is known as its *displacement*. The displacement represents the volume of air/fuel sucked into the intake valve before the compression begins.

We often talk about the displacement of the entire engine, which is the cylinder's dis-

placement times the number of cylinders. The displacement of an engine can give you a good idea of how much power it can produce.

For motorcycles, the displacement is often part of the name. For example, the Kawasaki Ninja 650 has about 650 cubic centimeters of displacement.

## 181.7 The Ford Model T and Ethanol

The Ford Model T was the first mass-produced car. It came out in 1908 and remained in production until 1927. It had a four-cylinder engine that would run on ethanol, benzene, or kerosene. For the purposes of this exercise, let's assume you are running yours on ethanol.

A molecule of ethanol has 2 carbon atoms, 6 hydrogen atoms, and 1 oxygen atom. The oxygen in the atmosphere is $O_2$. When one molecule of ethanol combines with 3 molecules of $O_2$, 2 molecules of $CO_2$ and 3 molecules of $H_2O$ are created. A large amount of heat is also created — 1330 kilojoules for every mole of ethanol burned.

The engine block is usually very hot once the engine has been running. That is important because the ethanol will be completely vaporized at that temperature.

In any sample of air, 21

## *Exercise 480* Fuel Mix for the Model T

*Working Space*

On the Model T, a carburetor mixed the fuel and air before it went into the cylinder. The question to answer in this exercise is: How rich should the mix be at sea level (100 kPa)?

On the Wikipedia page for the Model T, we see the following facts:

- Cylinder bore: 9.525 cm
- Piston stroke: 10.16 cm

You can assume that the air/fuel mixture is 80° C before the pre-burn compression starts. (Thus, the ethanol, which boils at 78° C, is in its vapor phase.)

The questions, then, are:

- What is the displacement of a single cylinder?
- How many moles of gas (80° C and 100 kPa) will get sucked through the intake valve?
- How many moles of vaporized ethanol should be part of that?
- How many moles of $CO_2$ and $H_2O$ are created in each burn?
- How much heat is created in each burn?

(This exercise has a lot of steps, but nothing you don't already know. You will need the Ideal Gas Law to figure out how much many moles of air gets dragged into the cylinder.)

*Answer on Page 2074*

## 181.8 Compression Ratio

Most of the inefficiency of a motor is heat that escapes through the exhaust valve. If your piston stroke were long enough, you could keep increasing the volume (which would cool the gases inside) until the gases inside were the same temperature as the outside world. Then there would be no wasted heat in the exhaust.

For this reason, generally, engines with a higher compression ratios tend to waste less energy through the heat of the exhaust. The Model T had a compression ratio close to 4:1. Modern car engines typically have compression ratios between to 8:1 and 12:1.

Cars with especially high compression ratios often require fuels with a lot of kilojoules per mole — we say *high octane*. If the fuel does not have enough energy, the engine makes loud knocking noises, as the pistons do not have enough energy to push through their entire stroke.

It turned out that an easy way to boost the octane of the gasoline was to add a chemical called tetraethyl lead. Gasoline containing tetraethyl lead was known at "Leaded Gasoline", and was intended to prevent the knocking. It is difficult to overstate the damages caused by putting large amounts of lead in the air. Gradually, starting with Japan in 1986, every country in the world has banned leaded gasoline.

## 181.9 The Choke and Direct Fuel Injection

Most cars built before 1990 will have a carburetor, which ensures that the ratio between fuel and oxygen is constant, regardless of the amount of fuel released by the throttle.

If you go to start an old car on a cold morning, the cold engine will not vaporize the fuel properly, and the engine may not have enough power to start. For this reason, most carburetors have a *choke value* that makes the mix richer. (If you pull the choke valve, be sure to push it back after the engine warms up.)

The carburetor was a common source of engine problems and inefficiencies. Starting in the 1990s, car engines started using direct fuel injection. Air still came in through the intake manifold, but fuel was sprayed directly into the cylinder by a fuel injection system.

In modern cars, the fuel injection system is controlled by a computer (an *Engine Control Module* or ECM), which delivers the fuel at the perfect time with the perfect amount based on environmental variables like the temperature of the engine and the barometric pressure (usually related to that altitude at which the engine is operating).

# Chapter 182

# u-Substitution

U-Substitution, also known as the method of substitution, is a technique used to simplify the process of finding antiderivatives and integrals of complicated functions. The method is similar to the chain rule for differentiation in reverse.

Suppose we have an integral of the form:

$$\int f(g(x)) \cdot g'(x)\, dx \tag{182.1}$$

The u-substitution method suggests letting a new variable $u$ be equal to the inside function $g(x)$, i.e.,

$$u = g(x) \tag{182.2}$$

Next, the differential of $u$, $du$, is given by:

$$du = g'(x)\, dx \tag{182.3}$$

Substituting $u$ and $du$ back into the integral gives us a simpler integral:

$$\int f(u)\, du \tag{182.4}$$

This new integral can often be simpler to evaluate. Once the antiderivative of $f(u)$ is found, we can substitute $u = g(x)$ back into the antiderivative to get the antiderivative of the original function in terms of $x$.

The method of u-substitution is a powerful tool for evaluating integrals, especially when combined with other techniques like integration by parts, partial fractions, and trigonometric substitutions.

**Example**: Find $\int 2x^2 \cos\left(x^3 - 3\right) dx$.

**Solution**: Integrating $\cos\left(x^3 - 3\right)$ isn't so straightforward, so let's try the substitution

$u = x^3 - 3$. Then:

$$du = 3x^2 \, dx$$

We don't have $3x^2$ in the integral, but we do have $2x^2$:

$$\frac{2}{3} du = 2x^2 \, dx$$

Substituting:

$$\begin{aligned} \int 2x^2 \cos\left(x^3 - 3\right) dx &= \int \frac{2}{3} \cos(u) \, du \\ &= \frac{2}{3} \sin(u) + C \end{aligned}$$

Now that we have the antiderivative of $f(u)$, we can back-substitute in for $u$:

$$\frac{2}{3} \sin(u) + C = \frac{2}{3} \sin\left(x^3 - 3\right) + C$$

We can check our answer by taking its derivative: we should get the original integrand back:

$$\begin{aligned} \frac{d}{dx}\left[\frac{2}{3} \sin\left(x^3 - 3\right) + C\right] &= \frac{2}{3} \cos\left(x^3 - 3\right) \cdot \left[\frac{d}{dx}\left(x^3 - 3\right)\right] \\ &= \frac{2}{3} \cos\left(x^3 - 3\right) \cdot \left(3x^2\right) = 2x^2 \cos\left(x^3 - 3\right) \end{aligned}$$

Sometimes, the right substitution takes a little thinking. Consider the following example:

**Example**: Find $\int \sqrt{x^2 - 1} \, x^5 \, dx$.

**Solution**: We can guess that $u = x^2 - 1$ could be an appropriate substitution, as that is what is under the square root. What to do with $x^5$? First, let's look at how the u-substitution for $x^2 - 1$ works out:

$$\begin{aligned} u &= x^2 - 1 \\ du &= 2x \, dx \\ \frac{du}{2} &= x \, dx \end{aligned}$$

Then we will need to use one of the x's in $x^5$ for the square root u-substitution. What can we do with the remaining $x^4$? Well, we see that if $u = x^2 - 1$, then $u + 1 = x^2$ and therefore $(u + 1)^2 = x^4$. Substituting this all in:

$$\int \sqrt{x^2 - 1} \, x^5 \, dx = \int x^4 \sqrt{x^2 - 1} \, x \, dx$$

$$= \frac{1}{2}\int (u+1)^2 \sqrt{u}\, du$$

We can expand this to find the antiderivative:

$$= \frac{1}{2}\int \left(u^2+2u+1\right) u^{1/2}\, du = \frac{1}{2}\int u^{5/2}+2u^{3/2}+u^{1/2}\, du$$

$$= \frac{1}{2}\left[\frac{2}{7}u^{7/2}+\frac{4}{5}u^{5/2}+\frac{2}{3}u^{3/2}\right]+C$$

$$= \frac{1}{7}u^{7/2}+\frac{2}{5}u^{5/2}+\frac{1}{3}u^{3/2}+C$$

$$= \frac{1}{7}\left(x^2-1\right)^{7/2}+\frac{2}{5}\left(x^2-1\right)^{5/2}+\frac{1}{3}\left(x^2-1\right)^{3/2}+C$$

## *Exercise 481* Indefinite Integrals and u-substitution

*Working Space*

Use u-substitution to evaluate the following indefinite integrals. Confirm your answer by taking the derivative of the result.

1. $\int \sin x \sqrt{1 + \cos x}\, dx$
2. $\int \frac{\cos(\pi/x)}{x^2}\, dx$
3. $\int 2x^2 \left(9 - x^3\right)^{2/3}\, dx$
4. $\int 3x^2 \sqrt{1 + x}\, dx$
5. $\int \frac{3x^2}{x^3 - 1}\, dx$

*Answer on Page 2076*

## 182.1 The Substitution Rule for Definite Integrals

How to we use u-substitution for definite integrals? We will apply the fundamental theorem of calculus to answer this question. We define $f$ and $F$ such that $F$ is the antiderivative

of $f$. Then:

$$\int_a^b f(g(x)) \cdot g'(x)\,dx = F(g(x))|_a^b = F(g(b)) - F(g(a))$$

This represents the method of finding the indefinite antiderivative and evaluating from the original limits of integration.

We can also see that:

$$F(g(b)) - F(g(a)) = F(u)|_{g(a)}^{g(b)} = \int_{g(a)}^{g(b)} f(u)\,du$$

Therefore, if $g'$ is continuous on $[a, b]$ and $f$ is continuous on the range of $u = g(x)$, then:

$$\int_a^b f(g(x)) \cdot g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

This represents *changing the limits of integration* into the new variable, $u$, then evaluating the integral. While the second method is preferable, both methods yield the same answer.

**Example**: Evaluate $\int_0^5 \sqrt{3x+1}\,dx$ using both methods outlined above.

**Solution**: We start with the first method. We will use the substitution $u = 3x + 1$, and therefore $du/3 = dx$:

$$\int_0^5 \sqrt{3x+3}\,dx = \frac{1}{3}\int_{x=0}^{x=5} \sqrt{u}\,du$$

(We write the limits as $x = \cdots$ to remind us the limits are for $x$, not $u$.)

$$\frac{1}{3}\int_{x=0}^{x=5} \sqrt{u}\,du = \frac{1}{3}\left[\frac{2}{3}u^{3/2}\right]_{x=0}^{x=5}$$

Now we substitute back in for $u$ and evaluate:

$$\frac{2}{9}(3x+1)^{3/2}|_0^5 = \frac{2}{9}(16)^{3/2} - \frac{2}{9}(1)^{3/2}$$

$$= \frac{2}{9}(64-1) = \frac{2\cdot 63}{9} = 14$$

Let's compare this to the second, preferred method. We already know the $u$-substitution we'll make, so next we need to find $g(0)$ and $g(5)$ (recall that we choose $u$ such that $u = g(x)$):

$$g(x) = 3x + 1$$

$$g(0) = 1$$
$$g(5) = 16$$

Now we can make our substitution *and* change the limits of integration:

$$\int_0^5 \sqrt{3x+1}\,dx = \frac{1}{3}\int_1^{16} \sqrt{u}\,du$$
$$= \frac{1}{3}\left[\frac{2}{3}u^{3/2}\right]_1^{16} = \frac{2}{9}\left[16^{3/2} - 1^{3/2}\right]$$
$$= \frac{2}{9}(64-1) = 14$$

With the second method, we get the same answer in fewer steps.

## *Exercise 482* Definite Integrals and u-substitution

Use u-substitution to evaluate the following definite integrals.

*Working Space*

1. $\int_0^{\pi/2} \cos x \sin(\sin x)\, dx$
2. $\int_0^{13} \frac{1}{\sqrt[3]{(1+2x)^2}}\, dx$
3. $\int_1^2 \frac{e^{1/x}}{x^2}\, dx$
4. $\int_0^{\pi/6} \frac{\sin x}{\cos^2 x}\, dx$
5. $\int_0^4 \frac{x}{\sqrt{1+2x}}\, dx$

*Answer on Page 2078*

Chapter 183

# Calculus with Polar Coordinates

We have been working in Cartesian coordinates, which are rectangular, with $x$ representing the horizontal position and $y$ representing the vertical position. As we have talked about, you can use polar coordinates to represent circular points. To review another way to represent a position in 2D space is with **polar coordinates**. The first number and dependent variable is $r$, which represents how far the point is from the origin. The second number is $\theta$, which represents the degrees of rotation from the $x$ axis (see Figure-183.1).

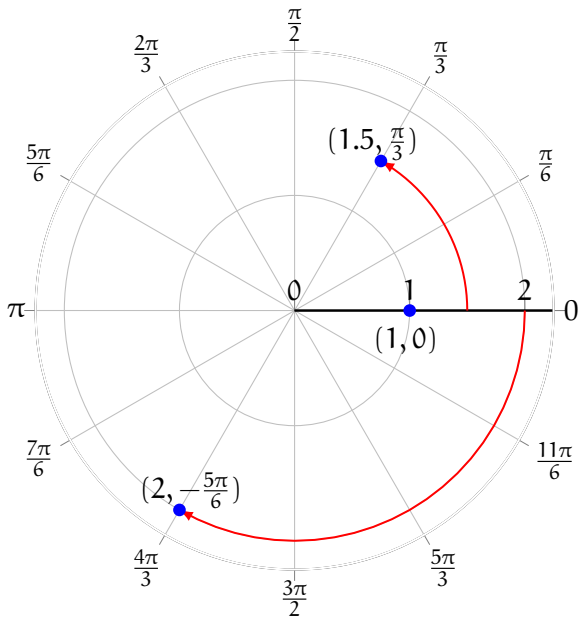



Figure 183.1: Polar coordinates give a degree of rotation, $\theta$, and a distance from the origin, $r$, in the form of $(r, \theta)$

## 183.1 Derivatives of Polar Functions

Consider the cardioid $r = 2 + \sin\theta$ (see Figure 183.2). What is the slope of the line tangent to the curve at $\theta = \frac{\pi}{2}$?

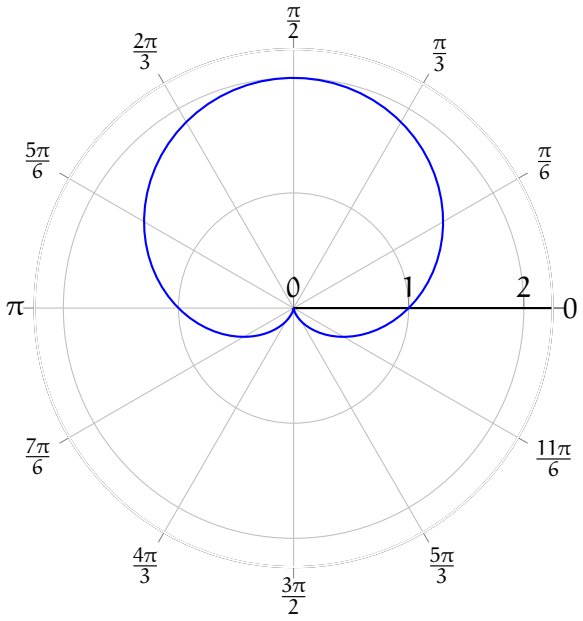



Figure 183.2: $r = 2 + \sin\theta$

From a visual inspection, we can guess that the slope of the tangent line is zero. Let's prove this mathematically:

First, recall that to convert polar coordinates to Cartesian coordinates, we can use the trigonometric identities:

$$x = r\cos\theta$$

$$y = r\sin\theta$$

So, we can write the parametric equation:

$$x = [2 + \sin\theta]\cos\theta$$

$$y = [2 + \sin\theta]\sin\theta$$

Recall from parametric equations that we can use implicit differentiation to find $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Finding $\frac{dy}{d\theta}$ and $\frac{dx}{d\theta}$:

$$\frac{dy}{d\theta} = \frac{d}{d\theta}\left(2\sin\theta + \sin^2\theta\right) = 2\cos\theta + 2\sin\theta\cos\theta$$

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(2\cos\theta + \sin\theta\cos\theta) = \cos^2\theta - \sin^2\theta - 2\sin\theta$$

Substituting $\theta = \frac{\pi}{2}$, we find that:

$$\frac{dy}{d\theta} = 2(0) + 2(1)(0) = 0$$

$$\frac{dx}{d\theta} = (0)^2 - (1)^2 - 2(1) = -3$$

Therefore,

$$\frac{dy}{dx} = \frac{0}{-3} = 0$$

Which is the result we expected from examining the graph of $r = 2 + \sin\theta$. So, in general for polar equations,

**Tangent to a Polar Function**

For a polar function, $r = f(\theta)$, the slope of a tangent line is given by:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Where $y = r \cdot \sin\theta$ and $x = r \cdot \cos\theta$.

### *Exercise 483*

[This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC exam.] What is the slope of the line tangent to the polar curve $r = 1 + 2\sin\theta$ at $\theta = 0$?

*Working Space*

*Answer on Page 2079*

### Exercise 484

Find the slope of the tangent line to the given polar curve at the value of $\theta$ specified. Use this to write an equation for the tangent line in Cartesian coordinates.

1. $r = \frac{2}{3}\cos\theta$, $\theta = \frac{\pi}{6}$
2. $r = \frac{1}{2\theta}$, $\theta = \frac{\pi}{2}$
3. $r = 2 + 3\cos\theta$, $\theta = \frac{2\pi}{3}$

*Working Space*

*Answer on Page 2079*

## 183.2 Integrals of Polar Functions

Similar to Cartesian functions, an integral of a polar function tells us the area within the function. We say "within" as opposed to "under" because a polar function describes how far from the origin the graph is based on the angle. Consider the graph of $r = 2\sin\theta$ (figure 183.3). Geometrically, we expect the area inside the curve to be $\pi r^2 = \pi$. However, this is not the result we get from directly integrating the function (we only integrate from $\theta = 0$ to $\theta = \pi$ because the circle is complete when $\theta$ reaches $\pi$):

$$\int_0^{\pi} 2\sin\theta \, d\theta = -2\cos\theta\Big|_{\theta=0}^{\theta=\pi}$$

$$= -2\left[\cos\pi - \cos 0\right] = -2\left[-1 - 1\right] = 4 \neq \pi$$

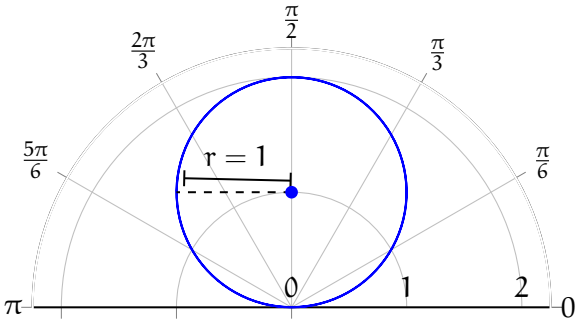



Figure 183.3: The graph of $r = 2\sin\theta$ is a circle of radius 1 centered at $(1, \frac{\pi}{2})$

Clearly something else is happening here. We can just take the integral of a Cartesian function because the area of a rectangle is the base times the height. When integrating Cartesian functions, the base is given by the $dx$ and the height by the function, $f(x)$. In polar coordinates, the integral sweeps across a $\theta$ interval, making a wedge, not a rectangle.

Let us consider a generic polar function, shown in figure 183.4

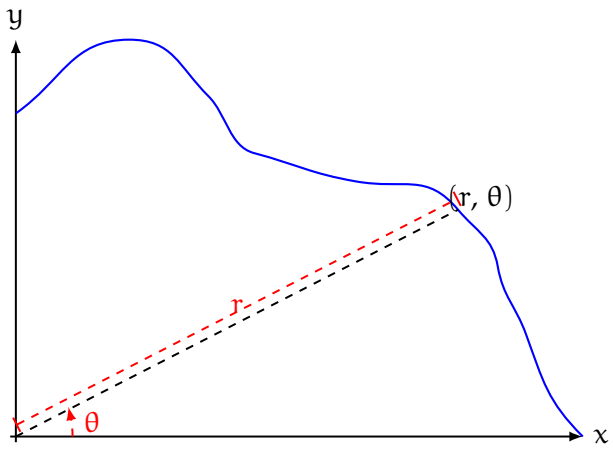



Figure 183.4: A generic polar function

Suppose we are interested in a specific region, bounded by $a \leq \theta \leq b$ (see figure 183.5).

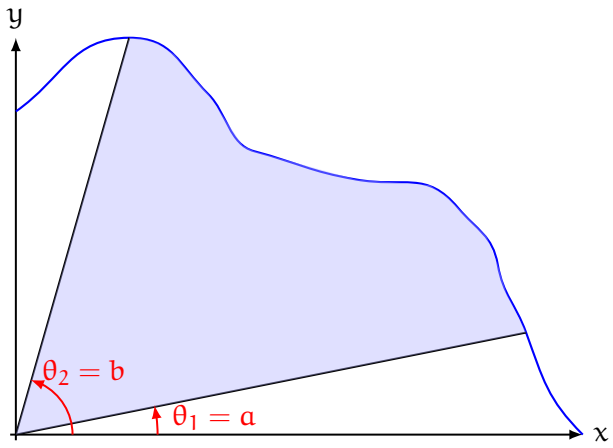



Figure 183.5: A generic polar with a region from $\theta = a$ to $\theta = b$ highlighted

We can divide the region into many small sectors. Then, each small sector has a central angle $\Delta\theta$ and a radius $r(\theta_i^*)$, where $\theta_{i-1} < \theta_i^* < \theta_i$ (see figure 183.6).

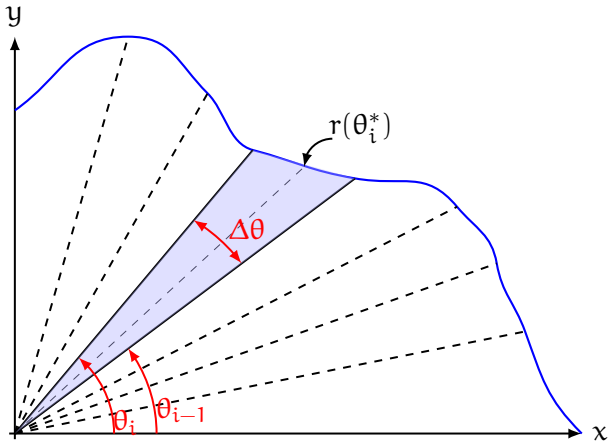



Figure 183.6: A single sector from $\theta_{i-1}$ to $\theta_i$

What is the area of the $i^{th}$ sector? Recall from the chapter on circles that the area of a sector with angle $\theta$ and length $r$ is $A = \frac{1}{2}r^2\theta$. Substituting, we see the area of the $i^{th}$ sector is:

$$A_i = \frac{1}{2}\left[r(\theta_i^*)\right]^2 d\theta$$

Therefore, the total area of the whole sector from $\theta = a$ to $\theta = b$ is the limit as the number of sectors approaches infinity of sum of the areas of all the small sectors:

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{2}\left[r(\theta_i^*)\right]^2 \Delta\theta$$

Does this look familiar? It is the definition of an integral!

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{2}\left[r(\theta_i^*)\right]^2 \Delta\theta = \int_a^b \frac{1}{2}\left[r(\theta)\right]^2 d\theta$$

**Area of a Polar Function**

The area of a polar function is given by the integral

$$\int_a^b \frac{1}{2}r^2 d\theta$$

Where $r$ is a function of $\theta$.

We can check this with the example from the beginning of the section. Recall that the

polar function $r = 2\sin\theta$ graphs a circle with a radius of 1. Therefore, we expect the area enclosed by the graph of $r = 2\sin\theta$ from $\theta = 0$ to $\theta = \pi$ to be $\pi$:

$$A = \frac{1}{2}\int_0^{\pi} [2\sin\theta]^2 \, d\theta$$

$$A = 2\int_0^{\pi} \sin^2\theta \, d\theta = 2\int_0^{\pi} \left[\frac{1-\cos 2\theta}{2}\right] d\theta$$

$$A = \int_0^{\pi} [1-\cos 2\theta] \, d\theta = \left[\theta - \frac{1}{2}\sin 2\theta\right]_{\theta=0}^{\theta=\pi}$$

$$A = [\pi - 0] - [0 - 0] = \pi$$

Which is the expected result, confirming our formula for the area within a polar function.

**Example**: The graph of $r = 3\sin 2\theta$ is shown below. What is the total area enclosed by the graph?

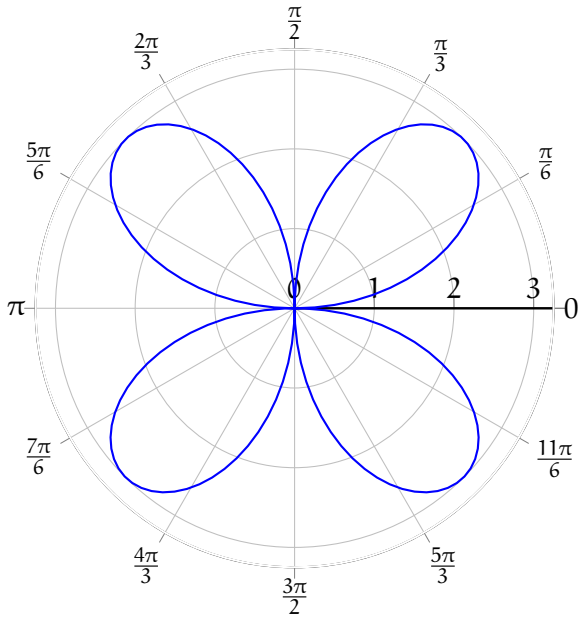



Figure 183.7: $r = 3\sin 2\theta$

**Solution**: Since each lobe is symmetric to the others, we can find the area of one lobe and multiply it by four. To find the area of one lobe, we need to determine an interval for $\theta$ that defines one lobe. You can imagine each lobe being draw out from the center and then back in. So, we will find where $r = 0$:

$$0 = 3\sin 2\theta$$

$$\sin 2\theta = 0$$

$$2\theta = n\pi$$
$$\theta = \frac{n\pi}{2}$$

Taking the first two solutions, $\theta = 0$ and $\theta = \frac{\pi}{2}$, as our limits of integration, we see that the area of one lobe is:

$$A_{lobe} = \frac{1}{2}\int_0^{\pi/2} [3\sin 2\theta]^2 \, d\theta$$
$$A_{lobe} = \frac{9}{2}\int_0^{\pi/2} \sin^2 2\theta \, d\theta$$

Applying the half-angle formula $\sin^2\theta = \frac{1-\cos 2\theta}{2}$, we see that:

$$A_{lobe} = \frac{9}{2}\int_0^{\pi/2} \frac{1-\cos 4\theta}{2} \, d\theta = \frac{9}{4}\int_0^{\pi/2} 1 - \cos 4\theta \, d\theta$$
$$= \frac{9}{4}\left[\theta - \frac{1}{4}\sin 4\theta\right]_{\theta=0}^{\theta=\pi/2} = \frac{9}{4}\left(\frac{\pi}{2} - 0\right) - \frac{9}{4}\left(\frac{1}{4}\right)(\sin 2\pi - \sin 0)$$
$$= \frac{9\pi}{8} - \frac{9}{16}(0) = \frac{9\pi}{8}$$

Since the area of one lobe is $\frac{9\pi}{8}$, the area of all four lobes is $\frac{9\pi}{2}$.

### 183.2.1 Area between polar curves

Consider the circle $r = 6\sin\theta$ and the cardioid $r = 2 + 2\sin\theta$. How can we find the area that lies inside the circle, but outside the cardioid (see figure 183.8)? First, let's find where these curves intersect. This will determine the limits of any integrals we take.

$$6\sin\theta = 2 + 2\sin\theta$$
$$3\sin\theta = 1 = \sin\theta$$
$$2\sin\theta = 1$$
$$\sin\theta = \frac{1}{2}$$
$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

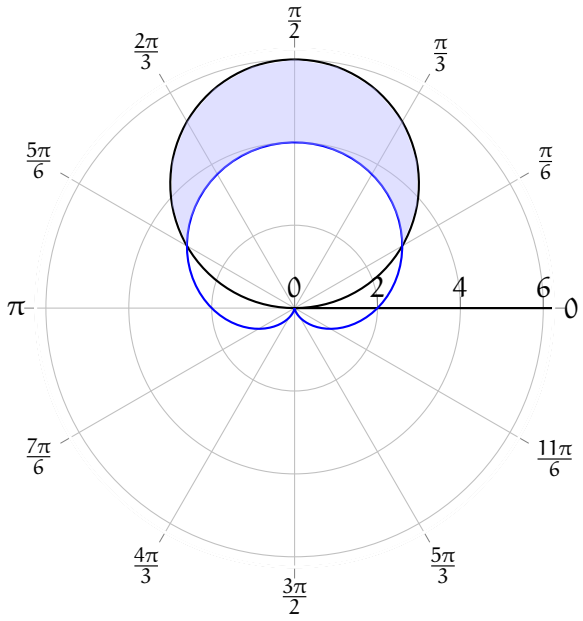



Figure 183.8: The area inside $r = 6\sin\theta$ and outside of $r = 2 + 2\sin\theta$ is highlighted

Recall that for Cartesian functions, to find the area between two curves, we subtract the area under the lower curve from the total area under the higher curve. In polar coordinates, we want to subtract the area in the inner curve from the total area in the outer curve. In this case, the outer curve is $r = 6\sin\theta$ and the inner curve is $r = 2 + 2\sin\theta$. We have already found our limits of integration ($\frac{\pi}{6} \le \theta \le \frac{5\pi}{6}$), so we set up and evaluate our integral:

$$A_{between} = \frac{1}{2}\int_{\pi/6}^{5\pi/6} [4\sin\theta]^2 \, d\theta - \frac{1}{2}\int_{\pi/6}^{5\pi/6} [2 + 2\sin\theta]^2 \, d\theta$$

$$= \frac{1}{2}\int_{\pi/6}^{5\pi/6} \left[16\sin^2\theta - 4 - 8\sin\theta - 4\sin^2\theta\right] \, d\theta$$

$$= \int_{\pi/6}^{5\pi/6} \left[6\sin^2\theta - 4\sin\theta - 2\right] \, d\theta$$

$$= \int_{\pi/6}^{5\pi/6} [3(1 - \cos 2\theta) - 4\sin\theta - 2] \, d\theta$$

$$= \int_{\pi/6}^{5\pi/6} [1 - 3\cos 2\theta - 4\sin\theta] \, d\theta$$

$$= \left[\theta - \frac{3}{2}\sin 2\theta + 4\cos\theta\right]_{\theta=\pi/6}^{\theta=5\pi/6}$$

$$= \left[\frac{5\pi}{6} - \frac{\pi}{6}\right] - \left[\frac{3}{2}\sin\left(2\cdot\frac{5\pi}{6}\right) - \frac{3}{2}\sin\left(2\cdot\frac{\pi}{6}\right)\right] + \left[4\cos\frac{5\pi}{6} - 4\cos\frac{\pi}{6}\right]$$

$$= \frac{4\pi}{6} - \left[\frac{3}{2} \cdot -\frac{\sqrt{3}}{2} - \frac{3}{2} \cdot \frac{\sqrt{3}}{2}\right] + \left[4 \cdot -\frac{\sqrt{3}}{2} - 4 \cdot \frac{\sqrt{3}}{2}\right]$$

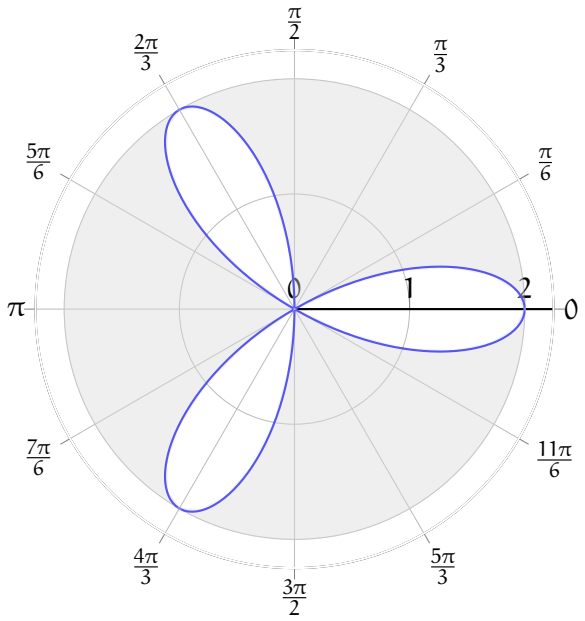
$$= \frac{2\pi}{3} + \frac{3\sqrt{3}}{2} - 4\sqrt{3} = \frac{2\pi}{3} + \frac{3\sqrt{3} - 8\sqrt{3}}{2} = \frac{2\pi}{3} - \frac{5\sqrt{3}}{2}$$

(*Note*: Because these polar functions are symmetric about the y-axis, we could have also taken the integral from $\theta = \frac{\pi}{6}$ to $\theta = \frac{\pi}{2}$ and doubled the result. We leave it as an exercise for the student to show this works.)

## *Exercise 485*

*Working Space*

[This question was originally presented as a multiple-choice, calculator- allowed problem on the 2012 AP Calculus BC exam.] The figure below shows the graphs of polar curves $r = 2\cos 3\theta$ and $r = 2$. What is the sum of the areas of the shaded regions to three decimal places?




*Answer on Page 2080*

### Exercise 486

Find the area of the region bounded by the given curve and angles.

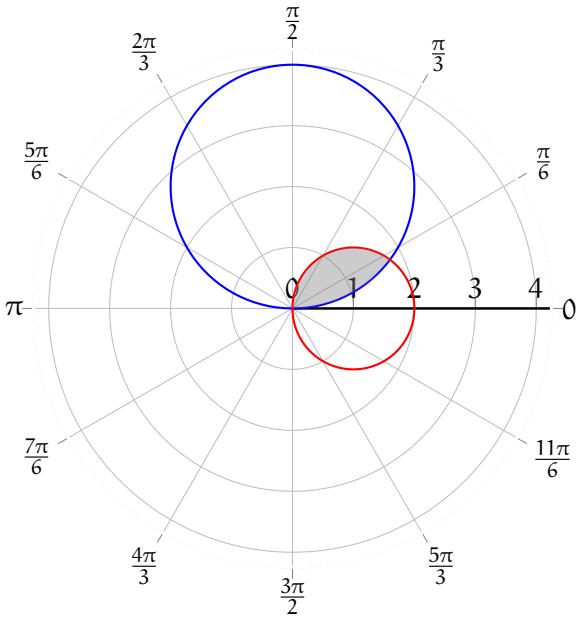
1. $r = e^{\theta/2}$, $\pi/4 \leq \theta \leq \pi/2$
2. $r = 2\sin\theta + \cos 2\theta$, $0 \leq \theta \leq \pi$
3. $r = 4 + 3\sin\theta$, $-\pi/2 \leq \theta \leq \pi/2$

*Working Space*

*Answer on Page 2081*

## Exercise 487

Find the area of the region that lies between the curves $r = 4\sin\theta$ and $r = 2\cos\theta$. A graph is shown below.




Working Space

Answer on Page 2082

CHAPTER 184

# Differential Equations

Differential equations are equations involving an unknown function and its derivatives. They play a crucial role in mathematics, physics, engineering, economics, and other disciplines due to their ability to describe change over time or in response to changing conditions.

## 184.1 Ordinary Differential Equations

An ordinary differential equation (ODE) involves a function of a single independent variable and its derivatives. The order of an ODE is determined by the order of the highest derivative present in the equation. An example of a first-order ODE is:

$$\frac{dy}{dx} + y = x \tag{184.1}$$

Here, $y$ is the function of the independent variable $x$, and $\frac{dy}{dx}$ represents its first derivative.

A real-world example of the application of differential equations is an oscillating spring (or any harmonic motion). When a spring is stretched, the restoring force (the force pulling or pushing it back to its neutral position) is proportional to the distance by which the spring has been stretched (see Figure 184.1). Mathematically, we say that

$$\text{restoring force} = -kx$$

where $k$ is the positive spring constant (the stiffer a spring, the greater $k$).

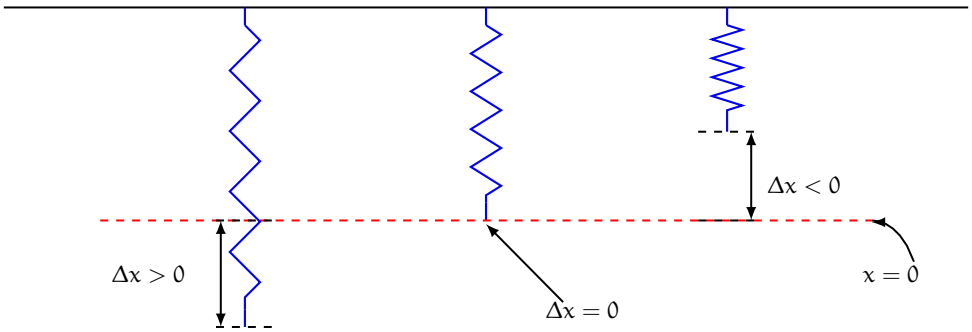



Figure 184.1: A spring can have a positive or negative displacement

Recall that Newton's Second Law tells us that force is equal to mass times acceleration, and that acceleration is the second derivative of position. We can then write the differential equation:

$$m\frac{d^2x}{dt^2} = -kx$$

This is called a **second-order differential equation**, because it involves second-order derivatives. The order of a differential equation is the same as the highest order of derivative in the equation. We can further rewrite the equation to isolate the second derivative:

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

In everyday language, this is saying that the second derivative is proportional to the original function, just negative. There are two trigonometric functions that have this property, take a second to see if you remember and write down your guess.

The sine and cosine functions both have the property $\frac{d^2x}{dt^2} \propto -x(t)$ (recall that $\propto$ means "proportional to").

**Example**: Assuming $x(t)$ is a sine function, solve the second-order differential equation $\frac{d^2x}{dt^2} = \frac{-k}{m}x$.

**Solution**: Let $x(t) = \sin Ct$. Then $\frac{dx}{dt} = C\cos Ct$ and $\frac{d^2x}{dt^2} = -C^2 \sin t$. This implies that $C^2 = \frac{k}{m}$ and $C = \pm\sqrt{\frac{k}{m}}$. So, a solution to the differential equation $\frac{d^2x}{dt^2} = \frac{-k}{m}x$ is $x(t) = \sin\sqrt{\frac{k}{m}}t$.

### 184.1.1 Population Growth

Another real-world application of differential equations is modeling population growth. Under ideal conditions (unlimited food, no predators, disease-free, etc.), the population of a species grows at a rate proportional to the current population size. We can identify two variables:

$$t = \text{ time (the independent variable)}$$

$$P = \text{ the number pf individuals in the population (the dependent variable)}$$

So, what is the rate of growth? Recall that a rate is change over time. In that case, the rate of growth is given by $\frac{dP}{dt}$. If the rate of growth is proportional to the population, then we can write a first-order differential equation:

$$\frac{dP}{dt} = kP$$

where k is a proportionality constant. This is called **natural growth** or **logarithmic growth**. To find a solution, we must answer the question: What function's derivative is a constant multiple of itself? Recall that we have seen that the derivative of the exponential function $e^{kt}$ is $ke^{kt}$. Setting $P(t) = Ce^{kt}$ (where C is some constant), we see that the derivative is $\frac{dP}{dt} = kCe^{kt} = kP(t)$ (see figure 184.2). You can determine C from initial conditions.

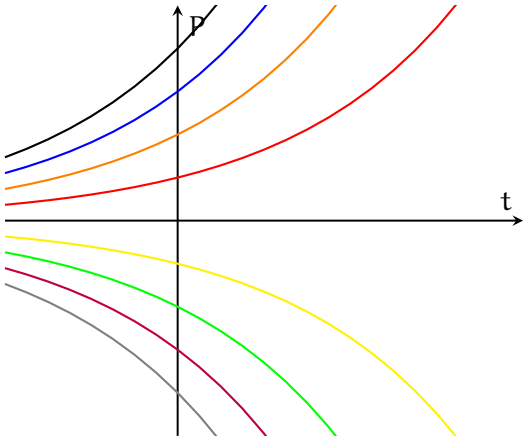



Figure 184.2: Several solutions to $\frac{dP}{dt} = kP$

**Example**: Suppose a population of bacteria has an initial population of 100 bacteria. If the bacteria's growth rate is given by $\frac{dP}{dt} = 2P$ (where t is in hours), how many bacteria are present after 4 hours?

**Solution**: We have seen that the solution to $\frac{dP}{dt} = 2P$ is $P(t) = Ce^{2t}$. We can then use the given initial condition to find C:

$$P(0) = 100 = Ce^{2\cdot 0} = C \cdot 1 = C$$

Which means that the complete solution is:

$$P(t) = 100e^{2t}$$

To answer the question, we need to find $P(4)$:

$$P(4) = 100e^{2\cdot 4} = 100e^{8} \approx 298096$$

As stated above, this model works well for populations under specific, ideal conditions. However, there are very few environments in which these conditions are met. Real animals suffer from disease, are hunted by predators, and have limited food supplies. Most environments have a maximum number of animals they can support, which ecologists call a **carrying capacity**. Let us call the carrying capacity of an environment M. So, the

population growth can be modeled by the logistic differential equation:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$

This is called a **logistic differential growth model**. Notice that if $P$ is small, then $\frac{dP}{dt} \approx kP$. This makes sense: If the population is very small compared to the carrying capacity, the conditions are nearly ideal, and so growth should be nearly ideal too. On the other hand, if the population ever goes *above* the carrying capacity, the $\frac{dP}{dt} < 0$ and the population will decrease back below the carrying capacity (see figure 184.3). Notice that if the initial population is $P_0 = M$, then $\frac{dP}{dt} = kP(1-1) = 0$ and the population is stable at $P(t) = M$. We call this an **equilibrium solution**. Can you logically find the other equilibrium solution?

If there are no animals to begin with, then there are none to reproduce, and $P(t) = 0$. This is the other equilibrium solution. Notice that when the population is in equilibrium, then the rate of change is zero. Mathematically, to find equilibrium solutions, we can set $\frac{dP}{dt} = 0$ and solve for $P$.

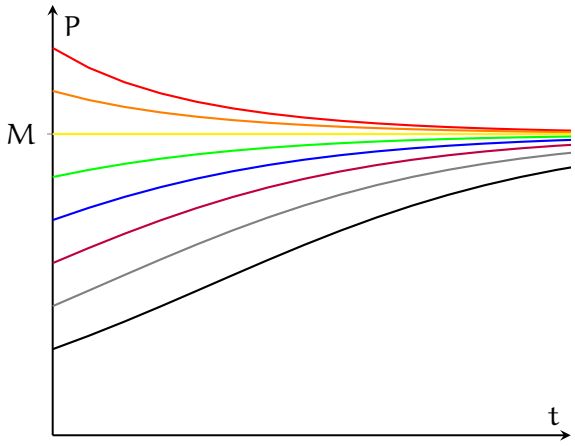



Figure 184.3: Several solutions to $\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$

### Exercise 488

A population is modeled by the differential equation $\frac{dP}{dt} = 1.2P\left(1 - \frac{P}{4200}\right)$.

1. What is the carrying capacity of the environment?
2. For what values of $P$ is the population increasing?
3. For what values of $P$ is the population decreasing?
4. What are the equilibrium solutions?

*Working Space*

*Answer on Page 2083*

### Exercise 489

[This problem was originally presented as a calculator-allowed, free response question on the 2012 AP Calculus BC exam.] Let $k$ be a positive constant. Which of the following is a logistic differential equation?

(a) $\frac{dy}{dt} = kt$
(b) $\frac{dy}{dt} = ky$
(c) $\frac{dy}{dt} = kt(1 - t)$
(d) $\frac{dy}{dt} = ky(1 - t)$
(e) $\frac{dy}{dt} = ky(1 - y)$

*Working Space*

*Answer on Page 2084*

### 184.1.2 Separable Differential Equations

Sometimes, differential equations can be explicitly solved. A first-order differential equation is separable if $\frac{dy}{dx}$ can be written as a function of $x$ times a function of $y$. Symbolically, a differential equation is separable if it takes the form

$$\frac{dy}{dx} = g(x)f(y)$$

The equations may be solvable by separating the $x$ from the $y$ and integrating each side. For our generic form, we can separate the variables thusly if $f(y) \neq 0$:

$$\frac{dy}{dx}\frac{1}{f(y)} = g(x)$$

$$\frac{1}{f(y)}dy = g(x)dx$$

Integrating both sides:

$$\int \frac{1}{f(y)}\,dy = \int g(x)\,dx$$

.

Let's look at the example $\frac{dy}{dx} = \frac{x^2}{y}$. We can separate the variables by multiplying both sides by $ydx$:

$$ydy = x^2dx$$

Integrating both sides:

$$\int y\,dy = \int x^2\,dx$$

$$\frac{1}{2}y^2 + C_1 = \frac{1}{3}x^3 + C_2$$

We can combine the constants by defining $C = C_2 - C_1$. Making this substitution and solving for $y$, we find:

$$y^2 = \frac{2}{3}x^3 + 2C$$

$$y = \sqrt{\frac{2}{3}x^3 + 2C}$$

Noting that $2C$ is also a constant (which we will call $K$ for convenience), we find the general solution is

$$y = \sqrt{\frac{2}{3}x^3 + K}$$

A graph showing the solution for several values of K is in figure 184.4.

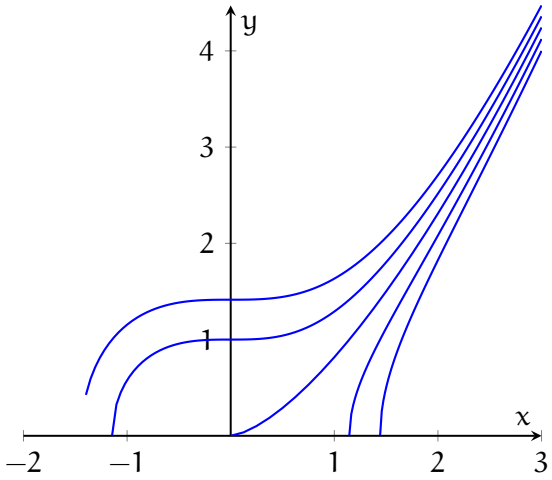



Figure 184.4: Several possible solutions to $\frac{dy}{dx} = \frac{x^2}{y}$

It is not always possible to solve for $y$ explicitly in terms of $x$. The practice problem below is an example of this.

### *Exercise 490*

Solve the differential equation $\frac{dy}{dx} = \frac{3x^2}{2y+\sin y}$.

*Working Space*

*Answer on Page 2084*

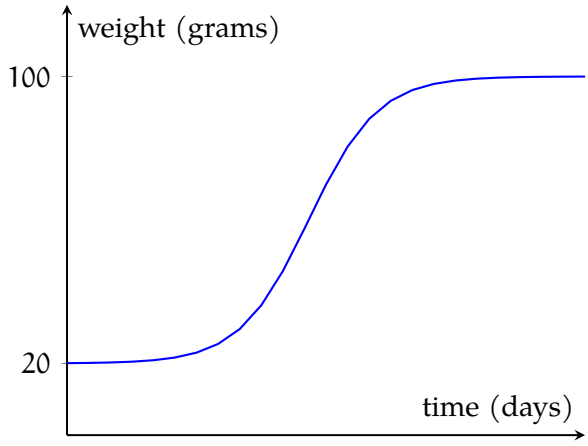
### Exercise 491

[This problem was originally presented as a calculator-allowed, free response question on the 2012 AP Calculus BC exam.] The rate at which a baby bird gains mass is proportional to the difference between its adult mass and its current mass. At time $t = 0$, when the bird is first weighed, its mass is 20 grams. If $B(t)$ is the mass of the bird, in grams, at time $t$ days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

Let $y = B(t)$ be the solution to the differential equation with initial condition $B(0) = 20$.

1. Is the bird gaining mass faster when it masses 40 grams or when it masses 70 grams? Explain your reasoning.
2. Find $\frac{d^2B}{dt^2}$ in terms of $B$. Use it to explain why the graph of $B$ cannot resemble the graph shown below.
3. Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.




*Working Space*

*Answer on Page 2084*

### *Exercise 492*

[This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC exam.] If $P(t)$ is the size of a population at time $t$, which of the following differential equations describes *linear* growth in the size of the population?

(a) $\frac{dP}{dt} = 200$
(b) $\frac{dP}{dt} = 200t$
(c) $\frac{dP}{dt} = 100t^2$
(d) $\frac{dP}{dt} = 200P$
(e) $\frac{dP}{dt} = 100P^2$

*Working Space*

*Answer on Page 2084*

## 184.2 Partial Differential Equations

Partial differential equations (PDEs), on the other hand, involve a function of multiple independent variables and their partial derivatives. An example of a PDE is the heat equation, a second-order PDE:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \tag{184.2}$$

In this equation, $u = u(x, t)$ is a function of the two independent variables $x$ and $t$, $\frac{\partial u}{\partial t}$ is the first partial derivative of $u$ with respect to $t$, and $\frac{\partial^2 u}{\partial x^2}$ is the second partial derivative of $u$ with respect to $x$.

Chapter 185

# Slope Fields

While separable differential equations are solvable, most differential equations are not separable. In fact, it is impossible to obtain an explicit formula as a solution to most differential equations. How do computers solve these, then? They start with a given quantity (usually initial conditions) and perform many small calculations to estimate the behavior of the solution. We can do this graphically with slope fields (also called direction fields), which allow us to visualize the family of solutions to the differential equation.

## 185.1 Drawing Slope Fields

When a differential equation is in the form

$$y' = f(x, y)$$

we can use the coordinates $(x, y)$ to determine the slope of a solution to the differential equation at that coordinate. Take $y' = x + y$ as an example. According to this differential equation, a solution that passes through the point $(1, 1)$ would have a slope of 2. We can represent this with a small tick of slope 2 at the $(1, 1)$ (see figure 185.1).

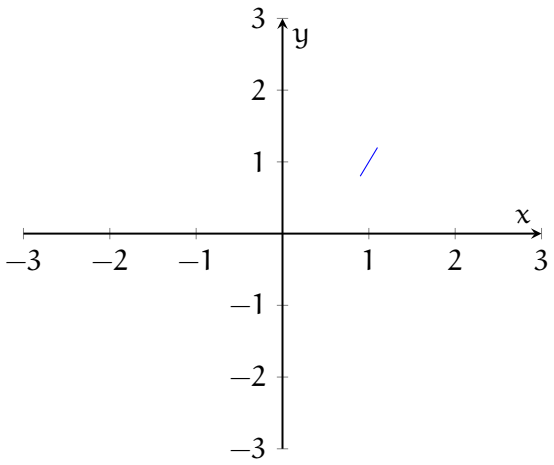



Figure 185.1: A solution to $y' = x + y$ that passes through $(1, 1)$ will have a slope of 2 at that point

Continuing, we want to choose coordinates that are easy to determine the slope. Notice that $y' = 0$ when $-x = y$, so let's go ahead and fill those ticks in (see figure 185.2):

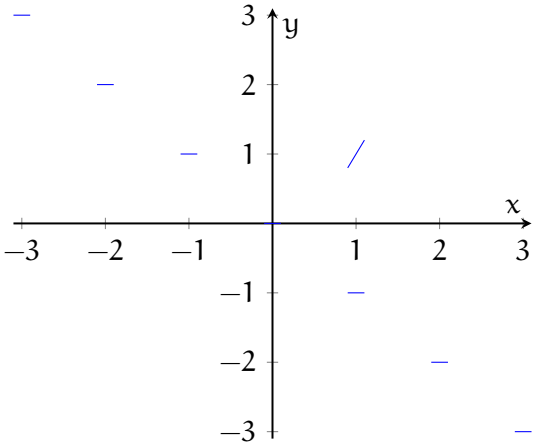



Figure 185.2: Solutions to $y' = x + y$ that lie on the line $y = -x$ will have a slope of 0.

We can repeat this process for all the coordinates shown, resulting in a slope field (see figure 185.3).

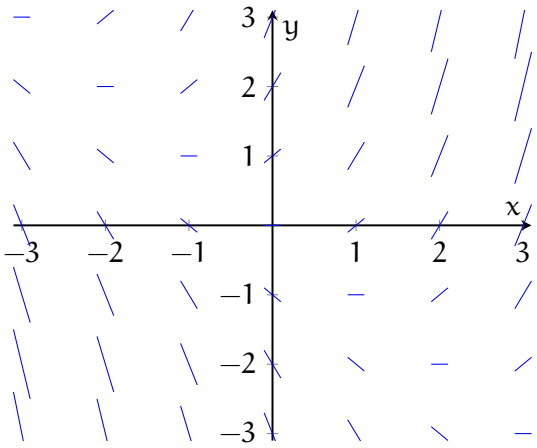



Figure 185.3: Slope field of $y' = x + y$

## 185.2 Sketching solutions on slope fields

If you are given an initial condition or a known point in the solution to the differential equation, you can begin sketching a curve on the slope field. Start at the given point and draw parallel to the nearby slopes. For example, suppose we know that particular solution to $y' = x + y$ passes through the point $(1, 0)$. Begin by extending the dash at $(1, 0)$ (see figure 185.4), changing the slope of your sketched solution to be approximately parallel to the nearby slopes (see figure 185.5).

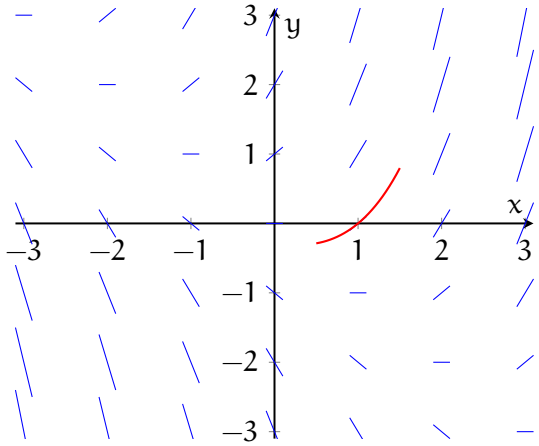



Figure 185.4: To begin sketching a solution to the differential equation, start at the point given as part of the solution

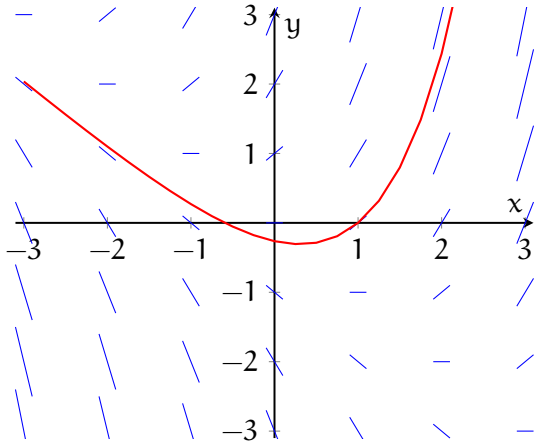



Figure 185.5: To sketch a solution to the differential equation, draw a function parallel to the nearby slopes that passes through the given point in the particular solution

While this method doesn't yield an exact, formulaic solution to the differential equation, it does allow us to visualize solutions and generally describe the behavior of any solutions. Sketching solutions in this way is logically similar to Euler's method for finding numerical approximations of solutions to differential equations, which we will discuss more in the next chapter.

## 185.3 Example: Application of Differential Equations to Electronics

Think back to the chapter on DC circuits. You learned that Ohm's Law relates voltage (electromotive force), current, and resistance for simple DC circuits:

$$V = IR$$

Simple resistors have a constant resistance, so once the voltage source (battery) is connected, the current is constant. There are other electronic components, such as inductors and capacitors, that behave differently. When current changes in an inductor, a voltage drop is induced across the inductor. This is described by the differential equation:

$$V = -L\frac{dI}{dt}$$

Where L is inductance, measured in henries (H), of the inductor. Consider, then, a circuit consisting of a constant-voltage battery, a fixed resistor, and an inductor (shown in figure 185.6). Since Kirchoff's Law states that the sum of the voltage drops across each component must equal the voltage supplied by the battery, we can write a differential equation to describe the circuit:

$$V = L\frac{dI}{dt} + RI$$

Where the current, I, is a function of time, t.

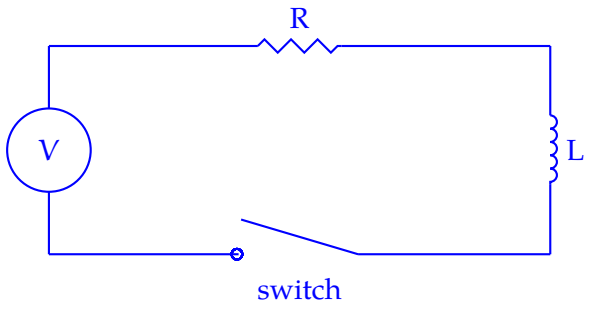



Figure 185.6: A simple circuit with a battery, resistor, inductor, and switch

**Example**: If the resistor is $12\Omega$, the inductance is 4H, and the battery supplies a constant voltage of $60V$:

1. Draw a slope field for the differential equation describing the current in the circuit.
2. Describe the expected behavior of the current over a long period of time.
3. Identify any equilibrium solutions.
4. If the initial current at $t = 0$ is $I(0) = 0$, sketch the particular solution to the differ-

ential equation on the slope field.

**Solution**: Substituting the given values into the differential equation and rearranging to isolate $\frac{dI}{dt}$, we get $\frac{dI}{dt} = 15 - 3I$. Notice that the current is not dependent on time. When the slope is only dependent on the value of the function (as in this case), we call this an **autonomous differential equation**. This means that the slope will be the same of all values of t for a given I. The slope field is shown in figure 185.7.

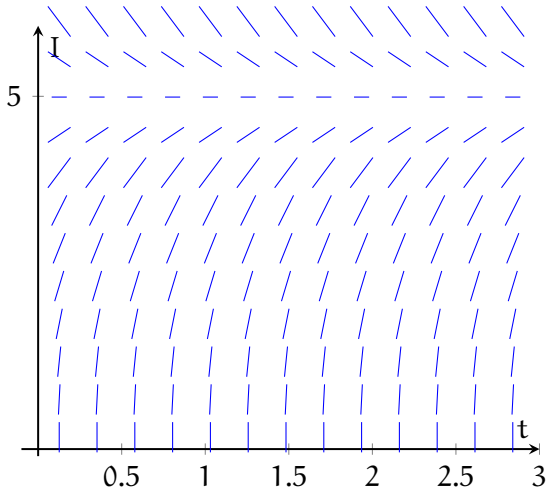



Figure 185.7: Slope field for the differential equation $\frac{dI}{dt} = 15 - 3I$

Examining the slope field, we see that the solutions tend towards $I(t) = 5$, which suggests that over an extended period of time, the current will approach 5 amperes. Similarly, if the initial current were 5 amperes, then the current would be constant at 5 amperes. Therefore, $I(t) = 5$ is an equilibrium solution. A sketch of the solution with $I(0) = 0$ is shown in figure 185.8.

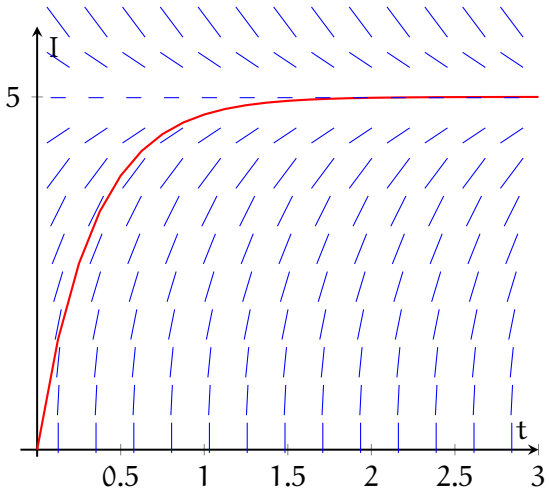



Figure 185.8: Slope field for the differential equation $\frac{dI}{dt} = 15 - 3I$

## 185.4 Practice

### *Exercise 493*

Sketch the slope field for the differential equation $y' = x + y^2$. Use your slope field to sketch a solution that passes through the point $(0, 0)$.

*Working Space*

*Answer on Page 2085*

# Chapter 186

# Euler's Method

How do computers approximate the solution to a differential equation that cannot be explicitly solved? Let's consider the differential equation

$$\frac{dy}{dx} = x + y \text{ with initial condition } y(0) = 1$$

.

This means the solution passes through the point $(0, 1)$. Additionally, the slope of the solution is $\frac{dy}{dx} = 0 + 1 = 1$ at that point. This means we can approximate the solution with the linear function $L(x) = x + 1$ (see figure 186.1). As you can see, near $(0, 1)$ the approximation is good, but as $x$ increases, the divergence between the actual solution and the approximation grows.

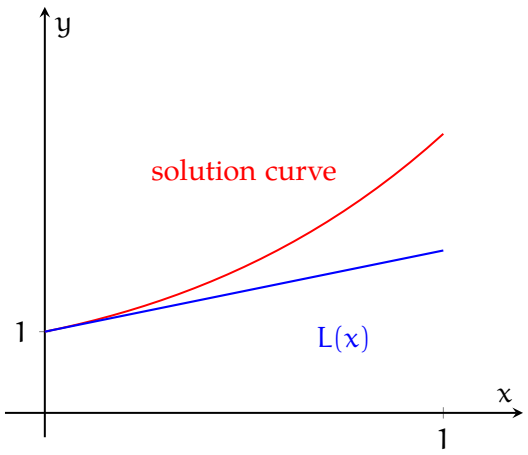



Figure 186.1: A first Euler approximation

How can we make a better approximation? Suppose we stop the first approximation at $x = 1.5$, re-evaluate $\frac{dy}{dx}$, and use that to make a second linear approximation. When $x = 0.5$, $L(x) = 0.5 + 1 = 1.5$. Taking the point $(0.5, 1.5)$, then $\frac{dy}{dx} = 0.5 + 1.5 = 2$. We can then write a second linear approximation, $L_2(x) = 2(x - 0.5) + 1.5 = 2x - 1 + 1.5 = 2x + 0.5$. As you can see (figure 186.2), this new approximation is closer than our first approximation. We call this an approximation with a step size of 0.5.

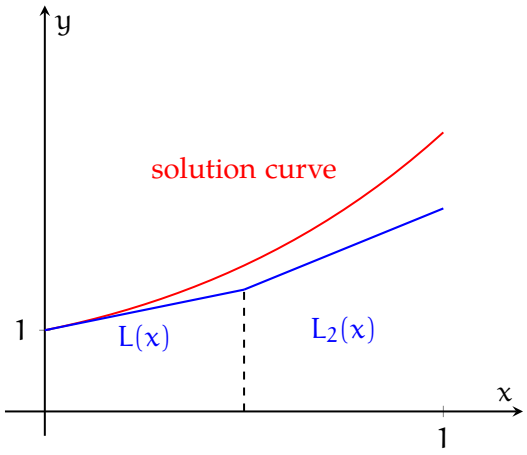



Figure 186.2: An Euler approximation with step size 0.5

We can improve this further by taking a step size of 0.25 (see figure 186.3). As the step size decreases and the step number increases, the approximation gets closer and closer to the true solution.

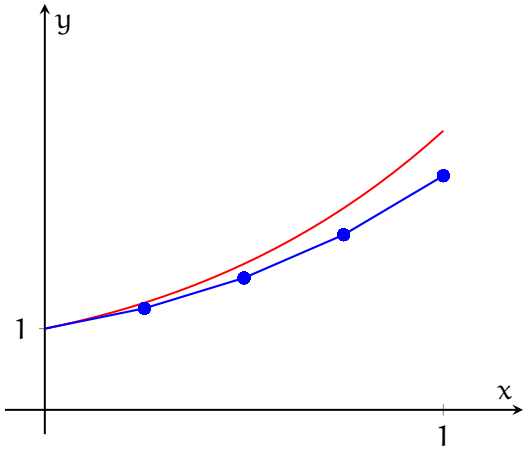



Figure 186.3: An Euler approximation with step size 0.25

In general, Euler's method is a numerical process similar to sketching a solution on a slope field. One begins at the given initial value, proceeds for a short step in the direction indicated by the slope field. You adjust the slope of your approximation based on the value of the slope field at the end of each step.

For a first-order differential equation, let $\frac{dy}{dx} = F(x, y)$ and $y(x_0) = y_0$. If we have step size $h$, then our successive $x$-values are $x_1 = x_0 + h$, $x_2 = x_1 + h$, etc. The differential equation tells us that the slope at $x_0$ is $F(x_0, y_0)$. So, $y_1 = y_0 + hF(x_0, y_0)$ (see figure 186.4).

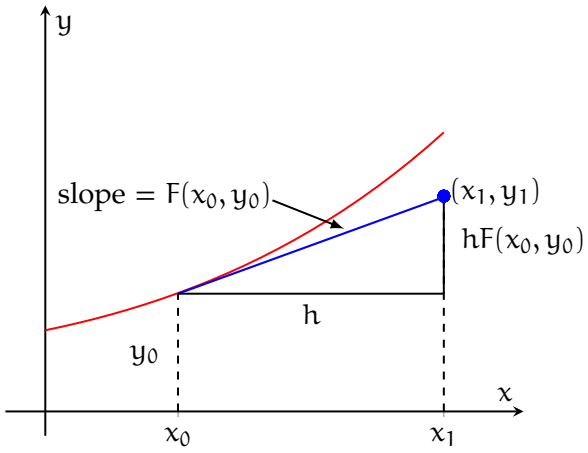



Figure 186.4: Visualization of Euler's method

Continuing, once we have found $y_1$, we can then define $x_2 = x_1 + h$ and $y_2 = y_1 + hF(x_1, y_1)$. And in general, for an initial-value problem when $\frac{dy}{dx} = F(x, y)$ and $y(x_0) = y_0$, we can make an approximation with step size $h$ where:

$$y_n = y_{n-1} + hF(x_{n-1}, y_{n-1})$$

where $n = 1, 2, 3, \cdots$.

**Example**: Use Euler's method with a step size of 0.2 to approximate the value of $y(1)$ if $\frac{dy}{dx} = 2x + y$ and $y(0) = 1$.

**Solution**: We are given $h = 0.2$, $x_0 = 0$, $y_0 = 1$, and $F(x, y) = 2x + y$. This means we will need 5 steps to reach $x_5 = 1$. So, we know that:

$$y_1 = 1 + 0.2[2(0) + 1] = 1 + 0.2[1] = 1.2$$

$$y_2 = 1.2 + 0.2[2(0.2) + 1.2] = 1.2 + 0.2(1.6) = 1.52$$

$$y_3 = 1.52 + 0.2[2(0.4) + 1.52] = 1.984$$

We can continue in this manner. The results are shown in the table:

| n | $x_n$ | $y_n$ | $F(x_n, y_n)$ |
|---|---|---|---|
| 0 | 0 | 1 | 1 |
| 1 | 0.2 | 1.2 | 1.6 |
| 2 | 0.4 | 1.52 | 2.32 |
| 3 | 0.6 | 1.984 | 3.184 |
| 4 | 0.8 | 2.6208 | 4.2208 |
| 5 | 1 | 3.46496 | – |

Therefore, $y(1) \approx 3.4696$.

**Example**: This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC exam. Let $y = f(x)$ be the solution to $\frac{dy}{dx} = x - y$ with initial condition $f(1) = 3$. What is the approximation of $f(2)$ obtained using Euler's method with two steps of equal length starting at $x = 1$?

**Solution**: The question asks that we use Euler's method two steps. The step size should be $h = \frac{x_2 - x_0}{2} = \frac{2-1}{2} = \frac{1}{2}$. Taking $x_0 = 1$ and $y_0 = 3$, we find that:

$$y_1 = y_0 + h\,[x_0 - y_0]$$

$$y_1 = 3 + \frac{1}{2}\,[1 - 3]$$

$$y_1 = 3 + \frac{1}{2}\,[-2] = 3 - 1 = 2$$

So our intermediate point is $(x_1, y_1) = (\frac{3}{2}, 2)$. Finding $y_2$:

$$y_2 = y_1 + h\,[x_1 - y_1]$$

$$y_2 = 2 + \frac{1}{2}\left[\frac{3}{2} - 2\right]$$

$$y_2 = 2 + \frac{1}{2}\left[\frac{-1}{2}\right] = 2 - \frac{1}{4} = \frac{7}{4}$$

So the approximate value of $f(2)$ is $\frac{7}{4}$.

## Exercise 494

*Working Space*

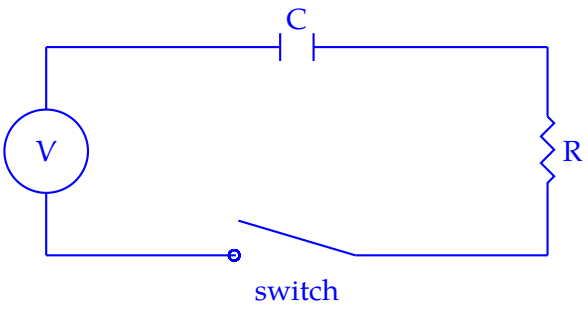
In the previous chapter on slope fields, we discussed the behavior of inductors in electronic circuits. As you may recall, capacitors also exhibit more complex behavior than regular resistors. Consider a circuit with a resistor and capacitor (see figure below). Let the resistor have resistance R ohms and the capacitor have capacitance C farads. By Kirchhoff's law, we know that:

$$RI + \frac{Q}{C} = V$$

where Q is the charge on each side of the capacitor and $\frac{Q}{C}$ is the voltage drop across the capacitor. Recall that current is the change in charge over time. Therefore, $I = \frac{dQ}{dt}$, and we can write the differential equation:

$$R\frac{dQ}{dt} + \frac{1}{C}Q = V$$

When the switch is first closed, there is no charge (that is, $Q(0) = 0$). If the resistor is $5\Omega$, the battery is 60V, and the capacitor is 0.05F, use Euler's method with a step size of 0.1 to estimate the charge after half a second.




*Answer on Page 2085*

### Exercise 495

[This problem was originally presented as a calculator-allowed, free-response question on the 2012 AP Calculus BC exam.] The function $f$ is twice- differentiable for $x > 0$ with $f(1) = 15$ and $f''(1) = 20$. Values of $f'$, the derivative of $f$, are given for selected values of $x$ in the table below. Use Euler's method, starting at $x = 1$ with two steps of equal size, to approximate $f(1.4)$. Show the computations that lead to your answer.

| $x$ | 1 | 1.1 | 1.2 | 1.3 | 1.4 |
|---|---|---|---|---|---|
| $f'(x)$ | 8 | 10 | 12 | 13 | 14.5 |

*Working Space*

*Answer on Page 2088*

CHAPTER 187

# Sequences in Calculus

We introduced sequences in a previous chapter. Now, we will examine them in more detail in the context of calculus. You already know about arithmetic and geometric sequences, but not all sequences can be classified as arithmetic or geometric. Take the famous Fibonacci sequence, {1, 1, 2, 3, 5, 8, ...}, which can be explicitly defined as $a_n = a_{n-1} + a_{n-2}$, with $a_1 = a_2 = 1$. There is no common difference or common ratio, so the Fibonacci sequence is not arithmetic or geometric. Another example is $a_n = \sin \frac{n\pi}{6}$, which will cycle through a set of values.

Sequences have many real-world applications, including compound interest and modeling population growth. In later chapters, you will learn that the sum of all the values in a sequence is a series and how to use series to describe functions. In order to be able to do all that, we first need to talk in depth about sequences.

Some sequences are defined explicitly, like $a_n = \sin \frac{n\pi}{6}$, while others are defined recursively, like $a_n = a_{n-1} + a_{n-2}$.

**Example**: Write the first 5 terms for the explicitly defined sequence $a_n = \frac{n}{n+1}$.

**Solution**: We can construct a table to keep track of our work:

| n | work | $a_n$ |
|---|---|---|
| 1 | $\frac{1}{1+1}$ | $\frac{1}{2}$ |
| 2 | $\frac{2}{2+1}$ | $\frac{2}{3}$ |
| 3 | $\frac{3}{3+1}$ | $\frac{3}{4}$ |
| 4 | $\frac{4}{4+1}$ | $\frac{4}{5}$ |
| 5 | $\frac{5}{5+1}$ | $\frac{5}{6}$ |

So, the first five terms are $\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}$, and $\frac{5}{6}\}$.

### *Exercise 496*

Write the first five terms for each sequence.

*Working Space*

1. $a_n = \frac{2^n}{2n+1}$
2. $a_n = \cos \frac{n\pi}{2}$
3. $a_1 = 1$, $a_{n+1} = 5a_n - 3$
4. $a_1 = 6$, $a_{n+1} = \frac{a_n}{n+1}$

*Answer on Page 2086*

## 187.1 The Idea of Approaching

### 187.1.1 Law of Large Numbers Revisited

Recall that, in the limits chapter, we talked about the *Law of Large Numbers*. The Law states that as the number of tests (in the case from the limits chapter, coin flips) increases, the number of successes (heads) approaches closer to the expected value or mean.

Let's bring back our flip sequence from that chapter

$$H, T, T, H, T, H, T, T, H, T$$

We want to make a sequence from this. Let's have the

$$a_n = \frac{\text{heads so far}}{\text{flips so far}}$$

The first few terms would be

$$a_1 = \frac{1}{1} = 1$$

$$a_2 = \frac{1}{2} = 0.5$$

$$a_3 = \frac{1}{3} \approx 0.333$$

$$a_4 = \frac{2}{4} = 0.5$$

And after all 10 flips,

$$1, \frac{1}{2}, \frac{1}{3}, \frac{2}{4}, \frac{2}{5}, \frac{3}{6}, \frac{3}{7}, \frac{3}{8}, \frac{4}{9}, \frac{4}{10}$$

or as decimals,

$$1,\ 0.5,\ 0.333,\ 0.5,\ 0.4,\ 0.5,\ 0.429,\ 0.375,\ 0.444,\ 0.4$$

We represent this with the equation

$$\bar{X}_n \to \mu$$

the average value of X approaches the true mean, $\mu$.

The Law of Large Numbers says that, over the long run, this sequence tends to settle near 0.5. In calculus, this kind of behavior has a name. When the terms of a sequence approach a fixed value as $n$ gets larger, we say the sequence *converges*. If the terms do not approach one fixed value, we say the sequence *diverges*. Let's look at this with graphs and actual equations.

The coin-flip sequence is random, so now we will look at sequences whose rules are known exactly.

### 187.1.2 Convergence and Divergence

You can visualize a sequence on an $xy$-plane or a number line. Figures 187.1 and 187.2 show visualizations of the sequence $a_n = \frac{n}{n+1}$. To visualize this on the $xy$-plane, we take points such that $x = n$ and $y = a_n$, where $n$ is a positive integer. What do you notice about this sequence? It isn't as easy to identify as our coin example. As $n$ increases, $a_n$ gets closer and closer to 1.

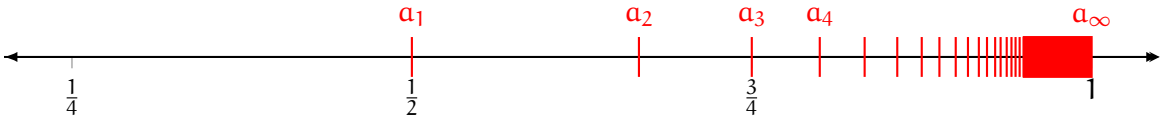



Figure 187.1: $a_n = \frac{n}{n+1}$ on a number line

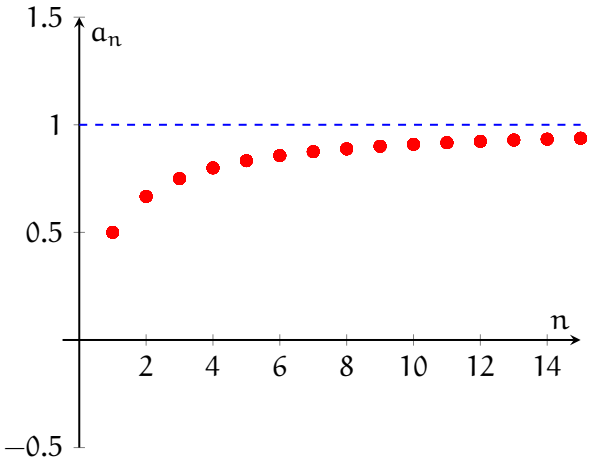



Figure 187.2: $a_n = \frac{n}{n+1}$ on an xy-plane

Because $a_n$ approaches a specific number as $n \to \infty$, we call the series $a_n = \frac{n}{n+1}$ *convergent*. We prove a sequence is convergent by taking the limit as $n$ approaches $\infty$. If the limit exists and approaches a specific number, the sequence is convergent. If the limit does not exist or approaches $\pm\infty$, the sequence is divergent.

We can see graphically that $\lim_{n\to\infty} \frac{n}{n+1} = 1$, so that sequence is convergent. What about $b_n = \frac{n}{\sqrt{10+n}}$? Is $b_n$ convergent or divergent?

$$\lim_{n\to\infty} \frac{n}{\sqrt{10+n}} = \lim_{n\to\infty} \frac{n/n}{\sqrt{\frac{10}{n^2} + \frac{n}{n^2}}}$$

$$= \lim_{n\to\infty} \frac{1}{\sqrt{\frac{10}{n^2} + \frac{1}{n}}} = \infty$$

Therefore, the sequence $b_n = \frac{n}{\sqrt{10+n}}$ is divergent.

Here is another example of a divergent sequence: $c_n = \sin\frac{n\pi}{2}$. The graph is shown in figure 187.3. As you can see, the value of $c_n$ oscillates between 1, 0, and -1 without approaching a specific number. This means that $c_n$ does not approach a particular number as $n \to \infty$ and the sequence is divergent.

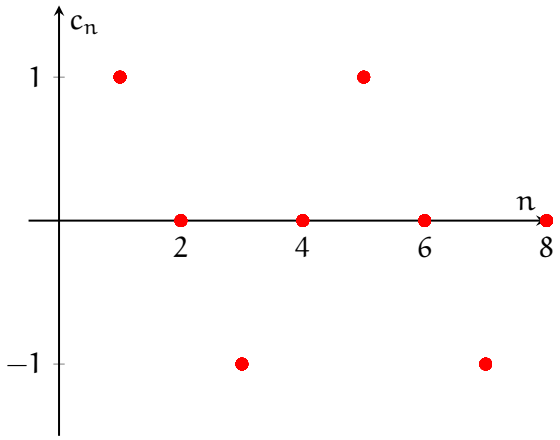



Figure 187.3: $c_n = \sin \frac{n\pi}{2}$ on an xy-plane

### *Exercise 497*

Classify each sequence as convergent or divergent. If the sequence is convergent, find the limit as $n \to \infty$.

*Working Space*

1. $a_n = \frac{3+5n^2}{n+n^2}$
2. $a_n = \frac{n^4}{n^3-2n}$
3. $a_n = 2 + (0.86)^n$
4. $a_n = \cos \frac{n\pi}{n+1}$
5. $a_n = \sin n$

*Answer on Page 2086*

## 187.2 Evaluating limits of sequences

Recall that a sequence can be considered a function where the domain is restricted to positive integers. If there is some $f(x)$ such that $a_n = f(n)$ when $n$ is an integer, then $\lim_{n\to\infty} a_n = \lim_{x\to\infty} f(x)$ (see figure 187.4). This means that all the rules that apply to the limits of functions also apply to the limits of sequences, including the Squeeze Theorem and l'Hôpital's rule.

**Example**: What is $\lim_{n\to\infty} \frac{\ln n}{n}$?

**Solution**: First, we will try to compute the limit directly:

$$\lim_{n\to\infty} \frac{\ln n}{n} =$$

$$\frac{\lim_{n\to\infty} \ln n}{\lim_{n\to\infty} n} = \frac{\infty}{\infty}$$

This is undefined, but fits the criteria for L'Hospital's rule:

$$\lim_{n\to\infty} \frac{\ln n}{n} = \lim_{n\to\infty} \frac{\frac{d}{dn} \ln n}{\frac{d}{dn} n}$$

$$= \lim_{n\to\infty} \frac{\frac{1}{n}}{1} = 0$$

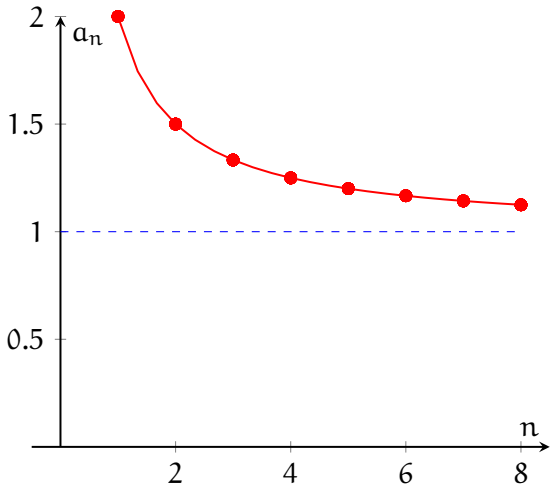



Figure 187.4: The limit of the function is the same as the limit of the sequence

**Example**: Is the sequence $a_n = \frac{n!}{n^n}$ convergent or divergent?

**Solution**: First trying to take the limit directly, we see that:

$$\lim_{n\to\infty} \frac{n!}{n^n} = \frac{\infty}{\infty}$$

which is undefined. Because the factorial cannot be described as a continuous function, we can't use l'Hôpital's rule. We can examine this sequence graphically (see figure 187.5) and mathematically. We examine it mathematically by writing out a few terms to get an

idea of what happens to $a_n$ as $n$ gets large:

$$a_1 = \frac{1!}{1^1} = 1$$

$$a_2 = \frac{2!}{2^2} = \frac{1 \cdot 2}{2 \cdot 2}$$

$$a_3 = \frac{3!}{3^3} = \frac{1 \cdot 2 \cdot 3}{3 \cdot 3 \cdot 3}$$

$$\cdots$$

$$a_n = \frac{n!}{n^n} = \frac{1 \cdot 2 \cdot 3 \cdots \cdot n}{n \cdot n \cdot n \cdots n}$$

From examining the graph in figure 187.5, we can guess that $\lim_{n\to\infty} a_n = 0$. Let's prove that mathematically. We can rewrite our expression for $a_n$ as $n$ gets larger:

$$a_n = \frac{n!}{n^n} = \frac{1 \cdot 2 \cdot 3 \cdots \cdot n}{n \cdot n \cdot n \cdots n} = \frac{1}{n}\left(\frac{2 \cdot 3 \cdots n}{n \cdot n \cdots n}\right)$$

The expression inside the parentheses is less than 1; therefore, $0 < a_n < \frac{1}{n}$. Since $\lim_{n\to\infty} 0 = 0$ and $\lim_{n\to\infty} \frac{1}{n} = 0$, by Squeeze Theorem, we know that $\lim_{n\to\infty} \frac{n!}{n^n} = 0$. Therefore, the sequence $a_n = \frac{n!}{n^n}$ is convergent.

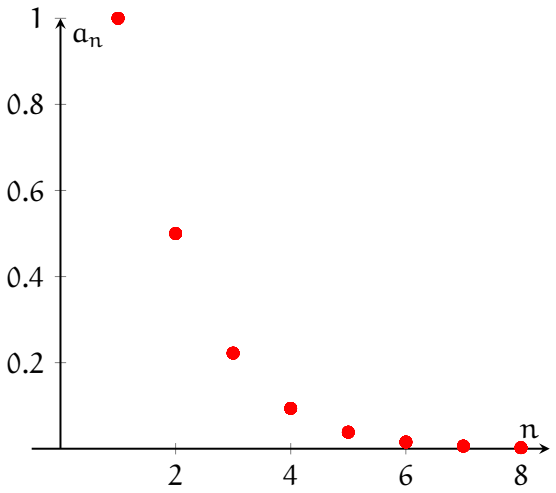



Figure 187.5: $a_n = \frac{n!}{n^n}$

[[FIX ME intro]] If $\lim_{n\to\infty} a_n = L$ and the function $f$ is continuous at $L$, then $\lim_{n\to\infty} f(a_n) = f(L)$. For example, what is $\lim_{n\to\infty} \sin\frac{\pi}{n}$? Well, we know that $\lim_{n\to\infty} \frac{\pi}{n} = 0$ and that the sine function is continuous at 0. Therefore, $\lim_{n\to\infty} \sin\frac{\pi}{n} = \sin\lim_{n\to\infty} \frac{\pi}{n} = \sin 0 = 0$.

## 187.3 Monotonic and Bounded sequences

Just like functions, sequences can be increasing or decreasing. A sequence is increasing if $a_n < a_{n+1}$ for $n \geq 1$. Similarly, a sequence is decreasing if $a_n > a_{n+1}$ for $n \geq 1$. If a sequence is strictly increasing or decreasing, it is called *monotonic*.

The sequence $a_n = \frac{1}{n+6}$ is decreasing. We prove this formally by comparing $a_n$ to $a_{n+1}$:

$$\frac{1}{n+6} > \frac{1}{(n+1)+6} = \frac{1}{n+7}$$

**Example**: Is the sequence $a_n = \frac{n}{n^2+1}$ increasing or decreasing?

**Solution**: First, we find an expression for $a_{n+1}$:

$$a_{n+1} = \frac{n+1}{(n+1)^2+1} = \frac{n+1}{n^2+2n+2}$$

Since the degree of $n$ is greater in the denominator, we have a guess that the sequence is decreasing. To prove this, we check if $a_n > a_{n+1}$ is true:

$$\frac{n}{n^2+1} > \frac{n+1}{n^2+2n+2}$$

We can cross-multiply, because $n > 0$ and the denominators are positive:

$$(n)(n^2+2n+2) > (n+1)(n^2+1)$$

$$n^3+2n^2+2n > n^3+n^2+n+1$$

Subtracting $(n^3+n^2+n)$ from both sides we see that:

$$n^2+n > 1$$

Which is true for all $n \geq 1$. Therefore, $a_n > a_{n+1}$ for all $n \geq 1$ and the sequence is decreasing.

A sequence is *bounded above* if there is some number $M$ such that $a_n \leq M$ for all $n \geq 1$. A sequence is *bounded below* if there is some other number $m$ such that $a_n \geq m$ for all $n \geq 1$. If a sequence is bounded above and below, then it is a *bounded sequence*.

Not all bounded sequences are convergent. Take our earlier example of $a_n = \sin\frac{n\pi}{6}$. This sequence is bounded, since we can say that $-1 \leq a_n \leq 1$ for all $n$. However, $a_n = \sin\frac{n\pi}{6}$ is divergent because $\lim_{n\to\infty} \sin\frac{n\pi}{6}$ does not exist (see figure 187.6). Additionally, not all

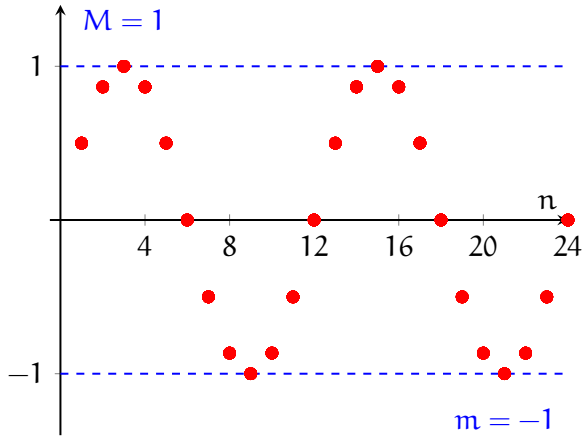



Figure 187.6: The sequence $a_n = \sin \frac{n\pi}{6}$ is bounded and divergent

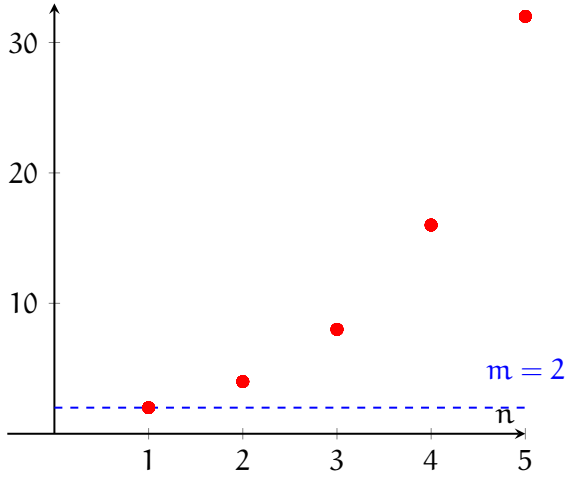



Figure 187.7: The sequence $b_n = 2^n$ is bounded below, monotonically increasing, and divergent

monotonic sequences are convergent. Consider $b_n = 2^n$ (shown in figure 187.7). This is monotonically increasing (that is, $b_n > b_{n-1}$ for all $n$), but $\lim_{n\to\infty} 2^n = \infty$ and the sequence is divergent.

A sequence must be convergent if it is **both** monotonic and bounded. Why is this? Recall that to be bounded, a sequence must be bounded above and below, which means there is some $m$ and some $M$ such that $m \leq a_n \leq M$ for all $n$. If the sequence is increasing, the terms must get close to but not exceed $M$. Likewise, if the sequence is decreasing, the terms must get close to, but not be less than $m$.

**Example**: Is the sequence given by $a_1 = 4$ and $a_{n+1} = \frac{1}{2}(a_n + 7)$ bounded above, below, both, or neither?

**Solution**: We start by calculating the first several terms:

| Term | Work | Value |
|---|---|---|
| $a_1$ | $a_1 = 4$ | 4 |
| $a_2$ | $= \frac{1}{2}(4+7)$ | 5.5 |
| $a_3$ | $= \frac{1}{2}(5.5+7)$ | 6.25 |
| $a_4$ | $= \frac{1}{2}(6.25+7)$ | 6.625 |
| $a_5$ | $= \frac{1}{2}(6.625+7)$ | 6.8125 |
| $a_6$ | $= \frac{1}{2}(6.8125+7)$ | 6.90625 |
| $a_7$ | $= \frac{1}{2}(6.90625+7)$ | 6.953125 |
| $a_8$ | $= \frac{1}{2}(6.953125+7)$ | 6.9765625 |

The sequence is increasing, so it is bounded below by the initial term, $a_1 = 4$, and we can state that $a_n \geq 4$. Examining the computed terms, we see that $a_n \to 7$ as $n$ grows larger. We can guess that this sequence is bounded above, with $a_n \leq 7$. We can prove this by induction. Suppose that there is some $k$ such that $a_k < 7$ (which is true for $a_1$, etc.). Then,

$$a_k < 7$$

$$a_k + 7 < 14$$

$$\frac{1}{2}(a_k + 7) < \frac{1}{2}(14)$$

$$a_{k+1} < 7$$

Therefore, $a_n < 7$ for all $n$ and the sequence is bounded above. Because the sequence is monotonic and bounded, we know the sequence is convergent and, therefore, that the limit of $a_n$ as $n \to \infty$ exists.

## 187.4 Applications of Sequences

### 187.4.1 Compound Interest

You previously learned about compound interest and modeled the accumulation of compound interest by $P_n = P_0(1+r)^n$, where $P_0$ is the principal investment, $r$ is the yearly interest rate, and $n$ is the number of elapsed years. This sequence describes the value of an investment accumulating interest, but most people add to their savings on a regular schedule. We can write a sequence to model the value of a savings account that the owner makes regular deposits into.

**Example**: Suppose you open a savings account with an initial deposit of $3,000 and you plan to deposit an additional $1,200 at the end of every year. If your savings account has an annual interest rate of 3.25%, how long will it take you to save $10,000?

**Solution**: We can write a recursive definition for the sequence. At the end of each year,

the account will gain the interest on the entirety of the previous year's balance plus \$1200:

$$P_n = P_{n-1}(1 + 0.0325) + \$1200$$

With an initial investment $P_0 = \$3000$. We can write out the first few terms to find how many years it will take to save \$10,000:

| Year | Savings |
|---|---|
| 0 | \$3,000 |
| 1 | \$4,297.50 |
| 2 | \$5,637.17 |
| 3 | \$7,020.38 |
| 4 | \$8,448.54 |
| 5 | \$9.923.12 |
| 6 | \$11,445.62 |

The accumulation of interest with deposits is better described by a sequence than a function. That iss because the deposits are happening at discrete times, not continuously.

### *Exercise 498*

You invest \$1500 at 5%, compounded annually. Write an explicit formula that describes the value of your investment every year. What will your investment be worth after 10 years? Is the sequence convergent or divergent? Explain.

*Working Space*

*Answer on Page 2086*

### 187.4.2 Population Growth

Sequences can be used to model a reproducing population that is being occasionally culled from or added to. Similar to compound interest, a population of living things (plants, animals, fungi, etc.) reproducing at a rate $r$ can be modeled with an exponential function:

$$P_n = P_0(1 + r)^n$$

where $P_0$ is the initial population, $r$ is the yearly reproductive rate, and $n$ is the number of years elapsed.

**Example**: Suppose the population of deer in a national park is estimated to be 6,500. If the deer reproduce at a rate of 8% per year and wolves hunt and kill 500 deer per year, how many deer will be in the park in 5 years?

**Solution**: We can write a recursive sequence:

$$P_n = P_{n-1}(1 + 0.08) - 500$$

$$P_0 = 6500$$

And calculate $P_5$ (we round to the nearest whole number because half of a deer is not a living deer):

| Year | | Population |
|---|---|---|
| 1 | $6500(1.08) - 500$ | 6520 |
| 2 | $6520(1.08) - 500$ | 6542 |
| 3 | $6542(1.08) - 500$ | 6565 |
| 4 | $6565(1.08) - 500$ | 6590 |
| 5 | $6590(1.08) - 500$ | 6617 |

There will be 6617 deer in the park after 5 years.

## *Exercise 499*

A farmer keeps his pond stocked with fish. If the fish are eaten by predators at a rate of 5% per month and the farmer can afford to restock the pond with 10 fish every 6 months. If the farmer starts with 100 fish, how many total fish will he have lost to predation after 4 years?

*Working Space*

*Answer on Page 2087*

# Chapter 188

# Series

When writing a number with an infinite decimal, such as the Golden Ratio (also known as the Golden Number):

$$\phi = 1.618033988\cdots$$

The decimal system means we can rewrite the Golden Ratio (or any irrational number) as an infinite sum:

$$\phi = 1 + \frac{6}{10} + \frac{1}{10^2} + \frac{8}{10^3} + \frac{0}{10^4} + \frac{3}{10^5} + \cdots$$

You might recall from the chapter on Riemann Sums that we can represent the addition of many (or infinite) with big sigma notation:

$$\sum_{i=1}^{n} a_i$$

where $i$ is the index as discussed in Sequences and $n$ is the number of terms. For infinite sums, $n = \infty$.

This notation when included with an infinite number of terms is called a *series*.

## 188.1 Partial Sums

Let's quickly define a *partial sum*. A partial sum is where we only look at the first $n$ terms of a series. This usually consists of $i = 1$ to a given $n$. For the general series, $\sum_{i=1}^{n} a_i$, the partial sums are:

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$\cdots$$

$$s_n = a_1 + a_2 + \cdots + a_n = \sum_{i=1}^{n} a_i$$

**Example**: A series is given by $\sum_{i=1}^{\infty} (\frac{-3}{4})^i$. What is the value of the partial sum $s_4$?

**Solution**: $s_4$ is the sum of the first 4 terms:

$$(\frac{-3}{4})^1 + (\frac{-3}{4})^2 + (\frac{-3}{4})^3 + (\frac{-3}{4})^4$$

$$= \frac{-3}{4} + \frac{9}{16} + \frac{-27}{64} + \frac{81}{256} = \frac{-75}{256}$$

## 188.2 Reindexing

Sometimes it is necessary to re-index series. This means changing what $n$ the series starts at . In general,

$$\sum_{n=i}^{\infty} a_n = \sum_{n=i+1}^{\infty} a_{n-1} \text{ and } \sum_{n=i}^{\infty} a_n = \sum_{n=i-1}^{\infty} a_{n+1}$$

In other words, to increase the index by 1, you need to replace $n$ with $(n-1)$ and do decrease the index by 1, you need to replace $n$ with $(n+1)$. Let's visualize why this is true (see figure 188.1). Notice that for each series, the terms are the same. This is similar to shifting functions: to move the function to the left on the x-axis, you plot $f(x+1)$, and to move it to the right, $f(x-1)$.

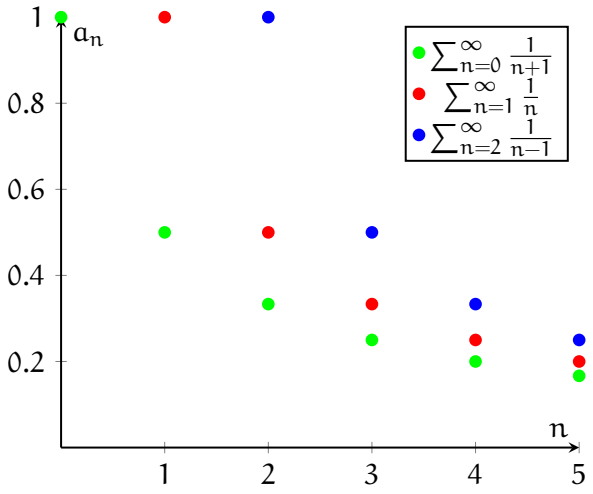



Figure 188.1: $\sum_{n=0}^{\infty} \frac{1}{n+1} = \sum_{n=1}^{\infty} a_n = \sum_{n=2}^{\infty} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots$

We can also prove each reindexing rule mathematically. Recall that

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

We also know that

$$\sum_{n=2}^{\infty} a_{n-1} = a_{2-1} + a_{3-1} + a_{4-1} + \cdots = a_1 + a_2 + a_3 + \cdots$$

Therefore, $\sum_{n=1}^{\infty} a_n = \sum_{n=2}^{\infty} a_{n-1}$.

Similarly,

$$\sum_{n=0}^{\infty} a_{n+1} = a_{0+1} + a_{1+1} + a_{2+1} + \cdots = a_1 + a_2 + a_3 + \cdots = \sum_{n=1}^{\infty} a_n$$

**Example**: Reindex the series $\sum_{n=3}^{\infty} \frac{n+1}{n^2-2}$ to begin with $n = 1$.

**Solution**: We are decreasing the index, so we will use $\sum_{n=i-1}^{\infty} a_{n+1} = \sum_{n=i}^{\infty} a_n$. We will apply this rule twice, to decrease the index from 3 to 1:

$$\sum_{n=2}^{\infty} \frac{(n+1)+1}{(n+1)^2-2} = \sum_{n=2}^{\infty} \frac{n+2}{(n+1)^2-2}$$

$$\sum_{n=1}^{\infty} \frac{(n+1)+2}{[(n+1)+1]^2-2} = \sum_{n=1}^{\infty} \frac{n+3}{(n+2)^2-2}$$

It is easier and faster to be able to reindex a series by more than one step at a time. Using the example above, we can write an even more general rule for reindexing:

$$\sum_{n=i}^{\infty} a_n = \sum_{n=i+j}^{\infty} a_{n-j}$$

where $i$ and $j$ are integers. (Then, to decrease the index, you would choose a $j$ such that $j < 0$.)

## 188.3 Convergent and Divergent Series

Just like sequences, series can also be convergent or divergent. Consider the series $\sum_{i=1}^{\infty} i$. Given what you already know about the meaning of "convergent" and "divergent", guess whether $\sum_{i=1}^{\infty} i$ is convergent or divergent.

Let's determine the first few partial sums of the series (shown graphically in figure 188.2):

| n | Terms | Partial Sum |
|---|---|---|
| 1 | 1 | 1 |
| 2 | 1+2 | 3 |
| 3 | 1+2+3 | 6 |
| 4 | 1+2+3+4 | 10 |

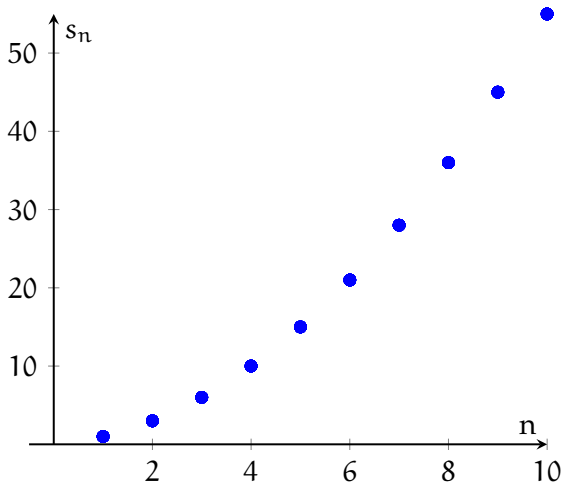



Figure 188.2: For the divergent series $\sum_{i=1}^{n} i$, the value of the partial sum increases to infinity as $n$ increases

As you can see, as $n$ increases, the value of the partial sum increases without approaching a particular value. We can also see that the value of the first $n$ terms summed together is $\frac{n(n+1)}{2}$. This means that as $n$ approaches $\infty$, the sum also approaches $\infty$ and the series is divergent.

Obviously, for a series to not become overly large, the values of the terms should decrease as $i$ increases (that is, each subsequent term is smaller than the one before it). Take the series $\sum_{i=1}^{\infty} \frac{1}{2^i}$. As $i$ increases, $\frac{1}{2^i}$ decreases. Let's look at the first few partial sums of this series (shown graphically in figure 188.3):

| n | Terms | Partial Sum |
|---|---|---|
| 1 | $\frac{1}{2}$ | $\frac{1}{2}$ |
| 2 | $\frac{1}{2}+\frac{1}{4}$ | $\frac{3}{4}$ |
| 3 | $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}$ | $\frac{7}{8}$ |
| 4 | $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}$ | $\frac{15}{16}$ |

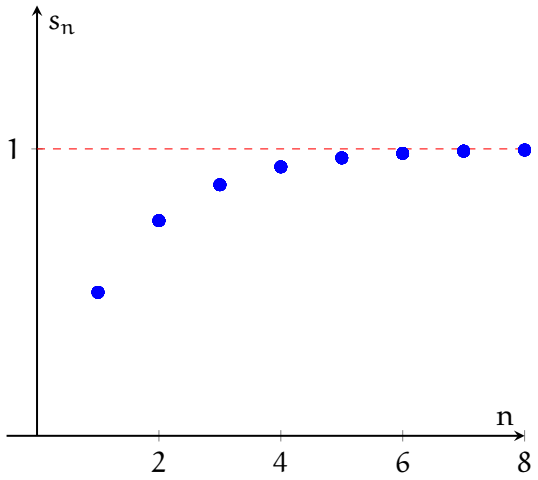



Figure 188.3: For the convergent series $\sum_{i=1}^{n} \frac{1}{2^i}$, the value of the partial sum approaches 1 as $n$ increases

Do you see the pattern? The $n^{th}$ partial sum is equal to $\frac{2^n-1}{2^n} = 1-\frac{1}{2^n}$. And as $n$ approaches $\infty$, the partial sum approaches 1. The series $\sum_{i=1}^{\infty} \frac{1}{2^i}$ is convergent.

Let's define the sequence $\{s_n\}$, where $s_n$ is the $n^{th}$ partial sum of a series:

$$s_n = \sum_{i=1}^{n} a_i$$

.

If the sequence $\{s_n\}$ is convergent and $\lim_{n\to\infty} s_n$ exists, then the series $\sum_{i=1}^{\infty} a_i$ is also convergent. And if the sequence $\{s_n\}$ is divergent, then the series $\sum_{i=1}^{\infty} a_i$ is also divergent.

**Example**: Is the harmonic series, $\sum_{n=1}^{\infty} \frac{1}{n}$ convergent or divergent?

**Solution**: You may think that the series is convergent, since $\lim_{n\to\infty} \frac{1}{n} = 0$. Let's see if we can confirm this. We begin by looking at the partial sums $s_2$, $s_4$, $s_8$, and $s_{16}$:

$$s_2 = 1 + \frac{1}{2}$$

$$s_4 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = 1 + \frac{2}{2}$$

$$s_8 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) = 1 + \frac{3}{2}$$

$$s_{16} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) >$$

$$1+\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}\right)+\left(\frac{1}{16}+\cdots+\frac{1}{16}\right)=1+\frac{4}{2}$$

Notice that, in general, $s_{2^n} > 1+\frac{n}{2}$ for $n > 1$. Taking the limit as $n \to \infty$, we see that $\lim_{n\to\infty} s_{2^n} > \lim_{n\to\infty} 1+\frac{n}{2} = \infty$. Therefore, $s_{2^n}$ also approaches $\infty$ as $n$ gets larger and the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

This example shows a very important point: A series whose terms decrease to zero as $n$ gets large is not necessarily convergent. What we can say, though, is that if the limit as n approaches infinity of the terms of a series does not exist or is not zero, then the series is divergent (i.e., not convergent). This is called the **Test for Divergence**, and we will explore it further in the next chapter.

### 188.3.1 Properties of Convergent Series

We just saw that if $\lim_{n\to\infty} a_n \neq 0$ then the series $\sum_{n=1}^{\infty} a_n$ diverges. The contrapositive statement gives a property of convergent series:

$$\text{If the series } \sum_{n=1}^{\infty} a_n \text{ is convergent, then } \lim_{n\to\infty} = 0$$

If a series is made of other convergent series, it may be convergent. Recall, if a series is convergent, this means the $\lim_{n\to\infty} \sum_{i=1}^{n} a_i = L$. By the properties of limits, we can also say that the series multiplied by a constant is convergent:

$$\sum_{n=1}^{\infty} ca_n = c \cdot L = c\sum_{n=1}^{\infty} a_n$$

Suppose there is another convergent series such that $\lim_{n\to\infty} \sum_{i=1}^{n} b_i = M$. In this case, the sum of those series is also convergent. That is:

$$\sum_{n=1}^{\infty} (a_n + b_n) = L + M = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

Similarly, the difference of the series is convergent:

$$\sum_{n=1}^{\infty} (a_n - b_n) = L - M = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$$

## 188.4 Geometric Series

A geometric series is the sum of a geometric sequence, and has the form:

$$\sum_{n=1}^{\infty} ar^n \text{ or } \sum_{n=1}^{\infty} ar^{n-1}$$

Where $a$ is some constant and $r$ is the common ratio. For $\sum_{n=1}^{\infty} ar^{n-1}$, $a$ is also the first term.

**Example**: Write the series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ in sigma notation.

**Solution**: We see that the first term is $a = 1$ and the common ratio is $\frac{1}{2}$, so we can write the series:

$$\sum_{n=1}^{\infty} 1(\frac{1}{2})^{n-1} = \sum_{n=1}^{\infty} (\frac{1}{2})^{n-1}$$

When are geometric series convergent? First, let's consider the case where $r = 1$. If this is true, then $s_n = a + a + a + \cdots + a = na$. As $n$ approaches $\infty$, the sum will approach $\pm\infty$ (depending on whether $a$ is positive or negative), and the series is divergent.

When $r \neq 1$, we can write $s_n$ and $rs_n$:

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1}$$

$$rs_n = ar + ar^2 + ar^3 + \cdots + ar^n$$

Subtracting $rs_n$ from $s_n$, we get:

$$s_n - rs_n = (a + ar + ar^2 + \cdots + ar^{n-1}) - (ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n)$$

$$= a - ar^n$$

Solving for $s_n$, we find:

$$s_n = \frac{a(1 - r^n)}{1 - r}$$

We take the limit as $n \to \infty$ to determine for what values of $r$ the series converges:

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \frac{a(1 - r^n)}{1 - r}$$

$$= \lim_{n\to\infty} \left[\frac{a}{1 - r} - \frac{ar^n}{1 - r}\right] = \frac{a}{1 - r} - \left(\frac{a}{1 - r}\right) \lim_{n\to\infty} r^n$$

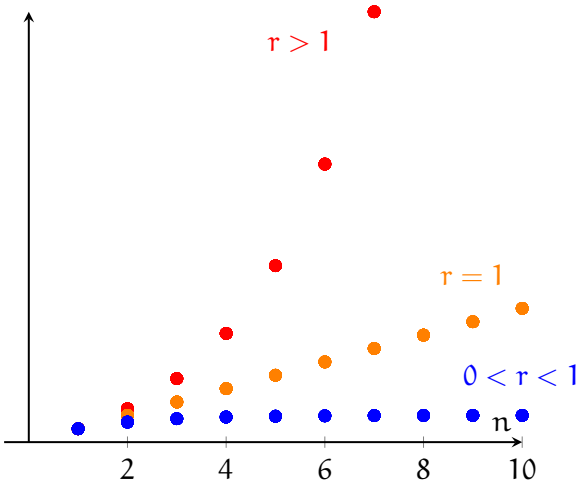



Figure 188.4: Geometric sequences are divergent if $r \geq 1$

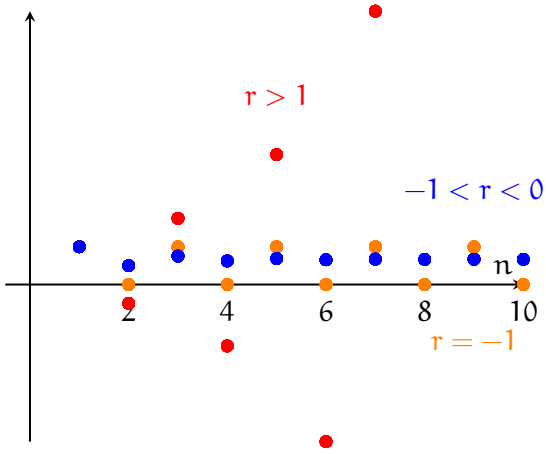



Figure 188.5: Geometric sequences are divergent if $r \leq 1$. Notice that for $r = -1$, the partial sums alternate between the initial term and zero.

This introduces the question: When is $\lim_{n\to\infty} r^n$ convergent? From the sequences chapter, we know this limit converges if $|r| < 1$ (that is, $-1 < r < 1$). If this is true, then $\lim_{n\to\infty} r^n = 0$ and

$$\lim_{n\to\infty} s_n = \frac{a}{1-r}$$

(see figures 188.4 and 188.5 for a visual)

**Example**: Find the sum of the geometric series given by $2 - \frac{2}{3} + \frac{2}{9} - \frac{2}{27} + \cdots$.

**Solution**: The first term is $a = 2$, and each common ratio is $r = \frac{-1}{3}$. Since $|r| < 1$, we know that the series converges. We can calculate the value of the sum using the geometric series

formula:

$$\sum_{i=1}^{\infty} a(r)^{i-1} = \frac{a}{1-r}$$

$$\sum_{i=1}^{\infty} 2(\frac{-1}{3})^{i-1} = \frac{2}{1-\frac{-1}{3}} = \frac{2}{\frac{4}{3}} = \frac{6}{4} = 1.5$$

We can confirm this graphically (see figure 188.6). You can also write out the first several partial sequences. You should find the sums approach 1.5 as $n$ increases.

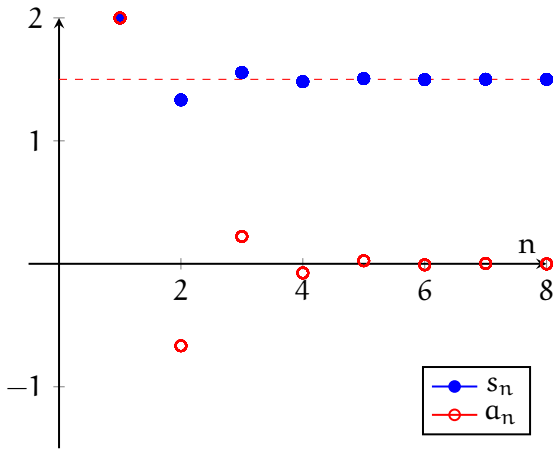



Figure 188.6: the $n^{th}$ term and partial sums of $\sum_{i=1}^{n} 2(\frac{-1}{3})^{i-1}$

**Example**: What is the value of $\sum_{n=1}^{\infty} 2^{2n}5^{1-n}$

**Solution**: The key here is to re-write the series in the form $\sum_{n=1}^{\infty} ar^{n-1}$ so we can use the fact that convergent geometric series sum to $\frac{a}{1-r}$.

$$\sum_{n=1}^{\infty} 2^{2n}5^{1-n} = \sum_{n=1}^{\infty} \left(2^2\right)^n \left(\frac{1}{5}\right)^{n-1}$$

$$= \sum_{n=1}^{\infty} 4 \cdot (4)^{n-1} \left(\frac{1}{5}\right)^{n-1} = \sum_{n=1}^{\infty} 4 \cdot \left(\frac{4}{5}\right)^{n-1}$$

Which is in the form $\sum_{n=1}^{\infty} ar^{n-1}$ with $a = 4$ and $r = \frac{4}{5}$. Since $|r| < 1$, the series converges to

$$\frac{a}{1-r} = \frac{4}{1-\frac{4}{5}} = \frac{4}{\frac{1}{5}} = 20$$

## Exercise 500

Determine whether the geometric series is convergent or divergent. If it is convergent, find its sum.

1. $3 - 4 + \frac{16}{3} - \frac{64}{9} + \cdots$
2. $2 + 0.5 + 0.125 + 0.03125 + \cdots$
3. $\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n}$
4. $\sum_{n=1}^{\infty} \frac{e^{2n}}{6^{n-1}}$

Working Space

Answer on Page 2087

## Exercise 501

Find a value of c such that $\sum_{n=0}^{\infty}(1 + c)^{-n} = \frac{5}{3}$.

Working Space

Answer on Page 2088

## Exercise 502

For what values of p does the series $\sum_{n=1}^{\infty} \left(\frac{p}{2}\right)^n$ converge?

Working Space

Answer on Page 2088

## 188.5 p-series

A p-series takes the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$ and converges if $p > 1$ and diverges if $p \leq 1$. We won't prove this here, since it requires the application of a test you will learn about in the next chapter.

**Example** Write the series $1 + \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{3}} + \frac{1}{\sqrt[3]{4}} + \cdots$. Is it convergent or divergent?

**Solution**: We see that $a_n = \frac{1}{\sqrt[3]{n}}$, so the infinite series is

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}}$$

We see that this is a p-series with $p = \frac{1}{3}$. Since $p < 1$, the series is divergent.

## Exercise 503

Euler found that the exact sum of the p-series where $p = 2$ is:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

And that the exact sum of the p-series where $p = 4$ is:

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

Use this and the properties of convergent series to find the sum of each of the following series:

1. $\sum_{n=1}^{\infty} \frac{n^2+1}{n^4}$
2. $\sum_{n=2}^{\infty} \frac{1}{n^2}$
3. $\sum_{n=3}^{\infty} \frac{1}{(n+1)^2}$
4. $\sum_{n=1}^{\infty} \left(\frac{3}{n}\right)^4$
5. $\sum_{n=1}^{\infty} \left(\frac{4}{n^2} + \frac{3}{n^4}\right)$

Working Space

Answer on Page 2088

### *Exercise 504*

For what values of k does the series $\sum_{n=1}^{\infty} \frac{1}{n^{2k}}$ converge?

*Working Space*

*Answer on Page 2089*

## 188.6 Alternating Series

An alternating series is one in which the terms alternate between positive and negative . Here is an example:

$$-\frac{1}{2}+\frac{2}{3}-\frac{3}{4}+\frac{4}{5}-\frac{5}{6}+\cdots=\sum_{n=1}^{\infty}(-1)^n\frac{n}{n+1}$$

Alternating series are generally of the form

$$a_n=(-1)^n b_n \text{ or } a_n=(-1)^{n-1}b_n$$

Where $b_n$ is positive (and therefore, $|a_n|=b_n$).

An alternating series is convergent if (i)$b_{n+1}\leq b_n$ and (ii)$\lim_{n\to\infty} b_n=0$. In other words, we say that if the absolute value of the terms of a series decrease towards zero, then the series converges. This is called the **Alternating Series Test**.

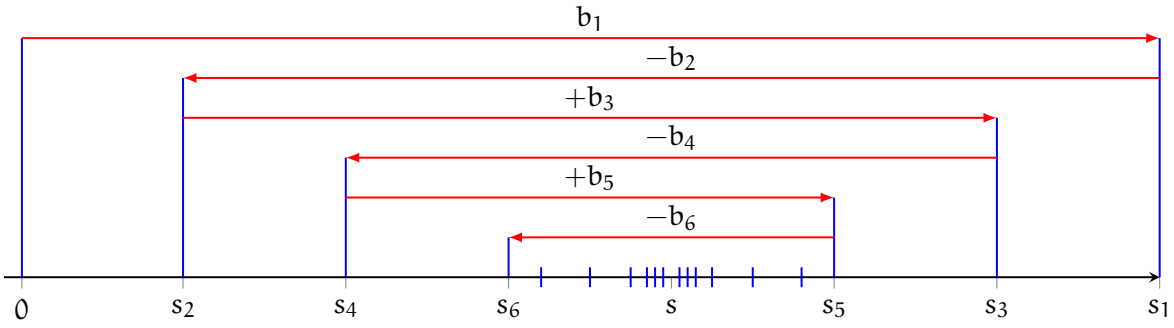



Figure 188.7: As $n$ increases, $s_n$ approaches $s$

**Example**: Is the alternating harmonic series $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}$ convergent?

**Solution**: The Alternating series test states that an alternating series is convergent if

$|a_{n+1}| < |a_n|$:

$$\left|\frac{(-1)^{n-1+1}}{n+1}\right| < \left|\frac{(-1)^{n-1}}{n}\right|$$

$$\frac{1}{n+1} < \frac{1}{n}$$

Since $|a_{n+1}| < |a_n|$ and the series is alternating, $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is convergent.

## Exercise 505

Test the following alternating series for convergence:

*Working Space*

1. $\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n-1}$
2. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3+1}$
3. $\sum_{n=1}^{\infty} (-1)^{n-1} e^{2/n}$

*Answer on Page 2089*

Chapter 189

# Convergence Tests for Series

## 189.1 Test for Divergence

Recall from the previous chapter that if the terms of a series do not approach zero as $n$ approaches infinity, then the series is divergent. This is the Test for Divergence, and there are two possible outcomes. For a series $\sum_{n=1}^{\infty} a_n$:

$$\text{If } \lim_{n\to\infty} a_n \neq 0, \text{ then the series diverges}$$

$$\text{If } \lim_{n\to\infty} a_n = 0, \text{ then the test is inconclusive}$$

It is important to remember that the Test for Divergence cannot tell us conclusively that a series converges. Rather, it only identifies series that are divergent.

**Example**: Apply the Test for Divergence to the series $\sum_{n=1}^{\infty} \sqrt{n}$ and $\sum_{n=1}^{\infty} \frac{1}{n}$

**Solution**: $\lim_{n\to\infty} \sqrt{n} = \infty \neq 0$. Therefore, the series $\sum_{n=1}^{\infty} \sqrt{n}$ is divergent. $\lim_{n\to\infty} \frac{1}{n} = 0$. Therefore, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ may be divergent or convergent. This is the harmonic series, which we proved to be divergent in the previous chapter. This is a good example that demonstrates that just because $\lim_{n\to\infty} a_n = 0$ does not mean the series is convergent.

## 189.2 The Integral Test

We were able to determine the exact value of some infinite series because it was possible to write the $n^{th}$ partial sum, $s_n$, in terms of $n$. For example, we determined that the $n^{th}$ partial sum of $\sum_{i=1}^{n} \frac{1}{2^i}$ is $s_n = 1 - \frac{1}{2^n}$. However, it is not always possible to do this. How can we estimate the value of an infinite series in cases where we can't explicitly write $s_n$ in terms of $n$?

Consider the series $\sum_{i=1}^{\infty} \frac{1}{i^2}$. The first few terms are:

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \cdots$$

The series is decreasing, but is it convergent? Let's plot this series on an $xy$-plane (see figure 189.1).

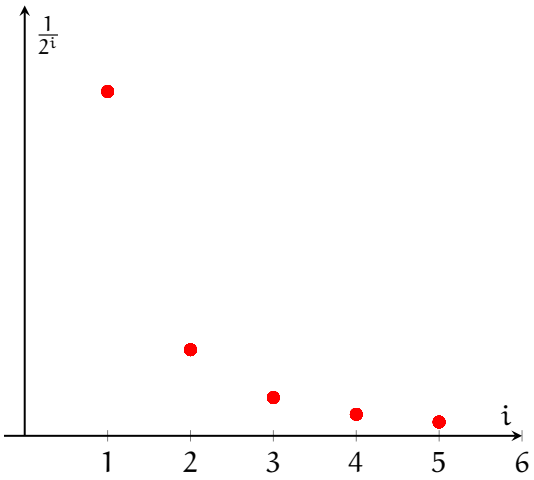



Figure 189.1: The first 5 terms of $\sum_{i=1}^{\infty} \frac{1}{2^i}$

We can overlay the function $y = \frac{1}{2^x}$ (figure 189.2). We can draw rectangles of width 1 and height $\frac{1}{x^2}$ (see figure 189.3). The area of the first $n$ rectangles is equal to the $n^{th}$ partial sum.

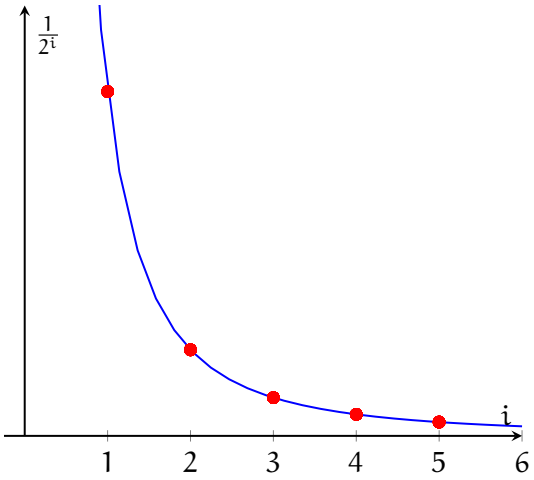



Figure 189.2: The first 5 terms of $\sum_{i=1}^{\infty} \frac{1}{2^i}$ lie on the curve $y = \frac{1}{x^2}$

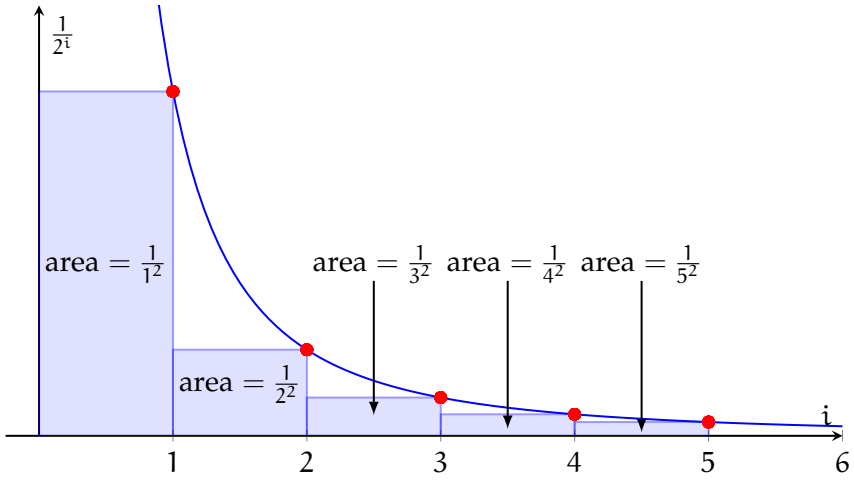



Figure 189.3: The partial sum $\sum_{i=1}^{n=5} \frac{1}{2^i}$ is equal to the area of the rectangles

This should remind you of a Riemann sum. Since the total area of the rectangles is less than the area under the curve, we can state:

$$\sum_{i=1}^{\infty} \frac{1}{2^i} < \int_0^{\infty} \frac{1}{x^2}\,dx$$

We can exclude the first rectangle and also state that:

$$\sum_{i=1}^{\infty} \frac{1}{2^i} < 1 + \int_1^{\infty} \frac{1}{x^2}\,dx$$

We can evaluate this integral:

$$\int_1^{\infty} \frac{1}{x^2}\,dx = \lim_{t\to\infty}\left[\int_1^{t} \frac{1}{x^2}\,dx\right]$$

$$= \lim_{t\to\infty} \frac{-1}{x}|_{x=1}^{t} = \lim_{t\to\infty}\left(\frac{-1}{t}\right) - \frac{-1}{1} = 0 - (-1) = 1$$

Therefore:

$$\sum_{i=1}^{\infty} \frac{1}{2^i} < 1 + 1 = 2$$

This means the series $\sum_{i=1}^{\infty} \frac{1}{2^i}$ is bounded above. Since the series is also monotonic (each term is positive, so the value of the sum increases as $n$ increases), we can state that the sum is convergent!

Let's look at a divergent example: $\sum_{i=1}^{\infty} \frac{1}{\sqrt{x}}$. Again, we will make a visual, but this time we will draw rectangles that lie above the curve $y = \frac{1}{\sqrt{x}}$ (see figure 189.4). In this case, $\sum_{i=1}^{\infty} \frac{1}{\sqrt{x}} > \int_1^{\infty} \frac{1}{\sqrt{x}}\,dx$. Let's evaluate the integral:

$$\int_1^{\infty} \frac{1}{\sqrt{x}}\,dx = \lim_{t\to\infty}\left[\int_1^{t} \frac{1}{\sqrt{x}}\,dx\right]$$

$$= \lim_{t\to\infty}\left[2\sqrt{x}\right]_{x=1}^{t} = \lim_{t\to\infty}\left(2\sqrt{t}\right) - 2\sqrt{1} = \infty - 2 \to \text{divergent}$$

Since the integral diverges to infinity and the series is greater than the integral, the series must also diverge to infinity. This is another case where a monotonic decreasing series is not convergent!

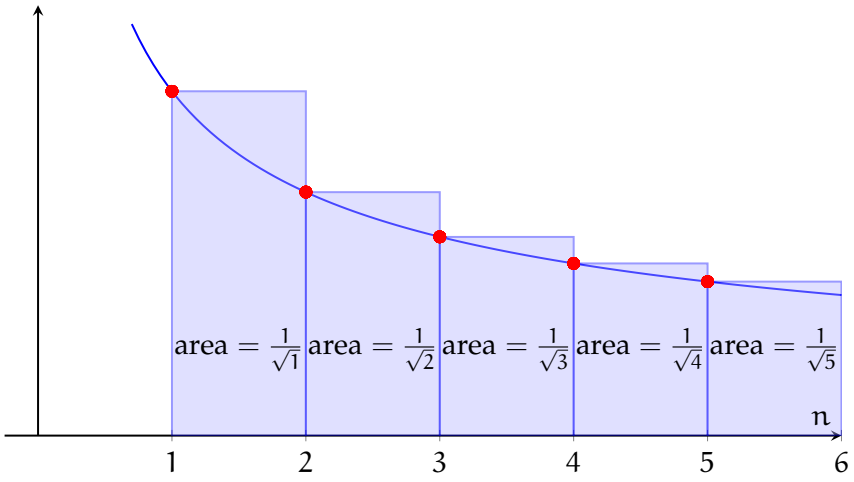



Figure 189.4: $\sum_{i=1}^{\infty} \frac{1}{\sqrt{x}} > \int_1^{\infty} \frac{1}{\sqrt{x}}\,dx$

**The Integral Test**

This leads us to the **Integral Test**. If $f$ is a continuous, positive, decreasing function on the interval $x \in [1, \infty)$ and $a_n = f(n)$, then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if $\int_1^{\infty} f(x)\,dx$ is convergent. Subsequently, if $\int_1^{\infty} f(x)\,dx$ is divergent, then the series is also divergent.

**Example**: Is the series $\sum_{i=1}^{\infty} \frac{1}{n^2+1}$ convergent or divergent?

**Solution**: To apply the integral test, we define $f(x) = \frac{1}{x^2+1}$, which is a positive, decreasing function on the interval $x \in [1, \infty)$.

$$\int_1^{\infty} \frac{1}{x^2+1}\,dx = \lim_{t\to\infty} \int_1^{t} \frac{1}{x^2+1}\,dx$$

$$= \lim_{t\to\infty} [\arctan x]_{x=1}^{t} = \lim_{t\to\infty} (\arctan t) - \arctan 1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

Because the integral $\int_1^\infty \frac{1}{x^2+1}\,dx$ converges, so does the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$.

## Exercise 506

Use the integral test to determine if the following series are convergent or divergent.

1. $\sum_{n=1}^{\infty} 2n^{-3}$
2. $\sum_{n=1}^{\infty} \frac{5}{3n-1}$
3. $\sum_{n=1}^{\infty} \frac{n}{3n^2+1}$

*Working Space*

*Answer on Page 2089*

### Exercise 507

Apply the Integral Test to show that p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ are convergent only when $p > 1$ (hint: consider the cases $p \leq 0$, $0 < p < 1$, $p = 1$ and $p > 1$).

*Working Space*

*Answer on Page 2090*

### 189.2.1 Using Integrals to Estimate the Value of a Series

Recall that $\sum_{i=1}^{\infty} a_i = a_1 + a_2 + a_3 + \cdots = s$ and that the $n^{th}$ partial sum, often represented as $s_n$, is $s_n = a_1 + a_2 + \cdots + a_{n-1} + a_n$. We can then define the $n^{th}$ remainder $R_n = s - s_n$. Expanding $s$ and $s_n$, we see that:

$$R_n = [a_1 + a_2 + \cdots + a_{n-1} + a_n + a_{n+1} + \cdots] - [a_1 + a_2 + \cdots + a_{n-1} + a_n]$$

$$R_n = [a_1 - a_1] + [a_2 - a_2] + \cdots + [a_{n-1} - a_{n-1}] + [a_n - a_n] + a_{n-1} + a_{n-2} + \cdots$$

$$R_n = a_{n+1} + a_{n+2} + a_{n+3} + \cdots$$

Just like the integral test, suppose there is some continuous, positive, decreasing function, such that $a_n = f(n)$. We can then represent $R_n$ as the right Riemann sum with width $\Delta x = 1$ from $x = n$ to $\infty$. Since the rectangles are below the curve (see figure 189.5), we can state that $R_n \leq \int_n^{\infty} f(x)\, dx$.

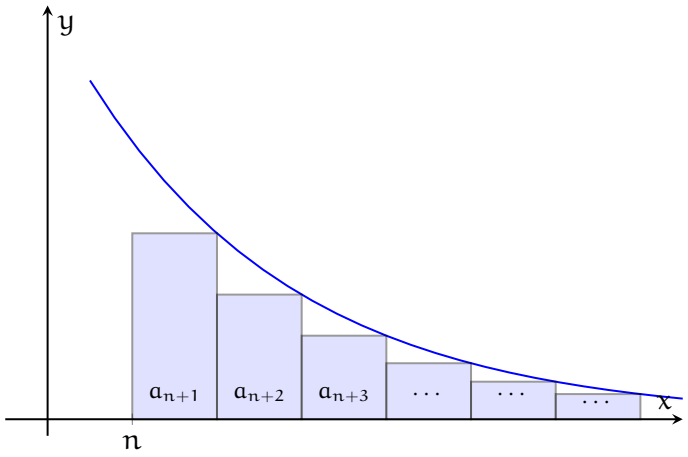



Figure 189.5: $R_n \leq \int_n^\infty f(x)\,dx$

Similarly, we can represent $R_n$ as the left Riemann sum with width $\Delta x = 1$ from $x = n+1$ to $\infty$. This time, the rectangles are above the curve (see figure 189.6), and we can state that $R_n \geq \int_{n+1}^\infty f(x)\,dx$. Putting this all together, we have an estimate for the remainder, $R_n$, from the integral test:

Suppose there is a function such that $f(k) = a_k$, where $f$ is a continuous, positive, decreasing function for $x \geq n$ and $\sum a_n$ is convergent. Then, $\int_{n+1}^\infty f(x)\,dx \leq R_n \leq \int_n^\infty f(x)\,dx$, where $R_n$ is $s - s_n$.

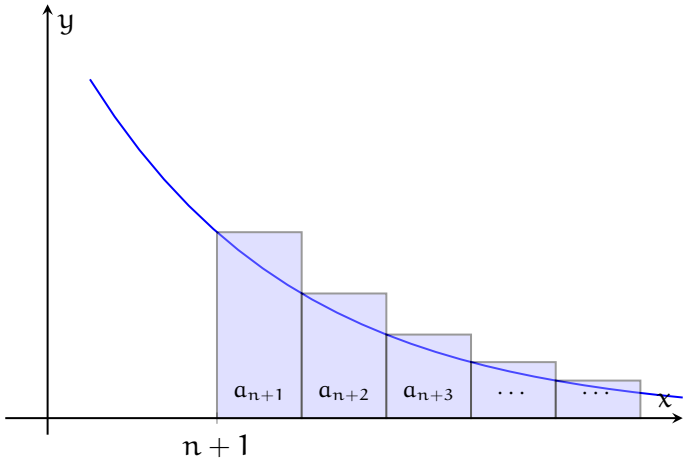



Figure 189.6: $R_n \geq \int_{n+1}^\infty f(x)\,dx$

**Example**: Approximate the sum of the series $\sum_{n=1}^\infty \frac{3}{n^3}$ by finding the $10^{\text{th}}$ partial sum. Estimate the error of this approximation.

**Solution**: Using a calculator, you can find the $10^{th}$ partial sum:

$$\sum_{n=1}^{10} \frac{3}{n^3} = \frac{3}{1^3} + \frac{3}{2^3} + \frac{3}{3^3} + \cdots + \frac{3}{10^3} \approx 3.593 = s_{10}$$

Recall that the remainder, $R_{10}$, is the difference between the actual sum, $s$, and the partial sum, $s_{10}$. Using the integral test to estimate the remainder, we can state that:

$$R_{10} \leq \int_{10}^{\infty} \frac{3}{x^3}\, dx = \frac{3}{2(10)^2} = \frac{3}{200} = 0.015$$

Therefore, the size of the error is at most 0.015.

**Example**: How many terms are required for the error to be less than 0.0001 for the sum presented above?

**Solution**: We are looking for an $n$ such that $R_n \leq 0.0001$. Recalling that $R_n \leq \int_n^{\infty} \frac{3}{x^3}\, dx$, we need to find an $n$ such that $\int_n^{\infty} \frac{3}{x^3}\, dx \leq 0.0001$.

$$\int_n^{\infty} \frac{3}{x^3}\, dx \leq 0.0001$$

$$\frac{-1}{6x^2}|_{x=n}^{\infty} \leq 0.0001$$

$$\lim_{x\to\infty} \frac{-1}{6x^2} - \frac{-1}{6n^2} \leq 0.0001$$

$$0 + \frac{1}{6n^2} = \frac{1}{6n^2} \leq 0.0001$$

$$1 \leq 0.0006n^2$$

$$1667 \leq n^2$$

$$40.8 \leq n \rightarrow n = 41$$

Therefore, $s - s_{41} \leq 0.0001$ and the partial sum $\Sigma_{n=1}^{41} \frac{3}{n^3}$ is less than 0.0001 from the value of the infinite sum $\sum_{n=1}^{\infty} \frac{3}{n^3}$.

### *Exercise 508*

*Working Space*

1. Find the partial sum $s_{10}$ of the series $\sum_{n=1}^{\infty} \frac{1}{n^4}$.
2. Estimate the error from using $s_{10}$ as an approximation of the series.
3. Use $s_n + \int_{n+1}^{\infty} \frac{1}{x^4}\,dx \leq s \leq s_n + \int_n^{\infty} \frac{1}{x^4}\,dx$ to give an improved estimate of the sum.
4. The actual value of $\sum_{n=1}^{\infty} \frac{1}{n^4}$ is $\frac{\pi^4}{90}$. Compare your estimate with the actual value.
5. Find a value of $n$ such that $s_n$ is within 0.00001 of the sum.

*Answer on Page 2090*

## 189.3 Comparison Tests

In comparison tests, we compare a series to a known convergent or divergent series. Take the series $\sum_{n=1}^{\infty} \frac{1}{3^n+3}$. This is similar to $\sum_{n=1}^{\infty} \frac{1}{3^n}$, which is a geometric series that converges to $\frac{1}{2}$. Notice that:

$$\frac{1}{3^n+3} < \frac{1}{3^n}$$

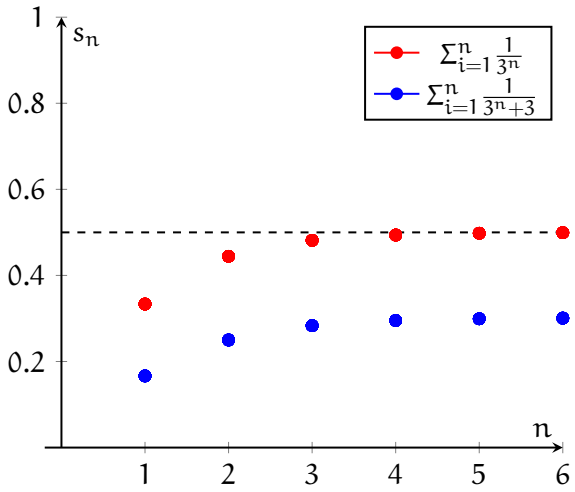



Figure 189.7: $\sum_{i=1}^{n} \frac{1}{3^n+3} < \Sigma_{i=1}^{n} \frac{1}{3^n}$ for all $n$

Which implies that

$$\sum_{n=1}^{\infty} \frac{1}{3^n+3} < \Sigma_{n=1}^{\infty} \frac{1}{3^n}$$

Since $\sum_{n=1}^{\infty} \frac{1}{3^n}$ is convergent, it follows that $\sum_{n=1}^{\infty} \frac{1}{3^n+3}$ is also convergent (see figure 189.7). As you can see, since $\sum_{n=1}^{\infty} \frac{1}{3^n}$ approaches $\frac{1}{2}$, $\sum_{n=1}^{\infty} \frac{1}{3^n+3}$ must be $\leq \frac{1}{2}$ and therefore convergent.

### 189.3.1 The Direct Comparison Test

For the **Direct Comparison Test**, we compare the terms $a_n$ to $b_n$ directly. Take $\sum a_n$ and $\sum b_n$ to be series with positive terms. Then,

1. If $a_n \leq b_n$ and $\sum b_n$ is convergent, then $\sum a_n$ is also convergent.
2. If $a_n \geq b_n$ and $\sum b_n$ is divergent, then $\sum a_n$ is also divergent.

We already discussed above why the first part is true. The second part follows a similar argument: If $a_n$ is greater than $b_n$, then you can imagine that as $\sum b_n$ grows and diverges, it is pushing upwards on $\sum a_n$, meaning that $\sum a_n$ must also diverge. Consider the series $\sum_{n=1}^{\infty} \frac{2 \ln n}{n}$. For $n \geq 2$, $2 \ln n > 1$, and therefore if $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, then $\sum_{n=1}^{\infty} \frac{2 \ln n}{n}$ must also diverge. We recognize the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. Therefore, $\sum_{n=1}^{\infty} \frac{2 \ln n}{n}$ is also divergent (see figure 189.8).

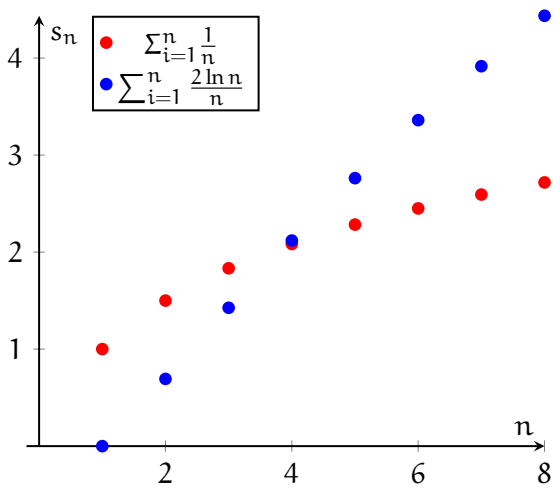



Figure 189.8: $\sum_{i=1}^{n} \frac{2 \ln n}{n} > \sum_{i=1}^{n} \frac{1}{n}$ for $n \geq 4$

### 189.3.2 The Limit Comparison Test

Consider the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$. We may want to compare this to the convergent series $\sum_{n=1}^{\infty} \frac{1}{2^n}$. The direct comparison test isn't helpful here, since $\frac{1}{2^n - 1} > \frac{1}{2^n}$, so $\sum_{n=1}^{\infty} \frac{1}{2^n}$ doesn't put a cap on $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ like our earlier example (see figure 189.7). In a case such as this, we can use the **Limit Comparison Test**, which states that:
If $\sum a_n$ and $\sum b_n$ are series with positive terms and $\lim_{n \to \infty} \frac{a_n}{b_n} = c > 0$, then either both series converge or both series diverge.

Let's apply this to the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$. We know that $\sum_{n=1}^{\infty} \frac{1}{2^n}$ converges, since it is a geometric series with $r < 1$.

$$\lim_{n \to \infty} \frac{\frac{1}{2^n - 1}}{\frac{1}{2^n}} = \lim_{n \to \infty} \frac{1}{2^n - 1} \cdot \frac{2^n}{1}$$

$$= \lim_{n \to \infty} \frac{2^n}{2^n - 1} = \lim_{n \to \infty} \frac{1}{1 - 1/2^n} = \frac{1}{1 - 0} = 1 > 0$$

Therefore, by the Limit Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ converges.

In general, comparison tests are most useful for series resembling geometric or p-series. When choosing a p-series to compare the unknown series to, choose p such that the order of your p series is the same as the order of the unknown series.

**Example**: What p-series should one compare the series $\sum_{n=1}^{\infty} \frac{\sqrt{n^3 + 1}}{3n^3 + 4n^2 + 2}$ to?

**Solution**: We can determine the order of $\frac{\sqrt{n^3 + 1}}{3n^3 + 4n^2 + 2}$ by looking at the highest-order terms

in the numerator and denominator:

$$\frac{\sqrt{n^3}}{n^3} = \frac{n^{3/2}}{n^3} = \frac{1}{n^{3/2}}$$

So, we should compare $\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{3n^3+4n^2+2}$ to the convergent p-series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$.

**Example**: Is $\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{3n^3+4n^2+2}$ convergent or divergent?

**Solution**: We have already determine that we should compare this series to $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$. To apply the limit test, we need to evaluate

$$\lim_{n\to\infty} \frac{\frac{\sqrt{n^3+1}}{3n^3+4n^2+2}}{\frac{1}{n^{3/2}}}$$

$$= \lim_{n\to\infty} \frac{n^{3/2}\sqrt{n^3+1}}{3n^3+4n^2+2}$$

$$= \lim_{n\to\infty} \frac{\sqrt{n^6+n^3}}{3n^3+4n^2+2} = \frac{1}{3} > 0$$

Therefore, by the Limit Comparison Test, $\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{3n^3+4n^2+2}$ is convergent because the p-series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ is convergent.

### *Exercise 509*

Use the Comparison Test or the Limit Comparison Test to determine if the following series are convergent or divergent.

1. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+1}}$
2. $\sum_{n=1}^{\infty} \frac{9^n}{3+10^n}$
3. $\sum_{n=1}^{\infty} \frac{n \sin^2 n}{1+n^3}$

*Working Space*

*Answer on Page 2091*

## 189.4 Ratio and Root Tests for Convergence

### 189.4.1 Absolute Convergence

Suppose there is a series $\sum_{n=1}^{\infty} a_n$, then there is a corresponding series $\sum_{n=1}^{\infty} |a_n| = |a_1| + |a_2| + |a_3| + \cdots$. If $\sum_{n=1}^{\infty} |a_n|$ is convergent, then the series $\sum_{n=1}^{\infty} a_n$ is called **absolutely convergent**.

**Example**: Consider the alternating series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} + \cdots$$

Is this series absolutely convergent?

**Solution**: We examine the corresponding series where we take the absolute value of each term:

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n^2} \right| = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

We can identify $\sum_{n=1}^{\infty} \frac{1}{n^2}$ as a convergent p-series. Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent, we can state that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$ is absolutely convergent.

**Example** Is the convergent series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ absolutely convergent?

**Solution** We consider the sum of the absolute values of the terms:

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$$

You should recognize this as the harmonic series, which is divergent. When a series is convergent but the corresponding series of absolute values is not, we call it **conditionally convergent**.

We won't prove the theorem here, but it is useful to know that if a series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then it is convergent. You can prove this yourself using the Comparison Test.

## *Exercise 510*

*Working Space*

Is the series given by

$$\frac{\cos 1}{1^2} + \frac{\cos 2}{2^2} + \frac{\cos 3}{3^3} + \cdots$$

convergent or divergent?

*Answer on Page 2091*

### Exercise 511

Determine whether each of the following series is absolutely or conditionally convergent.

1. $\sum_{n=1}^{\infty} \frac{(-1)^n}{3n+2}$
2. $\sum_{n=1}^{\infty} \frac{\sin n}{4^n}$
3. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{2n}{n^2+4}$

*Working Space*

*Answer on Page 2091*

## 189.4.2 The Ratio Test

The ratio test compares the $(n+1)^{th}$ term of a series to the $n^{th}$ term and takes the limit as $n \to \infty$ of the absolute value of this ratio:

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$$

There are three possible outcomes of the ratio test:

1. If $L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (and therefore convergent).
2. If $L = 1$, then the ratio test is inconclusive and we cannot draw any conclusions about whether $\sum_{n=1}^{\infty} a_n$ is convergent or divergent.
3. If $L > 1$ or $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

**Example**: Apply the ratio test to determine if $\sum_{n=1}^{\infty}(-1)^n\frac{n^3}{3^n}$ is convergent or divergent.

**Solution**:

$$\lim_{n\to\infty}\left|\frac{\frac{(-1)^{n+1}(n+1)^3}{3^{n+1}}}{\frac{(-1)^n n^3}{3^n}}\right| = \lim_{n\to\infty}\frac{(n+1)^3}{3^{n+1}}\cdot\frac{3^n}{n^3}$$

$$= \lim_{n\to\infty}\frac{(n+1)^3}{n^3}\cdot\frac{3^n}{3\cdot 3^n}$$

$$= \lim_{n\to\infty}\left(\frac{n+1}{n}\right)\cdot\frac{1}{3} = \frac{1}{3}\lim_{n\to\infty}\left(\frac{n+1}{n}\right) = \frac{1}{3}\cdot 1 = \frac{1}{3}$$

Since $L < 1$, the series $\sum_{n=1}^{\infty}(-1)^n\frac{n^3}{3^n}$ is absolutely convergent.

The ratio test is most useful for series that contain factorials, constants raised to the $n^{th}$ power, or other products.

## *Exercise 512*

[This question was originally presented as a multiple-choice, no-calculator problem on the 2012 AP Calculus BC exam.] Which of the following series are convergent?

1. $\sum_{n=1}^{\infty}\frac{8^n}{n!}$
2. $\sum_{n=1}^{\infty}\frac{n!}{n^{100}}$
3. $\sum_{n=1}^{\infty}\frac{n+1}{(n)(n+2)(n+3)}$

*Working Space*

*Answer on Page 2092*

### *Exercise 513*

[This question was originally presented as a multiple-choice, calculator-allowed problem on the 2012 AP Calculus BC exam.]If the series $\Sigma_{n=1}^{\infty} a_n$ converges and $a_n > 0$ for all $n$, which of the following statements must be true? Explain why.

*Working Space*

1. $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 0$
2. $|a_n| < 1$ for all $n$
3. $\sum_{n=1}^{\infty} a_n = 0$
4. $\sum_{n=1}^{\infty} n a_n$ diverges
5. $\sum_{n=1}^{\infty} \frac{a_n}{n}$ converges

*Answer on Page 2092*

#### 189.4.3 Root Test

The root test examines the behavior of the $n^{th}$ root of $a_n$ as $n \to \infty$. Similar to the ratio test, there are three possible outcomes:

1. If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, and therefore convergent.
2. If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L > 1$ or $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.
3. If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L = 1$, then the Root Test is inconclusive.

The root test is best when there is a term or terms raised to the $n^{th}$ power. Consider the series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2}\right)^n$:

**Example**: Is the series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2}\right)^n$ convergent or divergent?

**Solution**: Since $a_n$ consists of terms raised to the $n^{th}$ power, we will apply the root test

for convergence:

$$\lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{2n+3}{3n+2}\right)^n\right|} = \lim_{n\to\infty} \frac{2n+3}{3n+2} = \frac{2}{3} < 1$$

Therefore, by the root test, the series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2}\right)^n$ is convergent.

### *Exercise 514*

Use the Root Test to determine whether the following series are convergent or divergent.

*Working Space*

1. $\sum_{n=1}^{\infty} \left(\frac{3n^2+1}{n^2-4}\right)^n$
2. $\sum_{n=1}^{\infty} \frac{(-1)^n}{(\ln n)^n}$
3. $\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2}$

*Answer on Page 2093*

## 189.5 Strategies for Testing Series

When testing series for convergence, we want to choose a test based on the form of the series. While you may by tempted to try each test one-by-one until you find an answer, this quickly becomes cumbersome and time-consuming. Additionally, if you plan to take an AP Calculus exam, you need to be able to quickly choose an appropriate test as to conserve the time you have available for the exam. Here are some tips:

1. Check if the series is a p-series ($\sum_{n=1}^{\infty} \frac{1}{n^p}$). If so, then if $p > 1$, the series converges.

Otherwise, the series diverges.

2. If the series is not a p-series, check to see if you can write it as a geometric series ($\sum_{n=1}^{\infty} ar^{n-1}$ or $\sum_{n=1}^{\infty} ar^n$). Recall that geometric series are convergent if $|r| < 1$ and divergent otherwise.

3. If the series can't be written as a p-series or geometric series, but has a similar form, consider the comparison tests (the Direct Comparison Test and the Limit Comparison Test). When choosing a p-series to compare your series to, follow the guidelines outlined in the Comparison Tests section above.

4. If you can see at a glance that $\lim_{n\to\infty} a_n \neq 0$, then apply the Test for Divergence to show the series is divergent. REMEMBER: $\lim_{n\to\infty} a_n \neq 0$ implies the series $\sum_{n=1}^{\infty} a_n$ is divergent, but $\lim_{n\to\infty} a_n = 0$ **does not** necessarily imply the series $\sum_{n=1}^{\infty} a_n$ is convergent.

5. If the series is alternating (has $(-1)^n$ or $(-1)^{n-1}$ in the term), the Alternating Series test may provide an answer.

6. The Ratio Test is excellent for series with factorials, other products, or constants to the $n^{th}$ power. Remember that the Ratio Test will be inconclusive for p-series, rational functions of $n$, and algebraic functions of $n$.

7. If $a_n$ is of the form $(b_n)^n$, use the Root Test.

8. If $a_n = f(n)$ where $f(n)$ is continuous, positive, and decreasing and you can evaluate $\int_1^{\infty} f(x)\,dx$, use the Integral Test.

You don't need to treat this as a checklist, where you check for every condition. Rather, you should use this as a guide to quickly determine the convergence test most likely to be useful.

## Exercise 515

Choose an appropriate test to determine if the series is convergent of divergent. Apply the test and classify the series as convergent or divergent.

*Working Space*

1. $\sum_{n=1}^{\infty} \frac{e^n}{n^2}$
2. $\sum_{n=1}^{\infty} \frac{3^n n^2}{n!}$
3. $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}$
4. $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$
5. $\sum_{n=1}^{\infty} \left(\sqrt[n]{2} - 1\right)^n$

*Answer on Page 2093*

Chapter 190

# Power Series

Consider the function $f(x) = \frac{1}{1-x}$. This looks similar to the value of convergent geometric series, $\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$. If we let $a = 1$ and $r = x$, then we see that $\sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x}$. We can reindex to begin at $n = 0$ and see that:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

This is a **power series**. We use power series in place of functions for many applications, such as integrals where an explicit antiderivative don't exist, solving differential equations, and computer scientists representing functions on computers. Consider $f(x) = \frac{1}{1-x^2}$. What is $\int f(x)\,dx$? We can't directly use u-substitution, and this is not a derivative of any inverse trigonometric function. [You may have realized we could integrate this explicitly by using partial fractions, but this is not true for other functions, and we are using this as a demonstration anyway.] One way to evaluate this integral would be to represent $f(x)$ as a power series, then integrate the series. This is easier, since we know how to take the integral of any polynomial ($\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C$). First, we discuss what power series are further.

## 190.1 Power Series

Power series are series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots + c_n x^n$$

for some fixed $x$. Depending on $x$, the series may converge or diverge. For example, the power series $\sum_{n=0}^{\infty} x^n$ converges for $-1 < x < 1$ and diverges for all other values of $x$. This is because $\sum_{n=0}^{\infty} x^n$ is essentially a geometric series with $r = x$, which we already know converges for $|r| < 1$.

The form given above is for a power series centered on 0, but a power series can be centered on any value, $a$. In that case, it looks like this:

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n$$

Which we say is a *power series in* $(x-a)$, or a *power series centered at* $a$, or a *power series about* $a$.

**Example**: Find a power series representation for $f(x) = \frac{2x-4}{x^2-4x+3}$.

**Solution**: Since we know $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, we will use partial fractions to decompose the function into two fractions. (The process is left as an exercise for the student.) We find that:

$$\frac{2x-4}{x^2-4x+3} = \frac{1}{x-1} + \frac{1}{x-3}$$

Noting that $\frac{1}{x-1} = (-1)\cdot\frac{1}{1-x}$, we can say that:

$$\frac{1}{x-1} = (-1)\cdot\sum_{n=0}^{\infty} x^n$$

Now, let's look at $\frac{1}{x-3}$. We can show that:

$$\frac{1}{x-3} = \frac{\frac{1}{3}}{\frac{x}{3}-1} = \frac{1}{3}\frac{1}{\frac{x}{3}-1} = \frac{-1}{3}\frac{1}{1-\frac{x}{3}}$$

Substituting $\frac{x}{3}$ for $x$ into $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ we see that:

$$\frac{1}{1-\frac{x}{3}} = \sum_{n=0}^{\infty}\left(\frac{x}{3}\right)^n$$

Therefore:

$$\frac{1}{x-3} = \left(\frac{-1}{3}\right)\sum_{n=0}^{\infty}\left(\frac{x}{3}\right)^n$$

Adding the terms, we see that:

$$\frac{1}{x-1} + \frac{1}{x-3} = (-1)\sum_{n=0}^{\infty} x^n + \left(\frac{-1}{3}\right)\sum_{n=0}^{\infty}\left(\frac{x}{3}\right)^n$$

## 190.2 Power Series Convergence

Sometimes, you will be asked to find one or more values of $x$ for which a power series converges. To do this, choose a test to apply, then find $x$ such that the test is passed.

**Example**: For what values of $x$ is the series $\sum_{n=0}^{\infty} n!x^n$ convergent?

**Solution**: We will apply the Ratio Test (since there is a factorial in the series) and find $x$

such that $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$.

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)!x^{n+1}}{n!x^n}\right|$$

$$= \lim_{n\to\infty}\frac{(n+1)n!x\cdot x^n}{n!x^n} = \lim_{n\to\infty}\frac{(n+1)\cdot x}{1}$$

which converges to 0 when $x = 0$ and diverges for all other values of $x$. Therefore, $\sum_{n=0}^{\infty} n!x^n$ converges if $x = 0$.

**Example**: For what values of $x$ does the series $\sum_{n=1}^{\infty}\frac{(x-4)^n}{2n}$ converge?

**Solution**: We will use the Ratio Test again. We are looking for an $x$ such that:

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$$

$$\lim_{n\to\infty}\left|\frac{\frac{(x-4)^{n+1}}{2(n+1)}}{\frac{(x-4)^n}{2n}}\right| < 1$$

$$\lim_{n\to\infty}\left|\frac{(x-4)(x-4)^n}{2n+2}\cdot\frac{2n}{(x-4)^n}\right| < 1$$

$$\lim_{n\to\infty}\left|\frac{(x-4)(x-4)^n(2n)}{(x-4)^n(2n+2)}\right| < 1$$

$$\lim_{n\to\infty}\left|\frac{(x-4)(2n)}{2n+2}\right| < 1$$

$$\lim_{n\to\infty}\left|\frac{2(x-4)}{2+\frac{2}{n}}\right| < 1$$

$$2\cdot\lim_{n\to\infty}\left|\frac{x-4}{1+\frac{1}{n}}\right| < 1$$

$$2\cdot|x-4| < 1$$

$$|x-4| < \frac{1}{2}$$

Which is true when

$$-\frac{1}{2} < x-4 < \frac{1}{2}$$

$$3.5 < x < 4.5$$

We are not done yet, though! We know the series converges for $3.5 < x < 4.5$ and diverges for $x < 3.5$ and $x > 4.5$. What about when $x = 3.5$ and $x = 4.5$? (These are the cases

where the Ratio Test is indeterminate, because $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1$.) We need to test each case. Substituting $x = 3.5$ into the series yields:

$$\sum_{n=1}^{\infty}\frac{(3.5-4)^n}{2n} = \sum_{n=1}^{\infty}\frac{\left(\frac{-1}{2}\right)^n}{2n}$$

$$= \sum_{n=1}^{\infty}\frac{(-1)^n}{n\cdot 2^{n+1}}$$

This is an alternating series, so we apply the alternating series test. First, we check that $|a_{n+1}| < |a_n|$:

$$\frac{1}{(n+1)\cdot 2^{n+2}} < \frac{1}{n\cdot 2^{n+1}}$$

Which is true for all $n > 0$. Next, we check if $\lim_{n\to\infty}|a_n| = 0$:

$$\lim_{n\to\infty}\frac{1}{n\cdot 2^{n+1}} = \frac{1}{\infty} = 0$$

Therefore, $\sum_{n=1}^{\infty}\frac{(-1)^n}{n\cdot 2^{n+1}}$ is convergent and $\sum_{n=1}^{\infty}\frac{(x-4)^n}{2n}$ is convergent for $x = 3.5$. Next, we test $x = 4.5$ for convergence:

$$\sum_{n=1}^{\infty}\frac{(4.5-4)^n}{2n} = \sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n\cdot\frac{1}{2n} = \sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^{n+1}\frac{1}{n}$$

This series is less than the harmonic series $\sum_{n=1}^{\infty}\frac{1}{n}$ for all $n$. We know the harmonic series diverges, therefore, by the direct comparison test, $\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^{n+1}\frac{1}{n}$ must also diverge. So, our final answer to the original question is that the series is convergent fort $3.5 \le x < 4.5$.

### 190.2.1 Radius of Convergence

There are three possible outcomes when testing a power series $\sum_{n=0}^{\infty}c_n(x-a)^n$ for convergence:

1. The series only converges for $x = a$
2. The series converges for all $x$
3. The series converges if $|x-a| < R$ and diverges for $|x-a| > R$, where $R$ is some positive number

We call $R$ the **radius of convergence** . If we rearrange $|x-a| < R$, we can see why this is called a radius (see figure 190.1):

$$a - R < x < a + R$$

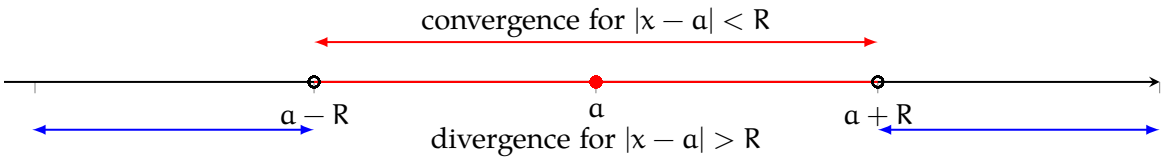



Figure 190.1: $R$ is called the radius of convergence because it is half the width of the window of convergence

When $x = a \pm R$, the series could be convergent or divergent. You will need to test the endpoints of the window of convergence to determine if the interval is open or closed. Thus, there are four possibilities for the interval of convergence:

1. $(a-R, a+R)$
2. $[a-R, a+R)$
3. $(a-R, a+R]$
4. $[a-R, a+r]$

It is important to check each endpoint to determine the interval of convergence individually.

In the example of $\sum_{n=1}^{\infty} \frac{(x-4)^n}{2n}$ (shown above), $a = 4$ and $R = 0.5$, and we found that the power series is convergent for $x \in [3.5, 4.5)$.

**Example**: For what values of $x$ is the Bessel function $J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$ convergent?

**Solution**: Because there is a factorial, we will apply the Ratio Test:

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$$

$$\lim_{n\to\infty} \left| \frac{\frac{(-1)^{n+1} x^{2(n+1)}}{2^{2(n+1)}((n+1)!)^2}}{\mathrm{frac}(-1)^n x^{2n} 2^{2n} (n!)^2} \right| < 1$$

$$\lim_{n\to\infty} \frac{x^{2n} x^2 2^{2n} n! n!}{2^{2n} 2^2 (n=1)!(n+1)! x^{2n}} < 1$$

$$\lim_{n\to\infty} \frac{x^2 n! n!}{2^2 (n+1) n! (n+1) n!} < 1$$

$$\lim_{n\to\infty} \frac{x^2}{4(n+1)^2} = 0 < 1$$

Because $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = 0$ for all $x$, the Bessel function $J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$ is conver-

gent for all real values of $x$, and the interval of convergence is $(-\infty, \infty)$.

**Example**: Find the radius and interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}$.

**Solution**: Again, we apply the ratio test to find values of $x$ such that $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$:

$$\lim_{n\to\infty} \left| \frac{(-3)^{n+1}x^{n+1}}{\sqrt{n+1+1}} \cdot \frac{\sqrt{n+1}}{(-3)^n x^n} \right| < 1$$

$$\lim_{n\to\infty} \left| \frac{(-3)x}{\sqrt{n+2}} \cdot \frac{\sqrt{n+1}}{1} \right| < 1$$

$$\lim_{n\to\infty} \left| (-3)x\sqrt{\frac{n+2}{n+1}} \right| < 1$$

$$3|x| \lim_{n\to\infty} \sqrt{\frac{n+2}{n+1}} < 1$$

$$3|x| \lim_{n\to\infty} \sqrt{\frac{1+2/n}{1+1/n}} = 3|x|(1) < 1$$

$$3|x| < 1$$

$$|x| < \frac{1}{3}$$

Therefore, the radius of convergence is $\frac{1}{3}$. We need to test the endpoints, $x = \frac{-1}{3}$ and $x = \frac{1}{3}$, to determine the interval of convergence. First, we will test if $\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}$ when $x = \frac{-1}{3}$:

$$\sum_{n=0}^{\infty} \frac{(-3)^n \left(\frac{-1}{3}\right)^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{1^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n+1}}$$

This is a p-series such that $p < 1$, so it is divergent, and our original series does not converge for $x = \frac{-1}{3}$. Next, we test $x = \frac{1}{3}$:

$$\sum_{n=0}^{\infty} \frac{(-3)^n \left(\frac{1}{3}\right)^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$$

which is an alternating series that converges by the alternating series test. Therefore, the interval of convergence for $\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}$ is $x \in \left(\frac{-1}{3}, \frac{1}{3}\right]$.

### Exercise 516

[This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC exam.] What is the radius of convergence of the series $\sum_{n=0}^{\infty} \frac{(x-4)^{2n}}{3^n}$?

*Working Space*

*Answer on Page 2094*

### Exercise 517

[This problem was originally presented as a no-calculator, multiple-choice question on the 2012 AP Calculus BC exam.] A power series is given by $\frac{x}{3} - \frac{x^3}{5} + \frac{x^5}{7} - \frac{x^7}{9} + \cdots$. Write the series in sigma notation and use the Ratio Test to determine the interval of convergence.

*Working Space*

*Answer on Page 2094*

## 190.3 Calculus with Power Series

You can integrate and differentiate power series. Let $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$. Recall that $f(x)$ is just a very long polynomial and that the derivative of a polynomial $x^n$ is $n \cdot x^{n-1}$. We can then state that:

$$\frac{d}{dx} f(x) = \frac{d}{dx}\left[c_0 + c_1(x-a)^1 + c_2(x-a)^2 + \cdots + c_n(x-a)^n\right]$$

$$f'(x) = 0 + c_1 + 2c_2(x-a)^1 + \cdots + nc_n(x-a)^{n-1}$$

$$f'(x) = \sum_{n=1}^{\infty} c_n(x-a)^{n-1}$$

which is true when $x$ is in the interval of convergence for the series.

Similarly, we know $\int x^n \, dx = \frac{1}{n+1}x^{n+1}$. We can then say that:

$$\int f(x) \, dx = \int \left[c_0 + c_1(x-a)^1 + c_2(x-a)^2 + \cdots + c_n(x-a)^n\right], dx$$

$$\int f(x) \, dx = C + c_0(x-a) + \frac{c_1}{2}(x-a)^2 + \frac{c_2}{3}(x-a)^3 + \cdots \frac{c_n}{n+1}(x-a)^{n+1}$$

$$\int f(x) \, dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1}(x-a)^{n+1}$$

Where $C$ is the integration constant. Again, this is true when $x$ is in the interval of convergence for the series.

**Example**: Express $\frac{1}{(1-x)^2}$ as a power series by differentiating $\frac{1}{1-x}$.

**Solution**: Recall that $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ when $|x| < 1$.

$$\frac{1}{1-x} = \sum_{n=1}^{\infty} x^n$$

Differentiating both sides:

$$\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{d}{dx}\sum_{n=0}^{\infty} x^n$$

$$(-1) \cdot \frac{1}{(1-x)^2} \cdot \frac{d}{dx}(1-x) = \sum_{n=1}^{\infty} nx^{n-1}$$

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$$

Reindexing to begin at $n = 0$

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)x^n$$

Because $\sum_{n=0}^{\infty}$ has a radius of convergence of 1, so does $\sum_{n=0}^{\infty}(n+1)x^n$. We can confirm our series makes sense by plotting the partials sums for $n = 3, 5, \text{and} 7$ with the original function (see figure 190.2).

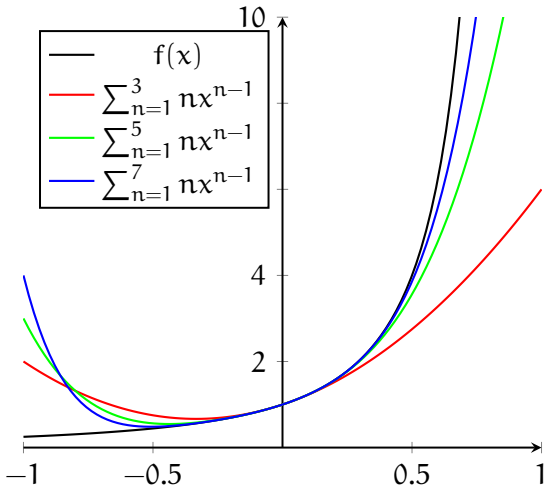



Figure 190.2: The function $f(x) = \frac{1}{(1-x)^2}$ is equal to the power series $\sum_{n=1}^{\infty} nx^{n-1}$

**Example**: Find a power series representing $\ln(1+x)$.

**Solution**: We know that $\frac{d}{dx}\frac{1}{1-x} = \ln(1-x)$. Replacing $x$ with $-x$, we see that:

$$\frac{1}{1-(-x)} = \frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n$$

which converges for $|x| < 1$. We can then integrate both sides:

$$\int \frac{1}{1+x}\,dx = \int \left[\sum_{n=0}^{\infty} (-x)^n\right] dx$$

$$\ln(1+x) = \int \left(1 - x + x^2 - x^3 + \cdots\right) dx$$

$$\ln(1+x) = C + x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots + (-1)^{n-1}\frac{x^n}{n}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1}\frac{x^n}{n} + C$$

when $|x| < 1$. To find $C$, substitute $x = 0$ and solve:

$$\ln(1+0) = C + \sum_{n=1}^{\infty} (-1)^{n-1}\frac{0^n}{n} = C + 0$$

$$C = \ln 1 = 0$$

So, our final answer is $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1}\frac{x^n}{n}$.

### Exercise 518

Find a power series representation for $f(x) = \arctan x$.

*Working Space*

*Answer on Page 2095*

Chapter 191

# Population Proportion Statistics

Let's say that you are trying to get a candidate elected. The candidate asks you, "What proportion of the voting population is going to vote for me?" So, you go out and ask a random sample of 12 voters. 11 say that they are going to vote for your candidate. What can you tell the candidate?

## 191.1 Sample Probabilities from Population Proportion

To address these sorts of questions (and there are a lot of them), we start with the opposite question: If we knew what the proportion was in the entire population, what sort of results should we expect in a random sample of just 12?

For example, let's say that 62% of the entire population plans to vote for your candidate. You ask 12 people, "Will you vote for my candidate?" How many will say "Yes"? You don't know — it depends on the sample. For example, there is some chance that you will just happen to choose all 12 from the 48% of the population that does not plan to vote for your candidate.

We can compute the probability of each outcome using the binomial distribution. Let $r$ be the probability that a random person will say, "I plan to vote for your candidate." Let $n$ be the number of people you ask. The probability that exactly $k$ people will say, "I plan to vote for your candidate" is given by:

$$p(k) = \binom{n}{k} r^k (1-r)^{n-k}$$

Note that even though most people support your candidate, there is some chance that no one you ask will say that they will vote for your candidate.

Using $r = 0.62$, we can compute the probability of each outcome:

| k | p(k) |
|---|---|
| 0 | 0.000009 |
| 1 | 0.000177 |
| 2 | 0.001593 |
| 3 | 0.008663 |
| 4 | 0.031801 |
| 5 | 0.083017 |
| 6 | 0.158024 |
| 7 | 0.220996 |
| 8 | 0.225358 |
| 9 | 0.163418 |
| 10 | 0.079989 |
| 11 | 0.023729 |

Looking at this, the most likely outcome is that 8 people will say "Yes." However, although it is the most like, there is still less than a 1 in 4 chance of that outcome. It is very unlikely that less than 2 people will say "Yes." Here is a bar chart of the data:

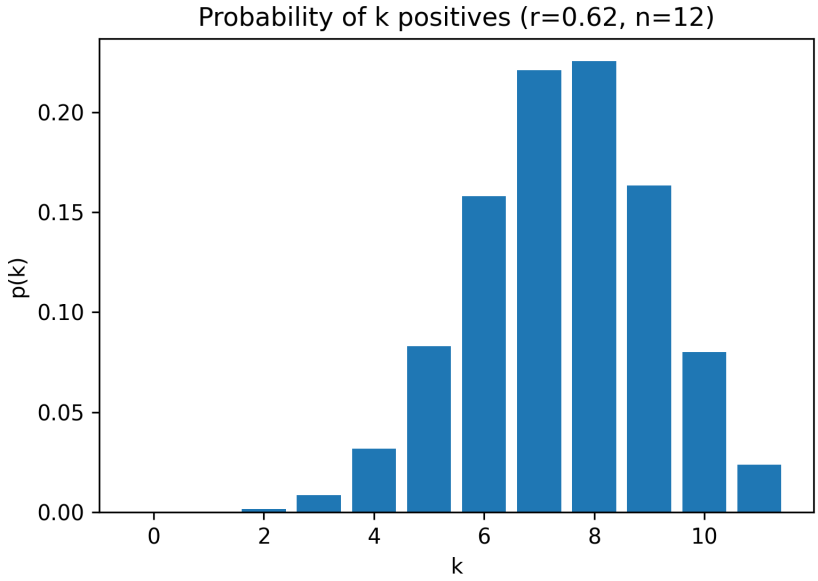



Figure 191.1: Binomial distribution of k positive votes.

In this section, we knew the proportion of the population (p) and used that to find the probability of each possible number of positives in a random sample (k). Now, we are going to go the other way: You know k, and you are finding the probability of possible values of p.

## 191.2 Population Proportion from Sample

You ask 12 people if they will vote for your candidate. 9 say "Yes."

Next, you do a thought experiment: "If only 10% of the population were going to vote for my candidate, what is the probability that I would see this outcome?"

$$p(9) = \binom{n}{k} r^k (1-r)^{n-k} = \binom{12}{9} (0.1)^9 (0.9)^{12-9} = 0.00000016$$

This outcome would be quite unusual. What if 70% of the population were going to vote for your candidate? What is the probability that you would see this outcome?

$$p(9) = \binom{n}{k} r^k (1-r)^{n-k} = \binom{12}{9} (0.7)^9 (0.3)^{12-9} \approx 0.2397$$

In this case, the observed outcome would be a lot less unusual.

So, you decide to plot out the likelihood of this outcome for every possible value of r:

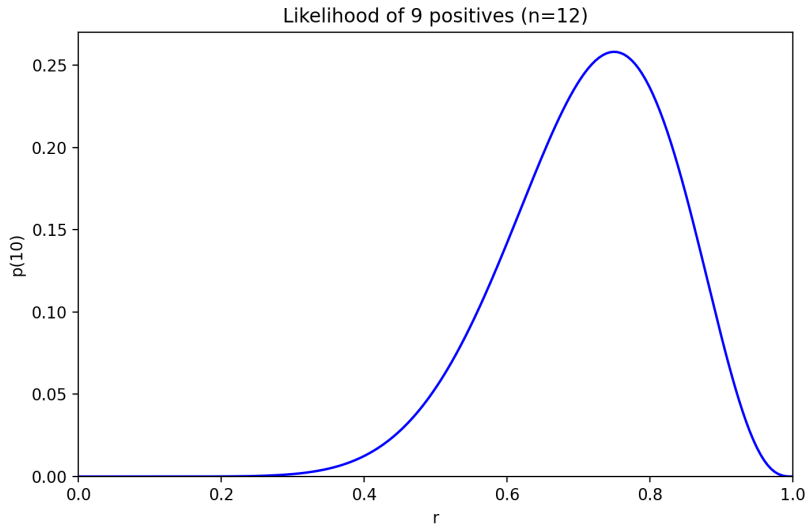



Figure 191.2: Likelihood of 9 positives.

This looks very similar to a probability distribution, but *it is not*. The area under the curve does not integrate to 1.0 — it is significantly less. This is a called a *likelihood*.

However, it still tells us something, right? The maximum likelihood estimator is $9/12 = 0.75$.

## 191.3 From Likelihood to Probability Density Function

How can we make this likelihood into a probability density function? We use Bayes' Law for continuous probability:

$$p(r|k) = \frac{p(k|r)p(r)}{\int_{r=0}^{1} p(k|r)p(r)dr}$$

In other words, given that we had $k$ positive responses, what is the probability that the proportion of the population that will vote for your candidate is $r$? The numerator of the fraction is the likelihood scaled up or down by our prior belief about the value of $r$. What is the denominator of the fraction? For this to be a probability distribution, we need it to integrate to 1. This is taken care of by the denominator.

Let's say we have no prior belief about the value of $r$. That is, $p(r)$ is the continuous uniform distribution between 0 and 1; thus, $p(r) = 1$ for all possible values of $r$. Our formula becomes:

$$p(r|k) = \frac{p(k|r)}{\int_{r=0}^{1} p(k|r)dr}$$

That is the likelihood scaled up so that it integrates to 1. If we plot this, we get:

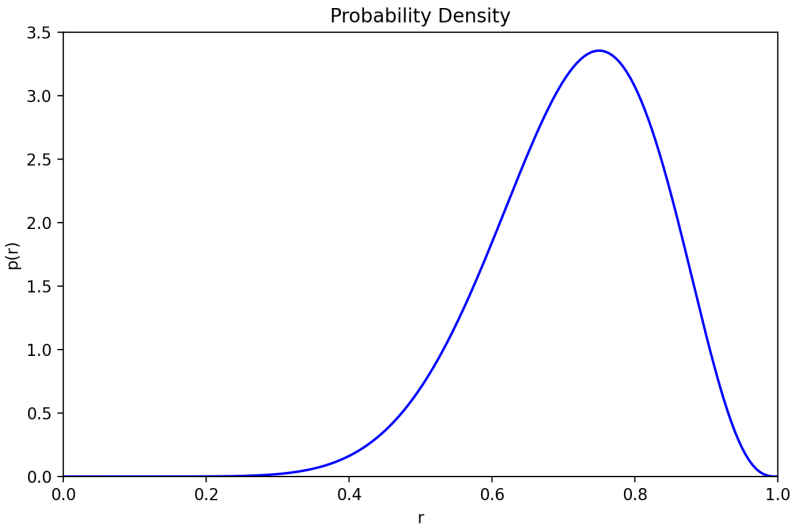



Figure 191.3: PDF of sampled likelihood function.

Here, then, is your report to your candidate: "I asked 12 voters if they were going to vote

for you. 9 said yes. Using a uniform prior, here is what I believe about your support in the general population." You also include this graph.

What happens to this graph if you ask 120 voters and 90 say yes?

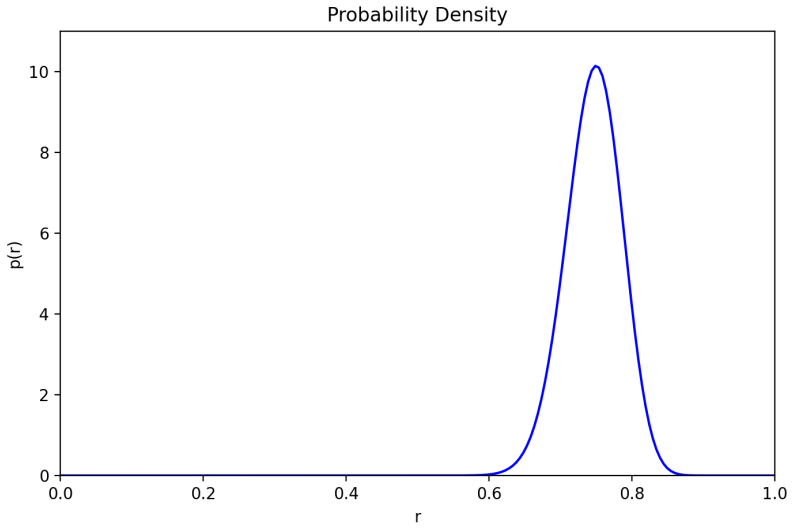



Figure 191.4: Bayes shifted for 90 positive votes from 120 voters.

The MLE (0.75) is the same, but because of the much larger sample size, you are more confident when you say "It is probably close to 75%."

## 191.4 Beta Distribution

This probability distribution that you discovered is actually pretty common. It is known as the *beta distribution*.

The beta distribution has two parameters $a$ and $b$ that determine its shape. If you get $k$ positives out of $n$, then use $a = k + 1$ and $b = n - k + 1$.

When you make your report to your candidate, they will look at your probability distribution with quiet awe and ask "Based on your sample of 12 people, what is the probability that at least 50% of the population will vote for me?" So, you'd fill in the region and say, "This area represents that probability."

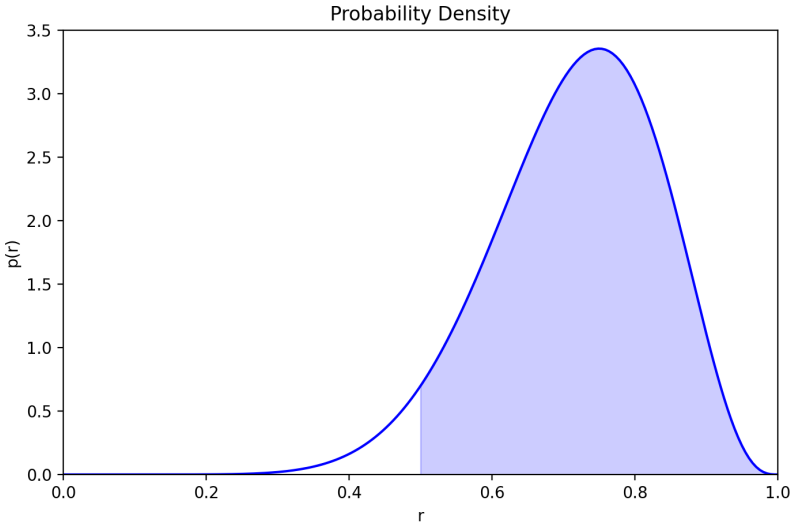



Figure 191.5: PDF for the beta distribution at least 50% positives.

Once again, there will be a long silence, followed by a simple question: "Can you give me a number?" Here is the Python code:

```
import numpy as np
from scipy.stats import beta

# Constants
K = 9
N = 12

# What is the probability r <=0.5?
p_less = beta.cdf(0.5, K +1, N - K +1)

# What is the probability r > 0.5?
p_more = 1.0 - p_less
print(f"I'm {p_more * 100.0:.2f}% sure you will win.")
```

This will give you:

```
I'm 95.39% sure you will win.
```

Chapter 192

# The Normal Distribution

The Normal distribution, also known as the Gaussian distribution, is a type of continuous probability distribution for a real-valued random variable. It is one of the most important probability distributions in statistics due to its several unique properties and usefulness in many areas.

## 192.1 Defining the Normal Distribution

The Normal distribution is defined by its mean ($\mu$) and standard deviation ($\sigma$). The probability density function (pdf) of a Normal distribution is given by:

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

where:

- $x$ is the point up to which the function is integrated,
- $\mu$ is the mean or expectation of the distribution,
- $\sigma$ is the standard deviation,
- $\sigma^2$ is the variance.

## 192.2 Importance of the Normal Distribution

There are several reasons why the Normal distribution is crucial in statistics:

- **Central Limit Theorem:** One of the main reasons for the importance of the Normal distribution is the Central Limit Theorem (CLT). The CLT states that the distribution of the sum (or average) of a large number of independent, identically distributed variables approaches a Normal distribution, regardless of the shape of the original distribution.
- **Symmetry:** The Normal distribution is symmetric, which simplifies both the theoretical analysis and the interpretation of statistical results.

- **Characterized by Two Parameters:** The Normal distribution is fully characterized by its mean and standard deviation. The mean determines the center of the distribution, and the standard deviation determines the spread or girth of the distribution.
- **Common in Nature:** Many natural phenomena follow a Normal distribution. This includes characteristics like people's heights or IQ scores, measurement errors in experiments, and many others.

Given its properties, the Normal distribution serves as a foundation for many statistical procedures and concepts, including hypothesis testing, confidence intervals, and linear regression analysis.

### 192.2.1 Example

A set of course grades is modeled by a normal distribution with mean $\mu = 75$ and standard deviation $\sigma = 8$. Using a 90–80–70–60 grading scale, find the percentage of students who earned a B (scores from 80 to less than 90) using Excel and Python.

**Solution**

Let $X$ be a student's grade. Then

$$X \sim N(75, 8).$$

A B corresponds to the interval $80 \leq X < 90$, so we want

$$P(80 \leq X < 90).$$

Excel computes cumulative probabilities using

```
NORM.DIST(x, mean, standard_dev, TRUE)
```

Therefore,

$$P(80 \leq X < 90) = P(X < 90) - P(X < 80).$$

In Excel, enter:

```
=NORM.DIST(90,75,8,TRUE) - NORM.DIST(80,75,8,TRUE)
```

Excel returns approximately

$$0.2356.$$

So about 23.56% of students earned a B.

Now compute the same probability in Python.

```
from scipy.stats import norm

mu = 75
sigma = 8

probability = norm.cdf(90, mu, sigma) - norm.cdf(80, mu, sigma)

print(probability)
print(probability * 100)
```

Python returns approximately

$$0.2356$$

which confirms the Excel result.

Both Excel and Python are computing the same cumulative distribution function.

### Exercise 519

*Working Space*

A batch of parts has diameters modeled by a normal distribution with mean $\mu = 10$ mm and standard deviation $\sigma = 2$ mm:

$$X \sim N(10, 2).$$

A part is considered *acceptable* if its diameter is between 6.72 mm and 12.28 mm.

(a) Write the probability statement that represents the fraction of acceptable parts.

(b) Use Excel to compute this probability using `NORM.DIST`.

(c) Convert your probability to a percentage.

*Answer on Page 2095*

Chapter 193

# Change of Variables

Let's say that we are making ice spheres by filling water into balloons, and we tell you that the radius of the ice spheres is normally distributed with a mean of 0.7 cm and a standard deviation of 0.08 cm. You can then draw the probability distribution and cumulative distribution for that:

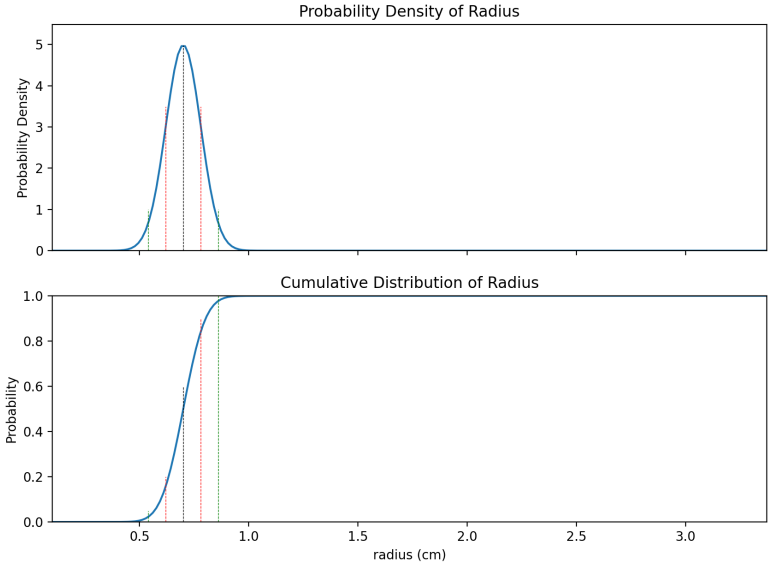



Figure 193.1: Ice sphere PDF and cumulative density function.

This includes lines indicating the mean and two standard deviations on each side.

Now, let's say we ask you what the cumulative distribution is for the *mass* of the balloons. A cubic centimeter of ice weighs about a gram, so if you know the radius of a particular ice sphere, it is easy to compute the mass of it:

$$m = \frac{4}{3}\pi r^3$$

So, for example, if a sphere has a radius of 5cm, its mass in grams is $\frac{4\pi(0.7^3)}{3} \approx 1.44$ g.

Thinking about the graph of the cumulative distribution, if half the balloons have a radius

less than 5 cm, then half of the balloons have a mass less than 523.6 g. For each point on the cumulative graph, we can use the radius of that point to compute the corresponding mass — the CDF gets stretched out:

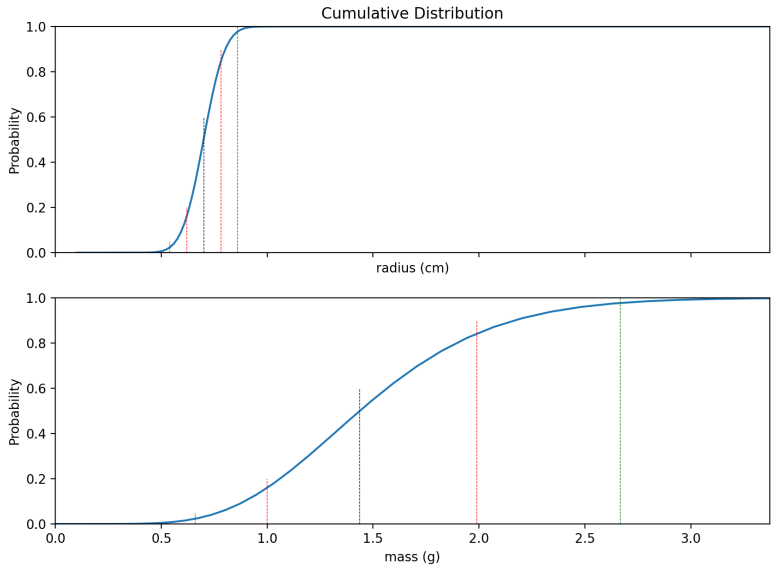



Figure 193.2: Cumulative distribution of both the radius and mass of the ice spheres.

If $F$ is the original cumulative distribution function, and $g$ is the function that maps the new variable (mass, in this case) to the old one (radius), then the new cumulative distribution function $H$ is given by

$$H(m) = F(g(m))$$

In this case, $F$ is the cumulative function for the normal distribution, with mean $0.7$ and standard deviation of $0.08$. $g$ maps the mass to the radius:

$$g(m) = \left(\sqrt[3]{\frac{3}{4\pi}}\right) x^{\frac{1}{3}}$$

## 193.1 Making a Probability Density Function

Now, we know how to calculate a new cumulative distribution function using the new variable. However, we usually want a probability density.

Here is the CDF and the PDF of the mass of the ice spheres:

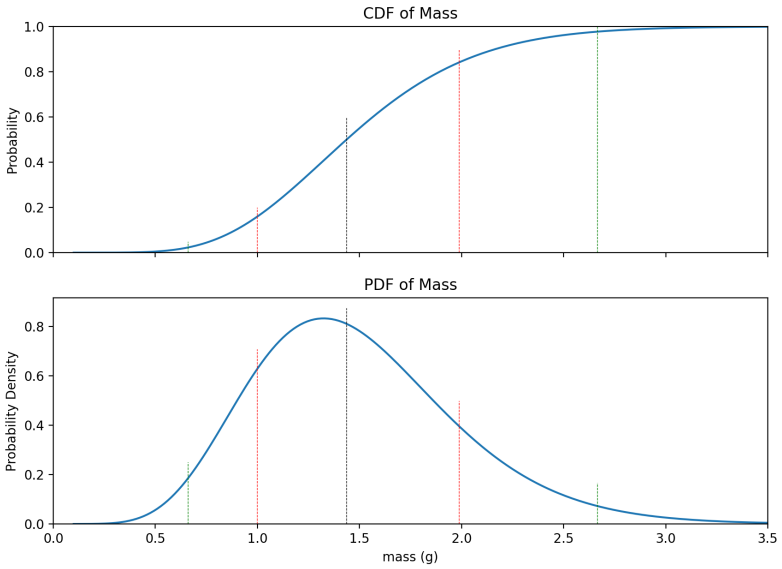



Figure 193.3: CDF and PDF of the masses.

Reminder: The probability density function is the derivative of the cumulative distribution function. We know the CDF is

$$H(m) = F(g(m))$$

By the chain rule:

$$H'(m) = F'(g(m))g'(m)$$

The function $F$ is the cumulative distribution for the normal distribution with mean $0.7$ and standard deviation of $0.08$. So, we know its derivative:

$$F'(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$$

where $\mu = 0.7$ and $\sigma = 0.08$.

We've already said that

$$g(m) = \left(\sqrt[3]{\frac{3}{4\pi}}\right) m^{\frac{1}{3}}$$

Which is easy to differentiate:

$$g'(m) = \left(\frac{1}{3}\right)\left(\sqrt[3]{\frac{3}{4\pi}}\right) m^{-\frac{2}{3}}$$

Here, then, is the code to generate that last plot:

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

# Constants
MEAN_RADIUS = 0.7
STD_RADIUS = 0.08

# Range to plot
MIN_MASS = 0.1
MAX_MASS = 3.5

# Number of points to plot
N = 200

# Needed for radius_for_mass and d_radius_for_mass
C = np.power(3 / (4 * np.pi), 1/3)

# In these three functions, x can
# be a number or a numpy array

def mass_for_radius(x):
  return 4 * np.pi * np.power(x, 3) / 3

def radius_for_mass(x):
  return C * np.power(x, 1/3)

# Derivative of radius_for_mass()
def d_radius_for_mass(x):
  return (C/3) * np.power(x,-2/3)

# Compute mean and 2 standard deviations in each direction
m_mean = mass_for_radius(MEAN_RADIUS)
m_minus_std = mass_for_radius(MEAN_RADIUS - STD_RADIUS)
m_plus_std = mass_for_radius(MEAN_RADIUS + STD_RADIUS)
m_minus_2std = mass_for_radius(MEAN_RADIUS - 2 * STD_RADIUS)
m_plus_2std = mass_for_radius(MEAN_RADIUS + 2 * STD_RADIUS)
```

```
38
39  # Make N possible values for mass
40  m_values = np.linspace(MIN_MASS, MAX_MASS, N)
41
42  # Compute g(m) for each of these masses
43  # That is: What is the radius for each of these masses?
44  r_values = radius_for_mass(m_values)
45
46  # Compute F(g(m)) for each of these masses
47  # That is: What is the cumulative distribution for each those radii?
48  cdf_values = norm.cdf(r_values, loc=MEAN_RADIUS, scale=STD_RADIUS)
49
50  # Compute g'(m) for each of these masses
51  dg_values = d_radius_for_mass(m_values)
52
53  # What is F'(g(m))g'(m)?
54  pdf_values = norm.pdf(r_values, loc=MEAN_RADIUS, scale=STD_RADIUS) * dg_values
55
56  # Sanity check: It should integrate to a little less then 1.0
57  dx = (MAX_MASS - MIN_MASS)/N
58  area_under_curve = pdf_values.sum() * dx
59  print(f"Integral from {MIN_MASS:.2f} to {MAX_MASS:.2f}: {area_under_curve:.3f}")
60
61  # Make a figure with two axes
62  fig, axs = plt.subplots(nrows=2, sharex=True, figsize=(10, 7), dpi=200)
63
64  # Draw the CDF on the second axis
65  axs[0].set_title("CDF of Mass")
66  axs[0].set_ylim(bottom=0.0, top=1.0)
67  axs[0].set_xlim(left=0.0, right=MAX_MASS)
68  axs[0].set_ylabel("Probability")
69  axs[0].plot(m_values, cdf_values)
70
71  # Add lines for mean, mean-std, and mean+std
72  axs[0].vlines(m_minus_2std, 0, 0.05, "g", linestyle="dashed",lw=0.5)
73  axs[0].vlines(m_minus_std, 0, 0.2, "r", linestyle="dashed",lw=0.5)
74  axs[0].vlines(m_mean, 0, 0.6, "k", linestyle="dashed",lw=0.5)
75  axs[0].vlines(m_plus_std, 0, 0.9, "r", linestyle="dashed",lw=0.5)
76  axs[0].vlines(m_plus_2std, 0, 1.0, "g", linestyle="dashed",lw=0.5)
77
78  # How high does the pdf go?
79  max_density = pdf_values.max()
80
81  # Draw the PDF on the second axis
82  axs[1].set_title("PDF of Mass")
83  axs[1].set_ylim(bottom=0.0, top=max_density * 1.1)
84  axs[1].set_xlim(left=0.0, right=MAX_MASS)
85  axs[1].set_xlabel("mass (g)")
86  axs[1].set_ylabel("Probability Density")
87  axs[1].plot(m_values, pdf_values)
88
89  # Add lines for mean, mean-std, and mean+std
```

```
axs[1].vlines(m_minus_2std, 0, max_density * .3, "g", linestyle="dashed",lw=0.5)
axs[1].vlines(m_minus_std, 0, max_density * .85 , "r", linestyle="dashed",lw=0.5)
axs[1].vlines(m_mean, 0, max_density * 1.05, "k", linestyle="dashed",lw=0.5)
axs[1].vlines(m_plus_std, 0, max_density * .6, "r", linestyle="dashed",lw=0.5)
axs[1].vlines(m_plus_2std, 0, max_density * .2, "g", linestyle="dashed",lw=0.5)
fig.savefig("pdf.png")
```

## 193.2 Decreasing Conversions

The last case (mass and radius) is relatively straightforward, because the function g is always increasing. What if we have a change of variables where g is decreasing? For example, $V = IR$ so $\frac{V}{R} = I$.

Let's say that you work at a lightbulb factory and you sample the lightbulbs to see what their resistance is. You find the resistances of the lightbulbs are normally distributed with a mean of 24 ohms and a standard deviation of 3 ohms. The voltage will be exactly 12 volts. What is the PDF of the currents that will pass through the lightbulbs?

$$I = \frac{12}{R}$$

so

$$g(x) = \frac{12}{x}$$

is the function that will convert the current to resistance. Taking the derivative, we get:

$$g'(i) = -\frac{12}{x^2}$$

FIXME finish this section with graphs and code.

Chapter 194

# Volumes with Integrals

Suppose we wanted to know the volume of a theoretical irregular shape (we stipulate theoretical because, if you had this object and a large enough container, you could use displacement to determine the volume of the object). [fixme better intro]

## 194.1 Volume of a Sphere

Below, we will prove the volume of a sphere is given by $\frac{4}{3}\pi r^3$ using the integral method. Suppose we have a sphere of radius $r$ centered at the origin (see figure 194.1).

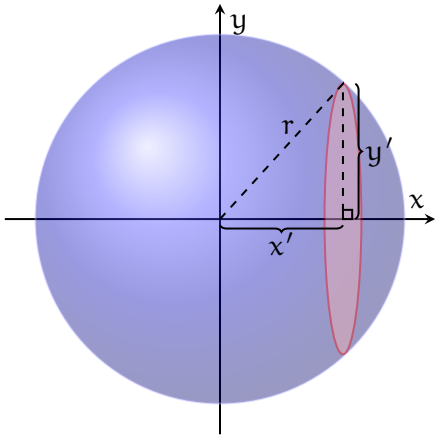



Figure 194.1: A vertical cross-section of a sphere

We begin by taking very thin vertical cross-sections. The radius of the cross-section is the height, $y$, of the sphere at the horizontal position, $x$. Since the edges of the cross-section lie on the sphere, we know the edge of the cross-section is distance $r$ from the origin. Applying the Pythagorean theorem, we see that $r^2 = x^2 + y^2$, which implies that $y = \sqrt{r^2 - x^2}$. So, the area of the cross-section is given by $\pi y^2 = \pi(r^2 - x^2)$. If we imagine each cross section as having a width, $dx$, and taking the sum of all the cross sections from $x = -r$ to $x = r$, we can write an integral equal to the volume of the sphere:

$$V_{\text{sphere}} = \int_{-r}^{r} \pi(r^2 - x^2)\, dx$$

We can then evaluate that integral:

$$V_{sphere} = \pi \int_{-r}^{r} r^2 \, dx - \pi \int_{-r}^{r} x^2 \, dx$$

$$V_{sphere} = \pi \left[ r^2 x \right]_{x=-r}^{x=r} - \frac{\pi}{3} \left[ x^3 \right]_{x=-r}^{x=r}$$

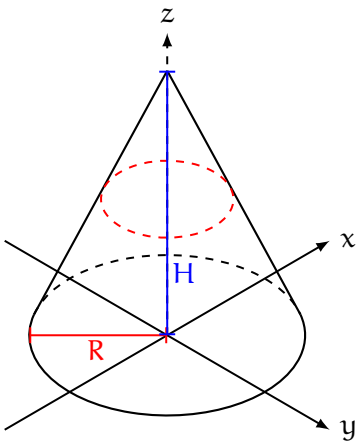
$$V_{sphere} = \pi \left[ r^3 - (-r^3) \right] - \frac{\pi}{3} \left[ r^3 - (-r^3) \right]$$

$$V_{sphere} = 2\pi r^3 - \frac{2\pi}{3} r^3 = \frac{4}{3} \pi r^3$$

### *Exercise 520*

Prove the volume of a regular cone is $\frac{\pi}{3}R^2H$, where R is the radius of the base and H is the height of the cone. (Hint: A cone is a series of decreasing circles stacked on top of each other; see figure below.)

*Working Space*




*Answer on Page 2096*

## 194.2 Volumes of Solids of Revolution

We can also find the volume of solids made by revolving a graph about the x or y-axis. Suppose the graph $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{2}$ were rotated vertically about the x-axis to form a solid. How could we find the volume of that solid? Well, we can imagine a

rectangle of width $dx$ and height $y$ (see figure 194.2)

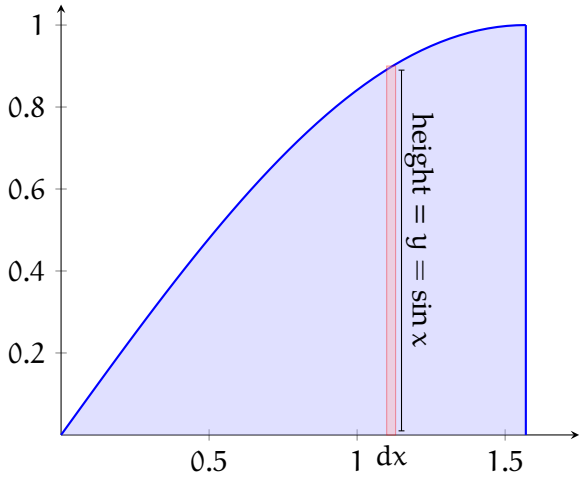



Figure 194.2: A cross section has width $dx$ and height $y = \sin x$

If we rotate the plot vertically about the $x$-axis, the rectangle becomes a cylinder with radius $y = \sin x$ and height $dx$ (see figure **??**). Therefore, the volume of each cylindrical slice is $V_{slice} = \pi r^2 dx = \pi \cdot \sin^2 x dx$.

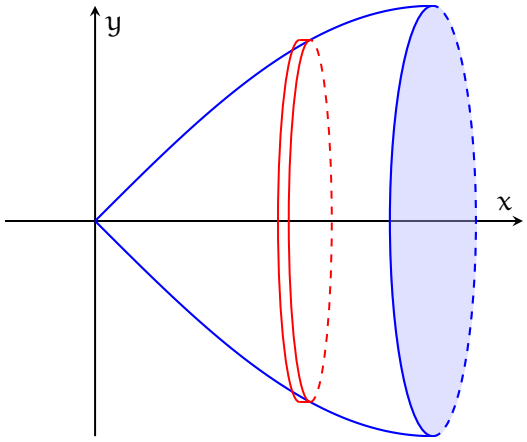



Figure 194.3: When rotated, the cross-section becomes a cylinder with radius $\sin x$ and width $dx$, which has a total volume of $\pi \sin^2 x dx$

We can find the total volume by integrating from 0 to $\pi/2$:

$$V = \pi \int_0^{\pi/2} \sin^2 x \, dx$$

Recall the half angle formula, $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$. Substituting, we see that:

$$V = \frac{\pi}{2}\int_0^{\pi/2} (1 - \cos 2x)\ dx$$

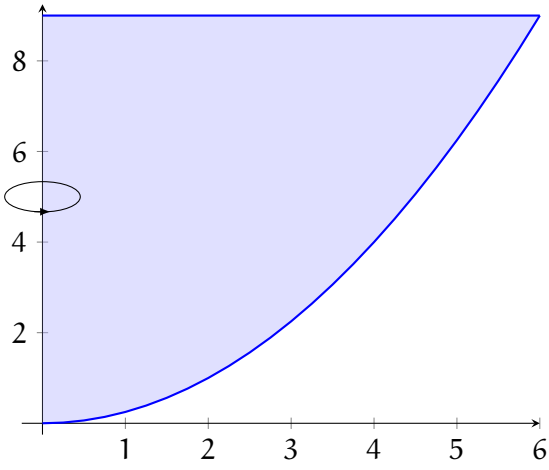
$$V = \frac{\pi}{2}\left(x - \frac{1}{2}\sin 2x\right)\Big|_{x=0}^{x=\pi/2}$$

$$V = \frac{\pi}{2}\left[\left(\pi/2 - \frac{1}{2}\sin\pi\right) - \left(0 - \frac{1}{2}\sin 0\right)\right]$$

$$V = \frac{\pi}{2}\left[\pi/2 - 0 - 0 + 0\right] = \frac{\pi^2}{4}$$

## *Exercise 521*

Find the volume of a solid created by rotating the region bounded by $x = 2\sqrt{y}$, $x = 0$, and $y = 9$ about the $y$-axis. A graph is shown below.




*Working Space*

*Answer on Page 2097*

### Exercise 522

Let $f(x) = (ax^3 + bx^2 + cx + d)\sqrt{1 - x^2}$. Bird's eggs of various sizes can be modeled by rotating $f(x)$ about the x-axis, with different values of $a$, $b$, $c$, and $d$ defining different sizes and shapes of eggs. For a domestic chicken, $a = -0.02$, $b = 0.03$, $c = 0.12$, and $d = 0.454$. For a mallard duck, $a = -0.06$, $b = 0.04$, $c = 0.1$, and $d = 0.54$. Use a calculator, such as a TI-89 or Wolfram Alpha, to determine which species lays a bigger egg.

*Working Space*

*Answer on Page 2097*

### 194.2.1 Using donuts for solids of revolution

Sometimes there is space between the region we are rotating and the line we are rotating it about. Consider the region bounded between $y = 2x$ and $y = x^2$ (see figure 194.4):

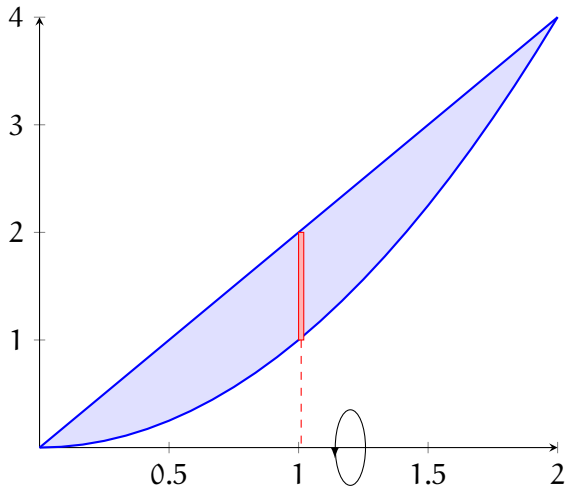



Figure 194.4: When rotated, the slices will become donuts with outer radius $2x$ and inner radius $x^2$

When rotated, the slices will take the form of donuts (or washers), the volume of which is $\pi\left(R^2 - r^2\right) dx$, where $R$ is the outer radius and $r$ is the inner radius. Therefore, in this case, the total volume of the rotated region is given by:

$$V = \int_0^2 \pi \left[ (2x)^2 - \left(x^2\right)^2 \right] dx$$

$$V = \pi \int_0^2 4x^2 - x^4 \, dx = \pi \left[ \frac{4}{3}x^3 - \frac{1}{5}x^5 \right]_{x=0}^{x=2}$$

$$V = \pi \left[ \frac{4}{3}2^3 - \frac{1}{5}2^5 \right] = \pi \left[ \frac{32}{3} - \frac{32}{5} \right]$$

$$V = \frac{64\pi}{15}$$

### *Exercise 523*

What is the volume of the region bounded by $y = x^2$ and $y = 2\sqrt{x}$ when rotated about the y-axis?

*Working Space*

*Answer on Page 2098*

## 194.3 Volumes of Other Solids

You can also model a solid as a base defined by a function with cross-sections of specific shapes. Consider the function $y = x^2$ from $x = 0$ to $x = 2$ ( see figure 194.5). Suppose the area between the curve, the y-axis, and the line $y = 4$ defines a base and each vertical cross-section is a square. So, the width of the each cross section is dx, the length is $4 - x^2$, and (because they are squares) the height in the z-plane is also $4 - x^2$. The volume of each cross-section is $V_{slice} = \left(4 - x^2\right)^2 dx$ and the total volume of the solid is:

$$V = \int_0^2 \left(4 - x^2\right)^2 dx$$

$$V = \int_0^2 \left(16 - 8x^2 + x^4\right) dx$$

$$V = \left[16x - \frac{8}{3}x^3 + \frac{1}{5}x^5\right]_{x=0}^{x=2}$$

$$V = 16(2) - \frac{8}{3}(2)^3 + \frac{1}{5}(2)^5 = \frac{256}{15} \approx 17.067$$

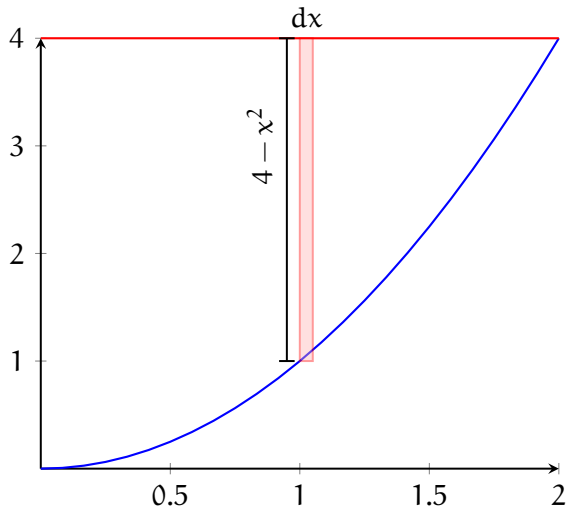



Figure 194.5: $y = x^2$ with a vertical cross-section

You can use a similar method for triangular, semi-circular, or any other shape cross-section. The trick is writing everything in terms of $x$ (when you cross sections are vertical and have width $dx$) or $y$ (when your cross section are horizontal and have length $dy$).

## Exercise 524

[This question was originally presented as a multiple-choice, calculator-allowed question on the 2012 AP Calculus BC exam.] Let $R$ be the region in the first quadrant bounded above by the graph $y = \ln(3-x)$, for $0 \leq x \leq 2$. $R$ is the base of a solid for which each cross section perpendicular to the $x$-axis is square. What is the volume of the solid? Give your answer to 3 decimal places.

*Working Space*

*Answer on Page 2099*

### Exercise 525

Find the volume of a solid whose base is defined by the ellipse $9x^2 + 16y^2 = 25$ and is made up of isosceles-triangular cross-sections perpendicular to the x-axis (with the hypotenuse in the base of the solid).

*Working Space*

*Answer on Page 2099*

Chapter 195

# Double Integrals over Rectangular Regions

In this chapter, we extend this powerful idea into higher dimensions using the tools of multiple integration. While single integration enables us to calculate the area under a curve or the volume under a surface, multiple integration allows us to calculate volumes in three dimensions, and even hypervolumes in higher dimensions.

First, we will discuss double integration, which allows us to find the volume under a surface in three dimensions. This method involves slicing the solid into infinitesimally small columns, and summing the volumes of these columns.

In a subsequent chapter, we will cover triple integration, a tool that lets us find the volume of more complicated solids in three-dimensional space. The idea is similar to double integration.

To properly implement these techniques, we will also discuss the different coordinate systems that can be used in multiple integration, such as rectangular, cylindrical, and spherical coordinates, and when it's advantageous to use one system over another.

By the end of this chapter, you will have a deeper understanding of the techniques of multiple integration and how to apply them to find the volumes of various types of solids. The methods we study here will serve as a foundation for many topics in higher mathematics and physics, including electromagnetism, fluid dynamics, and quantum mechanics.

## 195.1 Double Integrals

Double integrals extend single-variable integration to functions of two variables, allowing us to calculate quantities like area, volume, and mass over a two-dimensional region. By integrating a function across a specified domain in the xy-plane, they help analyze how a quantity changes in both dimensions. Common in physics, engineering, and economics, double integrals involve setting up limits for the region and performing two successive integrations, often tailored to the region's geometry. We begin by discussing double integrals over rectangular regions, then extending that discussion to regions of any general shape. Finally, we discuss applications of double integrals.

### 195.1.1 Over Rectangular Regions

Suppose there is some function, $z = f(x, y)$, that is defined over the rectangular region, $R$, defined by $R = [a, b] \times [c, d] = \{(x, y) | a \leq x \leq b, c \leq y \leq d\}$, and $f$ is such that $f \geq 0$ for all $(x, y) \in \mathbb{R}$. So, the graph of $f$ is a surface that lies above the rectangular region, $R$ (see figure 195.1).

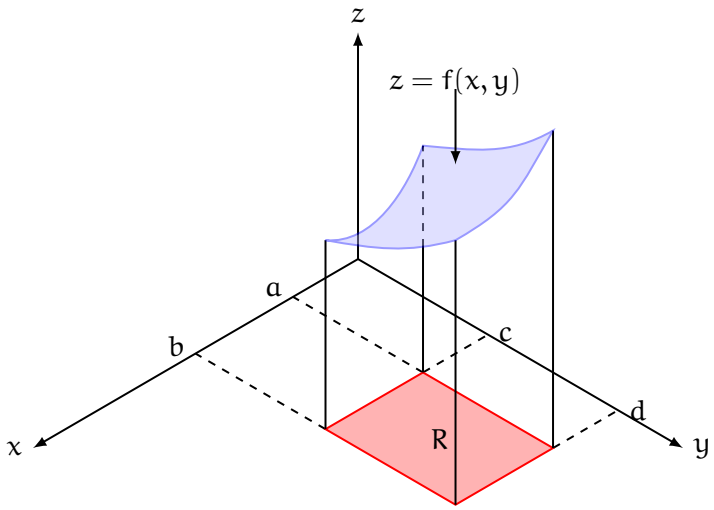



Figure 195.1: The graph of $f$ over the region $R$

Let's call the solid that fills the space between the $xy$-plane and the surface $z = f(x, y)$ $S$. Formally, this is written as

$$S = \{(x, y, z) \in \mathbb{R}^3 | 0 \leq z \leq f(x, y), (x, y) \in \mathbb{R}\}$$

How can we find the volume of the solid, $S$? We will apply what we learned about Riemann sums and definite integrals in two dimensions to this three-dimensional problem.

First, we divide $R$ into rectangular subregions. We do this by dividing the interval $[a, b]$ into $m$ subintervals with width $\Delta x = (b - a)/m$ and the interval $[c, d]$ into $n$ subintervals with width $\Delta y = (d - c)/n$. Drawing lines through these divisions parallel to the $x$- and $y$-axes, we create a field of subrectangles, each with area $\Delta A = \Delta x \Delta y$ (see figure 195.2).

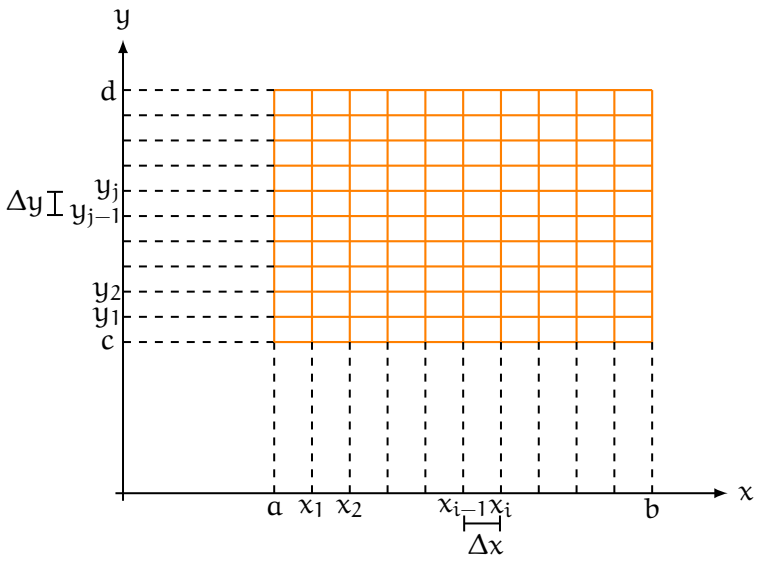



Figure 195.2: The region, $R$, on the $xy$-plane divided into subrectangles

Each subrectangle is defined by:

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] - \{(x, y) | x_{i-1} \leq x \leq x_i, y_{j-1} \leq y \leq y_j\}$$

Since $f(x, y)$ in continuous over the $R$, there is some point, $(x^*_{ij}, y^*_{ij})$, equal to the average value of $f(x, y)$ over the subrectangle. We can approximate the volume between the $xy$-plane and $z = f(x, y)$ over the subrectangle as a column with base area $\Delta A$ and height $f(x^*_{ij}, y^*_{ij})$ (see figure 195.3) and the volume of the column is given by:

$$V_{ij} = f(x^*_{ij}, y^*_{ij}) \Delta A$$

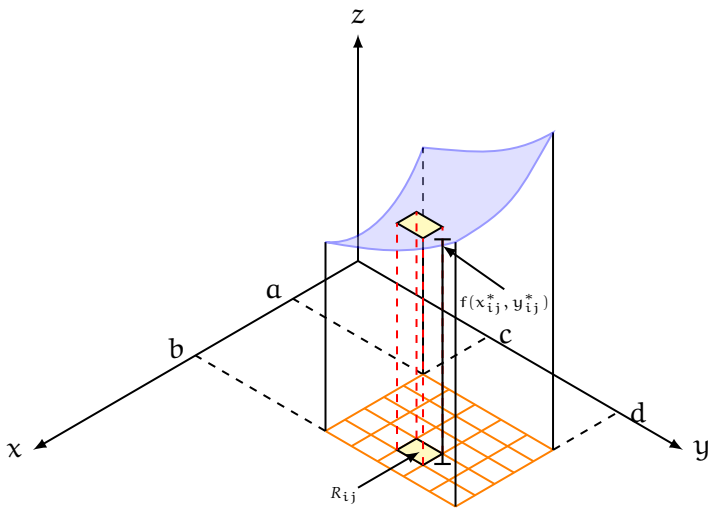



Figure 195.3: A single column with base $\Delta A$ and height $f(x^*_{ij}, y^*_{ij})$

Therefore, the approximate volume of the solid, $S$, that lies between the region, $R$, and $z = f(x, y)$ is the sum of all the columns over $i$ and $j$:

$$V_S \approx \sum_{i=1}^{n} \sum_{j=1}^{n} f(x^*_{ij}, y^*_{ij}) \Delta A$$

Just like with the area under a curve, we get the true volume by taking the limit as $n \to \infty$, which becomes a **double integral**:

**Volume of a Solid over a Region**

$$V_S = \lim_{n\to\infty} \sum_{i=1}^{n} \sum_{j=1}^{n} f(x^*_{ij}, y^*_{ij}) \Delta A = \iint_R f(x, y)\, dA$$

## 195.2 Iterated Integrals

To be able to evaluate the double integral as outlined above, we must first discuss iterated integrals. Iterated integrals happen when you evaluate two single integrals, one inside the other. Consider some function, $g(x, y)$. We could integrate that function from $x = q$ to $x = r$ thusly:

$$\int_q^r g(x, y)\, dx$$

Notice that we are integrating with respect to $x$, so $y$ terms will be treated as constants (recall partial differentiation: this is the opposite process). Let's call the result of this first

integral $A(y)$:

$$A(y) = \int_q^r g(x, y)\, dx$$

We can then integrate the resulting function, $A(y)$, from $y = s$ to $y = t$:

$$\int_s^t A(y)\, dy = \int_s^t \left[\int_q^r g(x, y)\, dx\right] dy$$

This is called an **iterated integral**. When evaluating iterated integrals, we work from the inside out. You can also write it without the brackets:

$$\int_s^t \int_q^r g(x, y)\, dx\, dy$$

**Example**: Evaluate the iterated integral $\int_0^3 \int_1^2 xy^2\, dy\, dx$.

**Solution**: We can re-write this to more explicitly show the inner and outer integrals:

$$\int_0^3 \left[\int_1^2 xy^2\, dy\right] dx$$

As you can see, the inner integral is with respect to $y$. Let's isolate and evaluate the inner integral:

$$\int_1^2 xy^2\, dy = x\int_1^2 y^2\, dy = x\left[\frac{1}{3}y^2\right]_{y=1}^{y=2}$$

$$= \frac{x}{3}\left[2^3 - 1^3\right] = \frac{x}{3}[8 - 1] = \frac{7x}{3}$$

We were able to move $x$ outside the integral because when we are integrating with respect to a specific variable (in this case, $y$), other variables are treated as constants. Now, we can substitute $\int_1^2 xy^2\, dy = \frac{7x}{3}$ into the iterated integral:

$$\int_0^3 \left[\int_1^2 xy^2\, dy\right] dx = \int_0^3 \left[\frac{7x}{3}\right] dx$$

$$= \frac{7}{3}\left[\frac{1}{2}x^2\right]_{x=0}^{x=3} = \frac{7}{6}\left[3^2 - 0^2\right] = \frac{7 \cdot 9}{6} = \frac{21}{2}$$

## *Exercise 526* Order of Evaluating Iterated Integrals

Show that $\int_0^3 \int_1^2 xy^2 \, dy \, dx = \int_1^2 \int_0^3 xy^2 \, dx \, dy$.

*Working Space*

*Answer on Page 2100*

## *Exercise 527* Evaluating Iterated Integrals

Evaluate the following iterated integrals.

*Working Space*

1. $\int_0^1 \int_1^2 (x + e^{-y}) \, dx \, dy$
2. $\int_{-3}^3 \int_0^{\pi/2} (2y + y^2 \cos x) \, dx \, dy$
3. $\int_0^3 \int_0^{\pi/2} t^2 \sin^3 \theta \, d\theta \, dt$

*Answer on Page 2100*

## 195.3 Fubini's Theorem for Double Integrals

Fubini's Theorem states that for a function, $f$, that is continuous over the rectangular region, $R$, the double integral of $f$ over the region $R = \{(x,y)|a \leq x \leq b, c \leq y \leq d\}$ is equal to the iterated integral of $f$ with respect to $x$ and $y$. This is expressed mathematically below:

**Fubini's Theorem**

If $f$ is continuous on the rectangle $R = \{(x,y)\,|a \leq x \leq b, c \leq y \leq d\}$, then

$$\iint_R f(x,y)\ dA = \int_a^b \int_c^d f(x,y)\ dy\ dx = \int_c^d \int_a^b f(x,y)\ dx\ dy$$

### *Exercise 528* Applying Fubini's Theorem

Rewrite the following double integrals as iterated integrals.

*Working Space*

1. $\iint_R \frac{xy^2}{x^2+1}\ dA,\ R = \{(x,y)|0 \leq x \leq 1, -3 \leq y \leq 3\}$
2. $\iint_R \frac{\sec\theta}{\sqrt{1+t^2}}\ dA,\ R = \{(\theta,t)|0 \leq \theta \leq \frac{\pi}{4}, 0 \leq t \leq 1\}$

*Answer on Page 2101*

## Exercise 529

Evaluate the double integral.

1. $\iint_R \frac{xy^2}{x^2+1}\,dA,\ R = \{(x,y) \mid 0 \le x \le 2,\ -3 \le y \le 3\}$
2. $\iint_R \frac{\tan\theta}{\sqrt{1-t^2}}\,dA,\ R = \{(\theta,t) \mid 0 \le \theta \le \pi/3,\ 0 \le t \le \frac{1}{2}\}$
3. $\iint_R x\sin(x+y)\,dA,\ R = [0,\pi/6] \times [0,\pi/3]$

*Working Space*

*Answer on Page 2101*

Chapter 196

# Double Integrals Over Non-Rectangular Regions

Now that we've seen how to evaluate double integrals over rectangular regions, let's consider non-rectangular regions. Suppose we are interested in the integral of a function, $f(x, y)$, over a region, $D$, which exists such that it can be bounded by inside a rectangular region, $R$ (see figure 196.1). We can then define a new function:

$$F(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \text{ is in } D \\ 0 & \text{if } (x, y) \text{ is in } R \text{ but not in } D \end{cases}$$

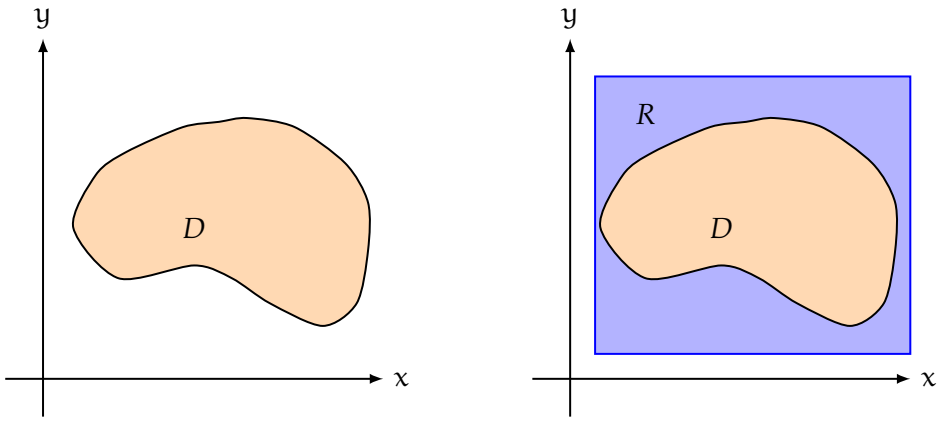



Figure 196.1: We can find a rectangular region, $R$, that completely encloses $D$

Then, we can see that:

$$\iint_D f(x, y)\, dA = \iint_R F(x, y)\, dA$$

This makes sense intuitively, since integrating over $F$ outside of $D$ doesn't contribute anything to the integral, and the integral of $F$ inside $D$ is equal to the integral of $f$ inside $D$. In general, there are two types of regions for $D$. A region is **type I** if it lies between two continuous functions of $x$ and can be defined thusly:

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

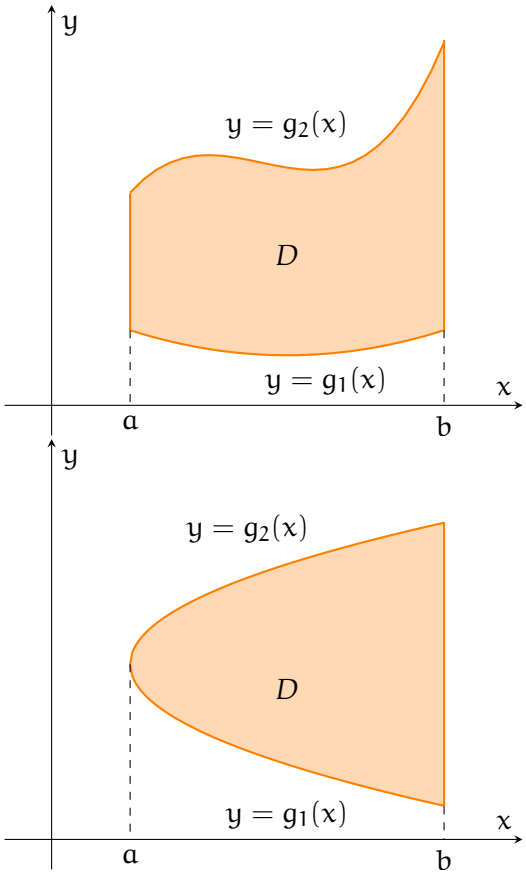



Figure 196.2: Two examples of type I domains

Some type I regions are shown in figure 196.2. To evaluate $\iint_D f(x,y)\,dA$, we begin by choosing a rectangle $R = [a,b] \times [c,d]$ such that $D$ is completely contained in $R$. We again define $F(x,y)$ such that $F(x,y) = f(x,y)$ on $D$ and $F = 0$ outside of $D$. Then, by Fubini's theorem:

$$\iint_D f(x,y)\,dA = \iint_R F(x,y)\,dA = \int_a^b \int_c^d F(x,y)\,dy\,dx$$

Since $F(x,y) = 0$ when $y \leq g_1(x)$ or $y \geq g_2(x)$, we know that:

$$\int_c^d F(x,y)\,dy = \int_{g_1(x)}^{g_2(x)} F(x,y)\,dy = \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy$$

Substituting this into the iterated integral above, we see that for a type I region $D = \{(x,y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$,

$$\iint_D f(x,y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

Another way to visualize the double integral over a type I region is shown in figure 196.3. For any value of $x \in [a, b]$, we know that $g_1(x) \leq y \leq g_2(x)$. The inner integral represents moving along one blue line from $y = g_1(x)$ to $y = g_2(x)$ and integrating with respect to $y$. Next, for the outer integral, we integrate with respect to $x$, which is represented by moving the line from $x = a$ to $x = b$.

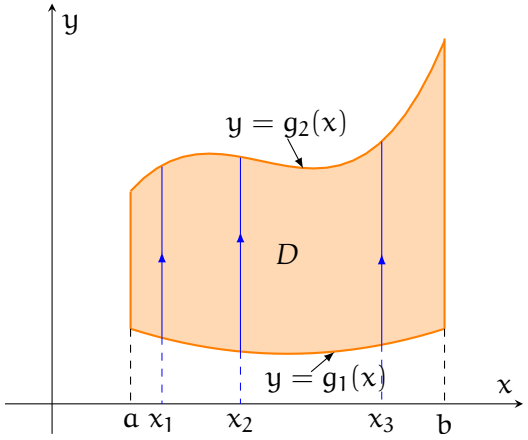


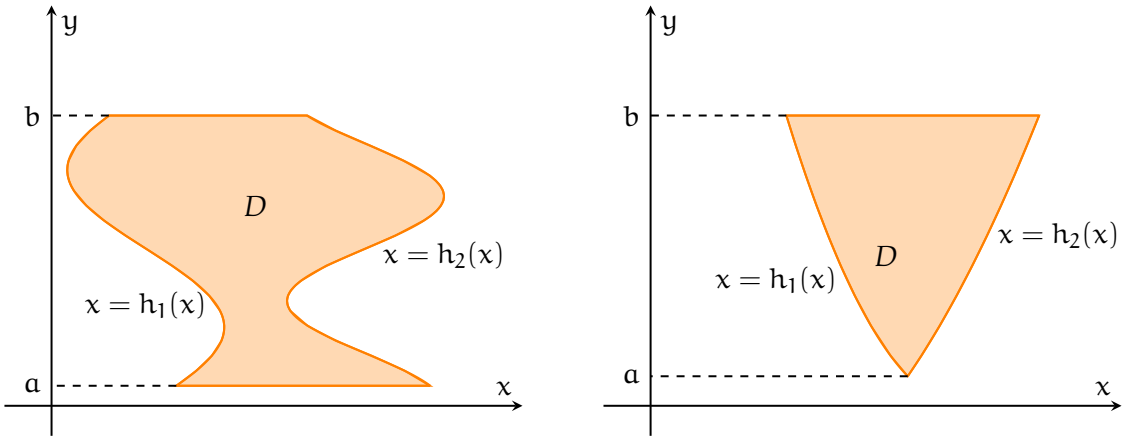

Figure 196.3: On type I domains, for a given value of $x$, $g_1(x) \leq y \leq g_2(x)$

A **type II** region is a region such that we can define the limits of $x$ in terms of $y$ (see figure 196.4). In other words, a type II region can be defined as:

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y0 \leq x \leq h_2(y)\}$$

In a similar manner to above, we can show that:

$$\iint_D f(x, y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy$$




(a) Example 1 (b) Example 2

Figure 196.4: Two examples of type II domains

You can annotate type II regions with horizontal lines to show that, for a given $y$ values, all $x$ values in the region are contained in $h_1(y) \leq x \leq h_2(y)$ (see figure 196.5).

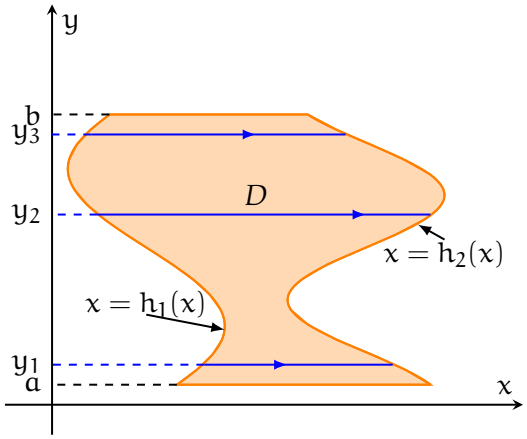



Figure 196.5: On type II domains, for a given value of $y$, $h_1(y) \leq x \leq h_2(y)$

## 196.1 Determining Region Type

Many regions can be described as either type I or type II. Consider the region between the curves $y = \frac{3}{2}(x-1)$ and $y = \frac{1}{2}(x-1)^2$ (see figure 196.6).

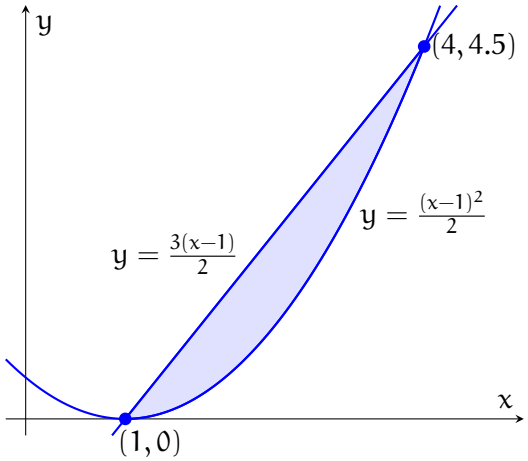
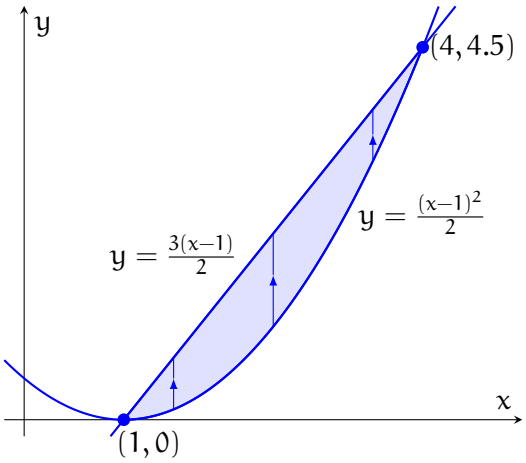
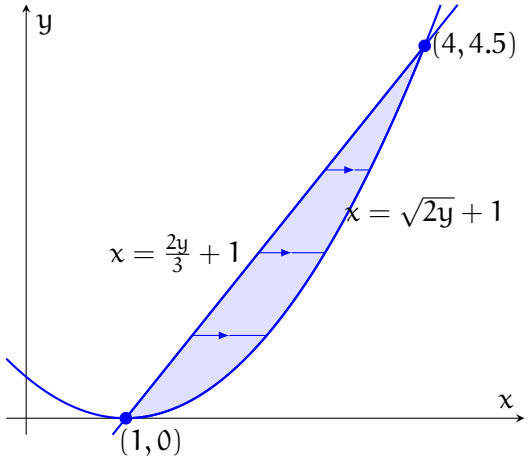



Figure 196.6: The region that lies between $y = \frac{(x-1)^2}{2}$ and $y = \frac{3(x-1)}{2}$ can be classified as type I or type II

We could classify this as a type I (see figure 196.7a) or a type II domain (see figure 196.7b).



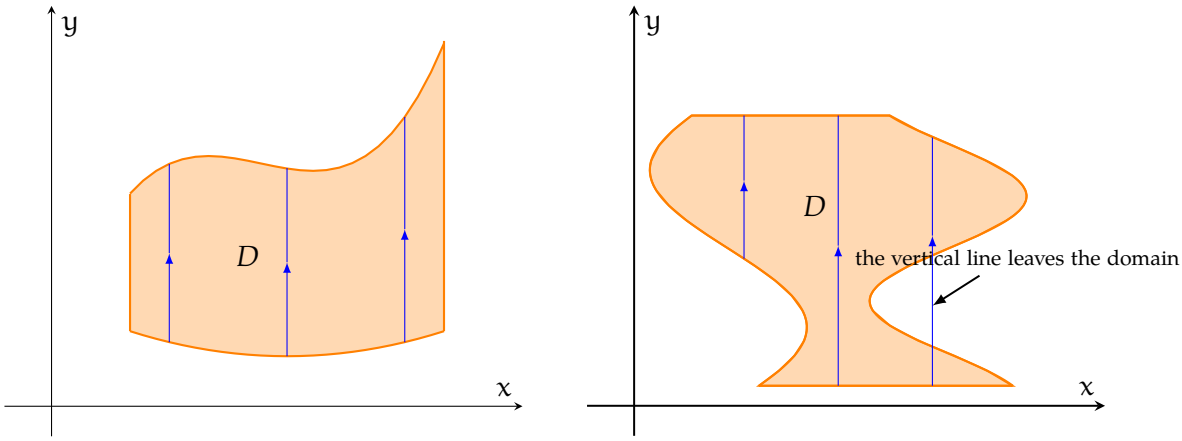

(a) Type I region: $\frac{(x-1)^2}{2} \le y \le \frac{3(x-1)}{2}$




(b) Type II region: $\frac{2y}{3} + 1 \le x \le \sqrt{2y} + 1$

Figure 196.7: Comparison of Type I and Type II descriptions of the same region.

However, not all domains can be classified as both type I or type II. A region can be classified as type I if you can take a vertical line ($x = c$, where $c$ is some number in the domain of the region) and move it across the region without any gaps. Consider the two regions shown in figure **??**. The top is type I, and if you put vertical lines, the line traverses the entire region without leaving it. On the other hand, the lower region is not type I. There are vertical lines where the line exits the lower region before traversing the entire region.

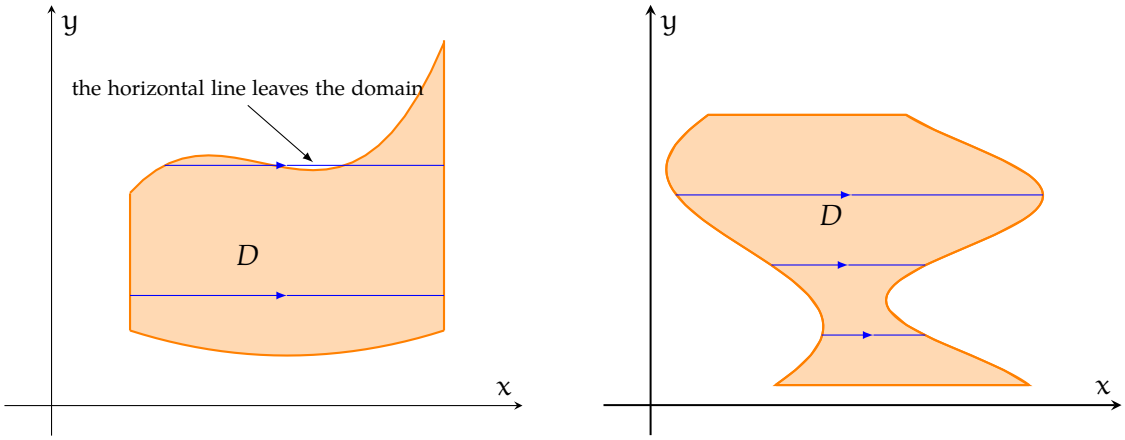



(a) A domain that *is* Type I

(b) A domain that is *not* Type I

Figure 196.8: Comparing two regions: one that satisfies the Type I condition and one that does not.

To determine if a region can be classified as type II, you can use a horizontal line. If you can move a horizontal line ($y = c$, where $c$ is in the domain of the region) across the region and the line always traverses the entire region without leaving and re-entering it, then the region is type II (see figure **??**).




(a) A domain that is *not* Type II

(b) A domain that *is* Type II

Figure 196.9: Comparing two regions: one that fails the Type II condition and one that satisfies it.

**Example**: Evaluate $\iint_D (2x+y)\,dA$, where $D$ is the region bounded by the parabolas $y = 3x^2$ and $y = 2 + x^2$. Region $D$ is shown in figure 196.10.

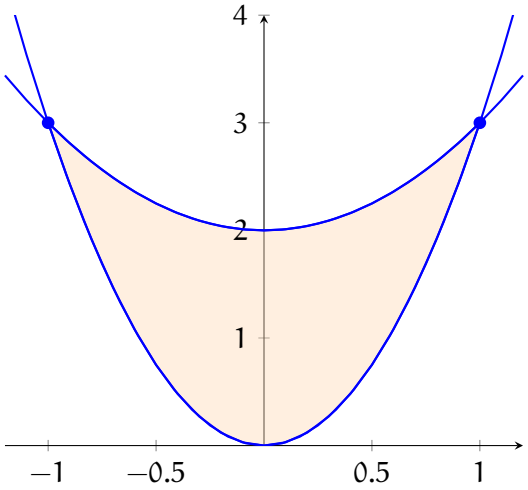



Figure 196.10: Region $D$ is bounded above by $y = 2 + x^2$ and below by $y = 3x^2$

**Solution**:This is a type I region, since for a given $x$, $y \in \left[3x^2, 2 + x^2\right]$. We can define region $D$ as $D = \{(x, y) \mid -1 \leq x \leq 1, 3x^2 \leq y \leq 2 + x^2\}$. Therefore,

$$\iint_D (2x + y)\, dA = \int_{-1}^{1} \int_{3x^2}^{2+x^2} (2x + y)\, dy\, dx$$

$$= \int_{-1}^{1} \left[ \int_{3x^2}^{2+x^2} 2x\, dy + \int_{3x^2}^{2+x^2} y\, dy \right] dx$$

$$= \int_{-1}^{1} \left[ 2xy|_{y=3x^2}^{y=2+x^2} + \frac{1}{2} y^2 |_{y=3x^2}^{y=2+x^2} \right] dx$$

$$= \int_{-1}^{1} \left[ 2x\left(2 + x^2 - 3x^2\right) + \frac{1}{2}\left((2 + x^2)^2 - (3x^2)^2\right) \right] dx$$

$$= \int_{-1}^{1} \left[ 2 + 4x + 2x^2 - 4x^3 - 4x^4 \right] dx$$

$$= \left[ 2x + 2x^2 + \frac{2}{3}x^3 - x^4 - \frac{4}{5}x^5 \right]_{x=-1}^{x=1}$$

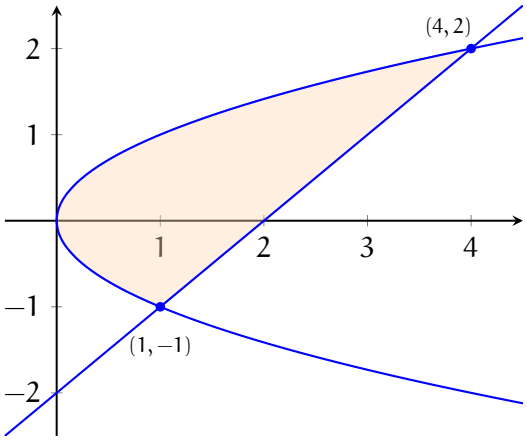
$$= \left(2 + 2 + \frac{2}{3} - 1 - \frac{4}{5}\right) - \left(-2 + 2 - \frac{2}{3} - 1 + \frac{4}{5}\right)$$

$$= 4 + \frac{4}{3} - \frac{8}{5} = \frac{56}{15}$$

**Example**: Set up integrals to evaluate $\iint_D xy\, dA$ if D is the region bounded by $y = x - 2$ and $x = y^2$ as both a type I and type II region. Which method will be easier to evaluate?

Evaluate the easier double integral.

**Solution**: Let's begin by visualizing D:




This region could be classified as type I and type II. To set up an integral as if D were a type I region, we need to describe $y$ in terms of $x$. However, we run into a little problem. For $0 \le x \le 1$, the region is bounded by the parabola $x = y^2$, and for $1 \le x \le 4$, the region is bounded by both the parabola and the line. So, we will have to split the integral into to parts: $0 \le x \le 1$ and $1 \le x \le 4$:

$$\iint_D xy \, dA = \int_0^1 \int_{-\sqrt{x}}^{\sqrt{x}} xy \, dy \, dx + \int_1^4 \int_{x-2}^{\sqrt{x}} xy \, dy \, dx$$

To set up an integral as if D were a type II region, we need to describe $x$ in terms of $y$. This time, we will not need to split the integral, because $y + 2 \le x \le y^2$ over the entire region:

$$\iint_D xy \, dA = \int_{-1}^2 \int_{y+2}^{y^2} xy \, dx \, dy$$

It is easier to evaluate the integral with D as a type II region, since we do not need to split the integral. Evaluating:

$$\int_{-1}^2 \int_{y+2}^{y^2} xy \, dx \, dy = \int_{-1}^2 \frac{y}{2} \left[x^2\right]_{x=y+2}^{x=y^2} \, dy$$

$$= \int_{-1}^2 \frac{y}{2} \left[\left(y^2\right)^2 - (y+2)^2\right] \, dy = \frac{1}{2} \int_{-1}^2 y \left[y^4 - \left(y^2 + 4y + 4\right)\right] \, dy$$

$$= \frac{1}{2} \int_{-1}^2 y^5 - y^3 - 4y^2 - 4y \, dy = \frac{1}{2} \left[\frac{1}{6}y^6 - \frac{1}{4}y^4 - \frac{4}{3}y^3 - \frac{4}{2}y^2\right]_{y=-1}^{y=2}$$

$$= \frac{1}{2}\left[\frac{1}{6}(2)^6 - \frac{1}{4}(2)^4 - \frac{4}{3}(2)^3 - 2(2)^2 - \left(\frac{1}{6}(-1)^6 - \frac{1}{4}(-1)^4 - \frac{4}{3}(-1)^3 - 2(-1)^2\right)\right]$$

$$= \frac{1}{2}\left[\frac{32}{3} - 4 - \frac{32}{3} - 8 - \frac{1}{6} + \frac{1}{4} - \frac{4}{3} + 2\right] = -\frac{45}{8}$$

## *Exercise 530* Double Integrals over Non-Rectangular Regions

Evaluate the double integral.

*Working Space*

1. $\iint_D e^{-y^2}\,dA$, $D = \{(x, y) \mid 0 \le y \le 3, 0 \le x \le 2y\}$.
2. $\iint_D x \sin y\,dA$, $D$ is bounded by $y = 0$, $y = x^2$, $x = 2$.
3. $\iint_D (2y - x)\,dA$, $D$ is bounded by the circle with center at the origin and radius 3.

*Answer on Page 2103*

## 196.2 Double Integrals in Other Coordinate Systems

Consider a region composed of a semi-circular ring (see figure **??**). Describing the region in Cartesian coordinates is complicated; you would have to split it into three regions (see figure ...). However, in polar coordinates, we can describe the whole region in one statement:

$$D = \{(r, \theta) \mid 1 \leq r \leq 4,\ 0 \leq \theta \leq \pi\}$$

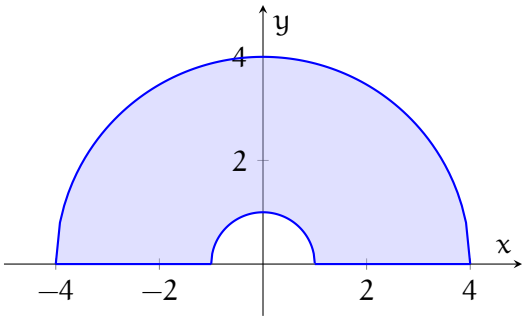



Figure 196.11: A semi-circular ring

There are many instances where a region is simpler to describe in polar coordinates, so how do we take double integrals in polar coordinates? Suppose we want to integrate some function, $f(x, y)$, over a polar rectangle described by $D = \{(r, \theta) \mid a \leq r \leq b,\ \alpha \leq \theta \leq \beta\}$ (see figure 196.12). Similar to Cartesian coordinates, we can divide this region into many smaller polar rectangles, with each subrectangle defined by $D_{ij} = \{(r, \theta) \mid r_{i-1} \leq r \leq r_i,\ \theta_{i-1} \leq \theta \leq \theta_i\}$. The center of each subrectangle has polar coordinates $(r_i^*, \theta_j^*)$, where:

$$r_i^* - \frac{1}{2}(r_{i-1} + r_i)$$

$$\theta_j^* = \frac{1}{2}(\theta_{j-1} + \theta_j)$$

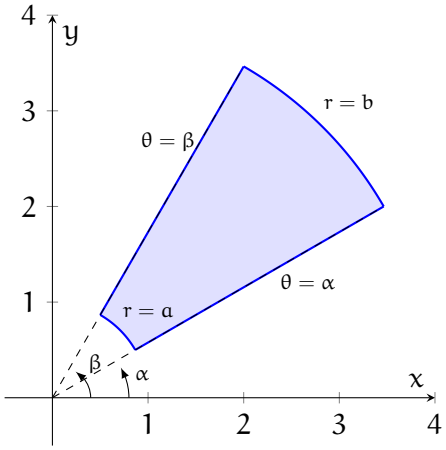



Figure 196.12: A polar rectangle described by $D = \{(r, \theta) \mid a \leq r \leq b,\ \alpha \leq \theta \leq \beta\}$

Each subrectangle is a larger radius sector minus a smaller radius sector, each with the same central angle, $\Delta\theta = \theta_j - \theta_{j-1}$. The total area of each subrectangle is given by:

$$\Delta A_i = \frac{1}{2}(r_i)^2 \delta\theta - \frac{1}{2}(r_{i-1})^2 \Delta\theta = \frac{1}{2}\left(r_i^2 - r_{i-1}^2\right)\Delta\theta$$

Substituting $\left(r_i^2 - r_{i-1}^2\right) = (r_i + r_{i-1})(r_i - r_{i-1})$, we see that:

$$\Delta A_i = \frac{1}{2}(r_i + r_{i-1})(r_i - r_{i-1})\Delta\theta$$

Recall that we have defined $r_i^* = \frac{1}{2}(r_{i-1} + r_i)$. Additionally, $\Delta r = r_i - r_{i-1}$. Substituting this, we find a simplified expression for the area of each subrectangle:

$$\Delta A_i = r_i^* \Delta r \Delta\theta$$

Therefore, the Riemann sum of $f(x, y)$ over the region is:

$$\sum_{i=1}^{n}\sum_{j=1}^{n} f(r_i^* \cos\theta_j^*, r_i^* \sin\theta_j^*)\Delta A_i$$

(Recall that to convert from Cartesian to polar coordinates, we use $x = r\cos\theta$ and $y = r\sin\theta$). Substituting for $\delta A_i$:

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} f(r_i^* \cos\theta_j^*, r_i^* \sin\theta_j^*) r_i^* \Delta r \Delta\theta$$

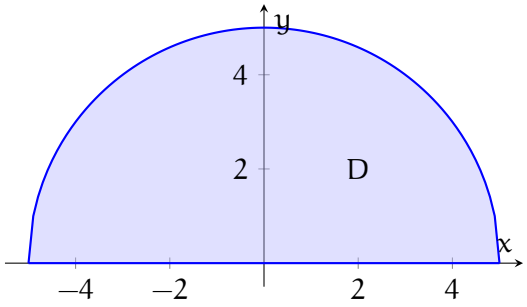
Taking the limit as $n \to \infty$, the Riemann sum becomes the double integral:

$$\int_{\alpha}^{\beta} \int_{a}^{b} f(r\cos\theta, r\sin\theta) r \, dr \, d\theta$$

Therefore, if $f$ is continuous on the polar rectangle $a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, then:

$$\iint_D f(x,y) \, dA = \int_{\alpha}^{\beta} \int_{a}^{b} f(r\cos\theta, r\sin\theta) r \, dr \, d\theta$$

**Example**: Evaluate $\iint_D x^2 y \, dA$, where $D$ is the semi-circle shown below.




**Solution**: Since the region is a semi-circle with radius 5, we can describe $D$ as $D = \{(r,\theta) \mid 0 \leq r \leq 5,\ 0 \leq \theta \leq \pi\}$. Therefore,

$$\iint_D x^2 y \, dA = \int_0^{\pi} \int_0^5 (r\cos\theta)^2 (r\sin\theta) \, r \, d\theta \, dr$$

$$= \int_0^{\pi} \int_0^5 r^4 \cos^2\theta \sin\theta \, dr \, d\theta$$

$$= \int_0^{\pi} \cos^2\theta \sin\theta \left[\frac{1}{5} r^5\right]_{r=0}^{r=5} d\theta$$

$$= \int_0^{\pi} \cos^2\theta \sin\theta \frac{5^5}{5} \, d\theta = 625 \int_0^{\pi} \cos^2\theta \sin\theta \, d\theta$$

Using u-substitution, let $u = \cos\theta$. Then $-du = \sin\theta d\theta$, therefore:

$$625 \int_0^{\pi} \cos^2\theta \sin\theta \, d\theta = 625 \int_{\theta=0}^{\theta=\pi} -u^2 \, du$$

$$= -625 \frac{1}{3} u^3 \Big|_{\theta=0}^{\theta=\pi} = -625 \frac{1}{3} \left(\cos^3\theta\right) \Big|_{\theta=0}^{\theta=\pi}$$

$$= -\frac{625}{3}\left[(-1)^3 - (1)^3\right] = -\frac{625}{3}(-2) = \frac{1250}{3}$$

## *Exercise 531* Changing to Polar Coordinates

*Working Space*

Evaluate the following iterated integrals by converting to polar coordinates:

1. $\int_0^2 \int_0^{\sqrt{4-x^2}} e^{-x^2-y^2}\,dy\,dx$
2. $\int_0^{1/2} \int_{\sqrt{3}y}^{\sqrt{1-y^2}} xy^2\,dx\,dy$
3. $\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2+y^2}\,dy\,dx$

*Answer on Page 2103*

## *Exercise 532* Using Polar Coordinates in Multiple Integration

*Working Space*

Find the volume of the solid that lies under the surface $z = 4 - x^2 - y^2$ and above the $xy$-plane.

*Answer on Page 2105*

## *Exercise 533* The volume of a pool

A circular swimming pool has a 40-ft diameter. The depth of the pool is constant along the north-south axis and increases from 3 feet at the west end to 10 feet at the east end. What is the total volume of water in the pool?

*Working Space*

*Answer on Page 2106*

Chapter 197

# Applications of Double Integrals

## 197.1 Total Mass and Charge

Suppose there is a generic, thin layer (called a *lamina*) with a variable density that occupies an area $B$ (see figure 197.1). Further, let the density of the lamina be described by a function, $\rho(x, y)$, which is continuous over $B$. For some small rectangle centered at $(x, y)$, the density is given by:

$$\rho(x, y) = \frac{\Delta m}{\Delta A}$$

where $\Delta m$ is the mass of the small rectangle and $\Delta A$ is the area. Then the mass of the rectangle is given by:

$$\Delta m = \rho(x, y)\Delta A$$

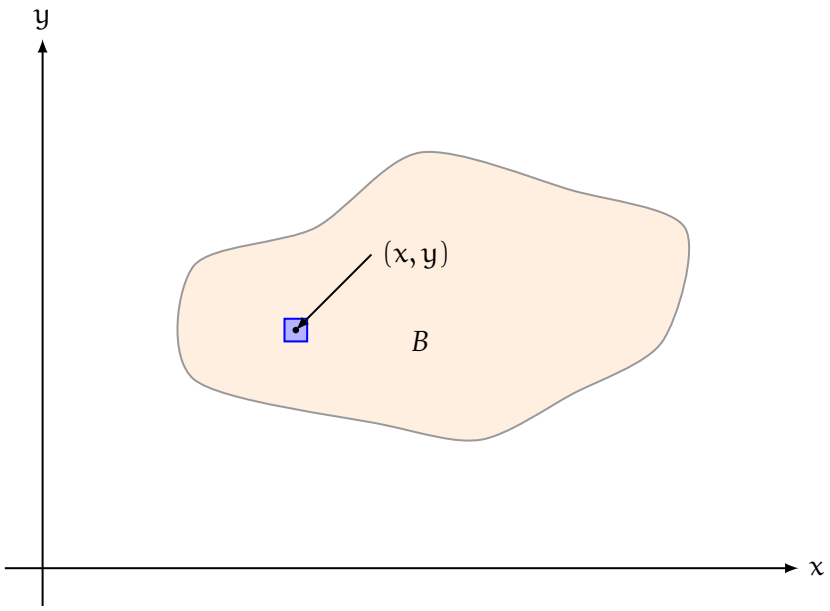



Figure 197.1: A generic lamina that occupies the region $B$

We can find the mass of the entire lamina by dividing it into many of these small rectangles and adding the masses of all the rectangles (see 197.2). Just like in previous examples, there is some point $(x^*_{ij}, y^*_{ij})$ in each rectangle, $R_{ij}$, such that the mass of the part of the

lamina that occupies $R_{ij}$ is $\rho(x_{ij}^*, y_{ij}^*)\Delta A$. Adding all these masses yields:

$$m_{total} \approx \sum_{i=1}^{m} \sum_{j=1}^{n} \rho(x_{ij}^*, y_{ij}^*)\Delta A$$

Taking the limit as $m, n \to \infty$ increases the number of rectangles to yield the true total mass:

$$m_{total} = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} \rho(x_{ij}^*, y_{ij}^*)\Delta A = \iint_B \rho(x, y)\, dA$$

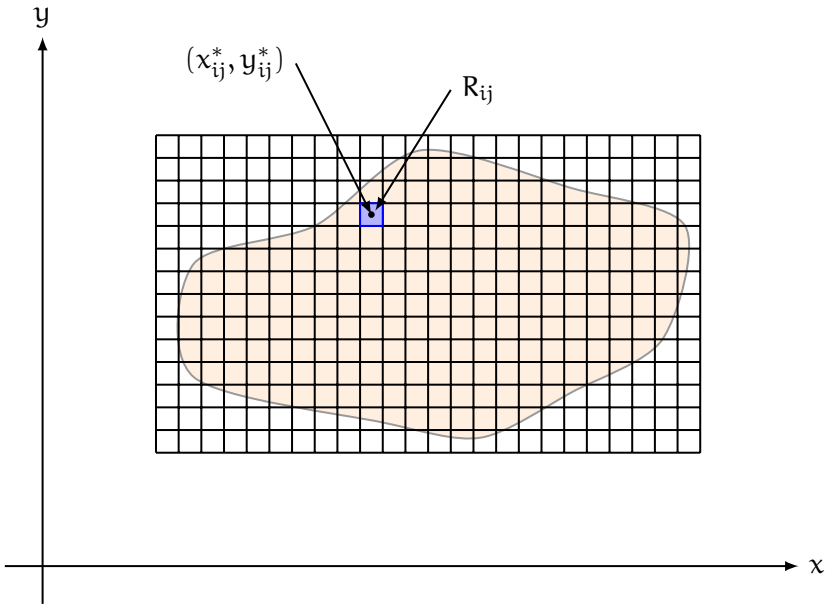



Figure 197.2: A generic lamina divided into many rectangles

**Example**: Find the total mass of a lamina that occupies the region $D = \{(x, y) \mid 1 \leq x \leq 3, 1 \leq y \leq 4\}$ with a density function $\rho(x, y) = 3y^2$.

**Solution**: We know that the total mass is given by:

$$\iint_D 3y^2\, dA$$

Applying Fubini's Theorem, we see that:

$$\iint_D 3y^2\, dA = \int_1^3 \int_1^4 3y^2\, dy\, dx$$

$$= \int_1^3 \left[y^3\right]_{y=1}^{y=4} dx = \int_1^3 \left[4^3 - 1^3\right] dx$$

$$= \int_1^3 63\, dx = 63x\Big|_{x=1}^{x=3} = 126$$

## *Exercise 534* Finding Total Mass

Find the mass of the lamina that occupies the region, $D$, and has the given density function, $\rho$.

1. $D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq 3\}$; $\rho(x, y) = 1 + x^2 + y^2$
2. $D$ is the triangular region with vertices $(0, 0)$, $(2, 1)$, $(0, 3)$; $\rho(x, y) = x + y$

*Working Space*

*Answer on Page 2107*

This method applies not only to mass density, but any other type of density. Some examples could include animals per acre of forest, cells per square centimeter of petri dish, or people per city block. A density physicists are often interested in is charge density (that is, the amount of charge, Q, per unit area). Charge is measured in coulombs (C). Often, charge density is given by a function, $\sigma(x, y)$, in units of coulombs per area (such as $cm^2$ or $m^2$). If there is some region, $D$, with charge distributed across it such that the charge density can be described by a continuous function, $\sigma(x, y)$, then the total charge, Q, is given by:

$$Q = \iint_D \sigma(x, y)\, dA$$

**Example**: Charge is distributed over the region $B$ shown in figure 197.3 such that the charge density is given by $\sigma(x, y) = xy$, measured in $C/m^2$. Find the total charge.

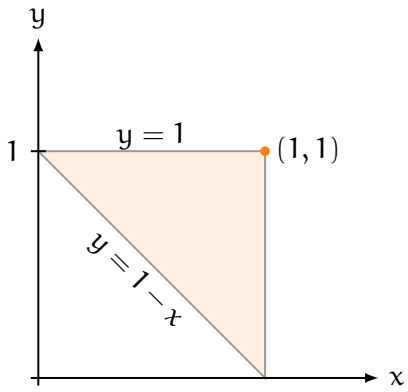



Figure 197.3: A triangular region over which charge is distributed such that $\sigma(x, y) = xy$

**Solution**: We know that total charge is given by:

$$Q = \iint_B xy \, dA$$

Examining figure 197.3, we see that:

$$\iint_B xy \, dA = \int_0^1 \int_{1-x}^1 xy \, dy \, dx$$

$$= \int_0^1 \frac{x}{2} \left[ y^2 \right]_{y=1-x}^{y=1} dx = \int_0^1 \frac{x}{2} \left[ 1^2 - (1-x)^2 \right] dx$$

$$= \frac{1}{2} \int_0^1 x \left( 1 - 1 + 2x - x^2 \right) dx = \frac{1}{2} \int_0^1 x \left( 2x - x^2 \right) dx$$

$$= \frac{1}{2} \int_0^1 2x^2 - x^3 \, dx = \frac{1}{2} \left[ \frac{2}{3}x^3 - \frac{1}{4}x^4 \right]_{x=0}^{x=1} = \frac{1}{2} \left( \frac{2}{3} - \frac{1}{4} \right) = \frac{5}{24} C$$

## 197.2 Center of Mass

For a thin disk (lamina) of variable density in the $xy$-plane, the coordinates of the center of mass, $(\bar{x}, \bar{y})$, are given by:

$$\bar{x} = \frac{1}{m} \iint_D x\rho(x, y) \, dA$$

$$\bar{y} = \frac{1}{m} \iint_D y\rho(x, y) \, dA$$

where $m$ is the total mass and $\rho$ is the density of the lamina as a function of $x$ and $y$.

**Example**: Find the center of mass of a triangular lamina with vertices at $(0,0)$, $(2,0)$, and $(0,1)$ and a density function $\rho(x,y) = 2 + x + 3y$.

**Solution**: We begin by visualizing the region so we can determine if it is type I or type II:

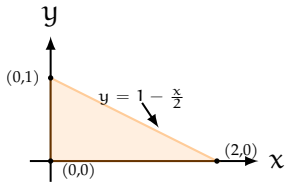



Figure 197.4: Region of integration for the triangular lamina.

Recall that the total mass is given by $m = \iint_D \rho(x,y)\,dA$. As shown above, we can define $D = \{(x,y)\,|0 \le x \le 2, 0 \le y \le 1 - \frac{x}{2}\}$:

$$m = \int_0^2 \int_0^{1-x/2} (2 + x + 3y)\ dy\,dx = \int_0^2 \left[2y + xy + \frac{3}{2}y^2\right]_{y=0}^{y=1-x/2} dx$$

$$= \int_0^2 \left[2(1-\frac{x}{2}) + x(1-\frac{x}{2}) + \frac{3}{2}(1-\frac{x}{2})^2\right] dx = \int_0^2 \left[\frac{7}{2} - \frac{3x}{2} - \frac{x^2}{8}\right] dx$$

$$= \left[\frac{7x}{2} - \frac{3x^2}{4} - \frac{x^3}{24}\right]_{x=0}^{x=2} = \frac{7(2)}{2} - \frac{3(4)}{4} - \frac{8}{24} = 7 - 3 - \frac{1}{3} = \frac{11}{3}$$

Finding $\bar{x}$:

$$\bar{x} = \frac{1}{m} \iint_D x\,(2 + x + 3y)\ dA$$

$$\bar{x} = \frac{3}{11} \int_0^2 \int_0^{1-x/2} \left[2x + x^2 + 3xy\right]\ dy\,dx$$

$$\bar{x} = \frac{3}{11} \int_0^2 \left[2xy + x^2y + \frac{3}{2}xy^2\right]_{y=0}^{y=1-x/2} dx$$

$$\bar{x} = \frac{3}{11} \int_0^2 \left[\frac{7x}{2} - \frac{3x^2}{2} - \frac{x^3}{8}\right] dx$$

$$\bar{x} = \frac{3}{11} \left[\frac{7x^2}{4} - \frac{x^3}{2} - \frac{x^4}{32}\right]_{x=0}^{x=2}$$

$$\bar{x} = \frac{3}{11} \left[\frac{7(4)}{4} - \frac{8}{2} - \frac{16}{32}\right] = \frac{3}{11} \left(7 - 4 - \frac{1}{2}\right) = \frac{3}{11} \left(\frac{5}{2}\right) = \frac{15}{22}$$

We can similarly find $\overline{y}$:

$$\overline{y} = \frac{1}{m} \iint_D y\,(2 + x + 3y)\ dA$$

$$\overline{y} = \frac{3}{11} \int_0^2 \int_0^{1-x/2} \left[2y + xy + 3y^2\right]\ dy\,dx$$

$$\overline{y} = \frac{3}{11} \int_0^2 \left[y^2 + \frac{x}{2}y^2 + y^3\right]_{y=0}^{y=1-x/2} dx$$

$$\overline{y} = \frac{3}{11} \int_0^2 \left[\left(1 - \frac{x}{2}\right)^2 + \frac{x}{2}\left(1 - \frac{x}{2}\right)^2 + \left(1 - \frac{x}{2}\right)^3\right]\ dx$$

$$\overline{y} = \frac{3}{11} \int_0^2 \left[2 - 2x + \frac{x^2}{2}\right]\ dx = \frac{3}{11}\left[2x - x^2 + \frac{x^3}{6}\right]_{x=0}^{x=2}$$

$$\overline{y} = \frac{3}{11}\left[2(2) - 2(2) + \frac{8}{6}\right] = \frac{3}{11}\left(\frac{4}{3}\right) = \frac{4}{11}$$

Therefore, the center of mass $(\overline{x}, \overline{y})$ is $(\frac{15}{22}, \frac{4}{11})$.

## *Exercise 535* Center of Mass

Find the center of mass of:

1. A lamina that occupies the area enclosed by the curves $y = 0$ and $y = 2\sin x$ from $0 \leq x \leq \pi$ if its density is given by $\rho(x, y) = x$.
2. The region $D$ if $D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq 3\}$; $\rho(x, y) = 1 + x^2 + y^2$
3. The triangular region $D$ with vertices $(0, 0)$, $(2, 1)$, $(0, 3)$; $\rho(x, y) = x + y$

*Working Space*

*Answer on Page 2108*

## 197.3 Moment of Inertia

We can also use double integrals to find the **moment of inertia** of a lamina about a particular axis (we will extend this to three-dimensional objects in the next chapter on triple integrals). Recall that the moment of inertia for a particle with mass $m$ a distance $r$ from the axis of rotation is $mr^2$. Dividing a lamina into small pieces, we see that the moment of inertia of each piece about the x-axis is:

$$\left(y_{ij}^*\right)^2 \rho\left(x_{ij}^*, y_{ij}^*\right) \Delta A$$

Where $x_{ij}^*$ and $y_{ij}^*$ are the x- and y-coordinates of the small piece. The moment of inertia of the entire lamina about the x-axis is then the sum of all the individual moments:

$$I_x = \sum_{i=1}^{n} \sum_{j=1}^{n} \left(y_{ij}^*\right)^2 \rho\left(x_{ij}^*, y_{ij}^*\right) \Delta A = \iint_D y^2 \rho(x, y)\ dA$$

Similarly, the moment of inertia of a lamina about the y-axis is:

$$I_y = \sum_{i=1}^{n} \sum_{j=1}^{n} \left(x_{ij}^*\right)^2 \rho\left(x_{ij}^*, y_{ij}^*\right) \Delta A = \iint_D x^2 \rho(x, y)\ dA$$

**Example**: Find the moment of inertia of a square centered at the origin with side length $r$ and constant density $\rho$ about the x-axis.

**Solution**: We can describe the square as the region bounded by $D = \{(x, y) | -r/2 \leq x \leq r/2, -r/2 \leq y \leq r/2\}$ with density function $\rho(x, y) = \rho$. Therefore, the moment of inertia about the x-axis is given by:

$$\begin{aligned}
I_x &= \int_{-r/2}^{r/2} \int_{-r/2}^{r/2} \rho y^2 \, dy \, dx \\
&= \rho \int_{-r/2}^{r/2} \left[\frac{1}{3} y^3\right]_{y=-r/2}^{y=r/2} dx \\
&= \frac{\rho}{3} \int_{-r/2}^{r/2} \left[\frac{r^3}{8} - \left(-\frac{r^3}{8}\right)\right] dx \\
&= \frac{\rho}{3} \int_{-r/2}^{r/2} \left[\frac{r^3}{4}\right] dx = \frac{r^3 \rho}{12} \int_{-r/2}^{r/2} 1 \, dx \\
&= \frac{r^3 \rho}{12} [x]_{x=-r/2}^{x=r/2} = \frac{r^3 \rho}{12} \cdot r = \frac{r^4 \rho}{12}
\end{aligned}$$

We can also find the moment of inertia about the origin, $I_o$. This is the moment of inertia

for an object rotating in the $xy$-plane about the origin. The moment of inertia about the origin is the sum of the moments of inertia about the $x$- and $y$-axes:

$$I_o = I_x + I_y = \iint_D \left(x^2 + y^2\right) \rho(x, y)\, dA$$

**Example**: Find the moment of inertia about the origin of a disk with density $\rho(x, y) = b$, centered at the origin, with a radius of $a$. Show that this is equal to the expected moment of inertia, $\frac{1}{2}MR^2$, where $M$ is the total mass of the disk and $R$ is the radius of the disk.

**Solution**: Since we are examining a circle about the origin, the region can be described in polar coordinates as $D = \{(r, \theta) \mid 0 \leq r \leq a,\ 0 \leq \theta \leq 2\pi\}$. Converting from Cartesian coordinates to polar coordinates:

$$I_o = \iint_D \left(x^2 + y^2\right) b\, dA = \int_0^a \int_0^{2\pi} r\left(r^2\right) b\, d\theta\, dr$$

$$= \int_0^a r^3\, dr \cdot \int_0^{2\pi} b\, d\theta = \frac{a^4}{4} \cdot (2\pi b) = \frac{\pi a^4 b}{2}$$

The total mass of this disk is the density, $b$, multiplied by the area, $\pi a^2$. Therefore,

$$R = a$$

$$M = \pi a^2 b$$

Substituting into the result of our double integral, we see that:

$$\frac{\pi a^4 b}{2} = \left(\pi a^2 b\right) \cdot \left(\frac{a^2}{2}\right) = M \cdot \frac{R^2}{2} = \frac{1}{2}MR^2$$

### 197.3.1 Radius of Gyration

When modeling rotating objects, it can be helpful to have a simplified model. A spinning, continuous object can be modeled as a point mass by using the lamina's *radius of gyration*. The radius of gyration of a lamina about the origin is a radius, $R$, such that:

$$mR^2 = I_o$$

where $m$ is the mass of the lamina and $I$ is the moment of inertia of the lamina. Essentially, we are finding a radius such that if the lamina were shrunk down to a point mass and rotated about the axis at that radius, the moment of inertia would be the same.

We can also find radii of gyration about the $x$- and $y$-axes:

$$m\bar{\bar{y}}^2 = I_x$$

$$m\bar{\bar{x}}^2 = I_y$$

About the origin, $R = \sqrt{\bar{\bar{x}}^2 + \bar{\bar{y}}^2}$.

**Example**: Find the radius of gyration about the $y$-axis for a disk with density $\rho(x, y) = y$ if the disk has radius 2 and is centered at $(0, 2)$.

**Solution**: We are ultimately looking for a radius such that $m\bar{\bar{x}}^2 = I_y$, so we need to know the mass, $m$, and the moment of inertia about the $y$-axis, $I_y$. First, let's find the total mass, $m$, of the disk. We can describe the disk in polar coordinates as $D = \{(r, \theta) \mid 0 \leq r \leq 4\sin\theta, 0 \leq \theta \leq \pi\}$, and therefore the mass is given by:

$$m = \iint_D y \, dA = \int_0^\pi \int_0^{4\sin\theta} r\,(r\sin\theta)\; dr\, d\theta$$

$$= \int_0^\pi \sin\theta \int_0^{4\sin\theta} r^2 \, dr\, d\theta = \frac{1}{3}\int_0^\pi \sin\theta\,[4\sin\theta]^3 \; d\theta$$

$$= \frac{64}{3}\int_0^\pi \sin^4\theta \, d\theta = \frac{64}{3}\int_0^\pi \left(\frac{1-\cos 2\theta}{2}\right)^2 d\theta = \frac{64}{3}\left(\frac{1}{2}\right)^2 \int_0^\pi \left(1 - 2\cos 2\theta + \cos^2 2\theta\right)\; d\theta$$

$$= \frac{16}{3}\left[(\theta - \sin 2\theta)_{\theta=0}^{\theta=\pi} + \int_0^\pi \frac{1+\cos 4\theta}{2}\, d\theta\right] = \frac{16}{3}\left[\pi + \frac{1}{2}\left(\theta + \frac{1}{4}\sin 4\theta\right)_{\theta=0}^{\theta=\pi}\right]$$

$$= \frac{16}{3}\left[\pi + \frac{\pi}{2}\right] = \frac{16}{3}\left(\frac{3\pi}{2}\right) = 8\pi$$

Now that we have found the mass, let's find the moment of inertia, $I_y$:

$$I_y = \iint_D x^2 y \, dA = \int_0^\pi \int_0^{4\sin\theta} r\,(r\cos\theta)^2\,(r\sin\theta)\; dr\, d\theta$$

$$= \int_0^\pi \left[\cos^2\theta\sin\theta \int_0^{4\sin\theta} r^4\, dr\right] d\theta = \int_0^\pi \cos^2\theta\sin\theta\left[\frac{1}{5}r^5\right]_{\theta=0}^{\theta=4\sin\theta} d\theta$$

$$= \frac{1024}{5}\int_0^\pi \cos^2\theta\sin^6\theta\, d\theta = \frac{1024}{5}\int_0^\pi \left(\frac{1+\cos 2\theta}{2}\right)\left(\frac{1-\cos 2\theta}{2}\right)^3 d\theta$$

$$= \frac{1024}{5}\left(\frac{1}{2}\right)^4 \int_0^\pi (1+\cos 2\theta)\,(1-\cos 2\theta)\,(1-\cos 2\theta)^2 \; d\theta$$

$$= \frac{64}{5}\int_0^\pi \left(1 - \cos^2 2\theta\right)\left(1 - 2\cos 2\theta + \cos^2 2\theta\right)\; d\theta$$

$$
\begin{aligned}
&= \frac{64}{5}\int_0^{\pi} 1 - 2\cos 2\theta + \cos^2 2\theta - \cos^2 2\theta + 2\cos^3 2\theta - \cos^4 2\theta \, d\theta \\
&= \frac{64}{5}\int_0^{\pi} 1 - 2\cos 2\theta + 2\cos 2\theta \left(1 - \sin^2 2\theta\right) - \left(\frac{1+\cos 4\theta}{2}\right)^2 d\theta \\
&= \frac{64}{5}\int_0^{\pi} 1 + 2\cos 2\theta \sin^2 2\theta - \frac{1}{4}\left(1 + 2\cos 4\theta + \cos^2 4\theta\right) \, d\theta \\
&= \frac{64}{5}\int_0^{\pi} 1 + 2\cos 2\theta \sin^2 2\theta - \frac{1}{4} - \frac{\cos 4\theta}{2} - \frac{1}{4}\left(\frac{1+\cos 8\theta}{2}\right) \, d\theta \\
&= \frac{64}{5}\left[\frac{5\theta}{8} + \frac{1}{3}\sin^3\theta - \frac{\sin 4\theta}{8} - \frac{\sin 8\theta}{64}\right]_{\theta=0}^{\theta=\pi} = \frac{64}{5}\cdot\frac{5\pi}{8} = 8\pi
\end{aligned}
$$

We have found that $m = 8\pi$ and $I_y = 8\pi$. Substituting to find the radius of gyration:

$$m\bar{\bar{x}}^2 = I_y$$

$$(8\pi)\bar{\bar{x}}^2 = 8\pi$$

$$\bar{\bar{x}} = 1$$

Therefore, the radius of gyration about the $y$-axis is $\bar{\bar{x}} = 1$

## *Exercise 536* Moments of Inertia and Radii of Gyration

Find the requested moment of inertia and radius of gyration of the lamina with the given density function.

1. about the $x$-axis, $D = \{(x, y) \mid 1 \leq x \leq 4,\ 0 \leq y \leq 3\}$, $\rho(x, y) = xy$.
2. about the $y$-axis, $D$ is enclosed by the curves $y = 0$ and $y = 2\cos x$ for $-\pi/2 \leq x \leq \pi/2$, $\rho(x, y) = x$.
3. about the origin, $D = \{(r, \theta) \mid 1 \leq r \leq 2,\ 0 \leq \theta \leq \pi\}$, $\rho(r, \theta) = r$.

*Working Space*

*Answer on Page 2109*

## 197.4 Surface Area

We have already seen how to find the areas of surfaces of revolution using single-variable calculus. Now, we will use multivariable calculus to find the surface area of a generic, two-variable function, $z = f(x, y)$. Suppose a surface, $S$, is defined by the continuous, partially differentiable function, $f(x, y)$, over a rectangular region, $R$ (see figure 200.4).

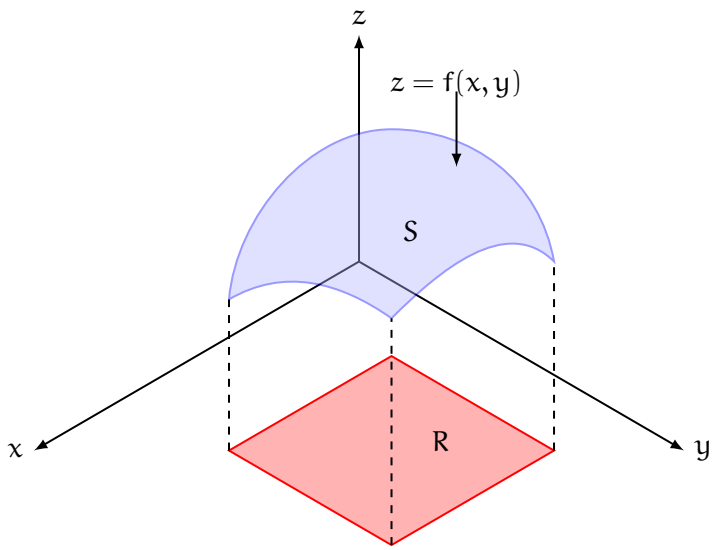



Figure 197.5: The graph of $f$ over the region $R$ creates a surface, $S$

We begin by dividing the region, $R$, into sub-rectangles, $R_{ij}$, each with area $\Delta A = \Delta x \Delta y$. Then, projecting upwards from the point closest to the origin, $(x_i, y_j, 0)$, we find a point on the surface, $P_{ij} = (x_i, y_j, f(x_i, y_j))$. Next, there is a small plane, $\Delta T_{ij}$, tangent to the surface at $P_{ij}$, and the area of the tangent plane is approximately the same as the area of the surface over the sub-rectangle $R_{ij}$ (see figure 197.6).

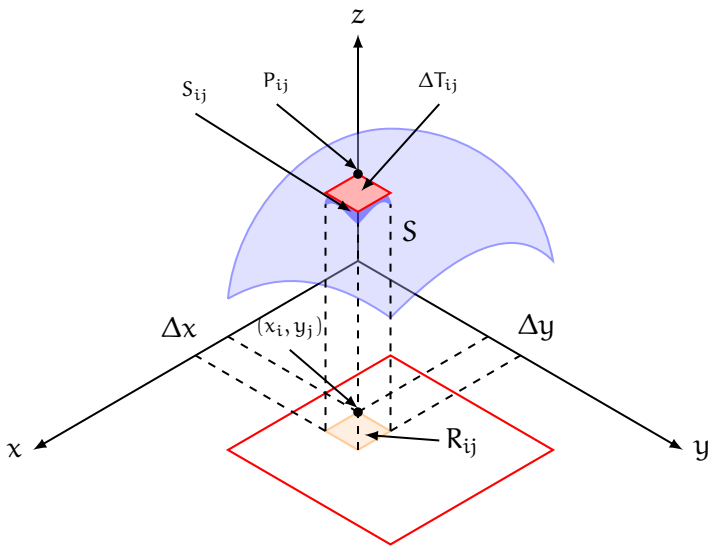



Figure 197.6: The tangent surface, $\Delta T_{ij}$, is approximately the same surface area as the surface, $S_{ij}$, over the sub-rectangle, $R_{ij}$

It follows that the total surface area of the surface, S, is the sum of all these little tangent surfaces as the number of tangent surfaces approaches infinity:

$$A(S) = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} \Delta T_{ij}$$

How can we find an expression for $\Delta T_{ij}$? We will define two vectors, **a** and **b**, that are equal to the sides of $\Delta T_{ij}$ (see figure 197.7). Geometrically, the area of $\Delta T_{ij}$ is the absolute value of the cross product of the two vectors. Mathematically,

$$\Delta T_{ij} = |\mathbf{a} \times \mathbf{b}|$$

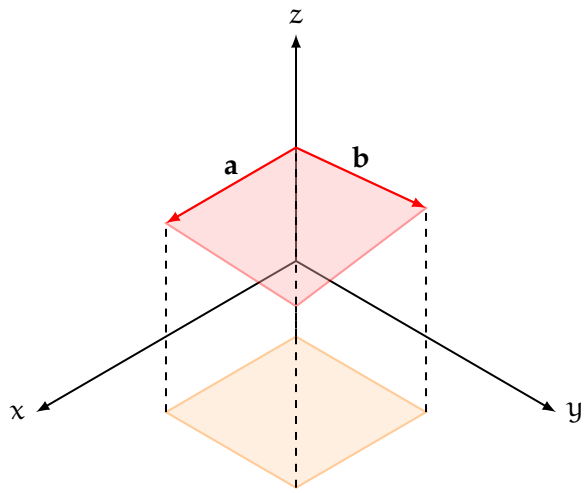



Figure 197.7: The vectors **a** and **b** define the sides of the tangent surface $\Delta T_{ij}$

Recall the three unit vectors: **i** in the x-direction, **j** in the y-direction, and **k** in the z-direction. We can then describe **a** and **b** in terms of **i**, **j**, and **k**:

$$\mathbf{a} = \Delta x\mathbf{i} + f_x(x_i, y_j)\Delta x\mathbf{k}$$

$$\mathbf{b} = \Delta y\mathbf{j} + f_y(x_i, y_j)\Delta y\mathbf{k}$$

(Recall that $f_x$ is the partial derivative of $f(x, y)$ with respect to x, and $f_y$ is the partial derivative with respect to y.) This is true because the partial derivative of $f_x$ gives the slope of a tangent line parallel to the x-axis, and $f_y$ parallel to the y-axis. We then find an expression for $|\mathbf{a} \times \mathbf{b}|$ (we've omitted some details here):

$$\mathbf{a} \times \mathbf{b} = -f_x(x_i, y_j)\Delta x\Delta y\mathbf{i} - f_y(x_i, y_j)\Delta x\Delta y\mathbf{j} + \Delta x\Delta y\mathbf{k}$$

Substituting $\Delta A = \Delta x\Delta y$:

$$\mathbf{a} \times \mathbf{b} = [-f_x(x_i, y_j)\mathbf{i} - f_y(x_i, y_j)\mathbf{j} + \mathbf{k}]\,\Delta A$$

To find the area of $\Delta T_{ij}$, we need to find the length of $\mathbf{a} \times \mathbf{b}$. Recall that we can use the Pythagorean theorem to find the length of a vector. For a 3-dimensional vector $\mathbf{v} = r\mathbf{i} + s\mathbf{j} + t\mathbf{k}$, it's length is given by:

$$|\mathbf{v}| = \sqrt{r^2 + s^2 + t^2}$$

Applying this, we find the length of $\mathbf{a} \times \mathbf{b}$ (which is the same as the area of $\Delta T_{ij}$) is:

$$\Delta T_{ij} = |\,[-f_x(x_i, y_j)\mathbf{i} - f_y(x_i, y_j)\mathbf{j} + \mathbf{k}]\,\Delta A|$$

$$= \sqrt{(-f_x(x_i, y_j)\Delta A)^2 + (-f_y(x_i, y_j)\Delta A)^2 + (\Delta A)^2}$$
$$= \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1}\Delta A$$

So, the area of the entire surface over region $R$ is:

$$A(S) = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1}\Delta A$$

This is the definition of a double integral; therefore, the surface area of a two-variable function, $f(x, y)$ over a region, R, where $f_x$ and $f_y$ are continuous, is:

$$A(S) = \iint_R \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1}\, dA$$

Using the notation of partial derivatives, this is also expressed as:

$$A(S) = \iint_R \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA$$

**Example**: Find the surface area of the part of the surface $z = 2 - y^2$ that lies over the triangle whose vertices are at $(0, 0)$, $(0, 4)$, and $(3, 4)$.

**Solution**: We can define $R = \{(x, y) \mid 0 \leq x \leq 34y,\ 0 \leq y \leq 4\}$. Additionally,

$$\frac{\partial z}{\partial x} = 0$$
$$\frac{\partial z}{\partial y} = -2y$$

Therefore, the area of the surface that lies above $R$ is:

$$A(S) = \iint_R \sqrt{1 + 0^2 + (-2y)^2}\, dA = \int_0^4 \int_0^{\frac{3}{4}y} \sqrt{1 + 4y^2}\, dx\, dy$$
$$= \int_0^4 \sqrt{1 + 4y^2}\, [x]_{x=0}^{x=\frac{3}{4}y}\, dy = \frac{3}{4} \int_0^4 y\sqrt{1 + 4y^2}\, dy$$

Let $u = 1 + 4y^2$, then $du = (8y)dy$ and $(y)dy = \frac{du}{8}$. Substituting:

$$A(S) = \frac{3}{4} \int_{y=0}^{y=4} \frac{1}{8}\sqrt{u}\, du = \frac{3}{32} \left[\frac{2}{3}u^{3/2}\right]_{y=0}^{y=4}$$

$$= \frac{1}{16}\left[\left(1+4y^2\right)^{3/2}\right]_{y=0}^{y=4} = \frac{1}{16}\left[(65)^{3/2}-1\right] \approx 32.69$$

## *Exercise 537* Surface Area of Two-Variable Functions

Find the area of the surface.

*Working Space*

1. The part of the plane $9x+6y-3z+6=0$ that lies above the rectangle $[2,6]\times[1,4]$.
2. The part of the paraboloid in the circle $z=2x^2+2y^2$ that lies under the plane $z=32$.
3. The part of the surface $z=3xy$ that lies in the cylinder $x^2+y^2=4$.

*Answer on Page 2111*

## 197.5 Average Value

Recall that the average value of a one-variable function over the interval $x \in [a,b]$ is given by:

$$f_{ave} = \frac{1}{a-b}\int_a^b f(x)\,dx$$

For a two-variable function, the average value over a region, R, is given by:

$$f_{ave} = \frac{1}{A(R)}\iint_R f(x,y)\,dA$$

Where $A(R)$ is the area of the two-dimensional region.

**Example**: Find the average value of $f(x,y)=xy^2$ over the rectangle with vertices at $(-2,0)$, $(-2,4)$, $(2,4)$, and $(2,0)$.

**Solution**: The rectangular region has an area of $(2-(-2))\cdot(4-0)=4\cdot4=16$. Therefore,

the average value is given by:

$$f_{ave} = \frac{1}{16}\iint_R xy^2\,dA = \frac{1}{16}\int_{-2}^{2}\int_{0}^{4} xy^2\,dy\,dx$$

$$= \frac{1}{16}\int_{-2}^{2}\frac{x}{3}y^3\Big|_{y=0}^{y=4}dx = \frac{1}{16}\int_{-2}^{2}\frac{x}{3}\left(4^3\right)dx = \frac{4}{3}\int_{-2}^{2}x\,dx$$

$$= \frac{4}{3}\left(\frac{1}{2}\right)x^2\Big|_{x=-2}^{x=2} = 0$$

## *Exercise 538* Average Value

Find the average value of the function over the region $D$:

*Working Space*

1. $f(x,y) = x\sin y$, $D = [0,2]\times[-\pi/2,\pi/2]$
2. $f(x,y) = x+y$, $D$ is the circle with radius 1 centered at $(1,0)$
3. $f(x,y) = xy$, $D$ is the triangle with vertices at $(0,0)$, $(2,0)$, $(2,2)$

*Answer on Page 2113*

Chapter 198

# Multivariate Distributions

The world of probability and statistics doesn't limit itself to the study of single variables. Often, we are interested in the interconnections, relationships, and associations among several variables. In such a scenario, the univariate distributions that we have studied so far become inadequate. To comprehend the joint behavior of these variables and to uncover the underlying patterns of dependency, we must turn to the realm of multivariate distributions.

This chapter aims to introduce the reader to the concept of multivariate probability distributions. These are probability distributions that take into account and describe the behavior of more than one random variable. We will start our exploration with a discussion on the joint probability mass and density functions. These functions extend the concepts of probability mass and density functions for one variable to the situation where we have multiple variables.

Next, we will explore important properties of joint distributions, including the concept of marginal distribution and conditional distribution, which allow us to explore the probability of a subset of variables while conditioning on, or ignoring, other variables. We will also introduce the idea of independence of random variables in the multivariate context.

Subsequently, we will discuss some commonly used multivariate distributions such as the multivariate normal distribution, and the multivariate Bernoulli and binomial distributions. These specific distributions will provide us with practical tools for modelling multivariate data.

Finally, we will delve into covariance and correlation, two key measures that give us a sense of how two variables change together. Understanding these measures is critical for capturing the relationships in multivariate data.

Chapter 199

# The Multivariate Normal Distribution

Let's say that you are gathering statistics on a species of snail. You have found 89 snails. Every snail has been weighted and had its shell measured. You have created a table like this:

| Weight | Diameter |
|---|---|
| 4.4769 grams | 2.2692 cm |
| 5.4755 grams | 2.1973 cm |
| 4.1183 grams | 2.52928 cm |
| ... | ... |
| 3.0522 grams | 1.7822 cm |

You have been told that for any particular species of snail, the weight and shell diameter are typically normally distributed. So, you compute the mean of each:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Your snails have a mean mass off 4.25 grams and a mean shell diameter of 2.35 centimeters. Now you know where the center of each normal distribution will be.

To know how wide each normal distribution is, you need to compute the variance:

$$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}$$

(Reminder: The standard deviation is the square root of $\sigma^2$)

Plotting this data is shown in Figure 199.1.

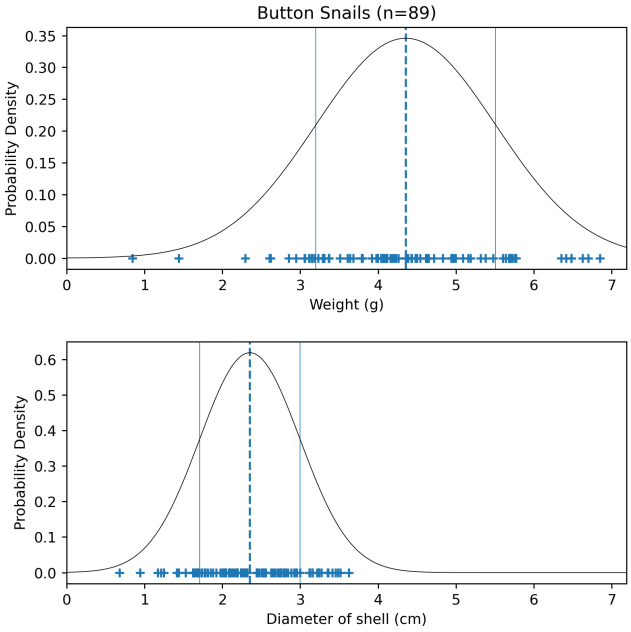



Figure 199.1: Probability density functions of weight and diameter.

Nice! However, then you think: ”Hmm. Maybe the mass and the diameter of the shell are related. It seems like a heavy snail might tend to have a larger shell.” So, you make a scatter plot:

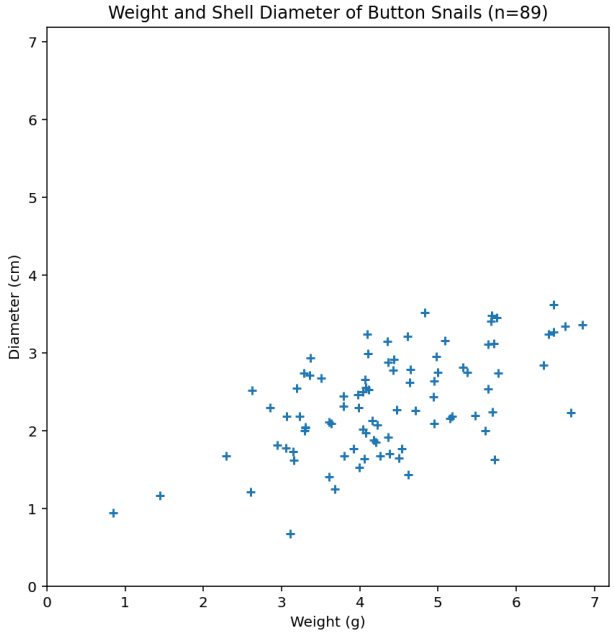



Figure 199.2: A scatter plot with weight of the snail horizontally and diameter of the shell on the y-axis.

Sure enough, there is some correlation between the mass of the snail and the diameter of its shell.

There is a form of the normal distribution that can deal with multiple variables and their correlations. This is known as the *multivariate normal* or the *Gaussian distribution*.

In a multivariate normal distribution, the mean (often denoted $\mu$) is a vector containing the mean of every variable. For your snails, $\mu = [4.25, 2.35]$.

Just as in the single-variable normal distribution, measurements near the mean are what you are most likely to observe. In a single-variable normal distribution, the standard deviation tells us how fast that likelihood falls off as we move away from the mean. In multivariate normal distributions, we need something a little more expressive: a matrix.

## 199.1 The Covariance Matrix

We can think of the data as a list of vectors. For example, if we have $n$ snails and $d$ properties that we have measured, we would get a list like this:

$$\begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,d} \\ x_{2,1} & x_{2,2} & \dots & x_{2,d} \\ \dots & \dots & & \dots \\ x_{n,1} & x_{n,2} & \dots & x_{b,d} \end{bmatrix}$$

Each row represents one snail. Each column represents one property.

Note that to make $\mu$ (the mean vector), you just take the mean of each column of this data matrix. (For convenience, we will use $\mu_i$ to refer to the mean of column $i$.)

We usually use $\Sigma$ (the uppercase Greek sigma) to represent a covariance matrix. $\Sigma$ is a $d \times d$ matrix. The entries on the diagonal of $\Sigma$ are just the variance of each property. So, we can calculate the entry at row $j$, column $j$ like this:

$$\Sigma_{j,j} = \frac{\sum_{i=1}^{n}(x_{i,j} - \mu_j)^2}{n}$$

(Yes, we are using $\Sigma$ to represent both summation and the covariance matrix here. It can be confusing, but you will be able to tell from context how it is being used.)

The other entries, $\Sigma_{j,k}$ where $j \neq k$, are the *covariance* between property $j$ and property $k$. If the covariance is a positive number, property $j$ and property $k$ are correlated. For example, in your snails, weight and diameter have a positive covariance: If the snail is

heavier than average, it tends to have a diameter that is larger than average.

If the covariance is a negative number, when property j is greater than average, property k tends to be less than average. For example, the number of times a restaurant mops the floor in a week has a negative covariance with how much bacteria lives on the floor.

How do we compute the covariance?

$$\Sigma_{j,k} = \frac{\sum_{i=1}^{n}(x_{i,j} - \mu_j)(x_{i,k} - \mu_k)}{n}$$

Note that $\Sigma_{j,k} = \Sigma_{k,j}$, so the matrix is symmetric.

### 199.1.1 Computing Mean and Covariance in Python

If you have a numpy array where each row represents one sample and each column represents one property, it is really easy to compute $\mu$ and $\Sigma$:

```
# Read in the data matrix, each row represents one snail
snails = ...

# Compute mu
mean_vector = snails.mean(axis=0)
print(f"Mean = {mean_vector}")

# Compute Sigma
covariance_matrix = np.cov(snails, rowvar=False)
print(f"Covariance = {covariance_matrix}")
```

## 199.2 Multivariate Normal Probability Density

Now that we have good estimates of $\mu$ and $\Sigma$, how can we compute the probability density?

When you were working with just one variable, $x$, you computed the probability density using the mean $\mu$ and the variance $\sigma^2$ like this:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The probability density function of a d-dimensional multivariate normal distribution with a mean of $\mu$ and a covariance matrix of $\Sigma$ is given by:

$$p(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})\right)$$

If we draw lines to show where $\boldsymbol{\mu}$ is and contours to show lines of equal probability density, we get a plot shown in Figure 199.3.

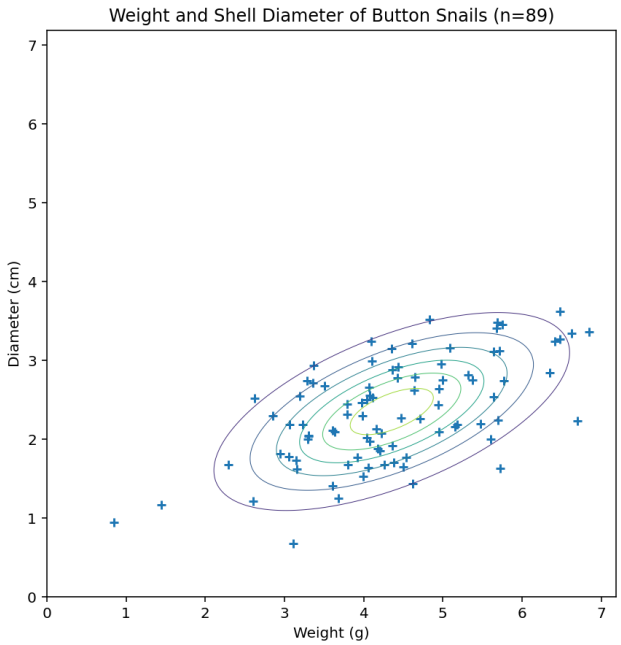



Figure 199.3: Contour plot of the snail data.

Or, we can do a 3D plot, as shown in Figure 199.4.

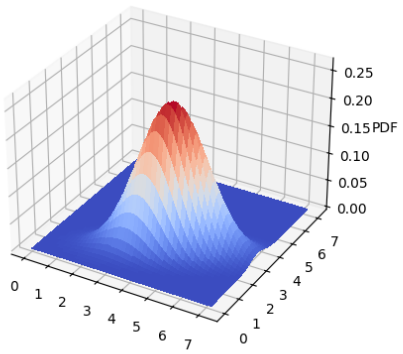



Figure 199.4: 3D plot of the probability density function.

A probability density always integrates to 1.0, so the volume under the surface must be 1.0.

### 199.2.1 Multivariate Normal Probability Density in Python

The scipy library has a class that represents the multivariate normal. Here is how you could compute the probability density for a particular snail:

```
import numpy as np
from scipy.stats import multivariate_normal

...Compute mean_vector and covariance_matrix...

# Get the probability density at [3 grams, 2 cm]
x = np.array([3.0, 2.0])
pd = multivariate_normal.pdf(x, mean=mean_vector, cov=covariance_matrix)
print(f"The probability density at {x} is {pd}")
```

Alternatively, maybe you would like to generate weights and diameters for a fictional population of 700 snails:

```
new_snails = multivariate_normal.rvs(mean=mean_vector, cov=covariance_matrix,
↪  size=700)
print(f"My fictional snails:\n{new_snails}")
```

Chapter 200

# Numerical Double Integration

In an earlier chapter, we gave an example of the multivariate normal distribution: the mass and shell diameter of a population of snails.

Starting with a table of measurements, we estimated that the mean vector $\mu$ was $\begin{bmatrix}4.25\text{g} & 2.35\text{cm}\end{bmatrix}$.

We then computed the covariance matrix:

$$\Sigma = \begin{bmatrix}1.33 & 0.443 \\ 0.443 & 0.416\end{bmatrix}$$

Plotting the data points, the mean, and the equal-density contours looked like this:

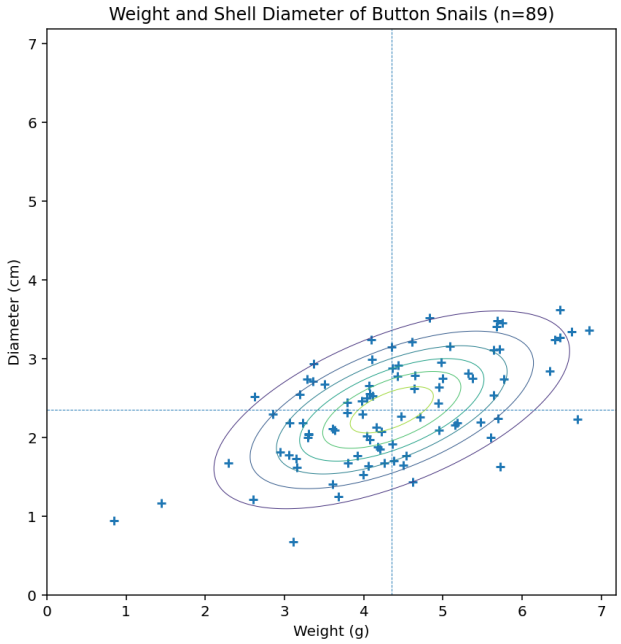



Figure 200.1: Equal density contours plotted with the snail data.

We have a great formula for computing the probability density for any mass/diameter combination:

$$p(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})\right)$$

Can you answer the following question? "If I pick a random snail off the floor of the ocean, what is the probability that its mass is between 3 and 4 grams and its diameter is between 1.5 and 2.0 centimeters?"

If we think of the probability density as a surface, this question is really "What is the volume under the surface in the rectangular patch $3 <= x_1 <= 4$ and $1.5 <= x_2 <= 2.0$?" Which you should recognize as a double integration problem:

$$P = \int_{1.5}^{2.0} \int_{3}^{4} \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})\right) dx_1 dx_2$$

Sadly, however, no one has ever been able to find the antiderivative of the multivariable probability density function, so no one can solve this problem.

Instead, we use Riemann sums to find an approximate solution. This is known as *numerical integration*.

(After spending so much time learning the techniques for integration, it may be disappointing to hear this: For a lot of real-world problems, there is no way to find an antiderivative, so we end up doing numerical integration much more often than most people realize.)

## 200.1 Riemann Sums on 2-Dimensional Domains

When doing Riemann sums on function that takes a single real number, you summed the area of the rectangles under the function to approximate the integral:

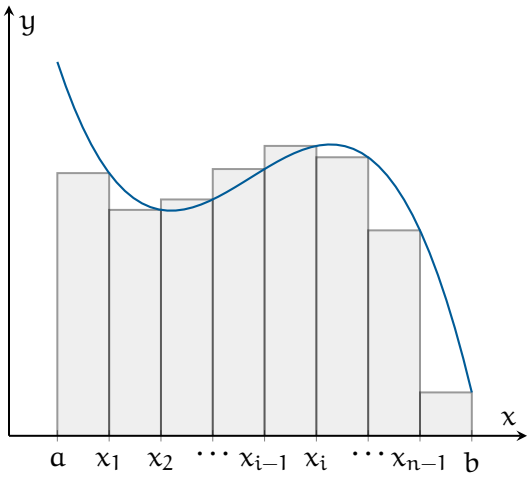



Figure 200.2: A representative function divided into $n$ rectangles of equal width, with rectangle height determined by the right endpoint of the subinterval

Here you are finding the volume under a function that takes two variables (the probability density is based on the mass and diameter of the shell).

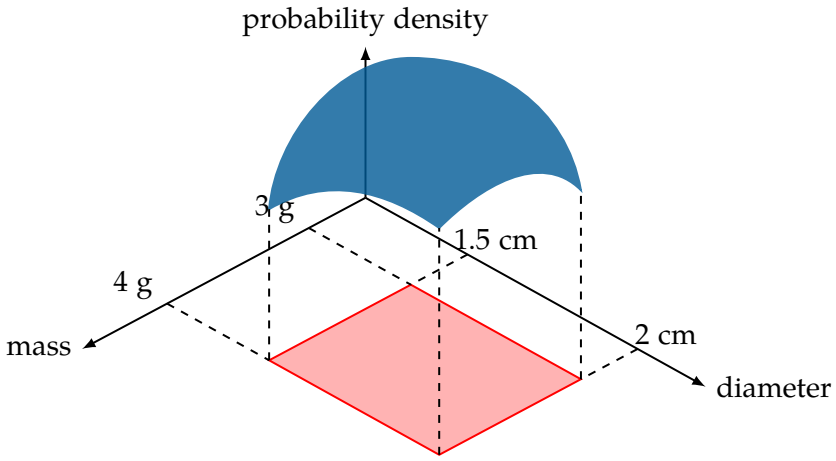



Figure 200.3: The probability is the volume under a surface for a region

To do the Riemann sum, we can break the range of the mass into $n_1$ equally sized intervals and the range of the diameter into $n_2$ equally sized intervals. For the diagram, we will just break each range into three, but you will get more accurate estimate as the intervals get smaller.

Now you will be calculating the volume of rectangular solids and summing up those volumes. Here are a few of the rectangular volumes:

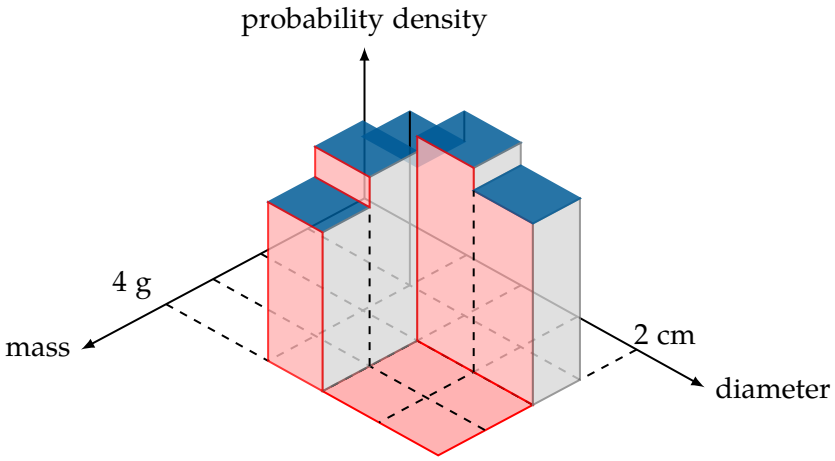



Figure 200.4: Riemann Sum

## 200.2 Numerical Integration in Python

Here's the code:

```
import numpy as np
from scipy.stats import multivariate_normal

# pip3 install scipy
weight_lower_limit = 3.0
weight_upper_limit = 4.0
weight_slices = 100

diameter_lower_limit = 1.5
diameter_upper_limit = 2.0
diameter_slices = 100

# What's the average weight and diameter
mean_vector = np.array([4.35559489,2.3526593 ])
print(f"Mean [weight, diameter] = {mean_vector}")

# Do heavier snails tend to have bigger shells?
covariance_matrix = np.array([[1.33099714,0.44309754],
 [0.44309754,0.41603925]])
print(f"Covariance = \n{covariance_matrix}")

# Create a multivariate normal distribution
rv = multivariate_normal(mean_vector, covariance_matrix)

delta_weight = (weight_upper_limit - weight_lower_limit) / weight_slices
delta_diameter = (diameter_upper_limit - diameter_lower_limit) / diameter_slices

sum = 0.0
```

```
# Step though each different weight
for i in range(weight_slices):
    # What is the weight in the middle of this slice?
    current_weight = weight_lower_limit + (i + 0.5) * delta_weight

    for j in range(diameter_slices):
        # What is the diameter in the middle of this slice?
        current_diameter = diameter_lower_limit + (j + 0.5) * delta_diameter

        # What is the probability density there?
        prob_density = rv.pdf((current_weight, current_diameter))

        # What is the volume under that for this tiny square
        sum += prob_density * delta_weight * delta_diameter

print(
    f"\nThe probability that the weight is between {weight_lower_limit} and
    ↪ {weight_upper_limit}"
)
print(
    f"and that the diameter is between {diameter_lower_limit} and
    ↪ {diameter_upper_limit}"
)
print(f"is about {sum * 100.0:.4f}%")
```

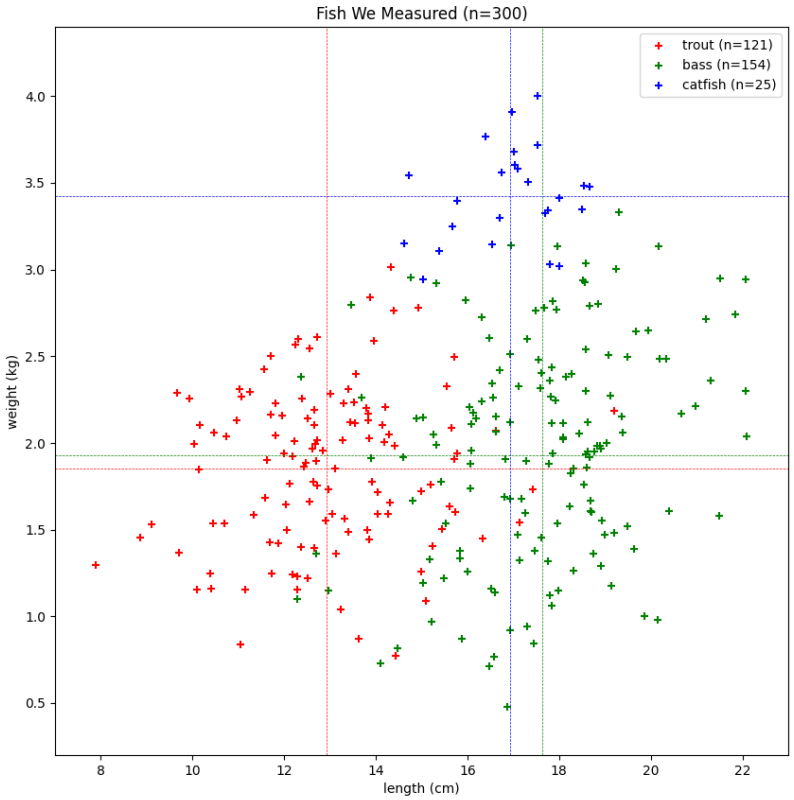
This should get you the following output:

```
> python3 num_integration.py
Mean [weight, diameter] = [4.35559489 2.3526593 ]
Covariance =
[[1.33099714 0.44309754]
 [0.44309754 0.41603925]]

The probability that the weight is between 3.0 and 4.0
and that the diameter is between 1.5 and 2.0
is about 7.8316%
```

# Chapter 201

# Bayesian Classifiers

You drop a net in several places in a lake. You measure, weigh, and identify every fish to be one of the following: `trout`, `bass`, or `catfish`. Here is a scatter plot of what you find:




(Data: `make_data.py`, Graph: `1_scatter_fish.py`)

The vertical and horizontal lines represent the mean length and width (respectively) of each type of fish.

You are writing a system that someone on the lake could use to identify their fish.

## 201.1 The Prior

Even before the person measures and weighs the fish, they can make a decent guess at what any randomly selected fish is:

| Fish | Count | Percent |
|---|---|---|
| trout | 121/300 | 40.3% |
| bass | 154/300 | 51.3% |
| catfish | 25/300 | 8.3% |

If you know nothing about the fish, there is a 51.3% chance it is a bass. The probability without any observations is known as the *prior*:

$$\mathrm{P}(\text{fish} = \text{trout}) = .403$$
$$\mathrm{P}(\text{fish} = \text{bass}) = .513$$
$$\mathrm{P}(\text{fish} = \text{catfish}) = .083$$

However, now the user makes an observation: The fish is 16 inches long and weighs 2.1 kg. Can we update our beliefs based on this? Sounds like a job for conditional probability.

By the definition of conditional probability:

$$\mathrm{P}(\text{fish} = \text{trout}|\text{length} = 16, \text{weight} = 2.1) = \frac{\mathrm{p}(\text{fish} = \text{trout}, \text{length} = 16, \text{weight} = 2.1)}{\mathrm{p}(\text{length} = 16, \text{weight} = 2.1)}$$

At this rate, we are going to run out of space, so we will use $\mathrm{F}$ to represent fish, $\mathrm{L}$ to represent length, and $W$ to represent weight.

We can calculate the denominator by summing up all the possibilities. Now, the right-hand side is:

$$\frac{\mathrm{p}(\mathrm{F}{=}\text{trout}, \mathrm{L}{=}16, \mathrm{W}{=}2.1)}{\mathrm{p}(\mathrm{F}{=}\text{trout}, \mathrm{L}{=}16, \mathrm{W}{=}2.1) + \mathrm{p}(\mathrm{F}{=}\text{bass}, \mathrm{L}{=}16, \mathrm{W}{=}2.1) + \mathrm{p}(\mathrm{F}{=}\text{catfish}, \mathrm{L}{=}16, \mathrm{W}{=}2.1)}$$

We can calculate the probability of all three possibilities (which will add up to 1.0).

## 201.2 Naive Bayes

The assumption in naive Bayes is that the probability distributions of length and weight are conditionally independent given the breed of the fish. That is:

$$\mathrm{p}(\mathrm{F}{=}\text{trout}, \mathrm{L}{=}16, \mathrm{W}{=}2.1) = \mathrm{p}(\mathrm{L}{=}16|\mathrm{F}{=}\text{trout})\mathrm{p}(\mathrm{W}{=}2.1|\mathrm{F}{=}\text{trout})\mathrm{P}(\mathrm{F}{=}\text{trout})$$

Within a particular species, things like length and weight, which represent the sum of a lot of genetic and environmental factors, are often normally distributed.

What if we assume that length has a normal distribution for each kind of fish? What is the most likely estimator for the mean of each trait?

$$\mu = \frac{1}{n} \sum_{i=1}^{n} x_i$$

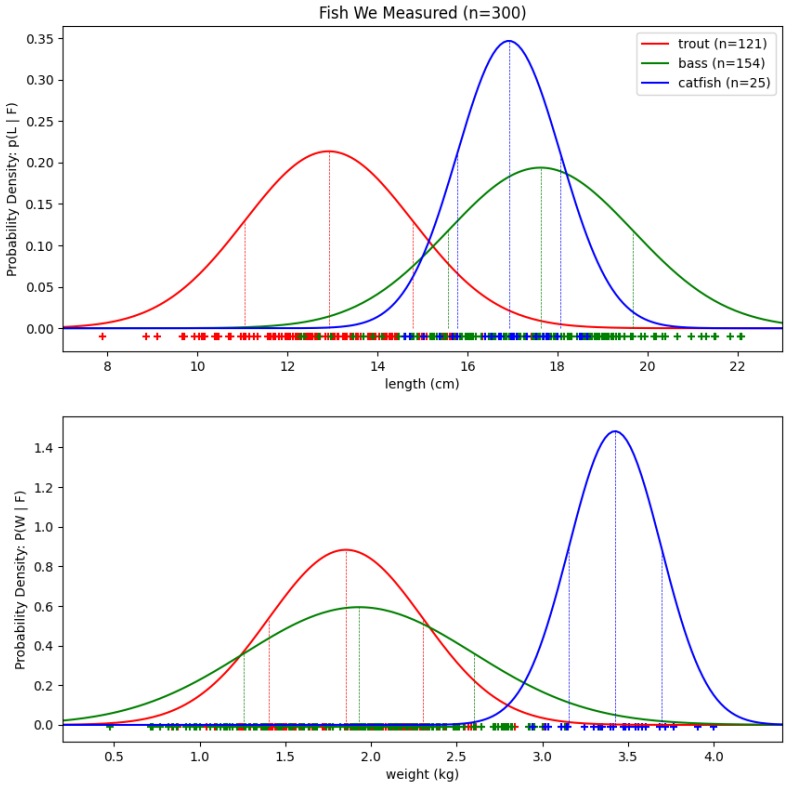
And variance?

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^{n} (\mu - x_i)^2$$

We can compute those pretty easily:

| Fish | Trait | $\mu$ | $\sigma^2$ |
|---|---|---|---|
| Trout | Length | 12.91522571 | 3.49054363 |
| | Weight | 1.85290281 | 0.20387574 |
| Bass | Length | 17.62628327 | 4.23881747 |
| | Weight | 1.92960088 | 0.45092986 |
| Catfish | Length | 16.92187387 | 1.32301935 |
| | Weight | 3.42423185 | 0.07244546 |

If we plot those probability distributions out:




(Source:2_univariates.py)

Note that the area under each one of those curves is exactly 1.0. They are probability distributions.

Now we can use the formula for the normal distribution:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Thus,

| Given | $p(L = 16\|F)$ | $p(W = 2.1\|F)$ |
|---|---|---|
| F=Trout | .0546 | .7607 |
| F=Bass | .1418 | .5753 |
| F=Catfish | .2516 | 0.01 |

Using the naive assumption, we can now compute the joint probability for any type of fish:

$$p(F = ?, L = 16, W = 2.1) = p(W = 2.1|F = ?)p(L = 16|F = ?)p(F = ?)$$

| For | $p(L = 16, W = 2.1, F = ?)$ |
|---|---|
| F=Trout | .016763 |
| F=Bass | .041886 |
| F=Catfish | 0.00001 |

The sum of these is $p(L = 16, W = 2.1)$: .0586.

By the definition of conditional probability:

$$p(F = ?|L = 16, W = 2.1) = \frac{p(F = ?, L = 16, W = 2.1)}{p(L = 16, W = 2.1)}$$

So,

| For | $p(F = ?|L = 16, W = 2.1)$ |
|---|---|
| F=Trout | .286 |
| F=Bass | .714 |
| F=Catfish | $\approx 0$ |

If you pull a random fish out of the lake and its length is 16 and its weight is 2.1, there is a 71.4% chance that it is bass. There is a 28.6% chance it is a trout.

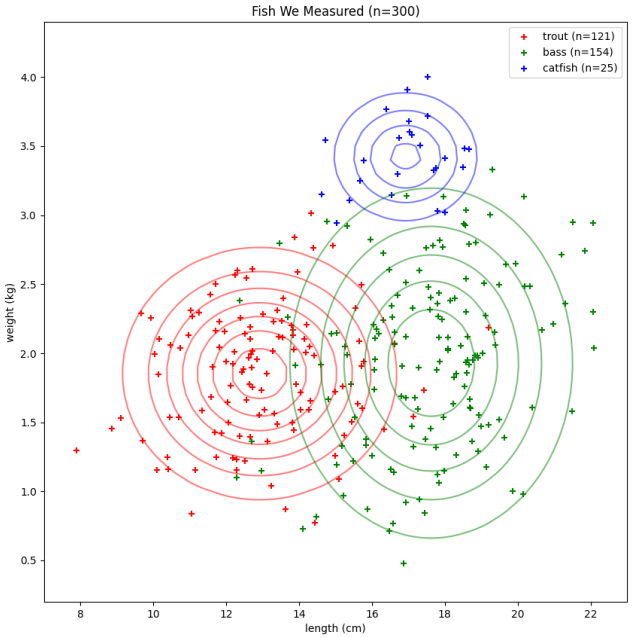
## 201.3 Using the multivariate Gaussian distribution

Given the kind of fish, length and width are independent!? That makes no sense. Longer fish tend to be heavier, right?

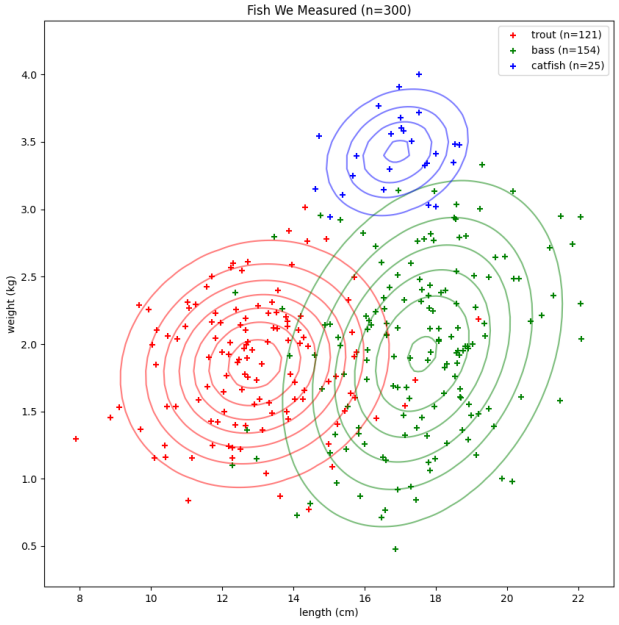
The multivariate Gaussian distribution is a generalization of the normal distribution::

- It deals with data of any dimension d.
- It can include the idea of covariance. For example, length and width are not independent.

If we use your naive Bayes, we can get two-dimensional density plots that would look like this:




With independence, there are ovals that are circles, ovals that go up, or ovals that sideways. If we use the multivariate Gaussian distribution, we get:




When there is covariance, the ovals can slant in any direction.

If we discard the naive assumption, we should get slightly more accurate results.

The multivariate Gaussian distribution uses vectors and matrices. The mean $\vec{\mu}$ is a vector

of dimension d. The variance of each variable and its covariance with the other variables is captured in a $d \times d$ matrix called *the covariance matrix* is usually called $\mathbf{\Sigma}$. (Yes, this is also the symbol for summation. This creates some confusion at times, but you can usually figure out which is intended by the context.)

When you know $\vec{\mu}$ and $\mathbf{\Sigma}$, the probability density for any vector $\vec{x}$ is:

$$p(\vec{x}) = (2\pi)^{-d/2} \det(\mathbf{\Sigma})^{-1/2} \exp\left(-\frac{1}{2}(\vec{x} - \vec{\mu})^T \mathbf{\Sigma}^{-1} (\vec{x} - \vec{\mu})\right)$$

If you have several data points $\vec{x_1}, ..., \vec{x_n}$, how do you compute the values of $\vec{\mu}$ and $\mathbf{\Sigma}$ for which the observations $\vec{x_1}, ..., \vec{x_n}$ would be most likely?

$$\vec{\mu} = \frac{1}{n} \sum_{i=1}^{d} \vec{x_i}$$

And the covariance matrix? (Here, we think of each $\vec{x_i}$ and $\vec{\mu}$ as column vectors.)

$$\mathbf{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} (\vec{x_i} - \vec{\mu})(\vec{x_i} - \vec{\mu})^T$$

Notice that the diagonal entries of the matrix are the variance of each variable. Notice also that this is a symmetric matrix.

Applying this to our data we get:

| Fish | $\vec{\mu}$ | $\mathbf{\Sigma}$ |
|---|---|---|
| Trout | $\begin{bmatrix} 12.91522571 \\ 1.85290281 \end{bmatrix}$ | $\begin{bmatrix} 3.51963149 & 0.09975744 \\ 0.09975744, & 0.20557471 \end{bmatrix}$ |
| Bass | $\begin{bmatrix} 17.62628327 \\ 1.92960088 \end{bmatrix}$ | $\begin{bmatrix} 4.26652216 & 0.39553268 \\ 0.39553268 & 0.45387711 \end{bmatrix}$ |
| Catfish | $\begin{bmatrix} 16.92187387 \\ 3.42423185 \end{bmatrix}$ | $\begin{bmatrix} 1.37814515 & 0.07281453 \\ 0.07281453 & 0.07546403 \end{bmatrix}$ |

We can use these to compute the likelihood of seeing a fish with length = 16 and weight = 2.1 for each type of fish:

| Given | $p(L = 16, W = 2.1 \mid F = ?)$ |
|---|---|
| F=Trout | .04578 |
| F=Bass | .07732 |
| F=Catfish | .000001 |

Multiplied by the prior to get the joint probability:

| Given | $p(L = 16, W = 2.1, F = ?)$ |
|---|---|
| F=Trout | .01847 |
| F=Bass | .03969 |
| F=Catfish | $\approx 0$ |

We sum those to get $p(L = 16, W = 2.1) = .0582$.

| Given | $p(F = ? \mid L = 16, W = 2.1)$ |
|---|---|
| F=Trout | .318 |
| F=Bass | .682 |
| F=Catfish | $\approx 0$ |

Given a randomly selected fish that is 19 inches long and 2.1 kg in weight, you would guess that it is a bass with a confidence of 68.2%.

Appendix A

# Answers to Exercises

## Answer to Exercise 1 (on page 8)

1. C:H = 1:4
2. Cu:S:O = 1:1:4
3. C:H:O = 6:12:6 = 1:2:1

## Answer to Exercise 2 (on page 17)

1. balanced
2. unbalanced; oxygen
3. unbalanced; barium, sulfur, oxygen, and carbon
4. balanced

## Answer to Exercise 3 (on page 20)

1. $v = \sqrt{2gh} = \sqrt{2\left(9.8 m/s^2\right)(3.7 m)} = 8.5 m/s$
2. $h = \frac{v^2}{2g} = \frac{\left(15 \frac{m}{s}\right)^2}{2\left(9.8 \frac{m}{s^2}\right)} = 11.5 m$. You can hit your little brother, since you could shoot the pebble as high as 11.5 m and he is only 5.0 m above you.
3. We know that $E_{p,initial} = E_{k,final} + E_{losttofriction}$. Therefore, $E_{lost_to_friction} = E_{p,initial} - E_{k,final} = mgh - \frac{1}{2}mv^2 = (63 kg)\left(9.8 \frac{m}{s^2}\right)(8 m) - \frac{1}{2}(63 kg)\left(11.2 \frac{m}{s}\right)^2 = 4939.2 J - 3951.4 J = 988 J$.

## Answer to Exercise 4 (on page 23)

1. The human body is an open system because matter can enter (food, water, oxygen) and leave (waste, carbon dioxide, sweat) your body.
2. The Earth is an open system because matter can enter (asteroids falling, spaceships returning) and leave (space vehicles and astronauts). On the other hand, the Earth can be well-approximated as a closed system. Before the 20th Century, humans had no way to deliberately expel matter from the Earth, and the mass of asteroids that are pulled in by the Earth's gravity is negligible compared to the Earth. Therefore, in the right circumstances, it would be appropriate to model the Earth as a closed system. (It is closed because energy in the form of sunlight is constantly entering the system.)
3. A cell phone is a closed system - you don't put any matter in or take it out of your phone, but it constantly uses battery and then is recharged, showing that energy enters and leaves your phone.
4. Since the cooler is described as well-insulated and the lid is closed, it can be approximated as an isolated system. Scientific equipment, like bomb calorimeters, rely on this approximation.
5. With the lid open, matter can enter and leave and therefore the cooler is an open system (yes, even though it is well-insulated).
6. A sealed bottle of soda is a closed system - the soda and carbon dioxide can't escape, but energy in the form of heat can be transferred in and out of the system (the contents of the bottle will lose heat if you put it in the fridge and gain heat if you leave it in the sun).

## Answer to Exercise 5 (on page 27)

(a)

$$\begin{aligned} F &= G\frac{m_1 m_2}{r^2} \\ &= (6.674 \times 10^{-11})\frac{(6.0 \times 10^{24})(8.0 \times 10^{22})}{(4.0 \times 10^{6})^2} \\ &= 2.0022 \times 10^{24} \end{aligned}$$

(b) Now we have a new distance, $r' = 8.0 \times 10^{6}\,\text{m}$, double the original. Notice that gravitational force is inversely proportional to the square of the distance:

$$F \propto \frac{1}{r^2}$$

So if distance doubles,

$$r' = 2r \quad \Rightarrow \quad F' = \frac{F}{4}$$

Which results in $5.0055 \times 10^{23}$ N

## Answer to Exercise 6 (on page 28)

$$F = G\frac{m_1 m_2}{r^2} = (6.674 \times 10^{-11})\frac{(6.8 \times 10^3)(6 \times 10^{24})}{(10^5)^2} = 6.1 \times 10^6$$

About 6 million newtons.

## Answer to Exercise 7 (on page 30)

The gravitational acceleration is 9.8 $m/s^2$, no matter the mass of the object.

## Answer to Exercise 8 (on page 31)

The mass of the coin is $\frac{.0198 \text{ N}}{9.8 \text{ m/s}^2} = 0.002$ grams

## Answer to Exercise 9 (on page 33)

We can set up a balanced center of mass equation for this problem. Our problem is only in one dimension, so there is only one distance to be checked. If we set our $x_{CM} = 0$ where the fulcrum or pivot of a seesaw would be by assuming the balance scenario, we can solve our problem:

$$\begin{aligned} x_{CM} &= \frac{\Sigma m_i x_i}{\Sigma m_i} \\ 0 &= \frac{(30)(40) + (50)(-20) + (15)\,d}{30 + 50 + 15} \\ &= \frac{1200 - 1000 + (15)\,d}{85} \\ &= \frac{200 + (15)\,d}{85} \\ d &= -13\tfrac{1}{3}\text{m} \end{aligned}$$

Since our answer is negative, it means to the left of the fulcrum (given we chose left to be negative). Thus, $M_3$ should be placed $13\frac{1}{3}$ m to the left.

## Answer to Exercise 10 (on page 41)

1. Acceleration: the satellite is moving in a circle, therefore changing direction and accelerating.
2. Not acceleration: the car isn't changing speed or direction.
3. Acceleration: the child is changing speed.
4. Acceleration: the roller coaster is changing direction.
5. Acceleration: the roller coaster is changing speed.
6. Not acceleration: the book isn't changing speed or direction.

## Answer to Exercise 11 (on page 43)

If you accelerate to 20 m/s in 120 s, the acceleration is:

$$a = \frac{v_f - v_i}{t} = \frac{20\ \text{m/s} - 0\ \text{m/s}}{120\ \text{s}} = \frac{1}{6}\frac{\text{m}}{\text{s}^2}$$

To achieve this acceleration, you will need to apply a force of:

$$F = m \cdot a = 2400\ \text{kg} \cdot \frac{1}{6}\frac{\text{m}}{\text{s}^2} = 400\ \text{N}$$

## Answer to Exercise 12 (on page 45)

C, A, E/F, B, D

$$a_A = \frac{F_{net,A}}{m_A} = \frac{4\text{N} - 3\text{N}}{1\text{kg}} = \frac{1\text{N}}{1\text{kg}} = 1\frac{\text{m}}{\text{s}^2}$$

$$a_B = \frac{F_{net,B}}{m_B} = \frac{3\text{N} - 4\text{N}}{1\text{kg}} = \frac{-1\text{N}}{1\text{kg}} = -1\frac{\text{m}}{\text{s}^2}$$

$$a_C = \frac{F_{net,C}}{m_C} = \frac{4\text{N} - 3\text{N}}{0.5\text{kg}} = \frac{1\text{N}}{0.5\text{kg}} = 2\frac{\text{m}}{\text{s}^2}$$

$$a_D = \frac{F_{net,D}}{m_D} = \frac{3\text{N} - 4\text{N}}{0.5\text{kg}} = \frac{-1\text{N}}{0.5\text{kg}} = -2\frac{\text{m}}{\text{s}^2}$$

$$a_E = \frac{F_{net,E}}{m_E} = \frac{5\text{N} - 5\text{N}}{0.5\text{kg}} = \frac{0\text{N}}{0.5\text{kg}} = 0\frac{\text{m}}{\text{s}^2}$$

$$a_F = \frac{F_{net,F}}{m_F} = \frac{5N - 5N}{1kg} = \frac{0N}{1kg} = 0\frac{m}{s^2}$$

## Answer to Exercise 13 (on page 48)

3, 1/2, 6, 5, 4 The velocity does not affect the apparent weight, only the acceleration. If the elevator is accelerating upwards, the apparent weight increases. If the elevator is accelerating downwards, the apparent weight decreases. In fact, for scenario 4, the elevator is in free-fall and the person has no apparent weight. When the elevator is not accelerating, the person's apparent weight is their true weight.

## Answer to Exercise 14 (on page 52)

Kinetic energy? $E = \frac{1}{2}mv^2 = (.5)(55)(11^2) = 3,327.5\frac{kgm^2}{s^2} = 3,327.5$ joules.

Frictional force? $F = \mu N = (0.7)(55)(9.8) = 377.3$ newtons.

Distance? $D = \frac{3,327.5}{377.3} \approx 8.8$ seconds. (found by using $W = F_f D$ and $W = \Delta E$)

## Answer to Exercise 15 (on page 54)

The empty sled is pushing directly down on the floor with a force of $(50)(9.8) = 490$ N.

The force to overcome the static friction is:

$$270 = 490\mu_s$$

Thus, $\mu_s = 0.551$

The force to match kinetic friction is:

$$220 = 490\mu_k$$

Thus, $\mu_k = 0.449$

Once you are on the sled, it is pressing directly down on the floor with a force of $(50 + 55)(9.8) = 1,029$ N.

The force to overcome the static friction is:

$$F = (1,029)(0.551) = 567 \text{ N}$$

Once the sled is moving, friction is counteracting some of your force. How much?

$$F_f = (1,029)(0.449) = 462 \text{ N}$$

All of your acceleration is due to the remaining $567 - 492 = 75$ N.

We know that $F = ma$. In this case $F = 75$ N and $m = 105$ kg. So

$$a = \frac{75}{105} = 0.714 \text{ meters per second per second}$$

## Answer to Exercise 16 (on page 61)

$$4.5 \text{ kWh} \left(\frac{3.6 \times 10^6 \text{ joules}}{1 \text{ kWh}}\right)\left(\frac{1 \text{ calories}}{4.184 \text{ joules}}\right) = \frac{(4.5)(3.6 \times 10^6)}{4.184} = 1.08 \times 10^6 \text{calories}$$

## Answer to Exercise 17 (on page 62)

$$\frac{0.1 \text{ gallons}}{2 \text{ minutes}}\left(\frac{3.7854 \text{ liters}}{1 \text{ gallons}}\right)\left(\frac{1000 \text{ milliliters}}{1 \text{ liters}}\right)\left(\frac{1 \text{ minutes}}{60 \text{ seconds}}\right) = \frac{(0.1)(3.7854)(1000)}{(2)(60)} \text{ ml/second} = 3.1545 \text{ ml/second}$$

## Answer to Exercise 18 (on page 69)

1. 12.011 amu + 4(1.008 amu) = 16.043 amu

2. 63.546 amu + 32.06 amu + 4(15.999 amu) = 159.602 amu
3. 6(12.011 amu) + 12(1.008 amu) + 6(15.999) amu = 180.156 amu

## Answer to Exercise 19 (on page 71)

| Substance | num. of particles | num. of moles | grams |
|---|---|---|---|
| $NaHCO_3$ | $2.509 \times 10^{23}$ | 0.4166 | 35.00 |
| HCl | $7.53 \times 10^{23}$ | 1.25 | 45.58 |
| $KH_2PO_4$ | $12.5 \times 10^{24}$ | 20.8 | 2820 |

$$\frac{35.00 \text{ g } NaHCO_3}{} \times \frac{1 \text{ mol } NaHCO_3}{84.007 \text{ g } NaHCO_3} = 0.4166 \text{ mol } NaHCO_3$$

$$\frac{0.4166 \text{ mol } NaHCO_3}{} \times \frac{6.02214076 \times 10^{23} \text{ molec } NaHCO_3}{1 \text{ mol } NaHCO_3} = 2.509 \times 10^{23} \text{ molec } NaHCO_3$$

$$\frac{1.25 \text{ mol HCl}}{} \times \frac{36.46 \text{ g HCl}}{1 \text{ mol HCl}} = 45.58 \text{ g HCl}$$

$$\frac{1.25 \text{ mol HCl}}{} \times \frac{6.02214076 \times 10^{23} \text{ molec HCl}}{1 \text{ mol HCl}} = 7.53 \times 10^{23} \text{ molec HCl}$$

$$\frac{12.5 \times 10^{24} \text{ molec } KH_2PO_4}{} \times \frac{1 \text{ mol } KH_2PO_4}{6.02214076 \times 10^{23} \text{ molec } KH_2PO_4} = 20.8 \text{ mol } KH_2PO_4$$

$$\frac{20.8 \text{ mol } KH_2PO_4}{} \times \frac{136.086 \text{ g } KH_2PO_4}{1 \text{ mol } KH_2PO_4} = 2820 \text{ g } KH_2PO_4$$

## Answer to Exercise 20 (on page 72)

From the last exercise, you know that 1 mole of water weighs 18.01528 grams, meaning 200 grams of water is about 11.1 moles. So you need to burn 11.1 moles of methane.

What does one mole of methane weigh? Using the periodic table: $12.0107 + 4 \times 1.00794 = 16.04246$ grams.

$16.0424 \times 11.10 = 178.1$ grams of methane.

## Answer to Exercise 21 (on page 76)

The average hydrogen atom has a mass of 1.00794 atomic mass units.

The average oxygen atom has a mass of 15.9994.

$2 \times 1.00794 + 15.9994 = 18.01528$ atomic mass units.

## Answer to Exercise 22 (on page 80)

| Element Name | Atomic Symbol | Protons | Most Common Number of Neutrons | Electrons |
|---|---|---|---|---|
| Francium | Fr | 87 | 136 | 87 |
| Arsenic | As | 33 | 42 | 33 |
| Erbium | Er | 68 | 99 | 68 |
| Cadmium | Cd | 48 | 64 | 48 |

## Answer to Exercise 23 (on page 84)

Equilibrium will be achieved when the box has displaced 10 kg of water. In other words, when it has displaced 0.01 cubic meters.

The area of the base of the box is 0.12 square meters. So if the box sinks $x$ meters into the water it will displace $0.12x$ cubic meters.

Thus at equilibrium $x = \frac{0.01}{0.12} \approx 0.083$ m. So the box will sink 8.3 cm into the water before reaching equilibrium.

## Answer to Exercise 24 (on page 86)

$$p = dgh = (900)(3.721)(5) = 16,744.5 \text{ Pa}$$

## Answer to Exercise 25 (on page 89)

$$E_C = (1200)(0.4)(T - 10) = 480T - 4800$$

Total energy stays constant:

$$0 = (12600T - 252000) + (900T - 72000) + (480T - 4800)$$

Solving for T gets you $T = 23.52°$ C.

## Answer to Exercise 26 (on page 90)

During the 3 minutes, you want the coffee to give off as much of its heat as possible, so you want to maximize the difference between the temperature of the coffee and the temperature of the room around it.

You wait until the last moment to put the milk in.

## Answer to Exercise 27 (on page 92)

In the case, the force you exert must balance the weight of the book, so $F = mg = 2(9.8) = 19.6$N. Since this force is straight upward and the displacement is in the same direction, $W = F \cdot d = 19.6 \cdot 3 = 59.8$ J.

## Answer to Exercise 28 (on page 95)

(a) Since all kinetic energy is transferred into thermal energy, we can say:

$$Q = KE = \frac{1}{2}(1200)(22)^2 = 290,400 \text{ J}$$

(b) The rise in heat is given by $\Delta T = \dfrac{Q}{mc} = \frac{290,400}{15 \times 460} \approx 42°$ C

## Answer to Exercise 29 (on page 97)

At the top of the ladder, the cannonball has $(9.8)(5)(3) = 147$ J of potential energy.

At the bottom, the kinetic energy $\frac{1}{2}(5)v^2$ must be equal to 147 joules. So $v^2 = \frac{294}{5}$. This means it is going about 7.7 m/s.

(You may be wondering about air resistance. Yes, a tiny amount of energy is lost to air resistance, but for a dense object moving at these relatively slow speeds, this energy is negligible.)

## Answer to Exercise 30 (on page 97)

The initial total gravitational potential energy is $mgh = 4(10)(2.5) = 100$ J. The final kinetic energy is $\frac{1}{2}mv^2 = \frac{1}{2}(4)(4.2)^2 = 35.28$. So, the rest of the energy must be lost due to friction: $100 - 35.28 = 64.72$ J.

## Answer to Exercise 31 (on page 98)

The total useful energy can be calculated as $0.65(1.2 \times 10^5) = 7.8 \times 10^4$ J. Then, we can set the final KE $\frac{1}{2}mv^2 = 7.8 \times 10^4$ and solve for $v$

$$\begin{aligned} \frac{1}{2}(150)v^2 &= 7.8 \times 10^4 \\ (150)v^2 &= 15.6 \times 10^4 \\ v^2 &= 1,040 \\ v &\approx 32.249 \text{ m/s} \end{aligned}$$

## Answer to Exercise 32 (on page 99)

The force acting on the ball is its weight, $F_w = mg$, and we will designate this as negative since weight points downwards. Using the work-energy theorem,

$$\Delta KE = F \cdot d = (mg) \cdot d = (0.5kg)\left(-9.8\frac{m}{s^2}\right)(1m) = -4.9J$$

Therefore, the ball loses 4.9 joules of kinetic energy for every 1 meter it moves upwards (the fact it is *losing kinetic energy* is represented by the result being negative). Note that

this is also the formula for potential energy, so it gained 4.9 J of potential energy.

## Answer to Exercise 33 (on page 99)

$$F_f \cdot d = \Delta KE = KE_f - KE_i$$

You'll reach the maximum distance you can slide when you stop moving, so we will use a final velocity of zero, which means a final kinetic energy of zero:

$$F_f \cdot d = -KE_i = \frac{1}{2}mv^2$$

Since the force of friction is 0.3 times your weight, we know that:

$$F_f = 0.3F_w = 0.3mg$$

Substituting and canceling the mass:

$$(0.3mg) \cdot d = \frac{1}{2}mv^2$$

$$0.3g \cdot d = \frac{1}{2}v^2$$

Since we know g and v, we can solve for d:

$$d = \frac{v^2}{0.6g} = \frac{\left(10\frac{m}{s}\right)^2}{0.6\left(9.8\frac{m}{s^2}\right)} \approx 1.7m$$

So, if you want to reach home base, you should start your slide no more than 1.7 m from home.

## Answer to Exercise 34 (on page 100)

Since all the cars need to stop in the same distance, the cars with the greatest kinetic energy will take the most force to stop. Calculating the kinetic energies (we won't change the units from kilometers per hour to meters per second, since we're just comparing the values):

| Car | Mass [kg] | Max speed [kph] | KE [kg (kph)$^2$] |
|---|---|---|---|
| A | 2000 | 100 | $1 \times 10^7$ |
| B | 2400 | 120 | $1.728 \times 10^7$ |
| C | 3200 | 240 | $9.216 \times 10^7$ |
| D | 2400 | 140 | $2.352 \times 10^7$ |
| E | 3200 | 120 | $2.304 \times 10^7$ |
| F | 2000 | 240 | $5.67 \times 10^7$ |

The correct ranking is A, B, E, D, F, C.

## Answer to Exercise 35 (on page 103)

(a) $500 \times 0.82 = 410$ bolts

(b) $\frac{\$24}{410} \approx \$0.06$

## Answer to Exercise 190.2 (on page 1779)

(a)

$$W = F \cdot d = 450 \cdot 1.8 = 810 \text{ J}$$

(b)

$$P = \frac{W}{t} = \frac{810}{.75} = 1080 \text{ W}$$

Alternatively,

$$v = \frac{d}{t} = \frac{1.8}{0.75} = 2.4 \text{ m/s}$$
$$P = F \cdot v = 450 \cdot 2.4 = 1080 \text{ W}$$

## Answer to Exercise 37 (on page 104)

$$P = \frac{E}{t}$$
$$E = P \cdot t = 120(0.4) = 48 \text{ J} \qquad = 48\text{J}$$

So 48 J of energy is stored into the spring. We can find the spring's PE converted to

distance by:

$$\begin{aligned} E &= \frac{1}{2}kx^2 \\ 48 &= \frac{1}{2}(800)x^2 \\ x^2 &= \frac{48}{400} \\ x &= \sqrt{0.12} \approx 0.346 \text{ m} \end{aligned}$$

So the spring had to have been compressed 34.6 cm.

## Answer to Exercise 38 (on page 105)

(a) Electrical energy input can be calculated as $E = Pt = 18 \text{ kW} \times 1000 \text{ J/s} \times 12 \text{ sec} = 216,000 \text{ J}$.

(b) The elevator travels 18 m upwards, so the total gravitational potential energy is $mgh = 750(9.8)(18) = 132,300 \text{ J}$.

(c) The efficiency can be calculated as $\frac{\text{Useful Output}}{\text{Total Input}} = \frac{132,000}{216,000} \approx 61.1\%$

## Answer to Exercise 39 (on page 113)

The weight of the barrel is $136 \text{ kg} \times 9.8 \frac{\text{m}}{\text{s}^2} = 1332.8 \text{ N}$.

Let L be the length of the inclined plane. The force needed to push the barrel up is related by:

$$300 \text{ N} = \frac{2 \text{ m}}{L} \times 1332.8 \text{ N}$$

Solving for L, we find $L = \frac{2 \text{ m} \times 1332.8}{300} \approx 8.885 \text{ m}$.

## Answer to Exercise 40 (on page 114)

Paul exerts a force of $70 \text{ kg} \times 9.8 \frac{\text{m}}{\text{s}^2} = 686 \text{ N}$ at a distance of 4 meters from the fulcrum, creating a torque of $686 \text{ N} \times 4 \text{ m} = 2744 \text{ N} \cdot \text{m}$. Jan exerts a force of $50 \text{ kg} \times 9.8 \frac{\text{m}}{\text{s}^2} = 490 \text{ N}$.

Let r be the distance from the fulcrum to Jan's seat. To balance the torques:

$$490\ \text{N} \times r = 2744\ \text{N} \cdot \text{m}$$

Solving for $r$, we find $r = \frac{2744}{490} \approx 5.6$ meters.

## Answer to Exercise 41 (on page 118)

The equation relating these quantities is:

$$583 = 70 \times 2.2 \times \frac{53}{n}$$

Solving for $n$, we find $n = 14$ teeth.

## Answer to Exercise 42 (on page 120)

We are solving for the radius $r$ of the piston. The area of the piston is $\pi r^2$, so the pressure is:

$$\text{Pressure} = \frac{12}{\pi r^2}$$

Setting the pressure equal to 2,500,000 pascals:

$$2,500,000 = \frac{12}{\pi r^2}$$

Solving for $r$, we find:

$$r = \sqrt{\frac{12}{\pi \times 2.5 \times 10^6}} \approx 0.00124 \text{ meters.}$$

## Answer to Exercise 43 (on page 123)

There are 4 segments supporting the net weight total of 420 N. Each segment, then, splits the weight into $\frac{420\ \text{N}}{4} = 105$ N. The input force required to move both weights is 105 N,

which moves it 4 m. The output work matches as well: 420 N is moved 1 m.

## Answer to Exercise 44 (on page 124)

The mechanical advantage of the wedge is $\frac{12\text{ cm}}{3\text{ cm}} = 4$.

## Answer to Exercise 45 (on page 125)

A knife cuts better than a screwdriver tip because its edge is much thinner, giving it a greater mechanical advantage. This allows the applied force to be concentrated over a very small area, producing much higher pressure. As a result, less force is needed to cut the material. A screwdriver, on the other hand, has a very thick edge (in comparison), so its pressure and mechanical advantage are much less.

## Answer to Exercise 46 (on page 128)

$$\mu = \frac{1}{6}(87 + 91 + 98 + 65 + 87 + 100) = 88$$

## Answer to Exercise 47 (on page 130)

The mean of your grades is 88.

The variance, then, is

$$\sigma^2 = \frac{1}{6}\left((87-88)^2 + (91-88)^2 + (98-88)^2 + (61-88)^2 + (87-88)^2 + (100-88)^2\right) = \frac{784}{6} = 65\frac{1}{3}$$

The standard deviation is the square root of that: $\sigma = 8.083$ points.

## Answer to Exercise 48 (on page 131)

In order the grades are 65, 87, 87, 91, 98, 100. The middle two are 87 and 91. The mean of those is 89. (Speed trick: The mean of two numbers is the number that is halfway

between.)

## Answer to Exercise 49 (on page 144)

The formula for the RMS is "=SQRT(SUMSQ(A2:A1001)/COUNT(A2:A1001))".

## Answer to Exercise 50 (on page 156)

| Variable | Value | Data Type |
|---|---|---|
| a | 10 | `int` |
| b | 3.5 | `float` |
| c | "10" | `str` |
| d | True | `bool` |
| e | [1, 2, 3] | `list` |

## Answer to Exercise 51 (on page 157)

```
1 False
```

## Answer to Exercise 52 (on page 159)

| Line Executed | n (value, type) | m (value, type) |
|---|---|---|
| After line 2 | (5, int) | ("3", str) |
| After line 3 | (6, int) | ("3", str) |
| After line 4 | (6, int) | ("31", str) |

## Answer to Exercise 53 (on page 161)

The code will raise a `TypeError` because `x` is a string and cannot be added to the integer `5`. The error message will be similar to:

```
1 TypeError: can only concatenate str (not "int") to str
```

To fix this, we need to cast `x` to an integer using `int()`:

```
x = int(input("Enter a number: "))
print(x + 5)
```

## Answer to Exercise 54 (on page 166)

| Input | Output |
|---|---|
| 5 | Small |
| 10 | Ten |
| 15 | Large |

## Answer to Exercise 55 (on page 166)

```
for i in range(0, 21, 2):
    print(i)
```

## Answer to Exercise 56 (on page 171)

The start index is omitted and the end index is omitted, which means they are 0 and `len(seq)` $= 4$ respectively. Since the bounds are 0 and $4 - 1 = 3$, every index is included, in the given order. Therefore, `seq[:]` returns a copy of the given string, `[a, f, j, k]`

## Answer to Exercise 57 (on page 171)

```
c
m
r
t
put
```

## Answer to Exercise 58 (on page 173)

| Line Execution | `items` content |
|---|---|
| After Line 1 | `items = ["pen", "book", "eraser"]` |
| After Line 3 | `items = ["pen", "book", "eraser", "ruler"]` |
| After Line 4 | `items = ["pen", "book", "eraser"]` |
| After Line 5 | `items = ["pen", "eraser"]` |
| After Line 6 | `items = ["pen", "eraser", "marker"]` |

## Answer to Exercise 59 (on page 183)

| A | B | not (A or B) | ( not A) and ( not B) |
|---|---|---|---|
| F | F | T | T |
| F | T | F | F |
| T | F | F | F |
| T | T | F | F |

Notice that the two expressions are equivalent!

DeMorgan's Rule says "not (A or B)" is equivalent to "(not A) and (not B)".

It also says "not (A and B)" is equivalent to "(not A) or (not B)".

## Answer to Exercise 60 (on page 191)

*Proof.*

**Case 3.** *Suppose* $x \geq 0$. *By the definition of absolute value,* $|x|$ *represents the distance of* $x$ *from* $0$ *on the real line. Since* $x$ *is nonnegative, this distance is simply* $x$. *Hence,* $|x| = x$. $0$

**Case 4.** *In this case,* $x$ *is negative, so its distance from* $0$ *is given by its additive inverse. Thus,* $|x| = -x$.

□

Notice how explicit we are being with our proof, using words and mathematical syntax.

## Answer to Exercise 61 (on page 207)

$V = IR$ so $I = \frac{V}{R} = \frac{24V}{6\Omega} = 4A$.

## Answer to Exercise ?? (on page 209)

There is a total resistance of $8\Omega$, so your 16V will push 2A of current around the circuit.

2A going through a $5\Omega$ resistor represents a 10V drop.

2A going through a $3\Omega$ resistor represents a 6V drop.

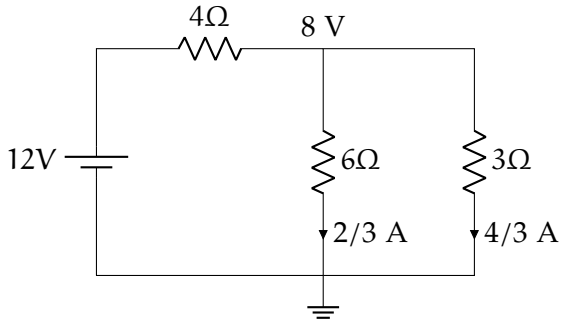
## Answer to Exercise 63 (on page 211)

The effective resistance of the $6\Omega$ and the $3\Omega$ is $2\Omega$ because

$$\frac{1}{R_T} = \frac{1}{6} + \frac{1}{3} == \frac{1}{2}$$

This means the battery experiences a resistance of $4\Omega + 2\Omega = 6\Omega$. A 12V will push 2A through a resistance of $6\Omega$.

The voltage drop across the $4\Omega$ resistor is $2A \times 4\Omega = 8V$. Thus there will be a 4V drop across the two resistors in parallel. So 2/3 A will flow through the $6\Omega$ resistor. 4/3 A will flow through the $3\Omega$ resistor.



## Answer to Exercise 64 (on page 213)

Let's set up the equation. Here are the entering conductors:

$$i_1, i_3$$

And leaving: $i_2, i_4, i_x$ we are assuming $i_x$ is leaving for the calculation.

$$i_1 + i_3 = i_2 + i_4 + i_x$$

Inputting the values gives us:

$$5 + 2 = 3 + 9 + i_x$$

$$7 = 12 + i_x$$

$$i_x = -5\,\text{A}$$

The magnitude of the current is 5 A. Because the result is negative, the actual direction is *entering* the node, based on our leaving assumption.

## Answer to Exercise 65 (on page 222)

On earth, holding a 100 kg man aloft requires 980 Newtons of force.

$980/700 = 1.4$, so you need a cable with a cross-section area of 1.4 square millimeters.

$$\pi r^2 = 1.4$$

$r = \sqrt{1.4/\pi} \approx .67$ millimeters. This means the cable would have to have a diameter of at least 1.34 millimeters.

## Answer to Exercise 66 (on page 229)

The function returns `3 * 2`, which is `6`, so the program prints:

```
1 6
```

The value of `a` is still `3` because the function does not change `a`; it only uses its value to compute a result.

## Answer to Exercise 67 (on page 229)

```
def is_even(n):
    return n % 2 == 0

print(is_even(7))
```

This prints `False` because 7 is not divisible by 2.

## Answer to Exercise 68 (on page 234)

The initial value is 5. The `increment()` method is called twice, so the value becomes 7. The program prints:

```
7
```

## Answer to Exercise 69 (on page 234)

```
class Book:
    def __init__(self, title, author):
        self.title = title
        self.author = author

    def __str__(self):
        return f"{self.title} by {self.author}"
```

## Answer to Exercise 70 (on page 243)

1. **TypeError**: The `input()` function returns a string, so when the user inputs an index, it is treated as a string. Attempting to use this string as an index for the list will raise a `TypeError`. To fix this, we can cast the input to an integer using `int()` and handle the potential `ValueError` if the input is not a valid integer.

2. Pencil is not in the list, so attempting to remove it will raise a `ValueError`. To fix this, we can use a try-except block to catch the `ValueError` and print a message indicating that the item was not found in the list.

## Answer to Exercise 71 (on page 247)

```
def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n - 1)
```

## Answer to Exercise 72 (on page 252)

1. $2\pi$
2. $60°$
3. $\frac{5\pi}{4}$
4. $135°$
5. $\frac{\pi}{6}$
6. $\frac{\pi}{4}$

## Answer to Exercise 73 (on page 256)

$180° - (92° + 42°) = 46°$

## Answer to Exercise 74 (on page 257)

$360°$

## Answer to Exercise 75 (on page 258)

$(7, 7, 10)$: two sides match, so this is isosceles. $(9, 40, 41)$: $9 + 40 = 49 > 41$ and no two sides match, so this is scalene. $(3, 4, 9)$: $3 + 4 = 7$, which is less than 9, so this is not a triangle.

## Answer to Exercise 76 (on page 260)

a: 10 because $6^2 + 8^2 = 10^2$

b: 12 because $5^2 + 12^2 = 13^2$

c: 8 because $8^2 + 15^2 = 17^2$

d: $3\sqrt{2} \approx 4.24$ because $3^2 + 3^2 = \left(3\sqrt{2}\right)^2$

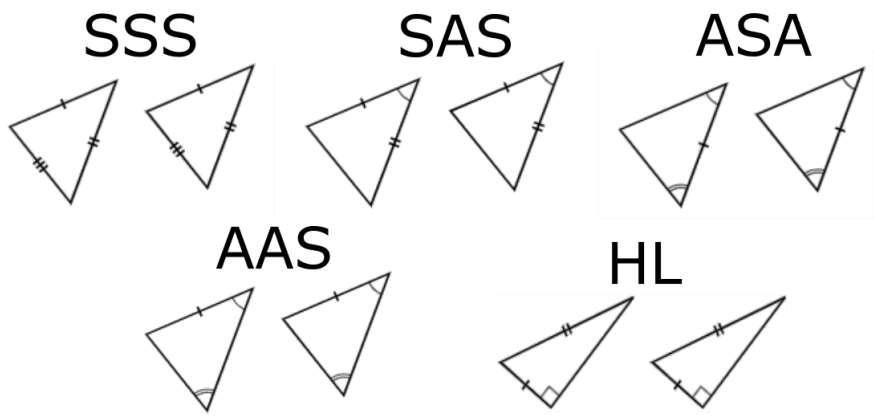
## Answer to Exercise 77 (on page 267)

Congruent by the Side-Side-Right Congruency Test.

Congruent by the Side-Side-Side Congruency Test.

Congruent by the Side-Angle-Angle Congruency Test.

We don't know if they are congruent. The measured angle is not between the measured sides.




## Answer to Exercise 78 (on page 272)

The table has a radius of 3 meters.

So the area of its top is $3^2\pi \approx 28.27$.

$$28.27 \text{ square meters} \left( \frac{1 \text{ liter}}{6 \text{ square meters}} \right) = 4.72 \text{ liters}$$

## Answer to Exercise 79 (on page 273)

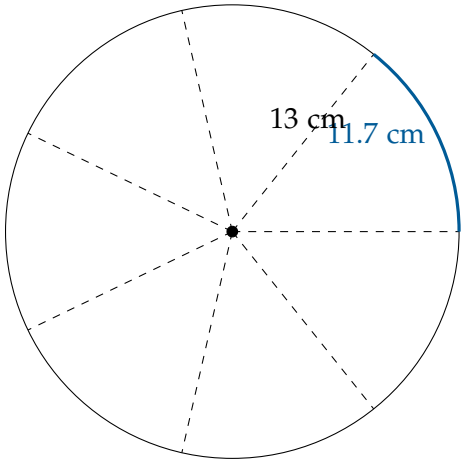
The diameter is

$$\frac{c}{\pi} = \frac{64}{\pi} \approx 20.37 \text{ centimeters}$$

## Answer to Exercise 80 (on page 273)

The circumference of the pie is $26\pi \approx 81.7$ centimeters.

The length of the crust for each piece would be about $\frac{81.7}{7} = 11.7$ cm.




## Answer to Exercise 81 (on page 275)

If you roll forward by 10 cm, that means you move along the edge of your tire such that the arc length is 10 cm. So, we are looking for a central angle such that $r\theta = 10$ cm. Substituting $r = 25$ cm and solving for $\theta$: $\theta = \frac{10 \text{ cm}}{25 \text{ cm}} = 0.4$ radians.

## Answer to Exercise 82 (on page 276)

1. $\frac{\pi}{4} \cdot 2 \text{ cm} = \frac{\pi}{2} \text{ cm}$
2. $\pi \cdot 1 \text{ cm} = \pi \text{ cm}$
3. $\frac{\pi}{10} \cdot 5 \text{ cm} = \frac{\pi}{2} \text{ cm}$
4. $\frac{3\pi}{4} \cdot 3 \text{ cm} = \frac{9\pi}{4} \text{ cm}$

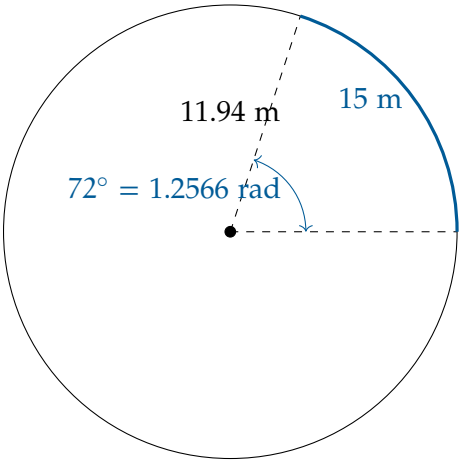
Therefore, from longest to shortest are 4, (1,3), 2 (1 and 3 are the same length).

## Answer to Exercise 83 (on page 277)

$$72 \text{ degrees} \left( \frac{2\pi \text{ radians}}{360 \text{ degrees}} \right) \approx 1.2566 \text{ radians}$$

$$15 = 1.2566r$$

$$r = 11.94 \text{ meters}$$

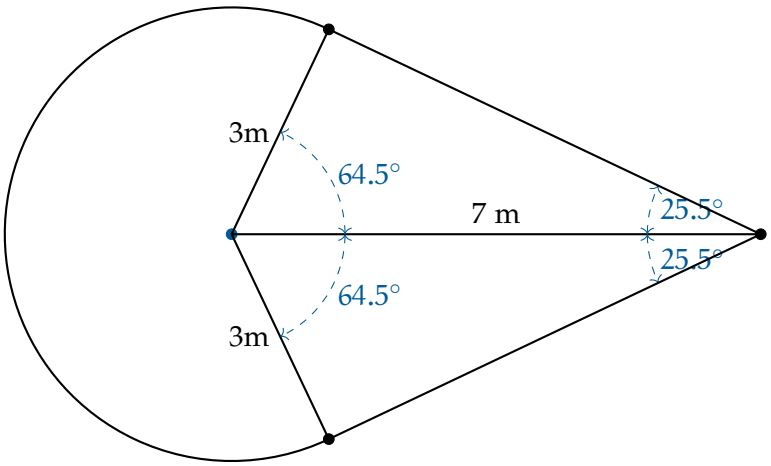



## Answer to Exercise 84 (on page 278)

If the logo is 5 meters wide, the diameter is 5 meters and the radius is 2.5 meters. Using the formula for the area of a sector: $A_{\text{sector}} = \frac{1}{2}\frac{3\pi}{4}(21.5 \text{ m})^2 \approx 7.363 \text{ m}^2$. Since a liter covers 6 $\text{m}^2$, you will need $\frac{7.363}{6} \approx 1.227$ L of paint.

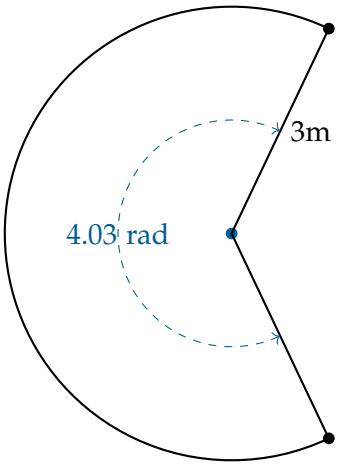
## Answer to Exercise 85 (on page 281)

The trick here is to take advantage of the fact that the tangent is perpendicular to the radius to make right triangles:



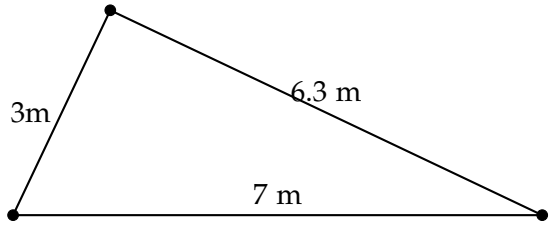

The sector has radius 3 and represents $360 - 2(64.5) = 231° \approx 4.03$ radians.

We are finding the area of this piece:




The area of this piece is $(4.03)(3^2) = 36.27$ square meters.

If a right triangle has a hypotenuse of 7m and one leg is 3m, the other leg is $\sqrt{7^2 - 3^2} = 2\sqrt{10} \approx 6.3$ m.




A right triangle with legs of 3m and 6.3m has an area of 9.45 square meters.

There are two of them, so the total area is $36.27 + 2(18.9) = 74.07$ square meters.

Six square meters per liter, so you need $\frac{74.07}{6} = 12.35$ liters of paint.

## Answer to Exercise 86 (on page 284)

You can only take the square root of nonnegative numbers, so the function is only defined when $x - 3 \geq 0$. Thus, the domain is all real numbers greater than or equal to 3.

## Answer to Exercise 87 (on page 286)

The graph of this function is a line, its slope is $-3$, and it intersects the y axis at $(0, 3)$.

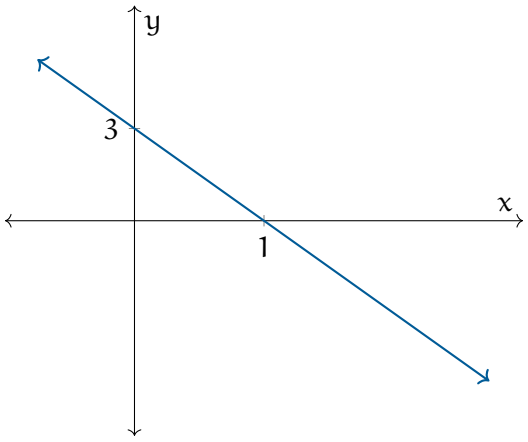



Figure 33.3: The resulting graph of $f(x) - 3x + 3$.

## Answer to Exercise 88 (on page 289)

This graph is the graph of $y = x^2$ that has been moved to the right by three units and up

two units:

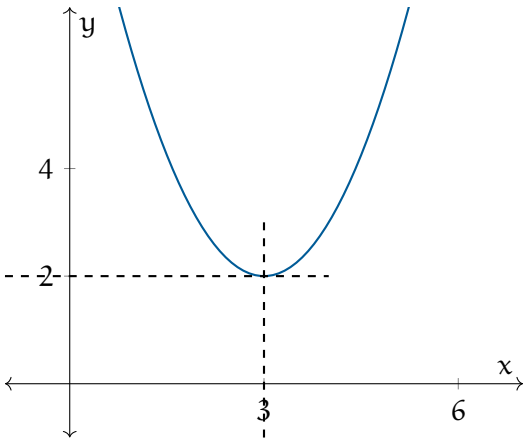



Figure 33.9: The graph of $f(x) = (x-3)^2 + 2$.

To prevent any horizontal line from containing more than one point of the graph, you would need to use the left or the right side — either $\{x \in \mathbb{R} | x \leq 3\}$ or $\{x\mathbb{R} | x \geq 3\}$. Most people will choose the right side; the rest of the solution will assume that you did too.

To find the inverse we swap $x$ and $y$: $x = (y-3)^2 + 2$

Next, we solve for $y$ to get the inverse: $y = \sqrt{x-2} + 3$

You can take the square root of nonnegative numbers. So the function $f^{-1}(x) = \sqrt{x-2}+3$ is defined whenever $x$ is greater than or equal to 2.

## Answer to Exercise 89 (on page 290)

1. This function is not one-to-one. From $x = 3$ to $x = 4$, the function increases from 3.6 to 5.3, which means it must pass through $f(x_1) = 4.0$. From $x = 4$ to $x = 5$, the function decreases from 5.3 to 2.8, which means it must pass through $f(x_2) = 4.0$ again.
2. This function is not one-to-one by a similar argument in the above solution
3. This function is not one-to-one, because it fails the horizontal line test
4. This function is one-to-one, because it passes the horizontal line test
5. $f(t)$ would not be one-to-one because the football must pass through each height (except the peak height) both on the way up and on the way back down
6. $v(t)$ would be one-to-one because a falling object only speeds up. Therefore, every time has a unique speed.

## Answer to Exercise 90 (on page 292)

It prints `-6 0 0`. The first value says the graph crosses the $y$-axis at $(0, -6)$. The two zeros say $x = 3$ and $x = -2$ are the $x$-intercepts, the points where the parabola crosses the $x$-axis. Both match the Desmos graph.

## Answer to Exercise 91 (on page 293)

This array skips 0. The two entries closest to it are $x \approx -0.0606$ and $x \approx 0.0606$, with $y \approx -16.5$ and $y \approx 16.5$. Matplotlib connects neighbors, so it draws a nearly vertical segment from $(-0.0606, -16.5)$ up to $(0.0606, 16.5)$. That segment belongs to neither branch of the graph. Plotting each side of the asymptote separately removes it. (With 101 points instead of 100, the array contains 0 exactly; NumPy then warns `divide by zero` and stores `inf` for that entry.)

## Answer to Exercise 92 (on page 294)

NumPy refuses to raise a negative decimal number to a fractional power, so every negative entry of `x**(1/3)` becomes `nan` ("not a number"). For example, it returns `nan` for $(-8)^{1/3}$ even though the real cube root is $-2$. Matplotlib skips `nan` points, which erases the left half. The function `np.cbrt` accepts negative inputs and returns $-2$ for $-8$. The TikZ version of Figure 33.6 solves the same problem by graphing the left half as $-(-x)^{1/3}$.

## Answer to Exercise 93 (on page 296)

It prints `2 0 0`. An even function needs $h(-1) = h(1)$, but $0 \neq 2$. Oddness requires $-h(-1) = h(1)$, which fails for the same reason. A single input that breaks a definition is enough to rule it out.

## Answer to Exercise 94 (on page 297)

For `np.abs(x)`, the dashed $h(-x)$ curve lies exactly on the solid curve, and the tests print `True` then `False`: $|x|$ is even. With `x**3`, the dotted $-h(-x)$ curve lies on the solid curve, and the tests print `False` then `True`: $x^3$ is odd.

## Answer to Exercise 95 (on page 299)

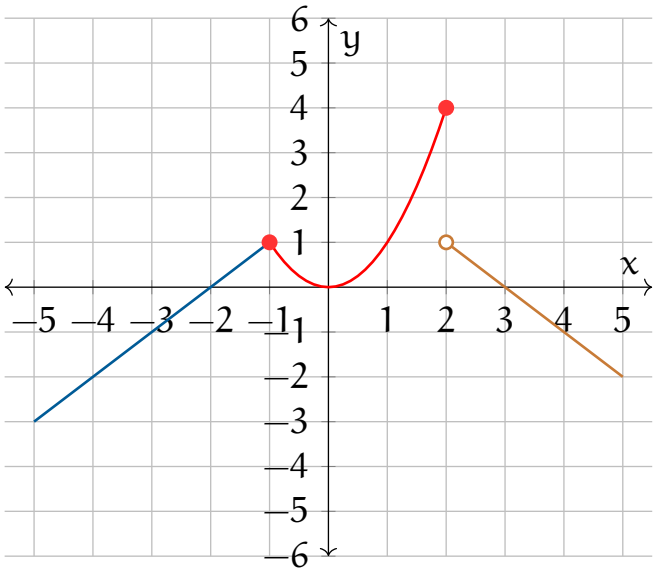
From $x \in [0, \infty)$, we say that $|x| = x$. For $x < 0$, we say $|x| = -x$, since we want to invert the magnitude. So,

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

## Answer to Exercise 96 (on page 300)

First, let's create a table of values for each integer from -5 to 5:

| $x$ | $-5$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $f(x)$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $0$ | $1$ | $4$ | $0$ | $-1$ | $-2$ |




## Answer to Exercise 97 (on page 302)

1.

$$C(x) = \begin{cases} 400x & 0 < x \leq 5 \\ 2000 + 350(x-5) & 5 < x \leq 15 \\ 5500 + 300(x-15) & x > 15 \end{cases}$$

2. $C(12) = 2000 + 350(12 - 5) = \$4450$
3. The average cost is \$360 when $\frac{C(x)}{x} < 360$. This will only happen during the 3rd case, when $x \geq 15$.

$$C(x) = 5500 + 300(x - 15) = 5500 + 300x - 4500 = 1000 + 300x$$

Now,

$$\frac{1000 + 300x}{x} < 360$$

$$300 + \frac{1000}{x} < 360$$
$$\frac{1000}{x} < 60$$
$$1000 < 60x$$
$$x > \frac{1000}{60} \approx 16.67 \to 17$$

There must be 17 TVs sold for the average cost to fall to \$360

## Answer to Exercise 98 (on page 304)

The outputs for $x = -5, -4, \ldots, 5$ are $-3, -2, -1, 0, 1, 0, 1, 4, 0, -1, -2$, matching the table. At $x = -1$ the first condition fails ($-1 < -1$ is false), so Python uses $x^2$ and returns 1. At $x = 2$ the second condition includes 2, so Python returns $2^2 = 4$. Both are the closed circles on the graph.

## Answer to Exercise 99 (on page 304)

The input 4 satisfies `x >= 0`, so Python returns $4^2 = 16$ and never reaches the `elif`. Python does not flag overlapping cases; it takes the first match. Putting the special case first fixes it:

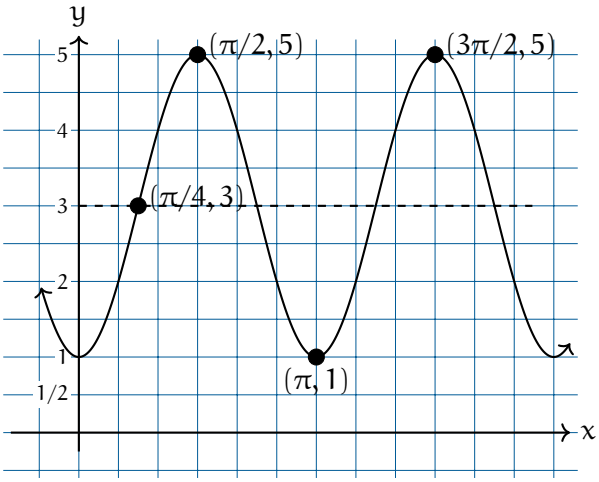
```
def q(x):
    if x == 4:
        return 10.2
    elif x >= 0:
        return x**2
```

Now `q(4)` returns `10.2` and `q(3)` returns `9`. Ordering the checks this way does the same job as the condition $x \geq 0,\ x \neq 4$ in the revised definition.

## Answer to Exercise 100 (on page 305)

It prints `-3 -2 9 7`. Flooring $-2.5$ moves down to $-3$, the greatest integer less than or equal to $-2.5$; rounding toward zero would give $-2$, which is wrong here. Its ceiling is $-2$. An input of 8.00001 rises to 9, and an integer such as 7 comes back unchanged.

## Answer to Exercise 101 (on page 313)




This wave has an amplitude of 2; its baseline has been translated up to 3.

This wave has wavelength of $\pi$. A sine wave usually has a wavelength of $2\pi$, so we need to compress the $x$ axis by a factor of 2.

The wave first crosses its baseline at `pi/4`. The sine wave starts by crossing its baseline, so we need to translate the curve right by $\pi/4$.

$$f(x) = 2\sin(2x - \frac{\pi}{4}) + 3$$

## Answer to Exercise 102 (on page 319)

We start by writing out the limit:

$$\frac{d}{dx}\cos x = \lim_{h\to 0} \frac{\cos x + h - \cos x}{h}$$

Applying the sum formula for $\cos(x+h)$, we get:

$$= \lim_{h\to 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h}$$

Rearranging to group the $\cos x$ and applying the Difference Rule:

$$= \lim_{h\to 0} \frac{\cos x \cos h - \cos x}{h} - \lim_{h\to 0} \frac{\sin x \sin h}{h}$$

Applying the Constant Multiple Rule:

$$= \cos x \lim_{h\to 0} \frac{\cos h - 1}{h} - \sin x \lim_{h\to 0} \frac{\sin h}{h}$$

Recalling that $\lim_{h\to 0} \frac{\sin h}{h} = 1$,

$$= \cos x \lim_{h\to a} \frac{\cos h - 1}{h} - \sin x \cdot 1$$

Recalling that $\lim_{h\to 0} \frac{\cos h - 1}{h} = 0$:

$$= \cos x \cdot 0 - \sin x = -\sin x$$

Therefore, $\frac{d}{dx} \cos x = -\sin x$

## Answer to Exercise 103 (on page 319)

1. $\frac{\sec x(\tan x - 1)}{(1+\tan x)^2}$
2. $\sec t[\sec^2 t + \tan^2 t]$
3. $\frac{4 - \tan\theta + \theta \sec^2\theta}{(4-\tan\theta)^2}$
4. $2\sec t \tan t + \csc t \cot t$
5. $\frac{2}{(1+\cos\theta)^2}$
6. $\cos^2 x - \sin^2 x$

## Answer to Exercise 104 (on page 323)

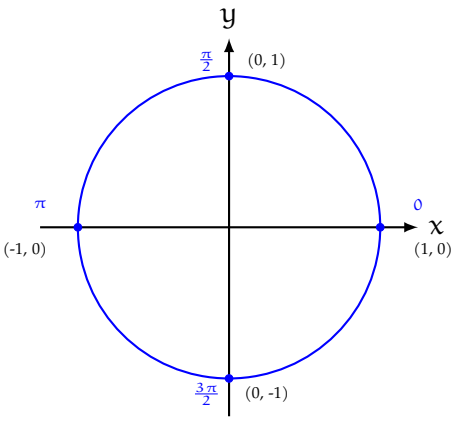
1. $\sec x + C$

## Answer to Exercise 105 (on page 326)

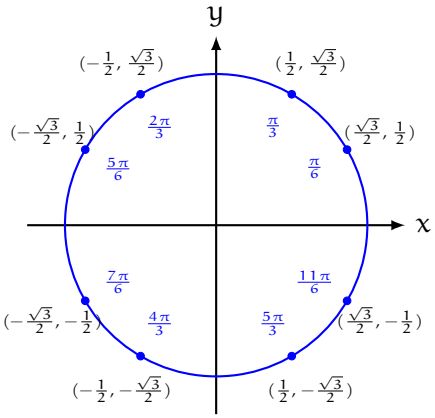
We know that for a right triangle, $\cos\theta = \frac{\text{adjacent leg}}{\text{hypotenuse}}$. For a right triangle inscribed in the Unit Circle, the adjacent leg is parallel to the x-axis and has the same length as the x-value of the coordinate point on the circle. Additionally, the length of the hypotenuse is 1. Therefore, $\cos\theta = \frac{x_o}{1} = x_o$.

## Answer to Exercise 106 (on page 327)

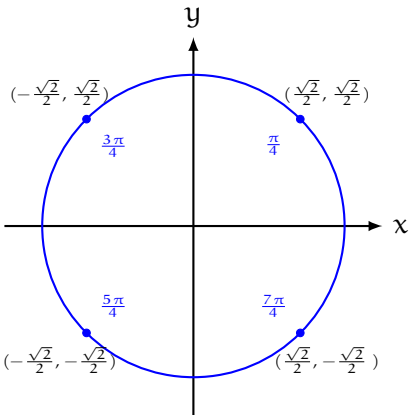
1. $\sin\frac{\pi}{2} = 1$
2. $\cos\frac{3\pi}{2} = 0$
3. $\sin\pi = 0$
4. $\cos -\pi = -1$ (Negative angles are measured clockwise from the x-axis, so $\theta = -\pi$ is at the same angle as $\theta = \pi$.)



## Answer to Exercise 107 (on page 329)




## Answer to Exercise 108 (on page 330)




## Answer to Exercise 109 (on page 331)

1. 0
2. $\sqrt{2}/2$
3. $-1/2$
4. $-1/2$
5. $\sqrt{2}/2$
6. $1/2$

7. $\sqrt{2}/2$
8. $-1$
9. $-\sqrt{3}/2$
10. $1/2$

## Answer to Exercise 110 (on page 333)

1. $\sin(\pi/12) = \sin(\pi/3 - \pi/4) = \sin\frac{\pi}{3}\cos\frac{\pi}{4} - \cos\frac{\pi}{3}\sin\frac{\pi}{4} = \left(\frac{\sqrt{3}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{1}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{\sqrt{6}-\sqrt{2}}{4}$
2. $\cos(7\pi/12) = \cos(4\pi/12 + 3\pi/12) = \cos(\pi/3 + \pi/4) = \cos\frac{\pi}{3}\cos\frac{\pi}{4} - \sin\frac{\pi}{3}\sin\frac{\pi}{4} = \left(\frac{1}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\cdot\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} = \frac{\sqrt{2}-\sqrt{6}}{4}$
3. $\tan(13\pi/12) = \frac{\sin(13\pi/12)}{\cos(13\pi/12)}$ First, we will find $\sin(13\pi/12)$: $\sin(13\pi/12) = \sin(3\pi/12 + 10\pi/12) = \sin(\pi/4 + 5\pi/6) = \sin\frac{\pi}{4}\cos\frac{5\pi}{6} + \cos\frac{\pi}{4}\sin\frac{5\pi}{6} = \left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{-\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{1}{2}\right) = \frac{-\sqrt{6}}{4} = \frac{\sqrt{2}}{4} = \frac{\sqrt{2}-\sqrt{6}}{4}$. Next we find $\cos(13\pi/12) = \cos(\pi/4 + 5\pi/6) = \cos\frac{\pi}{4}\cos\frac{5\pi}{6} - \sin\frac{\pi}{4}\sin\frac{5\pi}{6} = \left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{-\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\cdot\left(\frac{1}{2}\right) = \frac{-\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{-\sqrt{6}-\sqrt{2}}{4}$. And therefore $\tan(13\pi/12) = \frac{\sin 13\pi/12}{\cos 13\pi/12} = \frac{\sqrt{2}-\sqrt{6}}{4}\cdot\frac{4}{-\sqrt{6}-\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}+\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}+\sqrt{2}}\cdot\frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}-\sqrt{2}} = \frac{6-2\sqrt{12}+2}{6-2} = \frac{8-4\sqrt{3}}{4} = 2 - \sqrt{3}$

## Answer to Exercise 111 (on page 334)

$\sin 2\theta = \sin\theta + \theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$

## Answer to Exercise 112 (on page 335)

Similar to $\cos(\alpha/2)$, we begin with the double angle formula for cosine, but another version:

$$\cos 2\theta = 1 - 2\sin^2\theta$$

Substituting $\theta = \alpha/2$:

$$\cos\alpha = 1 - 2\sin^2(\alpha/2)$$

And rearranging to solve for $\sin(\alpha/2)$:

$$\sin(\alpha/2) = \pm\sqrt{\frac{1-\cos\alpha}{2}}$$

## Answer to Exercise 113 (on page 339)

The volume of the sphere (in cubic meters) is

$$\frac{4}{3}\pi(1.5)^3 = 4.5\pi \approx 14.14$$

The mass (in kg) is $14.14 \times 7800 \approx 110,269$

The kinetic energy (in joules) is

$$k = \frac{110269 \times 5^2}{2} = 1,378,373$$

About 1.4 million joules.

## Answer to Exercise 114 (on page 340)

In your mind, you can disassemble the tablet into a sphere (made up of the two ends) and a cylinder (between the two ends).

The volume of the sphere (in cubic millimeters) is

$$\frac{4}{3}\pi(2)^3 = \frac{32}{3}\pi \approx 33.5$$

Thus, the cylinder part has to be $90 - 33.5 = 56.5$ cubic mm. The cylinder part has a radius of 2 mm. If the length of the cylinder part is $x$, then

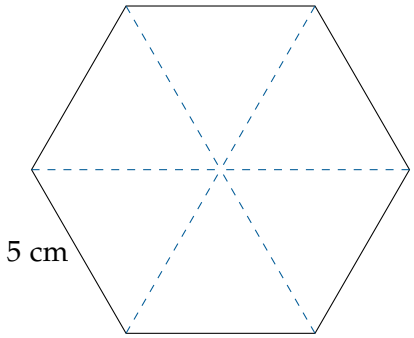
$$\pi 2^2 x = 56.5$$

Thus $x = \frac{56.5}{4\pi} \approx 4.5$ mm.

The cylinder part of the table needs to be 4.5mm. Thus the entire tablet is 8.5mm long.

## Answer to Exercise 115 (on page 343)

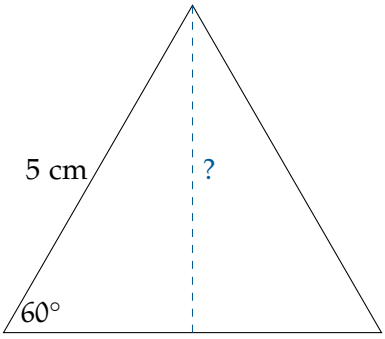
First, you need to find the area of the base, which is a regular hexagon:




All the angles in this picture are $60°$ or $\frac{\pi}{3}$ radians. This means each line is 5 cm long.

This tells us that we need to find the area of one of these triangles and multiply that by six.

Every triangle has a base of 5cm. How tall are they?




$$5 \sin 60° = 5\frac{\sqrt{3}}{2}$$

Which is about 4.33 cm.

So, the area of single triangle is

$$\frac{1}{2}(5)\left(5\frac{\sqrt{3}}{2}\right) = 25\frac{\sqrt{3}}{4}$$

And the area of the whole hexagon is six times that:

$$75\frac{\sqrt{3}}{2}$$

Thus, the volume of the pyramid is:

$$\frac{1}{3}hb = \frac{1}{3}13\left(75\frac{\sqrt{3}}{2}\right)$$

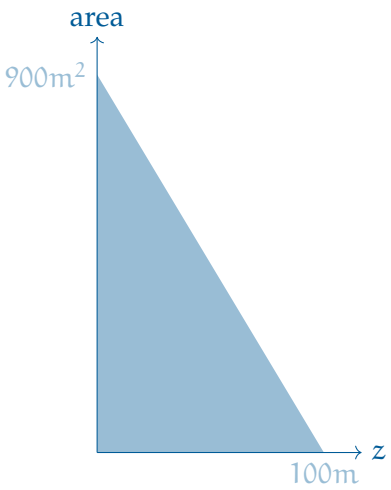
About 281.46 cubic centimeters.

## Answer to Exercise 116 (on page 345)

The area at height $z$ is given by:

$$a = \frac{1}{2}w^2 = \frac{1}{2}\left(30\sqrt{1-\frac{z}{100}}\right)^2 = \frac{1}{2}900\left(1-\frac{z}{100}\right)$$

If we plot that, it looks like this:




What is the area of the blue region? $\frac{1}{2}(900)(100) = 45,000$

The building will be 45 thousand cubic meters.

## Answer to Exercise 117 (on page 354)

- $[1,2,3]+[4,5,6]=[5,7,9]$
- $[-1,-2,-3,-4]+[4,5,6,7]=[3,3,3,3]$
- $[\pi,0,0]+[0,\pi,0]+[0,0,\pi]=[\pi,\pi,\pi]$

## Answer to Exercise 118 (on page 354)

To get the net force, you add the two forces:

$$F=[4.2,5.6,9.0]+[-100.2,30.2,-9.0]=[-96,35.8,0.0]\text{ newtons}$$

## Answer to Exercise 119 (on page 355)

- $2\times[1,2,3]=[2,4,6]$
- $[-1,-2,-3,-4]\times-3=[3,6,9,12]$
- $\pi[\pi,2\pi,3\pi]=\pi^2,2\pi^2,3\pi^2]$

## Answer to Exercise 120 (on page 360)

- $|[1,1,1]|=\sqrt{3}\approx 1.73$
- $|[-5,-5,-5]|=|-5\times[1,1,1]|=5\sqrt{3}\approx 8.66$
- $|[3,4,5]+[-2,-3,-4]|=|[1,1,1]|=\sqrt{3}\approx 1.73$

## Answer to Exercise 121 (on page 365)

At its apex, the hammer's velocity is $0\text{m/s}$, so we will find the time where $v=0\text{m/s}$:

$$0\frac{m}{s}=12\frac{m}{s}-9.8\frac{m}{s^2}\cdot t$$

$$t=\frac{12\frac{m}{s}}{9.8\frac{m}{s^2}}\approx 1.22s$$

## Answer to Exercise 122 (on page 380)

1. D, B/C/F, A, E; displacement is the difference in position between the starting and ending points. Object D moves backwards and therefore has a negative displacement. Objects B, C, and F all end in the same position they started and therefore have zero displacement. Objects A and E both move forward and have positive displacement. Object E moves forward 4 units while Object A only moves forward by 1 unit. Therefore, object E has a greater positive displacement than object A.

2. C, D, B/F, A, E; instantaneous velocity is given by the slope of a position-time graph at the indicated time (in this case, $t = 7.5s$). Since the graphs show 10 seconds of motion and there are 4 tick marks on each x-axis, each unit on the x-axis represents 2.5 seconds of time. Therefore, $t = 7.5s$ is the third tick mark on the x-axis. At $t = 7.5s$, the graphs of objects C and D have negative slopes and therefore negative velocities. Since the slope for object C is steeper than the slope for object D, object C's *speed* is greater than object D's, and object C's *velocity* is more negative than object D's. The graphs of object's B and F are horizontal at $t = 7.5s$ and therefore their velocities are zero. Graphs A and E have positive slopes, and since E is steeper, object E has a greater speed and more positive velocity than object A.

3. B/F, A, D, E, C; distance is a scalar and therefore always positive. B and F do not change position, and therefore travel a distance of 0. A moves forward 1 unit. D moves backwards 3 units (for a *distance* of 3). E moves forward 4 units. C moves forwards 3 units, then backwards 3 additional units, for a total of 6 units of distance.

## Answer to Exercise 123 (on page 387)

(a) To find how far from the base of the cliff the rock lands, we can use the horizontal motion equation:

$$\Delta x = v_{0x} t$$
$$\Delta x = (8\tfrac{m}{s})(3s) = 24m$$

(b) To find the height of the cliff, we can use the vertical motion equation. Since $v_{0y} = 0$, and the initial vertical position is at the top of the cliff (we can call this $y_0 = 0$), we have:

$$y_f = v_{0y} t + \frac{1}{2}(-g)t^2$$
$$y_f = 0 + \frac{1}{2}(-10)(3)^2$$
$$y_f = -45m$$

We get a negative value for $y_f$ because the rock is falling downward from its initial position, the top of the cliff which we selected as the origin. Therefore, the height of the cliff is 45 meters.

So the rock lands 24 meters from the base of the cliff, and the cliff is 45 meters high.

## Answer to Exercise 124 (on page 390)

Let's first find the horizontal and vertical components of the initial velocity:

$$\begin{aligned} v_{0x} &= (25)\cos(40^\circ) \approx 19.15 \text{ m/s}, \\ v_{0y} &= (25)\sin(40^\circ) \approx 16.07 \text{ m/s}. \end{aligned}$$

(a) To find the peak height, we can use the vertical motion equations. Taking it in two steps, let's first find the time to reach the peak height, where the vertical velocity $v_y = 0$:

$$\begin{aligned} v_y &= v_{0y} + (-g)t \\ 0 &= 16.07 - 10t \\ t_{peak} &= \frac{16.07}{10} \approx 1.607 \text{ s} \end{aligned}$$

Now, we can use this time to find the peak height using the vertical displacement equation:

$$\begin{aligned} \Delta y &= v_{0y}t + \frac{1}{2}(-g)t^2 \\ \Delta y &= (16.07)(1.607) + \frac{1}{2}(-10)(1.607)^2 \\ \Delta y &\approx 12.91 \text{ m} \end{aligned}$$

(b) To find the total horizontal distance traveled, we first need the total time of flight. Since the motion is symmetric, the total time will be twice the time to reach the peak height:

$$t_{total} = 2t_{peak} \approx 2(1.607) \approx 3.214 \text{ s} = 3.21 \text{ s}$$

Since there is no horizontal acceleration, we can use the horizontal motion equation to find the horizontal distance:

$$\Delta x = v_{0x}t_{total} \approx (19.15)(3.21) \approx 61.5 \text{ m}$$

So, the peak height of the pumpkin is approximately 12.91 m, and the horizontal distance it travels before hitting the ground is approximately 61.5 m.

## Answer to Exercise 125 (on page 390)

To maximize the horizontal distance of a projectile being launched, the vertical and horizontal components must be proportionally equal. This is only possible at a launch angle of 45°. Let's prove this mathematically.

Let the initial launch speed be $v$ and the launch angle be $\theta$. The horizontal and vertical velocity components are

$$v_x = v\cos\theta, \qquad v_y = v\sin\theta.$$

The vertical motion kinematics equation can be solved for:

$$0 = t\left(v\sin\theta - \tfrac{1}{2}gt\right)$$

This has two solutions: $t = 0$ (the time at launch) and

$$t = \frac{2v\sin\theta}{g}.$$

Thus, the horizontal range is

$$R_x = v_x \cdot t = v\cos\theta \cdot \frac{2v\sin\theta}{g} = \frac{v^2}{g}\sin(2\theta).$$

The function $\sin(2\theta)$ reaches its maximum value of 1 when $2\theta = 90°$, or $\theta = 45°$. Therefore, the optimal launch angle for maximum horizontal distance is 45°.

The range equation above can result can result in two different values: $\theta$ and $90° - \theta$. There are two ways to get a certain horizontal launch distance, one at an angle less than 45° and one at an angle greater than 45°. For example, a projectile launched at 30° will travel the same horizontal distance as one launched at 60°, assuming the same initial velocity. However, the projectile launched at 30° will spend less time in the air and have a lower peak height than the one launched at 60°.

## Answer to Exercise 126 (on page 391)

Using the horizontal distance equation, R, with $v_0 = 14$ m/s and $R = 100$ m, we have:

$$R = \frac{v^2 \sin(2\theta)}{g}$$
$$\sin(2\theta) = \frac{Rg}{v^2}$$
$$\sin(2\theta) = \frac{(100)(10)}{14^2} \approx 5.10$$

Since the sine of an angle cannot be greater than 1, it is impossible to hit the target with a launch speed of 14 m/s.

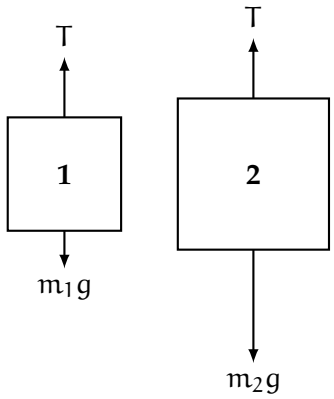
Instead, we can calculate the minimum launch speed needed to hit the target at a 45° angle, while satisfying the 100 m requirement. Assuming the $\sin(2\theta) = 1$ for a 45° launch angle, we can rearrange the range equation to solve for $v$:

$$v = \sqrt{\frac{Rg}{\sin(2\theta)}}$$
$$v = \sqrt{\frac{(100)(10)}{1}} = \sqrt{1000} \approx 31.62 \text{ m/s}$$

So at an assumed launch angle of 45°, a minimum launch speed of approximately 31.62 m/s is needed to hit the target 100 meters away.

## Answer to Exercise 127 (on page 396)

1. Each block is acted on by gravity and the tension of the string. The force of gravity on block 2 is 3 times that of block 1, and the tension vector should be between the lengths of the gravity vectors:




2. Block 1 will have a positive acceleration with a magnitude less than $g$. Block 2 will have a negative acceleration with a magnitude greater than $g$.

3. Since the blocks are connected, they will move as a system and therefore have the same magnitude acceleration. From this, the free body diagrams, and Newton's Second Law, we know that:

$$m_1 a = T - m_1 g$$

$$m_2 (-a) = T - m_2 g$$

We know $m_1$ and $m_2$, so the two unknowns are $T$ and $a$. One way to solve this system of equations would be to subtract equation 2 from equation 1:

$$m_1 a + m_2 a = m_2 g - m_1 g$$

$$a = \frac{m_2 - m_1}{m_1 + m_2} g$$

Substituting for the masses, we see that:

$$a = \frac{9-3}{9+3} g = \frac{6}{12} g = \frac{1}{2} g$$

Therefore, Block 1 is accelerating at $+\frac{1}{2}g$ and Block 2 is accelerating at $-\frac{1}{2}g$.

## Answer to Exercise 128 (on page 401)

(a) The normal force must be equal to the perpendicular component of the gravitational force, since there is no acceleration in the y-direction of our rotated y-axis:

$$F_N = mg\cos(\theta) = (30\,\text{kg})(9.8\,\text{m/s}^2)\cos(30^\circ) \approx 254.6\,\text{N}$$

(b) The frictional force can be found using the coefficient of kinetic friction and the

normal force:

$$F_F = \mu_k F_N = (0.3)(254.6\,\text{N}) \approx 76.4\,\text{N}$$

(c) The acceleration can be found by setting up the equation for the sum of forces in the x-direction:

$$\sum F_x = F_F - mg\sin(\theta) = ma_x$$

Solving for $a_x$:

$$\begin{aligned} a_x &= \frac{F_F - mg\sin(\theta)}{m} \\ &= \frac{76.4\,\text{N} - (30\,\text{kg})(9.8\,\text{m/s}^2)\sin(30^\circ) -}{30\,\text{kg}} \\ &= \frac{76.4 - 147}{30} \\ &\approx \frac{-70.6}{30} \\ &\approx -2.35\,\text{m/s}^2 \end{aligned}$$

## Answer to Exercise 129 (on page 402)

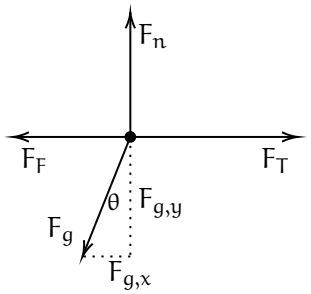
For a box to be on the verge of sliding, the frictional force (static) must equal the component of gravitational force parallel to the incline. Thus, we can set up the following equation:

$$\begin{aligned} \sum F_x &= mg\sin(\theta) - F_F = 0 \\ mg\sin(\theta) &= F_F \\ mg\sin(\theta) &= \mu_s F_N \\ mg\sin(\theta) &= \mu_s mg\cos(\theta) \\ \tan(\theta) &= \mu_s \end{aligned}$$

Solving for $\theta$, we get $\theta = \tan^{-1}(\mu_s)$. Plugging in the value of $\mu_s$, we get $\theta = 24.89^\circ$

## Answer to Exercise 130 (on page 402)

Let's create free body diagrams for both objects.




FBD Block $m_1$

FBD Block $m_2$

Note that the dashed lines are components of forces, not forces themselves. We can see that the common force uniting the two objects is $F_T$. We can solve for the acceleration in the x-direction of the block of mass $m_1$ and the acceleration in the y-direction of $m_2$.

$$\begin{aligned} F_{net,m_1,x} &= F_T - m_1 g \sin\theta - \mu_k m_1 g \cos\theta \\ &= m_1 a_x \\ F_{net,m_2,y} &= m_2 g - F_T \\ &= m_2 a_x \\ F_{net,m_1,x} + F_{net,m_2,y} &= m_2 g - m_1 g \sin\theta - \mu_k m_1 g \cos\theta = (m_1 + m_2) a \\ a &= \frac{m_2 g - m_1 g \sin\theta - \mu_k m_1 g \cos\theta}{m_1 + m_2} \end{aligned}$$

Notice that both masses share the same magnitude of acceleration. $m_2$ accelerates vertically, $m_1$ accelerates horizontally, but their magnitude is identical. It is *not* the same as the net force on $m_2$ alone.

## Answer to Exercise 131 (on page 404)

The momentum of the first car is 12,000 kg m/s in the north direction.

The momentum of the second car is 24,000 kg m/s in the east direction.

The new object will be moving northeast. What is the angle compared with the east?

$$\theta = \arctan\frac{12,000}{24,000} \approx 0.4636 \text{ radians } \approx 26.565 \text{ degrees north of east}$$

The magnitude of the momentum of the new object is $\sqrt{12,000^2 + 24,000^2} \approx 26,833$ kg m/s

Its new mass is 2,5000 kg. So the speed will be $26,833/2,500 = 10.73$ m/s.

## Answer to Exercise 132 (on page 406)

The original forward momentum was 1.2 kg m/s. The original kinetic energy is $(1/2)(0.4)(3^2) = 1.8$ joules.

Let $s$ be the post-collision speed of the ball that had been at rest. Let $x$ and $y$ be the forward and sideways speeds (post-collision) of the other ball. Conservation of kinetic energy says

$$(1/2)(0.4)(s^2) + (1/2)(0.4)(x^2 + y^2) = 1.8$$

Forward momentum is conserved:

$$0.4\frac{s}{\sqrt{2}} + 0.4x = 1.2$$

Which can be rewritten:

$$x = 3 - \frac{s}{\sqrt{2}}$$

Sideways momentum stays zero:

$$(0.4)\frac{s}{\sqrt{2}} - 0.4y = 0.0$$

Which can be rewritten:

$$y = \frac{s}{\sqrt{2}}$$

Substituting into to the conservation of kinetic energy equation above:

$$(1/2)(0.4)(s^2) + (1/2)(0.4)(\left(3 - \frac{s}{\sqrt{2}}\right)^2 + \left(\frac{s}{\sqrt{2}}\right)^2 = 1.8$$

Which can be rewritten:

$$s^2 - \frac{3}{\sqrt{2}}s + 0 = 0$$

There are two solutions to this quadratic: $s = 0$ (before collision) and $s = \frac{3}{\sqrt{2}}$. Thus,

$$y = \frac{3}{2}$$

and

$$x = 3 - \frac{3}{2} = \frac{3}{2}$$

So, both balls careen off at 45° angles at the exact same speed.

## Answer to Exercise 133 (on page 410)

Because both carts collide elastically, the Conservation of Kinetic Energy and Conservation of Momentum both apply.

We can rearrange the Conservation of Kinetic Energy in the following ways:

$$\begin{aligned}
\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 &= \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2 \\
\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 - \frac{1}{2}m_1v_{1f}^2 - \frac{1}{2}m_2v_{2f}^2 &= 0 \\
m_1v_{1i}^2 + m_2v_{2i}^2 - m_1v_{1f}^2 - m_2v_{2f}^2 &= 0 \\
m_1v_{1i}^2 - m_1v_{1f}^2 + m_2v_{2i}^2 - m_2v_{2f}^2 &= 0 \\
m_1(v_{1i}^2 - v_{1f}^2) + m_2(v_{2i}^2 - v_{2f}^2) &= 0 \\
m_1(v_{1i} - v_{1f})(v_{1i} + v_{1f}) + m_2(v_{2i} - v_{2f})(v_{2i} + v_{2f}) &= 0 \qquad (1)
\end{aligned}$$

And rearranging Conservation of Momentum:

$$\begin{aligned} m_1 v_{1i} + m_2 v_{2i} &= m_1 v_{1f} + m_2 v_{2f} \\ m_1 v_{1i} + m_2 v_{2i} - m_1 v_{1f} - m_2 v_{2f} &= 0 \\ m_1(v_{1i} - v_{1f}) + m_2(v_{2i} - v_{2f}) &= 0 \\ m_1(v_{1i} - v_{1f}) &= -m_2(v_{2i} - v_{2f}) \end{aligned} \tag{2}$$

Inputting Equation 2 into Equation :

$$\begin{aligned} m_1(v_{1i} - v_{1f})(v_{1i} + v_{1f}) + m_2(v_{2i} - v_{2f})(v_{2i} + v_{2f}) &= 0 \\ -m_2(v_{2i} - v_{2f})(v_{1i} + v_{1f}) + m_2(v_{2i} - v_{2f})(v_{2i} + v_{2f}) &= 0 \\ m_2(v_{2i} - v_{2f})\left[(v_{2i} + v_{2f}) - (v_{1i} + v_{1f})\right] &= 0 \\ (v_{2i} + v_{2f}) - (v_{1i} + v_{1f}) &= 0 \text{ assuming } v_{2i} \neq v_{2f} \\ v_{2f} - v_{1f} &= v_{1i} - v_{2i} \end{aligned} \tag{3}$$

Equation 3 tells us that, in an elastic collision, the relative speed of separation is equal to relative speed of approach. This allows us to do the following plug-ins:

$$\begin{aligned} m_1 v_{1i} + m_2 v_{2i} &= m_1 v_{1f} + m_2(v_{1f} + v_{1i} - v_{2i}) \\ m_1 v_{1i} + m_2 v_{2i} &= (m_1 + m_2) v_{1f} + m_2 v_{1i} - m_2 v_{2i} \\ (m_1 - m_2) v_{1i} + 2m_2 v_{2i} &= (m_1 + m_2)\, v_{1f} \\ v_{1f} &= \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i} \end{aligned}$$

The same process results in $v_{2f}$:

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i} + \frac{m_2 - m_1}{m_1 + m_2} v_{2i} \tag{45.7}$$

Equations **??** and 45.7 can be very useful but take a while to derive. Make sure you understand the process to solve for them. Let's plug in the values:

$$v_{1f} = \frac{2-1}{2+1}(3.0) = \frac{1}{3}(3.0) = 1.0 \text{ m/s}$$

$$v_{2f} = \frac{2(2)}{2+1}(3.0) = \frac{4}{3}(3.0) = 4.0 \text{ m/s}$$

## Answer to Exercise 134 (on page 411)

Using the Conservation of Momentum theorem:

$$\begin{aligned} m_A v_{A,i} + m_B v_{B,i} &= m_A v_{A,f} + m_B v_{B,f} \\ (1000)(20) + (1500)(-10) &= (1000)(5) + (1500)v_{B,f} \\ 20000 - 15000 - 5000 &= (1500)v_{B,f} \\ 0 &= (1500)v_{B,f} \\ v_{B,f} &= 0\text{m/s} \end{aligned}$$

The final velocity of Car B is 0 m/s. The collision must be inelastic as both velocities change and kinetic energy is not conserved.

## Answer to Exercise 135 (on page 412)

(a) Using the conservation of momentum, we can say that:

$$\begin{aligned} m_{\text{bullet}}(v_{\text{bullet}}) + m_{\text{block}}(0) &= (m_{\text{bullet}} + m_{\text{block}})v_f \\ (0.01)(500) + (2)(0) &= ((0.01) + (2))v_f \\ v_f &= \frac{(500)(0.01)}{2.01} \\ v_f &\approx 2.49\text{m/s} \end{aligned}$$

The *block-bullet system* moves at a speed of 2.49 m/s, in the same direction the bullet was fired.

(b) To find the initial kinetic energy, we use the initial velocity:

$$\text{KE}_i = \frac{1}{2}(0.01)(500)^2 = 1250$$

And, to find the final kinetic energy, we use the summed masses and the calculated final velocity

$$\text{KE}_f = \frac{1}{2}(2.01)(2.49)^2 \approx 6.23$$

So the net energy lost is the difference:

$$\Delta\text{KE} = \text{KE}_i - \text{KE}_f \approx 1250 - 6.23 \approx 1243.7\,\text{J}$$

(c)

$$\text{Fraction lost} = \frac{\Delta\text{KE}}{\text{KE}_i} = \frac{1243.7}{1250} \approx 0.99496$$

So 99.5% percent of the energy is lost. It is impossible for this to be an elastic collision, so this has to be an inelastic collision. Because *more* than 95% of the energy is lost, the collision is perfectly inelastic (by the problem guidelines).

## Answer to Exercise 136 (on page 414)

The total momentum before immediately before the explosion can be separated into two components:

$$p_{i,x} = m_i v_{i,x} = (6)(20) = 120 \text{ m/s} \qquad p_{i,y} = m_i v_{i,y} = 0$$

Let's analyze Fragment A:

$$p_{Ax} = m_A v_{Ax} = 2(30\cos 40^\circ), \qquad p_{Ay} = m_A v_{Ay} = 2(30\sin 40^\circ)$$

And Fragment B,

$$p_{Bx} = m_B v_{Bx} = 1(25\cos 60^\circ), \qquad p_{By} = m_B v_{By} = 1(-25\sin 60^\circ)$$

Since the Conservation of Momentum applies to the components:

$$\begin{aligned} p_{x,\text{final}} &= p_{Ax} + p_{Bx} + p_{Cx} = 120 \text{ kg·m/s} \\ p_{Cx} &= 120 - (p_{Ax} + p_{Bx}) \\ &= 120 - 2(30\cos 40^\circ) - 1(25\cos 60^\circ) \\ v_{Cx} &= \frac{120 - 2(30\cos 40^\circ) - 1(25\cos 60^\circ)}{3.0} \\ &= 20.51 \text{ m/s} \end{aligned}$$

And same for the Y-components

$$\begin{aligned} p_{y,\text{final}} &= p_{Ay} + p_{By} + p_{Cy} = 0 \text{ kg·m/s} \\ p_{Cy} &= -(p_{Ay} + p_{By}) \\ &= -(2(30\sin 40^\circ) - 1(-25\sin 60^\circ)) \\ v_{Cy} &= \frac{-(2(30\sin 40^\circ) - 1(-25\sin 60^\circ))}{3} \\ &= -5.64 \text{ m/s} \end{aligned}$$

So Fragment C moves at $\sqrt{20.51^2 + (-5.64)^2} = 21.3$ m/s with a direction of $\theta = \tan^{-1}\left(\frac{-5.64}{20.51}\right) = -15^\circ$ or $15^\circ$ below the horizontal.

## Answer to Exercise 137 (on page 472)

$$p = 101,332 \times \left(1 - 2.25577 \times 10^{-5} \times h\right)^{5.25588}$$

and $h = 9,144$. Thus,

$$p \approx 30.1\text{kPa}$$

## Answer to Exercise 138 (on page 483)

For what $t$ is $-4.9t^2 + 12t + 2 = 0$? Start by dividing both sides of the equation by -4.9.

$$t^2 - 2.45t - 0.408 = 0$$

The roots of this are at

$$x = -\frac{b}{2} \pm \frac{\sqrt{b^2 - 4c}}{2} = -\frac{-2.45}{2} \pm \frac{\sqrt{(-2.45)^2 - 4(-0.408)}}{2} = 1.225 \pm 1.382$$

We only care about the root after we release the hammer $(t > 0)$.

$1.225 + 1.382 \approx 2.61$ seconds after releasing the hammer, it will hit the ground.

## Answer to Exercise 139 (on page 490)

1. not arithmetic
2. arithmetic, common difference is 3
3. arithmetic, common difference is -4
4. not arithmetic

## Answer to Exercise 140 (on page 492)

The first five terms are {-4, 1, 6, 11, 16} and an explicit formula is $a_n = -4 + 5(n-1)$.

## Answer to Exercise 141 (on page 493)

The common difference is $\frac{3\pi}{2} - \pi = \frac{\pi}{2}$. The recursive formula is $a_n = a_{n-1} + \frac{\pi}{2}$ with $a_1 = \pi$. The explicit formula is $a_n = \pi + \frac{\pi}{2}(n-1)$.

## Answer to Exercise 142 (on page 494)

1. not geometric
2. geometric sequence with common ratio $r = \frac{1}{2}$
3. geometric sequence with common ration $r = -5$
4. not geometric

## Answer to Exercise 143 (on page 495)

The first five terms are $\{1, \frac{3}{2}, \frac{9}{4}, \frac{27}{8}, \frac{81}{16}\}$. An explicit formula for this sequence is $a_n = 1(\frac{3}{2})^{(n-1)}$.

## Answer to Exercise 144 (on page 495)

The common ratio is $\frac{a_n}{a_{n-1}} = \frac{2}{-4} = -\frac{1}{2}$. A recursive formula would be $a_n = a_{n-1} \times -\frac{1}{2}$ with $a_1 = -4$. An explicit formula would be $a_n = (-4)(-\frac{1}{2})^{(n-1)}$.

## Answer to Exercise 145 (on page 505)

Using 56.1 with $P_0 = 1500$, $r = 0.05$, and $n = 8$:

$$P_8 = 1500(1 + 0.05)^8 = 1500(1.05)^8 \approx \$2216.18$$

Alternatively, in Sheets/Excel, we can use `=FV(0.05, 8, 0, -1500)` to say that we want the *future value* of 1500 deposited (hence, negative), after 8 years compounded once per year.

## Answer to Exercise 146 (on page 505)

Using $P_n = P_0(1 + \frac{r}{n})^{nt}$ with $P_0 = 3000$, $r = 0.06$, and $n = 5$ years:

Account A (monthly, $t = 12$):

$$P = 3000\left(1 + \frac{0.06}{12}\right)^{12\times 5} \approx \$4046.55$$

Account B (daily, $t = 365$):

$$P = 3000\left(1 + \frac{0.06}{365}\right)^{365\times 5} \approx \$4049.48$$

| | A | B | C | D |
|---|---|---|---|---|
| 1 | **Principal** | $ 3,000.00 | | |
| 2 | **Annual Rate** | 6% | | |
| 3 | | | | |
| 4 | **Year** | **Account A** | **Account B** | |
| 5 | 0 - initial | $3,000 | $3,000 | |
| 6 | 1 | $3,185.03 | $ 3,185.49 | |
| 7 | 2 | $3,381.48 | $ 3,382.46 | |
| 8 | 3 | $3,590.04 | $ 3,591.60 | |
| 9 | 4 | $3,811.47 | $ 3,813.67 | |
| 10 | 5 | $4,046.55 | $ 4,049.48 | |

Account B has the higher balance, by about $2.93. This makes sense: the more frequently interest is compounded, the closer the balance gets to the continuous compounding limit, $P_0e^{rt}$.

## Answer to Exercise 147 (on page 516)

1. By the chain rule, $f'(x) = 2\arctan x \times \frac{d}{dx}\arctan x = 2\arctan x \frac{1}{1+x^2}$
2. By the Product rule, $f'(x) = x\frac{d}{dx}\text{arcsec}\,(x^3) + \text{arcsec}\,(x^3)$. Further, by the chain rule, $\frac{d}{dx}\text{arcsec}\,(x^3) = \frac{1}{(x^3)\sqrt{(x^3)^2-1}} \times \frac{d}{dx}(x^3) = \frac{3x^2}{x^3\sqrt{x^6-1}}$. Therefore, $f'(x) = \frac{3}{\sqrt{x^6-1}} + \text{arcsec}\,(x^3)$
3. By the chain rule, $f'(x) = \frac{1/x}{\sqrt{1-(1/x)^2}} \times -\frac{1}{x^2} = -\frac{1}{x^3\sqrt{1-\frac{1}{x^2}}}$

## Answer to Exercise 148 (on page 551)

1. $\sum_{i=1}^{7} X_i = 22 + 19 + 25 + 21 + 18 + 23 + 20 = 148.$
2. $(2) + (-1) + (5) + (1) + (-2) + (3) + (0) = 8$. The sum decreased by $7 \times 20 = 140$, which is $n$ times the constant subtracted. Shifting every value by a constant shifts the total by $n$ times that constant.

## Answer to Exercise 149 (on page 554)

1. $\bar{X} = \dfrac{12 + 15 + 9 + 11 + 14 + 8 + 13 + 52}{8} = \dfrac{134}{8} = 16.75$ books.
2. Sorted: 8, 9, 11, 12, 13, 14, 15, 52. $l = \dfrac{9}{2} = 4.5$, so $M = \dfrac{12 + 13}{2} = 12.5$ books.
3. The mean is more affected. It uses every value in the calculation, so the outlier of 52 pulls it upward. The median depends only on the middle two values, which are unaffected by how extreme the highest value is.
4. Without 52: $\bar{X} = \dfrac{82}{7} \approx 11.71$ books. Sorted: 8, 9, 11, 12, 13, 14, 15; median = 12. The mean dropped by about 5 books; the median changed by only 0.5.

## Answer to Exercise 150 (on page 555)

1. Mean. Symmetric data with no outliers means the mean and median are close, and the mean is preferred for further calculations.

2. Median. The CEO's salary is an extreme outlier that would pull the mean well above what a typical employee earns. The median better represents the center of the bulk of the data.
3. Median. The long right tail indicates right skew; the mean would be pulled toward the older ages and away from where most people cluster.
4. Mean. Once the outlier is removed and the distribution is symmetric, the mean is the appropriate choice.

## Answer to Exercise 151 (on page 559)

1. $s^2 = \frac{743.21}{14-1} = \frac{743.21}{13} \approx 57.17$ cars$^2$.
2. $s = \sqrt{57.17} \approx 7.56$ cars.
3. $s$ is measured in cars. $s^2$ is measured in cars$^2$, which has no natural interpretation. This is one reason we prefer the standard deviation over the variance for reporting.
4. The second dealership. Both have the same mean, but a smaller standard deviation means the daily sales are more tightly clustered around the average.

## Answer to Exercise 152 (on page 561)

1. Standard deviation. Symmetric data with no outliers is exactly the situation where the mean and standard deviation are appropriate.
2. IQR. The extreme high values would inflate the standard deviation substantially, making it a misleading measure of typical spread. IQR focuses on the middle 50% and is unaffected by the luxury properties.
3. Standard deviation. Once the distribution is symmetric and outlier-free, the standard deviation is the preferred measure and will now accurately reflect the spread of the remaining data.

## Answer to Exercise 153 (on page 565)

Multiplying every value by $b = 1.02$:

| Measure | Original | After multiplying by 1.02 |
|---|---|---|
| Mean | 12,000 | $1.02(12{,}000) = 12{,}240$ |
| Median | 8,000 | $1.02(8{,}000) = 8{,}160$ |
| Range | 30,000 | $1.02(30{,}000) = 30{,}600$ |
| Std dev | 5,000 | $1.02(5{,}000) = 5{,}100$ |
| $Q_1$ | 5,000 | $1.02(5{,}000) = 5{,}100$ |
| $Q_3$ | 14,000 | $1.02(14{,}000) = 14{,}280$ |
| IQR | 9,000 | $1.02(9{,}000) = 9{,}180$ |

Unlike adding a constant, multiplying affects the spread measures too. Every distance in the distribution is stretched by the same factor.

## Answer to Exercise 154 (on page 567)

1. $z = \dfrac{24 - 14.64}{7.56} \approx 1.24$. The best day was about 1.24 standard deviations above the mean. It was a notably good day but not extreme.
2. $z = \dfrac{2 - 14.64}{7.56} \approx -1.67$. The worst day was about 1.67 standard deviations below the mean and further from typical than the best day was above it.
3. Second dealership: $z = \dfrac{31 - 20}{4} = 2.75$. The second dealership's best day was more unusually high at 2.75 standard deviations above its mean, compared to 1.24 for the first dealership.
4. Still $z \approx 1.24$. Adding a constant to every value shifts the mean by the same constant and leaves the standard deviation unchanged. The relative position of every observation in the distribution stays the same, so z-scores are unaffected.

## Answer to Exercise 155 (on page 570)

1. $l = \dfrac{(14 + 1)(50)}{100} = 7.5$. The 50th percentile falls halfway between the 7th observation (14) and the 8th (17): $P_{50} = \dfrac{14 + 17}{2} = 15.5$. This matches the median exactly.
2. $Q_3 = P_{75} = 21$, so 75% of days had 21 or fewer cars sold.
3. $Q_1 = 9$ and $Q_3 = 21$. The value 17 lies between $Q_2 = 15.5$ and $Q_3 = 21$, placing it in the third quarter (between the median and $Q_3$). The consultant is correct.
4. Days with 2 or fewer cars: only the day with 2 cars sold. That is 1 out of 14 days, or about 7%, consistent with the bottom 10% cutoff of 2.5.

## Answer to Exercise 156 (on page 575)

The nodes in this graph are unique words. The edges represent a valid one letter operations that can transform one word into another word. These operations include adding a letter, removing a letter, or changing a letter into another. For example, the word `cat` would be connected to `cot` (change), `cart` (add), and `cut` (change), but not to `dog`. Usually this will be represented as a tree, which is a special type of graph we will cover later.

Here is an example of a word graph:

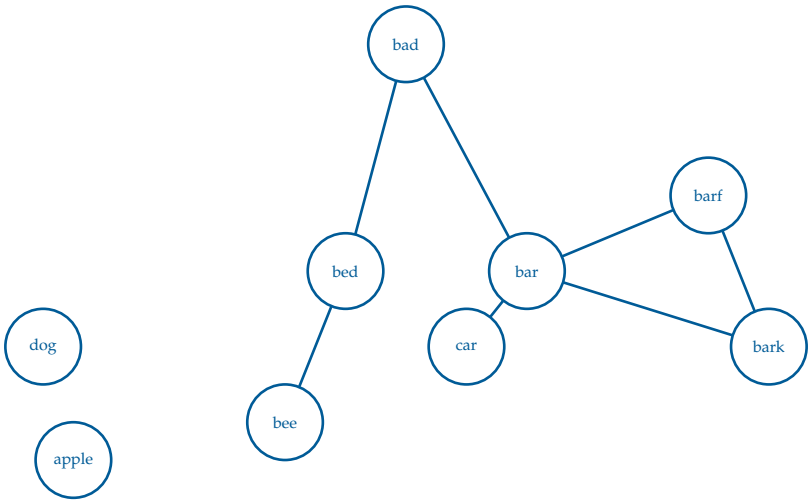



Figure 63.4: A word graph where the nodes are unique words and the edges represent a one letter operations between two words.

## Answer to Exercise 157 (on page 577)

In a weighted graph, each node would have to store a set of neighbors and their respective weights. The best way to do this would be a tuple. For example,

- A: [(B, 4), (C, 3)]
- B: [(A, 4), (D, 8)]
- C: [(A, 3)]
- D: [(B, 8)]

## Answer to Exercise 158 (on page 581)

- $[1,2,3]\cdot[4,5,-6] = 4+10-18 = -4$
- $[\pi, 2\pi]\cdot[2,-1] = 2\pi - 2\pi = 0$
- $[0,0,0,0]\cdot[10,10,10,10] = 0+0+0+0 = 0$

## Answer to Exercise 159 (on page 586)

- $[1,0]\cdot[0,1] = 0$. The angle must be $\pi/2$.
- $[3,4]\cdot[4,3] = 24$. $|[3,4]||[4,3]|\cos(\theta) = 24$. $\cos(\theta) = \frac{24}{(5)(5)}$. $\theta = \arccos(\frac{24}{25}) \approx 0.284$ radians.
- $[2,-1,2]\cdot[-1,2,-2] = 4-2-4 = -2$. $|[2,-1,2]| = \sqrt{4+1+4} = \sqrt{9} = 3$. $|[-1,2,-2]| = \sqrt{1+4+4} = \sqrt{9} = 3$. $3(3)\cos\theta = -2$. $\theta = \arccos(-2/9) \approx 1.795$ radians.
- $[-5,0,-1]\cdot[2,3,-4] = -10+0+4 = -6$. $|[-5,0,1]| = \sqrt{25+0+1} = \sqrt{26}$. $|[2,3,-4]| = \sqrt{4+9+16} = \sqrt{29}$. $\sqrt{26}(\sqrt{29})\cos\theta = -6$. $\theta = \arccos(\frac{-6}{\sqrt{26}\sqrt{29}}) \approx 1.791$ radians.

## Answer to Exercise 160 (on page 587)

Power is the dot product of the force vector and the velocity vector:

$$P = \vec{F}\cdot\vec{v}$$

Since the force and velocity are at an angle, we can use:

$$P = |\vec{F}||\vec{v}|\cos\theta$$

Here, $|\vec{F}| = 40$ N, $|\vec{v}| = 3$ m/s, and $\theta = 30°$:

$$P = 40(3)\cos(30°)$$

Since $\cos(30°) \approx 0.866$:

$$P \approx 120(0.866) \approx 104 \text{ W}$$

Only the part of the force pointing in the direction of motion transfers power to the skate-

board. The angled part still matters physically, but it does not all help move the skateboard forward. So the power transferred to the skateboard is about 104 watts.

## Answer to Exercise ?? (on page 588)

The work done is the dot product of the force vector and the displacement vector:

$$W = \vec{F} \cdot \vec{d}$$

Here, the force is $\vec{F} = [12, 5]$ N and the displacement is $\vec{d} = [3, 0]$ m. So, we use vector multiplication but multiplying each component:

$$W = [12, 5] \cdot [3, 0] = 12(3) + 5(0) = 36 \text{ J}$$

The horizontal part of the force does work because the box moves horizontally. The vertical part of the force does no work because the box does not move vertically. So the total work done is 36 joules.

## Answer to Exercise 162 (on page 605)

A is 440 Hz. Each half-step is a multiplication by $\sqrt[12]{2} = 1.059463094359295$ So the frequency of E is $(440)(2^{7/12}) = 659.255113825739859$

## Answer to Exercise 163 (on page 612)

1. A value of $r = 1.3$ is impossible. The correlation coefficient must satisfy $-1 \leq r \leq +1$, so the student made a calculation error.
2. The researcher using the full population uses $\rho$. The researcher using a sample uses $r$.
3. Both are possible. An $r$ of 0.5 would indicate a moderate positive linear relationship. The points follow an upward trend but with noticeable scatter. An $r$ of $-1$ would indicate a perfect negative linear relationship, meaning all points fall exactly on a downward-sloping line.

## Answer to Exercise 164 (on page 613)

| $x$ | $y$ | Sign of $r$ |
|---|---|---|
| Hours studied | Exam score | Positive |
| Temperature | Hot drinks sold | Negative |
| Shoe size | Number of pets owned | Zero |
| Age of a car | Resale value | Negative |

## Answer to Exercise 165 (on page 614)

| $r$ | Direction | Strength |
|---|---|---|
| $+0.92$ | Positive | Very strong |
| $-0.34$ | Negative | Weak |
| $+0.07$ | Positive | No correlation |
| $-0.88$ | Negative | Very strong |

## Answer to Exercise 166 (on page 616)

## Answer to Exercise 167 (on page 618)

1. $e = 40.1 - 42.3 = -2.2$, or $-\$2{,}200$.
2. The salary was over-predicted. The observed salary was \$2,200 less than the model expected.

## Answer to Exercise 168 (on page 619)

1. $3 + (-3) + 2 + (-1) + (-1) = 0$. The positive and negative residuals cancel out, making the sum zero even though the line does not fit perfectly.
2. $3^2 + (-3)^2 + 2^2 + (-1)^2 + (-1)^2 = 9 + 9 + 4 + 1 + 1 = 24$.
3. A line that over-predicts some values and under-predicts others by equal amounts would have a residual sum of zero but it would be a poor fit. Squaring forces every error to contribute positively, so the minimization is meaningful!

## Answer to Exercise 169 (on page 620)

1. $b = 0.91\left(\frac{5.8}{3.2}\right) = 0.91 \times 1.8125 \approx 1.649$
2. $a = 44 - 1.649(6) = 44 - 9.897 \approx 34.103$
3. $\hat{y} = 34.103 + 1.649x$
4. Substituting $x = \bar{X} = 6$: $\hat{y} = 34.103 + 1.649(6) = 34.103 + 9.897 = 44.000 = \bar{Y}$. ✓

## Answer to Exercise 170 (on page 621)

1. For every additional year of teaching experience, the predicted starting salary increases by \$1,490, on average.
2. The interpretation is missing the phrase "on average" and the word "predicted." The slope is an estimate from sample data, not a guarantee that every teacher's salary increases by exactly \$1,490 per year.

## Answer to Exercise 171 (on page 622)

1. The predicted starting salary for a teacher with zero years of experience is \$34,100.
2. Yes, it is meaningful in this context. A teacher with no prior experience is a realistic case, so $X = 0$ is within the plausible range of the data.

## Answer to Exercise 172 (on page 624)

1. $$b = r\left(\frac{S_y}{S_x}\right) = 0.9964\left(\frac{5.013}{3.169}\right) \approx 1.576$$

   For every additional year of teaching experience, the predicted starting salary increases by \$1,576, on average.

2. $$a = \bar{Y} - b\bar{X} = 37.9 - 1.576(4.6) \approx 30.650$$

   The predicted starting salary for a teacher with no prior experience is \$30,650. Since

$X = 0$ represents a newly hired teacher with no experience, this interpretation is meaningful in context.

3. The equation of the least-squares regression line is:

$$\hat{y} = 30.650 + 1.576x$$

where $x$ is years of experience and $\hat{y}$ is the predicted starting salary in thousands of dollars.

## Answer to Exercise 173 (on page 624)

1.

$$\hat{y} = 30.650 + 1.576(5) = 30.650 + 7.880 = 38.530$$

The predicted starting salary for a teacher with 5 years of experience is \$38,530.

2.

$$\hat{y} = 30.650 + 1.576(8) = 30.650 + 12.608 = 43.258$$

$$e = y - \hat{y} = 43 - 43.258 = -0.258$$

The residual is $-\$258$. The teacher's actual starting salary was \$258 less than the model predicted. The salary was slightly over-predicted.

3.

$$R^2 = r^2 = (0.9964)^2 \approx 0.9928$$

About 99.28% of the variation in starting salaries is explained by the linear relationship with years of experience. This is an exceptionally strong fit.

4. About $100 - 99.28 = 0.72\%$ of the variation is unexplained. This is very small, but it may be due to factors such as the teacher's level of education, the subject area taught, or the specific school within the district.

## Answer to Exercise 174 (on page 628)

| Task | Part |
|---|---|
| Fix a number that was typed wrong | Spreadsheet |
| Choose an analysis | Analysis ribbon |
| Read the mean | Results panel |

## Answer to Exercise 175 (on page 629)

| Column | Measure type |
|---|---|
| `student_id` | ID |
| `breakfast` | Nominal |
| `sleep_hours` | Continuous |
| `sleep_quality` | Ordinal |

## Answer to Exercise 176 (on page 630)

1. Breakfast: 13 points. No breakfast: 11 points.
2. Students who ate breakfast had the higher median score.

## Answer to Exercise 177 (on page 631)

1. Yes.
2. Students who ate breakfast scored higher on the quiz, and the difference is too large to blame on chance alone.

## Answer to Exercise 178 (on page 631)

$9.55 - 0.44 \times 6 = 6.91$ hours of sleep.

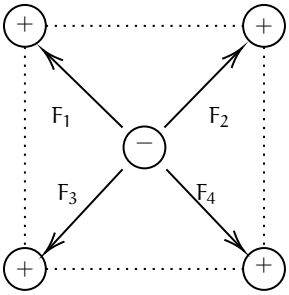
## Answer to Exercise 179 (on page 632)

1. An .omv file.
2. Right-click the table, choose Copy, and paste it into the report.

## Answer to Exercise 180 (on page 634)

$$F = K\frac{|q_1 q_2|}{r^2} = (8.988\times10^9)\frac{(-5\times10^{-9})(-5\times10^{-9})}{0.12^2} = \frac{224.7\times10^{-9}}{0.0144} = 15.6\times10^{-6}\,\text{N or} 15.6\ \text{micronewtons}$$

## Answer to Exercise 181 (on page 636)




The attractive forces of $F_1$ and $F_3$ cancel each other out. Similarly, the attractive forces of $F_2$ and $F_4$ also cancel each other out. The net charge on the negative point charge is zero.

## Answer to Exercise 182 (on page 638)

We noted above that the electric field caused by a point charge is defined as $E = \frac{kq}{r^2}$. We need to find the electric fields for each charge and then subtract their strengths.

Inputting known variables as constants for X from point P:

$$E = \frac{kQ}{d^2}$$

And doing the same for Y:

$$E = \frac{2kQ}{(2d)^2} = \frac{kQ}{2d^2}$$

Because the fields are in opposite directions, we subtract the smaller magnitude from the larger one. Let's define *to the right* as the positive direction:

$$E_{net} = E_X - E_Y \implies \frac{kQ}{d^2} - \frac{kQ}{2d^2}$$

To subtract, we find a common denominator:

$$E_{net} = \frac{2kQ}{2d^2} - \frac{kQ}{2d^2} = \frac{kQ}{2d^2}$$

The correct answer is C.

## Answer to Exercise 183 (on page 643)

The amount of charge dQ on a small *surface* section of the belt is $dq = \sigma\, dA$. For a belt of width $w$, the area passing the wire in time dt is $dA = w(v\, dt)$. Substituting this in:

$$I = \frac{\sigma(w \cdot v \cdot dt)}{dt} = \sigma \cdot w \cdot v \tag{70.12}$$

Plugging in our values:

$$I = (5.0 \times 10^{-3}\ \mathrm{C/m^2}) \cdot (0.60\ \mathrm{m}) \cdot (2.5\ \mathrm{m/s})$$
$$I = 7.5 \times 10^{-3}\ \mathrm{A} = 7.5\ \mathrm{mA}$$

## Answer to Exercise 184 (on page 644)

Since the field is uniform, the charge has a constant electric force felt on it: $\vec{F}_E = q\vec{E}$. Since q is positive, $\vec{F}_E$ points in the same direction as $\vec{E}$. Because the force and the displacement r are in the same direction, the work $W_E$ done by the electric field is positive:

$$W_E = F_E \cdot r = qEr$$

Thus the change in electric potential energy is

$$\Delta U_E = -qEr$$

It is *negative* because a doing positive work results in negative change in potential energy.

## Answer to Exercise 185 (on page 645)

Using our equations, we can derive an equation for the potential difference V:

$$\begin{aligned} \Delta U_E &= q\Delta V \\ W_E &= \Delta U_E \\ W_E &= qV \\ \text{gain in kinetic energy} &= \text{work done by field} \\ \frac{1}{2}mv^2 &= qV \\ V &= \frac{1}{2}\frac{mv^2}{q} \end{aligned}$$

Since our we are given the speed, and the charge and mass of the electron, we can plug in our values:

$$\begin{aligned} V &= \frac{\frac{1}{2}(9.11 \times 10^{-31})(9.4 \times 10^{6})^2}{1.60 \times 10^{-19}} \\ &= \frac{\frac{1}{2}(9.11 \times 10^{-31})(8.836 \times 10^{13})}{1.60 \times 10^{-19}} \\ &= \frac{\frac{1}{2}(8.05 \times 10^{-17})}{1.60 \times 10^{-19}} \\ &= \frac{\frac{1}{2}(8.05 \times 10^{-17})}{1.60 \times 10^{-19}} \\ &= \frac{4.03 \times 10^{-17}}{1.60 \times 10^{-19}} \\ &= 252\,\text{V} \end{aligned}$$

So each coulomb of charge is accelerated by a potential difference of 252 volts. And since the electron has a charge of $1.6 \times 10^{-19}$ C, the total energy gained by the electron is all transferred into kinetic energy.

## Answer to Exercise 186 (on page 647)

First off, why do we know that the electron will hit a plate? Because the electric field between the plate exerts a force on the electron. The electron is attracted to the **positively** charged upper plate, so it will accelerate upwards and eventually hit the upper plate. Let's first find. the acceleration caused by the electric field. The electric field strength E between two parallel plates is given by:

$$E = \frac{V}{d} = \frac{30}{0.040} = 750\,\text{V m}^{-1} \tag{70.20}$$

Converting this to solve for the electron's acceleration:

$$a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.6 \times 10^{-19})(750)}{9.11 \times 10^{-31}} \approx 1.32 \times 10^{14}\,\mathrm{m\,s^{-2}} \tag{70.21}$$

To find the horizontal distance travelled, we use the vertical motion to find time, and then use that time to find the horizontal distance travelled. The vertical motion is a simple kinematics problem with constant acceleration:

$$\begin{aligned} y &= v_{y0}t + \frac{1}{2}a_y t^2 \\ 0.02 &= 0 + \frac{1}{2}(1.32 \times 10^{14})t^2 \\ t^2 &= \frac{0.04}{1.32 \times 10^{14}} \\ t &\approx 1.74 \times 10^{-8}\,\mathrm{s} \end{aligned}$$

Then, putting this into a horizontal motion equation:

$$\begin{aligned} x &= v_{x0}t \\ &= (9.4 \times 10^{6})(1.74 \times 10^{-8}) \\ &\approx 0.16\,\mathrm{m} \end{aligned}$$

So the electron travels a horizontal distance of about **0.16 meters** before hitting the upper plate.

## Answer to Exercise 187 (on page 658)

The change in magnetic flux is:

$$\Delta\Phi_B = \Delta B \cdot A = (5\,\mathrm{T} \times 0.2\,\mathrm{m^2}) = 1\,\mathrm{Wb}$$

The induced emf is:

$$\mathcal{E} = -N \cdot \frac{\Delta\Phi_B}{\Delta t} = -100 \cdot \frac{1\,\mathrm{Wb}}{0.1\,\mathrm{s}} = -1000\,\mathrm{V}$$

The magnitude of the induced emf is 1000 V.

## Answer to Exercise 188 (on page 669)

The force of gravity is $9.8 \times 45 = 441$ newtons.

At any speed $s$, the force of wind resistance is $0.05 \times s^2 = 0.05s^2$ newtons.

At terminal velocity, $0.05s^2 = 441$.

Solving for s, we get $s = \sqrt{\frac{441}{0.05}}$

Thus, terminal velocity should be about 94 m/s.

## Answer to Exercise 189 (on page 671)

$$\frac{300 \times 10^6}{5.66 \times 10^{14}} = 530 \times 10^{-9} = 530 \text{ nm}$$

## Answer to Exercise 190 (on page 682)

Assuming the mirror is truly vertical and the floor is truly horizontal, the new cut off should be exactly the same as the old one: It should be below your chin the same amount that your eyes are above your chin.

*Illustration Here*

## Answer to Exercise 191 (on page 683)

*Are there white photons?* No. What we call "white" is a blend of photons that are several different colors.

Some people like to say white light is the combination of all visible colors of photons in equal amounts. That seems oddly specific and unusual.

Maybe it is better to imagine it from the human experience of white light. In our eyes, we have three different types of color-sensing cones, which generally correspond to the red, green, and blue regions of the spectrum. When all three are excited about equal amounts, humans experience that as white. On your computer screen, for example, what you see as white is just a blend of three colors of photons: a red, a green, and a blue.

*Are there black photons?* No. What we call "black" is an absence of photons in the visible range.

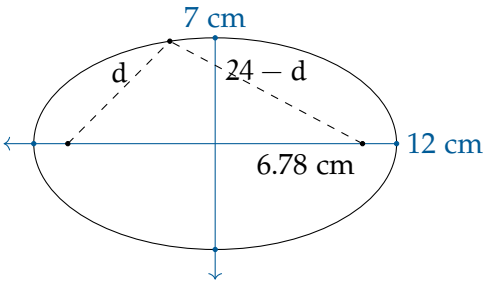
*Are there yellow photons?* Yes! There is a region of the color spectrum that is yellow. It has a wavelength of about 527 nm. Photons at this energy level excite both our green-sensitive and red-sensitive cones (we well expand on this in the eye chapter!). Your computer

monitor does not actually create light with a 527 nm wavelength. Instead, it creates red light and green light, which our eyes interpret as yellow.

## Answer to Exercise 192 (on page 686)

The length of the string is easy: $2 \times 12 = 24$ cm.

The distance from the center to the focal point is $\sqrt{12^2 - 7^2} \approx 6.78$ cm.




## Answer to Exercise 193 (on page 690)

We need the distance from the center out to each of the three axes. We know that $a = \left(\frac{1}{2}\right) 30 = 15$ cm.

We can calculate the b and c (which are equal) using the circumference given: $2b\pi = 60$, so $c = b \approx 9.55$ cm.

The volume, then is

$$V = \frac{4}{3}\pi(15)(9.55)(9.55) \approx 5,730 \text{ cubic centimeters}$$

The mass would be $5,730 \times 11.34 = 64,973$ grams or about 65 kg.

## Answer to Exercise 194 (on page 718)

1. The population is all households in the city.
2. The frame is the city's water billing records.
3. The sample is the 200 households selected and interviewed.

4. No. A census would require interviewing every household in the city. Only 200 were selected here.

## Answer to Exercise 195 (on page 719)

- **Library volunteers:** Census. With only 8 volunteers, asking all of them takes almost no effort.
- **Battery lifespan:** Sample. Testing can destroy the batteries, so a census would leave nothing to sell.
- **Commute times:** Sample. The population of 40 million is far too large for a full census to be practical.
- **Dietary restrictions:** Census. With only 22 guests, collecting complete information is straightforward and important for planning.

## Answer to Exercise 196 (on page 721)

1. Number the 300 drivers from 001 to 300. Enter a random number table at any point and read three-digit numbers. Select any driver whose number falls between 001 and 300, skipping values from 301 to 999 and any numbers that have already been selected, until 25 drivers have been chosen. Alternatively, use `randInt(1,300,25)` on a TI-84, ignoring repeats.
2. This is a convenience sample. Drivers who are easy to reach at shift changes may work particular routes or shifts that differ systematically from others. For example, they may have different safety exposure than drivers on overnight or rural routes.
3. No. An SRS requires that every possible group of 25 drivers from all 300 has an equal chance of being chosen. This procedure cannot produce a sample where all 25 come from the same depot, so not all groups of 25 are possible. This is a stratified sample, not an SRS.

## Answer to Exercise 197 (on page 725)

- **Supermarket:** Cluster sampling. The stores are clusters, which means entire clusters are included once selected.
- **University:** Stratified random sampling. The four faculties are strata. An SRS is taken from each.

- **Water utility:** Systematic sampling. Random start followed by every 25th unit.
- **Film festival:** Simple random sample. Every possible group of 90 attendees is equally likely.

## Answer to Exercise 198 (on page 728)

1. Nonresponse bias. The 92% who did not respond are unlikely to be satisfied in the same proportions as those who did. Highly satisfied and highly dissatisfied subscribers are both more motivated to respond than those with neutral feelings, but the pattern of non-response is unknown and the reported figure cannot be trusted.
2. Undercoverage bias. Residents of lower-income neighborhoods are excluded from the frame entirely. Recycling behavior may differ systematically by neighborhood type, so the results cannot generalize to the full population.
3. Wording effect bias. The question assumes the office has improved productivity and asks only "how much." Respondents are not given a neutral opportunity to say productivity has not changed or has declined.
4. Response bias. The formal appearance of the interviewers creates social pressure to give a health-conscious answer. Respondents adjust their reported behavior to match what they believe the interviewers expect.

## Answer to Exercise 199 (on page 732)

```
SELECT bike_id FROM bike WHERE purchase_price > 330 AND brand='Trek'
```

## Answer to Exercise 200 (on page 747)

The two triangles are similar; one is 2 m and 3m, the other is x cm and 3 cm.

The image of the cow is 2 cm tall.

## Answer to Exercise 201 (on page 749)

The pixels will remain dark, or black, because the energy of the incoming photons is less than the work function of the sensor material. Calculating $E = \frac{1240}{1200} = 1.033\,\text{eV}$, which is

less than the work function of 1.1 eV.

## Answer to Exercise 202 (on page 774)

1. Underflow. $6.63 \times 10^{-34}$ is far below $10^{-4}$.
2. Neither. $3.00 \times 10^{8}$ is exactly at the upper boundary.
3. Any number greater than $10^{8}$.

## Answer to Exercise 203 (on page 775)

1. $4 \times 10^{6}$ bytes; $4 \times 10^{6} \times 8 = 3.2 \times 10^{7}$ bits.
2. $500 \times 10^{9} \times 8 = 4 \times 10^{12}$ bits.
3. $2^{8} = 256$ values for one byte; $2^{64} \approx 1.8 \times 10^{19}$ values for 64 bits.

## Answer to Exercise 204 (on page 776)

- $0.5050 \times 10^{0} = 0.5050$
- $0.5050 \times 10^{2} = 50.50$
- $0.5050 \times 10^{-1} = 0.05050$
- $0.5050 \times 10^{5} = 50{,}500$

For $505{,}000{,}000 = 0.5050 \times 10^{9}$, the exponent needed is 9.

## Answer to Exercise 205 (on page 778)

1. Base 10.
2. 4 significant digits.
3. Smallest: $m = -3$; largest: $M = 4$.
4. $12{,}345 = 0.12345 \times 10^{5}$. The exponent 5 exceeds $M = 4$, so this causes **overflow**. The system would need to round to $0.1235 \times 10^{5} = 12{,}350$ and even then the exponent

is out of range.

5. $0.00001 = 0.1000 \times 10^{-4}$. The exponent $-4$ is below $m = -3$, so this causes **underflow**.

6. Any number in the range $10^{-3}$ to $(1 - 10^{-4}) \times 10^{4}$ works, for example $0.5000 \times 10^{2} = 50.00$ or $0.9999 \times 10^{4} = 9999$. Overflow example: any number with magnitude above 9999, such as 100,000.

## Answer to Exercise 206 (on page 780)

1. $16 - 1 - 10 = 5$ bits for the exponent.
2. $M = 32$.
3. Less precise than single precision. Single precision has $p = 23$ significand bits; this system has only $p = 10$, so it can store fewer digits and will produce larger roundoff errors.

## Answer to Exercise 207 (on page 781)

1. $u = \frac{1}{2} \times 10^{1-6} = \frac{1}{2} \times 10^{-5} = 5 \times 10^{-6}$.
2. The worst-case relative error is $u = 5 \times 10^{-6}$, meaning about 6 significant digits of accuracy.
3. $u = \frac{1}{2} \times 10^{1-12} = 5 \times 10^{-12}$. Doubling the precision shrinks machine epsilon by a factor of $10^{6}$.

## Answer to Exercise 208 (on page 784)

$\boxed{-2x^3 + \frac{1}{2}x + 3.9}$

$\boxed{(4.5)x^2 + \pi x}$

$\boxed{7}$

$2x^{-10} + 4x - 1$

$x^{\frac{2}{3}}$

$\boxed{3x^{20} + 2x^{19} - 5x^{18}}$

## Answer to Exercise 209 (on page 784)

Standard form would be $-x^3 + 21x^2 - 1000x + \pi$. The degree is 3. The leading coefficient is $-1$

## Answer to Exercise 210 (on page 785)

$4^3 - (3)(4^2) + (10)(4) - 12 = 64 - 48 + 40 - 12$. So $y = 44$

## Answer to Exercise 211 (on page 788)

```
favorites[3] = "Gloves"
```

## Answer to Exercise 212 (on page 791)

```
pn2 = [2.5, 5.0, -2.0, -7.0, 4.0]
y = evaluate_polynomial(pn2, 8.5)
print("Polynomia 2: When x is 8.5, y is", y)

pn3 = [-9.0, 0.0, 0.0, 0.0, 0.0, 5.0]
y = evaluate_polynomial(pn3, 2.0)
print("Polynomial 3: When x is 2.0, y is", y)
```

## Answer to Exercise 213 (on page 793)

The polynomial crosses the y-axis at 6. When x is zero, all the terms are zero except the last one. Thus, you can easily tell that $x^3 - 7x + 6$ will cross the y-axis at $y = 6$.

Looking at the graph, you tell that the curve crosses the y-axes near -3, 1 and 2. If you plug those numbers into the polynomial, you would find that it evaluates to zero at each one. Thus, $x = -3$, $x = 1$, and $x = 2$ are roots.

## Answer to Exercise 214 (on page 796)

$3x^3 - 7x^2 + x - 18$ and $3x^5 - 7x^3 + x^2 - 12$

## Answer to Exercise 215 (on page 797)

$x^3 - 3x^2 + 5x$ and $x^5 - 3x^3 + 5x^2 - 2x + 6$

## Answer to Exercise 216 (on page 798)

```
def add_polynomials(a, b):
    degree_of_result = max(len(a), len(b))
    result = []
    for i in range(degree_of_result):
        if i < len(a):
            coefficient_a = a[i]
        else:
            coefficient_a = 0.0

        if i < len(b):
            coefficient_b = b[i]
        else:
            coefficient_b = 0.0

        result.append(coefficient_a + coefficient_b)
    return result
```

## Answer to Exercise 217 (on page 800)

```
def subtract_polynomial(a, b):
    neg_b = scalar_polynomial_multiply(-1.0, b)
    return add_polynomials(a, neg_b)
```

## Answer to Exercise 218 (on page 802)

$(3x^2)(5x^3) = 15x^5$

$(2x)(4x^9) = 8x^{10}$

$(-5.5x^2)(2x^3) = -11x^5$

$(\pi)(-2x^5) = -2\pi x^5$

$(2x)(3x^2)(5x^7) = 30x^{10}$

## Answer to Exercise 219 (on page 803)

$(3x^2)(5x^3 - 2x + 3) = 15x^6 - 6x^3 + 6x^2$

$(2x)(4x^9 - 1) = 8x^{10} - 2x$

$(-5.5x^2)(2x^3 + 4x^2 + 6) = 11x^5 - 22x^4 + 33x^2$

$(\pi)(-2x^5 + 3x^4 + x) = -2\pi x^5 + 3\pi x^4 + \pi x$

$(2x)(3x^2)(5x^7 + 2x) = 30x^{10} + 12x^4$

## Answer to Exercise 220 (on page 805)

$(2x + 1)(3x - 2) = 6x^2 - x - 2$

$(-3x^2 + 5)(4x - 2) = -12x^3 + 6x^2 + 20x - 10$

$(-2x - 1)(-3x - \pi) = 6x^2 + (4 + 2\pi)x + \pi$

$(-2x^5 + 5x)(3x^5 + 2x) = -6x^{10} + 12x^6 + 10x^2$

## Answer to Exercise 221 (on page 805)

The degree of the product is determined by the term that is the product of the highest degree term in $p_1$ and the highest degree term in $p_2$. Thus, the product of a degree 23 polynomial and a degree 12 polynomial has degree 35.

## Answer to Exercise 222 (on page 812)

1. $f'(t) = 3t^2 - 6t - 4$
2. $g'(t) = (\frac{-3}{4})2t^{-3/4-1} = \frac{-3}{2}t^{-7/4}$
3. $F'(r) = \frac{-15}{r^4}$
4. First, we expand the function by multiplying out the two binomials: $(3u-1)(u+2) = 3u^2 + 6u - u - 2$. Therefore, $H(u) = 3u^2 + 5u - 2$, and we can differentiate using what we have learned about differentiating polynomials. $H'(u) = 6u + 5$. In a later chapter, you will learn the Product rule, which will allow you to differentiate this function without multiplying out the binomials.

## Answer to Exercise 223 (on page 813)

```
def derivative_of_polynomial(pn):

    # What is the degree of the resulting polynomial?
    original_degree = len(pn) - 1
    if original_degree > 0:
        degree_of_derivative = original_degree - 1
    else:
        degree_of_derivative = 0

    # We can ignore the constant term (skip the first coefficient)
    current_degree = 1
    result = []

    # Differentiate each monomial
    while current_degree < len(pn):
        coefficient = pn[current_degree]
        result.append(coefficient * current_degree)
        current_degree = current_degree + 1

    # No terms? Make it the zero polynomial
    if len(result) == 0:
        result.append(0.0)

    return result
```

## Answer to Exercise 224 (on page 815)

(a) Velocity is the first derivative of the position function, $s'(t) = 4t^3 - 6t^2 + 2t - 1$. Acceleration is the derivative of the velocity function, $s''(t) = 12t^2 - 12t + 2$.

(b) $s'(1.5) = 4(1.5)^3 - 6(1.5)^2 + 2(1.5) - 1 = 2$ We should note that this is a measurement

and needs units to make sense. Since $s$ is in meters and $t$ is in seconds, our velocity should have units of $\frac{m}{s}$, so our final answer is $s'(1.5s) = 2\frac{m}{s}$.

(c) $s''(1.5) = 12(1.5)^2 - 12(1.5) + 2 = 11$. Similarly to part (b), our answer needs units. The units for acceleration are the units for velocity divided by the unit for time (because acceleration is a rate of change of velocity), and our final answer should be $s''(1.5s) = 11\frac{m}{s^2}$.

(d) When velocity and acceleration are occurring in the same direction (i.e. have the same sign), the speed (the absolute value of velocity) is increasing. Since $s'(1.5s)$ and $s''(1.5s)$ are both $> 0$, the speed of the object is increasing.

## Answer to Exercise 225 (on page 826)

$(2x-3)(2x+3) = 4x^2 - 9$

$(7+5x^3)(7-5x^3) = 49 - 25x^6$

$(x-a)(x+a) = x^2 - a^2$

$(3-\pi)(3+\pi) = 9 - \pi^2$

$(-4x^3+10)(-4x^3-10) = 16x^6 - 100$

$(x+\sqrt{7})(x-\sqrt{7}) = x^2 - 7$

$x^2 - 9 = (x+3)(x-3)$

$49 - 16x^6 = (7+4x^3)(7+4^3)$

$\pi^2 - 25x^8 = (\pi + 5x^4)(\pi - 5x^4)$

$x^2 - 5 = (x+\sqrt{5})(x-\sqrt{5})$

## Answer to Exercise 226 (on page 828)

$(x+1)^2 = x^2 + 2x + 1$

$(3x^5+5)^2 = 9x^10 + 30x^5 + 25$

$(x^3-1)^2 = x^6 - 2x^3 + 1$

$(x-\sqrt{7})^2 = x^2 - 2x\sqrt{7} + 7$

## Answer to Exercise 227 (on page 829)

$(x + \pi)^5 = x^5 + 5\pi x^4 + 10\pi^2 x^3 + 10\pi^3 + x^2 + 5\pi^2 x + \pi^5$

## Answer to Exercise 228 (on page 830)

```
"""Return the nth row (0-indexed) of Pascal's Triangle."""
n = int(input("Enter row index n: "))
row = [1]
for k in range(1, n + 1):
    row.append(row[-1] * (n - k + 1) // k)
return row
```

## Answer to Exercise 229 (on page 832)

1. The GCD of 18 and 15 is 3, and the least degree is 1, so $18x^3 - 15x = (3x)(6x^2 - 5)$.
2. The GCD of 4, 12, and 16 is 4, and the least degree is 1, so $4x^3 - 12x^2 + 16x = (4x)(x^2 - 3x + 4)$.
3. The GCD of 18, 24, and 30 is 6, and the least degree is 8, so $18x^{50} - 24x^{20} + 30x^8 = (6x^8)(3x^{42} - 4x^{12} + 5)$.

## Answer to Exercise 230 (on page 833)

1. 3 and 11 multiply to 33 and add to 14, so $x^2 + 14x + 33 = (x + 3)(x + 11)$.
2. $-6$ and $-10$ multiply to 60 and add to $-16$, so $x^2 - 16x + 60 = (x - 6)(x - 10)$.
3. 9 and $-6$ multiply to $-54$ and add to 3, so $x^2 + 3x - 54 = (x + 9)(x - 6)$.
4. $-5$ and 2 multiply to $-10$ and add to $-3$, so $x^2 - 3x - 10 = (x - 5)(x + 2)$.

## Answer to Exercise 231 (on page 835)

```
6
```

```
0
```

At $x = 2$: $4 + 6 - 4 = 6$. At $x = -4$: $16 - 12 - 4 = 0$. So $-4$ is a root and 2 is not. This matches the factorization $x^2 + 3x - 4 = (x + 4)(x - 1)$.

## Answer to Exercise 232 (on page 836)

At $x = 0$ both sides equal the constant term, $-10$, because $-5$ and 2 do multiply to $-10$. A single match proves little here. The polynomial has degree 2, so all three values must match, and two of them fail. The numbers $-5$ and 2 add to $-3$ instead of 3, so the signs are swapped. The correct factorization is $(x + 5)(x - 2)$.

## Answer to Exercise 233 (on page 837)

In Python, `^` does not mean "to the power of." It is a different operator, so `c` computes something other than $x^2 + 5x + 6$. The fix is to write `x**2`. Notice that the bad version still matches at $x = 1$ by chance, which is another reason to check several values.

## Answer to Exercise 234 (on page 837)

Pull out $2x$ to get $(2x)(x^2 - 3x - 4)$. The numbers $-4$ and 1 multiply to $-4$ and add to $-3$, so $2x^3 - 6x^2 - 8x = (2x)(x + 1)(x - 4)$.

```
def e(x):
    return 2*x**3 - 6*x**2 - 8*x

def h(x):
    return (2*x)*(x + 1)*(x - 4)

for x in [0, 1, 2, 3]:
    print(x, e(x), h(x), e(x) == h(x))

for r in [0, -1, 4]:
    print(r, e(r))
```

The polynomial has degree 3, so four values are enough, and all four lines say `True`. The roots are 0, $-1$, and 4, and each one prints 0.

## Answer to Exercise 235 (on page 839)

```
(x - 3)*(x + 3)
```

The polynomial $x^2 - 9$ is a difference of squares, so it factors into $(x - 3)(x + 3)$.

## Answer to Exercise 236 (on page 839)

Pull out 3 to get $3(x^2 + 4x - 5)$. The numbers 5 and $-1$ multiply to $-5$ and add to 4, so $3x^2 + 12x - 15 = 3(x + 5)(x - 1)$.

```
from sympy import symbols, factor, expand

x = symbols('x')
print(expand(3*(x + 5)*(x - 1)))
print(factor(3*x**2 + 12*x - 15))
```

```
3*x**2 + 12*x - 15
3*(x - 1)*(x + 5)
```

Expanding gives back the original polynomial, and SymPy's factors match yours with the binomials in a different order.

## Answer to Exercise 237 (on page 856)

x-intercept: $(-5/2, 0)$; y-intercept: $(0, 5/4)$; horizontal asymptote: $y = 2$; vertical asymptote: $x = -4$

## Answer to Exercise 239 (on page 859)

The correct answer is E. To explain, we examine the behavior of each function as $x \to \infty$.
A. $\lim_{x\to\infty} \frac{\sin 5x}{x} = \pm\infty \neq 5$
B. $\lim_{x\to\infty} 5x = \infty \neq 5$
C. $\lim_{x\to\infty} \frac{1}{x-5} = 0 \neq 5$ (this function does have a *vertical* asymptote at $x = 5$).
D. $\lim_{x\to\infty} \frac{5x}{1-x} = \frac{5}{-1} = -5 \neq 5$ (this function has a horizontal asymptote at $x = -5$).
E. $\lim_{x\to\infty} \frac{20x^2-x}{1+4x^2} = \frac{20}{4} = 5$.

## Answer to Exercise 239 (on page 859)

Factored form: $\frac{x^2(x+2)}{x(x+1)}$; hole: $(0,0)$; vertical asymptote: $x = -1$; oblique asymptote: $y = x + 1$

## Answer to Exercise 240 (on page 863)

1. $x^2 + 4x + 12 \frac{30x-27}{x^2-3x+2}$
2. $2 + \frac{6x^2-4x+13}{x^3-3x^2+2x-4}$
3. $3x - 2 + \frac{8x^2-6x+1}{x^3-3x}$

## Answer to Exercise 241 (on page 867)

1. $\frac{10}{7(x+6)} + \frac{-3}{7(x-1)}$
2. $\frac{1}{x^2+1} + \frac{x}{(x^2+1)^2}$
3. $\frac{-2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2}$

## Answer to Exercise 242 (on page 872)

$$\lim_{x\to -6^-} p(x) = -\infty, \lim_{x\to -6^+} p(x) = \infty$$

$$\lim_{x\to -5^-} p(x) = \lim_{x\to -5^+} p(x) = \lim_{x\to -5} p(x) = 1$$

$$\lim_{x\to -3^-} p(x) = \lim_{x\to -3^+} p(x) = \lim_{x\to -3} p(x) = \frac{1}{3}$$

$\lim_{x\to\infty} p(x) = 0$ called simply a limit, although it is a left-hand limit

## Answer to Exercise 243 (on page 875)

$$\lim_{x\to-\infty} 3^x + 1 = 1;\ \lim_{x\to4^+} \log_2(x-4) = -\infty;\ \lim_{x\to\infty} 2^{1-x} = 0;\ \lim_{x\to0^-} \log_{10}(-2x) = -\infty$$

## Answer to Exercise 244 (on page 877)

$\lim_{x\to-\infty} \tan^{-1}x = -\frac{\pi}{2}, \lim_{x\to\infty} \tan^{-1}x = \frac{\pi}{2}; \lim_{x\to-\infty} \frac{1}{1+e^{-x}} = 0, \lim_{x\to\infty} \frac{1}{1+e^{-x}} = 1$

## Answer to Exercise 245 (on page 879)

1. $\lim_{x\to-1_-} h(x) = 2$ and $\lim_{x\to-1_+} h(x) = 2$, therefore the limit exists and $\lim_{x\to-1} h(x) = 2$

   $\lim_{x\to0_-} h(x) = 3$ and $\lim_{x\to0_+} h(x) = 3$, therefore the limit exists and $\lim_{x\to0} h(x) = 3$

   $\lim_{x\to1_-} h(x) = 2$ and $\lim_{x\to1_+} h(x) = 2$, therefore the limit exists and $\lim_{x\to1} h(x) = 2$

2. $\lim_{x\to-1_-} f(x) = 2$ and $\lim_{x\to-1_+} f(x) = 2$, therefore the limit exists and $\lim_{x\to-1} f(x) = 2$.

   $\lim_{x\to0_-} f(x) = 3$ and $\lim_{x\to0_+} f(x) = 0$, and because $\lim_{x\to0_-} f(x) \neq \lim_{x\to0_+} f(x)$, the limit does not exist.

   $\lim_{x\to2_-} f(x) = -2$ and $\lim_{x\to2_+} f(x) = -2$, therefore the limit exists and $\lim_{x\to2} f(x) = -2$.

3. $\lim_{x\to-2_-} g(x) = -1$ and $\lim_{x\to-2_+} g(x) = -1$, therefore the limit exists and $\lim_{x\to-2} g(x) = -1$.

   $\lim_{x\to0_-} g(x) = 1$ and $\lim_{x\to0_+} g(x) = 1$, therefore the limit exists and $\lim_{x\to0} g(x) = 1$

   $\lim_{x\to1_-} g(x) = 2$ and $\lim_{x\to0_+} g(x) = 1$, and because $\lim_{x\to1_-} g(x) = 2 \neq \lim_{x\to0_+} g(x)$, the limit does not exist.

   $\lim_{x\to2_-} g(x) = 0$ and $\lim_{x\to2_+} g(x) = 0$, therefore the limit exists and $\lim_{x\to2} g(x) = 0$

## Answer to Exercise 246 (on page 882)

1. Never true. If a function is continuous at $a$, then $f(a) = \lim_{x\to a} f(x)$.
2. Never true. If a function is continuous at $a$, then $\lim x \to a^+ f(x) = \lim x \to a^- f(x)$
3. Always true. This is the definition of continuity.

4. Sometimes true. The derivative of $f$ at $x = 3$ exists for $f(x) = x^2$ but not for $f(x) = |x - 3|$.
5. Sometimes true. This statement is true for $f(x) = -(x - 3)^2$ but not for $f(x) = 4x$.

## Answer to Exercise 247 (on page 882)

1. $f(x)$ is not defined at $x = 3$. Therefore, it is also discontinuous at $x = 3$. As we learn about the continuity of polynomials, we will see why $f(x)$ is continuous everywhere else.
2. Here, $f(0)$ is defined, so we need to check if $\lim_{x\to 0} f(x) = f(0)$. The left and right limits as $x$ approaches 0 are the same $(\infty)$, so the limit exists. However, $f(0) = 1 \neq \lim_{x\to 0} f(x)$. Therefore, the function is discontinuous at $x = 0$.
3. In this function, $f(3)$ is defined, so we need to check if the limit equals the function value. The limit of $f(x)$ as $x$ approaches 3 is:

$$\lim_{x\to 3} \frac{3x^2 - 8x - 3}{x - 3} = \lim_{x\to 3} \frac{(3x + 1)(x - 3)}{x - 3} = \lim_{x\to 3} 3x + 1 = 10$$

So the limit exists, but $\lim x \to 3f(x) \neq f(3)$, and we see that the function is discontinuous at $x = 3$.

## Answer to Exercise 249 (on page 886)

1. True, $h(x)$ approaches 2 from the left and right, therefore the limit exists
2. False, $h(x)$ approaches 5 from the left and right, therefore the limit exists
3. False, $h(x)$ approaches 2 from the left and 4 from the right, therefore the limit does not exist
4. True, $\lim_{x\to 5} h(x) = h(5)$, therefore $h(x)$ is continuous at $x = 5$
5. True, $\lim_{x\to 4} h(x)$ does not exist, therefore $h(x)$ is discontinuous at $x = 4$
6. False, $h(2) = 1 \neq 2 = \lim_{x\to 2} h(x)$

## Answer to Exercise 249 (on page 886)

1. From the quotient law, we know that:

$$\lim_{x\to 3}\frac{f(x)}{g(x)}=\frac{\lim_{x\to 3} f(x)}{\lim_{x\to 3} g(x)}$$

From the graph, we see that:

$$\lim_{x\to 3} f(x)=-3$$

and that:

$$\lim_{x\to 3} g(x)=1$$

Substituting these values, we get:

$$\lim_{x\to 3}\frac{f(x)}{g(x)}=\frac{-3}{1}=-3$$

2. From the Sum Law, we know that:

$$\lim_{x\to 2}[f(x)+5g(x)]=\lim_{x\to 2} f(x)+\lim_{x\to 2} 5g(x)$$

and applying the Constant Multiple Law, we see that:

$$\lim_{x\to 2}[f(x)+5g(x)]=\lim_{x\to 2} f(x)+5\lim_{x\to 2} g(x)$$

Examining the graph of $f(x)$ and $g(x)$, we can determine that

$$\lim_{x\to 2} f(x)=-2$$

and

$$\lim_{x/to2} g(x)=0$$

Substituting these values, we get:

$$\lim_{x\to 2}[f(x)+5g(x)]=-2+5\cdot 0=-2$$

3. From the quotient law, we see that:

$$\lim_{x\to -1}\frac{3g(x)}{f(x)}=\frac{\lim_{x\to -1} 3g(x)}{\lim_{x\to -1} f(x)}$$

Applying the Constant Multiple Law, we get:

$$\lim_{x\to -1}\left[\frac{3g(x)}{f(x)}\right] = \frac{3\lim_{x\to -1} g(x)}{\lim_{x\to -1} f(x)}$$

From the graph, we see that:

$$\lim_{x\to -1} f(x) = 2$$

and

$$\lim_{x\to -1} g(x) = 2$$

Substituting, we get:

$$\lim_{x\to -1}\left[\frac{3g(x)}{f(x)}\right] = \frac{3\cdot 2}{2} = 3$$

4. Applying the Product and Constant Multiple Laws, we get:

$$\lim_{x\to 0}[f(x)\cdot 5g(x)] = \lim_{x\to 0} f(x)\cdot 5\cdot \lim_{x\to 0} g(x)$$

Examining the graphs, we see that $\lim_{x\to 0} f(x)$ does not exist and $\lim_{x\to 0} g(x) = 1$. Because $\lim_{x\to 0} f(x)$ does not exist, $\lim_{x\to 0} f(x)\cdot 5\cdot \lim_{x\to 0} g(x)$ also does not exist.

5. Applying the Difference and Constant Multiple Laws, we see that:

$$\lim_{x\to -1}[f(x) - 3g(x)] = \lim_{x\to -1} f(x) - 3\cdot \lim_{x\to -1} g(x)$$

Examining the graphs, we see that:

$$\lim_{x\to -1} f(x) = 2$$

and

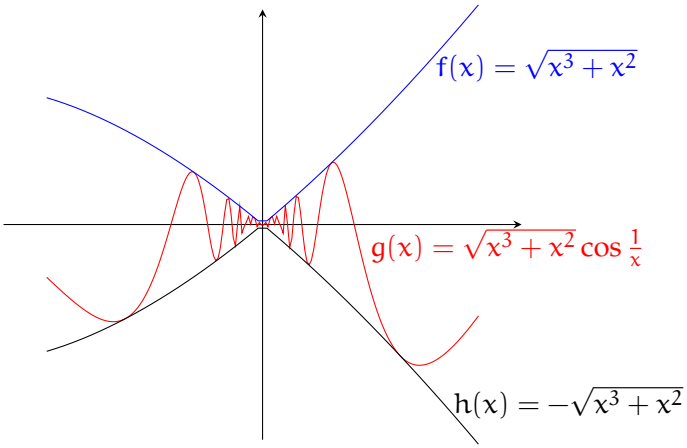
$$\lim_{x\to -1} g(x) = 2$$

Substituting, we get that:

$$\lim_{x\to -1}[f(x) - 3g(x)] = 2 - 3\cdot 2 = 2 - 6 = -4$$

## Answer to Exercise 250 (on page 889)

Let $f(x) = -\sqrt{x^3 + x^2}$ and $h(x) = \sqrt{x^3 + x^2}$. Near 0, $f(x) \leq \sqrt{x^3 + x^2} \cos \frac{1}{x} \leq h(x)$. Additionally, $\lim_{x \to 0} f(x) = \lim_{x \to 0} h(x) = 0$. Therefore, by the Squeeze Theorem, we can state that $\lim_{x \to 0} \sqrt{x^3 + x^2} \cos \frac{1}{x} = 0$. Plotting all three functions, we can confirm our answer:




## Answer to Exercise 251 (on page 890)

The question tells us that the function in question, $f(x)$, is between two other functions. $\lim_{x \to 2} 2x + 3 = 2(2) + 3 = 7$ and $\lim_{x \to 2} x^2 - 2x + 7 = 2^2 - 2(2) + 7 = 7$. Since the limits are equal, by Squeeze Theorem we can also know that $\lim_{x \to 2} f(x) = 7$.

## Answer to Exercise 252 (on page 890)

Note that we can only evaluate the limit from the right, as the domain for this function is $x \geq 0$. Since the range of the sine function is $[-1, 1]$, we can state that

$$-1 \leq \sin \frac{\pi}{x} \leq 1$$

and therefore

$$\frac{1}{e} \leq e^{\sin \frac{\pi}{x}} \leq e$$

. Because we assume the positive root, it is also true that

$$\frac{\sqrt{x}}{e} \leq \sqrt{x} e^{\sin \frac{\pi}{x}} \leq \sqrt{x} e$$

. Taking the limits of the border functions, we see that

$$\lim_{x\to 0^+} \frac{\sqrt{x}}{e} = \lim_{x\to 0^+} \sqrt{x}e = 0$$

Therefore,

$$\lim_{x\to 0^+} \sqrt{x}e^{\sin\frac{\pi}{x}} = 0$$

## Answer to Exercise 253 (on page 892)

1. define $f(x) = 2x^4 + x - 12$, $a = 1$, $b = 2$, and $N = 0$. Calculate $f(a)$ and $f(b)$:

   $$f(1) = -9 \text{ and } f(2) = 22$$

   Since $f(x)$ is a polynomial, it is continuous on the interval $x \in [1, 2]$ and we see that $f(a) < 0 < f(b)$. Therefore, there exists some $c \in [1, 2]$ such that $f(c) = 0$.

2. First, we can rearrange the equation we are considering and define $f(x) = \ln x - 3x + 4\sqrt{x}$, and realize we are looking for values where $f(x) = 0$. Both $\ln x$ and $\sqrt{x}$ are only continuous for $x > 0$. The interval we are interested in, $x \in [2, 3]$, is in the domain of continuity for both $\ln x$ and $\sqrt{x}$. Defining $a = 2$, $b = 3$, and $N = 0$, we find that $f(a) = 0.35 > N > -0.973 = f(b)$. Since $N \in [f(b), f(a)]$, there must exist some $c$ such that $f(c) = N = 0$ and there is a solution to the equation $\ln x = 3x - 4\sqrt{x}$ on the interval $x \in (2, 3)$.

3. Similar to above, define $f(x) = 2\sin x - 3x^2 + 2x$, $a = 1$, $b = 2$, and $N = 0$. Calculate that $f(a) = 0.683$ and $f(b) = -6.181$. Since $f(x)$ is continuous on the interval $x \in [1, 2]$ and $f(b) < N < f(a)$, there exists some $c \in (1, 2)$ such that $f(c) = 0$. Therefore, there is a solution to $\sin x = 3x^2 - 2x$ on the interval $x \in (1, 2)$.

## Answer to Exercise 254 (on page 896)

To sketch the curve, we begin by choosing several values of $t$ and computing the corresponding $x$- and $y$-values.

| $t$ | $x$ | $y$ |
|---|---|---|
| $-2$ | $-1$ | 4 |
| $-1$ | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 2 | 1 |
| 2 | 3 | 4 |

Plotting these points produces a parabola opening upward. As $t$ increases, the $x$-values increase and the curve is traced from left to right.

Although the rectangular equation of the curve is

$$y = (x-1)^2,$$

the parametric equations determine the direction in which the curve is drawn, which cannot be seen from the rectangular form alone.

## Answer to Exercise 255 (on page 898)

To sketch the curve, we begin by evaluating the parametric equations at several values of $t$.

| $t$ | $x$ | $y$ |
|---|---|---|
| 0 | 4 | 3 |
| 1 | 2 | 5 |
| 2 | 0 | -1 |
| 3 | -2 | -15 |

Plotting these points produces a curve that opens downward. As $t$ increases, the $x$-values decrease, indicating that the curve is traced from right to left.

The Python animation confirms this behavior by showing the curve being drawn progressively as $t$ increases. The moving point illustrates both the direction of motion and how the curve is traced over time, reinforcing the hand-drawn sketch.

## Answer to Exercise 256 (on page 901)

First compute the derivatives:

$$\frac{dx}{dt} = 3t^2 - 3, \quad \frac{dy}{dt} = 2t.$$

Horizontal tangents occur when $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$.

$$2t = 0 \Rightarrow t = 0.$$

At $t = 0$, the point on the curve is

$$x = 0, \quad y = 0.$$

Vertical tangents occur when $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$.

$$3t^2 - 3 = 0 \Rightarrow t = \pm 1.$$

At $t = 1$, the point is

$$(-2, 1),$$

and at $t = -1$, the point is

$$(2, 1).$$

## Answer to Exercise 257 (on page 902)

First compute the derivatives with respect to t:

$$\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = 2t.$$

The derivative of the curve is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t}{2} = t.$$

At $t = 1$,

$$\frac{dy}{dx} = 1.$$

Thus, the slope of the tangent line at the corresponding point on the curve is 1.

## Answer to Exercise 258 (on page 905)

Two sets of parametric equations can represent the same graph but describe different curves because the parameter controls how the graph is traced, not just the shape of the graph itself.

Different parametric equations may generate the same set of points in the plane while differing in one or more of the following ways:

- the direction of motion along the curve,
- the speed at which the curve is traced,
- the starting point of the curve,
- or the number of times the curve is traced.

## Answer to Exercise 259 (on page 916)

The radius is 5 m, because the coefficients of both the $x$ and $y$ functions is 5.

Recall that the velocity in each direction is given by the derivative of the position functions:

$$v_x(t) = \frac{d}{dt}\left[5\cos(3t)\right] = -15\sin(3t)$$

$$v_y(t) = \frac{d}{dt}\left[5\sin(3t)\right] = 15\cos(3t)$$

The overall *speed* can be found from the $x$ and $y$ components of the velocity:

$$\begin{aligned}|v| &= \sqrt{v_x^2 + v_y^2} \\ &= \sqrt{[-15\sin(3t)]^2 + [15\cos(3t)]^2} \\ &= \sqrt{15^2\left[\sin(3t)^2 + \cos(3t)^2\right]} \\ &= \sqrt{15^2} = 15\frac{m}{s}\end{aligned}$$

Notice that this is the coefficient for both components of the velocity.

To complete the path, the particle must travel $10\pi$ m. If the speed is $15\frac{m}{s}$, then the time it takes is:

$$t = \frac{d}{v} = \frac{10\pi \text{ m}}{15\frac{m}{s}} = \frac{2\pi}{3}s \approx 2.09 \text{ s}$$

## Answer to Exercise 260 (on page 922)

$$\frac{120 \text{ km}}{1\text{hour}} = \frac{1000 \text{ m}}{1 \text{ km}}\frac{120 \text{ km}}{1\text{hour}}\frac{1 \text{ hour}}{3600 \text{ seconds}} = 33.3 \text{ m/s}$$

$$F = \frac{mv^2}{r} = \frac{0.4(33.3)^2}{200} = 2.2 \text{ newtons}$$

## Answer to Exercise 261 (on page 923)

Givens:

$$T_{max} = (20\text{kg}) \cdot \left(9.8\frac{m}{s^2}\right) = 196\text{N}$$
$$r = 0.35\text{m}$$
$$m_{whistle} = 165\text{g} = 0.165\text{kg}$$

Unknown:

$$v = ?$$
$$f = ?$$

Equation(s):

$$F = ma$$
$$a = \frac{v^2}{r}$$
$$v = rf$$

Solution: First, we find the tangential speed if the tension in the string is the maximum tension:

$$T_{max} = m_{whistle}a = m_{whistle}\frac{v^2}{r}$$
$$v = \sqrt{\frac{T_{max}r}{m_{whistle}}}$$
$$v = \sqrt{\frac{196\text{N}\,(0.35\text{m})}{0.165\text{kg}}} \approx 0.645\frac{m}{s}$$

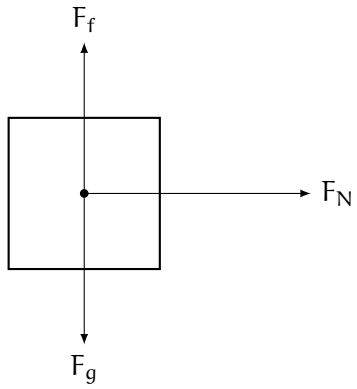
Therefore, the maximum tangential speed of the whistle before the string breaks is $0.645\frac{m}{s}$. Now finding the equivalent frequency (rotations per second):

$$f = \frac{v}{r} = \frac{0.645\frac{m}{s}}{0.35\text{m}} \approx 1.842\text{Hz}$$

So, to break the string, the lifeguard would have to spin it at nearly 2 rotations per second, which is achievable. The lifeguard could possibly break the string.

## Answer to Exercise 262 (on page 923)

There are three forces acting on the rider: gravity, friction with the wall, and the normal force with the wall:




The FBD lets us write equations for Newton's Second Law in each dimension:

$$(1)\ \ ma_x = F_N$$

$$(2)\ \ ma_y = F_f - F_g = \mu F_N - mg$$

Because the rider doesn't fall down, we know that $a_y = 0\frac{m}{s^2}$ and therefore equation (2) becomes:

$$\mu F_N - mg = 0 \rightarrow F_N = \frac{mg}{\mu}$$

Having solved for $F_N$, we substitute for it into equation (1):

$$ma_x = \frac{mg}{\mu} \rightarrow a_x = \frac{g}{\mu}$$

Since we know $g$ and $\mu$, we can calculate $a_x$:

$$a_x = \frac{9.8\frac{m}{s^2}}{0.32} = 30.625\frac{m}{s^2}$$

This is the minimum acceleration needed to keep the rider from slipping down. We can now use the relationship between centripetal acceleration, tangential velocity, and the radius to find the angular velocity:

$$a = \frac{v^2}{r} = \frac{(\omega r)^2}{r} = \omega^2 r$$

$$\omega = \sqrt{\frac{a}{r}} = \sqrt{\frac{30.625\frac{m}{s^2}}{10\ m}} \approx 1.75\frac{rad}{s}$$

## Answer to Exercise 263 (on page 926)

Givens:

- Mass of tetherball: $m = 0.5$ kg
- Length of string: $L = 2.5$ m
- Angle with vertical: $\theta = 38°$

The components of the tension force are separated as follows:

$$T\cos\theta = mg, \qquad T\sin\theta = \frac{mv^2}{r}$$

Dividing the second equation by the first equation gives:

$$\tan\theta = \frac{v^2}{rg} \rightarrow v = \sqrt{rg\tan\theta}$$

But we are not given $r$. We can find $r$ using trigonometry:

$$r = L\sin\theta$$

Now we can find $v$:

$$v = \sqrt{(L\sin\theta)g\tan\theta}$$

Solving this gives us:

$$v = \sqrt{(2.5\ m\sin 38°)(9.8\frac{m}{s^2})\tan 38°} \approx 3.4\frac{m}{s}$$

The speed of the tetherball is approximately $3.4\frac{m}{s}$.

## Answer to Exercise 264 (on page 930)

- At the top of the loop: At the top of the loop, both the gravitational force and the normal force point downward. Therefore, we can write:

$$F_{net} = F_N + F_g = \frac{mv^2}{r}$$

Solving for the normal force gives:

$$F_N = \frac{mv^2}{r} - F_g$$

Substituting for $F_g = mg$ and $r = \frac{d}{2} = 25$ m:

$$F_N = \frac{(1200 \text{ kg})(25\frac{m}{s})^2}{25 \text{ m}} - (1200 \text{ kg})(10\frac{m}{s^2})$$

$$F_N = 30000 \text{ N} - 12000 \text{ N} = 18000 \text{ N}$$

- At the bottom of the loop: At the bottom of the loop, the gravitational force points downward while the normal force points upward. Therefore, we can write:

$$F_{net} = F_N - F_g = \frac{mv^2}{r}$$

Solving for the normal force gives:

$$F_N = \frac{mv^2}{r} + F_g$$

Substituting for $F_g = mg$ and $r = \frac{d}{2} = 25$ m:

$$F_N = \frac{(1200 \text{ kg})(25\frac{m}{s})^2}{25 \text{ m}} + (1200 \text{ kg})(10\frac{m}{s^2})$$

$$F_N = 30000 \text{ N} + 12000 \text{ N} = 42000 \text{ N}$$

## Answer to Exercise 265 (on page 936)

The correct answer is: The frequency would increase by a factor of $\sqrt{2}$.

The frequency of oscillation for a mass-spring system is given by the formula:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

where $k$ is the spring constant and $m$ is the mass attached to the spring. Since frequency is inversely proportional to the square root of the mass, if the mass is halved, the new

frequency $f'$ can be calculated as follows:

$$\begin{aligned} f' &= \frac{1}{2\pi}\sqrt{\frac{k}{m/2}} \\ &= \frac{1}{2\pi}\sqrt{\frac{2k}{m}} \\ &= \sqrt{2}\cdot\frac{1}{2\pi}\sqrt{\frac{k}{m}} \\ &= \sqrt{2}\cdot f \end{aligned}$$

Therefore, the frequency increases by a factor of $\sqrt{2}$ when the

## Answer to Exercise 266 (on page 938)

First, we find the angular frequency $\omega$ using the formula:

$$w = 2\pi f = \sqrt{\frac{k}{m}} = \sqrt{\frac{400\,\mathrm{N/m}}{4\,\mathrm{kg}}} = \sqrt{100} = 10\,\mathrm{rad/s}$$

Therefore, the general equation for the position of a mass-spring system undergoing simple harmonic motion is:

$$x(t) = A\cos(\omega t + \phi)$$

becomes $x = 6\sin(10t + \phi)$ since the initial velocity is 0, meaning it starts at maximum displacement. Solving for $\phi$ using the initial condition $x(0) = 6$ cm:

$$6 = 6\sin(\phi) \implies \sin(\phi) = 1 \implies \phi = \frac{\pi}{2}$$

So the base equation becomes:

$$x(t) = 6\sin(10t + \frac{\pi}{2})$$

Which is option C.

## Answer to Exercise 267 (on page 939)

To find the speed of the ball as it leaves the spring, we can use the principle of conservation of mechanical energy. The net energy in the system remains constant, meaning that the potential energy stored in the compressed spring is converted into the kinetic energy of the ball when it is released. The potential energy stored in the spring when compressed

is given by:

$$U_s = \frac{1}{2}kx^2 = \frac{1}{2}(500)(-0.24)^2 = 14.4\,\text{J}$$

The kinetic energy of the ball as it leaves the spring is the same amount, so we set the potential energy equal to the kinetic energy:

$$\frac{1}{2}mv^2 = U_s$$

Solving for $v$:

$$v = \sqrt{\frac{2U_s}{m}} = \sqrt{\frac{2(14.4)}{0.2}} = \sqrt{144} = 12\,\text{m/s}$$

The speed of the ball as it leaves the spring is 12 m/s.

## Answer to Exercise 268 (on page 941)

The correct differential equation is $x'' + 100x = 0$. This is derived from the equation of motion for any simple harmonic oscillator: $mx'' + kx = 0$, where $m = 2\,\text{kg}$ and $k = 200\,\text{N/m}$. Dividing by $m$, we get $x'' + \frac{k}{m}x = 0$, which simplifies to $x'' + 100x = 0$.

## Answer to Exercise 269 (on page 942)

- This system is damped. Specifically, it is underdamped because the amplitude of oscillation decreases over time.
- To make this system undergo simple harmonic motion, we would need to remove the damping term from the equation of motion. This means setting the damping coefficient $c$ to zero in the differential equation, which would eliminate the exponential decay factor in the position function. The equation becomes $x(t) = 0.15\cancel{e^{-0.40t}}\cos(4t + 0.30) \to x(t) = 0.15\cos(4t + 0.30)$

## Answer to Exercise 270 (on page 943)

1. Notice that the system equation is linear as it is defined with an $x''$ and $x$.The general solution to the differential equation $x'' + 16x = 0$ is of the form $x(t) = A\cos(4t) + B\sin(4t)$. This comes from the idea that the characteristic equation $r^2 + 16 = 0$ has roots $r = \pm 4i$, where the $\beta$ term is 4. Using the initial conditions $x(0) = 0.20$ and $v(0) = x'(0) = 0$, we find our velocity equation to be $v(t) = x'(t) = -4A\sin(4t) + 4B\cos(4t)$.

   From $x(0) = A\cos 0 + B\sin 0 = A(1) + B(0) = A = 0.20$, we find that $A = 0.20$.

From $v(0) = -4A\sin 0 + 4B\cos 0 = -4(0.20)(0) + 4B(1) = 4B = 0$, we find that $B = 0$. Therefore, the solution is:

$$x(t) = 0.20\cos(4t)$$

2. To find the acceleration, we double differentiate $x(t)$ with respect to time: To find the velocity, we differentiate $x(t)$ with respect to time:

$$v(t) = x'(t) = -0.80\sin(4t)$$

$$a(t) = v'(t) = -3.20\cos(4t)$$

Since cosine and sine functions oscillate between -1 and 1 (regardless of angular frequency $\omega = 4$), the maximum velocity is 0.80 m/s and the maximum acceleration is $3.20\,m/s^2$.

## Answer to Exercise 271 (on page 945)

We know that the restoring force for a pendulum is given by $F = -mg\sin\theta$. For small angles, we can approximate $\sin\theta \approx \theta$ (in radians), leading to $F \approx -mg\theta$. For any point on the circle, we can relate the angular displacement $\theta$ to the arc length $x$ using $x = L\theta$, where $L$ is the length of the pendulum. Thus, we can rewrite the restoring force in terms of the second derivative, which represents angular acceleration:

$$a_{tan} = \frac{d^2x}{dt^2} = L\frac{d^2\theta}{dt^2} \tag{107.9}$$

Using Newton's Second Law, we have:

$$mL\frac{d^2\theta}{dt^2} = -mg\sin\theta \tag{107.10}$$

$$L\frac{d^2\theta}{dt^2} + g\sin\theta = 0 \tag{107.11}$$

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0 \tag{107.12}$$

But for small angles, we can approximate $\sin\theta \approx \theta$, leading to:

$$[label = eq : linearpendulum]\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0 \tag{107.13}$$

This is a second-order linear differential equation with constant coefficients, similar to the mass-spring system problems discussed earlier. The angular frequency $\omega$ of the pendulum can be identified as:

$$\omega = \sqrt{\frac{g}{L}} \tag{107.14}$$

Therefore, the period T of the pendulum is given by:

$$T = 2\pi\sqrt{\frac{L}{g}} \tag{107.15}$$

## Answer to Exercise 272 (on page 946)

The correct answer is: Approximately 4.4 seconds.

The period T of a simple pendulum is given by the formula:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

where L is the length of the pendulum and g is the acceleration due to gravity.

On Earth, the period is 2 seconds, so we can set up the equation:

$$2 = 2\pi\sqrt{\frac{L}{g_{\text{Earth}}}}$$

This shows that T is inversely proportional to the square root of g. We can rearrange this equation to solve for L. So if g decreases by a factor of 6, then T will increase by a factor of $\sqrt{6}$.

Now, on the Moon, the acceleration due to gravity is approximately $\frac{1}{6}g_{\text{Earth}}$. We can substitute this into the period formula:

$$T_{\text{Moon}} = 2\pi\sqrt{\frac{L}{g_{\text{Moon}}}} = 2\pi\sqrt{\frac{L}{\frac{1}{6}g_{\text{Earth}}}} = 2\pi\sqrt{\frac{6L}{g_{\text{Earth}}}} = \sqrt{6}\cdot 2\pi\sqrt{\frac{L}{g_{\text{Earth}}}} = \sqrt{6}\cdot T_{\text{Earth}}$$

Since our original period on Earth is 2 seconds, we have:

$$T_{\text{Moon}} = \sqrt{6}\cdot 2 \approx 4.9 \text{ seconds}$$

Which is equal to option (c).

## Answer to Exercise 260 (on page 954)

$$v = \sqrt{3.721(3.4\times 10^6)} = 3,557 \text{ m/s}$$

The circular orbit is $2\pi(3.4 \times 10^6) = 21.4 \times 10^6$ meters in circumference.

The period of the orbit is $(21.4 \times 10^6)/3,557 \approx 6,000$ seconds.

## Answer to Exercise 274 (on page 958)

Using equation **??**:

1. $$\begin{aligned} v_{esc} &= \sqrt{\frac{2GM}{R}} \\ &= \sqrt{\frac{2(6.674 \times 10^{-11}\ \text{N m}^2/\text{kg}^2)(6.42 \times 10^{23}\ \text{kg})}{3.4 \times 10^6\ \text{m}}} \\ &\approx 5,027\ \text{m/s} \end{aligned}$$

2. Now the radius is $3.4 \times 10^6 + 11,000 = 3.411 \times 10^6$ meters, because the satellite is 11 km above the surface. Recalculating:

$$\begin{aligned} v_{esc} &= \sqrt{\frac{2GM}{R}} \\ &= \sqrt{\frac{2(6.674 \times 10^{-11}\ \text{N m}^2/\text{kg}^2)(6.42 \times 10^{23}\ \text{kg})}{3.411 \times 10^6\ \text{m}}} \\ &\approx 5,013\ \text{m/s} \end{aligned}$$

## Answer to Exercise 275 (on page 962)

The torque on the door is equal to $rF\sin\theta$. Since our force is tangential, the sin component is at its max: $\sin 90 = 1$, our torque simplifies to $\tau = rF = 0.5 \cdot 38 = 19$ Newton-Meters.

## Answer to Exercise 276 (on page 963)

1. Using $\tau = rF\sin\theta$, we can state that the torque is $\tau = (2.0)(10)\sin(30°) = 20(0.5) = 10$ N·m.
2. Because The force has an upward component at the right end, so the beam tends to rotate counterclockwise.

## Answer to Exercise 277 (on page 965)

1. Since the beam center is 2.0 m from the left end, and the pivot is at 1.0 m from the left end, so the weight is $W = mg = 20(9.8)$,

$$W = 196 \text{ N}, \quad \text{acting 1.0 m right of pivot}$$

2. We can choose our own torque direction, so let's set counterclockwise be positive. The left end is 1.0 m to the left of the pivot and right end is 3.0 m to the right of the pivot. The beam's weight acts 1.0 m to the right of the pivot.

   Since there are no angles and all pivots forces are perpendicular, there are three torques:

$$\tau_{50} = (1.0)(50) = +50 \text{ N} \cdot \text{m} \qquad \text{CCW}$$
$$\tau_W = -(1.0)(196) = -196 \text{ N} \cdot \text{m} \qquad \text{CW}$$
$$\tau_F = -(3.0)(F) \text{ N} \cdot \text{m} \qquad \text{CW}$$

   So equilibrium is solved by:

$$50 - 196 - 3F = 0$$

3. 

$$\begin{aligned} 50 - 196 - 3F &= 0 \\ -146 - 3F &= 0 \\ -3F &= 146 \\ F &= -48.7 \text{ N} \end{aligned}$$

   The negative informs us that the torques act opposite the direction we assumed, so the force acts $F = 48.7$ N upward at the right end.

## Answer to Exercise 278 (on page 966)

You can pick your own pivot, so let's choose the point that A and B stem from as our pivot. This eliminates those forces, as the $r$ component of the torque would equal zero. Recall that the net torque is $\sum \tau = \tau_{CCW} - \tau_{CW}$. Analyzing the remaining torques:

- C is 1 m from the pivot, so the torque is $\tau_C = (1)C \sin -90°$. It acts downwards, so our torque is clockwise.
- D acts directly parallel the rotating rod, adding no additional torque on the rod.
- E is 2 m from the pivot, and acting at a $-135°$ angle from the horizontal. Our torque can be calculated from $\tau_E = (2)E \sin -135°$. Note that using $-45°$ provides the same

answer. This torque is also clockwise.

Our net torque is $C \sin -90^\circ + (2)E \sin -135^\circ = 0$

## Answer to Exercise 279 (on page 967)

Using Equation 109.8, we can rearrange to solve for $\alpha$ as $\alpha = \frac{rF \sin \theta}{I}$.

$$\alpha = \frac{rF \sin \theta}{I} = \frac{0.40 \cdot 25 \sin 60}{0.8} \approx 10.825 \text{ rad / s}^2$$

## Answer to Exercise ?? (on page 968)

We choose counterclockwise torques to be positive. The net torque is

$$\sum \tau = r_1 F_1 \sin \theta_1 - r_2 F_2 \sin \theta_2$$

Using the rotational form of Newton's second law,

$$\sum \tau = I\alpha$$

Solving for I,

$$I = \frac{r_1 F_1 \sin \theta_1 - r_2 F_2 \sin \theta_2}{\alpha}$$

Substituting values,

$$I = \frac{(0.50)(30) \sin 90^\circ - (0.30)(20) \sin 45^\circ}{8.0}$$

$$I = \frac{15 - 4.24}{8.0} \approx 1.35 \text{ kg·m}^2$$

## Answer to Exercise 281 (on page 969)

$$\begin{aligned} I_{end} &= I_{CM} + MD^2 \\ &= \frac{1}{2}ML^2 + M\left(\frac{L}{2}\right)^2 \\ &= \frac{1}{2}ML^2 + M\left(\frac{L^2}{4}\right) \\ &= \left(\frac{1}{12} + \frac{1}{4}\right)ML^2 \\ &= \left(\frac{4}{12}\right)ML^2 \\ &= \frac{1}{3}ML^2 \end{aligned}$$

## Answer to Exercise 282 (on page 974)

As we can see, each unit square is equivalent to 1 torque $\cdot$ time. Counting the partial and full squares on $t = [2, 6]$, we get approximately 8 unit squares. This comes from the (approximately whole) 4 squares under $y = 2$, combining all squares from $y = [2, 4]$ to form 2 squares, and the additional partial area forming approximately another 2 squares, resulting in an angular impulse of approximately 8 $N \cdot m \cdot s^{-1}$.

This is a practice in simple integration, which is a method of finding areas under given functions. Integrating the given function from $x = [0, 8]$ gives us 7.51988482389 $N \cdot m \cdot s^{-1}$, a value pretty close to our estimate.

## Answer to Exercise 283 (on page 1012)

The Earth and 1 kg on the surface would attract each other with a force of:

$$F_g = \frac{\left(6.67430 \times 10^{-17}\right)\left(5.97219 \times 10^{24}\right)(1)}{6,371^2} = \frac{3.98583 \times 10^8}{4.0590 \times 10^7} = 9.7987 \text{ N}$$

Thus, if the Earth were still and alone in the universe, the oceans would form a perfect sphere.

## Answer to Exercise 284 (on page 1013)

$$F_c = \frac{(1)(465)^2}{6,371,000} = 0.03373 \text{ N}$$

So the spinning of the Earth is trying to throw you into space, but the force of gravity is about 289 times more powerful.

This centripetal force decreases as you move from the equator to the north pole. In fact, at the north pole, there is no centripetal force. Thus, the spinning of the Earth makes the oceans an oblate ellipsoid instead of a perfect sphere: the diameter going from pole-to-pole is shorter than a diameter measured at the equator.

You should feel a teensy-tiny bit lighter on your feet at the equator than you do at the north pole: 0.34% lighter.

## Answer to Exercise 285 (on page 1014)

Overhead, the moon is $384,467 - 6,371 = 378,096$ km from your 1 kg mass.

$$F_g = \frac{gm_1m_2}{r^2} = \frac{\left(6.67430 \times 10^{-17}\right)\left(7.347673 \times 10^{22}\right)(1)}{378,096^2} = \frac{4.9040574 \times 10^6}{1.42956585216 \times 10^{11}} = 3.43058{\times}10^{-5} \text{ N}$$

This is a very small force: The force due to Earth's gravity is nearly three hundred thousand times stronger.

Underfoot, the moon is $384,467 + 6,371 = 390,838$

$$F_g = \frac{gm_1m_2}{r^2} = \frac{\left(6.67430 \times 10^{-17}\right)\left(7.347673 \times 10^{22}\right)(1)}{390,838^2} = \frac{4.9040574 \times 10^6}{1.52754 \times 10^{11}} = 3.2103{\times}10^{-5} \text{ N}$$

The force due to the moon's gravity is about 6% stronger when the moon is overhead than when it is underfoot.

## Answer to Exercise 286 (on page 1015)

If we let $r$ be the distance (in km) from the center of the Earth to the center of mass, the distance from the center of the mass to the center of the moon is $384,467 - r$.

To find the balance point, multiply each mass by how far it is from the center of mass:

$$\left(5.97219 \times 10^{24}\right) r = \left(7.347673 \times 10^{22}\right) (384,467 - r)$$

Solving for $r$:

$$r = \frac{4,730.15}{1 + 0.0123} = 4,673 \text{ km}$$

The point on the Earth closest to this? It is where the moon is directly overhead. The it is $6,371 - 4,673 = 1,698$ km from the center of mass.

The point on the Earth farthest from this? It is where the moon is directly underfoot. The it is $6,371 + 4,673 = 11,044$ km from the center of mass.

## Answer to Exercise 287 (on page 1016)

First, lets figure out $\omega$. It travels through $2\pi$ radians in 27.3 days. 27.3 days = 2,358,720 seconds. $\omega = \frac{2\pi}{2,358,720} = 2.663811435 \times 10^{-6}$

$$F_c = (1)(11,044,000)(2.663811435 \times 10^{-6})^2 = 7.8365 \times 10^{-5}$$

Now the weakest:

$$F_c = (1)(1,698,000)(2.663811435 \times 10^{-6})^2 = 1.20512 \times 10^{-5}$$

## Answer to Exercise 288 (on page 1017)

Closest to the moon, the gravitational force of the moon and the centripetal forces are in the same direction: toward the moon.

$$F_{total} = 1.20488 \times 10^{-5} + 3.43045 \times 10^{-5} = 4.6356 \times 10^{-5} \text{ N}$$

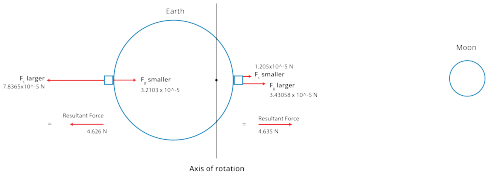
Farthest from the moon, the gravitational force of the moon and the centripetal forces are in opposite directions:

$$F_{total} = 7.8367 \times 10^{-5} - 3.2104 \times 10^{-5} = 4.62604^{-5}\text{N}$$

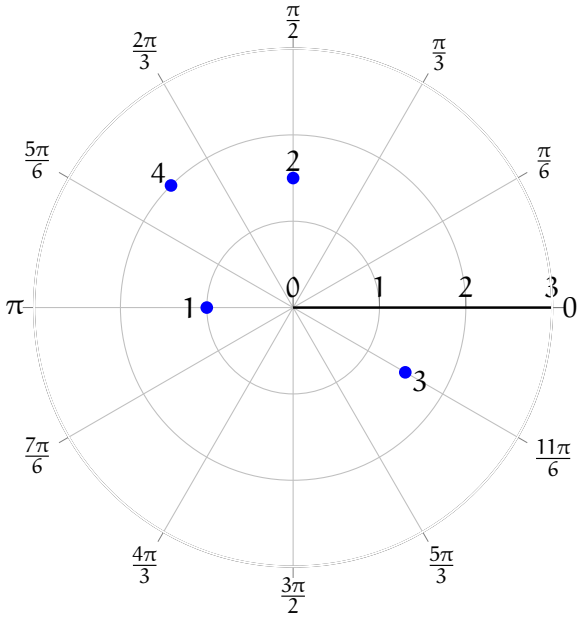
This is great conclusion: The two forces are basically equal: one pulls the water closest to the moon toward the moon, the other pulls water farthest from the moon away from the moon.

Both forces are pretty small: The force due to Earth's gravity is about $211,000$ times more than either.

And that is why there are two basically equally large high tides every day.



## Answer to Exercise 289 (on page 1022)




## Answer to Exercise 290 (on page 1026)

1. $(0, -2)$. $x = 2 \cdot \cos \frac{3\pi}{2} = 2 \cdot 0 = 0$ and $y = 2 \cdot \sin \frac{3\pi}{2} = 2 \cdot -1 = -2$.
2. $(-1, 1)$. $x = \sqrt{2} \cdot \cos \frac{3\pi}{4} = \sqrt{2} \cdot -\frac{\sqrt{2}}{2} = \frac{2}{2} = -1$ and $y = \sqrt{2} \cdot \sin \frac{3\pi}{4} = \sqrt{2} \cdot \frac{\sqrt{2}}{2} = \frac{2}{2} = 1$.
3. $(\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2})$. $x = 3 \cdot \cos -\frac{\pi}{4} = 3 \cdot \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{2}$ and $y = 3 \cdot \sin -\frac{\pi}{4} = 3 \cdot -\frac{\sqrt{2}}{2} = -\frac{3\sqrt{2}}{2}$.
4. $(-\frac{3}{2}, -\frac{3\sqrt{3}}{2})$. $x = (-3) \cdot \cos \frac{\pi}{3} = (-3) \cdot \frac{1}{2} = -\frac{3}{2}$ and $y = (-3) \cdot \sin \frac{\pi}{3} = (-3) \cdot \frac{\sqrt{3}}{2} = -\frac{3\sqrt{3}}{2}$.
5. $(0, -2)$. $x = 2 \cdot \cos -\frac{\mathrm{pi}}{2} = 2 \cdot 0 = 0$ and $y = 2 \cdot \sin -\frac{\pi}{2} = 2 \cdot -1 = -2$.

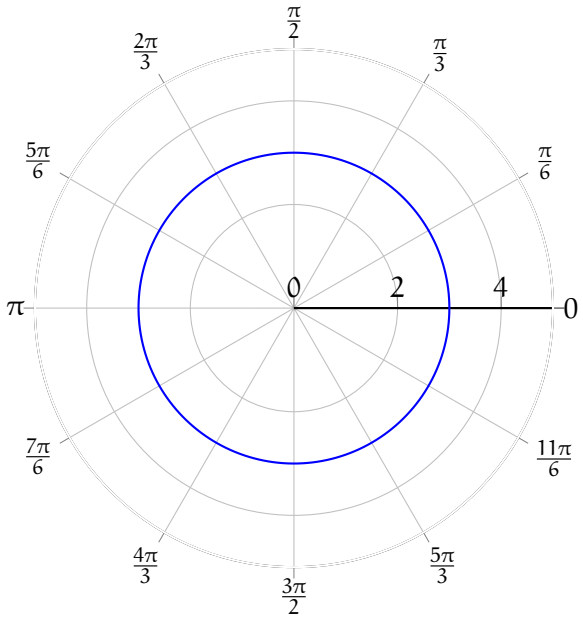
## Answer to Exercise 291 (on page 1027)

1. $(4\sqrt{2}, \frac{3\pi}{4})$. $r = \sqrt{x^2 + y^2} = \sqrt{32} = 4\sqrt{2}$. $\arctan \frac{y}{x} = \arctan \frac{4}{-4} = \arctan -1 = -\frac{\pi}{4} + n\pi$. We take $\theta = \frac{3\pi}{4}$ to satisfy the domain restriction and be in the correct quadrant.
2. $(6, \frac{\pi}{3})$. $r = \sqrt{3^2 + \left(3\sqrt{3}\right)^2} = \sqrt{9 + 27} = \sqrt{36} = 6$. $\arctan \frac{3\sqrt{3}}{3} = \arctan \sqrt{3} = \frac{\pi}{3} + n\pi$. We take $\theta = \frac{\pi}{3}$ to satisfy the domain restriction and be in the correct quadrant.

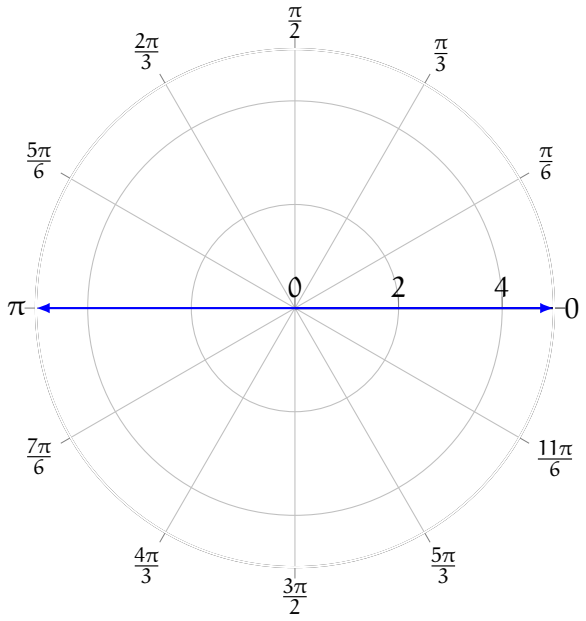
3. $(2, \frac{11\pi}{6})$. $r = \sqrt{\sqrt{3}^2 + (-1)^2} = \sqrt{3+1} = 2$. $\arctan \frac{-1}{\sqrt{3}} = -\frac{\pi}{6} + n\pi$. We take $\theta = \frac{11\pi}{6}$ to satisfy the domain restriction and have the point in the correct quadrant.
4. $(6, \pi)$. $r = \sqrt{(-6)^2 + 0^2} = 6$. $\arctan \frac{0}{-6} = \pi + n\pi$. We take $\theta = \pi$ to satisfy the domain restriction.
5. $(2\sqrt{2}, \frac{5\pi}{4})$. $r = \sqrt{(-2)^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}$. $\arctan \frac{-2}{-2} = \arctan 1 = \frac{\pi}{4} + n\pi$. We take $\theta = \frac{5\pi}{4}$ to satisfy the domain restriction and be in the correct quadrant.

## Answer to Exercise ?? (on page 1031)

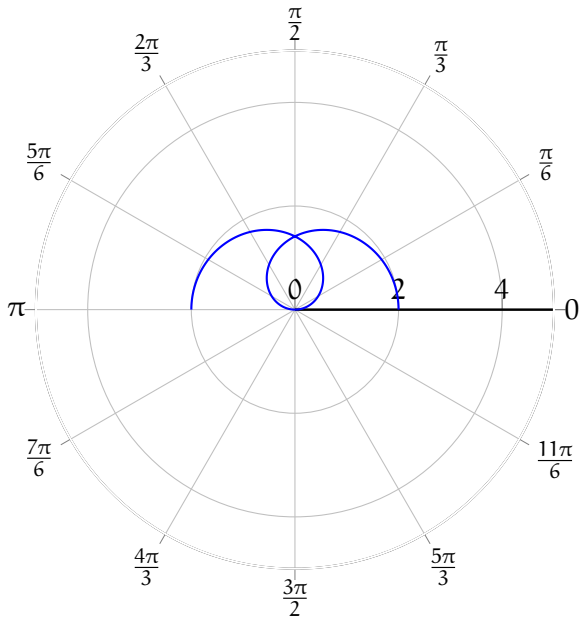
1. $r = 3$



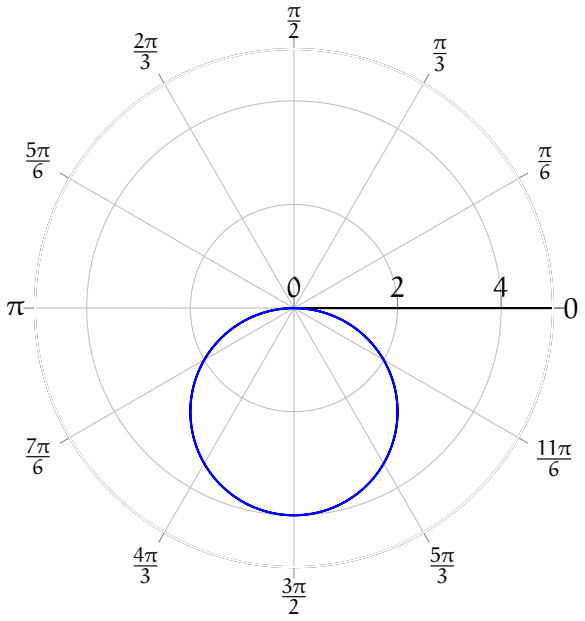

2. $\theta = \pi$ Because $r$ includes all real numbers, negative $r$ is possible and the line $\theta = \pi$ extends in both directions



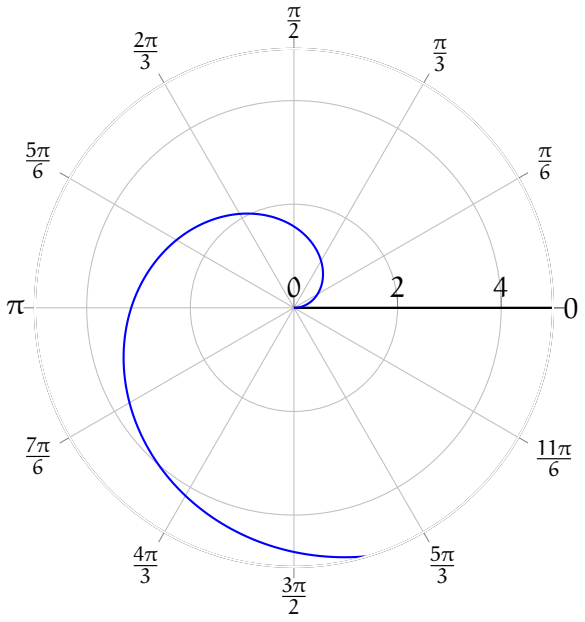

3. $r = 2\cos\frac{\theta}{2}$




4. $r = -4\sin\theta$




5. $r = \theta$ (The spiral continues, but is beyond the boundary of the graph)




## Answer to Exercise 293 (on page 1036)

1. True. $f(2)$ exists and $\lim_{x\to 2^+} f(x) = \lim_{x\to 2^-} f(x) = f(2) = 0$.
2. False. Because of the absolute value, there is a corner in the graph of $f$ at $x = 2$.

$\lim_{x\to 2^+} f'(x) < 0$ and $\lim_{x\to 2^-} f'(x) < 0$. Therefore there is a discontinuity in $f'(x)$ at $x = 2$ and $f(x)$ is not differentiable at $x = 2$.

3. True. $\sqrt{|2-2|} = \sqrt{0} = 0$.

4. False. $f(2)$ is defined at $x = 2$.

## Answer to Exercise 294 (on page 1041)

To estimate the slope at $t = 12$, we can use the data at $t = 9$ and $t = 15$. The slope of the line connecting those points is approximate of the slope at $t = 12$.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{67.9 - 61.8}{15 - 9} = \frac{6.1}{6} = 1.017$$

The units for the numerator are degrees Fahrenheit and for the denominator are minutes. Therefore, the estimated slope has units of degrees Fahrenheit per minute. This represents the change in temperature of the water in the tub. When $t = 12$, the water in the tub is increasing in temperature at about 1 degree Fahrenheit per minute.

## Answer to Exercise 295 (on page 1043)

1.
$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{[m(x+h) + b] - [mx + b]}{h}$$
$$f'(x) = \lim_{h\to 0} \frac{mx + mh + b - mx - b}{h} = \lim_{h\to 0} \frac{mh}{h} = \lim_{h\to 0} m = m$$

2.
$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\sqrt{16 - x - h} - \sqrt{16 - x}}{h}$$
$$f'(x) = \lim_{h\to 0} \frac{\sqrt{16 - x - h} - \sqrt{16 - x}}{h} \cdot \frac{\sqrt{16 - x - h} + \sqrt{16 - x}}{\sqrt{16 - x - h} + \sqrt{16 - x}}$$
$$f'(x) = \lim_{h\to 0} \frac{(16 - x - h) - (16 - x)}{h\left(\sqrt{16 - x - h} + \sqrt{16 - x}\right)}$$
$$f'(x) = \lim_{h\to 0} \frac{-h}{h\left(\sqrt{16 - x - h} + \sqrt{16 - x}\right)} = \lim_{h\to 0} \frac{-1}{\sqrt{16 - x - h} + \sqrt{16 - x}}$$
$$f'(x) = \frac{-1}{\sqrt{16 - x} + \sqrt{16 - x}} = \frac{-1}{2\sqrt{16 - x}}$$

3.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\frac{(x+h)^2-1}{2(x+h)-3} - \frac{x^2-1}{2x-3}}{h}$$

$$f'(x) = \lim_{h\to 0} \left(\frac{1}{h}\right)\left[\frac{x^2+2xh+h^2-1}{2x+2h-3} - \frac{x^2-1}{2x-3}\right]$$

$$f'(x) = \lim_{h\to 0} \left(\frac{1}{h}\right)\left[\frac{x^2+2xh+h^2-1}{2x+2h-3}\left(\frac{2x-3}{2x-3}\right) - \frac{x^2-1}{2x-3}\left(\frac{2x+2h-3}{2x+2h-3}\right)\right]$$

$$f'(x) = \lim_{h\to 0} \left(\frac{1}{h}\right)\left[\frac{\left(x^2+2xh+h^2-1\right)(2x-3) - \left(x^2-1\right)(2x+2h-3)}{(2x-3)(2x+2h-3)}\right]$$

$$f'(x) = \left(\frac{1}{h}\right)\left[\frac{2x^3+4x^2h+2xh^2-2x-3x^2-6xh-3h^2+3\left(2x^3+2x^2h-3x^2-2x-2h+3\right)}{(2x-3)(2x+2h-3)}\right]$$

$$f'(x) = \lim_{h\to 0} \left(\frac{1}{h}\right)\left[\frac{2x^2h+2xh^2-6xh-3h^2+2h}{(2x-3)(2x+2h-3)}\right]$$

$$f'(x) = \lim_{h\to 0} \frac{2x^2+2xh-6x-3h+2}{(2x-3)(2x+2h-3)} = \frac{2x^2-6x+2}{(2x-3)^2}$$

## Answer to Exercise 296 (on page 1046)

First, let's confirm that l'Hôpital's rule applies here:

$$\lim_{x\to 0} \frac{\tan x - x}{x^3} = \frac{0-0}{0} = \frac{0}{0}$$

Therefore, we can apply l'Hôpital's rule:

$$\lim_{x\to 0} \frac{\tan x - x}{x^3} = \lim_{x\to 0} \frac{\frac{d}{dx}(\tan x - x)}{\frac{d}{dx}x^3}$$

$$= \lim_{x\to 0} \frac{\sec^2 x - 1}{3x^2} = \frac{1-1}{0} = \frac{0}{0}$$

which is an indeterminate form. We apply l'Hôpital's rule again:

$$\lim_{x\to 0} \frac{\tan x - x}{x^3} = \lim_{x\to 0} \frac{\frac{d}{dx}(\sec^2 x - 1)}{\frac{d}{dx}3x^2}$$

$$= \lim_{x\to 0} \frac{2\tan x \sec^2 x}{6x} = \frac{2(0)(1^2)}{6\cdot 0} = \frac{0}{0}$$

which is also an indeterminate form. We apply l'Hôpital's rule again:

$$\lim_{x\to 0}\frac{\tan x - x}{x^3} = \lim_{x\to 0}\frac{\frac{d}{dx}(2\tan x\sec^2 x)}{\frac{d}{dx}6x}$$

$$= \lim_{x\to 0}\frac{2\sec^2 x[2\tan^2 x + \sec^2 x]}{6} = \frac{2\cdot 1\cdot[2\cdot 0+1]}{6}$$

$$= \frac{2}{6} = \frac{1}{3}$$

## Answer to Exercise 297 (on page 1047)

1. $\lim_{x\to 3}\frac{x-3}{x^2-9} = \frac{0}{0}$, so we apply l'Hôpital's rule. $\lim_{x\to 3}\frac{x-3}{x^2-9} = \lim_{x\to 3}\frac{1}{2x} = \frac{1}{6}$
2. $\lim_{x\to 1/2}\frac{6x^2+5x-4}{4x^2+16x-9} = \frac{0}{0}$, so we apply l'Hôpital's rule. $\lim_{x\to 1/2}\frac{6x^2+5x-4}{4x^2+16x-9} = \lim_{x\to 1/2}\frac{12x+5}{8x+16} = \frac{11}{20}$
3. $\lim_{x\to 0^+}\frac{\ln x}{\sqrt{x}} = \frac{-\infty}{0} = -\infty$. This limit does not require l'Hôpital's rule because it is evaluable
4. $\lim_{x\to 1}\frac{x\sin x-1}{2x^2-x-1} = \frac{1\cdot\sin 1-1}{2(1)^2-1-1} = \frac{0}{0}$, so we apply l'Hospital's rule: $\lim_{x\to 1}\frac{x\sin x-1}{2x^2-x-1} = \lim_{x\to 1}\frac{\frac{d}{dx}(x\sin x-1}{\frac{d}{dx}(2x^2-x-1} = \lim_{x\to 1}\frac{x\cdot\cos x-1+\sin x-1}{4x-1} = \frac{1\cdot\cos 0+\sin 0}{4-1} = \frac{1\cdot 1+0}{-3} = \frac{-1}{3}$.

## Answer to Exercise 298 (on page 1049)

The speed of a car must a continuous, differentiable function, since your car can't "jump" from one speed to another: it must smoothly accelerate from one speed to another. Therefore, the Mean Value Theorem applies. The average acceleration from 3:30 PM to 3:40 PM is given by:

$$\frac{\text{change in speed}}{\text{change in time}} = \frac{50\frac{\text{mi}}{\text{hr}} - 30\frac{\text{mi}}{\text{hr}}}{3:40\text{PM} - 3:30\text{PM}}$$

Simplifying and converting minutes to hours, we see the average acceleration is:

$$\frac{20\frac{\text{mi}}{\text{hr}}}{\frac{1}{6}\text{hr}} = 120\frac{\text{mi}}{\text{hr}^2}$$

Therefore, by MVT, there must be some time between 3:30 and 3:40 PM where the car's acceleration is exactly $120\frac{\text{mi}}{\text{hr}^2}$.

## Answer to Exercise 299 (on page 1050)

(a) For the domain given, $f(x)$ is defined and differentiable. Finding the slope of the secant line connecting the endpoints:

$$\frac{f(b)-f(a)}{b-a}=\frac{\sqrt{4}-\sqrt{0}}{4-0}=\frac{2}{4}=\frac{1}{2}$$

So we are looking for some number $c$ such that $f'(c)=\frac{1}{2}$. Let's find $f'(x)$:

$$f'(x)=\frac{d}{dx}\sqrt{x}=\frac{1}{2\sqrt{x}}$$

Setting this equal to $\frac{1}{2}$ to find $c$:

$$f'(c)=\frac{1}{2\sqrt{c}}=\frac{1}{2}$$
$$\sqrt{c}=1$$
$$c=1$$

(b)For the domain given, $f(x)$ is defined and differentiable. Finding the slope of the secant line connecting the endpoints:

$$\frac{f(2)-f(0)}{2-0}=\frac{e^{-2}-e^{0}}{2}=\frac{1-e^{2}}{2e^{2}}\approx -0.432$$

And find $f'(x)$:

$$f'(x)=-e^{-x}$$

According to MVT, there must be some $c$ such that $f'(c)\approx -0.432$:

$$-e^{-c}\approx -0.432$$
$$e^{-c}\approx 0.432$$
$$-c\approx \ln 0.432$$
$$c\approx -\ln 0.432\approx 0.839$$

(c) For the domain given, $f(x)$ is defined and differentiable. Finding the secant line connecting the endpoints:

$$\frac{f(b)-f(a)}{b-a}=\frac{\ln 4-\ln 1}{4-1}=\frac{\ln 4}{3}\approx 0.462$$

And find $f'(x)$:

$$f'(x) = \frac{1}{x}$$

According to MVT, there must be some $c$ such that $f'(c) \approx 0.462$

$$f'(c) = \frac{1}{c} \approx 0.462$$

$$c \approx \frac{1}{0.462} = 2.164$$

## Answer to Exercise 300 (on page 1051)

Velocity is the derivative of position. Therefore, $v(t) = s'(t) = 3t^2 - 12t + 6$.

## Answer to Exercise 301 (on page 1051)

$$v(2) = 3(2)^2 - 12(2) + 6 = -6\frac{m}{s}$$
$$v(4) = 3(4)^2 - 12(4) + 6 = 6\frac{m}{s}$$

## Answer to Exercise 302 (on page 1051)

When the particle is at rest, $v(t) = 0$.

$$3t^2 - 12t + 6 = 0$$

$$3(t^2 - 4t + 2) = 0$$
$$t^2 - 4t + 2 = 0$$

This is not easily factorable, so we will use the quadratic formula:

$$t = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(2)}}{2(1)}$$

$$x = \frac{4 \pm \sqrt{16 - 8}}{2} = 2 \pm \sqrt{2} \approx 0.586, 3.414$$

Therefore, the particle is at rest at 0.586s and 3.414s.

## Answer to Exercise 303 (on page 1054)

1. $\frac{dy}{dx} = \frac{d}{dx}[x \sin x] = x\frac{d}{dx}\sin x + \sin x \frac{d}{dx}x = x(-\cos x) + \sin x(1) = \sin x - x\cos x$
2. By the chain rule, $f'(x) = 5(x^3 - \cos x)^4 \cdot \frac{d}{dx}(x^3 - \cos x) = 5(x^3 - \cos x)^4 \cdot (3x^2 + \sin x)$
3. By the chain rule, $f'(x) = \frac{d}{d(\sin x)}[\sin^3 x] \times \frac{d}{dx}\sin x = 3\sin^2 x \cdot \cos x$

## Answer to Exercise 304 (on page 1054)

$f'(x) = \frac{d}{dx}(7x) - \frac{d}{dx}(3) + \frac{d}{dx}(\ln x) = 7 - 0 + \frac{1}{x} = 7 - \frac{1}{x}$ and $f'(1) = 7 - \frac{1}{1} = 6$

## Answer to Exercise 305 (on page 1055)

The particle is at rest when $x'(t) = y'(t) = 0$. First, we find each of the derivatives:

$$x'(t) = 3t^2 - 6t$$
$$y'(t) = 12 - 6t$$

We can solve $y' = 0$ for $t$ and find that the $y$-velocity is 0 when $t = 2$. Substituting $t = 2$ into our expression for $x'$, we find $x'(2) = 3(2)^2 - 6(2) = 0$. Therefore, the particle is at rest when $t = 0$. To find the $xy$-coordinate, we substitute $t = 2$ into $x(t)$ and $y(t)$:

$$x(2) = (2)^3 - 3(2)^2 = 8 - 12 = -4$$
$$y(2) = 12(2) - 6(2) = 24 - 12 = 12$$

Therefore, the particle is at rest when it is located at $(-4, 12)$.

## Answer to Exercise 306 (on page 1055)

$f(g(x)) = \sqrt{(3x-2)^2 - 4} = \sqrt{9x^2 - 12x}$ and $\frac{d}{dx}f(g(x)) = \frac{18x-12}{2\sqrt{9x^2-12x}}$. Substituting $x = 3$, we find $f'(g(x)) = \frac{18(3)-12}{2\sqrt{9(3)^2-12(3)}} = \frac{42}{2\sqrt{45}} = \frac{21}{3\sqrt{5}} = \frac{7}{\sqrt{5}}$

## Answer to Exercise 307 (on page 1055)

First, recall that the velocity of a particle is the derivative of its position function. There-

fore, $v(t) = x'(t) = \frac{d}{dt}[(t-a)(t-b)]$. Applying the Product Rule for derivatives, we see that $v(t) = (t-a)(1) + (t-b)(1) = 2t - a - b$. To find the time(s) when the particle is at rest, we set $v(t) = 0$ and solve for $t$.

$$0 = 2t - a - b$$

$$2t = a + b$$

$$t = \frac{a+b}{2}$$

## Answer to Exercise 308 (on page 1056)

The question is asking when the derivative of $f$ is $\frac{1}{2}$. We will take the derivative and set it equal to $\frac{1}{2}$.

$$f'(x) = \frac{(x+2)(1) - x(1)}{(x+2)^2} = \frac{2}{(x+2)^2}$$

$$\frac{2}{(x+2)^2} = \frac{1}{2}$$

$$4 = (x+2)^2$$

$$\pm 2 = x + 2$$

$$x = 2 - 2 = 0 \text{ and } x = -2 - 2 = -4$$

## Answer to Exercise 309 (on page 1056)

(a) Let $t_0$ be the time at which the particle is first at rest. The velocity of the particle is given by $v(t) = x'(t) = \cos t + \sin t$. Setting $v(t) = 0$, we find:

$$\cos t = -\sin t$$

which is true for $t = \frac{3\pi+4n}{4}$, where $n$ is an integer. Therefore, the first time the velocity is 0 is $t_0 = \frac{3\pi}{4}$.
(b) To find the acceleration at $t = \frac{3\pi}{4}$, we take the derivative of the velocity function to yield the acceleration function.

$$a(t) = v'(t) = -\sin t + \cos t$$

. Substituting $t = \frac{3\pi}{4}$, we find the acceleration is $-\sin\frac{3\pi}{4} + \cos\frac{3\pi}{4} = \frac{-\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2}$

## Answer to Exercise 310 (on page 1056)

First, we find the $x$ such that $y = 0$

$$0 = e^{\tan x} - 2$$

$$2 = e^{\tan x}$$

$$\ln 2 = \tan x$$

$$x = \arctan(\ln 2) = \arctan 0.693 \approx 0.606$$

Then, we find the slope of the function at $x = 0.606$ by finding $y'(0.606)$

$$y' = e^{\tan x}(\sec x)^2 = \frac{e^{\tan x}}{(\cos x)^2}$$

$$y'(0.606) = \frac{e^{\tan 0.606}}{(\cos 0.606)^2} = 2.961$$

## Answer to Exercise 311 (on page 1057)

(a) Apply the chain rule to find $f'(x)$

$$f'(x) = \frac{1}{2\sqrt{25 - x^2}} \cdot (-2x) = \frac{-x}{\sqrt{25 - x^2}}$$

.

(b) First, substitute $x = -3$ into $f'(x)$

$$f'(-3) = \frac{-(-3)}{\sqrt{25 - (-3)^2}} = \frac{3}{\sqrt{16}} = \frac{3}{4}$$

This is the slope of the line. To complete an equation for the tangent line, we need a point. We know the tangent line touches $f(x)$ at $x = -3$, so the tangent line must pass through the point $(-3, f(-3))$.

$$f(-3) = \sqrt{25 - (-3)^2} = 4$$

We use $m = \frac{3}{4}$ and the coordinate point $(x_1, y_1) = (-3, 16)$ to complete the equation $y - y_1 = m(x - x_1)$

$$y - 16 = \frac{3}{4}(x + 3)$$

## Answer to Exercise 312 (on page 1057)

$$a(t) = v'(t) = -\frac{\pi}{6}\sin\frac{\pi}{6}t$$

$$a(4) = -\frac{\pi}{6}\sin\frac{2\pi}{3} = -\frac{\pi}{6}\cdot\frac{\sqrt{3}}{2} = -\frac{\pi\sqrt{3}}{12}$$

## Answer to Exercise 313 (on page 1057)

Recall that the rate of change of a function is given by the derivative of that function. Therefore, we are looking for the interval(s) where $f'(x) > g'(x)$. First, we find each derivative:

$$f'(x) = e^x$$

$$g'(x) = 4x^3$$

We are looking for x-values, such that $e^x > 4x^3$. This inequality can be restated as $e^x - 4x^3 > 0$. Using a calculator, you should find that $e^x - 4x^3 = 0$ when $x \approx 0.831$ and $x \approx 7.384$. We will check values on either side of and in the interval $x \in (0.831, 7.384)$ to determine the sign value of $e^x - 4x^3$. We know that when $x = 0$, $e^x - 4x^3 > 0$, when $x = 5$, $e^x - 4x^3 < 0$, and when $x = 10$, $e^x - 4x^3 > 0$. Therefore, $f'(x)$ is greater than $g'(x)$ on the open intervals $x \in (-\infty, 0.831) \cup (7.384, \infty)$.

## Answer to Exercise 314 (on page 1062)

Recall that critical values are values of x where $g'(x) = 0$ or is undefined. We need to find an expression for $g'(x)$, set it equal to zero when $x = \frac{2}{3}$, and solve for k.

$$g'(x) = x^(2)[k * \exp kx] + \exp kx[2x]$$

$$g'(\frac{2}{3}) = (\frac{2}{3})^2[k * \exp\frac{2k}{3}] + \frac{4}{3}\exp\frac{2k}{3} = 0$$

$$\frac{4k}{9}e^{\frac{2k}{3}} + \frac{4}{3}e^{\frac{2k}{3}} = 0$$

$$(\frac{4k}{9} + \frac{4}{3})e^{\frac{2k}{3}} = 0$$

There are no real values of k such that $e^{\frac{2k}{3}} = 0$, therefore, we will examine the other factor:

$$\frac{4k}{9} + \frac{4}{3} = 0$$

$$\frac{4k}{9} = \frac{-4}{3}$$
$$\frac{k}{3} = -1$$
$$k = -3$$

Therefore, $g(x)$ has a critical value at $x = \frac{2}{3}$ when $k = -3$.

## Answer to Exercise 315 (on page 1064)

First, we will find $f'$ and set it equal to zero:

$$f'(x) = 300 - 3x^2 = 0$$

$$300 = 3x^2 \rightarrow x = \pm\sqrt{100} = \pm 10$$

(Note: $f'(x) = 3(10 - x)(10 + x)$, which implies roots at $x = \pm 10$. Now, we will evaluate the value of $f'(x)$ for $x < -10$, $-10 < x < 10$, and $x > 10$.

| Value of x | (10-x) | (10+x) | $f'(x)$ | $f(x)$ behavior |
|---|---|---|---|---|
| $x < -10$ | positive | negative | negative | decreasing |
| $-10 < x < 10$ | positive | positive | positive | increasing |
| $x > 10$ | negative | positive | negative | decreasing |

Therefore, the function is increasing on the interval $x \in [-10, 10]$ because $f'(x) > 0$ for $x \in [-10, 10]$.

## Answer to Exercise 316 (on page 1064)

Given $f(x) = x^3 - 3x^2 - 9x + 4$, it follows that $f'(x) = 3x^2 - 6x - 9$. Factoring, we find that $f'(x) = 9(x - 3)(x + 1)$ and $f'(x) = 0$ when $x = 3$ and $x = -1$. We construct our table to help us analyze the value of $f'(x)$ and behavior of $f(x)$ on the whole domain of the function:

| Value of x | $(x - 3)$ | $(x + 1)$ | $f'(x)$ | $f(x)$ behavior |
|---|---|---|---|---|
| $x < -1$ | negative | negative | positive | increasing |
| $-1 < x < 3$ | negative | positive | negative | decreasing |
| $x > 3$ | positive | positive | positive | increasing |

So, $f(x)$ is increasing for $x \in (-\infty, -1) \cup (3, \infty)$ and decreasing for $x \in (-1, 3)$. Since $f'(-1) = 0$ and changes from positive to negative, $f(x)$ has a local maximum at $x = -1$. And since $f'(3) = 0$ and changes from negative to positive, $f(x)$ has a local minimum at $x = 3$.

## Answer to Exercise 317 (on page 1066)

First, we identify any critical numbers:

$$f'(x) = \frac{x * (\frac{1}{x}) - \ln x * 1}{x^2} = \frac{1 - \ln x}{x^2}$$

Recall that critical numbers are values where $f'(x) = 0$ or does not exist. We might identify $x = 0$ as a critical number, but the presence of $\ln x$ limits the domain of the function to $x \in (0, \infty)$, excluding $x = 0$. For all $x \in (0, \infty)$, $f'(x)$ exists. So, we look for values where $f'(x) = 0$.

$$\frac{1 - \ln x}{x^2} = 0$$
$$1 - \ln x = 0$$
$$1 = \ln x$$
$$x = e$$

Finding the value of $f(x)$ at $x = e$:

$$f(e) = \frac{\ln e}{e} = \frac{1}{e}$$

Because the domain of $f(x)$ is on an *open interval*, instead of checking the endpoints directly, we'll take the limits as $x$ approaches 0 and $\infty$.

$$\lim_{x \to 0} \frac{\ln x}{x} = -\infty < \frac{1}{e}$$

$$\lim_{x \to \infty} \frac{\ln x}{x} = 0 < \frac{1}{e}$$

Therefore, the absolute maximum values of $f(x) = \frac{\ln x}{x}$ is $\frac{1}{e}$ at $x = e$.

## Answer to Exercise 318 (on page 1067)

1. $f'(x) = 4 - 2x$ and to find the critical numbers, we set $f'(x) = 0$:

$$4 - 2x = 0$$
$$x = 2$$

We evaluate $f(x)$ at $x = 0, 2, 5$:

$$f(0) = 12 + 4(0) - 0^2 = 12$$

$$f(2) = 12 + 4(2) - 2^2 = 12 + 8 - 4 = 16$$
$$f(5) = 12 + 4(5) - 5^2 = 12 + 20 - 25 = 7$$

Therefore, the global maximum is $f(2) = 16$ and the global minimum is $f(5) = 7$.

2. $f(t) = \dfrac{\sqrt{t}}{1+t^2}$, $[0, 2]$.

We write $f(t) = t^{1/2}(1+t^2)^{-1}$ and differentiate:

$$f'(t) = \frac{1 - 3t^2}{2\sqrt{t}(1+t^2)^2}.$$

Critical points occur when the numerator is zero (and $t$ is in the interval):

$$1 - 3t^2 = 0 \quad \Rightarrow \quad t = \frac{1}{\sqrt{3}}.$$

We evaluate $f(t)$ at $t = 0, \frac{1}{\sqrt{3}}, 2$:

$$f(0) = 0, \qquad f\left(\frac{1}{\sqrt{3}}\right) = \frac{3^{3/4}}{4}, \qquad f(2) = \frac{\sqrt{2}}{5}.$$

Since $\dfrac{3^{3/4}}{4}$ is the largest of these and $0$ is the smallest, the global maximum is

$$f\left(\frac{1}{\sqrt{3}}\right) = \frac{3^{3/4}}{4},$$

and the global minimum is

$$f(0) = 0.$$

3. $f(t) = 2\cos t + \sin(2t)$, $\left[0, \frac{\pi}{2}\right]$.

We differentiate:

$$f'(t) = -2\sin t + 2\cos(2t).$$

Set $f'(t) = 0$:

$$-2\sin t + 2\cos(2t) = 0 \quad \Rightarrow \quad \cos(2t) = \sin t.$$

Use $\cos(2t) = 1 - 2\sin^2 t$:

$$1 - 2\sin^2 t = \sin t \quad \Rightarrow \quad 2\sin^2 t + \sin t - 1 = 0.$$

Let $u = \sin t$:

$$2u^2 + u - 1 = 0 \quad \Rightarrow \quad u = \frac{1}{2} \text{ or } u = -1.$$

On $\left[0, \frac{\pi}{2}\right]$, $\sin t = \frac{1}{2}$ gives $t = \frac{\pi}{6}$ (and $\sin t = -1$ is not in this interval).
Evaluate $f(t)$ at $t = 0, \frac{\pi}{6}, \frac{\pi}{2}$:

$$f(0) = 2\cos 0 + \sin 0 = 2,$$

$$f\left(\frac{\pi}{6}\right) = 2\cos\left(\frac{\pi}{6}\right) + \sin\left(\frac{\pi}{3}\right) = 2 \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2},$$
$$f\left(\frac{\pi}{2}\right) = 2\cos\left(\frac{\pi}{2}\right) + \sin(\pi) = 0 + 0 = 0.$$

Hence the global maximum is

$$f\left(\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{2},$$

and the global minimum is

$$f\left(\frac{\pi}{2}\right) = 0.$$

4. $f(x) = \ln(x^2 + x + 1)$, $[-1, 1]$.

   Differentiate:

   $$f'(x) = \frac{2x + 1}{x^2 + x + 1}.$$

   To find critical points, set the numerator equal to 0:

   $$2x + 1 = 0 \quad \Rightarrow \quad x = -\frac{1}{2}.$$

   Evaluate $f(x)$ at $x = -1, -\frac{1}{2}, 1$:

   $$f(-1) = \ln(1 - 1 + 1) = \ln 1 = 0,$$
   $$f\left(-\frac{1}{2}\right) = \ln\left(\left(-\frac{1}{2}\right)^2 - \frac{1}{2} + 1\right) = \ln\left(\frac{1}{4} - \frac{1}{2} + 1\right) = \ln\left(\frac{3}{4}\right),$$
   $$f(1) = \ln(1 + 1 + 1) = \ln 3.$$

   Since $\ln\left(\frac{3}{4}\right) < 0 < \ln 3$, the global minimum is

   $$f\left(-\frac{1}{2}\right) = \ln\left(\frac{3}{4}\right),$$

   and the global maximum is

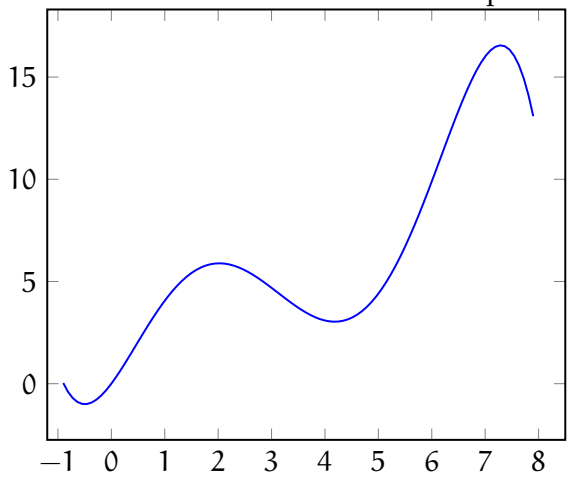
   $$f(1) = \ln 3.$$

## Answer to Exercise 319 (on page 1070)

[Your answers are meant to be estimates; anything within $\pm 0.1$ of the given answers are reasonable estimates.]

1. $f(x)$ is increasing on the intervals $x \in (-0.5, 2.2) \cup (4, 7.3)$. $f(x)$ is decreasing on the intervals $x \in (-\infty, -0.5) \cup (2.2, 4) \cup (7.3, \infty)$.
2. $f(x)$ has local maxima at $x = 2.2, 7.3$ and local minima at $x = -0.5, 4$.

3. Your sketch should show the maxima and minima identified in part 2. One possible solution is shown below.




## Answer to Exercise 320 (on page 1073)

Noting that $f''(2) = 0$, we examine the value of $f''$ around $x = 2$. For $0 < x < 2$ , $f'' < 0$, which indicates $f$ is concave down in the domain $x \in (0, 2)$. For $x > 2$, $f'' > 0$, which indicates $f$ is concave up. Therefore, there is an inflection point at $x = 2$ for $f$. Recalling that $f(x) = x^4 - 4x^3$, we find the coordinate of the inflection point by substituting $x = 2$:

$$f(2) = 2^4 - 4 * 2^3 = 16 - 4 * 8 = 16 - 32 = -16$$

Therefore, $f(x)$ has an inflection point at $(2, -16)$.

## Answer to Exercise 321 (on page 1074)

According to the graph, $g'$ is positive and increasing. Therefore, $g$ is increasing (because $g'$ is positive) and concave up (because $g'$ is increasing, and therefore $g''$ is positive).

## Answer to Exercise 322 (on page 1074)

Since the question asks about concavity, we need to examine the second derivative:

$$f''(x) = \frac{d}{dx} f'(x) = \frac{d}{dx}\left[e^{(x^4-2x^2+1)} - 2\right]$$

$$f''(x) = \left(x^4 - 2x^2 + 1\right) e^{(x^4-2x^2+1)} \left(4x^3 - 4x\right)$$

The second derivative equals zero when $x^4 - 2x^2 + 1 = (x^2 - 1)^2 = 0$ or $4x^3 - 4x = 4(x)(x^2 - 1) = 0$, which gives roots $x = 0$, $x = 1$, and $x = -1$. So the intervals we need to test are $(-1.5, -1)$, $(-1, 0)$, $(0, 1)$, and $(1, 1.5)$. To test $x \in (-1.5, -1)$, we will substitute $x = -1.25$ into $f''(x)$:

$$f''(-1.25) = -3.85928 < 0$$

Therefore, $f(x)$ is concave down on the interval $x \in (-1.5, -1)$. Next, we test $x \in (-1, 0)$:

$$f''(-0.5) = 2.63258 > 0$$

So, we eliminate $x \in (-1, 0)$. Next, we test $x \in (0, 1)$:

$$f''(0.5) = -2.63258 < 0$$

And $f(x)$ is concave down on the interval $x \in (0, 1)$. Finally, we test the interval $x \in (1, 1.5)$:

$$f''(1.25) = 3.85928 > 0$$

Which eliminates that interval. Therefore, $f(x)$ is concave down on the intervals $x \in (-1.5, -1)$ and $x \in (0, 1)$.

## Answer to Exercise 323 (on page 1075)

1. False. For $x \in (-2, 0)$, the slope of $f(x)$ is negative, which implies that $f'(x) < 0$ for $x \in (-2, 0)$.
2. False. The graph comes to a point at $x = 0$, therefore $\lim_{x \to 0^+} f'(x) \neq \lim_{x \to 0^-} f'(x)$, which means the limit does not exist and $f$ is not differentiable at $x = 0$.
3. True. The graph of $f(x)$ is concave up for $x \in (0, 2)$, which means the second derivative is positive.
4. True. Recall that critical values are where derivatives equal 0 or do not exist. Since we have established that $f(x)$ does not exist at $x = 0$, then there is a critical value at $x = 0$.

## Answer to Exercise 324 (on page 1076)

1. True. $f'(3) = 0$ and $f'$ has a positive slope, which means there is a local extreme and $f$ is concave up at $x = 3$. Therefore, there is a local minimum at $x = 3$.
2. False. Though it appears that $f'' = 0$ at $x = -2$, the slope of $f'$ is positive before and after. Therefore, $f''$ does not cross the x-axis and there is not an inflection point at $x = -2$.

3. True. For $0 < x < 4$, the slope of $f'$ is negative, which means $f''$ is negative, which means $f$ is concave down.

## Answer to Exercise 325 (on page 1077)

The graph of $f$ has inflection points at $x = -1$ and $x = 1$. Since $f(x) = f(-x)$, we can expand the table to include the entire window we are investigating:

| | $-2 < x < -1$ | $-1 < x < 0$ | $0 < x < 1$ | $1 <$ |
|---|---|---|---|---|
| $f(x)$ | Negative | Positive | Positive | Ne |
| $f'(x)$ | Negative | Negative | Negative | Ne |
| $f''(x)$ | Positive | Negative | Negative | Po |

Recall that inflection points occur when $f''$ changes from positive to negative or from negative to positive. Examining the table, we see that the sign of $f''$ changes at $x = -1$ and $x = 1$.

## Answer to Exercise 326 (on page 1080)

- **Seedlings:** Experiment. Fertilizer was randomly assigned. Causal conclusion is justified.
- **Screen time and attention:** Observational study. No assignment was made. Causal conclusion is not justified; many confounding variables (parenting style, diet, sleep) could explain the relationship.
- **Customer service training:** Experiment. The training was randomly assigned. Causal conclusion is justified.
- **Libraries and literacy:** Observational study. No assignment was made. Causal conclusion is not justified; wealthier cities may have had both more libraries and higher literacy for unrelated reasons.

## Answer to Exercise 327 (on page 1083)

1. Number the 60 seedlings from 1 to 60. Use a random number generator to select 30 for the music group. The remaining 30 form the no-music group.
2. Control means growing all seedlings under identical conditions except for the music. Variables to hold constant include: soil type and quantity, pot size, amount and frequency of watering, light exposure, temperature, and humidity.
3. Replication. With only 5 seedlings per group, a single unusually fast or slow grower could distort the group average. With 30 per group, individual variation averages

out and a real effect of the music is more detectable.

4. Yes. The no-music group serves as the control group, establishing the baseline growth rate without the treatment. Without it, there is nothing to compare the music group's growth against.

## Answer to Exercise 328 (on page 1085)

1. (1) Yes, random assignment. (2) No, volunteers are self-selected. Conclusion: the noisy room likely caused lower comprehension scores, but this result cannot be generalized to all university students or beyond. Volunteers may differ from the broader student population in ways that affect how noise impacts their reading.

2. (1) No random assignment. The researchers observed existing behavior and imposed no treatment. (2) Yes, random selection from voter rolls. Conclusion: the association between exercise and lower stress can be generalized to the adult population of the city. We cannot conclude that exercise caused lower stress because no treatment was assigned. A confounding variable such as income or work conditions could explain both.

3. (1) Yes, random assignment at the school level. The district used chance to decide which 10 of its 60 schools received the new curriculum, so the schools are the experimental units. Students were not individually assigned. (2) No random selection from a larger population. Every elementary school in the district took part, so the study covers the district's schools and makes no attempt to sample from other districts. Conclusion: the new curriculum likely caused the higher scores in this district's elementary schools. The result cannot be extended to schools in other districts.

4. (1) Yes, random assignment to app or printed plan. (2) No, volunteers are recruited through social media. Conclusion: the app likely caused greater endurance improvement compared to the printed plan. However, this result cannot be generalized to all people interested in fitness. Social media recruits may be younger, more comfortable with mobile apps, or more motivated than the broader population.

## Answer to Exercise 329 (on page 1088)

1. Experimental units: the 60 volunteers. Factor: warm-up routine. Levels: three (static stretching, dynamic stretching, no warm-up). Treatments: the three levels, since there is only one factor.

2. Number the volunteers from 01 to 60. Use `randInt(1,60,20)` on a TI-84, ignoring repeats, to select 20 volunteers for group 1 (static stretching). From the remaining

40, use `randInt` again to select 20 for group 2 (dynamic stretching). The remaining 20 form group 3 (no warm-up).

3. Yes. Because treatments were randomly assigned, a causal conclusion is appropriate, provided the experiment was otherwise well-controlled (same track, same time of day, same measurement method, etc.). Random assignment is what licenses the claim of causation.

## Answer to Exercise 330 (on page 1090)

1. Blocking factor: prior reading level (above grade level vs. at or below). If above-level readers respond better to one method regardless of which method it is, mixing them randomly with below-level readers would make it harder to detect the true method effect. Blocking ensures each method is compared within similar groups.
2. Blocking factor: steel mill source (mill A vs. mill B). Differences in base composition could affect how well coatings adhere, confounding the coating comparison. Blocking by mill source ensures all four coatings are tested on panels from each mill.

## Answer to Exercise 331 (on page 1092)

1. Randomized block design. There are three treatments and each GPA block contains more than two units. It is not a matched-pairs design because the block size exceeds two and there are more than two treatments.
2. Matched-pairs design. There are exactly two treatments and each block (pair of farms) contains exactly two units, one receiving each treatment.
3. Matched-pairs design (repeated measures). Each participant receives both treatments in random order and serves as their own pair of size two.

## Answer to Exercise 332 (on page 1094)

1. Completely randomized design. The volunteers are similar in relevant ways (comparable computing experience, no prior exposure to either interface). There is no identified variable that would consistently affect how they respond to one interface versus the other. Randomly assign 40 volunteers to the new interface and 40 to the existing interface.
2. Matched pairs (repeated measures). Each participant has their own baseline sleep

patterns that differ from others. Using each person as their own pair eliminates individual differences in baseline sleep quality from the comparison. Randomly determine which temperature each volunteer experiences first, include a washout night between conditions, then compare each person's cool-night and warm-night sleep quality directly.

3. Randomized block design, blocking on school level (primary vs. secondary). Prior research indicates that the two groups respond differently to professional development formats. If teachers are randomly assigned to workshop formats without blocking, differences in how primary and secondary teachers respond could mask or distort the true effect of format. Block by school level and randomly assign teachers to the three formats within each block.

4. Randomized block design or matched pairs, blocking on baseline skin dryness level. Because baseline dryness varies widely, a participant who starts very dry might show a large reduction regardless of which moisturizer they use. Grouping participants by baseline dryness (for example, low, medium, high) and assigning both moisturizers within each group removes this variability from the comparison. Alternatively, if each participant can use each moisturizer on separate occasions, a matched-pairs repeated measures design would control individual differences even more completely. Both designs earn credit here because each one accounts for baseline dryness; the matched-pairs version is stronger when repeated use is practical.

## Answer to Exercise 333 (on page 1100)

First, take the derivative of the function:

$$f'(x) = 2x - 6.$$

Set the derivative equal to zero to find the critical point:

$$2x - 6 = 0.$$

Solving this equation gives

$$x = 3.$$

Next, take the second derivative:

$$f''(x) = 2.$$

Since the second derivative is positive, the function is concave up, and the critical point corresponds to a minimum.

Finally, substitute $x = 3$ back into the original function:

$$f(3) = 3^2 - 6(3) + 5 = -4.$$

Therefore, the function has a minimum value of $-4$ at $x = 3$.

## Answer to Exercise 334 (on page 1101)

Because only three sides require fencing, the total amount of fencing is given by

$$x + 2y = 60.$$

Solving this equation for $x$ gives

$$x = 60 - 2y.$$

The area of the enclosure is

$$A = xy.$$

Substituting for $x$ yields

$$A(y) = y(60 - 2y) = 60y - 2y^2.$$

Take the derivative:

$$A'(y) = 60 - 4y.$$

Set the derivative equal to zero:

$$60 - 4y = 0.$$

Solving for $y$ gives

$$y = 15.$$

The second derivative is

$$A''(y) = -4.$$

Since the second derivative is negative, this critical point corresponds to a maximum area.

Substituting $y = 15$ into the constraint gives

$$x = 60 - 2(15) = 30.$$

Therefore, the enclosure has dimensions 30 units by 15 units.

## Answer to Exercise 335 (on page 1104)

1. For $f(x) = 20x - x^2$,

   $$f'(x) = 20 - 2x.$$

   Setting $f'(x) = 0$ gives $20 - 2x = 0$, so $x = 10$.

   $$f''(x) = -2.$$

   Since $f''(10) < 0$, the function is concave down at the critical point, so the critical point is a maximum.

2. For $f(x) = x^2 - 6x + 5$,

   $$f'(x) = 2x - 6.$$

   Setting $f'(x) = 0$ gives $2x - 6 = 0$, so $x = 3$.

   $$f''(x) = 2.$$

   Since $f''(3) > 0$, the function is concave up at the critical point, so the critical point is a minimum.

3. For $f(x) = x^3 - 3x$,

   $$f'(x) = 3x^2 - 3 = 3(x^2 - 1).$$

   Setting $f'(x) = 0$ gives $x^2 - 1 = 0$, so $x = -1$ and $x = 1$.

   $$f''(x) = 6x.$$

   At $x = -1$, $f''(-1) = -6 < 0$, so there is a local maximum at $x = -1$. At $x = 1$, $f''(1) = 6 > 0$, so there is a local minimum at $x = 1$.

   The second derivative test is not inconclusive for this function at the critical points, because $f''(-1)$ and $f''(1)$ are not zero.

4. For a quadratic function of the form $f(x) = ax^2 + bx + c$ with $a \neq 0$, there is exactly one critical point at

   $$x = -\frac{b}{2a}.$$

   Also,

   $$f''(x) = 2a.$$

   If $a > 0$, then $f''(x) > 0$ and the critical point is a minimum. If $a < 0$, then $f''(x) < 0$

and the critical point is a maximum.

## Answer to Exercise 336 (on page 1110)

Using implicit differentiation, we see that:

$$\frac{d}{dx}\arcsin x = \frac{d}{dx}\ln y$$

$$\frac{1}{\sqrt{1-x^2}} = \frac{1}{y}\frac{dy}{dx}$$

Multiplying both sides by $y$ to isolate $\frac{dy}{dx}$, we find that:

$$\frac{dy}{dx} = \frac{y}{\sqrt{1-x^2}}$$

## Answer to Exercise 337 (on page 1110)

First, we need to find $\frac{d^y}{dx^2}$:

$$\frac{d}{dx}\frac{dy}{dx} = \frac{d}{dx}x^2 + \frac{d}{dx}y$$

$$= 2x + \frac{dy}{dx} = 2x + x^2 + y$$

At $(-1,-1)$, $\frac{dy}{dx} = (-1)^2 + (-1) = 0$ and $\frac{d^2y}{dx^2} = 2(-1) + (-1)^2 + (-1) = -2 < 0$. Since the slope of $y$ is zero and the graph of $y$ is concave down, $(-1,-1)$ is a local maximum. At $(1,-5)$, $\frac{dy}{dx} = 1^2 + -5 = -4 \neq 0$ and $\frac{d^2y}{dx^2} = 2(1) + 1^2 + (-5) = -2 \neq 0$. Since neither the first nor second derivative of $y$ are zero, $(1,-5)$ is neither a local extrema nor an inflection point.

## Answer to Exercise 338 (on page 1119)

1. $f_x(x,y) = \frac{\partial}{\partial x}\left[3x^4 + 4x^2y^3\right] = 12x^3 + 8y^3$ and $f_y(x,y) = \frac{\partial}{\partial y}\left[3x^4 + 4x^2y^3\right] = 12x^2y^2$
2. $f_x(x,y) = \frac{\partial}{\partial x}\left(xe^{-y}\right) = e^{-y}$ and $f_y(x,y) = \frac{\partial}{\partial y}\left(xe^{-y}\right) = -xe^{-y}$
3. $f_x(x,y) = \frac{\partial}{\partial x}\sqrt{3x+4y^2} = \left(\frac{1}{2\sqrt{3x+4y^2}}\right)\left(\frac{\partial}{\partial x}\left(3x+4y^2\right)\right) = \frac{3}{2\sqrt{3x+4y^2}}$ and $f)y(x,y) = \frac{\partial}{\partial y}\sqrt{3x+4y^2} = \frac{1}{2\sqrt{3x+4y^2}}\left(\frac{\partial}{\partial y}\left(3x+4y^2\right)\right) = \frac{8y}{2\sqrt{3x+4y^2}} = \frac{4y}{\sqrt{3x+4y^2}}$

4. $f_x(x,y) = \frac{\partial}{\partial x}\sin\left(x^2y\right) = \cos\left(x^2y\right)\left(\frac{\partial}{\partial x}\left(x^2y\right)\right) = 2xy\cos\left(x^2y\right)$ and $f_y(x,y) = \frac{\partial}{\partial y}\sin\left(x^2y\right) = \cos\left(x^2y\right)\left(\frac{\partial}{\partial y}\left(x^2y\right)\right) = x^2\cos\left(x^2y\right)$
5. $f_x(x,y) = \frac{\partial}{\partial x}\ln\left(x^y\right) = \frac{\partial}{\partial x}\left(y\ln x\right) = \frac{y}{x}$ and $f_y(x,y) = \frac{\partial}{\partial y}\left(y\ln x\right) = \ln x$

## Answer to Exercise 339 (on page 1121)

1. Finding $f_x$:

$$f_x = \frac{\partial}{\partial x}\left[\sin\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\right] = \cos\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\left[\frac{\partial}{\partial x}\left(x^2-y^2\right)\right]$$

$$f_x = 2x\cos\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)$$

Finding $f_y$:

$$f_y = \frac{\partial}{\partial y}\left[\sin\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\right] = \cos\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\left[\frac{\partial}{\partial y}\left(x^2-y^2\right)\right]$$

$$f_y = -2y\cos\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)$$

Finding $f_z$:

$$f_z = \frac{\partial}{\partial z}\left[\sin\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\right] = \sin\left(x^2-y^2\right)\left(-\sin\sqrt{z}\right)\cdot\left(\frac{\partial}{\partial z}\sqrt{z}\right)$$

$$f_z = \frac{-\sin\left(x^2-y^2\right)\sin\left(\sqrt{z}\right)}{2\sqrt{z}}$$

2. Finding $q_t$:

$$q_t = \frac{\partial}{\partial t}\sqrt[3]{t^3+u^3\sin(5v)} = \frac{1}{3\left(t^3+u^3\sin(5v)\right)^{2/3}}\left(\frac{\partial}{\partial t}\left(t^3+u^3\sin(5v)\right)\right)$$

$$q_t = \frac{t^2}{\left(t^3+u^3\sin(5v)\right)^{2/3}}$$

Finding $q_u$:

$$q_u = \frac{\partial}{\partial u}\sqrt[3]{t^3+u^3\sin(5v)} = \frac{1}{3\left(t^3+u^3\sin(5v)\right)^{2/3}}\left(\frac{\partial}{\partial u}\left(t^3+u^3\sin(5v)\right)\right)$$

$$q_u = \frac{u^2\sin(5v)}{\left(t^3+u^3\sin(5v)\right)^{2/3}}$$

Finding $q_v$:

$$q_v = \frac{\partial}{\partial v}\sqrt[3]{t^3+u^3\sin(5v)} = \frac{1}{3\left(t^3+u^3\sin(5v)\right)^{2/3}}\left(\frac{\partial}{\partial v}\left(t^3+u^3\sin(5v)\right)\right)$$

$$q_v = \frac{u^3\cos(5v)}{3\left(t^3+u^3\sin(5v)\right)^{2/3}}\left(\frac{\partial}{\partial v}(5v)\right) = \frac{5u^3\cos(5v)}{3\left(t^3+u^3\sin(5v)\right)^{2/3}}$$

3. Finding $w_x$:

$$w_x = \frac{\partial}{\partial x}\left(x^z y^x\right) = \left(x^z\right)\cdot\left(\frac{\partial}{\partial x}y^x\right) + \left(y^x\right)\cdot\left(\frac{\partial}{\partial x}x^z\right)$$

$$w_x = \left(x^z\right)\left(\ln(y)y^x\right) + \left(y^x\right)\left(zx^{z-1}\right) = \left(x^{z-1}y^x\right)\left(x\ln(y)+z\right)$$

Finding $w_y$:

$$w_y = \frac{\partial}{\partial y}\left(x^z y^x\right) = \left(x^z\right)\left(\frac{\partial}{\partial y}y^x\right) = x^z\left(xy^{x-1}\right)$$

$$w_y = x^{z+1}y^{x-1}$$

Finding $w_z$:

$$w_z = \frac{\partial}{\partial z}\left(x^z y^x\right) = \left(y^x\right)\left(\frac{\partial}{\partial z}x^z\right) = \left(y^x\right)\left(\ln(x)x^z\right)$$

$$w_z = \ln(x)y^x x^z$$

## Answer to Exercise 340 (on page 1123)

1. $f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left(e^{2xy}\sin x\right)\right] = \frac{\partial}{\partial y}\left[\left(e^{2xy}\right)\left(\frac{\partial}{\partial x}\sin x\right) + (\sin x)\left(\frac{\partial}{\partial x}e^{2xy}\right)\right] = \frac{\partial}{\partial y}\left[e^{2xy}\cos x + 2ye^{2xy}\sin x\right] = \frac{\partial}{\partial y}\left(e^{2xy}\cos x\right) + \frac{\partial}{\partial y}\left(2ye^{2xy}\sin x\right) = 2xe^{2xy}\cos x + (2y)\left(\frac{\partial}{\partial y}e^{2xy}\sin x\right) + \left(e^{2xy}\sin x\right)\left(\frac{\partial}{\partial y}2y\right) = 2xe^{2xy}\cos x + 4xye^{2xy}\sin x + 2e^{2xy}\sin x$

   $f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}f(x,y)\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial y}\left(e^{2xy}\sin x\right)\right] = \frac{\partial}{\partial x}\left(2xe^{2xy}\sin x\right) = (2x)\left[\frac{\partial}{\partial x}\left(e^{2xy}\sin x\right)\right] + \left(e^{2xy}\sin x\right)\left(\frac{\partial}{\partial x}2x\right) = (2x)\left[\left(e^{2xy}\right)\left(\frac{\partial}{\partial x}\sin x\right) + (\sin x)\left(\frac{\partial}{\partial x}e^{2xy}\right)\right] + 2e^{2xy}\sin x = 2xe^{2xy}\cos x + 4xye^{2xy}\sin x + 2e^{2xy}\sin x = f_{xy}$

2. $f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left(\frac{x^2}{x+y}\right)\right] = \frac{\partial}{\partial y}\left[\frac{(x+y)(2x)-x^2(1)}{(x+y)^2}\right] = \frac{\partial}{\partial y}\left[\frac{x^2+2xy}{(x+y)^2}\right] = \frac{(x+y)^2(2x)-\left(x^2+2xy\right)(2(x+y))}{(x+y)^4} = \frac{\left(x^2+2xy+y^2\right)(2x)-\left(x^2+2xy\right)(2x+2y)}{(x+y)^4} = \frac{2x^3+4x^2y+2xy^2-2x^3-2x^2y-4x^2y-4xy^2}{(x+y)^4} = \frac{-2x^2y-2xy^2}{(x+y)^4}$

   $f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}f(x,y)\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial y}\left(\frac{x^2}{x+y}\right)\right] = \frac{\partial}{\partial x}\left[\frac{-x^2}{(x+y)^2}\right] = \frac{(x+y)^2(-2x)-\left(-x^2\right)(2(x+y))}{(x+y)^4} = \frac{\left(x^2+2xy+y^2\right)(-2x)+x^2(2x+2y)}{(x+y)^4} = \frac{-2x^3-4x^2y-2xy^2+2x^3+2x^2y}{(x+y)^4} = \frac{-2x^2y-2xy^2}{(x+y)^4} = f_{xy}$

3. $f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left(\ln(2x+3y)\right)\right] = \frac{\partial}{\partial y}\left[\frac{2}{2x+3y}\right] = \frac{-2(3)}{(2x+3y)^2} = \frac{-6}{(2x+3y)^2}$

$$f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}f(x,y)\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial y}\left(\ln\left(2x+3y\right)\right)\right] = \frac{\partial}{\partial x}\left(\frac{3}{2x+3y}\right) = \frac{-3(2)}{(2x+3y)^2} = \frac{-6}{(2x+3y)} = f_{xy}$$

## Answer to Exercise 341 (on page 1124)

1. $f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(x^5y^2 - 3x^3y^2\right)\right] = \frac{\partial}{\partial x}\left(5x^4y^2 - 9x^2y^2\right) = 20x^3y^2 - 18xy^2.$
   $f_{xy} = f_{yx} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left(x^5y^2 - 3x^3y^2\right)\right] = \frac{\partial}{\partial y}\left(5x^4y^2 - 9x^2y^2\right) = 10x^4y - 18x^2y.$
   $f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial y}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}\left(x^5y^2 - 3x^3y^2\right)\right] = \frac{\partial}{\partial y}\left(2x^5y - 6x^3y\right) = 2x^5 - 6x^3.$
2. $v_{pp} = \frac{\partial}{\partial p}\left(\frac{\partial}{\partial p}v(p,q)\right) = \frac{\partial}{\partial p}\left[\frac{\partial}{\partial p}\left(\sin\left(p^3+q^2\right)\right)\right] = \frac{\partial}{\partial p}\left(\cos\left(p^3+q^2\right)\left(3p^2\right)\right) = \cos\left(p^3+q^2\right)\cdot \frac{\partial}{\partial p}\left(3p^2\right) + 3p^2\cdot\frac{\partial}{\partial p}\left(\cos\left(p^3+q^2\right)\right)$
   $v_{pq} = v_{qp} = \frac{\partial}{\partial q}\left(\frac{\partial}{\partial p}v(p,q)\right) = \frac{\partial}{\partial q}\left[\frac{\partial}{\partial p}\left(\sin\left(p^3+q^2\right)\right)\right] = \frac{\partial}{\partial q}\left(\cos\left(p^3+q^2\right)\left(3p^2\right)\right) = \cos\left(p^3+q^2\right)\frac{\partial}{\partial q}\left(3p^2\right) + 3p^2\frac{\partial}{\partial q}\cos\left(p^3+q^2\right) = 0 + 3p^2\left(-\sin\left(p^3+q^2\right)\right)\left(\frac{\partial}{\partial q}\left(p^3+q^2\right)\right) = -6p^2q\sin\left(p^3+q^2\right)$
   $v_{qq} = \frac{\partial}{\partial q}\left(\frac{\partial}{\partial q}v(p,q)\right) = \frac{\partial}{\partial q}\left[\frac{\partial}{\partial q}\left(\sin\left(p^3+q^2\right)\right)\right] = \frac{\partial}{\partial q}\left[2q\cos\left(p^3+q^2\right)\right] = 2q\left[\frac{\partial}{\partial q}\cos\left(p^3+q^2\right)\right] + \cos\left(p^3+q^2\right)\left[\frac{\partial}{\partial q}\left(2q\right)\right] = (2q)\cdot\left[-2q\sin\left(p^3+q^2\right)\right] + 2\cos\left(p^3+q^2\right) = 2\cos\left(p^3+q^2\right) - 4q^2\sin\left(p^3+q^2\right)$
3. $T_{rr} = \frac{\partial}{\partial r}\left(\frac{\partial}{\partial r}T(r,\theta)\right) = \frac{\partial}{\partial r}\left[\frac{\partial}{\partial r}\left(e^{-3r}\cos\theta^2\right)\right] = \frac{\partial}{\partial r}\left(-3e^{-3r}\cos\theta^2\right) = 9e^{-3r}\cos\theta^2$
   $T_{\theta r} = T_{r\theta} = \frac{\partial}{\partial \theta}\left(\frac{\partial}{\partial r}T(r,\theta)\right) = \frac{\partial}{\partial \theta}\left[-3e^{-3r}\cos\theta^2\right] = 3re^{-3r}\sin\theta^2\left(\frac{\partial}{\partial\theta}\theta^2\right) = 6r\theta e^{-3r}\sin\theta^2$
   $T_{\theta\theta} = \frac{\partial}{\partial\theta}\left(\frac{\partial}{\partial\theta}T(r,\theta)\right) = \frac{\partial}{\partial\theta}\left[\frac{\partial}{\partial\theta}\left(e^{-3r}\cos\theta^2\right)\right] = \frac{\partial}{\partial\theta}\left[-e^{-3r}\sin\theta^2\left(\frac{\partial}{\partial\theta}\theta^2\right)\right] = \frac{\partial}{\partial\theta}\left(-2\theta e^{-3r}\sin\theta^2\right) = \left(-2\theta e^{-3r}\right)\left(\frac{\partial}{\partial\theta}\sin\theta^2\right) + \left(\sin\theta^2\right)\left[\frac{\partial}{\partial\theta}\left(-2\theta e^{-3r}\right)\right] = \left(-2\theta e^{-3r}\right)\left(\cos\theta^2\right)\left(\frac{\partial}{\partial\theta}\theta^2\right) + \left(\sin\theta^2\right)\left(-2e^{-3r}\right) = -4\theta^2e^{-3r}\cos\theta^2 - 2e^{-3r}\sin\theta^2$

## Answer to Exercise 342 (on page 1127)

1. We are looking for $dz/dt$ only:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$= \frac{\partial}{\partial x}\left[\sin x\cos y\right]\cdot\frac{d}{dt}\left[3\sqrt{t}\right] + \frac{\partial}{\partial y}\left[\sin x\cos y\right]\cdot\frac{d}{dt}\left[2/t\right]$$

$$= \left(\cos x\cos y\right)\cdot\left(\frac{3}{2\sqrt{t}}\right) + \left(-\sin x\sin y\right)\cdot\left(-\frac{2}{t^2}\right)$$

$$= \frac{3\cos x \cos y}{2\sqrt{t}} + \frac{2\sin x \sin y}{t^2}$$

Substituting for $x$ and $y$:

$$\frac{dz}{dt} = \frac{3\cos\left(3\sqrt{t}\right)\cos\left(2/t\right)}{2\sqrt{t}} + \frac{2\sin\left(3\sqrt{t}\right)\sin\left(2/t\right)}{t^2}$$

2. We are looking for $dz/dt$ only:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$= \frac{\partial}{\partial x}\left[\sqrt{1+xy}\right]\cdot\frac{d}{dt}\left[\tan t\right] + \frac{\partial}{\partial y}\left[\sqrt{1+xy}\right]\cdot\frac{d}{dt}\left[\arctan t\right]$$

$$= \left(\frac{y}{2\sqrt{1+xy}}\right)\cdot\left(\sec^2 t\right) + \left(\frac{x}{2\sqrt{1+xy}}\right)\cdot\left(\frac{1}{t^2+1}\right)$$

Substituting for $x$ and $y$:

$$\frac{dz}{dt} = \frac{\tan t \sec^2 t}{2\sqrt{1+\tan t \arctan t}} + \frac{\tan t}{2\sqrt{1+\tan t \arctan t}\,(t^2+1)}$$

$$= \frac{\tan t}{2\sqrt{1+\tan t \arctan t}}\left(\sec^2 t + \frac{1}{t^2+1}\right)$$

3. Finding $\partial z/\partial s$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$= \frac{\partial}{\partial x}\left[\arctan\left(x^2+y^2\right)\right]\cdot\frac{\partial}{\partial s}\left[t\ln s\right] + \frac{\partial}{\partial y}\left[\arctan\left(x^2+y^2\right)\right]\cdot\frac{\partial}{\partial s}\left[se^t\right]$$

$$\left(\frac{2x}{(x^2+y^2)^2+1}\right)\cdot\left(\frac{t}{s}\right) + \left(\frac{2y}{(x^2+y^2)^2+1}\right)\cdot\left(e^t\right)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial s} = \left(\frac{2\left(t\ln s\right)}{\left[(t\ln s)^2+(se^t)^2\right]^2+1}\right)\cdot\left(\frac{t}{s}\right) + \left(\frac{2\left(se^t\right)}{\left[(t\ln s)^2+(t\ln s)^2\right]^2+1}\right)\cdot\left(e^t\right)$$

$$= \left(\frac{2}{\left[t^2(\ln s)^2+se^{2t}\right]^2+1}\right)\cdot\left(\frac{t^2\ln s}{s}+se^{2t}\right)$$

Finding $\partial z/\partial t$:

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= \frac{\partial}{\partial x}\left(\arctan\left(x^2+y^2\right)\right)\cdot\frac{\partial}{\partial t}\left(t\ln s\right)+\frac{\partial}{\partial y}\left(\arctan\left(x^2+y^2\right)\right)\cdot\frac{\partial}{\partial t}\left(se^t\right)$$

$$= \left(\frac{2x}{\left(x^2+y^2\right)^2+1}\right)\cdot\left(\ln s\right)+\left(\frac{2y}{\left(x^2+y^2\right)^2+1}\right)\cdot\left(se^t\right)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial t} = \left(\frac{2}{\left[\left(t\ln s\right)^2+\left(se^t\right)^2\right]^2+1}\right)\cdot\left[t\left(\ln s\right)^2+s^2e^{2t}\right]$$

4. Finding $\partial z/\partial s$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$= \frac{\partial}{\partial x}\left(\sqrt{x}e^{xy}\right)\cdot\frac{\partial}{\partial s}\left(1+st\right)+\frac{\partial}{\partial y}\left(\sqrt{x}e^{xy}\right)\cdot\frac{\partial}{\partial s}\left(s^2-t^2\right)$$

$$= \left[\frac{e^{xy}\left(2xy+1\right)}{2\sqrt{x}}\right]\cdot\left(t\right)+\left[x^{3/2}e^{xy}\right]\cdot\left(2s\right)$$

$$= \left[\frac{e^{xy}}{\sqrt{x}}\right]\cdot\left(\frac{\left(2xy+1\right)t}{2}+x^2\left(2s\right)\right)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial s} = \left[\frac{e^{(1+st)(s^2-t^2)}}{\sqrt{1+st}}\right]\cdot\left(\frac{\left(2(1+st)(s^2-t^2)+1\right)t}{2}+(1+st)^2\left(2s\right)\right)$$

Finding $\partial z/\partial t$:

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= \frac{\partial}{\partial y}\left[\sqrt{x}e^{xy}\right]\cdot\frac{\partial}{\partial t}\left[1+st\right]+\frac{\partial}{\partial y}\left[\sqrt{x}e^{xy}\right]\cdot\frac{\partial}{\partial t}\left[s^2-t^2\right]$$

$$= \left[\frac{e^{xy}\left(2xy+1\right)}{2\sqrt{x}}\right]\cdot\left(s\right)+\left[x^{3/2}e^{xy}\right]\cdot\left(-2t\right)$$

$$= \left[\frac{e^{xy}}{\sqrt{x}}\right]\cdot\left[\frac{\left(2xy+1\right)s}{2}-2tx^2\right]$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial t} = \left[\frac{e^{(1+st)\left(s^2-t^2\right)}}{\sqrt{\left(1+st\right)}}\right]\cdot\left[\frac{\left(2\left(1+st\right)\left(s^2-t^2\right)+1\right)s}{2}-2t\left(1+st\right)^2\right]$$

## Answer to Exercise 343 (on page 1132)

1. $z(1,-1) = (1)^2 e(-1/1 = 1/e$. Therefore, we are looking for tangent lines through the point $(1,-1,1/e)$. Finding a tangent line parallel to the x-axis: $\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\left(x^2 e^{y/x}\right) = x^2\left(\frac{\partial}{\partial x} e^{y/x}\right) + e^{y/x}\left(\frac{\partial}{\partial x} x^2\right) = x^2 e^{y/x}\left(\frac{\partial}{\partial x}\frac{y}{x}\right) + 2xe^{y/x} = x^2 e^{y/x}\left(\frac{-y}{x^2}\right) + 2xe^{y/x} = (2x-y)\,e^{y/x}$ and $z_x(1,-1) = (2(1)-(-1))\,e^{-1/1} = (3)\,e^{-1} = 3/e$. So, the slope of a line tangent to the surface at $(1,-1,1/e)$ parallel to the x-axis is $3/e$ and an equation for that line is $z = 3/e\,(x-1) - 1/e$.

   Finding a tangent line parallel to the y-axis: $\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}\left(x^2 e^{y/x}\right) = xe^{y/x}$ and $z_y(1,-1) = (1)e^{-1/1} = 1/e$. So, the slope of a line tangent to the surface at $(1,-1,1/e)$ parallel to the y-axis is $1/e$ and an equation for that line is $z = 1/e\,(y+1) - 1/e$.

   The function is changing faster in the x-direction.

2. $z(\pi,\pi/2) = \cos(\pi) + \frac{\pi}{2}\sin(\pi/2) = \frac{\pi}{2} - 1$. Therefore, we are looking for tangent lines through the point $(\pi,\pi/2,\pi/2-1)$. Finding a tangent line parallel to the x-axis: $\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(\cos x + y\sin y) = -\sin x$ and $z_x(\pi,\pi/2) = -\sin\pi = 0$. So, the slope of a line tangent to the surface at $(\pi,\pi/2,\pi/2-1)$ parallel to the x-axis is 0 and an equation for that line is $z = \pi/2 - 1$.

   Finding a tangent line parallel to the y-axis: $\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(\cos x + y\sin y) = y\left(\frac{\partial}{\partial y}\sin y\right) + \sin y\left(\frac{\partial}{\partial y} y\right) = y\cos y + \sin y$ and $z_y(\pi,\pi/2) = \left(\frac{\pi}{2}\right)\cos\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right) = 1$. So, the slope of a line tangent to the surface at $(\pi,\pi/2,\pi/2-1)$ parallel to the y-axis is 1 and an equation for that line is $z = (y-\pi/2) - (\pi/2-1) = y - \pi + 1$.

   The function is changing faster in the y-direction.

3. $z(3,2) = 3^2(2) - 3(3)(2^2) = 18 - 36 = -18$. Therefore, we are looking for tangent lines through the point $(3,2,-18)$. Finding a tangent line parallel to the x-axis: $\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\left(x^2 y - 3xy^2\right) = 2xy - 3y^2$ and $z_x(3,2) = 2(3)(2) - 3(2)^2 = 0$. So, the slope of a line tangent to the surface at $(3,2,-18)$ is 0 and an equation for that line is $z = -18$

   Finding a tangent line parallel to the y-axis: $\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}\left(x^2 y - 3xy^2\right) = x^2 - 6xy$ and $z_y(3,2) = 3^2 - 6(3)(2) = 9 - 36 = -27$. So, the slope of a line tangent to the surface at $(3,2,-18)$ is $-27$ and an equation for that line is $z = -27(y-2) + -18 = -27y + 54 - 18 = 36 - 27y$.

   The function is changing faster in the y-direction.

## Answer to Exercise 344 (on page 1140)

1. First, we define $\mathbf{u}$ such that $|\mathbf{u}| = 1$ and $\mathbf{u}$ is in the same direction as $\mathbf{v}$:

$$\mathbf{u} = k\mathbf{v} = [-3k, -4k]$$

$$\sqrt{(-3k)^2 + (4k)^2} = 1$$

$$\sqrt{9k^2 + 16k^2} = \sqrt{25k^2} = 5k = 1$$

$$k = \frac{1}{5}$$

Therefore, we define $\mathbf{u} = [-3/5, 4/5]$ and the directional derivative is given by:

$$D_u(x, y) = \left(\frac{-3}{5}\right) \frac{\partial}{\partial x} f(x, y) + \left(\frac{4}{5}\right) \frac{\partial}{\partial y} f(x, y)$$

$$= \left(\frac{-3}{5}\right) \frac{\partial}{\partial x} \left[e^{3x} \sin 2y\right] + \left(\frac{4}{5}\right) \frac{\partial}{\partial y} \left[e^{3x} \sin 2y\right]$$

$$= \left(\frac{-3}{5}\right) \left(3e^{3x} \sin 2y\right) + \left(\frac{4}{5}\right) \left(2e^{3x} \cos 2y\right)$$

And substituting for $(x, y) = (0, \pi/6)$:

$$D_u(0, \pi/6) = \left(\frac{-3}{5}\right) \cdot \left[3e^{3 \cdot 0} \sin\left(\frac{\pi}{3}\right)\right] + \left(\frac{4}{5}\right) \cdot \left[2e^{3 \cdot 0} \cos\left(\frac{\pi}{3}\right)\right]$$

$$D_u(0, \pi/6) = \left(\frac{-3}{5}\right) \cdot \left[3 \cdot \frac{\sqrt{3}}{2}\right] + \left(\frac{4}{5}\right) \cdot \left[2 \cdot \frac{1}{2}\right]$$

$$d_u(0, \pi/6) = \left(\frac{-3}{5}\right) \cdot \left(\frac{3\sqrt{3}}{2}\right) + \left(\frac{4}{5}\right) \cdot (1)$$

$$D_u(0, \pi/6) = \frac{-9\sqrt{3}}{10} + \frac{8}{10} = \frac{8 - 9\sqrt{3}}{10} \approx -0.759$$

2. We can express $\mathbf{v}$ as $\mathbf{v} = [2, -1]$. And we define $\mathbf{u}$ such that $|\mathbf{u}| = 1$ and $\mathbf{u}$ is in the same direction as $\mathbf{v}$:

$$\mathbf{u} = k\mathbf{v} = [2k, -k]$$

$$\sqrt{(2k)^2 + (-k)^2} = 1$$

$$\sqrt{4k^2 + k^2} = \sqrt{5}k = 1$$

$$k = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

Therefore, we define $\mathbf{u} = \left[2\sqrt{5}/5, -\sqrt{5}/5\right]$ and the directional derivative is given by:

$$D_u(x, y) = \left(\frac{2\sqrt{5}}{5}\right) \frac{\partial}{\partial x} f(x, y) + \left(\frac{-\sqrt{5}}{5}\right) \frac{\partial}{\partial y} f(x, y)$$

$$= \left(\frac{2\sqrt{5}}{5}\right)\frac{\partial}{\partial x}\left[x^2y + xy^3\right] + \left(\frac{-\sqrt{5}}{5}\right)\frac{\partial}{\partial y}\left[x^2y + xy^3\right]$$

$$= \left(\frac{2\sqrt{5}}{5}\right)\left[2xy + y^3\right] + \left(\frac{-\sqrt{5}}{5}\right)\left[x^2 + 3xy^2\right]$$

And substituting $(x, y) = (2, 4)$:

$$D_u(2,4) = \left(\frac{2\sqrt{5}}{5}\right)\left[2(2)(4) + 4^3\right] + \left(\frac{-\sqrt{5}}{5}\right)\left[2^2 + 3(2)(4^2)\right]$$

$$D_u(2,4) = \left(\frac{2\sqrt{5}}{5}\right)[80] + \left(\frac{-\sqrt{5}}{5}\right)[100]$$

$$D_u(2,4) = 32\sqrt{5} - 20\sqrt{5} = 12\sqrt{5} \approx 26.833$$

3. We define $\mathbf{u}$ such that $|\mathbf{u}| = 1$ and $\mathbf{u}$ is in the same direction as $\mathbf{v}$:

$$\mathbf{u} = k\mathbf{v} = [k, k, k]$$

$$\sqrt{k^2 + k^2 + k^2} = 1$$

$$\sqrt{3}k = 1$$

$$k = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

Therefore, we let $\mathbf{u} = \left[\sqrt{3}/3, \sqrt{3}/3, \sqrt{3}/3\right]$ and the directional derivative is given by:

$$D_u(x, y, z) = \left(\frac{\sqrt{3}}{3}\right)\frac{\partial}{\partial x}f(x, y, z) + \left(\frac{\sqrt{3}}{3}\right)\frac{\partial}{\partial y}f(x, y, z) + \left(\frac{\sqrt{3}}{3}\right)\frac{\partial}{\partial z}f(x, y, z)$$

$$= \left(\frac{\sqrt{3}}{3}\right)\left[\frac{\partial}{\partial x}\ln\left(x^2 + 3y - z\right) + \frac{\partial}{\partial y}\ln\left(x^2 + 3y - z\right) + \frac{\partial}{\partial z}\ln\left(x^2 + 3y - z\right)\right]$$

$$= \left(\frac{\sqrt{3}}{3}\right)\left[\frac{2x}{x^2 + 3y - z} + \frac{3}{x^2 + 3y - z} + \frac{-1}{x^2 + 3y - z}\right]$$

$$= \left(\frac{\sqrt{3}}{3}\right)\left[\frac{2x + 2}{x^2 + 3y - z}\right] = \frac{\sqrt{3}\,(2x + 2)}{3\,(x^2 + 3y - z)}$$

And substituting $(x, y, z) = (2, 2, 1)$:

$$D_u(2,2,1) = \frac{\sqrt{3}\,(2(2) + 2)}{3\,(2^2 + 3(2) - 1)} = \frac{\sqrt{3}\,(6)}{3\,(9)} = \frac{2\sqrt{3}}{9} \approx 0.385$$

## Answer to Exercise 345 (on page 1143)

1. $z = f(50, 50) = 3000 - 0.01(50)^2 - 0.02(50)^2 = 2925$

2. A south-pointing unit vector is $\mathbf{u} = [0, -1]$. To find the rate of change, we find the directional derivative in the direction of $\mathbf{u}$ at $(50, 50)$:

$$D_u f(x, y) = (-1)\left[\frac{\partial}{\partial y}\left(3000 - 0.01x^2 - 0.02y^2\right)\right]$$

$$D_u f(x, y) = (-1)(-0.04y) = 0.04y$$

   And at $(50, 50)$, $D_u f(50, 50) = 0.04(50) = 2 > 0$. Therefore, if you walk south, you will ascend.

3. A northwest-pointing unit vector is $\mathbf{u} = \left[-\sqrt{2}/2, \sqrt{2}/2\right]$. To find the rate of change, we find the directional derivative at $(50, 50)$ in the direction of $\mathbf{u}$:

$$D_u f(x, y) = \left(\frac{-\sqrt{2}}{2}\right)\left[\frac{\partial}{\partial x} f(x, y)\right] + \left(\frac{\sqrt{2}}{2}\right)\left[\frac{\partial}{\partial y} f(x, y)\right]$$

$$D_u f(x, y) = \left(\frac{-\sqrt{2}}{2}\right)[-0.02x] + \left(\frac{\sqrt{2}}{2}\right)[-0.04y]$$

$$D_u f(x, y) = 0.01\sqrt{2}x - 0.02\sqrt{2}y$$

$$D_u f(50, 50) = 0.01\sqrt{2}(50) - 0.02\sqrt{2}(50) = \frac{-\sqrt{2}}{2} \approx -0.707$$

   The rate of elevation change walking northwest is approximately $-0.707$, so you will descend and your rate of elevation change would be less than if you walked south.

4. To find the direction of maximum elevation gain, we find the direction the gradient vector points in:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$

$$\nabla f = [-0.02x, -0.04y]$$

   And at $(50, 50)$,

$$\nabla f(50, 50) = [-0.02(50), -0.04(50)] = [-1, -2]$$

   Therefore, the rate of greatest elevation change is in a south-by-southwest direction indicated by the vector $[-1, -2]$ and the rate of elevation change is $|\nabla f(50, 50)| = \sqrt{(-1)^2 + (-2)^2} = \sqrt{5}$. Notice this is greater than the other two rates of change we have found.

## Answer to Exercise 346 (on page 1145)

1.
$$\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}=\frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(x^2-y^2\right)\right]+\frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}\left(x^2-y^2\right)\right]$$
$$=\frac{\partial}{\partial x}[2x]+\frac{\partial}{\partial y}[-2y]=2-2=0$$

Therefore, $f(x,y)=x^2-y^2$ is a solution to Laplace's Equation.

2.
$$\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}=\frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}(\sin x\cosh y+\cos x\sinh y)\right]+\frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}(\sin x\cosh y+\cos x\sinh y)\right]$$
$$=\frac{\partial}{\partial x}[\cos x\cosh y-\sin x\sinh y]+\frac{\partial}{\partial y}[\sin x\sinh y+\cos x\cosh y]$$
$$=-\sin x\cosh y-\cos x\sinh y+\sin x\cosh y+\cos x\sinh y=0$$

Therefore, $f(x,y)=\sin x\cosh y+\cos x\sinh y$ is a solution to Laplace's Equation.

3.
$$\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}=\frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(e^{-x}\cos y-e^{-y}\cos x\right)\right]+\frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}\left(e^{-x}\cos y-e^{-y}\cos x\right)\right]$$
$$=\frac{\partial}{\partial x}\left[-e^{-x}\cos y+e^{-y}\sin x\right]+\frac{\partial}{\partial y}\left[-e^{-x}\sin y+e^{-y}\cos x\right]$$
$$=e^{-x}\cos y+e^{-y}\cos x-e^{-x}\cos y-e^{-y}\cos x=0$$

Therefore, $f(x,y)=e^{-x}\cos y-e^{-y}\cos x$.

## Answer to Exercise 347 (on page 1147)

1. Finding the partial derivatives:
$$\frac{\partial^2 f}{\partial t^2}=\frac{\partial}{\partial t}\left[\frac{\partial}{\partial t}(\cos(kx)\cos(akt))\right]=\frac{\partial}{\partial t}[-ak\cos(kx)\sin(akt)]$$
$$\frac{\partial^2 f}{\partial t^2}=-a^2k^2\cos(kx)\cos(akt)$$
$$\frac{\partial^2 f}{\partial x^2}=\frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}(\cos(kx)\cos(akt))\right]=\frac{\partial}{\partial x}[-k\sin(kx)\cos(akt)]$$

$$\frac{\partial^2 f}{\partial x^2} = -k^2 \cos(kx)\cos(akt)$$

And we see that:

$$a^2\frac{\partial^2 f}{\partial x^2} = -a^2k^2 \cos(kx)\cos(akt) = \frac{\partial^2 f}{\partial t^2}$$

Therefore, $f(x,t) = \cos(kx)\cos(akt)$ satisfies the Wave Equation.

2. Finding the partial derivatives:

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{\partial}{\partial t}\left(\sin(x-at) + \ln(x+at)\right)\right]$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[-a\cos(x-at) + \frac{a}{x+at}\right] = -a^2\sin(x-at) + \frac{-a^2}{(x+at)^2}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(\sin(x-at) + \ln(x+at)\right)\right]$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\cos(x-at) + \frac{1}{x+at}\right] = -\sin(x-at) + \frac{-1}{(x+at)^2}$$

And we see that:

$$a^2\frac{\partial^2 f}{\partial x^2} = -a^2\sin(x-at) + \frac{-a^2}{(x+at)^2} = \frac{\partial^2 f}{\partial t^2}$$

Therefore, $f(x,t) = \sin(x-at) + \ln(x+at)$ satisfies the Wave Equation.

3. Finding the partial derivatives:

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{\partial}{\partial t}\left(\frac{t}{a^2t^2-x^2}\right)\right] = \frac{\partial}{\partial t}\left[\frac{\left(a^2t^2-x^2\right) - t\left(2a^2t\right)}{\left(a^2t^2-x^2\right)^2}\right]$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{a^2t^2-x^2-2a^2t^2}{\left(a^2t^2-x^2\right)^2}\right] = \frac{\partial}{\partial t}\left[\frac{-a^2t^2-x^2}{\left(a^2t^2-x^2\right)^2}\right]$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{\left(a^2t^2-x^2\right)^2\left(-2a^2t\right) - \left(-a^2t^2-x^2\right)\left(2\left(a^2t^2-x^2\right)\left(2a^2t\right)\right)}{\left(a^2t^2-x^2\right)^4}$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{\left(a^2t^2-x^2\right)\left(-2a^2t\right) - \left(-a^2t^2-x^2\right)\left(4a^2t\right)}{\left(a^2t^2-x^2\right)^3}$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{-2a^4t^3 + 2a^2tx^2 + 4a^4t^3 + 4a^2tx^2}{\left(a^2t^2-x^2\right)^3}$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{2a^4t^3 + 6a^2tx^2}{\left(a^2t^2-x^2\right)^3} = 2a^2t\left(\frac{a^2t^2+3x^2}{\left(a^2t^2-x^2\right)^3}\right)$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(\frac{t}{a^2t^2 - x^2}\right)\right] = \frac{\partial}{\partial x}\left[\frac{\left(a^2t^2 - x^2\right)(0) - t\,(-2x)}{\left(a^2t^2 - x^2\right)^2}\right]$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{2tx}{\left(a^2t^2 - x^2\right)^2}\right] = \frac{\left(a^2t^2 - x^2\right)^2(2t) - (2tx)\left(2\left(a^2t^2 - x^2\right)(-2x)\right)}{\left(a^2t^2 - x^2\right)^4}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\left(a^2t^2 - x^2\right)(2t) - (2tx)(2)(-2x)}{\left(a^2t^2 - x^2\right)^3}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{2a^2t^3 - 2tx^2 + 8tx^2}{\left(a^2t^2 - x^2\right)^3} = \frac{2a^2t^3 + 6tx^2}{\left(a^2t^2 - x^2\right)^3} = 2t\left(\frac{a^2t^2 + 3x^2}{\left(a^2t^2 - x^2\right)^3}\right)$$

And we see that:

$$a^2\frac{\partial^2 f}{\partial x^2} = 2a^2t\left(\frac{a^2t^2 + 3x^2}{\left(a^2t^2 - x^2\right)^3}\right) = \frac{\partial^2 f}{\partial t^2}$$

Therefore, $f(x,t) = \frac{t}{a^2t^2 - x^2}$ satisfies the Wave Equation.

## Answer to Exercise 348 (on page 1149)

1. The marginal utility of labor is given by $\partial P/\partial L$:

$$\frac{\partial P}{\partial L} = \frac{\partial}{\partial L}\left[1.01L^{0.75}K^{0.25}\right] = 0.7575L^{-0.25}K^{0.25}$$

2. The marginal utility of capital is given by $\partial P/\partial K$:

$$\frac{\partial P}{\partial K} = \frac{\partial}{\partial K}\left[1.01L^{0.75}K^{0.25}\right] = 0.2525L^{0.75}K^{-0.75}$$

3. Finding the marginal utility of labor in 1916:

$$\frac{\partial P}{\partial L} = 0.7575\,(382)^{-.25}\,(126)^{0.25} \approx 0.574$$

And finding the marginal utility of capital in 1916:

$$\frac{\partial P}{\partial K} = 0.2525\,(382)^{0.75}\,(126)^{-0.75} \approx 0.580$$

4. Since the marginal utility of capital is greater, I would invest in capital. This would yield a greater increase in production than the same investment in labor.

## Answer to Exercise 349 (on page 1166)

1. $3 \times 5$
2. $1 \times 2$
3. $4 \times 3$

## Answer to Exercise 350 (on page 1166)

1. The matrix should have 1 row and 3 columns. For example,

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

2. The matrix should have 2 rows and 4 columns. For example,

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

3. The matrix should have 4 rows and 3 columns. For example,

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}$$

## Answer to Exercise 351 (on page 1168)

1. $\boldsymbol{A} + \boldsymbol{B} = \begin{bmatrix} -2 & 7 & -2 & 10 \end{bmatrix}$. $\boldsymbol{A} - \boldsymbol{B} = \begin{bmatrix} 2 & 1 & 2 & 0 \end{bmatrix}$. $\boldsymbol{B} - \boldsymbol{A} = \begin{bmatrix} -2 & -1 & -2 & 0 \end{bmatrix}$
2. $\boldsymbol{A} + \boldsymbol{B} = \begin{bmatrix} 9 & -4 & -3 \\ -4 & -6 & 3 \\ -10 & 6 & -4 \end{bmatrix}$. $\boldsymbol{A} - \boldsymbol{B} = \begin{bmatrix} -1 & -4 & -1 \\ 6 & 0 & 7 \\ 0 & 0 & 4 \end{bmatrix}$. $\boldsymbol{B} - \boldsymbol{A} = \begin{bmatrix} 1 & 4 & 1 \\ -6 & 0 & -7 \\ 0 & 0 & -4 \end{bmatrix}$.
3. $\boldsymbol{A} + \boldsymbol{B} = \begin{bmatrix} -7 & -3 & -2 & -6 \\ 5 & 1 & 0 & 0 \end{bmatrix}$. $\boldsymbol{A} - \boldsymbol{B} = \begin{bmatrix} 3 & 1 & -8 & 4 \\ 5 & -9 & 8 & 6 \end{bmatrix}$. $\boldsymbol{B} - \boldsymbol{A} = \begin{bmatrix} -3 & -1 & 8 & -4 \\ -5 & 9 & -8 & -6 \end{bmatrix}$.

## Answer to Exercise 352 (on page 1169)

It is possible to compute $\mathbf{a} \cdot \mathbf{d}$, $\mathbf{b} \cdot \mathbf{e}$, and $\mathbf{c} \cdot \mathbf{f}$:

1. $\mathbf{a} \cdot \mathbf{d} = 1(-5) + 2(-1) = -5 + (-2) = -7$
2. $\mathbf{b} \cdot \mathbf{e} = -3(1) + 3(-5) + 5(3) + -5(1) = -3 + (-15) + 15 - 5 = -8$
3. $\mathbf{c} \cdot \mathbf{f} = 1(4) + 2(1) + -1(-3) = 4 + 2 + 3 = 9$

## Answer to Exercise 353 (on page 1171)

1. $\begin{bmatrix} -2 & -1 & 5 \end{bmatrix}$
2. $\begin{bmatrix} 0 & 5 & 1 \\ 0 & 25 & 5 \\ 0 & -25 & -5 \\ 0 & 20 & 4 \\ 0 & 5 & 1 \end{bmatrix}$
3. $\begin{bmatrix} -9 & -17 & 11 \\ -6 & 40 & -4 \\ 15 & 7 & -13 \\ 9 & -8 & -1 \end{bmatrix}$

## Answer to Exercise 354 (on page 1172)

1. $A \times B = \begin{bmatrix} 8 & -6 & 10 & 4 \\ -8 & 6 & -10 & -4 \\ -4 & 3 & -5 & -2 \\ 8 & -6+1- & 4 & \end{bmatrix}$ and $B \times A = \begin{bmatrix} 13 \end{bmatrix}$
2. $A \times B = \begin{bmatrix} -2 & 16 & 16 \\ 5 & -24 & -8 \\ -4 & 22 & 12 \end{bmatrix}$ and $B \times A = \begin{bmatrix} 8 & 6 \\ -12 & -22 \end{bmatrix}$
3. $A \times B = \begin{bmatrix} -22 & 1 \\ 15 & -4 \end{bmatrix}$ and $B \times A = \begin{bmatrix} 12 & 0 & 15 & 3 \\ -4 & 0 & 1 & -15 \\ -16 & - & -17 & -11 \\ -14 & 0 & -10 & -21 \end{bmatrix}$

## Answer to Exercise 355 (on page 1172)

$$A \times B = \begin{bmatrix} -3 & 5 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} -1 & 1 \\ 4 & -3 \end{bmatrix} = \begin{bmatrix} -3(-1)+5(4) & -3(1)+5(-3) \\ -1(-1)+0(4) & -1(1)+0(-3) \end{bmatrix} = \begin{bmatrix} 23 & -18 \\ 1 & -1 \end{bmatrix}$$

$$B \times A = \begin{bmatrix} -1 & 1 \\ 4 & -3 \end{bmatrix} \cdot \begin{bmatrix} -3 & 5 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1(-3)+1(-1) & -1(5)+1(0) \\ 4(-3)+-3(-1) & 4(5)+-3(0) \end{bmatrix} = \begin{bmatrix} 2 & -5 \\ -9 & 20 \end{bmatrix}$$

## Answer to Exercise 356 (on page 1179)

$$A = A^t = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

## Answer to Exercise 357 (on page 1183)

1. We are looking for $a_1$ and $a_2$ such that:

$$a_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Which creates the system of equations:

$$a_1 - 3a_2 = 4$$

$$2a_1 + a_2 = 5$$

We can multiply the first equation by $-2$ and add it to the second to solve for $a_2$:

$$-2(a_1 - 3a_2) + 2a_1 + a_2 = -2(4) + 5$$

$$6a_2 + a_2 = -8 + 5$$

$$7a_2 = -3$$

$$a_2 = -\frac{3}{7}$$

Substituting $a_2$ back into an equation and solving for $a_1$:

$$a_1 - 3\left(-\frac{3}{7}\right) = 4$$

$$a_1 + \frac{9}{7} = 4$$

$$a_1 = \frac{19}{7}$$

Therefore, $\frac{19}{7}\begin{bmatrix}1\\2\end{bmatrix} - \frac{3}{7}\begin{bmatrix}-3\\1\end{bmatrix} = \begin{bmatrix}4\\5\end{bmatrix}$.

2. We are looking for $a_1$ and $a_2$ such that:

$$a_1\begin{bmatrix}9\\4\end{bmatrix} + a_2\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}-5\\3\end{bmatrix}$$

Which creates the system of equations:

$$9a_1 = -5$$

$$4a_1 + a_2 = 3$$

We can find $a_1$ from the first equation:

$$a_1 = -\frac{5}{9}$$

Substituting for $a_1$ back into the second equation and solving for $a_2$:

$$4\left(-\frac{5}{9}\right) + a_2 = 3$$

$$a_2 - \frac{20}{9} = 3$$

$$a_2 = \frac{47}{9}$$

Therefore, $-\frac{5}{9}\begin{bmatrix}9\\4\end{bmatrix} + \frac{47}{9}\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}-5\\3\end{bmatrix}$.

3. We are looking for $a_1$ and $a_2$ such that:

$$a_1\begin{bmatrix}7\\-2\end{bmatrix} + a_2\begin{bmatrix}-8\\4\end{bmatrix} = \begin{bmatrix}6\\-2\end{bmatrix}$$

Which yields the system of equations:

$$7a_1 - 8a_2 = 6$$

$$-2a_1 + 4a_2 = -2$$

Doubling the second equation and adding it to the first:

$$7a_1 - 8a_2 + 2\left(-2a_1 + 4a_2\right) = 6 + 2\left(-2\right)$$

$$7a_1 - 8a_2 - 4a_1 + 8a_2 = 6 - 4$$
$$3a_1 = 2$$
$$a_1 = \frac{2}{3}$$

Substituting for $a_1$ back into the second equation and solving for $a_2$:

$$-2\left(\frac{2}{3}\right) + 4a_2 = -2$$
$$-\frac{4}{3} + 4a_2 = -2$$
$$4a_2 = -\frac{2}{3}$$
$$a_2 = -\frac{1}{6}$$

Therefore, $\frac{2}{3}\begin{bmatrix} 7 \\ -2 \end{bmatrix} - \frac{1}{6}\begin{bmatrix} -8 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$.

## Answer to Exercise 358 (on page 1190)

$$A = \begin{bmatrix} -4 & 9 & -8 \\ -1 & 0 & 6 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$
$$A\vec{x} = \vec{b} \implies \begin{bmatrix} -4 & 9 & -8 \\ -1 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

## Answer to Exercise 359 (on page 1201)

Notice immediately that $R_2 = 2R_1$, which automatically means redundancy.

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 1 & 1 & 0 & 2 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix}$$
$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow -R_2} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \boxed{\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}}$$

The RREF is:

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This gives us the system:

$$x_1 + x_3 = 1,$$
$$x_2 - x_3 = 1.$$

So,

$$x_1 = 1 - x_3, \qquad x_2 = 1 + x_3$$

making $x_3$ free. We can write this as: $\vec{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \qquad x_3 \in \mathbb{R}$ Equivalently, as a set:

$$\left\{ \begin{pmatrix} 1 - x_3 \\ 1 + x_3 \\ x_3 \end{pmatrix} : x_3 \in \mathbb{R} \right\}$$

Since $x_3$ can be anything, there are infinitely many solutions. Equivalently, noticing that the row of all zeros implies infinitely many solution is a valid step to come to an answer.

## Answer to Exercise 360 (on page 1202)

$$\begin{bmatrix} 1 & 0 & 2 & -1 \\ 1 & 1 & 2 & -1 \\ 0 & 0 & 3 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 \leftarrow \frac{1}{3} R_3} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{3} \end{bmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_3} \boxed{\begin{bmatrix} 1 & 0 & 0 & -\frac{5}{3} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{3} \end{bmatrix}}$$

This gives us a unique solution:

$$x_1 = -\frac{5}{3}, \quad x_2 = 0, \quad x_3 = \frac{1}{3}$$

or as a vector $\vec{x} = \begin{bmatrix} -\frac{5}{3} \\ 0 \\ \frac{1}{3} \end{bmatrix}$

## Answer to Exercise 361 (on page 1202)

Notice that one row of the matrix states $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$, which implies $0x_1 + 0x_2 + 0x_3 = 1$, which is not possible. Therefore the matrix represents an inconsistent system and there is no solution.

## Answer to Exercise 362 (on page 1202)

The RREF of this matrix is

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Note that there are two rows of zeroes, and the first column has a 1. This implies two dependent rows ($R_2 = 2R_1 \qquad R_3 = 3R_1$), and with only 1 pivot column, $\text{rank}(A) = 1$.

## Answer to Exercise 363 (on page 1203)

By definition in this chapter:

$$\text{rank}(A) = \text{number of pivots} = 3$$

A $4 \times 6$ matrix has $n = 6$ columns, so

$$\#\text{ free variables} = n - \text{rank}(A) = 6 - 3 = 3.$$

So, free variables $= 3$

## Answer to Exercise 364 (on page 1203)

There are $n = 4$ variables $(x_1, x_2, x_3, x_4)$.

1. Pivot columns are columns $1, 2, 4$, so the pivot columns are $x_1,\ x_2,\ x_4$.

2. Free variables: the only non-pivot column is column 3, so the free variable is $x_3$
3. The dimension of the solution set is the number of free variables: 1. Equivalently, dimension $= n - \text{rank} = 4 - 3 = 1$

## Answer to Exercise 365 (on page 1210)

$$Av = (11 \ 37 \ -43)$$

## Answer to Exercise 366 (on page 1211)

The different methods can be represented in a matrix:

$$\begin{bmatrix} 0.20 & 0.50 & 0.15 & 0.15 \\ 0 & 0.50 & 0.20 & 0.30 \\ 0 & 0 & 0.4 & 0.6 \end{bmatrix}$$

And Suzy's individual grades can be represented by a vector:

$$\begin{bmatrix} 65 \\ 30 \\ 80 \\ 75 \end{bmatrix}$$

To see the results of the three different methods, we can multiply the matrix and the vector:

$$\begin{bmatrix} 0.20 & 0.50 & 0.15 & 0.15 \\ 0 & 0.50 & 0.20 & 0.30 \\ 0 & 0 & 0.4 & 0.6 \end{bmatrix} \cdot \begin{bmatrix} 65 \\ 30 \\ 80 \\ 75 \end{bmatrix} = \begin{bmatrix} 0.2(65) + 0.5(30) + 0.15(80) + 0.15(75) \\ 0(65) + 0.5(30) + 0.2(80) + 0.3(75) \\ 0(65) + 0(30) + 0.4(80) + 0.6(75) \end{bmatrix}$$

Which yields:

$$\begin{bmatrix} 51.25 \\ 53.5 \\ 77 \end{bmatrix}$$

Since method C yields the highest grade, the professor will post a final grade of 77.

## Answer to Exercise 367 (on page 1214)

Two vectors are linearly dependent if one is a scalar multiple of the other.

1. Dependent, since $\begin{bmatrix} 4 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.
2. Dependent, since $\begin{bmatrix} -6 \\ -2 \end{bmatrix} = -2 \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.
3. Independent, since neither vector is a scalar multiple of the other.
4. Dependent, since $\begin{bmatrix} 1 \\ -2 \\ -3 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}$.

## Answer to Exercise 368 (on page 1215)

Any set containing the zero vector is linearly dependent, since a nontrivial linear combination can equal the zero vector.

1. Dependent.
2. Dependent.
3. Dependent.

## Answer to Exercise 369 (on page 1216)

1. The vectors are dependent when $\begin{bmatrix} 1 \\ k \end{bmatrix}$ is a scalar multiple of $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$. This occurs when $k = 2$.
2. The vectors are dependent for all values of $k$, since

$$\begin{bmatrix} 2k \\ 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} k \\ 1 \\ 2 \end{bmatrix}.$$

## Answer to Exercise 370 (on page 1219)

We see that $\frac{\mathbf{a}}{-4} = -\frac{1}{4}\begin{bmatrix}-4\\1\\4\end{bmatrix} = \begin{bmatrix}1\\-\frac{1}{4}\\-1\end{bmatrix} = \mathbf{d}$. Additionally, $\frac{3}{2}\mathbf{a} = \frac{3}{2}\begin{bmatrix}-4\\1\\4\end{bmatrix} = \begin{bmatrix}-6\\\frac{3}{2}\\6\end{bmatrix} = \mathbf{f}$. Therefore, vectors $\mathbf{a}$, $\mathbf{d}$, and $\mathbf{f}$ are linearly dependent.

We also see that $\frac{1}{2}\mathbf{c} = \frac{1}{2}\begin{bmatrix}2\\-4\\6\end{bmatrix} = \begin{bmatrix}1\\-2\\3\end{bmatrix} = \mathbf{e}$. Therefore, vectors $\mathbf{c}$ and $\mathbf{e}$ are linearly dependent. Vector $\mathbf{b}$ is not linearly dependent to any of the other vectors.

## Answer to Exercise 371 (on page 1227)

a Solving $A\vec{x} = \vec{0}$ gives us two equations:

$$\begin{cases} x_1 + 2x_4 - x_5 = 0 \\ x_2 - x_3 - x_4 - 3x_5 = 0 \end{cases} \implies \begin{cases} x_1 = -2x_4 + x_5 \\ x_2 = x_3 + x_4 + 3x_5 \end{cases}$$

So we establish $x_3$, $x_4$, and $x_5$ as bound or fixed variables. So, every vector in the nullspace has the form

$$\vec{x} = \begin{bmatrix} -2x_4 + x_4 \\ x_3 + x_4 + 3x_4 \\ x_3 \\ x_4 \\ x_4 \end{bmatrix}$$

or, equivalently

$$\vec{x} = x_3\begin{bmatrix}0\\1\\1\\0\\0\end{bmatrix} + x_4\begin{bmatrix}-2\\1\\0\\1\\0\end{bmatrix} + x_4\begin{bmatrix}1\\3\\0\\0\\1\end{bmatrix}$$

So a basis can be formed by the set

$$\left\{\begin{bmatrix}0\\1\\1\\0\\0\end{bmatrix}, \begin{bmatrix}-2\\1\\0\\1\\0\end{bmatrix}, \begin{bmatrix}1\\3\\0\\0\\1\end{bmatrix}\right\}$$

b We are given a vector $\vec{x} = \begin{bmatrix} -3 \\ x_2 \\ 1 \\ 2 \\ x_5 \end{bmatrix}$.

Recall that we have $x_1 = -2x_4 + x_5$, so inputting our knowns we can say $-3 = -2(2) + x_5 \implies = x_5 = 1$.

We can then solve $x_2 = x_3 + x_4 + 3x_5$ for $x_2$:

$$x_2 = 1 + 2 + 3(1) = 6$$

These two variables give us the $\vec{x}$

$$\vec{x} = \begin{bmatrix} -3 \\ 6 \\ 1 \\ 2 \\ 1 \end{bmatrix}$$

## Answer to Exercise 372 (on page 1231)

a

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

b Columns 1, 2, and 3 are pivot columns (containing a pivot). Column 4 is a non-pivot or free column. This tells you that there are 3 fixed variables, and 1 free one. The nullspace will be 1 dimensional.

c We have the system of equations:

$$\begin{aligned} x_1 + x_4 &= 0, \\ x_2 + x_4 &= 0, \\ x_3 &= 0. \end{aligned}$$

Letting $x_4$ be free, $\vec{x} = x_4 \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$. The $N(A) = \left\{ \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}$ with a dimension of 1.

d The row space is the subspace spanned by the rows of $A$. A convenient basis for the

row space is given by the nonzero rows of rref$(A)$. Thus a basis is

$$\left\{\begin{bmatrix}1 & 0 & 0 & 1\end{bmatrix}, \begin{bmatrix}0 & 1 & 0 & 1\end{bmatrix}, \begin{bmatrix}0 & 0 & 1 & 0\end{bmatrix}\right\}.$$

These three rows are linearly independent, and they span the row space because row reduction did not create any new row space, just simplified the set. The rowspace has rowspace$(A) = 3$.

e For the column space, we must take columns from the *original* matrix $A$. The usual rule is: pivot columns of the original $A$ form a basis for $C(A)$. Since the pivot columns are $1,\ 2,\ 3$, a basis is the set of the first three columns of $A$

$$\mathbf{c}_1 = \begin{bmatrix}1\\2\\1\\0\end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix}2\\4\\1\\1\end{bmatrix}, \quad \mathbf{c}_3 = \begin{bmatrix}-1\\-2\\0\\1\end{bmatrix}$$

and the column space is the span of each original pivot column:

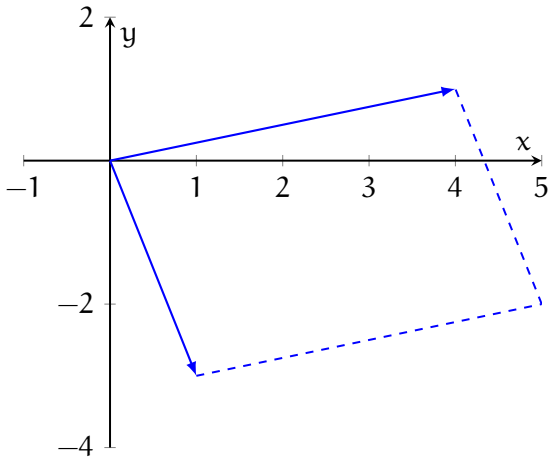
$$C(A) = \text{span}\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$$

## Answer to Exercise 373 (on page 1236)

1. Since the second vector is a scalar multiple of the first, the span of $S = \{[1,2,4], [-2,-4,-8]\}$ is a *line*.
2. Since the second vector is not a scalar multiple of the first, the span of $S = \{[2,0,0,], [0,1,3]\}$ is a *plane*.
3. None of the three vectors are scalar multiples or linear combinations of the other two. Therefore, the span of $S = \{[3,0,0], [0,3,3], [3,3,2]\}$ is $\mathbb{R}^3$.

## Answer to Exercise 374 (on page 1243)

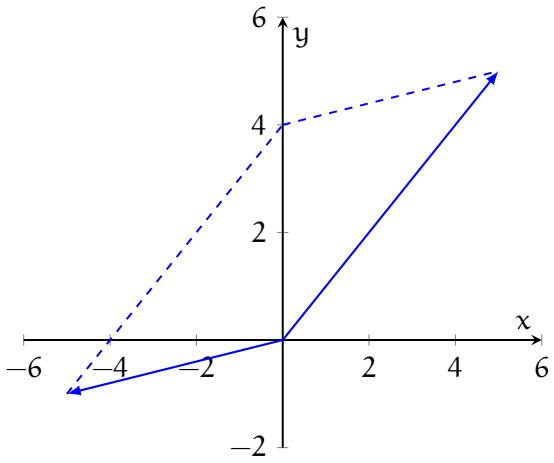
1. Our two vectors from the columns of the matrix are $[1,-3]$ and $[4,1]$. Plotting:




The area of this parallelogram is the same as the determinant of the matrix:

$$\det\left(\begin{bmatrix} 1 & 4 \\ -3 & 1 \end{bmatrix}\right) = 1 \cdot 1 - (4 \cdot -3) = 1 + 12 = 13$$

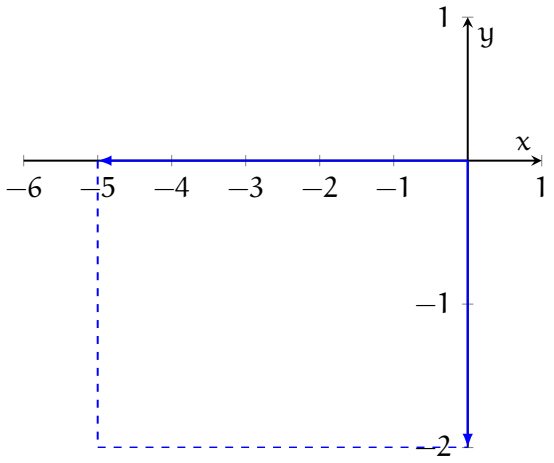
2. Our two vectors from the columns of the matrix are $[5, 5]$ and $[-5, -1]$. Plotting:




The area of this parallelogram is the same as the determinant of the matrix:

$$\det\left(\begin{bmatrix} 5 & -5 \\ 5 & -1 \end{bmatrix}\right) = 5 \cdot -1 - (-5 \cdot 5) = -5 + 25 = 20$$

3. Our two vectors from the columns of the matrix are $[0, -2]$ and $[-5, 0]$. Plotting:




This is a rectangle, and we can see the area is $5 \cdot 2 = 10$. However, the determinant is:

$$\det\left(\begin{bmatrix} 0 & -5 \\ -2 & 0 \end{bmatrix}\right) = 0 \cdot 0 - (-5 \cdot -2) = 0 - 10 = -10$$

We will discuss this unusual response in a future chapter.

## Answer to Exercise 375 (on page 1248)

First, find the determinant of $A$:

$$\det(A) = (1)(-1) - (2)(0) = -1$$

So our inverse is given by:

$$A^{-1} = -1\begin{bmatrix} -1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

## Answer to Exercise 376 (on page 1249)

First, check that $A$ has an inverse:

$$\det A = 1\begin{vmatrix} 1 & 1 \\ 4 & -3 \end{vmatrix} - 2\begin{vmatrix} 0 & 1 \\ 2 & -3 \end{vmatrix} - 2\begin{vmatrix} 0 & 1 \\ 2 & 4 \end{vmatrix} = 1(-3-4) - 2(0-2) - 2(0-2) = -7+4+4 = 1$$

Since $\det(A) \neq 0$, $A$ must have an inverse. We find it's inverse by the augmented matrix

method:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 4 & -3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -3 & 0 & 2 \\ 0 & 1 & 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -7 & -2 & 4 \\ 0 & 1 & 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right].$$

This gives us our inverse on the right side, $A^{-1} = \begin{bmatrix} -7 & -2 & 4 \\ 2 & 1 & -1 \\ -2 & 0 & 1 \end{bmatrix}$

To check our answer, we can use $AA^{-1} = I_3$:

$$AA^{-1} = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & 1 \\ 2 & 4 & -3 \end{bmatrix} \begin{bmatrix} -7 & -2 & 4 \\ 2 & 1 & -1 \\ -2 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-7) + 2(2) + (-2)(-2) & 1(-2) + 2(1) + (-2)(0) & 1(4) + 2(-1) + (-2)(1) \\ 0(-7) + 1(2) + 1(-2) & 0(-2) + 1(1) + 1(0) & 0(4) + 1(-1) + 1(1) \\ 2(-7) + 4(2) + (-3)(-2) & 2(-2) + 4(1) + (-3)(0) & 2(4) + 4(-1) + (-3)(1) \end{bmatrix}$$

$$= \begin{bmatrix} -7 + 4 + 4 & -2 + 2 + 0 & 4 - 2 - 2 \\ 0 + 2 - 2 & 0 + 1 + 0 & 0 - 1 + 1 \\ -14 + 8 + 6 & -4 + 4 + 0 & 8 - 4 - 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

## Answer to Exercise 377 (on page 1249)

No. Each row is a scalar multiple of the first row:

- Row 2 = 2 × Row 1
- Row 3 = 4 × Row 1
- Row 4 = 8 × Row 1

So the rows are *linearly dependent*, meaning the matrix has rank 1, and its determinant is 0. A matrix with determinant 0 is singular, so no inverse exists.

## Answer to Exercise 378 (on page 1254)

Setting up the matrix, we get:

$$\begin{bmatrix} \vec{\mathbf{i}} & \vec{\mathbf{j}} & \vec{\mathbf{k}} \\ 4 & 5 & 6 \\ 3 & 7 & -8 \end{bmatrix}$$

$$(5(-8) - (7)(6))\,\vec{\mathbf{i}} - ((4)(-8) - (6)(3))\vec{\mathbf{j}} + ((4)(7) - (5)(3))\vec{\mathbf{k}}$$

$$(-40 - 46)\vec{\mathbf{i}} - (-32 - 18)\vec{\mathbf{j}} + (28 - 15)\vec{\mathbf{k}}$$

$$-86\vec{\mathbf{i}} + 50\vec{\mathbf{j}} + 13\vec{\mathbf{k}} \implies \langle -82, 50, 13 \rangle$$

## Answer to Exercise 379 (on page 1270)

The matrix that represents this transformation is:

$$A = \begin{bmatrix} 1.5 & 0 \\ -2 & -1 \end{bmatrix}$$

We find this by first noticing that the house and door components are flipped over the x-axis, which means that the y-values of the points are flipped. This gives us a negative one value in the $e_2$ basis of the matrix.

Next we can see that the house is stretched horizontally in the x-direction. The orange dots end at 4.5, which is 1.5 times the original length of 3, so we know that the x-values are scaled by a factor of 1.5. This gives us the value of 1.5 in the first row of the matrix.

Finally, the roof is slanted, which means that there is a shearing component to this trans-

formation. Notice that the roof (door corner) is moved up, so the $e_1$ also has a negative value y-component. This gives us the value of -2 in the second row, first column of the matrix.

## Answer to Exercise 380 (on page 1275)

1. The original matrix for an x-shear with a shear factor of $k$ is:

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

2. The inverse matrix for this transformation is:

$$\begin{bmatrix} 1 & -k \\ 0 & 1 \end{bmatrix}$$

(the determinant is 1, and the inverse is found by swapping the diagonal elements and negating the off-diagonal elements, which in this case results in negating $k$).

## Answer to Exercise 381 (on page 1282)

First, write the $A - \lambda I$ equation

$$\begin{bmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix}$$

Our goal is to get the determinant of $A-\lambda I$ to be equal to zero. For a $2\times 2$, the determinant is $ad - bc$:

$$\begin{aligned} (1-\lambda)(-2-\lambda) - 4 &= 0 \\ (-2+2\lambda-\lambda+\lambda^2) - 4 &= 0 \\ \lambda^2 + \lambda - 6 &= 0 \\ \lambda_1 = 2, \quad \lambda_2 &= -3 \end{aligned}$$

Then we solve for the two eigenvectors, $\mathbf{e}_n$, using $A - \lambda I$

For $\lambda_1 = 2$:

$$\lambda_1 = 2 : A - 2I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}$$

We solve the following using row-column multiplication (rows of numbers and columns

of $x_i$)

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} -x_1 + 2x_2 = 0 \\ 2x_1 - 4x_2 = 0 \end{cases}$$

We notice, $x_1 = 2x_2$. This means every eigenvector is a multiple of $\mathbf{e}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (using $x_2 = 1$) .

For $\lambda_2 = -3$:

$$\lambda_2 = -3 : A + 3I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} 4x_1 + 2x_2 = 0 \\ 2x_1 + x_2 = 0 \end{cases}$$

This results in $x_2 = -2x_1$, meaning $\mathbf{e}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

## Answer to Exercise 382 (on page 1288)

Compute dot product of **a** and **b**:

$$1 * -4 + 3 * 6 = -4 + 18 = 14$$

Compute the dot product of **b** and **b**

$$16 + 36 = 52$$

$$14/52 * (-4, 6) = (-1.076, 1.61)$$

## Answer to Exercise 383 (on page 1294)

The first vector of the orthogonal subspace is:

$$v_1 = x_1 = (1, 1, 1)$$

The second vector of the subspace is a projection of $x_2$ onto $v_1$.

$$v_2 = x_2 - \frac{x_2 v_1}{v_1 v_1} v_1$$

Substitute the values:

$$v_2 = (0,1,1) - \frac{(0,1,1)(1,1,1)}{(1,1,1)(1,1,-1)}(1,1,1)$$

$$v_2 = (0,1,1) - (2/3)(1,1,1)$$

$$v_2 = (-2/3, 1/3, 1/3)$$

Normalize:

$$v_1 = v_1/\sqrt{|v_1|}$$

$$v_1 = (1,1,1)/\sqrt{|v_1|}$$

$$v_1 = (0.577, 0.577, 0.577)$$

$$v_2 = v_2/\sqrt{|v_2|}$$

$$v_2 = (0,1,1)\sqrt{|v_2|}$$

$$v_2 = (-0.816, 0.408, 0.408)$$

## Answer to Exercise 384 (on page 1302)

$$U = \begin{bmatrix} -0.70710678 & -0.70710678 \\ -0.70710678 & 0.70710678 \end{bmatrix}$$

$$Singularvalues = [3.464101623.16227766]$$

$$V^T = \begin{bmatrix} -0.408 & -0.816 & -0.408 \\ -0.894 & 0.447 & 0.0 \\ -0.183 & -0.365 & 0.9129 \end{bmatrix}$$

## Answer to Exercise 385 (on page 1309)

Yes. Its eigenvalues are

$$2, 2$$

## Answer to Exercise 386 (on page 1309)

The answer depends on the 3 by 3 matrix you chose.

## Answer to Exercise 387 (on page 1345)

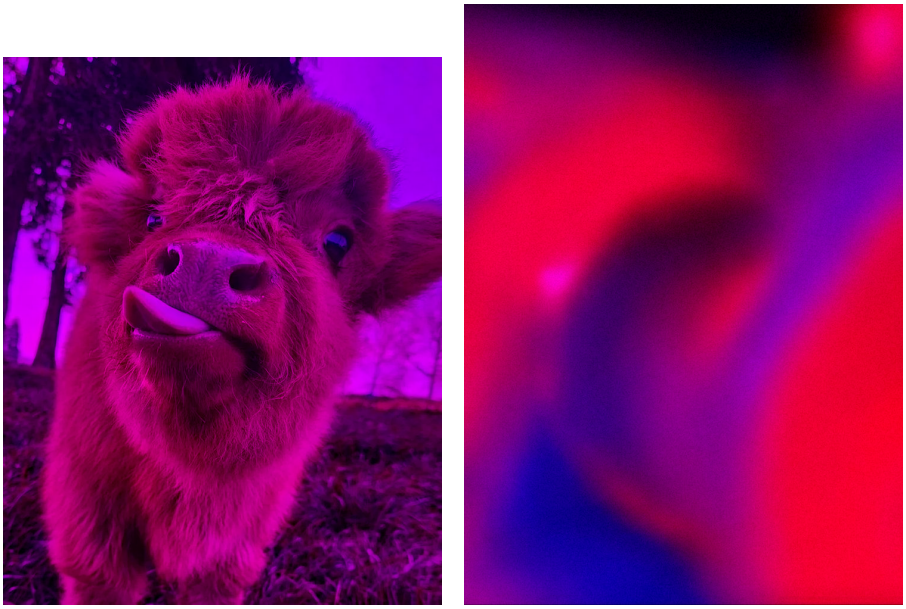
The only thing that changes in our for loop is the writing of the output pixel.

```
// CODE ABOVE STAYS THE SAME
for (int x = 0; x < w; x++)
{
    for (int y = 0; y < h; y++)
    {
        int idx = y * rowSize + x * 3;

        PIXEL p;
        BYTE B = data[idx];
        BYTE G = data[idx + 1];
        BYTE R = data[idx + 2];
        p = { B, G, R };

        out[idx + 0] = p.b;
        out[idx + 1] = 0; // NOTE THE CHANGE HERE
        out[idx + 2] = p.r;
    }
}
```

What is happening here?

In an RGB image, each pixel's final color is the combination of red, green, and blue intensities. If the green value is removed, every pixel changes from (R,G,B) to (R,0,B). Colors that relied heavily on green—such as greens, yellows, and many skin tones—lose a major part of their intensity and appear much darker or shifted in hue. Pure green areas become black, yellow areas (red + green) become red, and cyan areas (green + blue) become blue.

Visually, the image often takes on a magenta or purplish tint (see Figures 142.3a and 142.3b), because magenta is the combination of red and blue with no green. Overall brightness decreases, since green contributes significantly to the brightness in human vision. Refer to Figures 142.2a and 142.1a for the original bitmaps.



(a) Output bitmap - cow. (b) Output bitmap - gradient.

Figure 142.3: Running `bitmap_no_green.cpp` on both provided bitmaps.

## Answer to Exercise 388 (on page 1350)

1. How many bytes does each pixel use? - 3 `bytes per pixel`
2. How many bytes are required for one row before padding? 4 `pixels`$\times$3 `bytes per pixel` = 12 `bytes`
3. How many padding bytes are required per row? 0 `padding bytes`
4. What is the total size of the pixel data? 12 bytes per row $\times$ 2 `rows` = 24 `bytes`
5. What is the total file size, including header structs? 14 `bytes for BFH`+40 `bytes for BIH` = 54 `bytes` $\implies$ $24 + 54 = 78$ `bytes`

Remember that we need to write the rows bottom-up, so the second row is written first followed by the first row.

Let's establish the colors needed. Remember that we need to swap to BGR order:

- Black = (0, 0, 0)
- Red = (0, 0, 255)
- Green = (0, 255, 0)

- Blue = (255, 0, 0)
- Yellow = (0, 255, 255) (R=255,G=255,B=0 $\implies$ BGR = 0,255,255)
- Magenta = (255, 0, 255) (R=255,B=255 $\implies$ BGR = 255,0,255)
- Cyan = (255, 255, 0) (G=255,B=255 $\implies$ BGR = 255,255,0)
- White = (255, 255, 255)

Overall the program looks like this:

```
#include <cstdio>
#include <cstdint>
#include <cstring>


#pragma pack(push, 1)
struct BITMAPFILEHEADER {
    uint16_t bfType = 0x4D42; // 'BM'
    uint32_t bfSize;
    uint16_t bfReserved1 = 0;
    uint16_t bfReserved2 = 0;
    uint32_t bfOffBits = 54;
};

struct BITMAPINFOHEADER {
    uint32_t biSize = 40;
    int32_t  biWidth = 4;
    int32_t  biHeight = 2;
    uint16_t biPlanes = 1;
    uint16_t biBitCount = 24;
    uint32_t biCompression = 0;
    uint32_t biSizeImage;
    int32_t  biXPelsPerMeter = 0;
    int32_t  biYPelsPerMeter = 0;
    uint32_t biClrUsed = 0;
    uint32_t biClrImportant = 0;
};
#pragma pack(pop)

int main(int argc, char const *argv[])
{
    FILE* f = fopen("fourbytwo.bmp", "wb");

    BITMAPFILEHEADER bfh;
    BITMAPINFOHEADER bih;

    const int width = 4;
    const int height = 2;
    const int bytesPerPixel = 3;
    const int rowSize = ((width * bytesPerPixel + 3) / 4) * 4; // 12
```

```
    const int imageSize = rowSize * height;                    // 24

    bih.biSizeImage = imageSize;
    bfh.bfSize = bfh.bfOffBits + imageSize;

    fwrite(&bfh, sizeof(bfh), 1, f);
    fwrite(&bih, sizeof(bih), 1, f);
    uint8_t row[rowSize];

    // ----------------------------
    // Write bottom row first:
    // [ Black ] [ Red ] [ Green ] [ Blue ]
    // ----------------------------
    std::memset(row, 0, rowSize);

    // Black
    row[0] = 0;   row[1] = 0;   row[2] = 0;
    // Red (BGR = 0,0,255)
    row[3] = 0;   row[4] = 0;   row[5] = 255;
    // Green (BGR = 0,255,0)
    row[6] = 0;   row[7] = 255; row[8] = 0;
    // Blue (BGR = 255,0,0)
    row[9] = 255; row[10]= 0;   row[11]= 0;

    fwrite(row, rowSize, 1, f);

    // ----------------------------
    // Write top row:
    // [ Yellow ] [ Magenta ] [ Cyan ] [ White ]
    // ----------------------------
    std::memset(row, 0, rowSize);

    // Yellow (BGR = 0,255,255)
    row[0] = 0;   row[1] = 255; row[2] = 255;
    // Magenta (BGR = 255,0,255)
    row[3] = 255; row[4] = 0;   row[5] = 255;
    // Cyan (BGR = 255,255,0)
    row[6] = 255; row[7] = 255; row[8] = 0;
    // White (BGR = 255,255,255)
    row[9] = 255; row[10]= 255; row[11]= 255;

    fwrite(row, rowSize, 1, f);

    fclose(f);
    return 0;
}
```

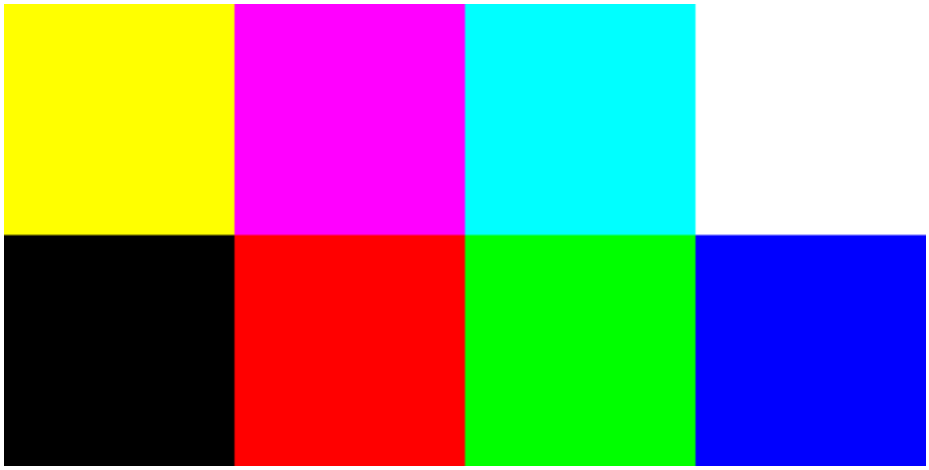


Figure 142.5: Exercise Output of a 4 by 2 bitmap.

## Answer to Exercise 389 (on page 1373)

The path to `data.csv` is

```
./Documents/Projects/project2/data.csv
```

The depth is the number of edges from the root to the node.

In this case, it is the number of subdirectories needed to reach `data.csv`:

(1) `./` → `Documents`

(2) `Documents` → `Projects`

(3) `Projects` → `project2`

(4) `project2` → `data.csv`

So `data.csv` is at a depth of **4**.

## Answer to Exercise 390 (on page 1381)

**Algorithm: DFS (Pre-order) run recursively on a tree**

```
procedure DFS(root)
    if root = null then
        return
```

```
    end if
    visit(root)                                  ▷ V
    DFS(root.left)                               ▷ L
    DFS(root.right)                              ▷ R
end procedure
```

This algorithm defines a depth-first search on a binary tree. Because the node is visited before its subtrees, it corresponds specifically to a pre-order traversal.

## Answer to Exercise 391 (on page 1382)

```
procedure DFS(root)
    if root = null then
        return
    end if
    pre(root) = counter
    counter = counter + 1
    DFS(root.left)
    DFS(root.right)
    post(root) = counter
    counter = counter + 1
end procedure
```

## Answer to Exercise 392 (on page 1382)

| Node | Pre-order Number | Post-order Number |
|---|---|---|
| Alan | 1 | 14 |
| Brad | 2 | 7 |
| Dean | 3 | 4 |
| Emily | 5 | 6 |
| Connie | 8 | 13 |
| Faith | 9 | 10 |
| Gordon | 11 | 12 |

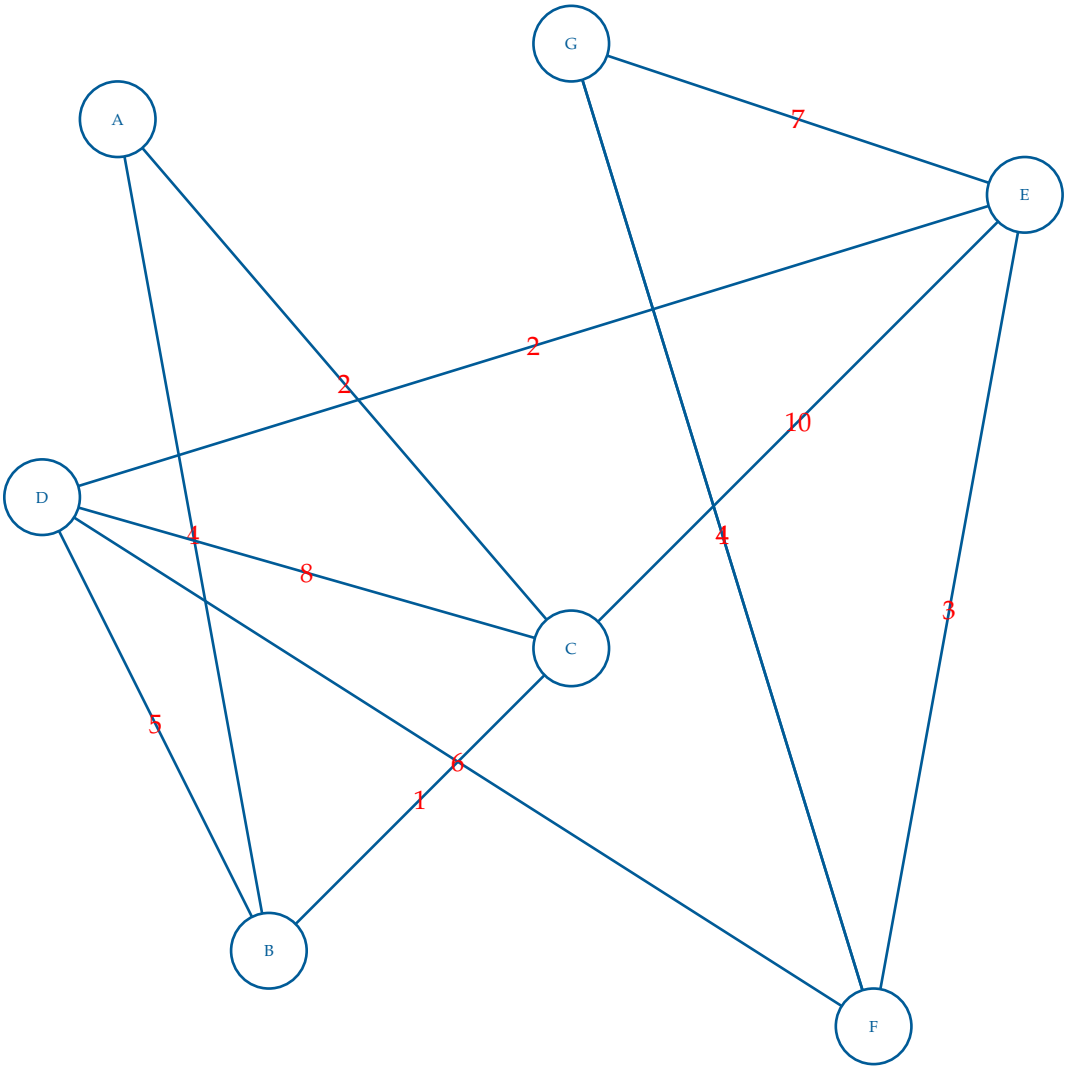
## Answer to Exercise 393 (on page 1386)

Los Angeles $\rightarrow$ Bakersfield is not chosen because it does not give the best known cost to Bakersfield. The `prev` pointer for Bakersfield goes to San Diego instead, since \$21 $<$ \$22.

The tree that is generated is called a *predecessor tree* or *explored tree* of the resulting edges

formed by only the previous edges. We can use this to track shortest paths from a given origin node.

## Answer to Exercise 394 (on page 1395)




## Answer to Exercise 395 (on page 1399)

1. Injective, but not surjective. No two inputs map to the same output, but $d$ is never reached.
2. Surjective, but not injective. Every element of the codomain is reached, but both $1$ and $4$ map to $a$.

3. Bijective. Every element of the codomain is reached exactly once.

## Answer to Exercise 396 (on page 1405)

Assuming `FindOdd` exists, we can make the following pseudocode.

`IsBipartite`(G)
**Require:** G, a graph on which DFS has been run
**Ensure:** **true** if G is bipartite, **false** otherwise
1: **for** all back edges $e$ **do**
2: **if** FindOdd$(G, e) =$ **true then**
3: **return false**
4: **end if**
5: **end for**
6: **return true**

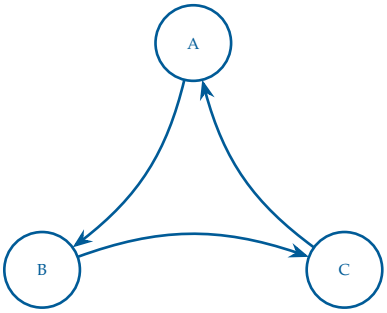
## Answer to Exercise 397 (on page 1406)

Remove the edge $e = (u, v)$ from G.
Run BFS or DFS starting from $u$.
**if** $v$ is reached **then**
Return True.
**else**
Return False.
**end if**

## Answer to Exercise 398 (on page 1407)

*Proof.* Run DFS on given graph G. Find a leaf, $l$ in the output DFS tree (sometimes called the explore tree). Remove $l$ from the graph. Run DFS on the graph $(G' = G/\{l\})$. If the DFS tree is connected, then we are done. If not, then there is a vertex $v$ in the DFS tree that is not connected to the rest of the graph. This means that $v$ was only connected to $l$ in the original graph. But this contradicts the fact that $l$ was a leaf in the DFS tree, since $v$ would have been explored before $l$. Since a leaf must exist, removing it from the explore tree does not create a new tree. Therefore, there must be a vertex whose removal leaves the graph connected. □

## Answer to Exercise 399 (on page 1407)




## Answer to Exercise 400 (on page 1436)

```
<!DOCTYPE html>
<html>
<head>
  <title>Chocolate Chip Cookies</title>
</head>
<body>
  <h1 id="recipe-title">Chocolate Chip Cookies</h1>
  <p class="intro" title="A short recipe description">
    This recipe makes soft and sweet chocolate chip cookies.
  </p>
</body>
</html>
```

## Answer to Exercise 401 (on page 1439)

```
<body>
  <div class="recipe-header">
    <h1 id="recipe-title">Chocolate Chip Cookies</h1>
    <p class="intro" title="A short recipe description">
      This recipe makes soft and sweet chocolate chip cookies.
    </p>
  </div>

  <div class="ingredients-section">
    <p>Ingredients</p>
  </div>

  <div class="steps-section">
    <p>Steps</p>
```

```
  </div>
</body>
```

## Answer to Exercise 402 (on page 1442)

```
<body>
  <header>
    <h1 id="recipe-title">Chocolate Chip Cookies</h1>
    <p class="intro" title="A short recipe description">
      This recipe makes soft and sweet chocolate chip cookies.
    </p>
  </header>

  <main>
    <section class="ingredients-section">
      <p>Ingredients</p>
    </section>

    <section class="steps-section">
      <p>Steps</p>
    </section>
  </main>

  <footer>
  </footer>
</body>
```

## Answer to Exercise 403 (on page 1443)

```
<main>
  <section class="ingredients-section">
    <h2 id="ingredients">Ingredients</h2>
  </section>

  <section class="steps-section">
    <h2 id="steps">Steps</h2>
  </section>
</main>
```

## Answer to Exercise 404 (on page 1445)

```
<footer>
  <p><strong>Important Note:</strong> Make sure not to overbake the cookies, or
  ↪ they will become hard.</p>
</footer>
```

## Answer to Exercise 405 (on page 1449)

```
<section class="ingredients-section">
  <h2 id="ingredients">Ingredients</h2>
  <ul>
    <li>2 1/4 cups flour</li>
    <li>1 teaspoon baking soda</li>
    <li>1 teaspoon salt</li>
    <li>1 cup butter</li>
    <li>3/4 cup sugar</li>
    <li>3/4 cup brown sugar</li>
    <li>2 eggs</li>
    <li>2 cups chocolate chips</li>
  </ul>
</section>

<section class="steps-section">
  <h2 id="steps">Steps</h2>
  <ol>
    <li>Preheat the oven to 375°F.</li>
    <li>Mix the dry ingredients in one bowl.</li>
    <li>Mix the butter, sugar, brown sugar, and eggs in another bowl.</li>
    <li>Stir the dry ingredients into the wet ingredients.</li>
    <li>Add the chocolate chips.</li>
    <li>Scoop the dough onto a baking sheet.</li>
    <li>Bake for 9 to 11 minutes.</li>
  </ol>
</section>
```

## Answer to Exercise 406 (on page 1451)

```
<header>
  <h1 id="recipe-title">Chocolate Chip Cookies</h1>
  <p class="intro" title="A short recipe description">
    This recipe makes soft and sweet chocolate chip cookies.
  </p>

  <nav>
```

```
    <ul>
      <li><a href="#ingredients">Ingredients</a></li>
      <li><a href="#steps">Steps</a></li>
    </ul>
  </nav>
</header>
```

## Answer to Exercise 407 (on page 1455)

```
<section>
  <h2>Recipe Facts</h2>
  <table border="1">
    <caption>Chocolate Chip Cookie Recipe Facts</caption>
    <tbody>
      <tr>
        <td>Prep Time</td>
        <td>15 minutes</td>
      </tr>
      <tr>
        <td>Bake Time</td>
        <td>10 minutes</td>
      </tr>
      <tr>
        <td>Servings</td>
        <td>24 cookies</td>
      </tr>
    </tbody>
  </table>
</section>
```

## Answer to Exercise 408 (on page 1460)

```
<header>
  <h1 id="recipe-title">Chocolate Chip Cookies</h1>
  <p class="intro" title="A short recipe description">
    This recipe makes soft and sweet chocolate chip cookies.
  </p>

  <nav>
    <ul>
      <li><a href="#ingredients">Ingredients</a></li>
      <li><a href="#steps">Steps</a></li>
    </ul>
  </nav>
```

```
13
14    <img src="cookies.jpg" alt="Fresh chocolate chip cookies on a plate"
      ↪  width="500" height="300">
15  </header>
```

## Answer to Exercise 409 (on page 1475)

probability of all 5's $= \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^3 = \frac{1}{216} \approx 0.0046$

## Answer to Exercise 409 (on page 1475)

probability of at least one heads = 1.0 - probability of all tails $= 1.0 - \left(\frac{1}{2}\right)^5 = 1.0 - \frac{1}{32} = \frac{31}{32} = \approx 0.97$

## Answer to Exercise 411 (on page 1488)

1. $(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$, valid for $|x| < 1$

2. $\sqrt{1-2x} = 1 - x - \frac{1}{2}x^2 - \cdots$

## Answer to Exercise 412 (on page 1489)

$$\text{P(exactly 4 makes)} = \binom{6}{4}(0.6)^4(0.4)^2 = 0.31104$$

## Answer to Exercise 413 (on page 1500)

$df = n - 1 = 24$. A 95 percent confidence interval splits the remaining 5 percent evenly between the two tails, leaving 2.5 percent, or 0.025, in the upper tail. Reading the t-table at $df = 24$ and upper-tail area 0.025 gives $t^* = 2.064$.

## Answer to Exercise 414 (on page 1501)

The $n \geq 30$ threshold is not a hard requirement for using a t-interval. It is one way to satisfy the normality condition when you cannot otherwise verify the shape of the population. If the population itself is known to be normally distributed, or if a dotplot or stemplot of the sample shows no strong skew or outliers, the normality condition is satisfied regardless of how small $n$ is, and a t-interval is entirely appropriate. The student conflated a fallback rule with a strict cutoff.

## Answer to Exercise 415 (on page 1502)

(a) Matched pairs. The same twenty subscribers are measured twice, so the two sets of measurements are linked subscriber by subscriber, and the natural next step is to subtract to get one sample of twenty differences.

(b) Two independent samples. The comedy-watching group and the drama-watching group are made up of entirely different subscribers with no pairing between them, so there is no meaningful way to subtract one group's values from the other's.

## Answer to Exercise 416 (on page 1503)

$df = 15$. A 98 percent confidence interval leaves 1 percent, or 0.01, in each tail, giving $t^* = 2.602$. The standard error is $\frac{6.2}{\sqrt{16}} = 1.55$, so the margin of error is $2.602 \times 1.55 \approx 4.03$ minutes. The interval is $28.4 \pm 4.03 \Rightarrow (24.37,\ 32.43)$ minutes.

## Answer to Exercise 417 (on page 1505)

This is matched pairs, so the sample is the twelve differences, and everything proceeds exactly as it did for a single mean. $df = 11$, and a 90 percent confidence interval leaves 5 percent in each tail, giving $t^* = 1.796$. The standard error is $\frac{4.5}{\sqrt{12}} \approx 1.299$, so the margin of error is $1.796 \times 1.299 \approx 2.33$ minutes. The interval is $-3.2 \pm 2.33 \Rightarrow (-5.53,\ -0.87)$ minutes. Because the entire interval is negative, and the differences were defined as before minus after, this suggests call-handling time decreased after training, by somewhere between about 0.87 and 5.53 minutes on average.

## Answer to Exercise 418 (on page 1505)

The true mean $\mu$ is a fixed, though unknown, number. It is not a random variable, so it makes no sense to assign it a probability of falling anywhere. What is random is the interval itself, since a different sample would have produced a different $\bar{x}$ and a different interval. The corrected interpretation is that the analyst is 95 percent confident that the true mean watch time per session falls between 43.75 and 51.85 minutes, meaning that if this sampling process were repeated many times, about 95 percent of the resulting intervals would capture the true mean.

## Answer to Exercise 419 (on page 1507)

$H_0 : \mu = 500$, $H_a : \mu \neq 500$. Conditions are satisfied, so use a t-test with $df = 15$. The standard error is $\frac{6.4}{\sqrt{16}} = 1.6$, giving $t = \frac{498.2 - 500}{1.6} \approx -1.125$. For a two-sided test with $df = 15$, this gives a p-value of about 0.278. Because $0.278 > 0.05$, fail to reject $H_0$. There is not sufficient evidence that the true mean loaf weight differs from 500 grams.

## Answer to Exercise 420 (on page 1507)

Since the differences are defined as before minus after, a decrease in heart rate shows up as a positive mean difference, so $H_0 : \mu_d = 0$, $H_a : \mu_d > 0$. This is matched pairs, so $df = 14$. The standard error is $\frac{6.1}{\sqrt{15}} \approx 1.575$, giving $t = \frac{4.8}{1.575} \approx 3.048$, with a one-sided p-value of about 0.0043. Because $0.0043 < 0.05$, reject $H_0$. There is sufficient evidence that the regimen decreased mean resting heart rate.

## Answer to Exercise 421 (on page 1508)

Hypotheses are statements about the population parameter, $\mu$, never about the sample statistic, $\bar{x}$. The sample mean is a single computed number from one sample; it is not something you test a hypothesis about, since it is already known exactly once the sample is collected. The correct hypotheses are $H_0 : \mu = 500$ and $H_a : \mu \neq 500$.

## Answer to Exercise 422 (on page 1509)

$SE = \sqrt{\frac{1.8^2}{15} + \frac{2.1^2}{17}} \approx 0.690$. With $df = 29$, $t^* \approx 1.699$, so the margin of error is $1.699 \times$

$0.690 \approx 1.17$ hours. The point estimate is $11.2 - 9.8 = 1.4$ hours, giving $1.4 \pm 1.17 \Rightarrow (0.23,\ 2.57)$ hours.

## Answer to Exercise ?? (on page 1509)

No. The interval $(0.23, 2.57)$ does contain values above 1, but it also contains values below 1, so it does not provide convincing evidence that the difference exceeds one full hour specifically. The interval does support the weaker claim that Model A's mean battery life is greater than Model B's, since the entire interval is positive, just not the stronger one-hour claim.

## Answer to Exercise 424 (on page 1510)

Pooling assumes the two population standard deviations are equal, $\sigma_1 = \sigma_2$. Here, the two samples show clearly different spreads, which is evidence against that assumption, not for it. Even setting that aside, the current AP Statistics CED does not include the pooled, equal-variances procedure at all, only the unpooled two-sample t-interval using $\sqrt{s_1^2/n_1 + s_2^2/n_2}$ with technology-computed degrees of freedom. The student should use the unpooled formula regardless of how similar or different the spreads look.

## Answer to Exercise 425 (on page 1511)

$H_0 : \mu_1 - \mu_2 = 0$, $H_a : \mu_1 - \mu_2 < 0$. The standard error is $\sqrt{\dfrac{2.1^2}{22} + \dfrac{2.8^2}{25}} \approx 0.717$, so $t = \dfrac{8.3 - 9.6}{0.717} \approx -1.813$, with $df \approx 44$. The one-sided p-value is about $0.0383$. Since $0.0383 < 0.05$, reject $H_0$. There is convincing evidence that Store 1's mean wait time is shorter than Store 2's.

## Answer to Exercise 426 (on page 1513)

This would be a Type I error. The researcher rejected $H_0$ (concluding Store 1's mean wait time is shorter), but if the true means are actually equal, $H_0$ was in fact true, so rejecting it was a mistake. Because the test used $\alpha = 0.05$, this kind of error would occur no more than about 5 percent of the time across many repeated studies like this one.

## Answer to Exercise 427 (on page 1513)

(a) Quantitative, paired. The same 30 members are measured twice, before and after, so the natural approach is to subtract and work with the sample of differences.

(b) Categorical. Proportions of an email being opened or not are categorical counts, not measurements, so this belongs to the tools from *Inference for Proportions*.

(c) Quantitative, one sample. A single random sample is used to estimate one population mean, with no comparison to a second group or a second measurement per subject.

## Answer to Exercise 428 (on page 1517)

1. $X$ is the amount of fertilizer applied (grams per plant); $Y$ is the resulting plant height (centimeters).
2. $Y = \beta_0 + \beta_1 X + \varepsilon$, where $\beta_0$ is the population intercept, $\beta_1$ is the population slope relating fertilizer amount to height, and $\varepsilon$ is random error.
3. If $\beta_1 = 0$, then fertilizer amount has no linear relationship with plant height in the population; the population regression line is flat, and knowing how much fertilizer a plant received tells you nothing about how tall it will grow.
4. $\beta_1$ describes the relationship for the entire population of plants the horticulturist could ever grow, under all possible conditions, not just the ones she happened to collect data from. Even collecting data from every plant she currently owns would still leave $\beta_1$ unknown, since her plants are themselves a sample of all plants that could exist under those growing conditions. $\beta_1$ is a parameter; only a $b_1$ calculated from data is ever directly observed.

## Answer to Exercise 430 (on page 1523)

1. Equal variance. The spread of the residuals is clearly not constant across values of $X$; it grows as car weight increases.
2. The standard error formula used to build $s_{b_1}$ assumes the same amount of scatter around the line at every value of $X$. If the scatter actually changes with $X$, as it does here, the standard error is not accurately estimated, so the resulting confidence interval and p-value cannot be trusted, even if every calculation is performed correctly.

## Answer to Exercise 431 (on page 1530)

(a) Row 1.3, column .07: 0.9147.

(b) The table gives $P(Z \leq -0.84) = 0.2005$, so $P(Z > -0.84) = 1 - 0.2005 = 0.7995$.

(c) $P(Z \leq 0.60) - P(Z \leq -1.25) = 0.7257 - 0.1056 = 0.6201$.

## Answer to Exercise 432 (on page 1530)

The closest entry to 0.90 in the body of the table is 0.8997, at row 1.2 and column .08. So $z \approx 1.28$.

## Answer to Exercise 433 (on page 1532)

(a) Row 24, column C = 95%: $t^* = 2.064$.

(b) The table has no row for $df = 45$, so use $df = 40$. Column C = 99% gives $t^* = 2.704$. The exact value for $df = 45$ is 2.690, slightly smaller, which confirms that rounding df down widens the interval.

## Answer to Exercise 434 (on page 1532)

In row 15, $t = 2.30$ falls between 2.249 (tail probability 0.02) and 2.602 (tail probability 0.01). So $0.01 < P < 0.02$. Technology gives $P = 0.0181$, inside these bounds.

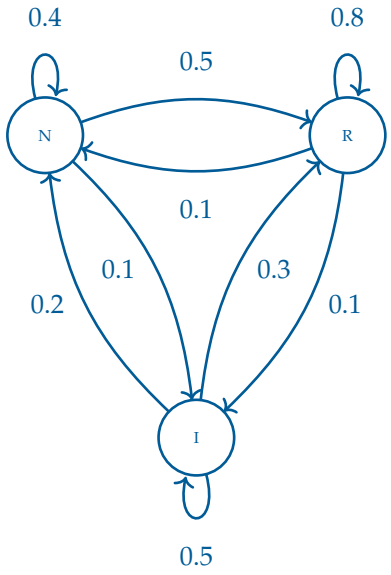
## Answer to Exercise 435 (on page 1534)

(a) Row 12, column .05: 21.026.

(b) In row 4, 9.80 falls between 9.488 (tail probability 0.05) and 11.143 (tail probability 0.025). So $0.025 < P < 0.05$. Technology gives $P = 0.0439$.

## Answer to Exercise 436 (on page 1547)

1. The state space of this Markov chain is $\mathcal{S} = \{\mathrm{N}, \mathrm{R}, \mathrm{I}\}$ where N represents new customers, R represents regular customers, and I represents inactive customers.




2.

3.
$$\mathrm{P}(\mathrm{X}_{n+1} = \mathrm{R} \mid \mathrm{X}_n = \mathrm{I}, \mathrm{X}_{n-1} = \mathrm{R}, \mathrm{X}_{n-2} = \mathrm{N}) = \mathrm{P}(\mathrm{X}_{n+1} = \mathrm{R} \mid \mathrm{X}_n = \mathrm{I})$$
Probability: 30%

## Answer to Exercise 437 (on page 1550)

1. Regular $\to$ Inactive: $\mathrm{P}(\mathrm{X}_{n+1} = \mathrm{I} \mid \mathrm{X}_n = \mathrm{R}) = 0.20$.
2. Inactive $\to$ Regular after two visits: We need to find $\mathrm{P}(\mathrm{X}_{n+2} = \mathrm{R} \mid \mathrm{X}_n = \mathrm{I})$. This can be calculated by two multiplications and summing over all possible intermediate states:
   - $\mathrm{I} \to \mathrm{N} \to \mathrm{R} = 0.10 \times 0.30 = 0.03$
   - $\mathrm{I} \to \mathrm{R} \to \mathrm{R} = 0.15 \times 0.65 = 0.0975$
   - $\mathrm{I} \to \mathrm{I} \to \mathrm{R} = 0.75 \times 0.15 = 0.1125$

   $$0.03 + 0.0975 + 0.1125 = 0.24$$
3. New $\to$ Inactive after three visits: We need to find $\mathrm{P}(\mathrm{X}_{n+3} = \mathrm{I} \mid \mathrm{X}_n = \mathrm{N})$. This can be calculated by *three* multiplications and summing over all possible intermediate states:
   - $\mathrm{N} \to \mathrm{N} \to \mathrm{N} \to \mathrm{I} = 0.50 \times 0.50 \times 0.20 = 0.05$

- $N \to N \to R \to I = 0.50 \times 0.30 \times 0.20 = 0.03$
- $N \to N \to I \to I = 0.50 \times 0.20 \times 0.75 = 0.075$
- $N \to R \to N \to I = 0.30 \times 0.15 \times 0.20 = 0.009$
- $N \to R \to R \to I = 0.30 \times 0.65 \times 0.20 = 0.039$
- $N \to R \to I \to I = 0.30 \times 0.20 \times 0.75 = 0.045$
- $N \to I \to N \to I = 0.20 \times 0.10 \times 0.20 = 0.004$
- $N \to I \to R \to I = 0.20 \times 0.15 \times 0.20 = 0.006$
- $N \to I \to I \to I = 0.20 \times 0.75 \times 0.75 = 0.1125$

$$0.050 + 0.030 + 0.075 + 0.009 + 0.039 + 0.045 + 0.004 + 0.006 + 0.1125 = 0.3705$$

## Answer to Exercise 438 (on page 1551)

To find $P^3$, we need to compute the matrix multiplication of P with itself three times. First, we compute $P^2$:

$$P^2 = \begin{bmatrix} 0.2(0.2) + 0.8(0.1) & 0.2(0.8) + 0.8(0.9) \\ 0.1(0.2) + 0.9(0.1) & 0.1(0.8) + 0.9(0.9) \end{bmatrix} = \begin{bmatrix} 0.12 & 0.88 \\ 0.11 & 0.89 \end{bmatrix}$$

Then find $P^3$ from $P^2 \cdot P$

$$\begin{aligned} P^3 &= \begin{bmatrix} 0.12 & 0.88 \\ 0.11 & 0.89 \end{bmatrix} \begin{bmatrix} 0.2 & 0.8 \\ 0.1 & 0.9 \end{bmatrix} \\ &= \begin{bmatrix} 0.12(0.2) + 0.88(0.1) & 0.12(0.8) + 0.88(0.9) \\ 0.11(0.2) + 0.89(0.1) & 0.11(0.8) + 0.89(0.9) \end{bmatrix} \\ &= \begin{bmatrix} 0.112 & 0.888 \\ 0.111 & 0.889 \end{bmatrix} \end{aligned}$$

So

$$P^3_{\text{Blue,Green}} = P(X_{n+3} = \text{Green} \mid X_n = \text{Blue}) = 0.111$$

, which means that if the chemical is currently blue, there is an 11.1% chance it will be green after 90 minutes.

## Answer to Exercise 439 (on page 1553)

1. We want to solve for $P\mathbf{x} = \mathbf{x}$. Let $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$. Then we have

$$\begin{bmatrix} p & 1-q \\ 1-p & q \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

   Then this gives us the following system of equations:

$$\begin{aligned} px + (1-q)y &= x \\ (1-p)x + qy &= y \end{aligned}$$

$$x = 1-q, \qquad y = 1-p$$

   Therefore, the eigenvector corresponding to the eigenvalue 1 is

$$\begin{bmatrix} 1-q \\ 1-p \end{bmatrix}$$

2. Normalizing it gives us $(1-q) + (1-p) = 2-p-q$. Therefore, the steady-state distribution is

$$\begin{bmatrix} \frac{1-q}{2-p-q} \\ \frac{1-p}{2-p-q} \end{bmatrix}$$

   Which for $A$ is $\frac{1-q}{2-p-q}$.

3. Plugging in $p = 0.3$ and $q = 0.6$ gives us $\frac{1-0.6}{2-0.3-0.6} = \frac{0.4}{1.1} = \frac{4}{11} = 0.364$ of the proportion of time spent in state `A`.

## Answer to Exercise 440 (on page 1557)

By the Markov property,

$$P(N \to V \to A) = P(V \mid N)\, P(A \mid V).$$

Substituting the values from the transition matrix,

$$P(N \to V \to A) = (0.5)(0.2) = 0.10.$$

Therefore,

$$\boxed{P(N \to V \to A) = 0.10.}$$

Thus, according to the model, the probability of observing the tag sequence

$$\text{Noun} \rightarrow \text{Verb} \rightarrow \text{Adjective}$$

is 10%.

## Answer to Exercise 441 (on page 1563)

Since we are finding the antiderivative of $\sin x$, we will define $f(x) = \sin x$. We are looking for a $F$, such that $F'(x) = \sin x$. The derivative of $\cos x$ is $-\sin x \neq f(x)$. However, the derivative of $-\cos x = \sin x = f(x)$. Since $\frac{d}{dx}[-\cos x] = \sin x$, the antiderivative of $\sin x$ is $-\cos x$.

## Answer to Exercise 442 (on page 1565)

First, we will find $v(t)$ by taking the antiderivative of $a(t)$ and using the initial condition $v(0) = -6$:

$$\int 6t + 4 \, , dt = 3t^2 + 4t + C = v(t)$$

$$v(0) = 3(0)^2 + 4(0) + C = -6$$

$$C = -6$$

Therefore, the velocity function is $v(t) = 3t^2 + 4t - 6$. Next, we repeat the process to find $s(t)$:

$$\int 3t^2 + 4t - 6 \, , dt = t^3 + 2t^2 - 6t + C = s(t)$$

$$s(0) = (0)^3 + 2(0)^2 - 6(0) + C = 9$$

$$C = 9$$

Therefore, the position function is $s(t) = t^3 + 2t^2 - 6t + 9$.

## Answer to Exercise 443 (on page 1565)

The antiderivative of $\sin x$ is $-\cos x$; therefore, the general solution is $f(x) = -2\cos x + C$. We use the given condition, $f(\pi) = 1$ to find $C$:

$$f(\pi) = -2\cos\pi + C = 1$$

$$C = 1 + 2\cos\pi = 1 + 2(-1) = -1$$

Therefore, the specific solution is $f(x) = -2\cos x - 1$

## Answer to Exercise 444 (on page 1565)

1. By the power rule, the antiderivative of $x^2$ is $\frac{1}{3}x^3$, the antiderivative of $2x$ is $x_2$, and the antiderivative of 4 is $4x$. So the general antiderivative of $f(x)$ is $\frac{1}{3}x^3 + x^2 - 4x + C$
2. We can rewrite $g(x)$ to more clearly see the powers of $x$. $g(x) = x^{\frac{2}{3}} + x^{\frac{3}{2}}$. Applying the power rule, we find the general antiderivative of $g(x)$ is $\frac{3}{5}x^{\frac{5}{3}} + \frac{2}{5}x^{\frac{5}{2}} + C$.
3. Recalling that the antiderivative of $\frac{1}{x}$ is $\ln|x|$, the general antiderivative of $h(x)$ is $\frac{1}{5}x - 2\ln|x| + C$
4. The antiderivative of $\sin\theta$ is $-\cos\theta$ and the antiderivative of $\sec^2\theta$ is $\tan x$. Therefore, the general antiderivative of $r(\theta)$ is $-2\cos\theta - \tan\theta + C$

## Answer to Exercise 445 (on page 1566)

1. The antiderivative of $\sin\theta$ is $-\cos\theta$, and the antiderivative of $\cos\theta$ is $\sin\theta$. The general form of $f$ is $f(\theta) = -\cos\theta + \sin\theta + C$. Substituting $\theta = \pi$, we find that $f(\pi) = -\cos\pi + \sin\pi + C = 1 + 0 + C = 2$, which implies $C = 1$. Therefore, $f(\theta) = -\cos\theta + \sin\theta + 1$.
2. The general antiderivative of $f''$ is $f'(x) = 4x^3 + 3x^2 - 4x + C_1$. We don't have a condition for $f'$, so we continue to find $f$. The antiderivative of $f'$ is $f(x) = x^4 + x^3 - 2x^2 + C_1x + C_2$. We can find $C_2$ with the condition $f(0) = 4$. $f(0) = C_2 = 4$, so we know $f(x) = x^4 + x^3 - 2x^2 + C_1x + 4$. Using the condition $f(1) = 1$, we find that $C_1 = 3$. Therefore, the specific solution is $f(x) = x^4 + x^3 - 2x^2 + 3x - 4$.

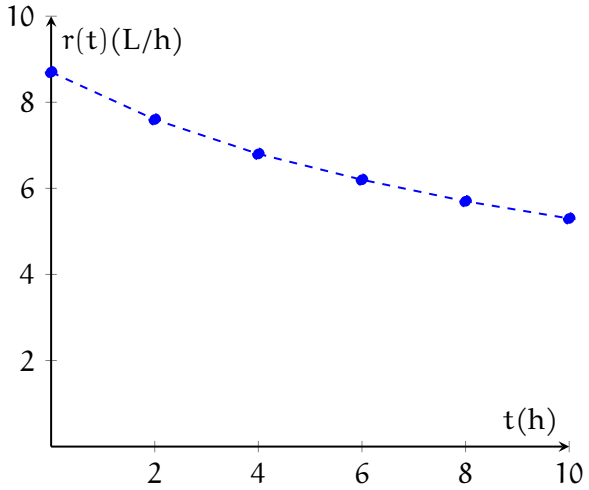
## Answer to Exercise 446 (on page 1569)

The units on the $x$-axis are $s$, and the units on the $y$-axis are $\frac{m}{s^2}$. The area would then have units of $s \times \frac{m}{s^2} = \frac{m}{s}$. Based on the units, the area represents a net change in velocity. The area above and below the axis are equal $(4.5\frac{m}{s})$; therefore, the total area is 0. This means the object's starting and ending velocity are the same.

## Answer to Exercise 447 (on page 1570)

The units of the area will be $\text{days} \times \frac{\text{deaths}}{\text{day}} = \text{deaths}$. The area under the curve represents the total number of people who died of SARS in Singapore during the time period represented [from March 1 to May 24 (if you took the time to do the math for the dates)].

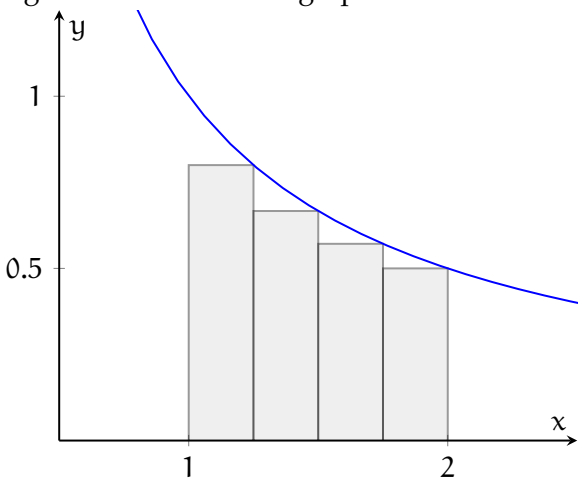
## Answer to Exercise 448 (on page 1570)



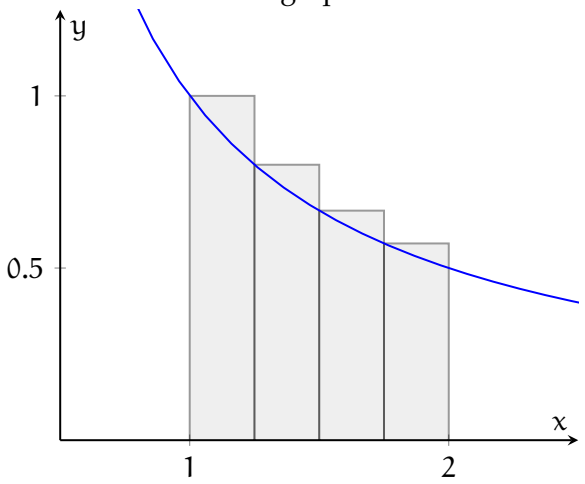

Based on the units, the area under the data would represent the total oil lost. One way to estimate this area would be to create rectangles, but there are other valid methods.

## Answer to Exercise 449 (on page 1574)

Right-determined sum graph:



The area of the right-determined sum is $0.25 \times (0.8 + 0.6667 + 0.5714 + 0.5) = 0.4202$. This is an underestimate of the actual area.
Left-determined sum graph:




The area of the left-determined sum is $0.25 \times (1 + 0.8 + 0.6667 + 0.5714) = 0.7595$. This is an overestimate of the actual area.

## Answer to Exercise 450 (on page 1581)

1. $\Delta x = \frac{3-1}{n} = \frac{2}{n}$ and $x_i = 1 + i\frac{2}{n} = 1 + \frac{2i}{n}$. Substituting, we get $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{2(1+\frac{2i}{n})}{(1+\frac{2i}{n})^2+1} \cdot \frac{2}{n}$
2. $\Delta x = \frac{7-4}{n} = \frac{3}{n}$ and $x_i = 4 + \frac{3i}{n}$. Substituting, we get $\lim_{n\to\infty} \sum_{i=1}^{n} [(4 + \frac{3i}{n})^2 + \sqrt{1 + 2(4 + \frac{3i}{n})}]\frac{3i}{n}$
3. $\Delta x = \frac{\pi-0}{n} = \frac{\pi}{n}$ and $x_i = \frac{i\pi}{n}$. Substituting, we get $\lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{\sin \frac{i\pi}{n}} \frac{\pi}{n}$

## Answer to Exercise 451 (on page 1582)

1. $\Delta x = \frac{3}{n}$, which implies $b - a = 3$. We could interpret $\sqrt{1 + \frac{3i}{n}}$ two ways: either $f(x) = \sqrt{1+x}$ and $x_i = \frac{3i}{n}$ or $f(x) = \sqrt{x}$ and $x_i = 1 + \frac{3i}{n}$. In the first case, we can find that $a = 0$ and $b = 3$, so the limit of the sum represents the area under $f(x) = \sqrt{1+x}$ from $x = 0$ to $x = 3$. For the second case, we can find that $a = 1$ and $b = 4$, so the limit of the sum represents the area under $f(x) = \sqrt{x}$ from $x = 1$ to $x = 4$.
2. $\Delta x = \frac{\pi}{4n}$, which implies $b - a = \frac{\pi}{4}$. We can see that $x_i = \frac{i\pi}{4n}$, which implies $a = 0$ and therefore also that $b = \frac{\pi}{4}$. Therefore, the limit of the sum represents the total area under $f(x) = \tan x$ from $x = 0$ to $x = \frac{\pi}{4}$.

## Answer to Exercise 452 (on page 1586)

| Number of Intervals | Calculated Area |
|---|---|
| 10 | 1.7820 |
| 50 | 1.6419 |
| 100 | 1.6256 |
| 500 | 1.6126 |
| 1000 | 1.6110 |
| 5000 | 1.6098 |

The area approaches the natural log of the endpoint, $\ln 5 \approx 1.6094$.

## Answer to Exercise 453 (on page 1587)

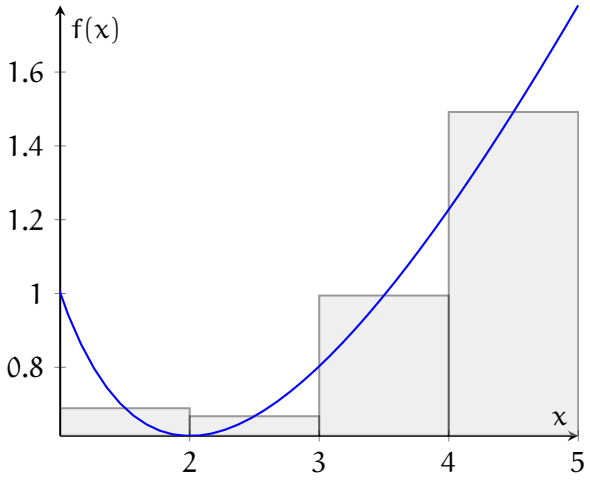
The volume of water will be the amount of water at 4 hours (50 liters) plus the area under the graph of $R(t)$ from $t = 4$ to $t = 15$. We will estimate this area with a right Riemann sum. The approximate volume added from $t = 4$ to $t = 7$ is $(7 - 4) * (6.2) = 18.6$ liters. The approximate volume added from $t = 7$ to $t = 12$ is $(12 - 7) * (5.9) = 29.5$ liters. The approximate volume added from $t = 12$ to $t = 15$ is $(15-12)*(5.6) = 16.8$ liters. Therefore, the approximate total volume of water in the tank at $t = 15$ is $50+18.6+29.5+16.8 = 114.9$ liters.

## Answer to Exercise 454 (on page 1587)

We will divide the area from $x = 1$ to $x = 5$ into four intervals at $x = 2$, $x = 3$, and $x = 4$. Next, we will find the value of $f(x)$ at the midpoint of each interval:

| Interval | Midpoint | Value of $f(x)$ at midpoint |
|---|---|---|
| 1 | 1.5 | $\approx 0.68907$ |
| 2 | 2.5 | $\approx 0.66742$ |
| 3 | 3.5 | $\approx 0.99447$ |
| 4 | 4.5 | $\approx 1.49185$ |

Using the values in the table, we can make a possible sketch of $f(x)$:

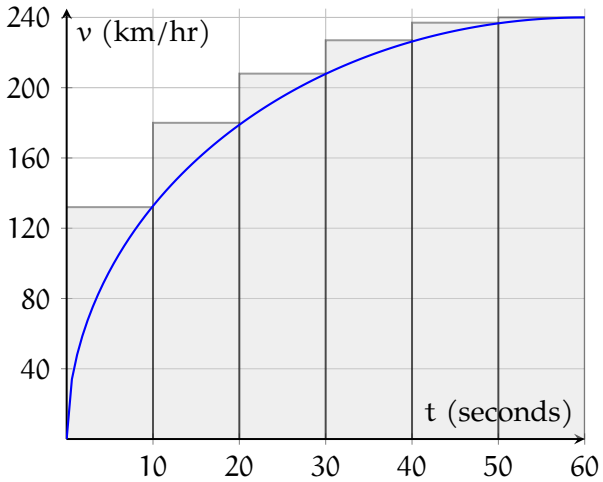



And we calculate the total area in the rectangles:

$$1 \times (0.68907 + 0.66742 + 0.99447 + 1.49185 = 3.84281$$

## Answer to Exercise 455 (on page 1588)

The question allows the student to choose the type of sum (left, right, or midpoint) and the number of intervals. A possible solution is given, but there are many ways to answer the question.
The tricky part here is noticing the units! In order to have a solution in kilometers, we will need to convert km/hr to m/s when we calculate the areas. A possible solution is to divide the graph into 6 intervals (one every 10 seconds) and use a right Riemann sum.




We can use the graph to *estimate* the height of each rectangle. Some reasonable estimates are $f(10) = 130\frac{km}{hr} \approx 36.1\frac{ms}{s}$, $f(20) = 180\frac{km}{hr} = 50\frac{m}{s}$, $f(30) = 210\frac{km}{hr} \approx 58.3\frac{m}{s}$, $f(40) = 230\frac{km}{hr} \approx 63.9\frac{m}{s}$, $f(50) = 235\frac{km}{hr} \approx 65.3\frac{m}{s}$, and $f(60) = 240\frac{km}{hr} \approx 66.7\frac{m}{s}$. [Any values within

$\pm 5$ of the listed values are reasonable.] Noting that each interval is 10 sec wide and usingß the estimates of $f(x)$ listed, we can estimate that the distance traveled is $10\ sec \times (36.1\frac{m}{s} + 50\frac{m}{s} + 58.3\frac{m}{s} + 63.9\frac{m}{s} + 65.3\frac{m}{s} + 66.7\frac{m}{s}) = 3403$ meters.

## Answer to Exercise 456 (on page 1592)

Following the structure shown in the formal definition of a definite integral, we can set $f(x) = x^3 + x\sin x$ and rewrite the limit of the sum as $\lim_{n\to\infty} \Sigma_{i=1}^{n} f(x)\Delta x = \int_0^{\pi} f(x)\,dx$. Therefore, the full definite integral would be written as $\int_0^{\pi}(x^3 + x\sin x)\,dx$.

## Answer to Exercise 457 (on page 1602)

By property 6, we know that

$$\int_0^1 (5 - 6x^2)\,dx = \int_0^1 5\,dx - \int_0^1 6x^2\,dx$$

By property 5, we know that

$$\int_0^1 5\,dx - \int_0^1 6x^2\,dx = \int_0^1 5\,dx - 6\int_0^1 x^2\,dx$$

By property 3, we know that

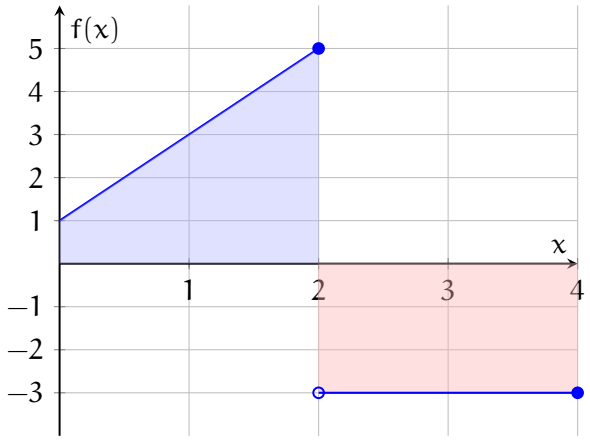
$$\int_0^1 5\,dx = 5(1-0) = 5$$

Putting it all together, we see that

$$\int_0^1 (5 - 6x^2)\,dx = 5 - 6(\frac{1}{3}) = 5 - 2 = 3$$

## Answer to Exercise 458 (on page 1602)

We can break the integral into two parts: from $x = 0$ to $x = 2$ (shaded in blue), and from $x = 2$ to $x = 4$ (shaded in red). The blue portion is a trapezoid, so it has a total area of $\frac{1}{2}(b_1 + b_2)(h) = \frac{1}{2}(1+5)(2) = 6$. Because it is above the x-axis, the area is positive. The red portion is a rectangle and has a total area of $2 \times 3 = 6$ and is *negative*, because it lies below the x-axis. Therefore, the total area is $6 + -6 = 0$.




## Answer to Exercise 459 (on page 1603)

Using the properties of integrals, we can rewrite $\int_{-1}^{9} 3g(x) + 2\,dx$ as $3\int_{-1}^{9} g(x)\,dx + 2(9 - (-1))$. From the graph, we can determine $\int_{-1}^{9} g(x)\,dx = 2.5$. Therefore, $\int_{-1}^{9} 3g(x) + 2\,dx = 3(2.5) + 2(10) = 27.5$.

## Answer to Exercise 461 (on page 1608)

(B). If $f'(x) > 0$ for all $x$, then $f$ must be increasing for $4 < x < 7$. Since (C) decreases from $x = 5$ to $x = 7$, we can eliminate it. We can also eliminate (D), since $f(4) = f(5) = f(7)$, which implies either the slope of $f$ is zero or changes from positive to negative. If $\int_4^7 f(x)\,dx =$, then some portion of $f(x)$ lies above the $x-axis$, while some other portion lies below (we must have positive and negative areas for the sum to be zero). This eliminates (A) and (E), since the integral of (A) would have a negative value and the integral of (E) would have a positive value. This leaves (B).

## Answer to Exercise 461 (on page 1608)

(D) 2. First, we try to compute the limit directly: $\lim_{x\to 1} \frac{\int_1^x g(t)\,dt}{g(x)-6} = \frac{\int_1^1 g(t)\,dt}{6-6} = \frac{0}{0}$, which is undefined. Because $g$ is continuous and differentiable, we can apply L'Hospital's rule. $\lim_{x\to 1} \frac{\int_1^x g(t)\,dt}{g(x)-6} = \lim_{x\to 1} \frac{d}{dx}\left[\frac{\int_1^x g(t)\,dt}{g(x)-6}\right] = \lim_{x\to 1} \frac{g(x)}{g'(x)} = \frac{g(6)}{g'(6)} = \frac{6}{3} = 2$.

## Answer to Exercise 462 (on page 1609)

1. $g'(x) = \sqrt{x + x^3}$
2. $F(x) = -\int_0^x \sqrt{1 + \sec t}\, dt$ and therefore $F'(x) = -\sqrt{1 + \sec x}$
3. setting $u = e^x$ and noting $\frac{du}{dx} = e^x$, then $h'(x) = \frac{d}{du}\int_1^u \ln t\, dt(\frac{du}{dx})$ Taking the derivative and substituting for $\frac{du}{dx}$, we find $h'(x) = \ln u \cdot e^x = \ln(e^x) \cdot e^x = x \cdot e^x$
4. $y = -\int_{\frac{\pi}{4}}^{\sqrt{x}} \theta \tan\theta\, d\theta$. Setting $u = \sqrt{x}$ and noting that $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$, we see that $y' = -\frac{d}{du}[\int_{\frac{\pi}{4}}^{u} \theta \tan\theta\, d\theta]\frac{du}{dx} = u\tan u \cdot \frac{1}{2\sqrt{x}} = -\sqrt{x}\tan\sqrt{x} \cdot \frac{1}{2\sqrt{x}} = \frac{-\sqrt{x}\tan\sqrt{x}}{2\sqrt{x}} = -\frac{1}{2}\tan\sqrt{x}$

## Answer to Exercise 463 (on page 1611)

1. The antiderivative of $t^{-3/2}$ is $\frac{-2}{\sqrt{t}}$. Therefore, the integral is equal to $[\frac{-2}{\sqrt{t}}]_1^4 = \frac{-2}{\sqrt{4}} - \frac{-2}{\sqrt{1}} = -1 + 2 = 1$.

## Answer to Exercise 464 (on page 1611)

According to FTC, $h'(x) = f(x)$ and $h''(x) = f'(x)$. Examining the graph, we see that the curve lies below the x-axis for $0 < x < 6$, which means that $h(6) = \int_0^6 f(t)\, dt < 0$. $h'(6) = f(6) = 0$ and $h''(6) = f'(6) > 0$. Therefore, $h(6) < h'(6) < h''(6)$.

## Answer to Exercise 465 (on page 1612)

We know that $f(2) = \int_{-5}^2 f'(x)\, dx + f(-5)$. Examining the graph, we know that $\int_{-5}^2 f'(x)\, dx = frac12(3)(2) - \frac{1}{2}\pi(2^2)$ (the area of the triangle above the x-axis less the area of the semicircle below the axis). Therefore, $f(-5) = f(2) - \int_{-5}^2 f'(x)\, dx = 1 - (3 - 2\pi) = 2\pi - 2$

## Answer to Exercise 466 (on page 1612)

11.71. The particle's velocity will be given by its initial velocity plus the integral of its acceleration over the time period. Therefore, $v(3) = v(0) + \int_0^3 a(t)\, dt = 5 + \int_0^3 \frac{t+3}{\sqrt{t^3+1}}\, dt \approx 5 + 6.71 = 11.71$.

## Answer to Exercise 467 (on page 1613)

We begin by setting up the integral for average value of a function with $a = 0$, $b = \frac{\pi}{2}$, and $f(x) = \sqrt{\cos x}$:

$$\frac{1}{\pi/2 - 0} \int_0^{\pi/2} \sqrt{\cos x}\, dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} \sqrt{\cos x}\, dx$$

There's not an obvious way to evaluate this integral by hand. Luckily, this question allows for the use of a calculator. Entering this integral into a calculator (such as a TI-89 or Wolfram Alpha), we find that:

$$\frac{2}{\pi} \int_0^{\pi/2} \sqrt{\cos x}\, dx \approx 0.763$$

## Answer to Exercise 468 (on page 1621)

1. $L = \int_1^3 \sqrt{1 + \frac{1}{x^2}}\, dx$
2. $L = \int_0^{\pi} \sqrt{1 + \cos^2 x}\, dx$
3. $L = \int_1^4 \sqrt{1 + (x^2 - \frac{1}{4x^2})^2}\, dx$
4. $L = \int_0^{\frac{\pi}{3}} \sqrt{1 + (\frac{1}{\cos x} \times -\sin x)^2}\, dx = \int_0^{\frac{\pi}{3}} \sqrt{1 + \tan^2 x}\, dx$

## Answer to Exercise 469 (on page 1622)

From looking at the structure of the given arc length integral, we can see that $f'(t) = \sqrt{3t + 4}$. Taking the antiderivative, we find that $f(x) = \frac{2}{9}(3x + 4)^{3/2} + C$. Substituting $f(0) = 2$ , we can solve for $C$.

$$2 = \frac{2}{9}(3(0) + 4)^{3/2} + C$$

$$2 = \frac{2}{9}(4)^{3/2} + C$$

$$2 = \frac{2}{9}(2)^3 + C$$

$$2 = \frac{16}{9} + C$$

$$\frac{18}{9} = \frac{16+9C}{9}$$
$$18 = 16+9C$$
$$2 = 9C$$
$$C = \frac{2}{9}$$

Therefore, $f(x) = \frac{2}{3}(3x+4)^{3/2}+\frac{2}{9}$. To find the coordinate point where $s(x) = 3$, we first note that the antiderivative of $\sqrt{3t+5}$ is $\frac{2}{9}(3t+5)^{3/2}+C$. Therefore, $s(x) = \frac{2}{9}(3x+5)^{3/2}-\frac{2}{9}(5)^{3/2}$. Setting $s(x) = 3$ and solving for $x$, we find that $x = 1.159$

## Answer to Exercise 470 (on page 1622)

Recall that arc length is given by $\int_a^b \sqrt{1+[f'(x)]^2}\,dx$. Since $f(x) = \ln x$, then $f'(x) = \frac{1}{x}$. Taking $a = 1$, $b = 2$, and $f'(x) = \frac{1}{x}$, the integral that gives the length of the curve $y = \ln x$ on the specified interval is $\int_1^2 \sqrt{1+\frac{1}{x^2}}\,dx$.

## Answer to Exercise 471 (on page 1622)

Since we are told the vector-valued velocity of the ship, we know that $\frac{dx}{dt} = -1412\sin t$, $\frac{dy}{dt} = 1412\cos t$, and $\frac{dz}{dt} = t$. The distance traveled in the first 60 seconds is given by $\int_0^{60} \sqrt{(-1412\sin t)^2+(1412\cos t)^2+t^2}\,dt$. Using a calculator, the integral evaluates to 84745 meters. The distance traveled in the second 60 seconds is given by $\int_{60}^{120} \sqrt{(-1412\sin t)^2+(1412\cos t)^2+t^2}\,dt$. Using a calculator, this integral evaluates to 84898 meters.

## Answer to Exercise 472 (on page 1633)

```python
from scipy.stats import norm

# Constants
MEAN = 164.7
STD = 7.1

# What is the cutoff for the top decile?
cutoff = norm.ppf(0.9, loc=MEAN, scale=STD);
print(f"To be in the top 10 percent, you must be at least {cutoff:.2f} cm")

# What proportion of women are between 160cm and 165cm?
shorter_than_160 = norm.cdf(160, loc=MEAN, scale=STD)
```

```
shorter_than_165 = norm.cdf(165, loc=MEAN, scale=STD)
between = shorter_than_165 - shorter_than_160
print(f"{between * 100.0:.2f}% of adult women are between 160 and 165 cm.")
```

When run, this will give you:

```
> python3 women.py
To be in the top 10 percent, you must be at least 173.80 cm
26.29% of adult women are between 160 and 165 cm.
```

## Answer to Exercise 474 (on page 1641)

First, we convert the temperatures into Kelvin:

- Dawn: $12 + 273.15 = 285.15$
- Noon: $28 + 273.15 = 301.15$

So, the temperature T has increased by a factor of $\frac{301.15}{285.15} \approx 1.056$.

Thus, the volume of the air mattress has also increased by a factor of 1.056.

This means that the air mattress that had a volume of 1000 liters at dawn will have a volume 1056 liters at noon.

## Answer to Exercise 474 (on page 1641)

What is the pressure in kPA?

- Before squeezing: 100 kPa
- While squeezing: 120 kPa

So, the pressure P has increased by a factor of $\frac{120}{100} = 1.2$

$$1/1.2 \approx 0.833$$

The air in the bottle had a volume of 1 liter before squeezing, so it has a volume of 833 milliliters while being squeezed.

## Answer to Exercise 475 (on page 1642)

First, let's convert the known values to the right unit:

- Radius = 0.12 m
- Length = 0.5 m
- T = 20 + 273.15 = 293.15 degrees Kelvin
- P = 600 kPa = 600,000 Pa

The volume of the cylindrical chamber is $V = \pi r^2 h = \pi(0.12)^2 0(0.5) \approx 0.0226$.

The Ideal Gas Law tell us that $PV = nRT$. We are solving for $n$.

$$n = \frac{PV}{RT} = \frac{(600,000)(0.0226)}{(8.31446)(293.15)} \approx 5.68 \text{ moles of } O_2$$

## Answer to Exercise 476 (on page 1646)

$E = C_{V,m}(3 \text{ moles })(20 degrees Celsius) = (12.47)(3)(20) = 748$ Joules

$E = C_{V,m}(3 \text{ moles })(20 degrees Celsius) = (20.8)(3)(20) = 1247$ Joules

## Answer to Exercise 477 (on page 1648)

10 degrees Celsius is 283.15 degrees Kelvin. 30 degrees Celsius is 303.15.

For any gas:

$$E_K = C_{V,m} nT$$

And $C_{V,m} = 12.47$ for all monoatomic gases.

So, the energy at 10 degrees Celsius:

$$E_1 = (12.47)(3)(283.15) = 10,594 \text{ Joules}$$

The energy at 30 degrees Celsius:

$$E_2 = (12.47)(3)(303.15) = 11,342 \text{ Joules}$$

The difference?

$$E_2 - E_1 = 11,342 - 10,594 = 748 \text{ Joules}$$

Which is consistent with your earlier exercise.

## Answer to Exercise 478 (on page 1662)

When one mole of water goes from 33° to 17°, it will give off $(75.38)(33 - 17) = 1,206$ Joules or 1.206 kJ.

Tom needs 300,000 kJ, so he needs $300,000/1.206 = 248,739.72$ moles of water.

How many liters is that? $248,739.72 * 0.018 = 4,477.31$ liters.

How many barrels is that? $4,477.31/159 = 28.16$ barrels. He will need 29 barrels.

## Answer to Exercise 479 (on page 1663)

When one mole of mirabilite goes from 33° to 17°, it will give off $(550)(33-17)+82,000 = 90,800$ Joules or 90.8 kJ.

Tom needs 300,000 kJ, so he needs $300,000/90.8 = 3,304$ moles of mirabilite.

How many liters is that? $3,304 * 0.22 = 726.9$ liters.

How many barrels is that? $726.9/159 = 4.57$. He will need 5 barrels.

## Answer to Exercise 480 (on page 1675)

The pre-compression temperature is 80irc C $+ 273.15 = 353.15°$ K.

The radius of the cylinder is $9.535/2 = 4.7625$ cm.

The area of a cross section of the cylinder is $\pi r^2 = \pi(4.7625)^2 \approx 71.26$ ml.

So, the change in volume between the minimum and maximum volume is $(71.26)(10.16) = 724$ ml, or 0.724 liters.

(With four cylinders, the total displacement of a Model T is thus $(4)(724) = 2,896$ cc.)

Now we use the ideal gas to figure out how many moles of gas will fit into 0.724 liters at 100 kPa and 353.15° K.

$$n = \frac{PV}{rT} = \frac{(100)(0.724)}{(8.314)(353.15)} = 0.02466 \text{ moles of air+fuel}$$

So, if we suck $n_a$ moles of air and $n_e$ moles of vaporized ethanol in to the cylinder:

$$n_a + n_e = 0.02466$$

So

$$n_a = 0.02466 - n_e$$

21% of $n_a$ is $O_2$:

$$n_{O_2} = 0.21 n_a = 0.21(0.02466 - n_e) = 0.005178 - 0.21 n_e$$

For a clean burn, we need 3 times as many $O_2$ molecules as ethanol molecules. Thus:

$$3n_e = n_{O_2} = 0.005178 - 0.21 n_e$$

Solving for $n_e$:

$$n_e = \frac{0.005178}{3.21} = 0.001613 \text{ moles of ethanol}$$

For every molecule of ethanol that burns we get 2 molecules of $CO_2$: 0.003226 moles.

For every molecule of ethanl that burns we get 3 molecules of $H_2OI$: 0.004839 moles.

How much heat? $(0.001613)(1330) = 2.145$ kilojoules from each burn.

## Answer to Exercise 481 (on page 1682)

1. Let $u = 1 + \cos x$. Then $du = -\sin x\ dx$ and $-du = \sin x\ dx$. Substituting:

$$\int \sin x\sqrt{1+\cos x}\,dx = \int -\sqrt{u}\,du = -\frac{2}{3}u^{3/2} + C$$

$$= -\frac{2}{3}(1+\cos x)^{3/2} + C$$

Taking the derivative:

$$\frac{d}{dx}\left[-\frac{2}{3}(1+\cos x)^{3/2} + C\right] = -\frac{2}{3}\left[\frac{d}{dx}(1+\cos x)^{3/2} + \frac{d}{dx}C\right]$$

$$= -\frac{2}{3}\left[\frac{3}{2}(1+\cos x)^{1/2}\cdot\frac{d}{dx}(1+\cos x)\right] = -1\sqrt{1+\cos x}\cdot(-\sin x) = \sin x\sqrt{1+\cos x}$$

2. Let $u = \pi/x$. Then $du = (-\pi/x^2)dx$ and $-du/\pi = (1/x^2)dx$. Substituting:

$$\int \frac{\cos(\pi/x)}{x^2}\,dx = -\frac{1}{\pi}\int \cos u\,du = -\frac{1}{\pi}\sin u + C = -\frac{1}{\pi}\sin(\pi/x) + C$$

Taking the derivative:

$$\frac{d}{dx}\left[-\frac{1}{\pi}\sin(\pi/x) + C\right] = -\frac{1}{\pi}\left[\frac{d}{dx}\sin(\pi/x) + \frac{d}{dx}C\right]$$

$$= -\frac{1}{\pi}\left[\cos(\pi/x)\cdot\frac{d}{dx}\left(\frac{\pi}{x}\right)\right] = -\frac{1}{\pi}\left[\cos(\pi/x)\cdot\left(\frac{-\pi}{x^2}\right)\right]$$

$$= \frac{\cos(\pi/x)}{x^2}$$

3. Let $u = 9 - x^3$. Then $du = -3x^2\ dx$ and $-\frac{2}{3}du = 2x^2\ dx$. Substituting:

$$\int 2x^2\left(9-x^3\right)^{2/3}\,dx = -\frac{2}{3}\int (u)^{2/3}\,du$$

$$= -\frac{2}{3}\left[\frac{3}{5}u^{5/3} + C\right] = -\frac{2}{5}\left(9-x^3\right)^{5/3} + C$$

Taking the derivative:

$$\frac{d}{dx}\left[-\frac{2}{5}\left(9-x^3\right)^{5/3} + C\right] = -\frac{2}{5}\left[\frac{d}{dx}\left(9-x^3\right)^{5/3}\right] + \frac{d}{dx}C$$

$$= -\frac{2}{5}\left[\frac{5}{3}\left(9-x^3\right)^{2/3}\cdot\frac{d}{dx}\left(9-x^3\right)\right] = -\frac{2}{3}\left(9-x^3\right)^{2/3}\left(-3x^2\right) = 2x^2\left(9-x^3\right)^{2/3}$$

4. Let $u = 1+x$. Then $du = dx$. Additionally, $u-1 = x$ and $x^2 = (u-1)^2$. Substituting:

$$\int 3x^2\sqrt{1+x}\,dx = \int 3(u-1)^2\sqrt{u}\,du = 3\int\left(u^2-2u+1\right)\sqrt{u}\,du$$

$$= 3\int u^{5/2}-2u^{3/2}+u^{1/2}\,du = 3\left[\frac{2}{7}u^{7/2}-\frac{2\cdot 2}{5}u^{5/2}+\frac{2}{3}u^{3/2}+C\right]$$

$$= \frac{6}{7}u^{7/2}-\frac{12}{5}u^{5/2}+2u^{3/2}+C$$

$$= \frac{6}{7}(1+x)^{7/2}-\frac{12}{5}(1+x)^{5/2}+2(1+x)^{3/2}+C$$

Taking the derivative:

$$\frac{d}{dx}\left[\frac{6}{7}(1+x)^{7/2}-\frac{12}{5}(1+x)^{5/2}+2(1+x)^{3/2}+C\right]$$

$$= \frac{6}{7}\left(\frac{7}{2}(1+x)^{5/2}\right)-\frac{12}{5}\left(\frac{5}{2}(1+x)^{3/2}\right)+2\left(\frac{3}{2}(1+x)^{1/2}\right)$$

$$= 3(1+x)^{5/2}-6(1+x)^{3/2}+3(1+x)^{1/2}$$

$$= \left[3(1+x)^2-6(1+x)+3\right](1+x)^{1/2}$$

$$= \left[3\left(1+2x+x^2\right)-6-6x+3\right]\sqrt{1+x}$$

$$= \left[3+6x+3x^2-6-6x+3\right]\sqrt{1+x} = 3x^2\sqrt{1+x}$$

5. Let $u = x^3-1$. Then $du = 3x^2\,dx$. Substituting:

$$\int\frac{3x^2}{x^3-1}\,dx = \int\frac{1}{u}\,du = \ln u + C$$

$$= \ln\left(x^3-1\right)+C$$

Taking the derivative:

$$\frac{d}{dx}\left[\ln\left(x^3-1\right)+C\right] = \frac{d}{dx}\ln\left(x^3-1\right)+\frac{d}{dx}C$$

$$= \frac{1}{x^3-1}\cdot\left[\frac{d}{dx}\left(x^3-1\right)\right] = \frac{3x^2}{x^3-1}$$

## Answer to Exercise 482 (on page 1685)

1. Let $g(x) = u = \sin x$. Then $du = \cos x\, dx$. Additionally, $g(0) = \sin 0 = 0$ and $g(\pi/2) = \sin(\pi/2) = 1$. Substituting and changing the limits of integration:

$$\int_0^{\pi/2} \cos x \sin(\sin x)\, dx = \int_0^1 \sin u\, du = -\cos u|_0^1 = \cos 0 - \cos 1 = 1 - \cos 1$$

2. Let $g(x) = u = 1 + 2x$. Then $du = 2\, dx$ and $\frac{du}{2} = dx$. Additionally, $g(0) = 1$ and $g(13) = 1 + 2(13) = 27$. Substituting and changing the limits of integration:

$$\int_0^{13} \frac{1}{\sqrt[3]{(1+2x)^2}}\, dx = \frac{1}{2}\int_1^{27} \frac{1}{\sqrt[3]{u^2}}\, du = \frac{1}{2}\int_1^{27} u^{-2/3}\, du$$

$$= \frac{1}{2}\left[3u^{1/3}\right]_1^{27} = \frac{3}{2}\left[\sqrt[3]{27} - \sqrt[3]{1}\right] = \frac{3}{2}(3-1) = 3$$

3. Let $g(x) = u = 1/x$. Then $du = (-1/x^2)dx$ and $-du = dx/x^2$. Additionally, $g(1) = 1$ and $g(2) = 1/2$. Substituting and changing the limits of integration:

$$\int_1^2 \frac{e^{1/x}}{x^2}\, dx = -\int_1^{1/2} e^u\, du = \int_{1/2}^1 e^u\, du$$

$$= e^u|_{1/2}^1 = e - \sqrt{e}$$

4. Let $g(x) = u = \cos x$. Then $du = -\sin x\, dx$ and $-du = \sin x\, dx$. Additionally, $g(0) = \cos 0 = 1$ and $g(\pi/6) = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$. Substituting and changing the limits of integration:

$$\int_0^{\pi/6} \frac{\sin x}{\cos^2 x}\, dx = -\int_1^{\sqrt{3}/2} \frac{1}{u^2}\, du = \int_{\sqrt{3}/2}^1 \frac{1}{u^2}\, du$$

$$= -\frac{1}{u}|_{\sqrt{3}/2}^1 = \frac{2}{\sqrt{3}} - 1 = \frac{2\sqrt{3}-3}{3}$$

5. Let $g(x) = u = 1 + 2x$. Then $du = 2\, dx$ and $\frac{du}{2} = dx$. And if $u = 1 + 2x$, then $x = \frac{u-1}{2}$. Additionally, $g(0) = 1$ and $g(4) = 9$. Substituting and changing the limits of integration:

$$\int_0^4 \frac{x}{\sqrt{1+2x}}\, dx = \frac{1}{2}\int_1^9 \frac{\frac{u-1}{2}}{\sqrt{u}}\, du = \frac{1}{4}\int_1^9 \frac{u-1}{\sqrt{u}}\, du$$

$$= \frac{1}{4}\int_1^9 \sqrt{u} - \frac{1}{\sqrt{u}}\, du = \frac{1}{4}\left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^9$$

$$= \frac{1}{4}\left[\frac{2}{3}\left(9^{3/2} - 1^{3/2}\right) - 2\left(\sqrt{9} - \sqrt{1}\right)\right] = \frac{1}{4}\left[\frac{2}{3}(26) - 2(2)\right]$$

$$= \frac{1}{4}\left[\frac{52}{3} - 4\right] = \frac{1}{4}\left[\frac{52-12}{3}\right] = \frac{1}{4}\left[\frac{40}{3}\right] = \frac{10}{3}$$

## Answer to Exercise 483 (on page 1690)

Recall that for a polar function, $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$. We also know that $x = r\cos\theta$, which equals $[1 + 2\sin\theta] \cdot \cos\theta = \cos\theta + 2\sin\theta\cos\theta$ in this case. We also know that $y = r \cdot \sin\theta$, which equals $[1 + 2\sin\theta] \cdot \sin\theta = \sin\theta + 2\sin^2\theta$ in this case. Taking the derivative with respect to $\theta$:

$$\frac{dy}{d\theta} = \frac{d}{d\theta}\left[\sin\theta + 2\sin^2\theta\right]$$
$$\frac{dy}{d\theta} = \cos\theta + 4\sin\theta\cos\theta$$

And

$$\frac{dx}{d\theta} = \frac{d}{d\theta}\left[\cos\theta + 2\sin\theta\cos\theta\right]$$
$$\frac{dx}{d\theta} = -\sin\theta - 2\sin^2\theta + 2\cos^2\theta$$

Evaluating each at $\theta = 0$:

$$\frac{dy}{d\theta} = \cos 0 + 4\sin 0\cos 0 = 1 + 0 = 1$$

$$\frac{dx}{d\theta} = -\sin 0 - 2\sin^2 0 + 2\cos^2 0 = 0 - 0 + 2 = 2$$

Therefore, $\frac{dr}{d\theta} = \frac{dy/d\theta}{dx/d\theta} = \frac{1}{2}$

## Answer to Exercise 484 (on page 1691)

1. Answer: slope $= -\frac{\sqrt{3}}{3}$ and an equation for the tangent line is $y - \frac{\sqrt{3}}{6} = -\frac{\sqrt{3}}{3}\left(x - \frac{1}{2}\right)$.

   Explanation: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{d}{d\theta}r\cdot\sin\theta}{\frac{d}{d\theta}r\cdot\cos\theta} = \frac{\frac{d}{d\theta}\left(\frac{2}{3}\cos\theta\sin\theta\right)}{\frac{d}{d\theta}\left(\frac{2}{3}\cos^2\theta\right)} = \frac{\frac{2}{3}\left(\cos^2\theta - \sin^2\theta\right)}{\frac{2}{3}(-2\cos\theta\sin\theta)} = \frac{\sin^2\theta - \cos^2\theta}{2\cos\theta\sin\theta}$

   Substituting $\theta = \frac{\pi}{6}$: $\frac{dy}{dx} = \frac{\sin^2\pi/6 - \cos^2\pi/6}{2\cos\pi/6\sin\pi/6} = \frac{(1/2)^2 - \left(\sqrt{3}/2\right)^2}{2\left(\sqrt{3}/2\right)(1/2)} = \frac{1/4 - 3/4}{\sqrt{3}/2} = \frac{-1/2}{\sqrt{3}/2} = \frac{-1}{2}\cdot\frac{2}{\sqrt{3}} =$

$-\frac{\sqrt{3}}{3}$

To write an equation for a line, we need a Cartesian point. First, we find $r$ at $\theta = \frac{\pi}{6}$: $r = \frac{2}{3}\cos\left(\frac{\pi}{6}\right) = \frac{2}{3}\cdot\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{3}$. So the point the tangent passes through is the polar coordinate $\left(\frac{\sqrt{3}}{3}, \frac{\pi}{6}\right)$. We convert this to Cartesian coordinates: $x = r\cos\theta = \frac{\sqrt{3}}{3}\cdot\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{3}\cdot\frac{\sqrt{3}}{2} = \frac{3}{6} = \frac{1}{2}$ And $y = r\sin\theta = \frac{\sqrt{3}}{3}\cdot\sin\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{3}\cdot\frac{1}{2} = \frac{\sqrt{3}}{6}$

So, an equation for a line with slope $-\frac{\sqrt{3}}{3}$ that passes through Cartesian coordinate $\left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right)$ is: $y - \frac{\sqrt{3}}{6} = -\frac{\sqrt{3}}{3}\left(x - \frac{1}{2}\right)$

2. Answer: slope $= \frac{2}{\pi}$ and an equation for the tangent line is $y - \frac{1}{\pi} = \frac{2}{\pi}x$

Explanation: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{d}{d\theta}\left(\frac{\sin\theta}{2\theta}\right)}{\frac{d}{d\theta}\left(\frac{\cos\theta}{2\theta}\right)} = \frac{\frac{\theta\cos\theta - \sin\theta}{2\theta^2}}{-\frac{\theta\sin\theta + \cos\theta}{2\theta^2}} = \frac{\sin\theta - \theta\cos\theta}{\theta\sin\theta + \cos\theta}$.

Substituting $\theta = \frac{\pi}{2}$: $\frac{dy}{dx} = \frac{\sin\frac{\pi}{2} - \left(\frac{\pi}{2}\right)\cos\frac{\pi}{2}}{\left(\frac{\pi}{2}\right)\sin\frac{\pi}{2} + \cos\frac{\pi}{2}} = \frac{1 - \left(\frac{\pi}{2}\right)\cdot 0}{\left(\frac{\pi}{2}\right)\cdot 1 + 0} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$

To write an equation for a line, we need a Cartesian point. First, we find $r$ at $\theta = \frac{\pi}{2}$: $r = \frac{1}{2\theta} = \frac{1}{2\frac{\pi}{2}} = \frac{1}{\pi}$. So the tangent line passes through the point with polar coordinates $\left(\frac{1}{\pi}, \frac{\pi}{2}\right)$. We convert this to Cartesian coordinates: $x = r\cdot\cos\theta = \frac{1}{\pi}\cdot\cos\frac{\pi}{2} = \frac{1}{\pi}\cdot 0 = 0$ and $y = r\cdot\sin\theta = \frac{1}{\pi}\cdot\sin\frac{\pi}{2} = \frac{1}{\pi}\cdot 1 = \frac{1}{\pi}$.

So, an equation for a line with slope $\frac{2}{\pi}$ that passes through Cartesian coordinate $\left(0, \frac{1}{\pi}\right)$ is $y - \frac{1}{\pi} = \frac{2}{\pi}x$

3. Answer: slope $= -\frac{5}{\sqrt{3}}$ and an equation for the tangent line is $y - \frac{\sqrt{3}}{4} = \left(-\frac{5}{\sqrt{3}}\right)\left(x + \frac{1}{4}\right)$.

Explanation: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{d}{d\theta}[(2+3\cos\theta)\cdot\sin\theta]}{\frac{d}{d\theta}[(2+3\cos\theta)\cdot\cos\theta]} = \frac{\cos\theta(2+3\cos\theta) - 3\sin^2\theta}{(2+3\cos\theta)\cdot(-\sin\theta) + \cos\theta(-3\sin\theta)} = \frac{\cos\theta(2+3\cos\theta) - 3\sin^2\theta}{-2\sin\theta(1+3\cos\theta)}$.

Substituting $\theta = \frac{2\pi}{3}$: $\frac{dy}{dx} = \frac{\cos\frac{2\pi}{3}\left(2+3\cos\frac{2\pi}{3}\right) - 3\sin^2\frac{2\pi}{3}}{-2\sin\frac{2\pi}{3}\left(1+3\cos\frac{2\pi}{3}\right)} = \frac{\left(-\frac{1}{2}\right)\left(2+3\left(-\frac{1}{2}\right)\right) - 3\left(\frac{\sqrt{3}}{2}\right)^2}{-2\left(\frac{\sqrt{3}}{2}\right)\left(1+3\left(-\frac{1}{2}\right)\right)} =$

$\frac{\left(-\frac{1}{2}\right)\left(2-\frac{3}{2}\right) - 3\left(\frac{3}{4}\right)}{-\sqrt{3}\left(1-\frac{3}{2}\right)} = \frac{\left(-\frac{1}{2}\right)\left(\frac{1}{2}\right) - \frac{9}{4}}{-\sqrt{3}\left(-\frac{1}{2}\right)} = \frac{-\frac{1}{4} - \frac{9}{4}}{\frac{\sqrt{3}}{2}} = -\frac{\frac{10}{4}}{\frac{\sqrt{3}}{2}} = -\frac{10\cdot 2}{4\cdot\sqrt{3}} = -\frac{5}{\sqrt{3}}$

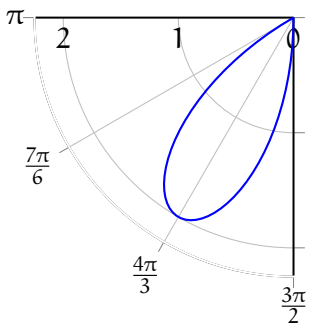
To write an equation for a tangent line, we need a Cartesian point. First, we find $r$ at $\theta = \frac{2\pi}{3}$: $r = 2 + 3\cos\frac{2\pi}{3} = 2 + 3\left(\frac{-1}{2}\right) = 2 - \frac{3}{2} = \frac{1}{2}$. So the tangent line passes through polar coordinate $\left(\frac{1}{2}, \frac{2\pi}{3}\right)$. We convert this to Cartesian coordinates: $x = r\cos\theta = \frac{1}{2}\cos\frac{2\pi}{3} = \frac{1}{2}\left(-\frac{1}{2}\right) = -\frac{1}{4}$ and $y = r\sin\theta = \frac{1}{2}\sin\frac{2\pi}{3} = \frac{1}{2}\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{4}$.

So, an equation with slope $-\frac{\sqrt{5}}{3}$ that passes through the Cartesian coordinate $\left(-\frac{1}{4}, \frac{\sqrt{3}}{4}\right)$ is: $y - \frac{\sqrt{3}}{4} = \left(-\frac{5}{\sqrt{3}}\right)\left(x + \frac{1}{4}\right)$

## Answer to Exercise 485 (on page 1698)

We know the area of the circle is $\pi r^2 = \pi(2)^2 = 4\pi$. To find the area of the shaded regions,

we need to subtract the area of the trefoil from the area of the circle. The trefoil has three equal areas. We can find the area of the leaf that is formed on the interval $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$ (see figure below).




The area of one leaf of the trefoil is given by $\frac{1}{2}\int_{\pi/6}^{\pi/2}[2\cos 3\theta]^2\,d\theta$. Using a calculator, the area of one leaf is $\approx 1.0472$. The area of the circle is given by $\pi r^2 = \pi(2)^2 \approx 12.5664$. The area of the shaded region is the area of the circle minus three times the area of a single leaf: $12.5664 - 3 \cdot 1.0472 = 9.4248 \approx 9.425$.

## Answer to Exercise 486 (on page 1699)

1. Answer: $A = e^{\pi/8}\left(e^{\pi/8} - 1\right)$

   Explanation: $A = \frac{1}{2}\int_{\pi/4}^{\pi/2}\left[e^{\theta/2}\right]\,,d\theta = \frac{1}{2}\cdot 2\left[e^{\theta/2}\right]_{\theta=\pi/4}^{\theta=\pi/2} = e^{\pi/4} - e^{\pi/8} = e^{\pi/8}\left(e^{\pi/8} - 1\right) \approx 0.712$

2. Answer: The area is $\frac{1}{2}\left[e^{\pi/2} - e^{\pi/4}\right] \approx 1.309$

   Explanation: $A = \frac{1}{2}\int_{\pi/4}^{\pi/2}\left[e^{\theta/2}\right]^2\,d\theta = \frac{1}{2}\int_{\pi/4}^{\pi/2} e^{\theta}\,d\theta = \frac{1}{2}e^{\theta}\big|_{\theta=\pi/4}^{\theta=\pi/2} = \frac{1}{2}\left[e^{\pi/2} - e^{\pi/4}\right] \approx 1.309$

3. Answer: $A = \frac{41}{4}\pi \approx 32.201$

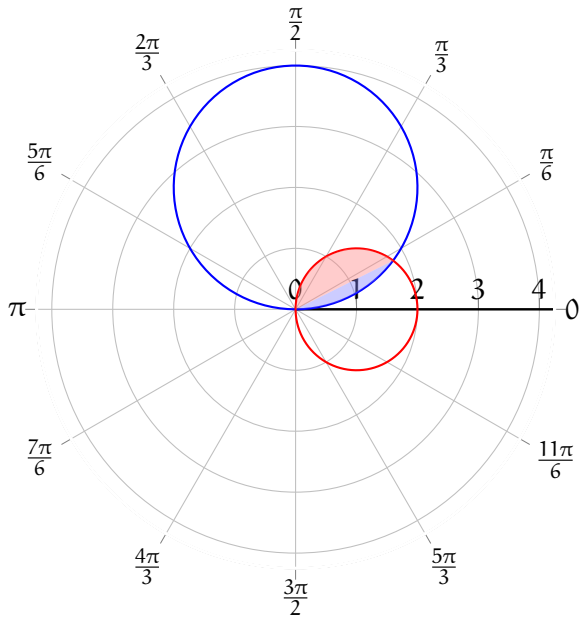
   Explanation: $A = \frac{1}{2}\int_{-\pi/2}^{\pi/2}[4 + 3\sin\theta]^2\,d\theta = \frac{1}{2}\int_{-\pi/2}^{\pi/2}\left[16 + 24\sin\theta + 9\sin^2\theta\right]\,d\theta = \int_{-\pi/2}^{\pi/2} 8\,d\theta + 12\int_{-\pi/2}^{\pi/2}\sin\theta\,d\theta + \frac{9}{2}\int_{-\pi/2}^{\pi/2}\sin^2\theta\,d\theta = [8\theta]_{\theta=-\pi/2}^{\theta=\pi/2} + 12\left[-\cos\theta\right]_{\theta=-\pi/2}^{\theta=\pi/2} + \frac{9}{2}\int_{-\pi/2}^{\pi/2}\frac{1-\cos 2\theta}{2}\,d\theta = 8\left[\left(\frac{\pi}{2}\right) - \left(\frac{-\pi}{2}\right)\right] + 12\left[\left(-\cos\frac{\pi}{2}\right) - \left(-\cos\frac{-\pi}{2}\right)\right] + \frac{9}{4}\int_{-\pi/2}^{\pi/2} 1\,d\theta - \frac{9}{4}\int_{-\pi/2}^{\pi/2}\cos 2\theta\,d\theta =$

$8\pi+12\,(0-0)+\frac{9}{4}\,[\theta]_{\theta=-\pi/2}^{\theta=\pi/2}-\frac{9}{4}\,\left[\frac{1}{2}\sin 2\theta\right]_{\theta=-\pi/2}^{\theta=\pi/2}=8\pi+\frac{9}{4}\,\left[\left(\frac{\pi}{2}\right)-\left(-\frac{\pi}{2}\right)\right]-\frac{9}{8}\,\left[\sin\left(2\cdot\frac{\pi}{2}\right)-\sin\left(2\cdot-\frac{\pi}{2}\right)\right]=$
$8\pi+\frac{9}{4}\pi-\frac{9}{8}\,[\sin(\pi)-\sin(-\pi)]=\frac{41}{4}\pi-\frac{9}{8}\,[0-(-0)]=\frac{41}{4}\pi\approx 32.201$

## Answer to Exercise 487 (on page 1700)

Answer: The area between the circles is approximately 0.96174.

Explanation: Examining the graph, we see that the region we are interested in is the area within $r=4\sin\theta$ from $\theta=0$ to $\theta=\theta_i$ plus the area within $r=2\cos\theta$ from $\theta=\theta_i$ to $\theta=\frac{\pi}{2}$, where $\theta_i$ is the angle where the two curves intersect. Examine the graph below to see why this is true.




Setting the equations equal to each other to find $\theta_i$:

$$4\sin\theta_i=2\cos\theta_i$$
$$\frac{\sin\theta_i}{\cos\theta_i}=\tan\theta_i=\frac{2}{4}$$
$$\theta_i=\arctan 1/2\approx 0.464$$

So, the total area between the circles is:

$$\frac{1}{2}\int_0^{\theta_i}[4\sin\theta]^2\,d\theta+\frac{1}{2}\int_{\theta_i}^{\pi/2}[2\cos\theta]^2\,d\theta$$

$$= 8\int_0^{\theta_i} \sin^2\theta\,d\theta + 2\int_{\theta_i}^{\pi/2} \cos^2\theta\,d\theta$$

$$= 8\int_0^{\theta_i} \frac{1}{2}(1-\cos 2\theta)\,d\theta + 2\int_{\theta_i}^{\pi/2} \frac{1}{2}(1+\cos 2\theta)\,d\theta$$

$$= 4\int_0^{\theta_i} (1-\cos 2\theta)\,d\theta + \int_{\theta_i}^{\pi/2} (1+\cos 2\theta)\,d\theta$$

$$= 4\left[\theta - \frac{1}{2}\sin 2\theta\right]_{\theta=0}^{\theta=\theta_i} + \left[\theta + \frac{1}{2}\sin 2\theta\right]_{\theta=\theta_i}^{\theta=\pi/2}$$

$$= 4\left[(\theta_i - 0) - \frac{1}{2}(\sin 2\theta_i - \sin 0)\right] + \left[\left(\frac{\pi}{2} - \theta_i\right) + \frac{1}{2}(\sin\pi - \sin 2\theta_i)\right]$$

$$= 4\left[\theta_i - \frac{1}{2}\sin 2\theta_i\right] + \frac{\pi}{2} - \theta_i - \frac{1}{2}\sin 2\theta_i$$

$$= 4\theta_i - 2\sin 2\theta_i + \frac{\pi}{2} - \theta_i - \frac{1}{2}\sin 2\theta_i = 3\theta_i - \frac{5}{2}\sin 2\theta_i + \frac{\pi}{2}$$

Substituting $\theta_i = \arctan 1/2 \approx 0.464$:

$$= 3(0.464) - \frac{5}{2}\sin 0.927 + \frac{\pi}{2} \approx 0.96174$$

## Answer to Exercise 488 (on page 1705)

1. 4200
2. Logically, we can say that the population will increase if it is below the carrying capacity (that is, $P < 4200$), but we can also prove it mathematically: $\frac{dP}{dt} < 0 \rightarrow 1.2P\left(1 - \frac{P}{4200}\right) < 0 \rightarrow P\left(1 - \frac{P}{4200}\right) < 0$. Since we are talking about population, we can assume that $P > 0$ and continue: $1 - \frac{P}{4200} < 0 \rightarrow 1 < \frac{P}{4200} \rightarrow 4200 < P$, which is the result we expected.
3. Similarly, we know the population should be decreasing when $P$ is greater than the carrying capacity of 4200.
4. The equilibrium solutions can be found by setting $\frac{dP}{dt} = 0$ and solving. The solutions are $P(t) = 0$ and $P(t) = 4200$.

## Answer to Exercise 489 (on page 1705)

Recall that logistic differential equations are of the form $\frac{dy}{dt} = ky(1 - \frac{y}{m})$ where $y$ is a function and $t$ is the independent variable. (e) is the only logistic differential equation, with $m = 1$.

## Answer to Exercise 490 (on page 1707)

$$\frac{dy}{dx}dx = \frac{3x^2}{2y + \sin y}dx$$

$$(2y + \sin y)(dy) = \frac{3x^2}{2y + \sin y}(2y + \sin y)(dx)$$

$$(2y + \sin y)dy = (3x^2)dx$$

$$\int 2y\, dy + \int \sin y\, dy = \int 3x^2\, dx$$

$$y^2 - \cos y = x^3 + C$$

## Answer to Exercise 491 (on page 1708)

1. Since $\frac{dB}{dt}$ depends only on B, we can use the given masses to find the rate of growth for each mass. $\frac{dB}{dt}(40) = \frac{1}{5}(100 - 40) = \frac{1}{5}(60) = 12$ and $\frac{dB}{dt}(70) = \frac{1}{5}(100 - 70) = \frac{1}{5}(30) = 6$. Since $\frac{dB}{dt}$ is greater when $B = 40$, the baby bird is gaining mass faster when it has a mass of 40 grams.

2. $\frac{d^2B}{dt^2} = \frac{d}{dt}\left(\frac{dB}{dt}\right) = \frac{d}{dt}\left[\frac{1}{5}(100 - B)\right] = \frac{1}{5}\left(-\frac{dB}{dt}\right) = \frac{-1}{5}\left[\frac{1}{5}(100 - B)\right] = -\frac{1}{25}(100 - B)$. For $20 < B < 100$, $\frac{d^2B}{dt^2} < 0$ and the graph of B should be concave down. The graph shown has a concave up portion, so it cannot represent $B(t)$.

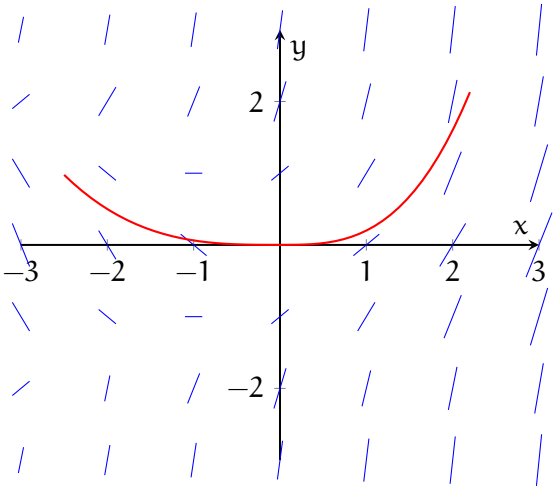
3. $\frac{dB}{dt} = \frac{1}{5}(100 - B) \to \frac{dB}{100-B} = \frac{1}{5}dt \to \int (100 - B)\, dB = \int \frac{1}{5}\, dt \to -\ln 100 - B = \frac{t}{5} + C \to e^{\frac{-t}{5}+C} = 100 - B \to ke^{\frac{-t}{5}} = 100 - B \to B(t) = 100 - ke^{\frac{-t}{5}}$. Setting $B(0) = 20$ to find k: $20 = 100 - ke^0 \to 20 = 100 - k \to k = 80$. So, the particular solution is $B(t) = 100 - 80e^{\frac{-t}{5}}$

## Answer to Exercise 492 (on page 1710)

(A). (a), (b), and (c) are all separable equations. But only the solution to A is linear

$(P(t) = 200t + C)$. (d) is logarithmic, or natural growth and (e) is also not linear.

## Answer to Exercise 493 (on page 1716)




## Answer to Exercise 494 (on page 1721)

Substituting the given values, we find that $(5)\frac{dQ}{dT} + \frac{1}{0.05}Q = 60$. Solving for $\frac{dQ}{dt}$:

$$(5)\frac{dQ}{dt} + (20)Q = 60$$

$$\frac{dQ}{dt} + 4Q = 12$$

$$\frac{dQ}{dt} = 12 - 4Q$$

We also know that $Q(0) = 0$. Using Euler's method with step size $h = 0.1$, $Q(0.1) \approx Q(0) + h\,[12 - 4Q(0)] = 0 + 0.1\,[12 - 4(0)] = 1.2$. And $Q(0.2) \approx Q(0.1) + h\,[12 - 4Q(0.1)] = 1.2 + 0.1\,[12 - 4(1.2)] = 1.92$. And $Q(0.3) \approx Q(0.2) + h\,[12 - 4Q(0.2)] = 1.92 + 0.1\,[12 - 4(1.92)] = 2.352$. And $Q(0.4) \approx Q(0.3) + h\,[12 - 4Q(0.3)] = 2.352 + 0.1\,[12 - 4(2.352)] = 2.6112$. And finally, $Q(0.5) \approx Q(0.4) + h\,[12 - 4Q(0.4)] = 2.6112 + 0.1\,[12 - 4(2.6112)] = 2.76672$. Because we are finding a charge, the unit it Coulombs (C), so our final answer is $Q(0.5) \approx 2.77C$.

## Answer to Exercise 503 (on page 1748)

We are given $x_0 = 1$ and $x_2 = 1.4$. Therefore we will use step size $h = \frac{1.4-1}{2} = \frac{0.4}{2} = 0.2$.

Taking $x_0 = 1$ and $y_0 = f(1) = 15$, we find $y_1$: $y_1 = y_0 + h \cdot f'(x_0) = 15 + 0.2 \cdot f'(1) = 15 + 0.2(8) = 15 + 1.6 = 16.6$. And then $y_2 = y_1 + h \cdot f'(x_1) = 16.6 + 0.2 \cdot f'(1.2) = 16.6 + 0.2(12) = 16.6 + 2.4 = 19$. Therefore, $f(1.4) \approx 19$.

## Answer to Exercise 496 (on page 1726)

1. $\frac{2}{3}, \frac{4}{5}, \frac{8}{7}, \frac{16}{9}, \frac{32}{11}$
2. 0, -1, 0, 1, 0
3. 1, 2, 7, 32, 157
4. 6, 3, 1, $\frac{1}{4}, \frac{1}{20}$

## Answer to Exercise 497 (on page 1729)

1. convergent, 5
2. divergent
3. convergent, 2
4. convergent, -1
5. divergent

## Answer to Exercise 498 (on page 1735)

Out principal is $P = 1500$ and the interest rate is $r = 0.06$. After $n$ years, your investment will be worth $a_n = 1500(1.06)^n$. For $n = 10$, your investment will be valued at $a_{10} = \$1500(1.06)^{10} = \$2686.27$ (that's over \$1000 in interest!). To determine if the sequence is convergent or divergent, we examine the limit as $n \to \infty$:

$$\lim_{n\to\infty} 1500(1.06)^n = 1500 \cdot \lim_{n\to\infty} (1.06)^n = 1500 \cdot \infty = \infty$$

The sequence is divergent.

## Answer to Exercise 499 (on page 1736)

The number of fish in the pond is:

$$P_n = P_{n-1}(0.95)^6 + 50$$

$$P_0 = 100$$

where $n$ is the number of 6-month periods that have passed. The four-year period is given by $1 \leq n \leq 8$. The amount lost to predation every 6 months is given by $P_{n-1}(1 - 0.95^6)$.

| $n$ | Fish Population | Lost to Predators |
|---|---|---|
| 0 | 100 | |
| 1 | 84 | 26 |
| 2 | 71 | 22 |
| 3 | 62 | 19 |
| 4 | 56 | 17 |
| 5 | 51 | 15 |
| 6 | 48 | 14 |
| 7 | 45 | 13 |
| 8 | 43 | 12 |

Adding up all the fish lost to predators, we find that over 4 years, the farmer loses 138 fish.

## Answer to Exercise 500 (on page 1746)

1. We need to identify $a$ and $r$. If we use the form $\sum_{n=1}^{\infty} ar^{n-1}$, then $a = 3$. To find the common ratio, we can evaluate $\frac{a_{n+1}}{a_n} = \frac{-4}{3}$. We can then write the series as $\sum_{n=1}^{\infty} 3\left(\frac{-4}{3}\right)^{n-1}$. In this case, $r = \frac{-4}{3}$ and $|r| \geq 1$, and therefore the series is divergent.
2. Following the process outlined above, we see that $a = 2$ and $r = \frac{1}{4}$. Therefore, the series is $\sum_{n=1}^{\infty} 2\left(\frac{1}{4}\right)^{n-1}$. Since $|r| < 1$, the series converges to $\frac{a}{1-r} = \frac{2}{1-1/4} = \frac{2 \cdot 4}{3} = \frac{8}{3}$
3. We need to rewrite the series into a standard from in order to identify $a$ and $r$:

$$\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n} = \sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4(4)^{n-1}} = \sum_{n=1}^{\infty} \frac{1}{4}\left(\frac{-3}{4}\right)^{n-1}$$

So $r = \frac{-3}{4}$ and $|r| < 1$. Therefore, the series converges to $\frac{1/4}{1-(-3/4)} = \frac{1}{4} \cdot \frac{4}{7} = \frac{1}{7}$

4. We need to rewrite the series into a standard from in order to identify $a$ and $r$:

$$\sum_{n=1}^{\infty} \frac{e^{2n}}{6^{n-1}} = \sum_{n=1}^{\infty} \frac{(e^2)^n}{6^{n-1}} = \sum_{n=1}^{\infty} \frac{(e^2)(e^2)^{n-1}}{6^{n-1}} = \sum_{n=1}^{\infty} e^2 \left(\frac{e^2}{6}\right)^{n-1}$$

Therefore, $r = \frac{e^2}{6} \approx 1.232$. Since $|r| > 1$, the series diverges.

## Answer to Exercise 501 (on page 1746)

We want to rewrite this as a geometric series of the form $\sum_{n=i}^{\infty} ar^{n-1}$, so we can use the fact that the sum of a convergent geometric series is $\frac{a}{1-r}$. $\sum_{n=0}^{\infty}(1+c)^{-n} = \sum_{n=0}^{\infty}\left(\frac{1}{1+c}\right)^n = \sum_{n=1}^{\infty}\left(\frac{1}{1+c}\right)^{n-1}$. This is a geometric series with $a = 1$ and $r = \frac{1}{1+c}$. So, the value of the series is $\frac{1}{1-\frac{1}{1+c}} = \frac{1}{\frac{c}{c+1}} = \frac{c+1}{c}$. Setting this equal to $\frac{5}{3}$ and solving for $c$, we find that $c = \frac{3}{2}$.

## Answer to Exercise 502 (on page 1746)

$-2 < p < 2$ Let's rewrite this geometric series into standard form: $\sum_{n=1}^{\infty}\left(\frac{p}{2}\right)^n = \sum_{n=1}^{\infty}\frac{p}{2}\left(\frac{p}{2}\right)^n$ which means $a = \frac{p}{2}$ and $r = \frac{p}{2}$. We know that geometric series converge if $|r| < 1$, so we set up an inequality and solve for $p$:

$$\left|\frac{p}{2}\right| < 1$$
$$-1 < \frac{p}{2} < 1$$
$$-2 < p < 2$$

## Answer to Exercise 503 (on page 1748)

1. Separating the terms, we see that $\sum_{n=1}^{\infty}\frac{n^2+1}{n^4} = \sum_{n=1}^{\infty}\left(\frac{n^2}{n^4}+\frac{1}{n^4}\right) = \sum_{n=1}^{\infty}\frac{1}{n^2} + \sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{\pi^2}{6} + \frac{\pi^4}{90}$
2. Notice that this series starts at $n = 2$. By the properties of series, we know that $\sum_{n=1}^{\infty} a_n = a_1 + \sum_{n=2}^{\infty} a_n$. Therefore, $\sum_{n=2}^{\infty}\frac{1}{n^2} = \sum_{n=1}^{\infty}\left(\frac{1}{n^2}\right) - \frac{1}{1^2} = \frac{\pi^2}{6} - 1$
3. We can begin by reindexing this series: $\sum_{n=3}^{\infty}\frac{1}{(n+1)^2} = \sum_{n=4}^{\infty}\frac{1}{n^2}$. Similar to the previous problem, we also know that $\sum_{n=4}^{\infty}\frac{1}{n^2} = \sum_{n=1}^{\infty}\left(\frac{1}{n^2}\right) - \left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}\right) = \frac{\pi^2}{6} - \frac{49}{36}$
4. We can rewrite this series as $\sum_{n=1}^{\infty}\left(\frac{3}{n}\right)^4 = \sum_{n=1}^{\infty}(3^4)\frac{1}{n^4} = 81\sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{81\pi^4}{90} = \frac{9\pi^4}{10}$

5. We can re-write the series as $\sum_{n=1}^{\infty}\left(\frac{4}{n^2}+\frac{3}{n^4}\right)=\sum_{n=1}^{\infty}\frac{4}{n^2}+\sum_{n=1}^{\infty}\frac{3}{n^4}=4\sum_{n=1}^{\infty}\frac{1}{n^2}+3\sum_{n=1}^{\infty}\frac{1}{n^4}=\frac{4\pi^2}{6}+\frac{3\pi^4}{90}=\frac{2\pi^2}{3}+\frac{\pi^4}{30}$

## Answer to Exercise 504 (on page 1749)

This is a p-series where $p=2k$. We know that p-series converge for $p>1$: $2k>1\rightarrow k>\frac{1}{2}$.

## Answer to Exercise 505 (on page 1750)

1. The series is convergent if $\left|\frac{(-1)^{n+1}3(n+1)}{4(n+1)-1}\right|<\left|\frac{(-1)^n3n}{4n-1}\right|$ if $\frac{3n+3}{4n+4-1}<\frac{3n}{4n-1}$ and if $\frac{3n+3}{4n+3}<\frac{3n}{4n-1}$ if $(3n+3)(4n-1)<(3n)(4n+3)$ if $12n^2+12n-3n-3<12n^2+9n$ if $-3<0$ which is true. Therefore, $\sum_{n=1}^{\infty}\frac{(-1)^n3n}{4n-1}$ is convergent.

2. The series is convergent if $\left|(-1)^{n+1+1}\frac{(n+1)^2}{(n+1)^3+1}\right|<\left|(-1)^{n+1}\frac{n^2}{n^3+1}\right|$, which is true if $\frac{(n+1)^2}{(n+1)^3+1}<\frac{n^2}{n^3+1}$ if $(n+1)^2(n^3+1)<(n^2)((n+1)^3+1)$ if $(n^2+2n+1)(n^3)<(n^2)(n^3+3n^2+3n+1+1)$ if $n^5+2n^4+n^3<n^5+3n^4+3n^3+2n^2$, which is true for all $n\geq 1$. Therefore, $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{n^3+1}$ is convergent.

3. The series is convergent if $\left|(-1)^{n-1+1}e^{2/(n+1)}\right|<\left|(-1)^{n-1}e^{2/n}\right|$, which is true if $e^{2/(n+1)}<e^{2/n}$, which is true if $\frac{2}{n+1}<\frac{2}{n}$ which is true for all $n\geq 1$. Therefore, $\sum_{n=1}^{\infty}(-1)^{n-1}e^{2/n}$ is convergent.

## Answer to Exercise 506 (on page 1755)

1. The function $2x^{-3}$ is positive and decreasing for $x\in[1,\infty)$. $\int_1^{\infty}2x^{-3}\,dx=\lim_{t\rightarrow\infty}\int_1^t 2x^{-3}\,dx=\lim_{t\rightarrow\infty}\left[-x^{-2}\right]_{x=1}^{t}=\lim_{t\rightarrow\infty}(-t^{-2})--(1)^{-2}=0+1=1$. Since the integral $\int_1^{\infty}2x^{-3}\,dx$ converges, the series $\sum_{n=1}^{\infty}2n^{-3}$ is also convergent.

2. The function $\frac{5}{3x+1}$ is positive and decreasing for $x\in[1,\infty)$. $\int_1^{\infty}\frac{5}{3x-1}\,dx=\lim_{t\rightarrow\infty}\int_1^t\frac{5}{3x-1}\,dx$ Using u-substitution to evaluate the integral, we set $u=3x-1$ and find that $du=3dx\rightarrow dx=\frac{du}{3}$. Substituting, $\int_1^t\frac{5}{3x-1}\,dx=\int_{x=1}^{x=t}\frac{5}{3}\frac{1}{u}\,du$. Evaluating the integral, $\int_{x=1}^{x=t}\frac{5}{3}\frac{1}{u}\,du=\frac{5}{3}\ln u|_{x=1}^{x=t}=\frac{5}{3}\ln 3x+1|_1^t$. Substituting this back into the limit, $\int_1^{\infty}\frac{5}{3x-1}\,dx=\lim_{t\rightarrow\infty}\frac{5}{3}\ln 3x+1|_1^t=\lim_{t\rightarrow\infty}[\frac{5}{3}\ln 3t+1]-\frac{5}{3}\ln 4=\infty-\frac{5}{3}\ln 4=\infty$. Therefore, the integral $\int_1^{\infty}\frac{5}{3x-1}\,dx$ is divergent and so is the series $\sum_{n=1}^{\infty}\frac{5}{3n-1}$.

3. The function $\frac{x}{3x^2+1}$ is positive and decreasing for $x\in[1,\infty)$. $\int_1^{\infty}\frac{x}{3x^2+1}\,dx=\lim_{t\rightarrow\infty}\int_1^t\frac{x}{3x^2+1}\,dx$.

Applying the substitution $u = 3x^2+1$ and $\frac{du}{6} = x\,dx$, we see that $\lim_{t\to\infty}\int_1^t \frac{x}{3x^2+1}\,dx = \lim_{t\to\infty}\int_{x=1}^{x=t}\frac{1}{6u}\,du = \lim_{t\to\infty}\frac{1}{6}\ln u|_{x=1}^{x=t} = \lim_{t\to\infty}\frac{1}{6}\ln 3x^2+1|_1^t = \lim_{t\to\infty}\left[\frac{1}{6}\ln 3t^2+1\right] - \frac{1}{6}\ln 4 = \infty$. Therefore, the integral $\int_1^\infty \frac{x}{3x^2+1}\,dx$ is divergent, and so is the series $\sum_{n=1}^\infty \frac{n}{3n^2+1}$.

## Answer to Exercise 507 (on page 1756)

1. If $p \le 0$, then $\lim_{n\to\infty}\frac{1}{n^p} \neq 0$, and the series fails the Test for Divergence. Therefore, a p-series is divergent if $p \le 0$.
2. If $p > 0$, then $f(x) = \frac{1}{x^p}$ is continuous, positive, and decreasing on the interval $x \in [1,\infty)$, and we can apply the integral test. So, we want to know, when is $\int_1^\infty \frac{1}{x^p}\,dx$ convergent? When $p = 1$, $\int_1^\infty \frac{1}{x^p}\,dx = \ln x|_{x=1}^{x=\infty} = \lim_{t\to\infty}\ln t - \ln 1 = \infty$ and the integral and p-series are both divergent.
3. What about when $0 < p < 1$? In this case, the integral $\int_1^\infty \frac{1}{x^p}\,dx = \lim_{t\to\infty}\int_1^t x^{-p}\,dx = \lim_{t\to\infty}\frac{1}{1-p}x^{1-p}|_{x=1}^{x=t} = \lim_{t\to\infty}\frac{1}{1-p}\frac{1}{x^{p-1}} = \left(\frac{1}{1-p}\right)\left[\lim_{t\to\infty}\left(\frac{1}{t^{p-1}}\right) - 1\right]$. When $0 < p < 1$, then $1-p > 0$ is positive and $\lim_{t\to\infty}\frac{1}{t^{p-1}} = \lim_{t\to\infty} t^{1-p} = \infty$ and the integral diverges. Therefore, p-series are divergent for $0 < p < 1$.
4. When $p > 1$, then $\int_1^\infty \frac{1}{x^p}\,dx = \left(\frac{1}{1-p}\right)\left[\lim_{t\to\infty}\left(\frac{1}{t^{p-1}}\right) - 1\right]$. When $p > 1$, $p-1 > 0$ and $\lim_{t\to\infty}\frac{1}{t^{p-1}} = 0$. Therefore, $\int_1^\infty \frac{1}{x^p}\,dx$ converges to $\frac{1}{p-1}$ when $p > 1$, and therefore the p-series is convergent when $p > 1$.

## Answer to Exercise 508 (on page 1759)

1. $s_{10} = \frac{1}{1^4} + \frac{1}{2^4} + \cdots + \frac{1}{10^4} \approx 1.082037$.
2. $R_{10} \le \int_{10}^\infty \frac{1}{x^4}\,dx = \frac{-1}{3x^3}|_{x=10}^\infty = \lim_{x\to\infty}\frac{-1}{3x^3} - \frac{-1}{3\cdot 10^3} = \frac{1}{3000} = 0.000333$. Therefore, the error is less than 0.000333.
3. Given $s_{10} \approx 1.082037$, we can say that $1.082037 + \int_{n+1}^\infty \frac{1}{x^4}\,dx \le s \le 1.082037 + \int_n^\infty \frac{1}{x^4}\,dx$. Using a calculator to evaluate each integral, we see that: $1.082037 + 0.000250 \le s \le 1.082037 + 0.000333$ and therefore the sum is between 1.082287 and 1.082370.
4. Writing the actual value as a decimal, $\frac{\pi^4}{90} \approx 1.082323$, which is in the estimate window from the previous part.
5. We are looking for an $n$ such that $\int_n^\infty \frac{1}{x^4}\,dx \le 0.00001$. $\lim_{x\to\infty}\frac{-1}{3x^3} - \frac{-1}{3n^3} = \frac{1}{3n^3} \le 0.00001$. $100,000 \le 3n^3$. $33,333.33 \le n^3$. $32.183 \le n$. Since $n$ must be an integer,

$n = 33$ gives $R_n \leq 0.00001$.

## Answer to Exercise 509 (on page 1763)

1. This is similar to $\sum_{n=1}^{\infty} \frac{1}{n}$, which is divergent. Unfortunately, $\frac{1}{n} > \frac{1}{\sqrt{n^2+1}}$, so we can't use the direct comparison test. We will try the limit comparison test:

$$\lim_{n\to\infty} \left( \frac{\frac{1}{\sqrt{n^2+1}}}{\frac{1}{n}} \right) = \lim_{n\to\infty} \left( \frac{1}{\sqrt{n^2+1}} \cdot \frac{n}{1} \right) = \lim_{n\to\infty} \frac{n}{\sqrt{n^2+1}} = \lim_{n\to\infty} \frac{1}{\sqrt{1+1/n^2}} = \frac{1}{1+0} = 1 > 0$$

   Therefore, since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, so does $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+1}}$.

2. This series is similar to the convergent geometric series $\sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n$. Given that:

$$\left(\frac{9}{10}\right) = \frac{9^n}{10^n} < \frac{9^n}{3+10^n}$$

   Since $\frac{9^n}{3+10^n} < \left(\frac{9}{10}\right)^n$ and $\sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n$ is convergent, by the direct comparison test, $\sum_{n=1}^{\infty} \frac{9^n}{3+10^n}$ is also convergent.

3. We can compare this to the convergent p-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Noting that $\sin^2 n \leq 1$:

$$\frac{n \sin^2 n}{1+n^3} < \frac{n \sin^2 n}{n^3} \leq \frac{n}{n^3} = \frac{1}{n^2}$$

   Because $\frac{n \sin^2 n}{1+n^3} \leq \frac{1}{n^2}$ for all $n \geq 1$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent, we can state by the direct comparison test that $\sum_{n=1}^{\infty} \frac{n \sin^2 n}{1+n^3}$ is also convergent.

## Answer to Exercise 510 (on page 1764)

We can write the series as $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$. Since $n$ is real, we know that $n^2 > 0$ and we can say that $\sum_{n=1}^{\infty} \left|\frac{\cos n}{n^2}\right| = \sum_{n=1}^{\infty} \frac{|\cos n|}{n^2}$. Additionally, $|\cos n| \leq 1$ for all $n$, and therefore $\frac{|\cos n|}{n^2} \leq \frac{1}{n^2}$. We know the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. And since we have shown that $\sum_{n=1}^{\infty} \frac{|\cos n|}{n^2} \leq \sum_{n=1}^{\infty} \frac{1}{n^2}$, by the comparison test $\sum_{n=1}^{\infty} \frac{|\cos n|}{n^2}$ is convergent. Therefore, $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ is absolutely convergent and therefore convergent.

## Answer to Exercise 511 (on page 1765)

1. Conditionally Convergent. $\sum_{n=1}^{\infty}\left|\frac{(-1)^n}{3n+2}\right| = \sum_{n=1}^{\infty}\frac{1}{3n+2}$ Applying the integral test to this sum: $\int_1^{\infty}\frac{1}{3x+2}\,dx = \lim_{t\to\infty}\int_1^t\frac{1}{3x+2}\,dx = \left[\frac{1}{3}\ln 3x+2\right]_{x=1}^{t} = \lim_{t\to\infty}\left[\ln 3x+2\right] - \ln 3(1)-2 = \infty - 0 = \infty$. Since $\int_1^{\infty}\frac{1}{3x+2}\,dx$ is divergent, $\sum_{n=1}^{\infty}\frac{1}{3n+2}$ is divergent, and $\sum_{n=1}^{\infty}\frac{(-1)^n}{3n+2}$ is conditionally convergent.

2. Absolutely Convergent. $\sum_{n=1}^{\infty}\left|\frac{\sin n}{4^n}\right| \le \sum_{n=1}^{\infty}\frac{1}{4^n}$. Applying the integral test to $\sum_{n=1}^{\infty}\frac{1}{4^n}$: $\int_1^{\infty}\frac{1}{4^x}\,dx = \lim_{t\to\infty}\frac{-1}{4^x\ln 4}\Big|_{x=1}^{t} = \lim_{t\to\infty}\left[\frac{-1}{4^t\ln 4}\right] - \frac{-1}{4^1\ln 4} = 0 + \frac{1}{4\ln 4} = \frac{1}{4\ln 4}$. Since $\int_1^{\infty}\frac{1}{4^x}\,dx$ is convergent, the series $\sum_{n=1}^{\infty}\frac{1}{4^n}$ is also convergent. And since $\sum_{n=1}^{\infty}\left|\frac{\sin n}{4^n}\right| \le \sum_{n=1}^{\infty}\frac{1}{4^n}$, $\sum_{n=1}^{\infty}\left|\frac{\sin n}{4^n}\right|$ is also convergent, which shows that $\sum_{n=1}^{\infty}\frac{\sin n}{4^n}$ is absolutely convergent.

3. Conditionally Convergent. We are asking if the series $\sum_{n=1}^{\infty}\left|(-1)^{n-1}\frac{2n}{n^2+4}\right|$ is convergent. $\sum_{n=1}^{\infty}\left|(-1)^{n-1}\frac{2n}{n^2+4}\right| = \sum_{n=1}^{\infty}\frac{2n}{n^2+4}$ We will apply the Limit Comparison test and compare this series to the known, divergent series $\sum_{n=1}^{\infty}\frac{1}{n}$. $\lim_{n\to\infty}\frac{\frac{2n}{n^2+4}}{\frac{1}{n}} = \lim_{n\to\infty}\frac{2n^2}{n^2+4} = 2 > 0$. Therefore, by the Limit Comparison test, $\sum_{n=1}^{\infty}\left|(-1)^{n-1}\frac{2n}{n^2+4}\right|$ is divergent AND $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{2n}{n^2+4}$ is conditionally convergent.

## Answer to Exercise 512 (on page 1766)

Series 1 and 3 converge

1. We apply the ratio test: $\lim_{n\to\infty}\left|\frac{\frac{8^{n+1}}{(n+1)!}}{\frac{8^n}{n!}}\right| = \lim_{n\to\infty}\frac{8\cdot 8^n}{(n+1)(n!)}\cdot\frac{n!}{8^n} = \lim_{n\to\infty}\frac{8}{n+1} = 0$. Therefore, the series converges.

2. We apply the ratio test: $\lim_{n\to\infty}\left|\frac{\frac{(n+1)!}{(n+1)^{100}}}{\frac{n!}{n^{100}}}\right| = \lim_{n\to\infty}\frac{(n+1)n!}{(n+1)^{100}}\cdot\frac{n^{100}}{n!} = \lim_{n\to\infty}\left(\frac{n}{n+1}\right)^{100}\cdot(n+1) = \lim_{n\to\infty}\frac{n^{100}}{(n+1)^{99}} = \infty$. Therefore, the series diverges.

3. We apply the comparison test: $\frac{n+1}{(n)(n+2)(n+3)} = \frac{n}{(n)(n+2)(n+3)} + \frac{1}{(n)(n+2)(n+3)} = \frac{1}{(n+2)(n+3)} + \frac{1}{(n)(n+2)(n+3)} = \frac{1}{n^2+5n+6} + \frac{1}{n^3+5n^2+6n} \le \frac{1}{n^2} + \frac{1}{n^3}$. The series $\sum_{n=1}^{\infty}\frac{1}{n^2}$ and $\sum_{n=1}^{\infty}\frac{1}{n^3}$ are both convergent, because they are p-series with $p > 1$. Having established that $\sum_{n=1}^{\infty}\frac{n+1}{(n)(n+2)(n+3)} \le \sum_{n=1}^{\infty}\frac{1}{n^2}+\frac{1}{n^3}$ and that $\sum_{n=1}^{\infty}\frac{1}{n^2}+\frac{1}{n^3}$ converges, by the comparison test we can state that $\sum_{n=1}^{\infty}\frac{n+1}{(n)(n+2)(n+3)}$ converges.

## Answer to Exercise 513 (on page 1767)

1. This is not necessarily true. For a convergent series, the result of the ratio test is $L < 1$, so the limit could be $\neq 0$.
2. This is not necessarily true. Consider the geometric series $\sum_{n=1}^{\infty} 2(\frac{1}{2})^{n-1}$. This series is convergent because the common ratio is less than one, but the first term is $2(\frac{1}{2})^0 = 2 > 1$.
3. This is not necessarily true. Again, consider the geometric series $\sum_{n=1}^{\infty} 2(\frac{1}{2})^{n-1}$, which converges to $4 \neq 0$.
4. This is not necessarily true. Consider the p-series $\sum_{n=1}^{\infty} \frac{1}{n^4}$. Then the series $\sum_{n=1}^{\infty} n\frac{1}{n^4} = \sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent.
5. This must be true. By the comparison test, $\sum_{n=1}^{\infty} \frac{a_n}{n} \leq \sum_{n=1}^{\infty} a_n$. Since $\sum_{n=1}^{\infty} a_n$ converges, so much $\sum_{n=1}^{\infty} na_n$.

## Answer to Exercise 514 (on page 1768)

1. $\lim_{n=1}^{\infty} \sqrt[n]{\left|\left(\frac{3n^2+1}{n^2-4}\right)^n\right|} = \lim_{n=1}^{\infty} \frac{3n^2+1}{n^2-4} = 3 > 1$. Therefore, the series $\sum_{n=1}^{\infty} \left(\frac{3n^2+1}{n^2-4}\right)^n$ is divergent.
2. $\lim_{n\to\infty} \sqrt[n]{\left|\frac{(-1)^n}{(\ln n)^n}\right|} = \lim_{n\to\infty} \sqrt[n]{\frac{1}{(\ln n)^n}} = \lim_{n\to\infty} \frac{1}{\ln n} = \frac{1}{\infty} = 0 < 1$. Therefore, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(\ln n)^n}$ is convergent.
3. $\lim_{n\to\infty} \sqrt[n]{\left|\left(1+\frac{1}{n}\right)^{n^2}\right|} = \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e > 1$. Therefore, $\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2}$ is divergent.

## Answer to Exercise 515 (on page 1770)

1. Divergent. Since there is a constant to the $n^{\text{th}}$ power and an algebraic function of $n$, we will try the Ratio Test. $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{e^{n+1}}{(n+1)^2} \cdot \frac{n^2}{e^n} = \lim_{n\to\infty} \frac{e^n \cdot e}{e^n} \cdot \left(\frac{n}{n+1}\right)^2 = \lim_{n\to\infty} e \cdot \left(\frac{n}{n+1}\right)^2 = e \cdot 1^2 = e > 1$. Therefore, $\sum_{n=1}^{\infty} \frac{e^n}{n^2}$ is divergent.
2. Convergent. Since there is a factorial, we will try the Ratio Test. $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{3^{n+1}(n+1)^2}{(n+1)!} \cdot \frac{n!}{3^n n^2} = \lim_{n\to\infty} \frac{3 \cdot 3^n}{3^n} \cdot \frac{n!}{(n+1)n!} \cdot \left(\frac{n+1}{n}\right)^2 = \lim_{n\to\infty} \frac{3(n+1)^2}{(n+1)n^2} = \lim_{n\to\infty} \frac{3(n+1)}{n^2} = 0 < 1$. Therefore, the series $\sum_{n=1}^{\infty} \frac{3^n n^2}{n!}$ is convergent.
3. Divergent. Since $\int_2^{\infty} \frac{1}{x\sqrt{\ln x}}\,dx$ can be integrated, we will apply the integral test.

$\int_2^\infty \frac{1}{x\sqrt{\ln x}}\,dx = \lim_{t\to\infty}\int_2^t \frac{1}{x\sqrt{\ln x}}\,dx$. Setting $u = \ln x$, then $du = \frac{dx}{x}$ and $\frac{1}{x\sqrt{\ln x}}dx = \frac{1}{\sqrt{u}}du$. Then we can say that $\int_2^\infty \frac{1}{x\sqrt{\ln x}}\,dx = \lim_{t\to\infty}\int_{x=2}^{x=t}\frac{1}{\sqrt{u}}\,du = \lim_{t\to\infty}\left(\frac{-1}{2}\right)\sqrt{u}\Big|_{x=2}^{x=t} = \lim_{t\to\infty}\left(\frac{-1}{2}\right)\sqrt{\ln x}\Big|_2^t = \left(\frac{-1}{2}\right)\lim_{t\to\infty}\sqrt{\ln t} - \left(\frac{-1}{2}\right)\sqrt{\ln 2} = \infty$. Since the integral diverges, so does the series.

4. Convergent. Since this series has terms to the $n^{th}$ power, we will try the Root Test. $\lim_{n\to\infty}\sqrt[n]{\left|\left(\frac{n}{n+1}\right)^{n^2}\right|} = \lim_{n\to\infty}\sqrt[n]{\left(\frac{n}{n+1}\right)^{n+2}} = \lim_{n\to\infty}\left(\frac{n}{n+1}\right)^{\frac{n^2}{n}} = \lim_{n\to\infty}\left(\frac{n}{n+1}\right)^n = \lim_{n\to\infty}\left(\frac{1}{1+\frac{1}{n}}\right)^n = \frac{1}{\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n} = \frac{1}{e} < 1$ Therefore, by the root test, the series is convergent.

5. Convergent. This series also has terms raised to the $n^{th}$ power, we will try the Root Test again. $\lim_{n\to\infty}\sqrt[n]{\left|\left(\sqrt[n]{2}-1\right)^n\right|} = \lim_{n\to\infty}\sqrt[n]{\left(\sqrt[n]{2}-1\right)^n} = \lim_{n\to\infty}\left(\sqrt[n]{2}-1\right)^{n/n} = \lim_{n\to\infty}\left(\sqrt[n]{2}-1\right) = \lim_{n\to\infty} 2^{1/n} - 1 = 1 - 1 = 0 < 1$. Therefore, the series converges.

## Answer to Exercise 516 (on page 1777)

Since this sum has terms to the $n^{th}$ power, we will apply the Root Test, which states a series is convergent if $\lim_{n\to\infty}\sqrt[n]{|a_n|} < 1$.

$$\lim_{n\to\infty}\sqrt[n]{\left|\frac{(x-4)^{2n}}{3^n}\right|} < 1$$

$$\lim_{n\to\infty}\left|\frac{(x-4)^{2n/n}}{3^{n/n}}\right| < 1$$

$$\lim_{n\to\infty}\frac{\left|(x-4)^2\right|}{3} < 1$$

$$(x-4)^2 < 3$$

$$|x-4| < \sqrt{3}$$

Therefore, the radius of convergence is $\sqrt{3}$.

## Answer to Exercise 517 (on page 1777)

We see that the series is alternating, so we know it involves $(-1)^n$ (we will begin indexing at $n = 0$). The powers of $x$ are given by $x^{2n+1}$ and the denominators are given by $2n+3$. Therefore, the sum in sigma notation is $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+3}$. Applying the ratio test, the

series is convergent when:

$$\lim_{n\to\infty}\left|\frac{(-1)^{n+1}x^{2(n+1)+1}}{2(n+1)+3}\frac{2n+3}{(-1)^n x^{2n+1}}\right|<1$$

$$\lim_{n\to\infty}\left|\frac{x^{2n+3}}{2n+5}\frac{2n+3}{x^{2n+1}}\right|<1$$

$$\left|x^2\right|\lim_{n\to\infty}\frac{2n+3}{2n+5}<1$$

$$\left|x^2\right|<1$$

$$|x|<1$$

So, we know that the series is convergent on the open interval $x\in(-1,1)$. We check the endpoints, $x=-1,1$ for convergence.

$$\sum_{n=0}^{\infty}\frac{(-1)^n(-1)^{2n+1}}{2n+3}=\sum_{n=0}^{\infty}\frac{(-1)^{3n+1}}{2n+3}$$

When $x=-1$, the series is an alternating series such that $|a_{n+1}|<|a_n|$ and $\lim_{n\to\infty}a_n=0$. Therefore, the series converges for $x=-1$.

$$\sum_{n=1}^{\infty}(-1)^n\frac{(1)^{2n+1}}{2n+3}=\sum_{n=1}^{\infty}\frac{(-1)^n}{2n+3}$$

which is also an alternating series such that $|a_{n+1}|<|a_n|$ and $\lim_{n\to\infty}a_n=0$. Therefore, the series converges for $x=1$ and the interval of convergence is $x\in[-1,1]$, which can also be written as $-1\le x\le 1$.

## Answer to Exercise 518 (on page 1780)

Recall that $\frac{d}{dx}\arctan x=\frac{1}{1+x^2}$. Replacing $x$ with $-x^2$, we see that $\frac{1}{1+x^2}=\frac{1}{1-(-x^2)}=\sum_{n=0}^{\infty}\left[-x^2\right]^n=\sum_{n=0}^{\infty}(-1)^n x^{2n}$. Then we can also say that $\arctan x=\int\frac{1}{1+x^2}\,dx=\int\left[\sum_{n=0}^{\infty}(-1)^n x^{2n}\right]\,dx$. Evaluating the integral, $\int\left[\sum_{n=0}^{\infty}(-1)^n x^{2n}\right]\,dx=\sum_{n=0}^{\infty}(-1)^n\int x^{2n}\,dx=C+\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1}$. Knowing that $\arctan 0=0$, we find that $C=0$ and $\arctan x=\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1}$.

## Answer to Exercise 519 (on page 1790)

(**a**) The fraction of acceptable parts is

$$P(6.72\le X\le 12.28).$$

(**b**) Since `NORM.DIST(x, mean, sd, TRUE)` returns $P(X \leq x)$, we compute

$$P(6.72 \leq X \leq 12.28) = P(X \leq 12.28) - P(X \leq 6.72).$$

In Excel, enter:

```
=NORM.DIST(12.28,10,2,TRUE) - NORM.DIST(6.72,10,2,TRUE).
```

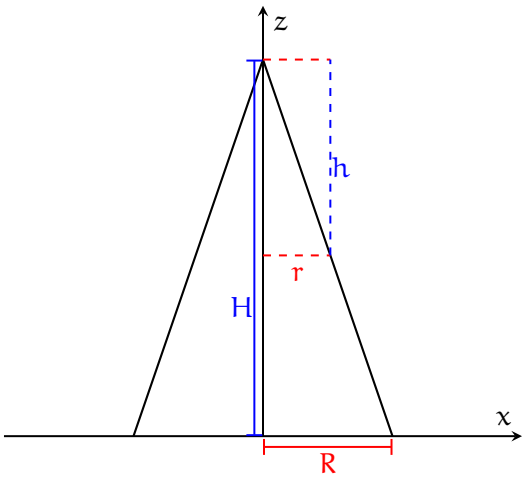
(**c**) Excel returns approximately 0.8224, so the percentage acceptable is

$$82.24\%.$$

## Answer to Exercise 520 (on page 1801)

Imagine a side view of the cone (see figure below), an isosceles triangle with height $H$ and base $2R$. If we take horizontal cross-sections, then each cross-section is a circle $h$ from the top with a radius $r$. Because the triangles are similar (FIXME: better wording/explanation here), we also know that $\frac{H}{h} = \frac{R}{r}$. Therefore, we can define $r$ in terms of $h$: $r = \frac{hR}{H}$ and the volume of each subsequent cross-section is $\pi r^2 dh = \pi \frac{h^2 R^2}{H^2} dh$. We start with $h = 0$ and end with $h = H$:

$$\begin{aligned} V_{cone} &= \int_0^H \pi \frac{h^2 R^2}{H^2}\, dh = \pi \frac{R^2}{H^2} \int_0^H h^2\, dh \\ &= \pi \frac{R^2}{H^2} \left[\frac{1}{3} h^3\right]_{h=0}^{h=H} = \pi \frac{R^2}{3H^2} \left[H^3 - 0^3\right] \\ &= \pi \frac{R^2}{3H^2} H^3 = \frac{\pi}{3} R^2 H \end{aligned}$$



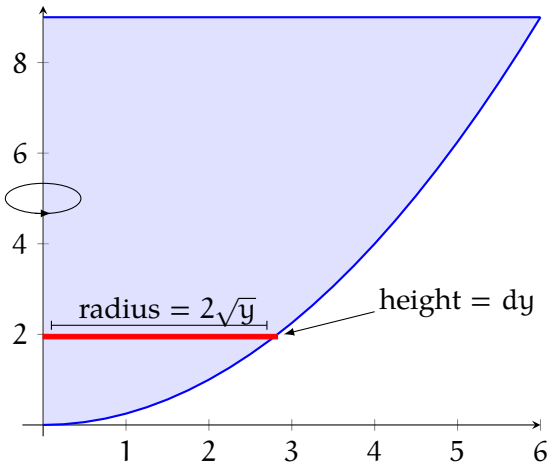
## Answer to Exercise 521 (on page 1803)

If we are rotating about the $y$ axis, we should make our slices horizontal, so their width is $dy$ (see graph below). Then, the volume of each cylinder is given by $V = \pi r^2 dy$ and the total volume is given by:

$$V = \int_0^9 \pi \left[2\sqrt{y}\right]^2 \, dy$$

$$V = 4\pi \int_0^9 y \, dy = 2\pi y^2 \Big|_{y=0}^{y=9}$$

$$V = 2\pi (9)^2 = 162\pi$$




## Answer to Exercise 522 (on page 1804)

Since the graph is rotated around the $x$-axis, we will take vertical slices with width $dx$, and rotate them to make cylinders with radius $f(x)$ and height $dx$. The volume of each egg is given by:

$$\int_{-1}^{1} \pi \left[f(x)\right]^2 \, dx$$

To determine our limits of integration, we note that $\sqrt{1-x^2} = 0$ (and therefore, $f(x) = 0$) when $x = \pm 1$.

For the chicken:

$$V_{\text{chickenegg}} = \pi \int_{-1}^{1} \left[\left(-0.02x^3 + 0.03x^2 + 0.12x + 0.454\right)\sqrt{1-x^2}\right]^2 \, dx$$

For the mallard duck:

$$V_{duckegg} = \pi \int_{-1}^{1} \left[ \left( -0.06x^3 + 0.04x^2 + 0.1x + 0.54 \right) \sqrt{1 - x^2} \right]^2 \, dx$$

Using a calculator, we find that $V_{chickenegg} \approx 0.897$ and $V_{duckegg} \approx 1.263$. Therefore, mallard ducks lay larger eggs than chickens do.

## Answer to Exercise ?? (on page 1806)

First, since we are revolving around the $y$-axis, we know our slices will have width $dy$. We will rewrite the functions as $x$ in terms of $y$:

$$x = \sqrt{y}$$

$$x = \frac{y^2}{4}$$

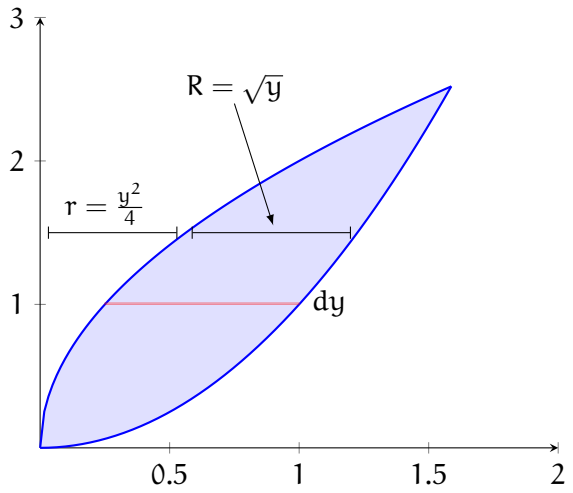
Setting them equal to each other to find the $y$-value at which they intercept:

$$\sqrt{y} = \frac{y^2}{4}$$

$$4 = \frac{y^2}{\sqrt{y}} = y^{3/2}$$

$$y = \sqrt[3]{4^2} = 2\sqrt[3]{2}$$

Examining a graph (shown below), we see that the outer radius is $x = \sqrt{y}$ and the inner radius is $x = \frac{y^2}{4}$.



So, the total volume of the solid of revolution is given by:

$$V = \pi \int_0^{2\sqrt[3]{2}} (\sqrt{y})^2 - \left(\frac{y^2}{4}\right)^2 \, dy$$

$$V = \pi \int_0^{2\sqrt[3]{2}} \left[y - \frac{y^4}{16}\right] \, dy$$

$$V = \pi \left[\frac{1}{2}y^2 - \frac{1}{80}y^5\right]_{y=0}^{y=2\sqrt[3]{2}}$$

$$V = \pi \left[\frac{6}{5}2^{2/3}\right] \approx 5.9844$$

## Answer to Exercise 524 (on page 1808)

If each cross section is a square, then the volume of each cross section is given by $s^2 dx$, where $s$ is the side length of the square. Since the side length is equal to the distance between the graph of $y$ and the $x$-axis, we can see that $s = y = \ln(3 - x)$. And, therefore, the total volume of all the cross sections is given by $\int_0^2 [\ln(3-x)]^2 \, dx$. Using a calculator, this integral evaluates to $\approx 1.029$.

## Answer to Exercise 525 (on page 1809)

Since the cross-sections are perpendicular to the $x$-axis, they will have width $dx$ and we will integrate across the domain of the ellipse. Setting $y = 0$ to find the domain of the ellipse:

$$9x^2 = 25 \to x^2 = \frac{25}{9} \to x = \pm\frac{5}{3}$$

A right isosceles triangle with hypotenuse $h$ has area $\frac{1}{4}h^2$. In this case, each triangle's hypotenuse is given by the distance between the top and bottom of the ellipse. The top of the ellipse if defined by $y = \frac{1}{4}\sqrt{25 - 9x^2}$ and the bottom by $y = -\frac{1}{4}\sqrt{25 - 9x^2}$. Therefore, the length of each hypotenuse is $\frac{1}{2}\sqrt{25 - 9x^2}$.

Then, each cross-section has a total volume of $\frac{1}{4}h^2 dx = \frac{1}{4}\left(\frac{1}{2}\sqrt{25 - 9x^2}\right)^2 dx$ and the volume of the solid is:

$$V_{\text{solid}} = \int_{-5/3}^{5/3} \frac{1}{4}\left(\frac{1}{2}\sqrt{25 - 9x^2}\right)^2 \, dx$$

$$= \frac{1}{4}\int_{-5/3}^{5/3} \frac{1}{4}\left(25 - 9x^2\right)\,dx$$

$$= \frac{1}{6}\int_{-5/3}^{5/3} \left(25 - 9x^2\right)\,dx = \frac{1}{16}\left[25x - 3x^3\right]_{x=-5/3}^{x=5/3}$$

$$= \frac{1}{16}\left[\left(25\left(\frac{5}{3}\right) - 25\left(\frac{-5}{3}\right)\right) - \left(3\left(\frac{5}{3}\right)^3 - 3\left(\frac{-5}{3}\right)^3\right)\right]$$

$$= \frac{1}{16}\left[\frac{250}{3} - \left(\frac{375}{27} + \frac{375}{27}\right)\right] = \frac{1}{16}\left[\frac{250}{3} - \frac{250}{9}\right] = \frac{1}{16}\left[\frac{750}{9} - \frac{250}{9}\right] = \frac{1}{16}\left[\frac{500}{9}\right] = \frac{125}{36}$$

## Answer to Exercise ?? (on page 1816)

We have already shown that $\int_0^3 \int_1^2 xy^2\,dy\,dx = \frac{21}{2}$. We will evaluate $\int_1^2 \int_0^3 xy^2\,dx\,dy$ and see if we get the same result.

$$\int_0^3 xy^2\,dx = y^2\int_0^3 x\,dx = y^2\left[\frac{1}{2}x^2\right]_{x=0}^{x=3}$$

$$= \frac{y^2}{2}\left[3^2 - 0^2\right] = \frac{9y^2}{2}$$

Substituting this back into the iterated integral:

$$\int_1^2 \int_0^3 xy^2\,dx\,dy = \int_1^2 \frac{9y^2}{2}\,dy = \frac{9}{2}\int_1^2 y^2\,dy$$

$$= \frac{9}{2}\left[\frac{1}{3}y^3\right]_{y=1}^{y=2} = \frac{9}{2}\cdot\frac{1}{3}\left[2^3 - 1^3\right]$$

$$= \frac{3}{2}(8-1) = \frac{21}{2}$$

## Answer to Exercise 527 (on page 1816)

1. Answer: $\frac{5}{2} - \frac{1}{e}$. Solution: $\int_0^1 \int_1^2 \left(x + e^{-y}\right)\,dx\,dy = \int_0^1 \left(\frac{1}{2}x^2 + xe^{-y}\right)\Big|_{x=1}^{x=2} dy = \int_0^1 \left(2 - \frac{1}{2} + 2e^{-y} - e^{-y}\right)\,dy = \int_0^1 \left(\frac{3}{2} + e^{-y}\right)\,dy = \left[\frac{3}{2}y - e^{-y}\right]_{y=0}^{y=1} = \left(\frac{3}{2}(1) - e^{-1}\right) - \left(\frac{3}{2}(0) - e^{0}\right) = \frac{5}{2} - \frac{1}{e}$

2. Answer: 18. Solution: $\int_{-3}^3 \int_0^{\pi/2} \left(2y + y^2\cos x\right)\,dx\,dy = \int_{-3}^3 \left[2xy + y^2\sin x\right]_{x=0}^{x=\pi/2} dy = \int_{-3}^3 \left[(\pi y + y^2) - (0+0)\right]\,dy = \int_{-3}^3 \left(\pi y + y^2\right)\,dy = \left[\frac{\pi}{2}y^2 + \frac{1}{3}y^3\right]_{y=-3}^{y=3} = \left(\frac{\pi}{2}(9) + \frac{1}{3}(27)\right) - \left(\frac{\pi}{2}(9) + \frac{1}{3}(-27)\right) = 9 - (-9) = 18$

3. Answer: 6. Solution: $\int_0^3 \int_0^{\pi/2} t^2 \sin^3\theta \, d\theta \, dt = \left(\int_0^3 t^2 \, dt\right) \times \left(\int_0^{\pi/2} \sin^3\theta \, d\theta\right) = \left[\frac{1}{3}t^3\right]_{t=0}^{t=3} \times \left(\int_0^{\pi/2} \sin\theta \sin^2\theta \, d\theta\right) = 9\int_0^{\pi/2} \sin\theta \left(1-\cos^2\theta\right) \, d\theta = 9\left[\int_0^{\pi/2} \sin\theta \, d\theta - \int_0^{\pi/2} \sin\theta \cos^2\theta \, d\theta\right] = 9\left[(-\cos\theta)|_{\theta=0}^{\theta=\pi/2} + \left(\frac{1}{3}\cos^3\theta\right)|_{\theta=0}^{\theta=\pi/2}\right] = 9\left[-(-\cos 0) + \left(-\frac{1}{3}\cos^3 0\right)\right] = 9\left(1-\frac{1}{3}\right) = 9\left(\frac{2}{3}\right) = 6$

## Answer to Exercise 528 (on page 1817)

1. $\int_0^1 \int_{-3}^3 \frac{xy^2}{x^2+1} \, dy \, dx$ OR $\int_{-3}^3 \int_0^1 \frac{xy^2}{x^2+1} \, dx \, dy$
2. $\int_0^{\pi/4} \int_0^1 \frac{\sec\theta}{\sqrt{1+t^2}} \, dt \, d\theta$ OR $\int_0^1 \int_0^{\pi/4} \frac{\sec\theta}{\sqrt{1+t^2}} \, d\theta \, dt$

## Answer to Exercise 529 (on page 1818)

1. $\iint_R \frac{xy^2}{x^2+1} \, dA$, $R = \{(x,y) \,|\, 0 \le x \le 2, -3 \le y \le 3\} = \int_0^2 \int_{-3}^3 \frac{xy^2}{x^2+1} \, dy \, dx = \int_0^2 \frac{x}{x^2+1} \, dx \cdot \int_{-3}^3 y^2 \, dy$. To evaluate the integral with respect to $x$, we use the $u$-substitution $u = x^2+1$, $(x)dx = \frac{1}{2}du$: $\int_0^2 \frac{x}{x^2+1} \, dx \cdot \int_{-3}^3 y^2 \, dy = \int_{x=0}^{x=2} \frac{1}{2}\frac{1}{u} \, du \cdot \int_{-3}^3 y^2 \, dy = \frac{1}{2}\ln|u||_{x=0}^{x=2} \cdot \frac{1}{3}\left[y^3\right]_{y=-3}^{y=3} = \frac{1}{2}\left[\ln\left(2^2+1\right) - \ln\left(0^2+1\right)\right] \cdot \frac{1}{3}\left[3^3 - (-3)^3\right] = \frac{1}{2}\ln 5 \cdot \frac{1}{3}\left(27-(-27)\right) = \frac{\ln 5}{2}\frac{54}{3} = 9\ln 5$
2. $\iint_R \frac{\tan\theta}{\sqrt{1-t^2}} \, dA$, $R = \{(\theta,t) \,|\, 0 \le \theta \le \pi/3, \, 0 \le t \le \frac{1}{2}\} = \int_0^{\pi/3} \int_0^{1/2} \frac{\tan\theta}{\sqrt{1-t^2}} \, dt \, d\theta = \left[\int_0^{\pi/3} \tan\theta \, d\theta\right] \cdot \left[\int_0^{1/2} \frac{1}{\sqrt{1-t^2}} \, dt\right]$. Recall that $\frac{d}{dt}\arcsin t = \frac{1}{\sqrt{1-t^2}}$. Applying FTC, then $\left[\int_0^{\pi/3} \tan\theta \, d\theta\right] \cdot \left[\int_0^{1/2} \frac{1}{\sqrt{1-t^2}} \, dt\right] = \left[\int_0^{\pi/3} \tan\theta \, d\theta\right] \cdot [\arcsin t]_{t=0}^{t=1/2} = \left[\int_0^{\pi/3} \tan\theta \, d\theta\right] \cdot \left[\arcsin\frac{1}{2} - \arcsin 0\right] = \left[\int_0^{\pi/3} \tan\theta \, d\theta\right] \cdot \left[\frac{\pi}{6}\right] = \frac{\pi}{6}\int_0^{\pi/3} \frac{\sin\theta}{\cos\theta} \, d\theta$. To evaluate this final integral, we use the $u$-substitution $u = \cos\theta$ and $-du = \sin\theta d\theta$: $\frac{\pi}{6}\int_0^{\pi/3} \frac{\sin\theta}{\cos\theta} \, d\theta = -\frac{\pi}{6}\int_{\theta=0}^{\theta=\pi/3} \frac{1}{u} \, du = -\frac{\pi}{6}\ln u|_{\theta=0}^{\theta=\pi/3} = -\frac{\pi}{6}\left[\ln(\cos\theta)\right]_{\theta=o}^{\theta=\pi/3} = \frac{\pi}{6}\left[\ln(\cos 0) - \ln\left(\cos\frac{\pi}{3}\right)\right] = \frac{\pi}{6}\left[\ln 1 - \ln\frac{1}{2}\right] = \frac{\pi}{6}\ln\frac{1}{1/2} = \frac{\pi}{6}\ln 2$
3. $\iint_R x\sin(x+y) \, dA$, $R = [0,\pi/6] \times [0,\pi/3] = \int_0^{\pi/6} \int_0^{\pi/3} x\sin(x+y) \, dy \, dx$. Recall the sum formula for sine:

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

We can substitute this into our iterated integral:

$$\int_0^{\pi/6} \int_0^{\pi/3} x\sin(x+y) \, dy \, dx = \int_0^{\pi/6} \int_0^{\pi/3} x\left[\sin x \cos y + \cos x \sin y\right] \, dy \, dx$$

$$= \int_0^{\pi/6} \left[ \int_0^{\pi/3} x \sin x \cos y \, dy + \int_0^{\pi/3} x \cos x \sin y \, dy \right] dx$$

Let's designate $\int_0^{\pi/3} x \sin x \cos y \, dy$ as integral **A**, and $\int_0^{\pi/3} x \cos x \sin y \, dy$ as integral **B**. First, we will evaluate integral **A**:

$$\int_0^{\pi/3} x \sin x \cos y \, dy = x \sin x \int_0^{\pi/3} \cos y \, dy$$

$$= x \sin x \, [\sin y]_{y=0}^{y=\pi/3} = x \sin x \left[ \sin \frac{\pi}{3} - \sin 0 \right]$$

$$= x \sin x \left( \frac{\sqrt{3}}{2} \right) = \frac{x\sqrt{3}}{2} \sin x$$

Next, we evaluate integral **B**:

$$\int_0^{\pi/3} x \cos x \sin y \, dy = x \cos x \int_0^{\pi/3} \sin y \, dy$$

$$= x \cos x \, [-\cos y]_{y=0}^{y=\pi/3} = x \cos x \left[ -\cos \frac{\pi}{3} - (-\cos 0) \right]$$

$$= x \cos x \left[ -\frac{1}{2} - (-1) \right] = \frac{x}{2} \cos x$$

Substituting back in for integrals **A** and **B**:

$$\int_0^{\pi/6} \left[ \int_0^{\pi/3} x \sin x \cos y \, dy + \int_0^{\pi/3} x \cos x \sin y \, dy \right] dx = \int_0^{\pi/6} \left[ \frac{x\sqrt{3}}{2} \sin x + \frac{x}{2} \cos x \right] dx$$

$$= \frac{\sqrt{3}}{2} \int_0^{\pi/6} x \sin x \, dx + \frac{1}{2} \int_0^{\pi/6} x \cos x \, dx$$

Again, we will designate $\int_0^{\pi/6} x \sin x \, dx$ as integral **C** and $\int_0^{\pi/6} x \cos x \, dx$ as integral **D**. We start by using integration by parts to evaluate integral **C**:

Let $u = x$ and $dv = \sin x dx$. Then, $v = -\cos x$ and $du = dx$. Therefore:

$$\int_0^{\pi/6} x \sin x \, dx = [x \, (-\cos x)]_{x=0}^{x=\pi/6} - \int_0^{\pi/6} (-\cos x) \, dx$$

$$= \left[ \frac{\pi}{6} \left( -\cos \frac{\pi}{6} \right) \right] - [0 \, (-\cos 0)] + \sin x|_{x=0}^{x=\pi/6}$$

$$= -\frac{\pi}{6} \cdot \frac{\sqrt{3}}{2} - 0 + \sin \frac{\pi}{6} - \sin 0 = \frac{1}{2} - \frac{\pi\sqrt{3}}{12} = \frac{6 - \pi\sqrt{3}}{12}$$

Next, we will use integration by parts to evaluate integral **D**. Let $u = x$ and $dv =$

$\cos x\,dx$. Then $du = dx$ and $v = \sin x$ and therefore:

$$\int_0^{\pi/6} x\cos x\,dx = [x\sin x]_{x=0}^{x=\pi/6} - \int_0^{\pi/6} \sin x\,dx$$

$$= \left[\frac{\pi}{6}\sin\frac{\pi}{6} - 0\sin 0\right] - (-\cos x)\,|_{x=0}^{x=\pi/6} = \frac{\pi}{6}\cdot\frac{1}{2} + \cos\frac{\pi}{6} - \cos 0$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1 = \frac{\pi + 6\sqrt{3} - 12}{12}$$

Substituting back in for integrals **C** and **D**:

$$\frac{\sqrt{3}}{2}\int_0^{\pi/6} x\sin x\,dx + \frac{1}{2}\int_0^{\pi/6} x\cos x\,dx = \frac{\sqrt{3}}{2}\left(\frac{6-\pi\sqrt{3}}{12}\right) + \frac{1}{2}\left(\frac{\pi+6\sqrt{3}-12}{12}\right)$$

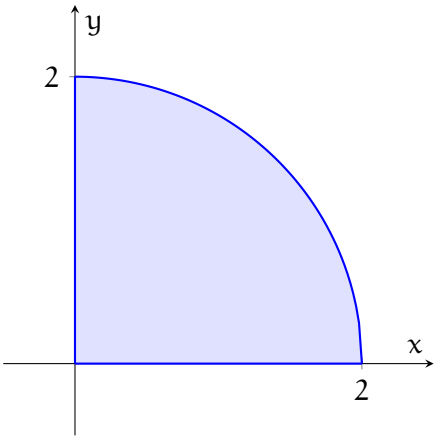
$$= \frac{6\sqrt{3} - 3\pi + \pi + 6\sqrt{3} - 12}{24} = \frac{6\sqrt{3} - 6 - \pi}{12}$$

## Answer to Exercise 530 (on page 1827)

1. $\iint_D e^{-y^2}\,dA = \int_0^3\int_0^{2y} e^{-y^2}\,dx\,dy = \int_0^3 \left[e^{-y^2}x|_{x=0}^{x=2y}\right]\,dy = \int_0^3 2ye^{-y^2}\,dy = -e^{-y^2}|_{y=0}^{y=3} = 1 - e^{-9} \approx 0.9999$
2. $\iint_D x\sin y\,dA = \int_0^2\int_0^{x^2} x\sin y\,dy\,dx = \int_0^2 x\int_0^{x^2}\sin y\,dy\,dx = \int_0^2 x\,[-\cos y]_{y=0}^{y=x^2}$ $= \int_0^2 x\left(\cos 0 - \cos x^2\right)\,dx = \int_0^2 \left(x - x\cos x^2\right)\,dx = \left[\frac{1}{2}x^2 - \frac{1}{2}\sin x^2\right]_{x=0}^{x=2} = \frac{1}{2}(2)^2 - \frac{1}{2}\left(\sin 2^2 - \sin 0\right) = 2 - \frac{1}{2}\left(\sin 4 - 0\right) = 2 - \frac{\sin 4}{2} \approx 2.378$
3. We can describe the region as $D = \{(x,y) \mid -3 \le x \le -3, -\sqrt{9-x^2} \le y \le \sqrt{9-x^2}\}$. Therefore, $\iint_D (2y - x)\,dA = \int_{-3}^3\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (2x - y)\,dy\,dx = \int_{-3}^3 \left[2xy - \frac{1}{2}y^2\right]_{y=-\sqrt{9-x^2}}^{y=\sqrt{9-x^2}}\,dx$ $= \int_{-3}^3 \left[2x\left(\sqrt{9-x^2} + \sqrt{9-x^2}\right) - \frac{1}{2}\left(9 - x^2 - (9-x^2)\right)\right]\,dx = \int_{-3}^3 4x\sqrt{9-x^2}\,dx$. Let $u = 9 - x^2$, then $du = -2x$ and $4x = -2du$. Substituting, $\int_{-3}^3 4x\sqrt{9-x^2}\,dx = \int_{x=-3}^{x=3} -2\sqrt{u}\,du = -2\cdot\frac{2}{3}u^{3/2}|_{x=-3}^{x=3} = -\frac{4}{3}\left[(9-x^2)\right]_{x=-3}^{x=3} = 0$

## Answer to Exercise 531 (on page 1832)

1. Let's visualize the region in the $xy$-plane:




The region is a quarter-circle that can be described with $D = \{(r,\theta) \mid 0 \le r \le 2,\ 0 \le \theta \le \pi/2\}$. We can then rewrite the integral in polar coordinates:

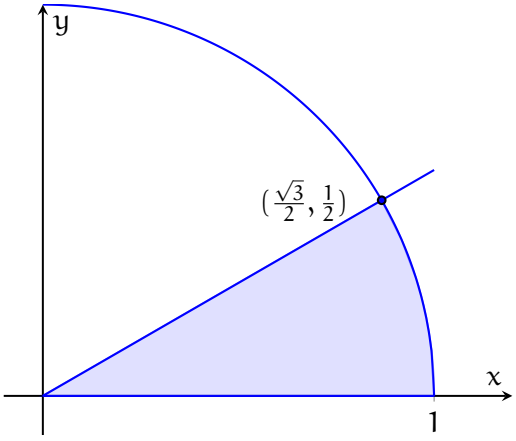
$$\int_0^2 \int_0^{\sqrt{4-x^2}} e^{-x^2-y^2}\,dy\,dx = \int_0^{\pi/2} \int_0^2 re^{-r^2}\,dr\,d\theta$$

$$= \int_0^{\pi/2} \left[-\frac{1}{2}e^{-r^2}\right]_{r=0}^{r=2} d\theta = \int_0^{\pi/2} \left(-\frac{1}{2}\right)\left[e^{-4}-1\right]\,d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2} 1 - e^{-4}\,d\theta = \frac{1}{2}\left(1-\frac{1}{e^4}\right)\int_0^{\pi/2} 1\,d\theta$$

$$= \frac{1}{2}\left(1-\frac{1}{e^4}\right)\theta\Big|_{\theta=0}^{\theta=\pi/2} = \frac{\pi}{4}\left(1-\frac{1}{e^4}\right)$$

2. The region is bounded by the x-axis, the line $y = x/\sqrt{3}$, and the circle $x^2 + y^2 = 1$:

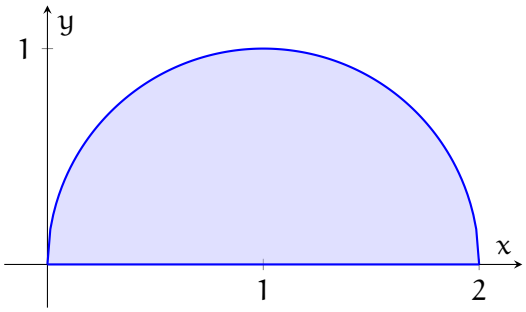



We see that the region defined in polar coordinates is $D = \{(r,\theta) \mid 0 \le r \le 1,\ 0 \le \theta \le \pi/6\}$ . And therefore:

$$\int_0^{1/2} \int_{\sqrt{3}y}^{\sqrt{1-y^2}} xy^2\,dx\,dy = \int_0^{\pi/6} \int_0^1 r\,(r\cos\theta)\,(r\sin\theta)^2\,dr\,d\theta$$

$$= \int_0^{\pi/6} \left[\cos\theta \sin^2\theta\right] \, d\theta \cdot \int_0^1 r^4 \, dr$$
$$= \left(\frac{1}{3}\sin^3\theta\Big|_{\theta=0}^{\theta=\pi/6}\right) \cdot \left(\frac{1}{5}r^5\Big|_{r=0}^{r=1}\right)$$
$$= \frac{1}{15} \cdot \left(\frac{1}{2}\right)^3 = \frac{1}{120}$$

3. Visualizing the region:

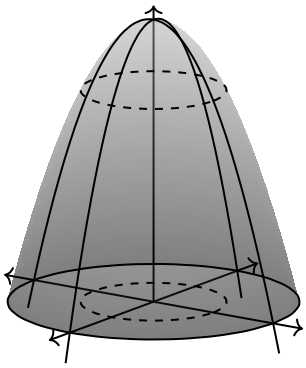



We see that the region is the top half of a circle of radius 1 centered at $(1, 0)$. In polar coordinates, this region is $D = \{(r, \theta) \mid 0 \le r \le 2\cos\theta, 0 \le \theta \le \pi/2\}$. And therefore:

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2+y^2} \, dy \, dx = \int_0^{\pi/2} \int_0^{2\cos\theta} r\sqrt{r^2} \, dr \, d\theta$$
$$= \int_0^{\pi/2} \int_0^{2\cos\theta} r^2 \, dr \, d\theta = \int_0^{\pi/2} \frac{1}{3}\left[r^3\right]_{r=0}^{r=2\cos\theta} d\theta$$
$$= \frac{8}{3}\int_0^{\pi/2} \cos^3\theta \, d\theta = \frac{8}{3}\int_0^{\pi/2} \cos\theta\left(1-\sin^2\theta\right) \, d\theta$$
$$= \frac{8}{3}\left[\int_0^{\pi/2} \cos\theta \, d\theta - \int_0^{\pi/2} \cos\theta \sin^2\theta \, d\theta\right]$$
$$= \frac{8}{3}\left[(\sin\theta)_{\theta=0}^{\theta=\pi/2} - \left(\frac{1}{3}\sin^3\theta\right)_{\theta=0}^{\theta=\pi/2}\right]$$
$$= \frac{8}{3}\left[(1-0) - \frac{1}{3}\left(1^3-0^3\right)\right] = \frac{8}{3}\cdot\frac{2}{3} = \frac{16}{9}$$

## Answer to Exercise 532 (on page 1833)

We are finding the volume of the solid that lies under the surface $z = 4 - x^2 - y^2$ and above the $xy$-plane.



We can use polar coordinates to simplify the double integral. In polar coordinates, $x = r\cos(\theta)$ and $y = r\sin(\theta)$, so $x^2 + y^2 = r^2$. The volume under the surface and above the $xy$-plane is given by

$$V = \int\int (4 - r^2) r \, dr \, d\theta, \tag{196.1}$$

where $r$ ranges from 0 to 2 (since $4 - r^2 \geq 0$ if $0 \leq r \leq 2$) and $\theta$ ranges from 0 to $2\pi$.

Hence,

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^2 (4r - r^3) \, dr \, d\theta \\ &= \int_0^{2\pi} \left[ 2r^2 - \frac{1}{4} r^4 \right]_0^2 d\theta \\ &= \int_0^{2\pi} (8 - 4) \, d\theta \\ &= \int_0^{2\pi} 4 \, d\theta \\ &= [4\theta]_0^{2\pi} \\ &= 8\pi. \end{aligned}$$

So, the volume of the solid is $8\pi$ cubic units.

## Answer to Exercise 533 (on page 1834)

Let's describe the footprint of the pool as a 20-foot radius circle centered at the origin (that is, a region $D = \{(r, \theta) \mid 0 \leq r \leq 20, \ 0 \leq \theta \leq 2\pi\}$). Further, let's take north-south as

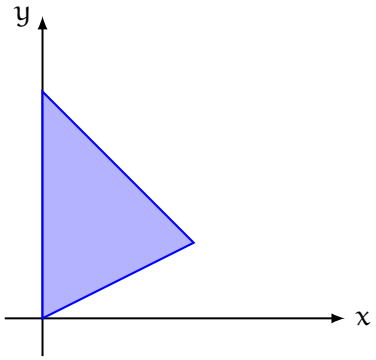
parallel to the y-axis and east-west as parallel to the x-axis. So, the depth of water is then given by $z = f(x,y) = \frac{7}{40}x + \frac{13}{2}$ over the footprint of the pool. The total volume of water is given by:

$$\begin{aligned}
&\int_0^{2\pi}\int_0^{20} r\left(\frac{7}{40}r\cos\theta + \frac{13}{2}\right)\,dr\,d\theta \\
&= \int_0^{2\pi}\int_0^{20}\left[\frac{7}{40}r^2\cos\theta + \frac{13}{2}r\right]\,dr\,d\theta \\
&= \int_0^{2\pi}\left[\frac{7\cos\theta}{40}\int_0^{20} r^2\,dr + \frac{13}{2}\int_0^{20} r\,dr\right]\,d\theta \\
&= \int_0^{2\pi}\left[\frac{7\cos\theta}{40}\left(\frac{1}{3}r^3\right)_{r=0}^{r=20} + \frac{13}{2}\left(\frac{1}{2}r^2\right)_{r=0}^{r=20}\right]\,d\theta \\
&= \int_0^{2\pi}\left[\frac{1400}{3}\cos\theta + 1300\right]\,d\theta = \left[\frac{1400}{3}\sin\theta + 1300\theta\right]_{\theta=0}^{\theta=2\pi} \\
&= 2600\pi \text{ cubic feet}
\end{aligned}$$

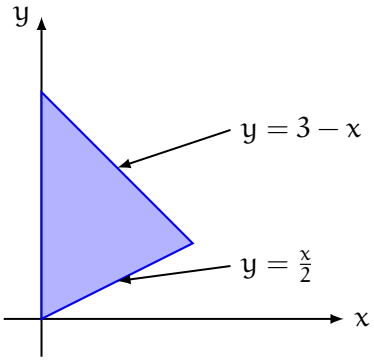
## Answer to Exercise 534 (on page 1837)

1. $\iint_D (1 + x^2 + y^2)\,dA = \int_0^4\int_0^3 (1 + x^2 + y^2)\,dy\,dx = \int_0^4 \left[y + x^2y + \frac{1}{3}y^3\right]_{y=0}^{y=3} dx = \int_0^4 \left[3 + 3x^2 + \frac{1}{3}(3)^3\right]$ $= \int_0^4 (12 + 3x^2)\,dx = \left[12x + x^3\right]_{x=0}^{x=4} = 12(4) + 4^3 = 112$

2. First, let's visualize this region, since it isn't a rectangle:




Let's divide the triangle horizontally and write equations for each of the sides that do not lie on the y-axis.




We see that we can describe region $D$ as $D = \{(x,y) \mid 0 \le x \le 2, \frac{x}{2} \le y \le 3 - x\}$. Therefore $\iint_D (x+y)\,dA = \int_0^3 \int_{x/2}^{3-x} (x+y)\,dy\,dx = \int_0^3 \left[xy + \frac{1}{2}y^2\right]_{y=x/2}^{y=3-x} dx =$
$\int_0^3 \left[(x(3-x)) - (x(x/2)) + \frac{1}{2}\left((3-x)^2 - (x/2)^2\right)\right] dx$
$= \int_0^3 \left[\left(3x - x^2 - \frac{x^2}{2}\right) + \frac{1}{2}\left(9 - 6x + x^2 - \frac{x^2}{4}\right)\right] dx = \int_0^3 \left[-x^2 - \frac{x^2}{2} + \frac{x^2}{2} - \frac{x^2}{8} + 3x - 3x + \frac{9}{2}\right] dx$
$= \int_0^3 \left(-\frac{9x^2}{8} + \frac{9}{2}\right) dx = \left[\frac{9x}{2} - \frac{3x^3}{8}\right]_{x=0}^{x=2} = \frac{9(2)}{2} - \frac{3(8)}{8} = 9 - 3 = 6$

## Answer to Exercise 535 (on page 1841)

1. First, we find the total mass: $m = \int_0^\pi \int_0^{2\sin x} x\,dy\,dx = \int_0^\pi [xy]_{y=0}^{y=2\sin x}\,dx = \int_0^\pi 2x\sin x\,dx$.
   We apply integration by parts to evaluate the integral: $\int_0^\pi 2x\sin x\,dx = (-2x\cos x)\,|_{x=0}^{x=\pi} +$
   $\int_0^\pi 2\cos x\,dx = [-2\pi(-1)] - (0) + \sin x|_{x=0}^{x=\pi} = 2\pi + \sin\pi - \sin 0 = 2\pi$
   Now that we know $m = 2\pi$, we can find $\bar{x}$ and $\bar{y}$: $\bar{x} = \frac{1}{2\pi}\int_0^\pi \int_0^{2\sin x} x \cdot x\,dy\,dx$
   $= \frac{1}{2\pi}\int_0^\pi x^2 y|_{y=0}^{y=2\sin x}\,dx = \frac{1}{2\pi}\int_0^\pi x^2(2\sin x)\,dx = \frac{1}{\pi}\int_0^\pi x^2\sin x\,dx$.
   Applying integration by parts: $\frac{1}{\pi}\int_0^\pi x^2\sin x\,dx = \frac{1}{\pi}\left[x^2(-\cos x)\,|_{x=0}^{x=\pi} - \int_0^\pi 2x(-\cos x)\,dx\right]$
   $= \frac{1}{\pi}\left[(-\pi^2\cos\pi) + 2\int_0^\pi x\cos x\,dx\right] = \frac{1}{\pi}\left[\pi^2 + 2\int_0^\pi x\cos x\,dx\right] = \pi + \frac{2}{\pi}\int_0^\pi x\cos x\,dx$.
   Applying integration by parts again: $\pi + \frac{2}{\pi}\int_0^\pi x\cos x\,dx = \pi + \frac{2x\sin x}{\pi}|_{x=0}^{x=\pi} - \frac{2}{\pi}\int_0^\pi \sin x\,dx$
   $= \pi - \frac{2}{\pi}\int_0^\pi \sin x\,dx = \pi + \frac{2}{\pi}[\cos x]_{x=0}^{x=\pi} = \pi + \frac{2}{\pi}[\cos\pi - \cos 0] = \pi + \frac{2}{\pi}(-1-1) = \pi - \frac{4}{\pi} = \bar{x}$
   And finding $\bar{y}$: $\bar{y} = \frac{1}{2\pi}\int_0^\pi \int_0^{2\sin x} y\cdot x\,dy\,dx = \frac{1}{2\pi}\int_0^\pi \left[\frac{1}{2}xy^2\right]_{y=0}^{y=2\sin x} dx = \frac{1}{4\pi}\int_0^\pi x[2\sin x]^2\,dx$
   $= \frac{1}{4\pi}\int_0^\pi 4x\sin^2 x\,dx = \frac{1}{\pi}\int_0^\pi x\sin^2 x\,dx = \frac{1}{\pi}\int_0^\pi x\frac{1-\cos(2x)}{2}\,dx = \frac{1}{\pi}\int_0^\pi \frac{x}{2}\,dx - \frac{1}{\pi}\int_0^\pi x\cos(2x)\,dx$
   $= \frac{1}{2\pi}\left[\frac{x^2}{2}\right]_{x=0}^{x=\pi} - \frac{1}{\pi}\left[\frac{1}{2}x\sin(2x)|_{x=0}^{x=\pi} - \frac{1}{2}\int_0^\pi \sin(2x)\,dx\right] = \frac{1}{2\pi}\left(\frac{\pi^2}{2}\right) - \frac{1}{2\pi}\left[\pi\sin(2\pi) - 0 + \frac{1}{2}\cos(2x)|_{x=0}^{x=\pi}\right]$
   $= \frac{\pi}{4} - \frac{1}{2\pi}\left[\frac{1}{2}(\cos 2\pi - \cos 0)\right] = \frac{\pi}{4}$
   Therefore, the center of mass is found at $(\bar{x},\bar{y}) = \left(\pi - \frac{4}{\pi}, \frac{\pi}{4}\right)$

2. We know from a previous question that the total mass of this lamina is 112 (see *Finding Total Mass*).
   Finding $\bar{x}$: $\bar{x} = \frac{1}{112}\int_0^4 \int_0^3 x\left(1 + x^2 + y^2\right) dy\,dx = \frac{1}{112}\int_0^4 \int_0^3 \left(x + x^3 + xy^2\right) dy\,dx$

$= \frac{1}{112}\int_0^4 \left[xy + x^3y + \frac{x}{3}y^3\right]_{y=0}^{y=3} dx = \frac{1}{112}\int_0^4 \left[3x + 3x^3 + 9x\right] dx = \frac{3}{112}\int_0^4 \left[4x + x^3\right] dx =$
$\frac{3}{112}\left[2x^2 + \frac{x^4}{4}\right]_{x=0}^{x=4} = \frac{3}{112}\left[2(4)^2 - 2(0)^2 + \frac{4^4}{4} - \frac{0^4}{4}\right] = \frac{3}{112}\left[32 + 64\right] = \frac{3\cdot 96}{112} = \frac{3\cdot 2\cdot 2\cdot 2\cdot 2\cdot 2\cdot 3}{2\cdot 2\cdot 7\cdot 2\cdot 2} =$
$\frac{18}{7}$

Finding $\bar{y}$: $\bar{y} = \frac{1}{112}\int_0^4\int_0^3 y\left(1 + x^2 + y^2\right) dy\, dx = \frac{1}{112}\int_0^4\int_0^3 \left[y + x^2y + y^3\right] y\, dx$

$= \frac{1}{112}\int_0^4 \left[\frac{y^2}{2} + \frac{x^2y^2}{2} + \frac{y^4}{4}\right]_{y=0}^{y=3} dx = \frac{1}{112}\int_0^4 \left[\frac{3^2}{2} + \frac{3^2x^2}{2} + \frac{3^4}{4}\right] dx = \frac{1}{112}\int_0^4 \left[\frac{99}{4} + \frac{9}{2}x^2\right] dx$
$= \frac{1}{112}\left[\frac{99}{4}x + \frac{3}{2}x^3\right]_{x=0}^{x=4} = \frac{3}{224}\left[\frac{33}{2}(4) + 4^3\right] = \frac{3}{224}\left(66 + 64\right) = \frac{3\cdot 130}{224} = \frac{3\cdot 65}{112} = \frac{195}{112}$

Therefore, the center of mass of the rectangular region $D$ is $\left(\frac{18}{7}, \frac{195}{112}\right)$

3. We know from a previous question (see *Finding Total Mass*) that the total mass of $D$ is 6, and it can be described as $D = \{(x, y) \mid 0 \le x \le 2, \frac{x}{2} \le y \le 3 - x\}$

   Finding $\bar{x}$:

   $\bar{x} = \frac{1}{6}\int_0^2\int_{x/2}^{3-x} x\left(x + y\right) dy\, dx = \frac{1}{6}\int_0^2\int_{x/2}^{3-x} \left(x^2 + xy\right) dy\, dx = \frac{1}{6}\int_0^2 \left[x^2y + \frac{x}{2}y^2\right]_{y=x/2}^{y=3-x} dx$
   $= \frac{1}{6}\int_0^2 \left[x^2\left(3 - x - \frac{x}{2}\right) + \frac{x}{2}\left(\left(3 - x\right)^2 - \left(\frac{x}{2}\right)^2\right)\right] dx = \frac{1}{6}\int_0^2 \left[3x^2 - x^3 - \frac{x^3}{2} + \frac{x}{2}\left(9 - 6x + x^2 - \frac{x^2}{4}\right)\right] dx$
   $= \frac{1}{6}\int_0^2 \left[3x^2 - \frac{3}{2}x^3 + \frac{x}{2}\left(9 - 6x + \frac{3}{4}x^2\right)\right] dx = \frac{1}{6}\int_0^2 \left[3x^2 - \frac{3}{2}x^3 + \frac{9}{2}x - 3x^2 + \frac{3}{8}x^3\right] dx =$
   $\frac{1}{6}\int_0^2 \left[\frac{9}{2}x - \frac{9}{8}x^3\right] dx = \frac{1}{6}\left[\frac{9}{4}x^2 - \frac{9}{32}x^4\right]_{x=0}^{x=2} = \frac{1}{6}\left[\frac{9\cdot 4}{4} - \frac{9\cdot 16}{32}\right] = \frac{1}{6}\left[9 - \frac{9}{2}\right] = \frac{1}{6}\cdot\frac{9}{2} = \frac{9}{12} = \frac{3}{4}$

   And finding $\bar{y}$:

   $\bar{y} = \frac{1}{6}\int_0^2\int_{x/2}^{3-x} y\left(x + y\right) dy\, dx = \frac{1}{6}\int_0^2\int_{x/2}^{3-x} \left(xy + y^2\right) dy\, dx = \frac{1}{6}\int_0^2 \left[\frac{x}{2}y^2 + \frac{1}{3}y^3\right]_{y=x/2}^{y=3-x} dx$
   $= \frac{1}{6}\int_0^2 \left[\frac{x}{2}\left(\left(3 - x\right)^2 - \left(\frac{x}{2}\right)^2\right) + \frac{1}{3}\left(\left(3 - x\right)^3 - \left(\frac{x}{2}\right)^3\right)\right] dx$
   $= \frac{1}{6}\int_0^2 \left[\frac{x}{2}\left(9 - 6x + x^2 - \frac{x^2}{4}\right) + \frac{1}{3}\left(27 - 27x + 9x^2 - x^3 - \frac{x^3}{8}\right)\right] dx$
   $= \frac{1}{6}\int_0^2 \left[\frac{x}{2}\left(9 - 6x + \frac{3x^2}{4}\right) + \frac{1}{3}\left(27 - 27x + 9x^2 - \frac{9x^3}{8}\right)\right] dx$
   $= \frac{1}{6}\int_0^2 \left[\frac{9}{2}x - 3x^2 + \frac{3}{8}x^3 + 9 - 9x + 3x^2 - \frac{3}{8}x^3\right] dx = \frac{1}{6}\int_0^2 \left[9 - \frac{9}{2}x\right] dx = \frac{1}{6}\left[9x - \frac{9}{4}x^2\right]_{x=0}^{x=2}$
   $= \frac{3}{6}\left[3(2) - \frac{3}{4}(2)^2\right] = \frac{1}{2}\left(6 - 3\right) = \frac{1}{2}\cdot 3 = \frac{3}{2}$

   Therefore, the center of mass is $(\bar{x}, \bar{y}) = \left(\frac{3}{4}, \frac{3}{2}\right)$

## Answer to Exercise 536 (on page 1846)

1.
$$I_x = \iint_D y^2\rho(x, y)\, dA = \int_1^4\int_0^3 y^2(xy)\, dy\, dx$$
$$= \int_1^4\int_0^3 xy^3\, dy\, dx = \int_1^4 x\left[\frac{1}{4}y^4\right]_{y=0}^{y=3} dx = \frac{1}{4}\int_1^4 81x\, dx$$
$$= \frac{81}{4}\left[\frac{1}{2}x^2\right]_{x=1}^{x=4} = \frac{81}{2}\left(4^2 - 1^2\right) = \frac{81}{2}\cdot 15 = \frac{1215}{2}$$

To find the radius of gyration, first we need to find the total mass:

$$m = \iint_D \rho(x,y)\,dA = \int_1^4 \int_0^3 xy\,dy\,dx$$

$$= \int_1^4 \frac{x}{2}\left[y^2\right]_{y=0}^{y=3} dx = \frac{9}{2}\int_1^4 x\,dx = \frac{9}{2}\cdot\left(\frac{1}{2}\right)\cdot\left[x^2\right]_{x=1}^{x=4} = \frac{9}{4}\left[16-1\right] = \frac{135}{2}$$

Finding the radius of gyration about the x-axis:

$$I_x = m\bar{\bar{y}}^2$$

$$\frac{1215}{2} = \left(\frac{135}{2}\right)\bar{\bar{y}}^2$$

$$9 = \bar{\bar{y}}^2$$

$$\bar{\bar{y}} = 3$$

2.

$$I_y = \iint_D x^2\rho(x,y)\,dA = \int_{-\pi/2}^{\pi/2}\int_0^{2\cos x} x^2 y\,dy\,dx$$

$$= \int_{-pi/2}^{\pi/2} x^2\left[\frac{1}{2}y^2\right]_{y=0}^{y=2\cos x} dx = \frac{1}{2}\int_{-\pi/2}^{\pi/2} x^2\,(2\cos x)^2\,dx$$

$$= 2\int_{-\pi/2}^{\pi/2} x^2\cos^2 x\,dx = 2\int_{-\pi/2}^{\pi/2} x^2\left(\frac{1+\cos 2x}{2}\right)dx = \int_{-\pi/2}^{\pi/2} x^2\,dx + \int_{-\pi/2}^{\pi/2} x^2\cos 2x\,dx$$

$$= \frac{1}{3}\left[x^3\right]_{x=-\pi/2}^{x=\pi/2} + \frac{1}{2}x^2\sin 2x\Big|_{x=-\pi/2}^{x=\pi/2} - \int_{-\pi/2}^{\pi/2}\frac{1}{2}\sin 2x\,(2x)\,dx$$

$$= \frac{1}{3}\left[\left(\frac{\pi}{2}\right)^3 - \left(\frac{-\pi}{2}\right)^3\right] - \int_{-\pi/2}^{\pi/2} x\sin 2x\,dx$$

$$= \frac{1}{3}\left(\frac{2\pi^3}{8}\right) - \left(\left[-\frac{1}{2}x\cos 2x\right]_{x=-\pi/2}^{\pi/2} - \int_{-\pi/2}^{\pi/2}(-\frac{1}{2}\cos 2x)\,dx\right)$$

$$= \frac{\pi^3}{12} + \left(\frac{1}{2}\left(\frac{\pi}{2}\right)\cos(\pi) - \frac{1}{2}\left(\frac{-\pi}{2}\right)\cos(-\pi)\right) - \left[\frac{1}{4}\sin 2x\right]_{x=-\pi/2}^{\pi/2}$$

$$= \frac{\pi^3}{12} + \frac{\pi}{4}(-1) + \frac{\pi}{4}(-1) = \frac{\pi^3}{12} - \frac{\pi}{2}$$

In order to find the radius of gyration, we need to first know the total mass:

$$m = \iint_D \rho(x,y)\,dA = \int_{-\pi/2}^{\pi/2}\int_0^{2\cos x} y\,dy\,dx$$

$$= \int_{-\pi/2}^{\pi/2} \frac{1}{2} y^2 \Big|_{y=0}^{y=2\cos x} \, dx = \frac{1}{2} \int_{-\pi/2}^{\pi/2} 4\cos^2 x \, dx = \int_{-\pi/2}^{\pi/2} 1 + \cos 2x \, dx$$

$$= \left[ x + \frac{1}{2} \sin 2x \right]_{x=-\pi/2}^{\pi/2} = \pi$$

We can then find the radius of gyration about the y-axis:

$$m\bar{\bar{x}}^2 = I_y$$

$$\pi\bar{\bar{x}}^2 = \frac{\pi^3}{12} - \frac{\pi}{2}$$

$$\bar{\bar{x}} = \sqrt{\frac{\pi^2}{12} - \frac{1}{2}}$$

3.

$$I_o = \iint_D \left(x^2 + y^2\right) \rho(x, y) = \int_1^2 \int_0^{\pi} r(r^2) r \, d\theta \, dr$$

$$= \int_1^2 r^4 \theta \Big|_{\theta=0}^{\theta=\pi} \, dr = \pi \int_1^2 r^4 \, dr = \frac{\pi}{5} r^5 \Big|_{r=1}^{r=2} = \frac{\pi}{5} \left(2^5 - 1\right) = \frac{\pi}{5}(31) = \frac{31\pi}{5}$$

We find the total mass:

$$m = \iint_D \rho(x, y) \, dA = \int_1^2 \int_0^{\pi} r^2 \, d\theta \, dr = \int_1^2 r^2 \theta \Big|_{\theta=0}^{\theta=\pi} \, dr$$

$$= \pi \int_1^2 r^2 \, dr = \frac{\pi}{3} r^3 \Big|_{r=1}^{r=2} = \frac{\pi}{3} \left(2^3 - 1\right) = \frac{\pi}{3}(7) = \frac{7\pi}{3}$$

To find the radius of gyration about the origin:

$$mR^2 = I_o$$

$$\left(\frac{7\pi}{3}\right) R^2 = \frac{31\pi}{5}$$

$$R^2 = \frac{31}{5} \cdot \frac{3}{7} = \frac{93}{35}$$

$$R = \sqrt{\bar{\bar{x}}^2 + \bar{\bar{y}}^2} = \sqrt{\frac{93}{35}}$$

## Answer to Exercise 537 (on page 1851)

1. Rearranging the formula for the plane, we find that $z = 3x + 2y + 2$. Therefore,

$\partial z/\partial x = 3$ and $\partial z/\partial y = 2$. Then the surface area is given by:

$$A(S) = \int_2^6 \int_1^4 \sqrt{1+3^2+2^2}\, dy\, dx = \int_2^6 \sqrt{14} y|_{y=1}^{y=4}\, dx$$

$$= \int_2^6 3\sqrt{14}\, dx = 3\sqrt{14}x|_{x=2}^{x=6} = 12\sqrt{14}$$

2. The paraboloid intersects the plane when $2x^2+2y^2 = 32$, which is the circle of radius 4 centered at the origin. So, we are looking for the area of the surface $z = 2x^2 + 2y^2$ that lies above the region $R - \{(r,\theta) \mid 0 \le r \le 4,\ 0 \le \theta \le 2\pi\}$. The surface area is:

$$A(S) = \iint_R \sqrt{1+(4x)^2+(4y)^2}\, dA = \int_0^4 \int_0^{2\pi} r\sqrt{1+(4r\cos\theta)^2+(4r\sin\theta)^2}\, d\theta\, dr$$

$$= \int_0^4 \int_0^{2\pi} r\sqrt{1+16r^2}\, d\theta\, dr = \int_0^4 r\sqrt{1+16r^2}\, [\theta]_{\theta=0}^{\theta=2\pi}\, dr$$

$$= \int_0^4 2\pi r\sqrt{1+16r^2}\, dr$$

Let $u = 1+16r^2$, then $du = 32r(dr)$ and $r(dr) = du/32$. Substituting:

$$A(S) = \frac{2\pi}{32}\int_{r=0}^{r=4} \sqrt{u}\, du = \frac{\pi}{16}\left(\frac{2}{3}\right)\left[u^{3/2}\right]_{r=0}^{r=4}$$

$$= \frac{\pi}{24}\left[\left(1+16r^2\right)^{3/2}\right]_{r=0}^{r=4} = \frac{\pi}{24}\left[(257)^{3/2}-1\right] \approx 6470.15$$

3. The region, $R$, we are interested in is the circle of radius 2 centered at the origin of the $xy$-plane, described by $R = \{(r,\theta) \mid 0 \le r \le 2,\ 0 \le \theta \le 2\pi\}$. Noting that $\partial z/\partial x = 3y$ and $\partial z/\partial y = 3x$, we see that the surface area is given by:

$$A(S) = \iint_R \sqrt{1+(3y)^2+(3x)^2}\, dA = \int_0^2 \int_0^{2\pi} r\sqrt{1+9r^2\sin^2\theta+9r^2\cos^2\theta}\, d\theta\, dr$$

$$= \int_0^2 \int_0^{2\pi} r\sqrt{1+9r^2}\, d\theta\, dr = 2\pi\int_0^2 r\sqrt{1+9r^2}\, dr$$

Let $u = 1+9r^2$, then $du = 18r(dr)$, which means that $r(dr) = du/18$. Substituting:

$$A(S) = \frac{2\pi}{18}\int_{r=0}^{r=2} \sqrt{u}\, du = \frac{\pi}{9}\left(\frac{2}{3}\right)\left[u^{3/2}\right]_{r=0}^{r=2}$$

$$= \frac{2\pi}{27}\left[(1+9(4))^{3/2}-1\right] = \frac{2\pi}{27}\left[(37)^{3/2}-1\right] \approx 52.14$$

## Answer to Exercise 538 (on page 1852)

1. The area of $D$ is $2\pi$. Therefore, the average value is:

$$\frac{1}{2\pi}\iint_D x\sin y\,dA = \frac{1}{2\pi}\int_0^2\int_0^\pi x\sin y\,dy\,dx$$

$$= \frac{1}{2\pi}\int_0^2 -x\cos y\Big|_{y=0}^{y=\pi}dx = \frac{1}{2\pi}\int_0^2 -x\left(\cos\pi-\cos 0\right)\,dx$$

$$= \frac{1}{2\pi}\int_0^2 (-x)(-1-1)\,dx = \frac{1}{2\pi}\int_0^2 2x\,dx$$

$$= \frac{1}{2\pi}x^2\Big|_{x=0}^{x=2} = \frac{2}{\pi}$$

2. Since $D$ is a circle of radius $r=1$, the area is $A=\pi r^2=\pi$. $D$ can be described with $D=\{(r,\theta)\mid 0\le r\le 2\cos\theta,\ -\pi/2\le\theta\le\pi/2\}$. Therefore, the average value of $f(x,y)=x+y$ over $D$ is:

$$f_{ave} = \frac{1}{\pi}\iint_D (x+y)\,dA = \frac{1}{\pi}\int_{-\pi/2}^{\pi/2}\int_0^{2\cos\theta} r\cdot(r\cos\theta+r\sin\theta)\,dr\,d\theta$$

$$= \frac{1}{\pi}\int_{-\pi/2}^{\pi/2}(\cos\theta+\sin\theta)\left[\int_0^{2\cos\theta} r^2\,dr\right]d\theta$$

$$= \frac{1}{\pi}\int_{-\pi/2}^{\pi/2}(\cos\theta+\sin\theta)\cdot\left[\frac{1}{3}r^3\right]_{r=0}^{r=2\cos\theta}d\theta$$

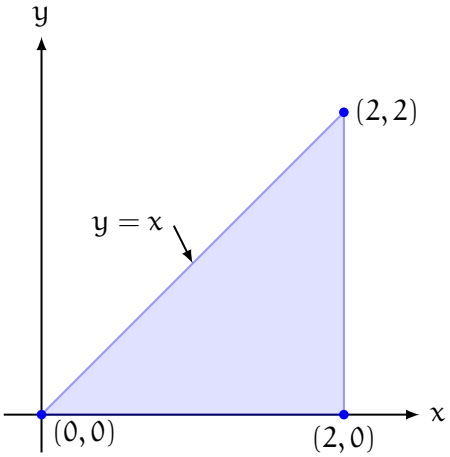
$$= \frac{8}{3\pi}\int_{-\pi/2}^{\pi/2}(\cos\theta+\sin\theta)\cos^3\theta\,d\theta = \frac{8}{3\pi}\int_{\pi/2}^{\pi/2}\left(\cos^4\theta+\sin\theta\cos^3\theta\right)d\theta$$

$$= \frac{8}{3\pi}\left[\int_{-\pi/2}^{\pi/2}\left(\frac{1+\cos 2\theta}{2}\right)^2 d\theta - \left[\frac{1}{4}\cos^4\theta\right]_{\theta=-\pi/2}^{\theta=\pi/2}\right]$$

$$= \frac{2}{3\pi}\int_{-\pi/2}^{\pi/2}\left(1+2\cos 2\theta+\cos^2 2\theta\right)d\theta = \frac{2}{3\pi}\left[(\theta+\sin 2\theta)_{\theta=-\pi/2}^{\theta=\pi/2} + \int_{-\pi/2}^{\pi/2}\frac{1+\cos 4\theta}{2}\,d\theta\right]$$

$$= \frac{2}{3\pi}\left[\pi+\frac{1}{2}\left(\theta+\frac{1}{4}\sin 4\theta\right)_{\theta=-\pi/2}^{\theta=\pi/2}\right] = \frac{2}{3\pi}\left[\pi+\frac{1}{2}(\pi)\right] = \frac{2}{3\pi}\left(\frac{3\pi}{2}\right) = 1$$

3. Let's visualize $D$:

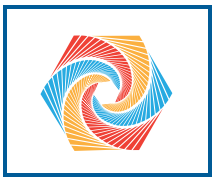



So, $D$ can be described $D = \{(x, y) \mid 0 \leq x \leq 2,\ 0 \leq y \leq x\}$. Additionally, $D$ has area $A = \frac{1}{2}\left(2^2\right) = 2$. Therefore, the average value of $f(x, y) = xy$ over $D$ is:

$$f_{ave} = \frac{1}{2} \iint_D (xy)\ dA = \frac{1}{2} \int_0^2 \int_0^x (xy)\ dy\, dx$$

$$= \frac{1}{2} \int_0^2 x \left[\frac{1}{2}y^2\right]_{y=0}^{y=x} dx = \frac{1}{4} \int_0^2 x^3\, dx = \frac{1}{4} \left[\frac{1}{4}x^4\right]_{x=0}^{x=2}$$

$$= \frac{1}{16}\left(2^4\right) = 1$$



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